

The statement ledger

Companion volume: every published statement, its commitment, and its zero-knowledge proof

Aristotle (Harmonic)

Artifacts generated 2026-08-25T09:46:44Z

This volume accompanies *Decomposition and framing of F-bundles, machine-checked: a Lean 4 formalization published with its proofs withheld*. It reproduces the published statement ledger in full: for each of the 692 declarations of the library, its statement in clear (the Lean type, printed with full names and universes), the SHA-256 of that printed text, the SHA-256 of the kernel serialization of the type, the Pedersen commitment to the SHA-256 digest of the *withheld* kernel proof term, and the complete Fiat-Shamir compiled Schnorr transcript proving knowledge of the opening of that commitment, bound to the statement.

No proof term, and no fragment of one, appears in this volume. What is recorded of a proof term is its size in bytes, its axiom closure, the verdict of the kernel replay, and a commitment to its digest. Together the withheld terms are 159 288 447 bytes (159.3 MB).

The verification recipe — the hash encoding, the Fiat-Shamir context, the challenge and the verification equation $g^z h^w = t C^c \bmod p$ — is in §6.1 of the main document, and the group parameters are in §6.2 there. The leaves below fold to the ledger Merkle root `5d6125712e3e7c4a499dd11ad876f34f760c7eae34a4029325442de0cd27eb3c`, signed under the issuer key `6f996285844038e962d600147c4faa74f6b7eb30b490865a72433dea415322ae`.

Entries are grouped by module and ordered by name within a module.

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1 RequestProject.BlockPartition

FBundlePoint.IsBlockDiagOn definition, BlockPartition:50

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {ι : Type u_2} → {κ : Type u_3} → (ι → κ) → Matrix.{u_2, u_2, u_1} ι ι R → Prop
```

SHA-256 of the statement above

931a62c08cb16bb139e02c785586b47934e4620a85ecfda7d748849249a1eef9

SHA-256 of the kernel type

ab6c71b3c6a88b9d8b76ba93103370b9514d5e6f2fc98c750179c32e7286196a
commitment C

1240f4d5db058821a73858d5a0332c9f89eb46b6e544e4585e8c6b290fb6c0c9
f4b5ace063eeae1d1f18dbacf273eaa69939d036915c35cc379443e709983770
59ac96f54fcc8cf8cbbbf557b7ec2ac5e064181b252a97dc337a8d88eb21d
c5ce2986f8d947770c7bbda2446614cb854694c3927e42c72951632749c700b7
372896ef76dfc5cc7545e34ed0b18e3e96dab1cc711b7f3eb3a96257c729c73c
fd1f210ca93161a31c0e51d9bb5822f9d4437d6318d2f2a7188aad1a6e731eec
e54fb20077bc741b4ca9d60d6ed7b7deb32c0f08dafd15e5fd0ddd6166479c15
8458cf8673cd07197a15cd75eced6b4ea65d0d7456ef06d2b003854a2b3204d5
first message t

dfe1866f816dc44e1b3b770ddd3244d0d36630d68a1a72b7f72ade8e2eef709
dc6805c17c92395eed5a05cf7428ac0e8279060cb56b6d4cc408a69f90eda850
49f4eed3f1c33c0822be28ba1bb16ae525ae977a64c5c6d1eec7e16c45f5fd0c
e37c94a0937d5ad9a94197296141492e6ff3490a506ff79d8f4487f705dd6e9c
44ed57ca60aef2c849a3ecfec920636e55b7dc78c0f2e78de08dee5c566e337c
b58d16446748deace616eb2758a66da5890d558f18b465cbebc83b719b7bb43a
079a048e899020879aaf3679b75edce81a004b49ec0df47759fcb0685bd2a0dc
3aa52311524ca5e1be70a795584390cb3f6f0e4414edefc7c191a58398c7a57d
response z

68dd2998f986a65bbdab2b43ed4883b3c01011abb31f1ee1ac5e202076b10604
61db22f4aeeded00ba4c6f264cfcc7612dfe5ff17088efd75c5d91cc0a42eed3
dd3a80dc7667a382e8e6c7bc038dd9660c2fee3a0bc5dde937c2ec81a682f901
bfec71e3ae1538a573ac544eb144ede770e273be17c6a44865113698027a5c66
9775bfec5f3990de0d8588ce74f20daa886d9fcf9c91e01adcb1474d1070ed2
0969908f91405adfb20311e9e379889bc886ee49afe2cc26217df3a897633540
93108c07dd3ecd59c5ca03147af3d933eaa27a8e2241e593c3342af116425574
46b7aafb31a1bfb1f593e517d8cebe7c4e10e433c96a77ea6e854b2e5490043e
response w

6a488a2357ce6342cd5259893f4946aab06c06d4ab842a221efee3e63209dcf1
219d6ac2aea358ebcd8af177a3601c8ba3566529ec2b0bf1c8b8d4bcbeb3753a
5e358754270cc09db79c9a1e252f426c5a4bf99f8cf465128b8bee5c614bbd4c
e55e410e4184e6d9623dbbdcf721c0dd7006abdba156aba7d60ada41f323f831
35b6b211b99b4d559ceec7b29bd1b18b1c64e30ea46c5d0754a834e56439643
05293a86d2db262a2315b82ec18089bb47d705ab46b4dd7acd587c4edbedeb27
b6196d576c9254c61cfe14ac7bfcalf1241e270e7197a6b8cab8264d1d1884cd6
c59968fbf952cdc4dc4b1e78f0fd64b0a9a19f376a179f4cf1bc91b31fbc39d6
Fiat-Shamir context

2b4830a9170eeca84d11d7356334e682dbd3ee93d632c2b1a08c343987412b5d

withheld proof term: 699 bytes • axioms: • kernel replay: yes

FBundlePoint.IsBlockDiagOn.add theorem, BlockPartition:117

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{l : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} \{\text{part} : l \rightarrow \kappa\}$
 $\{X \ Y : \text{Matrix}\{u_2, u_2, u_1\} l R\},$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part } X \rightarrow$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part } Y \rightarrow$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part}$
 $(\text{HAdd.hAdd}\{ \max u_1 u_2, \max u_1 u_2, \max u_1 u_2 \} X Y)$

SHA-256 of the statement above

df90c9117cabf53acc498aa4b8ee8c461fc8dba0af804ca927784eae3466f27a

SHA-256 of the kernel type

4f41f2b9243627660335e33a9497b293997619258d8095f64b4da14144bf369d
commitment C

da448f17ec8ce0eb3e6d84fa6531f4aa534fc3fd6499381913990c77b91c8f1e
0b8c7007d6672ba325d82da062c0a39f9da5e7c760808fb319ebfa4c7e5805a0
919966efe5ef3b3b48c0d12d5a811652162d7b09d29bedee4d572bd90cc841db
6eae969fd6e6aba691aba26cda868c9cb1ffb3a532aceee675132dcb2ac4f29c
aae77d523ae7e8314d850b3cb1096abb897998c563a5c0edfe29ddeda50bed2
51583ae242b1bd092c14c4fd743261173141bfdd5df2133408d181ff379ab6eb
a043386cd3e1ca7f0c2a50c6a951cfac28454da2fff586186151a27bee28d8d4
c626def9c36341e5182817c4c7be7c5ed9466a63bc19aa2fcf452b27995988b
first message t

f5e46b632c72c825d6017369acfe633e91d552afa51af653cd123ea252c802ce
a6b5182daef21611dd400e2d02cb74e319f940e2dca243b0f9bbb5aef2dceb6e
1937f9312654b894156bd991d31d63c5601147e16de6badee70e05f9bb0ef405
1409be7a74992a2952fe698be3782bab3a646a38a260970926787b7ad03973fa
8d7b45b2b3c8656134f88f3d25d4c8a79b81b82eec841d9f8fc955efb3e06e59
1696567e7145992a39c0188496dc5e2b78b668ee55fe7befac5cd5c59fa0e3d0
ce36837c8341cd9bceab495a6123aed2d4568682e1fc141ded11919521de63e3
36c80dd0904341438a7868df40ac3aa790938aa7c23c60cc52b58cbd309b191a
response z

39bb54a7ef5e6846ad32306f3ed8f6ef58eeab89b37ed0c47ef5d51a11cf3972
46fad7592fe4970946121998ace4d9ba8de99cdc338248fcc9563a567035bab4
cf5af1e7c00fe4782ce3899be4dc502b08c323951c64f8797149dd4e1e887fb2
8cf8b3789e0603a9b73f700c13e060905b9d9c3c49adb0f6168f0f3f70f8b0b2
c7c274ee3e98cbcd5cda8943e333b0b58e6a0ef8253eff327ba8ae2fad2f6df
282a89acdafbc538d46e4b1b7cf197794e5afedb8f14416e97f7e33e82d8fc77
95a3b88d029f66312e87da1005b27bbb6ff157f8f963a05e63acc293f3c96fb4
0fa19417f6536005ff2727e9d29ba4b0e3febaf28f03dfe64ee1ec9b8f00156
response w

8d4fad0db3b4e1a11e38fdc11924e1c1d886f3f2e3c8d2ca55e259c69e75bdc1
ba8ed872d390142bf71690a8d52403fae7f86a85cd2447be96ef2e8e2bdb6cd7
10020bf5ed4414a23c70c1ce565a54b1d5d178c4c8ee06e46312b10e17adab9b
b0826058806370deaff9f19ba4ce30ae60a69a723dea652b29c28580a54d206a
4552b87bd233486e756a677a6321e58a1d115234d7f961187d542a60079d296f
b1a4a5580f044a7c2904ef3e224af825ff16342b108a0dc468d3208387b78460
1488b086297cf45358d49052db05f21e58781683f20530f2f92bf774c1ef5477
6a2bfabb1b17e44fa247338735ad59187cbd76fe720a9bcf669b3cc25a067f0

Fiat-Shamir context

5185f2e9fd2cd33805169da6cfc082a9e5b58b694f57b94b3dd14a6c65bbb0a0

withheld proof term: 11770 bytes • axioms: propext • kernel replay: yes

FBundlePoint.IsBlockDiagOn.mul theorem, BlockPartition:246

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{\iota : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} [\text{inst}_1 : \text{Fintype}\{u_2\} \iota]$
 $\{\text{part} : \iota \rightarrow \kappa\} \{X Y : \text{Matrix}\{u_2, u_2, u_1\} \iota \iota R\},$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{ part } X \rightarrow$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{ part } Y \rightarrow$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{ part}$
 $(\text{HMul.hMul}\{ \max u_1 u_2, \max u_1 u_2, \max u_1 u_2 \} X Y)$

SHA-256 of the statement above

1154fbabe41376c8df6574421a794cce1d1a0dbd5c6eafd82f794e8765532a07

SHA-256 of the kernel type

69b6c35f60598e31185002de07027d0c7c5c18f291cbb1faa59019551f812c7d

commitment C

d07ae49a8c1cd562426daa7d909c9001a6ad871e162807dfa5f2341f7dabf4f1

796846c20b7c834bc4d5a70b5f15bfa79265d69758fec24673ce830ef39d0396

6653f6fb12f392760c1d61015c2451b5af31c06274d24b27a70f02ee4ac14222

ea323bbb35dcef993ab0949dc4886b53cb68f2e458462f8689acd6dc2ca45ed1

b220087d466150ffd2da5465716dcd97e39208595f76b7e169480e3856ff4251

20ff85cedf0ddd000dc9bd6b2fd9d25b1203ab6653b63dc4f3c2ab175a9c0288

3bf74660df0d4da804e0e38eba743e12b52ecc7815b72db53261a3ca70d3c33c

1db4aa68b43ad2ff12c4fb95d0865a31dc7f3474f028a0377ed97a79995f7b94

first message t

6d14d5994f955ca14d83221abfb52a58e1a6a8d24e49dd70259ca15521f61b6e

a981d358fd49e67fad89b3426ec4635ef3ad9e32531c56c48d43fdb1f5cf2f09

686be8cddb5be888251c8f0ba631fad36c32ba698e685e41fef99026dcd314a

5657b6d54c9bf242426c6ff73f9dfafd03d0fff02f6d3dcc6b2684eeb891aef6

bd4e70ae59df95e44df73d79c6787272d43c74aa94633eabce4c8c846a6b91b1

470824fab643aeae40b9d47e2f756bdbbf8df4cbbdb550271c26a1f8b580f867

82b5f6592bf0a911b8ea16a57477fdded9901ca0302a8e0df55ef8825c9ff5a03

0f8a27304af268884b1f91ce71e0ecf6569342292a92cc776f3e248d957e0c4d

response z

15d711c104dd0a0bc78b01df25e6d7760c81f932c256cd73e6954769e2db973e

5cfa930fc111c8a40b427c31edca6fe29c11217b8c2d9daf6e514bba5fddccc9

b1f71473e65cc39fccac984243b2eb90d91b22a55d719b79fd1d365cb32dee1b

8e1fa04c24994d8f032ccd364ce54ea23f1ea41f2291f497643ecc3049539600

0a614cc75f14d09f937b0ff2c658f5330f10b85caa730239cab28bac7094b286

2570fc34fd6ea27c5bbe369dc21fdc3f1f917f45c06b821a51cbd2080500e29

26ae516c19af3c1f96ba4cae716e073a47cf5012922e35d7c9782d8de6d301b6

4b88d52387ad810b129cc9f21e87d7bbb4cf2f7bbe2a0db5d8af7f5d36fc966b

response w

73fbe49d7d97078a65832163a0f8fe2d3ca6872133c540ea32699f97752784b5

99200e87655c77523e89c1eac4d404240f36c11be58d1e05643d4442ba5202b6

fa19765049925fc250064a78392cd34905fb29aacd3112c07ccc79bafee5cdca

6426d732f464d852e5ce63247aea700cc815db999269e5ce0a64145ef30ad229

ad0bae92e14ab138265c667ab2185d6c70874f949f9319baef5fb2191489c85c

9c828d12dd941f360f08dfc36d6fe667ac2512a0a2339d5eac2d59d5802e93c2

d7c4f02f34a41e6b233567e1b2583e86a709be50e9b386a42cbf6dd085f8723b

bb5d49aa023e55f8a7f9c1086937892e456a5f3b1eac33943a3ad1a5b868a92a

Fiat-Shamir context

a88a4e0038a7d13e339c39a0c5308a71bd61db9fd76086f64cb9aeba8faf4ff6

withheld proof term: 38415 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.IsBlockDiagOn.neg theorem, BlockPartition:123

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{l : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} \{\text{part} : l \rightarrow \kappa\}$
 $\{X : \text{Matrix}\{u_2, u_2, u_1\} l l R\},$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part } X \rightarrow$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part } (\text{Neg.neg}\{\max u_1 u_2\} X)$

SHA-256 of the statement above

355be95fad87dbb313372526c8837268bd67d6a1f6d0a6a4f6ca5ef649460936

SHA-256 of the kernel type

d8869df7d7390f3ecb2414bdf78b35ac4dba9d7a1538327a073fec78ec0c867a
commitment C

a274867f19aef086e36286a6cc94b260ebaf03e44a881fdda3fabb64fa73d6db
1a13a801bf09dcfb671a339380a2c6873a02397dc7061e643715c65e2c42d352
1de950d264d875d57206ba59bf616b7c150da085ce5532639f6162e4b9a5a960
5c81afc3b6aa564958ea5051eb032a5d1cd5750a7f94b05a356b7ecc5c332ac4
3e629b972050f8beeadd4cbc0133206b3556bf4fbf062f4ab74ed97ec9637fe5
f814c77bf9b096feb57af86bcda4cee9a814706d811fd4232e7bcfd1e1b9dfc0
6922357387c359e2695e731b6028899d4076ece3d44355a5da141fc89da99405
d8d06109a81d26299ea0b6cc43ebb79adb18ff5c82a0815365f55521524e89a
first message t

3770570eaa74db162eae8046f39394102cf13e6ca2f52d5e3382ca34312abce
0fc147cb4dc7388d72aa8dcd6e4f5d829b47be89f87668df7996daee51c02ff2
4d2c48ca60a7720211d5e459d7d7b22c3fb3a8a05410ccb5f7e4725bf3318e24
e4e7dac2c5b5656f73db4d62178a2acaae997aaba5bb57940573729506f33c6
157f46fd8b79fda9290d5718d580d354a4a4738f219435459e464b88842c568
dfe71262cc5fbf0169c15b79a20be6b2f3c403712fc60013e2c08d643243e1b3
8ff9c59fd6d7a9f4213061afbc71b4de3a03a9d50f033243cdc30d5736e3f64f
799ce5bd664c7d0d4e707d900b2ca9ad67856eab91eff0ccba9bc96f1e14c9cf
response z

151f1dbc7343c5ee8a85773cda51a4a1e29b6fc80e8459154482452d3e0459e8
3f4e4e7561f2a348a9e30a01da1d5ca815013c0fcd41d4caca20841317bae4f3
db4bfc8ac3af6abb50c863dba297fee643e1ab9eeae8c215b2bb36aef8bb12b
88909a2cdec5abe4f4b11455f9bfff218cee19d27053212b316df73bc185ea4a4
6c9bcef13603401b6b7e0e35b087c948a518c1b7bf91ad0b89d0e3758d0c3727
c5a02c2daca267dd649c9333d2fe4358b683b35ddec0b1f31f0479ddd5ea2d76
67fe9838a7da0721258a6e2883e49db08fe542b62f4b248f53fa8784ff8db311
c59dc7dace3bf165319a61bc21e9c2a57414cbfdac606dcb37f08ce9ee268417
response w

90c888a25eb0dbd95f1da77a18c15338c12acf201d9fd6d4771c1a0aec0a86a8
18c7600f5deed3b18ec3f82e4f6057384b66fb7fbf3f3e0e0c9f78836574d116
80c25106ee37f1542cb7cc69422bada1d48afc4a2b4a096872931a491b9ca7e6
cc06fd1e79429c9e0e1f5b3a078c70339acd7a90b6be26ccc9c01342a3931b79
8732f560022d9f0154a53a9b59bda06f166508c0525d9954eb28a9394d460858
bd21d3134f5fa286fd4e5479e635882acb2771e2c7266717550125e294d4a281
4490c2b5a7bc855eab6a451c0222a1752408fdb653babd6b2e87e151b1b7b34c
ee31d522b0a11172836c191bd42eacd12c4004205d2f44e39541ae907047d9e

Fiat-Shamir context

0ea8b88dc0d315925d8077542fb2eea3807c34cdcd54b268567a95b1bd36b982

withheld proof term: 8 242 bytes • axioms: propext • kernel replay: yes

FBundlePoint.IsBlockDiagOn.sub theorem, BlockPartition:97

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{\iota : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} \{\text{part} : \iota \rightarrow \kappa\}$
 $\{X Y : \text{Matrix}\{u_2, u_2, u_1\} \iota \iota R\},$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part } X \rightarrow$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part } Y \rightarrow$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part}$
 $(\text{HSub.hSub}\{ \max u_1 u_2, \max u_1 u_2, \max u_1 u_2 \} X Y)$

SHA-256 of the statement above

58504b09707aa716559791f2b76996a8f56d1af2948747c1ed33d62564cdb730

SHA-256 of the kernel type

119a9c216b4e4c715b7f197450251beebce1053fa6315d8590b23d2b1d64ce5fc
commitment C

b7cb9a724bd144c97abfe8d1a53cce3a5ded929c7a0f35ca70497b7539b9fed6
332242c423aaecbf188a24c11db7fd7bbe242d3ea64220cdf415e5f3118031e5
b37752d2f259583a5c5238ae9c82d233f6ec5e543e42c2b78d2966df06bd7117
d236707a32aae5d920d160e98233e4210ddf0498e8392165e9096c9a10a3c1d1
bdfb930643606464ebfac30e1c0394057c2d167d40d84fcbcb39e1cf840257874
13aa8b9f7ae231be24495edabbdf2d11af44dcc6dd37604fde2d47e200ea7194
ffc75d4e19a385ce8f785dbc10937595bdcf323eff32f9208bdc0ec93a1771b3
8d29faf2555cf673b6497cf7ea3a77abb52c0f62eb18818161be5431de1c3a02
first message t

ff5f6fc779e747568a2252b9b92c91a9567e5d7aa2001e3af5826c8e4e75d3fe
b64599f37ee97ddb53daee94b98a357573fb0f6c63d5471c1f7a98dfa56ccb
627a91d7e2db360a20bd4037f722fa22ed0df5a49dd926f133329b6de2f352d4
4fece62eb04c0a564ab47a4d21ecdac26b36ed964a8a226ba4010d6c4b26ff59
93b1bd7e013db3adeec352f49132af57aad3bec99d36ac0ded444fd28353d96a
ba3fb831885ab1b6f6459871479f27dba01e0217712736d3fe998e7b76b71d68
65d095c1ffe863914ad9c85e0e46cb57ebce3bf2bdc983cd75529b82c6b0739b
6d67cee9e71d186db19cf3484f760f1bfd65e7ffb74af23e21057996508c5918
response z

61ebc3945ecc2a9a2acf2e1eef8abf150598afcf1432be0aadb2ecdff81e87b2a
ae20f1386614c631cca8fd5d7ec21e722835a5ca20621a94b139403b02956b6a
4cc3d8dc5aa7aaca51e09f9a8219db9af7d2d172e37d1bd9134ffe69861076b2
c06098c1ff300d2d03dac2223062825fd092b1ed587687917cb33158ff62569b
827eddc260baf39008a76239b55b3a39934c50c0b3da6474af1b03be905a15a5
afb83a6471a6ebf4cc0b5dac41f6f6cedbc7bd49f767c9019ff21ddd1884777
c98709128c44304613b1dfe961963281f110fcf603be5fe60f71a4122aaf4797
006364173182e890e1577f7e6688f004d6a26b3626c5c5afa91128ad16dd102
response w

358271c71775f293cf041a1f37bc35b1a4fbd46365665d7010d7f079baba7c4
cc875817c58058c00246a2ba6f8ee9b2291450b953017fb7d9dc4f473ba5d581
750b932c570d865b9482206d122e51a84357deb45af53e6e8d9a638ce14dadcd5
7adc59339dda3e228b1a59fca11aa680f3c1725e42f400449e9d8bf317aa26a1
4128570bdd6af6cd7d623c304d5e5f3a135df93dae87741fb0158fd7789516b2
2d862be5ef52a50a8d339211dafda3be638f809131e54548eb8f319e0ee82c1b
02e73b8e21cf283905b1b243176985ff1259daf4649570cefbdb8b6f9ddcdedaf
3ebd0cf588206275cb0e26c717416b56c5a50f3b8740720d6908a5220a7e9877

Fiat-Shamir context

fea1ed727261cb8497787839b846e05754d7b3c7d701b160f7962e101bf4e587

withheld proof term: 10336 bytes • axioms: propext • kernel replay: yes

FBundlePoint.IsBlockDiagOn.sum theorem, BlockPartition:129

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{l : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} \{\text{part} : l \rightarrow \kappa\}$
 $\{\alpha : \text{Type } u_4\} \{s : \text{Finset}\{u_4\} \alpha\} \{X : \alpha \rightarrow \text{Matrix}\{u_2, u_2, u_1\} l l R\},$
 $(\forall (a : \alpha),$
 $\quad \text{Membership.mem}\{u_4, u_4\} s a \rightarrow \text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part } (X$
 $\quad a)) \rightarrow$
 $\text{FBundlePoint.IsBlockDiagOn}\{u_1, u_2, u_3\} \text{part } (\sum a \in s, X a)$

SHA-256 of the statement above

868fd4ea9ebf9b2e4e24311b777c0a502c07e52d4dafc33557e2f89b2cc18eb3

SHA-256 of the kernel type

9f0e3e78071bef9df095421bf2399ffd3ca5fefdfeddf983d59e59255b56517a
commitment C

915201869b43c18971a5b43f7acf8ddca9f148a6839e48257b6d5f26d98a9480
f60826398ed3e75091795bc5a97ec4270ec137e8b9b0906e130cea0537e65168
cf21361b5da204a9a8ea2525f7cdec1ce0b336b539b0dcec63fb628b70c2acf5
1a7f947fb668ff5999f81347724ddfb00844f379fcb3fe9909a237f19c7ecb11
f4947bbb5e20243489f2c5f6a322b3e773398f63b37fb94106aa9fb724b5efbb
a397ad141789243624e7bbe09ada31d892666b67bcb154acb2ee243dd75afd18
03ca9c93041e5860c12bbb6aa4a35827c8ea714caa0e46a4190d598003cb6637
3fedc44d57a9f8a15630a938cfb629311a1ab33c71c9f3e08d87649341088e21
first message t

54ed8d8d069c5ef801c1c7fe5abeac611abe2e3accad07cc6e1ea3d6157d5016
b302d7343f85dda0c73a9f19dd28025893d40300f8c27f85654963250a068cc7
ec24d3f28edbe7da2b23f08144c2fd23e3609ebd6fcd1b929d7da101a0577654
9deade0235f7f4e65dc80a35ea5894611e4cc51c0bf4a77a43d84693e8ec28b
13ad4f83cea37430220b02991d10789d22e4ade72a5e1d0284141e443f528262
0d75fc7fc63588b0d0ca6742e8e8e5779ea51d11fe679ad4f768ec1fba6f2c8
65c2cb040fcfb4757197397bb41f2df7e2b190b58f211b0b572c39be1e11fe4f
2ca50d7ce1040bf0c29edd80282c0af5585eb8d3f5b06a212fabd3af2b23ff44
response z

78ee64f4e18a4050c74218440ce38a47a6a886e72dd4ddbb2a94cce474849601
0d029b74fe48c8e2095214cc3fed7d8fb5414a0d0f280aa558aade4567a0e9ef
1dca2e2d5bf3c4dae9b846c041b125ed9e728c1b1d40bcdccadd6b059a04d252
135b01180d0ac80ac000b75e4488229600219c6d23ecb939a261c3f00be8f425
a34f521936078eb5ecd8027c3f74aa4ba1533d40f93366f2b72e73f4636eb26b
f53e67d4c14389132186c66da2cd3c7392273c572c749ff52b2f7c6018882d66
ff07bf5fb795def726efa011cfc04a0c94f06b1593bf23146a3156c9caaff5b9
bd63d1580a0dde735c27c97c719dd6bb7129976b783bdcc6e12bfd19d1c3a23e
response w

1d7f60fb5b36c255a538345cbb0dcac60d90a4f9572c6b4ee87a5f6265f60496
b5837de763a23e1c6723fbd13acb780b0ec23901b238701168f54ac55da11d7f
4e32492087f04832a00232a1d97b84fecfaae451437f0acf90c17e8703a3c6bb
f2497a1c1523c4a546c85998d36d86af9ba95d650f2e9141670fb0cb17ec7e3a
55267a6a9036ae97f341cd6ed40a93853a6a2bcdeaab214678e52b8ed1eddc1
1f12f12f090c06a7b928a5b2721111cfff9117d1dacda003a2c21357f7c5ba4a3
ca92ef8d2899aae38973222806a213d98408ce1670af3d64ec9649b94bd4957b
5b63857e757152a8be967d9e3d0ca2974e829c647cbe7835debcae2c8cd1ee46
Fiat-Shamir context

5d5cec550cf7d69227c688e5b82964602a5403d5cdb3d201ea282625156eff9d

withheld proof term: 12111 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.IsBlockOffDiagOn definition, BlockPartition:55

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {ι : Type u_2} → {κ : Type u_3} → (ι → κ) → Matrix.{u_2, u_2, u_1} ι ι R → Prop
```

SHA-256 of the statement above

931a62c08cb16bb139e02c785586b47934e4620a85ecfda7d748849249a1eef9

SHA-256 of the kernel type

d827e7a75fa31c68a68e08d4c46367aff2654487d375c118f44e3d047135e45b
commitment C

bf189c0213d6330b2e84df657a47ba6f8f0fc1b1d3f1fdc6295d187b282e90f1
da9ddf1b4e2cdee6284d33d28f66daef0aea61f0226169e83a51b284a32d4a0c
cdc9aaf8b30a2cad4dc39ed911849b917e6ad646c89b9283db240f66d4610c83
684f0edf3db1594a8e38882c68303fba059b182ce8a7f413bfe6ba5fbbfb23b
60455b08077e0662df8000889d49be26a699731a7073c9b896e62a677a884fc6
a4d31e9793875097a55a59e73dee9151b8b4fd694e3fcdcae7be2f62d963bd1d
d4592735039e2b1d3760adb79e4cfaa2e92190e70feb95dfab6cf7af6993b111
d3af087c2171411d474d93dd4bb145bc8ffffdea31ac8af807467cdaa707a9e2
first message t

e2d4a8a37f2dda2c00f5cc8e494d17a25a751e62d2cfb97f0dbb364217025c3d
bb4fff776c0f91c921165af85a435db4c9d0446758fe4b146f74324b6cc7f2f0
8a56eb3df7c9f1427d5f2c1def5740efdf028e9baae469051c779961ca944d7c
289cdccde06054c6dd387eb9d7df718c7b3e7f6fc4a676473aff4b5364d61813
d8af3efbe910b8a658766ef95fab2d76fd4fc7ab5a6d63244fb3bf4a05e97b5e
3b34ba10e8f8808cdf4e3e22ddfa6de0625e303446e86ec7b80618bb26c16f33
84f0b427fca81987caad5b0997d8f93f576305c9808b2e302ddba9f706e827e3
ee903589a38b44dbdb81b8aa4a8d7613c85ed1ffb93a006076bf128798749f86
response z

2ce193b8d6e12f118d2c1c079bb0b87a9404951c798008261f581f0f480b4ace
0ac365d750630cd59f41af41ef63e7616419d1c570af8484adc616b7c6f0143d
1c5b027c290fff22ea6a006f3490d8d032ef33a08caade0fbb9028ba8e32acc4
9656c9e5138e97acd764438d7ec2385e256db7c8e6adb27e8719200bd387913c
4f7f57d596aa43dd8edb8b157caf5fff9321808734bc6ed5b8ca0df78a9ea39f
a644f9380f51da4d785db615eaaae08ced6581425181b59ad86bbf4a210dd698
c85b6c11b7661d221168a8b744786d5b5ce24aa30c163be53b0445229aa88b1f
5a74b1ba50ea9bbb9b25410c73e4d8a41e6834d64b29d5d615095110cb0c15b2
response w

d7dbe9ddf8303a32208d9171244f618bf32279ef8b1dbcbe9f0156940be5b3b1
21df2ed0e0275fa47a1c42c094551d4674994533b61e2203c9b1f8b885e375be
1707d6e4fb1476623743c24e7a6ee9fed1ef6d1770abe899cedd2b4ddbe90025
2d92c3ce0e1098b418f9eeffcab4374be20eb4d96381dbdbc04da215ad133e0c
7f4fc09b8e5cd9b635f9dea70b272f42f8321cb04c5e56b092febd2407d577f8
2802b7cd1ac5a819ee8916d9e2511059f925db71371f4552d4b75133c69656a5
2de3882af4920ec82dd635af3edf678432ffc4faa7f4f0bb32f50c9cd2f39323
4970b74469a122b9361a4d36acbb01cb3084204e8c94a2d5cd541fc6d3a8b89

Fiat-Shamir context

04a88665ea14bc3d8643c3fd43f01b014aa090dd17daced38d95b32f1f3c37f8

withheld proof term: 697 bytes • axioms: • kernel replay: yes

FBundlePoint.IsBlockOffDiagOn.sub theorem, BlockPartition:103

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{l : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} \{\text{part} : l \rightarrow \kappa\}$
 $\{X \ Y : \text{Matrix}\{u_2, u_2, u_1\} \ l \ R\},$
 $\text{FBundlePoint.IsBlockOffDiagOn}\{u_1, u_2, u_3\} \text{ part } X \rightarrow$
 $\text{FBundlePoint.IsBlockOffDiagOn}\{u_1, u_2, u_3\} \text{ part } Y \rightarrow$
 $\text{FBundlePoint.IsBlockOffDiagOn}\{u_1, u_2, u_3\} \text{ part}$
 $(\text{HSub.hSub}\{ \max u_1 \ u_2, \max u_1 \ u_2, \max u_1 \ u_2\} X \ Y)$

SHA-256 of the statement above

8d0fd2340667fa9d86ac28c9d8315833e735dc6677a7ae4916b7b398446a5f68

SHA-256 of the kernel type

b1f24f3c6a78601c2de98e281fb8157da3fe39b04d3e260f212613a6853985fe
commitment C

d69474be5151300d7e5e87114cd46c1907ba0dae381a67fbb9a9eb29b612db1f
1ba51138c6bafddfed377d428fbb382e0daf350898f14db20625451f69a3e8
90d7f5202fb6cd1441ba3b1b46e60d58ff26364e58c13ada3ceea7afc9b199ea
3d8c725ded7092036bbdb2751fbf3d6c86f2add502cca8153b069be4e58ac9a
2befc856832905b08660c2918d4c5f0508275e28538d1215f03777835fba7ed2
4a554e433446d254362b4d89634dfadca5c25a86ad06207fab877bcc3d18bc37
dd125aab9c21147fc961bdcfbf2aec79aaa20c05bf2124d2e536fd86caef4a14
8b5723b32aa8feb743435a20b1c26d082706ae271b6c29e42ee93c3171ee98eb
first message t

af0fedc9e27380890712630cf1760e5767271d788701c87cdfc39de614bef31e
ce03c10c80eaeef3c0fd02da17d348a340a989ff462e34e425818c62f33d853cc
0556666c1237981eeb1ba63e08bf112e86ed9f3f3e7c6e88ed0f1c3922b850d3
9475402fb4bcfe556a02bc0dfa17bde70934d1f939221b0bd135b99e8c788cee
469d27721b41b3d05e5e969a438880018d674a3d784bea1d989ece9589a97add
3e69af339c35bda10f3c7c53c88592c351d4a04d1d6d31254817a714df8ac8f7
38b7cbca67bb8000864c3a8164a9e740520d8b592fd91311fe4e81bf1b1ae71a
37bca0b58e44a102182315063d30ba4303802f330401fc5b5629eb876568e5ad
response z

b691ec3bbbea18c60e2eea509137cff6bce114cf288ae32038cb743e79a890d4
1c46b7a49c4c590857f070c3dccc35c0872e173ab5ecadb7da0117f353036cc
249fb422bd11ec4eded4ab696b0561e785d528434be39d6ec91f675869e14dd8
cae4958e04e6dd8bfc07e89511cb55139c6d75e6beac4a2d08a77c7945f053c1
e801400af2424c2b693e16e74e9933bd4d50a4f150d6293d063add88c0b46e8e
6fe3073661542d2e986f30fb174629760e0c027efb78aafec74abd22fd567f64
24083362ed8cba461fe3dd2d78106bcac5bce43d556454ce07c55edcf9410be4
1510695a20fede6bcf3d310420edc328d00226d38244438df9f7247d53adbf1
response w

6c2625eeac3cb73bc9c69b700480d5cebd2a12af280a874eefa30a5b05dc810c
f7ef42ad18aac4a4577f31f5fe6c2ef874eb599d2b785d37935f9cfb3d1e0fc6
814c99cba41e2209fa5f2644dba882243b881237fa009488cb01b3bc146c460f
30af819fe703c93737e38ace453aaa9c1240efb8201a8793ca44fe1d4dc6e60f
a45b60054d486cbf3e023a34d6402e0db051c00c91251ffe47c20e5e5edef8c8
5f0adb5ba5d626d02965defe3d87b5f1283a1b450deb6309d67f53ff814c0983
d47dad0a3ab93129135a5c7c5e3ebc89b52dcdf4f49552e51b2d99628ae1edab
f83e381d6bd63071f539c017fc9f25ed1bb2ca4a4840bd0ddcb986339acb31c7

Fiat-Shamir context

c447ae1746f2aa7dc5e69b76e3e1b5f72eec2275dccc85fdd182f1b8b87eddd0

withheld proof term: 10342 bytes • axioms: propext • kernel replay: yes

FBundlePoint.blockOf definition, BlockPartition:60

```
{R : Type u_1} →
  {ι : Type u_2} →
    {κ : Type u_3} →
      (part : ι → κ) →
        Matrix.{u_2, u_2, u_1} ι ι R →
          (k l : κ) →
            Matrix.{u_2, u_2, u_1} (Subtype.{u_2 + 1} fun i => Eq.{u_3 + 1} (part i) k)
              (Subtype.{u_2 + 1} fun j => Eq.{u_3 + 1} (part j) l) R
```

SHA-256 of the statement above

1f894ff1f2744ecd623f75c73037e1db74c5caa2f2f2bae97df70ff0cf5e9241

SHA-256 of the kernel type

2598cadd7f9d08a98401cc808a1f7662d36dbddf8525af483ac6a9932811445f
commitment C

c83790204df51c995159daf2170d160ef4ad857530e90d1060c1fec726b65c6c
9ae11c30c6f33d13877c8f8f1a35f72efa4866d8dc4d31403cac5f29e3ce2573
23ccb736a82a650636a9ebef2ed13dc2f3b386800eb68e3bcc5862e2b73bc4a4
93a65888181c3f10bb90506f875c5935f45ad1a1a0f5314028ba168de8f98355
0ffabe4612739cde4494a942d836415f31d70cdfc71a69ab27b142e7ee701ab8
0e3720bd625549caec6e477e2c96e81735d6966330526111c6f64d1d9e2291df
12d96172994015a190d5b60aaa73ca169cac7de1a7b8512a2d84a4818c99e986
722fb751da78f517ed71ff744ba7941fa26903135cca5ed18c3ad4577fac444d
first message t

46904aa08c139d3379960b90c1c1eb85fec3d5e3790855c1b86b5fecb96b5865
59bf7b29c22b6deef0cd4a1a6825ec007659043f9d65ed6740ddf49842faaa48
da078ca23f2ac5e3f2b2a0c8e8f3e878372e8924f636adc9d997850694be7458
32b996616ca22fc9ecd1581ccb131134475ddb9ab7b1aeb94ec510ef3ea279a3
af846383d7f84d58fdd3bd4aafb47971bed31aa25b1aeec23b84b49332114947
998ca137ed18ee76bca93cbf573702405024c0dc84a44f4e2d65ce7ad00090ce
04b32bf3c69a49ebc352317cf21e809afaf6ef80d7dac2668f0280057fd66a1a
4d805d56641f6134a63a622deec6bdec93b0cfb2e1e3bc753d56652a5c0c8361
response z

74803630180696ec8accb9162bcaa39ddbba9d36d733b7b719a25ba61240572c
29c60a00a21237ca59cc8fbf2aafc99feb3468b93ed9b0a1d9a0a244895cf8a0
c6f991966b2f9f3469fb3114cf43e7bc1cac954a02a4efa9b690debc1d1573
f3a146f10cfcbb50d15eda15f9562f4af3e820fd42a264d243db465d0d6bb36b
f303c87acab4dd5623f59bd66c12733c185a5a948698d356571fe890dbea94db
e8ed85f7ead67588cd72624a3e46d5cc8618116bd8143ba746f97b7418577589
693f058425199d6189a53a77f3f687e65c07a779cba506a2032503506fac47d9
b9061b760b21c93e3017d8f5f6f7d0b3aa082cc2a0ec886c897d0fa062bfbfb4
response w

427ffdaa74a82b0ac77fde96dce30b596d3e065bd7a864a6fc8db36f8462775e
8868cb4e355683cd99f8f89811bde345a91c722fc3e82b467bf8267447ca6338
ef80c2f29781d81ec1c9e8b50cf9840706b4e1b335509ef743bd6f68bbe3f080
43dd88ce89342be7f9e9a6e7f9ddf2b802d85c2f67e59b0217f7b41f735b2d3d
d083d83d7adff1f55473887586054327694e5544aeb75d3dff3af98dcd473bf39
19a410354dfd49e89ea8ac779ab3e2dd849db2e3a394b47bdab82cfee1f774c4
49815850e4f143d9478402afbddae3fb78dd9a46da6e96dda1c25149fadcf1b
f1f08b974c2d39dbf0d7e001e40f074e88632eb9e4e4257f19bec70796fa5

Fiat-Shamir context

b82e81382334fd539d613aec9fe2810682f376dc7935a3faea47549d478bf31c

withheld proof term: 473 bytes • axioms: Quot.sound • kernel replay: yes

FBundlePoint.blockOf.eq_1 theorem, BlockPartition:0

$\forall \{R : \text{Type } u_1\} \{\iota : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} (\text{part} : \iota \rightarrow \kappa) (X : \text{Matrix}\{u_2, u_2, u_1\} \iota$
 $\iota R)$
 $(k \ 1 : \kappa),$
 $\text{Eq}\{(\max u_2 u_1) + 1\} (\text{FBundlePoint.blockOf}\{u_1, u_2, u_3\} \text{ part } X \ k \ 1)$
 $(\text{Matrix.submatrix}\{u_1, u_2, u_2, u_2, u_2\} X \text{ Subtype.val}\{u_2 + 1\} \text{ Subtype.val}\{u_2 +$
 $1\})$

SHA-256 of the statement above

34f484deaaf1ca2abc8ab431db1a01328816c65ea1700bce1caf1465ec4f3885

SHA-256 of the kernel type

a5441bd0220ee0600b4c8b39f4c6e750f640ab8335ed0c0c153e152534f3b4b4
commitment C

750d33ff52aba7ad1c150b0dd2733b37d1775cdf34259d27872fc2cbc15b43f9
096ccc1b8411793ba648e92f66116dc14d007b46c7ef4937da833fa57447a459
76e8bc9188bfd09ec68f51e32992461ad128b3f47636754d92eda721d8ab9093
859fbf2e2e0cfcff6a3279ae04888c2a8830492e3d2417739ab03bc218364c30
9538e1f9fce45f950b020a5272c64a5eb2c2a49cbb9a5e6ac371b74a89374516
4dfe50f53cf9f2fc8bdcf83571fe1ebbd9ea3f0dad51d51c5f6450c14c073f03
a3bdc669e4988b8454b8cc3cb81b6a3de35c020459d2a198b5a10f311c9d4740
d770528b0b96d856f33e0266870008abd4d6b0e400be046539adfa0c32e3885
first message t

55d45aac626fc57e41f62af7614c70171a5084d699113bdd2f99f7b7591c99e9
8fdcaa3bba0d2c8dea92c296858a28c4db6b0a9294ec715bd947fb2cd3889a16
11c0f03b29bba198e343c71571f30bfa9d331318ed53ba061a39b79f0d5f6c3d
e9d89eb70066778e00a1ecb733580fb16c1db5a0f36c4fab6bb293e75bdd061
eb385294d867e7c8f444003734345acc4fd06ed94f3ac19a8b5d999ff8c655f7
adeb3d8fd18a6f0d7686487a910fc42c4c25f6df7dbc54b2dd2384b83a30e689
54bf3287ba711a168ac99b4660a118651c15301ad2e48046886a1702c4aa0438
95b6f19f3238d04d5a7d6b51447f72421481e66e72ca51789b3bc7e99bfb22f2
response z

2aebc3b7563fabff71c8b293bac94d833d175a9cdc6ead4b16bd1c8ce9dcedd
0f3007147fba6f7f5949e6d397895a04476758d22d0000718a99a12b1c063d0d
1697a12a2445da5429f623020b706808e64fe61341319b50a74c100c243eece9
dbbca8b058ce9f8bd930deb5433942c3b0ac1c91f065632b9d8139941e7f843a
04f440fca46aa30d967856f408bcb7576b9fc30fe219494c09e156b413e84729
f0e2f635640cc082c53f7f414d8c7afe67d31f0085ca1198c9bbd9dc7f362ab3
ea9e49288a662604c0fb4d91c09ede3892b3d7b6e7c00be7d5b3b9a21944e5fe
beb0b71e90cf7ae16ea811369caf5b5b2622608b903d528d908afff43c78f0fb
response w

5d188ed1669deedba6433daf3ecb03bf484aecc17f9f6892069c06813d474e9
6cc6a33f312773863ff855df4afffd80ec1cdd33eef81113d8a9dab0d5f2fcd
de07878ef302cb5934c69495fc70f715a7d60fcdeeb1e0ce1a875b6bf9aeb506
d7a9223f52be717930f80bc8e095f90fe192f521c541ced172ef738f409e0bb6
2d14f25c4fd7836e1a4e3d8b3fa41bb5c322e27a3dde2a2e731328de24c9cf9e
95a565fbd9644dfdd7505ef7d737f573a0df0a84e6242c0a642ac678c706b05f
87bffe1f73fb995144d31ee87840b112f7812ff3c7b769957d3fe183e1818e9
3ab2a8bea6f8232a41b93557fddb0e8dc89b85ee97edc62a50a7ab1416571a39
Fiat-Shamir context

efc9eec243b8a4c5f069c839e649f5e5f64be44039e5a3d3e5c752c1cbc1657d

withheld proof term: 386 bytes • axioms: Quot.sound • kernel replay: yes

FBundlePoint.blockOf_apply theorem, BlockPartition:66

```
∀ {R : Type u_1} {ι : Type u_2} {κ : Type u_3} (part : ι → κ) (X : Matrix.{u_2, u_2, u_1} ι R)
  (k l : κ) (i : Subtype.{u_2 + 1} fun i => Eq.{u_3 + 1} (part i) k)
  (j : Subtype.{u_2 + 1} fun j => Eq.{u_3 + 1} (part j) l),
  Eq.{u_1 + 1} (FBundlePoint.blockOf.{u_1, u_2, u_3} part X k l i j) (X ↑i ↑j)
```

SHA-256 of the statement above

2cd212cc8adad7b4686a5785e7c84b9dc5e421bfa08f8b1ed8bd4977de9a5468

SHA-256 of the kernel type

9f26cb0c298ce1d375e0e30ea640ef935b5a4dec24cafad4d90884b900f9f21a

commitment C

716be8c8f7db6359e2d8319ac2a2de7a064feaf3774abdba003a4bf8e6f8e153

dfdfa34c761e258113c904b9251a648bea04be7e40dc3440a728307aa7dc697c

08b86ece5a4898107d6096c9d9314487f30942aa6443044d95adbd2d9ef6453c

6d7a5b244337b46b4c30c9cfe39e75087c5b35a468c0f6c33d854ff95e55f11d

e6d1fea658867e567c74b4d391d48dc62227369cac3b658cefed1d70e62570eb

6ab5999f0880dbcfee5ddf9cc573e0ecac735d04b318003f53ecd66c6d96d0d5

5a62b061d4656805a00fb5746deae1d7dff52ed9b3514834ce357c721bb63729

fecb42dd56d74ca5521c7265e53971a01cee4333d52ccb117a0bb779e2374754

first message t

da716f5a4b15bb20f5d8ff33ddf86cc16d9d57c466ef592122d118b07ffe137

9fdfeee34c9d6fad68e587fe971ba3bc11c4cadd5031009e474425a403287037

2a8cfba25096a77aa5c1ea9a420796d5a3e5028aedb7e8f5bb5c0281a5f600da

60ea47dae1ff0a416185e9aae3cb887c7677311fc9d7fddda2883e804b409226

2a2661a709dc864d721f6a448b25b54280e8437a9e12a74fb1f7a3145e94131c

976d2804148ed4c9dcbeb38608913e628fc359dafe3f58e408bdebf71844a766

6735ffa2715c9a97b3d93493d06811766f41ae288645684642184d07b6f3cac6

7e8ee36892471af2b87ee5f998fc41866480e177a2e1439383b89c5b4fdbf216

response z

73ec690eb07ac24acaa71af61aa358ebeabe2fd4355c6e5eaea6db6709cc753f

89f52545b51de027e4b1b1a752844fd849d9b2ce692957dd154fd4010c5de669

505e31dac368f092c5e1116c4e412996041280d199355a3ebe2071afb6ea541

1cc5643f4c38e34c51c2858454fe3cf7f5127ac74be0e529115f23aebae68bd5

caa182242944d20c88dde79175a7337e1b928c8ddfe81900f2b268e9b6ab953b

ad5c3a4508dea51bd171e3ce5e27a5d561a75297f19055065fa3aa8dd0ba3488

38f8e21e3602c49bd96ab105dc48d5e9424ca1f69f40dd59d392b8c429d92bb3

393e82a9bd8f0dc665b596ceae40624ee17371da550910f3baf856ca33782cd

response w

524922d2a731219509c6cb4f38fc8bd919569a9cf605dbbdfac30365b1f7702

b7150bf90546c5684a4d331ec27bfc7b883b0e4b9e6bd7fa5071e97349928e56

b1276ffe79f3b0f7c5e20b19e9188a7d4b1c8a9cb57ebcd6aa116c0af302138

62f3bca1abd7fe9d287643ac2afb201d9ec993338612b28ecfa746355bf85cc7

8b34f223d3484b10f40a9a2d3e9330c3138dc6cc114a3a93ab4ce3878249114e

21e25d2a1b5baa543c9fbc3aaac3478a4b56a4f780ca407686b8969108f0b643

ec39aebae7423143d97d5ac46a9f32a7441f3054c69bd540553c27ed3b1f7b59

5c2c01cd96e2eae2a0644fc2d0b845d490dd17cdfbbe652ef6dfd3a44dd6f026

Fiat-Shamir context

7b9f4d72d8f5448b3291fa8b9badaa024af7207a53174d09320248bdd5065e4e

withheld proof term: 359 bytes • axioms: Quot.sound • kernel replay: yes

FBundlePoint.blockOf_commutator theorem, BlockPartition:194

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_3 + 1} κ] {part : ι → κ} {U : Matrix.{u_2, u_2, u_1} ι ι R},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →
  ∀ (X : Matrix.{u_2, u_2, u_1} ι ι R) (k l : κ),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.blockOf.{u_1, u_2, u_3} part
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X)
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U))
      k l)
    ((FBundlePoint.sylvesterMap.{u_1, u_2, u_2}
      (FBundlePoint.blockOf.{u_1, u_2, u_3} part U k k)
      (FBundlePoint.blockOf.{u_1, u_2, u_3} part U l l))
      (FBundlePoint.blockOf.{u_1, u_2, u_3} part X k l))
```

SHA-256 of the statement above

0bc41aca2f7802777510b5d78e865543b87a964b9231be4677d6e4328dca5ecf

SHA-256 of the kernel type

6b468d8d20b8cc82b676c1a38c1dd76e06caa03ce0256340d094b51cb9f62a30
commitment C

ebcf53857a95b1f1919c48bb20161c23fa324c858ef525c57dc1218c3ae7c113
74b3641eeb97e5e4394a6dc0e09fd59ae54937a8b480f74fe6fd159a753448d4
f1bb609e8bec360549d02e70bfc0826199cf468aabce39ebac51aff05bf3056
611f6f6ea020be0b5e1d00b52d11b1b7f252774f3d744f01a68852f53162c1a
95f3527a0f34521bee39362094018873c0d81e0f6db1eb3d5396ac9888287deb
c0e4ca21207b5ee519dfccfadf73f1360514b12c98d540a652bd1de81e0900da
77966b8bf0e85c7d2adb7acddbacc9e10596b0663005c5fd30a92191356e94
a132d5cd68ea214d7193d6df707611320a109608c3e9b0768b3338953064ea47
first message t

8d2bb19b6909de7e4d302c161ac081fc590273c59e1e54c24bd09753c81583d9
c95abc28d5b92715a4c618a09a5937a3b2c137371f8fcc4ab0f1adc475e76b78
208149f63858a4e09c7a86705d3eb44a521a11538a01266ae9f4fdd36772635c
7e74fb78903753cc4b3286c100f27d5e26ab9c107d79e7ab8f570b8e164bd494
440e2718fa91b81e567a0fcb16b9f14e254d4af7bea990e377259316104c45b5
4ca8ca559e8fd29169687490bc75ad5f3c3e6df31dfe59ecac35da719ff80386
ceb0ee24538ffb317832e009f025589f4ae971ef12e1a1945197cd240c545ccd
1476648e96a514e0b1d395f78cf5a1b44a6d396b8f032dc464fa8abe08c34470
response z

492c873df01a52344587c52ac05e8d5aaa7986848eaba7702864cfc6ac4385b4
a759ae87665f39280428d5f0a2fefe45e063fc1f1bd4f22169198126648588c9
7d6a0a41a0dd3079d8234bd497ac224edbf077608cf6165b1593945c6582daec
a9be6d76b74e102be57caf4c8c752ae4c1d06ed2e0d4810a435b67270772a41d
ab24bb81ba4590841fdc75e32f973b7ebaca62945ad161134d382f4d818c25e7
6d882c35917516cec635a94ff50e7d6bc68d05fb930d230bff092db2e00fe328
8eee1b6113762c24c678f425f095316540eafda15e021ecd9781c20e06150ab4
fd55fd0cda763af497265d43b489823582f266726109e984dfcb65ab38638857
response w

36f399866e75390c640cc4486c034df94ab51776d5a45880e082c93f3126bf37
68463db754abe8fc890cf0c8d1660fd0593962430e1a8b0aee08df67c7b3d0d1
e5c6120ccd602f2ebc3d8bfd3a127a78581ed973de14f0d8a3f4eba3880212d5
2a6a5a568f56be7e5cadf6a8dce1112e0090d7f79cbfb7941d54e42805a827d8
f69c848d208cba62f159ece233a4739a3983fbd3de56413c98510f6ced310124
256af076584e71a78088783e32f6a3417b5d4eca962ce45ffe0390860da94f16
788e92990ef8e93caa9e660d9bb11f077fae9ef9f537c7d9be8088f947fb9890
901daa039a8957921928687b18d05bd30e6d26654a05c5828a31ded9bbaac7f7
Fiat-Shamir context

1fc2646e897030701491923beb973f5e6ba30fad9d3ce84d8aaa71f5a5c43147

withheld proof term: 284513 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.blockOf_diagonal theorem, BlockPartition:238

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {ι : Type u_2} {κ : Type u_3}
  [inst_1 : DecidableEq.{u_2 + 1} ι] (part : ι → κ) (d : ι → R) (k : κ),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.blockOf.{u_1, u_2, u_3} part (Matrix.diagonal.{u_1, u_2} d) k k)
    (Matrix.diagonal.{u_1, u_2} fun x => d ↑x)
```

SHA-256 of the statement above

cf9a32dda72de06d153c031ae86360c3df1b9bbf949ed2586e71e743ae5eb6f4

SHA-256 of the kernel type

b2374fa5677f03dbe6f5f3dd882cb2a5b12bb76663b0d05d23b60c43c12d8cf8

commitment C

6de9fc76483be9816902d1bef94caf16ba7fdd690875ba25707c26918a1ee275
f12149034f556c2a6b610ecdcb5672809df8b08a68ffe365d26e8a2fcb894146
615e790cc61afb74340e03a8ef21d242dc57bfbdd27f697648bec0faae503cb
6e14e43f3a72be33ab06827becd035064513e3755b56031e82de5f53f12d9393
53ffe9910a8d07ba1f5204753c5e903d727d5ac94ebdb737dba369153cd07049
734aad88b7342c8366e9a3ca28886bbe732cdceb1ff1d141a292e62b40834fd7
e0e61b7ccc4a596653cdc8140a9bd1f4e9961a42c605f4f3febc52dfcdc96d92
dddd3fd8be915e4628836db9e373307ae052a3f46a4ee21f283d80528e767fd

first message t

e28107e8d10aea4f51f2b6ae0f7858c8b1a4cae7342d0a5216ec6710eb21a7e0
0c21acd84c816a6ae3d76b106868e25dbd6a8199b3955af7771e046dad8b8945
ab2cc962f24b2bf50fabe5be79972498fc32bd83ca455ae3a2dbbc6805b32a3e
6b8347f4d7dd802006c5c220d40c6d3302017c4dd4bba334e830b8e8e533f92d
bb43b1809d22ae9149450fcb48a8964dd94c5aaa60de8b5eb2743f5f351d05de
14467e18b14a1f1d14646910c5dd036c70697828fd289c2b26fb226498cc6e8f
5dc4192f4f765c9d487d6f9ecf139162cd2b7aa23e1b0bbb268b675fd92114f2
29026811f3047c7dff7ab5272c54245f49a033271c062d2af2b655d6801392c3

response z

664492b9ad820852b3a0de5cdabb87f885c18441445404868621c0dc9716c9514
ec5b8d68b2806c7800c3a2ece235fa73ca7f66c650749eae8a8729f5f584ffeb
08fb6f01fcf14998807f1ec87e5f5d5eda9eda76bb0d199372cb64c1c25d6c16
f04196a5294bf9e30a541e237b88e6e66c72aa9d6ae9b2b70c4b79509c28e31a
a19011a7cd49f45f420c2ebf3276b389f24aa125d676556f038772c22c449602
d46af10cfd3b18df32ab57d3a7e96b4a6c31510a9b69a6d904eaedb951f2b8de
aaaf378a8d210f5c35433c240eb872564e26d3efb779a254aab3eb5bd88dffb0
de2253566f2b953701491e23afa43b3dc20083b79bdeb6067fcea76adc0e95b

response w

710cc0f66db00d4241c38c419e22c7e9941e53b1a9e944174000ed1e0e53c56c
3996d54af62c18da191714fc1b898155c703497a08d279ce6e50d28cf6e2f22f
e5eab0015d550498ca6eaf2201e80fde60fc36f1ce30f17c3d97334e8bde4519
d80e7f82e07728b242e197514af4e7922ef55b4c96b36dc78337c2c76e634398
62a9be12a0cb29b69cc2d141bf2049711726bd734214f835ebdbff8ee95f2a0b
d8fb5af9837ccc4f5433115cdfd0fe48a74f763b97f5543a3851d93e124aff0b
315e834bba8bce29355275ac23b60a7d561333f8c41bd44ffb6d9a8bbb70dc0c
91b4ba9648d0800fd3cb4220c77e6ac7d299e75bb684e8fe50c397698b3a44dc

Fiat-Shamir context

d04a932c6dbf97057083628ac1b5a63c9cb2ace81b5bb7facff0c877b8f42043

withheld proof term: 13991 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.blockOf_mul_of_isBlockDiagOn_left theorem, BlockPartition:170

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_3 + 1} κ] {part : ι → κ} {U : Matrix.{u_2, u_2, u_1} ι ι R},  
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →  
  ∀ (X : Matrix.{u_2, u_2, u_1} ι ι R) (k l : κ),  
    Eq.{max (u_1 + 1) (u_2 + 1)}  
      (FBundlePoint.blockOf.{u_1, u_2, u_3} part  
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X) k l)  
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}  
        (FBundlePoint.blockOf.{u_1, u_2, u_3} part U k k)  
        (FBundlePoint.blockOf.{u_1, u_2, u_3} part X k l))
```

SHA-256 of the statement above

0793fa61e156c09a4e0553466074e0296f5cc97c81c359c314cd582a2a8aeefb

SHA-256 of the kernel type

dbb83c0c2ce793076501a0028bfb95d465cc137bf2af83d0d2de99e6ccfa7404
commitment C

d80fe8b50c47aba9367fb7e060e53a44b94d7891027877019de9b3c9080da958
cb16323d14c313022f086b905d0af68b4492e8e0f7ddf7ea3d501d7d03292151
9e6b5fce12ca69a440890b7c6504d72a2065247651d1deb02abede26a8fe9356
96626166f5fa3b1530f916516ef8dc76c58b5ad9a881d4ba4ef561c339edbd2a
10595baf628be131b422b73b4f01492725744932281928156972b585fe589456
bc7ae466eacfa131202057ae3292956d1100e14b69a9dd050867eecd3d246ada
bec69efba3a5b3abfbfa79387215d5de5e9eb9e8a3296f065c8a4f81b1cd2248
acc173593498a352c6b03d0ece913aafc7188fc53308c99c2404be7eb841e522
first message t

9b6e234ea72afe358ec306255fb36ac8a2940068208ed5ef78da838197ea49fb
40fe6a0a772cd34930323748d8546d3791d0a4be44559101dc206fb8c499ce75
4a1b68a66118621446cf3ee8344b0c14ea4e489ae54a68ad41bbdb9f2843fdea
96a3cb7a120f0c17a5291dc1f23a6004e8554f1332bd7dcd6fbfcdad07cbb65a8
a7d5b6f0dc7161cdd3fc64501db487f8d5182794829f80f27b02b489d66fbf62
59eb790342fc26edf492b93e876721af6724a9bc61e077d4e3c56efa0d6df9d4
6439b85ef2be16565e0bed861362e1cf6081ee2e86a2ad9c3aee3c46dda51640
5cf6beff0564ea89f176978848c7d97e8542b65b292eb8732ca521c5446c654f
response z

3fec524748ec04a93b659b88f2a7213a62aa29df39a788b1210c538584f6377d
00a6347a47d5aa5bbaba67d1f59b44e360a4179a9e1cbd02f088a397bdc06937
7f4be7db0b4540d223e9b1b5abae4fa408979dac25628fca1181b7d671dd58e
2c02ababf96ec75072f5240460d32362ca7f36fe899e54e1922022b5c170a5b8
453112fac032fbec5ef3c296419d4c26f24391ce98f4cb076bac3034d0383667
5c7614eb548c307612b547d0c171a188bab5ab3be156f1a35610a853f9dab3b3
2f2d1c2ed0ddbc09df1bb4a1ef6ee6431d3622398ced90400d46cdbb3ec7333e
f1efcd13998b1c90c31ce25b6dd623c030c7dfe8041b0a957d2370e68898f6ab
response w

1878d80ddc7f0a17924e7056e33d63e6045c9cfffaf4c645b2b050460ca714fee
003d50aaf85e1675b9fef1ea9b5a8f76bff10425a75201a502459a511b29fda6
92a4a9f4217cc0e22072a32ae3bfc8875ca455e9500f1a6a36e48aabc8e605fa
ec3fdc77d8c39994a392baaddf331212e3bc52e00454fd6f1cf178afe8e7b107
d7a7ffc3586aa5e88ff7de93ea39c6011eb842785cbbf80131a7d6a8c3f72e7b
2cf18f49bf04b9e3d6a7a1323bc34bb1b128d172f08008a19f8568f8be153ab8a
2ea2aeddac5613afcbe8ce1175c882aeb1d1a46f026facb3ddde854741b75cec
190550c9d1e4109b32c4bfc41038ac61aa4f3f80602d166787998b5db1f909bf

Fiat-Shamir context

6520699bd39688bb10d73764408e719db5e61dfcf2826112f4913d66a39e9e57

withheld proof term: 39352 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.blockOf_mul_of_isBlockDiagOn_right theorem, BlockPartition:182

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_3 + 1} κ] {part : ι → κ} {U : Matrix.{u_2, u_2, u_1} ι ι R},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →
  ∀ (X : Matrix.{u_2, u_2, u_1} ι ι R) (k l : κ),
    Eq.{max (u_1 + 1) (u_2 + 1)}
      (FBundlePoint.blockOf.{u_1, u_2, u_3} part
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U) k l)
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (FBundlePoint.blockOf.{u_1, u_2, u_3} part X k l)
        (FBundlePoint.blockOf.{u_1, u_2, u_3} part U l l))
```

SHA-256 of the statement above

364d1023a83e6cef061f2dda5c3a4f3ff2cebf7d43cdfd911ae3f8e20ed0d308

SHA-256 of the kernel type

8763fca63590de861309b8a158c191ec4029a90e197287b34181005750a4fa48
commitment C

7fb92d0b122b41ede4bdb6b3def4a6e289a759608a848304660d2ba81d6c1cca
fe0f635a6a29f928e1c7ee23ec966ca1c6e994d867e4973991766306172203b4
d931757d708d3904c02225c7fdd0821578310f61ec16d570ea13c9dcb2548a6e
38a0af3aa868f9e75621c667659cbf5742166e08381ac036684fffb353085d0f
14f57ecb760b1a2ec499a1766b4daf63d269b27617308a2dc915c93f51490706
4f72381bcd8a8a76b2d00246003ae0c314b0c899d498510d6741d08841a2a23c
e20465b0d48f1227ad3ed437672351cba9e0c71bf8650e6d1be957c0a036a912
d392c68a723e0304a1b71b6dd8b408e73db0c945fb04cea0afa00773b6078e91
first message t

95bde8ca0617c1bb554beea5afe85921302f48580d18581c73751b846694e360
8e0557336e6f0803f41b4741870dbb18f91b99362fd5871fa9c534cea2dea1e2
e1f2eaacb9e72c15326f2943179ba11bdfef5091c1fd83e605d65220d26e76f8
94fbd8defa077f82d5b88da98ce3bddc97c61d090e284c697efb0db7cbd74c5f
ec23b82a4375748bf047036f37cc9b4edc39e16cce2c9ff3bd12dbeafd5256e0
4aa46ddd05c47c0be789cfc2532de67964f4e38c4852c88f514ff7daf79360c1
bd681365abefb84b24c376bfb2ecbe55bd4588a6587878571ec187782d9cb92
c1ea72b405b1f17e40c12b7cd249ae70a5bffe9fda3d99a127b5d67daa8304

response z

27409cbe20ec1e5ebd97a299dc6816e4286e23aa6cbaacecab5e290ff03f19a8
7de378ad35aea884573353b36eb37a95df5be6d588a8fbb236081a17db3e736
3f1289b125f8f6a17022ff4492ddfcc0a770163221b002f2caf0d11e99f05a7c
d9ea2f831998c4329d2dcc2927bfa06766a43e11ba028e095c46cef1ee83e49c
89f3dac9d91cc84a29ac977608bacf38cb538fe1be081c11a3e72248cc7e67a2
83a3d845a3637beb7092c01ffbad7b14139205302fc59c4879f56831a6ada950
148f7886e76858f8be6e7260ab6cb0a9c7c235549963dde71f90f3677bb9c4b6
fc3d227fa66ba9960338de73108dbb21f5c55a17de94ad67bea83499c57f5e60

response w

4cde3c06f908b29658f9c5b24127896a7ec4ad7b147b831e795d644ddc3d41e2
c55b861ce57e4000267d245e91f7e4f221a79b3d7cc44494f67d5990faeef922
ffd2cd8e5d8abf2bdd724beba403319b0868a71946b2cde25a2e1b895b9e0da2
88b56993d39353c00ac99128d22f63d01edc69e23053a9374b47f2a2cf7f4fc5
d825b41aad9319bbaa60c061142a67df9ca0edcae99a001d2fc05d4a52f28e3
6c63e42c62720fcf5073791d379ed6668fea77b99fa7ed7acca2f608f03dc275
31162c45887753b22c1d0e631833cf33b780dd824e9c116c86ff6236d3353a50
6de0dd72476b8ca9cfc7e8636dd00582417699023ff26c8413f9e5d7e11d476c

Fiat-Shamir context

9e18eb43de7c70d0b2b7f78dd720400e4a34b3a2817ae847c9cbec25e956e908

withheld proof term: 39369 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_zero_of_isBlockDiagOn_of_isBlockOffDiagOn theorem, BlockPartition:109

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{\iota : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} \{\text{part} : \iota \rightarrow \kappa\}$
 $\{X : \text{Matrix}.\{u_2, u_2, u_1\} \iota \iota R\},$
FBundlePoint.IsBlockDiagOn. $\{u_1, u_2, u_3\}$ part X $\rightarrow$
FBundlePoint.IsBlockOffDiagOn. $\{u_1, u_2, u_3\}$ part X $\rightarrow \text{Eq}.\{\max (u_1 + 1) (u_2 + 1)\} X 0$

SHA-256 of the statement above

28fa68a6d98354e80dba34879b2d3583cff53d442a6bf19ded4a06ff6be1b4b0

SHA-256 of the kernel type

171396c7f5f4c2feb7bfcd4801c8144a6eb6db066e667d5c936e073c05c93b83

commitment C

ee026829688d727515e5476f3bc49d24702888cae8827f868315efd93f00d0fb
4fca8b20c5bc2657dfb36aa49b8595b5613b7e950255f65ac04a09bac337c62
8447c1ede61b2a218016cc81afb7fb6d97ea9be7eedf25fe6324577fd9fc1c90
7062e441c270dc13c7e48360c36caa792728602279c86138c93976ed12de173d
ff2fcfa767b9797e471465ede36c20e37c87ccf0dde744a3010e440f7aa19607
15f4748bf5a49e8a5104571d7e2d900009aadd53f8dcfce1520b31f573ce8695
3536ee5cd00a29802502184b2d662cbe20ced3ea58fc220f2c482bd823a28eb9
b9f04514cf102e921d0c4d4ba2ced745a2b7017ad8205e7e512799a125ec4fbb

first message t

4b9769181fe3421682d3a5a1c11fb7ca3bc4f5d08b75806116937d86cd1e3d90
6071e75eb3b70c0eece48212adc5abf44a4a9476b87a653a00dd2ce0455b8d8f
7243044a51f23bac42ad5d5288e17c4028881f2140c3c0b29f82d38e370108e4
5102cb3e80b8d50bdb5dd5aa8d250b89d838b88fd2cb1fa3fb24927b1df1f04a
3a86fe536389028d49cebd46e1e03b7cb117004ba81c740dfb3ba9a51fdb0ee
399d5066ea6361150534f4b9cda1340dfd5facd4086d7aed011b92b6e1299077
1b94f4bca8d385f0e33016863a40ff5dec558bdeed17b487f2f82d9b1cb5b0e2
e98a4b19cf93ac8512eff3fded1c5b8e826c648d916c61bde7e713e76b18048

response z

3c1ae06745d0025798a85758f075fa130f906a83555d2807418ed0a8218a23cd
6b75351b7f57abe0705dea10d13c8aba9db4b5451298f4117bda9bccdfc66347
b41cda91d8aaf21efbdae6017d6b0940b3f1a69fb04ce416386a361d70db7152
f3cd998a492bb7a10d0b66b7579f3d8b8546d7f9acefbcd17b0bd259cd8818da
93e3870037c4c8a254c8d3b8db48055b2ca42021f441fca85b1f81653bbf091
2319ab405ef7457ab673d42245f7c1202a7bc36695f05bcf3c2d7c2f5328fca7
a51a844d2eadcc44168221eb975fd8c3f83a9cbf51567e9f013fcb4a084c530e
8ff3a8bf2f55a4f63c6a0f7b4a6e18bbaf7b6d53ec5837b17971e45cae3a98fb

response w

3b83e332af192a4bc1e6d03aba35f12e69ab74564a96c714a2e5c566c7b76415
0aa873f6e7a177954fae77595208252e55f0ec1f2da82913aa364e6c593002de
d0aacf7e4151e9802a6486789129a89e8ae7265f15a37a3a4110e77559543590
84b9babd7def098a2612641ad0ecbc968ee4dc7d001da176c96ab8590ba468f3
2ad094499aebc22f5bcfce97130bfd4dba4d15b43a66645d2c528578eafe3962
8db6179e13f109c32c83346150c12f8972c9657e3bf2c2767c67732041473b6b
c4ecef88bd2925c6edf69e444362b3ae0bd167bb1bffd5561b073ec05165476
1ff6bf6aab8baa305679fb565606e5eeb434cc0a480d326e7efeae41ea8127a7

Fiat-Shamir context

ebdb20e2d2879475b3595be6732b2592ca39d678fa137be23b584e31074492d8

withheld proof term: 8902 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.existsUnique_commutator_isBlockOffDiagOn theorem, BlockPartition:314

```

∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{U : Matrix.{u_2, u_2, u_1} ι ι k},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →
  (∀ (k 1 : κ),
    Ne.{u_3 + 1} k 1 →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U k k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U 1 1)))
    →
  ∀ {C : Matrix.{u_2, u_2, u_1} ι ι k},
  FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part C →
    ExistsUnique.{max (u_1 + 1) (u_2 + 1)} fun X =>
      And (FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part X)
        (Eq.{max (u_1 + 1) (u_2 + 1)}
          (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
            (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X)
            (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U)))
        C)

```

SHA-256 of the statement above

243c5c213bdb3e4fa91467ffb72abf8fa655bcf6e4edbf5158c13ffff8a84f79

SHA-256 of the kernel type

739536fc8c2e6ebfc08a38898547089596933751d597536115c53866b76f6728

commitment C

10c36a19425d967027dcb9063ff171fbc4e3d49f8a038873ff4f0cc386a14d3f
 09aff0d642c562aa0d68848b84da6e6a5d62f1f24e533b3666d3661ea576887f
 032fd81ab6b4a900fbce9e42c8a7a2c3e7e69fc63cdcc352130a9a1d9a14c794
 5b8266b219347eaaa70d7debd5782b79c3867c97077b45cd3f1c1dff7f218d4e
 3c64aba2e191f1ff71041f07caa23aa8305127289bf6b375a9036d4156b7919e
 1bafb1d70353d78b17b6c4cb0034612402927380d1cbbef53cee6a08d26c727
 a37cc20c13eb55a96e3f0edbc1cc4b80556f0661b70716b65b6d78fc3faf7aab
 12048e4ef6c246ebbc4b6691b84d56627d41e41d4b577410322d72a988375216

first message t

ec9bb17aca0ddff20681f7778e7c33f12a40b927c7424e4ed6a817abd19d25dc4
 ebdf49bf1a69c34177b171d72295a70ffb5190c00bbabba72e6d05d5fedb96
 11c88afec26cbf3183920874d0c0725c044b095f8028077b37ce899bce514045
 0c8c5c92269195b3ee8e5a1068838fb4f58601cf2046be47c954355adebb3edb
 3704a7aed7c919c2bfa009581329cbd93922f26c8873f494f00fa7eaa686ca3
 81f7ab0efe3a58ea9da68be34320d84e886533f3bef4b1bff87a7bc0f8ff0431
 e0890cd40c4bd73fff44df5ee5c007a9c9400ee4c6097e432928e32cea5cd6dd
 78cba70565271d7efc567095c69ac73a29a70814f8a092d3b09bce5d5b414dce

response z

a2e16d214b7864f2fec1a5b261ba14260408ed21625dd9a0c4f223b49544c6bd
 85030999db1263907b01b917dff43ac27da0d79415c4b212606326e3f2abca05
 4224d3cbbcb24abddca2c8b269567e8e76aaca40388a2b91ee376102a7e9c3693
 3420445be36a550eaf30af430549188dbe9f5dbe5e698b0ed2160fc417bb1dd6
 b7eeca3b2a2af616100b5ad96ffc3a378b018578a338c13b304db9ea8007a0ba
 481db2e4923d0e9b4d12e691a800474e1129d00c7d89c07990f3885599d8e6df
 815f25c445a534b1aed9ede3f09e58b0a4bf955847a77e85dd4b4b6b0311fdab
 8a4d6ee03a22ba0e1ecec9ff8137ceb628290a462683d36b2c44176ca935b30

response w

1562486bb85ed950a9850468335b88d0a26d5d4cac47c00bb3137d3d28c9f887
 9965452ed8c4ead153e7e6bdb3c840046e870f8ecb4044ae70f64f46f6ba2f1d
 0fa6ddf8384b323582d1de7068841673ccdeab6eb561c24e719dd25ffde564aa
 e1a20e66b0c4167bf3ea5a97dcaf4e0ef7be7ca0f3bab06c62585e215c572a06
 5d9e5778b1f65a1543712aa2341c2b5ecc87a8660934878cd747be3b28a50c95
 e1725965081eedf0c3a0b45eea8b983620c0afdaa23f68fea80ef67dcd201a0
 463cf725657ec60408a27fc6b42399e9924c6fd58c4a53d9a95844532362dece
 385db028b175b3725739cf08dbf4ad60ef3ec563778d00763e5ed8de7574a98e

Fiat-Shamir context

c219d071a12ab149ba5940106cf4a72f470cad75f687f3f68548266724d3205d

withheld proof term: 833341 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_sub_commutator_isBlockDiagOn theorem, BlockPartition:362

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{U : Matrix.{u_2, u_2, u_1} ι ι κ},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →
  (∀ (k 1 : κ),
    Ne.{u_3 + 1} k 1 →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U k k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U 1 1)))
    →
  ∀ (V : Matrix.{u_2, u_2, u_1} ι ι κ),
  ∃ T,
    FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2} V
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} T U)
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U T)))
```

SHA-256 of the statement above

ce0017aa4979f62ed47060f4b816e735af78dea4e2fd27977979a48de3424172

SHA-256 of the kernel type

6e2385d219eab98d4a35c06c87610fe7d63e1827dbb601a2e9ebb3d2f8b2d3d8

commitment C

131b560d3902b8b5be5ae43ce87eb4568a02340fe6de21ef46ec4ae10dd26dd3
6771b052c4705464df2f84254fb5e1df347c75e0bd9375b1ba7205f3892ba509
eb97ffa73eeef6ee2768ed1cf4c04b342046ecd7ca0a15b927be8ec2b16c7b2f
8091f4f9993ac6d9b96f1e78716a75714416a25093801546314d3a3b65696583
6806e69f5bf3ba860afcd07dd3dea90ab46bb69cb3e22f1c60ab41ec8220d841
e93dc3083d1f26dfe9902b76479ed7018810daacd372f9bb8a390539525332b8
297e98dcbec82d1c2b66cdc1495cedf5a45f8b018f60ae0483f73eea9f546efa
fcf3cb185641fcee8c4a813236f5eaf26a51a73b945c2cba4b6cce1178526d1c

first message t

66955211f153ffb9c0db7f6550aeb399c67997e9431c955c32d8e7239aa6b3be
b3fce76992e1da4174925db33ef84d1f9a8d4da5029d2d04498a859cdb9b9988
f4f86665b5c6935f16871ad46c151d20c118262b1dc5808654b02a36d16b1eef
0675098f8ab048037da0dc047fcbd9fe46ab07fbbce113a14a09d31e888df129
08f8433c4cddb9e4749feff847b254045f2b72550b3e311cd8274d211cd8b056
7f6aea70fdacf5f8a267798ba7fb4a137e29a93fb1fafaab7baed53891e09c1c
383c52bb7b91672a24a4239b1a79ec02fc9e518a95282c50b9648fe86c80850d
614479d5c31a4be59f2e5803c06c9bb4920f0297f0ae118daa98dffe3659c6e4

response z

4465dac71e7b7a715b074e36a9b89a339c9c5edab0d3305b066f0af56a3c542
70f8d5e13241ab1c79305a24d9746c197c9ec91f3a5f4fc93a84f076ac33d59b
ddb04f944cee55ee94bcb6a746c0a5a0f8360b4f753a50f44492a5e623ca6add
d9f9b8f755114ba0a7a2d7e2f48b84a6a2514c357aaa5e3d747d1c4956629155
6b7c5ec43f7cf105fe0ef4fcc353eb52efcefc0e3f8286239243c4362ccf142d
900d72a48b6d9a09b831d37fccd0c7ad2db4294762af66d206a65a50dfceeb78
de31b02f4ec06fdb0ede192ba0bbec9eba0de864db57b79acbb33ab91f2a0228
76c93dc3d27547670478ddcfb50fa73e80794a211a8e32d9aa475deb7782667c

response w

503724b0482082a1614f855f5854a44bddf680f90bb2c6bcc38237bad7a5d274
215a187b2a5e07745870cc6685ba80e5da081904b434c0f947f72dc9690a6f5c
c3865bf6f62357a51817c78ca33957e3f60ae983ed211999e5ae9c8e5f8cbbcf
c74008d58a6bc2a273dc859a84bda717e22979a6f74957f03ebe2859f3d12879
b655df9c11614b3b800b5ea924c754ab7aa43a434b232ccec48c5ec06426efce
f6807d4dcacf2a9efe1ee8aa4378f452b48e734b459e57cf746aaf0f91bfa1e03
71a99781b27eab077a47ee43c4fe640870010d0c636b8c13635cd14ab29cd495
586e879b32b461a27c22d0a198fa5d71ac3be3204dfc5a526b621a0451a7d227

Fiat-Shamir context

0358153bed0f886530b6e1c828544d45be859713554379068882a45887c6d32f

withheld proof term: 321074 bytes • axioms: Classical.ch20.e, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiagOn_diagonal theorem, BlockPartition:230

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{\iota : \text{Type } u_2\} \{\kappa : \text{Type } u_3\}$
[inst_1 : DecidableEq.{u_2 + 1} ι] (part : $\iota \rightarrow \kappa$) (d : $\iota \rightarrow R$),
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (Matrix.diagonal.{u_1, u_2} d)

SHA-256 of the statement above

312758193c3793f9fa07d1fe238e14c21857269785b868c9eba0da993ecb4c8e

SHA-256 of the kernel type

be81a8b1d4d4e101adc47bb89dfada8f058a21ae8d606e20f4d52c890ff464e5
commitment C

174e68f0b851781706ded88300482d8a1cc147ea56e7f36e2bd0b09a50e9f87a
aee0ee4a284cf42bf12ec6abdfc3bdd69befb1b270af66cf64a4c3f6627e3ead
b4d2afeb424726b0bdd5f1b697d8914a1a9095dd9ec6c5443f2068029ec4020e
276823a5a481e21950eac54be81a18e9163b4337feba9b78066710aa4d5d3cbc
28cb6432680235e7a8312284cb67eeca7a8afdbae0b8e01fe9d696a9cea5b1b
d71da0d712f04b8fd709249cf590ff277b9c6bedcc0fb8df5930ef7cf4f33135
d4e177852b43baf7b681091032a652b1bab9969360d38b2548126cd20ddb527
7a46b9cd8fc5607e5c11ec48f2fa503d5db95d4d8e6e575b191364eb4c2cfb88
first message t

f7f1a6dfa03f449a1eb5381d08c97061b78b9413335d302b155c3aee30c4f887
8cd5e429f5704acc3feab0b1e653bd81e192293c6bc2888c06dbeecaecbc78ed
635a84d95484e2d198d86cbb2118d46862634cc122ac44af204685c650c45bbc
c3dafff06930cddbeeb3bf4cfd32b0f3175743adae14827bb35e567ed521bb49
6f0fef6ee8f223f1100e4a013991d9020ce074a9d0fb329ab0e7f6d1bd66bd67
3c9fbef68b1a64833b50a3db4ccf98f6e7c0718d79ebfc712eb25e8031a712f6
1cc30e30bdf24d3beee2478e6781f19a8c0dd5fdcaaae4ffac0e870dfdfb0dd
0f23ecd0f34166f583b87008d472619753770a80f43d5f90ee85d9090c86b60
response z

304c251466e1b67bdfb3bc0e83af19f0df9706ebad9e8c982b83a0e90274c94c
c2c53170824ebadf713610902db0992787573a7357d970bab02c6d7216cfd1a4
83f7c69ec20deb52a90a28c4d6a588e384a472b51311573e747aa9232cf71e9c
c4d44134148346d73173f79f7cef4dada97f1b58bdfad143deee5918a9e52615
6797254e962c5aa9ab013cac915f8b18a7a8f33e6afdcbbd3b8dda5f202bd631
d871db4ff220fbce67d4495b3b2c5423f1eb45d038a8d753b1fa472768e5d458
c9caaa80ee43ff010d3c4c89a3ccda388c2b529638c3235c383afcc76a5c7924
c6da4ec67e8e53516b38b930a4d2520a80c4a7b0adfeac539dd330c5b69124e1
response w

1afabdee9b42c5b98b579ddb8a47f79f6a5dbe6b4cf31e4e9333314eb8cc098e
84e0ef298eb0e38b888b1623a520d35c5876cb86d442f002dd5c808186f18447
1d9cd026e1b34917a9cbc23fd2b4d60c3d3eb471e35c0e34a905922f9f59387c
841d610dd34d761622602dc4c6d8af8e63f6822006d9f92c1d3f6f6414365855
837962bbae5ff81abfb3701f7006fd9cd50537ae426f999ff7069610032c8612
6ab4e5460ac6147dadebfd0fe068704d77e68c15fa22da5a55e84b8ed41c94f6
36e79c1d363bbcf962f2754dfe044224a6a1c8da6d5a17072e4ba65f8fd821a
110670095baa1f642c4d6b110b0d76485f146242acd85cfd9cdd2833440fabf3

Fiat-Shamir context

81d2dd3881e8fba54d48b8f870e79cf18cc4878bb83e3fbda0818e44a08a37c6

withheld proof term: 6727 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiagOn_iff_blockOf theorem, BlockPartition:71

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{l : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} (\text{part} : l \rightarrow \kappa)$
 $(X : \text{Matrix}\{u_2, u_2, u_1\} l l R),$
Iff (FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part X)
($\forall (k l : \kappa),$
Ne.{u_3 + 1} k l $\rightarrow$
Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.blockOf.{u_1, u_2, u_3} part X k l) 0)

SHA-256 of the statement above

a2402fbd9a40d35ce4fa5776b7959d79040574081f37700969b248037f52a37f

SHA-256 of the kernel type

dde31aee1967564de5f78dda982397f9cd208a530351f31113c1715c8ba7460c
commitment C

789040a395c33bccdbfaf097c3445832ac243f06cd7ee39c9d09db95bd4be6c4
4b2843bfba6641b773feba9bc00bd10ed09c7da3207f9eff32e1a668754fb34b
da3d4f07c3d830c43476762d8b74e0cd6e82fe045753997619b000e4a7f71545
7ff8be811bd283f80e8e21a8d7de5a00236a104ad6505fe3f38cb8148d5690a7
4d3b27edb000971386b53eb6b55dff47524170d8c6e3218f6c47c18bb79c5a9f
d25e8a24492acc07b1b8215dc632912b46a9055ce4e034ac11713398c847744b
93c540d31054876ea150b7f0977622aabc38bc75b46aa2275fd99ac1f62265b
3d69b5de538342064f664402a83592dd83a9294f88c9a9737d142d79ea4e5eca
first message t

cdbd3965bae0db8136eaa95545f95fa4b420cd1b8218d02ea68cbcd6af78dc7
7c32cd85d0d85630f9cef37ca6b0a2661286db5c5ad823dcf30fe5519f4d8c1b
53e941c89607973607f9b2514370719019de7cbf13c6d1f6de3aae31affe609d
c32930cbb16f6f406fd2cb54d1cc3ed523f8492892c38ae28fb2047222da3cf4
30f651a91f08758de643dcd285cc013fae720dfeba1c499b97e500e84207a127
0ca5882bf6d4dc1a3ee501b8419596bab59f38b720d7c1f1668e1187b5555bcb
27df512eb9d71e89dbeeb0f9f0427e1d6ac0b96e34591e068a4ebb0b84add80d
9e039d57de850e173f96a1f31e804ee7e183fdc9e6a4556ccf4826ee3f14eac5
response z

56204c91cec5bcd9fa04d6f806057b75f7fec94b4710d8d399e94999ce3e445
57c468a5f80bcf14d9d444b0e5a49d9ac43eabcf90bf67a95d860d0bb5603c98
1da23f24c82358134352f74b6ac13cc5dfb74fc24166cd29196a2bd4bf4833e4
974ff4a163904c68b549469a5df1a4045d415f353ba33c5d50808e6ca711d877
f5c9941558155711add2891e0632c3c9ef309906e2e4cc182cfef3cb9d4be02a
69e91876189c3bbaa70be92df1e00fdd0cac95f6df4ac23b46bbc5f42b820cbd
26bb2311a5f138e52bb8d228c9b0f7cb48b71e9aba82436f69fe96267be9964c
7c001a0f6e4f2e51195ed70269b8e4aab497105fdaeefc02556e8775f0ad0fdb5
response w

29d406d0849e227942d04a6fb36cba6fa16153e07d97d42ddeb3fe3fb0a09940
b9b247a67025d6777eb03fb7a636a7884a2ad832b5cbaf8a809bd4a7cc5360b3
fb28acc1f628e34165e48399f84bca46989b41f292dbae22613cc5674ae7c14
f2423d45e83298da29ebe24bc7ff09ff48ca80ee3b470f804535bfe3325500ea
4eb13b36ea989e7ec9250505cd22d283f960b9ab6247fbf184f784e992722052
12a4a6ce62ac600de8acfdbe434e2f976c2f56648993326d6c8609aec224b97
70a9573eb5e13076dc789a35852180f2bf76382eb5c73c3d3d0f092e39b32386
d50fe135c037dfa17eca88d4513396f8dec17cccbd9d68177839e581ed84cbd3
Fiat-Shamir context

5daf23340850071059c5af3516b2c4b4c10e39d2e99277b3ee77ea519c171afd

withheld proof term: 10 434 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiagOn_of_commutator_isBlockDiagOn theorem, BlockPartition:206

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}  
{U X : Matrix.{u_2, u_2, u_1} ι ι R},  
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →  
  (∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
      IsCoprime.{u_1}  
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U k k))  
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U l l)))  
    →  
      FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}  
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X)  
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U))) →  
    FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part X
```

SHA-256 of the statement above

5ba85481019d06a0363c9233d4edfc229a4dcb77162859042b4ad774b13c8b38

SHA-256 of the kernel type

d4e474c65143ecb69212cc5dd33c5794dee70436cb7ca99455438401c88f7a87

commitment C

8585ffe2c5d00cc5032638cfe5ef04c18bfb3be4348cbeef4a57d06569ed7113

123516b61dc67ccf2c9775da2120a89523da311bf7903db1bda5ae67e4463add

4c4d7c8118208aa75c0dbab26dd3469e39ead1063bf6a0ea16f10ffaa0067772

31327d2a9974393eb9233e29eb49362e375b5bf23ef28bfefb1c3575cc15bd34

983dd88a6bdebb80a60db69bba822dacb374536713a6db5f5425c68996c10629

ee95aacc4e3cd77669b4371997a91ca49cc9be683762f252485e1885c3211fdb

79e8e01c4170b1460237e42661efeb9aff68695487775e9c0890d9e9b1954c7f

cb45e40f539c3e18a1d6105fced37085ebc1a77e0609668a7d1191915342d386

first message t

10c877ac5d2f096ff59a07e919997bc8c45756886e4a96d392a33fe7e7f7eb24

a912bdfa750d4833ba244b1a504f6e2122d5c4f26e283119bc58869ecb22a7cb

052e05451a5f04d213a2a322e4bcd34456f4d13fa49a99c9fffb772bd1456a68f

35ad7dc199642d008b9e15c578a2b184870b8ff649938922dd68c455da237146

2162bf95d4a3075eaeaf7d614b1ecec4fef5e37f2dd6017070d29f81e55019444

e124bb60c292ff07a09f37f8510457b7c9ebd148a2bfd85d2c2a89a28c6c1bd2

79a7075cab2629afa2b71c3e90eed8883a220a0d990964614cbb4c7fe044db12

25e3a4d5c80f642633e5fa088b28b8962f09b9a9af17f09fffd8f181ba5cb9b6

response z

44e21110ca93b6e10ffe8b9f6fa91be6890adb61366bc810d40e0cf097c18e6

c95fa990ea1930784c2ee1959600bd8047591f5714ae3eb0789f30f734414ea6

bb929e247f17078d00aa5e2aaecf8af5838d2a5e17a08605b3f9b3b71ad07a75

559203de7e73d1bdc4018e8a2b213200a4828105cd677ad6911cf5b64ed08ca5

84cd77032ecd9278d7a54fa30344951ca9588f47650b52305bc06af6ce09904c

ea7b609812bc9d98f9becb921e2efb43deb8b4175293139a1aa8f1fb9b633e69

5609df2bb96fc2c6bb5406e45ad0b4e3acaaa9a3ff347ba6acdb8c17dd2e224f

c7111853ff8f69e54b5cb63ecf223b0fa7a3573977f3a1a31567ff77b1da7d67

response w

68bff8b0503bfdbcf6fd272535012c254489ec3742d2a2c6951b8a59849698045

79e2cfb06b65122406f2ee34ce4a4555848a5eef3f3e534cae677143d34423aa

87ee98eb4e911fb0e90aa005aefeb7bd2d0cd61761a9aaac846e396213d174da

ba75d3fc6611e386b4a571683445aa3c6bdc4a52b07bd6091d5173fba3d26d2e

f03f47d8603b298ca9f97c123cc7023758baae9a77135628ab5937269363c3d

6d8d1df535809b6a1344685e30c377ebb6092f29122838742f7d2bbf7ce5a61b

4e2f1d9907b3f0e5cdabc4c9ca4756ccf0eb37a6d234b53599f5f60efcc33fdae

ff84451c7a7a8fd1546c944b39ae45eaf7b77f511bbe8bb3d676a4dfa6c80961

Fiat-Shamir context

72e9a4d6251293055ad64aaa2c74421182accd89eaf6591544ec7eff8db560e2

withheld proof term: 223 446 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiagOn_of_commutator_of_disjoint_spectrum theorem, BlockPartition:297

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] [IsAlgClosed.{u_1} κ] {ι : Type u_2} {κ : Type u_3}
  [inst_2 : Fintype.{u_2} ι] [inst_3 : DecidableEq.{u_2 + 1} ι] [inst_4 : DecidableEq.{u_3
    + 1} κ]
  {part : ι → κ} {U X : Matrix.{u_2, u_2, u_1} ι ι κ},
  FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →
  (∀ (k l : κ),
    Ne.{u_3 + 1} k l →
    ∀ (x : κ),
      Membership.mem.{u_1, u_1}
        (spectrum.{u_1, max u_1 u_2} κ (FBundlePoint.blockOf.{u_1, u_2, u_3} part U
          k k))
        x →
        x ∉ spectrum.{u_1, max u_1 u_2} κ (FBundlePoint.blockOf.{u_1, u_2, u_3} part
          U l l)) →
    FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X)
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U)) →
    FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part X
```

SHA-256 of the statement above

5854d62da921112d6f1e1dd9d4328b9a351c59a16ab0983e3eaa78affb5a5235

SHA-256 of the kernel type

08fbc8ee7b6d59a032839ed75deb985216c9a5e34e632c49edef2c5e079654f0

commitment C

d6288f5469f4dd4c6bce230cf65d1a8a76165a8a58a434e02544bd8a72444f81
1749031bebd6018eac79d51f3c28f49081c416222e6d6d84d7f2f478bd7de602
e7f27508495d8d17f4850b461f567cf3b80fbbdd3025f59f664d348631c22dd0
3f304c007be3eb72fe17d5431203c5ac8773cd6839021cc1221bdf58f51f7260
78dde9c224088ae0dbe1161e250779ffa66bba53368fffc62a58d55410f7761
b6bff593cef93eec475f8816ab44e7e2a3935539b3313e08d07d6bfd1c1dc960
d780939368c297d4ef0a6a0bb483a2b69874093a1e58149d9665e4dd2d67b718
ac6a04c80e62e5799eb542856411a553322becf079d4b588703abb6db5ef7a1

first message t

c6838283cfbfd92f211a4a955a63803222a32fd18c5e37a1a80ed11cc9161399
c4b62255ab0849d0a8f4e48e256771133446de97dbaea4c5a43f3c10f93df401
aa785f8dbe0d9ca38b86a3fde119d2bcd4f1f32db3683e2fb7bf39dd3426337d
539e42a11d45f4903e981dc248d6148493e3988603d2bc1a296b58651024e50e
1706a4febbaa60342b75436af3f880cb865de4496f66377e006ceb5ef83035c8
9839fb77d60574329423a14d1a3cecd71b2377d2dd2ba8e482807a478e679ae
0a9c19eaa139bd3677a7d7cbfa6da4c7aa2a80161f96a5384b01782cbde961ba
f31f8a073eb6860c2e0b93ed9ca4a87fa9f6b62fc19b87d1f09e211f37d86933

response z

6156d307afc57c5daaae916405efc3e4310470a45c32953b1c9016106631e479
6b04b4c1a41f22cfb0ab85b8b13fd066e8af71d6445e7ec07c826fd49b46cee6
f403177f2f24323c76fcb9491e1054d69cd5f5adfc8be9fe79624bf9f39668
78af55105f66fcbb1f0fce653ab25b0f3b31fe68bdccd5cc16726a3c5ce15182
eba97207d61717525a99cc380def6cd7eaba4c38920616ca2889981693284bd3
26368cb0ba93c55c535b33fe31ced6f6badd5d1bd6b76c9a8a134964a22b56f
ddcc433ee488b28404a5bf35d52d6a29607f3f0c65923e5cddeb89159c99dc5e
e68d150ec0d97926111b40c5f5f206a4cff6d2a8035b9e897845e2ab79e79920

response w

205771b519a617c3ddd23ba20860ce70565f4aab921ced5243a91c3c2353041c
f023db2771c70d2c9a8248319057c21ed945ddf8def36f229d4f5e945439a662
a47c19f231754320706c898a631a5e3048a5f0de52388522370f442a7d27916c
aaaa4bea0e48efb9714873153aeef3f9e9aa772c62d29eb2c6169a7e7faaec2
e5b43d34faabebf4972259625cec53b2b772f649a8d9c1184369b7735a6d0911
0ae05665f17f88ca0d78cbc63dc009f6dc71a200da04cfc10cd184a23a8ab5c6
73c7cb5be4db07360c0a195ca551733fb59af2b67f81ade72fab5873ef4de0b3
aa8f7502873106bec76a75541a4434e975a17117b2546416a9c7a07a3ee7a6e5

Fiat-Shamir context

b1c1db714e4276bcf2c7b15ffedf92ade3b055402a74bda1565855a1d33e546f24

withheld proof term: 17 695 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiagOn_of_commute theorem, BlockPartition:220

```

∀ {R : Type u_1} [inst : CommRing.{u_1} R] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{U X : Matrix.{u_2, u_2, u_1} ι ι R},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →
  (∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U k k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U l l)))
    →
      Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X)
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U) →
        FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part X

```

SHA-256 of the statement above

cfa16ab5344c7876debd4df2fd73d0bc47d2e9b2afde465005c13f35b8ec68c6

SHA-256 of the kernel type

d6b98d2990cf7826be589ae4d41eef1d27f80d8de2e2e3e557106db8ef22fc36

commitment C

a0b0a2198e08049337f60d4a2ace5ad16b9c8163e68d3305fa70e3674878c614
 0213d06977f108594f9fc628c43943ab1d104f66324832db656dc3f8a7dec74f
 5fb5beb4157ef19c6816366a05ac43a2d04f72e078bd61286e87904ff83799aa
 60f40b583df863285bd9ad2268842251cf5e20908a043a644ba41007670d524b
 b7243b26fa2805e2fd1f3a05fe571ce2bd80042eb1785e91c599e077d6c6c4ed
 2439ead93bca393b55e5821b4adbc36791e76ce1b53a8f8dbf08230000290c28
 9e91370219bf0a0949f0c229cda819872344f669f499e02fac35d352352be1e7
 bb3290834a807d3aaa6a1af7ec2b849c92323d0436bae13e0ee8c8fad62502cc

first message t

ae83480f8b454ea34a8acd7bf22205a6b4775bec0cefd515c7c016be0dd72bdc
 11881626132901561cc7ca7931e1d5fd1781e341b34da24770ca8e0ece3fa8c6
 bd2335ee1cc95fa771d1eae6a6941f025c59b4409aaedde1e0d447014b908a9c
 97aaa116042e1e7956e31390e742dbd4811d4f448b33ad0475c1991209ee97c
 f719d5b3299c3d02166339db289c9eb3bff7c6b61340993e40e291a685556311
 1bf50938f0279a0e7b6eeba5317b5d3d190d517492ea535bbba0d668395bade4
 dd08a8afe5291d466bd2ebf0d17817713b458debe4113d5156d6d2126290c867
 c254a504acbd78c15fe38e4a64a55c390422016ea0e55fa3b7af6a994375309a

response z

db4c92ce7460ca6569fcfeb8a7ed9c9f7f4eeca15e5339d3df515ffdcf1e58d9
 e6c020303359d5668485651f88410830798edab305ea4e1a5110e5ec5d697930
 b78ed68f107f6aad1cb780a5a3972288f74f31b63e74bdc3b637824ee9744854
 de73006105d2dc5f5ba609f7da59610045e6b9b1f117f9fa787cbb84f0fc13bb
 118f171815f52b68e4b6f7999b7fdb1942fac141d016e5bd54e975437e4c9ad8
 962a2a93003f3542048c0c4cdf30e3663f30ee76e39ead4b3b53ce974a9b768a
 ea40da56e0e814b2af3cdbc6888ab68a030512f4552e339e04961454206d688f
 a40ed3a6874066ef43ccd6f284b6740d4e0b5028306d77f83da97f0dc252cf9

response w

4de280a60adfedec4b12afaede822bd6764cbd6e7e324286f24da0c43cbc0e80
 ac9fd84e270f9eba4a55ab3c11a80f78a81582c77493b153660122e2c8ed3637
 d48d054b89be95edb740c6ed1898cf791c99fb546589055935a770d886bffed7
 45ddc99b50340b43e647adf96c833a7f60787ff436c3b8c31acaef972ecd57ea
 d66a591daa68249688066ef71b02b0767916fddf2dcca87af195fe8d9f7fe945
 f33384b7515b706fc0fa2614ce66972cbb180815b7bcb041dac67d7088c81d8b
 7a108e5a4e8f2544712dfc08f191e521c57fac776fa4f3907451535fb915b1ee
 e3d39e6b4b548936feab36114464679827e0708e31a3983297721c0b350e9d9f

Fiat-Shamir context

49470f9e25e8427e8c29bfe2d23ecd577d603733dff9e46676234587704a5633

withheld proof term: 30 227 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiagOn_of_flatness theorem, BlockPartition:256

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{U T S : Nat → Matrix.{u_2, u_2, u_1} ι ι R},
(∀ (l : Nat), FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (U l)) →
  (∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part (U 0) k
          k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part (U 0) l
          l))) →
  (∀ (k : Nat), FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (S k)) →
  (∀ (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)}
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (T k) (U 0))
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (U 0) (T k)))
      (HAdd.hAdd.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (S k)
        (∑ j ∈ Finset.range k,
          HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
            (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (T j)
              (U (HSub.hSub.{0, 0, 0} k j)))
            (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (U (HSub.hSub.{0, 0,
              0} k j))
              (T j)))))) →
    ∀ (k : Nat), FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (T k))
```

SHA-256 of the statement above

a1e831378a902bdcedf465651c5e8d7cf7598a8179a3dd29f43b288626da501e

SHA-256 of the kernel type

6ae0d1c3de086ed9d8b5071e8b9ff5350a706627a719d1152eccc662de474c6c

commitment C

bfa2a075d2a1cf2eab20362c07cdd2939d6d071ea5419c18aee7ea945d581118

fa33cd600df2ebf685c639d15b578d2a5a0ae9dde07acb082ef4604343c02b1b

915f672c18b4c69f70dff3e7689d9e67ff014dbf3fdfac59cb0a2f33ce5ed4cb

fdffed445402fa385361514b8cf2b5188b2d79079c3ee01ac0aafc0b683f36db

e1f193364f3c0f94a1d62d7952038db9e211f2f08230efe5deef9cd20d5d08cb

1740097edf215513ddf40383aa45e735c9e82faa5e5bd85734903b38a68ba26d

d4419562f134dbfafc1a3800104b3cacb3c8a867563ebd8df7dca7d649f5bca9

9ddeee84a1b1cd85da6a72ab30396414d6fd4f6c0fcd264ec354925d2baba147a

first message t

a9caa8e19a8031eacaa05226bf7c70b7cdca5add3e5f3f3bbc9c293512c64d3a

9b5afd8b28bbe044c75a9c5d42be2ab882b67860c452ae9b54dd81cc04287298

e4285c099047bfd3abd2d4f2eb82916479ac37591b3dbaf3b20cb233aed841a3

dcfd673587b2bcb743907ec31961759d5a0a8b4cea3963e2407966e6511d606

1de9d2f2cb6bab8383dbafb84aff4469c52bfff929ceb6e4dc201df395cbc2067

f541766861f8d59003981e22eb84c72c6b0b138e326e663f158ca2748f170965

d23310d183361ec7e8e2c7d593d3d6915c6d6d2cca0ff6ea581e29cf234e8640

b9da89c6a2282a2ed31970670595ccffabbc91db8ec739d6ae6b6e86871b0dbe

response z

43a81dd839a3b92192f024cdb56fdee4ad9316fbd86c0735d32ea57b7965e67

d19412fc6619cecdf62fa3af7f6db89023c74c4ac5d93b5cb7888c9ff3aee2c6

bc364247f2a0ad81b8728c0a34715eeafb7484924811d73c6f641c805fc50ab

646ffe425bb747b816ce7917f1443810c27e6d32cfdc443ee2cb50a11a0c7b07

54ccc97ec9cd86135a40a83ff654641e237929f3e324f505a38e726aadd4923

be3f15a84453324d09704fc4274efcc49c99e2a21be8162fd5080e98f041d59a

2cdaf5bd38281eafa86c46e729369cdb51f5d98b7807c4fc5169666ed76c6d82

20231f30f9f1cc913d4a4ccb33d836908c8e45dba8309260834524393c4b2a88

response w

234955382499bd1cd08a084fec1a2175983bd69516bb365c3eaf2eee4a4fe22b

59b68b8fe67bc93c829fd6a551691023f519d4d2af0d9aa70a961893637ddf5f26

7cbf977f76fcb9ff1efe35575ffc554c70b843069e4b4b80f83719da128845c3

9d7243804c8645df94f868c5bd5eda9c7ec4b80044a78abe63d0cdc4b801f8a7

FBundlePoint.isBlockDiagOn_sub_offDiagPart theorem, BlockPartition:147

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{l : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} (\text{part} : l \rightarrow \kappa)$
 $(X : \text{Matrix}\{u_2, u_2, u_1\} l l R),$
 FBundlePoint.IsBlockDiagOn. $\{u_1, u_2, u_3\}$ part
 (HSub.hSub. $\{\max u_1 u_2, \max u_1 u_2, \max u_1 u_2\}$ X
 (FBundlePoint.offDiagPart. $\{u_1, u_2, u_3\}$ part X))

SHA-256 of the statement above

259cab76941bd3a306a9f2e3763cb50980919698cb0302fba10e1b421e647261

SHA-256 of the kernel type

29a5215c80f936c20d332af1040084d6cba86507d18f5805270e8f1099d95b7a

commitment C

7e101baba87856411b21709fdb04c9466100b95d72b8e0da37ca77ab811e99ba
b7ab1bc3e9becd232ad1faaffc19332f488fd487765a9c4278f69206fafe651b
39b49759e6dc6c1aa42a42f35a80e6000ff9b49262ed405819564786b554d237
b06ba132e7cf475634d58bd414827289c0b700a5e7119f01d076e2976b764a9c
260d7b8c0efdc082ccb8200634e01e9932fb7ec0c107f0fcedcd34774a7e7ee2
3a478bf777c27258af3c8d28c6c6673964b946ed778e5d3896d428ad2063c01a
ae04716d88cdda3f7d0999c510e41c83ad0953bdb67c70b4626f0195e731264f
cd711c474ac2ebef8f33da0bb51dc1d3426b4b5eb443ffdad65ef3b4e3dd8991

first message t

fb3b9db89a3a620fe8345226524c6984850e46823deaeca544a654af1780a4c0
81fb7dd95d29b02b30da2101ea99fcfa2469689c1ef9b6b57225142cf46db2bf
30ecb3bc557ed7ed881daa0e77490556efd13a393ff72af3ee50e07dbe469389
7420bbfbd6adbdad1ba1d059fcda72ccfa5afe5d1fa412baf062d1d9a9ada0a
7d2574e1be2134f885795a3e4633ffbe9de7d40f42af087c5ac8244623b1ad47
5dbbe282f8b59b7f43a7bd3e7c51693d009be4821f2f553826ee511b02eb3973
d26242d30b85bb4d0f834677344ceebf7c1c70b146706719318bda467da97e5c
b5920c91e0c923539b7b58bc879ac2ff6db3bff95f90999e620a882319990e08

response z

630b7391f370e81c97cef4e8e735256b608c41ed573a4468bfbea92670a40aa2
ac31a943a4f09595d9c8fcb1aa56c1ffb19fd9f357948ee178e8ced76e1d9ef
773372f6f6a559ed4996073fbec67f61b16e5797b2c27ab7a234bda066052480
9c19b15165af36b6b7d35410a8e3fa4496bf6513ec04cb503885cd0a12f799ec
0652c190d6807837cf2fe76f7f31544ffe4e472c8ca53a98eeef836f43a62e5d
54afd5a0c581759a5f58f4d7f20315b29be29de7ab794c3242501b8d637dcc5d
36f3083c46f5dc60e8fd4f11ce95efab7ebca1600847d646846f510d7df103b0
ddc08b88a8ac23a8df848ca353db29544c461ae17987c78b670cdc1dd9d90699

response w

4cb3f7c190a2a7b5fb8fa766fee569c0d8ae34bb71f261ac2f93c873fb34c4f7
60527cc07f6c4133b287ec85aae01eecf02d4894d66a5a2d02aaabade9b88b8b
0fd3b96ae31c6b0390aad9ea4f5990ff5d81c0df3c5198a491c7dc672f3d5beb
02c827c5321b3aaeea109509b99e53eba9ca9d65826040f98dbcb76ee743723d
87a7f439f6ab02d8aa63291fb71ea7fd985efd355cb81e4c1bc9c923a8bc43e9
89b5f5dded1f152e493bc6c7db28bb5353c2031998f89b626eda8d6fa02e219c
b5f9db023eb505b5fd42a4fb02ea3c79475a9b3231eec14b0ec3fa5b8739d742
39e1acf8eda09c9b0778578ab2a96bb5768b0950691209edad796693ceab4ac6

Fiat-Shamir context

65b08b7e0c3fbe7783c9896c79f7328f9393ef8249a1eca0f4f5e8f7be6c31cd

withheld proof term: 8344 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiagOn_zero theorem, BlockPartition:93

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{\iota : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} (\text{part} : \iota \rightarrow \kappa),$
FBundlePoint.IsBlockDiagOn. $\{u_1, u_2, u_3\}$ part 0

SHA-256 of the statement above

cc3f607f9df5063e1f44c33020c466690885a21d584bef5b5b541233d8ababd6

SHA-256 of the kernel type

fc4a50c1d636ed28fb8a5a351e87427f90ef7a57bf6949281bb440dcedcaf36

commitment C

9279cf29592efeb17ca70022e0e78181f0e336c360c99c07acfd14b3aaa632f3
e40d7b2784fb84e17a6523bfa5c9027610cd86aeab9187010d3da4ec1123885f
8dc5fa026fd4120ea30f49a08a87ef8c108d70f57a7a417b8399efcbcd97b714
a9e1d81952b629da0b5d47eb3082f0c6b76e92eadf2ca5e78d5d3532675f7207
98719b304b6dd7ee850435c81c2462308a8c97c42f4bebd939152e9029cb6a5
5c354e0bd6945d194074f72ad7825b7eb15279ad7ac38317750c2773dfe7235f
d47ece730916afca01beb78bfac58a876b056c712c1ca9df78d0a7a800b10a8f
aad5092b02d88903d3f922dfcc5e7a2e7bb70f96cb68fa298313925726398085

first message t

80fc4cf6c7bad5e6175570bdeb9c4e1ddfd8ae5dc1c684dc2bbb660f97a0158f
1ebd491a77e923f879d58cdd223ff728062f3b6d82f3d627441e2598e3484f6
4758ae2cfa2e2f770b9ae49be8522fd12fa1cb1ab3ad970675173163720a1ad8
ea995472e3706365a5511556fa402d7309076499fd05df280f63d158b2050b43
e050728d92050b2ee99908fa35a2a55023b093db8e28987342a0d636cc5efdba
a4e9e5a8e770785ade94a906d382faadf4fa3c239883b14f8e7752891201ecf6
6877142abd1411e7611b0c8052f679778be69229fec74e28e7aa44749aaee931
f650765dfdf6649340e0cbd94481c5914fcd0c61cc80809d403860e6cc4c1623

response z

78cd4328cd737adb2b3f362304fbd30a4ad948df52deb273a47f0d08e340e466
504897cae1585518f5187971efc43c3b1fe52fcb07d0820003ea7abe150d75a0
7ad292a80643b57f2656def5a501051cf5c5a89bcf6ffe3b646fcbba98029add
6e46640f187870a0fd73d01861ef3b87bb5667ca9ff18ab301210f10b85357a0
09b4de467d311aebf94e876f875d904edbcee352c44e22b5593458d6b2b633f3
5ba412aced3f170b90aba49920f3fb66dc6b8c48d7cf115aa45faa8a19663e06
98cbe0dc122f99d5e3fddf6928be99446c31a0dc294351ac4cfbc2ff1f945be7
5c78c78687c5dd12a8590094caa02f42e2ba374d45f19a96f5688f421ae8d5df

response w

5dd88f5d9d6ba38e1561f47f513f5613a433ad7f39e94cefa8efab9647033005
af59c573e02bd4b9fe3164270a03735cc07de8ceb8c92425ce7f616eede97528
b13885c2c66608c258612e2c617c66ca231a6da67ca3ab86fb823e613cb08582
833c927e73ae87f8d52eaf0783a297afd4fe61a709c91a130c7b5eb8468440c6
50aa5442138d96ee5da9b8eccdeba37d7260ed75cddc03276f39a5e177cbc438
f7f335892066c79028451bf894980eb0eda4a9015e56740ecf3669099f17c5bd
9b01a248c5790594657c62631aa850e20448d868db2fab2fdd003ebccadff4d3
56fe69ef80611067b07c233a7d32db050a6fc6bd24f85be921161a7f6149d357

Fiat-Shamir context

ec617648efeb3f9f35c1a23b4cd36881e88d249e9d928ecefca4a12a7e9ded1e

withheld proof term: 1 186 bytes • axioms: • kernel replay: yes

FBundlePoint.isBlockOffDiagOn_commutator theorem, BlockPartition:281

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[DecidableEq.{u_3 + 1} κ] {part : ι → κ} {U X : Matrix.{u_2, u_2, u_1} ι ι R},  
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →  
  FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part X →  
    FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part  
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}  
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X)  
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U))
```

SHA-256 of the statement above

4d11ceca1f75a2c446079a5f3fb4909b398ff24639c30e949fd116ba6a0b7f66

SHA-256 of the kernel type

b14b8f3a55767d011b9acfcad1a46108dd64d543fea3d7ca0707e8d8a30f1095

commitment C

ba0c8e2ffa96de4417c088d5353ba50123beb7b66cac37b942b1157c4731e995
fd3077b6c819fb2b58824cf20c197cb9d2ba722295bd9cdc7f2fcb1a7d81aaaf
8da8ad25c74e5e123bf08ddd83fc5764b2341f6a9f534f10ec2bfa0231516bdc
04485e176f77ca01bb0a4499e20fa47f66b780318cbd69b52c4df074444524c4
c807a5060402ca37d644bea3ee267b32eb11b7b55a09bca9c0a419a394aee8b4
c5e2870d21102d8355beaf5a40afcd50fbd12c682f8985ccd555ac3249204f1
a079d009e21b849b119cb1d9972c5d7834e2345130b324a1843402ff4ad7dbfb
ceec9688fc20ee257057990679c4b5569713ca1534927d33e70b8c96071f1e9
first message t

e4a5be65e674cf686e626d812155d8417d7ff6bda2667dc22590a8e45752582a
78a2d1c34bf5f89020739fb7b1ef4e9cb6f73ba571793fd193a1af16595ce536
d91bd159fb6346755a253ad12a143e5b35dce04427830897967361ee8c13e94b
899af82d67f4501cf1624d40f914340b7baf92b7ab87cd676f1b95a25a4d0400
ab8af47a35582b6ae4fe37d3658f56e74d2c34fa9a2d2c017b06297694e7e0dc
8d792416546b0da01f64beff209f2ab16109481aea52de8a725c6100db0b907b
5f90ad2a8a61674866af46b346f94a16e9775c47b900acca251382d59d644ada
72193b43a0627e1bfd025d90b47d8cf02e95b5342001aad9285923437d3ef821
response z

6aaf5441de9b2d47328b5beaa57645c8224e079373430d6dbf77b1a9b35499f8
8c31f052b525f66fcd6b5fadec18639b999a515c0f7647e8b1e3985f7591ca1f
3bf90f2359e422048d85d7fe96cdc6069f4b5ba0748690e125662859874afc06
54206df9b73abfc5012b33e9a50e5b79c655e3ee453b73c3934f244a969532cd
fc11b9675510b651ea91b7b223f5abd1d8aa6c031cb42a05ea43314e70936a59
7af7f72e3217b860225fa74a002be44564aef8fc7cb643970e27a0290e15f133
85d66361086b7119a8dd726cb61d2635d3d8bee14b90022cfd2989106a76dcf8
db17eb8a21b3c71d6bb7f818f5bbb3c75e4e30a54baab88fe70804a510520135
response w

49e435a59871589e0793738ee70ada3e7fdb72e343367a43fc9fa27f9ab45561
94b7c2dc36bcc70d5ffecdb0c8b6be307e267df8de14718c0867af70939b700b
0b8f43da16a88d549e479f252677d7091d63894085b4a872ec8d0717f7dfc197
991f1bf7b3bb14a0ceeff2fa296afba63108bbabf9549c9d17fcf159c69c1f12
f15e4c1a30ee635da4e3a706210abd72e527303a42f3660ab266f4d0bda9dade
cfa08ec8a46035e006dd55040d3fc8d2d6845734499333aaa741083e5270ec2f
592cae9e4a46e44b2c570df671b6b64aca9c22adb2f67999831ccc9a59b0d8d4
afdb19d61426418976fbef2555dff10abbe91f5f66e821ba2861771e18fb9f86

Fiat-Shamir context

3c9338323749b90aa53696b9a99a88b1918903c0ae9aa8a35f3f55d571fdbfb5

withheld proof term: 204673 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockOffDiagOn_iff_blockOf theorem, BlockPartition:82

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{\iota : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} (\text{part} : \iota \rightarrow \kappa)$
 $(X : \text{Matrix}\{u_2, u_2, u_1\} \iota \iota R),$
 Iff (FBundlePoint.IsBlockOffDiagOn. $\{u_1, u_2, u_3\}$ part X)
 $(\forall (k : \kappa), \text{Eq}\{\max (u_1 + 1) (u_2 + 1)\} (\text{FBundlePoint.blockOf}\{u_1, u_2, u_3\} \text{part } X \ k$
 $k) \ 0)$

SHA-256 of the statement above

ed8d9f1e56bafcbff2bd78770ca0860d68528364aaead8f250405af604f7226c

SHA-256 of the kernel type

f84d3b986352d3901f58e48fcf57a510c7d2679a521b29606113c54c59507d09

commitment C

d00c9f85c77103b9e9dc96de5cb57a6f42d675a34559902a27f2500b1841e9cb

239cf58891e0bd22ebee285541397acd92659f8ca00c813abc85ca239c5d6b31

d9b76ae4aa47e6270f4fa19ee1514f0b090a0bb1d7f3f7aa4e73437651bf0afa

2dc1bcbe77d3d60aa39a32f1d58d67fb512060ce706682c92f67e373a80caecc

63c0218da7806c7ccf0a6c73585fd9aa296098d9de778609a8964e5e496b5bd1

36dd99f8a0053d4efb0ffcd01ca61a5fc1e6aa91764508a1d06e1f7d3f1d6377

f39c000c10167b684ce61e41a752f893727fa5a811b94a4db5af5f0df38e9b22

b7980a4a10cbbf14219aedc9b830d2685fc65dceb7f3ffed0ecb642dc9ed6d0d

first message t

e357c25b6e8dc083f3446e8061ca90c5f90831b3ac89fd6938ebdd3946aa4688

ed4066cd12a9cb7dd17a83ede87e44025dc1643e728bec2a8b01cc03e997dcc9

964829603b4bf64d21118fd54a02d62327d68e6dc48463de69fcfac878e47e43

39c5230931739acb4b4ff14afbcb45488c93fcd0b3fcf06b4798b85900fd6192

85090224a8ef7f799101b8431ce9c23319a7301c25e1ece05bb0fa858e04ad80

3e6086357da006e803fb76014f3d3a87efb87f3298b3b701a5287924a99962d0

c5ccbd73d28edd897f0857e04fae35159ad5b7fb7670ede3dd8b812d62c5c6a1

4072d00b50e7eaf76a8300ba3fe2766462700c8128ae1ccb40c6a8c201066d7d

response z

554171f999d5cb51572be3b83ed12e6755aa058180a80500d252e51f7e868907

3db0916fef6df30c98ebc27effe118218d8257a40ed4db744b233815bd09bb1e

c7e72024a8ad4af7c31f8de33bbe3ee8ea7d74ec204a8d427e90374e4c1ccf4

70530d4f53d435cf4550662f5c55085469dfa1afcd2cbffca06185452a9b7a1e

897c3b5b8a2ba5518a2163c99acbc5bb60b1654585563a60b3f9b20e5035a222

8ee84f71b05e37b916243de32f2e6325e62c1e5930df91fcc34227f6dfad17f

dac07651bc8fd458187ba3d6b12dac8ab249a8b7cd6837cc425d156856e91575

532e7d0d63f42cb897549d944a961ae1609c6f7cbc4345f1eafc6f4f13c5a69a

response w

12915ce9123b65a30fe4fb89adbb542c66ab6e2ee8d943134229dd74e9728b72

7b9eddbb3a661b1aa20a0ebf39a0ca8190e4ca7e0354af001eaecf480d51103

aadf96bc2d22252ef51c463830f4bc4c9f2ed1763fee4b3b834daabd05c1674d

4bfadcb03438093f37164d79a32b2c58cb944711452a459781c0ffde946e8319

2ece66f3ae380b059458de8f6139ba6452402205473ad6a50e4b9db160e89ba6

1e7106cbceb594b32d71747f36bf15fa6d6e262d1bd9c7cabb05db92d95bc024

51620f881fbc27c772936a9ab3aa5f93fd753d49209dab64ee98a9b90e047f78

9bc454730b4216cc95d32b59f176200ab365e04fd786496d39d1f98d095abd63

Fiat-Shamir context

d3f96d8833ccd3f0763aeb300833da6492c5668e4b6c3e5ab9ad78e6ec5697b2

withheld proof term: 10 298 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockOffDiagOn_offDiagPart theorem, BlockPartition:141

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{l : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} (\text{part} : l \rightarrow \kappa)$
 $(X : \text{Matrix}\{u_2, u_2, u_1\} l l R),$
 $\text{FBundlePoint.IsBlockOffDiagOn}\{u_1, u_2, u_3\} \text{part}$
 $(\text{FBundlePoint.offDiagPart}\{u_1, u_2, u_3\} \text{part } X)$

SHA-256 of the statement above

300c7df7f6b7456a8585538c59758edb44bcef7f87ccfe227f6d150eb4b8ef10

SHA-256 of the kernel type

51ef93ecd1c29e07a02350f6f44d70919d2fa5d8f9dc8c073ae446c2d2f3d5d3
commitment C

5f434d3f50404485cce46f79dcf7d907952434e12fa0e76479888ac481eebc7a
552f23366de5f35f3a278825139e28d852b2e64f74ec20578425ff86d65bc145
da2dc234affaca44b1681de18e815e9e62e22feb409325083b7503e3e98ad19c
e0764240d4411187ef151094cabbbaa8fcb8af6f1c6eb91275b24faf3e61f57d
3e56d94ccac33fb2d8fbc4be9260cb600a81bbac4001daee174d36585ceecc41
d2d2b6ae9b54495c904ec9ea37f4ea0b2ea55409413fad47549b5f9e67aed0da
0651d398cb9d049ebb574c15c2920af26d7b361818c4a1208b2bc1cfb638b0e2
a12a50f081de602662bb42061b4dfed9917503e59bb507ac84c65feb11534816
first message t

d8f702398fbfe2fa38da6da0851479875392fc957fb54c4baea1d9a4080b13b7
36d4f41ada7b48c4a685ee53d669c3aebca285e7d452cc9ca930acff6ebd4b4e
1bdab13150763a108a772c098b6e82ced0711a651f955490be10aaa29dda67c1
0055ba45a882e7838f57f2265b2244b801b6f2550baead03b0f3a36a513d1fc6
7ea48b5f2b8c4b09851d1bfa49d214e287f3dfdaf430320675dde833ac8f7086
975f1fb32a10e788a6a65531671fcb5b2e0f623db6e9389686297d8bc196be9b
46d68e13c1471c6126a1a29926cde7e48d83a50802ef28b3384f915bf45bc38a
7bc09d77aef6b82ff93346ee06c4d52a0c7807e5627f6c3ae03aed4c918291ab
response z

54425cab8f19a013af626f0e414f05efd686788237d4e7f2d15675648851f648
edd434dc3234d052020d55d36d5e42b432b2475d8264afc6caaf3fcef9a80483
a7b54a1c0a44bfb636788f9faf55c98cce7de69c61211fe2b57e683c4734f72f
0aef57e98f6c34807ba03a8fb62145792bf4b34df5054a38d34bad7a801f7337
8d7ac4ef519dc5ec00189f9092cf77a0605acbc9f85cbba19b43b6b9d0d236db
89d44086eac3ecd16143c50ebf0961ed34edbb31d3b0b177b0a21c55f11540e5
3047e6f8fc8c47ef51a084c9ca25f41503aa703353c09493e3bb6ed6483660d9
8b899042237a8016b8ca7a8c39246fe4dae97fe5a58070908c76a54e1c657a1d
response w

35da21ffba2d7e66577ba6ebe60eda2c6b9f251a1e7bd8471307fd759f0f198a
83ff60e2e1605fb254dbe6f5c987b5347f329d1a97c8d6c38195cffc770ab8f4
b0ba562697605da1658e410f04c86eb45a160b37dfa3e53e976c464a1126dcf7
92af8ec31a3f06a3cc2e54eba61eef55254c4fa52fe518d4f3a249bfcaa1a3f9
3d72ca3a83d3259b2df192ae09736f0783273b3ee999291415b08a5f241346c4
d7f8ec0033d65975b8854739e838881677c1925f89271c6da4153bdb05f4f317
5f80d0c8e14fcf5495ad9ee96fe287d505af5ad57a44c2b441a5efd879d7b6b7
bd9a6cd4fab9f6396e882deda78c8ba41c5d4b5113b41650bdaa4c7df72a8c28

Fiat-Shamir context

0d48204fccfd81d13b0342efbcb15065abb0fb7b6157e25897435646abcda927

withheld proof term: 5114 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.offDiagPart definition, BlockPartition:136

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {ι : Type u_2} →  
      {κ : Type u_3} → (ι → κ) → Matrix.{u_2, u_2, u_1} ι ι R → Matrix.{u_2, u_2, u_1} ι ι R
```

SHA-256 of the statement above

c12134d08da9b3416c712058c67bed8206c71cd4e4639ab427bc29f59daff6ea

SHA-256 of the kernel type

c3acbe576b7939f9c52a17df67722ccd2375353957aea2955004d4f5522685be
commitment C

6b6fcd49235fd11bc52942bc4fd757b66c767ce7501d496d1d69bd1ca721a6
9bbab22035db7be18b6d32a95f7aa6215130c59c65175bcaf26dfbdf7d0a38a5
0c402cada7672db1b3f4b7bb90b6ec9fc28ead007d706aaa6f74f98811467c19
7e4c68bf97ce028af6c7f39aa534f39f1d0e2449bd64a4accae5118c648414fd
3db478e0ed5f9b8cbdac305610431f847a4b1866589031cfd96129b25c3955e
0fccce08f57668d58434c82a9e90213fb270c8eb8e3e4ee7af22ee72f8816197
4f7576d71a93497589b71b0c04678fa6ba7d8e3032ebb6d0c25815400ee4aa5d
6e89662b442db152a3da87df5391b7e1605dc76991a4df0b3261209a69074ea8
first message t

feda90f72eae91b7e7199a5741405a63ba116c59a04c6c5540475d07e673ec4c
2b44110459676bd84b2256c442423a04d218f981b2f11a21cd2f991bbfb557cb
e18c1e607e7ed9bddae0919216a5508fa36af3c69771f2d25ff20380ad0af2c1
a179646f80e73435d2781dbbab1506d9d335a98345f4423a25ec9718753c408b
b14e90ec691752f5806d98d26e19c3cc02bd87c24c308d6bea380621a01c0823
094858e630df384e637e401fbb12105044ef9150c5b60b9eecb407c6b042e3fa
560095899482a073f0a3b82ef88c8aea3d009f037f7e068200628cb55818774b
788b54bec9aa53548c26302b2219fd9d468c03d62fc546dd7f1a5d5888361758
response z

15b617daab09cf3ff3c9edd65f9e944e19872265aa6cbfda6c3fabe41980caf3
9b514a189cf6109cd2e584192453cfec9d284ffd4c34d4aa7ff345e2c558a740
0e2268780a229b489d76ce10cfdcbe00266268d7ddf32a1ea3a0bd89db2dd5e9
96a7e9a82901a3f04c26cce0cfab0ee644d9e766943075403a8d6f987f38b3fa
1a75c9803820e74836d3d75e4b47fda788799b6e150e30eb9d407d6e3bfb1ca0
253f94ddc27286026a8dafe05769c3a825b00e43b2baceb3a309b7b8a04f3842
6e6c01993a09e3f3b359bddcade4d959d7af05a74dbece3b4ca7649e0691bd00
84b2da9c8ba2fffb222b64bfe2ffae22258449215783e1920df7ed3e2e42f8304
response w

5fd24ceffd7cdacf1e86e353b3da8e2e679989c9fb90304565a187dcecc32200
f35ffb42c314a63edcf253a11b627a1d771016d3827b5e012d2fe2c654d0bbf9
f90c5a9aa2189e2adb1b7045f20ed866b0f253b2c9ace185eede3b6f7dbd874b
128739e247037b3b4a05183b41bd47cd72279921d67ec6f88b57b86173821e6b
53f6eeb69ef231711028c897d82196b02db325473debf107c997bd2c79629c50
127119c15b9afbc43a0158e5850628a98aacded745d5826d85a06583c38d49
1889893ab02839a7081e718c5ec469e2a0d565678a8e487e93cafcbad0823365
fdee9a92564cab34a5a6f979d5d02f448b3c08e76fd627655a96996d85621d80

Fiat-Shamir context

b45d15aee882f7898c9f5fdefafe6c4b016eb767d85ecd4e2973cb38704e534c

withheld proof term: 750 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.offDiagPart.eq_1 theorem, BlockPartition:0

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{l : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} (\text{part} : l \rightarrow \kappa)$
 $(X : \text{Matrix}.\{u_2, u_2, u_1\} l l R) (i j : l),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.offDiagPart}.\{u_1, u_2, u_3\} \text{part } X i j)$
 $(\text{ite}.\{u_1 + 1\} (\text{Eq}.\{u_3 + 1\} (\text{part } i) (\text{part } j)) 0 (X i j))$

SHA-256 of the statement above

b557058d1076de5162bc16ce6f13c0b3c468c7d8c605526f96a5d5aeb420d9e5

SHA-256 of the kernel type

b759b9a93c35d38d4784ae29238ea16ad2bbb961387711c3270fc51b62404a7e

commitment C

fba75fe155d147ac7b12e9ce48ec4a3fe40ec66c966cced0121e2d71dd606893
f4262a33aefd5c4334b51266204a118c86af0f1c08076c67a99c937c7e67bd1c
be552858d5bb0be3529adf99fcfda11e378cbda38e682853060d1b4efaea00f3
61bbde1d1dac27457d3f9b67e4f4b57ee77f21697782fdc6c40bd9906bbe3e51
78235f1efaf282c426adb6421f7ea41fb48167f2a759d9dddbaf24ce7c2ff79a
2a67ee31d0e60996f8d4a2697d3bac28409cf8c95b67d580537e41544c7c2599
8d6dc9881d5d34c916068b9efec0f027361ebadb976dac831a6ed48885beaf47
a4aac1b1c2002f3812a2bff97f1338b6cd38c2630701f22175e554e37de1d7bb
first message t

cf36863a515b88a5e6fd0c39b288090e551570849b8eb2ced13587c9b7d6e7c7
cd2d3172d2371b308d7880ae10f0a06e82079d8e61347c9b87e9ba7a3463edd0
a3264afd6d5e1164703591a6bf3a87fe87fbc3ef3199bbae633bac4feb6a110a
608e9116983c9a9d07c84e2188aeb490c34d0009c8dd92ba568b86382f058910
b1c111682be4cf8c478c092a11f55e7d54fd7b1bb46bff1eeff4a60a21bea770
a482b237e134c471087372abb7229f3931a919f877c55694df76f5f4a9cac6dd
ae13163090d8ec034eb84366cb03a0d110ca15ac49876707c1a24beeddea78e7
fbe0103cd45c382af6317de6b04e1a4fc0189333cdc554e7cab9daa595f1bea8
response z

5d7c6f93d1068143cfb10eb7b160d7e8c6c33cb654a04820d4dbc87b3245342d
f282df2c6ef295bdfbf0a42e42b46fdd0c9de3d6ffcd1999538b3d78c4bbff4
80a22a41832264df0ef684f4aec8a630289a62aa35705b1ab0c4359965c0c12c
a7b78682d74da98dd7962dbda98b9ad18012ca2a3f709b766f2765445a0fb35c
b1b2e8a189acffa874ecb0225bea5abbe0f5d19e0a06496ee9d9d7614f632c8a
55bf3668e79d056ed2855298e3608ed60da493352a26a3113e2ddf61667a1a40
a628714cfa3b04f9f771336e81d036aad236ae2bdb6d75213a0d98bd29f2eb84
6bc5c29a625e7cc5dfa8ea3256668ddce9be540fe94a100fa806fca421665049
response w

7d4b181e9b2d8a955526bb29def915fc80933f32793a0d176c50d077809627b4
73f7c9d324a8da578cf8a8cb1b16a3d1cf649791a40cc44c4ed2772eae4d635b
dee8630531c9912ee5502baf15d4d80f77534a032646ddaff79ff52dd54fcc5d
746b35f2a15e671bd610cb719ee95e3e9e1200f0886b3b67531ce1893b81adeb
36ab4c7ca345df1d92935d488a9c782aefe935fd79562156be56b05b58dff4bb
83501d8ca9507e872f7ca8e497e2450798b92438dbda0d4e9ff4490ad0eb042
94364bac4c66c9aa4f87c038ea2f8722e010a3752b40b00f801397cc1fa5d862
6f2bafef76668bb7cb288e6f7a29863a67e00079e70d163877357b0cd43196bba

Fiat-Shamir context

893584377eb874085b1ef78453ede09c2e2e72fe205531169abd1e852920adc3

withheld proof term: 246 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sum_eq_sum_fibre theorem, BlockPartition:160

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{\iota : \text{Type } u_2\} \{\kappa : \text{Type } u_3\} [\text{inst}_1 : \text{Fintype}\{u_2\} \iota]$
 $[\text{inst}_2 : \text{DecidableEq}\{u_3 + 1\} \kappa] (\text{part} : \iota \rightarrow \kappa) (k : \kappa) (f : \iota \rightarrow R),$
 $(\forall (p : \iota), \text{Ne}\{u_3 + 1\} (\text{part } p) k \rightarrow \text{Eq}\{u_1 + 1\} (f \text{ } p) 0) \rightarrow \text{Eq}\{u_1 + 1\} (\sum p, f \text{ } p) (\sum p, f \text{ } \uparrow p)$

SHA-256 of the statement above

68f5a113938600c337f0b966ece464573d1ad7045b4c2f4769a8a45c2df63a3c

SHA-256 of the kernel type

f10b55820c94558cec0a8d365bccd26a33ba5db986ff6c41c9f3937e9c92bed4
commitment C

6ad599a21d23161882a1047b61e4a8fef4ad7ecedcbe40d0533b555ea875e207
f85a02b4a266a4582dd0d3c3e9407e0aeca1cc991501b60de2a78b8a8d1776b87
8fae892dd431b223c6bbbd200eace3e1400312908d5b24a507591c534461344d
7ba61c6855a065030721afb06b2a113a244ae856dde8e101867c8f66f29ba69e
214cff4cc50acb4b372741d6fb8d65990ab309f2fb2f2a8f3ce3e8dd4d98548a
0900251911413db15b53c07f647a7fa3ce72516234ba56d9f2d5fccfbd446916
54b5bd53f2e2fc74cd513f6452179f31ec403dac61f41d8c1cee92f8cf9a0e1d
a9d6341f173fb98cb2812e7c9696e4edde577562d023fd2f4ee7552e1757449c
first message t

3991c5829f24823b4d306c56ca32b24ce58996bef0b6660f0c6c093ae76d3b37
56024d533c4750eb3a18e5fd0ed7f303b71e00607856c612b8671546f603fe77
feff021614021efd4075aa06910b1d2faba0e484323d34046a19b1c0e672a8db
943538a209171b06a76f6e40e35c8a23b9ef0b3e5d1dacc4dc297928a508ffa7
9e03881b00c980a426f18960a38424d49f3316511d398cefa60786e86fede116
7f1beb56964907133b17f76f1ab05d74e629fe1963620f46ba123ab1a0ac68c4
597e1bd71fcc20c970dce7b04ad3ca7013efd7be3bb4a4496cff71507f6cf95e
95e7b34c2e42eed3af9ed6b2f60a36d2804b774f73bedce5fc505d3aa8dfa41a
response z

269b9db6b2b49e88ba320b35c5b4f8fe0673aed830e2a2d2f65de4f586000ebf
c59bbc636ab2bde6c07dd8a38043d142463221e9461d86cd373384177b7334b5
3efc549a4f5a941a2b1310b8415ae6680d0e3522b2cbe069c3634db255a51baa
4f4f3b974bcc0e9fb573ae3988a55d79a0742989d906ac27f15ed2ba324016f0
23fcd4720788273e52aaf099d83c27b53e0f0eee987c1755ee23801c3c67b622
2e9f0cec028b1de1342c379d3d5054b6b1da31e3b1e51ff42937042ef0c967b3
32f22b48bd50c8b569e4f40dedb532ad65e557972ccf3c3b888ea772dc055ce3
3c274c19575ab3dfcef716db8596fb8a22388214cf5915c706fe804bdf779037
response w

799da7c630d50fd97dcfb8b7feaecbaa2fdbf6082732ecebb77ab7ac9e1f4f42
a6653a91802f918c03d06ebd41bbf45079ceede0aa0adf87bf30650a7451caa1
4c342be9f2cb38bcf01e1d6fae9fa90fdf952d4ff3b8928c86464fe8e18c097c
1874c69cb1501dec130fe8894069f06ceff3ca58834491119645b543a2cf1a9e
c67fa0e6c5a3787680001e624561f17fedb3b505864785c989ea7298dec738c5
fe2f0c1f6b4650328dbf3d5bb01ae58b844d118a43a9617eefda7ddadc29d2fa
9eealade261c548e08e20c73d33dfec77ef4149a0e79366ef11e4f2cf6a92608
d44b9f6a0a89555e6f1605c7b14f89160953671fd142c66a401810d10b7d316b

Fiat-Shamir context

a0f9b7008590804f0112618156440bb8ae0837eecd8046c1590074d1af4c5ec3

withheld proof term: 20 200 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

2 RequestProject.CompanionDiagonalization

FBundlePoint.castSucc_add_one theorem, CompanionDiagonalization:256

$\forall \{m : \text{Nat}\} (j : \text{Fin } m), \text{Eq.}\{1\} (\text{HAdd.hAdd.}\{0, 0, 0\} (\text{Fin.castSucc } j) 1) (\text{Fin.succ } j)$

SHA-256 of the statement above

63fb27b1d297a716d1bd155209e17c88c63763dbb7b598c5e53c7f7f9e89b3f8

SHA-256 of the kernel type

305b6df00cf01882ffcb2d448cef2b508868119beed8c08c6de77c574c4ecc7a
commitment C

263086d3f8465c7fca4128506ee617a487716fae66a2d2a11e20862d3671f3aa
92dfa082ebefbdd02c624ff27669bafd856507abaccbd79efc982ceddffcebe4
591a26fc3e5ad89db4fe85fad01869564dc9e7bc880f431053c5d77888127085
7c76f444a751e11d5d57493a9425114df687b80b2795c3df0baccac77274a00b
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bf32145eae535a434e17fee9c136bea4e2e122a13da9fce69b7db824e1db2029
49215d8cb283e034b02941442e7e630e6df6079cd6e51a15cf2c82420f6f0c97
fb091d4ecc1b30b4387cb4e7d36bdd69b70fccce09868e83cc411173c434ad8da
first message t

d15b78f27e7466abc63a23d3fd58e282b86609dd92f8e3a72776773fed45ed81
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183badab19837b78f8317f244ba493d35e73996c892f6b5eaaca951e41a831aa
f37daad879b0d0f086abf62aa249f69cae316eace244a73ba76adaf1bc8e29e8
f29c37cbcee19bdfcd029058c579274f00b22668adf794661e04e83d474ed7a
c6abfe5e19ddf4a9d0d8de8b26e9aa2844fcd45e2dba62722358c7e09f5fa65c
6691f8101deea49636d54f9a417067bfa5cd14eb989d6924d8511a66f63ca986
f20038e43937eaffef3bbf6f55a7882a37fc84955ace471efc879c5dbd817d20
response z

8b1056655c3d4011e1df2e4c4d1d28823112b12a8385bd698284fb9448ddae7b
970e03540067e984cdd3e5bde94337aa3dd25495b1eae092e5cc57d04a438766
a042949c82e40af7fe77829c7ccea60bf0c465d3198d9637dda3876c465006f5
33ed9038ccbb0b86625d16f28f821fb5875ef00232e186f7cad87d40db7ae3da
58eaf9ee558d59337be0800fab3ddc9c6555792c0d9a8c4eb260b6059f90226e
1fb6014a933217a17e6483fb806f583ffa2a4e14dd9b363f240411bb83182eea
c3f97b2bd04e0407e685940b87bebbcc8a7b0084ea226a63b23a8a7eb85a6e1
e1a8148332c40ce6adf5696be713f3d2ae441f7c44ca155a6cc8bd6b7ef2dfa
response w

62e0bc9ea5ee0581e7bd9857819e55d70241e1423eaa58d9aa859033cb63b76a
9b230a5727a6210bba9810b7f40c43706666a66fe472b4d48495c4fc6344dbee
6f928d9dd410cfb275f3a90ab5352c14f707dd9170984cd070710ead3d0983d5
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f2eacdf4d6505452b828d443e83e7125108c6af5dc165a61e7c891ec62f1045d
543f5051361ae0ba25f1bb06d515314ce3ffaca43080a31f89b7c7344dac79c0
70ca0533417135b012a6ad2d2a7a6ef96c839e3912b899546baea4ef8432d635
e0e441f7278cb9778c156c833e1b43d4dbe91f073b16ecf3eddb9d930e4f7d3

Fiat-Shamir context

a9c946fbd775b6a78070d7779f14f4cdc58d59b120f8b9243a6a675af4862d35

withheld proof term: 6958 bytes • axioms: propext • kernel replay: yes

FBundlePoint.companionK_mul_eigenMat theorem, CompanionDiagonalization:230

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b lam : Fin (HAdd.hAdd.{0, 0, 0} m 1)
  → A),
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Eq.{u_1 + 1} (Polynomial.eval.{u_1} (lam i) (FBundlePoint.companionPoly.{u_1} b)) 0) →
    Eq.{u_1 + 1}
      (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.companionK.{u_1} b)
        (FBundlePoint.eigenMat.{u_1} b lam))
      (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.eigenMat.{u_1} b lam) (Matrix.diagonal.{u_1,
        0} lam)))
```

SHA-256 of the statement above

d88c8b87abb3a516d8244690f863e48a7f6e8a9c1ce9e6941b60811ca3a306ef

SHA-256 of the kernel type

2db3bcaec27771766f0d04be4a8961f7674fa046e1ea19cb69b05069d755814e

commitment C

13da30c1712192e3cc724767b81ddd07630f75e2bbe50cd7182aabd1e721936b

18af4f46883bbf8c839353a0c103b7857e448195433b9744e72977b5304c99ce

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c6d984061171ba309dc894ca2c1ff560369facb721dc6fbb13c6f0e61164a454

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c838cd4e52916de33081c69a4d83c927fd7d30e919b74e5dfd55dfbdddfe107

0abd65270ae37e32bd126fd05cfb740d01cb45a56a16524865b90afe5bacf21a

first message t

f3c5ba03a8e84553dd76ab5ac1f97d05d16fb13ab0303368e9ee5fd547e8f7c7

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7e42993d13511cb36a8b7589e1283fb9a7df03da3d2615fc96eca7ae73d89717

8b7ef137af5e820dd08f071f0bd916545974fefc001a2c52ea46657942c496d6

d07289d1a47cdce323374dbd4feb336efb3853b68d912b343e900842d7ea950

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5dafa14f182b0c2833ca7d3679a27176b363d73ab786fac7ba507d3588675079

366c24f2ec9bf13752ee83231cddf4cf8eb429b0a29db29e2cd286ae4c340e54

response z

3266c7c100281cf773f6c3e2f9d720bf95bb66c64484847783b8721142f374ea

d17878d72fbfc89e1582a7407dc825d7421edbaaf4562988f0abdeed2cdd4407

8b43f519fb4b28e46d97a41883b74c43c8a4bc7c5c47468a2542daf9506cadd

8a59cfc7b886512b736101d659711837b373b39f7e97a24eae591671b287240e

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747f63d9448378b8db9b3bf5ad54a8e3d820f46c68407d7ff9df6297863ca398

1a7245508de36ea7cc678016730c78829a6b48616ba1e3fb887cd11c2c3c2e60

bd6f9c5bc879a995babb1584fc36c60e4a7ba50f711ad8e0ed39c0a48c7e986a

response w

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e6b2856c2cec7dc67475fb75d9ad3c6a1d5b285bc54ca88a48f610ef8b9045e1

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bea6634d0ae763663871be22d6e05dee32b7b77d87bb57323c13cd1f47542942

b29dff294ff763915f3d1807a9b948089c43b995715eeab7288d97c37cadc274

e3c1b974cb4a124e7b9e1df13f3d8d49bb0d09b432fbd313e5fcbe97403e0318

92551c362bf56a7b3ed6c74e281eb30e46aeb6dfdeb569bebb1f923f139e7c

Fiat-Shamir context

c1c0a2c8abb7d311324c6342332ca9b217844b9e9d3d1818caef43c3bb6c3f12

withheld proof term: 877 660 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionPoly definition, CompanionDiagonalization:96

```
{A : Type u_1} →  
  [inst : CommRing.{u_1} A] → {m : Nat} → (Fin (HAdd.hAdd.{0, 0, 0} m 1) → A) →  
    Polynomial.{u_1} A
```

SHA-256 of the statement above

efc1a4ba26710cb409082664fdb7e8c210e8699f50e624e26051cf511f153df

SHA-256 of the kernel type

ffe9b73fba1eb107d4173d288935663688baa20b32e8d00fc6e42f393d0c0197
commitment C

4b254bd7e9bada92cc90bd90e04a8895ad32942302b3983317a8ed68e29961c9
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93bad3007d5dfbe9af76859a5823ef9fbb81cd5597ab7444ed11ca47b0dd7fa9
ce299e353e203e0c800679cfa540191c213187882495d581f56deb3119faf60
1a9f4e53ea4b491f23c408313e63c9b477452d70991d1b26962808ceac4ebde8
3329e2a6274aa3d53281503ddc7b3cfb0c8af4b82fac4efe9e9263fd3d109249
b60fd230452f43ff0fe633c5f9ae4b9c19726f145253e44244b775bdb2dd61cd
e18b7e7e709015a4a3e4b489f4638f611857140f3cb700814ec49f489f39a04
first message t

7a127ad8bee7aba5decba4acb2c61b946feecda969ace0f533ce9d71512487b8
fdbea965202822f7ce8007ba9cb211c67a41809908433e7d331d4d8bbec9a728
018118ff4335de681733d6c660121eb1908d82d9021494044d9afd742b2d28b9
8c67dc00e8ad8ff3cdc0c059717236cdb5e2936b55f44e3946b6deb33a69f441
3cd048623964b615586aeb4623175c1634dac883c793fb19044b7bedd0e1c758
92e348b20271ac27cf0432d36a5e1ec45e5a68c3f9047b3a92501748c9d1a0d5
cd95bbfde91032e47f5edda116c6f35ef5a8b24216a7cdf122dff6ab21899184
9cd3402eac48811f2503e8752e44a1b4533e27419ae54c23a36b2973928da12b
response z

4c15792fab8a25127791ca42876780b258b6caf7e1941662609787f80e78bc6a
a5a149a6262c9f2628ae7157233c9c6e891300ef5529715147af565461b72d52
e8a26f6d2aae64236fa836ebc18d60a48ae26b06256e7bc2ba26757c627ff6e6
8e70f028fa02eb82a044674f18fc5e02b8a7144c39047bd4661b008c9c11fa7c
b0349feb07abceff196c5ca30aec9c3da1899979ed8e585426b14a5bf7988d08
1e9a82d492a89f6b91ab89d4e3d10823cfffddd6614e9c3f9642d205a5048290
a47e0e4ae7e1fb5c9450cb5acc6524aca17a73e5ba475b2d0896b1faed294d40
0dd0cdb19f45c85f9ea98500010d10a6c493190bb790e1d274070cf47ec1c254
response w

685f46e3e51774faafaf227dd9aa104c01a99a1c8ab7a4fbc39b2dcfb8adcc46
c5331af0d09c45f117b846f98530ded1a51a1192b995ed3e9ffdac8ca24295b7
21e61c97bb239576b3bd9d5351e9f725acaa2d549234046cbdc9b63048d35aa6
561434f56eae7cdf28ff0140e78642af3ba7750586c9304f84bcbf72b439b701
f762631dd21229a4a6654108401a56a61f5c81a7d5db0dedeb9b4898de11b5c1
e3671b7eb70b5a598f8f93f188a9a10f01d9ba5d98f5f05952b147b45f5640c0
d6dabfcb77e540cd68343e36bcaaba63f2ec372f29dc88ed7004df3f24a1961b
31e05db6e12eed757d5e3596b8c5ad7b0cc38b9cfff2b7ef11ab2de5d2f04ae

Fiat-Shamir context

1599f9a629902f759236bac1e5a8917d5925f9a743bf868f8afe844005d0111dc

withheld proof term: 10 286 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionPoly.eq_1 theorem, CompanionDiagonalization:0

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
Eq.{u_1 + 1} (FBundlePoint.companionPoly.{u_1} b)
(HSub.hSub.{u_1, u_1, u_1}
(HAdd.hAdd.{u_1, u_1, u_1}
(HPow.hPow.{u_1, 0, u_1} Polynomial.X.{u_1} (HAdd.hAdd.{0, 0, 0} m 1))
(∑ k,
HMul.hMul.{u_1, u_1, u_1} (Polynomial.C.{u_1} (b k))
(HPow.hPow.{u_1, 0, u_1} Polynomial.X.{u_1} ↑k)))
1)
```

SHA-256 of the statement above

df8c10e5561f70eccc0347089a8aea0639018a7be27d5f0ea6e6cddb738f8021

SHA-256 of the kernel type

60973eb8276b56ca518636e8147c171f537437d79c2438d4ff175f859ea30fb3

commitment C

5995e8be1bf4f79d533298982c78e87e8844fd39f644664b672f93595758dea7
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fe5b36019156ac767f0e52597079fe62ea5121289c9dec856d5e68efed3aa9a1
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453c8a042d050012d152ee10d3cd3ec4dc7e90c2b07bfd12549a7f343ff535c6
511436d6f477d109650e1092bf24189aada14bad16e618f2b88d03763f3d4ba6
2b38d6736907df458e5c0f4efd79d60a5f6d27fa376942bdeb722aef7a355d9b
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first message t

c6fb9da77f8fc3c01cad08995e2e6b376ad3c08d24e3ef4ffd939c6308180cc2
9a32e8b0712679e0867747810337c494852a6b12e5f17e92ae4c137577b833
2b815cf854d207d1d51af6f2f1fc1502734aa11abd609abb55f115ba7659e478
b529815cd0b1bfb4af5a5fee24fd9d995d809718f3d538d746ff04c1f89c03dc
3ee6a8ef0cff52a60add159d3ec2bfd0e4e7b406935bdb0b70909d093ac466aa
9429b9b398f12509bab8d1f1b6e3ba9740b0c7ec44cd32ebf89b2953ce8a4fff
674258064ddc541816e36c7701ba9185317a6e0e465f7ab9a72a17c73c8e3e5b
3c5dde09ad6d966839b6cd2bc7e08c4520def88f6a0ef5832434fbc731d57e60

response z

4dd74b8175109cfbd01b2190adad492a7721758bb3639c251c8c66c548acb076
ea67aee8616895d1849631d6600e932d0d9faa786447b34d0fea56444b38295c
dd05692a1556fbef18466b6f444af371c60476efb4600c66d3e238f6b520adb0
31c8d6e1932ae51bf10ea58b854e6bf2ed785932386fbbcb1ff0ba767725bb4c6
205bc7778bbd3f0110cbf5a3e764c5e9fc9fd8e44f9d04486d59f0142f36cb5d
ffa0fbc6d0e4211e3ee668da16021de3b545a6b67539d742536ecea12030a43
45b3256f14f373f84c1dafd649e4d8130dfe5de173ac17018f93a275e4eab91c
7bb5712000af3180c498a8ee67020fd3d55ba5543ae84f6a34508a7f5904e1bd

response w

609f1f887c2c26ac080debc8a28be62b709bf6033dad421824afd6359bb32a38
7c758927a8b0bb19fe6c2fefa63ae058144163839182f4d4f553a205b19d911
776e0910bb24d5ac8ca497c10fcc69cad8160d24b8f7e8451f410d102ac6702c
42753e3e3dba0f46fdbcb428469c75a1a2f5ea1a22f777cd1dc7315f672ebde1e
720e92c12587b2f95480390c6198a05bc5aea68213f766c8dd667d507a6143bf
06bdef3e23e9781a7e218a8fe429016293305bc8a3d8b68923674fb707ec1ae5
f542530ae6891b1f3b2f635069d492eb50ce79f37fa5949b796dd52da7523847
71bae7d3021738c101a127eb2da3b128e962044d0710ca7ceeaf3d07ad3c5455

Fiat-Shamir context

ea0b6c341c0c3d83952e4c647c0a1eac6701a63869f3fca3807925867dadac54

withheld proof term: 342 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionPoly_derivative_eval theorem, CompanionDiagonalization:121

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (x : A),
  Eq.{u_1 + 1}
    (Polynomial.eval.{u_1} x (Polynomial.derivative.{u_1} (FBundlePoint.companionPoly.{u_1}
      b)))
    (HAdd.hAdd.{u_1, u_1, u_1}
      (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑m) 1) (HPow.hPow.{u_1, 0,
        u_1} x m))
      (∑ k,
        HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (↑↑k) (b k))
        (HPow.hPow.{u_1, 0, u_1} x (HSub.hSub.{0, 0, 0} (↑k) 1))))
```

SHA-256 of the statement above

bb0d8a0d14c68551b27d4c0d735eb1e576ae86807392303cd607acd73efdfb81

SHA-256 of the kernel type

acca4235938da72a0f72027d73e888e58a39ebf3a41903207aae55b1c31907af
commitment C

646a136a193dc9c0566078fdaddbe6d7e6d0ad0639269615f56bc82efaf110ab
a623f3d6ba5db92d2e4bee29586aedd7d79499c77fe69a58d9f9c0c1e5b3c6f1
f71976bf5145269d95ce2019e87f03dca3ef310184df0bf561771e111028139b
296ae86ae004e1f4c2b25e10a901fd54bd56736d0c98e616401e908562a6c13f
b01b1e107ba2e99f52ca5fd1e14728ac39ae7595836d0ac0667ed4c9ef4258c6
842392d5bd7caae668c5742d7ef4240e6d67243c81cd58b257cc441d4ed17461
c94fc53739613b4ac8ece881af39e31989c3eeb1391f6116c339be4dd3e73738
6f7b74a2f87a89b13117001c0a6b2b2b6a55975b24e33c28f3dd8bd565c6a7
first message t

4f5f4c6347d9f83ab2a752fcb06453d1111bd6fde3459d904e991db973343e
6bb82e7fb33b7b2654490c5e91fae4a7dbb1879eb05f22f8cbe65203730fdc99
d169bb641f5393025cc792f3d75ec0f047b37b2cff556d70451d47313b1ff259
f1e9ef2ffe1f1b58569b19951b3c74a4791997b995f2a106ae3c9db9fb5a363a
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b3c7e4537d30589fe53455acf51530d6a6fac42aa83305bba9765c66b6f73e35
713a95feb44de0a026643f8a0d636619a16a0f1bd29ce05f37a547793c59ff16
44848b452050b90c51aa5fdd524bd096d74257bbe92ea2f8ace245e6548c2c39
response z

4f6b54e4ebf6dba8edef7a6a953434538ab2a63e4b2e744235d872d8fcf36747
9a7c2107a48a169f2bd695066e1926a058f05d53bad0ac03987a0634b1ba3c1a
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cbb447d2f21768914f50d347affcabd23ee1ca932ef5e51c867514f2dbad4e43
629a281490f42f3ac4ab05c6f3333300a614072798214136db388c0a5a615eba
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response w

77ba498edea67d59e949ce3ab66aa4e7ea61a6e509a8254d199d29d29ba2b820
ff72e8f3a60ed5cd5ec9891155ead8bdc601d8eb4fcfe9965bb23fc6481fc3d2
a17ba92fc398c9d3fe07ce0a9c42d92f90876b3e1558e413d58a68d9e1d43e0b
a1f3df391c00b9ae743c58d0c5a56aea002d31ee4fe90f456d1ac2c01b24a6d7
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Fiat-Shamir context

9c78e069e27730164e03ff46fb6db180b4de74db52d9e862c1cf3580a1946cf1

withheld proof term: 2 290 736 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionPoly_eval theorem, CompanionDiagonalization:100

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (x : A),
  Eq.{u_1 + 1} (Polynomial.eval.{u_1} x (FBundlePoint.companionPoly.{u_1} b))
    (HSub.hSub.{u_1, u_1, u_1}
      (HAdd.hAdd.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} x (HAdd.hAdd.{0, 0, 0} m 1))
        (∑ k, HMul.hMul.{u_1, u_1, u_1} (b k) (HPow.hPow.{u_1, 0, u_1} x ↑k)))
      1)
```

SHA-256 of the statement above

8f9791f44d53728df204bb926127f8d6159e3780e42d77caa6c52e4adcf21b

SHA-256 of the kernel type

97779ed67e79c0d2aca488e3d834667f2eb2cdc45a6e1c2e5093df752c3a2830

commitment C

182f5da7c7ede02df1d1122e4ac61250564ac43ea149b40fcef03c095f838d4f

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first message t

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response z

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1d92b371e55a13b603b5fff3fd8f05a8bf6769c61615a4d42771f7ef7796ca82

response w

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Fiat-Shamir context

1e8ed3553c80e8a8adaca5aa03de1c767c1f17c6ebe6be7343cf6b0c178535c23

withheld proof term: 180 306 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionPoly_firstChern_eval theorem, CompanionDiagonalization:486

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (c x : A),
  Eq.{u_1 + 1}
    (Polynomial.eval.{u_1} x
      (FBundlePoint.companionPoly.{u_1} (FBundlePoint.firstChernCol.{u_1} c)))
    (HSub.hSub.{u_1, u_1, u_1}
      (HAdd.hAdd.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} x (HAdd.hAdd.{0, 0, 0} m 1))
        (HMul.hMul.{u_1, u_1, u_1} c (HPow.hPow.{u_1, 0, u_1} x m))))
    1)
```

SHA-256 of the statement above

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SHA-256 of the kernel type

179e76544dc83b26ead74db07f947765bdb26175b39bcdee95155a17bf5793c3

commitment C

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first message t

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response z

1d26acf0b44bf8cf94b54541832be85f06d5a9209eda585eb7c61f0067ae8c2a

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response w

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b52f325a98cbda577deedff54f8f5e54468169614425e841e6f45e33d85fe9d

Fiat-Shamir context

954c530ec23411ba5b8de06899a29c2679fc52d52985a33c45b117f438254ef0

withheld proof term: 79 616 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionPoly_firstChern_root theorem, CompanionDiagonalization:496

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (c d zeta : A),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} d d) 0 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑m) 1) d) c →
      Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} zeta (HAdd.hAdd.{0, 0, 0} m 1)) 1 →
        Eq.{u_1 + 1}
          (Polynomial.eval.{u_1} (HSub.hSub.{u_1, u_1, u_1} zeta d)
            (FBundlePoint.companionPoly.{u_1} (FBundlePoint.firstChernCol.{u_1} c)))
          0
```

SHA-256 of the statement above

a789d4f1ee92ec2c6ddbe424aa25b3095199ba31ddd8fe975891fb56ae8a1b23

SHA-256 of the kernel type

378ffad7953162e61322f0786e6d50ad2bae6adc1f68ddd708b2a9955fbba6d3

commitment C

a1e69d7a333a70a458c7f95ff209ea9abf80741465e65d616905df0294fc043b

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first message t

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response z

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response w

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Fiat-Shamir context

8565ae832fb901716a4aabbfd703719d32f4b47577967877dfeb06c8795cab0

withheld proof term: 3 003 639 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionPoly_monic theorem, CompanionDiagonalization:105

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} A] \{m : \text{Nat}\} (b : \text{Fin } (\text{HAdd.hAdd}\{0, 0, 0\} m 1) \rightarrow A),$
Polynomial.Monic. $\{u_1\}$ (FBundlePoint.companionPoly. $\{u_1\}$ b)

SHA-256 of the statement above

75ebd916386dcc69dc74a065a605c8b83326faa7bdfa9fca9974f18b4328f281

SHA-256 of the kernel type

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commitment C

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first message t

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response z

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response w

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19c5153e24ad8fae9833b91a29d3c9adbf31eb3fabc6ba8f6076428df8125346
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4b68b159b1a0ad990514766c22f89307300df8c5ca8abfef96085f2522c6666d
fc3074bfb0696885eb0ca82833f18672b8fb01b358bbbe5ad3286301dd3205ed
b945ccfb070575050ff3048d5e54ddd7444d99476743ca6c5a5d53109415397c

Fiat-Shamir context

cb88bf2c0a58bbc75783e7dfafe259a4926e10f22f580b94b0518606b67495e3

withheld proof term: 174933 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.diag_conj_diagonal_mem_of_nilpotent theorem, CompanionDiagonalization:441

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (I : Ideal.{u_1} A) {N : Nat},
  Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} I N) Bot.bot.{u_1} →
  (∀ (k : Fin (HAdd.hAdd.{0, 0, 0} m 1)), Membership.mem.{u_1, u_1} I (b k)) →
  ∀ (zeta : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} (zeta i) (HAdd.hAdd.{0, 0, 0} m 1)) 1) →
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    IsUnit.{u_1}
      (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑m) 1)
        (HPow.hPow.{u_1, 0, u_1} (zeta i) m))) →
  (∀ (i j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Ne.{1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (zeta i) (zeta j))) →
  ∀ (mu : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  Membership.mem.{u_1, u_1} I (∑ i, mu i) →
  ∃ P Q,
    And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1)
      (And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1)
        (∀ (k : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
          Membership.mem.{u_1, u_1} I
            (HMul.hMul.{u_1, u_1, u_1}
              (HMul.hMul.{u_1, u_1, u_1} Q (Matrix.diagonal.{u_1, 0} mu)) P
                k k)))
```

SHA-256 of the statement above

f5daf0cf6ddf71bc64c12d480d0bbc1bab7cfc7c5b60a2535bd8d47f9bc3bd95

SHA-256 of the kernel type

e717e85ca450239215ac9907b4f99e8c3f8229c9732ccb0f128f99e6a87adbc4

commitment C

9f466353d449b58062a44c8d761334819b24d6ff1a68a6421abaf6b51d50f83f
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first message t

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response z

1b1dae8b9a333ae8b390926e82253de62a069c9c21cc1a050ce818d5f37faf4
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response w
6127235479251a4ce84bd5fe6ff955141e1200d256190c44c46deeadc1232d40
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FBundlePoint.eigenCol definition, CompanionDiagonalization:133

```
{A : Type u_1} →  
  [CommRing.{u_1} A] →  
    {m : Nat} → (Fin (HAdd.hAdd.{0, 0, 0} m 1) → A) → A → Fin (HAdd.hAdd.{0, 0, 0} m 1) → A
```

SHA-256 of the statement above

c6194b0d6a2f8af7b07d5245231de57bd5bf073ba6e79252e8581a6963167a6d

SHA-256 of the kernel type

14cbad6b44f030e8836c8c5c85d6b40d5e4c1bbbf8bd971e89dae62b248540f6
commitment C

4bfd0488ad35109f48813d408f70245cb37dff6cee34d7fdd6a5a2e69da3bd78
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first message t

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dbfa1f6549520852baa06e856ceb472e7e059ab3b3a1bea7a609eb3abd5c1167
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ad54662b4a1b0fddeb84da11fbcb81de7431254a5c02d58d97ac53faaced7b0
response z

1149906839e60229f9de4fa8c74910169b2676894394cde73a1995cdd7b75f96
214f12fd65f5c14f507c4258580fe2ba049361201b0f9365d2aa0247fcd0bce7
3c3852d33f31f46cf65c2d661574cb10bed904ce66dfa24e53dd06d5adee2f3c
742fd31225f38b0a4b52185515745572545ba576af5f6918ba90fbffc3aa5c34
004c00b9f99a845a1548890462d1ce1938ce960f9d08cda89eb3d5c22257d504
14b097aecf87ccc426eb6239c5bc2153e947be244004fc71c89143d8c1e1c824
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response w

51f073cb7c653fc6e9b4bd7094d0a30b3e849f13db6dba3f6a87f7b24cb25168
782b66112a58c48765cc32f5de5fa9ffb0624c191eb2257daff1c515b42d80a6
1d3d0ad4c6db65a3b6737ecf0343b5d2a726a381ca096ef2ec21fc0a1affa54c
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Fiat-Shamir context

fa0cd693c17f8091a16a485fb6714107889259db9068d56210fced8eefac0869

withheld proof term: 4146 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eigenCol.eq_1 theorem, CompanionDiagonalization:0

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) →
  A) (x : A)
  (j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
Eq.{u_1 + 1} (FBundlePoint.eigenCol.{u_1} b x j)
  (HAdd.hAdd.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} x (HSub.hSub.{0, 0, 0} m ↑j))
    (∑ i with LT.lt.{0} ↑j ↑i,
      HMul.hMul.{u_1, u_1, u_1} (b i)
        (HPow.hPow.{u_1, 0, u_1} x (HSub.hSub.{0, 0, 0} (HSub.hSub.{0, 0, 0} ↑i ↑j 1))))))
```

SHA-256 of the statement above

62c25a26636307b5338df4c9c9685a01a4df3f5f576cc7929435b24d443c702e

SHA-256 of the kernel type

24f79f889f5e1e3ab61a789224815f06b83c6ebc7b2a624a6d7e3fc20f1e17c7

commitment C

6c123dbfb2df640ec6126d0d8ab6b4e2c9f52f5350d538753d1b6d5409b6b34d

14ad8a0afe05302bc3ef2ff822b3c99d1a0a10cc4195aabfc06a474e22c1febb

8621e1f1d9112eb66f54819597dcd2cb7ce267f19a4040258e7658d31e38ae6f

5c347d95a0adb3bf79f672bb65c9056d52d52bheadb0d7a2d9bdf0040c6ce22

be62eedd398b5944241c2f664bb4e780289b922ca68ec34bd52cf0cdaff9c001

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first message t

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5e9e99de29a6857812c9c542794be97287e03b27dcb3babd8f368129a1646c5e

97096d63f78bc350ae79731cb724de7b001db516577b2dff90b52ca3e37f8a20

ed41c90945b9dc3bd7f3aaf05549d56f3ee88617460ddaf744c30c573305e5e8

b985fcb0dcdc610ef36b4491b98bffd81b14a06b541a7f96f4f01ec1925d14e9

dcad33464e06e5e7bbe8c5e3ee64f731fe6934e5892b23bf6525495b15b92c5a

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response z

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response w

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21cd30f797dfb01f07fdfa5cb7f81014df77cad9c8ce40ca7862597400f7a86c

1773fec2f09a022949b13993041a8ea9d62b85c914134f14aaaf9b00337440a

2c352c8968032de884f0ef0cb10c883ddc45746893be08e9acc83b0a7e3bb2f0

Fiat-Shamir context

dcc22833f495d3283727336c82e4fc90180605342c34fbf3499c76095589e72c

withheld proof term: 374 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eigenCol_last theorem, CompanionDiagonalization:139

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} A] \{m : \text{Nat}\} (b : \text{Fin } (\text{HAdd.hAdd}.\{0, 0, 0\} m 1) \rightarrow A)$
 $(x : A), \text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.eigenCol}.\{u_1\} b x (\text{Fin.last } m)) 1$

SHA-256 of the statement above

29c596a802021cd56460bae8178d6cce76bd6fd3fff52e4afdc3e3959499093b

SHA-256 of the kernel type

a4da18be9cb5de6b9ccd27093fc5f936064cbb152c672ad916f3b21e5ca987ad

commitment C

c8f102bae1d8a750be93620a3a9511029b428742109454124f93c88618df9cda

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first message t

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response z

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response w

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72be1c4baed8202130e4c77c80827b3184b07d45b3d3f7b0a8a5d40dc6e57c52

Fiat-Shamir context

4fd0c60b2a6930eb635f433d9a3160ad7a5d6e9cb66f32df03e3c3c8165ef8c1

withheld proof term: 48 425 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eigenCol_of_lt theorem, CompanionDiagonalization:148

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) →
  A) (x : A)
  (j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
  LT.lt.{0} (↑j) m →
  Eq.{u_1 + 1} (FBundlePoint.eigenCol.{u_1} b x j)
    (HAdd.hAdd.{u_1, u_1, u_1}
      (HMul.hMul.{u_1, u_1, u_1} x (FBundlePoint.eigenCol.{u_1} b x (HAdd.hAdd.{0, 0, 0}
        j 1)))
      (b (HAdd.hAdd.{0, 0, 0} j 1)))
```

SHA-256 of the statement above

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SHA-256 of the kernel type

8a4e600d45440ecdcd3c1be5bea9585aab7e8e2eab8ce499f775309318a4c42

commitment C

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b0c48adac91c1b39d23ad21f0fc7c2af24a78253b652eeb10d098d54a2657eb1
e74b6e1cfd4b4ebd840a1d89dd72f862a827b3993c8b6850e9369bec47eec5d

first message t

1b7cba88a3a513ebbb7d06c4b6d8e10eff4f608015597b35768b67baefadf5fb
a6791ac8f7ac21322de61298a88172d776af5f0e4daa6a5a0ca42a95e0c14851
1fc54e047c19c38a56b44b63f73cb24189829f34351fbcfc4cbd5f48a4fe9f90
75f096c560effc32336230fe09c5edf8f416591057d8685877bb562b9e80cd54
8c976a844cf1c1370616ee5214abf12816d62f8c3e3d0b8fc4c548e456a0c4a6
961fe9639a18f759e33792dbe34a2f9cd87520707aabb934fbfd13b91e63440
4cd65645cc598284656560ac14639ac6da2b5aa062d011a7032334902d1a8223
59d1bc51fe257c4e4db4d8fbfca58ae7e4b6db8e6d697617e69ecc2569a77d05

response z

4d1fca704c74302eccbb694dd10357144f921a4b1de39b4ca5cd2daa631d72ab
47dd454037a1e1546fcf20dd866e1af7d71b6afe0156a1da52aff2caf116fb99
e0e741cd5015b7b84fe5dac48a6c56f9fe2ecf51f2ca9f51f244bd7099310c1d
8460001e25b3af596870257d772081791d63bed96c086e0999184c83c3e50210
2f147523623b9d1df018cec6ed145acfc67353582ddb6f8c6a39b12c2d43b68
9ad5afa17ca63298876a8fad3497d2c143df349b2641e5def250239d3adcc41a
fdb999d589eba9abf86b5674b3458a3b33cc7ea4b974b4c381ad35ffed0681d4
4f294b1db7623dca5ce5f44779df096501db50104c954049f15a242cd238f71e

response w

40a48eed922de74639aa48f0ebe9e68172747c56d302f7d5647fc6618410cd7e
e06958b727d7c6019aa055ef7f23db3b7d2a12867a1579551b26fcd331c19c24
5f1f607883b02df3c1d4fbec1e4ac070fbbf8b2adfe0d633ca911531c93d9db5
dff20856989a3930b6fa8518eb32adc2f7fce37c2915f319679ad45cdf0dad17
9ecab3c6508edf1dd0a44feeb740aceee6339e47531cef9502a751267e02d9cc
41116214e9dce0ff01621b009d783b0c01cf94c4dca202b72c794c3a7a371c87
54dc356975e1f84ec5ce8a26a098b70df45acd9e8d87fa5f4c4f6ec9197b66bc
cca44692594f235e15443907e1b6793307385519f16e7dd3eba10d416ee5ef7e

Fiat-Shamir context

afa9a73c90a33463ed5c1f27d7e3290811e210e50fce9a5d8a44311dd155a080

withheld proof term: 1 293 598 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eigenCol_zero theorem, CompanionDiagonalization:186

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (x : A),
  Eq.{u_1 + 1}
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} x (FBundlePoint.eigenCol.{u_1} b
      x 0))
      (b 0))
    (HAdd.hAdd.{u_1, u_1, u_1} (Polynomial.eval.{u_1} x (FBundlePoint.companionPoly.{u_1}
      b)) 1)
```

SHA-256 of the statement above

46809071217f222ed4f3e39e9f162d9df61c4e54af1bae1ee7326e0db3715c50

SHA-256 of the kernel type

e6ed14a411b1355025ea85619b18d54bb983d31b4cb5e7d00fe423edd348eb3b
commitment C

3af25619f541ceb22e8ca3239c07132047fe37a1aa54bea1b1c78ceec1dc0665
3b2ff7b74550547c8a36b6eac6e496d5deb3d4aa28cd47e8ce960c9a8331e0d
37f1bbbe14f22f8bddae1ffd37890c0eaefc2edf293bf1bfd4698e6fbc34baee
1e5db4191635c9fd09f4603d43182eff434afc855588d2c7a2108e79ca0bf765
d21ba5c29f63d21ecf3cbe8cc81ae87ed77138ed3c8481302a3c2762d25d99ad
5f7b0a3bf7005124d7f66886c97ad277cd1232e04637f80abc02953cbc936716
bca1ac786c16b68eed48800422f48943a4e8e0c55dbef93b2d8f3220be921451
6b594d8e8985a6b09f7555f23452e0c1c88c150156e6db3ca7dac0d8c038969

first message t

9defd3379b5c50b9d453393d957f2acbb5b840d87920d5cd8b60c9714059fcd
b90e598c8afc897417611cfadbab427e1a7b40a6082d0a6da63342809a1acc00
4c1c8769c2e7ddb6fa9c9405016eb913089cf62fb53a60625a7d15477271a0b8
96004f695cabcf7f0f1625763b8e2f9992712d05ffb2c5ae40c730c910a6b1da0
0fff82cae522083229fc80d79a0905d82f862ee205c57d592c1d821941056c9f
988dea0bf0756e5e364cd90848102882b53ca2e440a6826c2fc3de4f0561b39a
420cbf157841d1f822bd337fc29ad571cb3b1ee2005abaf16b7341f351b52723
c15456b0289a928d61f14694decfffb0c85f53cbd4fce2d15b3cf27ee9a303663

response z

7abd0436d0a44bffa9baf4becb8b2c38f2016a4b2fde2c3e5116195a91b5c6afc
fa28c3e1f21dd3272f8d50c9a159dc596ef393c13f1f526815e2f523b349d45a
5097a7a6ace022ce52956c8957a4b6fa317afd392d9a7801aa59d7d985d52c05
61182af72f62e9b0b6884c1e4c656ef19eaccacf13b2529cb1193ebf5635d4cf
87bd66285c9b1e686d43444680399ade3d748fe68f02bee47518dd371e599b2f
537e24074b98596db209f1f6eaa6b79da6d0953adb2f4848638bcc9916d2f74e
326725db3dee365e06dcb0679c77deb06d730c87f2e8c5553f8874a8f04daa77
d32d0b3392453e01436adcf7f9f6392df70e782fc7494d0ce1a0501cd2e9c4f9

response w

5144cc9adc1ee2de1b34528f6a7a16d18d30f7102177ed6b402913668bbc9e77
c08897590b174d10a03c416db5621277243970608f42ed94c8978acde0c60c30
0c1c13f52317a2384206658d2acffb8e34a604a029271c39e545b3f7b52a5f70
5b937bc0f024bf24c708bcf3bf2bbfa491fc4ef0e2f0dcee34e9fa1600be52de
b47f2fb1ca26d68202f4948541748b233668b51c072a7e4a8cba00e8f8752f72
147b4d0f9712cfd28d0f6976ed53eb4af8d6d52a90a91d462972a16cf00e0771
38d179705948228a133353ebabe3cdaa0bfc585701a39d1d16397eb1d132fe90
1e8ade15da0954727d94b14642f4306e3403cbc2428fe0287fc6c6dcf31d5ad9

Fiat-Shamir context

dd17d89c880ed1b30ef120e6b50ee9592ae1dbc6e7fba934170ef1fc0ff2621f

withheld proof term: 1 203 910 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eigenMat definition, CompanionDiagonalization:217

```
{A : Type u_1} →
[CommRing.{u_1} A] →
{m : Nat} →
(Fin (HAdd.hAdd.{0, 0, 0} m 1) → A) →
(Fin (HAdd.hAdd.{0, 0, 0} m 1) → A) →
Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m
1)) A
```

SHA-256 of the statement above

b218848a9f8c32c984dab7d0dcbbf4fc02b183ea3da4c9587f972929d3e5442c

SHA-256 of the kernel type

ea0dddc553328a5a5da04cde1e8914a0d15f8654f2643f3b49cbafdf3a27d7da

commitment C

6f5ea51651ac67379e1c889e9430210bb9f0b99bf5d94daf5b0a93e509d69301
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59b72776656d17c4729f1f6a7f8bc890906d5a411ca0be3c93bc1739a6b468aa
79803f89ef6a3b2b14008377e920a2535eedafa53a287ab5bdacd12c249c10cd
772d17beac9310b419cc4a896f80170d9ece278dcf1b3b90bdd1b6a285c3a56e
0be94546aabdb0ed41b04882564ae0502e71a98cf61277802723403bcbeaa
6b2e73df4c2e13a6f7646e0191611c0e24dc4ff9f1eaf7a5368c953b6650fd16
17870deba95aea2db0f067f5f04f2433705f83aa44639b026ab6d3b8d24ee67d
first message t

ef67ce4331390e37d2b08fba8cbb107978a76e9eff842b02d3a8f34788da8963
ce23153f61f82f1641a7b1fdd8663a077f1e587272fdcc57dba3a9026367ccaf
ad7249912a3cb82265aee0a3b10fc85dc35c1e1a7c66cf67d6573a6461543265
0ba6cf3f2c5fe89d8b287b1438ff7ad9f584b2151eaf5759d1639da8a37afe22
6c8973bd174e5d69dd0aa800bfb7a1c2a0b864c2c7e3ef6faefd613e196729dd
248db519bef0162003dcd226b6d365276f028a844baeafc319e140a3e98d30ec
31c4bfe117e2b178d9b5a0359996056f58bf963a754b8083a53d164faed18423
6808e3465b8f2b778ed6423e745b8f874bf9e31a1c4b859e2f5b1d42496a36f2
response z

55df423b4d2e9b7f08ad0f77f8450a45df3fc3c85286368220bfb9078f376894
8e3593aae6786c11c226b30239ecf736a97a21afdc171afacf4b853bd933d13
b17d42393ebb8151f9b11cf63e07fd91c3fdf3f2051c3bb645398c304dc123e3
c18a19fc58eb19807521487e8531364e4908c3848ca079c467a3ef19f1e61766
c9b9d4a802e8f561ad3222d9d4c4fe569d79d74b4602c4c78e23702345d549a5
c6389c0e2d0130bd885db2d460dce90fd863bc229f83f4e9f63dbfda0dbce4a4
4a092465192f417e1a2451dbbddd71380ddc4626c15331b63970d8df515d33d
4ec00a2669afecce08a3550cd48858cc59535e24cd62d442955a1b0688a3559c
response w

41a2cb2bfb8ef33d6ede43aa0521b55089c46a2e78c3e5e5d620934eca0dddf66
96684e7e9b4988c3eefe0d749d30a59842f8bad825cd71376ec1bedbb6894db3
3fce983b3591262705f84dadbf1c01f77b1efd43cff1a44f2ec15c7348e91020
e35d6bd01eb153f79915467c5790a03bf447ea4dbe8b7963b8568b10ea3c3d00
c27ce0f82817c5696dc81b0850c079c92e3e51a3e9c4989525ff36cbd7505782
959e5980f697c1d31fbc35288fc607c76cfe3df1ef6864543f3623057fd87cdb
96fdeb780a7490c25999ebf86f847bef5b741625aaabf5c46c8a8f7b89dab3d
45101660acf7d31ab93b87b214ac564df7c7628b58b48260420ea2fd76363dfb

Fiat-Shamir context

700f66a37d78d47f63bdd00bcc8dd2e25430235243f19f1e021ea926f7c4c5ed

withheld proof term: 4014 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eigenMat.eq_1 theorem, CompanionDiagonalization:0

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b lam : Fin (HAdd.hAdd.{0, 0, 0} m 1)
  → A),
  Eq.{u_1 + 1} (FBundlePoint.eigenMat.{u_1} b lam)
    (Matrix.of.{u_1, 0, 0} fun j i => FBundlePoint.eigenCol.{u_1} b (lam i) j)
```

SHA-256 of the statement above

ee2932a2425aa9f7274cbe843c9420d33007e4be323a4025d39c8f461ba9847b

SHA-256 of the kernel type

a174a8702be89ef5c075e4463e7d692b59d8e2901b332fb5e9fc68878c4a4cd9
commitment C

4e89eeb6c8d2f018ea87645931c07482cb535d4ae0e30b7791e96b61ec3d0942
78d64aa283202e21c9270f3fd5d2baf7f7be324879aca64ca45aa2b4d7fa4d698
46744e9337e9c5139356b34d5b4cbb4225ddce75aff62b2fd7fc6e8c51982515
43693feb90660995f423557fd9b3a59dfe125ed50e0f40d94720d6706d722dbc
ac12dba7e7894b2ac9abc754b865a017c6c290fafa88eefc3337c4571475ee9b
502bcabe6e005478862d900e6be56aca0a26e00603687483c30b95b83763da23
c70a2a05ff840923c362d60d9174dda3c8073e16f917ba8ce0c7203499781b0f
66c36f475c31cf74ca7cb163eb97d7c0917b222f6b6e35c680d93c3addccb15
first message t

e5897b28b30da447a38ecce9cc592c9aacfa5dbff7932b38d0e1675a4301684
15a3edf07562f435f980f76f5c92afd2c45bbe3fa1355dbfbb0c8f0e16e7826c
078ea50adc7a4623ad0c6e59a90e25b9c7d2455947e48ebd3997287606f10ebc
873976a7f0fc6a3f2c63a29ee76cbdedb552515015addeeca673d71b12757888
985b1c8b545c72aeb8bfc0ca5e0df11155321ff0d294ba7ba84195a7cde3590
5985d565071ea754babb9c93f66276aa6216eb0f46538595efedc92aba068a0f6
e675eb036547c6bd53b4eeb1cf49d15143c84734564683ce9cb63a58d81dc724
e3111f72e40fd24a74b393c99495bd48734181f2d448b412e25e5eac0173e4b
response z

5155c6875902433f09a1b5a0956ecd47e0f4485036e17fd1cc620dca2b6ec961
18468ef6eafa18720915cfaacdab68d53ff3fde1d3b2a203f27e22e25c8cce82
cca77ac53404aaa2a5ab4b597221735ad2afa267e3542a1853f0f6bc6fcd96e9
193ac9fd81d7575795cc1d2b8988aa7a03fd2e061fea6d7f509f1d4714ee4493
d956d719e85605b4104a4a7029399748c45544211fca4e5688ba3826abc4a7f2
7cf7e30e9eb96049004e4f180a048054c694fec94f3cce2b6dbce00733db57cb
49f089271610321f76de3b40f53fa71e1c6e2c4281efd5867cb46587f5efde89
5a215a2f74dd5872c32e0510b1ffca025ff02b2aa087af28467fa6c4022e742e
response w

4eba184137eada20ccc2b6242f0b52a6780abaf57f7d9da21fca2c1b938474f
a92cca844b906bb23363dddc062bf513f4468585c821137082d1d7edef05b8fa
76c715198825c89167aa308487eaa9b5611e4cf3037c7f9fdce6aa61d75ee198
63715f66832462ce941747ec7fec30d0e9dc48b37c8dccbe0bbc48205b0bbb9
0ef5cb978805c1c9d0ec2058b43cebac8319aac7e5c1c7940becf10d8760b09a
c565df53037379cb35521f9aa90370390c354390fa41ec0211313ba865b1e15d
d6f69735bfdee785ae27d34e7a41c5fde0ed1d91f33358b295429788c989d406
cffba620db73ddcd0adb17c8305ad77bc7aa07497cdbcc49eddb13da5d4e4d51

Fiat-Shamir context

0ae47735df138945ede81b5593978f310421be6febfdadec89408612b1e6dfe3

withheld proof term: 622 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eval_sub_mem_of_sub_mem theorem, CompanionDiagonalization:372

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (I : Ideal.{u_1} A) (f : Polynomial.{u_1} A) {x
  y : A},
  Membership.mem.{u_1, u_1} I (HSub.hSub.{u_1, u_1, u_1} x y) →
    Membership.mem.{u_1, u_1} I
      (HSub.hSub.{u_1, u_1, u_1} (Polynomial.eval.{u_1} x f) (Polynomial.eval.{u_1} y f))
```

SHA-256 of the statement above

f409011350a844e00b0f58cbfb6675315c8e85f40d4f9817a228229d9b440daf

SHA-256 of the kernel type

389d68e782dba222e4cd477ebf18d384fe0b4b3122931ecdfb621f12f39ba972

commitment C

6f448cb39d51c9a2a822a0261765f32553a35cbca7bb4fab5334dc6bb3037bb4

3ddb5ea5237e0facd9fb507b0c91defb520cfb92b3558806749f0c8e1e9555b9a

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first message t

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4475f557a0fd43f9ef8d8de979d857ab74eab02ae17badc470237327c98d15ca

6d27575f0eb9a587132459b170dba665383511970c140c78ac418b07a9bd7483

b4bf86680abe8ad7965b70c21c0014cdd9f35886f7dcd00ff0e56afc65f040dc

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74ee5a8dfee7e64e14bcbe1d8b33d302fdb2a9e6bb7ae94251929a02c6067895

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response z

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6205e08809144583625adbfa694d73ea8e2bac0bed906e63b42bf1842c614d0

fb982f42b4492d172b8aca3cbf50001e25a22acc0c7991dbf85023f20062f0fe

72262d24d4032ae548cf6f278876a9eae72f39cd50e380f2090da16fbc450313

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9d2079c7ff601fcacc7eab6da801fbc15dc6f83886115d0d1613747007592750

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response w

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ca84d06b1e2532e7e84d10527a5c358a8ec8b9511569ac65690267fa0c8d98ab

bb63763a58a8b6f3bed816b06b1b6f80bfca784c07dffe6c63857767c3358c21

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6db05827c591dbe757a552b3633388dbb0f2b71774c713625b5ce5b84463a8d0

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Fiat-Shamir context

0f82547d3af80bf34a7e33c6f895b34e4c8238444590664876b99e96f192fda7

withheld proof term: 17 220 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_diagonalization_companionK theorem, CompanionDiagonalization:424

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (I : Ideal.{u_1} A) {N : Nat},
  Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} I N) Bot.bot.{u_1} →
  (∀ (k : Fin (HAdd.hAdd.{0, 0, 0} m 1)), Membership.mem.{u_1, u_1} I (b k)) →
  ∀ (zeta : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} (zeta i) (HAdd.hAdd.{0, 0, 0} m 1)) 1) →
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    IsUnit.{u_1}
      (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑m) 1)
        (HPow.hPow.{u_1, 0, u_1} (zeta i) m))) →
  (∀ (i j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Ne.{1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (zeta i) (zeta j))) →
  ∃ lam P Q,
  And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1)
    (And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1)
      (And
        (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1}
          (FBundlePoint.companionK.{u_1} b) P)
          (HMul.hMul.{u_1, u_1, u_1} P (Matrix.diagonal.{u_1, 0} lam)))
        (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
          Membership.mem.{u_1, u_1} I (HSub.hSub.{u_1, u_1, u_1} (lam i)
            (zeta i))))))
```

SHA-256 of the statement above

ead5cd9235a72130b117ddaf024f2023755aaee4da6cb563e3f8b8bd24ce5b1e

SHA-256 of the kernel type

9d2f26597300cc5f6a07af52456c678f6ea7022f5315bf6a42a58b4b2080ad51

commitment C

689261c851462ec242bdf83aeb34c6b6032a7f167abc8ddf5e7dc67fd5745b0e
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121141457484d143930a8cf9409ca91052c015e8338ac81051591c0e62969e72
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first message t

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320fa69a4f690caae1029d054a000789ea793927302061722e51e1a711b4e68e
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083456672a617c402d7dbe7caf4ab2d34a246493521987ed8a05997956342caa
cd340c2144edfbbf24f6f502b163b7224b959a47811252416555619edd83cbf86
9432242d8aac04cc493b21f21452c756eac030d8f8cc2701f9dc68dcff541e6a

response z

6c6469c730ea546d44c1fdbcd1d6c38a27c22e1adf6e44696c20a3499682a92a6
f31c9c0ac1a5dda2eb3d242ae579a66fce2d0eaac575a073bb3c7f6a7f7af769
3963a0564023b18b1892eefd573bfb4554b586350766651d94265ef784e7eb4c
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4b2efbe0d5174cfce09a0e9836bd4cf80b20cb294b7afb1a7d28fab20be3692a
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1dc9a367c66097ec4cc4cf70742ce480e60991324bf26f9af31dbbcefa50a6ee
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response w
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FBundlePoint.exists_diagonalization_firstChern theorem, CompanionDiagonalization:510

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (c d : A),
Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} d d) 0 →
Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑m) 1) d) c →
  ∀ (zeta : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} (zeta i) (HAdd.hAdd.{0, 0, 0} m 1)) 1) →
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    IsUnit.{u_1}
      (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑m) 1)
        (HPow.hPow.{u_1, 0, u_1} (zeta i) m))) →
  (∀ (i j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Ne.{1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (zeta i) (zeta j))) →
  ∃ P Q,
  And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1)
    (And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1)
      (Eq.{u_1 + 1}
        (HMul.hMul.{u_1, u_1, u_1}
          (FBundlePoint.companionK.{u_1} (FBundlePoint.firstChernCol.{u_1}
            c)) P)
        (HMul.hMul.{u_1, u_1, u_1} P
          (Matrix.diagonal.{u_1, 0} fun i => HSub.hSub.{u_1, u_1, u_1} (zeta
            i) d))))
```

SHA-256 of the statement above

c8a1883caf2e2ef55d46dae6555111e9b48f5b0b2091a3bc4445f1aac3d178d8

SHA-256 of the kernel type

e7ddf9a7bb481ccd5cd7cb1fe860b0b6f43e73802ecfeaa657f6efa590ea

commitment C

88d01144494cb67156944a178271f66f8a027f3b6d5c27206d57a45e68d453a0
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first message t

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d728fcfbf7ebdefca64385f6395f0d3ff42cc40fa8632faf31c7e08b07612f37
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5cb0db766abcbb54057eaec4e6d328136ed59db1762e9515c274e9b0dd321a6
5d253ed55350808fddcce5063c2c3d4a9b42b51cbd0f205665d64ddb8ad0c740
response z

536c3003ae5f40c09a8d54148f07ea59bb96b74043711e830f4539925e994e0d
53d5cb5bfd8c3b05c4d3eee36458e0e924dc114e3c9663f60b0c7f85f77d5f0f
c10e8d1b1bd8873db86fda9b506c6b9710a089966ade2a1d1dcc25befce0847c
7994696af5b10fffe4838a153e47cb283278b79d389aeea82af8caebcc92042d
7510deac27efb6648c59fe25bf1418da3c5c624f7dc4aaebaac998aca5c8456d
e3f9868482ef2b78c567f0990007b7ac05ab1b5ca6222b57d447670cd80c4087
1f1e1c7833cc80c0498ce96dbc17b75e950f2aadf2a6040c63a21358545660c6
0bae9f019e7386102157589e99df4d69996cd39a3c86ae6d0fa7850937b3c97e
response w

3c7bad5050a204582e6fb5564830af2574ed5aa72297807c684d0f1a274747e5
67a862c9df1fe57e1cb77c7831d79d2f93dc349ef9c72acd3e45f196b0b4cf82
9c351297fbbf9b59ad1c0d56bc667fb52adf28b100ce0e8bf59f52c363f35ee1
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b11bbf9088964c02b0d723aed4390d7fa78252dde9d606bfff700017e4e08485
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fdcaed05eee8b9038c259c700b326cc5f019085b6e9b53e7f0442bd03d55836854
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Fiat-Shamir context

FBundPoint.exists_diagonalizing_of_roots theorem, CompanionDiagonalization:335

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b lam : Fin (HAdd.hAdd.{0, 0, 0} m 1)
  → A),
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Eq.{u_1 + 1} (Polynomial.eval.{u_1} (lam i) (FBundPoint.companionPoly.{u_1} b)) 0) →
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    IsUnit.{u_1}
      (Polynomial.eval.{u_1} (lam i)
        (Polynomial.derivative.{u_1} (FBundPoint.companionPoly.{u_1} b)))) →
  (∀ (i j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Ne.{1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (lam i) (lam j))) →
  ∃ P Q,
    And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1)
      (And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1)
        (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (FBundPoint.companionK.{u_1} b) P)
          (HMul.hMul.{u_1, u_1, u_1} P (Matrix.diagonal.{u_1, 0} lam))))
```

SHA-256 of the statement above

a9f928db04fdbbe1309774556f15630cedd7315709b1555b4382316fa9cdd3910

SHA-256 of the kernel type

3e4fc59c4edbd1d8cd0639debe6ef68db24c7192ef7c01e13ca6f7e564084d022
commitment C

daf382189064bf298d3a04faa451f430bc0c65b83ab6d43a45c3fbf5c9645986
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b611d9a9f9c1f87c36ee2659e316605c57ff8576e1487b1d545a3d2814e88c7f
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efdc3e8d31851f9c525bd7672c6402cd9c0a86f5eb9a9a2300df6716136ac92
dc8a845fc8ae6a0b9338f2bc51141efbcca05a287fc80a7ccd83275c08dc8b9
first message t

39e2a70842cf74f56d813e81cb92eeb4be0d3df3fcf62c215dec5943d9ba6d21
4cef32cbd6d575f01c4d2bace2c643e7f5bc32d62dac0b30cca7ac041165d448
8180ad23c232f5e12bad61685c6d1e3375e2ee7f80bf4ef4151fa037b3473a9c
ef4a7a196cc0cee20211e81ccf8c8e9834f3d4094353dfc4f25d467430d4eef6
8b06f8dc96a4f64262c6b5b241866e93675c42285d5a990761942c57db510b8d
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ff89e193048166db809ed6d23cb3c5732fede533b880fee75ce96668d1a22c2f
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response z

648d55ce0e3680421a86d409f2668b7dd310e356826f5f08f3f3c0af54628e32
d8bc7bf07a4ac363f3467d796427f09198bd037b050ac03c06748190b2384c79
954295c127e17c43110bffe148381a5652e8636d6283c1abf1898f8f2d661e76
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f096addebd69c163ecd3404b9217f60fe1bfee6217da8127865a02c91dcfd42a
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response w

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dcc6ebb1b0cb5eb4c237954a562b247123db5a34e6776c2a38ab01c3251a762b
c6becc702d47063a8cd66bf2634e1cc153248d2dc521cdbffa286fdefc8b8509
b9504c51d9a711c4f08b95a6030f3d41a5187ce77f85942b89db95e8afe59c98
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89b3e08f3cb2e62d743cd12462ba7a33498e3787188b8244178c0fddc1b915c8
a7a1a18c898414f2d7694dd639c191702f18df4112c072fb1d0265a49ed1738d
Fiat-Shamir context

7baa292dd126a4a7620f0006376f1038e03b196c4966b404b39aa485b896cbf9

withheld proof term: 423 228 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_isRoot_of_pow_eq_bot theorem, CompanionDiagonalization:74

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (I : Ideal.{u_1} A) {N : Nat},
  Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} I N) Bot.bot.{u_1} →
    ∀ (f : Polynomial.{u_1} A),
      Polynomial.Monic.{u_1} f →
        ∀ (a : A),
          Membership.mem.{u_1, u_1} I (Polynomial.eval.{u_1} a f) →
            IsUnit.{u_1} (Polynomial.eval.{u_1} a (Polynomial.derivative.{u_1} f)) →
              ∃ r,
                And (Polynomial.IsRoot.{u_1} f r)
                  (Membership.mem.{u_1, u_1} I (HSub.hSub.{u_1, u_1, u_1} r a))
```

SHA-256 of the statement above

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SHA-256 of the kernel type

671b2e638bb062d3018b7f969601ac9ed586e05ba916d43af02a2876c383f32c

commitment C

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first message t

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response z

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d80d5861a250fafcb3a46987389c7569fd06a2c3f74fa54efa999cd58e6f9209
33439e3badd33a86394f15577c50ae11cbb638c956b60807301181b85ef178f4
b66b28a97c628682462ae0ecc640b642fd4c75d961ae605ec2debf9152842fe6
afa56deaf6c3a1a6378d8117713faf45de2c852f5cf61fffa20a0e697eadc734
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response w

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Fiat-Shamir context

59bd91419ab646c654597986cc1c1a1c60b346bc17d9fab636fd1b9e90376ed7

withheld proof term: 34530 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_roots_companionPoly theorem, CompanionDiagonalization:385

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (I : Ideal.{u_1} A) {N : Nat},
  Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} I N) Bot.bot.{u_1} →
  (∀ (k : Fin (HAdd.hAdd.{0, 0, 0} m 1)), Membership.mem.{u_1, u_1} I (b k)) →
  ∀ (zeta : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} (zeta i) (HAdd.hAdd.{0, 0, 0} m 1)) 1) →
  (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    IsUnit.{u_1}
      (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑m) 1)
        (HPow.hPow.{u_1, 0, u_1} (zeta i) m))) →
  (∀ (i j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
    Ne.{1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (zeta i) (zeta j))) →
  ∃ lam,
  And
    (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
      Eq.{u_1 + 1}
        (Polynomial.eval.{u_1} (lam i) (FBundlePoint.companionPoly.{u_1} b))
          0)
    (And
      (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
        Membership.mem.{u_1, u_1} I (HSub.hSub.{u_1, u_1, u_1} (lam i) (zeta
          i)))
      (And
        (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
          IsUnit.{u_1}
            (Polynomial.eval.{u_1} (lam i)
              (Polynomial.derivative.{u_1} (FBundlePoint.companionPoly.{u_1}
                b))))
        (∀ (i j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
          Ne.{1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (lam i) (lam
            j)))))
```

SHA-256 of the statement above

5fa8b9e4558f8fd554e119b05bdc07dbad734b3e41084f303ea346efb7483e0c

SHA-256 of the kernel type

2ec090267e01f09cf4639a4a30c15ad85672430f0b400dd9e0a2004c3c77c712

commitment C

88e37833a8f2ee59086e3c56ca377990f40b9b7efb7bc5e6855f3060fa993011

f3274373c83ddf3e3e9ccfba4319075e7e3098b4071e792b46d261b1b85e990c

78d3cf006edef7d9bfa83bcf207b01cd87b502ef8bfdbc1bd65312d5177c5862

48f32db5d972985424501e4814537203fd268d7d4ba0f7d0920f2fc5eeeee99e

7cc0f4a6d187e066d1351dde661f18d79881b5553a6a8c1461d3f0baa5a48dd9

fc84fb4161253a8d06b7485de270103395d56158e99ab422cd6f892529322a5d

67e9934d9f6a0dfe0c88f41cd4f6b154cefe02e50dc613d351c522d2a4b37342

64772fac8693cf9e735cecc3d8df4aa9bf41a7e700bfda1964bc85e4be151f1c

first message t

f189badaa9c3dc990ecc41b54762fb6c5c084256aa58522ead1ae7eb11373a2

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a7ab259fe07d55580bf87b59d118be4282cc27372f7b2aa9d64c51195cea63ee

ae7350e65f90d2d7810f852e179317be529b25e848876a6164cb736e73aadbb89

88f473e5fd825520b1d0570ffd1b7bc210e603bc079157af371a66c4fd5857b9

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c0171cf0d4fd63990d4ff5f51b64b90588f1260786f6cbbc7d192d502d658b2a

response z

2303e9552fa9e6993007196daa10a507b8a40034162ad9147237e1a38769929e

3e9155b7d8554c50eb6a85145a8fb560b90015dfd9f6338303ae3ebf499a74f4

c541b629a3d5976edfe5c8501df41bd108f812c8b309a2780fc9381dfa2767de

386617ec7d9bbd6cc43b84b6d91a6b407435428396d826dd6897c1ebca91778a

82f56e1eb916ce0ef11968d6fe6612b9a38e9288db3a93b2e0d9a9e8602df03b

0fcff3586d1fbbd8e1a3bb4e1972d9bc2a598da25222a52aa80b023590a0847657

7d42509e8a87d45ce7e8207bbca9b46e2bb734e7c68cfc14df94b8b4636b943e

b30150bf40db702558afed24838ecd8d84ead2e62a4752c114d0f53029c7185

FBundlePoint.firstChernCol definition, CompanionDiagonalization:482

{A : Type u_1} → [CommRing.{u_1} A] → {m : Nat} → A → Fin (HAdd.hAdd.{0, 0, 0} m 1) → A

SHA-256 of the statement above

53c8c75c7a8f299ee9b70368d91c2d7cb95d80b5717f16904fcd3dadf21b96c4

SHA-256 of the kernel type

4116825f508e81e91e244b81bbf1e9019494094ff72507112c5d91422aaec3b1
commitment C

d6207fa25f58bf0df4ec3a65cceecf7d8e9d40f5e986f9e960fdd8296d3ab9cd
73ba5b150bf95c8c2f30c23ff4efdcaf99f647e2b24ac5c75ad39e03f0ff7ad1
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c1fd209c0098a487e066b342fe5691ea163aca9c1caae422ba6b79ae219085c6
83de7784af4072cd9d2562bd0913efbc7353d6fc7b150468e619bb4046eebd80
eb98d41be65ee4ca1f8856c6ff00bc3957ee629ffb16a837f5aef27116064fa6
62e7d8b91d595238a749c66caa962265f8ce2b3f468c29d311f8b2f94353d9e0
cd62a6f656e0b575475251262f35d00752dc5349d2fb9852325a2c59fb913e24
first message t

a76aba359955581829846b1d90243950e8661241b4313105d9d2ff50cf790429
b2260191025ae811730e56597bf0c29d9274acb4b9c7a71884d5a06bdd29438b
3701c5d0866bbddf97c961d91505937a9344173d02c6e9aad32bfb3b1384ec67
eelf357916f2b231dfba7e574255c102dc355965e9036798f912307a7bc0fb28
59ff6a6d03eaf97481e1495df5cd52c6371554b830629f4dbc918b498e517844
ed13d3257b4bb0c229ec82b00644cdd9d561b725a978d2d6d40505d26cdaa3bf
3c3aee1bbdc5e7999cd6ba01c7464c29ce6631caa7fe6d09a6614db11ab46def
595607b69aa3910d4255fa41f4a6ddb715394b78083eb8bc1369a98c214dd710
response z

380ffbb4e2fc72e405815ac1e7994362fc7e06b242f57286c419b6aff4e75da9
d5b5d2af22f51b246e8ed86ff4e30428bfb67ca6320960e861e83829583c76b5
a29a411533caeb41ca5f3962451281d1421bb67e4bdf29c8b39100875f7d2135
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d088478d6dcf9756e43575f8ac7d82fdc3de8ae32f99bf8ae8baeec1f6349cd5
da96057f3052684a15b059aeead30722139f7e0b2c18a05de642491f07e003db
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response w

1a208997dd7e9292b328410ae11e3dcb5fe4f1b599e75251c3bece89457940a5
ae68839d5c72eb0abff1b90dfcb89b57e384ac3b936bbe79143189a8a18754f7
eb55ef3a1b7cff38d17cbe1cb7f1c4fbc81df58a11197dfec0fd1766c2cefd07
937bcf5920ed334a3dfe03a5d2ce0da7b048628c526980464e197e9fcad65105
43d50e8d9deb3f7cc801d458078593d19eefd4ddcc282136c20650066beae509
ab26db679d4b0a25a06bead991038a8ff24417c6227ad1e039f7eca9d6f5f821
a1225a96e2edbcefa64f175cdd9f482bde46a40500a5365e30c410b77d9ee638
b6af02a8f8e78b2dffbf1a36573b407c9bf2bde20b6f45cffd4be35a0394eeeb
Fiat-Shamir context

7a392ef982c06cfd3c77213c70223194509948e941de4f4cc672f887affd0a6d

withheld proof term: 970 bytes • axioms: • kernel replay: yes

FBundlePoint.firstChernCol.eq_1 theorem, CompanionDiagonalization:0

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (c : A) (k : Fin (HAdd.hAdd.{0, 0, 0}
  m 1)),
  Eq.{u_1 + 1} (FBundlePoint.firstChernCol.{u_1} c k) (ite.{u_1 + 1} (Eq.{1} k (Fin.last
    m)) c 0)
```

SHA-256 of the statement above

022421509d0b93dfc3c5d9a5e5cea0f6a9bb47cf9bdc344be8b7472be029e18f

SHA-256 of the kernel type

d0b66f6e66ca8a5a3bdf38688e5579549cb0b7bafcaf677d50888d7ccd37270

commitment C

f9b328eccdc1e0c91450818f938e640c17939f4d9eabb2beaacc0baa45d30689
428df09329bbdad87baa1f262e731164db20f82a67d0fd6a5eb613cabbd9ee93
874792c152d6345d7add2b0b69d61769d5e881f847f7eee5be7a5a7582312761
c0d20f10710fb5861cb4d733e795a79b0917ebf58d5942627f76025470b3a22f
d8f3fec75b4f24f687b9b507bd036a2e4b06118759b4f5bbb0386b297aeabe73
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b88a2436e211e86458eaf41b9580c31b84b308ba9d2e52becebe1615f7154d0
9a022df24793da961c7c7b1f351a2496ee116d531691a1ed4cd0830707cb9979
first message t

f5d718488c52f4b578fca9a5436d450a49ccdb4d1a26b1584eff50e963679d74
1ea3f01316b58fa91285f9f4805128460b883f0e8dfac3357f79675e4ac51e50
784a3785b42d6d983e123d3e72221234cf9485a65b3fd526423d0c8c05548c12
f14c94151996a754a129cab4d5b24713a7b53065a068d53bc7531e1bc4c0cc42
ecc85fb02e3873bf40bf31af976a064b432d83b44c659494bccb144e525b3ffb
18b983e48dc5c29eac367a1ab7887183be2b73932459b8788bc83c515e0f524e
436c2ed1e316c1149ac24ea78d27c43696bd2d8b3d95e62e79db05092ba7be06
cb5a641ee07a12d7db5a8127f3e34eeb51b02b1ce68740d837dfbdb0e8f25eab
response z

49d2cbb151cc1a53cb276a4462ded173ff63a5a886af58e5a19beaad6745979
39ab338f5b7d9381102dc61e41a9f3e5462ed3569c2fb60ea6052114ed14c5d6
127df3a290c17a46f68ad6dd271a1d577171d66b2d4fca78c655c98fc5a8f25c
b4197fd0e19873d3ff18a6d8b5668e474ce550af107a679daa05918e00b26557
4879ef4d7894cf993f03e55be9429422ab187ba6171bb303194a53cf75e9df8e
f0ddfd43e3441e594bd4ebd8e8c7fc7c20e82804f0bcfb1cc9d6575ced2a41e
8588352e1f29bfd5f8d43bdb6cbc0ff01d758586864e182d5b3b8d6c169a95ac
1e5dc10ba889313e368c5a2e22c617fb9cf573aa5d6e28a22fe9ba1b5cb1b3cd
response w

5b71b86e8f75f6b5f76e714adeca6f8367a8ca072915d9bebf17a37b35685860
2db0f708bf825ecb01aca7f7a140d7ba5a77565ea25ad387d2e5aedbedfde2d7
f958789f9bb28e6e6d27944650317c47194e853ff2f90c929cf8f8c45e22f5a8
708771d2cc813c1b22681ac5670a88f08330b7240278969d75f04e13b3228884
ef31b9eacd2d164967b684066434712d59462f86b8099ce0b7e4ed316306640d
e3cc129804496414b05e88e08d04000c59a9b584308f912cd7c569781ed561c5
034bbb10b8efe06a0feadb9617d7285277a9b52d87a6e6bfdb182d8c43ef403
a7a8eaffb3220181f817f8391a7cf20e6360e8ee90e6c4f21077ae08192654d1

Fiat-Shamir context

808ef11b93cd79d3dea9a15bae8dcf41cd3ca3c3eb2789ef4512f7e9592e70c6

withheld proof term: 253 bytes • axioms: • kernel replay: yes

FBundlePoint.isAdicComplete_of_pow_eq_bot theorem, CompanionDiagonalization:51

$\forall \{A : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} A] (I : Ideal.\{u_1\} A) \{N : \text{Nat}\},$
 $\text{Eq.\{u_1 + 1\} (HPow.hPow.\{u_1, 0, u_1\} I N) Bot.bot.\{u_1\} \rightarrow IsAdicComplete.\{u_1, u_1\} I A}$

SHA-256 of the statement above

ca2ce41b1928f1a18ad3407f02242233e91b51a81db0cd1038358d4de4185024

SHA-256 of the kernel type

b18b1e58b19948a9155dba8b8bb13151d39561c26a748c5ce7ada86e96b39bb0

commitment C

45b62ed380b501c0471f7b63917fd97fe23ef663f1183136342b500795f19e85

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feb5ef3c0a3f4eed99d60a07a7d8bf40d79098a363c2cca99d448f04a6dd794e

631e3236d4e73fa54221f01160a58c40a593d7c2a621df748493b2a031d5e2fa

first message t

f7851fc14885cd7bb139f5802d9922770f1c590fa7811fdf028fd5019e653b7e

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30732bedb2a7ccfbdb814009f1c337edd417b5208c9bafa524f7a3c18ab24e914

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b492d3de04c892c9ccc7996ffc36bda6450fc4b496b9603cc604a5d378cab162

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a102642461762c5e1d94d91d8269fe0b77a6d78625e824496e37715ceb2059b4

response z

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467cb18cbf0d6a61820c131f9d38aee9a43797ba42db1ff1205d95080e7e53b5

0e4d0a40a77d76010bd44fae349ccbd046233ab1ff7961c041c15da433b479e7

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cc9910887fff92c14d8ab628ba7e93cf84cac200aef712ce774df9cb89deca46

response w

7a4c423f8a534a8ea9d1305f3af31ab88cdcb38603ad7093aa7594108a62ac4c

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833142f77c25c3b70aba8e553ec97507d46552982eadc8d26a3694ffae92ec98

3a2606b72c3fc6bf9cf8ede114803ade1185fabcd267e23157c53b75e88eb4f6

Fiat-Shamir context

140c416a32f0ad74dcc3157a26c07546aca38463d9007458543e6b4b1fd41b2e

withheld proof term: 398619 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isNilpotent_of_mem_of_pow_eq_bot theorem, CompanionDiagonalization:82

$\forall \{A : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} A] \{I : Ideal.\{u_1\} A\} \{N : \text{Nat}\},$
Eq.\{u_1 + 1\} (HPow.hPow.\{u_1, 0, u_1\} I N) Bot.bot.\{u_1\} $\rightarrow$
 $\forall \{x : A\}, \text{Membership.mem.\{u_1, u_1\} I } x \rightarrow \text{IsNilpotent.\{u_1\} } x$

SHA-256 of the statement above

9dadce3f9eef2eedb0a3a0af3af41e975e5adf0c480acb7f6b5ecce0d56966f2

SHA-256 of the kernel type

99f623a4973161b8346d7cfd558df399c2b7f0cf0d5ec62c3cb67ef212cec37a
commitment C

8b839c548045a8a48d306f3d61e5fcd1e4c64b7047f9ac9152b569fa7a935bf7
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3797bd4882c906f7e1c6b3114030486bf5b2079837ade0daebcd255620908b1
40c9f9caa38303d32d44d9f8d4ce419ea347128fe09e2a9962e5bb162271eaae
43f4b27cdd5c05853d8e604e38a1975c3f29d2efea9da112ca7573726baea6e5
8647978281f22a1315a65336e76edb74f1cd6266812e294ed9984adb39790ae8
9d36ea94fa52182e2b22032a0ceb25852f4411e70c9642477e93dbab2952b990
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first message t

c89deac7d30135605c27293412d769bd52e533c04e292265ef99fb4c8bdce537
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2ef339746da820ac40725ffd9c3630ec69c3fe6041e0b67e102606335c672415
5603c443ca77cea1cb567d13039a63947ab9115d6d4327274ba709ccf0a616d3
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f7f83c4f77964eeb6910b2e6c70e34a5ed64ff4685a2a6ea721e08bd77712ad8
3305e18db4d94329cc94b172698cfe7edf29e1b7ff4ef00636a68ad3c9a6ab0
84a3d3bdfad89df03100437d057dc75deee8b9d142151fdbabe5230bb995659b

response z

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323ae7fb3c2ce685254f5cfd2b05c237c0ab44229900722f4a4490cbd6106b09
9d870bd8cdccccdee4c08b7c9c76e114ed3189fc36e017def0040d27f9e1ae9c
6c5f7c83fbc7bedfe5385ab70c04722ee00bc3c434b51498efe73c8a2c2857a4
85e20591199dffa760d7f96e8d3f9c02f40e54c369dbd6f197d85eda237c165
5e990c44a82ad927ad6c74f6180fb7860affad68fcee4ae0c0b8bf88167c7ed0
7d135fc35871f76123a1a2abaa57492f25b1b531285ec10c7be018ffbf6598cd

response w

5ed7b7a890009c9149b762d8e77f0c7352e29a2b5f11fd10f7f4c4175fee3641
c953b9a5ec94373423e3828e6d30fbb24788cc4788af9865eb3b6f245fce212b
23ace3078954ee59b7509ebb2fa7d0fad3c18332dd3a3b4700b23adda61580d4
7a05ec3e3a5552fc775498479524a7473891c87d33f1ff434fed416ebb6352a5
1555048f0793e7c834fdf07c31ef9291964e53c067f03253929c61215edec072
c342a519e2b53457d2767a01fe23fa7c774a0267871697004c6be6195faa4a43
1fc8be18c58fbe0f82e14457110b181535f70bfe634e606f32718bfc9c8a8df4
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Fiat-Shamir context

7970ebab7e9f0c6b9d03239dad27d4d64b1e8f65b185648fe69bb96bcfc54bac

withheld proof term: 20 186 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isUnit_add_of_mem_of_pow_eq_bot theorem, CompanionDiagonalization:379

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {I : Ideal.{u_1} A} {N : Nat},
  Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} I N) Bot.bot.{u_1} →
    ∀ {u n : A},
      IsUnit.{u_1} u → Membership.mem.{u_1, u_1} I n → IsUnit.{u_1} (HAdd.hAdd.{u_1, u_1,
        u_1} u n)
```

SHA-256 of the statement above

ebb24a53f48501bb79d055294fd7b3b036ad3fab2ad1d7f29f674c6b83efaac1

SHA-256 of the kernel type

6e0781f308f842587c07d8b7099dfbb308f8419531ccd7f1cd7d2b0b343503f2

commitment C

882b97ccea57a3f4d53c1cabeeefb835654251f28ed8f58cde769eab38dc68e4

52a561d3aa8a9c47aceb269d42553ef6b4d8a5b3239802bb291b5d73b2e393ef

6931f0387516f7de1d168031e647ec6bf501bab1743a03edd74e0f8ed9c70b83

75550f6af59c295f4ca2e6d18349602631ed91105a3acb3688f88018b75ca617

8d0e3520198b045d347fb24d7a4558c6241aa2e612083bed27d74bf1ebbbe673

863b431632646bc3afcdadd528084d01c11f5d18d461417fa377a1c03fd785a

f4fb9670fa33d10347fe18b0a78b7f5d164789c3f7a320779ecbed91a924b9c4

3dde2a0aab092531c3fad3ed26921122ec9b9d53868036511f05730386fbf6f2

first message t

720abbfb6dc719443eb1fdff2f2fbb4f2f47457c384fe92be407187c9aae97c2

a5fb71fe596a41260382943c98f7b56becfe813c69931a60e90eedb4c550035b

4a579e0ef6b8b78fed253e64d404b1680594060c092f7bf69e50ed4f5102e4d5

addc898440234f307e6936f9fa4e3bdf41398cc92d68a9cb7f627e90ace67472

97995073d1ffa3d2b9973bcc3e37d3b9e233eb651fbc3da56744789925e038de

f0759f2312fa7f4d0a7544a0825ed735caa057d50dc6994daed22b1cf98f7f4e

f04e4d1ecc4e8479b371974d04c3178dc4879ea6d36329eac02cb666522328

c1e9a7707712e3ffdb2653a678e60ae030453b87475554b3d234a5e106bcab7c

response z

4df1ce8cb9746cfada82e4080d6b6b6fdd23d17d4ebfea1fad585e38d624301d

579cb8a4ac84df2eb9f2df973aab88eb526ddcf6d8d1a24dee11964dc010a84f

c9d1a965d443b867b06ca70fd86349950fd247351dfe6b86ee380104a12e0984

f942b3ce43781ff4d7c87506800a8b927534822c76fc1c310f53d03309eb45cf

42548d4ffbfbb4bf08ae1ba2d74b6a217bccbd946dda103310e903a90984cdd04

7dca7e7e924c1295056e528d483e43e0e242b815a1396247079745d5231e7306

ff5b476479725e1aa1088a3cd2c3878d0347e4b096e653624f242c0bd1123c1d

50cac3975e77547beb5b9cb1b28448c1b0d393d66c4b0e8e83b1b22525edda11

response w

3910833ce8b156142c9992d0d6dede09acdd8d385245b559d73b4bed938abe88

3ed4dc9b8d325ed758f7ad6b785bd050cd8f895abe19b9770d75b85a40a9c27e

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378711a7a603f2c24018a69327d825432a4fe6331342beae302c168268126835

6e6f59c73286a99b2b6ecdef5a259e973ea15ff827a6b890771a03b3789534db

37e9d0ff5d589e2e4f243124bb3cee980c06561fbbe3b7e0f00ee7aefba471a0

09ebd07ea0971ed95971fef8f949f39a3ac5054bcd8cb4904a4c33cddbccc17ba

Fiat-Shamir context

342417365be69b79a86b21086a9ea89e9d5c701f565714fae7e62b1c5d473c5f

withheld proof term: 4607 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isUnit_two_dual theorem, CompanionDiagonalization:555

IsUnit.{0} 2

SHA-256 of the statement above

1e3c7441099d5d016be06ac03d3a9d43f9e512627ce664f7914962c097400f4e

SHA-256 of the kernel type

dc454e113c6963eab0e1821acce7a70bccc940d84786b0f73b89a7265ca44c35
commitment C

ecd4bc6d7ed8477102831559eea0082b797ece4c01e29dcf5716ceb60906f4a3
8d37b6e4a37be2605f3a88b8620e4fab5d597625bcf73696de2548c951a11525
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bbe0f55f09a726907788b243ca0ae3aeecc27315735ca7ef59315ed7156543ada
e850d9e61b24a4bff8c0c7d0bf1d34533906199003a749a01823e8adf74860b9
1c84de13e2e0f9700962c878709b3ec7ceedde83e9ec62ca2fdb56e698046fb3
2dd9133bd510cc1500323fa71e5b6250f24b1d3e2d2cbfbc30bd5be098ceeac
first message t

ec44607358cc5f4533a0147e92f076d634210cd6366d4faf202a55c0eeab2ab0
5cc7cc60abe074a02e0619b32848485eb483b89ea3e3b892ec04279153c53234
8fdf7595843a7fb1848792d69127e41a27fc5ee40e762bce23342562644cc84e
8009a98a4d5e2443fb16711780052aec8f6186cc7ad88dd2ef2d1ffd987b1352
bbc5f8d2e7cf9c8a9df601d69029589203b27d6abdc922cad989da0074986f56
fdc9209fc30caf7b859199513b1dd20d9a0cc592383a2672b7cd508e74f083b7
6b8ff183f43ce62554c5e29f9ccee1f4c74df4b2176a60a7a870510221201106
69cfa7ca328468f351268f52b3eae6814af3ff2583afbf5f5aee6d3479ad5ced
response z

559d83a65a5fead9ded35fba2ca455abee450e38480cbf2b3c2eaa209fe8c279
dd926adf32071e8b9ce23d50c106f7f0ade6233bf292b8716310512e1c182a8c
0ea9caa5faa34ae1100d6e356d29e401ca7021e202b02a08404d8a58d3958982
1d32f320eb3665c983c2586414f1e1016e665b24dd1258a96ec0ed6d813df793
6de8accdd679ec6678ab9984ae5680d9be6a73cd7f4136a1c7cb3aeae65cebe6
95fc41d81f1df3756cb6ab7149943f6b4f071041a30b547801c925a2ec5634f5
b1f547fa564c6d171a7c146d4eb63b8fb72437ed41a5fa40910840a6a2859e46
f9b97f34b22a9b5cf8036bf050f49e9e15aa38520a868259f81031a354189e6c
response w

1f47343081e2b48a8a420ec88bc0c5612f703f58dec91f14b6746a25c473acd9
1d4a5a5949f92085937367df73bfff7cf1bd983f2fa211641826750627a55e593
fcf226441e7c168565b1ac829625801b9b8914ce7268a87dbde74cab5813f070
98cd0773d383a65b072a83125243e915c1b047a6558740c739b2b19d1e553a5b
7c79856c9ce1a2d8e397b31efac2e7f8dbc3bf2db51259d644d6fd6143c50415
3301e3c390115b703d2e2a12c8b05f41e98ebc20393a923b9d43111c148f1b61
2ffacbf9e8759a9538484d90f871755411727daaf211645cb06158618e5827d5
28c1569d7e34797d8b91a4cbe2a70298e3e3d33550cdaa6e901a3fcd69775f02

Fiat-Shamir context

6581879da07f21d19e809f1c7d9529bbb9d0a1459d15a0400cdf3792ac5919c6

withheld proof term: 13646 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.lastCol_mul_apply theorem, CompanionDiagonalization:222

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (P : Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m 1)) A)
  (k i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.lastCol.{u_1} b) P k i)
    (HMul.hMul.{u_1, u_1, u_1} (b k) (P (Fin.last m) i))
```

SHA-256 of the statement above

c454a7adff72cd217d13f31789f21c64c82943031e2b220b11bdc9327ef4b5eb

SHA-256 of the kernel type

f41db852bb356fce04c789c78c23ff30b0466748dc1f513fec0efb2bd24a8b9d

commitment C

db8e427a0b89e4b7d441c3652586aed2a88974040b022169eade2ca97b4719e9

5236c9c48ea84429787d279316e4cd9a2e8a6b61bdeb2fa3b3ad094a36dba9c6

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first message t

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885fc62657916a759d05335c53ac4e0868bcf20e8a7d4b8ddb59f425be645d63

response z

717d3ed4c2a07ab13974c5420773696ef290a238b66c6cae736be47456ee71e0

10a325b60f1cec5f72e0000424453064b8550ff4111fbb740c88da54c3d56610

9f8d04be8fef138a944e8ef169ecffa8f7a30fc9a88fc9edfd99d17142004c4b

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response w

7009aa2aa31946cde6c40ccc2e3648cf4e825a8b681391403fdc27c99f608f2d

9ed781eb051a7c14b40456479ed2dacb36050a0723f0c4e78afed743039d2817

cb0ca0849d940b09a425257ff8d352740746bf7783d5cbf3c033acb7190d10b9

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3097157ce27caedf4e2dd58ad2cddec34f792c165b6f3f226e324e1d46712892

9a53e85835aaec01766f69b733438535ae70fd1a48493a1847b1d3865465444a

Fiat-Shamir context

c8b49977023c9452c9e617cbd959d4f841c4ba615262e4985b0ecb36ea7374f5

withheld proof term: 48300 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mul_pow_sub_of_mul_eq_zero theorem, CompanionDiagonalization:473

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c x d : A),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} c d) 0 →
    ∀ (k : Nat),
      Eq.{u_1 + 1}
        (HMul.hMul.{u_1, u_1, u_1} c (HPow.hPow.{u_1, 0, u_1} (HSub.hSub.{u_1, u_1, u_1} x
          d) k))
        (HMul.hMul.{u_1, u_1, u_1} c (HPow.hPow.{u_1, 0, u_1} x k))
```

SHA-256 of the statement above

df98ae75410210484f3df936e60c864aeb8d17ff0c6606f040c8bfe191b60b2

SHA-256 of the kernel type

80266e9dc05978e24e5bfbdb2448369f55adcaff5337d01d082e34bfb7a58e4762

commitment C

df91952df266573445ccb2018a6d4735d5a122edae0526373da4b31ed5540576
35e762bab33ab3d358c76efe8425f0dbeaa184f82d6b241aa212055a33a3b7b9
cc299a1ca9c4aa262233699300eb66d4a90295f5c8189fd3209343fc6fe0e3fc
2e97ff7415501624bb050829002f2c3777a6cffe5b940ece70aa9a93de08ba3c
8c30413fc9fa68e62a5f16dfb675ed034a1faef9e6f56baee61863b80b8f2495
d1dc821194d58281ca3b1047fd460154e373f65ee407a7dd8dc86ed7593bd4e9
e6c055a2a3f8a64ca52b4c5dbd574acb3de226961d4da613d14e49c9e84afef7
dbe5d8c3a5307a261c0b76f979efa2d89d06a8d9653045623680c59a780032be
first message t

d8fa80fd530c4ce6663a9fe34f93337e0f8ff793f893f1b1f73a6429fd3dd37e
b92d8c5ec1d16cbb9c0eec9d60296d99dfe578eb3f6cbba7d02d6693eae45446
ae2e245ba3d24d0595c1e20a65ea79ecac0a95ea86e23187b9092cb8ff176616
5ab823c3977f054182611a931339780ef29b50682c1b71b92a2413ffe917a190
66deb7f710001d3f1920c0d40b202b52032a92e886de1d1b4542391703579c58
0ebad5a437da42a393ec7302d165d1b5f581731216916fdf640f650aefccaa7b
0de196542355c4b0c0b0b8693c3456813e843e897a9319dc4b8e834ec65c117a
2f7e80ad9fbd4017986251a91cb03dd3f01f0a2c23864f5a6f56ffd359141d04
response z

443c5e1ea7b4f742d7e67fbba7947116087322327ec6b05c6bc12970d761cba6
fb2b0090ffc7f417ee66c849be6bf8ab25644737776cfc83e95f17de986bee59
d0dd7eafc4fc0478fd18b7f775b48ee6f14fc3678e3d06baa7574b9071c15890
38020e03e63f87d21fd49dbab94dc1763699f58f526befff31dfda292d95fc83
9cd0fc4de8c5ba490e0284180a8a34b409b3a6a1bd4377858707b190d8c0ba3f
4b4e5049329af2c251024287ffa971c1ce267823a6c463ad428d318cc58a99f3
6a06a2f1cb38cd7e0bf51e6cef0521ac243f72acb9d8edb35a07f08308826d49
ff6583c83baf2aaccad17550d9365f9955375d2eed33afed40d495f8c706e1
response w

4c3514185d5afb6b5012f31bcd3a02778f0c7b00735dbd96be17932fbc38eb8e
c1ad92cce572a98a32cc796bc01e0ae3e511e8f0bfa20fb6e710f63035e0e851
cfe892e5607352961160bb59e175737a52583784cbd85d5e48cb46ae4216c697
55ff72eef3d587937f4ddfa750aae156b0c433f1a906b8d5fcecba254f74627b
1d585ce42644fac88f535933c24c2b881cad951cf3e856c362dc5c5877550cbc
e6bfacba7cd08fa5c27060c9d7d3fa063508d80222f5506aafc4f78a2732d9
51530fca60b6263c7e64c984863e616a359814b62906ac0ff8752339c7b4f9a7
2787095f3db371a25c7ec1c24777a3b9854aa5ea35e11cd5faf27ec8814c265e
Fiat-Shamir context

28d48d9a51c252f626b5e4cfaf143311d1ff3ecc7bf08c3f1c0d05f8425afe8b

withheld proof term: 1013992 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.mul_pow_sub_of_mul_eq_zero.match_1_1 definition, CompanionDiagonalization:0

$\forall$ (motive : $\text{Nat} \rightarrow \text{Prop}$) (x : Nat),
 ($\forall$ (a : Unit), motive 0) $\rightarrow$ ($\forall$ (k : Nat), motive ($\text{Nat.succ } k$)) $\rightarrow$ motive x

SHA-256 of the statement above

3c06ee6afb10101d7a1fd893ba47f2f6a7b644856bddb0fd1f4d0a03c2754e86

SHA-256 of the kernel type

48b92f67a70ca137599f408c9a4ae2526ab9bfaad472c8533c1eda4e5f9db9a5

commitment C

ace8551d03e9a8e3e9be6a0f1517623171c4a93ed95d115358cb8052fee4c465

5cb73c576f31f9c68809650d57fffc24840514ec366698ef8666f45fe68630c9

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first message t

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response z

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response w

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7b6aab3699fe91b9286b6ed947bcc94fce463008b25afade89d10b3f3ff10674

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bb9456829a68f03ea1169cfd10c527442c3b6aeabb95fdb35680d98d2226496

Fiat-Shamir context

df59b96d6c17e1dfbc5e2dfda9650920d0fea1ea5dbb94e0a97f92b0c4643e6f

withheld proof term: 636 bytes • axioms: • kernel replay: yes

FBundlePoint.pow_succ_add_of_sq_zero theorem, CompanionDiagonalization:462

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (x y : A),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} y y) 0 →
    ∀ (k : Nat),
      Eq.{u_1 + 1}
        (HPow.hPow.{u_1, 0, u_1} (HAdd.hAdd.{u_1, u_1, u_1} x y) (HAdd.hAdd.{0, 0, 0} k 1))
        (HAdd.hAdd.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} x (HAdd.hAdd.{0, 0, 0} k 1))
          (HMul.hMul.{u_1, u_1, u_1}
            (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑k) 1)
              (HPow.hPow.{u_1, 0, u_1} x k))
            y))
```

SHA-256 of the statement above

e2302c3815819e9944e18a432ea45372c0e783c613733e2cb79653a875458109

SHA-256 of the kernel type

24756ca03523eca07f0a7e4f2317e2f81dba774ba9c230f57cf30be615b6f915

commitment C

3efbce4e97e845afb226da3c565fe88ef42d1ca56b2cc1ff99f05e8661e4c529
66bea20e2291396d54adcf0f57920d0d80877d67b59fa1dfba2c0b6df6162da4
f16eabe837fe30305fe37c0685b2d252aff5fc19024f87fd0497a7e2d62fea6c
783c5fd1a2827bdcab201b8d7e3503f186bd4efde372f9781100eead314743c9
f5b29e5b10d310d8868dda6314c7b0c1b0b95dc8b4e48d4ab7dd719d549b416a
053e36f940bdaecfbce444b2e14ad6cde76d74a1ae32455bab8db149d32cde71
f2928c939a353c7d6d72937dcf4cb7235bc9a8d084e09d069b66884ab0b660dd
2b6c5fc596474d90b2f007d632d402d6ba6691c65faccdcad3d10470bb2436d0
first message t

c4d31204e97946a653eb98d02535a07a081049112aaa6748e4ca0b9849952725
72ec5b0f2b04e6c59f4d5b02001ecbe2ae7a88710f9caa304e3a06584e5a63ae
55ac85dcbe657ee32b1edda180a33ab407c7ab9c85c7373e683d5ea1aa685686
0c8e91c6a69103b0f9380517fe16272599ae5d99104387936b1fd38867a02954
3a372a95c30eeabc3fe571b1b342a9f574eb44ee2a8d1f8c43f825587a30e0da
7e7125bcb772cda7430413f520a834a590dc628089dc80bf13509e181b5a025f
775ebe7aa9bdf47dac6f78c4f08618080910d468dbe8d730e7d4c724f8337cab
47729eb494cf3e918d616a6898694dd460d4c8ee5b3b27903f25fe97d433f38c
response z

56c81a434bc0c326908f7d45655f9616db92e60f80421f984c701d334dae140d
e74d9b1a81f5e17001be5bb5fcf5587c0a96c516f41fdacad76e96afa7964810
016474b426befc6cba9d7979342301e76acc5195f3ad540072a5a9ba4f501cc4
82174ffcc12594d22dca2f315557204d656dad1d39a97aa28e191caaa962ac7e
a49424e0d0667fbccf7828f41887d61a6f2f5c9594979f574d5eb71140ff09be
9055d092490245e34e21999c6af7145f76e703a26afd84b81b7496f54df2d1d3
57c8a9b21712893849e63ed31e397e0334c5c060048b4bdc3832d254ea83537f
f54a1ab0128e805913fbb14b01bc3321d37d405ae1278604ff28da0c94a308ac
response w

7d6903da70cfead3e40f8fbcf105510ca02ca2e84cf75818f1247a078cfa7ccc
02b9511da4474628b741a7bba328933a773ae2c9eeaae28b60b7a581182049c2
4b2f63a7a4512210bdef34aa15445799ca1f017e823158e41d490681bcf1046a
61ecb2a6f1a425e60550cacd37cf2c7f88e7e05ba730abc7de73f68df121fd6b
1208b90a7d023f2a436eb7ea81c2279f78a9846815419b2cd728fd3bbd3868cb
2c640fc6e73faf5e2e2289523801b005358989dec28e25f8c9fc30e2548f299b
92c6d350e5250df20978bfc4e0fbd3356095eb22ca79f1976bd7ed9ddb5979ee
4efb1ed3df5460da5e9cfb46e8ca84c436132e9ef2a67e1648edec46368ce5cc

Fiat-Shamir context

1c7247e2e9e1fd6d3def759da1792c7ca9906c1e2c23b7c41af9f91e2bd2b2d0

withheld proof term: 3 236 660 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.pow_succ_add_of_sq_zero.match_1_1 definition, CompanionDiagonalization:0

∀ (motive : Nat → Prop) (x : Nat),
 (∀ (a : Unit), motive 0) → (∀ (k : Nat), motive (Nat.succ k)) → motive x

SHA-256 of the statement above

3c06ee6afb10101d7a1fd893ba47f2f6a7b644856bddb0fd1f4d0a03c2754e86

SHA-256 of the kernel type

f005be8fe43721aeedad341774edbc059434f6fa8d2930e2058882268fec170

commitment C

c5283d9a8ed71c5c13fd8922c546b39fdd8e5e16fafa28ac7f7b7b0286adc6b4

fd85cc5eb3675eb6ce98518bf2faaf4650c524c8c916c2bd6420cd42e0044f00

d208201fc21b4fc88df3eba0c969f04f883a789e2781bec91f0bcbf1f53c1fee

fc4b718719211cb1864e63bcc76233e002c16af562228bfc0214229068146ff9

a04715b161dd7507349d72cc9f4f5ab6f80236073db62936321b902acdd0fb8c

51fd99269854757dc15863c9f72eb94253b75c2510aecc669bc0c7923da2f0c4

2a624b65736474412959c956e99da24971761b027522ceb37d9fd97545c2a4c3

73605f5fe3f38c82fbfaa213a0d3bf48ed6d640fc517997fb3f43472a84cc077

first message t

5ee2f70993b05cc4b42be4f9590c794c565fe69f5d6ff646cee8dab71bd2d302

2435e16688bc665fb12d2bf6019596a28e766dd63ca1615f2d3757f8975ef0a2

fd36d7976089473d6011dac6cf3a07b6ebebfff90b9bc2fce29be3e204a0e63

b46c7b4e0d52b02d8988bcb99212cc04eb03d305a47fa133d02bb91a1d4f80ce

83646d302dcf53b51c8bad7ef27ab085a7ae3b14d3f30fd8a7c35f42822540b2

aecb79246fc4a3c80349d9bea9dcee42e10e2ee18b6d6edc0356e3ffe36a2703

33a81e917e045d37e16b7c14f45967379445e62fcc78ea36876769e692e241ce

77d3b9861c6d039707161d2fb1d52d201bf05e942888edd724ae964627a92cce

response z

588b54be931e27244f3626cfb590af8e7c523f46ae0b8fa42e065bd68fcf79b0

de5ada082da72555bbb947ef4f66bb7dc8b50db9685f950924752bebee316cf3

e8753dcce949f399492890fff2029e951c575f55d84e47c0af4880f46cecf63

42b4fffae5a1e85ac02abfa2655ac2ef658d7ab58f1a681c8578e18489aba80d

895107fb45c4a30d575d71a5dd350bf93b6560320d3f1b31402d018c2a6ade1d

a6b4348f6bb2ead1626913897f9178249c2757485078c07b8d06c9c9989ae04c

e6e779e2c6333acba6f323c507c971d57fcf4422280d1bd6690e0bbce2701948

32d398cdef4be6c20ad386230a1713871872cf06147b8e5d6fa3348b4b44bf5

response w

60b063538f4a48af242f1257d0691f473fabd6693ee7ac9f7709b2d4ccaca7ef

02914819593ef32deeb9e36b40c3f7bfb43e0d8baead086fcd0bd89cc86b03ed

cfa6ac86a71e452f3c8136d3054e8b776c00f854b89c74e5976c76ee6785e215

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1d0480b38bb3bc218981fbeb73f717b79d1b737d4b14c7edfea48acd6417a6

a5b498d3c95cba88c796fea47fd534f31e299bd5507fbed4b54dad350789b08d

e89ef8174b60edf398fa640842553461b84e337a134967585686daccf729f47c

dec8c40e1b3b3bd4117920f6d0a591486ef9a321f5dd47c0cd397ab86eab7457

Fiat-Shamir context

4ab2c0075c1ec92318e69659e0ce3d5b21152a7cfb4abf3998f976810965fcda

withheld proof term: 642 bytes • axioms: • kernel replay: yes

FBundlePoint.span_eps_sq theorem, CompanionDiagonalization:559

Eq.{1} (HPow.hPow.{0, 0, 0} (Ideal.span.{0} (Singleton.singleton.{0, 0}
DualNumber.eps.{0}))) 2)
Bot.bot.{0}

SHA-256 of the statement above

838dbae6f3fc586581455e027caeb88492dbe0d949757fe445f2505e8c7ab6e5

SHA-256 of the kernel type

6e56c830b1bb260082b972bbdd1f89741c830afb7a32b128f28fab5e6b5d37d4
commitment C

2de82c282a7e7a43e0091e41c6f572fa265e3cd4ca150f2f26daaffc6f30d775
1117f793d7a8a1448b5baa7f7434fcaa22064b95de090dee151a0af54b9ee97c
2cbdd93b690f5773e3799561f5fb2e5609d5439b7cde58460ada4fceab8bc5d
a917a9da7b2eb386304623f9e4bc6d3e6cae9271cfde7522faf55041dd234937
7c9101f1705ba83f82c76f4bf50459f72ab4e18c653c90ec6d353d9d35580ae8
382a09a88e60fa6aebf4f2642829ca3d98534485f3924f1d069e83174f8f0d0e
058e17cee1b1d72e1ff9e7346f61fcc62fe4ac06cc98f616e98607958f2be751
2b430e2b0c54dbc11d1d829af1424c5dab3e273d64af1af1b408bc8da99aa9b8
first message t

19c63a1d4cb1e528733da2012a45100cda1c9ea3b4ec551a0c2f6e06465f8986
9f0dfba85d23153b34f9ef62eda138a5383fefdb8f294c701d633e24e73cea94
ead17a77591b6ee8a4e8cfe5472631c1fedffd1c6426101dac385017c4b38890
cbcddebd0d4b04a82732003107c6b80279289b66759600f01bcb2a84cebd9a0
1f93020b77ffc09d401256def0fb5c999caa1be12ac07728b41e064defcb5888
1dd2f39bc9d8812d63bf1df7837f40f21eb2452fdd198ff8ce16cbcb4882888
2f2d2f6cc01a051adea233c6cf8d542d04cf60397b06b5f3125ce9978c4beb66
9a5d27fb1a093520d2162b02e8624f58e0eeb4c35eab73f582c60c80fe7b9c2a
response z

36b6e6eeb5f84f80d83a38803a58f7c45c64580617445a79387d409c83f42054
6d54defe5553159dd132e4837d64ed5b32f09322c60b5e96506b0191f3ff991c
cd8ff918b3b271d7222905e4cbc65dc29c65fa1be54372e77241e51d2f7bc1d5
107aee4bc29245537f1a12b0086ac3aff29e79058ce4a85e5b7195cec8f3a1a8
68d201fb6005f1728233114e68b36a05157178c6dc8ad9cc852404c97828a794
3547ed2c7438a7b671f7989bf11fa9cfc73ed8bc7d3364000ca26e84930ae77e
96d1ab1b8dcfc8e5c21f69746da7ddefbc4ef16c5eb19b977c634a516838da2d
f76340cf2b583020016d481702d5a7b9d1c032275589a0f2c7a1d9bf339effce
response w

4c95fa0862fd3aa10338c7e0f8e7ba63242499090325bbcf54ad41e1e1219e2f
d9b245fae76a3e81525422165c818495877ef958e7727f8bf57ad58cab4af35b
85c9f52ba578e060a364ab88476bd8ae32e106a02411f2f807290cdf784e1abe
5deb5a831164e0183bc682fb9292f51e788047955b6796865729c4f5f24da2a7
4723a4ee770b6ddff77f172f1dd0653cf451cc0e40c26d18425b3e920a475944
7f885a3b9a08d54720d83248dbb7206ba0e4d66d32d1439c384b42bcc9704336
4519c8cb2919e647b07cd36593ef6925528a3e6ee6a562c154bb5e0d63de7985
2a7f930e39afb7bd59dc28467a5a63a6c7160fea442e7e37ea01699712cb545c

Fiat-Shamir context

07b822a0488a59a92bf7845d5f1e907925009b12da17f3ea18b8556ed347604f

withheld proof term: 144 593 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sub_mul_sum_eigenCol theorem, CompanionDiagonalization:263

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (x y : A),
  Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1} (HSub.hSub.{u_1, u_1, u_1} x y)
      (∑ j,
        HMul.hMul.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} x ↑j)
          (FBundlePoint.eigenCol.{u_1} b y j)))
    (HSub.hSub.{u_1, u_1, u_1} (Polynomial.eval.{u_1} x (FBundlePoint.companionPoly.{u_1}
      b))
      (Polynomial.eval.{u_1} y (FBundlePoint.companionPoly.{u_1} b))))
```

SHA-256 of the statement above

f2789b125ab9209f9d937b0b6f704ea686a77978db486ec71663326bc07cfc07

SHA-256 of the kernel type

4ef318dcccac64bba64506640dc1b3b6ef75c8bb05d6957e9502fe0a3146e8648

commitment C

2296b91febe482f01c2cd47f6814756a3b50a054361dd1e996274bd23140c95f
5a72ee22be0e40d8686696ca2a11e425dc0940ce22aa56d920ff88cecea2a93a
38fdf381c13281c512d4fe4a9b885ccee8fb60f7aef0b6c3935333c1e4297a7
38995a551cfe66a1c9dcb90d5dd021df7773ee6bc5552dc0e5c05b4275cf0e52
4cbf6543cd2c9613d10435f6b2574bd993e1c6a184e19295329325a0acde0225
ed32df783ffdec4ee50057441a48ee83c30b9b900d12474183e274ebc199821c
a9124ada942959de4ac95de946aa351c7e91c80634a40eb0412cf5dd1d1beff3
219faba6db5e1bb2298bdca3ecf4760ae3be574b468a65b3315a6a4f5d1901bf

first message t

ca03956069c6071978c180b1399b6b8eb37d88f71378df7472ab01742bd0e319
025d8dddc55556de1aff30a356d27390961e83ec90fa0d850d794bcb4442a50
058d029ea5d66239c332cf039ad9e1e75aadf51679ef9bd341c8e77926530bde
b36cd5d4ccfb68e2b605b3a468a4d84c9182e2226259e450dba41786af7798ea
43d19bc9b1a884e4b38dc5fc07cedc03a93c544cc7225b422230e8a7a0e9cf2c
0cd34ccf9404d720d41df285e8e149f95636bc05fe84f6c7e19dbdacc23595ef
396718689a6a3b8e82000f76775198a9dfc18da87eda6696938e75255e5f547c
7b783a4c8b347ac27af6d3dc7fbf93f9437cd1bc3c9642facf84ed3610d93e86

response z

40f30b6671165e946af503ae07e1a860aa2a723f0b07cbd53de8868fe38f84c5
61363f8d6b3153bd52661f595cbabc2be8ff4faabf65947f97f13856eb368f86
cde5f714c757c89da9f18888439b3189e5257895fb4ce8d901087e8d2ba0b32f
e66d7d75275bf322c17ce1e3cf362ff259740285774a41135e58d6d6936377a6
5165dd19ba7f3c85bf683a06584c7d94e2a95dfc5e561f3ca65475511b1edf5d
61c113b492a917b1d501bcf1b2b8eedc92c9d48763ce2816f8e7056c2668401
1ce168def6886711a5610c852d18485ac7d1d6466d8c6ad82b8d0891e3bbd6b3
938b12fbf5ed034519425748fb32fb14c76d38e618ab4b6217933ee610a3d04b

response w

d9e76ff2d7c69f6dcb55e236dce14428476eab892d3648cafa0d705abfbae4a8
622859cd3ca2b44405d3e15203ab054531b657e820727fca3b93216381f090c8
1f5228f63bb5c2a9ab5a353fb1eaf131d5407c86490da87897778344ec169f99
123f75503f9b39b55fb19a3bb7971d286f84ee7602f27162465ae3ba1d8ee1c3
f2c27357ba1042516be50ad43726f94410907be911106655cf377e99ab3e0f4e
8cb43f8ee1a3203c2fb3ccd6176a7e6ede8b05f3b9852a054ba25c8972b39465
991120858ce78239ba83abe3b7584ea955236e5517368189c73e18d68062f0cd
5cbfb90ba52823804c11ae470258203ffb96f25d4084f4afe766091c4bcdcf0

Fiat-Shamir context

446154a6da6f96b215c20e9645264f9cf0367459402296ff5f929cb560525607

withheld proof term: 3847864 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sum_eigenCol_self theorem, CompanionDiagonalization:298

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (x : A),
  Eq.{u_1 + 1}
    (∑ j,
      HMul.hMul.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} x ↑j) (FBundlePoint.eigenCol.{u_1}
        b x j))
    (Polynomial.eval.{u_1} x (Polynomial.derivative.{u_1} (FBundlePoint.companionPoly.{u_1}
      b)))
```

SHA-256 of the statement above

3750aa0a7cc06a067bfffca027570fe4bf0e18aeb196ec2ae71b867d55d481f2

SHA-256 of the kernel type

b94a2f87d963e3c26df9c03513af9d28dbf3fa0bd71d9ce016dc0e5a6d728f06
commitment C

d9a01f55edbcf4ba895be6fb78edf1243b3ede105d3c711aa946fe70e8c06380
f6d1e123c120a22ce0cb46c13ad8d5458f192af9a26ff1e6c8424832bc65c161
a05388ab900f6a2281a405594c8703db3c2bf284b979992ebe6d95c0fe674f02
34abaea9d43576e253f8c7799ac08656e67465db1ec7d32c88db74e89f553af6
f7b6a87f1526f5dcbd575a58ed905c7a3bc9948f5c5728577aff7f79c250c152
e0797e890788b210b4c6818b97671038fea3ef66e05132a4176fe3b2d21ef46c
ed090d29a0ed74c20105a2b7afdde84eb107d1d7558e1556243d9ecd9e7a17fd
6af4c58430d73eb697cf18dec80f415b24dbdc8fba7ba0848599b2657d770e
first message t

820a1f7a4694693758d88c3301aac29d7a5cfce645f2f8b062d615a6e2fb3ec9
6f17cce0baf68985c2d8e1841ff65910ff71aa4a14237145786eb605550f780b
a492935ebc759685a0183df25961293845410d3d372b681582a8e44ed7cfe44c
ca21eb3bd263d96efb3ebe9916d853ce8f8ecd47fb5b7acfae72e07fe1d5c0e0
036efa94b181b3cdc2efe38437bc5f399ea38e332e8a4aed88bb9bdf915835d
dbb78ee48d13161ba474603a006274eab103ef38400952199aa6629ef22e83ce
14816d7ad773e69a508777a8b4c852d5b78462c25b392341a1302e8eebb93a27
b626034670eb51016983cdb3941027785f9a0c1d0e78917438c3420520471eb4
response z

6dae82d49a3fc9f60c5e11f62999a625035fba270fc164c6ec78ce70fc64598c
062857145736143e7187e1a00152bcb5227e1c0cc09b98ac2b20ed803418e5e
441ef9dd6f48e0d307bed5e9b5d9c0bcb4807f2d53ba78668b92ab8fd1ca74b3
510d99b0d3274cf94448031c13881f36b4486c344a33da3a6b194c7bdc9fa14c
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1b92ed4985a47cfa72a06d26e1dc8084e4e72f89da9c07c9d79bb38d6e130d2c
e1602749cb8a1959f7783ae9b0ecb33ffd4d6a6eca5328fff7e5cd073945dda1
50a2044212aa2c1dfc75d9d9a7770df96fe307fa7fa9aa82004e1b2742b8dc0d
response w

ee5bfce85def981d9d6670977e23e395fa59b7057b32cc9925becf5666a78461
a0711bc6006cb155e2213837ec0f9f81d62ae2b50ddc87e47010312388348deb
836c80f71a770966485362d4186ac4eeec5a9926d22ad958df23a9aea7babbf4
0ed1b42493038d0fd513ea396c33bb2737d196e9c08d723c85dae163ecd7513e
8f15250aff0b11866a615446db2afd81056b2f87092768f7b02ec8f4c8ec6103
c7afca8da0ea77da1c854688c07b1f0c429ac6c8900e07db8ac57872d952b819
a816aee4fc392b5c5fc37e21cde3cc40c6eda097a5ad11e0cb178345aea27e05
ac3e0d2125ef5e3acaf51348852174a1f315207ad06acb8459c9788c2db90c0

Fiat-Shamir context

3c73246a40648f51f49bf7e90ccc11131fedd956e7740cdfc2c61440d420ce04

withheld proof term: 1236812 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

3 RequestProject.Crypto.Homomorphic

ZKP.Paillier.enc definition, Crypto.Homomorphic:28

$(n : \text{Nat}) \rightarrow \text{Nat} \rightarrow \text{ZMod} (\text{HPow.hPow}\{0, 0, 0\} \ n \ 2) \rightarrow \text{ZMod} (\text{HPow.hPow}\{0, 0, 0\} \ n \ 2)$

SHA-256 of the statement above

22753e8eebf79b0514870006645c1f634dc8d457e1b496cd85cbfbf7cb11ef24

SHA-256 of the kernel type

9c5ee04257c4bdbd4780a1f8bcf0128805163a5de0de62250a75c4bddfff4158

commitment C

dffd627f5d2b446da25662f161a74d98522137d2d1ad37e4df1088fa2d966404

02b48466c59873f51cb0f9efa00417b374e78a83d6037ae135474c37110cd4fa

5b2f604d2de7abf4aed0e6a6eaa8efdaea41fef36b204f5234716a08c7130332

c97b8f03e34ad352b0fcab94592b61606f855d7d2e952ab732a1fa1c8f9d78e3

84e6bf871f16bdffc939bba10f68cb894d1b457e90de6466014640e94266ce95

4d65eb1067331b1a52ca188f95c1c6b2741a4a9a6f72146f02a6bfa80cc5e445

197d680867c411090c217fbddb52f381f0479f2e11e82083e611141d674f209b

e9aac38018b5b7e777da1265b2b80df044dc2bb115d067fe4e77f479e74f14f

first message t

d8b067a75957a8bdd1171e7741a38789a588545bc0b68801ec862feeb9e2f62b

8975be22cc37f5c387ecd285b6071d2ecbc3a6423e6efa9b53f668ba37092485

031b8a5c1a975a31d9dc89e734c4fe344d6d9478618316da7ddf5c73d3c87aff

3d66f1084325585b4408c4085c722543337f73935c6978f81d926dafa1c00bd5

0bb3a1830a307711a658e9e09d645df8efc6010171b4e3a68bd256d2bd508d5a

47c7ec31439f2d9272c825b36e479a2b2e007875c977fba6b5e9cd115e4b180c

428c7386602648415b38e5ed2e39ab74c9856e90cd88180e4aee07a11d742b62

ce5a7b35a514a565733cd0a033bd9abb37b25db0338e7b76f6c93eb918de0356

response z

1f5d2a09912922201c9b61f94c70a20bf0bf3d552afeacda5f237ce0a1ecf21a

c838dbc19075e886d4b785389a2674ba3b565570ec9831f67e71e6411a4388f5

04ac1a4f81dad0b567ce87c9ee575c8a5a9dc1856edcdf8af51d266e8a45931e

821bd1b94f284fd1b4d91dadaecbdd0315363a048ab2ff8bcd7eae3d2945df

1da635e565334940dec28ba6b356cd6fefb678c40217b4c8ed9ba14da8ef687f

59e14e4fe7661726db7657feec2f7a56a7b0fe44c38d159f3099314e867d8366

a1ae929f705da5c9ef0b4669789f553441f20015d2230003b5119a22a114a5cc

7da583ee9d8c78e96e0e69bdeec9f88f264498b8a3c20320b7002d6db4a9f8

response w

7d3f247b8744720a8b1590304c0c38448003ce79bdb79076fe4b31456de897b4

c527b690ed0669c7dbd1eaaee8a6d993dc434744cbdd0340625a01a1f5a833ca

44e86f721fd74c629b720f19ae6042ca6a01f9b5014283839b816fb02e512cd8

bf6b25801c0b61ef1a15768ce2f4e78785981cf61f933ab753fa80c6c5c6acab

56b2b58661eca3d7b82774cff979a1d4d290de67300508b2a3a479c95a587866

c6dd38520200d1cde50eab9db399d1bb03d962c66c67cdf74c99274cf5c4e70

22407922512f5194af285e18d400120401734402cabb0f31aa0b47f3c8b4f887

895cc175cdab6ff86d605e62b5c0402e2e26a0c51cb5905c5cdfbc7b4994c3af

Fiat-Shamir context

21e1ea50b3223a8a31f2a3474556f0673620d5e2899e0ee8418021a6889419e5

withheld proof term: 9524 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Paillier.enc.eq_1 theorem, Crypto.Homomorphic:0

$\forall (n\ m : \text{Nat}) (r : \text{ZMod} (\text{HPow.hPow}\{0, 0, 0\}\ n\ 2)),$
 $\text{Eq}\{1\} (\text{ZKP.Paillier.enc}\ n\ m\ r)$
 $(\text{HMod.hMod}\{0, 0, 0\} (\text{HPow.hPow}\{0, 0, 0\} (\text{HAdd.hAdd}\{0, 0, 0\}\ 1\ \uparrow n)\ m)$
 $(\text{HPow.hPow}\{0, 0, 0\}\ r\ n))$

SHA-256 of the statement above
4b900bdc82ef075f54816b0ee6ee52ca9c912e6e8aa0806c569c5118ffc41ea3

SHA-256 of the kernel type
b9208ada8598981f23531dfbd4fdbab97cdb67d81f4ceb93941128aff862411f

commitment C
a0b7659ddaead40048b26b901cb9aba4d7120757c0dcc227bf7f77b32e4b4ad0
2b453732597bfe89a0bfe7ab94353b6f4483b82221a407151675852c4f1a532d
35a2222a0d94cf0ab81f734a9843ef771a96ca2fcc39a89439fafa70ae5786e5
61a3674658ab14ec805d32f6cc126a148da576a02ae0490e2cd871836eff16e5
eb56e25cf986eeb37b99960dddfc93da5dcf61f074765377d51cb4516c35ac23
b76a32c6d6a2eee2db46b310dc0e8f7019fd59a9630aeca8eb2f21135ce150
8630ca5e0742eb0631ab6e1b23419e451d21d4bbe3ab3224b6a3eb573e39276b
ebe7831e598eb16a24bba4f0be59572c4e3727da2c99ab086906e7e9025afbd

first message t
a5114b7c5f9f1c5a5d66fe6a26c5fdcc4e45dc0e2db05cff95ff3c2664e1f9
e82b8b450a068dd83a99eed7813bfa3ea62c67a11b4d7520f592de82ca80fc75
d9f2d31bebcdd8d400d70ebcb9b41788a70095c898e9b3c5aa8d01022d89d8f2
bd6795e125d06f4477b8f9d2f7e5224e49176debb6b413751983d6e1b41cc691
9a0ef799f275cb9f0f923996c44d341cd2977c16615606dbca22518d665754b2
2f7e98c33dcf5a68724d45fb58840fc0e536792c18907e85fa7316369600c28f
ce4d6526c2ade5235bde9d29493bcb708df1d8ddbd2e2c29c2c7ec6048e5fb0
dfa9934d7d5f5ef2041279234b34c83b17969ea498dd0a13806bc4805d433e7c

response z
53c57b0522328658a1d246741c2981116e49d06427382ec62d6e6b2740517d45
38ed4b17b71f3a9587e365eb255fce8a81befb6b82e6d8a0d6a9af4390f454ad
c7603e643c9e445a9fd5480d9bc10f4bcebee756f6f54969c09ce2d7d9c8f788
931de8589972ca0e7eca98d2dc52e104720822c3597affae56ed0231f8aab443
2e184b141e9df50f9c25f286958afe0f5fb50303849394b1fa2d6e64f75e0787
e22742b48cb9c3c01f29250af99e008eb04bbcd8f1080819a37253e4c7eb85fe
61b0dd639e7bc2d4c2068e7df62c45c3820a1462750bf7a322ba486636cff608
1b6cf029c8de4ea53a7b9ad3b67c59629090787abe6ac5d5002375d5d30c29a6

response w
18b962ba0238f6e0f73a234a8eb3135f57b4e6602aec7c0da70f77e0fb0f36f8
54357a3f0d001413b35e81cd23f0916bd36bd9a1fd17eff80061f1aebb936ce0
db95a01c511cb74759526954d7a849ddaabceae15e66e17d2b9733250e351407
712b6815999932ff16c74c5e973488187b69f449a63445a0f97890b650c1363d
8287dbd884ca2de2b1f78e43995b5493293e3e090e713448fd2d447323d3d7e5
7299560828696833ae3d19985c5a94884a137d7b615acd2a068f655a4b40731b
1576361938059b2b40d0c2e3317ad350e312637a3a44146b50ce5522efa1d96c
f4cfc4a775b432cd16e2126847e08c9257e6cedeeb3ba4daebf7557d4c98f0db

Fiat-Shamir context
f50176b3ee25c70e5b24ccb62c2818ef953eddb29a2c8665472ab7bd01cffd49

withheld proof term: 369 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Paillier.enc_mul theorem, Crypto.Homomorphic:32

$\forall (n\ m_1\ m_2 : \text{Nat}) (r_1\ r_2 : \text{ZMod } (\text{HPow.hPow}\{0, 0, 0\}\ n\ 2)),$
Eq.{1} (HMul.hMul.{0, 0, 0} (ZKP.Paillier.enc n m₁ r₁) (ZKP.Paillier.enc n m₂ r₂))
(ZKP.Paillier.enc n (HAdd.hAdd.{0, 0, 0} m₁ m₂) (HMul.hMul.{0, 0, 0} r₁ r₂))

SHA-256 of the statement above

6fe777a19b7596cdf4e45062698cba7b16bf0df3db76859f30d05f636ad938e6

SHA-256 of the kernel type

a3c4954ca4d877b525e4a0accc3be0a37a0b05089e43e57951edd04c741ad670

commitment C

4aa16b67132ee2038b95893ecc7cd0da80bc1ae1b661dab1af018e97535080be
9693152e489d790ad61dea2e3c3a84d895beadfe254e26374e4e00f9823b9fe9
ee554d1622d980da94792a5162a3949bc8fb9a9d7d8aff588a17ed709361613e
843556daaad39c7ff833416596dddc08f75f7d51d7a123a8e0effdf76a99ff6
26e7830c54093769c7c712fb3040869e4d23b61ee7456d7ef9b47fc70ddb3893
71ad616ee27d5868396270ad7ea9ad64a95d21b0114b7bbb35256520a3d45dcc
34cf9123fb635edb46a5588f3c1ec03c215428f6544202935ff705421849c63
a089a13071f727b7bc707269aad14aec7ac05c660fa42828474da4dae8cc145e

first message t

901cbeb121d7af66d84406398f4953a5573f8251d74fd84b77baad715e720775
f241ca7b7e02ec80e886ec722dfdebcb4576238f71cf6fb1064b90a6ff12e4b
989369c79b6363e012d6359fd6d01668192f908eaac8a327d60c7111a4e64da6
47e90a11db15bbbf062f1c085242234809ed22f56913d36a182a2b77badb63a7
a76c145b42fef2b328be6873e101e2ab8a4621e1c6064a736320173644780499
29ec24aab36f9b7add654597f98bc3aaee76796f171a0ecb510f324e2201c9a2
c3d95d0a1c800b833eda84e60d9aaf44f6b54977727df602497b7d8b4e73f769
81483f25abc0d7feeb4901332fb7bd708dea7cfc7b9b742465b7207df8faef9d

response z

2ed59f52959b5102329b5b0df6a955ba052d0e31039484f3ff63fdd97de0faf8
fa79d11fea62f8dce8c2d24e534545c9bd925d63112c8e151b784632c30abba2
1b528292c4772c5b5d01a1f99db176090307c597cf6ae1caf4e65733935ebafb
8085e0972aba0fce61ea5510753b65259d59df16787dbdf65101e3bd75c1d921
42acaad6b2e36e4e484d73e63ce6e724347b97a509faa54c3ec4bc83468a9948
ba9027f61ff4df8d83687b02fc7b11673bb56104e1b15500d79fc98967f6d69f
1ecc5aab88210e26a7918d540845628cb060122c5071fa1d85b895dc0c7de56a
091b7bbf3a70bdef34d2f4cbf92e9f61440e48af1b3cc6ee7888066d29bea0

response w

73eb0515476ebf6befe3cfca3bbc9cc86f93f847bff002db7f4a782531b46726
aec36695ad5d97ebf15085b945ae3870b1221e9509518f62ebbe8ad4217ae725
130b698401b311755b2e360b9b7761222b068bed60b7c4e88bb0142d41cdb14f
6baa239d0cf3f523f11a1eacc28a743a0af64764ba8a7f4cb0f4fc8fa127d486
66ef07ad43010644c7926d56e553808dbb93221eef52fe5080c119782a179a7a
a6adde4f7701d53182c34d2e67303f0775a69ec9e6a67c3f6b2e0cc6943f73f9
d3203d43af53ff83b93eea788fd29ec00fa78d6b419aa4e7e255ec6be76ebde7
149c8b0205c6a4d6326942bcd06910afa08ae36cf51011b3372e70c89a76fde3

Fiat-Shamir context

c50543bc3a77d7c52514f63b884cf214961dc817c285c02dc3e39ecbc3767ae1

withheld proof term: 5712070 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Paillier.enc_one theorem, Crypto.Homomorphic:71

$\forall \{n : \text{Nat}\} (m : \text{Nat}),$

$\text{Eq.}\{1\} (\text{ZKP.Paillier.enc } n \ m \ 1) (\text{HAdd.hAdd.}\{0, 0, 0\} \ 1 \ (\text{HMul.hMul.}\{0, 0, 0\} \ \uparrow m \ \uparrow n))$

SHA-256 of the statement above

42e27c07ce2a3330126e689c6f594a1ae9a929fc4ec5fcb87be5748a8058776f

SHA-256 of the kernel type

373f6c80232ab742cefbf7b03cda0c9f81ee73bc6dac717982fd39e8d25845ee

commitment C

dad9ec63c03427b83f74a0869fc232dca86867aa96296c162d11df920717c25f

ab958d725afc4f17c1d1117bf924d55ea4e572cfa9bbf7ee0ddccd5665f422a3

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6b5f493e278fc6afbd50efa9d9de2bea0853f4e1864f3ea5024e27815896ea1f

1e9cadfa92a683b7c69aeeb49b3d83ddad0aa0b86018acde3af2983bb28b2a12

f914503794e1cc262f489d4f4116ffc37ab38869cfbb64d5c2059a7ecb22dcd5

5121c216d8e388c9b47f9b3647ba53c64f416d9e4b1f2b155f9bc8584f408285

first message t

dbbbd7f7b9c996a8a893a3051cbfefc61df0e785a880259423ad097234ff780e

c77472506a927fc653fad1283789637060b7ebcea9b2cc9e8dc6e13d19e95054

c51efb626e6a15ef6ca9d314f837f979cfc9dc48bf51115b84ba027bafcb4e41

0b0dddcdb8ab536fb8da6a3eb72a919aa145d24c64d15304f8d25fba05260b01

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4871649d1b750d769f7612210ecff11422df9ed7bd9c5a6e0f7e6cfe2e6b56a6

b764b3b4dd8a5ed93d3c5407308b256abb62c46e29fa8b75387bf0450eb1411d

70c020aca49286e1daeffa8a13025cdab50ede92c6871091e9f8769463717d74

response z

64a478c3f1689d7c510f9c8a0a4d3a1474981beae0ac86b2952942808364127b

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179611bb0dc98de6b65c75e7f3d15d8b46a481d2d1d9bd63e589f94f5f344290

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e93295ddafe450ef58ed271a6288ae7a134280ab5d42982f7404d07aa89c0be5

12e11a0fb07ff182d221e33fcef36112eeff12ecc51e70d13b6b70a591c72b3e

0c5cd10e6417c68b41a652894823fce89174d21b53b56bcb580a9385a95b6d4

response w

3ad5c4301a81fc90a6c7796a4d8408d9b9ddf538d10c0e3fc05a76fe17deaa4

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7e5ba19b4a22be13b90b377057c135cd89e6591a9d9ee991233d362f2caa10f2

ec48a8d79065df09aab9ae36a711492f51872739b039aa6a228a63c40919651b

9db884f4a69c96a5ecb7ce0ef94fcc8014a17f59632623989e0431c426c11b98

a9aecac63e6622d8c9c2f3fe0c1ae6e54b08be7c7a75c5ab7367fef4e6d2be75

Fiat-Shamir context

88a47923d9298e307a3d3da785d81b29108be3c938956204fc25dd5a33789f78

withheld proof term: 140933 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Paillier.enc_prod theorem, Crypto.Homomorphic:39

```
∀ (n : Nat) (l : List.{0} (Prod.{0, 0} Nat (ZMod (HPow.hPow.{0, 0, 0} n 2)))) ,
  Eq.{1} (List.prod.{0} (List.map.{0, 0} (fun p => ZKP.Paillier.enc n p.1 p.2) l))
    (ZKP.Paillier.enc n (List.sum.{0} (List.map.{0, 0} Prod.fst.{0, 0} l))
      (List.prod.{0} (List.map.{0, 0} Prod.snd.{0, 0} l)))
```

SHA-256 of the statement above

f1d45cb47be877d6d739a0c7908b22d706dab1ea1d7533864c0739df1e52b823

SHA-256 of the kernel type

4dcb3859ad1b75cbf61d1a975063399a6f62827007a3e2d3effba4e4694dc3ae
commitment C

558a8727a6e736aa2f044f272d184e389fbe0c8ecd9c383e26302b3c9d6921ef
fe738dfaab2661f62ce4d640e0b9f7bacee79e8dfcd9391016777c4ba8398b0b
b4f4e053df3d7cacd9da6a485021da69bb262fc86c176555204276d2bc85c085
12d92185121b8732479e8ec174c770491863d280de7e27ddfa4b6634c28c229a
aab06671d42484bdad9e32184ebb0b7037c7132ff6262e91bf89622627eaa08d
61912b43ee7a28b3b27fe275be4b0e1afbc6a4500010c1ae9d71505fb3771e70
870eec59b8b3f807cfb88ae946a73a911a1d41af856263edb0ec030dbabb2b28
2f2557738630110cb83e179c66cee71dd855b4880ae0253fde8c0bd16302a15e
first message t

9c04ec2450a25a7e1c8dc8af27813364eb62e08283128399485d29932b88040f
d541dc396b830bc6ca37a89d8016f6c907351a551c6ff807c6b392a154181b1f
984f949949ba21c3ba4249a44e02e79ef6d3837337d7e061cf4e4c6790d87c0e
39de0cb00cf281450740a01729e7d0d8f287f446ff4e381b976f6afe40c98f0c
73385766ca43b8029d4720855c620db82fa66cca659bbf32e709ee699e992cd
e178deaad336940e4c10c88da197007573093282621b2baf36c75e99d265ba19
2cce4ee60ffc2236d8d27e5e9062b4618d1222b687105e12f4b28a7d12c671f4
12bb4fbb3f06e89972ce8be8754970090f67a283ac43d12ce5bf7901cb52e04b
response z

af4333771e61f640525bdd3d6598624639ce74ea575f568a518dd232ecd15d5c
0311dd15d49a3e5c31b4cf0af06e4573a35e4ccb196ea52e9007d3b96bb4a9fb
c1bcb2b2b6dfff8e2c53074e88ee7b40f07ae39882c384dc916a5190566a35db
209d4b3168fbf3e1fa5286e7a709e45ee9dfc7e113e15b3abef548f6a6b077dd
c682cc537ff3197070159390748c53897a02719395b6284ce4882ffbab4c03d3
4cbcef0faf62596626728929e61078a27ef92a1b19e78137d4a1c97269df615f
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77181c82d9ba431db319c9045773770d8af2771d947328ad4e111ae87d4d0d
response w

7c5024d8100c66319c5430eba44bd432f823444832ad826bbc22876944be1df4
c0d48e7de0ded238e7992bc85fd5b3c5825341bcf529a364587f2d2f7e7c02ff
515017ffc2c158bcdcd5866302f72ce61b7ee7504fe09f2845ec7f14fd7adc55
0e916a932f8b34d7e50c0493a29e450cc447501ed6414570c5501ba77f3591a5
8d9a0d518b91cf772894974d3d434470644308b05b974024e5221084d1ec0be3
6aca40c94b40afa1858365cf56f9e977356e39e6d10450aa33e291c021315ecf
ab069614f0f177df0cca2f44ac2867b1cf36796e7631d12da53f82480fe8b6cb
bf5a0358e04f3e2efeaff327405088f80993d69d5bb51e982fa7bd0beb482766

Fiat-Shamir context

3761f3d9e929c188c617341b7a890f5d39cd9f798cff595c2d772b3275e25d81

withheld proof term: 171 606 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Paillier.msg_unique theorem, Crypto.Homomorphic:75

$\forall \{n : \text{Nat}\},$
LT.lt.{0} 0 n $\rightarrow$
 $\forall (a\ b : \text{Int}),$
Dvd.dvd.{0} (HPow.hPow.{0, 0, 0} ($\uparrow n$) 2)
(HSub.hSub.{0, 0, 0} (HMul.hMul.{0, 0, 0} a $\uparrow n$) (HMul.hMul.{0, 0, 0} b $\uparrow n$)) $\rightarrow$
Dvd.dvd.{0} ($\uparrow n$) (HSub.hSub.{0, 0, 0} a b)

SHA-256 of the statement above

ebba08b1a79fa19647848a870dc9713632627339364a58ba0befc3175d6957a9

SHA-256 of the kernel type

b9d26eaffc8b6b709c638f938b090c7b18a7fc9a2ee3b27ecd76dfe109ffc249
commitment C

9dfcf3c76c394a94f029bf7d32df4c52a3ba5c00d9410cd69e1805a2b663efb1
99a23fd648346509d89807d9d2d5469c981151fd752c753822ba08cf6252689f
b7af690948dccacf8147ab353a55ef910d2a9a47ddb630b88f467ef9d58abd40
8a236c8fcd177aea9f67f335e0d4218950909919a8e56f53424782ca0a90b78
99541d1804767d540292fe94ff2e5e06da7070fd8ab6e05e9d730989929be94c
ed9ac0396be391e7647dfdbd634669d532990f38b589c9724840a7c91946f91e
6b25faac8f9bff60798da1b07ef7b0ee792ee0ef8bdd557d3cb64dd52abb5aec
c23cbdd0d9dedeb298392201b9811722382c4c33b455da13ee019051675ab10a
first message t

7bd1c0f3f7da90487c94fe82d9472a047958c7873148a2ffd8c2b7ac4c081c87
4c3bd4c83612b1eeee5514531853bb9f57674b3000d6fdf0723e4044e54f42c1
52788cddfc8118be1ac5c1847762a6e89f066caceef917fb8b529f4c0f285872
398c6ef983d1e0c95ad1d159b56402a6719fd45201a886746773db61e1994c5f
b65069ea3d72f6238c3ba98c6ba19db42c1e3ee75c6ccff2aec31107172f0f36
d8ef4c9f54e22fd75210d17856f8daf330eaf57bbc8dce84ee6f6a516065e7db
70ec3935bf83d6aad4918f7dc81317061517d5a8d24effd4b174bfd782e5d3b2
1816c34dbda53b95c4323f533a3bdd77b193db366707b46fc58c772e7ec42d7a
response z

2958444e4ad71226bdec8644173dc46fcff0f4bf98865418155a1f5d5262a87a
3f2184fc8ee26caa1e3b5036e01e4a322df6352d73e911b536aa39ee1de046d4
810cb27cadb523aafbde291ee77094f0c8dc7fad3d3b1f4423e27b0693de9d51
f2e33f0dcff29e829bbdbf4adfb4d15eee649fa1cbebf669e78ffd2f0625e3d0
b80eee214f519c537687419996d414deaaee8fc9de88db9bff6c3305c83045c29
6ee9061d20605ea11ed76ad27314149aa354fb5a5538716de8b1a4a88f4ab2e3
901d92932c671a15d3d9ca63f420adb9ff2b22a7fb7949bbe875a66bc1b858a8
ff6f8996ea988c9770cbb396f1109d379f76d0bd49283938f2a318273e750315
response w

47f6a71680afbbf2ddbf005cc3d9b12bc0eb1d35754662eaeec2e275981c3211
fcd1e3d75c2af32287b2abed72f264a7a4d5760724bc17ffd54ce2e59580fac5
8d96979c81afa4a8eb88c932ae1feb5e2f4842c1918cd4044081ee01917cd218
5e0f7856c5897501c13d12cb11de6a2947aaa3531fcd5111f844f2e29bfcf54b
ccf2dfdcc935475802969549644e3772458c1a357b73512bb5aaf9e583563e07
d236e69c474ec5c59d3f45a019d00f67ea39af68a3913ad004898c7c37290c08
d9b0a748a816969011065126b7776fe9ad09b8ae89644f3c4dbc8e64199ec7eb
e5bfe9f89cc0f3a31396637d48903c219f2453592c46fd5733481fa8dfce5b07
Fiat-Shamir context

63b2d1003560b216f5bbe5e4da7cc2c8ce1940b69a3665367a65ca8675ded35d

withheld proof term: 197 410 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Paillier.nsq_eq_zero theorem, Crypto.Homomorphic:50

$\forall \{n : \text{Nat}\}, \text{Eq.}\{1\} \text{ (HPow.hPow.}\{0, 0, 0\} (\uparrow n) 2) 0$

SHA-256 of the statement above

b483529204a9093f9bb30b7a9b1a7c666535c6d1819b7b0df49e9d02ab7d3f43

SHA-256 of the kernel type

842c0cb4cc3f11c8a75b5b6620732bbdb660349f6ac4b45c09648a72cddcf69
commitment C

7371d0a57ebccaa790615a85c842780bf8e55254939af802e3cab3c3edb6a9b4
4113da3344f21f99d2b840284be7a1a0016af56b27aec4616803ea77bc919fd8
1665a63cbaa12185c80a049b9c131cad49dff605ef6a6d1ff88bb92182d37aec
f2db64792289313de3bbba545306c527c365f993713734dce751c5197bcf3e64
daf60678a33f5d701bb3b54e42a88bda70b0ecd1783a3f67117d42fafef766ae5
266dd4c9c3952281fcb7add0adc2184a58cdde11375c3e252b5793134e4618b8
eaf979e05f33709ba21bb1c7f70870ac6f415ae4ac725ed42cebc9151a81bb89
a458f3c50d4192ebaf3986bf3d8bfbcb6e630c709c5ecb607d0600cff69c2e62c
first message t

30db934f402e740bb64478d1e15430e9475580bc0f503d8b5e62799e2f82fd2f
f563261f6131e0d37edf42690ebb0aee84720dd9d401ba02777e22833acacc2a
263216c527dc719565d5997090833e6a09b1853f34f2af75cc771ee428434d4f
5ee640bf7daeb94ad0a0e1f2d92a31b90ff49ff61b6fd533a840749a8223ed0
661b12eae09b7a561713e395625e5f8157f5a2ff4e1b21f26e0759a76b1306c6
00fcc89cf0bb0fdda6eeddc11c1885c209f03f96bedbc49e15083af86410fae5
3728a352ae3603d222bb7df9e993bc6bc9168f20fdc027f47ab9a1b4507506
d162fcf5ceadaa8e9ae513d14738c85d437d4629710b72e352be127fabc477e7
response z

405557eaae8ee49b7ab97a987266d6f9eb7582164a9d695bdcfeed05c9bbd2d8
efd31a455ce05f3c6b8d5c469900f59987c69943fa184e399bc2ce5386a5951f
4be8ad6b822e010400cdf8bc7b2b505f306131ea434d969f2ef73e455e6a24d9
20b590ccb7fa2359d41e82451bad2bbcf4e83daad930bd20e712d6afbb524134
9e397f9ac07314ea71c47c851f2ef14d42d5071e0db05b72076d38db8ee8e69d
e58418343712b52209e154b06a94f28c280d3055f35b1eae01bbf102527e8006
1b8865a4f49575e6681c00e2f0f819f6c047c549b156dcd92e6f9a5cd140053
a29765aba31e805c721dd89cb2058f9ffb5887979ef5875b757da992c5badebc
response w

5064d275ba3ecc5a15d25339950a0e2f7bdb9d17b385154b3c1ea280ce5134ea
6a69a97ed8a3c9c352859651ce33ae11c70b79fabe7c08f523e3e394a778cb6d
1cd9811659efe8eb7e9e254e9041e2d2e9e1c885c52fbe340d4ac014abaf3337
29d6c4043b5376e6a25c78f384dfb993cd9708f15d3590e30c5d3f5d6af628a8
8e99fe5428e72e77a76ade16a7987fd2f73ceb2ae443ca290ebe8f0fec0f73fa
ef67abd7402fa7a355a27e2399b4947f61cdb9ef7d971392d3bfd3a1e57586d
5a829b657dcde5ba759b4be558653e0556dbcd961a72099fcd030e4e2b64fdd5
4d17bb25e43b691a40289666b626e53483fa5d48a7f0f6620ccfcb7f68a722f0

Fiat-Shamir context

3554288765d7cc66103408e5771c2f373754c1661fb3c9e0c683e17b4e70b24e

withheld proof term: 19 021 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Paillier.one_add_n_pow theorem, Crypto.Homomorphic:56

$\forall \{n : \text{Nat}\} (m : \text{Nat}),$
Eq.{1} (HPow.hPow.{0, 0, 0} (HAdd.hAdd.{0, 0, 0} 1 $\uparrow$ n) m)
(HAdd.hAdd.{0, 0, 0} 1 (HMul.hMul.{0, 0, 0} $\uparrow$ m $\uparrow$ n))

SHA-256 of the statement above

25b2aa8dd53e657ef151f0d56dadb7e5bf1e449709c08ceb1e52c602feaab571

SHA-256 of the kernel type

f79c4c7f9ff879a49f064cdb6410a776d4d449a8c77ad62c69423014ef25f09
commitment C

fbc00fa9ac1284046f7acf27cb32cc62db44df7ef839f5f9a0fd228194072e4c
f0d33f73e7885a72bae5906e93c3575ad2a3d5979f86f5b9604a7808f9646bc1
9c8a5617117b746d61c9b4dcfe762beee1db3910fa9663a3b39a71259efb83d7
e13f32703c5e70a7804421ec78a30124a57e181128b6b6560e5ba67f7df37fb1
d6c5f0245d2bbd020db9148c31d305a8a91db1490f35785d48b7b10c6232c5b7
a6bc16ea003e8923498208d27276104a85f02409d130af9d5d5c50e2049638c2
a66b2f978a59a12fdb9034304e45abb604ffc1b8c89ef25b0ebd7f0a7772f966
f36a9f73e720b7a65619f537009805fd1fe9f90f7ced6a848daa6cdfceb12c6
first message t

68b7d95f8df0aa22f641ecd5281b5906c4c6914e9d59466014009ac7c9cc3151
e1ce33c4c282e00bcb9ad783434ba38635664761431978a0374ecc5eab2ee860
7670737bc7c8e8706bbc485fdd83e4d84c5fe0ca24f1034caf4b2ed2646c443
4f2e425632eb9368fc9492d3e935dd6d37cf914d23d597d0010e997490b9e3e0
ec1e17294d4c66423776008027a016daca2b52335e1b85426b77786319735776
72fab0d92aad565cf10f794e7c81acf32365b7f6816d381c6931a3f6a5a20028
b2178db89b5b9804bbaba9f6a2403ca4dd7441a93da60cada10fcb6715affcb9
7840764e0254649a3ea9db0f08d55559f4b7f94dc06d45006987cf19e78f7105
response z

4eaf7ff132e38183a1e96a2debbea016c2d4489d099022bceb68e6879fa85926
c66502543699112df119a5374b87ddb8cf12664d2721eac53cc0eac881cf2da
a91794c6fa721427fe581a565a564053328cce324aa0f998f5f14bc82121e1b8
150b9898d445b376701e24f6a12d9b8298ce280a9031eccfeba98a0e3de18fa0
54ec3caa6d5c4e2fd821fe53e2dd119a8bd30443c029a3e76c268b7ba0616ae3
fede224de39072a3c7193236117fe3908b8cd09e8e443996c21804151b73fa63
07182d3db7efeeabd10b2184fd36514beac8e3fda7d94990f8b8481c984b8e93
f1ad1aa50d033f68e37600aad5f1c7672a6984e7c8f16492edd5d9fc1888290f
response w

17aa9231dc3916f81ee68a7c58a1c2f880b5a7490474cf3bcd553459b96828de
2872a55f602b353e3a0cc5b8c29d337846566411efa661c5fa9b703513c08308
728dde4d93fde0bc3339e574bea8aa961f9d0e09fdff6d14ce086baaf7b25f8
5bde98b7e76d3f7d78fa5486e34149d86f434050870c16a6485f71ec14e6c86b
8ddc48e8302e8cb8341dc94841117233ec2991d02bba707e42c35b1ffea9a1c5
ef164e2e5c5e39452d94283282e60d0f2e863b236942e68f16ba5b5beabef35c
aa7abdfef64e92cf845c4a97d0619eef5422ec32982b143a576f963ec8ec154aa
c69bf4413f8d39b7e99aafd483354ba62c2a38c1ea7cfafae047b0c93b903d69

Fiat-Shamir context

aa513ffba34e915148258b893ee5a6a3cd225e96ece3884af2c6f8341099ceb3

withheld proof term: 4 102 443 bytes • axioms: Quot.sound, propext • kernel replay: yes

4 RequestProject.Crypto.Ledger

ZKP.Ledger.Valid definition, Crypto.Ledger:65

```
{G : Type u_1} →  
  [CommGroup.{u_1} G] → {H : Type u_2} → {q : Nat} → G → G → ZKP.Ledger.View.{u_1, u_2} G H  
  q → Prop
```

SHA-256 of the statement above

a453f8bdfbf8664bec947fd9818ff4aba4d83e5300c7896351224002fa6fae8c

SHA-256 of the kernel type

a21b82334c25440a0371d4d77837b0c85062adda818519ff799e2673e72b6361
commitment C

89a9c619a8c6e7e80dd96d5371ea462d8cdcb128219d4b2c2d58aa138b1408a0
66bd263d3d15a32f02fe57a4eb905b9a9a2981850b04e0bd54a0d3097f34b8f0
9077b7df7ef95d0232876912469351e757de92f3ce6f7e2c157e48066b45b536
01a80893c41027f5202b82ce57720f4540820d40faabbc5fd479f0c46f0356d6
1f4cd1519a6ef87f59f01db95dec939a49ecac265f5666b2d91e5e1a64d55ac0
fe79400bb4338e33345de42606c2ced89ae4206626571941034c23c36d09b74a
cad051ac7ec8c4dfd1ac111aae53a9cf588d76c7dec5b8affd01839d76880a25
08a90cca614031bd9d9b1b8f6508ff1d4bc354e26c818ec36dd81acb7145f72f
first message t

3241078cbfb550b47bbf64879e4dca22d148fb5d79b3e1687e1fea0600d25796
080f12dd10cb6e64710a37dd932da0524d43775b9201ff249886d889137ec898
86072de4ccab09d444e6c42383e4d442d22689a40eb6b3447869942463abf29d
5ea94ab029ea5c0efd8391fff7c5148254d9769aeae129030cd76d796f115417
84e3d9bb96800bcb9a4a4027728d6cf95ccf825837b0a345d6b7c8bdd530ab64
80f974e903b4ee8b66bc480f65780c57bd0f8f4dc99bfe6558bcd855ce1ffd91
73b8ddada994cb6aaa2f33aec4220e0a2ec4108d0c502af2d6599335cd36bef4
cc0ea07d3c295b3bed556f655a55359807e0358cb992824bec610bf83322a3f2
response z

6625227a41f98c4764759684f64667b2a25c7899d588ca521cd4091f90f32770
9ad61672e17ad9fadc9b049a2992f1f9e874769bbddae077c186f157763809e8
fcb76ce373c305a911cde60902e7fb11d4f3c586e3a83701809fd322e91ab4dc
178cffc129d7346cb1f5de90233ac560e9428c5e0067b975aaff1834a451fd6d
32c70f645bc86f11e42d919e878e6dd085fb1a000a1decdbdcb68ba1f10840b0
be1d1c4427998b6eab8e9bb2570e4040a34c273cd24559db2b538679ac40c104
1e5b24f3bb1e6035cd581f977e6f8b22e1e2db12975f96fcd3cd1be9d6eb532
dffeb6d3fdbed42a407d4d8de686b0e1ad0663a248c90b3de811b26fd5960d0e
response w

750266bd502655fdb39ea44113e8ec0c12b549a8f2010df047d42db798b8382c
3ff5b62d1fb5353c0203b7cefea986ee8f85aed6743a41baa61f6c0e6ad274ce
ef98f62a572e9472f374b9bbd9647366e0e9480e13f563e6829b220176876d9a
64385ad308e260608816824752c504bab03d43a83be1590c726af2c88955d00f
886c614375b0b869af05dd0ec6f77bb6a00c7939cd62ef0346f5251cd70696f6
c9e478970cdb79166afb5dbcf2830479c222d252436f2fe3bbc6474dbf82de3d
1ca97353b6a1d04c4968f27438135689b0f8b89927161b9a7974f79f593bc52a
d41940835739688f4042ff6a7596b2ded169f02ee0a1c9736491d0a234c322b5

Fiat-Shamir context

450fd0977dc90c78e3a3a1076ead881d09561894f9ae6aa897a2b506dc51d183

withheld proof term: 509 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.casesOn definition, Crypto.Ledger:37

```
{G : Type u_3} →
{H : Type u_4} →
{q : Nat} →
{motive : ZKP.Ledger.View.{u_3, u_4} G H q → Sort u} →
(t : ZKP.Ledger.View.{u_3, u_4} G H q) →
((statement : H) →
(commitment first : G) →
(challenge respM respR : ZMod q) →
motive
(ZKP.Ledger.View.mk.{u_3, u_4} statement commitment first challenge
respM
respR)) →
motive t
```

SHA-256 of the statement above

0c556fe373d9061cde601badb58be52e9436fd9f64d013371f8ec0e0dcde9a6d

SHA-256 of the kernel type

dd8870c2bdd8c00bc720ceec7c271e0512b9fcfe58a99d8d3b25abf35b1ba8e
commitment C

f28a1115eb38de03a72cbc8e19bb2e925bf07363f9f19f0c74babad6f828b618
3bb6284f905efcea98995d08407f395ea247fddffa9ff5b89973c10015a629f1
386f88c6059ba9277350aa0f49fc49823531d57c826bb5a228eb1ebbb7b954dd
7a8ed02548f02f15841f57d7f43dbf9cc90747024c65a225884aeceee92ec382
149443a7aa988647251891229500c39f11e4b6f42f3fe82154b37b759ac50a80
7d88794323b105d8027ea90227d4f84235d9df2d3c70c5e48a520c574db4b91d
e56c8d05fec9aeea309673dc498caa6e85e71b747e4d1736b4d618494b5d478d
1d921a3abd2463df65e3814dc1e855a151922ee7b178ca7da3fc1185d83ccc48
first message t

7424119274a49e299c576862e792bbab21a992725b3727a33171c1cc82e73d44
532fb72b70baf577ae525da4f566e1aeded985970bbfe66933e9691e9aa2e03b
b234ac722058c23c7347d5db39fd00a39e604dc97789b0f60abff46475884d0f
eef8325684779dc67000d0a0014ccbc9833bc3892c7694f14bfca1306bb2dbb0
e12e8fe6a1a941830522e97d2197bd409ee0d5dc9ca700394ebb35665f6a4fa7
8a225155ec98d2d6c4e7c6fe5ede096fee50cbe7226445ba7d0119c882df9917
86fd83df8f9b023aa6b785e099e5bee793b8f07d33137a7556d364775bbb65e4
f06f56ebf109a33158fd160cb3ae973b2164101b88398987133128b09502ca3a
response z

7bcbbc1fe419ba880cbe84bfc2ee87ea6ded3ecf66344e84f9b68d5d374328a7
8798b2859f5acd4a0d3098eafdb53a01dae512a8911b0e31bda5bd7a5bb34417
88cccc4e95336c2e9df505e8e2ebda49d40bb132e0884393435a114fc083b8c8
8015716528de45cf2013a2736b42ac003769b3533ed4da2a19dd04d2ab7c326e
00332225b727eb083f1e0ffe1e1cdca53666c01da28d9a40a74fd8cb35ae7562
6f7cafe786b6cdfd97bd775fad2bb225a611e914eb081ec3ac7a3a47f57ce76
f5f90a045a2c894f59458b7c8fab954315a6b82124f854bedce5d3784f08d115
1042c69135b91a7d2c6378c366f79098dd7f195243fb1b925d21966b3e99acf0
response w

18644666ada963c0fd3333fce2fcaae1c46b3060c9b6a965f78b6e8c265635aa
805ce5c7bd45616994fc2d7f4cb355e157451a2dc27895ad379f3943deac2259
dc1d7ac7b74905d56540d8663b5ba84d136914869cd6bd0340878862838e71e8
a050ee2a5500f8ed12f4027b1fe59fb3b418addefc0fb41dd18e9b2312a42d59
5072aa4816cf03dd5b7f9e79ee11aaec4bab7b4e2b05996ed5938180c9586cbd
3ffd354cc84c987e4cb0b23993da086bab65d186f1f6457807e074caf11186eb
2b64d9be94c6127b921a059d2b5cc3c0f4fc0800381b713129650e23a476427c
8e18789792fd4e6320941f43cc71230c34e84d24e3db0be6975a1b71cd592e2e
Fiat-Shamir context

a89bb16a3ef1cdfd0323a46e3c8ac7ed3f9344080c05c30abbf04313bf6998ad

withheld proof term: 566 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.challenge definition, Crypto.Ledger:48

$\{G : \text{Type } u_3\} \rightarrow \{H : \text{Type } u_4\} \rightarrow \{q : \text{Nat}\} \rightarrow \text{ZKP.Ledger.View}\{u_3, u_4\} G H q \rightarrow \text{ZMod } q$

SHA-256 of the statement above

347bd0147a5b2b49888bef6e490cf6814b38be9dff5f009f58a90543f5167487

SHA-256 of the kernel type

73e8f85672bb5290ca866593aa2f8c2403d282e0607fe6312454aa1fbac03ce7
commitment C

6b997d0131fb84b93143d816db85024e9d72830f9dad2438be28c46699c4c2b6
afe08be24ef08fcf744a975bd768a1cb65090b58039b88563de90fa76b51aee5
5301a7a1494fa8a03020de500000bb032807f4b31fd95723a9656dcbdd236f5
828f0d8bd734ba502ea077d78a183579475125e69d06a22c413d53537a6107cd
17ec62c9d0dd3b0b7f61405df7596dd99a2980ab60616b57e7a351ab1d60fa8d
9843ec486f0e42b344ae4b49188eef8ae30088efce573a23e09582625b3205df
c7d97ed601cd78e156b75fa045c411c372c85465675cd4edb1b5db2d16bb58ab
62886f931d70e040ce4baee2e6539aec3bd5f9707ca8ec8926653692ee7ecae
first message t

a85747b0ecc79c88b64bb07f0490fcc8e2f9b83f25d3237aa7f11ce59bf00bc3
594dd9085b634751f39dc992a92dcf5ebfefcbdf285c7d0ab2ac8bb8f886c510
6b530cc7b19fbab5aa4dc4bcb88797e8f6e10fd620c1e488e39a5fbc3a326d4ee
d67c58a3133a2e2fdb60276cd7809cfcc06c80e128cbce0406cab376ab6f4b59
b49cb9484cf4cf565585b1465881b5e826dde8a909f7e04c529ddc62b7a9ecb2
57aa34051cabd2c254f4e0b962600d46ad754edce4ba3577f24b2e52d6870896
1e8016da4b196501f72a9ddad4f683c362d2ceb01a243aa5d6cfa3d47eb5c5e5
45cb5bd96e395098927b9095e5c3b751bc12e0c98c2db993982cf58648f0da97
response z

13c74532f976a54e0c0f8b21310e4cfe87b5af491b6535d42aa580c17bb79b0f
c0a9c22c4b83f208ca35549a6b0f9ec8e4902e04aec156eed9108a25a23b2122
906618e9652d1bfe16e854f357bb38481eefc6729dacb27f1a7e5c44019ab449
2abc2200d673e73e03160613b8760763990a848d423b4d6ff1a1f1d02924e6a3
bde7637a3b309a95400e8bf0cc1268cdc99115129cf8bfc35699d637c25c1830
29301264b66ce79411bcda54c21373ab3645f504f07a5a90f2f6b32df266f400
25ad3957c23e38c34064b578520d4302306757a361f80e4160e226483771c4a6
10b2c981c1bad684cb178f0ef35840836502b8ceb8e1af84a972e52d3ddea69c
response w

40b98f03e6ebafd7c2f9ba6b1106e1700238b065e5eb002b6b443edbf7b9a835
5363525822f37c074b210d1d6829ab6eb61c16095e78dae8dcb05f9bc99e6f3a
5146598e0e77e360f3966c0f8836faa6623e4e74a2c31139c782d28f32785630
8c919b51d13bb75bde7d8005f43bc337e00ee28f98eb0dff3af856846a74207
4bdc1b11d39934aaf31972bb010cc5acd3c1aa7d3f7d529f1218bbfcea730f
4e1a26068f86cf05c17015da543960385a46c527badd198e93dec40337de8143
2bafb3c40e0cab8c159934008199f4cdd5530bf78209e06be1954062e59c0398
0193d49589cfe26c10b5502f0a02f832757ccea389b67c9b2e0f8c43432fada2

Fiat-Shamir context

b5019f815b5b8a322fa480f442e8ce0f08be2a5d6448bfca8bf1e6ad575468fd

withheld proof term: 99 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.commitment definition, Crypto.Ledger:44

$\{G : \text{Type } u_3\} \rightarrow \{H : \text{Type } u_4\} \rightarrow \{q : \text{Nat}\} \rightarrow \text{ZKP.Ledger.View}\{u_3, u_4\} \ G \ H \ q \rightarrow G$

SHA-256 of the statement above

99ec958241b9165874b1e0bf8045a730de11512b298313a175870c0629f732c4

SHA-256 of the kernel type

5e8fa17c0798603165abe651c98546a4ffd61f437e311872da31f9e1f8542179
commitment C

b3cb5d99c34d2664cff8ba281a0f621277c7c99d95378d14782ed2e27d0a9e86
9c96e7723517faac50dde65ca14a218ac0541140021ed2d0c6c839cb85020d5f
001cc53cdda87415ce49db2e961f808d9397725838c271b4a56cc510c5172b79
60a2d4a190cba4a7b17910701a022e35d048827024ae8589e41365519aaffb11
505b5a94aa355ed17fb2dece75282cca8a952495441c35e95799679668d93e68
1b6b21d97c38704c88d40dce12c743cd3b43eb3713775804ad659c81a7a2d894
c7b81cb7d23517b62cf7b6586680809d94226c0def12940420f17e3a609902aa
cf6f8e61f68d344e205927322b74d995be0c9aadbbbffdbad5fccc5fda473711
first message t

909cc0d5aeb1c55b6aae1d08b1be17b83c306fc087eaf3a921a094813991f783
5da9fac5d14dca09518d5254b5b4fae2944d21f978077ade6674b9aedef7a9de8
aec1d571a676c5be08d7f9cec3f9f5a6d76e2fab7bea11ad59f80113989db484
ba6aa432ab1659a1743920157e7fc6e2b1845ad96d9dc869a32808c17e4eb88d
8ea851c3a6432664b366741904209be00506a32287edb264e00e56b632a4a101
b79f0a8b20d85c698f3e5166c8ab56496e8a94230b418c5c45adb5c907e3cb14
f37809289a04a81cd18dd400035a5015515ba1b75a1d2f363c2bbedd6047dad1
673ba1526b8ea730aec691788103fd2fb4d0fbde6a5e8ee54fcc8c27e540970
response z

3e2acda468fd3cac936948a878994b29344796f62ffd95f27ae8ae6d1692be6b
f65d5e7d691fae789471e3cc1b81a7187481a12da6c4abf2c254d0e6178a7ae8
2bb85082419e5986927bdc3a2a94ff28d104f3dbec449cdf86f62e9f2c676c32
fd21021417d88f36ba4a5a6f048b5b9506abe48f977dda8240997076c4ec9998
cdcd05c7c07d5b1e6a1a1b02a5e39ff2d22092781d199676fcc65a97a66baf10
2e4e846dee5d02ee1efc1ff0a5e4aff34d2ca9f8900f0b4a69306b2f412c2ade
2380378a2c7dd3fe7f924d8e98957c6ac49ec80925e522122ea4ab619804bf7c
3a18663a0bda92daac88a6fd85c048ff9532e76c416188c2c7713aa61f818616
response w

2ced2a931c0d60b39bc2991b3e0e8a31a26c33727db47c9350eb81d61f11ecf9
5bdafbfcc61b38ded36cedaa9f62bfb26f9ce0e98f9ab5cf6085cad263fddf52b
a0769670098d274658d60799f3d1630dc22376336cbf60ffa84e2f31338a2900
2fbe2f8917548f3d72f0915c283bc1072862e14959e838e90b4d650960dd2d8d
3baeefdb4d996dfb93b3f7bde8e8066265fa5783615ed36121d764cfe49f7adb
99887323245c09dc81096fdd8d47dc7bd0e59144f04d2885056a062a5ebd5bcb
b79df8153d776ffd1a6aa518fda7101346fdf42b2bc6fd3978f6ac4cc13e6fc9
4ae4605b6dfeba92f0145b4a58b8a826fc0f3a805c76796e3557223f3a0b5896

Fiat-Shamir context

b877e056fff2a41af863a6152ba3b73245ef6c64e7dac7f2de10a5a0f5ac8d2a

withheld proof term: 99 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.ctorIdx definition, Crypto.Ledger:0

$\{G : \text{Type } u_3\} \rightarrow \{H : \text{Type } u_4\} \rightarrow \{q : \text{Nat}\} \rightarrow \text{ZKP.Ledger.View}\{u_3, u_4\} \text{ G H } q \rightarrow \text{Nat}$

SHA-256 of the statement above

65b3b6610f0c85667a9652abc8dbeefc01268b46b9ec03324605e68db1e3bb8

SHA-256 of the kernel type

9a6efed9579354fe632fab6040457889d14ad52c3fd0dce8214b40b47bfbcec0
commitment C

61a4af8da7e63fbac5943e718545c7e3922f111f46725adb2937d78f9d365469
4af7b07de73e6e90137040d32fdb932004a8b3410081609fd2fe3b80cf7c43da
de30c9a35c8a51ebb723642cc3022ece9b8bb8f13730f92df120ca71363bcf1d
32b768452e3267ba676718a617a4d52e52dd6d11c7299a5708235392af329df2
6486921d755c964d3366f6ef3bb3b18d44efecb6ffeccc7d6e4b4ee728e758a9
d4368a3143bac3246f70d3228654a90d5b081178cc48481aefd10a50e7322ed4
bbddd3a174ed48c2b4b5d5584da03df3da515e435ce032ea5ceb6e88026bbff
81666d8966397303aa17c80e9741c13abfe183fb7ab4b7ab4c9b8e4ca8402940
first message t

bf7946f3d456fc49589bdec366c8ae52d27d965a0f5dfdf9399c24360a23e30d
164d0868ca617e38374f649fdea62887969c2c14e28e1cb4d5c519df7ea54494
9360f71921a1a3b6b9e217ec5a67ea5b757cef4ee85ae779fc8068cdbeafea1
62d8e9d022139b7ca43d3378f079d6472a8ab24407b2d15bca0a0e722a896f81
a31b2579cbf76f685513e5e96b643a9df678a732d8613068d0dc13d027510d5e
1e16aadaf02026b31297862f3e9d4069d7aa8a287038a69633f799bc55564133
a45237976006d07a28751d9898110a2e9d458f76a55198dacb655fcaec7862c9
3506edd657a5560922170b6104c8c599cb8e4dc430bfc122928394eba49d8b52
response z

56cacf6e134d208f0c78c0e129c7e48864c6a2d2cc051db25b13fd497d2da9f6
2f121210d3bad2e1f8478d76474a589b4f084b150f621ceeab8a9e43e273e47d
a9b62be9f12c56ce38cc3ba5c126d14d5c6752bef7dafc15505bd15d10082b51
121ab932babdf5168d241368268e0da97012269bc3ab8ff57847f821e6fa8955
92ff1f617f91cd6e89ed68bde569a06f6b5555a27229b6f44be27f2199c8a24b
d2664209dff4888bfd78dfe3fbbbe84c179198fb5b75ae16c9473a8310a810ce
35c79a30d64d520590fab9d1af768654abeb64c5265a4bf9e3ad66166776059b
de985d7d7c1e5fc98134f8d2c0ec19d91d6933691b1b287564eb0b536e70c1f1
response w

313c6c8055c5551766b4b34aa89f4c9b02d29e495d407b17966168117cecc52d
ba98a8769612c0f529725b706c4b3a046f6c5e3ddc46a6e20bcf86b6901ce0c6
131bf8a4e14b5839a34b432adb84d4380d0aea4642126a050665f89d44e7c9b9
9c042027cb87a963b5c0721a87b90a5e142690f9144920552ee1c33124bfd02d
5de6914d2852b4e577eae3131c6578e6e3160d92cf3412a3b7f851a637e7d98e
83b24f60249064a370629daa7b9c20a53a796abe2f66b35f85859ecea3dde744
9316ce409dd9aae685fa845f383fa53c3f3ddd8e7713e711e3cde2d5fb3be1ab
40e11d90c75095b1f6a0969e1cd14726dfde3bb46ec5767d87a919c679bd053f

Fiat-Shamir context

04d453cd1dcc11c2de96adf201f23a5cb20509436e662514eef4ee57b462a64d

withheld proof term: 91 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.first definition, Crypto.Ledger:46

$\{G : \text{Type } u_3\} \rightarrow \{H : \text{Type } u_4\} \rightarrow \{q : \text{Nat}\} \rightarrow \text{ZKP.Ledger.View}\{u_3, u_4\} \ G \ H \ q \rightarrow G$

SHA-256 of the statement above

99ec958241b9165874b1e0bf8045a730de11512b298313a175870c0629f732c4

SHA-256 of the kernel type

5e8fa17c0798603165abe651c98546a4ffd61f437e311872da31f9e1f8542179
commitment C

7ce750bcd2821c606608cb2e599a50c7eeaf52be088f33e0302bca4c63e56689
edc79c18fbfd29d4de7a13fa2e4e8ac39807f842e669f84c4801e52c248a26aa
adad5f7f2a6bed3fa048294c8a72c7b9a4f03ae65fbcc53b632c593d84c6d9bf
29846b4d4f1ca9f3a5552c6c5fa48e9923e034a8000d9af0fda93f94d35a37af
bebf1cd0ec7c3cce4c03e0ce3503705cead87dc374f918d7292e5df2d838b8d5
0bc250355e6c1759e87eeafbca2348a5c7308c822d65a0992c368ca654b6673e
fa90598f52de19719ef01b93a7245310415a1f54dc91e333b781537ab8332dcf
1489c883f1c0ab1770b78992389c4f917dfff2dfb76129829aac48b71cc9b63
first message t

34428e8bbae49fd381155cb76102446cc0d540a82928fc9cd8addccf3813a420
02c69cb870f03e281415e60dd16a74c81fcc038ea33ef0391928210c516a99d0
59c05f9bd35cd93471b7f114fc2c3bc31ee7924668b1c20c9ddb5f25fb4483dc
2372e2942f9b2f0760f5e1ed8e5388325d8f4dca2deb3dd499b961cc7c6dc2cf
7c6395c9392dec4110d743d03e1c6877e5b306156d3e4b6c66f5bffa48382a30
5f3e2f294f9af22b87809961346f61b39b6d439a343008969a840c92ee956ec9
ef09579f56e32be2ee5e3245eba126e71d1cb14d9e3b826321ebc345793e78c2
9bdf5c4ae4fb28ff5d4f683e988b6a38f9db50bcf42fc76021ff45b8dc97b24c
response z

448b7e63bfc6a8f0d77f5dfb5e85523571a2c8469d59ad938b6dbad658e2c94d
cfb07d88fa4f379e2128a39c250f1666636bc7c1b620655350adc964b76ca11a
05884112fc5f08a0ff41d7d703b2826aea61b3ae4d27d0a323353c81b634227f
4d482281fda1e40cfa5d06fa7099a1e201530d438273a1736022e89a93ebca67
70ffb50cedae8b8567c8420e6832913507fc85edd5f51baa7d5ca1ab2d00d20f
a4f598e89e2d4209c733d8f9a92d6f509b5a10fe83a1e684db9e1fd15e233e31
79e1ee776346cb26f632cf49c9edc14b01958f2281105179353f09c7643ca9c7
eac184a903392b3a2ac8ce371255bb9f65e9cbe0ca7ac79a0d057ae106ad70ba
response w

125d44e327758e900c4d54c8b708e7f6cbb420f4a8452f7ab3bdb0b9217415b7
9879570322e23d2210eb9da0ae907659ef9c9de60f7c5fef5057b7b642749157
74dbf3ac94b851e8a8678cd2fda2bdc8efb305f515dbd771e93af32ca15e240f
2b112e8551f57ad611663c4b41eaf52923f7a08847769f3b2b1b01194530b4ef
6d1f6c30f64bf76989e7655e062667150f3f36f0217bde03e9c08ff3178bfa09
0a4ccb83d606e496faa61990b716abf3a64bd0f58ba10f31bcd81b858ee98fef
be59300ee8327c208932adb6c3be1fdb3c7adb54a31cef0a2df5b397bc8bea3
1b4e4b8e6b7405ba8c3460b79fa5e7b3e34df23446d8fe81735dbe8c600b28cc

Fiat-Shamir context

8f20b3505fac9a2b5969ed7d279a1e94bbddc3c5af02f11fc19972194e23331c

withheld proof term: 99 bytes • axioms: • kernel replay: yes

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ZKP.Ledger.View.mk.inj theorem, Crypto.Ledger:0
∀ {G : Type u_3} {H : Type u_4} {q : Nat} {statement : H} {commitment first : G}
  {challenge respM respR : ZMod q} {statement_1 : H} {commitment_1 first_1 : G}
  {challenge_1 respM_1 respR_1 : ZMod q},
Eq.{max (u_3 + 1) (u_4 + 1)}
  (ZKP.Ledger.View.mk.{u_3, u_4} statement commitment first challenge respM respR)
  (ZKP.Ledger.View.mk.{u_3, u_4} statement_1 commitment_1 first_1 challenge_1 respM_1
    respR_1) →
And (Eq.{u_4 + 1} statement statement_1)
  (And (Eq.{u_3 + 1} commitment commitment_1)
    (And (Eq.{u_3 + 1} first first_1)
      (And (Eq.{1} challenge challenge_1) (And (Eq.{1} respM respM_1) (Eq.{1} respR
        respR_1))))))

SHA-256 of the statement above
e2a58eaeefba6ee728fb13d7736f8cdfc95a63195028fe21f2f793514fc1a9e0
SHA-256 of the kernel type
00dbb02b91ac936d99dc59db0c4020de834b1c36e1eeb2382d54ddb823abdd97
commitment C
705ce2a6c6fc33cafbce6953bc1920d00e7b9cda5356809ce4c166a62210019b
14875b904da62aeab06359f9b99d360e384b1103776018821080339f2174f8d8
10457531354deeeafa6ddd2cc716b04f4ba8891a411af5a61136d674a20da041a
305832f362a1f39b1426d4e81972b64e181eb5dade76c1d5464170875c6facad
64aff12e38c5d35d2d749194b6beb504698314f109366e852395f3f5b85bed39
9007492c7c1d129178950c2e04e1f4d4934a749a94552680cfa0e559f2e3f50b
ff2c126d08ee98a989e5e336db2e32644503cfd31deabaa5a104efdc50b7629d
95892adb1c45a8f27f9f679429e4eab7ccec14d854136c26a14a9194fb37bef8
first message t
5b089eca20b839dde7825cfb7293c5c509cb5087a01a8a9f41a514f2f159ca71
e94213ddf02739ff9b9faaeb86378de2e63d9e6a0307ea1478f09cbd5bf8755
af0b194758bccdd99b3cb5046204d7fa42b4013e3805c301b6fa4bafa4f67f938
0231ae06da7d109ccdd37ce87d0827c339ce206ac38f7439304c9a7717da191a
5add122d67afbbe7723879c75cba880bd0868e635e9205cbbf85e6c39ef70a58
e515295f9daa80561dffdd5f3521f1c97f7d1d96fd6c02bf5509aa22edf159c0
f416571bc15961c414061ac03cc3f1118f9eb540f4574561a5c8b6014fb29467
c827d6744fc4ea78f53113ce4649bbeecd64e403da7db17401e2b15e16adbca
response z
7fbce03ab0ea6ef29a09fa756877a6aad17e8655bbc68ca9d3a029aa377a3142
10b7f5e71a693125045b4e0099d25cd3a8352f66b80c6e6e3da059b2b5d817ea
aa4ad32a5f1566e573238c0547a2afa7976bf603992b0c109a0f701d9de5a9a4
226a930a316e106498a6350654a73c2b042ce70a74bbb206fd561214a2da561
939e4b26f04ce79f304a9eeb72c82d940df85617e138cae439da7ee6f839b836
0e6eec67ee955502e92d792be0e4c8ccaaa92967f893d48b00dce41b380a2b57
11cf2760b4be72c33087e49bbf1d16183d8effd4ca8f45245c8d7e2a6deffd34
ee37b0bb6a0361ce7a25d8a41bcba95d4649c7feb34f816b618735d88b09a8b
response w
777d3f4c21fe7742e2a6dd1c7f0034e662c54754cbabeaa021cce8103fb45bcc
1c38d97d01153bc6c538938f4828f4035d655fa63ab946aab58ff7aff5caeecd
9ec9e39757dd0e3d235a625c722e128fa8991b14a044d6b30c34dea5f19f9b91
ea7a14f16553df694e6481a0c93fba292b470ba8e60b85f78e0fba42c0888d3b
896453ee34eae8b005ab4072c67e02caf11bb46c103fb14710d64bdfc4d1b640
2041d08e3c514c029245a7344c73a69c69e660ade7e1371e22f492df0d754729
1bc5b6947a48f5f223c624923f6833f4a5bc74599457df955626af68a2e24fee
562ea02af336a0defb1a841f2c90865898e571971869a22f96701b127ca2c45e
Fiat-Shamir context
2b14b3fdd5deb94aec53ea1c6d5214501bee4a684ce8a45ef69ec07706a41614
withheld proof term: 3 247 bytes • axioms: • kernel replay: yes

```

```

ZKP.Ledger.View.mk.injEq theorem, Crypto.Ledger:0
∀ {G : Type u_3} {H : Type u_4} {q : Nat} (statement : H) (commitment first : G)
  (challenge respM respR : ZMod q) (statement_1 : H) (commitment_1 first_1 : G)
  (challenge_1 respM_1 respR_1 : ZMod q),
Eq.{1}
  (Eq.{max (u_3 + 1) (u_4 + 1)}
    (ZKP.Ledger.View.mk.{u_3, u_4} statement commitment first challenge respM respR)
    (ZKP.Ledger.View.mk.{u_3, u_4} statement_1 commitment_1 first_1 challenge_1 respM_1
      respR_1))
  (And (Eq.{u_4 + 1} statement statement_1)
    (And (Eq.{u_3 + 1} commitment commitment_1)
      (And (Eq.{u_3 + 1} first first_1)
        (And (Eq.{1} challenge challenge_1)
          (And (Eq.{1} respM respM_1) (Eq.{1} respR respR_1)))))))

SHA-256 of the statement above
0502ee4fe3c91ed2efe0611cfcab758196a3567965acf3a14d7746bcfb7cb71

SHA-256 of the kernel type
42484a48e901556d476bca07b75c0f3f706926a3cbc582e1b5e891bbb3ac9642
commitment C
c812f9a53c66ff643cba56df7d97970b637bbb786704be5553e2153035b845b1
a751625e450e69c37695330f6cb43027f327efa38455ff5b7e23fb46348059e3
3ee16d3d90b7008170d8541c2178831c157792853d4bdf6698e63cdc7848e75
fd06b860bc57151cff3db406d5ba0c24f5b6f43cdc68a92964086c360c107179
47a60a84b6acf06730dbbeead11a5737cc97e8db213c839bd88bca7b5d78977b
14b397583830a824da66b66cd702baa07806e54182498a5bd3e6ad578a9b461e
e7483060af2dbb0832a51c8ba137fd9ba058e295e30b7b57467654619092eec9
4f23d8491628a6e49f92cfe2cdc965b9291bede4e17b5ecfef3fa7ecbe4d7115
first message t
9ea102f18836f58d04bda4812ecabc39af5c03d8b2a37ee9aa36d5c47038b316
6e2d813d6b5941adad5dcc6b313a03b3bc27af55a014c55560911100eedb129f
22052aa12e9c8e8737f9ba2c4e3b815e34b5bda177860c576a342cfb96332be8
5f7124b71036bb3a5be718ea82ca4a65f173d7196a20e1289739843a3ab41227
8333ce140b0751b0e1fd47b0e0f0e4a245cd5e816c1b5136a9afd0dc5f9b49c
d9112ab8e933086f9e08979bbdb086c53a9c2f5f11fdaf969a67e85979f59f06
13b9610914dae23bc8c38f008d32f87f1c7bdb42e9107e70975e4093ab45bb53
47c1dc797fb3fc5280704271f9f513ef55a11fa4999245d3f971dc71978f8cff
response z
655288185571b68296f78f033a4a141d3b236b3060b303c1a666d138a37e5ac9
5388ae9cb00bc31b8bbdb9a68c2edab74b7f4fce61a083130283b96ff51d7901
c47a0b0d3f7ed5e148ff7aa22fdc947020980055bceb5ed5887e9be65be85677
f90cdb3a283b3014cd8f17c0ba0190ae2ef38877ec7f51bd6722ac96b3cdbbd0
df4743ed9815a6ab11366ac2fefa4c27dba71036c5603b43ce2f585ff2b0ea22
9fe7f87c44f2190996ae531dd865388f88fbeatbd40698fc8d100d760e499206
560467322553f182e31660bf7bed4d360ef16bc55fd163571bb56b7520874324
13815ab2ac682f6ae8c86f166d535223c8d9f112f7b86a1f17a335ab5ba3a004
response w
6e5740c5d0c34448ee86d90dc31f1c7acaf99c860acca3eab93d50d90ce4aa69
675ebd9e90b2b442d39e11160a2c3e9e3bedf3f7bfd6c16f6db32c6ac0657064
ee20a940263694ee81ea2c1ee22bbf3b418c6ffcb7701b519d838001aade3351
f509a6a58a5a2262ff349a7094a22bec883200c63a28386fde03ed4ad1f68c45
bea12ffd46cf0f8ce7933b537664ea9c49861a8844f21ebbb48d54f73a713e2b
4d82f2c9f58913e22bd167c7d0737909f73658fb69ff15201f7759767dc28924
e446504471a0809cf920ed4eac43d498da37a6ff44a61b9ecaacbecb2e02842c
85c53684ffad2a1a8a8efa474539273459ebff5d21966c639b33e0832cd90215
Fiat-Shamir context
7cec5a86e2389e5e1298646916851255a32a9f9c9416df08377db9f2826ee22b
withheld proof term: 9732 bytes • axioms: propext • kernel replay: yes

```

ZKP.Ledger.View.mk.noConfusion definition, Crypto.Ledger:40

```
{G : Type u_3} →
{H : Type u_4} →
{q : Nat} →
{P : Sort u} →
{statement : H} →
{commitment first : G} →
{challenge respM respR : ZMod q} →
{statement' : H} →
{commitment' first' : G} →
{challenge' respM' respR' : ZMod q} →
Eq.{max (u_3 + 1) (u_4 + 1)}
  (ZKP.Ledger.View.mk.{u_3, u_4} statement commitment first challenge
   respM
   respR)
  (ZKP.Ledger.View.mk.{u_3, u_4} statement' commitment' first'
   challenge'
   respM' respR') →
  (HEq.{u_4 + 1} statement statement' →
   HEq.{u_3 + 1} commitment commitment' →
   HEq.{u_3 + 1} first first' →
   HEq.{1} challenge challenge' →
   HEq.{1} respM respM' → HEq.{1} respR respR' → P) →
  P
```

SHA-256 of the statement above

bcd5c5716f1e4539d50c43bb53bc9459b1aab0d85814361d1dd9dc38e539986f

SHA-256 of the kernel type

c6fdd240afcfd788cf9fd362ec455cf9ec1b7b394cbb62a9c88f473d29b2c6a

commitment C

e613c861cebe3ae363b0216e47e68fb2efec50267fd07b9d82b0976e278d3831
48dc9bbbc7b7f5f53f75d7703b0d452e38586314cafd2869ce0da56a2a4bd86c9
833e0c0dabd47860cc8c5ce1dc8a8136a28c7dce51c31e5d87fd4858c8b363a4
30480765a17b385635026d1624945303a24cb419a22592f83fd43bef2ebc2a8c
ef3b34d4fe793523a2ba90aeb9f839df323f4a68b5a2cf5a7271cca74a4b33f4
095471c8b9a24bc4735d5a721098952b2f28a83dc909fde91a0126b661e057ac
65645c5a572bc4290c8855b3fc445f5239232d1eeabce8e700e8a9cc14d99514
997822e40285785beb5e29f55fb43ad0f4b60df8e385113523c9c025de84f1f9
first message t

365d904448d4b29ade3af43468ae4e062376d1aa9b60b63d7fb12667464807ef
1e99effbd51a6634e182282bc8ed4a67206800632e0560f6718a896a573aa905
92fdf94e4abf2d5f3d55ca1e742c1ad8a34be22c6e6d30881b34758b5df09792
2688522c5db8476059aab0433653d2cb76e6413f52f22f4a10f2c9c2d372b484
045c671b7333b3074d19fedbae0556c731577be5055a26e0283cf4b2b353b14f
e0400cc975f4b565282bc30c91555fbc2d53721d6fdf92e8348552dfb6d1773b
0fe913aa5f1aedacf32b0f626315fb4aba5a036e7aad922236e18014e9918e20
908e7c0e1748ea8354aed7c3019476a80ab90f27ee7b3d85e1cb438711a38323
response z

168dd5d019790a4ee6034a5b3090efb338ec4e43d6403d7c7cdb26f7bc6b1e8f
1d4ae892560e170d72cfabce2cd4fa6063c881a4a111953d227f9b83b8f01d4e
610a65dfc213a152dbb5113ea2a5e3ab3e8406b99dfcb74eca73d1f52137abf5
9d01bc74e3df1ade89a2e1981964973a3b7fbc8e8676126dc601e57ab83cc073
dc8b33fc7f049d6c28a3bfc0d5e6b24f5f2395fae49566a6d541bc6044a612b2
0ad0d8c434c6fd13cf7ae2b4afc0cad769ee66f29e02e3a1dbd4c48e753c7932
013b81dd1721890e956264d845d5f356b8d12b0947f8afe417d461420b95bf3b
9925a6ccdd442cda3c508b2d26283cac3a662f23f665bbb2b0365cba8d0d2d51
response w

13ca0cc31c36c7a72579e4ec31038909e761ffd4fe1e146d36e6390575b34106
5a2fdb7e8fe4dfe7d2b41836753bcd1d9576253b943b5823bfd4d734ae5b6e234
3cfc974b829c9cff629fec6aad3c492de1228d545bd5258dbc43c0bc6026eeb7
ac6794fafd784363f158215ac3dcdc2e43e8bd796470ca912789367d45608e77
641661e05176f5b2017412c707256d5bf1164129de3f3014bc7a0ea7cfcc09e9
fa1d3cd24c8a3a2ac4a896d9e685b31f832898c459ad3c3dc9ca116f4c8284fa88
9ef5f47bee5060ad85c1eecfaf4250bb1ba5a79108a4444c27aaae7c613ba5a1
81e6a444ef48e853bb89c8560ec3bcd4c8b8595e6c49d751d14c312704cc1e2a

ZKP.Ledger.View.mk.sizeOf_spec theorem, Crypto.Ledger:0

```
∀ {G : Type u_3} {H : Type u_4} {q : Nat} [inst : SizeOf.{u_3 + 1} G] [inst_1 : SizeOf.{u_4 + 1} H]
(statement : H) (commitment first : G) (challenge respM respR : ZMod q),
Eq.{1}
(SizeOf.sizeOf.{max (u_3 + 1) (u_4 + 1)}
  (ZKP.Ledger.View.mk.{u_3, u_4} statement commitment first challenge respM respR))
(HAdd.hAdd.{0, 0, 0}
  (HAdd.hAdd.{0, 0, 0}
    (HAdd.hAdd.{0, 0, 0}
      (HAdd.hAdd.{0, 0, 0}
        (HAdd.hAdd.{0, 0, 0} (HAdd.hAdd.{0, 0, 0} 1 (SizeOf.sizeOf.{u_4 + 1} statement)))
        (SizeOf.sizeOf.{u_3 + 1} commitment)))
        (SizeOf.sizeOf.{u_3 + 1} first)))
        (SizeOf.sizeOf.{1} challenge)))
        (SizeOf.sizeOf.{1} respM)))
        (SizeOf.sizeOf.{1} respR)))
```

SHA-256 of the statement above

4b0e3b28efeb9810e6b2b12543334361c2b23ffdd184c26108f1c7aa452131cf

SHA-256 of the kernel type

1bc115c52f675c45ae1f897bc0368a85a133992a158eb77a7cbd7e11346ae449
commitment C

48da625d3e8e81e9d0b16d7a87ed25a167b31a102ec37b368200887298683c24
2780cee3940500f09209443ae048dae545d9ca916aae36e4099c4e55adf726f0
95b9e1fb718c3387de186da2c299117ca57ac4198235f8fa48d3c06ace87f69b
015def00eddaef73d7667c6442cd5b85ae49380eec11e3e101caec51185d4331
b572ffd92b900a77a41546b96e2d2116d1e2f276b41afd52f2c7c0bab583a6c2
2b7da553b7216528830961880bc16ee5cd4edb6017eac79a7daf2ea821bf0192
d9ff26704fec2852c3f75dcc918a4dd03ac70d10aa7e04156184d2520e1b9fc0
72a974a9491b05dcc77236f9755955776aeb3c4d979095aac893229c333f89c1
first message t

b82a17aeb6eb24387e50e1ca4419517bc8bd5d37b72b5c38e8b9b7667ac289f9
9d44ba93b8bd11bd68c19fb4718ac91b1f6bab5a7f8a68c25693ab2d15b835c1
d7c505d18d3622289e9bccb25a4d40353206a65e5d22f59646a59ab1ea2583ec
20a70aa1e5584936be5ffaf206452cf182e0716fbb98368b676f518d77df93d7
68fa7e65a5293cfd298d6c4d7e25628c37dde5d948b79f4711e525f7940dbe85
8a1933b3294740497369264d93eeeddab4199264bca962823a68b46affcc145b
cbc34dbc3439b07e0be8f416f495c705f051bef8f33df8340feb38a082ece73c
eeee014a2c69741b65987f7ed82b78e0a4f2e582aefb8b273328655904b4672
response z

5012cb4e61a34de366e47ac8684f9e410dbf6f8d0ddac4b980fc1caaa5bb926e
de1252cd2b7eec6368ffca69af3efda8860f334a8d740838280672aadd32450f
6234e127dc4db5cdc469c5a54b7cb03eab49868f9122776bd66922bd0cc4c2f7
69e0b4ea9816ada479ae1b0e2517dbd7a0a8786b38e954b5bb73a94c674826f4
0c9456655fc3fd02266cf70d456881be81349a2c8863839a5bbd2309072520e3
7ca8b54b99bbf4a1ec0c4cea1193031be5c36da1b255052be84ef7ca244256b2
68c0acad55346b4c2c821ee217ca48f3da7f3faa92262c546bd55767b63aa271
3c8f8dff29d035f7379a7a542ac4279f2b428b134237b26acded41b31cdf8f2b
response w

f746b5779dcd48712a6600f43599292d0b5329724d0bcdef0537214265ffd05c
5c97025c4303f8597a72de0b9b1f5cc5d43e2e429470f9c15ce3732ee3c8af34
53eada75041e0eb12ffa9b03b230e18aee1699b27b872030bd7275b0f0bfdbff
083c7cadd3fcc5d1c3bb15298f20f15f52d3c120ba309711b753d3f89e6f7d13
f7ba43235ad133726517d5bb547ffb9c25847a2581e38c4084ee7d4902c49029
446c7f1ce3999a17341816b0e178a90527d376df1bd2c49ae958e02c29ca3d89
4897a0c9351aa335b9f27519cf3b953791e5bc19aee7825c14f410a5f069c6b5
0ab7d8f4a9f6d12bd14c1444a60c531a3c1c05b079016ab09e563eceed6a0d4
Fiat-Shamir context

87544ef3561c0ab1c195bc66f4564d3d9db47fe394708913f2e3f008a627f4ed

withheld proof term: 1 281 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.noConfusion definition, Crypto.Ledger:37

```
{P : Sort u} →
  {G : Type u_3} →
    {H : Type u_4} →
      {q : Nat} →
        {t : ZKP.Ledger.View.{u_3, u_4} G H q} →
          {G' : Type u_3} →
            {H' : Type u_4} →
              {q' : Nat} →
                {t' : ZKP.Ledger.View.{u_3, u_4} G' H' q'} →
                  Eq.{u_3 + 2} G G' →
                    Eq.{u_4 + 2} H H' →
                      Eq.{1} q q' →
                        HEq.{max (u_3 + 1) (u_4 + 1)} t t' →
                          ZKP.Ledger.View.noConfusionType.{u, u_3, u_4} P t t'
```

SHA-256 of the statement above

8f07e8f776a71f1e6b483fcecfe3bd63a5901bf0bd887f670159057520b093d0

SHA-256 of the kernel type

6a2a2431bd47b91d5d3afeda7251660cf55b35a8fe44c9b4cc428acb670a60ff
commitment C

50e3a55c4730cd3700d978343d6ed32bfbf6ded6ccf7ed8eea05ef4cef76197c
14af7f0e3f75aa582ee2cee7e5b42683cdca3ec19315359d49a7a0a6ef10752a
f1c6627b9cd7182d6dade573761f976b05400346a377aff6486fbb737b4de313
9f2f21ace437375c928fbbalebd821f6995da3650f797bd74a70c772d7d1a81f
48c16f5507efa75770bff4879f9940c696ac92d3dba581fe641099dc07ca5188
5e1763ce909f00a359142bf1c9d7ae9d95cfe01648354c4875ca9c01aa961ba6
59df194208847b94a425c25174836e54fe7d8b7c4c9fce12c093195da0d68569
e5c6442d8d222e4cda546e335c240a5bd9ac84df73d15cd61b22952db3b7bc67
first message t

45104cd96d004e224c1e088811ff7c8dd1719323d824d24262692b8b5ec9fc8
0cf495527adb3cf0eaf109a2f3c772033f24e692a88a02ccd259268bd8932386
f536689d391a6a8c672cebeaba69e35a5723f5bdf9825ed5f1ea438fb988359
07e6ae49576cd08ae740043d55745777c02d6c8f2a4226acfb004448efb09ed
ed23b68d2a6883559c4571308f7156c15f21d5eaa579207e883f5915f82e1588
4b80aa0df58feb461dd51784dc8df329a2094011894181e2cbee7e79bae6fb05
dbfae263413e98c136b91a5f896d3622c831e3d200671a1b38fb5fc330decdb
70a63bfbcb0a267c0d241b8673803e5fcf97d4dbf39cc2b17e8dad22dc46831e6
response z

576dd79cf06696d1018248b05c66f1dea575ec8f4cca5690a5938b359ce142b2
572b60313a298d86f1e2898aa0573244257defecf6e0abd2b44df56c8828498a
c08aa44bd7587e7f0f078ef6776679f3ced905ebad8186805d263daa7867d9bb
aa781e459276ddf239b4bfb5526d5b070ef0b8e932f1e92afc9d795fefde2663
51df688826d668347f7b3729f58e8de2280a3569fe64953a844e0a20b24fb367
293115b2155acc67b72f59e72148bf67a92f4311a6f28c591a4263b636c2e13
03e2039db6ab1df449c8e832271a85121009211537b9f26a2d5451048b1c66f3
d31b609ea4332499978166ef18395e1eaaa518efc24bc259d212acdbf8fefe
response w

5a2012833e1a67583db4fd4a934c3e501a7bf12fe82c788120ae5dd10846b0c2
21d7fbf00bb21d1fe687ab02d81b3958e24eb2369046ec6028e96b9cfbf763e5
bec15595eea97019b25c41301d1c9c4b8d384cd2a5c6b089c05d64fc3610b976
c1a70a607070b117ee173b261befa8876a8a4286924eae208fd4b9647aace505
be025f6e6e3b63d0e2698eb2529f56e931455a713afd042640de3b17832e0b69
ba1525dde596a64caccb96b961bd2926eac4c820ea26bf3f8f70c03e81b10eb8
babf8450423c9add99121fbd9dd7be4edc3f1f358af96be65cb70206497bb4dd
0d82624b1cee825dc7c00641aae68c8ab346ebf2e44aa79cc658b5d3a9843460

Fiat-Shamir context

fb2b9bc89fac9cb15e6870c0c0ca6597477c3a2f813f8f232a9ef78a0369471b

withheld proof term: 2973 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.noConfusionType definition, Crypto.Ledger:0

```
Sort u →
  {G : Type u_3} →
    {H : Type u_4} →
      {q : Nat} →
        ZKP.Ledger.View.{u_3, u_4} G H q →
          {G' : Type u_3} →
            {H' : Type u_4} → {q' : Nat} → ZKP.Ledger.View.{u_3, u_4} G' H' q' → Sort u
```

SHA-256 of the statement above

451c0a71b333507549296421509fa5d0f43cd0eb4c12638c54588a19fc04447d

SHA-256 of the kernel type

8b4a23814dde5214c4031736a0dc77fcc0235eb6a6fb1a01184fbb7b104a7343

commitment C

8c062a8ea6c7c325b0a0fee109b9cae36c60b8a04a7615308d3dff1465b23608
59d49ae111c4892c6be3909c62748b96816e77306902def3b99627af5ffc9e14
36fe4180b65149166658a1cd5c606adfb2872e54a6e71363d5182d1c2300421c
ebc76b2584f39e0316b9be79e5ab891b04dba89ff8bbf0499d8561a34e326410
bb42e1620bb5d240580f89fce137e2c66d7977ca794a0a4d2c70eef96208ee12
b9278b03beef13d6abd2569d75e82c6adfa341d24f15ed6404812235ce1d18c4
1ba589694c45af24f522ab0e53bf980118f65ff50d0696c6f93521de9dd72aad
f95835ebc3e4c7d7d53ebba6d036ed9e6b615deef44d58dea5a73d3c035592b9
first message t

cb56e6505a3fbe28d749955c32a25637ce48a6f9874784c884b2809cc3c4ac7b
fd322a12bba2c975f978c8bca154ea1a317103bc5b68ca5a9b3930e221aa7461
ddc4c9681662f3ac6d00904cda3793a2ca5aa0da6c2af880538a2bca88077115
e71812b75f195a3b2c2ca2d55d680e171ed88e70c05f72d9c4ddb2e51d2c8598
d783da5b84e8742996d140aca0ea48679ec1682d8e88cf8f4f9796bf2bd1ddb5
4d7f354d7af6f896dfca35b4f0e442d8873f3ad1964b0442964eba3895ffa25e
cc71ed146a994f6808884f7064e120bf62cbb4fe9edcec5ac4e20a323965dd51
43a96926bd0355241dcb0b4fb47c2661cb4d0b027f1d65eab2d8572afed85226
response z

6ab474d1821b12b84a16a20371634a89c4c0ba4730038e2fadb7cbcf96c2bed
66d97922b5d01884cb05b13bba8276d3f0b34543d8a53a99100d590d8fc7c99c
a34a8dc597a98534e2fa66a8f349cca47b0ea773fa9d81d0514b519b0639c7de
be5dcb02fab424d27f4e72c8928f3426cc36d4a3294b4f36c43facde455cc416
e242b7079ee96e67ed6495e768bc0df67d6b1212bd48b6b271b80142fbd7cb70
20a5d28b92e3ef3f8bcfdbb5f9951e84de92e940ac69acc1308eeeb3949da4
c35da4eeac496a7ee82a97939994bc37a12b90443958191e9e2b8da4b3fb6429
67602f9a80671159daf4f85da51e5ffcea38d6d4ce1de17dc0fe2fa70545ff12
response w

101128ca7ccc0796db227fc54a7d8749fe7390f2dd5a8319ff10d28657787e67
1dd9d1c3f56a44a0a119809839a89bca552ebccf1e2a895303a7facd8164a274
81f9c41b6d52e51af18ae60032ddf0db347a07c9700b3e2a9c727e5558697bc5
af5bb22a51eb17d6de7c77bf8d9ecc5b194a1f145b446776a6561607e90385cf
0a3a103dcd8c627d11eee8dd4227cd3150a48547c2b57e8b0a32f633c70a4f97
785de55b11b57e3db2605a9d3e6f62f8a5d8e09838f5029eafcad978b2dcb805
c31c422f623974b1a9558ab16d16b6bf3eda1cbb7c553c29758d8d6888879854
3781032d8db28fa107cecd4719cae8c833a9a0c93754897c598ca5f910745913

Fiat-Shamir context

f36eca80b534eb10a81ac1c3b0fc7de996f251d3479a8392cf006ce31ba85d1c

withheld proof term: 943 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.recOn definition, Crypto.Ledger:37

```
{G : Type u_3} →
{H : Type u_4} →
{q : Nat} →
{motive : ZKP.Ledger.View.{u_3, u_4} G H q → Sort u} →
(t : ZKP.Ledger.View.{u_3, u_4} G H q) →
((statement : H) →
(commitment first : G) →
(challenge respM respR : ZMod q) →
motive
(ZKP.Ledger.View.mk.{u_3, u_4} statement commitment first challenge
respM
respR)) →
motive t
```

SHA-256 of the statement above

0c556fe373d9061cde601badb58be52e9436fd9f64d013371f8ec0e0dcde9a6d

SHA-256 of the kernel type

dd8870c2bdd8c00bc720ceec7c271e0512b9fcfe58a99d8d3b25abf35b1ba8e
commitment C

6cb8783ea5a92e853a68812825f92d5bcd39202c03b0db89f2aea7200beb5790
8974d56b543c6a6e9a2b639e22ae9a50aeb499614e9a2e1d860ea359032da2f6
6544b2d02451f7966f687231091de546ea17fd6f9848e88b4c37f7bc2c26af51
a805f946ed56fc718d136ccac7f3fb4fcb84cf3cba64a36184460ee5d9c91e31
e440ca8e7f4ecdb380ed69d8eed02bce501c05915a9ce2cc8870b318f85d5c6
f4337e7feebd908d3a582e288de5b213f793fb0250fd632dcfa651c54063872a
99097d95a797791690ade1014b805486d9c18f82c6d52a3a82f705dfc5a20ecb
080d4008e353b446d2cfa5509714df5b445432598c2dfb4ac4b6e3139dd2baa6
first message t

9af55bd288125d56785e176f6ef6a403ff8836d3f4d427a9384a5461be33c7f9
11532c16507a428f7bb650dd48615fa8c15d445b2eb2fbd25d56a178e9fed20b
8ed9d531387a1fcbf5e247089c4e06d70eeff0e77b4997358286bb2b6d40f34
0482dac09455f0149461b53f9b12d562ad04453f3c9e2d75980da95758bbcef4
09dcda9aee9386748b03f4f4efef82e718be571bd916de2a2fa65daf68b74dc1
a01fb9bd01f81573acd0dadcd4e14d2aa0f1411ad4d845c7d7ba06151c5ebf14
1981554c727fe2f8bce959c78db7b349404c45dd419fcbfeacaf1dc01d33222
bae44a475500b577332ba6b0a1ea932e9213154c74672c212c42bad5965c26b3
response z

c8e825312622d24cd0cecb5c7502818a3150622f08df76a13cb56159805b38ea
ea06c6565ff565ef618d208c12477183ed3f00fca1942df6725a26fb5661c508
13853b40655aa72b2b5bf88828c0897f5bc92ece9d4ba2134ed88aa6bb6a2e69
94945977ca9eaa5f2d93c6342afc0fbf59f05c2abe183e323fff2a1f96b6a57f
6a0e56029fc4b05e52529c5840b6130af3bc6584bc53e0c5863c9dfaa188d420
8e9f88bc1d75604ea259d9e9ba4b12ee0538acd03c2c4b71de1f8565c3127eb1
e95a8868b65efd808beceaae6907a2849be61f4345d3a5a924be2641d89fe86d
4720c2eee2bfec104cc0deccb05f971e785a679357d124174620c81a07981ef
response w

6ce8c174ed0e40e55d9eaf13baf051c9cb43b0fd536084bbc8cc40e2764712b
f47e6ee830028b55e109c9f560fd0bce8993bf4acd6bbd56f1f9baa029fa3787
205009ea341b229161448ea93113a3b3f8b03e71c7b43c49db282eb4dd8e2642
a9e80536a7828e38c9fe5b553cacdd8178eb8e8450d149bd207ba41c1502a588
eacc92eb8b14fd74e5741e2f3365f9612412569a2373c1bbb6c2b5ab0675e24d
490a3509ed36159b77b8b650b702356ead57ec6add076d91ce7ab1fb287f51a9
1204d581fb5f288de16908ba40b50636d9b5fa7bd2b0519fed84e9f9a59dbab9
c8c52f2eb374fbc0580adb91f29d757385b5ba2f2769b0a07c487bc14b77fd8e

Fiat-Shamir context

b836c70c4f0d6ec85097e85a845fc42785d6db9f33a7530aed0922ae7e877215

withheld proof term: 408 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.respM definition, Crypto.Ledger:50

$\{G : \text{Type } u_3\} \rightarrow \{H : \text{Type } u_4\} \rightarrow \{q : \text{Nat}\} \rightarrow \text{ZKP.Ledger.View}\{u_3, u_4\} G H q \rightarrow \text{ZMod } q$

SHA-256 of the statement above

347bd0147a5b2b49888bef6e490cf6814b38be9dff5f009f58a90543f5167487

SHA-256 of the kernel type

73e8f85672bb5290ca866593aa2f8c2403d282e0607fe6312454aa1fbac03ce7
commitment C

8b98896ad310f254e888622333864cefaa74349cd458f458ec316b65a156f4d0
eafffdeae4bc645c1f69d9f79adb6a2f6cce08cb3a5e8734ae13ce8bad3973fb
a6308b5b06d841a11d5f71ed8c6ebe3dc74dff134277eb02ed1377abec0d952b
67733880b832bb3dd75d4060c4445a957b1cfb74d2d718077ad0a435012a97c6
2c9adde8a42ac0c991b4a64ce1654ff85b4769b0f1c953ccec179dfc1f48c7d7
dde03a0f080eb791a63819d753fbd357cc42622379a8de0b0b87fd54daa0cc1e
1c7eac645f6d4e7d8f76c2d12bb3e4f5b694cf1f0824238917f2a09d041ebe81
80ea2b1fe2eae287d2f06508330c8c4b8ae5db52a03f65ca7a6e089f6d6eabc2
first message t

86a9d4d74ab1b47c63e26ae888a800f2e5ec20c61e2082cf6b3b47d70dbe2087
78a484be89f7eea74593270490f2dbaec853d702f72e1e412c2601d4505f9496
8677dc2164e345f1746c666055ea0813361467ad279dd43c618157b166ea53ad
7d3d4d4c8206ab9464753ed2e0a1a84ecb7d0d15513fa411c43ae22e81ff934b
5360e5c155af5c4d704e9ffef287caaa8c01a2e7ad7a671202ff0a1b1ac027ce
03ff068d5ccc573ae4e05bb9e50df80ae40630bed7a538e32b51afe686d0eb39
7fbcf20c99719a3bee84a07761e35ce357f67c8f507d8ad9f10d4b8ccd3b9bd6
96ae40d968def1d9e43aef8198005da85406962fcbbbb24b38892c6139fb7d49
response z

5e5e80a3529de388fe74cfe6469d5a8f83ee38650b13a1152da994499bdf9a58
8fc145c474f1d49f7017f0bdaa3a1ea741d0a96ad6c6562b523927fb447a5d8f
b1b26d5ef54ffeed6741b5bec5fddbe491a659473f8c5e4aac8624ea0b182b41
caacde53ee1bf7e0dc46ee8dff873b5f87bb231a5891902c642576a6a4242c68
2f9cf5f898f6c5092b56e7ca6bd4ea086ef7e454d8d89142cc627ab00f2d2baa
ea64e59e45867d0e7d48c193993ab7bb7d1b442d7fe313ed5b22251d4348abf2
9996ebcd3e4b5a940b2f753babc1fd64cae689898b7c564737bc7e9d9701120a
e005e2c6fee7530a56cddb1303a1b755dc022622ba89c830a1f057f12bde2566
response w

6ae85bb0a03f53a257e6e9cfc22e549c220832477a2185efeabcee473cdca50c
1b425ba084362a5188fb4fdf855b5bf678b1f4669f48271819d8cd3a75922ac9
514494ca690bbe652a8a030dcaffabbdccf266a327d0800bc0b7ee18cceb7d
e485cc258a417690ec91d045d8ca81d654edef91951dd2f27d540c5b851586a0
27a3bcc30b7d24153a3c2295e275281f360d238d6f5323f77225ca3b61d9897
b9b2851ddf162a1fe27261e0d69b730ba0945e5a4d131cec202fa7f29305f4d
441da66b95ba70e4ccc96f0cdb536a55ea18f0714df7a6db323d5104fffc9e
d7cb12d4f5f9199d0f6769c1f6182658445de88c3b4e150519a6bdf4dbb1c035

Fiat-Shamir context

180cf7791adb93cf9c74ede7e7818cbce8fa4e0573e74fa2194f7cbdc78a75c2

withheld proof term: 99 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.respR definition, Crypto.Ledger:52

{G : Type u_3} → {H : Type u_4} → {q : Nat} → ZKP.Ledger.View.{u_3, u_4} G H q → ZMod q

SHA-256 of the statement above

347bd0147a5b2b49888bef6e490cf6814b38be9dff5f009f58a90543f5167487

SHA-256 of the kernel type

73e8f85672bb5290ca866593aa2f8c2403d282e0607fe6312454aa1fbac03ce7
commitment C

1daceb60eeba7af3e267bd3917ae65d275e130383f6266c9250be6728385a137
5d1c345c76170310b4a4a4ac0d5b0fd86d225d0da11d29879547b5368fdb85d8
456059f05fcecfc33384ee34ee81f60b8c934b7ae818088a10fe1b23a0bd98087
2699da9930c8172eac838d358f141e0e72a8f31740143730dfcc9074c99b944e
08843ac025a146049aa1e11eb7e6d9b7915ff9a6f5ede7e0b1059f4ebcba4d45
00c67e9cd365c29deb2cf2948167f0a07bbbd134a69eee7142a1d26e00bd45d8
a16c1f5848724581a22a5f9c90c53e688c27c107341936b35811ef9e2bedc99e
59fe73a00b7429f44c46cd7cc137d4146f61019b39d949ff0f53b2891ae30657
first message t

c55acc0260ad0281548aaf4adb27bf8e334d0c9e9bb18e0dd987ebd6ffec8710
a7b88175e39b434d0c05fbc456f66ada2ca62951e73e766f609fd202c7bb7f04
96e69de35e8019c1827ebbdde8c364a2394036a064be31287fb5ed70f0c2fdd6
cf0bc75e9d0c18bd424f6463e7aa8bb8142a0e0c6bfd3b02f8dff637b2ca361
ca1a463994f0b71de03e974446664b97281b6825ec89d9406f7b9a5a27b95877
a03af153fea9021a49a52887a33b3285e9edbcc3a42888a345f09a7841e20423
9a28ba0a927b9b0aa143e6627f59208d76a858d6b8ca221f4dec30acb9900f9a
979a0d5bc94acce55f2de0d5c701565043c77bc1925e61c719b336c4e2a31ac
response z

10f6f9eda4c374e9c6fd3dbb32a35b5027b57138db0f00b6fd9dd5292cd58867
8870787ba81d902f676b18fa624879f4994f722c361f7866111861632acd5725
331dc2c79bc3dfa4cc4cf36c7615f99e93e62d0b1183fc25ee7e48738470be36
0fbede3a5af135079b72ab62734d1def4c7ebed004a2d4917aa344bc06c15464
d9429df581fe287f52523c616a4b05a3203a0d1ab5cbebc890fd8c23f631e343
927539281c0b9030a0963bb78d96f72d3d3da4ee8814bdad12f5d3e9dbc1d8b7
a3ec2074e5f2487b11294eaa02a985a06231a90c7860e88fa4da0716c8883beb
62d8a581e1703881e664a302e6589578ed730ce2aafcebb94e798977f095c9971
response w

5137c3d6df67b61de5140afe6171bdc9dd3734b14b7432a0e20aa6f855818cb
e60a1572f2146f7f09d8e69ee81d4011d2891912236dd9e1b01479b9a406f977
451839d805308a85a2f85c35a92882d5d8bd412802945eceb0cae796bb6ff993
259ed5cc8d791c6e42b871de8d6f7e3109deea28e6111bd3515b16e313a2bc82
c011110a94702bf0288ac71120c9d18b7e387f4d71fc3d231ffdbaae53ed26f5
4507d789730b94f310c10d02debd9f2d5542f5ce830668688f5832fdaab58eca
4ca23d3e299c3a8e2b46afd9c6699536d5c80d3555030b7b0263c86afcaafe76
cafb399767969cbcd4cf7302a55e3ef4ce91a61e173a45f7f9e3b7b328fca695

Fiat-Shamir context

934f200338097da63095470dbca7a7b32844db3d82d316d412af978d7ee06f56

withheld proof term: 99 bytes • axioms: • kernel replay: yes

ZKP.Ledger.View.statement definition, Crypto.Ledger:42

$\{G : \text{Type } u_3\} \rightarrow \{H : \text{Type } u_4\} \rightarrow \{q : \text{Nat}\} \rightarrow \text{ZKP.Ledger.View}\{u_3, u_4\} \ G \ H \ q \rightarrow H$

SHA-256 of the statement above

ff3cc294450608b692e56ff18cd95b4257b7b00462a1a1caaa92acea02eea991

SHA-256 of the kernel type

547492a3e05fac565a367a30842cdf900a492c0500836d14166c1c51ce6d2607
commitment C

e489cabb5fe72a23c3e96d9d2a5c96f74dd0d5edf084ae22057b30b6d617eab
11f93f36e33cfe6a8c2f209089ffd3e8c402585ea738e96c22a199d019590715
72ae7472c3e01c848aee7967aad663a2ab71d64127ecf4590e54127522e6318c
30eaa3a67302e5d91b489bd50b38ba8d0410f185755b060fa88c356ee2ed7f7e
63509ff139dba06abd4cf2a8b3b6f7b925a25135cea7449d703ea43c2a89b3a7
7c19e002fbe8bfdadaa7e30e05174fe9aa25ef8a02c261ab9d4c270997d83bd4
15dea8add3b940a1962401759bf1fac5940aa1c7c7ca041c4a319ef996b23f65
9ed808542a56d3c942773edf1b7a06380337cd4cb99fb3bf17ec7463d25b775c
first message t

c030164960c7c0df29f98b51b4eedaba3d4d315317829274d409699d921638c4
c3234a8d1cd01141331bdae731ac85e0b1314516a4511a85ca0809c6ee2b59a
cd2836f4472bda391f69fdd596c50045d7105e6ec9025a36d9741089fb907499
0656771445326a95b02b6d811270bd0cd1165ee12f45b02205414b1b2f7eb76d
042e512c64cccd3eef35c374bfa7b9ad246923f6338c2f71514826d35141fb4e
e765081a1e86a59d8f8ad485c668678a9100ea9dd093d0e5492886eba575a8c8
4891cd8c8653d32eedf64c4b8e3934b9a1a6e4be7faa369c81ac2a5558e990a8
45ce73ffde2a65850553b033a0c06821144f9190341e57bb4c34fdb8dcd954e
response z

16c830a27292ae1f0ddcb326576fc890fe67f4c31fa9f2af746fc505da6d61a0
4b3f4ab65669d766e87d08c45079d0afa954ecb160ec6b70336f72766161da69
4aebaed80a13490fb59e8c8652d5dcfd0f1bb0d0ab32e4c54206ea1681313d24
e1dbc8632d0f273e683e49824ec17eb550cfe6d9c3d2ebc72180e1605e42ac9e
da88757c7b84c2f6cc4c63db9d3ef5f5f16830ad6d02f13186d217a9988e27bd
ece61f263a322f5011b96c2e3e64f0f6f479f3787567ebe93b2e1dd828749630
a0dc9cc9a6cad80513f5dbdf97982ed9eb2065011cdc63e2e8d9a6e3a26b122
0d869ba806d514108afe2615d08398c0cc14b008f60193a0c3c4c7874f728138
response w

6347d50d46e916e349aadb5aba26b4d9f551575df992ed26748295d053d5cadd
94b00d767c755e71a34761722f4a2550a2188897fb1e1c0fb77fef3d9a6231dc
40ce373b3480154b96c8749ef7863ce445e84d39e75563225217d278499fe8e3
e50ceab5d99bcf4cb81780a12c3b519fb99f4d811b7161c15d89fbeda5747d68
ae9bdee8fe7610fa7833b472e93e1e6c7057cb3da94da90bbf2191430bacfab5
95b3925003b9c071dd88cd823830d7db64fc022ed5614a4c54d6a11bfe77af80
d9d3179faaeb450214ad6e8beff5cba9de7dbbcc519043f125eb2578f19e1238
4d5fa4bf6b05ea104210ca6c04b51930496cb26bd6b138bd5c00f39958e2e34c

Fiat-Shamir context

153bcc9a1c328443c841e7150805d9599b71ce50aaf0dc396ced974fc66e55d2

withheld proof term: 99 bytes • axioms: • kernel replay: yes

ZKP.Ledger.collision_of_ledger_records theorem, Crypto.Ledger:171

$\forall \{H : \text{Type } u_2\} \text{ (hnode hpair : } H \rightarrow H \rightarrow H) \text{ (p p' : List.\{u_2\} (Prod.\{0, u_2\} \text{Bool } H))}$
 $(s_1 \ c_1 \ s_2 \ c_2 : H),$
 $\text{Ne.\{u_2 + 1\} (Prod.mk.\{u_2, u_2\} \ s_1 \ c_1) (Prod.mk.\{u_2, u_2\} \ s_2 \ c_2) \rightarrow}$
 $\text{Eq.\{1\} (ZKP.Merkle.dirs.\{u_2\} \ p) (ZKP.Merkle.dirs.\{u_2\} \ p') \rightarrow}$
 $\text{Eq.\{u_2 + 1\} (ZKP.Merkle.foldPath.\{u_2\} \ hnode (hpair \ s_1 \ c_1) \ p)}$
 $\text{(ZKP.Merkle.foldPath.\{u_2\} \ hnode (hpair \ s_2 \ c_2) \ p') \rightarrow}$
 Or
 $(\exists \ x_1 \ y_1 \ x_2 \ y_2,$
 $\text{And (Ne.\{u_2 + 1\} (Prod.mk.\{u_2, u_2\} \ x_1 \ y_1) (Prod.mk.\{u_2, u_2\} \ x_2 \ y_2))}$
 $\text{(Eq.\{u_2 + 1\} (hpair \ x_1 \ y_1) (hpair \ x_2 \ y_2)))}$
 $(\exists \ x_1 \ y_1 \ x_2 \ y_2,$
 $\text{And (Ne.\{u_2 + 1\} (Prod.mk.\{u_2, u_2\} \ x_1 \ y_1) (Prod.mk.\{u_2, u_2\} \ x_2 \ y_2))}$
 $\text{(Eq.\{u_2 + 1\} (hnode \ x_1 \ y_1) (hnode \ x_2 \ y_2)))}$

SHA-256 of the statement above

0529f63de028f3f06f1c9f652df040bddd1fd1992624d44bd911c541128735b

SHA-256 of the kernel type

d2196ca3b5b23fdd90cacb146db882119ab176341fd222466a9f2ec9436e811c
commitment C

a12c515e7c582bbac6186624eda12ddeb145089c3e963d22d71da66f2dbcc4d
f449f327a122866cbb1d942fa2a5d3989d27ea4314e0d5b6d8e4a4fb7b3d478
5fa360198c482c5a1858b5dc822786fdc89006e08256304f76ef2c9e6dd7d8fc
9a1eb4ee52a3ffca8b4028ade8b1a07954f8ad845b7bdec6f985d5ec1e7c3fd3
a36e3799c7da4d079f098e345a20a2ace0f923d3817f0c80556370cd253d23e9
4264801ab669f1952892e50ac6c638c27384d582b9938751285234bdd899eea2
fc742f19ef7c6e2853ea325650067e4c888919d71c717848d66643fa30180d63
6edd063414737cdb5948fd8e9c3727ecfc085f36f738822ca79555b236e684ef
first message t

b7d7f57e1fcc2d5ac0eca85ccdf2814b0e51702f344b6470efc16cb70e79189
53021c57e95a449366d2b32dc5db579e770d7858407da6518a9d40f335b9db78
a8e1a498a9c49b6cb80e1b0b4b415615506033d3aa57f637d11bbb22f8e9e489
8b81a8d4e58258a628a928fea02a5427e7d4b3d0b08fcc49562f8471eb87ec41
df680520d9b5894536202427d1abf594631ecc8e3740f5b6092e41c3b3c27348
acf88ed85bdd5e29bf104d65f3adced655f4de865f91173ce91c29dbcdc0a395
cb0a95d14f1cee735f2c57cb8f9c4714e7c6652f64df48c2fa0a4dd6c5affb7a
3700e3a501173072abc7d9c5253c9536e3d7a56262ea4cf070f88537bda6942d
response z

3af6e3c4061b1177c519b18e82ec6af6a9a6b0841ec7325d94ca4faeabdaf9aa
2d5f88d5bb5973cd41355fd9a7eba2675fa935969916bd86164361f2d2b33df4
6dbdd682c4c42d6cf6f6d15ddaa2a4694948a6eca27a2517faa825bf03e55e69
867f0dced0ee7b7dfbcfac9188be7e49017e19499d5270154b949a546f0df20b
db413de60e2c4a7d4b1b1bf9880788f8543e6b64627ea2d18cee2a9c29d31c7
a07691546c77e0b8a5571af789db20fa50661ef370f76999d0e1a84f0a51aaca
d65d3c02fb7903a54b775d183106a16762e287c69a63aa1dac3c3fd35c3e3b3b
064b263b58ea9301afd866ac21e72a9f1faeba3157beae5213b62c64e106b713
response w

30ab47634d62907a7cb4c6fc6ca5bd7c649b32b75ee2435e52dec5a9ab845d06
6dda144395c83d7dd96f25c15a0cfaeed6379feb4b2e5ab79cb9bc4f15d60253
f21ed2621e615a23cdd6ea9f0a911fc7914185f342bbad653e60765024cd762f
136d19a29b0a75165c08823d2cae6c2ecdd1f01f35dfecda4270014670e99285
67e4f736427791497c6b83ee751fa41d38961cc47c62fe4b6582b0131117cab4
e4ad93b5b26771d80de1c85c0cec0651d517c0745be8335a72b22d4763c6958e
53f9b1efad9db8089ac751a6417f19bb4620f69df004fb65df181856c95726d9
6af73a6e399ffdd0d5e7db0b25b100797db4c7c7b2dae5151ae7284c80b7e6a6
Fiat-Shamir context

dca207f67621ff9bbbcf38e278b70ddd2cdc35a1735bb132d923ba652e0298c5

withheld proof term: 4218 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Ledger.commit_div_eq_one theorem, Crypto.Ledger:81

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\} [\text{inst}_1 : \text{Fact } (\text{Nat.Prime } q)],$
 $(\forall (x : G), \text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x q) 1) \rightarrow$
 $\forall (g h : G) \{m_1 r_1 m_2 r_2 : \text{ZMod } q\},$
 $\text{Eq}\{u_1 + 1\} (\text{ZKP.Sigma.commit}\{u_1\} g h m_1 r_1) (\text{ZKP.Sigma.commit}\{u_1\} g h m_2 r_2) \rightarrow$
 $\text{Eq}\{u_1 + 1\}$
 $(\text{ZKP.Sigma.commit}\{u_1\} g h (\text{HSub.hSub}\{0, 0, 0\} m_1 m_2) (\text{HSub.hSub}\{0, 0, 0\} r_1$
 $r_2)) 1$

SHA-256 of the statement above

40022de351193bca8436411ecfc5328b675ee8476405db8268d3a5f2828a0075

SHA-256 of the kernel type

df04046d83fde0ee71c85a0db4d7bc0733458c906449d1e02c172f479cf46c99

commitment C

ff6521bb5305d51f5bb9834f7d2e45952a831a4573eedaee4d625983d49cc938
8c18a1b1a17ef851cd361357b75f214aa1355987e4aba9e5cacf4311c5b8cbea
318f571ff2e2824eea592ef181409f72d8303d706b76bda0b789fe3591abb874
f251c0cc1b341cc1e3f075cabdc293a627b427033f5d2c4ea5f055ab0bc23c9a
deb8cd983d961c325f68ce3cb8524a2117f6ffc7ab497565839fced002d247e5
7c4547fd55c2b2297a36e31ee1c519941be379d0d863a1f89d7982b502f1d0e4
cf1c4ed64610d9721df2276dc4e17585cac233405703d165ec4db03098f505c2
14e866334862806f63e8bdb5cf03cc2f26a5ac6ae23aecc424e41ca09c4375d0
first message t

58e35500980656d312ac6adac10e30a693233eab8d3271e750fe5db26d1b34fe
373af6c9a0828f6cde443d9083393e892c2072161b94431c733b1e8b2d4fe01
ec74305e23867f91574c873737b292c8f6a7c664c749e63662eb704ab502451e
1a27d60338fee7552f2deff6cceabf61da559f8d83d255a662a6b98d95102b4b
10dcf69193b536353676c144f6bf63c1a3a0f4af6118c0dc675466cb2e6ec697
1899d84168dc01f1ea823e915f1484d50a1225001cd72768a60419715725f05c
c3b4bad7dfeec8fd2c903c9a756267d1c189ca00e8ce3dab682c91ac3c1bd831
495c4fd3444c6216473f46132a882e02169e595dd855c5648764e9baa752ff03
response z

578a0da9fd5ae8cd31c908a904bf6ba650fd7a11e0300d281815f18e0c2d8e2e
388aa54fb6d9884263f0dc8e62ad74df153e7047a414a2c715b2494a66b44282
48917e8190f9017b7ef261bebd6c9f043c935a96bb06b5c0edc04647287bd7a7
c8b415f7626b26d41499fe76e089c4a34e5609cb665a804b93e99d625e6b7610
93e29d727c8ed8d6759315e4e73d7baedcf22b61fa59d9a371093f0c4ee35730
0f2a8cf11351e334850b1e8ba0605417010e9cec08628f0b4490538f9451a95a
95d563c5b9b1d96eccc1076adfb3fc266662f18254b780c4333bc03ec3e3e6e9
54c0603650068894b5e8147a58fd87768a3e9ce550c312fc04d991c6b051e57a
response w

84dc751df31f7996406f1ccba00b7a09075683f26e5d0d4ae31cfe95961c81c9
7ac3541d8da7b5bbb3fa4b4bd07c54704e8baf07ae77d8f35eacccbd9d5fde67
5b1cc670b943d9a6fb8ee677f22307f3810894ce48f3a1dc83a23b0dbd858f16
8a65dc979a537ac4660d65135744dc32b369d2037ed11126518a59f6c9bc2c80
9ccf1f8d2599c558bc3bf42d7578a71a126bba1a5dccc6625fc80823a8369f6a
b0f69e7575599a7b8a4fa4d50752610cae1ea0657be13feeff13995a5e0c2b8a
d2096493978b3fa9333d43f42e33008d436f03a5eb0f6b3f68934e0b73b265e8
96844493350fa231d1c2c6c9d0ebd4d2d0199d4c986ecd0764d0ea344b8828c

Fiat-Shamir context

589c25af7a47c9ae65ff7221fd612733b8c7bb182641eceb826ae9bdda3db97e

withheld proof term: 66 409 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Ledger.commit_pow theorem, Crypto.Ledger:75

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\},$
 $(\forall (x : G), \text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x q) 1) \rightarrow$
 $\forall (g h : G) (m r c : \text{ZMod } q),$
 $\text{Eq}\{u_1 + 1\} (\text{ZKP.Sigma.commit}\{u_1\} g h (\text{HMul.hMul}\{0, 0, 0\} c m) (\text{HMul.hMul}\{0, 0,$
 $0\} c r))$
 $(\text{ZKP.Sigma.gpow}\{u_1\} (\text{ZKP.Sigma.commit}\{u_1\} g h m r) c)$

SHA-256 of the statement above

c1069dd79931c35d38870e8198f47cc30849b9efe226ad83e433b14196c2d80d

SHA-256 of the kernel type

6be581f6ec301b5679a7a63a861d904f85557e185964778eb7730cbcbcc40f16
commitment C

f74a441a3250362f5998ba2d503563d27414b3759095db8a5e053d1d853c1e26
a52b879ba840bfa609f4ce2558b366abd9edbf6d2cc9ee90fb18ea91aa483fc
ae6ea01446affc1bca29e9342494d81f3419aa0c07b5b8bd57a79ff6fb63984
a85b000020df39a3ba8ec8752fe64329b80468ba476f811208a3f3d693b80d5
61c03dd1a1080a40cbb5a0dadb885dec102643fef89f78c54c2bb0ca0db515d2
9b1416509cc8b4f92ce4aeebfd38fa343fbb76dcfb2eddac338fa08b11c39909
7f554412ffea19a8a70b7ea95e73e187bc125ceafc1c6cdb695e1aefd662cd79
f1750dba1f05bd0d236628d66cea57001cd6ee7623ac3dce6fa1b82859940bd0
first message t

d2fec0b208a7dcf9d03f7ed51a68a137eb167c1d1ef3abd27a828bbd2d2c4a45
e77b71c8a1fd3c6c4c6249e8132cb744b4eb1852aabec2cb0b93dcb661870a90
5edf970bab67ec3de941e15999b32341cdae22eca2ed1bb0e754889f9203ab19
906df5cf8020d677e3f295f9dfbc4d12d1ab9dfcb31a42b73413d8a95b56df6
3ee24b52170cc3a64257a6839e3e3c22c2cc41d54b088297ff3ee04665ab35d8
91a993f34de2765d011fd13a3e11b4d0eec5ef639ce247bb3d3bbf1d8c13577d
a37d57f777f2f3bc5839df4e906758ca466674c3af866e21d97c8afbd33f6230
be565f580d5aa5de0d6da537bd4bf136028454ae5bf89e2f69f9ceee03fc35a
response z

7078e2d3f1eec43f96c02f94eea9d94cdf6d76dec637c85de1ec2aa1cccec7e0c
b981735ec5ba78bfdaac799186a1a14e31e210bd0f96e50ceea60100072be4ef
6c8f7afa200cabe93cd64c343c1cd3d00598ecc1fec433ef51555bc010d05aac
d7c0ed3df5e196785c9dadbc33cf93bd33d675d664e32270d56eded2e016bf6b
76fe89a330f8fedf762b70a77f72949cb17de31fd9274a1d4140b50da222ca80
a4eca02a4e9ff44f7edde75e43dcb6ef6a5f5cabd50e290f59e6c8fb819dfa89
311c5175631a96526053c7f02a0c04b94ca71c991ad59dec83c3b4f4a757c12e
d4fbc21b5684ff842e5e65b846f5c62a7e8a458bfaaf5dfed1eb687626222e1d
response w

2dbc2eba49cf2f7727eb518b3486ba56e0dbfca801d905e6e2e450a917e736f4
faa1a3357b0dc356dd8729e6bb2d2a5bb6cc603048c756baaf30b7fa327d653a
2f51712135793dcea5962ff6ecaece5b284ac65977a9b4a52019f61b1228fac0
34fde79100e80cf91b7858c07a772d2bdeacfd0119de30fd5aa6c6aa179ad29
b73c39685eaf7ba924951848952c2cbde41ac159e6bf2898658d8c7fe2a1451d
c7010a83b0ee282dfe3de6625cf7e825e5a6dc28c54f244c0d634ca9b29ab9a2
89556f325a4aaef0d1ec847d030001730dbd976160dc81c288e1dc50ce942e68
664ef78f23e0378306d87c92528765a57a5b068d515d757a97f1fdde1915a842

Fiat-Shamir context

c9d52d69d6064a3f89c1ba870f5140aad61028324d723ad77712b596d3837498

withheld proof term: 67 721 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Ledger.exists_unique_view_match theorem, Crypto.Ledger:125

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {H : Type u_2} {q : Nat} [inst_1 : Fact
  (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h : G) (s : H) {y : ZMod q},
  (Function.Injective.{1, u_1 + 1} fun a => ZKP.Sigma.gpow.{u_1} g a) →
  Eq.{u_1 + 1} h (ZKP.Sigma.gpow.{u_1} g y) →
  Ne.{1} y 0 →
  ∀ (m₁ r₁ m₂ a b c : ZMod q),
  ExistsUnique.{1} fun r₂ =>
    Eq.{max (u_1 + 1) (u_2 + 1)}
      (ZKP.Ledger.honestView.{u_1, u_2} g h s m₂ r₂
        (ZKP.Ledger.shift c m₁ r₁ m₂ r₂ (Prod.mk.{0, 0} a b)).1
        (ZKP.Ledger.shift c m₁ r₁ m₂ r₂ (Prod.mk.{0, 0} a b)).2 c)
        (ZKP.Ledger.honestView.{u_1, u_2} g h s m₁ r₁ a b c))
```

SHA-256 of the statement above

419d4e8acfaae874aefc3d43c433622e10d4df47e52583671b29eda6057039fb

SHA-256 of the kernel type

249366de226d567018770dacdb8ff08fc13023c21859c887ed3e84a45ff42fb3

commitment C

d6b29fe2ee1197ff9d1b54e27d06300af0bf84b11c731f0d024fcaea975c353a
974ee1ce1e33fa382814e5b11782e5233f7ce13b80ace482e8333a9c7f08f349
1a50dedfd4ea3a81e15a318b9a9e00753795fa5fddd1659944f3105b3c794597
ee1034feb34dad09a05e0469d9abadb1c8d2bae90d0b5f2a419d4e54ed81fff9
14680f5199a117f80e3d38db41a2178d47aa7db7383770fc8cedc5134550206e
e4d68891ac73ec103cbfc578d57694f4f40b1f46c6ff224f9d5f47ca93912083
422b23f4af4df6af4885e2208a2f57b79b40b9597a417b0020185e9cb7ebe852
0bcd3ff3d77f0a053ed14bddc8bbe28aa734af9a18805a97a7cead38852dc5d

first message t

b3b2b2beb2922ff87a4ebc3452d446aeb8cb374ba170fbb20c642d7cec0d099d
5efd2c127057bf781a9f9f8cfb1677e94529d6ebb893094fdd0e4c8b263fb31a
f2e612859d5fdc19460ea028c21456f96af541e08b95ab60f5345803aba5147d
c035e5375d0ed5962bb9df5ad027888a8078cea0bf2e007896ec44b3ca4a200
85ef11e3fa50df732d4cf03c570719a095c4965b20f34d3ea6e9483d771e11e7
4ff7a54f3eea5ee65a47b916d2f4822fa255c7de702b4686fe5bcb8b3fa25e7
79698fd80dd36aae477722b9d3ea6bfa5e9755f07ad6895bb6ba717ca6e2dcf7
51d4d0b00995232cb4fbb757a9c8db3ae6b3423d2a2cafa10ab4cfea1538f0fd

response z

2bdf905cb5096c7a3704912ff9fc19c26f6ba35398d49de8603378d17cacd44
0de64cb4a1a96e5fb7b33d90dc4c8522198061c8ee5d22791122fd14cbe6b6f7
d996cd1add5833a635dbda237f6a740ce31b2b61fd3589a12ff1f8c6b48888c
90c869735f60c53a2bd5bc0edf4b713b1be3924f088a0f6365648198bf294b4a
9f2ba3606b07fc1a49af1bec51b358638db58a3940a9bf684a0c8f659a826d72
cc2fa81e69f862eb382a6133f3cdd5cfa6bd0dbe1681fb13ad0140f18f53bee3
2c1cf5694e147dfa085350bb91e730f2bf7e996083588484d41bf02ed738809
23b27000215e4f2e64f712cd2908ad390c9ff6f2f81da3e85c616e6a9c2bc614

response w

78f7401b00235b7553353792d08b98419bee26cd4218f7f6bb57b41fb5cfa515
cda46be01374f70cb774431b7e661cb61cf62bb1612d9c533232d0cfa07612e5
2f903311589fa69ae66203affecb7e8fb4b7c91c29ad81222fed523738aca42b
f1b5d1308df84bb07b04d73769f6c0da529ec376a373dd8283d56d399dace202
1102909353e58db2a597e1dda133d417386828b9441cf2b696e25212b912a568
c7803447fc3f4bfd2f1992b334c8963bcb54ec89fe541e83561981b0ab6fb30
e5707d55673e59fc5de5ba43b1809aaea0201835a8a1737d833ce2e4bbbc4850
4a78636eb2d27d06b804b8a15317ad3afec08896e34c69fd3bcfffb00aad6b18f

Fiat-Shamir context

25c5cef14e40729c012fb079c805660c837dc8a82d5e55622e08f35b5a0118fa

withheld proof term: 13 073 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Ledger.extract_opening_of_two_views theorem, Crypto.Ledger:156

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {H : Type u_2} {q : Nat} [inst_1 : Fact
  (Nat.Prime q)],
(∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h : G) {v_1 v_2 : ZKP.Ledger.View.{u_1, u_2} G H q},
    ZKP.Ledger.Valid.{u_1, u_2} g h v_1 →
      ZKP.Ledger.Valid.{u_1, u_2} g h v_2 →
        Eq.{u_1 + 1} (ZKP.Ledger.View.commitment.{u_1, u_2} v_1)
          (ZKP.Ledger.View.commitment.{u_1, u_2} v_2) →
          Eq.{u_1 + 1} (ZKP.Ledger.View.first.{u_1, u_2} v_1)
            (ZKP.Ledger.View.first.{u_1, u_2} v_2) →
            Ne.{1} (ZKP.Ledger.View.challenge.{u_1, u_2} v_1)
              (ZKP.Ledger.View.challenge.{u_1, u_2} v_2) →
              Eq.{u_1 + 1} (ZKP.Ledger.View.commitment.{u_1, u_2} v_1)
                (ZKP.Sigma.commit.{u_1} g h
                  (HMul.hMul.{0, 0, 0}
                    (HSub.hSub.{0, 0, 0} (ZKP.Ledger.View.respM.{u_1, u_2} v_1)
                      (ZKP.Ledger.View.respM.{u_1, u_2} v_2))
                    (Inv.inv.{0}
                      (HSub.hSub.{0, 0, 0} (ZKP.Ledger.View.challenge.{u_1, u_2} v_1)
                        (ZKP.Ledger.View.challenge.{u_1, u_2} v_2))))
                  (HMul.hMul.{0, 0, 0}
                    (HSub.hSub.{0, 0, 0} (ZKP.Ledger.View.respR.{u_1, u_2} v_1)
                      (ZKP.Ledger.View.respR.{u_1, u_2} v_2))
                    (Inv.inv.{0}
                      (HSub.hSub.{0, 0, 0} (ZKP.Ledger.View.challenge.{u_1, u_2} v_1)
                        (ZKP.Ledger.View.challenge.{u_1, u_2} v_2)))))))
```

SHA-256 of the statement above

f447f078cef9a8da763bfb3a8b8d2e6e6250287d597648a1605c46f9ddb01f2a

SHA-256 of the kernel type

2e17d8f938ab3a133e01d376a76519e27bb945f204079d3ea83e04626a0366b4

commitment C

c6dd4cc04286ea688ab53787db495d9f9ea620f476023970cd05c86c8188348e
4bbb66a905ff446d39309a454e6ff5b17b885123b7564e773dc8be0f7a2293a8
0bfe5affe20114dfcc709233d9e8f488bbef18ee439483e3578d48da6530e8cb
b1d28fb33f93a8055b75ee55f0f724189869115124d336768b85107b23c47f25
013f20ef9f4a740abcb9f0cac148c5e1c1ce435af3006e1e4036a548a778a58f
cc7b1c0ef70030823f1739599d41d34c32bc7d9900f0651ead9b834b0362b8bd
6b14533ae2a13fda4194bae6a4fe27ef05a56cc596f9ffc9702889138e464dfc
bdcc00a20e1c7b6dd56e7259825568ee2dacaad9552dca83f6a4f29e01fc77d1

first message t

2ab941bebb38252196f7cfcb483321adae5df865ad11058f1f3659dbd1b6fcc
19fb78c4871017022005b799e585b25702192a4fe9f77524176d6fc4397f1763
b7cea68f86119d6111437dde6ac4b180fc078217d66e67f54f3c7efb3a4b3fc6
a98b604ee8115b80aec07cef7ef47e470458c0cbc748b0ac15cf7d4385971dce
e4bdfb235fb0850ed2d7c05292d039066834b94b8d9ba27f505c7730fcc7df4
1dffdd9493206cc926fb9d09d4ee225dc9943346a3713433cf38ff5319c79b20
0622ed59792d8060ca427b3e4eeb5251315788989652da5320ce90f37a4fca54
d897e1e91c8e00ad83c53c439cd9862542e7c4ec2e3f2ec95ea9be12efec1bb9
response z

456c2f8f20e1bd319285fad2347aa63167d1721980dcace9d0746df0fac1afe
54c25511471dff3bc6f121157fbfd85613ff6ff958dbb5a700d3cd8106a6202aa
dee587cb7d7ff648a10b1aff894d6559985b88de3cc9e4fd25ff147f10f6e82c
da9432dd429570ff3ebffe35f0d23eb5bcc30eec91663d7a099e62795de7f6a0
3ac92a5b808124d4be8ed3e6f6e5cb62c1458285de0188f6aada8dbfa92f39d7
d658bf7b49e194668cfcbcd2d38604f37407694a3c7c9db975f50b53944d8f06
134688ef3b0ab2558e51a49cf6f6c3805bc8c66376c6bce89cbe1627b5913006
7eeb6ff9e0c825500344c9b705f2fa9515ee3d94e40d2b445030849bc07da455
response w

582b100e0019fcbcb16cd3df19b968ba96ec5a24112784e7016cd30a2b2032445
563545a0a67fa9c2df2ae4e45ab6c2e47dc2f8854c2efa3aa2b3fa3cd8d196a9
619a51a1956afe485db47c9b0b89675aab7ab5b6dccc3b2a63336f9ae80b0cbb400
1cbad3a43e2eb3319523cc9aa374c72d60f6dff518682324242b29c2b2b7255f
4b08a0d4b9b12e12c4c78a7134c68e0958fdb67fac99f72482b2195a9465f574

ZKP.Ledger.honestLedger_eq_of_commit_eq theorem, Crypto.Ledger:144

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {H : Type u_2} {q : Nat} [inst_1 : Fact
  (Nat.Prime q)]
{ι : Type u_3},
(∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h : G) (s : ι → H) (m_1 r_1 m_2 r_2 a b c : ι → ZMod q),
  (∀ (i : ι),
    Eq.{u_1 + 1} (ZKP.Sigma.commit.{u_1} g h (m_1 i) (r_1 i))
      (ZKP.Sigma.commit.{u_1} g h (m_2 i) (r_2 i))) →
  Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
  (fun i =>
    ZKP.Ledger.honestView.{u_1, u_2} g h (s i) (m_2 i) (r_2 i)
      (ZKP.Ledger.shift (c i) (m_1 i) (r_1 i) (m_2 i) (r_2 i) (Prod.mk.{0, 0} (a i) (b
        i))))
    (ZKP.Ledger.shift (c i) (m_1 i) (r_1 i) (m_2 i) (r_2 i) (Prod.mk.{0, 0} (a i) (b
      i))))
    (c i))
  fun i => ZKP.Ledger.honestView.{u_1, u_2} g h (s i) (m_1 i) (r_1 i) (a i) (b i) (c
    i))
```

SHA-256 of the statement above

9addadfea13b7446820a484c712b56fb78666004a69943012fe2df0d0fff12f0

SHA-256 of the kernel type

c75b5da265d48bfce2ee5af79c90e190b303cc148a75af41304b5fe276015de4

commitment C

60790f7adc81a6fb72a713a831c34727c24f27609beab813efba1f292f98afcc
e037fad82ee6ace3de2c5d99927dd47cd4458c8a8e20c3fd78aeaeef04853a2b
f95ea379b2306217490b3534cf9d2f3172eb8fe6aba4a52160a6526986bbbf5
4b12bddf562b2ba7eacb033617e0c63979487e2e26778131965913bdb8edc822
8ddf5c53d0219d6e1e0723e0ed6813cc71e7289a506e56a1bd17644383c435aa1
bdc5afcc094dfa64aef9fb3bba2b53f83158d0a52a30ed5e04b7e890599bc0
9e0f2d5f31bc223f1c70f2575e7653fb54bae54ad2f816d195ef6e197f17ddbd
3ee5443f5971d7e7c01f849b38eacbd014e708995ee3afe47d4826e8d915acb3

first message t

ba1132d6aeb83bea716ceaed3e7b4a49db6a0b599e1c438bda9edc41c95a259a
b7c6efd95b67bb3114d1f5598ced93395a34777fa0291a25abbe586e5d2cee19
f67021fbdef85d4b275568c38a4429a0207b11f3528f1092f07fd8fec6fabaea
8a8a844a838c18b4e92fa7c500e50f1317f5a2bef5b0a6c59bdb0e795fcfc341
f87329fe0dcbb54a337753ebf04c1b2e4b645243d356a4f0ea2cd91156a09eb9
5fdaede41e807069a9ab06271faafbf0702a6d60074da352b68bb72469c99659
9f74755dfb82d10e4eda35c0d23fd708c456b6223af305311da012ea33861ae5
86240dbdd8e403d745f1d7f6cace56d59d5d9e1b3725d2c42c7b0de8d31f52a3

response z

2057bf12a48607b6b9a462eacc3d1f6a258723f91c9300f1798b6092564b11fc
23f65c140bec9fb74f8f561c5699cf2df39aa0e43e5374893abf293a57833c48
8fdea8c3536f4cd5e5f043e9f28d6fca5d13e9c359442f0e4e961f5299851911
826177b8a2ec38d756506756f8163cddb903720a9da4e0c5cefa8a769f9c4024a
c092dbc8dabe5c48e1b47896646766ceccaaeaba1995482b8af2b56a835aa02a
7f2c5c29696d9ef9b6ac8d3e58e3152d002fee49290df9cf2d1475dcf97ce3d5
8877c1082ac86296c2a40c5a59fd341216c7867116230244dead9b74620c97a3
e46d6aa74be34349e5270ebad003b253b809265dfe9a951fcb17d8d1ee387481

response w

5f9378cbcafffb28f05f69bacb94a72914891079be81dd0de0be00fd54023f8c5
4a554bc88bf52a17357aa7f4aa3c8ff16f7dff2fec24a3e4bc36112597271ae0
7e0e0caa02a84436ab2eadb01c7cf24552427bbc30d7d7e8b192ade470825fd1
0cfc49ad83000dcd9585378a4a4b24d26f501584569bbc2b35e646f86504106c
76538316ba20901fb782df1ba7a3198816704275a6dc300d2bc190abbd350be
cee7d54945d75553067d602fddd77797d7a55067af840a3ddbfd89e63a955d7a
2e6893c1570bc0ef2ded4e06d2bcf491124b3a88ea35ea181d64cbe69f51abee
1393c17071a203b92967153b98ae4bce141b177176ee9ce7a39b9b9f4abb3d06

Fiat-Shamir context

d9f9b419d5ab702ab68db05d196d6c1226dade49930219e5bfc6ce303b7419c4

withheld proof term: 2391 bytes • axioms: Classical.choicel101Quot.sound, propext • kernel replay: yes

ZKP.Ledger.honestView definition, Crypto.Ledger:54

```
{G : Type u_1} →
  [CommGroup.{u_1} G] →
    {H : Type u_2} →
      {q : Nat} →
        G → G → H → ZMod q → ZMod q → ZMod q → ZMod q → ZMod q → ZMod q → ZMod q → ZKP.Ledger.View.{u_1, u_2}
          G H q
```

SHA-256 of the statement above

114e873b33d800b5247a025b88998f65b8764c1e43fcf0802822fa9398cbb0fa

SHA-256 of the kernel type

4fb6193e0fdc389a0f67207b4929970a6268aa7a367ca40434f2df4f16206ff7
commitment C

a299f34ce841f802cf9a4986efb6c65e34c206e607d61d67c2df75e5d5aca05a
66da493cbf2181cceb2f0d68d644e11b9e4f03a4517cd58335dc55169b641a02
d999c76c11a34e423bead54ca92b9178ae35118d447a386e97446658b943316f
9914ab90136b2ef6173d7f664d146e4bfb454556527d52c58665c4c3b093a6f
6dbaf4917da3b2552e7cc39064dd8fb8eabf475d9b3c2bf5651cb30cbfb442ce
d4349226a2a6b2a405c94d5e23bb7037706bc627c0a518b5c10b66b98d9d5b54
1b7c7489f3df481c6f6f04d776520f23ed4b746fe17aa2fc167b6bc5a9a8b43f
8ee063031683d96323134f7b325939e44f55fecf9488ef1c2cd98bd43964257
first message t

100b123283a39bdb52cb223de1bc153bfc92d16809f426b555e4fffb1e0340c45
11503640a0223ca0484b8fc9b09014ab50d915b58e52cf6659e5064f7470ce86
af2177a5d54f6e5c7d209197f1753435294492f9590ab9e60e34af9aa7431091
b6f33ee8270582a1651c5c68bf79813ba31ac19b14eb169e61de5dd31439eca7
ca6a16e46497eefa72b711962fce8b03dae3dd8ad4a43e3d689a2381f41800c
e19f6ce39444887324211b3ca3fb015a58479824cdbee18956d5562798113ddf
c2988a2a43092e2d16edd7843fc72057f890aa152cf1f29b04d083ec7a2be937
9668f7ce7758568acc66daf9f6786e5cc9dad2664cbac77c6609e5427488713b
response z

402349e177284734c8029751e0173ff93799b821fc8ccc7f9fe08f0c2e9941d3
fab1377cf31b9403da26ed5f3a82cc0711a01aa60c7d8e3171274e12fd89d580
fa5f3ba79d1ebe4249fa45f8d587c58ae71a15560c64e3cbe1f9abdb5a51beb5
9f1c0b3ba8d81782821f10e3257a6b81fffec57185d80f8d4c12033c0abd0c87
46ac100551e1f7aca749ae230ac84f0a1214e979efca16e21867e65cab84dc
143a44c04ce3cd4e8f14027d0a9f4b49427fe191d1e0022ffe9e3bc3469dabca
431816ab44569f8c977a67dccf47f62f3a9d4758b7136d1d350c12c0cd7bd555
59359555de5a48b56ef40d40579f9733ba33ec2c357d34afd1202cb1df58f67
response w

7cee7f91f20835bef305e5e5540933c20f1b2d96a8ed0b93bc5a24b5597f8ed6
5d9b57edda2911b5a6aa5a2e9916e2c0805346116bb3071d752d5cb0608ac6b0
353cfceddcf08a449ffef153e7afc784e4a717e79cd7bff7f8a25cc08ad13036
b3a3049363bbd7847f260f148b8b4c962448f5e150c186a3fbd62aa8abaaaa1
c2e7af37d316de06c5a85aa2d1f5fa1961f6d79e9a2acb4651cbe78ce7639d15
24c24d8d9a24cfbc581d199462528dbb59a3fc23df5014a6b514399e55fea872
c89e58171f83035214fed44b77b308a41b1c57d20cf0f3670a473d3326505009
8f4d2e6b265f08e64b5f1e9703c88ecd7ba6157236fb47be6dac7d20d78986ef
Fiat-Shamir context

81871342c5e6d4526855e494d28eb1287da6de0616809cc8f6e9c2770888f554

withheld proof term: 2100 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Ledger.honestView.eq_1 theorem, Crypto.Ledger:0

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {H : Type u_2} {q : Nat} (g h : G) (s : H)
  (m r a b c : ZMod q),
  Eq.{max (u_1 + 1) (u_2 + 1)} (ZKP.Ledger.honestView.{u_1, u_2} g h s m r a b c)
    (ZKP.Ledger.View.mk.{u_1, u_2} s (ZKP.Sigma.commit.{u_1} g h m r)
      (ZKP.Sigma.commit.{u_1} g h a b) c (HAdd.hAdd.{0, 0, 0} a (HMul.hMul.{0, 0, 0} c m))
      (HAdd.hAdd.{0, 0, 0} b (HMul.hMul.{0, 0, 0} c r)))
```

SHA-256 of the statement above

354890090a89063363bc5ec80bc33f07a29cc3e4da6aff7bba01c19a80fd9ea1

SHA-256 of the kernel type

bbf9df05355bea6f1be7052286141ba5dcc4f09c42140216557e56c35804d0b6
commitment C

a944c2811c502e6b396af7268de29a50e41c74bf2bcc7a763d97ee67705fcfbf
477792717b771af33775a0165bd0dee322be78dfe9befddfc90a86232cf9b15
0c5656b9d83383ee4f71f2a7e6d11e01b94d4a7446b96c9f8467892f617b55f7
eb09f1084101583bc050a241f170f979c91c7e0002575687d09755cbe7d2838f
f14ab804c89a48e76ad58e69a6751819a97bae8fbc0674085b066f97678c265b
ffe7993fd9c7c838898c05ea0ae27468626240e60dcc50a34cb5d5dadb9bac0f
e7eb174e2a8bf29273e309965bb8be87631b0a42ddd9bb0f0c62c8c20885edb8
58b20c21f8fd9ee23784e4e153cd4e6b55d1a447b7516205b2525f8f06e32b9a
first message t

a611e2c0f0b7d03a206fa8757254e4b03d8232d7b9a57e0c25db226c7b4f0bc7
9cf37070f661b2f194beba98253124dc3aa69027e8e6e7202b56fb1c8eae598
578693599863fea3346046ee15e6e81732917af49c05188beeb752f986f0fc79
7cbdf3f87a5cc96650e0d62e4ba916c7e99f4ee8baef1f9924b426540cfef35f1
25e5cb34dc27639a5fcfac8035625667e10dbd9a6f82d032247f59cd64ca2c25
17522da1ff3712e6ff55aca10e07118b6d69aa55efe9cb73e95309a953de607c
f65eaf896fff162f8c27961646b0fa6aa5892bec5c4320407d5928dfb7b1c7bf
3793343c6b4d51420aa0ea0f8f9a846a23773b7a3f1ff86cd6da269d479c5e9a
response z

617e5722b2cfb24064ea844acfcc180fa45b089b5225045d4c44b140c21c8ee9
3168c3d6527365093a884b2ffbcc98b3ca76c76dd1896ef12fca02008b2c7c18
6853cbfde3ef2faa38f750db541503aa1c20f120a6afc8bb2a1e8a8fb12ec345
35e4b85de7d67013aeab7727147fee005c518e5960b7e9cfc0326d895a8d798f
862d12c79f1d4acb3902e635ee0cdd3d4e1c05eb9a140257f95915897fdd9f67
ff5361a3b4d46ae29eb59f22ea2cb31eaa7c2025824e0d4c9fdf1838c7ec3b24
fe0cd3705800ea3a4a62920d6aaeffaf86f40aa6aa41585918dfe0e889c53d21
fe09d6e099cfa4df3bafd25004dd8e3b12f2956ead5450f174211cc251ad8393
response w

7fb951d21e1672952b536d34ba85e8b2afdfed89bec3d8a0b4a660334fa0bbd2
df45efe0fcfc82c976640da3fd66d50ed7537ea762713e86f8180d2503b41c5d
382a35b0935fb662be6b1b6ed26775a865a1b730f9a1a237e9258bb60622ea2f
f1bbbc40d445a95df67bea10d537a428a843a6f875097f26c145ad668dae6f19
06058f2652cc74bfd9b7571cc31ba103e4533770246f06ce2139dab1de2509b6
849d9f30a7c3a72602bd0d19d322ce647ccd1fe3fae299372370a88d0a641799
5e6102c43021c63f1207e5a6f6a1fb583e3a479685e67e2bcdafac5512e38c7
4f38262b2a83b459001e0050f1aaa1caffc9efaef97306e10355b73c2d5925eb
Fiat-Shamir context

d5f7f821b538a41579ad70bb93e56131d8c2030c04d7ba114e2ed04c22e8f1b0

withheld proof term: 299 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Ledger.honestView_eq_of_commit_eq theorem, Crypto.Ledger:105

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {H : Type u_2} {q : Nat} [inst_1 : Fact
  (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h : G) (s : H) {m_1 r_1 m_2 r_2 : ZMod q},
    Eq.{u_1 + 1} (ZKP.Sigma.commit.{u_1} g h m_1 r_1) (ZKP.Sigma.commit.{u_1} g h m_2 r_2) →
    ∀ (a b c : ZMod q),
      Eq.{max (u_1 + 1) (u_2 + 1)}
        (ZKP.Ledger.honestView.{u_1, u_2} g h s m_2 r_2
          (ZKP.Ledger.shift c m_1 r_1 m_2 r_2 (Prod.mk.{0, 0} a b)).1
          (ZKP.Ledger.shift c m_1 r_1 m_2 r_2 (Prod.mk.{0, 0} a b)).2 c)
        (ZKP.Ledger.honestView.{u_1, u_2} g h s m_1 r_1 a b c))
```

SHA-256 of the statement above

4a8460f69bdf5a009b85cf06e5d9c0d1fd373302b4cdd2f5ce455360b04ef5f

SHA-256 of the kernel type

420ccf63e14e1013167e46c1da2823b888d75c1a40433f812f0c64ff2f118cfb
commitment C

e76eea69df708fd15b1cfa3d437835f62d952e2c28a73128dc7b31c067aad1c1
4b725b913366559f0b7d7c956afcce6f948f0d1e1f52f79a6d059849c0d7b482
de6e7e04a6cafbf18af0b57f35d7a3218cfb4ae518ceb3278fee1800d0f44192
78b904bc65fb41435b60cb7c07bfa1346869310b00cab05979e5c8367ef7511b
b87b81d1ceaa8dd6375354180bb97596877c045db20eeab932e1643b0d5ee8a8
fe72dc7545233ae8fa8db1c0566927df36885d8739d52753788b64287ab9bc1b
657ce4f7184524f7815e0ca8f761eb8d3463b4e7a83aee0a10452a8aa566fc80
a50a4ebbfac5ff78e6e60946c22239e9bf049024fd18d188912175f31edcbf4
first message t

a373b46997087908daf01d1616b5094e51dd5275974bc318c71200874266c976
86abb4ed66892e2a6a06c712762cb1051cf6d254423c01b3c235b4886f061e89
ebae18a9a45ccf5fb6a201e990e595ddbaad16d25eebedf8f2ea7b4d39761f32
18d42cb958e80b201d6603fd00b58128fba6e6999b1c5a0d6c2869874c053b5a
acece42cb256e230df91ee4f5db80fd99660138d03ee4a5cb9101c32edbbc4bc
99340160dbb8467093a65ea79301b819eebe9244550fcbf094ef9314845dad28
500b42ab9d6cfe702e51afebb77b290b677c6aad95f453486bf7b5d91f8f8535
2ce3575b06c97c53488da7301f685db168eb3d574b91a4203e1087b04b0ca8e9
response z

4ff2f9132e5831eac1df9933c125e4a614e5f79d7bc275e54c4fc68c07922ebf
b511f8a97061321a0037efb00f1c444dedc5056d91c65ee16c904ca100201944
02858d41075a204111a66a282bac08845d95ff94e28c60e2bd1b0ff089b91c31
626d677df4bdddcf7bfd7949d5f4dece8c7b7d7954c39b527200018e95194fa
a0553c3ce53655873f620c2486d680d9ee2c588a6654ae2e261543667ad6c03
7336cfbab2bd0e2509e922c156ffb2148e376f85cd1fca66062fb335a68c31ae
a064b4ef9c7ca074fcec724df12181ab063f6ff459267dd438ae1e2a1e108d9
76820e30a2b18484717d530aa5283522f726469acbe057e1e9ac71a5df3f711a
response w

7468ffa3366d32ac805a8705c0b452a479824f65760fbafa6fb61b6e11234d5b
cddb5671b5177f193092e5500ac7f07212b6ebee97a82c9c26f45838eca16d81
be65b0c1852ccb8f1f74d397d95fed11d91ea3445dcaef15399a997d34a9a4af
bb68d5bee9ec80d1b47b3129ebf4759f2b3cb1c5828b98ca351ff65d15a0d562
7b63798546568070a69dbf50fb8faf414c5772b9ed0f077290fdc8910bb25965
ef9e74e28e71f20aeb894ce8989db2738f12f9e7208caad275695ec41568e59a
90f5f8163589f4d77b4c0f74ae9b2f100a3ae36a5ec4153548581504fe88f8df
1f1ff6bdb2c45ef147b68b835c353fd144f0ca0bdef1fd8c8f8127b583e41257

Fiat-Shamir context

036b941b674889daf6a5c3ea95d56d3ef623181b44aa851b1eda7ed17147921a

withheld proof term: 1094073 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Ledger.honestView_valid theorem, Crypto.Ledger:69

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{H : \text{Type } u_2\} \{q : \text{Nat}\} [\text{Fact } (\text{Nat.Prime } q)],$
 $(\forall (x : G), \text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x q) 1) \rightarrow$
 $\forall (g h : G) (s : H) (m r a b c : \text{ZMod } q),$
 $\text{ZKP.Ledger.Valid}\{u_1, u_2\} g h (\text{ZKP.Ledger.honestView}\{u_1, u_2\} g h s m r a b c)$

SHA-256 of the statement above

e20ec6c39cb27c6912f964c5b1918c7b68ab22cbbdf4d5f6c97a11be6a840ff5

SHA-256 of the kernel type

5ac16f1ec16d9ead7b5b0313427290116e851744467631830a9ff3ce2f840bdd
commitment C

b572e68104e002232f2b3fc6d5f1ed86b41416c4d772e73b3f69c16fef3a6627
b254f921e912a67b74b8c0a29fc0f7cb39538da79224d8f3aaf64db753d6fb77
5aced9d336a4b9583f518c17cfe2782406647cf6e5ca9d2e364e719444b817f6
1d9fb7da29d75a7e8522c1f67495ed9cf44a03fd4f93f89ea4c71af610c8e727
63add0a3f826d09976ec0f8ce15f848f4068f65f3f2f43f7a1cfb58651c010cc
65cc22f9699bfe5e545ca8b9bcd57434d77eeac3bd74a030b09e8f4dd06d0490
5a7e1774891e18eaff828dbae4fe29951bd9ec1f5b40753c83147faacdaedbbc
3e29076f6cec7c97555546bef08cdfaa761656f3b5af3ad6cf72969930fc0024
first message t

87548f9bf0bb3fbaa92142fc820a268716d344c2cc4ff20d3b289760d03f8057
172218e1f01d718dbfebd3a36d667a7ca62a67eee25e966c342bcc16024e5c2a7
da5f831adfc1726b34737ca8c75461f7e91fd0957710dbe8ea905fb17e32a106
8aa1b03e6c91123a9c4b9d5b0b5d016878ad3745110671874fc94dc4eff440bf
ec9a72353b959f61db1f61bd498d122edd6b945c88ce2b9a3d3ab571350bd3b4
2737a29212cd569d0971dcb4d10317dd3556dd6ebf311265520887294c84f38e
ce4a6c42a01be86e77e9eaa86232242bfa60956e20c18346d2420d76d094fb00
774d40f068b6edd3ad8334488df00c1b8fc7790baa8d0c128b6f5e131762547b
response z

335a3996451149e5adbdb4c05e5900b22f5da72a4476cc9dbd416bd896ae0552
6c4a3dbad18e768fe884eda43334133fa699782dda779d5cb65e4fc082c55207
32a24ca807b3bf997da10edba38791ab9e267722b1b74e122ad169c301d1f64
10a48531a0930d682b8563c203248af6d85243f5b3974af01bb2a5c389e44ed4
a9398972b862db2a120eea4fec1eb98bffd57fa2aa81be583f7f92a23676428c
5b9ae91de0538c6806621ee9ff484a492bebc3cccf5630cf1c6e09c529067a48
aca7b92470a68fa2a82a4c0b4b47733f8d3c89f190d84bab0a7fca1034bb1611
0c3a80494c53f105ec3a65bf04c9ba05e9a34ff02b64d8d2bf9e5f5da7db16e
response w

3c72a0e6665092201ba4f852e3350bda4444dad9c48705e17919da8193bb4d44
1d1c82c6469b7e163546a891c5a52adc06de6705e3ff6600c348cec70c45a723
c63a1a6d861db1b8851d90830664d38b26801dc9e02540a24d57310da8d1cb5
4121b2d980cc79a9991f10f741c513a739ad87a7ab400d2319bfb4dbac0dbd
f2277e59a3af9eca941b0340c3d52507bcc4f016add2e42ffa9d76916868d40d
cd5696e233eb33831360a0ac66c274183fda9e36beed4f64b9228f76a50a7990
b885b93f8194433b9c2cbd2bbf29360fb0c569f949cf154e703dfb357e56dc99
72261857bb9d451d520015f1b0d8b9eab85dc28e93476e03cb556738ef22a259

Fiat-Shamir context

594565c1263c4f678dab5acd6b7c20454bd9b837774ec147461a5e90f96e9fc7

withheld proof term: 1 083 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Ledger.shift definition, Crypto.Ledger:90

```
{q : Nat} →  
  [Fact (Nat.Prime q)] →  
    ZMod q →  
      ZMod q →  
        ZMod q → ZMod q → ZMod q → Prod.{0, 0} (ZMod q) (ZMod q) → Prod.{0, 0} (ZMod q)  
          (ZMod q)
```

SHA-256 of the statement above

41a7fda205131351bdc572f6a25b948eb6dd702dee6e0c418fb28e4cb4e76191

SHA-256 of the kernel type

2664f2f00f8aed261010ebd94e08911d9feef2b2f8c759a5220626b0dfa52f98
commitment C

f513351e61cf83ab0955ed5072cdf8e6affac60b3919fbb3a8f9b4a95ae3e485
cdc0f43de3e2607971a062dd06565f71cf38007a459da39d5b173f0948cea182
b02a1eef1c774f241a5265af62fe401410c42a3bb96622521136849bdb6f2126
ef8da54de5efcd10c99d03257b80d2ca776feea8d2a737dd5ec8b4791dc94bc0
4efa963f58e95374b6071da8257fe880c1676622cfba74c6630bd3aff65f549d
8335fd22721480bd8a3775b19a385e5fe7ba58416beb36deda5eab29ad2d9707
cfdd237835da18c7b90877a089742d4a898bdba80c2196ce553965dd41303c84
aa029c505d04dfee46f388f4315d5cab4bd870516571901e86fa8a09cb682a13
first message t

feb745382c01c171b93ec10e96653c65ab69570abc110fbef0448c98280ea9b4
2c0f22048460594ce714da255a9bea0b063b12b2b4cc6c05bb3d7d72124efabc
073a3e07cd8258b54483c9b02dbab2b9644212e226a7c2a65825c477f3eea5b5
23bf7050208afe8f464f9825210bb3514269c915d1a2c390c9054b7af3d1d80a
68a7e7fceef48469c42488ccc915b365b0b3d9489f42a5ed9dcce11f1ae69872
e8fea4baaff320b3f14a5c6d1690e7c9b8bc5bfd5ec028ed2ba67ebc797b1767
f3e0f1dd1d7f0d893dde7a5ae9d0c99671a29631c2b99516c379da6927069124
3a67766f5086e87e8bee03cfed06670dd10469b6113aac45f429d1f1f346db8a
response z

354de100f9a6f8eb6792a4485e0ec96315d3b532c4355f9bf57dd0d026146d40
4b17719280e82ba29745cbb358b5de29bd11a5b3e4101df648f05034edb999fa
2123315dbfb8beb0aaea73bf010e5b938b50b60bf317536dfcb771b4dd808915
cfefb0dfc6f566701cb1f4c062132ca7e93f7a5be560c5768f5c27d2f3f3bf84c
7ed829f2469cae812eeffb2efd43cc58f3009e326b60757559cd21cf222c87e49
a2e10deb15f0438d460455b1a7ca001424a979494669f504ccb32c1d9b27aecc
8ab90684b758bae2f7074ffdb762d721b294bf87931ed8e594153e74387b465
3651a6b18f48375baa5974c9212b8ff40577c87ac09eea7dd93d8deb0d5698f5
response w

14c6788d8d0443a4e64cd4c85bbd8db2f6945ab5b5aacc94c285fb0c52c0b883
213d023bc0422a4b0fac6a32c05e7a4db03f34a57a811d388c58ede4b46304c2
12c2e0191bd7cf933d81f24773f78673290dab86d34f5c82e57e6c0ae7eb24ad
77be6701b9c8d71c519fc1b5fe1ea21244b838712123a723786da5fc38e4b609
aa83271df0d93037d37b0b6c8e5d43c535e478471ecee577fe04a54720df5db8
7acfa072b647fcd2525063e06f03520e273217a6a94f4b91a3b92b643d602ad7
7bb8a2df2d6d30ccebe0f4398178439ff8ff79754381b1cdc766f22c413acca6
5c65bae9bcf1d0093925ac21d7cbfb2d5f2b46e2339b77bb0c7f8faf85613caa
Fiat-Shamir context

2003acc37a8f516fcd645c8c4441109368af236e521ee00f91ca7a2b659742e9

withheld proof term: 2733 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Ledger.shift.congr_simp theorem, Crypto.Ledger:0

```
∀ {q : Nat} [inst : Fact (Nat.Prime q)] (c c_1 : ZMod q),
  Eq.{1} c c_1 →
    ∀ (m_1 m_1_1 : ZMod q),
      Eq.{1} m_1 m_1_1 →
        ∀ (r_1 r_1_1 : ZMod q),
          Eq.{1} r_1 r_1_1 →
            ∀ (m_2 m_2_1 : ZMod q),
              Eq.{1} m_2 m_2_1 →
                ∀ (r_2 r_2_1 : ZMod q),
                  Eq.{1} r_2 r_2_1 →
                    ∀ (a a_1 : Prod.{0, 0} (ZMod q) (ZMod q)),
                      Eq.{1} a a_1 →
                        Eq.{1} (ZKP.Ledger.shift c m_1 r_1 m_2 r_2 a)
                          (ZKP.Ledger.shift c_1 m_1_1 r_1_1 m_2_1 r_2_1 a_1)
```

SHA-256 of the statement above

4d4e1c4833a96a8ebd59e4225bb32227cb6ca6d52f5305776ef81edd62003d02

SHA-256 of the kernel type

32d751075256d9499b69a0de86506fd3503905f2b1e6feb7a2fdc8ea2c0d084b

commitment C

1d8a8808fce61babce85aef0ba8bedcf1154523f6344bf422a0a7116da69a2be
23b02f1a20c81eccfcc015f8044ee2cdf6010f484ed9290fc7ad5c28d93cab08
9460e2767bb8fb7d357f852735054e53cd555d186270e87232291f799741e818
8079de807b2832182cf45d4c21534feb5ab2217790182b5720d6f579530a16ca
20fc4669f11d53f803ce67b28aec52ae3d3c283afa5ee578fe57a57f31f0f27d
11c7d742e3afdbe10ce74b845398082c133d8f2d4a2b590036a0e064c1b649bb
1e1a6b63b546256da90991f42ae18bf068d3d9a912eab30221bbee0d6d2aab79
0f9ffffe0799cee6fff6a8dd4e286a5dc87fb59672873a817f9adfc1cc4f7198

first message t

f01bcdd30e3a9de11d5e9640ca176d1dec1e28ee575d58020c2c37b746d9dc45
fb885451402de50cd567617990f8d26468e5aeb492f030dd794fa58f9ef13abb
9a0b598173daf79159ca95cb8af46c11797612cbb87283125561a5440ab2481e
bd7d7beca1f3183b4870669c21ef3d90d2f84d336e589a88f24956e7f0c1b92b
887c05ccd3dc3b9ec9b78fdb3d551d8585eefd40dce9d522381d0647f48d750d
9c5f3db1f2af3c2990ea6aad87a8f8a8aef42a9156ece968c239e9e1f2e092b4
f63576f64f2741d65ebc09c167095a03df8c1585f1c178f23bc9b4b4166a5580
8d0115602cdf79981c9254fd2a14405b6b5432275ff85a5906ac1e851c5fae1a

response z

6220cef55ed05d5d64a06a54441151214506b23002d4cf47fd4ca62053d49215
5c20340a1d3d1442e9a6aac95434455e1d6862060df1459c94717e6f59bb1972
ccadc79d7e49b6c3e769f10b1ba72dcafe1b2080d06532f6ed8881dc334f8d12
18b2da87ae40933ea60cb4a85260d1bc865e66affdecf3fa3f3cabb23f64b426
befdea6dd86b05a60f50d490ec7f74703b876a041cfaef42a50ffd56b2278f90
bd2df6fe15c57e12ff06ceea8a0a68b619446e54e21a02cdfa063dd4b0f0933c
7e7ad64fd9b6b3e24875ad176a1c3cb24562266180968e7a1a37ca53ef64b5b9
7871538e2dfb43d5eeaf198cafe0b2e2534c97039b83e0c94a6b302040c1c218

response w

5df1299b529769bc01aea8d33bf00fad7570dc460885793916db7f6bd62b85b
6fbaad72d4f6fdffa32e7a6a8aa7c49c4426d7dbd56a3848e4d34812fb4fa6e7
e1e938d14451328268f3dbc0cc696bae10a25428983184575764139a0faba20f
ebb70a80c78a1180e2a4500393f02d50f229d7f487725f781b044a65d9a9d5d4
1a698365df860f385aaf53d25d5f06a69edccbbe689f62418dd302cfd71c7fe4
12dfc340f45d3733661ece88d0e9e61911f05f4c8dfba9aefc9b5d9712a7046f
51ecfa497e081c3983fb2675e364192cc79a1a84b0ea6b2f23f67de65c44e5be
d54072f561c6a8d1e182a7f0964ac9b3345e9d4943a7e9f3667dc7cfcd9bb281

Fiat-Shamir context

01bf8aa34d1090b6c92d197d5c6767014e40d447c0df65dbac3b310c6aea30b3

withheld proof term: 5365 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Ledger.shift.eq_1 theorem, Crypto.Ledger:0

```
∀ {q : Nat} [inst : Fact (Nat.Prime q)] (c m1 r1 m2 r2 : ZMod q)
  (ab : Prod.{0, 0} (ZMod q) (ZMod q)),
  Eq.{1} (ZKP.Ledger.shift c m1 r1 m2 r2 ab)
    (Prod.mk.{0, 0} (HAdd.hAdd.{0, 0, 0} ab.1 (HMul.hMul.{0, 0, 0} c (HSub.hSub.{0, 0, 0}
      m1 m2)))
      (HAdd.hAdd.{0, 0, 0} ab.2 (HMul.hMul.{0, 0, 0} c (HSub.hSub.{0, 0, 0} r1 r2))))
```

SHA-256 of the statement above

0155df605f61d6415ce7a3750d20755c84c1b3b04144073c1aa685db0b299450

SHA-256 of the kernel type

de05e77aaa90950833d81b4996a3a29d3a1cffe97d2861c7690b7e6b41817811
commitment C

773b96ba11b20c75cc1471fa751afae474d95349c58f832ebb5ad84cede31ada
a40569eb6a6d71884421f65fa701360c583543d0d422bfe67c92d1a385897670
79550d3858abbd9ec800c4c765d89453b1e0fbe45ded2d2b482b380299549ddb
850c7a95c707457a66bba8b88af7d6cac1892633e0864dec8c4f61365b5bb3ab
0521cc2ebb191ccf581ef7a88f879894d5e25279908569e9f9b566b1253ec926
a6d139644e41a2e3df027cbc2958d8a33c689384ba6f64febcb609072211ad052
e20c5138e0a8ee4be84a47a9d332e42e62f604fa5c4490bc8d154e31c287b6b2
2eed0abba8971d7934e3cde9a0c179cf4b1b0eaceee5a6f624d6d879b7f65c7
first message t

261e2f22249420e7caee794ecfe0a5528b770744c9330263cebbcb19a19ebca65
e3c6eefdb0d1bee0d6458391432724b162a90b8b00d6300857387767a13ce4a0
e3e6a525c62923595c2b6272d19e61f0a4ba87f990e2b48763472af254217274
e1900d27255b15fc33d3de13e286b501af86170547647d1cb0bc22ec0943b58d
ba24fff45e603e586ec24f35c03b11a953eb79452caf038ef7e0a7dcd7adea86
50c40f092b70ca7556d2a41362d2eb156222beccd35ad97c7a8b2c6b5e9dcfc5
e314cce583805129739297c9bc091fdd2cd2fd5caabc27d831fc6687d6554d4b
393dc5b2f231285181eb9f86532062df158a08ce4a5d7d4b5331f8f2bae7f81
response z

777d65518f22b1edecc188eaaa10f3a1d8b8a739c1d60368063ba9b97b877752
6f8300b16eea99cf4d56071f9808f4bdfc38b4e191d9ecec36152ab59becce4b
b6a92acbef56fd76bf57b6942ef26b6a9181f8a1bad614939f0769a2459f4576
d365ec193a22c264ad0372ec03fb3ad4fae6d24da394b606aa18715ded11050e
81f2b5863ed8fb912ab0fcdc4cb58024c11087a53cd13aa6bb9b5b47ab81d603
f8c3e52dbd55d39d0f9e5125a72500cdebdc8c7e400b08dd468162af6f2215163
63bc80fca343bcc845283428947ecac840c0f3b65bc74cf7aa71cf9ef7b17853
4289837ae6266ba9342d9735f568c4d00cea1d61b5cd81b471483605c8c170c
response w

3037bd30ededd12aadb20e4ccb410a19ab6bce3c77db8d12845be68992702e18
0db72ff45028fe5d7d8ac23e6d920d164ad84c412ea21339279b314b20b8291f
257e09900316cbbe1b4fa6f632d9342f847f6e9d09964cb0c06b396537310c93
24d7367caa44d8bc4ba78d4dd488c8fc6cd579960d3471b80f6a217387a232da
0d0e263ab227c9ce37ac5427e9a55af0f8285de600b8c5e55f21f818c2e2968a
44393662cb98cd2a7e59768f960ecb6e06d22b9b18ec3f55e5b3b571b69fd576
68fd7e75d1736f085f1ca642e096925e16e4b7758e97c1ffb2da82130f9d191f
7a4648591d6ea2a6e792284e65dc1e2abd8be23fe4a56caa85d2fc50a2c9e08d

Fiat-Shamir context

7457fb96c9285cdce981a12a25f563a9b345523392d81743a025f7f8d9206965

withheld proof term: 253 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Ledger.shift_bijective theorem, Crypto.Ledger:95

$\forall \{q : \text{Nat}\} [\text{inst} : \text{Fact} (\text{Nat.Prime } q)] (c \ m_1 \ r_1 \ m_2 \ r_2 : \text{ZMod } q),$
Function.Bijective.{1, 1} (ZKP.Ledger.shift c m₁ r₁ m₂ r₂)

SHA-256 of the statement above

2c32726388d55bcec25e724b63c5627b0df107402c8572da0b38def21a9de1c8

SHA-256 of the kernel type

c55d06f6f33c288b5e079c9b21506173ca6c8ed80318ca8eef12a4239a57d205

commitment C

341b970010da00679354a103ba5e50529923ca327e12bdaaf30fceaab3c3c061

188de5e6e8d581aa2a5cf7e6017d7f56a93dc411daa038a6c71f98fddee4f6a1

6c5a3d0e770f65b3b75a9184158f7c66bbe2d7368986da1b0ea0f551a980b5cc

b8b9b02d2d98bdd939c1afa393b963ebcd68fa2db78e66c6e510e1e81e088ab3

65b7ee1797a45b716817a0382f6231ea9f79495c4d13e65ee72e4c6e83d13863

78c681671db8454b203ac56a07c4b1e9ca224db239fbedd5b679d5f6595c5ac7

1c6227989b800ea23a0bc7cdc48cc24b0564573c57009a53da9dc9e77d8bd249

1d5cbee1fab22d3fcba01ed7c7fbb4dfc3ce18aa8f073922542702011a949cf

first message t

26eb041b0f719d00a70bf4be4969426a7d4cd746be6de7af5cc470324712f0ed

d5ef355e3beea0fff03c140f994d47019966ebcd245d1a839bb92c87e897796f

cf9a8120c084bdfc2cf9c0f264d653e049202d7f6b6e31f88469dad13f3ad327

e16d4721b6d3a2c638249e64b2d72ae12e7331bd7c655f449b8538803c4d18ba

8fbf2a3a54066bb0bbadabcf3c7c9355474e0b0b53d8dc6736142dc36689e2c

d5b0780066452a91842d13bd0785bbd3022792aae8aba4b8d9c87cb47f8e0095

5eb37467027955db1d3a4807e2e6f3981825edf9d43b4dd5e2bbb90718f2d62

108c7a0da81cbd3a432c8742aa4719052747b181c1b864def457b46dbd42120e

response z

592e64b9d923924ec54b7444acbeb05bd25df86a14c6f711ccf343cce71c89cf

5a0b8b990c045f0d3b0848040f316897ce6dc340c466570053ef08bb4fe28145

60e6e180d5089362fe61eabf554c3abfc11b2ca683c938fdbcca539038ceab76

17578d67832bbf474aad77dc88f93843b0f995d3d5d882b95ab765b9c4eb97c7

9312d6765072b453dffdb5d7d79414a812d46b597f8a3c1ecafc1820f38a6a4f

56d98303fc70e2f3bf528655c1d1229369e498fc78c4a897753adc38d126cbe6

e7cbb03efd3fe59bb9768a4f5bf0c975a6829ae62f850e3b7c0b2b0404edb0a

600ad6f9ebbc68354a0dbe9af1a7c4c8670dac83cc533bb2f32258b68664e22f

response w

7115eb42e9c7518269a49fbe3bb7a3a591b6f6c6e32c00a923ac400e2698531f

511a8cc156516765630ba7cc48f46e1ffeabaf51a247c99600895a6351a2d8b0d

f4d3ca3d8b78b75793747b4196d7e3c7fd825edebe00c25da4b149397b7206ab

703727b6af8fec1c7e917d4f050522a7e141abc0b356c08519284daeffad8b0f

f1520053baaf05d74582772e0c32f9b9ec52efdfbbe497970280ca663ed91678

fcc2e02c2521b4d4fe2aee3c3d095e06d30ce4981ba9c17c10e44c175ef57834

5fdb37e1623b9afa9bf24d9477263a13db6a853e3c7f9fd34a970cacd4da497c

e8e053d458b9e7c51d76938206579c604feed05527e9d90099493a2fddf98b529

Fiat-Shamir context

c66fb117852d85837d7cc4343a3768c75a68c357f40b5f1380a31e534dbcd455

withheld proof term: 82 690 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

5 RequestProject.Crypto.Merkle

ZKP.Merkle.MTree.below definition, Crypto.Merkle:22

```
{H : Type u_2} →  
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} → ZKP.Merkle.MTree.{u_2} H → Sort (max (u_2  
    + 1) u)
```

SHA-256 of the statement above

e3f6ee2c0d6577913b36b278dde17350bf98aade3bc0d3a42eac695c05570f90

SHA-256 of the kernel type

a63906b3eb8d127b10d5c108a3562cec6c2f35124f0f1f8d152f56607f54e9f6

commitment C

9a08731f1e97e1e36eab3038b7ce2ecd8162e181732cb41f7f90ce9e3bf0487

ee4a0f7694d55398f9d9bec4ffa24405cbd91bc38718fc07b574746088106f0

db6187be2a0e7365f9019d6907d1b7ff33c41b9890336355134e52f6788acc00

03e3f2dd377c366876ad0ba496cc39078132c5e94eb31640c96c88c657383216

28a95c59c0de3dfdd3763da8ef02fa014639fab8b52910b961f619235110e5f

349020fd7aeeadad068cdf7e82a8c92a7dc1030c6cc9fc14dfc66ce80e0255dfc

ff040387f37012c5e82846293d37d94b02974821e9be04b880939c2fa1aae524

b8b8a28803322e9ee31529c73e0014c9e40dcef4e394f68bee4c06d2e9c05fae

first message t

faf00af2cc781cefealcdb66fcf68409c9ec85230bf3da0f2813217b1fb2a6e

57c7df2b0a5177fada810bea95910c2275e78e4cc1620542133f027f0e0a698a

dfb0124af7e3dcfac11e33fbd08a7ef23bad26644477a761f52f8595c91af0df

a3482d0e6b63b3ac0652da6fdbbc20c90be127a0344846f89e838532e731717e8

793e6c965c32d40521d35e566f530649644d5893c2143a8102419dbc82feea3c

a962a231ebe29a488a88c907d52974df820dcec231dcfc3d340e845ec2aad312

287574cee2250c6c97df38c7e57530252b97c806e03e94695d616c81d26b4caa

70f3baaa84e95915be284b3beb3e5486245d17d0fb56773a5cb282dc18705ecd

response z

5ba2cf57ac1f610d648195ceb17d47fa53d318e5d87f825b7b6d990d14b95bb0

b3dcf14e3fb9fec930b45a6eb83999516c33927139a9ffd3f955273cf0195c1a

ecc1636be114e0d4de537ecc04527b974bd6c294196d9fd28f4b42868dc585fb

1f3b946722478d431187fd49fe011680d12fe92ba77c5416b79257ec9222820e

e3584571f46845f1e5e0e44c4526a9389d4ab20055b51e078f90e51d1d2e32f3

6fcd5f593a97d882478482c321e88839384f67e2e698a6c8f9b47ad44a8f2127

eb6a19f11758e9b5c1902103201be14aaa15ceeb5b1a8c090b66bee6cdf33e23

e06ab70dbcca6586110fa65a0253b0732b62aed5f6ef625c9c989c099cbf5d73

response w

1cba088fd0d88cdc135e18f621fe7498d79f50585944ace1283c1c52eaa771a2

a2b14ca5446de5e10ecd335215a8071596aae2d35e85c0e1c8ae791df3984ac4

0620ce551c22c102e2bd00b724aa964f34ae686f8199956695a0aa5b2fd390d7

fe7c590819ad73f8254fcacccf4799955b1f16ea735005fd1688cc222e951f6f4

4af0e9b674fae93b3e5d3f1acc422a8db3908528f4ffd01c0b0d1d117a609464

1e8891d449e9e1ceda846d558b59a13ae354beac86f8ee7d9140b56a4fd8c971

9ff4df0ad3bdc72ea66b1502cbf0c67db00cbd20213c4153474981a3f8a5a62a

8f72ba5644eaa59d2ad581343cceb519e7e889018a53509baf0a3564aa8f6704

Fiat-Shamir context

7bc5dd425273a1e28e56bb6e73fd590a7e00ecc2b2e64e825a4dbb09ce80f390

withheld proof term: 746 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.brecOn definition, Crypto.Merkle:22

```
{H : Type u_2} →
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} →
  (t : ZKP.Merkle.MTree.{u_2} H) →
    ((t : ZKP.Merkle.MTree.{u_2} H) → ZKP.Merkle.MTree.below.{u, u_2} t → motive t) →
      motive t
```

SHA-256 of the statement above

8c3278d894d105aaf5c8498a1dbaa52f2733ae2364c7eb5c96654275838b0150

SHA-256 of the kernel type

e0b1c6b3070c9de4e4262b05649af7cf8c1a96460c524508b2df31b185b26447

commitment C

7043c28a46c4d57c18d49c144c2b280ca2d9146131b6694ee9031cd31555a879

ab2c6d1916fb50fbefb6b5edd641a752f156ea883c4791844f9d28d95db98bddf

e73a9e28f50b641505e0dd3a11f46aaa848e68513c6ed66d6b51b3f6fc86bc64

7225d2c94fdd58af8c76357dce5e66a94fad4433acfd0abc91f8c583116670

71cb33af403a536788ff547c7560c890132b214001f6b610e64b94dc4c607218

c3e1d3fb02e8444d29539412b8af65a47e1b6cc89ce34186c328f00e3e5a1442

4f74d18c0231e7ccf9aa5a0c46970ed33f7beae299ec15288a5f97fe97c01a05

de6269c0e9505e9199420e80b8c9bfd6562928943909795b41dd580764b6c3e6

first message t

644ca5c3c7fc812297eda414f4445cd2ff1240b6b8f9d989ac891b5270b787c7

7750dea31ea88a599b0736bca4eb82eab3808f9764a88a37a96f70b804674a2f

e2b3df14a07f94d080734d708e48c82bf85047c33809a527096a4d07683a76ca

77537cc819b44f7b0c1adfd99a57b024110e64441df71a73d3208fff39af3fae

7f835546de8e95b7e6e9e2c7198415fa585f8e64dc51 added 51fdd3cf4e90fbced2f540

eb556cbfff92cccdaff92ae94750846a55d266e38422008c881754d2f1772647b

459e4a0993c6bfb6cb6c82937bd815a5ca51c592329c9268949f1948e3639dee

2a1cf64b93d483bfd277704fca4fc9ab878b28a972ee52d25171909c5fa4e04

response z

1e3138d1be85eb03e4782e2a2d95c7b570117e1618e8d2c1be30b2638ca8a626

97e2326210b62cc77c11d625e980753854dcc84a0d190c35cba6fc6386db70bf

86ce4259f461c367fc311c51f5a9d9eccc9069f8105e81a2c0737d30d5dad4c

d4437bddee264ae9642049722fb32c6869407b3022d308095b3574475e4b8093

f9e239a576f33ca38b16f980b2f78336ecc045f34843202c3c707abb707a2e2

7b3fff878a06a04549edf092bd1238cd4a3addb0dc020e247502c8fcaa0e5c25

41fafaf3756d04702398c35a6c364b6a8de70b5aa80b369b2823c7c51d31ae866

123d95362f8b0d7c7fb15a9f091b058c506f1f0dde6b3a035c7880505215f645

response w

623f58623131e165984712ebda544687d49167eaea8c70563f1373d985fcc947

c890da11aaaa587064fb08cd07e52eb1cd458ef523133e1fcdd3f1f7f5833403

b2ade00afb52e015e2c369a1631ab5e98a5732bd1c2d1ccd24a7c255e6b07a21

08dd3d2c919d15d8765360880cead1b9e1caa9a3cb822dd5d7b7e047d1a60c35

cc26e75614932133274f2d1645558e7d4901339fd8a0e74d4b774a4f9839d20a

0845a0de0d33936024eb7f5279673deed414cc149361a4868bc44024991589d0

3c744c37fc49d9eab968a5fe84a3224c54aac461d4f8ef1e615234474dbb6841

b424c3466bfac623c65c04fde2b370a48863cd9d1e8d646d9eeab45caf3d3a0d

Fiat-Shamir context

b48971418d6e152f2105ae1ed8d0444fb8dcb6af58a35aad7bd0f80d6bf1ee4a

withheld proof term: 265 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.brecOn.eq theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_2\} \{\text{motive} : \text{ZKP.Merkle.MTree}\{u_2\} H \rightarrow \text{Sort } u\} (t : \text{ZKP.Merkle.MTree}\{u_2\} H)$
 $(F_1 : (t : \text{ZKP.Merkle.MTree}\{u_2\} H) \rightarrow \text{ZKP.Merkle.MTree.below}\{u, u_2\} t \rightarrow \text{motive } t),$
 $\text{Eq}\{u\} (\text{ZKP.Merkle.MTree.brecOn}\{u, u_2\} t F_1)$
 $(F_1 t (\text{ZKP.Merkle.MTree.brecOn.go}\{u, u_2\} t F_1).2)$

SHA-256 of the statement above

39dacaf86fabcecbcd511abf0ab3df12d3a89ef6e92137747fb544a58054267a

SHA-256 of the kernel type

f818aa92dd9e0195b6a8e918d6e61f2fe3ce2f4c3e12fcf3e5f5be4d3580e9fef
commitment C

f7844a24eab6784fb7967da39b4e49a08f510dee6ed166dd23e5b1a0fec7a715
584e36eec3a27da97f341f1888a93b2c094b1a628c89141bfa89be002beb32b
9657e5aff458d26eedf1e812ff27ca40abd0102ccffb49feadf4f24f9468c03b
b732a580d7e362807a1bb25d2d162ea10d031df87a793edee635188700dc388e
8b4d771f5b3113fccc3ddc3ec0249651048ff220e85e1b0649e56f72e7d9a78c
b07bb2e5412c35364a452b4ab1b4b3ca500a3965e317fcfbace8646ac8e19856
578e68c827c648b4a44d445db6f71fdf3e81fb7cfa769d895162c4e6e286562d
05161096e7c754629ef264a8346c506be23dea1199f85f7b24d61f970aca33c4
first message t

3301f7ce273cc4b3fbb29f117cde010a7db4df437e7954d7261249154a31819c
792c84022df74863389a6c796e4dd1079ad94aac92ed5465a7960693f88462b2
5b3548c0f81afc75c1d0a8ce9eb7b4a05d954b76e549d60737f6d2989019a153
d0f4d4fba0cbb9aaec5dcffd5b9b05687f745cfc93a91bf389791fac18957c0c
fe998dc2992715669f1b2a09f42e2e052aa5d2377a41ff95e3e2dc95829fdddb
4c4ac1f0959036c8f9229d375f3b61b068a63738cec11f6e75c2ce3c00d4a442
315175020725b4dc8744fd9faba5edc17cd6eb9f17b2fd9ab34c6b4a08aef97c
384ebb4f53de9966b0ad5d7683d716c311849d3bab926f1b359991647e154217
response z

64a6652493ecc877009c744d35c92631d9b02bcae56dcb0c5a14e57194d06ca
9f83d8bfe5796b71c538ddc1c7a3011f59dd3e68b76087491f9532225f9e6b0c
358855cf76b0cbce6a02e8726127f6bfddf89fc14c1f379fb987dd6004ab2310
6e19fdfcac4a565928fe851d6fd13b1280ee745d1f289781ed97a3e00548cf3e
8f9cb0e851c6066295d10b8086ade7e8b3f6aa40143bcd9d4ed6762595dec64c
176eb20d78bd484e671fff987003dbb7e6f01c6d80ecc5bc53d742a3e4b64e52
7fab3c9f52549b060e84fb9ff7b4f0fcea1da3b0400b6832124c56e978135991
fb61f4eaf87ddb0ad6471cdfc633d5d7dc6c3328430b55c075ad044aa33dc49
response w

5cd0d5c36edea865327ef98cdc46ddfd976ae430455eba352e31483757b87341
50ffdab1a379f69d9340ce6f3f8ed2317e59204dffedd878b16499dec705b98
264c75fbb87514782dc617e0881a2b1937656869d0c96622782a71f6f244e3af
96a77d93d740bb9a5088486d638248c3f8231c7b702e635dc8992e7b08002f16
660fcbae5f1169f524c27f555a7df25f57a2999be631bde408a625b396f706b3
73a1490e2a7f3892e0f2870c45ff76cc31116b9fa26cadefea24bca5d1988d08
8004b2dc40615b014eeff40512b8d1223a1a9c6a0c0ed88661139e172f424e50
e049ba4ec3e5bd75c4c346f9c4ab3c39ac6f5bc1514e93f085359dd8241a34db

Fiat-Shamir context

570855ebbb0be9c008ef8417d1dd64f21307eef6f7efaf2e77c68490d2205e05

withheld proof term: 1 426 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.brecOn.go definition, Crypto.Merkle:0

```
{H : Type u_2} →
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} →
    (t : ZKP.Merkle.MTree.{u_2} H) →
      ((t : ZKP.Merkle.MTree.{u_2} H) → ZKP.Merkle.MTree.below.{u, u_2} t → motive t) →
        PProd.{u, max (u_2 + 1) u} (motive t) (ZKP.Merkle.MTree.below.{u, u_2} t)
```

SHA-256 of the statement above

8ad9c10e667a5624f8f8523e4961c02c145c194cd6e4cd28b397c1fb1305cccd

SHA-256 of the kernel type

ec21e18b2be7103955dce071aa67c2e18a96cb2214bd25cce9d669f768d7ffd

commitment C

567b295ce5472e7366ce44de8ba53fe5b8878b99ded9a97e0eb3ee97857af62f

7242d237d211e0372535f513f73525190a438ce4ceb29d2cb22ffdf33992e14f

efb0bf01438cf16794ddd67e03cb8c4c47d6a67c4cc3e1168f136dc43c0fcda6

a4784e27f35ee87b623f048ef8d182f333a90bbb1c1354dd0ff99bb6910287a1

b87730319f5c5b501e4e8b608aafc4004ed47f11a16429870cfda8012f2338ea

49ffbd16c2bb9e9b17773ed7310064c1a0cc32ecb777cb7dfacd1e504ce95356

f944ef53269f3c1ba6203b0df1e971443431c8cfd430a8d713dfc8351389a7c

060a8e2434b72ac1a8ca931f9e9ec188654d2dd267366f8ffc59b81361b42d29

first message t

e943ba683d538f1ae92c402f6ad28e5cbee7da33d48975a2acf8eb1a8e4f87a8

14f639f37a422af3a36c6d78ecbe77e104b69efa7fd47651399bb8a27ad70100

15445b829e8f57271bb2096f0a0a7c0ca1c73914b562b5c4f9fed9391c20dc36

58489860790490fcc53e38e952b2c65e275bea5ad9b8a4a54d1075a4ea4203e9

06c7fd4253014b8ea43745b07719fd0a3f3e155d6581386959d2e794ddd9a4f9

ca4f71f00e0d77b9425a7667f5803075ef7ae8ed29780d45fd38a75bc501231f

c428f8da9196d7c565f60ff6bdf0d46c06395c63adc81f5228697b60e91b8d9a

a5de921339074d62a947b4097d780fa8d70a52bfc0057667628c0a5a4c50f1ef

response z

2ca51042d951160feb995d0bc0d53a50f9169f371deec4b44f03ece6a9bbbc2c

84848376215b4159743879a1c99901509ee65de8ee925794f7bfa586659a0c3

dc9f1d1f8a82e46d92e197f684b7d2c2da01e2bf8002a817440fe9cacdea0f2c

775e4a10f89f5ba2b9850eaabfc89a249189e8136d3aeca55aab1b3a658e922f

2c69f91aa0c1cb88739a453eb4bd9a83d661b4ecaf026b7a6d0c923db5e9b479

82fccea26c4811f36cc7141c7ee86bb9dba723abac7254889280a74215d0af13

f3f44bdf94259c2c9dedb13235eaeef340bba44a5f74576e7e8e59c157c61ff2

e6bf78e19abb305070406ed86f3b87d76ac1d6e6f79ccb2aa0ec8cb024803976

response w

3b6929c43e8d39c9a825650c963b3173ac1406821abae48e7452897266ee1aa5

6dc149dc69b4d7194af73091b7c3bbba70ea65243f6c93af4a0cb031f52b055f

ef1fd46fb7e54c51d55f3331d23dfda8e29f5925dfc1c7ac51d9c4272ca42789

93513518f02711f0490a0be7c7696b3934bfb4e8c84d20747aeb8d6e9a269cee

f1c1d7d75eb74699cb7fe09def9018dc16acfa6694e6a06b2909ea42611d4227

ff6614384c6e2d7ffdd66ba3450f2906e69369fa323acb35d0c863e11636063d

c64a3281132504e3faaced6f6a8ed43da90b7a5513bbd6e9525fac81d56eb514

f959df39225ae3b166f636623ed04323359765e0bde0fcf29e22461dbcca7d2b

Fiat-Shamir context

62522f8b5af040d5a06226bdf449b20fa4cb9bee213abee7f0974e52961562ca

withheld proof term: 2371 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.casesOn definition, Crypto.Merkle:22

```
{H : Type u_2} →
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} →
  (t : ZKP.Merkle.MTree.{u_2} H) →
    ((a : H) → motive (ZKP.Merkle.MTree.leaf.{u_2} a)) →
      ((a a_1 : ZKP.Merkle.MTree.{u_2} H) → motive (ZKP.Merkle.MTree.node.{u_2} a a_1)) →
        motive t
```

SHA-256 of the statement above

3ce3630d8cc6ff78235f3237a1d33114f9c78d857f1dad6e8364396eb3857d

SHA-256 of the kernel type

9e57bcc513d45dadef4cc99237b841dc7d5f138ddece45abf6188d382de2e8
commitment C

89a63b698a8b5f11670845c6088240eff4774060c5b1027e9b85c81b4f3cc756
2ea058b0c3d8f6e37fc2c147d099c920732232aee5451bd2a242b11afcb8fa12
46bb1a0e8d01074e53d96358789ff86548bfd6d8ca7a3da4d989d6aafb368879
65d43e208762ca714c40e71d0d710dff3c79ba14eaf8ad4b8365bf41c1c574a0
d1e94c0006990b59b815eb9ea1bf191037993ecaf95fa86fe12b6e488da3ca8
b55a64150f4d093ba1f30100f7ba0ff0172e29a26aebaf65909fe4c53959a47
f1afb0eec4fea79713449b13ebf7c999335c7922bb616b3df118386c582c120f
3dd9de357d37bfaa7d354e0c55ee01115eee07eeca011492e835da94fb218bc3
first message t

c28f5c2d99677ef30d6eb11d979507e3d083dbaca6d06392cf7093f826472006
9dd9a9d0c38cf8a89ec3d3f0b4b40dac06aa835e84cbc8364c93064fc256abdb
311dc93eaf70047be8de5d0c806c333ea5d9bcfa3fc511cd4ad85015e0034e5e
26d95a68a252907366f7d9770b76f9e8cc8f77903c4eb69a68c6af4f34e21dad
de82e7286a8feef463ea807d08b9c23c66995da1c32716881f19a724b6a7b7b5
5ed1af5a044f82633e6403fd990f0694000b9294d6c11fc859cf79a2c6f8ebd8
60527d7e8c116c8a078367839aba89365d0d493f584d263c756fcee69999110
1bc1e5dbe04f9b6a4b57d5933d9e04420751e67e764a5dbeaa78ae0697d2798c
response z

43e63421c388ce85902f37864a5c236704f3ab8b2ab22c2c9df7a8983e38743e
eb8969ebb9ba42d749244543bccd434d460d9e3ba95d987fdbd90d5cc118fafa
8245d90f918bb5f702debcd8ec5edf2bd92b369db7ab6da3cfcf942dbcd53e25
ad298370a928c294576ea51f37165306fc9e10d1c2d21dd558a27ae8e9d6c3f8
5309aa43717b15e70cd858001d0da6729dcf0ccb571932430650db39237d63bf
d5cc790ea913f07d3e71c45322f524fc707d4c5dafd478404a50a1e9c2070d95
8592369618275f74b6bbce3b03b71a3f484b086144ac90f493da7b41c1828e2
5220a07d025f5bdc1162f8b09b1ed7ee423e196c6fb6f21b2a8067bdbc2b9d30
response w

1e9ad8740ac2de4b6360d9d7030a13f8a07b4b6b5d9fa282e6f852ccd8c8d15a
cb25bb8901fc475053a9a5fd624040179aabc95d31810f92834ebc17d957377e
7d6be65d6185d70f19ab3b5f0ff041e2bcd625eef20d87c5b9dc8e1547a1903
6ae7ae6aca144e2d0a8078f033de4a1e60e4ba47beaa4592f925c67033973b57
554fe4fe7f8c8dac3004283ddccf7e59da66e139dadd326233900a82194a2bc8
00a2e9c5cb4c0de4b0faf1bb9ea8e5f0920d7b0a5a12f57eca2f1b67943d7cd8
2205bde01fcd89dbf068a5d8823fc41127a460e6f9af05c01d1fe18bd6d6e9b6
f567a52134b2a6dc208310761c3391110af9ed8b761a8449c0dcb1828104d135

Fiat-Shamir context

551dd657c1b2c19eb7515a1eab5721d27574506e0fb8aef9ea2881bc827b6f01

withheld proof term: 803 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.ctorElim definition, Crypto.Merkle:22

```
{H : Type u_2} →
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} →
    (ctorIdx : Nat) →
      (t : ZKP.Merkle.MTree.{u_2} H) →
        Eq.{1} ctorIdx (ZKP.Merkle.MTree.ctorIdx.{u_2} t) →
          ZKP.Merkle.MTree.ctorElimType.{u, u_2} ctorIdx → motive t
```

SHA-256 of the statement above

cfe35e02c8c9fecc5494c0cea3c282cab7a6dd137b592fcd12778ba0fcc392b4

SHA-256 of the kernel type

6f648c4f8606f71030e180de7c35cea5bbb558470546767638b681fdae447222

commitment C

fd99c4d3d9f01b225e9ba39fb5af9409eef26cbaf7f8769df5f5e59c6fec7829
d3a1359619b2d216bf94c7dd5e2c3f1224e55e6fc6e6e4629f89e4307344fafb
802d2e3c5081f70b3ec1e2ac0cd0f192216b4965ef02ae33eedd684576153a0f
5f0ff9696cfc0566149cb7b54d0a95fa3fc7a38c536c42a8d390c1c64bb2b2a9
5dff434e577960913f1f37273599e05aeb775fd7262e2d74572b441e7cedd31a
2a5a53718848097c525114945e1b122d8a248e70a11d882b455f1f5ede790375
267416beaec47968b0d64c7b13752c558ddedf8d29e21061eff06c6a987e8afe
9dca9d4be915b0673e7ff2c3d3eb3a5747294d11ca51d17cb9905235433a1dc0
first message t

3233c73ba4bdab4b8405d90d0380e6c5412ca03c90821c374824f640702119e0
4ca0802a7ad5a4bb48f1e8c891322a0d19d58aef6f81d1f044f122b392cc0624
04a17949e6aeba79fba3ca4a0f8b4f1071ee7379702bffbca854ddaf64cd2c45
63ac95cd6ebaf57e0df2d8e23bbb8f52d2e32ceddcaa61256a32cafe24fd0f4e
cb2c6a2c29cea6a838b54b84202311667ebc5eee9a5f2dde569b2d97651ec74c
fda6343ba444abd4a18ca7692c0f0f34d228456613591d2f86bfc73c719e1ddc
daf669f3006dba498a1cf967fb43afb5c529c7373dbb473c94cecd4eb336e85d
e70af97c9ad83cf0273e5a9b6b5d72caf34bd6b2fa312bec5866ea5f3ea455f5
response z

264f8cfe446600482874c62e023b193a5146ea3393a6f71e29e13e1b8b0f7744
36abdaf7eed762759cd0110c2afc6bae4ad4f8c07a215f846f2141f6482006f0
7cbaa59e877d4db2198e92e7e97a342f571080b2af35ae6b2867d7c83126d457
3f1c713dc1a33ad3459e60b705c4ca7c23cd2a8232c25ca5272b6461ed2fdda0
c63336ca3c30e443b7267cff4cc30ea33d1d251ea4f7813fc3c8f355e993da50
6ac2e8e147ffa0769551ae1f01fc926131a01b30ff27edbc83ec93d6c713bf61
2a272fb68eb5645a94cb66da1f7c934358c8b341b414f15b320f451839ba84e8
538ec560bbe65fb40f6f2086cd539cb23427e6121a810740e62c1667a95141b
response w

1a492077d4d19cfc4dd6292557c06c3ad8ef57b0354f369f2552daf40c2ea97b
921264928d05c0e7a175cd712bf375691edc65b1e7134a75ed85016b0a81af25
7d3a170701c6b99c8501f4ebfc68b85be9a00627df7a0409776a0da08a1d2816
bd4260090d2fc8e0443c325c6094a6bd388675796718557a438208356dc6eae5
b3345b85b53ecc5ca0ed4507f34e38a52e750734be226681a7a0bce2c5d1a52d
ca8b2025ead7258774bb108d963818807ed33069dc39b1f95f5b7e9656cccd0c
9fd3e2e9ef999b0a384fc3c4074a511284e2667e922e7acba9d294d1b1f795e3
cfea2f9450657848174dd5a2c5ae5ae15c95307ff54d89b46b9e14224d542a65
Fiat-Shamir context

892760ad86b4dee4a45e9a0ccfa777556848f96217ef4813ac3f53a41de3f3ed

withheld proof term: 1435 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.ctorElimType definition, Crypto.Merkle:0

```
{H : Type u_2} →  
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} →  
    Nat → Sort (max 1 (imax (u_2 + 1) u) (imax (u_2 + 1) (u_2 + 1) u))
```

SHA-256 of the statement above

9731be1b4af2660ecb5a85bebe6a1c3f1cf522840016ef5de1597380b0405cc9

SHA-256 of the kernel type

19769d5b289058c641de028e411e5c2acce387117718e5b4e6997c65a110ca4
commitment C

b0d74745282b680df002324a19c109614b7cee1e1a76d06616ee968e6c873ffb
2f8d85aca5b71fa4fb5fff9069e40f89537103a19d4373220bc62e4e6a4b42f9
4df05897e3f8a82d517f23e6b372ae6bfaef4765703a366f3543011b52bbe329
075da5b7abf0d56ae63c0a82b62036d606b65d594f7f9e6fe40ebdb8cb3fb97f
76e0867b73b0f1ecd3a749f10862b66682693452b88cdcb08f566d965c6f1bc4
abf054ce6d03e171d474f5515c5932f0f0dc51d6366e01bfe591dc2738f261fe
f0daa964ea61aa8ff40fe7ff04a991a5cd2dea56d521c75f502e1e38611dcfa2
b318bb086489203e5f346305610db2d6e24c173f87ac0699455216fc886518b8
first message t

c7fc75ad35faa52382c53c70b5a7207be399ec484edb2986cb4627c28ef078fe
c72f02b3b7500188b7002dea68bf523871f9af58113a9b1d5124a06219422d3
5423dace669833200b032015dd3c3bd86f8a67f75a5a4ff210bba63a2afa3bb7
3d834570b8bf1f1d05bb0002abb1cd42d2e426ea3d69624cf0643531868d6246
c3a441fc70675c4be24139c550dbd37a707b25b3279b95c7bad640c9ee121f29
d6f690f5cb64ce0e1841f4353eda2a1170e4882bbe2bbc526226137a0959b3fe
5db84ded0ee3e44f77667d20a949d5ce331015af47e632be906bd1ea6cdf14b3
a0b001eae5bcf95f406d5b6ef16bc447948db2db53445964f8634aac63e1cf69
response z

129075262c366c4c62f8d47b9c2b709aed4c478e7d4be7f3b1b593af4bfcc3dd
23907bbb5a9d29e1f61bbe6563e6cbbdf3397d7df6168a784b9c71e6fe29baf4
fe23e6b9bce002513d78b550277877a8bd0056780f83fec42de6e27fa35a7b75
5299d863877e49be28e33d6e9fcf9b694e7b2016ca45cd6569301a96b75a186b
bae2a6ca929c1a332ca3ad881b58843a2a81ddb94037cb0cd77c0cafbf2570ac
df1e6b73c24bc58afb8e344bb79d8ec7874f65da21f746646d1a9d89b92672c1
b46233c025c4df9dc0af55009031265bf38270bc202d8d63438b2b31bd3ca348
b3dc8b2f6d25d82d7ff755dd9a40a90e608910b45e8048d5916302318920abc7
response w

1f98aa825627bd83e3644bbf091778b4519dc3f69f958150cbb5ac2121dd3a58
56d89b9999c2d81a979788db88c6c71cdf4752a75b4f0607c77fe5cea4f4a7f7
762cf7dab4700748138129f9da1f600bdfef4996eed52f897bfdbc54b19aa2cd
4b867622bcd3df337c20bb091938b2115f36e88fbb48db0ae4951a84e783f7b
2d6e4cf0f2076748055cc4ad84615d7b880cf0d049868c21eee1d88ced332f04
a6c23e5a14fc5df383ac510f8006900794ad435bc4422cfb757804071c0ba764
90d76979ba6a5228b8318823fc4b69a1f7e289cb61778753f115f6c54481d98f
1e1e8403b8a300e34af463b7a68dfd608cedf0744b41013ba536d374b412fdc1
Fiat-Shamir context

7e5eef252f39ceef7a4d76588c372155fa7c236705fbc74eea383841fcc45556
withheld proof term: 723 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.ctorIdx definition, Crypto.Merkle:0

$\{H : \text{Type } u_2\} \rightarrow \text{ZKP.Merkle.MTree}\{u_2\} H \rightarrow \text{Nat}$

SHA-256 of the statement above

cf43034e267864631bbe5c58d7f08897e34194bb10fc492dd325c247b53ca057

SHA-256 of the kernel type

5e3de7bf381b6113c38f474eede0336cda766ba7946edf7959c3d1d20eaa58de
commitment C

f7fbe286bf64550befc0674d45ed03e3bda9090da6df6429e4aba8dc999df237
eee34e14a5c227a54d17ef5625232c366dc21e8e97a16aefbbbc35911b33640d
af5352aab984b48247b70458e9d162df246a9a3f520daf747c55bc1f090ddb05
4b0176412fddb72f0afca83995ee9014a7e69a1afde503d4593f0188ee2fcdf8
d41e4e2a6137d753ac5857d733c2fcfc0186c0186945f9fecb51884fdc94d201
75479c5ec4adcaf7fe2e9322dcc3004192ef4886b2389481764402058fc5bdf5
afca8736228d1c892769695d3e87f104cd370f0ea864ad7ee4a4d91c0c12b230
13d42114eb6f091e4813f85ae545a7541e1072c06bfe34a4b372aab1ea757ef7
first message t

409e206fc08884e100b1121b6b9005831260ef4a2ce63c613970978b176ad22b
bdfcd4dac0caf8746eb78480a50051f4e665cc5687495b686e7761fe818168e9
3881b61831192404378e85d32cbb979faa992f481fa24e109100eff599157649
d044b3debfa02fa3fed8b9391a5b794026723db744d50ce835b1a8f8629b1bfa
d0c95e7fbf036d2131a00284c2aaa75ce5bbc521e6ba6aee4c1a5f54434603b1
7b69fb8f08dc928609e9b051c454b530ae3e293632a9dc95c78387624d1715d8
1f123c3154719d3091770de20b42d9eab5782561f4fe65afe3412e760c483d0d
7eba1263031151f5c4243ba944158c4937457490be5c94e172185074d5e831e2
response z

7f18fd1e3c809b06147b3c494573167cef3729891ebb310796ff31fead811a45
13b86f9dccb401bc31a6010009262f808285ad6d57383412aa8bd40011374ab
343c443322fb2ed27be3c45eefaf553e8ed0d8d2dc79965fd42868d862674af
559d27dc3586ea8b529233105013fddc63471e2812a7651588a3f38a34b8c4f98
2138f453b5945df58646d4aa31f7d67ed9557626612f2a1dd01a32d2a33dd1d9
fb3bec5aeafcac2433f48e0a7d5ccabe044febf1cf51cd07fab1f2029f6706b
80d95a72c46ae7c4b53f378787ccdf71a940b695713cef89bbdbb72d1a63b2f
9893bcc5a50290620396f6ccaa72de42f05dce41bda7a142c95e466e9eb4866a
response w

5cffd93b979a6435327aa3aeddc3beeb2c895cad533b58d3844b05e076933e49
ffd8aeba5d54bb3df9186e96f9fe79b7445ab7f4bb9745fd1667dbb22e7aee3e
945fa3afda1f2edb2c4b380b736cc98acae2cc26c8f8b61519a2838990978a84
1c9c83feaf7305e4084f1c7766e34e2a67cc3c65c3b1b45f274d87b8382807b8
c40ee439f9c251b940bfa75f07aa1a4f20e86af873247627172aed0aa7a23d8e
67017ef2be9747e2a773cff407b627489e40b985c5bb6efa56eb361ae85b5819
5baae53a552da98814fbddb098b3596322fb5a80097321b8242f20b2daec1746
10f140d6c89159202416d2ac95ec7da5df99ea31ca0f87c2103eb05c4c3f438d

Fiat-Shamir context

38c039e2175d29aeea12d558c7ae4c0c942f4ae7051767d50317b25e7381cff2

withheld proof term: 288 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.leaf.elim definition, Crypto.Merkle:24

```
{H : Type u_2} →
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} →
  (t : ZKP.Merkle.MTree.{u_2} H) →
    Eq.{1} (ZKP.Merkle.MTree.ctorIdx.{u_2} t) 0 →
      ((a : H) → motive (ZKP.Merkle.MTree.leaf.{u_2} a)) → motive t
```

SHA-256 of the statement above

ba8dd13bffa02eb8ac3b213d291171f6ed363f1cd878f77925c25d676f4c34147

SHA-256 of the kernel type

41bf86d60ca57a8aa7c87b3ea651e377f6d6219fda395f786eb734699937a047

commitment C

ac2562f68b2f2e96c4485602425a422ec2990d6a5145189119a4a7196fc37750
d801b09cbe57dd25d62f42dbf01eb0a7d89b57ca6062921a2c9cae188a3fb6a7
3518c6fbc4f29ed01818346911222c9a26ccd40c215ae808a6888793256d543
3820bd8f9e3337c28d7ce2fb9779534ce2f381397b1317ddd8f48f57665e1654
c7f8adfc7e7b8b082a1c5eafbaaacd049907bc67e851a846bbebe985bf01e760
aa31e92c741b1686474d007958fb3fd9b6cc5202596ede9185b6038df015256d
e75497161c77937fd12f9bf37ec02c9faabf08b3299cc09bccd4700cc334ab3c
146f34591302adf7b50a9859800f89c5de53ebf284d925a9518b2a3ee52ecbba
first message t

c32fa8ac5ea3d605f114427db630844ffde9f601edbd26236a80858865c45e06
8f5c88ff8ffdb669874ab6fc7253e6b4a672bb9d028bfc0a9f34074a6e339bf6
b0827c796934af05422ca540f3a3595f34ee71fc88f2efd3a4317f314464954a
4943819c1599cd0e1f6641ca489eed3048d8598c09987797f46ca71718c0bc39
094b89bdab234d36df65334f65345da407b42f1fd075ecac23bbeb6ab3abd473
18fcd1f6f46d08c7056c553fced40406e8bdcc32759942c9459cb9052716a591
5b6c3a94e2e1989b8208302a81d5a1fe7732930ec237546a6270477349c60a28
bce19aec14f3762ac62fe6642ee8af2ae72199c0469eac6a014a66441a083834
response z

54bc4671f53e6e28be2eb698f8e853184eb31e97cec797221e99ad0848936417
7a7e68ed8cfff1c890afd680c99f55ab7dbc3c8d61b63ec36774eb638982358e
b4ad912707030b58177910e49497c48ff1d1c8b90e5e4349749aad75aad8c143
60174850201fbc718c6e53237d63f872ddbc713a7fcbf1814875d0a87755dd26
d9f717d4f12372ec6b6758e4c18d1d257ad40b73657ed54658c91b0abd2af9fa
1432e66e676fb3c785118dbdf4d73c44e36a70417a6858cdb195f8520f6aeff4
1c4be3e8a2441704f819b7f49c5aa2f0d0f5b478905f89e30bb58af7de20e40e
7813e65cb9f9f0ec440242a3d7b9a76100ab2dd9a6c752f9f7807475d0c357da
response w

4e702404691ad41c7055bdaf126fbbfbf85d8d520c464ff12e627a38078acc77
df94787b5a5dce1dc804bdbad7853f8d83f46cf43ee609f4a2bf81fe6742a22c
46793db8bd9a1f4f6f42fa345bab4cd5d9c0c0f1492b99f2416dce54985183a3
d84e870c90fcc26f14cd9eb341e88300a816cc0362996b791823ff5d300c7cbe
a940f647b120e3491bc8f793a32446f1ddb4a85f80ad3a929c2d26f0984ba5a1
c0113d6644dc8a1316f0624ae09e013db79e7fac9e683f28e30f62ccaf42bd1
958654708375e0389a87af2ca8f35aab4e3bc6cc5dfdd9e3845d9f7467174d6a
5ad9701254dfa559a2beabf6920592eb618d9fd21fcb246bf06b86acf9d366bb

Fiat-Shamir context

f2cb90781dbc21a001fe8f6c270041b2031eb4062fc222d2188d1a6aae399d71

withheld proof term: 565 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.leaf.inj theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_2\} \{a \ a_1 : H\},$
Eq. $\{u_2 + 1\}$ (ZKP.Merkle.MTree.leaf. $\{u_2\}$ a) (ZKP.Merkle.MTree.leaf. $\{u_2\}$ a $_1$) $\rightarrow$
Eq. $\{u_2 + 1\}$ a a $_1$

SHA-256 of the statement above

8a406cea483f328066d7ccafefb8ac9c8c93fa3afb3e9c21080111058d4e87bd

SHA-256 of the kernel type

770dbed1356cccc8b1a476e137292164b190c841b87058a76c662b34eece7fe2
commitment C

70804065d8797d6b172d4f736f3f3b3b7e0f65acf76f1a23e2bf3c67cf1d02bd
947434247acad707abc9ac16ebb021947abc6f81f962c68904227eafa44a4ee5
41fad923d6b9aff896567881561da8426f0bb5af5c84ec8edde432fa16f886ce
fd9ca3037dfc418a6ed4071fa67edaf8080bf15a87b9932bc54cc006a653dd75
82bc35c6efa3b3e767696036cf977bb9462bac56544f6800527633ba2dd02c35
41864a203c8be2312f154b243009163f63bdf4cf82dd106e77735ed547044367
848050ada4c37e8bf54ce4c78ee69f8b698b8d352f9ffe22079377db9d3948bc
ce90dec835faefd54f1ef79ec6d91c2b3dfc384ed69fdf50db4b125f8d4412cf

first message t

704124052a9219b1d8c5099afde69993beb6ff43a35ca100341d348ee3c19a53
ff06e66cbde85cc690adcb526913ccda8ae21e9536428a5305a003484c1eb5b4
52028756acc5004c16a1b0effd7202a51a2fe5ea8a9113110f7d3a69f1a88262
711adc88643ebe2b7103b8d2c95217183a99eda03d132c210ca57b8068523c2e
198c2e8d2ec50b48c3e9e57fbf14e9a3d47732ca34195143c0066eb9c2cde216
95757e8094084da95ac3630ecbe8de67db65fb64d3f1dc4969d5d6c8a33fee8f
20bbd9c79a0c545d052461667b1fe963752c12aec4b5cb74915f1e3319551156
65e387bf12ae4fac9c0438cde3cb939df6d41e9dc95337ebc4b426c385b41b90

response z

2fcfb539d3d4ca58ac4caabb83a7bc382e74c82875f4c17d310e41c4343272b9
2b01f9373803de926d2032ad4e8c8f915783072abeb5ed5247b0bf44ba766c42
00dddec3c02fee5aa3bf0a70593fd1d95faf4a457ef8e41c3237bdc8d7eb43d3
7f9785a637fff142ef9786cfc32b9bd168d366d95432772d9974055397ac3bd
6e96778ff2c94ce22422ad9b98390fb21f595b4a42de19e73b3f7d74525620a2
0a9105dc966c6891670271aa211734561883facd6b64d6d3bf26160799011edb
b876eb8fbae407f0c6fb1fc33fe94ff8576bd0b9e25a949de9159bc3386f8033
f437f6f38410bc1959b263dd042b872d780ef534a6663cb457c2e2d0888aecff

response w

67fa2077aa3de4bf936a745efacbf4940a7a8dab4accb296ea12cccf72083d3e
7af472678c0c12945e376b5ed570d9855b2a0c6b39492f2d4003cc8056c2761c
5a960e4f94d3d9bd044368cdfa6d49c3d4b781bf2ac38d9478d2327b447c8f5c
206059d9e3c456b9aa7df817a74c0ad331e93e1a4c95eb4be46a01d2765a61f7
452f30680ffe16fe2ebcf69bc9b6dc78361e46d1c3ce40ac4af4dd2d051e19f2
5ac1dc4de685ade8b0fb6e3841c87a0792fc72c7bcbca67a6444dce58eca70c
855d668a9c202af114ca69edc569a5724fe61101d8dcb2bd4509a59f08c67bf4
c6d0b7d0a09dad47828288d3add6642538f955d875b7a713b76c813ba55bbfc0

Fiat-Shamir context

e401d56cde9fd48104e0f518bd90316f9692e0272cd362cfb38d5f30d4d19b9e

withheld proof term: 812 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.leaf.injEq theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_2\} (a \ a_1 : H),$
Eq.{1} (Eq.{u_2 + 1} (ZKP.Merkle.MTree.leaf.{u_2} a) (ZKP.Merkle.MTree.leaf.{u_2} a_1))
(Eq.{u_2 + 1} a a_1)

SHA-256 of the statement above

5c401bf73a2e8267b96a3e0e1e1bbc566cd6eb673e9a8969cd4e4df16c43e45f

SHA-256 of the kernel type

7dfee0561308fbc6c157bd6ffbed1f95d7049f4e2535b2bae35b87e2a1d95b9c
commitment C

dd31b235705bf2986bb49a8eb5888fe0bf3194fc1f5c01a6a7d6325f47cdc950
0cd9d0e00f02ba4ac175721b207e2f7b985c36824c6a32243c65c59bdd50bc56
fd2902f211feb532b6f152bdc0302cfba4d36b957021f46c1272ab11a9b96ab9
23544cf42bf1b690aece414b7ce3145ae15ca9a080a241a4b8360c2fcc22ce
87291ab51f1026fda61d3eea8b3f0a7735a35561f0d98c2caca773c58dc2246
a69a82056f9e97c5092932656d236158e138fc5e4189e4f2050cde8ef53d9474
a778344d4800f9608a0ffe200e8eaa7c4f8e807597ea7bd1baa49343855001e
1a423371fadcd16e6da031af0718236f9265a31a627fe551841fabe460b5a12d8
first message t

aeaf431d06e44d88c22c2a6936d54c82a41f203e69c43e04cfec7e2539e85c58
c09a8c978e23fbaa7533112928f54b4bf23e3de8f1eab0d451ef85515c83b61e
2c0a56b8a9d5e2f521485846113fc500902ac2adbc3e014e863119e5d8d34972
1a7d4844527dbe15f21a8d9a6b7830b8017a6ad540b9ad2a0a8ab31ff430d982
b3dcc5ec18280814ad46f89fec249d869be70dbf8a8471fd3327b0c5d0c65dcc
41a4aa20e229e53a7cb4334d5819fe21f25d3fb6a7a357fc9ea479f804ee6d29
729d1e78aaf8a68d902f25d5be93053f12379fc521573c4b25e8246ba1c19630
1939047e4ce7657f7366ed8af5f601f5b83622a0f04a803af763f62e80912e02
response z

49cf2d212e15b5d2116d0cf281787550baa32ff5909bd7469a48a7f4fe2736d3
3ec89db311aa0f20d35d532b533818e5183dd8ab35578ebdd35a005117adffd0
fd4c3599a7171a372c17dc13efe7e296e36c3faa68413e4fc5809cad6de277ae
f3225f72802890b15994788dae9dd377628b925e3edb6d8fb14fa75f70a715b1
0ed4b987bac6e6af2849336c8beeeaf49651ce9791d659f65e8d8b399d560a65
04d569a5d24e31c8aa7ac3c08eae44f1718c94774df3b610eede80eaa903dea
e772f7ddda41b640bf38860f172efb962729143d2c357f42dee12dcdf0a43a64
c4cf8f3a4ca5579e3844de7cb867b28e40bfc06abe24a565ca513d21e4babb3
response w

3f53ccdfc9bc8d3c436b81f6a63f9d2a2b0336ce3859f459fae2ff137a1c814f
f76d2f4d14b2b216d4b841088c701feb2c2b545b332a231f212ee6c138bd64dc
9e612a3fcd8cc0cb29dfa8d346a28bfe25b542f19ca3602d4819f4d2a093f1a2
078a04baf162a6debd74ab5760e51c7c4c493b312e8d93eab396a6967cabaa05
049886e62ccb3632c9df3b113a12d60fb9507da480c76a5a71671b1d196d9720
615de4bfb79adfb17272577df20ec14e9004a9ffa3e0c6d8cf58391dd90d1022
cfa133768681e4d88c1034521f74166105e41de324472c17cb610947a7b531ab
f105e8f0d025533af45042a54e089b235e2498a19a3eed513cabe9c34491b346

Fiat-Shamir context

dbcd8805267c9e508a8035e8e8d602457bd646a1836b67e8a50f934f682156b9

withheld proof term: 1659 bytes • axioms: propext • kernel replay: yes

ZKP.Merkle.MTree.leaf.noConfusion definition, Crypto.Merkle:24

```
{H : Type u_2} →
{P : Sort u} →
{a a' : H} →
Eq.{u_2 + 1} (ZKP.Merkle.MTree.leaf.{u_2} a) (ZKP.Merkle.MTree.leaf.{u_2} a') →
(HEq.{u_2 + 1} a a' → P) → P
```

SHA-256 of the statement above

0df732f74fb7faec8c58c22d6b634f8da97cc9d09fec713fdc82caf74eec8c8d

SHA-256 of the kernel type

b38e9bf102226b5ebd0b1819f70c9c55ed4a53c27d75451c0b5de5968b4132b

commitment C

c105ca08faaf64f5f7cc8c7c5c744fd15dc64a01148c4c7575561510842ee666

64fdbc6ab8e3f85480fe33fcca30f5babe07ccab20805e859e36c286beb82

d5fc7c0d5d78fd3197ede6e8d82a77c059ba686e33fee85912e07deaea05547d

d320384f09cda01902c8f4aabe2c104d9bc7e9754eed69381f417a532e415abd

1c23f69944cfe8e8ed01d3afc9cfe82090a0271dea6147aadff6aa148b39f7c8

1c5b296900b8e16fc04518ee47580f22f5be82198ad008a59c2f384e08bc0cec

02f0fb65bee23e02d10d3a52c5e1d3ee6312bfc706f59078a8f5ec86fd549809

41a65975b53bfdad64ccf230fa06d6635be14db5460021c7591f285b1d842429e

first message t

6aac9069ef6cb33e5c9bb94fadcf62361e42e6b04b230d4f1d3b9b0ef5d7e657

8186c2387023a95a8b9d4643a6410b0fc54d31ed56f3bb5dc794501650b12616

3e34a511b99aabd562178b79f80b058560476801a78927c847fb6c02d398791c

c56a34a49f81e1aacd07b7e87fd7640c46ec3f20c6b726257e9afb9ee39f45dd

f2ea6256aeef65911d276f99858473729936c537a334ded4e4949447adf1e5eb

aef80907c7b0e7bc81ba3377ca2aaaf58dccc830a21478cdf27b72bd0610730

6bb3b01472a767f8658a259951f2d3a313be2da36f4f2a0c172ce52a14752b3d

c0a706223e76c15a544862ebdaf461c9b1caa9f008f381bfd099e5b5baf6d88

response z

1a309284dff9eff59ef94632e06a189a5098ca0ba48d77e254ab55b422928f2e

0941d38988ba0030bbe4197556e3fde42b825f331364cb11324894ca5623c534

7f02545528c9f31beec2f7484d244fca7376b1a32f7f6c46ca6096f4e5b497cc

2534756e20175c3e34b21b65b7d8d2758a839809045e9010e4fe953a6516ce90

ead97f55c879ff0b497dcc087fee5950ecd2ccbeff512b49c09b504956c1b871

912d22652b50c7d05fe516a23cfa678db979c35203a234b3aff2bdde34f6292d

87d90342f727acc9d4cd544f6ccbdf5c1c90f30902f392102d33f7011fbc3e69

6c733a08102215ad10016e3f82f2910aea90b2e6aa91f15b4ae4289bb91417b2

response w

7dd05123f89541b42ea2b446940bdc951488a5d7a42268eb865ba330dce4c9c3

03aa8565ffdde0963b9034cb591671e81b16122806cde7f1682c5584fc048f96

3f97b6a13f6be3bed57459188c0ff66cf31268486bf7a9995f8ee8728b82957f

2c4378e6b3ee9e1ee2609db2671d305a315a7412d910c311db0901f43be7c614

a17441000b6c672ed66b7d5bb4684ca47ae30edc1e150ae2d85416b3af532d34

05e3b9415d253ba178c8cf53a91ee8a23754736876239f128783b3831ad90c64

378736c841a3b5f17b377f1bb348a359d86fe9e6c90dcb6a0e9d6a98712eb0b3

c3f2e2fa4ce643e12a3a8f85142772b8ab294da77317a8ca35f58152e48e4591

Fiat-Shamir context

dcc011bae7344e5b478c0dab0583629381fa06fb062c8ed20e6398bd872db2b8

withheld proof term: 698 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.leaf.sizeOf_spec theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_2\} [\text{inst} : \text{SizeOf}. \{u_2 + 1\} H] (a : H),$
Eq.{1} (SizeOf.sizeOf.{u_2 + 1} (ZKP.Merkle.MTree.leaf.{u_2} a))
(HAdd.hAdd.{0, 0, 0} 1 (SizeOf.sizeOf.{u_2 + 1} a))

SHA-256 of the statement above

8c4c119f8bc44acc68ac94ae219d0cc45c07df0c62af5e5ccf75296df0478c68

SHA-256 of the kernel type

156287ae2acd137e7b074fc9386d756b1fc6247bd4e259266d7cef19a362b69d
commitment C

2dbda1f57524351ce5c9f6fe181f42b7cf2a7e8407adab9648f7b2e179f5fb3
a14739a199c38d2c5196d28616dc2eafd90eec772b2c57fdb7d996075ae46a1e
8848cfb9169ddebdfffa3b6e8c8786eb935f7e9a2c3687fb62aa2dd56d20b397
2a781403b8e6fe899706e0949e0e58b8ed07ae691494a46d4a7272eb5cd7a4a0
35b41829ee05942082282242e374ee256763d2e37d994763c47ba510eddbcf4f5
d0331480cd6a5ac0dc6f80f5469f4101b117c48d352c8636cbd801a1ea7b1358
be6af60f4571070ff0d361e997b45732147a828067f073f86db2f568d5c9f41a
4f2f5aa31d099684dae771d4c8705bb3e6adb4166b877a6310b84a43cd6b21e1
first message t

3f46bc22a64d0da832ca2e7b8df8fa370ff3c5d0c92bcb3cb9c9a1831fd6b9b1
bd2e8cc01b0534ffdd1d34054e43271615938c386cac58dda82e929f70ab43465
090396ae422bbeed9f91328c979f40e012732f56a70b1539d9df603e6d3a6155
887bb0b6657475d1cb507cf1a7b56362e7d591d25f0dec379e1bf0db56f8a0c2
291f8fcadb2b5922c99832ba501617fb8e82238625fd21217b566441cb6f55e4
d74df1cccd1b389eaea18c0c51d6da4957a314ef48dcd9dac50f99600eaf2c9
6b96d05c8a3130f505134730e59641a2db804ad0134b01961084b02b5799b592
7848db6e0064648a86a4a274894806312740e3629687da2b10073204cbac502b
response z

35c0482800b6223539d646cf052934341901fdc25276e5318f00e578bde862a1
d919a92caf650f7982c1074516bdf92e906b75aad2002c5769354a72e4063cfb
186d301fbfc6ee1647e71df45e3f45d4b1ef9dabd02acce13b0ae7d27a1830e
8872c784391120abdc23fec65f01d22e922913c283da2e7eeaf0068064168a58
8c7a36bc8661cca7967f5cdcb322c3e242e434f46318d4c074be60d3b5ecc4c0
7967faa7ff20d625089e4688c3e42de065704961f23ab8cd4b0eb0817d6ddd25
570264b656983ee6d0925a85a2ae8003bb9c56696a2022a7cd5e84e46963d596
7b65590593120ae91fd125afb19db1a8781d44ac677349f400749da8c764a165
response w

3931fb8b909eb67bbdb1fc531bf8283f731b2e404845c9e507a28952f1be665e
0dad46f9388733fddfa59e432400ef03700c99ac4b0dcb787410d5c1a91741f4
2b98c8ce894ae40e9eaceb62c66e4c5858074e34578e3f38a2d90cf005c81f9f
48cba1c793c229d135f99713f9090be95fd492c641b6bd74c38d1c66fefdf6b
a9967057f6301ff241251d047ca77c649552c5b3a62d8e07d991b1816c49158c
276d3dea7fd2ac68eaa417fa40182215c6c0a95c06b0c7eb5ffd7691f97e5b3a
9a1bc3fc6b5889f296ce2e00c40ecad16cd4729903a85e5921d53d5122ba6435
2d78950ef6a7305c726373189286f387fa6ee229f56ac0641bc8c67b23049e02

Fiat-Shamir context

478637b09d6464cc8b78f2d44e8d93bc02358178bfb57c600f96ffa58bf697b8

withheld proof term: 375 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.noConfusion definition, Crypto.Merkle:22

```
{P : Sort u} →
  {H : Type u_2} →
    {t : ZKP.Merkle.MTree.{u_2} H} →
      {H' : Type u_2} →
        {t' : ZKP.Merkle.MTree.{u_2} H'} →
          Eq.{u_2 + 2} H H' → HEq.{u_2 + 1} t t' → ZKP.Merkle.MTree.noConfusionType.{u,
            u_2} P t t'
```

SHA-256 of the statement above

be27991dcdcf62aab9fe96963ae277dad19e836040c80991c2ff0bfb77beb028f

SHA-256 of the kernel type

cdcebd0e6c44b2eb83b3a1e0b86f30edf163a278ebed5f851d704d8a58fe55f4

commitment C

815435d6cb10b8a8c97a1581b43e48e49f27ad46c71e863ceacef92b8d928ea9

53a3f1bbc06d66feab4b5d710ea27934fd2d8a5193a9506a9f693272aaedaec4

ef8955b7c64cb4ab3baac1e0abe9a7ae469127efe9fbf885afd892aaf3d64951

a6b83823bb1c988b0cb293bc7feef7504d62893e4a7bf15ec1985049f94e9699

0a556b1f1ae90c3f284d6b72944555aab882daa49decf475efedb600dfb735ef

7b2642ee41b61618cbfd60985b6d3dcccfe245a432cec5afafc5791fb0e715f4c

82ef783a40f9543f5be5b80b2a1dbdf6cefbb8f20582a19e2770d9a38b7c81476

411466688fef0a67d929416cdfd8f52c7e7b19b402e21abddb561fdf006bd807

first message t

9841c56d86bf3cd962534d025a40c88e7e20ebbe01337af53f9ed755e93ade14

243f7b08ff1231078f4ff517b1e2da4c11a4d1bf351bc272f3f783c4c20f757b

280fe3c9aebacd85725d6daa313028e957f885ca6603d06e5bdf7c5332a2361

009a9a866eb6f7518a63be0e4b070e9c05b6e1aaedec54a4c472950ef2dafa77

ddb00a9893f79b4cb4640367264b4f4559553a12f58f96415b3726c83c3f50f4

d9c08f498ab3d8105f244df507f476920a344394d0b498c9e3fc19bc65cc770d

a7dd8a7922aa09361dde42df6a5b2d1f8ed21d6169d56da6dd9afd2df399085f

84a5ac2b0e8f756c376979be90fb11a2725f475b9b074de25b5fefcb3fe21357

response z

5d17b4a09eb8abf79d3c38f40905ca07a6f2038316168bc94443739f0a9980a6

f4bb7ca03892dc07132ef82749e2dc832b192d1e2c0589c8ebc7e9fe4ae1c00f

626c639ac245050fb6907d18fdbae5e8358773595c798b62bd16047e07b32e60d

26b10ee3fac7c2bd0540b00e4ea13e2a7c2fac0cdcc1a33951fc8252c6cb3482

ebabd56a27c5865ecb0faafbc97b44bafb3814405c2b679fc73c6116459f0688

9e9b62a112759c745f34a15845c10e645ed75c9471fec8267b94579e295fe561

22627bf045009bfbff711baf8255555e205122966562c161421af69863f0483b

1e41a5711195558df4e50c806d3b8a120645c8d7b0d4bd26adf2a5e0df4c6243

response w

8c71ffa3baa7fc697e1a4b89c0ec1a6742aaff59cb2fd3161198400edf85322b

13d4e094d144ae4b3a4f5a73bcfc8ed4446c84a0303c4803e4eb7401ab3727c2

8993d05b3e51a74fe53041cd4d69d19cdcbdeb725d3e03141500f0644c1047a

f9bd0193e5f050ac28442cc59db56c30cd38558e3eba49606bd4ecb0204ae706

894a87e56d2ac81c33331e26130155a1a38bfd17d8dccb07a9fc7e67f9ecd333

7fe3d71bb2615eeb060157773690448f008855b6198bcd2464b09e91252f519c

9385dae8a5c52143f43278086c828494b7f6f230dd40c4367f806b9bf82ad552

530a909bb7da9a106bf24b707b14736962eb7b5f676d7623508cd4707c85d84

Fiat-Shamir context

5284140071558bab8cde6b8e7be6116431944ebe3a9b3b4364ec32c7583950d

withheld proof term: 1662 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.noConfusionType definition, Crypto.Merkle:0

Sort u →

{H : Type u_2} → ZKP.Merkle.MTree.{u_2} H → {H' : Type u_2} → ZKP.Merkle.MTree.{u_2} H' →
Sort u

SHA-256 of the statement above

9a6b2aa100243fc172bb803c08a90aed9ec80b97c6d3abffa8fbfec93067b9d2

SHA-256 of the kernel type

334e66a459a01bd5c04bea3402f0524e1fb2a8e24b1a0a4b312d9189a75151f4

commitment C

6b7c540ba315f43d38f82d8b35e1fdb5905fcf7875fdc175fb54e43248b8ef64

689f5e6b2e6084fafdc983859e0006d5f6094845470ce579af022d27d4f2e919

2541c0c7b94a2397506eac6e19a513e2f83ec4662e96c0acd6039dabe936795b

f01b05b814f72b47678473ffae4e933ae0ebdf39f2ec5f91ce2a22843a291be5

43ba25f2fb7c07c2df6d05d3d4d23028af306abd8778124c9ccb3aab41fc502f

192121ddbdf9d2b9d8056d7516987197eef1ed2a159f06467f19d4f9b8145c1c3

191db6edd52630df14e6a158842a9253fd89b67798d30cde96ac4cc5a75120ef

94aefb5056cf9eb7cc55288e4a907f7c122c6a62da895086e52c835d70795f8a

first message t

e6bc1c8aeacda8103bbc8a31e030c3da91466701c8b6492a947285b3b13e899b

cb5682e6d0f31dc8706bf4690475813b7e5df275aeb0db13a8fe61deb692876c

c998354b8051fbcdba561b9bb3a1d2547746dfa16ad4519f722dc342e3272084

4908c776c0d0e8aab10f37c477f04b797b7af50469ec487461951968ce061d13

ff934e5d051fd0ba8d1dcd6372f1311aae10fe93f6be0e6fb3b4c8b0516e3b10

40139a44abd35bec5d02689b4f4f57c4d8879487201ac6135cc25fef4171e0ec

704330d81decab31ab0a9cabdd12ebd0066bb373a1aec8e1e16d535bb54bf305

d9e7f837d3e46d63e1d9547a0d80dd80ddf95eb049593b6be099860977acbef

response z

795b6a779f63a1642d9631a3b80213852f8d68fcbb02c93f1b44551007595a54

def56995fa8cbbd89b5f62f1a62412badf917c9a5facbd37932eaab5c9b25006

4212f4c94739014521171976343d82bb11f57de2a12f8f5dd6d5b034b6a90a13

d545bee5278b0b988612e8dfe52972fa7f632a4801ee0f26f0c56311785e793a

7553b71412c7daf3ecb12a80c6eb591a5b9598c5704bf7226ac958593a9ceaf5

6d976d5754964dd8b6646ea7a16dd25cf16fd525c3f14c9eef2d5aa1b8cde307

92dec591f2f1b2f4509f973c66fa41da9ef388c79feba1cab0217787e1eddd84

6b95e9891d613c9de075d0bdb5966b0ad32ad80e06e97412b14c92bd90e4f661

response w

5a7de6e9f4244e760874e6768bf6688611ccb349f02b67855d2560ee8f1e57bf

d26427cccb6b927e858dd35623bf45581e88aca4d8035412f3e20e0de5bd3f72

02f569030d832032d405aef4e00c853f1330d58d1c89ee76078636eef7693a5c

de3f73187a11e28ea992c612dd3d7d46891e51e4f8265e163ee94ecf844954b9

2e6a0b34253f766923af2ab30fad06bfa76719888561be3a88ea2ebaf24a7df1

5759df345215378b2db641276124e437457d10375e3d38b5431a8b8afaae60c7

50f2f02a164e7f0b21fd1eb2eb287120177bf6c4e808f7523411dd5094e1c1db

8e7fed3d331d650c673033c4cc45ae69ffb9c5643a23ab31371dd3c71945a34c

Fiat-Shamir context

0e67d61830b59a46ebe46d9fa2a6268a3ec12bbea16ab202579d82de3e3651ef

withheld proof term: 1 197 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.node.elim definition, Crypto.Merkle:25

```
{H : Type u_2} →
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} →
  (t : ZKP.Merkle.MTree.{u_2} H) →
    Eq.{1} (ZKP.Merkle.MTree.ctorIdx.{u_2} t) 1 →
      ((a a_1 : ZKP.Merkle.MTree.{u_2} H) → motive (ZKP.Merkle.MTree.node.{u_2} a a_1)) →
        motive t
```

SHA-256 of the statement above

5c815c2cff196021bfcc977dde5342d2913f3e880de9695332b5adb7ffca48ca

SHA-256 of the kernel type

e5b68cb8ab110420d5830fc04beb207976f0f3c6fbe34e91158f15a256c5b1ac
commitment C

f099cd907e8b8264f14c467a8a874df1df92661a3f7c05a57d4620220c0b083c
bd849c35b20b3830743bdf98269da8f99045ced4e5ff0290acdea6b7572ebcd7
71eb6238ce4a03bdc09ef08c566948cd2e51f948f095cb7c0e7601ccc5772be0
22c4232e72c7b87226ec0e6c1900c40547b61cf1055273a5d4de0cc02d546eec
5e789dceafa67b9af42647d094dcc2b3a93554ff5f11c5cf2be872aa1c0f1a66
02c13a03991c2dec69ecb2b13c78b59c66c1382e876e59368f7178a9fe60d17f
ddb3e0a7b73f6f1344171b6ddd325862b6286cff8f41940dd1486e1f7f771122
702304cdb9deb2a9369f31519273eb78295cc052c062875a16322a51841bb243
first message t

19995959643d586b2e06c90eb4239f01ec8a24c8f92dfc46192894054b2bf662
f8848814bce9580ec3c161c27c5b3b325997790de0b2b82d9faebd981e292f84
a82f93f98b0360b11276d1a031d41cb8b05fb8d63d1aaa2971e900d75eb14d71
376947362f235fbb1190e77e98317032a5b4eda6a3ed96ab34a564a02ef7f805
1057a8eb618a2939c0cdc432f02a7e2e2d921fd78e6680256a8a3f661a02a564
eea1698bf58a15eb6c6f2b754d1d2d703fd2d72d1f549b93dd1f809117bc024c
8a5a5a55a4eb041e44a0c835bdf6d94cc59405321341d46a3363d92b8e1f648f
1ec7b9dc272ca19a79eb72f88dc520fc927ffc6c1a37ca18ea814bba575f6d48
response z

4c6329b24dd82056fd95709fcacf0861ae60b7b165c50bb6ed39ad625a4063bb4
15267b5ee290a7ba59c4775d383cddbc33ba8c3f314a86d23de73a5ad6a021e3
21c5b4ef95427529a130f5f55c576f27499a81d8fafc2112975f58ef9a82e81e
93afc136d6e39bf1055d214f68a2d3423e5eb1112f9bd27c8ef25f4a573200ee
0dcd71c86285b132f68ec1b5cb3f4adb1f9d9cc102ed8c2e83ef29fe6a5f912c
040684a6b206a20216d7e7b47ca6922b407050973bc91b4cbafd44cc8fbd6421
f238d3868c0f5e9360d5c621fec783f156e751fa60c5e11abf06c0007bd231aa
b92c6ef1c03616c9e2388bebed09e7016e35e90cdc327ef625ed738c0c696d5e
response w

4e79b0391070c8fc509fd0b7138707327586340dbcf1be5ea97a4c9b7ab888bd
b420d398a11136a2ed62c2b8a624c812efc9e70bffa20f89f4c1d1158849bf8f
9009f39ad4f18a0e5a786cd305684f96fe444f737aa6aefec9657f8b6b263690
7418eb2335110400ea8ed482f8e033840ff7beeb516751d10f1a7b11ce5c76b2
eff12dcc7b0fc271f1a70b1ac2f4252f5efb5b966bb7a102042ccd46920e46dd
4fe86f414481752f4148c0877ae6f27d6aa350a6c3f18f92172a9a9f8e402d02
6aaf7586ae146eb803cfdc481a9c68f394784cb76f40984581c59a034cd3f5db
2693301a4c82548650e36fe59d981d5f3f080ac2d8ec4dfd99e5cb29634c8359

Fiat-Shamir context

9094f7a0127fb31438d3c71e79f88479e3b48d74113ef7064979f5ab9d8102d0

withheld proof term: 776 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.node.inj theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_2\} \{a \ a_1 \ a_2 \ a_3 : \text{ZKP.Merkle.MTree}\{u_2\} H\},$
 $\text{Eq}\{u_2 + 1\} (\text{ZKP.Merkle.MTree.node}\{u_2\} \ a \ a_1) (\text{ZKP.Merkle.MTree.node}\{u_2\} \ a_2 \ a_3) \rightarrow$
 $\text{And} (\text{Eq}\{u_2 + 1\} \ a \ a_2) (\text{Eq}\{u_2 + 1\} \ a_1 \ a_3)$

SHA-256 of the statement above

3259885dc7a15112ccd24ba598ad066bb02d33ee4264e45f0782e128ae00286

SHA-256 of the kernel type

40b967c6f841e6df20bd026b21a8954b180efc2bc22002075d9bf42a6a86d4f4

commitment C

c4d68ad2db4c783096bbaddd1619be62d349265f8941b5796c785a38f391d6b5

8801aab3497ce38e429ef613cf5ad71f2fdb2b4c27a92c7fcabc7bd1e9f2c2231

e0a939f8f56c2223f61002df07512507a4fae7a8856a81699aa1ca3245ee996d

033ab68a4ecc9122a287b9645e4114ecf4c60408b3cd2012cd38d05b8e935336

53701bad553d5f9308c2bab9d5213c9d87b7772b6a02e5e7ffd90c12682ab680

5dc71f1ea019ef638ae7031accff566b77a5c0194b46ce30e7f7b752b7c1dffe

077293e68ec45502d904d46cfeaa9004873cd7eea958ccc2ed95a323cf9b61f8

fc3e3a7dd33655b0bac348835e79be5380f8842fdab8929dc2ccf024a8c203f

first message t

ddc135badf3207ef759d893bb13456c2996d06cce5a44ae6d2363934050b791b

f1d1dd41d09819f0309da746d33f8e77c6888ddd2d939b0f4881044f1a5c8747

a2bb280805cb3020ce87b29df6c7540ac1e4b859eaa1aec29f33fd3880ea6e51

cc719fee367a112965874b53603725286d50700471522657db529f1a942347be

d3ca99ed915f476f6ca2868d11418f77149114943df1a25df96ad0cb3982d5b9

ee8d8ccf04e99d4c76310c5a6c69a9894abdeee19c08e748b1d23f7097cf227e

12c02caf53173f15f7a47c208ba601b0cbccc968ea9aefb614d547647049c4cd

276a52af7f3ac2207e26414cfade911bb341ca3d0c1353138cb110a1b14b7201

response z

f9265dcff53ebfc1f06f326fb1708fa71a6537163de25d8df7d2e13a6df25503

d6066262e2624bae0efc26cb7cec823ca1c9c2b8261b08fe4a51894785f5dad6

f6bda464d3d02c93d0c1663437a7bbc08f6ae67c9e45cdab779a6ce86e73d455

0851ce2f484451fd8740daf592b2f3d55244a38a4193d3078986371ee4bd437b

ea21207f8f19ba07e5a6d32a7ece44280d4ab03a58098834e45f31c8ccf5bd74

aba1f26c33dd8fe1c2d92d0d67300001adbee2eeafe90b518d7acd102ee21763

69c85160898c5f2c90c1c6c7fdf8c963e191c87b93811d8559797083e6703b28

4e48c0222d136c87079de9190898dee4bd476d3cb98cc3605ef2ea66e9b4057

response w

507d41bdd0a9f3430a3a86924905307d412527f2d5436014c3c275bfd22dff55

5c610a6151b95ed08c126fca19bf9135260b8aa30a2a71aaf9fcdff68cb4c8a3d

1fc03d1e0676ad73d2db872ee11cc1a761411a2ec2edafcb64104e589cf6096e

3b7b8805bf2fe76590da996dbe8d53a8f05655c15b175919a87f2df92205357f

cb247a72f04425109d9ac01e2b85d6c558c421fa001d590ebf713f80eec11ec1

ac7d1cdb1ba0c02828288b39f0bee6ee255055d86d67c99bd50f71f989c0f467

3be68c29bcc2f995720378a9226991a7eccb3cf84191a9b547f34c092779e3c8

313e0387bf1961ef1bb99ec7be5ff046e13b05d01bc498d6a5525c294c4d4226

Fiat-Shamir context

0f7f90a3658692b413afd309fb449c6880c0046907b31c93c2501478c30f6e17

withheld proof term: 1 820 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.node.injEq theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_2\} (a \ a_1 \ a_2 \ a_3 : \text{ZKP.Merkle.MTree}\{u_2\} \ H),$
Eq.{1} (Eq.{u_2 + 1} (ZKP.Merkle.MTree.node.{u_2} a a_1) (ZKP.Merkle.MTree.node.{u_2} a_2 a_3))
(And (Eq.{u_2 + 1} a a_2) (Eq.{u_2 + 1} a_1 a_3))

SHA-256 of the statement above

329ef4eef2a7d9e96a240b73fc9d939450a41ff449ea62f54987645124000468

SHA-256 of the kernel type

65f6fb98171e437edf8f9dcb3f3366da1b0969aa64eaaca2d25f072bd4b57b3

commitment C

c8e2ee2cb02c6cf72a34692b6902881aa6110aaaaef54552256c73500c8b7eec
c8a5f00c1a872882a0b5a913ef7fc9590f497ed7da08b14b88ecfa8454e9e6bf
bdd1f2785965c9b3c4c00743eabf54fb9cf0c2d182d5a5f961fd95e27e4887d8
c3c8eed071bcdcd7ea60a48f8f2333bea8579d48dd870419a8ee3157d16c3614
f5346beb1103f583e915a4364f3fe73f29ecf207fa1ba08fe14b07850ad71cae
524d8f1ce5266f1809cc9dcbdda5f038cc21622dc3f08b2c4f843d26a00c3574
a8d4c65f7d5e7d9fc442a208308f678bac0ed47c1176615bb55bc24f0b97d7fd
9998a7eb840ab5f98b4f30263ec227918c0b34ea1953314a12fbce7a30ed08fc
first message t

31efbc653387d188328bc3dcd3962c7d86873ffbfdf6790b158f461cf46fd5996
623378a30ca6eedb358a53d83dea328d45f149ce8bd8c697107ad6db64015beb
d5f3388408f91fec2bd851af7962920d1eb4529bac4562ad36ccb15876558dbb
c86a2484a9e1d3a85e7c0b0989c2e42e644596d3f48cc5be80536fccf5948e3c
094b27a1dbd4fd374e5ee386e6b5f3774f0b1f6c7a1a7c1768cb10bd73a7dee0
61b2e5cc7f08b229e66cec8f29d25202211c9dcbda9fa36ad5e51241e2a640
67fdbc68680e446707f27350007cd4910eaea171db6a4eb96a911dc0bffaafff
566602a467d03bc359db3108a974979d592e2a92c0dfbde1a3f3853733b7fc65
response z

3070052eca3a38dbcbcb94f976b42388b93241c975806776d831322b50928f24
c1966ce328bb0a611422be432dbc0cb2520b774d202dff30b33eca4441a9dbb2
8133757f119b5251dcee7450c2d1dc987cdf0247617c3541f6c339b68027d8c2
29100de078036266ea3c2b3c3c70d820701c0bee3384456423dea68c9401e00b
8dd1504967ff664106bd01d760cb5c96e95c2769a97915d1c5ae7f613da07329
dce85a8c660b5c6bea5b785902a69cc3649a7236a52790f51011be154b7af4c0
73d350fa0b1e23a698776729d5e68194f4b0fd9f848698b098317b07644a07f4
712abe732df984c3e8d1bda3dec3036d6a96e5b4464503ed7e7fb32d13efb5f5
response w

5b70850df11d2a425b90755f84237fb1646941096ca77f6f727ebdbe49ffc367
eb32167b616d817f1fbc9af53754731578e52f64c2026d337b78941526b868ea
3e3631041447fccd7c27a2c2bc6ca670ddadfbde5a26f1f2fcbd8994674131ca
6665ca00f7f0aff44c33e59b9e8d3db5261dd639b19873cb17b7a070d51a02a8
86fc9f318103785ce573d46afb5735830313a8fb953f685eeac6adcc4706d404
8314ff9333d38bf0ad361230490a8fba93e6e8c2182ff89ac287f825ff515cb6
10f504201c7f85342963ae6535644503b50787e755a2b9831058ab862692c5b7
fea2884c96ec702d037c3e5dbe4a013340632bfa049c73a74072e61fbdc337da

Fiat-Shamir context

5adf55cefcaef69338358b8a1f17e7335032d524329fdbc3b1354d09524d04e

withheld proof term: 4084 bytes • axioms: propext • kernel replay: yes

ZKP.Merkle.MTree.node.noConfusion definition, Crypto.Merkle:25

```
{H : Type u_2} →
{P : Sort u} →
{a a_1 a' a'_1 : ZKP.Merkle.MTree.{u_2} H} →
Eq.{u_2 + 1} (ZKP.Merkle.MTree.node.{u_2} a a_1) (ZKP.Merkle.MTree.node.{u_2} a'
a'_1) →
(HEq.{u_2 + 1} a a' → HEq.{u_2 + 1} a_1 a'_1 → P) → P
```

SHA-256 of the statement above

9067bc1b295e214dc867a5450f028b36c56f6c6f116d9c89c660034c22ad3abb

SHA-256 of the kernel type

d530268b8bf76099d708d91e6feae3239965e15c4c8836d1572e64409c8a80d6
commitment C

c38f1be64a34626893bb97df6010c8ddf27478ed0577dc2c4cc78a411798e137
a6f33fb87fd7682609047c18080ea7dc5e4c93a70c4b3cdd768cc4af6077f0ab
6ce4d8e0101b02913a0c1f1c029afe9f0b2b4ef6b92c34c9a3dc0ae403f92cbf
a183dcf630f49a2f1e551c05f24f26ba73903a5ebe3cb115b0190875a633af96
08714a78691d137c9e254b6d8770fdf4feb90a88bc5f0930e2a676e07f93717b
437ebf1f346bce69f2478808d5be5217d66dc0a05d9528221fcc044b1e22b0a1
45ab6ba27f6da34fc4bf6317bd0ef07762bf6fb0936f5e2306fcfd024eec0211
805c6473324cb2352aae5510087d50fe46fe70472b6ecbe48d3e6f0365a186f4
first message t

b09acfc3ddfe2dbc763ab5fc03bc70d6e6198d4b9445bc0ce70d9d33bf0a6ff0
cfa35132c1c9980d49e54f66325d7ffb1d9b40942e55cc3d0116d810c4b510fc
ba621e7efbc6e128ead83c2e2a1a6124f9069a5c3d8e779453e48a219eb0c7a2
f75235c888a3f8804dfd91fd91653693e1c10724d0386fa1a5a4229929d528ec
bf0574866238386415291afbd30c0c56f40db390bec88732fe82cf6d496aee5a
c5fe55a6c984117c98f449bc3a42aa059850988ad1816b3e9d48bc3228ae15ea
1b24c03754c1a0b6afd7c9034754d05ab6d66c963438e6f048c9755749d57be2
19baa81e5272a72bf8871d66b085d6261a37f3eb91d70466d29719659c241279
response z

4c911f0f3b98f297c0fc17d071ba187119b196c535da83e9275afaf345b74856
93af9e663b844a7fbf7512ab16448f40b84cf93d0317fdc329e1f71f2a82fad8
66988a2d0c7a2004734f51ea528ae4a9bb04db6022890b3f4adf65325d852bcb
0edfd2697bd30a6dc3a8ff0e64677778d036173210c234d2c72f1eda24726c10
05785effee33544d691bd72c44e3886887b34e6be086354403d8c95b2101148e
b3a5dc7db0e56095aa573d8af85e505a3a429aa11b5247e5a822d17a6e9f9c8e
6cf96bd28ee5f9e12e2452e0b1a244819ec77dbd3e99be994bd425affe00d25a
a57df668c54a6b61fd3eb150f1070b4eabaf4c1157c180cd6bc8bf92f932ff5a
response w

209df69a3224827008fffc45f142edc038caa236e3ef5c17089d8b260c40466
091a1b4319efa3680c93cd0fc9599f10782404dfc498b027a8ff193e707f1102
fe5ff0ee311389c0b08a36e2056d73985e675beb187907af3bc873e76de3ad68
7abc1d048ea094f0181b4c4de9f39f38703e8139ca6407837a0da4cbf85f1f89
7e64e335d12c7fb38c3a3147439d3fae416b6d50f52f795e388b7cf288239a27
1c8152fa01071266524f1bd4045eb6db2781a67c470842eff3a58a50e2f883ef
f6a8f477b9adcd71aad0722b44572828a25ede8af7e86eea34bb07942d69865d
91e6ebad9b36bae60f05c23f60b9927448e287caa73d3843862edcd2ee0cea73

Fiat-Shamir context

20d236aaf448c91e20b96ebe4287d68a78545f0d7a2d16e2bd29d4218577e4d2

withheld proof term: 1 171 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.node.sizeOf_spec theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_2\} \text{ [inst : SizeOf.\{u}_2 + 1\} H] \text{ (a a}_1 : \text{ZKP.Merkle.MTree.\{u}_2\} H),$
Eq.\{1\} (SizeOf.sizeOf.\{u}_2 + 1\} (ZKP.Merkle.MTree.node.\{u}_2\} a a_1))
(HAdd.hAdd.\{0, 0, 0\} (HAdd.hAdd.\{0, 0, 0\} 1 (SizeOf.sizeOf.\{u}_2 + 1\} a))
(SizeOf.sizeOf.\{u}_2 + 1\} a_1))

SHA-256 of the statement above

934ef64b73594efe40116b9444ebbeff6442ea7879e53da66b04d665c78c69ad

SHA-256 of the kernel type

2fd8f1de0c81c2a5e995687190e3eb1ac8a2b5d2bdae963e48fe24a7c6a1da46

commitment C

e3fdac28d03278318ac5b63f53d18b378dfbb7eaff68de8fc6f09309e57edda
db5c00c28d0c829a7032b1fe90f14636bab66ddca24a288e8fd456f4a3edb8f
567ab912e7b1361efb382fe1d62893e0b6762b760db84ab413fe8be25bfda425
0761af9dd73ce09b55741e05f0328cf8cfa36e66f6eef31d06a70d4f5f00a4
a4cf8ca311e68c8e9b6d09031270c55940cb861f93657b7616cf1ab0e7ff028c
769e8e482bc3d490614ebf17b8d1ec1a70a529322b06603c1723b8e1ea0409dd
25886b718f3d56e538b2ce5338dbd18bba1aada455a0f9128d56e180f85edfd9
fd0b569e556240eca1d2a8c7981ae089404cc25768d197448229abe3b578614c
first message t

e2d0ffd427d594d50ec852f36b3f50901d78cc0f6c0d8e108153b7a666f93aa3
577a08c86221766c16d13d8c13fae1d3c7459de85b9d9b285af6b81815d61fc0
d8c00476eeb82b9cd0270aca10937066bcacb70fec4e144e9702bc3fea1d77d
1201afd7b472e1db5230961b8f643599c54ff43f7fb6ab26c81d143be141ca3e
e1e457340bdd071c92e769088646277b8ca32140cfc7c7eb595be3885d8b4ae9
7a7abfed29f73ce384d8401c2dccfb6cf22353341c238a3ae861c205e8df1797
1ac10b1d117b8925ef18fe81e448e96eb29e71d80c638fc8fcef7e678161c2cc
658de8d82bf23e95d0e21ed458f20f08acf40ad87c37246cb399d0e20c7c30ee
response z

50959e9bd28ef52484db33bd79b336795808b158cc50cc8686b0753f76d47d2a
5015de0115ff67d4933814a49a8cf819afc5f4f867a724955db6fd19851840e7
9c2ca8bd6b918594ffb8885c280568ccaeaae00624442f95aef3be911dd2326
e7cfe706591b4805e05b663f45788c822cce1ff6e7ec76f3351c648d1928264
697aca7b6d29340fc27b3e6b121a21e99b44c806eaf9203a59b15b23ed5ac375
1e0a2a801ba07e98238ee800e4dfd2f5584e8840eb8f09d67a2e4f8bcf673d70
40a3e15b92b53db18d3c7b84da9b3418a9a229a7f4cd33016d98c8b3e608e49e
c185ca01ad3c4d517ec77fd27eeef97544809335858983506bde47d28465427c
response w

26f5143e78f218d611bc1ef695e63c1e34c9014b4633144dfc8b77c9fd3408ce
06e8dd786792f93d6b6ffabf21ae48318571d42f22d78b07c5936029484608c1
af97997c244e84eb89e86372396a353437df454b0917780445b8a5e3ae9104a0
9cf11b500ad0aca5ff67e44b26f71a80834b71d053931feae1248ce5f23794d9
378f05d5ccd1c1fb05ca7b157f1078436cfd8d278c22ab900dbcb236fa92c4eb
dc57dc334263ce707656a48327ee7413e2708094f4b79e22196646d401e79ea0
1cf852133ed783301d83a42c8bb7aded4dc3d2998125cb56d485cfc909b8fd21
aa0ec8da4f9ec6a3f13265772f5fa9d0cab457c613c72af1a6344f78487339cf

Fiat-Shamir context

8857b53ad322ec9b452c22dde723e33c4ae245d7987e8ac816fde8fa8ecadab5

withheld proof term: 762 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.recOn definition, Crypto.Merkle:22

```
{H : Type u_2} →
  {motive : ZKP.Merkle.MTree.{u_2} H → Sort u} →
  (t : ZKP.Merkle.MTree.{u_2} H) →
  ((a : H) → motive (ZKP.Merkle.MTree.leaf.{u_2} a)) →
  ((a a_1 : ZKP.Merkle.MTree.{u_2} H) →
    motive a → motive a_1 → motive (ZKP.Merkle.MTree.node.{u_2} a a_1)) →
    motive t
```

SHA-256 of the statement above

d430a4c04efc534d7666e30cc13c00c8f922397f725c6ce2bb68aa15e231f7ac

SHA-256 of the kernel type

9cc80e96c5d168029b43b165b2024f6db242945c5a59d22a87ab857402d2f58e

commitment C

c3122a15a2b59180d28033497b8032c75f5917a4b69198bfd3a27d8034a5c315

9d819be39e98247308708c6e2d65225aca53943599e86e2e880570b8098c2a74

25d8fc129282b6455fb4124188676fce68fef791e52eb70333b87fc994837473

88ee4b6546c7cde10e65f3a69d30334916c9931b3428686bfb23b635a1f09c93

05a42b26e48285fd5de7e17ea8378d1991d27d886086f5d5d346bd73ffd3e3e8

c764a265914959ed5df5dd57c664c6bb4fc881525023d9a5b3bfba2394b839f2

9c8ae25e7b4ddf634d22ca3859835b3822c886d4841f5b23eea39bbcf9ea1328

2359c14af580286a897ade05498a094c30a3f582cd49f992ea74d3ba7714356c

first message t

f95f07be743dbb4339b284066baa9849377df42ba06046a3abdac4cdcb62d3f9

25f6cd987169fc21a45124af5b950a05b3a73dea51d3dfd4f2f5a1329983d354

3ee21b8f8eb79db64bf9c664839668b56b61bed8fee8189934fb5e9b1899a208

146d5a99cc786eb56d51dec05fb748f7c96ba29aa570c4072abaf9e6e345a623

f40b86fce667e861e64f169db7ac8034af3d772844cb5bbd6114d82c09d21094

b57d4e8d0cee5d9636e5371dfdf11a6da71f871d382359064ed293f56d8366d6

2298a3251548fbdb25588881e0e3593bb06c8ca897a3171155f886289cf9aa3

e7bff8bb89b85e48e5d0956629a9591085ec4715167d48b074e9294d44434fb6

response z

7f6dbc8daf7847365e663f85cda71382335593d8e96b8e2b674a754b675b8e9d

9a68397e9581a88cb31b5125a7ed5393ba44b35200ef7c8dc14f87cb9b314203

488678ee23eb1a67606ed274739a4929d7f9c4278010c8c8eb466b8e6b34549b

d9f4ab5a9452931ef5454f8b6744042c07f42c1571721a3549a26b6465744602

0b832a7831f37a777d5c6bba0c1268d20a0e99109cbd9126b7f823a556964359

9d10ba5560bf675ca8a45bd8b31b64f3ed4f23bea66476cd73fc57ebd8fff6d0

9aadf856421fb3a2f755684fb2608c41ee86b7dee2a45bc8e9a3d22ec74d1d1c

bf499a0e828397fae2b110c071347729b9d1f79090104589ed4a30b2803af7d9

response w

341335145a08693399593475a30485694cef0799124fe3d89ad2ded963979248

f6532efb8912bee5468b3b8eee26fc5cb1ca05b09f509adc9246341ac00903e4

230ff2f9503bbbadd03d57beb55cbeafaa96c428c9e8b79753ac6e3858a1ef4d

2415db7aa065570bb32465eaab23d585c244c7641c39c34380e37c5ddcacd50f

fec22f806cfa3a187ea605981b5cd56ce291f9a2828fb1e9f8acb1e8e55e79ec

2f8854b610020cd93fbf0e3057c624849fb2f164fb01a9b2cf5d6255db524785

3378dfda457851c553b8c90c48add0bb81aae645ac708bc203e81ead45fd24b9

028ac1d22144d5a3f4c2d4025f76044211dce827bef544d61cd3321abf8bc8b

Fiat-Shamir context

9dd662a39a7f57d33e328e104546e63962f78c87ce9c91c6de2b71e5042896ea

withheld proof term: 539 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.root definition, Crypto.Merkle:29

$\{H : \text{Type } u_1\} \rightarrow (H \rightarrow H \rightarrow H) \rightarrow \text{ZKP.Merkle.MTree}\{u_1\} H \rightarrow H$

SHA-256 of the statement above

242872e13c02cb03dde816e96ec46d17f4fb51bd218e827de00a5d934a63e3aa

SHA-256 of the kernel type

b334608e28537593a9ba18e0507d1d3bbe76b823e2bed5aaf0e82c244a1b53d6
commitment C

ae6ed17de0a567c68616503a4987dbdff55e0500ed1ebed00ea2b3016dcdc9e3
adb3ea0df12e5269f8ceff9d5d30b20e6f429132342b7b16f3e687637a5d5926
bb63191ea0645d1d65686ff4efd7f6737f8d30293d631cfb8ce1ff463d473932
f980f6ef83dbc268d7d84d9ddb060aadbc8b47184b47c8e9c16e8c780dfca185
e821a1d0bd8eb25ac39c4d948f62a14d5acd8bd8754682ea89ce3003fa8b7d614
520fc178744a22616979d9aa0a3c93f76f0778f28fac3482fa1d0f629d80c9b2
306d83b9f796e0b262b0e522b5b0d0db3bad08b60656e0b96059388d6ec48fdf
a384653e323f474c3baa97adbe910389319c448e3bf8b1233b3e483f5c072df8
first message t

e4aa8690b07c103c98c83b678a8fdf3d0a5adf556c8c2c1efb3c8b8d5aa555a0
5101a067c3ab394daed7cf5fc51eb2cdc5d3bc2203d3fc13374e739c2807ca42
4b922970f2e9310dd8ad2c09e9f36ec4d5233f80c69235f920c54c9665f7d441
6d06fb4c83d4f878fd55ad908a4d7e4b9376f1024caff3b41adcd8bbf7930a3a
c354f2ad66a81cd718fd01972ee07e954dea4dc1863e723be92683f93e2b5167
df297e1fffbb721c8938dd6e6c922dd6767d1b6904d7130303d2d1eb90e04fa6
50541a050d8e18dd27f3b25cd875aa84915d1c999c903066d03ebcd391e3051b
8d8273896ce71d9e917a75f613a33fed3932ea5a8ec1e27ba0fceff88a83b594e
response z

1dfffb29cc63c2aec1babb7e3e518a36c7a83ad45c1fb1c626fd28f6cbb2831eb
52e154db459d50dbad9c6e73845eb1e967d658b06e639264da67b17513b19a97
02f5aca46d0b05412cec67ab8e24197f1fb6a31e757beb18f00a073df38aa787
4482a77f074492aa3e45896f660cdb14e1d85467204ac3aa5ee0ab11b5d6c0bd
dfe6daf8dae6f4b6ff0e66ced891c73be207fc19ef793d0ce7023b5ee5d4dcf
c63393c3415dc6b7340967fb59880fcf951320e693c18f24911572408c81400d
19280e881639334ee808d76e0963399a937e94d5847a170d36b1305eb6f655c6
e475aa09cd1003848f5fe7c09aa38b680d8e2f22bb145dfcc5714088b30db2fa
response w

4ca35ea3a683a15c09b776b6968b9387fcaaeec5662c36233222d7ab9c6d0510
e59069b59f278d6cce0d1bfd36df8ff1fe60a6e40d4539d998d10ce5fb76ba8e
73d8cf37af122fd727652473998ca06d26d7b09094a64075ee056a5e0c56f5cc
354211ca4a370a3c79aedd8418aa71f426ba6edcbbaf3e189bdc1ec9ec5299fe
42ae7efb999f390f5db96d37e068d700fca57557cefd0d386e3dc12087080adb
d65ca3ff56bcd3b151d866bd50b33fa81d3d83f388e6aba8afd2550fd76a0253
7584ba9d4cf4fc1182b587af3eae70c36156a10be2981e2209a7f3216f810866
820c406e746e2274ceda366ee15ea98e9e64085643cfeba6a8e89d1de1e6e8b6

Fiat-Shamir context

f401911a70d351c092c4eff0bddbf633d1dc4cfd55ea066a0fffb161828a0a1b

withheld proof term: 1 879 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.root.eq_1 theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_1\} \text{ (hnode : } H \rightarrow H \rightarrow H) \text{ (h : } H),$

$\text{Eq.}\{u_1 + 1\} \text{ (ZKP.Merkle.MTree.root.}\{u_1\} \text{ hnode (ZKP.Merkle.MTree.leaf.}\{u_1\} \text{ h)) } h$

SHA-256 of the statement above

f87ee92c18e47e225b718fd06d921b4646e39cd46feacd83da799ba0133484c4

SHA-256 of the kernel type

ae79e49ec7083f7f697a0c09709fed6a43284761e3195b0b4ba0846db73ec19

commitment C

175b21db6b249c93a2e2593744f8c6c74fa91b2a35b1b76059d97abaffeca452

2d90d6311bb3ec3344bcee38de88973bbeff7112c309bd1607e415e64599f958

69c1b07906d633087e398c7b5fa0662b42bcb43c25f5824d9e428bbad8513b72

58496c7c90090eb7a6173168244cb6cdb120dbe33067bd1edf7330fc18fbaca9

1cf8665ec20e91ec16a53876356a86cf4404feeeac61e64326f8a5511051cd06

45085a52d7bba8a534ff115b8147a5b53d8c8a1dbca7e79e6020bdfda11797fa

6c9a9f22965a6ad2aca8ccd4d04202da865c60b7f290ee265090f81c143c76fe

77988e3c535701202a250c98fd7add52530f81eb037b0c4c201296d0d224f55d

first message t

270ff0136e55bfc1af2f1b9215474374e3ddde6c787103045e56c54d28ad8429

226515c42f0d81be41871250318505700fc6754be3ebc17a7d3336e47acc9cdd

312fde143f3070dbb1f2d3021f32c9fdbc897d90c9d11e6df6be503f6a963ba3

687bcecd1d28ca6957933da89e8476a2517666de26c3667e7eaa3b7be4fa38ecb

c2ebf380ba0ebd13a0c5c6467c573e251cfb46e9bf11a444b465372d0b87fd00

a06f094cb85907f15b12abb26676be9ecf8b9765218537a550c4678293d13ad5

d9a8b11433f1667daabd52ab409e330012b04a56718b27f31b57646ed415433a

ce1d2cae9f1dedb7f846da18793e965dfc09dd03845a1e0b3e0c7aef96f7705

response z

566b72327535f6ab2dbd6439c6d934fc3817b9c4b42d7fd3f0b04499f5b01a52

199afe40eeef0c96a186b2f130735dbf37edd30e831004e242a655d2df65b902

222735e97c553065dadf613bfb85e8e245e6ea5946b9245d9f4cb49307a88131

bff3194e3e888b4726095d4e7a69c9d8ddbe0f7c17e01bd7082aed5462f14f40

42e43310497828563f985c3a797614abfcd5a8b1328ee69966521fa8a89ae06b

5fc8ef48fa22bde46a42c13b6fc82f0076096001ae9b76d790e5874104e19563

6bffbc11afcd386ee83555e6dd9cf9530378ece4c3f69b94412209831f5daad

136c830c9479c6cf8c892fefcf84326e2e9fa6ff9456cb7c4abb66620b35bb37

response w

384e165a32647a72d3f75ecc7afc40284136252d48d4ecd437d7f3d43d1c3b25

dc2382231a9a85dc85753604a6133ed74f416c6884df3a225e66c85dd41ac712

cd79f527cbd0ea3197ce7ebd4b8d4868df042bf50899006d8fa234fea0dc1932

3d70286f1098366cf393260a05902dc5acc5b4e3f29a7ab018a6695b447126b9

40c851229b0543ad5fa56c91b74092244258f7ffbd84a0d3316c421495579169

7bdb9eb43fc73f156d0a61c6c49bfb528b9e0cbdc7937ff300d81de0d30d9186

7d550a7b91cb5e388c1915a683b7b0dfd3e5acd280a80af938e3f47e63e9073c

0be1cd306f583e82d03cee613c24c322e39fa493abf5d367963f89eaa74f02e2

Fiat-Shamir context

aa38f469158635fbe2b4772bb54f41ea27747c0f9eacb1a1db50796f413ac830

withheld proof term: 146 bytes • axioms: • kernel replay: yes

ZKP.Merkle.MTree.root.eq_2 theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_1\} \text{ (hnode : } H \rightarrow H \rightarrow H) \text{ (l r : ZKP.Merkle.MTree.\{u_1\ } H),$
Eq.\{u_1 + 1\} (ZKP.Merkle.MTree.root.\{u_1\} hnode (ZKP.Merkle.MTree.node.\{u_1\} l r))
(hnode (ZKP.Merkle.MTree.root.\{u_1\} hnode l) (ZKP.Merkle.MTree.root.\{u_1\} hnode r))

SHA-256 of the statement above

f521e81d901db8d0c147c6ced5922953ef549b9d79341ae4a179ebd296d95c78

SHA-256 of the kernel type

beaafa6f0201d7d32199032863e39ba657ab9f2b1922043e70bc3ebfe00a032e
commitment C

7fbb844ea195234e93b0d6e4eb4a18a6119859316e8b929c40ade9b14e8d3ac8
b188d900aacdcde154841c96e43150593c4899a0d2fe4597aa65423b97d1b5c
83e5ec9b0acfab9e7aa7f09e02da972eec3ab8e41c7c0e6d9cf23b2357e1c5f0
4a479c07cabec6db792c3cee76e478297b070d6c3431c33894f42a843d17455a
f3c47b8e3225978dc5e00ee3946d2dbc6c715c0adb3de83cd1770d224a09a95d
2628c7bd3583361213abaa0e69c8320dd8ecfab1dbaa6cf808d8b62af6189b5e
7a38816f3f051d5c484f8f97c7f434dded6add86e184ce4bb14f4aae5387088a
31c0ef5acfa8da551dc38728ab39f170d77cad636ba723513c47111b0d97b4ce
first message t

8a11ef20ff5aa9bc110fb46d75d481e878af2257612db6c28f2e683ea8a1415e
a682ba1b60b0e986544cb0c0bf223ceab59ec0987fd6879bf152bf74834199ce
985f9999225b944113a7dfa1c0ab5e757c960a3354aa480378f2975595fdd72
7f3669355aed696ee557e8605f2133d64663e2283960d38cde9253b54c689dde
33e58051981a430f9aefe1886e280defa618053ef64e21fb2e793b8de6a73df7
ac5785deda88347cbfc55e117975488a3adf464c5bae918e73f2ae0f4d331504
58ca86beed766dfd82744432b565d20342e1570eff36622bf62eb59bd11ce819
81b2db0c2ca7187576329e20dda39f8a2b265ba81cc01037f5ad3d309f5b88ae
response z

165303d76549ca27d8ebb28777da43106e3871b6a7785799a12c86819e6131d2
88bc43a568d67b720f049dba3e625eaebcf7e39f817b02923c7048d8310fe9b4
00eb861cf679cba465ef6334e35e18daa1b4964bbcf6f459b52694c19c33a0d8
2b33cbcdffc55c83991b2d91dcee6deb9ced5a1998337f5dea7753bf6738164b
fb20a4241c7dc13f9fa021877e6849c88d52f19ed9fbbe12259f7282ae918e8e
6fbdd2e23a6566c277b77d42b61a340bdd157e5cbc8c306ddbbcf83849e6be73
6858571e7b5161a7ecbe8e82666cd203b51ed99638cff0f5e8bbac9bb1adebf2
845beccb70a15d15fb870fe3e2226435f3031cb853a4dd7ffddcc4c5ebd5ff9c
response w

7dd6131739c3406dabafc40d156c1733c3a560b41891625cadbe52c1e7d396c0
6665eb02fb5b85367dbbe2ae022f616dca0fcacdd65fb3c35f7e165e77b634e5
0c28b637dc04da653c982d9854780959ede55b85e032d71526e2967be3af0090
6c699219496ddf179dc8794a2bb642a0a3060da9770de94d2ac4e4a41d1c5ab2
5f9d9fca77074ced129950c5af5dbcf11301970dba3b17c4f85c80c81d090988
c99f5d84b2dfed3ddd0143a3fe8f18d3bddc9e5c06d54e12caead4afb915d55
e09eb57b96b3d5e1c109df24b362dede361f9572e1080651e1dcacc2e1b2652
28987fe8e3088022896cd569a06f79409289c847798e4580a12c1f2b6feb9d6b
Fiat-Shamir context

b2a7e7275675c38381bf27716c787c767ac4399f14144ad4b9b7df5c95146622

withheld proof term: 202 bytes • axioms: • kernel replay: yes

`ZKP.Merkle.MTree.root.eq_def` theorem, Crypto.Merkle:0

```
∀ {H : Type u_1} (hnode : H → H → H) (x : ZKP.Merkle.MTree.{u_1} H),
  Eq.{u_1 + 1} (ZKP.Merkle.MTree.root.{u_1} hnode x)
    (match x with
      | ZKP.Merkle.MTree.leaf.{u_1} h => h
      | ZKP.Merkle.MTree.node.{u_1} l r =>
        hnode (ZKP.Merkle.MTree.root.{u_1} hnode l) (ZKP.Merkle.MTree.root.{u_1} hnode r))
```

SHA-256 of the statement above

514c98dfbb0adbcab2f42f3a84ed414678742b9119c21895f2772db84aa4e127

SHA-256 of the kernel type

c9870bb99e53a662ee3bf44fc01e7f617eb80949507041c7d6624af5bab2c767

commitment C

d25ac96f5cd1e5760a68990381114784113a46c7eeaecbd3c072fe31085132d8
1cc94c9345b45d04dd91e69481d565ffbe6f8dfd0e0391fb9e18f96519ba946d
1e34e5bbb4c3554f21356ccc3bb27726a1a2d89547877ab82e4001bc0cfadc77
8e2bd896c34baeb5798e120adfcc172a51745865f188b46223ab0bd26891e21
38bb830dcbfef25eeced9b0d9361d8673f8a2c05ed43da4a826aec23a07e2e50
cf22b90ba14f214636aadb2f7bbc2c0270ca2deba99afb4ca36f126864cd4da
9fd32431fb3f65fcce55402c3ef719edf9e568abf88484b8262fa9d59184ed
705121ff88d86028a2b55cfc83f7e2d514c6fc7088e9184b7499ba243de371ac

first message t

54d12768fad3665de587f2ae59e4a5915dacd0bc4b9efalc5ab48f910e1de67
746cfe3658ffe9eab774c202fe2c737e78e9e801a5bf7f0bfa6d63d10a4466b1
cafff4b85a74f9862b7cb3c6818fac30ce596bd4ec9d4091b3c016ce06a7d888
0dc0e5a3c45152da964b8dc0a74aed65cd308c9b9f61a83f8deb80f559cefae1
b8aa5e8ba0993d1f89d388fa6399a928ed44756999a6931ac3f5fe3c08066abb
f64fc30dcf09fedfe1b3726c7f51bf444df82ba6af8f3b08656c240acc2f3ab5
2409dbcc6d924081113bb6855b7fb702b0b2bc78b4e01fb27017706234ce6520
42845dcf0a0e7ad6e05fe7b0ad174ea8cb1b24bdfc54c7644e0d59cc7f086fea

response z

7eaf17b161abbe3c2cfb8bb4df43bb6126436a6d166dd8c878129c2eff4ab8c4
19b0136a56e66c1e576c46de5f29a5c583e3c522690d3bdad59a8ace58866fa3
80bb94557132814e925e5bc86ff885414888e89c6cff7edaefed4817e466675e
b18805ec65ba8c7da6e796204e8651adca844c487aa4a4d9decc353fe6fb0276
a0b2b595089b7374a8be0db0871057b5d224c5892a3df33c6f4e4cc0f8a71d3b
c728f9d40c417d57f918c2b53dd1ddf9864b4ade537a396403a4b6e263993a37
2ee33936a458c88dee3a6dc16b4871cc2e4eddab43dacb1446e295a74f15b97e
f1d656a3cc1f05f484780f9a99d10f3e7a6f0ec258b200b55446086faf3700fd

response w

37b390b52997781e1345aabd80803b6ff9f96ca308b0fbf0ff5be8675d325ef8f
6ad5d5183aaa3f4268b178df2f67efb5150b99b85dda831b75363231e4d16adf
0a96a6732c979d7dd49f4857da0725081ad8bc7232c8a6b9ff58b0c863458bfa
d323132eb15655807f15c6dbb3e49a25627f1091111e23e24b61cf45ee1947ff
0ec046570765e4567bc619e6fe750be88eaf393d49e1e9cd9c2d2cd9248fc564
bf469819be13b568ec3f8a8e307b92b74dcd46a86eaf3316992d4d787dfd0900
6296b95d7989bd76e22d9043b21a529cf1878f0b90d161be9afec1d5717ae19d
aa377a9d29f4b0e5ae4a2796bc86029d109c22117f5b1a180e53ed6e7214f974

Fiat-Shamir context

bd289b580bbc661d91b9ff526a5c3d00b56dd341076107a6e4ea5921b8c7de05

withheld proof term: 45 189 bytes • axioms: • kernel replay: yes

`ZKP.Merkle.MTree.root.match_1` definition, Crypto.Merkle:0

```
{H : Type u_1} →
  (motive : ZKP.Merkle.MTree.{u_1} H → Sort u_2) →
    (x : ZKP.Merkle.MTree.{u_1} H) →
      ((h : H) → motive (ZKP.Merkle.MTree.leaf.{u_1} h)) →
        ((l r : ZKP.Merkle.MTree.{u_1} H) → motive (ZKP.Merkle.MTree.node.{u_1} l r)) →
          motive x
```

SHA-256 of the statement above

e09d168ff18696e65dabdf1ca295d45495ce8e6e625b79a7fcda0494e4f65798

SHA-256 of the kernel type

22c5c7bab365605e0c59109fbd77958f552e918ff1ab4a1da3db5bdfa18b0f59
commitment C

6a22d62625052029d4ba81a6c3ec40150d714c03a4c5f1f6fd6eb0755ab0514a
f23f8dff87ce12921a515f170580994ae7143c5bb869e008b1643f8bf9b714a9
58b105eaaed0714d15c43df27cb7e815074e26858464cc1208115d5101c32be8
f2c0f5503d38312a6706b6d0a180d22ff3ee03b3c0e50bbc3c7c2fa64b732775
149912a69b6ac0a8d93e68736e3683580929961df84e8aec1977cca40a4636d1
6376a436ecfb4ff1d8981688c63a1ca3cfe376d0c035caa34e7ab28a4fe3b317
0f7a3d8c6ac1a605b5cb228036d171a29bf42275cd03cd08bd7b18814eb8b3e8
be3a6c46c1bd2517cea9ab34a7e614c0aa5bd1da1f78a7c12090af8c1f468853
first message t

4b105e05d0d54b024c822d54c534f87e539374f0f5c367fbb84f6dcd8955875b
ab9ecb63217300f627dade2d8f7d8d26ce2475965f1e70592e7f4c4dfbe3f7ad
97bb100f20463dc94bcf8afc705669d7b0a55ffd51cb1c1fe70522601b66f679
3c988317a74cedd280224a43140bc56db5ccca04c4784aa9f9dcac1bc4bc306f
334828b13a77e5d839fe50e3528e713286a35e13e1bf061ae126770d99d303db
900301deedeb88f0d630b159a9929428746077cc09ab456fddd31cf17ac5e189
5e95f11f370f682152b3efb9960193cefa3c63bec6c8f21d7c26cdc8c27bfe50
dc802fa53e4b243f3204f8ea74cf81b9b678f673c91ae828af8c233ad377034f
response z

6309b836fadcd1227f21bc535aa8ae2499109b0e118032adc655194df90bc311f
1cd8da4891a3f90cfaf9357a9604d29ed0145311933cc452ec7f5cedea11f4
5d0dd6c2f97d334bd24d166d4d6e2c22bae9c10c1e9883554a37a794102eca1d
3c9042aef2ea644cdd8c14ff1bb984a1cad7298993f49807b1136bdde53289e1
cbeb22ebf24764274d7d63b1a1bcde39e25bb6b4f76d063464f9ca52896fd283
1dfda1cc78ec0cd385143c25e1664e5634ba3b450b4467e3ee41532c2d55ab0f
3122771413d9bcee6144880815f0058c895707fcc15f7b14fe5633b8e9c56e37
37d168b7617c56c847b754028906c7aac2581a8174d76c3c5e4a1b88fa2bf82e
response w

39e08d094d1fef0bc80d5f25d57641ebfe51164a06692d587a90b4705d503fc7
cfe691dd384037d2b318b2e45699591b82bc5ad4246eb9d8564ea161049026aa
a03ab978fc848cd05eccd013b9dd64c3303d91722f3b93516e78d2e2880dcb82
9a7995afab7d9b160c100edc37650d2f673ef5046fe7f70c67aaa0d7a46eebf5
91091889775c68f1b6f86b4b55348e6f16b1c6f9bfcfd69957de79d2cbcf3df
c690d197e210b247df903b9cb05de5567e8ce69962bd27e8ea727ebd764ba7ac
eeb63ca3cc6154ae190fb45f01b20be93eeb4a3a323d71c8bf6d3423e5e70450
4438671dd0ba2f91c7f5f7d1be8f5f962d80cbc8fffb2f8b04053ca6cc4aa916

Fiat-Shamir context

7f332d3e8a80f2aa962d89ff989bdb5b058a7791406c9ad3c9a2b2b0987e929f

withheld proof term: 1 129 bytes • axioms: • kernel replay: yes

ZKP.Merkle.collusion_of_foldPath_eq theorem, Crypto.Merkle:78

$\forall \{H : \text{Type } u_1\} \text{ (hnode : } H \rightarrow H \rightarrow H) \text{ (p q : List.\{u_1\} (Prod.\{0, u_1\} \text{Bool } H)) (a b : H),$
Ne.\{u_1 + 1\} a b $\rightarrow$
Eq.\{1\} (ZKP.Merkle.dirs.\{u_1\} p) (ZKP.Merkle.dirs.\{u_1\} q) $\rightarrow$
Eq.\{u_1 + 1\} (ZKP.Merkle.foldPath.\{u_1\} hnode a p) (ZKP.Merkle.foldPath.\{u_1\} hnode b
q) $\rightarrow$
 $\exists x_1 y_1 x_2 y_2,$
And (Ne.\{u_1 + 1\} (Prod.mk.\{u_1, u_1\} x_1 y_1) (Prod.mk.\{u_1, u_1\} x_2 y_2))
(Eq.\{u_1 + 1\} (hnode x_1 y_1) (hnode x_2 y_2))

SHA-256 of the statement above

8df4cd541795731c860e343897eb84120aa83dee73a8d1b137000beafe5434f2

SHA-256 of the kernel type

ad3535d24af541bdcf3d50262457d7be1a6916a24fbce952f812d8accee16ee4

commitment C

cb5e9f4dbeba96cc2573f0fa86b2f04d9be10f34d1de6bddf67be88aadbb1a0b
3973c35d5f05bbe77bdd67de063717152abefc4ab5a15c21d91dc97cf74feb0
e33c864163a0eba4424d31870efbb12eb8c6891fb7aba8994c8ede3e6e883845
c91e294b00cfc0affa1db471d915d491a5cc85be95c8dd48d600fec5952c86fa
4ac3667a6f101ecae406a4f02673a0426a4b49037752c5087ff437d23ab70975
6fa0bc2da4121f5ae5dcac5b66456740df04f93f2f2899afc714a70123a7b788
81f52803c6e4b94370bde8bc12772cd59eca7b92b4a2e218c0a12bb42a653f4d
9ef7d30af8ffcd660613d64265e4461a1e85ed9832c42f0c04f4db92641a42e9
first message t

2a9b6396dc6aaaad2a5d1ede6687972552abfa8b4e6e9e029aed4419c142d2a6
f39735ee5d202606453d816d665e9d9e0d85df3a8bb96beef2090ddd33d34925
b54fdb40db564116db4f4295f36952ba9757268f718a21336f744857a7ceae4bb
0f895fc13152131b27be9cfc632a4a468467e7aef64aade38ce9201a6c5bedf
50cd5d7d12213202058d43bca9326ad71fda646600a9b035f9d03d61c70c957c
0aca44d3c04a586bd1db0a45613071b98ff8da216e0c437c45fe7ed43c093483
7d0a056ff5eb65ac78f3824897e6240f44cdee2981ab48b71426373a0f88a499
63079722fdb08bcd0e3f568185ecf102864c97e0dc841dada5d086b147fa7b06
response z

39fd03535ae827e8b92c34f9418f88af5cc9fa730fcffbfbf2d67eb9a98cb3d2
d30aa3ce4e816e506e640433a9acb5cd1a7cbb9a058b6782994a96882ca737a
13a99ba5f93e81d82edaddb825eae3ce9d7603b84d3929c8b903b1ac669ccf8
cd10b6d4cf83ad8f982c53ba828180c89f0f7e14630f027eff0729bd4443bf76
d4b24d19da1601f6722936320706c0d17fcac3271f96ff9555b94ae93abdac3
a6f4e1d61e981540ba837892ad91e004a0a687c273596fe4dde16e50a1e481e2
f50f479d1b69259cb8aa3e7958c2be8e52b7ce85ea6944a25512d21e72e2884f
fc7b9c4383dfbe8a120b8d148fad94cdd16fca82f3a3e9b3308be041032bf88b
response w

4e2bf422d8d59d18089007f765935abb98d565d74d8ff15a0670c5d02ad29d5f
275e1d7d0183cd1f429751651690e62eca27531e5edea2d6319d0edec31d383d
85722a15b9905f88447cc44d0363ee7912f9205c33880e8c6bea1e8f8df71830
56eef9d54587db1cf541ca92066f8f23e378b83a5e5d5c4a5dd483a085384889
ab8c0b220dbf3ca7d46fb5f0769a4ea28ec762f0e7b0200885fbfe202a49df25
5ff8800fd2d7de2102a7688b4c87b04606753d626e7a8c6b2e7c2e26bd394289
3a6b4394f61e21e4cb14bf163a5ab3e2c1669816ba86e0db3db88f312c54d6f3
1cdd67acd4b5a069ab63a3fddccac8045228d50504d34c3bca28c7af12422214

Fiat-Shamir context

35783bf79f5423239b416dae20831579b8691083ff999db824e121632bda2bcd

withheld proof term: 27 566 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Merkle.dirs definition, Crypto.Merkle:75

$\{H : \text{Type } u_1\} \rightarrow \text{List}\{u_1\} (\text{Prod}\{0, u_1\} \text{ Bool } H) \rightarrow \text{List}\{0\} \text{ Bool}$

SHA-256 of the statement above

b1b025ebececf21f3a21a9f8fda34c1ec1bb5af543b92fc5cf458f1a6231a24f4

SHA-256 of the kernel type

c8d9fb174dd460a7d921442866815c53bcd66a3c6dbfdaf7e1a64f4ff7f583c

commitment C

d3c863e7c80d0f46c632e79836b66e4eb3794c90029f436dd9ca23592df5bb81

398f011b6bb242cd31af0a18e48324d3d8bf03622be3aec880862f3125c148b2

625953fbf96e61c8055a87a2d0510866e4f59d0e8fa7f1f0ec2427abaa6287b9

09dd0d3cb0707e33e4cc24ecf9d34ae0b651dcd1ee74cdc72284c64221d6d2b3

4877319844b6096eb85bdbc1463f47c6d5c6099dbdf9da2c9c06118ffa9b251b

d8da3203c59c7990f5c0c04fafbb77f3e60c8079de182ff6194ac58f276c3c32

8b51d70efd6a7eaf72488daf6b6376e332c86481d7458de709207bac5a478826

a2e55feca8f4be0fc9166efeede7d8954bedda8f9aadd8b082a965881ce2811

first message t

85db60f056b2f4bd77f9e47e4ca02664715c7bb796f1d966b53e3caec2a1ef2f

dc1f4aa225f6960c6db6c855f30ceed3045a504c714fe220398c45b7aa159236

ae78363bd0f298a3cbe5b83011f30cd3ecf11101f2f211626e58e66e1b21de0c

c8058f09162a1a4184cd2e234f20061d102ba1fda48f7395c47f1dc456afff02

5915789208eb3cc1ae5a3f962afc26609331480240a472e68d6a4cb0f93fafd2

7b7d841fa2f121e96ccce5e09b3e0271b8e0d53425b64d6e1a74b4f1bccbecee

1466e826c3e1e9bc32979e1911630e21b32f6367e25e7b0f9b5e2961cda0742b

27a119cb6a1e23c14690721b731cf7552124922bf8bd4a8cf3506699d1bdcc03

response z

5bbb8fc6e7b4047ace84c721e3c178d9c874cfa400e70addb7dbb06b62684c46

157864f1683b6818a69852fb184be7131736b7b6b2202ab1b05ec6dfc2503494

f435ae0101504db4cf3a85a3bd8b6e0773494b2390fb421fe46846292f42e1cd

172d26725f602a648279f5b400aa4d5fa88807797ffccbe78dd2bc8df65d7ee0

6f6f5deae5d7836b23fd2ec8b4094cbebad4a7f9c528c02b48740ba38e0d5600

707ebe0db76b9f3dd81e20e9678e848470b35acc6133f2ad1fd16c6f79d9d868

062d6cbd88c68a597f91a6da1ab4768c80902a925c64eb24ba76252e20214883

6bb528494ad24f645805887466b6540fe97945fe73a6b323eb92b79ac98884c3

response w

43eac8c21018a9458734ae33e28b96f12db965f79e9a6033bdf6fa5364418ab5

0dc0c55730a106fc6fa7c9b524528d24b073b37ed3676058ffe069c198e1f835

6ce888bd50f0995b74bfc1c6ece2dae2f3dc32a537183c0dd59ea161bd1a804a

bd7a0207dcf7acabf18e2ce5ff8eee24280ff42cd3804707ee213b9872832053

ef03f55a6579a99e5edf1d00f0cacfb45350c0a2df497730929b2c30cee943f6

d3d591eda5f866887c69ecd126174d96b979e2e478a3f6a9b359e913005bb19

5aa87d30646f2a9072ea0bc7c33e2208c06c231aa216e417ae6a7eae0b480f22

4d04fb9c3fc23ab52cbc334abb99080a861a15c5f6dd9382e6c366b36649cc8a

Fiat-Shamir context

e13f586af17e142ca203ff5023e17a32c02cc56d7256e153e31ae2571a13a350

withheld proof term: 138 bytes • axioms: • kernel replay: yes

ZKP.Merkle.dirs.eq_1 theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_1\} (p : \text{List}\{u_1\} (\text{Prod}\{0, u_1\} \text{Bool } H)),$
 $\text{Eq}\{1\} (\text{ZKP.Merkle.dirs}\{u_1\} p) (\text{List.map}\{u_1, 0\} \text{Prod.fst}\{0, u_1\} p)$

SHA-256 of the statement above

62552566cc5ffff408abcc3a50b3515384a679ff32986efe07457f61f9bfead4

SHA-256 of the kernel type

ffce92af6bae000f4c2b0c5cadd9904cbd44bc1293d2f1be0c3b2eab1187e4b5

commitment C

5855db8d3df2b5ccd08ed5f3d7dfa98357922396e43c1f7298c4edce249147c5

0d855729e43687d2da44893333fdf77f681421c1e081afb2746c61f13293ae41

f187f750098dfb48ef86dde6daf614204df3f2083b8de3a35c88ae47d82b20a1

6f202af9977467225798c9775724772bc724fc2a7149524f7a984fc6bf52aa3a

afcc9796f5b6f45c33ca7f4ed518f44d4f067185a0e89143ea0a93c8bfb35bf2

db94627cb937e8bb0e87760e88d6c8e1d2eb8c957ed5789399a17dd4aa13bad8

3f6f5f530bc86693a45f6a66b69bcb7cc7726aed1f675bdd782850d280c31382

d573f658d28799f070fc184903059ab087c2881ad801acd1f08e0985bc044ce

first message t

947c167c286ff4c7d4a3139916001489b52c5059417bf97a44799634668f8bf8

222d92848cda694c56d0cff629cbde1c746dfa2524f3d093e23f421432dd7fb0

845ea8f380568e3f584965977bc0e9873025e387355321a1891e2fae2f50dc98

d2d996d3a76583db816a25b966de5cedda15d7f5ae045d93bc238a61c0749871

06a3d76391f1e16f62806ba935dd929a8f8443e020def985752a85dec74a62e1

ac041589f1efc16840bf371704751d91a4a81708cd65a230ac50c8f240eb303e

baeb045ed55be2c46455d11000f9908e6e1e7a1ae83b8190f9941fd93cf0f847

019f3a020084ec2f21b8f64ebec970095e64512c943341dd9eab6aa827508ec8

response z

48218c863070cb543092d9470817bbb0b3110dbafc3dbc6676f4688ea769ff04

2f0fcfe301015d22a5e6e2851909c46044c98ea89238d36841efebfd3bafb937

6e459622a7257b05c0aa07bc403c58b6ef83241db532ed885c1754213c7c4efd

c7c23b990ff50421d420ea53401541590ba3c52a3498936580480d025fa929cd

013885e8aafc61cb29bb4c5a80c64218383b2f43f5dfd4ec926445186ddf7f35

7864350abd093a5e360e5b83aab191b82331c34c7a550d2c8d278d3a8d00bdbe

b32d1baadd7f3f14d46cdad9ea1a5f104b38b110e176ccab63a92b1398adf2c2

1f70db9a8a0ebbbe4f058a68bfb32b175e1faa22aa305ad6113c4a5f40499539

response w

31d7b49d90c0dc633a50b77c389b0e6470aef26b02c1b87f89d1c2193da625ad

d1c3c3ce4d6894d85de00faf397ee7f151879d5c3470306b2d9dbe75bf1e0205

087af7bc7d8f32ee15d97129e97d29a15115cc365bf88d7c9440289172601183

516ff30153108637b30006ee031c041588473753e2c11077c6900d601d17f942

028bce89fe9878468e8689e2f47b010a18db6397f57b708cc039430fbf8a8b6c

b0dfc4b81ad6407ca9a4615bce9f8fa5239f4ee00121176790fdf6f0b2d420f2

5ec7a3a543a5f271120bbc5fce8f9097dfb178e953b18f45d3ee7b89c05149f9

fec87632c9a4ffbf2a090bb0b44001865ad91dd5c0bb3dd4ff72db67a6ccb207

Fiat-Shamir context

458313c7bb431144c6afd192ad0b43a3e0b9adf7c34401a56ec52d498557c79d

withheld proof term: 119 bytes • axioms: • kernel replay: yes

ZKP.Merkle.foldPath definition, Crypto.Merkle:41

$\{H : \text{Type } u_1\} \rightarrow (H \rightarrow H \rightarrow H) \rightarrow H \rightarrow \text{List}\{u_1\} (\text{Prod}\{0, u_1\} \text{Bool } H) \rightarrow H$

SHA-256 of the statement above

2f536b3abfba3f015c34dbda2dcf23ec75076a5fed6c2181670132bed9c49ff4

SHA-256 of the kernel type

103d3274c68d32490ade6064f3908315fdbcb72b251d6d8d6aff614e0ee21fa7

commitment C

ea7823463ebc430d31b9dadce77d70098d543ebfe81cd18509ccf540e764a3d9

75ce29f59e32c8fd70f35283350863fd7acc6cab307ad6d7aac6ade2a13214a

d1e1ed5c767f4a2088681610ac8be2c39ffd690b31437c9340c9333844af095b

51c8d2c34c7c77e6fe9aa95f9368088dc2b5e3d4d30d64fa1eb2308cc5ad6d05

af09248a09d2844e74911657e125ccd7aaef2242f35b7020ed20ddbe0d5b5849

70f596af52379b308f8579ea195179f580c7e9c52cb9a93a9951f2e6c5710675

16bf64e6854083b2948ac5f0dc95b1a6a6d476ad8b26cf227c63fca4d3cba55c

9377a776afa9392f57dfb7357a391092f96dc24ce7d0e6c5971f139eca8ae300

first message t

873f0c4d9d0e5b9f857eb65da874edb3f970fefe9a93f62c365f21a741ce274f0

5838cb7e3fafa18b12ebf0ad6481847a5987e316e8f60b417bf286c3c1b12c01

1083019ec8b5411302976dd1ec5c52b5486a0cc84291bfd80321613a85db75a8

db9ea24adab32b7bb3e52a1f89dc3218c1469e614c06e4d6144d0f224a6da9f3

936402dd3acc7f9c33d0fbb95200f16fc8557dac738570723e548638357e0740

5b77990d4989b7dc31ebb08c0fc8dd1cabf62479140033525db5f6dde73623bb

ea6f884fe0669acebdcc8031fd746b7da8c47b92cf1073edf948abe986c7b3d5

f9106f53b35919e62501a979adde0d2129b264198e858032ae21339bc148efc0

response z

9349ac208d8cb67198891e6bd00abdddf2e8a658f0c47a64961b37cf9bee5fbc

d7998a9bb2b76b1a6069861436d15a398e3b68e648131f98444243c9156af1fa

bce7882c243e4c0fb981a85b608c0d299180a6bed076c4457c4548f97260b7c7

60178028cadcaace88143da6be0e0e1a32d88c1609d36072908df7da9b7083cd

cb5d0785cc9fc28953251b70d17293f734e5317226326e09e4a2d6d649075267

fdee15f8926cb5f9e65888e6ab43a27b1eca87cc461619115c84cc5d9edf4753

2063a1c727a02db75835b34b82143bf28b2f1a614174fa3df568aeb00ad3357d

be05d6bfe827e2df0aac6a006fef21a6cf4aa49ca604e00d656bd6d48edc626

response w

98d344467e36347261f5511f5e5c8110cd460228fa2d8a28a644bbf7c5185e62

f6cbc3071ded4df2449f51f84e6ae281f419f9f24aeca73142070932623606c7

eca4d270b54f9be59f43c91b4391f38f398592c33e18cab7a3da0e6423d1b532

7a742ad9a2538089ef886534e51c4de732b84d227bb883a638ab4a531bbe6bf3

22c944a33e3d4a188af8b8ba7d1868865679c597ba42f2e5a10d334a9af816e9

1d7c00ce9b2827d583bd58fd20eaa652c7f85fd0d66209aee34aa5b7ae192e1b

dbfac16599aff156cd5536de7c8ddd9240daf552e0420c07d380ac12dc6164fa

0683e5fea4da828a6b18121ac47505f9036dc9b745377f40ec93a8b72064073

Fiat-Shamir context

d7267a9e98a289510b13d58cf21aeb72acf1f9b47ca98666c38b8ff3d567c69c

withheld proof term: 2028 bytes • axioms: • kernel replay: yes

ZKP.Merkle.foldPath.eq_1 theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_1\} \text{ (hnode : } H \rightarrow H \rightarrow H) \text{ (h : } H),$
Eq.{u_1 + 1} (ZKP.Merkle.foldPath.{u_1} hnode h List.nil.{u_1}) h

SHA-256 of the statement above

8891dcB759762e1e8b214f5fab2b7577e3e0051ee65af8be0b81f8661e574dd2

SHA-256 of the kernel type

a1bc8861886fd1b5a3a8887b00502a53ac4ca3c42dfa0387babb35f07fdc7704

commitment C

c9c95d0b801a6ddcb95f8087c693862786b513d6a4b6713c099a06449dd6ab3f

acd498f5c86eba0dbd06a59c6d17b3fa809d3d6def32b391c9280c7aa76ca3fc

78b5e934824db0875b971575fc8a11fbecf927fdb10ea09ab05848b41ad1de2c

9e0dc87a222a01d1c99e8495f5de07a491da73e849abdf416695886769877f6c

5d12de72980a6d7c0be8beceff6b2c4091dea88a401363135f50b2ce261c0dfc

1df4932224d906a99372690822b02f88ac8726031b3d63cc479f4c4a22b7eb60

e2ad3c83eb5174646f170eed392d92f94ec8dbf9fd92a92b48f1a3fc1687b884

c8032c43d6e6ffe6389b5e560d7c5a5b7db4a11a4fc30c383bd4b15dc135bd73

first message t

5c68522e16aeb73de06cce8df4af29221a1924305308c11dfafacefdd8d9d28f3

547c63814c57fca538c2467f72418adfe0ff19cf4291107eed9efc4698c30c11

e41245ea10edcf2ad9d7b6271f61b3ef28dff641c6a0f5963b4f377960d534bf

d50c91416a21abcf1e9b58d2b53abc19d8ae95b73d5193731bd4b770f53068eb

881ef29c2cd447f38b7b441603a614ca2baaa84b9b3dbe6aa16862ff500791a8

dc9100fd2d5a5b60a8c5940bc9f8a5063581e42b0c427d371e04e4fdce8f4710

ae1646e7042bc67ce651330b35068c2097229eedd828dac279e0f27ec6e1dd7e

c7c53097e7784e1edce63b88799b37c72769542faf8b3fdf1ef99cf92182e414

response z

4bfea03e9fc4e50ef4a930e5b58b700853e280c345d7d3f9a0c44263f53a13ad

e8f4669edca6611bf4876a516e9dc5fe53e08f82402aa8260f6493c3a55fe70e

d9c71ef1efe9811a3c379c99aff12c99910b484e88d8248fac6896a93b1310da

1de9f679f3d8bb4d9769e45dc02a4efc1a78b2953d18d3de1adca8b209dd64ab

95682f2e7e5d78f3d43e4d4a267df785ed9236f5effd040f88e6dc8e7426b509

31e0e5381e4e6d385a1805752d827ab782172170393c968739a091ce56a131b9

e809d1b0b03ab3d382cea2bdf5f3e1d428e434bde93ca3c21ce730e1ce63754d

a2650c441385588c9636dfc749be4407c6c4832999e3d4e014d0396fd10e857c

response w

51f318acca66b52e5fd0d55f91a249d4874e9f3645c6911d90d984c3fb32fd13

d8d1ee6f55017295e32e9ec7b8118dcd6547df1d5152838076f3193a0fc31a1e

0c6608623f932fd7e9b552badb18fbd7470de8f7228e3cda9012fc77a1237cda

4c48e121e77c5f7a769695dc2b3430fb201ea4bc629c673c66aff8ca21191ffb

03aba42f68e7e48b1812302cbd7fd9990b379d59bd315b37a5b1bb361f35d61

a867df60d26f3b96ff044234341e8be2dba86d55e6ae168c1164c167ebef0f6d

6aa5e67b3f620b0751d9d50462de09c39db53219b45a8e7f2dbecfb0ef89da27

6217655d289006f8d41186268b586712a8ed0416b6bfa4ffd3d28f849117b966

Fiat-Shamir context

8965d19aa9294e3c2eb54c9bd3d03c0f686c369dd3e26630c9b0542e44149509

withheld proof term: 152 bytes • axioms: • kernel replay: yes

ZKP.Merkle.foldPath.eq_2 theorem, Crypto.Merkle:0

```
∀ {H : Type u_1} (hnode : H → H → H) (h : H) (b : Bool) (s : H)
  (p : List.{u_1} (Prod.{0, u_1} Bool H)),
  Eq.{u_1 + 1} (ZKP.Merkle.foldPath.{u_1} hnode h (List.cons.{u_1} (Prod.mk.{0, u_1} b s)
    p))
    (ZKP.Merkle.foldPath.{u_1} hnode (ZKP.Merkle.step.{u_1} hnode b s h) p)
```

SHA-256 of the statement above

22c8467b8a6726172e8ff000a0c9289754ba57da40820f92a2168b9724d32d04

SHA-256 of the kernel type

8f0281c839f017426421c8ccf8e7380cb2f26daa033d86f595172d75298994af

commitment C

641150817886e719abd4b5a2998c4a2428fddc4444cbbeec0b9ba90d6a7fbcf6
9e7cfec7d49e41114ed602c2b3a5dcd99ef615ea681bd601c8dfcc7493388631
f5618e17f3ee3c343be87cd54356c8942b41e94130ca2d36430a4e4ca1511176
4defacff147813dcd726bd0f035108cc42356f4d5459adae90d06a59e7e742b0f
91db4b2ab7a81eb503b6068350c69654a3eb837a3f88317cd76b35e54d5b80eb
b4e27c150f8ab380adf6e5cac6b2df961c7b013d7e3d286a9efa85550d4366e5
770b221e6046325d2c6a796bcc2188395e998b7d5c837f8f7d68831889e450ce
ac6e1e1cff5932eeba344a29588c6bf24ebbf98df82c9606771d04406d34674e

first message t

55d21f16ad314bbe0bd652fdbfd0f21185e7c3f0fb83b92865ca56b2ac168f40
357e62b5c092d49dc8421da46ed1a169db0ce55a97dc61eacd5f5dd2429b6109
6f740ef6c52a1fcf6e4a638088af98675e3f2daf8711a95b3dd162516f4f1556
d71fb650cfe188e23a8f883e873bf2ed427f5b58be00b8fc2f8ce6cc6ca1386
6d3e41e58c8a734a81e8ae893a96035ded564c44fe66e41dbd900f3bfb6f82fb
9d848b9b8fa73a91e10c9de322fe52791b845dd3dc2c69f015bc4ddb4bf99044
bac9e5ffb5bf5535cc07cec7297c141a2a742706de611a57788d0cc191f12c4d
fd336e646d1b9d631c1efecae2a216f2326d2f6979b332714fe85c856f8e742

response z

5ce1e07a6e90d23abfb0bc35f1db9cfc4fb2231fca23be1f68cc4e471f8edbf
7d42293086f494a3496023456abf0aa8e4dcafdbe3af2620f90f7369b28a28a
32da58f6d239473aad40329468d6bb8a35c2da22cdfea448cd458614a088a546
6c79ac2ab4efa3db3eee2660f62459389a9fe3d1cab0bb594a035fa5682528c6
1d448d38b4df08ed8e74012d422a8670005d2355902afbef9523cdbc9baf573b
1870979134522475cd201b2db75df83d097b66171a33ae9628fcb64bb915032
a71b859d6e906eaba83f5d78c0be1595026ac650e34387bcf953870d6068bbf6
247ba03068edb78c1c2c1f80fff59ba20399b0a0b7e9bd82d70f4eb874bf9fce

response w

15f57acda848b0cdaac0e0528b71fa445f7da9adcf0f6c0b1a84339bac2a988b
ad9ebefc742c926a740b54c8bb9ab452bd493cb8ef02b471b5eda717f131a5af
34fad00bd4abb3aace2ccffd25b4cb39978bc4c965777add7c7b053512a85289
c15eafb28a58890b3b6d597fd829d720e86b3c828952ce66765c521c769e9e87
0987c53378c5d8ec6f5fa6063de06762a23f281e52ca72c8345102e049f660e8
26b113c30e03a362a83f4bb27a403299d63ab6e5c1053d475f4d6d17071469c5
d715ce04380afadc5c0f5e54e65ca7418ab0eaf0379864faa90c0ad7d41fba2
c41695cab5f3d027bb1b42f81aec96dfff7aa603de3b7fa161cbf32342cbb47c

Fiat-Shamir context

90d8c83ef8a0cf69e540893622f52247eaa660fc12309bdf4c34ae4343f62298

withheld proof term: 244 bytes • axioms: • kernel replay: yes

ZKP.Merkle.foldPath.eq_def theorem, Crypto.Merkle:0

```
∀ {H : Type u_1} (hnode : H → H → H) (h : H) (x : List.{u_1} (Prod.{0, u_1} Bool H)),
  Eq.{u_1 + 1} (ZKP.Merkle.foldPath.{u_1} hnode h x)
    (match x with
    | List.nil.{u_1} => h
    | List.cons.{u_1} (Prod.mk.{0, u_1} b s) p =>
      ZKP.Merkle.foldPath.{u_1} hnode (ZKP.Merkle.step.{u_1} hnode b s h) p)
```

SHA-256 of the statement above

126ba0482892ddb5b71e76f350d041f537defda5e1361fe62ac109cf3e339

SHA-256 of the kernel type

0814b105cd8dfca5d59bca2df0b6ec04892cc135c2a0e58770c4bd07d2dad43b
commitment C

25c4309f941087f453c015775702ba0486d33c7f4bc068fa7603c06bce4f8654
7a486af92263655d0c38c5c754a531f17ce61098f10c618a1bc5a1f3b5574f57
3a9900b48882f7820f5143391d5f5e97f74b2d4b8adb3c3eef0c5ba21ac98f1d
2ce60e59b3bbd26edb45513926e2c792da4bdf20ac368f3da0e3846b1eae25a
f4f9bb7e2c0d20699acea5dc3a13e7149f4df2dc30495bfb76f2fd65392c8be3
810e15f4b36193cb249f70169651ac7d5f02e6e8fd807d9e958ca30cacde5835
02ede9de31cd0301174927943d4e56e0050c9b2169b4b969bce3ccde0f19e3c0
a8e7924503880d359414277b326aec5d5cf25da603693802f16418902937d733
first message t

76e3db2d8a63eff0607f7a03f0313a30966540b03852430914aae81716075fe7
42df85108537878abd0d67c7e6402f697a6237fc8faa71cfea4141ccc0bff6d1
2fbdcb081c385e110c065016cec92d9550235b8fb30a05c22ed69e58bf0df792
bcfc6599ece1d9cff343b2e5a3c3ecee72437809e81021e47a5ef003ade8ea30
08e57413485c8752d68a6d29dd6c1ad83ab5bfd735622b1b91e40153f196e7b1
b7398c334a169c5a285a4813419a5f2ea2cb43a01928374f5e06a2639f064547
529232e3dc7095be5a0c60a81c6d8831816057925b355d89bb52be2c5100c48b
3b95201288498132d4f2afbb653fc69aecc174f9ec4a5aa627895c1a590dc78e
response z

2d2620e6db4204837ab44a7e9c613a70fa4eba442af627140dbef12619a7a431
d878f1253fa6303298455bcc3aef62366e76fc3f0e91c472d567a1ca5b44de5e
cb9552e6164c1a94ef9738b647cec07b0e312ff8b5e5084be05ee6931285cbd3
4e1a5a78adfe72d5f1d9c9e5c41215501391999ad5cad7d1b3d7952ccdfa769
e62d4b975d462d800c421ba41243c831d7957ce8567e15de0e72f8392202ec39
70eeb60d12ad8aaa9df8e0752bda93590152ab5e8bbb9508b7d0d781335670c1
2b03154dd2db5a79924ee3b82e83b6e2551fec3c5750fc7bb9212143baece643
394056866e404028e921ad11a07740587694bc52585323b8242d0b5563787f51
response w

5f31ef764957d0e9343bbe3f6de19e168f9a5c6ab14979808a4704a1c55830c5
c19c68a60f54cea66d162d2aa304dcc848f84e8bd6e04801b626cf2a591901be
d267f331fc157daaf3134edbc9c67c63de702f2b0abb3cc907483adaed11f392
728cc82a6922dab1b8297779e588d4313b30159a99efbf9f9692bff3a517b4f9
466a3f2ace709713289e9bf30d3cd10c7b414b736129b9ae8200d4d3983f7349
ea235dd904527157cd053db4d23091c94c9587592c9cd47fd04b0c0220f97dfe
e83161efaed91553ab269499fa0433c450a131e5b31bd524d6210c61bd79cd41
dcea3928b01979f162066f3e0328834dc4fa194e8ce9d0c26ef00d924a87b784
Fiat-Shamir context

8b04fc2f70d27014b5e45110f1f03fe2680f5fe4e7efe5270f33b2aaf91a273e

withheld proof term: 42 527 bytes • axioms: • kernel replay: yes

ZKP.Merkle.foldPath.match_1 definition, Crypto.Merkle:0

```
{H : Type u_1} →
  (motive : List.{u_1} (Prod.{0, u_1} Bool H) → Sort u_2) →
  (x : List.{u_1} (Prod.{0, u_1} Bool H)) →
  (Unit → motive List.nil.{u_1}) →
  ((b : Bool) →
    (s : H) →
      (p : List.{u_1} (Prod.{0, u_1} Bool H)) →
        motive (List.cons.{u_1} (Prod.mk.{0, u_1} b s) p)) →
    motive x
```

SHA-256 of the statement above

f4a70dd55d09bbc5ca80a99d443956d8326dde245a3880b9f5a4b7584bb09f17

SHA-256 of the kernel type

700c333ab89c9e119bd4d1e3380fd56dab37b926092f77b5a4cd9f9caf2fdb7a

commitment C

d3fd7481ba5b7754641902e8c2606733ddef1937dcbbc5c5e74583fea71eb856

695d08e83f58203f9395774e7bcf807fa06061d129d3ce89d2648fd3626d8e18

5f13112835a2cbce997719007254a01431eed4a7c1cc41cc198f8f32d1c8e21f

b56b16198676303c500114663689396f1664f7f7feb348d32e36fef36934029b

59da3cd919a9c740cac626ad846894b0c1eed76b2b23393a9978092444adb3d3

e2a1b254b9f8055e5a61b75550b0db5bbb6b50f3b1f6ad679c045338f5be46fb

006fc3813c1db9007cd843fcbadd04bb27d37a9f6b511904ad73b24189646fb4

bc3945b61fc00e2b18e54cec04bf23120274a9b99059c373e7c8e51d73ab3ca1

first message t

aba1a41b75cbb5137b6efcb47da85bea25f0b9af423f519932b75914e267b04c

2113e2fcdca3723694e6c6615aaf30d37d6cb0e3a740896b21cd60778eb63b823

14a21d2218fd4c04ab8a3fc980b13c08e44640f1197c0c176b4771bb4b266800

c3885cda2b28a738fdbb66fbb38f620e007876158788f95a59b9199ab79a66c8

24a32ada5336c209783315f6c47b1d8fb71b2e55124ec3240496658c6e44b753

39c505a83b9121c7734f6bcc2cec9a6c49caa40e52424455aecf2dec0df462c

717cfd86daf02876ce9968354efcf702c02df03bbe76453590c467274386378a

313727875ae6b566ea321110fbc6743fa4a0fa6287b70ca5d7cd6da8e6c89a4

response z

5244fd6373d879dfb64caa319691de33975fd69a680dfb143d700b6e306c9526

bdd4845eca517745f3185d0db9d1c5c3534d598703db626fc5c48cccd3d3e778

19148922f4db547db6167bd7339a8e6479df041b3914ab215aeae2be3bfc98

891a0468fb33429840e35253d7c889b878aa1049e8cdac191730ef6a4910e881

443faf1fc04f5a687508dc740fec08a02a864b75541b7fcaac85a1246028a1fe

acb562d19cf6c3432337d8aa75eef053c3469135925679f612496246daff2b78

ba7fa58f7cb2391324fa89a12de2970dec81ebd8f167b5899ac60b9a7bafee71

18de8c134261d1e52dd5aafce22ca91222beffcb47bdb309a9d9fc613887aecd

response w

4d5ff829eabc21bc9f85c4da362ca659748880ff96460d4ba2cbe6f761499cfd

e689007ee5052d3bae070657175951c26c417a772c0ccb1cb531f355f12a9ee4

4dc4fd30d6c8163b95c31fa6c3f73616fa9accb08a0419a942ac2bd45e2a23de

8ab49d3c4987dbb5579fe827aab84dae489b226e17537bf1b555d6137e87909b

74a4067b9401de0b9b91c78e7660043e56327074b5fa39296fb3e1bdd3b0964c

ce1a4f6fda329932fffed46316b232a5d640ed3f7ff093ef22b981f613c7b2c7

1b24533de4d8f79eebe9e3c31c7e400ed2bab33d036ef1ba3c80e1757c2a2ab6

16b663d26de4fa9af16fd67ec7fea952da6de17792f696bbe9b7e4a7ac3756d

Fiat-Shamir context

d288fc43bba557749b012bb8df6b65e3decfd20cbdd9c172cecb5359a2213be

withheld proof term: 1 466 bytes • axioms: • kernel replay: yes

ZKP.Merkle.foldPath_append theorem, Crypto.Merkle:53

$\forall \{H : \text{Type } u_1\} \text{ (hnode : } H \rightarrow H \rightarrow H) \text{ (h : } H) \text{ (p q : List.\{u_1\} (Prod.\{0, u_1\} \text{Bool } H))},$
Eq.\{u_1 + 1\} (ZKP.Merkle.foldPath.\{u_1\} hnode h (HAppend.hAppend.\{u_1, u_1, u_1\} p q))
(ZKP.Merkle.foldPath.\{u_1\} hnode (ZKP.Merkle.foldPath.\{u_1\} hnode h p) q)

SHA-256 of the statement above

7f593bb53fcc89383993b3b5d2aa0dfe9e4c7ccbb5013a7afc8cfc63f6937cbf

SHA-256 of the kernel type

b216ea5edb9d9926ff73dc059d386b2d9299ca0002fa29991fadedb180051a34
commitment C

9740c6327092641f37f8dbb622277920b455b37b922687da72d0bbf0dcd4afd8
535edc2c78da8b3cb404117ab55f7f02a9151ea44bba3e5818ce6bb86f5e18c6
c91fdff26f953d795f38a32449f58e4f1df6b0ec67494580d06b2d987b4573d1
3ad52843ba5ef964d2943f65ce7cfb5f438a7f71fa1fa79bbc01703ebf7fe47e
50c66ae369b1f850625beb25ac324e281ce0d10404a8abc60eaf6fc8cf6e072f
af276a67840101aa922ddbcf1668a98603f35189fb23d8ca40a08088f6c0e463
a3f320f0659eac3d24773bba72c6cbe6deec06410bddff14703e7183b7806ee
8802b44867598077f3b2f122f744193bba1982314d298d4f950e0052835cc6f0
first message t

c5cf1efaf2538e44deecac404d1c2d15d8e4db1ff07a1bf8b0484e4befccefbc3
97dd24699f6ae0b076fb5eac88ec613e25b88a0bed00ff6de90baac40d913149
0c78a09c8ba9aa02e759c45642130a2f28690f84cdf6c557f7627069ab72cd5e
5d7c840795965aee2a524316ffb97a1fe8f592aaff0b0636f1522271fb1efd42
84a99d5ab2d25ca0a53c8a611f983a5a1f0305b725735dcfa9fa089eaabf8d91
ba0dbbaf0343a34f5a16449de1a03a1ca8b464366ceb9f37f4db59a92680042
5d1474f19f68019df19c5b0801a5e205745a88f66b4f2d8d5796b145692114f2
a47cbd9603e28b5f9ee884c5c4b39724e49754d0dfb25b56be82c0c1219dde22
response z

72abe946daaa7f0a27dbff71e81af0f2a594d2cd27c5a425cb1ba21477577a3f
1223b9558d857486260fbb95b99a04adf7e9b8935f73f737eae665c04ada444
f74e04df5689dd19d6085dce2618980267a267d0c94a7e7f845608417e2a20ac
bbfaad58eb7c8fdc9d61c8d81f2d1fbdab6a8e67fe6feab77f41ea837958826a
26058b341610a4e94311e1dd6e85d56dcb217e15ee571e94bf7996a6b0634098
0d6014cf5c12d038e4d3e059ec51a1649b2be5fbb98869030a468233f4e472ac
451c60b96940677fc4f087ba104ae9fef5c32ad864db0b32fa2e4b6b27d32079
72f17b31a5b89e7707bc7e4e1fc7d61de54d087b3437965bb62130d55781680f
response w

3ef8eae420d112591c5d5ebdb956e9ff84a0d4e3592e2018c9f75e6d8ad1cc9c
07ace24d480e425b11712e855b5ab5c4d863f976bba8a24c4ddc4bd320f3e7b1
02e77fcd07b3c02d224d0e7bfeffb76f3974048c51178463a5bc2d9aae8aeb78
9e85c66f5328f1fddebe38a456cce7fc0c21bf91151199cb1f1c3f45b7ab223c
28dc5b9f8c13c236ce1ab83f7cb84168635cd16455d2ba456cb67352c04b4123
f392f8e263662f5596b09539e61bad5cacbf314f920013dbb357e62ab3feedf5
e837bcd7b7c0e84b8b46c1451726bf9d52ee3cbf6294f25b695fc9d1bee305e8
765a2e8c9ce085e8ef69fde9d62720926c1e2c15b2ba4812e260cd69dce5d8e4

Fiat-Shamir context

0871ee8277bdec07a3f4e72001b83b65e363df8e4252a8061b7e5cda67d01fc2

withheld proof term: 3 729 bytes • axioms: • kernel replay: yes

ZKP.Merkle.foldPath_eq_root theorem, Crypto.Merkle:60

$\forall \{H : \text{Type } u_1\} \text{ (hnode : } H \rightarrow H \rightarrow H) \text{ (t : ZKP.Merkle.MTree.\{u_1\} H)}$
 ($\text{hp : Prod.\{u_1, u_1\} H (List.\{u_1\} (Prod.\{0, u_1\} \text{Bool } H))$),
 $\text{Membership.mem.\{u_1, u_1\} (ZKP.Merkle.paths.\{u_1\} \text{hnode t}) hp} \rightarrow$
 $\text{Eq.\{u_1 + 1\} (ZKP.Merkle.foldPath.\{u_1\} \text{hnode hp.1 hp.2}) (ZKP.Merkle.MTree.root.\{u_1\}}$
 $\text{hnode t})$

SHA-256 of the statement above

c82df551dc859d2e0e0eae9953d0a9088f4953bafd09f52daa979888b4ad401d

SHA-256 of the kernel type

3c45a749eb4895e7fb07aa703be76aad4f9672a507951e396b7433f5744228fb

commitment C

8b5bac4e8fc51b2d08210174b0cb38b97ced5027be4a0ee56579cd3bd139a2f3
d3b4587b4010ab66aba7c734b5efc5fe310c7dd05912411a017b8bf3cf6d11fa
b3b7da8b6df3e0da5645423946ec4b7a230ac96934acdd0844dcaf9ad5cd2989
e190d21138922dafabcbea6f8e8ecd8054f8057d2b7c29d9f95996692f31e241
49a3f8fcde4303d442b1b7eb76ab886accec4be8df20bb8ac241438134638512
c1836453050483140152246ae948637e89fb159df14185d29e5454c08dadbc2e
a385dd612d7f1c9ee5138800432cdc959d61999f69d51f188eadd92f807ad034
0d0839aa985f206f74908856f85ee67d7cd97ebb1aa6ce558b53aee1e049ce07

first message t

302cfbf4213296a4fc0e3d150a8c0531460c1d8c03dc37317246e5613e0fca1c
3cd75f24385115121de4989c3154ed807235b543c6b23494a28f360d20e15b44
d7e52d9de3e5eff62c086623b54616f9aa5107d31a6b495d41fafbf8ec40ce17
a6274497b942389142feae4238ecbde299a27a493a26285bac97e28de5f5ce5b
e5b5b67bf97ee74613a9c684f3cb533f53363ad526bee3a04e0841ec600c8003
08aa523bae83c917487fcd2a1acedb5a23cd300038860455c5f577f87891815
813728d176a492e3db3ef51ab02e1885a58a2ac6ea92fbd0c425fdd7285ea3a3
f753f65aeb105866d6421d2e12186801047b9fbbf9d8cd853c2413db8e9e096d

response z

765b11a569cbe4c218b0d70b47d15f593808a383638e6a6097875e5a313ce6c6
52163ae9083fa9e49184b4448b3405a628ae9b6d91aed52f7f1ea88366d3e76f
7f441bd5492daaeba5e48ad1fdc3290a0b9220ac764e840552dfcb77d63626fd
4811162019416f3c3e38834d123b5b55c402ae6d54c9cdab751baac7a65f015e
ab0b7ccf6bf084e01817336ee84be7feb9110c81fdcd46c6582c16c2fafe06f6
fabacb7314be912cc84198c9a03054a7791ef94aa54d3826940d8e265c4314d2
ec104b0e66dbeb3a0d917decbe503164eee216864a54b984541ec9970cfb77e1
cb62d08038b6b80900467a99be020d3dc14bc1d7eafe03664676d445a1939b37

response w

80d4380eb67d104c055324dc183c29b748ce0c071a37e25b5695f4c6395a0454
610f5df3075cc0bb620ab3f872d7726a773725612fae3ceebdfc5a433831d1fc
cddf318890a498432c942505adc7c10071d80811048a510dfe695a368ddc83b9
3dc22124dde94e674dfd17594318483957b1cea23e4559267bff452f6ca3b2fa
44ae9f8d3541dd52bcc188fc38c10718004d833c78a80d392a53954fef2de2b1
39892c9e488e1fb8ead959187e924812a71e61d9147b9e6e944847d65c39e6a2
6188a9a74bde991d3873fdf575f36c136d8a0d0b5e706bc3cd36fd8d2f8bb1f5
878a7bb3dab34584b87d47646879bc5f34bea934be5ee488b1c1763833d37f4

Fiat-Shamir context

559fb714baddf4543db1fcb3b8a6f6996af3a6a3c2e7b8d33ca7337c8d05851c

withheld proof term: 81 230 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Merkle.paths definition, Crypto.Merkle:46

```
{H : Type u_1} →
  (H → H → H) →
    ZKP.Merkle.MTree.{u_1} H → List.{u_1} (Prod.{u_1, u_1} H (List.{u_1} (Prod.{0, u_1}
      Bool H)))
```

SHA-256 of the statement above

2fe90d58106dccb32d5dd440d025a1c7edc64077fefdf4f7ad77e530cf5e014b

SHA-256 of the kernel type

1fd92e5deb7c9e139d27c4448e2654388d466b3afd29486f49177df472255520
commitment C

fc07e42b75f1973e4b936b641bd5c7514d6f78392217418a3c3470f80861d402
0dcfc0277f8891a7c7a5b0f9931eefbdf00ca100984f3e95ff5805243a07c409
d9e805d1b1c1c6ccdd8b056293a0a2ece8819dce425498b8130f82ba879cfd37
69fad441427b42bb4c4f21b745d789e85f3480c6fa3030cbcf9491f709e17886
418f18ba2732a3667ef321edad438dac9a4ec84adea3c1835113f52a3cbe2f87
83cbffba73884c2051446f04e24cce9eacdeeda60b603dc0a08088eb43fd9b6
2c73373d8cac116256ee11d3736b8da11337402e24ea41a289791108274477a6
4b5b072ee6015b246bec437c79dfa39f38b66ad700772d88d3eeaac28dae3e43
first message t

af7819685e31acb6cf3a4c292a541851b38bce1fdb4e9297c3031cd39d3ea368
272e755fad077ed659e38e747092c153db5de7e217b7bf752f80dc2c87dea992
0016d176b637ff29524d9f0104137d05c32c09f041d2d93dc02b8c37fc14ae2b
41eda061d18ded863a32e2b61e66992e14acf903b95941bc566d9826ae7ed7c4
46924165bf9c74abfd8e0bee125eccbfa69469417be7b09456396b984e42928
36770b2d2d7360517c32b155afd7cf58ab640f2fd959584e5c4eff6e8b52894e
e8de0e81eb748569aab04d7c2aa941e15deb9f89af2f31e62102ecf92c96f77
751d28e437d80dcb456d720b3802977341b516f97cb0c6257348f13b248fb489
response z

4fc1ace47506488ccd1fe13f6d48144144af1a0f32cf26d5c596a339749d9a7b
a3b972df8fe72aaf5ccbeaece114367fba2e7339d1d7a0e7c32d11c7ea69fc26
78b523dbd47001ce816d042df21a6d10d4e6b4e09a950e6b202b243e3576f34c
1b738bd0052bdcfb46028b99ab69f30c9a31da4dc258fa61deebced137b7d634
aa7c9f89ff8105558f80df090aa01bcd29c1833afd3a954c1f5c255ad4988b83
1891aa2ae5b2fa6b5160155abadb59c6b816225734477c336c9cd784bd09adc1
e48832c0c543c1d89fb77413d54504c910a454ee7bf063340f886e2c68e02827
faca10adf323babf06d828057de580b514428bfaf811086c32a670e3d5a60397
response w

fc6464f47e7f5d783b95f2ee7d55f9ee78130a4011f07cfb3da44519e5f3cad4
3289860978f7707fcc4d1ab602e8a79bc157e37bb7c29a7ae962a09259854cdd
596e59d9d6a00dbe1046bc86f0c93d7441bb188beb57494b4488677c9b3902d1
22d33f78aad360fd25e0edfd957fc587745966e1c30c737497234393295e45b2
1b04d0f5a9ba7f9bc348473506c52181ace87069bd8e014b952cff30bc1644ad
c6d2bb3ec883851a156779f765dc25477f75dba7e2882523961ee15977a8b610
3072742a1a3d6ad9f6e7a42eae8fa2c2f4dbf6adbd46f3f3f95cb21050d4136
c88e5ee9e1b51f05dacbddcc85afdb0f6bc8fec7f579b77ba6d63cecbfcd20d

Fiat-Shamir context

2d5ac86352cb057e905e93f8b8c7f13999b245b87a92cea58f5deddc1e743ac1

withheld proof term: 4518 bytes • axioms: • kernel replay: yes

ZKP.Merkle.paths.eq_1 theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_1\} \text{ (hnode : } H \rightarrow H \rightarrow H) \text{ (h : } H),$
Eq.{u_1 + 1} (ZKP.Merkle.paths.{u_1} hnode (ZKP.Merkle.MTree.leaf.{u_1} h))
(List.cons.{u_1} (Prod.mk.{u_1, u_1} h List.nil.{u_1}) List.nil.{u_1})

SHA-256 of the statement above

bfe7719d7c2785914e7d00301b3d19f38d75fa5ceff54c315e4bd207b8c34161

SHA-256 of the kernel type

65ab79036d1d91eeaed3a37b62e4feef1007de75c544b141b37959040953ba3
commitment C

37613f895e882c912c0d6dfd8a904f8872a0220918baeafbcbf4a962562179d
c4955fef721ddd980d4ca1fba3f59f5e1166a730cc5babdc84818d964357e069
efac72f11338e17b435ede84b4b7f0a9070c46f837cc9eb2025cb3096f158ca7
e95b36694b6eb693316485a49144ae994f0fa1b753e3a8ea670100cd01c4de90
db4c5d76c046bdd1a7a725ee08d4b955ee6fcca1bb719bd812a1d443759bd085
97b502a604d89b9fe3ebbf6f41be090a83db3d882dceb3c5e9e42318576269c1
8de79624a5f78637d9b9c0624c83c2bb52d20546749a2345c4a89ad17d44a59b
776cfa7e6b3b3cedc63ad14ae22b8de7a4088c7a95c320f5e418f746ef2a8f85
first message t

15c0af765b9944db89d784808003c659a086e40cd237cb9c28d1481b6bba30d1
64200b2d696458590237cde4fc9b51dc6113e200082d3d958387cdf2568b3918
238e1c7763a1e755998ef27ae14d318b4d460e107a34acb8b86d31cce438d829
7acabfa07528e7c039f107535680465e69263621b56fe998770381012587c9af
fd9d5bb30685a8772cde00a472b28d43ba889aaf0867f2a971cddb38b16b4cec
ed3b18d1d66098dfe0562f79ddff5f7927b8dc18dd9a9dfced33c37566526428
e55e53e327626214e2fc675d5cb34eb59818c6233fcb4ddc053b0d1192ca3810
efb499d5cae3cbe264d913f273a51dc078a870bffe7175ee16f919d660b69957
response z

69cb8a6bf2f068fe9433fb7996467e52070b29a5580ac7bff03ca04eb6e25596
949e91e67adc790e09892ec463c6f9ee0d9cdfa04abf99a70a4f6c1df12089ef
5f35af28327c945f20c0d0c09291f6e6c55b83f6555905cf5e7ef0b43e1fe584
59671848bd212d72f31f336888b0eda6cf762ef0c76676a8cf64f5b14ae9a2fa
3100d2cee79ff6817eaeaa8d9f63c6376bd91b3c309ab8094072e514c87b14b2
e02e698a2f11d363077290b192c356871cebe43a85d8115c2097261b40e281a9
a16b9ab2531bafedda61a1686bbeb74dbd35080e3d4876bb480a20897dd16074
a7aa6d4253c517a5f178217c83ad90679b16ab2aafe52cdf57946411e35fbddc
response w

7b5ce3da42e09a2977cdac6ad8ac7be8792862592377436914c734dfd99a02d2
274b1edfbcd490fd50e29167b4904a3811b46f1e1d2d7d6ec50200f5fd56eae2
c775b7f41a8e46a17146b3a06ee4e6443fe030c6ac37a0595432e2a3ef24cee0
6379b34cfa88d9a1c4e43cb037dbd46a86a9d318cc733f9e6dbc0afdcbe45c6f
e6ad73b32f5ba0933e8617e26033bb2f7a8c92245ed87b5efaeb91d081b44f7e
c04a3cfa5a15d3dbdb9615181033a6ee3d6dd90a402c99e5e1785b013d12182
688b4d77288d0c98386c4e1c597cff542344295617979bfe398e7cb03b1e824
a6aa4df0c6cd062dc3edb0c36e07a433784c207d39220bccfbd4fe059a624d3f

Fiat-Shamir context

07d7435e12353888a633b0bf3552a2c512837cbef2ff32c554c2da062f7ad0fd

withheld proof term: 208 bytes • axioms: • kernel replay: yes

ZKP.Merkle.paths.eq_2 theorem, Crypto.Merkle:0

```
∀ {H : Type u_1} (hnode : H → H → H) (l r : ZKP.Merkle.MTree.{u_1} H),
Eq.{u_1 + 1} (ZKP.Merkle.paths.{u_1} hnode (ZKP.Merkle.MTree.node.{u_1} l r))
  (Happend.hAppend.{u_1, u_1, u_1}
    (List.map.{u_1, u_1}
      (fun p =>
        Prod.mk.{u_1, u_1} p.1
          (Happend.hAppend.{u_1, u_1, u_1} p.2
            (List.cons.{u_1} (Prod.mk.{0, u_1} Bool.false (ZKP.Merkle.MTree.root.{u_1}
              hnode r))
              List.nil.{u_1}))))
        (ZKP.Merkle.paths.{u_1} hnode l))
    (List.map.{u_1, u_1}
      (fun p =>
        Prod.mk.{u_1, u_1} p.1
          (Happend.hAppend.{u_1, u_1, u_1} p.2
            (List.cons.{u_1} (Prod.mk.{0, u_1} Bool.true (ZKP.Merkle.MTree.root.{u_1}
              hnode l))
              List.nil.{u_1}))))
        (ZKP.Merkle.paths.{u_1} hnode r))))
```

SHA-256 of the statement above

d46863ab4ce3c36f9f97f443002465eef768d5aa6622b66e13bbdea5545cc05d

SHA-256 of the kernel type

ab00d4d64c8068565b5d9ee46cb2b228530dfcf253f7c25d69560fffd49ab6fe
commitment C

763e31e33b26b6e6017c91d393af1ce4eea6542e06dd7d682b58ff556ff59d4f
15c880acd3ae6915113ccb5dac1171a30fd9291d48c5b87dd9048d5050bdf5da
745a6d443a6e722bfd10fc86bcfb45515490c7233789ca152baffad37861513e
8e7f102fd7afe32a3fa62845065db8949fc16454c333c9f09f845df4d89a66ef
7e863d4ff2326346a995fa80ab50cea8669534f000759407203225bebc2dc5e9
999281f11bc91cf555c7c66d0c320f0f44c198fe56ea4d887b437339919d2cc5
c067cb852c1f6b8fd35c499c7f51f280464e539138883324390c994a5f598d51
42236433491d5476388d1af0896615afa79847038a44f5d568df68c7b25e9e6f
first message t

2495fee806484934db98b82d245547dd8464a17cb68b21b9cf70f388c5e5c192
e0901fe6ce96c0459a903d6cec828ff3b7831c9b6e1e835296464627c526ca48
506d161c7f1527e6612605f8b173241e105e2a154932507f225d1989fb436ec3
6a61d306e15b4eba8440e43259b978e70412fb70435ca6be528ca865f9679104
64a74a23d56acd72c54fff720a526cd0c7cf2d7751a049f23e99e4a1a7b2d824
98d2114d8557d1618d9458203dab38b9d03aca90854f99b7ad48d9c72eedab9f
264937fc28ec35a5a08f708a7f0469d03c24e35de37e51b967f0f023e4c9b470
06f29db431f3fe2120f33018d450366e28bd60ef194cae5c0a738fca9e509a47
response z

58ffa261de2e63098d9ff93f74c4cf6811c1a900b112c65315cbea382436b444
ad0d5f8af6ff29f3f9a3fcb1b20a4fa07141799fa2d44d8299473a4105f055b0
a47756f0fc5ac1b8e7e5cf74591bac5029854a357267752a33004c71c1b7493c
b97a7a8e5ee3f2376e87bdd547c223f0616ebac798b584608be1bdd645236388
be4753a0a84fa6d47516ca9e3744c6348cb450f03f19bfd8813a4ef8870142ce
e2c0bc70a0319eb165c843d8ff04d283dcfe381e75d1cc0c2e00c192f145b275
68a978ebc46f4f544d7cf2a583bab28b8a4dca70c97d722f2a35364dc4bcae26
f5527498a7559b3fa31e2c5a4ab901d8dfa87e48bd856617a811d6aef917ed93
response w

6c1b994884da8baed79b4f9c7315000483b56bd0f495aa0bc36b189fb6a67977
8b44dbcbaf4f5daba5b8095273db780abcb2e0a3fa454b5071a2bdfbab8b061e
b27c6610b7364692fae22aab7f4a110729f12c8ca81f1b2aac0d990163b7e278
6c7cd5593385ca3a0391034ce7f6d5106d0089eb36c120dce3b3a709a26716d1
defd7c109165528736a637b4499ec137395bd1499d02e0e35798342f276332fc
267cb9b19f7f81749a40001a855a7ed7d610fc20c1afa236840d4fa66889df83
0f4c55dbc1eb985f04d9f69e25d2ec061f91e0d254812772d246a4c997da1f01
9b2dcc2942fe30186cc7ddb283882fac16ba5eeefd4b7e89f8432ab28f5defe8
Fiat-Shamir context

02010bbb5dc992774b663e60059afa9de2ab1849b25101e8091a9247b1f27ff448

withheld proof term: 264 bytes • axioms: • kernel replay: yes

ZKP.Merkle.paths.eq_def theorem, Crypto.Merkle:0

```
∀ {H : Type u_1} (hnode : H → H → H) (x : ZKP.Merkle.MTree.{u_1} H),
Eq.{u_1 + 1} (ZKP.Merkle.paths.{u_1} hnode x)
  (match x with
  | ZKP.Merkle.MTree.leaf.{u_1} h =>
    List.cons.{u_1} (Prod.mk.{u_1, u_1} h List.nil.{u_1}) List.nil.{u_1}
  | ZKP.Merkle.MTree.node.{u_1} l r =>
    HAppend.hAppend.{u_1, u_1, u_1}
      (List.map.{u_1, u_1}
        (fun p =>
          Prod.mk.{u_1, u_1} p.1
            (HAppend.hAppend.{u_1, u_1, u_1} p.2
              (List.cons.{u_1} (Prod.mk.{0, u_1} Bool.false (ZKP.Merkle.MTree.root.{u_1}
                hnode r))
                List.nil.{u_1}))))
          (ZKP.Merkle.paths.{u_1} hnode l))
      (List.map.{u_1, u_1}
        (fun p =>
          Prod.mk.{u_1, u_1} p.1
            (HAppend.hAppend.{u_1, u_1, u_1} p.2
              (List.cons.{u_1} (Prod.mk.{0, u_1} Bool.true (ZKP.Merkle.MTree.root.{u_1}
                hnode r))
                List.nil.{u_1}))))
          (ZKP.Merkle.paths.{u_1} hnode r))))
```

SHA-256 of the statement above

9b58345a1ad2fdee8a4f2a949263f41eb3ded1377a5b213cf2e124722d07ad95

SHA-256 of the kernel type

be57cac1c2f83403be18b879285f81727d3da6ad74a6e17c566e51a0630cf5b6

commitment C

de33fe630b87e40a9996970588dcd38c97e8dc0545c22a1e7a93b1ef47111e3b
7baa9e2bad2291ce99d86a8a53838bc69cdb8095fe3d2e8d226d06bdfbc95ef6
a9f5abcaefa747ba96519982b5c0f3f3b3ad610799dd0dc7c1b3ec3645eea32b
ed27edd1d42fea37a3cd05fcbcff39b2bb348797da2eaefcd4349571a272bd5e
6fe0ba1b82c2f8e25a6272ba6468f73f333f5915949100b54a4ba880c6a2ee0e
a4bf78afc14fc9dc2c246c6edd9f137e723ebee3fc0d7f45646ad536aed0dc
5fafdb99770c2be10a1538cb7b3f2896e191947d7c08db2c06971f3bbc3dc995
b966ec887878bbe24059cbbd657aac0411a43b3a824747034a8f190396602e4

first message t

eec418a3e045a77cd5be24d4bfc98591fc3472862a4c21aca2026441a06d693e
1aa84e6e9694df995bc84b47e6f0a0136da893e58577da17395d38977d6d1fc1
dc076b61c7214486c20a43ce30505f4b4ff8bafcd4bc5ad8245382365ba8fdd31
1cd49035d274801363df1b546d2a1db94c8842da67508d0ff8a97a5a1e494187
56e8846e6140f5f967a71083a0cc2ced84cdcc661756a2c96c55c47204807f45
6c7b4ce27e8d6fc47b92378470b48274242314c0e8608b5ed463facebb95403a
82f302f7b0b674de8e765dbb8778c30184553ddd08dfa8dd87083bcc8e1ada40
1c565e90c05a36963db5505f8701e231a06a5b3adb9f79560d1c6cfc89d83a50

response z

36f44f295f34c44ce8ef37a2af2d654424c07a399e961b7b2398099a5387c14
2cc4469188329c738a78206406643d541a99aa8dc6d2756d592da45432b4d041
bb9e0be768aac8f99e3888d5c64d647cf9771beec176531b68daec56b0098aef
14379e3c36770bde9e6b030d757ce173ecad61609f825adfb4b6ceb0ce25729b
26c7e5b550e1a492f40cc8e4c1737ab8d1d8c7c7aa8a84530f555e954f777a14
8b7b65bca22e50ad81661dceb409618f01b167e3b962c9e4d16268a7f4c42aa3
f9bb41acbd0e86433239a32db3cfd92a21885fa7bd1b4f6c8f2aed3abe73b2
799dc0976a44c7a4859cce8df7e0b2660efad8377ef3b8c4ec6ab467fa0a2ed9

response w

1a210ad5b4cd2aca4107f82db5674661f5fb5e402e56cd896bfeee2749070ca3
cca736ba46412f6f19009b645a5ba0ffa20bfcc79b8f0bce6f5853edf4a91be4
554c2427dc4ee2bae0d0662de906dfb8671c12e722078ae4fb787b8530fb3632
2cd77f55f52d4626b5a4bdf7d982346b79f7dc9aa79946d65d1c1a468125ee27
160aba25b4e3bc8036b7156c6768884c9dbf6dcac57324a1653a785f0d77fe20
f2e7ebd8d0e538723d895fd57ebf7ff7e8ca8bfda1028ac1cae6b700345a76b449
8f9858383076729f8ce188c780b2dd8f5bb44a5a1b60451556ca0d73b2ef76c1
741071ebec3f102f93c9cd3a763f21bf6b4b36a80d9008430e93abfad4307dba

ZKP.Merkle.step definition, Crypto.Merkle:36

$\{H : \text{Type } u_1\} \rightarrow (H \rightarrow H \rightarrow H) \rightarrow \text{Bool} \rightarrow H \rightarrow H \rightarrow H$

SHA-256 of the statement above

788ec73c67bc11c1d0af33b920e081553ec7c725414682ec15c3bbbe8d052ebb

SHA-256 of the kernel type

5bf89ccfdad21e634ebfa32ae099830a64b58c9631dd0c2f08c9ec7e911f1b2e
commitment C

e28683396aa5189aaeef6004a224df0261b53f8768103af204cef7af8118bda7
2aba93aa0ad03b3d37d829d5da3594c2ab9c6a94643fd7a5945f2f8effb9213d
b0b818b0810c62c48b9389fa24ef0db12bad6afaf278665312019b1240629ea9
5a94226bf798467d1299fbb8b21ecd64a535cf5d01fe076cc6e9451c26fafac7
6c9da8e783fef630d10de1884f66311c8dff84a3ee8c7e696c85b63b6fabaa62
a66a3022885c462f78140a151b7541de78eab0203e8aa4fd82f1f04cbbe19a1e
217d6f85e0481d5b46e881eeb24073b114453befc5a910b8db610fa600f99178
8055d20118c4d119c0f7ed6ea53b8878aa2a4c0f45bdc4535488b9703c805d91
first message t

74f8bf1f52aabda428e8563de7787e039c6e040d4b7274fc19e360e907a9f752
0608395417c5c7b3ee59f1a884604274078aa1c1de78375a526af99a4a7d7a2b
fec16bcf199c7856aa9eb558a5b21e21053ec2fa4a0a0596d079a8b4a3990466
6005f5e2ce3a52b7b5a4f73bc1868f936a7e5735fec792a776679d9f9bef76205
828209c9c25fc0067002341f0d66d8e4cf394545e89d597eae265db367c6bebd
385601a1564e0a14386c33dfb28db815784c6c34e7a2f163a659ccb7fb5eb427
9caaa9f1118a6e805e11656d74eeb382d90453dcbbff3eb23e23f231a2de7da7
8e61874a2efb003762324091c8b8c9f74d891cd70c36dae91477ab8e735bedf1
response z

56b492cd22fef0d2869126527a85493771517d95081d2120946df070ae5c2722
d6eb33625dd36804062bbda5b4949a63e0ec3b574d687c13e02d43225f178512
bd43cd0042f89a0cc379d583e705ed1bcee39a5c591c394bdb1314576af65dc6
d85a66265c45d17f53cdcbe3d50c6c35737a2430bbaab3b7fe14205d3a5c0f11
5b183001d53f8902b22916fd92965fdb7fb86e87f9f018e649c0623307135cd9
af40fa88d54918e5bdc252a3a1d8e81fa0aa7eab598ee1771d69ead7ffb510a5
b34635f4f3e5a4d28170e851e165d824f9bec0a4386a8e0e8de997c055383557
8dc716e466356259e8598b824a40a404bcbe66c57d12c09fd0297e98ca3a29a1
response w

79bf761d4a8d1c42d910fb117c4556c27e16497bcc3adb94d8f3127670fb5f4e
9f9fc52d68942815264caf7b77860fdf80c340e1520c5c1468c6edb062eb8e5c
d43e52fb3f7461d59c885b7ee6b2fe4df7b91f7294441be36ced9e28702d2cf9
16f464060a7e5d35296b57563a5e099a101eef1dbf613436a10ac4521cc77d98
72e8e8d9d7dcfa56fab33955a74371286a3f11ba8b4721901350d74824522a67
b139a268b2638c2a27d664b9e4313abe11858171c4353f8c6cb99fc460ec14bb
743336a5c2169f9f6d436d56a0fd6d88d481e2ad7a5ae104b1d2d83720d20177
f7f2e4bc413e1f7aab075a2a5b871ce81c98c3ac1cba8d5d4c3671356ef1b02

Fiat-Shamir context

e0c83a29a21dfcb9d368f2b8fd7288da4ef387d6034af8cba9d23503bd11eaf3

withheld proof term: 209 bytes • axioms: • kernel replay: yes

ZKP.Merkle.step.eq_1 theorem, Crypto.Merkle:0

$\forall \{H : \text{Type } u_1\} \text{ (hnode : } H \rightarrow H \rightarrow H) \text{ (isRight : Bool) (sibling h : H),}$
Eq.{u_1 + 1} (ZKP.Merkle.step.{u_1} hnode isRight sibling h)
(ite.{u_1 + 1} (Eq.{1} isRight Bool.true) (hnode sibling h) (hnode h sibling))

SHA-256 of the statement above

c8801e0af8968ec85dee58e752a98e03068a2c9d41046c433a5dd9797a9304d1

SHA-256 of the kernel type

6f1542642bf3c12999cd22b129d0aaae26ece5b407a120cab172da092fe3398b

commitment C

a0b8b30495a8c90fbad050821fc002f63dcb3f3c5a28229de9bc300d8ca2648

5f4b0d145ed260d3621977c6d357c8b064bcd1082bef071f0e7873234d9f0772

039878f667095b99b8821292f4e893ec670fbe22f4ee5c21ebba33d4a5bc5a79

7970c9376fcea684b5787f6b2e2293f956a3266f51b3f7f8e55f753b16e96e5a

afdd6b85e5de327c0f4ae41c3a89ab31d8cc316ac8d7422e12f6493319bf127a

c4f39a58968f4fa83ac6eeaa56077cdc5586c6c6e4a2ea7ece7295b7f5dd4d54

f369f942099a08f904f6250df31dcd08f982962816945281e86eebcbfaa7242c

0d028540c05c634ec3e20a1094ec4a275f75e6f59d27b3617f9322587c4f370

first message t

92c61622aae00e2f1a6b317781c5ddb829509a535018c8c5ac504b8a9b804cf4

94154d2aae9a95b0bebd679179afcbab398782c9029c92b899b2f86c9082ad65

2ec9c124f7ec8bc76c0280d035c27f057536345c527c70bf0ad669f832d24326

b8c4c48a03a38e1eaf54bd4f989a14743de6a8a98f98793177f6a680d393731e

6c692ae617a8c0e2bf2b096cb5472d208e4189162c25d50398d333e5f0179400

558a9bd2dccc4b24f56c927db19c5bb9d2032a1719b6b72ca3e01bcdba5a501d

e2a1e9f980e020b12edce144bb3adf86db3db89683c124d63d6576b6b8fa05ef

c4864d84ee1fe27845c98ed46f89e1b52046e02181fa2553a9a3c5e31c57faa3

response z

476850083bb2c4ca5816e9681f5258ef7605acbbb9b180fdf3116c8d006fba25

f3091c42a42f96f301487182aa5c29c543b20af49e79ac58f2c2cb0535fda07e

272449d92db52dad722f976518c37454e73f48831492594be15e9023585a35a6

4153b741a145284ac0f24f3e1758ed882f9df970503a23ab05aaa66496ee4758

8b94056422d8afb539265a6dfb6666f2f8bd1659358ceaf4a17c69bc145cf2a

aa55c2291774b7e65c841d963b151808dc9fa791f18870a71225d8ac7996a8d0

f4678da6c3c710c25361dd6c2ed19a7fe5e863ffc0df2ac7891c0b39b02548da

497bb07d7396d0ce9016b9d05fe0fd7203b53c7125656bb93f66e0cfa3b3926a

response w

59ef3f860e13c065fcb63083493a998a98c9e53a2b951e0562a1ffb5929e3261

1e32c8f2ac8feb3048d3443e123aab68427776b3ae3f172725c363af48429f69

e7b70ccfc9d9cf5b53df7d6c947e032150673289c988822db36f7d12a0c30c7f

c93ced08a483aeceeedbd498201c22a69a9408158348f601c1fd7086571122e0

139db592a77cade2b64cea35f3feb3c8383d4deceb6e674c6c3cf1ea0db40283

4eb0a7edd96e8b57b6c5fac04616e42f84f5ffdbd482da8c1818ea02a3b77e83

032b85dd6d1463517b88b4d0dc170484446dfbde9367969c2e4b929701445fe0

e083fe608318c90550bac059b8eec67c61154b2044e4d05cabb9c256986896e8

Fiat-Shamir context

8754605dd254ed9c14781aa395036997a796d78b937285529f199e4acc0c1990

withheld proof term: 155 bytes • axioms: • kernel replay: yes

6 RequestProject.Crypto.SecretSharing

ZKP.deal definition, Crypto.SecretSharing:59

$\{w : \text{Nat}\} \rightarrow \text{BitVec } w \rightarrow \text{List}\{0\} (\text{BitVec } w) \rightarrow \text{List}\{0\} (\text{BitVec } w)$

SHA-256 of the statement above

b92911c954249cbea722673d0eafdaaa0917ec12cdfc388f00f0cbb94cc4e036

SHA-256 of the kernel type

dbf45195ee3834d59778a0f9ab699affe37c1907b6c78a9a10202ceae9c1f616

commitment C

c9df6a1eddd7d8b2b055c3e2761bd40807bcac02edd41e2221904bc629f74103

0bcc569f1f2dab61d52fa5ab47778c9acdc9ce03c8d392d3ab60897b37d8ca74

ade3b057a03eba617980497dae6c7ce2dd8b79492a8965ee3dfc25e8f571d1a

6285d03a83b546a0ce7e4b32857277817b3fbc250ced35d14c9ed3efd5068c38

aa4b9a251c784ca418ddd1f84319799eb9c2ee7ae3874687e0c434033ebe76f9

6f964fcf597850f86bbbed63c18d3d3992cda6311067481129310d88c26a464d4

c333325e954bb028a0dd8d7e61e9e293d1434aa302823389d62f4b0f8ffabf15

357565042da325c3747302244c81e869184e3d8eb932c912c5721dbb7f47d9cc

first message t

e90259bde89a71e1fe759b61ec112c6765f4074ea1ff0d551cc648ba10a15448

b102bea60380a2810ebc0acc97a88676adb621702890375a75fe58ad3789050b

fbff8f2414187fb870ade6b4d03639f3527edcf5e3b655e0c7b125587c689939

b247617886efa7b74a37d712de528f41342dd854676fa1687debbce0cec7b533

ff97c6ba4a5ab480762e56b1189b07b9f745eaf7c27842a68dab6342d7f481f3

4fd8bcdab547fbc46ef7d4d66512b96ae620b7dccd07c93fee3ef5b5cee8a0e6

66f2952806849626be33099bb34a40c28d70f2f4b73bcc096a9b232412804a82

75d8f32234ed084ce0bc20e25ce4b33e9f5f8dd9f5e4934391b81b4f34e08333

response z

13e3cceaf50a5474b1c7f0468c2bc6b042327fca3ea219932e086d8ad11f2218

067fc96bf5f9c8212168e5939ec4965a73032031d26d90f6a1e225e737c84e9

a4749580ec05e96a0a24a5b31b4418354bcdddaeb2066859ee201ea26de26600

aebcd822f430083a23adb7225bfaa8756331c0c269550764e3027af627da505d

3242eec5b07653e9ef5fba20a4fcd6117415ec04cbd0eb0993921f071076b1e6

9b0fbbd4ce02849d630d6dd40d88f95767175b33176c0120d21352e049e813f0

3095b2dc4022e7b05eb32ffb4528c90749505a12b318c451fd0983d886804d64

2268739ca03dc8434208760005af30cf97c07e66b20a4cd35126904ce4d45c68

response w

76ecb2f1a6128362a27445342477cc6116f46ccb7a28978eb375283af5f43454

debc9dc76d18876d96f960f2fc6bb62514c889abbba43e2a546b72188d62bfc2

8c10461da46938593801629613a1342747fbddc3db6659204beeb8002b1035af

195f7cd1452085f79b3775d7e9fb28f26b567125d104276b29342dbd963021eb

8c56c203de9915d5377b0d5af3746df9588cab5df91e96d3d490d162d034cf0a

3076bef6c16301348d281854c448a3225d35f19dd1e17629cfc17531d511a38a

447e6a4565093d10d494c5940e14788c00d4a4c9080ddc8ca573500923c86d16

aac71c793c830b5d69e34d5ec1eee5427e4700096087ae2091d3825fbd42f97b

Fiat-Shamir context

48a5eb6ffd08196339add03776f9231e5909bc019d90c0228335a0c6bde2baa6

withheld proof term: 423 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.deal.eq_1 theorem, Crypto.SecretSharing:0

```
∀ {w : Nat} (secret : BitVec w) (l : List.{0} (BitVec w)),
  Eq.{1} (ZKP.deal secret l)
    (HAppend.hAppend.{0, 0, 0} l
      (List.cons.{0} (HXor.hXor.{0, 0, 0} secret (ZKP.xorAll l)) List.nil.{0}))
```

SHA-256 of the statement above

a89ebbb078aa5d8d37dc11a3b9af2a32e7c16728b09de63c01b67e842768db4d

SHA-256 of the kernel type

1bfcd638d6553c01bd0c314f3702a024941a122b1d4637009a111a22b7d4146b

commitment C

eef53c911e16692b2f77a4ca6669d7ccefc88b4b8a2d30680bb05001b7fc4541
bb6004f4041f196714e13b9781b3982e9d4aaa64df74919f1f3862bd5db58dbd
91c1cda6bf569d5ceba7a81959c063c60f6521702b3e1422398d620e22f8578f
7f4a09c3aa8dd87a51a86b87416b17e9f19952b4793fe9c9b0d91fbf80cf43ec
675685b5e747c4e30666cfd29f4d5c03a5111ab25ff82552b1823ab7e8bbb59
61d4a5dfbb8b3efbc0c7f2d105b88277f279e9c3ad2343e297aba715ca2b008c
914cb49c060e8b0267e562d144ae8549c0b3df01b5deb8666daa56982afad62c
8ce424a5e377992218fa37c899ec2a458731bd1d18037e9d7243ec62d5f52757
first message t

f5fbae749efc3db93220f3dd23e7108113f4e5a35f968f3a72bc9db4508b002
904c5bfff22f712b3133e137eceb6fdd81d819560d7a02d2b26172461b4227c07
3bedf649f3038db03c5ec2a0c25974fe2b666285c6f5a820aa95c583833e8acb
22614e5abcf4f88cc00c038237c47c19f1a96d59b6593d02066c7f9bd8776deb
601997dba9ec6d9af5ad87a54d980da4326045c7dc628ee5e2f33cfff504bfa8
58961c9c2d2f6bcc1ff5fb1b1b2d081fb3fee553662bb316594e86800c82acbe
90a300e398822ac37da8e10d3963021ce766d53f4244d9fa3e93ddecc3743aaf
0aedc7a263024501ae776cfc81436ce9df2cfea24fdd4fc32b6bff336793aa5c
response z

387867f0fc2445070094294be6c9acf685f174662b3bcac793564be5820a4b47
b370c74ef63b25c84c2914f93c1198c12a9b7c4410f27f76c2bae78df8d3067c
101937971cb77cbd8f7862ce6f947e69acd95aefc592d19317753ede2e3bfe
0b2192c6c0755ea447076b1179866ac00621dcc167156590be32ff42261f28e1
eeff56a6a7c0817aa07b4f94d0c16773f7429c981b12fc846a0b4d1eb4f14b82
be9f791ab14beac21646af7b3ae987634320c8996c3f7acc020683f4508c0b98
2b3230d5e3439281fcba0e5b4d079118437f8f2153a4a8be6fb918d973ce4a8b
a1400d459c1d79f595479c0ee319b52cedb9bef882771a26f595e36af03a6dc2
response w

4d6cef2e51624946e27aacd3953609a8a763d1070e57d7ce8a4ea9adac85a672
23a874205b27dbd521a7232554088590e53cbfff2d93b80f03fd88eb29b11d89
c99781a9311b1e8c0068d89a6a8a9fd45ce9192de102b368cef3463c2d8f6776
1f4607a351b6523e2a63cbb24384a5deb899e7261e4430b6222703476cd852b9
ec8d08681803f4049a5f1c2c62eaf1f5f1b7597d3c52a502f033dda00b2e90e9
8ca6bebb73023a0573d6f88a4118d922bb402e91d3d8a9bb48a2cbade6837b63
68ca24cd6425ce306eb43b3ae8667f02153bb6a6ea355f49f0653a882c2d2a37
cc6c1f582633c1d9cc2693b430d2eb84d06f85619bb7a85a1f456919b7911450

Fiat-Shamir context

5dd1ec3b039e9cf5a78f83a51c969018fc81dde74fc809989a5fbae69fc40e5

withheld proof term: 118 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.dealAdd definition, Crypto.SecretSharing:108

$\{m : \text{Nat}\} \rightarrow \text{ZMod } m \rightarrow \text{List.}\{0\} (\text{ZMod } m) \rightarrow \text{List.}\{0\} (\text{ZMod } m)$

SHA-256 of the statement above

594c964c1dc746ce2e35927d008559e552763e5ee7d1906889a84601d3686a43

SHA-256 of the kernel type

de6d91861b87bfc8923780a7c3cd799ffe28b78dcc8727f6a65368f32a518a99
commitment C

9ab124ed8a5da636f7454d781621c3be6561583b2f4c5b054b02bae62627737b
5231d24c5d4fe8d9c94a2ad4a468f8eef8d6fb283d8f3eb0876ff80617a5f4b7
4ee0dc55785f4deb590799a9f88b63b6b4f1808c335cd5439941c2b0b5dc10b
3f505c81993086c79be11624f51b1901e74d9f75cfd61b92edbe9e18da415666
c51bc5ff647072fe527fdcf5ab660eaa77ebc6c8748462cd7316b38c0022b837
ac774120058522d5cdf348ee638ad1b93a271fcffe84591ed4c22182f72cc59c
6f52a7177747a003edbf601f19b0c877719c645c2f32a831852a3e4c3455c104
90a1b53cf0d3d4de64d4cba1705a52b1632bf1b8ac9911d59a0bbba0d2098b43
first message t

2b918f06ca43ead622adad114dd92b753696a64335e617129775b19a75ff7698
e114d8cb27b3aa508189bc3a646de8178547168c741ed53eac15c83eb8848f81
2ef969c9e804e8c17fd60b5083ed77119c5bb652e19de664e8870bfe9b838c24
f7429df87ecd29ddc9176742781ac6f168732f468d06feb9ccfe0ab6b6b32a13
c084ca1d7118d1b1eca6c54e073f84be177ec8e56d5b93244ba69cb7ce1fe89b
796eb85a9dde4a776506fe04848c2973aa7d66a18860fddfc1a889c35715223f
cff8994a58d61f4257d11f771fd98f9c1a76c40cf0b8aab69fc2c5826514f49c
54cefe22e0abca9fe4977b5b82453839c8d203f4067e8f9db15622148b1f619d
response z

507056cccc7e3e32b75ab64b262831f517be5d9d48af7c6a97ed40531f107570
b6b42ecb37c5d85981962b604f3add99354e6f105fa4729753e96990b988551d
5219311c08d3ccc2a095885c4c8594bf5795f0a45b0c2316892387b03d0ecea
592a81fc67b6343a5ce819c1f26ab360c41ffb8f59cdfc5dc6a2f947864d59af
4279a5ea88ebb59e77c80b273c27c013c37684e7ae7bae3e90d90865bc67ade7
1dc93ec9c1a5b29f585384ce9932fb7b6de7861acc4dfca37d7897ac538dbc6c
8a7ca4ea6c4d8b5bf4f5ef55bf30a074cc80b70b4b6a2ab02c1bfed2b4120260
5a25a4cf208cfb577295dbdaf14f6266b34e28de81dfc35ab0d28ca6e37e6c3
response w

4992abf5efd62962f6021227c990b9ead8f42bde1ef58ce5ec34ed5c7c122472
f665ad031c0ecda5b1657b5511e8f01c8aa18c245918eb850e2973cde3b281d8
9b3ae29dfa0b6ba93e2d642aa535858c76e7e593f40afda478b514576a4d9ba
4fbd54e93773eefc5b943a2324daae519b8b8f59ed2e2780c13c746f0f0d14e9
01f85f2d7c9d7ef7b1bb645c864827f9c9f2d440e214dbf18fbb15fdd0be341d
3bb3a6f918e08c7e5386ce9a458243c61bc3778d80ca01f7cececc4473471a1b
e3d4d8b754f0ea86f8a97580d1c239378ba6057bf12aec0e967be6b08149f239
076a7e01c4d54431940ef3c9c69121abf0ceed10dcff859713184da249beced6

Fiat-Shamir context

c350dc11178cff87dc14f37b258097262e7e324ff6ab74dd2df73c77ca3ccc3a

withheld proof term: 1 257 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.dealAdd.eq_1 theorem, Crypto.SecretSharing:0

```
∀ {m : Nat} (secret : ZMod m) (l : List.{0} (ZMod m)),
  Eq.{1} (ZKP.dealAdd secret l)
    (HAppend.hAppend.{0, 0, 0} 1
      (List.cons.{0} (HSub.hSub.{0, 0, 0} secret (List.sum.{0} l)) List.nil.{0}))
```

SHA-256 of the statement above

a9cbac9a076639e7fbd07ae5a30b0229f1c778b47560c127b0e0babaf7655b4

SHA-256 of the kernel type

60fce48e6e4753f69447940f381daafcffb0ae02e7c59738bafbb1f270d055df
commitment C

8e1873a2eaa0528e89838c26859957a2f5542f6ca8cd2df019c485d61e843ee6
f9b18fc7d1178d9eb296c33f89e53ba6f489719d6aa6b7e7f3bb8de708e45238
382c005109d7d4fba436f19603f9345ba1a764cd5ea4928db487dbb8af73f893
2a51e38075055f38440c083cff141ee08437d3a1a2bd9ae0bcd97bf2be0fa69f
49b5e305732647345099eb4385c531aaf8e0e602b441174d523344b90eaa9be3
14d690225180b4092b20b732163f62f9c332788c91938e67fa938e21a2f5f99e
1deab822aad408b1cea090b3d89d7fb19b4c653a83418c4236163afe3403fa74
ce9471cd6357d9e7f470ddd5e4429e59fd5953501d86d47c9801dd032927238
first message t

343f102a2c47c965ab8683a3b5d53137174b10adbacc55083a0c17e727fff547
69626e3f424ea377d3f7e23f2fbb430261b0171f433ed622345b0b9e5a19bdd1
17f146b4eb7bc23452c27f30d91f13d14c726b24f930adb05adbcf98f9e07ab5
40c593c6c52cb60d9f4376344c7de9184e5e20d15b2a1c14ca61744c576b0c98
3624278c32792473255ded17735e489b188074117d76f206e05bb1cad6ac1293
70bb88837b801b2fc226f5953d5f3e7ffb62231bc142f2994d9b747c0b9c8ddb
b7e8bb66b983ed032d7925c640d859f7b812a92e81f5aa4d3d24a717db86b8fa
a78b09f1a59800bc4ed3748f34801d8c02958c5af8454baa8ffc5cdb1cccc23f
response z

b7230bb304d3624093c3d74ac10021ad5f6eeeb75f89daec5cdfd2ddaf381df8
22882fcd23c7d77cdc4dde9fb116075b192ae403f5629943faea90707824df
549c7b4d0f8a878fb5c5b739210c7981783f32883dc10afe03c725a932057da8
c70136188f8db93b3b2fd10fcd476d92798465bf1f8ffc9978891ddb318fb0b9
f9c9b78a4fea2436a0bce81ca649e72d0963d2500b52ea2d60129be9433bd70b
d9d959ef75760afa5e128e680f1fef988b71953a61d72a4b7c855964a169083c
a02a52f68c3babba7e36a77752850dd6481f843fa3bd99611c41198a43687c3a
49e2eeda5c44c8cdf9fbd2b385f0f631a07fe68bdd20f7b05c316f1a6dc484
response w

51471f23f6036421df8647368d9969adbbff65df491d2994764b87da864de672
4afefaf7ca3b8ae1ffe504907e937a86eb33ef3c84fca1417b03e3c0964f2766
f471c2fe56ef922b83b02c08f82e978f39e8e6b647811895a2884b8ad097ed66
c033acd3d79456069767b231bbab412c7d82499ad66119c714cbcccece5c06ba3
75b925996cfb6d06d15e9a1643f0c3d9a6ca57b5dbe4d6f1b2f73792e2636de2
92bdf04067d4c2492c66feebcbcad9db2c1e3aae12223649e8afd630573d1cb
3ad8a9e18da7f76c51c72b523b5c8d07bf1e8e3b909ed5db126ae58098a0a210
60da62b3a584060fb66b14f9a3fe7fdc4d4d3d604a418253ce688f66b2aaa3a7

Fiat-Shamir context

c375ac364c0ae963e611908e806fa9eaf0b91e6b6bfe50b48f65f8279c3a3f92

withheld proof term: 115 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.existsUnique_missing_share theorem, Crypto.SecretSharing:75

```
∀ {w : Nat} (pre suf : List.{0} (BitVec w)) (secret : BitVec w),
  ExistsUnique.{1} fun x =>
    Eq.{1} (ZKP.xorAll (Happend.hAppend.{0, 0, 0} pre (List.cons.{0} x suf))) secret
```

SHA-256 of the statement above

e93732a5445805c8514ccf72f98965876039639fd4999143320a7ebd3c606cf8

SHA-256 of the kernel type

55c77c690c297e7fcad043fb2c2a286a073260e6fc6794c961d8eac3d0743afe
commitment C

23aad3c9c541c951d041a4d77cb7ee5714df9201afacafe356f13dbb49ea922
f822c3707cd331d9eece61285fc1da6f85729e1ac7ac164fa4ab0b414145fab4
5254f77962d5e283cbb99021c12a32143c486129c80bfa69e372bcadea6f6be4c
732de4eca626d60a98073f175b028fea61ea686b0c5f02f35c97704bcefe97ea
4624b8b7bb1e1e37fea5477f736075a680d2f01625ea1176345ff97a84b44064
196cfdff1b76ffb9cc9526136b8fd9f8a1c57f594af29c1e2bdccc86a783914
feadfb4a9e497dd4cd62496740c1f57c16d7e2f74a321f169377cb7ce41b76a
48d039d56abdfca03e9e77271e166c83a8a68304b1d435057b6e8406d6d4144e
first message t

ec93cae48d7aa2b3a615a43dee4170a6ad88d60656697b7af1ddd6500f681979
db828dfdb7f44b29efa98518458b2d1397b0b271290bcb600dd67b61c9e701da
2ec92c6a4c8de2894fea2e574b5a9d777f562eac50765b1c4a86ed393f3dc789
b5d472836ff45e8f19d667a2c34b929727df8d7eb58ad4428af06f2f28d74c55
724398636a4ad7e45cf09308c4366fd5bd5d42edaf61c36c89248e14a66a69e3
2f99c7a73623a030bb8f0a66e79ad92a2864d7826bf1f4d53827dea2d1c9ea50
25b33971bd23e76171207d6281bb33046296d69db5a382ed7c2d316e22d98dda
239733d53a22ff1e09097611d36c0e0257fafebc7160ce49097ab643d55f094f
response z

5cf8a7891970ce5335e5b53730c3be3c6ba65a81cddf7e1096dc696c874f3671
146c7b60eea0ebabdda0823b659c6eb003e81c8e4d1080dd7db663720f57f16a
9261e15cb7b62e4db6ff0d0d3921720a6177d4ddf9cd0010c4f2b719f21ef1b6
81ea682f5332e1134902f964eae6b151b741753f218aee53d0c823f3bb6151df
9852bad058b11cf9af28c16250d56f80abce9add1e056b3c0390586f841ccc9b
d274ea92ca558739ff2491d2139887c667fbbb86477432564645886b6b220995
11ad05d32e47e1c8012c5b28ad219ce33f78fd1ff01c1da5ba6ae964b5ee2e98
123bf7df75ed6520f62f244d059dca73ea38f37b0c74e49171a5b04544503a5c
response w

18df8d8a3be12d3c0126348dcbfe26d45640caaf3152416d82d70145aa6782c3
481d7f7dcf0723b99f5b58bfd17494d06169c70dbf02d76a0fb62f76a53f66d0c
18a1f1344e421506217bfd7b8865c554a93e4cb47df6528260151c8cbd60c6cb
e7aeb9193de0466701094b5689db5bbce8a284f2c5dbb0faf7e5c0c5c7cfe48
c8215738c9fa49a5e27032a224884f63a3a449f8971572df7a55d8a1e0916629
fe3438d96fdc11c023453d927e45ee613fa30b3927e0004803ab3d67ce009f65
4cf35a1b1c785112cfae749b1429a6709d4f5fbd8535b331d34baa565a056a91
37da52dfe03ff7d36c4e7e187f31aa4fa5a3cae6550182d27e06ef7f9c963067

Fiat-Shamir context

e07dc4057e66cdd364054c05ee4a0901f5636b85527379f3e07e2a9874d87a86

withheld proof term: 14245 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.existsUnique_missing_share_add theorem, Crypto.SecretSharing:117

```
∀ {m : Nat} (pre suf : List.{0} (ZMod m)) (secret : ZMod m),
  ExistsUnique.{1} fun x =>
    Eq.{1} (List.sum.{0} (HAppend.hAppend.{0, 0, 0} pre (List.cons.{0} x suf))) secret
```

SHA-256 of the statement above

62d9f6f0033034e77834e3e1185d69c72a132a39c909d08046618d72b47bc754

SHA-256 of the kernel type

fff135395a70172415172ca99ad169f16aeef7e647a41eda137df5a1706cca5e
commitment C

a2d8a51cf1fb74e01f7f76e226ca97b9d6e38af9a640e88181a259570a9f49ed
750197b3c1e69a2f35b2589d710dda4f9b28b60606014f6b9fd34cc68800c606
4add821ba83707b5b438ea1e2c28f9d490a2810f4aff63defb526f06d7c86187
7121d1274968a3db8f069f6aeb71be5052b2707f803192dd2d557fa00534b2d2
22a2f200e214f34878aaf4a2ff2af2183c2c767efe4ace896348c1bb531de680
4ff8a85a9332ec116a29daced8e11d4608e6b5e31a581d3d5d2b692c1977a6ec
5c6935b1f47e9ee344de5bb7604e24d71fdee51865981c3be03a38c9ac960ae5
b52b2d18d859c4e6e2d62bf816434b80cc291982625626c08aea78cdb18aac1a
first message t

406b6ad4961298c8f38baa64c694e626281883c6a635a56e7749f45df11ed7a6
9824f9bfb49335aa756b801ac057dc1a8fee4bb1668e1787152dc6c125426391
61c4cb93cc7885fcc15bcb95d678e2477843276db49769f43198b7f88b6bfcc4
a0259dcfb1ff46246c1d74c1f8e8e7614c9ead4e2d52339bc5f637946eed2869
1b7ea36453c6136946017779722828e3646b18ada93a076df0ee3f388de4c942
86383fa8fa641f3a7ebe27f670da5edebed7f2bdd1eff67a819359673eaf72b73
0cd5f17fcbf104989cddd1163cb33a726a29ea1fa6cdac650eaa7bcbfdd001637
95607d5fd029cb115dfa3c53894282a232d428456713ef7bbeb145b223375c28
response z

7fd21b650cabe69e19a403037c0f8803b867eac3e00781139d2c2b67e0b44564
d9d168b48466e900cab8cec4e449d7297dbfb083e85ee3578512c2e648ce3e9b
8e8ce85be29ff1e8f6130db171dbab55f265f7fafa4a12bf322f1f1a880ea19e
a2415ed1a5dcd988996bb73c0ccf52eca5419a15b0aafd73a6aa233099c6984
8d52a724c925db21cb6242cce5444d9c4ffa51fdd07a38cd9e2bd0536c56181c
d482161f65b371737a12b781131f7ce6f53e2fa4da2baa3ef9ac91f24341aa96
4a59e1e3553acd6b98dc4820b34e1b9bc7091bb39823ce08ff6b6e29bb139c6e
a8890b017bdb3d3fb258294189b1087b2d351acc97edd1d2fb60382fb8f385d6
response w

7434010524d4fc178acc2634931df0ded70a57f04634c637214637d71cc41a73
7a498e94633ef1a71200e899a86a85b44e31b8af7a6644b32267993a3a78a999
152fb7261614eebbdc876f8c5ffdaf79d5f52c658cab9722d6cd4fe5357b58c9
26f3cbbecf862482ef9dfb3975b6e354bdf626063fd43accf23f0db2f8766dca
e3deab477dc3c7a010f78fb95c287d1659d5dd0c0141ce264e7f11a85023d23d
d339270ecab9e5948fc212a7d5db2815eb979c118cfe33dd6051c22c2e762bf8
1f01a0f5369ccede9293b4d61653d1e8e922f4c0cea00800f29bd623fcab38e8
2b01d287a644f7ca20fc41b90866744789e639144ce27626a3921d20f350fc95

Fiat-Shamir context

ac752b09996b71f268b425b15afada34c74efdfbe4fc09c5a65f1c607fb02cd3

withheld proof term: 531 680 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.length_deal theorem, Crypto.SecretSharing:64

$\forall \{w : \text{Nat}\} (\text{secret} : \text{BitVec } w) (l : \text{List}\{0\} (\text{BitVec } w)),$
 $\text{Eq}\{1\} (\text{List.length}\{0\} (\text{ZKP.deal secret } l)) (\text{HAdd.hAdd}\{0, 0, 0\} (\text{List.length}\{0\} l) 1)$

SHA-256 of the statement above

c9e5e815f6a44e2152329e4c236e204343651ad7d500769f2137cb4433deec76

SHA-256 of the kernel type

f99a1fff3f4a13e0b3397255a1074adbab58f333ff45d6b8c06e2b138effc7b9

commitment C

a8a3950bf30c0ab007927744161057827c22ccdc7a28a196f9f95692adca9d72

dc6765fe9b99ebe44af740686c4d22a7a87d5dc72747a62026ef24876b401612

722a04fe8124332459dc080330c683520b34f3966b5137d0da42bf4d945f6cc4

1f1f7e8df03917a4d089484c7bea0ee50a6e2eba7f66d3c0138b852ca6335ac3

cb492abbbf0c346fb0a9414307023daa15f5366b37aa1a62d79a9ca10d4fc6e5

c16cbc1fec187bdbd7eca909a6266e74946b3325ef4424e416468730dd3a1699

d812f14103eb82b2f123dbb474ea17881c5df5db5c51d351cdef7d9dd9664d82

23413a6b795e9552f207f33f7b5005821dd9a3e899eced2d09b36ebd5dfad7c7

first message t

a3e222ab0b374a320f8afab9b4278b0a8ec3a47c1f513d877f1b35ca42063c0d

b4d7725f01fc3ea499582a5ee3bdaaa6946c4b51e9c31791f0bcb51fa834b89e

b669966c5026e185acf42d38a1e943b3038074720c1e31ecd153394d37304a2c

6971ea30f332991ff5736c92c90d69b8eec1a4198e1f763d3eb2cdc3e0c29769

bbc9b695ce764f8343ad745af66cee2c2efa7f3ae49959fb8f1270a943cbc711

555a40698aed8d3a1732589391b9de544e10a1b06a18747fb775ad43cd31f73a

bdc09d2dcafde2d3e0744a61d93387653660d5e07787a3bd8af11cc36600d0f5

4dafb7a70cc1bfe0538bf48cfc49a9d012ff09fd118e6b53e140f4cae19c13a0

response z

13698d5aea262cd648b7cdf12792c33ed77bb0a7e5d21654d2f830262d8578f2

989f835ac35b747302a89b8b8306f6dce67642a6ec367fa6962fe087f36e09

cc5effed8e66acd5e071e05f22f50dd54a549ec9d0e10eb24ed16abe815cbb5

5fe8da59db7c1e6b8000fc47ef26b771824fa24e050720df538bdd3b9f889090

6733749ceca6b18cd0065366cf19085b6d39a82fdf8cbcc54063750d15db719d

039250a105b1a6b836e325b916339701d3e976eb709321235ffb662142813ccc

48ed3b9e8eb94e26c2047aef59b401833613b71a332641d4914da5d68fb2c161

fe3d4ec0ad7e0f9ac67b9d6921d011b594907464c9edb7def6d755f9f23ed70d

response w

1e0ae223631585df270801d110ba6cca3af8b15f64f2317b37e6ca3ea3dbac37

daf1b2d9d649d9c4e800202a63e95850fb460abf22fbf74eae1b8f33a11b6eb0

8717232d6a25db7904ecc150da4a480cae8405b1bcb7e66e241f90a306caf273

b4ec29c328d7f6f8aa9c8d495aca06db2f98652be9104e261ae0dec9892a11f

dec99e6621f9a6ddaff8ef78ffc51901dbecff7fd89e6d99c767f5e6fc4cd43

fe6e46f458d46231d5b43972404af02bbc91ffe0b9fcf74e65e25abd8dbd3b8d

502bd94458f6e97a0d9dd13d4df82d2060f9a4a6cda8d0ca27f11e6c14d1e1c3

5eb2174e7b39ca1a4fc7a32d53e2179dcdb5cc6b13e66bedd682cf6b0e236c20

Fiat-Shamir context

2819c29ffa902749bfdbcbe9a235cf2e19ab4f3339ad05858690b248f0f16cd76

withheld proof term: 3 583 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.missing_share_bijective theorem, Crypto.SecretSharing:87

$\forall \{w : \text{Nat}\} (\text{mask} : \text{BitVec } w),$
Function.Bijective.{1, 1} fun secret => HXor.hXor.{0, 0, 0} secret mask

SHA-256 of the statement above

41a597c2b86afa3172bec2b816b2f6ba1771c3819fea8d3c46d5a858d4928479

SHA-256 of the kernel type

c861a18277569e140c00f625493e8fb3e6feb8843bc48ff06226cfaebe467e84

commitment C

57e5c07ac8001ef80d7a0938a0f31fdf1e650f1b2af7f7c9e0dcfa168b496505

ee50b9bffe250f668d0035440999e45f4bf252efbb17b783c363ac1df4cb4410

f79198ca1f6c6cfc6afa9e487efcd387a1de109671f47a0ec7d7becc1d6422e

afc8e7bc9524a2ba948323f70f5c7b056085fda090c728f6cbc6397ebd9bc56a

344eed0de5a2ef890ce3b1a5b2333120a28e781da99f72c2e5e0a65ff4bf62c5

ae6d2fc4f5999e0cff856d60d728ec6a5299d23a1e26597eed3b86862cef7f24

2271995526f3c530214818f767a0693ac1d8bb29a3a6eb6ea625bc3bf954c0c9

050fb58033f775c5b21585e5d481341d3c573536ec53511d81cabd148021c57

first message t

67e940806a6a19dd71d635c8ab8423d4a0f48081a7aa47ec259481c2bece4672

4a55d884ffd44faf8c86f681241ed61c36fc68fd08a8e90d199b5a99f70d20c89

44f5b8a07ceb7a085bc026cf9e88b48235f378368983299a96a547aa1d7fd175

28b1e86dc393085d3bb18d9d860244d0c26b2510dfd7531421af012f42e8574d

8e011468f8c70783c51b0f63bc39296455f9f971e6a05b66b2aae3c43251611f

4d24fc7c6c4c8bcb78ef526fc5c52e321d487184b059e00b757fa27f2ccb1d4d

bf7c577b5fd7b65f8e3faf5a2ea6b3b1cb3b136891de8907ccd229e4128af340

b30d04b523a95681d1a16056ecc3bc939cdece6888b5ca36b38c1e2fa0eadc

response z

7f5ec385f8b2db0896c5aa2cb9a7398a39c1a3e70a453abe3ff5bce7c55e23d9

d9f610e003129d163e7ae892080f25442afae293773d93d5dcab1abea9b7d024

fead93f69a12fd1bdd4b1e82ac41af2affc3bb5a4143d473596554272aebad16

12033630aa34c3b5ad72c16ef7476e302bb92c41f9f64b3d23164b9a02562713

1fe94ac001ebffada8f3c97bb670a810aaac6497a6564216604800de438667f

3592d11fc348f227b3ae24d3750708a631699dcfebf1a57c7971697eb565932c

df74612d7acfae4b6a0252e81b2a0c41b2ee746a4dfc67912868116241ebcf74

e58b7f9fbf3d4efa91fe65937c50d8bdb2dfdfa31a8c3d61264506cec35afd98

response w

6d1a6e54e2c0a644dffadde438113ecffee6a1d45258e1a11d94d9dab057d649

21704f643c05af084fd5a0b6299ba1234883cfafb5081538aa9ebe5559679172

ad8ba62c31f7f067cef70d337d6bdbb32d3d72362457220468e0319b30a70de4

3f2509c02d371e581ebff7521d271789d1627ea6ccab73e0f01220b331f8d2f5

c8c6bd7425803346b9234a36d355de2a52d4287dcd2412d3c8e017f8ae764368

fa0cb9d4ccd8d6f4236676dc9cc9a03b2f9b89f43ae79e85c0070cb24bbce374

885138d960e6fc450286f9bc2d4443917d519988f441863cf8a7e6963c091540

dba01dd47df1b0356fc5afe2a68503010df76053e0229be915f4680c38c0a0c0

Fiat-Shamir context

5dc92c53b50227424ad28e991afb85b6390fa4ea09d7c232c144c34c35dee54b

withheld proof term: 5374 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.numBundles definition, Crypto.SecretSharing:131

Nat

SHA-256 of the statement above

3d1917aa425fd7059394e1413a9cf36a1304374e10ea4b88200fd2d34ce749d4

SHA-256 of the kernel type

3d1917aa425fd7059394e1413a9cf36a1304374e10ea4b88200fd2d34ce749d4

commitment C

7900e9148c3575fb0b62291d41dd7ad31d4a327801aaab44c2b7277209288e4c

aeaa9f4182c99a635ad37b96696e3a77589fc94a400b14f3fc518b9ac569e6bb

57d4179de9830ffa42e88878f17a399e8655cd6c51d65036a26925df85b7652c

1654f81e119443b755416178f8f292db45ef4f1a1116cdce7e96f77d7a335ac5

622b4147f90289e9ee9341eba133b5f3104ee8e824110ed19cfbab70dcf7ebcf

9eb38b33ddee98476042162de5f2ae36741b4dfeeb01a117d938a73795a4287

9e495b302d03e7c9239bc31f24385899c4975a0281585e9e06d4e3d90f1e3fe1

071faf49215eff6821a3178c7b01e63bc7d99c3bafd32782158af7de59307cbf

first message t

7a39aef9a872966896446126f543dc1bfe7cdf3da04ac31d869f462b457a3d36

a871d840e9007ee0ceb4773404c623e7a6087489627e34aeb3abeae99866da47

54b9a3bfcddb2cba9967e6605f8919f7d86a3339a9f1642c8d9cef3ab4087009

564938e8414f8e8ff6c28c4bf52934dd2668de45aa761151c673ed6bb3f32030

9fe3b04e2f6846b5877653e8dfb4d5db4b274a227b0e8d929ba2c35d5c68a7a3

6d4330ebae3504871492550342ed137aa67826c3f83a81b58ec2724cb5ce7fb9

7d0aeaea967330b8ef0c57c9a9f6ec86ddc83ff92f22a099eae817d8b5d4d3b2

ae4e6626b02f6de5b458b00812dc4d4324f5f2b5ff60b3a9f4c5d658d713d8dd

response z

53b6879f193efdb1dbc302449fe7016b4f8cf9865bb84c694a2788b2a854d4f4

4b0ee44ed87464fda57b021af34980ee45d28c3a068965abdd7148cf013c4c2b

930dfdbb1b28886232e11637ac5aad6993105193207c2dc5efba766c377392ac

0cf0f0ae4ff0a6ff246f464117f5b1b7776af7aad57bb7c7a8f4c476b6cbc307

254a2809ac501e1e2babc930143e94a2a935430750666f5cbb9ed7fc48f5f8de

fe0e831d48b0acc879d614ef3569851324666cebf07542d7af12898fa1b8042b

f47fdef08b0012c315582c121edea336c4ffb00af531f17d1e428360f4816cef

c115d9d15deb35196f249e1535c798359f1a35f0d0eb92206050a33d48273ff6

response w

3f50242330b536c038cae44c66b10270be8fb81a2d4b11e43b7534c2eae6ea1d

2197bf333ca9fe46302137bb0b765576f0ad4648c3cccb47cfa9802291ea8b91

270d31e4fc8f03685ac3accaa1a8da900a98450a983038fbfe16bcbc85679cdf

83070e2cc78073a85a1f33bbb2341ec7cda052e07637cb2fea7169df7b2c9937

934b0aadd0f20f78c82723006fa76872a909767deddadf7d5f4a05cb61ed912d

ad5e38d334ecd10c4d7bddf965cd51732e502af7342bde591b1fdae273c4efb9

ab61224cd57c68cbdf4fe7eb0adb62224e6c3f40fc418f881996e25052ed579d

9797c9f31e341529a2b61c2ae67198cb26a0c438203c60a318df680504cbb32b

Fiat-Shamir context

f00fc2b6377dcab1e3eb031c453402c85f7c2f65cdfd29ecf346ad7d07729c

withheld proof term: 40 bytes • axioms: • kernel replay: yes

ZKP.sharing_24 theorem, Crypto.SecretSharing:134

```
∀ (secret : BitVec 256) (l : List.{0} (BitVec 256)),
  Eq.{1} (List.length.{0} l) 23 →
    And (Eq.{1} (List.length.{0} (ZKP.deal secret l)) ZKP.numBundles)
      (Eq.{1} (ZKP.xorAll (ZKP.deal secret l)) secret)
```

SHA-256 of the statement above

9e3e03f988099ced6efe89609da751c51698d979724015787a58c029794ad3aa

SHA-256 of the kernel type

cd017d0b885f29c0809f1e21d5c7bcb4cfd2428971a0928ebdaae4da4601eeec
commitment C

a65f6e64e91abccf4ddc287c0458680815164f4bb963a5506498f1d2d9e0224f
da98f89d4aba96f7e8ab5d698aec97f55b22832b13127994b0f0dec1ef2ae82e
34c8e89273163923289db057c25b6935a6cb3b8b7817b0351ad434a941ef0f23
4be6ad9190ad9d8768713f2e6d7bb43bb664fa88c94c4a69ed1c87a43d3328f5
9e7abb92ce47cb5142d48956b8a29933330463c97a445e229cdf831b22e856d1
5eb9e9c08db7528acf124da95922bc7648971f412abeb432442fbc6fe18689bd
6788ce4ee84f76206fcfaee0075f0e0ac8643593fb8fb7def515c2827e0b70c2
42364202d1406730f50f63dbc9ef1b0c18574714162055d5766fcf82ff531bbc
first message t

9d523d7a8ca395980c575e57e7983add76b55c3351104c788178d118c6938e99
bd00fe2e5add305427eddf1fc92f0b0f2e181dd1057ff600c98c468e19256f7e
85e6f0124e764494f1c6b2caefd2fce10db968473aa8c7bad460c061572de576
047acf31c498aa604506fb752361661e834b6f04674cd00af245d64ca992dd4d
9431eae7337b3b37ed7748d7c6241fabcf96f0e3c2d0c661c101b7bb08bee8d5a
711e7b377b3289f480e8df54d1efc1789d8868b0fcf4acc2d6f99471f7ef3f1c
fd53bf4cb5e4d111e25d04201f63483bdbdd3849554c5bcd823e519e1d5e3a79
3a60892723948f0c8075b7df9d126e1661076822a31d0a62bb120665d8d3ad37
response z

40ad567a4966b5f1518f159bdd42740efb4fe5cf87680a29a3a6acd9f6706a27
2c40dd7e9be7274f318602d576db0a12c80ff37a7df5d9c004b98aff2e1fc31f
c8a733b4be0dd3ecbfbee7731a20fd84595b0ad2e1c1a6eb83f97c1ed969a1b1
b489cc91a12647afce35a5b77b6629712612a632d7df15c35cfbd39b601b9659
788cc3c96410c4ffdc4161f607bf78413a9a9a34b0ed82c24ade94f8c5194933c
1b5b8a0ec8900759e8b983c624b3b856616d39e4a7083df959ceff8473b42bd1
e2139b914c225bfff578f126174c0475db40c1000b8b8e63bc94bfd1f3d121658
c107bd9cb5f47f8b7f08f4778d79613a2341f43d05ee92f4c181ef17d4b6f18e
response w

659843feb72b629995cb051594772f526fcc10578121419f582c049aa12d3ad4
6e1a20e1c950b346d861576a696a60285f991f28b785362af606bc15d44e72fb
0d2bf286ad5ad8b6cfdaef06a6442f9326333b99fa5c8befbdce360f7e1a8c49
0b7412f79059177a1b81f8dc2879cf5777d37ba7ab4ebcf25a539d15cb0ec0db
32e67d50681dc0ee442b49e980c383bb5ea435ba74da105016011516b700ed31
6baced3855d2b7a2d5afad2af9615dd00f3e9ad76bb32eab2eb13b1b2512c5a4
a70f880c82ce86d6f6e04aa1dd469405e8cd894a426361d2cc29ea7c6242e9f7
4371ee570335d1748e45810ed9f504c89bc550426bff38703c4274bd1d5b6320

Fiat-Shamir context

b0f1d2c678ae6ad07cbc0f68d70bc446ca5fcf905897018a7b7d9a5d863b8560

withheld proof term: 3 226 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.sum_dealAdd theorem, Crypto.SecretSharing:112

$\forall \{m : \text{Nat}\} (\text{secret} : \text{ZMod } m) (l : \text{List}\{0\} (\text{ZMod } m)),$
 $\text{Eq}\{1\} (\text{List.sum}\{0\} (\text{ZKP.dealAdd secret } l)) \text{ secret}$

SHA-256 of the statement above

75165eac80305aefdc959b9735fbecb3791a4d71972aa6eb1474f13e0a3bf5f0

SHA-256 of the kernel type

aca0641dfce210e0892d310db6569be496fa4c527be9276de433e836a95fe469
commitment C

450df4ef533cca9ae56a5b4d0b5bed348a246c2b91dcbf16a2ab2062a96a8904
2157f5f8cb19f067c79b7c1349777c05a56a3350d23aa0522f9264799ace23cf
718e56ab47b2a0adc096d23fa6877df802ab6b2b661b852cbbb31703cec57056
c472621e02a274decddc158caa3cc3ff3f05e8a3a0a5c65821abad0ef8afcf671
a57bf25bca8ccfbcfe5b9cd20433d45a3a046f033696f85244ec4b160ff91ed6
83e32228e10845b9292618734a90ff2609023951e64dcb0cf3451d1559e2c849
57d98cd42a874d409b0626b67aa1b55ced8b9f39f5a4aeba4d8d7e1922dbd4f6
eb43b5b728ade78ef50d805474a2f34e19181fd4f0a5a602cfc08371173faa2f
first message t

ef0fea661b6d289a91e9994e406ae10ecc90c9ff7e0fe3842794d1eca6d736e0
b06d384bd56924620ba78c1cb485c062602429354c46bdcc1556386a67eb0fd2
2cbe0e8d8b8e557e3e4406271eb417fbae3ae219070fcad5c2cea2ff5c3e9e24
1f6c4925ed0d8a14347fe93e50ebf46fab00888cbfabdca9db1347b33dd403f
0a3261ce79a7513852af2efaf207dd4b216a414e4f7036cdd6e4281fa03d8a9b
51059e4d3c8b95c0b25c13169c554c6cbbacd87897eb7b0abc314dbf546f06ba
a382e84730c7896913da6790bbe90121af690b72ca8767315095367ffad70f98
31e9f346ea0596234e8c5cfd8142e79d4bb86e9d80bff4a27136881df4f890c
response z

41ef8cfc6503a7eedb8b449fa8616cddbf133a8126b589366cfd6c299f4207ca
3b36d1ccd81d590eb9a43fdf68e455c2b1aa3bcb522a3a32d18acb1bf7018e95
27b026023e5450a562753c8bf1340ddb10f79d99b66d5cbff8dd254ccac12383
ffbc829e77c7009700bc80c4685c12db3045a698ba9a7fcd9c920daaa1cedb65
3411a622f2d20dd86164503eb88ce2d7a849e2e7ce43286abbc9f06450d476af
9a76a0e134115c6b6ce592aab4ad1b713099ab4951d3feb90f07bf7c1daf312f
1ed02125eb9f2db1f7374c55e156ddc45d442180a895ddf4bc4a2d577fa9001c
bc0462150c44ad81b55c9795b7bd8361e76d215bbfd1039c4c469ba95505bfe1
response w

19da97bd894e917d55a496bf593a3d0ef423ff17f1cad74edd8c9ab626fecce3
eabd87307a2ae150e1e9fc43a209be3971b99e6c01b7ddd33f6588c558b9ce17
34e098ba948dccc3e84bfedbc9639ba7da262173c3825e85c7fc8e3e7223419
63816ac94ee88dd4f061a4d91ccca8be4fba6161970d8fd3b5f89939e27803be
cdf61ac5a5689d2064afbdef64b6496bc61b5e85c28296bc5a870669f24f3e47
8726f09a18a2159780c4fc4c5e8e0d13ba07fe8a2603a5d10af825f7c0e29443
53b22cac7f784d06a91020b6c9542f406f2cb2cba3153fef561f010da495a259
b3d90e364b6f7d0cdd38cb791f6cc3fbdcc5823ce09b7e5c932656f09bd8d93dc

Fiat-Shamir context

0d79e4c974ce2c593d141a2a6f5e760d1a12b315610c8d7faf743f49fdfa2380

withheld proof term: 22 285 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xorAll definition, Crypto.SecretSharing;29

$\{w : \text{Nat}\} \rightarrow \text{List}\{0\} (\text{BitVec } w) \rightarrow \text{BitVec } w$

SHA-256 of the statement above

df4612cf68393578a4017ada0fa636eafdbab5b5b67a66a52442ecdffc4beee8

SHA-256 of the kernel type

9af04f703c781ea6bfe5c218cef776f9716c0dc711fbd36d55d257cff3a7ee58

commitment C

87fa3a9665ce2ec1b7f739c3e3b8ed2a8dce5acb2566d223a0eba89395806824

11b47e9811319b15f7b5e26aa0b5b867ae12a60d8845eac0e2a91bea7f915bf0

b827031e6de50e08f0284ef0aa683329a200c3aaed16ab5a735f96bef7a3ff09

e09dbdcdbde64510bbb1a085c15df9529c091bb6bdb5af2f644ba2fdfa4c5125f

bd15e249154231b5f05afc30a7649513c5809ecb27bbdefaa38fbd732f205617

c1c0e2e7cf20c3d3d40260a5fd16dd37915d0b92acbc1f393c317c70d4d0632

46916e38d1254fa09729eb511f58030be5313e94e28505b97218b44fd9aec851

c0b6431f3b790c1ca6df21dc1ea36d426e29a084efa930872dcfc5e5121afbe9

first message t

4e4d4dd4fbc94574fb68fd2cb4146f7d51aa8f5fb5a85022af97f8cb94bc8ab2

802ff435689d0d9f9d27a5275fdbbb41440bb60776e046b8f09885335281488f

d470ae5bbb56c6d27e434f600d907eb141e6f800adc8ddb82932ceffc4a5b735f

4a690b61fdd06484fc9b4ac62ef1b506c49fa6fb28e3a7964bd1b0c0f6bd5c0e

acb590e0f81d341a60a3fbf21040dada336ec3f7aed112b97e5c598854f2f0d7

4281915ff706eb1917c4b6648f16011189b0a6b528fb2a49c61a8f65898b996f

12a0b5f3942529d70e102c02656b3e877f5213e70f90f61cca22856b511b6f93

0aa773463ce06869064e1779c484b16866780ac00f7c64b04988c9eb0af8e411

response z

5bee97ef9983460c50410e052fe8f142e8741bb2472b3cc6723425bf46b0c2d0

d1210513f44c15f262051cbaa725aebf08f5a8f76b53fdb42a1640651faa7ded

a40a1f7c99868160d28f1058ded1d942592d764e4cef17adf84769c70b5d655c

88beedba6c18cac47b92d8af6e3a75dbbbac5bc6de658b90f24e3441b386f221

a3c17ae280701987c3c01178a7c63732757a997424c8f7c09a0b63e25b085abe

5502688520bc96f205a400a7ff9b78b59272fa60624b1db65e98317dedc72345

b4cedbdb09c9a7204d785ea4f010fdccde1408d5639c360a093983207f8fea02

e87088d4386943b33104f336a56ff532f77223b7b0b319050d580adb442acbc1

response w

2a087b4bfea71a742c05fe1875faef8db59f6633c81f063b98d8f6a497ca51e0

3a20ce0d087a7c743b54a3b688d31b749d69d7eb7f23884d41768298bfeed96d

a0e10138df584ac1fd7f8a9b1b79471dcc1fb1e75584f58b9a1005354516885f

ec426aea3e2c291c8a7fde072ca5b01541ded2cfa21187cfa09013383683ebe1

06f2ed389e72ff44de118f3b80120710c03f97ef0f0c44bf5307760b124849de

798dfc5e45bbe8faa36be5f0e296840232af28c3a389aa35ec1ae14c335e06e7

67128d254465f7f7fb3c1cd99ce23c89f66fa6b793b72bca13785661e882ad9

efed452bfb969927337002a86ca54ddd5e1d9d794e9d91d25c128664aacd4291

Fiat-Shamir context

0b7ae9a4a1475a0a48363c0fd5b213c841369e770a84c4dfbd03bdb8867077bc

withheld proof term: 556 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xorAll.eq_1 theorem, Crypto.SecretSharing:0

∀ {w : Nat} (l : List.{0} (BitVec w)),
Eq.{1} (ZKP.xorAll l) (List.foldr.{0, 0} (fun x1 x2 => HXor.hXor.{0, 0, 0} x1 x2) 0 l)

SHA-256 of the statement above

4986dec355587a594f0969c7c5a11635ff48997dec4f867eb78c87a3dde8df

SHA-256 of the kernel type

6f59a0578ccc8cb7e31eb0e8c1ded69221d3dfe7aafb5ce2cbb9569133e6dc26

commitment C

8fa6cc8bf23537005c705aa3d14a65dc662c7afdb7c2d92aaefb11df5c95b09

61d685d605e13efbfade537a9684838c0c2cf61c93c827b7a42e135ad24db5fc

646f9d99062bbcd40d060a0585903040ed991d7a289b89db0dd36f6e6687e69

489ef1239abc25882e4c21c43902754164a26912fdcccd21f45eb0aad0c98b2d

235bdd07841fbeb58e6890b9bd7ce10d3df767daa017cf6b68d7aid2b7ccb0

18a45db67a555e74610ca550544a8c7c1e357d5e339ed7426936fc93550bd493

cd5294c3c7ae0148d55f9716a374595013384469d3a1f379b972c1d1c00adce3

1afe2362dfe1fdab53f6af403e59f97081f356fff7096a60447fac995a61412e

first message t

a3b3478ec3acdce5b9fc34732df54a196eb0c6f7fb7a15c1093f9fbf6db51d3e

1b9c23c5fb6ad56b0c650ee5806e99bb2104bd51e5126214af18dd1488eef788

9d872bbff040199abf473a2d110d24210430f39bcf44f6d3dfca9db9b9b467eb

82e30741f8eeb5724f1969a5c0bd3c37494c8c6e5dd66ace03fbb01a44d78eb9

a714133fe28883592dcdceec132cb87df5a82da20a8085caa088d4cb1cf1a0d7

a695626f1c4833a858e8ad0a5a6592b12ebd111f4daa4a859610a56afbe86bfa

973cc569bd2df118f32c709b322101382bdd6d2fcb63391742cbf0394363fd38

4de98a974f04a92aec0415ea31b7e88b0ec05d664f3dca557552f820ad50e9

response z

15eb960eef757c810c2ab72a408065f7971aa73a869fcb91af0d912e94d698d

30dcb0910577dedacd158072e0743d4d7a8d087871deeb1e925866fe6b61544

9328c6d016d2129c356ffd4bb41ac2b96069ad98500365013d5297016126b1e1

6eaffc162a637bf6f875f5027b68b3fb3757b4e87ea2c963dacb3d9a2c659d09

ea321f4616b8f6e9ae1d1de0c2d7e8fa2e4f371041f837ec11583af3439c6f92

dade4915a2ab1a1cf9bad7900c41ebe7e2ec384b488bc71213794b3805e9b1ca

5a14a509a1d59e4eb353c0e19f1ced9bf7da4ff88b975f5bd726f3ce90101dd5

5b53c2243883d6494a8354c26bc355a3942f474815e0391c1ab9f56ff6d7e7a5

response w

78201f2afe1614ffb9a3454f01f5badacd3f80404f8f2558d193fc53be815ba5

dbfc596d7c9de56dde4b2e66ee2dee57e9b63840e93c4e8d729f48f1345ff248

1227f3544754003549a0d985907e62768fdb6d9c297c1d056f2427746225faf0

45aee5bf099a32a4fba45f59c0d5c3d0feb45d1a4bfa82c86427b98048d15c59

7bf25746bda3149e4c869da57deb3f1227d64dd5ee57101d1e684b2e9842766

8313704951b5d0e7ff6e3c588d307d9e06678d719821e460fec7ae54c5657ae4

399c7a5d9620b9a32341eec1cf36b0d9631a05f0e294998b4a083ccde3b37f9a

f94f6edbf939b0bc0e0a3e5cb46169e7d9cf968684680a5d4868d700a6280ef6

Fiat-Shamir context

90f02697d60a59e079b2a5c24a81164ea117e9c7c710c3b96c8b7b6f3226bc93

withheld proof term: 82 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xorAll_append theorem, Crypto.SecretSharing:45

```
∀ {w : Nat} (l1 l2 : List.{0} (BitVec w)),
  Eq.{1} (ZKP.xorAll (HAppend.hAppend.{0, 0, 0} l1 l2))
    (HXor.hXor.{0, 0, 0} (ZKP.xorAll l1) (ZKP.xorAll l2))
```

SHA-256 of the statement above

c015b4754ef5a8d9f63f714dbf99ff5f91fde8debfac6460fabec0e83a4fe20f

SHA-256 of the kernel type

bf4393d73380e431f2076592cbf38d12b6908e470ee99988e19b8c205b1c5b18
commitment C

423c49cbe3974da650ee083729ba1509e57d464c980172235f9772cf240e7319
405f63688e95e52a62a1085f0cde2f3e34d09fd265a8509de0ee0f30e4e4fb67
4be374709ade201f25df19eb2d21cb0862ff88fb7944d4f2cdccd9cbbcb051c2
d131199020f1826a6ac217d60d107efc0100d0b7b1fefe3f976fb9ec17c0396c
fae3576a04a6dc1ea4f0c45b496897399a1c439b8826a1279baa60f77db33b52
d80b4052f67519c0ce2af2664fc5e8769de5f1a40c44e19e56ccb6324f94f8bb
ae164c1fc899a4ac4f3a50b86e809aff655b4a69a6c16e3d67ebaca3af4fc5fb
c834a8cbae23d83edfd418408ad500e933571986c6554d3650c25d42f501bd60

first message t

a011d150b39a0b316a175333b6041f91bdb9bb438c09078d5a362b28bdbaa002
5f7ac5d96fec7bf6d56ce933fd72f0c1e85ffbe356e4a2fcc71cc2d5dbb0b7dc
8e81b7175d19bd74299b4978e6b44126d66acb896bc7abc727710e585dcc139
5c7484ed067d46e163b45596b474496504af7eed574cdc6b883791f3513c31b5
32d5677ce8c453bbe3afd721ae68d31f30d68adb38b42105244ab3593804ba3d
d8d016f3dc96e3da2cdeac248f97d877dbe58e51b255e2a8367ac6206bb01d65
7d69469bde45248e84c16849f4bcff971878d39d46a6d00ae402eb5b77515af5
7cc7370b987ef9ad30dc0abccf573d77b930229b339d53def7b0544055f0c79f

response z

525ad643951e88b501ac87ae346b051ecce2bdb1c39eb572534d270bbbeba65
077fa4ca90fe3b59e84db9ae33aac5871d5307187ac01cd32e9e3cddb362b0c8
831ce7817199121ad78cf2bc9d05087dc2aab4acde1c9c84237058399929a839
d29da7330b656eff5a02775fb3183ba0e0215174763bd4496c2b5f987f0ddd79
c7b498eaeefb1ce371139edda02ee6afbb3c693310b8f3486d4c6b77d10f22e7
9f18e00bad8f1464bd9b1504f9508ba9c3d53d44978eadd77373e757dd4509a4
f55f8ce2aeb59d144a17d4f1d80682620813af8e9737a6de97e1682d7218492d
a6104b873749d90251e8db0f89c38eb2bfd3a77454c16581b163efe29b483078

response w

21fb0c55ad01c0e800e812d66098c75a59c3f769a641f036dc885aa57b916c89
e828ef582222dd23005fc203b7d507561edd29e8a3644aece677be876a485edf
4b75bb9b1ef3bcb44dfa3d6f4482251a39a6bfb1edbf4a17041dc898affc4
a12d98a9525b092d51862e0ca9f846b02069c421f6925c4e28aa1949b74423ee
3fe2a7d34a4c7842878ae6de839e8829be3fad36ec0390dfb5fc43a989860f30
c72dd074e25e896c15de3ed5264af4fd4c819656f06acd48794828d640c76526
a86dfb7440991d24dc8f6d725e63d2da48b04c692b9d7568dca1612811d1c005
1522e6d15562d589131e75dcd9a74afbbfba099b8313f80770f1211ace7d3fcf

Fiat-Shamir context

7dcca317d90a876f99af891933f6630d05d3379a129f2e874d7cd5ddf98b380d

withheld proof term: 9610 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xorAll_cons theorem, Crypto.SecretSharing:36

$\forall \{w : \text{Nat}\} (x : \text{BitVec } w) (l : \text{List}\{0\} (\text{BitVec } w)),$
Eq.{1} (ZKP.xorAll (List.cons.{0} x l)) (HXor.hXor.{0, 0, 0} x (ZKP.xorAll l))

SHA-256 of the statement above

bf7b5721b7632137d1c36bb194282be2f81cab0dcaa97403b61ea9b084bd1ff5

SHA-256 of the kernel type

c7a58a29e5cd276a21b1970d8c5a3704f888076b54e9bd99299f4ad1ed025301

commitment C

4ad4f4c186dafd2870ed2f0335b199bd0298e7fa0e70a9b2e49772735c4a834b

b5be468a428125b0c72403aba4daf2b4f8f2305b073c2f7554a1b77c7386f782

f8ea13267ca1eaec7391adfea39fbc1762b27976aa91556c2e4ea6023603836a

c8074a0dd99215b8ebf3fc15f55b4470c11ed72f766291af2f06daa1b819a9c6

4c8b027df3cd78411839caa5adf19a29961d32c929d3848bbbd49b22018e85b5

ead9b1b5b3ab8bcc3bb07cae439c4a6140e8dfad9052af562be154a5870a006b

d28e514d8e23c23219cdf9712d9723e1ba8e5e954bda76d1ec1c8b9f00bbd4

a211448665b0ba7d369c824ecb9005de5dfc2c2ab31cd3c9d282381bc0f1e134

first message t

7959fd3a0a08905005dfb8d7c951a59ca35060368122267d248adec47a78d501

9ceff862b09c33957559f324974c6166433b2ceca673f8789f37e115c847c506

e7aa66588ba3575c543e06429e4df45eb9a00136eb7f7be920117d3d0b77a2f1

9d3db21ee40f6efc73c3a1b52aa5b7b41e985d08d603522d6cc144144f3d71f6

d5c7cb9c97165695a999f19526b85de0122dbd54e0e57a2c3575c7b0da289f97

4b41bbd54145adba5f50fd262ee8a34d58ab0c64db2720df0bffa4610e27ba10f

53552f7a97613943b4eff5b0649eb50f6de69eccc37c67f7465c618e4f1997

6974a8a3123c584aed945abf8ea27913ebb17511b3b32bea56fce8e1607ef1fa

response z

794f13eb49b9dd6779a76afc3f804ace653cf1146a357aea617e00239b5bb859

3172195961610d84381c152b6e0659e69893f8e9b466cf3bccf1c254ee640820

33f3ee49d58fd02db78b19a4b70b565eb13ecfe737a4f0378f667d03e95e6fbe

270f0d117c56f84cbf78f1e4e9a0a964bf8d425c603e81f32fc097998242eb3c

4c9624dd869e16637b48f3eefdfb1e052d2ee2cd3723c9c1735cbe506f1e2612

c056b8592664ecfa3efda84c79ff461b8fe01b5c87b9d506563e15f91e17abff

18ad80bbe5bec659413d79780b9e1885435370601d804378827385f3d0abb857

5a7e1eed7dae6b75ac06e0e2c246c4afcc205f3064fc7239dda8ef858cc10e33

response w

77a15eea315a5646000394ec662703f4db4d652230eb94fb18d1dc00c69be660

37689c18c2267b993358eecd833759ec979ec4f8457336242f38173718b833a3

090f8b8e106158ac9b1778e00b6280e4092fd3f9517268e07093f799535b5216

623760fa175c7f9b38c1519a395d514cd86f0138883699a666c615c083dd2f6

fb696b62004772e8c21a7368e5098e6c22ff41e2a43961e5538fa12891d89e2e

31e2ebef2a70640087b13dcfa2de1fbf6bf631bcf501d6193fff97a693455a60

1254cafcc631e5a6367ee4eedddc0e8b9f811d23cd6511b253a88653c9269275

c94e0045c7aca086bd3f41d3dbb9aa9faa7710534788d7fdcaed5848a241e49d

Fiat-Shamir context

a3d496af1913fce29269f03f94f3729aab27de9c30598a256a14d479dfb43836

withheld proof term: 122 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xorAll_deal theorem, Crypto.SecretSharing:69

$\forall \{w : \text{Nat}\} (\text{secret} : \text{BitVec } w) (l : \text{List}\{0\} (\text{BitVec } w)),$
 $\text{Eq}\{1\} (\text{ZKP.xorAll } (\text{ZKP.deal secret } l)) \text{ secret}$

SHA-256 of the statement above

f46082c4f747fdf660ef46ec81a971971db7e1871e3dfa5ae73ba116edb66d2b

SHA-256 of the kernel type

2217ab59b666350e3d82528eb5011000e1ae1dd34107637f89ce01b802d948c4

commitment C

a9e8aaef91f9840d1eb52f223af76a4c1097a50e7633d1089b95376e8cc8c742
cdc34614925854c7e17fb67588aa4bc398a86fc5b531276acab4e54d59a8b073
e7bb26a96395b254ba225245b109fcf8aaa030e7bcd90be452346df754d9ab01
893332d2fe7f2cfff2cf7e07b862806f93b1f94fb8842acc4537622d170238ff
754d8471bab6e6a37e8ac1b28485133b78568dbe26d3ec84369a4a795ea31651
6d902419afddcd68f98565020a6db0061ac7af23a2653266a7f1a39ac9600781
c1576f36112404523a593dbd1632705ecce6aaee914914c8ff69003bf9796d75
fc27b9f89326aec018d316193c18db39e36450beb7dfdb14993344cc588611e1
first message t

e2d24aec7d81d5d50ff93af3a0a7ddbe6cb8cf5dd9dec73836240e2203c77e95
ba3de3859bfcf564eae3efd259231804f51e2eed53df0a802ae2dd4233a0fd0ac
efeda845445f4d38df3a305a0094623a35a3ef12f7442a4b3c90743fe48bbcca
8103ee27f1b4a73e31924c195481924110165f6b5eb17c05acab3803d1689fad
cc6a203cbd748e4caf4eff70793f3efcadebeb95ac8dcb60d3d64b0473351449
5e9ae73304f4e3606cbdda63fc28a4b66f4357cc3178440ce2a2a6373a44b980
c0c328f3dd7a4921a976d42a936ed3bc88162becafc3aa908b6d7b3c984556f8
1204b9db965d0db748b5b553d35a7754f5f3a831f61182c336f1b1b643f957ca
response z

3637406ddf4ea356ddc7502a97918265e6be07b893a7399b872c161c4432eb40
a88314c53110978e05caa6fd70f2e59e053f3534ded89501230921a17f8419b7
c65ab6f1a42749c9aec5fe272c21777b4b4907ac3d6839202bf239edb902be5e
57f796a26d2eee76b8557403f3570c21bd155334fa99115f717ad1b2a58c7e00
48ccc1204a26c8f0e3fcc47a5ce50e7e9d270c62305fc93c6a72d2ff4a656f7a
852536fb5fdb1c5642116918a7670300c510b0708fcb2d56cefa54087cbf2c6a
2015c389a62df794c4280233caeabc08d9180f3bc37a7c86e1f76231b4cde2ab0
b3c4425e214f4a62cf8d88cbee1d744ed4814c414d352bc47d19d07057445069
response w

421e1b72481a6689b5e9d1718f1804758daf32e9ef4895d638f0037cbdd48675
41592a3d557243710dc6ba0372e7b1c805622b4d157faa05451b2495270f3568
5daef431615c8f75a1ab2e1aa6b8e93d1d5fe8ccf677b7152b5c6c262243da24
beb1d828c253dfba764e3aa0b010a69224aefb02573147eb8092c00950d89765
16a698e7680855890d0c21a595610a5aa67f8996ff748eb68a826a0f2b5483e4
2d260c46eef2cb23119c267a57962cc3fc4aeb58b3c4db2f996d59b3caf63a09
642fb7459a73d1e02a4b24ba4720ef113f5b984cfa9cf87be9dae1d638075fe9
ee9ae4987712911679c32f9340876a185991bdf792fd89b4f927444dbad3e94

Fiat-Shamir context

908aa076fffc433104ab8cf0bf6695dfea41b25b677d4712bd7730ff3f53ff47

withheld proof term: 12233 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xorAll_middle theorem, Crypto.SecretSharing:52

```
∀ {w : Nat} (pre suf : List.{0} (BitVec w)) (x : BitVec w),
  Eq.{1} (ZKP.xorAll (HAppend.hAppend.{0, 0, 0} pre (List.cons.{0} x suf)))
    (HXor.hXor.{0, 0, 0} x (HXor.hXor.{0, 0, 0} (ZKP.xorAll pre) (ZKP.xorAll suf)))
```

SHA-256 of the statement above

acf24ab9a4f8750320dac652d87941c73b746e2c6916ea898c4db5081493b10c

SHA-256 of the kernel type

2a71c76d1e1d44c3f4e8502d521ee7d99dd8f576bf2dc72beb9b896b3bd8886f
commitment C

437a0672dc6d98f8d6186d475e464a20a37e542ff9449de0e33d87597892ae0a
7e4d2ddda0cb49d927f01feb5c9d48a4c21f74d96b8abe9000e859a08b5e483e
9b5e0a831033e92c54d518d7172524caa8875aaf681e8c3d28fc6dafb455eed
70185a16e212a3b74becd169c65e5142a68f0aa05c0bcc63364bd52458fbd18
e8d6e59e2dbf13f96ac570f42921feb6793688da93d6e5bc713d8ae5c487d13d
e8a644f06928df34d2b25c14a58350ad0c7bd3f2bfe377e4c732e24057e176bd
cb47fd91c3208c7f7776b8a652f4053a21d61c938f5a0dea4871d7e5ec55712e
d301cb7e984d0b756e53fd866af45e62e6aabfcf621a54ad492c8ab760343343
first message t

f518d8daaecb005fd810c51e306b44dfd97455a006f11932c04caf4450be7405
50b6d0f705ab1a4d3e5840a5e188b97acc57632421675a1c2633cc99abdd0ae3
277a6f1456827becf766d365362957f19f6fc05e7a21b5b8a9018cd59eff50a7
4397e014ec831a5f7751cb54ffc6770d2acf4cec732fc4d5a833bd291526e958
ca58df7bb54c9ab372e4dd38507d4e1b52d3e69304d97e452dd1757d63e3ca62
be1233d28e1aa39ec133cb0646102894e30e2a106f2c070456080bb92854c8d3
468721fc7cebc9c41fad0849a40c4edc8ac06b4ec5b4ec955ecf8b7c444f4100
4abe3292664f754ac8c7f2c39c4b3fdd1373c8d430b5d1c058998c552496ee80
response z

70ca69e6cd3de6f878a2982626fb78debdd573faf8ae04b754a461c17b516acb
8a0601a2773a3e7e689f199261a0501091277655bc4eb9862613e5d7ec644b0f
82e0c32cc2d65b124322aaa0dcc374cd8346018751569d654c2741dde300cb9
97467191565f853be6929b25338d9c5d3f3b522483b36d155b8794e9f0831934
f25e36b9cbb5b006e3032503058d508f727b38e8eb665c574f936eb92f4738da
2a5112c00cc4e29a19780f8b99bed8b10069306a4af564c5ea44d31507b81db2
f0566646f7c3c6222748a4c9a8fe0adedc55e2e6b7c92006cbbc8060e30cb7b9
d891d2bba35bfa0e98e1387036730b26fc1decddd7fded3bdf8e92d25ae98de5
response w

325da8249c64f7480d2d410514e8b562abced3f824d1da816db5731d947a2c79
442e042f53644c3c80ec75b360989fc49c714abc7065f08d38a67668f30a474d
fcc6bacddbe3e2b4c6a88857094f07587fd2566df7a5889901621936f73d32c8
55a46a2bbf1f4df3b79d2b519b46316e53124bc93605b5ce61196ca70f4ec7da
81ee553cd31ca344b179c7af509f3063d58a4a8e1adaca9c380f4a909804a2a2
1fc07872cded4f6cab35ea84e606d10b13945339a6ad6b0a705d19c45dd31ee2
e6908ddb90f30d14f3e53ac5e0165573e8d13d4217bd306eba3f8c7fe287cccb
c525e3d6b4d16f6ff31b406d255d40c1a0fb5d09ca3d005ef065d6fef2425b6

Fiat-Shamir context

15460bfcad13b273d070d28d7753925bbcb9c5be0f1b3a15130aae942af2c118

withheld proof term: 15 486 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xorAll_nil theorem, Crypto.SecretSharing:33

$\forall \{w : \text{Nat}\}, \text{Eq}\{1\} \text{ (ZKP.xorAll List.nil.\{0\}) } 0$

SHA-256 of the statement above

b2c02956eaf8862e4eba25d1fa994565a921da37ab2ef3ed9a45b96e927ad91b

SHA-256 of the kernel type

fdb548acdf60c643d693e9406ba671a3da922209e448e08637d8f4c7fde3d1aa
commitment C

9994de318f629397117e60b3da5f5cc3b5c48cc69213290f4c567e23d9430c55
3907691f80eeb2520022afe8f3d809a3ec85ac411fbfb3ed566ad60f606ecb52
5cbc3d4f1e7efb10df3d96a3324ad9812faac6fe077f2633316faa4fcf7e76d3
c183e5887c8e80e9738f8097b7a67753a23900a8d26c80e7e3f136c3e7c1516f
7bcb338ad2daa03c901c5ff0e11f197be66257b12c6ae999edc220cd1d839e4b
a437e005db69976a7dfc76349ef06e3f4b522a84a692b7756dd173184c1e9dc8
0bae4572bdeb6ad1ac72af046c9ce95351d1b2f67d8c44d28dd325489c9d5a02
297e7dd86072c7b86b22979695b9917aa30dbf733df68c0425f5612f8b3d13af
first message t

feb5bc9ba508c2e8f9bb1393084a9d25abea0ede3a035e995b3b9b8f03e3207a
f06669d4894d99b323914b9f167ae748c1cca4dacb6f401d4c43fed119e92b2b
aa0c98e58caf6e71266f7e79cf8035e2cb57facf2a5773c24b5a058cae598b5a
ad5076afed734449f8f04c49eed54d9603b44eb42b88960f70ce145193bd91e
79eb171db3fb9a543095e7da5d7fb56179ae18edc6a6976ced376687505ba660
227a4a5e1ed809bebb96e6ce27eed6768df0e377953a671c8fa049b6f9439b1e
9aebe147469c8525e46209fe7566bb5b4a1184cb482562bb1ab8da1c29ebe2bb
e443cf306bae7b0c9c9a47a09fb66c49976c89da7d41a38cb0b7986a4c1ab000
response z

2108e1956b0cf258e035c149d9183a28e4e964cc040c2cbd9b30976a6ff5636e
0367ade4bbe0c32d5252fdc61c329bdc2d96180f6a18f889b2d5a96b08e157b
450cf9dd0d87dcd1b9cd1e65ae9153e58a94c0a76e6bc987c3f7e5d819c42ec
efd6fba92c8a54945d1294aeb53f5440b69c5b72452d87730b1f684256efb380
4e24d83333a2829153172d339ffa15c32585dac575d49bdd0c4d1be1094a140d
3360f3aa9914877e2092561e7816015109fb809348d68db724df3ffc0138bb86
b7447ea8c65404cad893a2f112c44bea793994c6087c7fd0df3e978e3c091421
8740af0ff4166817fbc7a0d5126e82a2ad411a2e5f36d18dd0071727b6696a6a
response w

6ac51abd13f12f1daa78d219010f257f8fae8e57683683ee42a7b6f9862cf9be
8bdbcdc748c47e8ab7abd2e6fa94bb121e1b2d8cf43feac481170cf702b721fa
ea2f187415f4e90639ac2ce0f88552de29e380b33bcd4894af149166926115e
014abbd47d37f420a4a284dd49d662d5f0a91195dc1f71bc77174e58448c803b
f656612c251d0c435cabdb6c703c0be137da8101562d1776cf4c7340694d73d9
bb623da05dafd3cd4da924707d33bfcdebb712fa12a25e59166ff0d3dcd63ef
f136360a3c17d44bdedca27d6b26f3ca8d1ee702d6303bcd0499dfaf57840a9f
f5a3a40a71f635af844d4ba5f9cf9db21b6688e8e6ea66081cdd5073741c9b0

Fiat-Shamir context

96d64f8ba28fc857b3088c8fc72ed950add626cb3a641f368e7b0260bdb990f3

withheld proof term: 76 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xorAll_singleton theorem, Crypto.SecretSharing:40

$\forall \{w : \text{Nat}\} (x : \text{BitVec } w), \text{Eq.}\{1\} (\text{ZKP.xorAll } (\text{List.cons}\{0\} \ x \ \text{List.nil}\{0\})) \ x$

SHA-256 of the statement above

31da677d32375f1b8ceae797be12808067bb66798c6580dacc3c03e513e87b8a

SHA-256 of the kernel type

82f74edbd13795905a4c0f2ed45d15c1e7f0ed1ebdcfd96149a94a2f28f9159
commitment C

4cb40d7ad5ca3e1e4428bc4ee0027f96d70b88babd342760c632e593ae353d51
e9f0b358c536143b40aed6e4958df3d0ddedc578d35663580f07f9548b33a5ec
e6bf18ea44211e3b6e31dad8bab243d01c74b61ab506a65fa509535c790d86d0
85acfa8f5f7dac6ec9c7c44bf682f2fc030d8e0c7d5cefe48a271dd65057d118
8979dfebea836dd1cd5fdf48a60550724e46c488486f4f4ccff896721741e1d3
c0892a5487c53f498f5001ab626128d173fb5725223b68a7a32e4619d127ffb
a41e160a74ef98b1528a5924690e1db47a46407c50edb04cdc782c66e6b85c38
1d16a6e35b2fc304430eec09f06e38757cab1fab7bbb5316583da02ba98b1d2a
first message t

e354a30c39b35765c37574c889b41b854a39ec607f5a122ff20a7bac8220cc2c
a2ac4f6aae01a7edfabeb6be3a4da74b03987b3d54d872a7459a8931b64182175
ac43be46e46232a21e2a37ed89b3b0276c36a5b286b918641fe0ef4a85dfeeb8
87135f3a94dcd2af472b8cb9daf67b444587a10204cee75f108860302317f09b
87038f2d9f922a85801512f5e5da6bed3d2e197b2542cb40e0f6743b752ffa19
98318bc8e2caca27eb3336d97faf0f05d437337863c776e5e3601095e8dc5dfb
adcb5abe5917b84072a10e0726bdcf6b52d30bc1b77f6252abf3be50e53ef5ff
6f164affa66a279dc8de50230795a8e993491ed133fa47abec065191308edbf2
response z

11010c1a64a719581b0c9ebf4ae0ff79eb13f00ede82f63b7ebd4eaacc40f1b8
c309007025119179a6557a95c2ffcc14d2f9b5879f25d625e6f01b237edf5427
81c5d54fafad3f8fd4189d38fa89c4391174521872466f9e356e0982de93230d
88b01c009ceb828354438dd26b8c393b04d5f4951ca507c9c905fe13f7bbd896
ee0482fb7f4602a92df33c878f92b7b8b4addb8fd51d2f301f1fd24e3b45b10b
e3cfe19edf564c8e7c8953d7ad055ead56769ebc4daa3d8a3e632fe390a20a35
ae1a64eed07f53185d570eb39a1b63c35158e0417aaae3139ab41a01bd160266
400e6e3ddb48403a9977d00dd6429b6923a246ee89fa6c09681a9244e8d4e16b
response w

2cc11f66c89a3f6ed2cc5ea72253c7d3f4662027f74a0be230caa1e9ba76fb4d
50dcbf7fd51b8232eba20fba732db4dc11d23d4d59ad7e273ffd303c207b2d8
e48cdb630674fd117e891652e379aed5eb38d7683b3294be1fb18ac440d6c087
c8885a91ab58d10dfd374c3d01af8eeFeb91771327dbe7270c798ba34a53297b
16d69dbaf3043b0c080124dc96a6eef3be0857eea812b6345edc83d68251bf88
6e98b22f3731dbaff78df16ce2a77d457d356fd9befa19ad5e83999d882c3b4a
d850f030ceb6cf656cc4028bd0ab7464615af6a1d0386e703c17430cf2cc3847
541d16d5f6b117249f4adc7cc307f9bc4b9ee3e3f721f614fb79bc7cf9dcdba2

Fiat-Shamir context

e1270a69b3b341ef14e02664a420faf4a98f591cae97bbf806cf58027d4a18ab

withheld proof term: 152 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.xor_xor_cancel theorem, Crypto.SecretSharing:25

$\forall \{w : \text{Nat}\} (a\ b : \text{BitVec } w), \text{Eq}\{1\} (\text{HXor.hXor}\{0, 0, 0\} (\text{HXor.hXor}\{0, 0, 0\} a\ b) b) a$

SHA-256 of the statement above

1e9e693d370507b75934b30c27b681b707fddf94d1a305e4c061f9d66d596582

SHA-256 of the kernel type

b2fce435e55db6a8eb7ebb79e33fa70c1a300e16b39aca0fdb422a7e59e8fe7c
commitment C

2d3a2699c921437bff798ded3a7354cd8c4ef6484dabf2fd5f4cc6c356064c4e
36489f29fdb9919906bfb3c7032036264a22dc416d9d9fcfeec4cc23f0f477ea
076e03abbe4abecba39ac64612676ffa0c653a9df2c6947f52e967d1cbd2759d
9fd62226caf36774dc5f85322880ab3f633f85d834d119e0a8da8618ca5bd678
2113bb43914ba8bb1a56bea36098b6fe0f72b4acbc862dcfc476b4551937932c
7dfb0974101c7ba2022fa56f6da8e54c635531d669f19d3ad27beeb2cafeceed
4572c950ca005b49447a299331ecf1d64bf5d43f6086e3826651cb308e72690d
a9f0da772bdf3edfb8af05749f25a411f93f16abc734be825cf5814885e434d7
first message t

9c9e2c1a12f6a15a28c112e9fdf62f4ec460e627c4fa7247cde42dab3655de71
d70aacd5b82e64b7f7dad2b2f442967e7a4ff4db7b7ba4713a2cf5d19dc0d683
89bdd24e18c2712a56c1e749dad2bec4cb3dae186437b14cb4181dada20404bc
3e444e4dc0292821cac776cdcda656a30e048394f21181a6dc30695f4a526732
c281e81884c90fcf35f50c9846953be8cab063d2bac676d75dec5f3619693c9
d5ab61142af184310a29b55b7baa088de9440448d1963a25aec42d66b357f718
b0402007aa2352e94b763447782513cec17a4a1ab0dff19e173c1f93db1f8f4e
53345818c0f73739c8b1f8e5678e26f1323d5ac18246265c461aa5f0429394f2
response z

4ad1bf8956b15e2fbf516f5156fb49656c5c84c2543453d58e969fac64c5cea4
7449d7e3529cd55a182c9de1c05da54a24752202f31a10e14fbc372aecddd9a3
ef0eee405325753b3fd14a3a74511393ebd1b5f8d7e3472b32c067b0f2b84f1c
c5a86509673291a754a025b3df98578a82c15f09e7b399419565d43fd8696c10
a6f2f921cac0df28a55b9e92bce1e2a33ffb1426de096738e4d28fae59b9b016
d914d8b33d5c4489cd30ef71a9b2acd3cf4694be5c7b5ae8c9752146f0f9385
e6d71ac2a56d76fb940eae28da8d97c55c14d2a589a7f1c653d58c4b79b145c3
fa6652ef62e1bc869cbe2811ee4b3f7cf39aae47efe3cc83b68dc563796f5f6d
response w

228f0698669564ae8e3a2e0eda412c24fcacf5399a55de56320b03a0bca5276eb
4ce06b15ef1a03641bbb135959f4c0af8e59b5a86f28e0a8263a91ce08976f36
a9dc63166dadd2a3494648282f7caa5f9ee167863d3f586edbf07a5773f06e79
1568a743c0ac5cb51d29dbf72b5d5afc16c3cddb02eadd56c7e9d7b877970f4e
30ffab73feefbe86f9d02e2a0bce7f80ca18e45fb088b9ca8bad1b3279a43228
3a6612d1824adc8764a42053930c48b3a30258e04485c5b4202dca8afc83db0a
a76d52228a1b4b6d124ad81b82a033741c67fffb60a94f475491b46f0c9eca40
6337436d1829357a7f8674124e02ff1529c1f67a9b23707b61dd05427563555c

Fiat-Shamir context

fb3fb81db716f3771a59a2df0164fb58245fc09a7a9bc98171dbb8fd792c46d

withheld proof term: 3 685 bytes • axioms: Quot.sound, propext • kernel replay: yes

7 RequestProject.Crypto.Sigma

ZKP.Sigma.Accepts definition, Crypto.Sigma:132

$\{G : \text{Type } u_1\} \rightarrow [\text{CommGroup}.\{u_1\} G] \rightarrow \{q : \text{Nat}\} \rightarrow G \rightarrow G \rightarrow G \rightarrow G \rightarrow \text{ZMod } q \rightarrow \text{ZMod } q \rightarrow \text{ZMod } q$
 $\rightarrow \text{Prop}$

SHA-256 of the statement above

22fa852be40eb55dd6d67586633a109766eef1593563cc122d72918832018667

SHA-256 of the kernel type

a7529b7b6c55f9ac8924869d70171af928c85addc926f6ef793902657f67a923

commitment C

743df34ad103c6687d83f72dcf03d29382eb924d8dd93d4e33d29a8e2b5917ad

4459fb958a896faae135dae75a038fb02c08ce53263e3493e426ed251e3c15e2

ac7fa70035020517109bdcece3238901a73142ff4cfbc39976b1cca5be06573a

8f0852f09760ea047468b4985015d39cfa8e316e3da5943e0e21acd6a3540335

06303b5b2550302c18605de2888ad94e8a37791b2874fd97c456cb25e5bbe073

6f446ea267a936ce3c8141002b2a14586a7ec6b70fd538a449eafddf5c933f

d509a9690b44e6583c38eef53d201894b0ba0cd1432a43f45704c0671a7af93

30097a1deb9d315b82a907a373726659ad3f17f2428b92a8c8b67f181b6b6f27

first message t

965cdcc183ffa441357ba5e66f08dc0087381c1885c899d614eabdc0929e5240

e3b116de2649bcb9a0a16c2fe9265b4334622a1d97fbc5c1db395f386cb8985c

7d909de37c58e121fc4ca060234aabd45817f482e36954f155dcb7cd4731ce33

2dba24a53700d5a399a01062b3779d8b3d97ad8fc65ba24233b1fd18e6e52e57

9a3f5ecb349248f299f1da4437b29d8a69d6c1d88c15c4d905583ecb724c7c06

2e663bed3390a64dbd7b706d33bcbfe394db08c670ff5b1c01fef99536168ddf

b12210c86d3ec53f355c8855ff04325e18e1ce25ee7056a21cb0bb2d58057bcd

e9e78cd5c1e940efbb8075737668b1a994777b01d40e08c51815bd2e359261a

response z

4f4f2379955f7771b86ae2dda9cf715091c747dbd2e6524a5cf94f41a26d33a0

96f29c3fe7a017c478edf3340107171b953e725e981ccf84465d42bb38d940ea

980647eaea1997c36b528cccf307edb8bfef957e5833ca087d6592758d928439

92d90dbb8a3a0034f56d12fe88aab21f8adf08828ea43b66904b6dfe4b00e4f8

fdba7814c9108c56cdbe65c4856a37ac155be0826af9860d4d5f7d6dfd8b04d0

ed4defd5957ca22ae2571d82cbc6c9584de5aada27d6e712a329fe2370729d88

58462123e795052daa11aec657685bd923613613ddb0a568157ab0d05d64920a

a351aa5753aad7c882f5c37eaf0c661728d9a8e6c9b6ee2c24363fdebd5d3df2

response w

3cb44a85dfa88ffa0ec1cda29ed3c1ad6c439588951d9d40c5c2499c4616d64b

8ef91e9933736a04bb1c3668eefd037b67928ad7a9ad61932619731d2ec19773

384497bd9a59359d330779dba4a92538fa96936950e633a3a88dd32c3b572822

c6273f32fa79d9df996295cbf378ad5546369fd52370ad97bd02923d0a8b2b65

7457f54abb5f0ca83671e3dd5d86a4e619d8746024980bd7dac23d82276df253

8ce63db6c1d0b1749f170073612e54c3422c924687b048db995456b522b5d15e

e4fa71b73876344b0dc96e9cd8f8531504ff132e7688bdc857196d846b4f0dcf

5b3ae719f860d8ebfde11c8b606ea4f8ab0e46df4565baf919e1ec89b150d339

Fiat-Shamir context

6f5915d036427b0536893d23dc3bc1edf23772d73a0830a64312c72a49a854d8

withheld proof term: 1 067 bytes • axioms: • kernel replay: yes

ZKP.Sigma.Accepts.eq_1 theorem, Crypto.Sigma:0

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\} (g \ h \ C \ t : G) (c \ z \ w : \text{ZMod } q),$
Eq.{1} (ZKP.Sigma.Accepts.{u_1} g h C t c z w)
(Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (ZKP.Sigma.gpow.{u_1} g z)
(ZKP.Sigma.gpow.{u_1} h w))
(HMul.hMul.{u_1, u_1, u_1} t (ZKP.Sigma.gpow.{u_1} C c)))

SHA-256 of the statement above

bba6d8f577190c99444bf29704a472e508d43136fa9b2066d8783a80cc1fc6d4

SHA-256 of the kernel type

ffe51f7a21f629254e1949004267698a03fa87b8adea26ee7b5a73d3321c9ff2

commitment C

f7979bd5cd7a0a5d0aa7262dd9ce17315b78a9acc1f86555ddc7de8cbd03f71d

9dc36547d319e15978a357ec918cef5f8e4f73d436b1c7f831f2ba6a53c95bf

15f144f8283a7f27206ca8bba8809b7fcb65397635f70662df60658501e806a4

7dc5f4bfaa0b74f1177cc902e2b964ca88989cd15aea456976b7f934da354ed4

683ad6a014d72e5bd1cfcfdbe36c4d9c2e3ae62a45cd7f9f3c9f304d21793b4d

7d4f3ad45f31db1778f4d4e37f173afc0424dc679bac1c5dfe54da016ab7c6e7

30bed8739e89b220fdf308306526472786431fd09f4d5485d06416fdd348ee20

95add09262e23e06e7cf8b395fa6dec9ef111138a81be638f2c66773a22f1082

first message t

b3bc2479eaa53731aff7ca90e44ea87f8e5374ece92bf3b8d02aff4dac288a5

cc3a9a340cf08ef81c62f2cb90454133c8537c03777ff77d8c2ce9fb4b2c4f38

a86c6a31caaf2adbbe289496c270c7cf729afb8b7624d801a800fcbdb0b5769c9

ccc7f0a77ee216bfd3bdc03a6717b0a3b3ddfae09e47aaeb557dc11f73bb6db

060b2e67f2002d8dad7a400663857e16cce6f3ca20006d9d9db4eb3addd39c17

4953d65e7791f348a317b6aef53a3420520b2c181efc7818e968ebdf24073c26

559a0cf837ad91689439f79b441b9a391039c88a74f42ade835255c5184b206b

7b53105aec029f4670c32bbbc04dee14b333e7bda1ba7958936117e003a5ff8c

response z

d1dfefbac48afdf8eea28a3e1eaba8e5e96aa6dee844949ad078d378fdef6116

a620b143ccbbc95936e2e9bdd05a3f83faeb0198592b959c1f0e4ba19db35ab2

0ed57ce27b5970c04f378ffa22de5be8a4727b707f8d19c8d490148a89d2c284

cc47015cfd7c4927431ef62c291803c9cfcf6b309efd9f1aacd493134226ec26

cc2c6915a7f48bc2b9817f2dc6d9efe0761ef65d8e1fb966f5a01b9753e8c937

144495a997b84df6eb4d05b51224f6152264ddd0aee9d21ad44b60fa487fab93

5af044cb08057fa1ab94928cd6ecb20368a9a2fe20e4a983b75492838f4ea100

ace84049482e8a2372c030a3bc3a563330ebfb7f558d1d34be98282da11e206

response w

235ca9e765c1740301e91c76ef17505186bd1fa3a049ca3e6c7e92e95c227a2a

0c63bbc5a1e1b742b057e61e81e11236955ab4d8b0175ddbfea668c4c8eda78b

7b5ae076d66e150082b29b68190c30c703b962473cf1fcd1d4b2c62e4c473a86

3bd2d9e13315113fb41fd0d7d39fd5dd6d9358c70fbc310909da0986a9cc42d2

e4724b0801b79657b1f983687ad948384a08c7c9e323063005b4c5e2806cef1b

12bf3683f06b8cd740ba8013e521c4277259d20452f63358c40c10081b5116b3

5c8e10ebf242e3bd02cd43bed2bd729d639ba83acf870619ef246c2a14959193

39fe7582c8ee5b69dc208901cb263b394686222461869e696e21e48106e9fdb2

Fiat-Shamir context

c3482e188280f5a05a2af7829b3401be2f7a6b56ea36a5ee5e117999386d7d60

withheld proof term: 197 bytes • axioms: • kernel replay: yes

ZKP.Sigma.accepts_first_message_unique theorem, Crypto.Sigma:177

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {q : Nat} (g h C : G) (c z w : ZMod q) {t : G},
  ZKP.Sigma.Accepts.{u_1} g h C t c z w →
    Eq.{u_1 + 1} t
      (HMul.hMul.{u_1, u_1, u_1}
        (HMul.hMul.{u_1, u_1} (ZKP.Sigma.gpow.{u_1} g z) (ZKP.Sigma.gpow.{u_1} h w))
        (Inv.inv.{u_1} (ZKP.Sigma.gpow.{u_1} C c)))
```

SHA-256 of the statement above

838bb894a70c4f3cb644c3eb5bd244153e08bb7d830152527c2940d18f8814f2

SHA-256 of the kernel type

014a70cb21283283180d41e4f97e84fd2a51f1d0554ad80b70c99e105d1fa04d
commitment C

3e8bf2360a09e8ebf863d641280648dfa8090c3e8506836896ebdfe61f64d5fa
9010c0d647368954b7a99e717b87817e9a0456c20da63dc492a53a0917229795
7b75fa6ced9c67f354f733cdc51a1baf0b79591dd38c43b67f06f13f41819f49
593ce9e363c747f9a8f1496fbf826eac4a41d11abe55a5604f5c14c246026cd9
09a6d9808c80266f2889a1cc19d6e0720318fde536063652c7a6f86ff965349d
8aa8fef7612f8359ceff75c0e769b5965613eb78742b1056e1f3489e8c2d7b9c
44c5dc5e6b25a157916f586ee1dab29c8ef5fa60f6157dfd3e8d41bc8f487980
2ebed330f68ae69636ecc3fcb02924d027aa2541c662f17b5284fb541daab65b
first message t

abbb039d7e618fe9fb1dc6f2ed615ac101fc3a3871fa786961dac6e1c038cf96
16e26fb95d66b126a043d1032f69432a09d9aa78662cff15e7f28e4299f4cf54
614e2ea54aa895312021b24c2084492ec79c607f76f1563f623a0a8cd855e33b
491049cecd0ace0e34e6f29a93e7a16730350ad809d4196beace36679bafb479
48c89c119940c803949e625c8c7b1db81a07477d6c8b08db19d09d8fb0190afa
a84c473bd431da4cd4e6fe5b324de6550d95a434578c19a184c4a665af3e98d4
c028a09d58505de09de3cfbe10b403d01a67cb2939b8dd9529edfb49cdef125f
c27e206be427ce112a75393d68b7cc852fcc5eb4e8b7be8677d9f6db2698159
response z

47ac953a0f8d6760e4332861a1ca8d3d0f06b8248af5f8b351dad1b46ed8d7f8
007d649b3aa298cb9616639801c9e8b2dd01c6ca81ecafcc3353d969bc2efc9ff
a5d29ae6b5008e1c83554a8f2c3756bb1964062ef2602ef91f4526c115700fd9
7168c2e76cc46f1786984f4d5488451a36098cd509b8d5bd8da5bcb30a77c48
27300599c88fdcd9af6b923bdb5f28afe40cffe83a5d9a125c75402ad15e98a
0f51c229362cdc7ea01f9f2c0d11584f8a6814e580a3430893d18af952d8e275
4c3f436967ec88914879a3ff88d86e1af935dac16c38e180fbb5636c9b9aaaf1
d9f754297b23e722472d21c89f822f7bf4bae492bf1ef5206417ebb9e519d65
response w

2e206354af6fe1ccbcfe8f25919a95edd3e943ff653a893e3ec749d32f8ce8ff
4e6520896a5cb4ec3857b171d7b7c749e618df8a828ff9f07587196f9a9c8da9
66d5ce5bffb62fff36bacd8706e121514b2aea0ce79caf90391b963d3810c672
3cdeda8eb340c66e23c9d7f032907924f3a6837b37db0beb57f3b1e98684c847
abe461ad937d43e0a621993413f0ff2db1481307198b73bef67926951ac7a9a3
1c1691847546b7ef0d16c2f18cb2fdd3b85d32ad4d9229ac4990716267478ab4
b4d5f3fb2a68d2cdc897e384a9322e23a0a62514d333b44b2f29b8ad9b864985
447e8826eafffdc36b908a8ad244458375fddb30da1d494e092fd5e39cf0a4be

Fiat-Shamir context

8c496dd887cd56995d3c1e5f4d48addb6499fadb25d95b781b505e4e16bfff7

withheld proof term: 12074 bytes • axioms: propext • kernel replay: yes

ZKP.Sigma.cancel_common theorem, Crypto.Sigma:38

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] (t x y : G),
  Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} t x)
      (Inv.inv.{u_1} (HMul.hMul.{u_1, u_1, u_1} t y)))
    (HMul.hMul.{u_1, u_1, u_1} x (Inv.inv.{u_1} y))
```

SHA-256 of the statement above

75bf60486f0a2e6a08e7cac01cd2075262718da96fcd054b54c24488fa1b403c

SHA-256 of the kernel type

5b17da2800c0861f8f7774930c9c09d6ca24f83bc118ae5a8b96b52c28a55e21
commitment C

1a91144f212d32b2541eddfef0edff75275d9ffe2c3aac8c266c14a47a3a8872a
88fd6084661a5dff9436a35b0ded6c7bc194599ac9cea6f6d40b8168f2fe432f
e0259119ca54d87f872ba689af4cdac8a312127b27fa135835eb492ce3ff311e
25458fca4fa9264d21e5b97bac91b20c8f02ed90e39727b62f7f55e856352b30
764491050a37a98bdccc669e2999401169a543a648d11d4dd1a5e5fa76e5fff4
3f81476527c54db9788d5625515ab1baee38928b156aceb6bcd157f0897bf5d0
b4b2f398290d47476afb8b5bd333a345f9dc315dad1856f3b30f5eac5e3dc2d8
569edd9b460794cfa7e44627303c96b661138ba4b5bb63eea4bcb0e38aed7f57
first message t

41dba35eaf57e6ad02c22d8078d98990e0f959dcb28b04bc89689332211cee14
07d2a02271ac70ac51b6be7bb0b826ff6d4ed95af68efb889f924b47ff3fb8e5
93e8e21e426a4f22a0c256b81dab8d2ac510cb9719b42f7fe09832f07e7db390
7699857c0f4e45fc0ab2a2330ce8be8b729e6101791b6011d46b8ef280adf7bb
05fb6f33a41ca15b8a03a0093e383757760635e3bc077dedacdf1e85c4124c7d
2b0c3dcb72d701d51a2da924584919f001d0893fb9213dd08af47ce987406798
6d12176f45a4239fd0f320545550e4adb24f7f9e1ec98cb7ce2b10a9ffa76122
300ac1a6474ee31b1a4264ac3242e814b2d85a3574b90b32eff7be5d5419f3be
response z

61dd3c82b6ed604bcb1eee2383fbe2c6f5fb9c7a541b5f37ee8736f35f4fa710
0a4ce3a5987846f1bc31013ee263d9dc1392a254f1c97ad03e98b3e96809a95f
3f3aa82ea7b21d359dc7db117ac9891271552e6f18ea71926ef544c1392c16f2
e86a10a5c7241033eb55b64c1d68a72c3d31400152877ef893be97a76ba86e30
de912a7e1c3c6c189e627f4c3b6ee6b3c8601032a78813836bdf5deea5fbb4a
c4ea08c422a3e6843cd8ab50d2b2f18b74b7e660f893c65ddeaa3ed38ee3da58
535834d1bd4f264d7197faddf97c144fd3b83f2e03da895b1d3c527ce2cb5ef8
c4ffce75bbc5db4edca27b27d32a68401939d6f3487707ebac52c3ba2605651
response w

3e4588ea17e4b28e628ed87e772026dbeb86bc10d4a12775febd28d404a6435b
a8de5549070b1ceb1bf027c68fcef94f1b276ad5f06d3bce3c61dd64ae13442
510b876aa55d0de7a1d0034257f86b84281bdb11f5fad4f60de8180e94910b9d
542cbfb346a939babc354129d03a2b0ff5ea4384b732c4563f45978e0fc97358
8b0e5a2a0da11ab098c219bdfda2116cbad1c20eb9ee3e8ea5929ccdc32f907
0a8edd4d081716c08eef93afa1ef6deb1ead8f05d62cb09cca1cfa3b2e064e18
152760778f8d05ec68c93cdd514aec7148afd190cb990537e3c7b217745e7f1a
b1b6352aff84befd064642f6bf7c79fd3f8b6d76fe3f16112be3041419c80a1e

Fiat-Shamir context

a9ac8c45b038a045ce8fd6197f3d3bd4bb4eb93ecb5a5c210e30f90c4d1e317d

withheld proof term: 52 468 bytes • axioms: propext • kernel replay: yes

ZKP.Sigma.cancel_right theorem, Crypto.Sigma:43

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] (x y b : G),
  Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} x b)
      (Inv.inv.{u_1} (HMul.hMul.{u_1, u_1, u_1} y b)))
    (HMul.hMul.{u_1, u_1, u_1} x (Inv.inv.{u_1} y))
```

SHA-256 of the statement above

7d9f1abd0c03d3373f4aeefb8692da728e8058a5ae54c9aa53db0a4dabd12724

SHA-256 of the kernel type

77bb4cbace992bef0470a4cc9a3e18098e5233d06bea4fd176b65e662af67f14

commitment C

54b91ad1b7ef715f2352910f4dcef976989ba7bdc7069271bac6f33eb29f1f5d
cf715e6859dfe2f9a5d84599aa00259e73af5b5db3750fc8823264e795809
4c8f69b200e247b3ff723c5d87be7881b274b6da3de6c60b9214e33f81f1a163
0a198407f549ec7bac35f4036816d8fb7c497cadb207b3db8921ce97977fc843
a407cce5b880749c3ce9e4619f996ab07929da7dc2770eab868a53ebbd26fbaa
72a91e27bb0258b6867ef18e6790280033ab96ffc0965d818e9b8f2dcfef08e5
3d94fecff16f962f93b1de0af84c6e877b4a780b9eecd21beea01f4b498ab109
728ef8f407ee8633f4544c863f8add29ff037da87fc320e11392573d9c13de3d

first message t

428152261384145ffd3b3b89a07cad464cae27fc159647cb0eb1b1dd6e94e3ed
bffd0709f79f85f456086aaac7d5c4bf6fa4e089ecd44e527639907d19930375
2b32a9cb73a9804258de9af0afdcde7147a30fcdcebf5ef069eae28d5e04c9c7
6e58c8128b56dcb18ceddb350749ed272c7359f0570f1405807e9e59da9a1710
5998e00cc6a40654e0bc2ae9e69e07ae83fa3b2da6e9a04c611cdfca19f60120
7d928811a8041741ddf7a9ac08485bee52b60275ad9ebc92ae6658c7621b9127
5c64ae016d99a95181e5a30577903a0c7bd8f74006be73666f80d5b531c73ce0
727a6def8bc60df42de966e7dbdc45cdfff213b8edb3c7f318dd51e910042d54

response z

3c8976f975c6d083090e0692230fc500a69130bcac8891fa2d32d305aeaf765f
603d17b21459e594ed8d36e20c5b91ab595a43c5ce90434aecbba3363469f554
4e86d023b77f3b3fda453fa55f510df3d67406d625839c11e8b733823098bfa7
36bb707454f91fbaf2b46936727d7ad3b4511faad1d6baea9b4b7f9f97fd59f0
26911b4657b8113ef0e3170868e45f7156ff0caea33e1b1b260c81a3b8a785bd
73616dbcad0f5e9187446d3e92dd3ccadb17a286b4217558c5229f3f720dfbdd
90c2948f6e362df3e7f149574a6bb740e6f3be75c47ef9faa1903521b1e2624d
b35f272e9766e3978cb3e4319625a339ce732f6f898f3338448054455aa33313

response w

673ad4b528efabd7110570491a4b060048d74a936de7f5af293fd8405623d35d
df3020cec0e9b21d75a9bb7b2de43d6947792e7166f4f5f1a7fd9c6dfdcf1c3e
14ecb00225306877d9d625c7741b8195922e3f8e5c7e571fb142b5c0ed66eba6
a8913ce1e7c2f6356cc4a6330466947ee6e471efcbb1b89e5be0f52a96ea8d53
a2b935eb0275c71251a3c1cc232d3145acc42e764ca86a9b68895c97a63759e7
965447bc9983793017e9edb8930d187d193d1efb1269df55a4e816202caf9c1e
9686819382e027d0cd65f8fd63f52c4e1a06c4e72080171b2ac5d19dcbc213da
ca5186a4abab6a2b64bc6998f3df9bfb5c5ba9ede680fd4e7e7a5f03df35851b

Fiat-Shamir context

30eba2d4bd84071ecc3b4e19b4b7e70dcf2af393b343815c7724f299accde75c

withheld proof term: 18 239 bytes • axioms: propext • kernel replay: yes

ZKP.Sigma.commit definition, Crypto.Sigma:107

{G : Type u_1} → [CommGroup.{u_1} G] → {q : Nat} → G → G → ZMod q → ZMod q → G

SHA-256 of the statement above

fe25330d79a88aa25bccbf1729dfee1d4f59122cbb6a42c49354897b73cf2977

SHA-256 of the kernel type

72ff5d7236b30dc89b1f537cdb55a9d89f542ddd422d316248cde0e61ef12cf2

commitment C

acd4c4205e1058d99706dde1dc763e6303e82e7a981337d9ccca0d13f6a39313

f490a3e64c82133ad596ebf0a2ed2cc20ec9c9f9b259a0f0ddefea00adc2a763

429e9a05e1943cbedef1082914c305465f20cf5000dad071cca2cafc21c39483

c328f66147b85a76ae0d473cf039a895eb4ca8e5710f653ccf381858ed351bbe

ff3c1fc74bb5d51f42f6d345d004da38fdb8494b639e699159d6b96139c67a40

d742d543e81d6cc728ef49b953b9225a0420867b14076e5c711af8897b66130

ec96fef86bd6c4e3e4a59a518d2e9b67c03b60de804b52a6e19ba5028372b3ae

d5d0f3955e88210b3fd70b4afcdccdd5f4c59398044051eeffe9862aab254173

first message t

ef21e3cb99a32d9355f6455dae8d4d0137bea8f4972cbcebddc10da27c2c0353

0e02f17c6ac4a20097ff83871e2f3ac0d9805422cf1d2bee1079c66e12c64165

76dabd5a15859eff15146155e996028e00a6cba6cbdf63c8d3e9b2125dd420f4

9a16ec2154090b6ba6f41d31f4be1a24fd6982b0abd61945d9fb7d64c7928af4

55c72fd1db3b155d59591773cf2dc97841c031b5a8523db9acc1f7447dc9e23d

7f76b5c543b9c4eb3581c70ab8abbd49de7f57f8237f302efb8229d661314cad

ce2591df119a2afb90361bc8637cc4a320e62e0d1295eeab9b6c90d676de4f89

9df7b03a04b8537df0d03ae625355f6d58a0ca31297e31b183aa20156f7a26b8

response z

4f6474c627e5cea68549476d43a53d569976ab98ac235caba00858ac3265e618

a51a21dc27dac97fb8021c876dc33d8a2b42608b7bcd76a2920ee1c151fc73f

b25705f24c717e6f276072acbeed851f6eaae1ba76840df0e6dd7422a50056a5

6d25597a22aa9e7bfb5e77ec00d1e71ddc5eff77238e0ee1ee02580bbb977dc

1795c4f95d0edbb1ece55e94d0d60b38f327841aaa6b70a4f669ba37a3f6cf70

a5a46b492572be4a604ca4c2e004dc4fb8f5287ddca40793a22f35d80a660543

6fe830c7735aa2b362b0605fc45285b0d785cd04f3ea5e7a7aa55a315effcae4

64a05c73e715044ceed86afcab9497eb28786eca4d6df6e5065ed575aeff5ca4

response w

8a1a69efcf14b5f8f03f1058e1dbafc2b7fa14642c9751a10ac68d36fe61d2a1

19821b2fa60c35c0c6e6b03a106430b144d47390606ef6a5c8fe8dc3d3bcef07

7fae6a12bf452d7bad1eb157eeacc14f0a7fd4082043a7cab33d807eb25eb81b

4fb7cce52a33f14810c106b3a09a29e197374a311d6cae881b02a828500a8e5b

fef135eb73e86bd1248b1adb902c77f7c976770fd91556680bbe062f2a80578a

db7526e918455f555db1e8604e61497f082047e7412c0f55091239b2ba4f0f12

305c88edff12820cbe99e3bd365a75383b5ccbc724c5ac017f8b045c0b7ab2a9

bcd7da94ee587a3743b9505d0ab7caa0c3ed7c891167240eeb514d7b105842f

Fiat-Shamir context

4a784af27b9e25da744647009e28fd18d1d02f075f28558a58c491291468d08a

withheld proof term: 634 bytes • axioms: • kernel replay: yes

ZKP.Sigma.commit.eq_1 theorem, Crypto.Sigma:0

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}.\{u_1\} G] \{q : \text{Nat}\} (g h : G) (m r : \text{ZMod } q),$
Eq. $\{u_1 + 1\}$ (ZKP.Sigma.commit. $\{u_1\}$ g h m r)
(HMul.hMul. $\{u_1, u_1, u_1\}$ (ZKP.Sigma.gpow. $\{u_1\}$ g m) (ZKP.Sigma.gpow. $\{u_1\}$ h r))

SHA-256 of the statement above

b8ea203462934224386c37765a9c5d387ee23341d8ef8a1019535c688546d5f3

SHA-256 of the kernel type

1f6e0801cfda09910fcd2fbb1ace8c4eb72f5228377dfb7702d0cfb7dbaf8e
commitment C

a40cd64bb88e51d761032f6f1291d9701f15e5f74b116164ac9b3347b49c6bc0
8df8f95071efcc8b95a414758b5fa1b796eda18a1b546fb2a61b7f2429e3c284
1e69bb44328fa8ba7d026be12e044c7b1678b1301f9c2c20fab52249d554ac0f
87c5fb1c3311d1d937d4fbe95e5c3089293ba46c5dc2d0b357950a60a02cfc6d
d35b27518271f5698b4f8030380f7ebb8c0a414e84170f3e18f09915bbb7dacc
dcbfc219c36355904610df0d29cbcc0f6982f35590743576e011b7638e3ee4d9
2281eb85d71c5ae81d622035ea65ca71c7afa0e4caf17be8131e6dfa46eff1e4
31542b0d54f8fd1dc5080aaf71f20e6516968cd89d7905fd7bcc098392d302f
first message t

29e1374771252892b8e162fc32856ba90e1caac24af4b665a3fd3d9fa0841260
368522dde641302cd728482fd528fb3ab6695fa4858871e143cbff23dea93
96c1f08cadfa1354cf267fbd171152d2b6c40a83c4b9767f89e8d3752f7ec7f0
688831e1776f99c33d2f83ead05c18ec9d545eeae540276b7ef8dd6945606a5
955ea2f7347396729a61308ef387ad0b8e082c0b0c7ad832ffc45d7c3cfe2187
a179683d297ab737e1e5a0fd7ca6c9a18440483ca5b36bcd261d5367600c158
f7b360a773140ab20d5342368ab9eb12ac08a820bbd497e446b4bba6882d6c4d
d00393b01d49174539dbac659509e19f256deaa058cf69b73fb233868a595b42
response z

68ec2fa22ef7d8c0e7275f1c3e96850bfaa08dfcac9ad0506889318f7b9f95c3
64fe1681b5753fdd5424e8fccf41727fac62c083851a4b6177e1906bef15ac2c
48721c800cab7f5acbff19867222cf36d1de46b599afb6c598a7420823ec09
4ad76ed77ced5949881e64b75419cea5eca18a76ce578e2715657eb88585ba0
0c4b8dfbfb7a17ca539061cc71a010be2ff5a140ff2d66fa2948d95a03c2bdd
768fec7ee4dcea309b45cfa90954fb81ce6d04234f414f143576c83bc2886b32
10572b44640110a7a1a45b37ba13169ef307c76a8e762ebfeaa1938103064eaf
550a7a3d29bf0555b40f4c5b56bfb4743a7dcb920ed6eb788b11c1401e6ae4c
response w

1341a3b2637fe80c336b61763069699a9cbcb768613a6e14a2c213e0bf1bba
b7e096a7e1051ddec45a588e33e11c4a7a427db5d6553f85d8abcca2ebca5c9c
682376e0229f75094ac6c2f207f205c000daeafe5b5bd2a7cdd3b65f4c3b801a
b51a1e7e608da2d9ac5e8db32a911d2aef1e28297bc73f36444270cc9415a634
ab44a7f0489dde926604664f76229ef9cd5eab62ec80f88593163bebab820bd5
88c3181123e555dabf813a0baf5b7b0452af83eaf5b18b7b0093df3c667c39a5
6d3e3dae79bcaa1afc3f4c9f865d2db808e1bf5d0e212b1b2e932e7fae799aa4
eabb90b302679f3fe002b041641dc911fb7b7f88002be6a5d7393315f5a6ccc

Fiat-Shamir context

f32bb395f83a59301ab5618709fa1faba8c7e68c82f03329a128ca9ca6722fb4

withheld proof term: 165 bytes • axioms: • kernel replay: yes

ZKP.Sigma.commit_add theorem, Crypto.Sigma:110

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {q : Nat} [Fact (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
    ∀ (g h : G) (m_1 r_1 m_2 r_2 : ZMod q),
      Eq.{u_1 + 1}
        (ZKP.Sigma.commit.{u_1} g h (HAdd.hAdd.{0, 0, 0} m_1 m_2) (HAdd.hAdd.{0, 0, 0} r_1 r_2))
        (HMul.hMul.{u_1, u_1, u_1} (ZKP.Sigma.commit.{u_1} g h m_1 r_1)
          (ZKP.Sigma.commit.{u_1} g h m_2 r_2))
```

SHA-256 of the statement above

4b8f7980619769ef8036cee3295e01143a84d68097c171643e5f9d2bd4e1e57c

SHA-256 of the kernel type

2d2aed61535ab6ace81bde53283f84d4e0e1d34df37b7e823f5b80afcdac4cc
commitment C

de1bb40683d440bcf3ce54e2668a1864d7fe2f940cd282c42c5cf3c1b8fec998
e89c038adc447abf97179fe8b6e3f5ec8d26f5f10f1f527bc43d4f46673dcdcb6
a86bff3b80cdf878de3cf41d0626c12cdda65ac0d97e987161580d5fc758e2a
83d3ba9e8a4be7f17deb1d4d193fe094c940f09da4d7954dec17ade3ed5caec0
5918a310b79c398daa9167afc282767b273e8807878f06ee2fc3ef3095bba621
7bd535729616c6c55044622b29a5822942441d565e8007e6372486f741b5975c
fad5c216a2dedd5403fdf9dd5e1e557017ffd3063977a9ff3efe1f5d1b4f7802
9912bb2bbe6631b6fe1b8497ed188a13d72279cc0073b322055f2e58d00724c6
first message t

72e402203b7d6472107141b3a406b2729340404198056b511a1f74a76a490be4
562f7feeca6381136e245e1155b2446b800aaead164ceb7ad8580042e552c4d
369b36676d1ea441dd3a88375fae7334e29fedb7d00ffb6790c7ca38156f23b9
bb25d19a60ccc403fa4998f16d6330b695456885077fac9052ac81ee9122fb4f
ec5f32ca0adb055cfc22c2e03a4334d06a32ed48cfb099f631443544a37d9f63
1a3fb857555ff7808711929889d702d1d6a26f4f4ced83b0d0011a3a46502f16
00e35ef5f03a81af3beba7fe2b0eb5c05b534440d765e1d4035a011799c16cbf
0cac2a599e416923d930e06805144c7c0b2482b183625e76b741ad0d96584639
response z

13542987278bfe2a7c554a9ef97db19ee19ea8bca9331d824bf16e358babb6f3
d3ad3bd175f08731d2bd665b5a4a859d1a0efefb6daa51600b87bd4d06c2814d
c30e3a7fffb84414ff0c65bce3988978e845999efc11ee52ff9c662959e281c4b
0ecc88915385790c47ab630971a49a1ff5f92f4bed18a8a362a9958f05fc4d4d
6de32da35dc7a0e465964543c4c105fae0932da542f585b467f34f5879d93f38
2c573f346ca69080ca2299d303b90f612c2c69a96a993ea96254b0438c9f4ee3
1e7454aa0c97becedf74fa82965eb5745c2f94ff2aa83fe749fc590712b7e20e
ebc02c422b1b6c15e9cf02526c20ef48bb0c4bad4f53fd55dfb8c2ae7e5a6dbd
response w

5242a73ca1835be40945d5bb97df3c8daf3c1fb7bda48e213976c289a72adf3f
d2e8194242eeab0f9bb34d4124c07dcc7efdaa28c2cffc2006a8207dd66260f3
1f18d068939eab1bff9258b2db37f4d0b36149e6ec294684a22067149289669c
3b2a04f8580556bd6dad0e34ca83bf1fcaaf4d4d23612aba050852e3a0f75fd2
bd68f7ad0dd6a67213db226b979bc66d55f7b8f46df2ba5eb52a3abb5e7d779b
c284cef41bf1b4a7608d06df73d2cb83fb1554c16f110384103272462ca6845e
a26715229e180c047364172274bb18af1e9a9cef34f63bda3690092c7e6c7ac3
b21bf66deeba6310cf977887c7737ab818c1a83483a04d8779ae927fce0e3a5b

Fiat-Shamir context

396aeb12c1628daa4ca36584997d6bdb63ecfd887243f4be97c1abaa2001101

withheld proof term: 19 525 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Sigma.commit_list_prod theorem, Crypto.Sigma:117

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {q : Nat} [Fact (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h : G) (l : List.{0} (Prod.{0, 0} (ZMod q) (ZMod q))),
    Eq.{u_1 + 1}
      (List.prod.{u_1} (List.map.{0, u_1} (fun p => ZKP.Sigma.commit.{u_1} g h p.1 p.2)
        l))
      (ZKP.Sigma.commit.{u_1} g h (List.sum.{0} (List.map.{0, 0} Prod.fst.{0, 0} l))
        (List.sum.{0} (List.map.{0, 0} Prod.snd.{0, 0} l)))
```

SHA-256 of the statement above

65563bde3624ba591c4c521ba5837f2a412f7462a8488af8fbde27dd559a3dae

SHA-256 of the kernel type

8be8b2303db97116b6f079c42273ed431264d83900bc02fcabdb8c40ea5c1d05

commitment C

90342cf99f1bd76364768a961705b4e70da9a42784e42e39f59ec35bd4eb677
c6ecaeb35dae012c19f689239d1142d3122d4068c8574922e6f353993cf2f3a4
c0d4cbd052cdda918f1436488df8c3e05febcbdcf50642010b8a43577bd45fd8
ea01bf43303bf991aab6c15d05eccc3fc944174033526a2c1374c6893e7f3d8f
e342a6fa5b6c785eb78e67ac813bd335e1e38dae9fec81dc8c46dcb234623a54
478220b5b9ce4049ec9d18253cad38534d775d5c662544d69e5b1a07d42ced41
c9618c0f1504159fc81be1a5ab083f21d72918bbc3787c74732a530beb29e3c0
0973cbf48c8c9c2dd901d3ec94234fcd08c03b583e2c000f390976464db68ec3
first message t

32f5eea4dba6efb0a143bbbf9b1a730bfb3cac6e6baa22f8f5ca5f4657e82015
099c44baadbebe517c3165a62ed957387fb0d3435f3b22fe67d21a4f33a6ec1b
2d421923f28602b704005a8079e81f6e711d3a8995f414e24a75877a4bfbda89
4b5869cc2878c0f162f16c5a1e22056c87c60349297e79a335c542f76f969b48
8f19ddd3140e0dce03972f1e9e633f44b45a2af5e6c56635bface502c19258d5
d05feb61ad31987cd0280da593b6e42f6f72a6940030a95e3c2574cde816d3e8
fafc1a74653fe58772fba74a828dd4013897f21655c9cb6896f7926634d3196a
e5e589c125de4e482581799f717ef6da3612fde5a80c70b58203ad5cde078a7
response z

10f325b11a11642cde48598dc5e6d1c0e3b557a18232d81fd112218aa7e32194
7fd5c999c37e5dbcd520c7a3f8c148245512a49c85d1053b2b5d401d07f909ec
9431900ba45a028a0d72a0231e09201a59dd7a7ac270c926244398980ae97141
d87ce7d56a1e06365cc1582c4738202a8337f85658459805574bfd12b590f44b
587777330e782cee1c9ff95403ecf70ee87a15c100fec1622d8abe4d22f221f5
58697883d2b770a918f8a669e127f133d6308ed040d3d9fd8b4470f2aee3dc54
5b889a9609672ca6144b4de04160d655a33fc163e84c0b4c3ba538ac8bf8923c
5cf8b3bd5aceaf6d9dd8380aa62e32e892a4dd43868c5d781eac7ad8c2974900
response w

512c0e5d7881cc2b6a658d3c07c36fde268af751046e9a0793b406f88c9cad22
c8075d1c810d6d140c73bbd4359de936edb3336cc49d78aebdd727d96829ee27
ccb1e864e506e73910a05a009487dd0139d87b3f111d9725966988a76ced509f
41932c1e12aea1deca044168e79a01c68906194e3e3e05a17b2a64c3f66ffcf6
ba3becb29f1c37a2a132665b5a4967dabb884b1b748806422b00b3fe0212adb7
d56ddf77f1ea5bff6220f4e0b5e6d1074b04658aed73a9b53a5957ac7bd7505d
cc4fbec9337a520015247a512691fe8a5954d2485ea57e6e8dc387d68ccdb2e7
4ba2bd9cff9fc16f7ad77c2c135c5ec039837ec79f554d808893fb86ca43ecbc

Fiat-Shamir context

8a3f60788e79fa46574ffa6bdb2cc721e708dc528fba577dd87bf3b8a309bc2f

withheld proof term: 55 488 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Sigma.gpow definition, Crypto.Sigma:56

{G : Type u_1} → [CommGroup.{u_1} G] → {q : Nat} → G → ZMod q → G

SHA-256 of the statement above

97efaad04a5114b1488b017bf83b3d9c7bd73047c6de86ad7e74735ed74c6a92

SHA-256 of the kernel type

9c0ab8cc537fa30d6e6896350777b01544cceec575f0ccddea6cb453214be4a2
commitment C

866c3803c7f4dad493ac554855d99fd5b34e887fda96f1a649e702902c8c6ec6
d1d6b14ad21cee4a7e3d0844b3034ec9335a36bb91c0e2050a238cc4897ff454
7ccdad2c32c038b7fce84ec10c151312dbac1fa5f6744aeb80a0fc63bef3f782
a65d24313e4b62dc81f1c133eed4a87091c25608b97d21141c08260d246adb23
9b385bc342e5af96699fce1ac1060d24167505b53fbc34d123c820f3d54a6c38
09002e6532b811824c6482914fcc9d036d6e8a07f27bb68b7bd38d8d85f00019
a31a8c25935ba131647b979c774ec8dec9eef457fd50f73a302e24063d55cb2b
14fa5871191e5bae8d19cf88ee729661e64b57c9c0c0a62707ff1d6659dc17cb
first message t

93a3ac0dc89a524c912129bff89c4fedca66f768f5d4d3830309ab97206661f3
2c377baf8865881ba39d08de77717196706d99632d9969183f917c3a366e7254
a3300191f101404275486d0d55a31a576c3b68b2ba1fb9a372cffb6f7c069680
fc50b7dce457278c9ff11f91e6b0b17a975e9c8058b26b04b9257569f8e11096
8b9413e0265071406c211e6766201e0d9d1169989ce420b6dedc888b295d02a3
776e683311b9385e2f2597d612690c6eac1b4c0539255a023f1ca7979364119c
0d134cfd8f8d8dadbbba0cb438254851fffac8971d0eccc76df0504f7529e3b27
9387b919b765c77a23dc20a902e73c1f634d8a010f77e4c3eafb8729c66ef6f2
response z

2bce970343e22b18e0f368469390880ead8ae24ae2fe02361cdbddd3fe10513b
0806b3c2dab0dff5bbbd27b1510ad3f6a94cf3bcd918ac192e5538d694e74da
2403a1068aaaa128c20de2288d8b35a495246d71d3c850fbad4a1907cc4e3eaf
42e6fa72275274531513d22ac2665bffbcbcc847537ab02570a0b4ba8be2ad70b
85e9dd11512452825590f58f53778c9401923edf6830ea15559859b8adae43c7
ca7ad72a05748b52cf73d114276b2ee2ab84686a9ef307b89e8eb6327c33c90d
4c08d5dc2fea7588354c5c9036fb184602c137f332b1e768ff7a0f2af0d14233
e5888d304e1e5c6f963ec28df6dad005c2b9210ba17f524415cd89bc2fbfb885
response w

2ac11f1541ec443ccb9e1e73ec3b7e533aa26c40c5b0aa46881a3fb50fc16778
aa902214e7962395e11675295fcb3df8b293e415f8cb1a291de3a3ffaa8288c4
c56110a2455db3753a6ae6e84333734d380a4de1b217940b1ed10f79895b6ca9
f88754021630f01012aad1a80d5d6d1e9a46bc502dc77cdbe87994f2bde97269
e2e37931f020658a913b48261872770b9103c0d18d6a63ef79ce6858acb56096
d407330747698ffeb99a596050d5006c741ca1633f2be586cd3ee4b0738d1e13
a17562df52323dbc0b643d018694b1f1dbd60b116454c4f138d18fa1a3d35b77
1e408d2ccb81ba98a3b040af788db8c2278194816e3c3a954230cf8281a0b66a
Fiat-Shamir context

668c277a5b46d4b90bfc610f3b12a70fe0e86e0bde3c1e0e5af66835c13f5be7

withheld proof term: 390 bytes • axioms: • kernel replay: yes

ZKP.Sigma.gpow.eq_1 theorem, Crypto.Sigma:0

$\forall \{G : \text{Type } u_1\} \text{ [inst : CommGroup.\{u_1\} G] \{q : \text{Nat}\} (x : G) (a : \text{ZMod } q),$
Eq.\{u_1 + 1\} (ZKP.Sigma.gpow.\{u_1\} x a) (HPow.hPow.\{u_1, 0, u_1\} x (ZMod.val a))

SHA-256 of the statement above

4dd61464a06a3027cafcd8cffa612c47b35f57e228baed0686ca4e51a98c3bdf

SHA-256 of the kernel type

cacc14fc2299ba940260d9643afc958f3cb38d8d6afd621a67adb038cb6dc6378

commitment C

5cdc91ddd6c7f24c59193af012ef4fb2d5fa39901d013eb028977c8311efa1b5

3cd87e4d587483b07c4492cd45e11d4de2391465fb8b0b2d25e01ba13c650c43

54d9a3bf006a67c61f1256860986af09424d83279f6429171e0c8e78ad67017e

27608d62901e98d6a8779f74e610aac592e5295249a504d538038aee5cea43c7

547e64d15958eb83f70a8566ab1598238b02334842057441cc6028e917cfe612

1ca5ed0b51455a9e562151c26ebd587687dab90e37f8ed2535b02225606f90de

ecbe8fd3d3394fb3defca3a2d181d209388c53e4c1f235b7e75f76d70de3161

3de3283b4d59592680a7c765718cc019e788c4007220dee69dfd383217442c47

first message t

16e39d6d55c9e5d7d4d4bb988b31c7b2adb46234d60164f6560b1cadffc9138b

96b0c20e1bb4ac4dccc25776cc2d1df27e72cd29ca49d17f2b0190c4fd20d8b6

b1970be11e74c89c2dfcbf8e41fba41bb8ac07e501f194331e1498983cf14af3

a36e50418f6e1807e11c028414d490e29913feba94f8f14800b25bec1398fab1

4c050260b73c2cd073b8024c7584fb383c5866e2f0855850f7988d5d93794c37

cf9c25bde5b7981d74f57b820776398e3bd30ace5d84f99c1a08c720ba7393f

400f1c99524e8e73e4c4cb02c95069b7b42e1701f4fd823af7c77025299a626a

91d21a7fc22b46a22b7a51f04bc563c3f58234cf74770b14dcf612cfe873c3ed

response z

694b696647ed31936c96e1235deea97c500aeafffd31e351e09eae9f5632314

71f5715472933844fa485247c4ea25e50562789ac598b8c3250008715cf55906

8d6ab25cfd82fbadeadb265fe78cf4d716c0d9fe0e51d5596465897febe2088

6c52294c9e6c335e42f9c7a195a2c0e89f8faf03afe96545f036b82774c29e4c

a9fe68fbb4ba9634f8721eaf0e67887057a1da9e2cf2a9207feca20d43afe048

6b546aebcfa28d2f1f8d29b7ba615dc79f97b3921c50d81ebb5e89548b430775

1d791f006b55620ed036658bf7fdc9cac5056f968ab83e26e571eb54ac1149ed

d801ddb9f01ef1fc4d6d8d577ecce047b9282e78da57b8dde00c7bfc4e51c523

response w

6f3034ea1e460fcc58140b9c8b454b2f63653f20801b0867b3158a46c3bcc91f

8d85c2f1296e77f26ad683ba7a383beca44e938e0fb87f8c9346aa01ca7fc80d

26c85014969281fc2cc4e3821df562e1c6ea54ea05e56bcbcb591d364976e0b7

ea5d0c0646018b2e41187e8c93706ca014d16fdec85ac09dfea6141aacb513a0

d323846fb5fcf29d711bf477af7cf96c9cf7e0b83f612ac8703219231651e829

9720d7740755cb3991c51f427554ff93a546ceb05fd091525fd48bc2334d9d8a

c6f82bfd17c6603a2343d1f18d53d4d9e24fb06ee0dc93ded1f7a34d92af8abd

02cb6eb2c96f24cccf24eb542d9130a36df438b1cc3ab18eb1bd295d5d51800a

Fiat-Shamir context

9b0ffb74f34944c763a64777e7ad6f426d4ea9bfa37e978171fb809c838ef54b

withheld proof term: 138 bytes • axioms: • kernel replay: yes

ZKP.Sigma.gpow_add theorem, Crypto.Sigma:76

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {q : Nat} [Fact (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
    ∀ (x : G) (a b : ZMod q),
      Eq.{u_1 + 1} (ZKP.Sigma.gpow.{u_1} x (HAdd.hAdd.{0, 0, 0} a b))
        (HMul.hMul.{u_1, u_1, u_1} (ZKP.Sigma.gpow.{u_1} x a) (ZKP.Sigma.gpow.{u_1} x b))
```

SHA-256 of the statement above

ed633ea2e30e2a413428c9cea48667dfb2494718c0fc04c73132f944c61660ea

SHA-256 of the kernel type

f70ba33a0ef3b0b8a904738606f39eefcad8ad8b9246a919f522dc4e1ca85efc
commitment C

31f75e7cd70284816a94e43c6a40533ea143a746b1a3face2d005c7589003587
19fad276edf2c506de8c65bbddab10c22c4adb4f187adc942795905186c804c8
194f909f82a91ead8567e6a35e0fe1c1e8b6371b305483da93a7ec1a90aee46
3afdb855b51b75adfed3c58e75dc6e73b898cf936619087de82de63ee7f52eb1
7c8c70188997f5c7b297503f12a66ebfc0282449d42e069cbc507cfc0bc9a612
3c7837a6395b1cb42b6f2936c91a8691bedd7c1efd380320fe9c235f2762f465
df5c448932b22a407eaf36cb8029f9d834eab5686463bfa386eb48638b8602d4
f7d2b8d3b4604f04e88b5f9d1450feb6c25f1da8a494df064269d2a6ea612421
first message t

6e8c8d6a073140f9be6de3fb7eb511ab9cd5ff46f2e65ded3f48d0f4f3325f0a
288f11a460846c822dcc8ce528367ec6d3bab7e4900b04f44fbc782cf57a533f
f95dc9b68383b00a9b532601e4556f22226ceb621fbdf778c8de1d19e7a57d5
6125d0e385c4ea3534e631a010291714a5f5880ee5838953f022de113b18bde5
a8db8f85ec0adbad21ca8a369bde504650c53999296aee9416a6fc483ab30d53
3c34b40b41fe7bcb7bf3ec3216c945fca69d7e6341814afbd839cf75715afa3c
f8d5722735e0a8f7552a7c27db181735c77692dc4e9e08c87ad60d1f05ef0917
78f95d2b3c08d2d1819086db77616bf247ebe9af4e560b8ad1b4c902eb27381
response z

5d0638e00b7a2f68bf4272271ef1a8fe915522819f732cfd81a898247f04f981
c36a2cd4544ecdecc5da421535ef93dd5c0a852e784cfcbdba46e16ab567c769
7f68ba3af74ddccc7cebd7c7832776bc70206023724f6bbfe710a25bd274d9da
81fbfead2d8e458edd0ff34fc1630b69058c0cef114df6e359d9d99b926c5e6e
c76c58ea5cf2c270614efe7a41cb280c0b33e89f9577d195e0548e87e45811e7
9de158b021e56d56f15e9f585f941663ba5877826522005e918036362fc91c2b
50704c4b83739a6e1c6f91ffaefd923d3a9e46ce3ac730c5dc2e3e600bb36fc
f37c286a9e03ca4fb5e02cff50c118851aab701c9af956198b6f3fe5f3d1130f
response w

44a9b7f0e23dbcd7c2a5ccc3736a488188f55b41c30e7e425bce5ebcec79d905
dd3eaf958676c17bdfc76011a81403e785ec44140dc0d5bdcfc500d904ccde51
2e3db7b2f6131f49cf2d294d2c3d6780dabb0be6f5483063d0cec2bfbe92f988
fd65047513fc839b5420f02ce9a9539d77a7d17982aed457cb7e7007ff9cfdaf
357fc4a9dc9a8b8e4abd17c2137ee20443e138d57e9a1616da3aca38d58ed267
50b5883effec572787b5635447527c9601b7dde7816f08f13117e76ecfd7979f
5f918790876455806b6f8c02eed1ebe85f87dedfe0b20c41fe0bd1346ee157ae
8a572e5d2b9cd7ebcd574996c9176b67a58fa5fd0cd7ad44a6a49f912b851d10

Fiat-Shamir context

169baed9b887cc649b9d143bea02b54c8a89c1e6eaa537c503d933e60798a310

withheld proof term: 45 340 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Sigma.gpow_mul theorem, Crypto.Sigma:83

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\},$
 $(\forall (x : G), \text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x q) 1) \rightarrow$
 $\forall (x : G) (a b : \text{ZMod } q),$
 $\text{Eq}\{u_1 + 1\} (\text{ZKP.Sigma.gpow}\{u_1\} x (\text{HMul.hMul}\{0, 0, 0\} a b))$
 $(\text{ZKP.Sigma.gpow}\{u_1\} (\text{ZKP.Sigma.gpow}\{u_1\} x a) b)$

SHA-256 of the statement above

74e5f93347f6a0bbdd791a56070f0e595161cf992e782c7747ec414cece87155

SHA-256 of the kernel type

f56a80ea9e31a3b923e71c96ba8b8f8e7ab51bca44e939bf5ba8177e8aaad572

commitment C

83144fc4b1acdbf9de2245b83ae20783d02be3e9638ddd1b3394b04851efdb6e
b0333013da5507fc42010f13c35f08ee5f97815e537312212a32bc1b0fe10440
00d328aed709836443e231e7c34ceb725040b78f52ff9c6fd39e3e15bcb8e1ed
257f2187f2d71759bdf2b29c4db22d33da8b11c583bf4e0c34aff0eea246c038
3a2c31498c37d33e21d579b5491ff493d787934d02869641eeb676251252a714
af25b5e912bab5b8951ce32bfe195eac5bbb79b9e29a35cbf384b7d9b7dae329
a510b7b30336bc99424166c5cc3cad7b0cabfb00a067955e8adbe6e93ba9e2ce
4781ff67e0de0a06143bb6b557997034d4e4ab3c0428710d13829d27e658ddf7

first message t

f45ec9f20689f820cc4570ce18f9d18a397e304b36bccb049ad210275b0245bb
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2006d4d5943337338f6bc13ddff9b58f0bc01ce71570b06305e14c8efaeab3d8b
0c3d73a7c87a9d7d17a7e94ddec4ab07365aa641e5cbae675628c43160fdfbcc
2384066584614bd1dd037326cd90522968bfd761ade9bb213ead50a669a6ab8a
89d87895b10b3b70e81e763dadbe845e3ec9948aaef82071aa1ee324ff0d7db
1b54f5268ac5bc8af946fcda03557193ba8fb7e29a7a46cb6141b493653f14ee
830c20a24a7516fdf1c54a82af6ac90e14906ed9e04a5e6fa3a16bf74fdf4d62

response z

3a3f46997dfecbe582469ca232bbb8df235b14d2b1ba40d7233992bbac5ce20
07953cf8c17318529c1b5b3d094f30c5c3d7970eaff46c40956840db964bc875
b07026edb156dba4acfa3240e943324e798e3eef26bf6e833c861540dbec9c7a
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1c6f3cfe0fcf456da12a03ef76fb63abab8dee3974e987803e427a1565cf8904
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6a722504138f229536469203ae877c536202fe61805a059902bc7e51f797a03f
8099f85d0872ad28c9a9b34223d6276c96971f3b3f05b6da59af28fee91ed0fe

response w

20ecde390a7bd8731a3a6fe7dd3f72ccecd1524a3415b23ea9c0ae8c9e02dc64
ea0834c990ceccc03e9d8059f4b1dc4a94e6f5ae9a019d2b71eab05a2688aa4
abd69dd8924df19efc90cf1bc7767157ae8aced27eb93bef6b446090861d2385
7700d529128888dee9a4ab7f15bf741d3d654a4291809fbda4d9572d28e0e70b
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ce349f319374db9813fb0e8542c719882659bbfac0a4c739f385ff41400df33f
d1b61aa4ede1cefb89ea7866efeb83aca3492f8d94ce72db18cd557b2414ad69
575abe3dbda840813691105ea4a087f7f51f6d41e549d7491f7e492d2d4f5cee

Fiat-Shamir context

01d74459bf953f777f37ed5a6043a83b54f2ccbdfeec89efe6660d768105f8baa

withheld proof term: 34629 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Sigma.gpow_mul_base theorem, Crypto.Sigma:89

$\forall \{G : \text{Type } u_1\} \text{ [inst : CommGroup.\{u_1\} G] \{q : \text{Nat}\} (x \ y : G) (a : \text{ZMod } q),$
Eq.\{u_1 + 1\} (ZKP.Sigma.gpow.\{u_1\} (HMul.hMul.\{u_1, u_1\} x y) a)
(HMul.hMul.\{u_1, u_1, u_1\} (ZKP.Sigma.gpow.\{u_1\} x a) (ZKP.Sigma.gpow.\{u_1\} y a))

SHA-256 of the statement above

87774b4dc5aa8a6a2d00da7ac924f2009799a686e1419ceb6339bb729dd6cda9

SHA-256 of the kernel type

95f884e346561dff7cc630f6b2370dbfd54e099cf98bc0d09b606d70c15e6f37
commitment C

51ff7e407c3f22655602b43c0d425d65513a57edace3b0b775bb0a580ac1f16f
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81f90a945bfbe51aa58555ca23e567626f12f3a8196417ea803ffbf192cc3b01
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dfe1e13b59c85fa156fdb49efc5a285e53ebb33b3501ba4e63e1317c503a8b1f
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9127049e83c054a2a665d6191336bbfadaf94c51b43b0b75a63873781eda91b9
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first message t

2ec37a61293fc213d893e115feb7a8f1bbca5fe39b115babe650b878c273813d
f89f0d8e9d3050a3312dcd6cc1edb108527c2b35487993998bbac8fe5770a60
5a4fb6b636f33f1ab2b372a2c96096580fb3c7b768211c0de6f7b5faef6a3c10
0b6eb12114cd7c43f9409106d6fcf1b872c5ae6d0bd06bd23767355894c8a4ef
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89516df017660a1b672788ecf7cd8bfeabac8806bc34ba35fab744683fc60717
f4e2be56459b30c6cf18dd88ddb6cb292dd6faf0d6edc9dc5e3af2ab7a00e1f
3c4f73c0e499dccabd2eb0b24b7171662b68f10706621e905eadd1c558004a8f
response z

4b28efa73a45e1f9a28ad596eeacfc0731ee0742ea085f13abc10cd9257ee169
c9195b2606d0e28e023ebc3a71b344a6bdaa4160200d1ab5c2b793f8ffefd6a5
e52d9f4960139b44c2f4d9a22a589cd16dabc792cf3d8962849708a8f649b980
bf83c6a1b9029e8467e9ecb3481a74885757a00e4b4f6d0aea1c99b28e02b034
afe7297c72a8cd87b9e093a724f45ac53121e7d365d85a25d8883a064952cdb
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response w

4747318d836f95417befde45a419c6cfdb7dbb3917847ab844c80b8158d3a3a7
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b39244cd3abc70bb3d11458bc6310dccb11841990fa06c1cb9473af627b38e87
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f379611e3b16400f9f206f3ca289b08840718962a4b3c0f8d88909c6c2b18411

Fiat-Shamir context

53146dd82467a9849cb7b9527f5b42de7e041d0d219deda24e4259b120513423

withheld proof term: 8156 bytes • axioms: propext • kernel replay: yes

ZKP.Sigma.gpow_neg theorem, Crypto.Sigma:93

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\} [\text{inst}_1 : \text{Fact } (\text{Nat.Prime } q)],$
 $(\forall (x : G), \text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x q) 1) \rightarrow$
 $\forall (x : G) (a : \text{ZMod } q),$
 $\text{Eq}\{u_1 + 1\} (\text{ZKP.Sigma.gpow}\{u_1\} x (\text{Neg.neg}\{0\} a))$
 $(\text{Inv.inv}\{u_1\} (\text{ZKP.Sigma.gpow}\{u_1\} x a))$

SHA-256 of the statement above

a1dcffbadc25dd89bbba94d389fa3ef65d5a348cabdeba0f86832876e24e2947

SHA-256 of the kernel type

50ae6b0d1e56367d3e613c9b8703d165e4aed63349890782587832f732103e2a

commitment C

67ac3285ac0fa550673b2ba247c8a20c538f2d9d06dd740205f593fe464c5d32

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5b191f05c5195e23218f3714242f46dae8614b71842d13f5dadcd27013fff1b2a

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first message t

a19cd20024b148db4c9ae81cb3c655b8757ad5f31b1a3effda110208e2880937

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6c150ebe370d72db9ab7eaa7d9a234b51aeae940072b5e145fc06b2bdb40bb11

response z

7594b786bcf036addcdf04e3b6ecee27de1eb63a30d11c7a0397efbf7ec06edb

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response w

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a3d022176ba2ab29cc83cd2263a1560331ab40e3117be4ad94c6a8700fe940b6

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727372c17c84860712206b647b7c3a585cc061388907f5e92b618d422f29563a

Fiat-Shamir context

077288567f6ff1b7f455f56703ea14460474e393a8fbdd92d9f2d76b3dd670b8

withheld proof term: 23883 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Sigma.gpow_one theorem, Crypto.Sigma:71

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\} [\text{inst}_1 : \text{Fact } (\text{Nat.Prime } q)] (x : G),$
Eq. $\{u_1 + 1\}$ (ZKP.Sigma.gpow. $\{u_1\} x 1$) x

SHA-256 of the statement above

9aeb696ea1b68a7213aad9025ee9e648f645d1c9e5a83850579ae51b9ec7203b

SHA-256 of the kernel type

181b4656e515e61436e3a8b0bc0a0d3cc30d0b9a95ea202b52a7c5e4882c9a01

commitment C

4baf1f8fcb44b3f980ea785378fe354653da91a0253adb3673522761d30bcfc9

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first message t

55c9c023bf13dfff2e8188e9123895c2cdce369bf98ac230ac89583e31dbef2a

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05448f5ebe8c0122cb34584e1a2246e3f148b8a25180b6643c3ffd804cd37928

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8f19739e93468675e29fc5413afd20cca575953708e74fbaf06d9a49afa290bd

b795e9a31542f54e24903c6885e7b64063be0c5219c68bc79f4f8247ff1dbe01

response z

476eae2fa89dd384d28d746591dce07c6ff1e33f47ff2ee5fe4a4b98bd882d80

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response w

17801dd3cba3a38ec5187ca052e2653b3ae3e72c740c2520759c6b150eb4a6fb

aa4b3fa4020af2f5abbafaa73f41881ddd0c222a0b46619a2923e70959ca4772

bfd8214627122d80ebf835b1b99a35acce08ef3af63b09fb0bcd2842c7072325

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Fiat-Shamir context

a1cdb8aed4a01ab4c8c5bfe053a5652bffffa04c7f779b16a6f6fea4cfb8e4d5c

withheld proof term: 4111 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Sigma.gpow_sub theorem, Crypto.Sigma:100

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {q : Nat} [inst_1 : Fact (Nat.Prime q)],  
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →  
    ∀ (x : G) (a b : ZMod q),  
      Eq.{u_1 + 1} (ZKP.Sigma.gpow.{u_1} x (HSub.hSub.{0, 0, 0} a b))  
        (HMul.hMul.{u_1, u_1, u_1} (ZKP.Sigma.gpow.{u_1} x a)  
          (Inv.inv.{u_1} (ZKP.Sigma.gpow.{u_1} x b)))
```

SHA-256 of the statement above

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SHA-256 of the kernel type

911ca240a1c6acbf44f8bac7ef4898aac808311309e07a39a84646effb43bc3f
commitment C

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first message t

bae5863ef4bfd83781f87572b84002d201cf2ab737da688d34b644ce0aec88e4
b339597420df234b930a7138bf5a31c0d762f688c0c11fb1c594680102375267
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880a916544dc96812b294be101a65f8f79e2f0448f9a5d8552bbbc94d3e1b933
e13b80d75bd0c5c9b1189577ea97e6664b1853978787294e8ebd57c500ef1f84
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response z

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response w

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Fiat-Shamir context

4a486d2a7c37d5d37855eef5f428c710f412b1b02e784085fef138d5d015d26e

withheld proof term: 27 989 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Sigma.gpow_zero theorem, Crypto.Sigma:67

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}.\{u_1\} G] \{q : \text{Nat}\} (x : G),$
 $\text{Eq}.\{u_1 + 1\} (\text{ZKP.Sigma.gpow}.\{u_1\} x 0) 1$

SHA-256 of the statement above

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SHA-256 of the kernel type

bc21997dba50eb7acb4870f9289628fec21c68ad818ec3c63f89b36e0198d3b9

commitment C

e2b251c7910744e114484eb92826cea8c98694b463998867609ab750281429f2

f47712d9bb3844146a2585d6c3861a4169c83fc413682888bbaef1cd7138d382

5700ff2fb3ef51ab094dfa9eea7e24508ccc8d7ec3b969caf69c9fce67b714d6

38211f9eb5e1895143ba20f6b7605f7d74541e38629e2d476e240d3224e7e075

36820704c722bdae705492d94f54d4b49e39997d076697712556760a557aca10

ed1990a069be839d9cc94057b74391526f2a63a0a16fa48c80e0eae23b0162e9

51174958539b23c11e2421379c5e35ee8811640720a93824b53410bfac60139d

3ff4d6ec0954786b6fc917cea414223e3f09089959de953f898ab7ae2849a9c0

first message t

208abccbdcd6b427f48631c8fdbcdcd127da41bfe1d5494c2919873499aed34d

a488a1b148fc7af1f7c410ee5ca95582f85315d2461a7f21bca75e931d918365

f27ee88c65b792cea4c81d1b3cf7c714f1f791375403baf8a1fa7c4f7380f787

34230b3a571df6e1d7cee8f1f1989609521c0e10aad312d91bac4cbbb2845ecb

7ed4250d20abad6b12f844a35be7e6aefd9b5c6b407cefb6e38af55d932fe9f0

3b26902bdf48bf09a0a182a27f54769840b4087031f05b0fd3a2dc3f2c4efdb7

aa0deb43052cf31441d9e3f5a58b90be8eb423a6a0695e1ef24375b20d36ab50

c30707dfac022acee117d388efc051397ab8b438290af03931113011c66cd2f8

response z

1c9fb269508c76d766548d6692238895137f9938020229dcba82d4e5e4ee1ee

5db945e8c6aa485744e6ee4956de4f294793bab42aebd0fddfd111838d65da0de

541fe961699aa73858b6a1e55b0d6dfeac9f875e4e1bdd763c754545583baee7

d730c03f6f5be558f8c5852cf44464bf1cc261d2d7e02e2eb3435ab07b10111c

d9b04d0855ae7e5b6dc03aece3e422e0295d09cd638cf40c6b93c3c0eeb5a8fd

17a73878d4d1beff3045fa8d5551f3ec0723564141f818f370b3042b55f8384d

5433ae5d5c5c234bb5b2d5f639049d040adf9c0239ed86f1019f25a079b6d85a

8db54e6478c0d4ca29a7f9541f12046a193601eb98908bf7bd3cbf5c4a13243d

response w

2cfc13d4de30325e8e973461d3a22c86e9a45e953c1f4c46c7098fcefbf76320

d4ee8c7fa24a8037d6f759fae1fd1621b0b084c0a2fe522f8072d29ffd530c5b

6c934434004731722a80499113248d0a0c544c068dc4ab241dfb6cc8448c107a

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0bc12920a8074285c773782771fcaf6919edba9fd584568ecd5887832d17c6cf

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f68908448c2408961ecefdae43f38f01fd0645b8a3addd26fbe95376418e3789

Fiat-Shamir context

0d405394eb8c345a74c43bb3163c521e7b331036f6910658efdeb9cbb2be8bb4

withheld proof term: 6 878 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Sigma.pedersen_binding theorem, Crypto.Sigma:200

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\} [\text{inst}_1 : \text{Fact } (\text{Nat.Prime } q)],$
 $(\forall (x : G), \text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x q) 1) \rightarrow$
 $\forall (g h : G) \{m_1 r_1 m_2 r_2 : \text{ZMod } q\},$
 $\text{Eq}\{u_1 + 1\} (\text{ZKP.Sigma.commit}\{u_1\} g h m_1 r_1) (\text{ZKP.Sigma.commit}\{u_1\} g h m_2 r_2) \rightarrow$
 $\text{Ne}\{1\} r_1 r_2 \rightarrow$
 $\text{Eq}\{u_1 + 1\} h$
 $(\text{ZKP.Sigma.gpow}\{u_1\} g$
 $(\text{HMul.hMul}\{0, 0, 0\} (\text{HSub.hSub}\{0, 0, 0\} m_2 m_1)$
 $(\text{Inv.inv}\{0\} (\text{HSub.hSub}\{0, 0, 0\} r_1 r_2))))$

SHA-256 of the statement above

951551294093c400a7397fee203fe8dec1f5df07e2524333fd4035a8a864019f

SHA-256 of the kernel type

b31fed4406a810b0ad7ce1fce83043f4d4f49b514237ab40fe8039e1b0359a18

commitment C

b75cf350d72b79b805ca184d3b8aad3f9a13177857a197a85dcd685e792452

3dd0f4d10d7d87536b181d82fa408fd0da74a4c5d89ca85ec25ca6ab1bbdeff9

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first message t

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c728cbcd80fc9309b7a391e81273c9a0d7583b564aee5e65d9ab3d2023aa114d

539e60f3c51e89d08dc9fb09adfd271ed578433bf7bc9eaa759357176ac08017

825924ae501c3758502c6fc506c96132f1222ad581ea91e38e2d5b54c94c5a6f

response z

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aa9aeebbf3c4b28fa371c27926dffa9830f4325828f8275499c3f3a9bef2551e

f8f5b60738048529ece3417faf8c0b22dc1bd181fba1e272744ed7e0ce0d6bb6

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f649e6002752e2657b540446874ca5627bef66d81b379dcbe8475dd2fda52ab2

71fde6b6f8f27909dfe9edf8939c6bfd00d9ead895cb39dc7c968070677d6f76

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response w

47cfb5abf0be25ceb5f4cdae141aaef067be4b5723b69abdf2f3d1f5ea4b7dcf

cbe70053986b766fb9a607c8df8beac337be9a7474f22b4bfbcb4c6a6ce523049

5a9309753e602824792b3943b1015b6ad630a2874a1f6f53dc3a61017a5059ef

e2a8913591fc5d4db1768799b7eb462646120252f8c1df31047350d0e5c98d75

c28a27acfa19376ebd358610e2d22c267f617effca8f0d23a8ae62d2a98cd756

b822b255eef3ad5620c68c0d32a6e2aaa442dfa1fbde2c584020f06e79f027a4

13c3e4e335971538b3d7f9163cdb7d915d439067f6f4e8f938e2a1687c8b2a01

236eedec1a5a9a393188cab8fe38c08fa435b867295a016af7567bc6193fffa4

Fiat-Shamir context

837ae4e45f168bbe792aeb7516c08d93bbb7cb605bdd07126568b61357d611a4

withheld proof term: 84779 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Sigma.pedersen_hiding theorem, Crypto.Sigma:218

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {q : Nat} [Fact (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
    ∀ (g h : G) {y : ZMod q},
      (Function.Injective.{1, u_1 + 1} fun a => ZKP.Sigma.gpow.{u_1} g a) →
        Eq.{u_1 + 1} h (ZKP.Sigma.gpow.{u_1} g y) →
          Ne.{1} y 0 →
            ∀ (m₁ m₂ r₁ : ZMod q),
              ExistsUnique.{1} fun r₂ =>
                Eq.{u_1 + 1} (ZKP.Sigma.commit.{u_1} g h m₁ r₁) (ZKP.Sigma.commit.{u_1} g h
                  m₂ r₂)
```

SHA-256 of the statement above

57be0a396f67325e826bbaa2704f0fe736a851a31f8cca85208245016736a81b

SHA-256 of the kernel type

6e02b58b52c11f241ce481ceeb3d976f58b6d1af01a022ff57d18f58757a4cfa

commitment C

3e11f92196e8af5197ebd1bca7f1157a269a33a8085be32fadeec76df8daea16
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3ed89248c48468c2f233905439e8aa3e0ead625bd15793fd9a8e53a714ad0
e9e60eda8686b0566b8a604ad2793108b620473fe08c0f068fcabe50d3786bef
ff6ce832196ad4b4be7a4a829a96cb2d35c90a1b99c320bb4ace4486be2f7336
e822fd6683878e76db9f3cb9d8d559ca891293b9926ac7ff9962a216090f5030
cb23194debd66cb3c8c2b444a7db84ac6b7a331a04bef659de8901394ba77232
1a759055db67e5d2a3ab2946d903420ca5574d76c18aa91df995b3bb6a26e27a
first message t

4b9fafbee4b9efba0d0b8e690f515467d664125991e73a36b2c69303c03a1fd8
593788cc21a3bd1fc61014869b0edd5116d03023f520cbc3bab7f045a2c6f5a1
46679fc8008587898193ad17c9a2c52a8744bcd85d92e18f252759bed558d68e
a08cbbbef810fe19ffd95a2c4ce4abe8dc750e04790c64137b7f03772466c410
f91df87f2d284401ac7462d7ce50601e8bfa9adac60fe0e7a1af8ca10997dcb
25cf7f601016090b48245c5f24bde6f2e126ce8f56016e2ade65641ce297d3b6
8a2bc1fac6fcd7cf8dd7ec8b73de77e151250df26f1e45d7e187c5edf971fa11
1f8e9bec5f971910c7dd7d1fabbc8da65437987a8cc1bb427d569e30012d3945
response z

60e9200b221d8383e32d353e836eace6bc51e08df99da570d0a22a6e2be8af0c
b1f8a18aef73de8bd08fb0ecf6725261a48674c665c6ad24b6acdc0e32421a85
4dd6a481e900e594405dfc43322ef41ac44cb9cd3dbd46cfbe76ae0bd83049d4
ea55b93ec026035bd4bd54bf0ba0de40ea052dcd23dc291b280b4613d8ec928a
a0ee979b1b059d98eca770670a5afade90a63c171fb554d4ebe3c0108a3e1da4
845529fe559459bbd63f60aec854e32b834a6c2a049e9a170836361c9af363c1
ea4460335e0c9161314d1446abd486293e57b923d6a5a6a9f0621858dbd8f164
d6308ea13486f0340e6695008c18305325764ae517c293528caf8435ca22340e
response w

369b859a1993d39f5456104b8f2b8cb0d1691884e289053044a4db3d88644ae9
0ab4a2a55c1ec5aa1fa3b40292dd856198577a07e277031a3b3ed3d02b543e1a
eae3f3db7be39d60468338fa5ed9207a99d53a134f51e416e086b171d04df502
4d4901bdc0535ef47d7741d665e9729e225d0d862f7bd6ce945034d8dff3a3f5
5a628268039c6e70f4c970b6ccc9490dfcb2cb1cadb88815e46ca9c0c1936579
a51c235180993254eee5e665b7f1c5d299dd7e04eaf0b3126e040f3e134d85d4
788e2a25d0dd1906fdf2930e9237d67be73961330eb657c2e5ef67e47600ae
f1b3f08d9c54a14484156d9d73003b2d583ed137ab8a8153a6fb6d7834d72e6e

Fiat-Shamir context

879bd52613d4441597cd06766fb7ce058bf7f6c45a2912efded2cb15f53cda67

withheld proof term: 1555249 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Sigma.pow_mod_exponent theorem, Crypto.Sigma:61

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\},$
 $(\forall (x : G), \text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x q) 1) \rightarrow$
 $\forall (x : G) (n : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x n)$
 $(\text{HPow.hPow}\{u_1, 0, u_1\} x (\text{HMod.hMod}\{0, 0, 0\} n q))$

SHA-256 of the statement above

23cf1186251062b9f3001fb9168f06ae4be80b19ab439e068db31120d48ce54e

SHA-256 of the kernel type

39add9ca7f02f68d15868f6b329d57f680170acff4b5d0389655c0e3539a5ed2

commitment C

4cca9ab436df3b1df27d7f7c57f13f333ae46b5a130d82adefc07b1bc3857749

6780a45fed4c1058afae2393f33912d89d9fe7afbf69e62f3184db01958f8fc1

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first message t

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70a667d3da7b5a50cdb149ced764c9c9680e79509a93364ae49253909cd4f4e3

03827ea7ab3b71084c79643dd45d96955fc69c1fe146721bd155e0c0ae639f41

6b12bf40604da870ad12c33c29fd1b50588677668f371f79088b0164d775ef61

9c04b45fe04224daf9c79879e15247a5aa43e5eb5855ea9d4a74a1be987cd5b

response z

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5f382d45a5528f2dc2f1574601fc8f66bd5ca7e0cb26f3f6e673d23d662796b4

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d67ba35f3ea031771eab7f863b13a48ee40b7079b34385dbc911621b0ace5c46

2784c1ab7c71d6711240071a2ace0df7a86e067a9edde23e0704a44eedda4d0

9eb0d103670e2b87559b2a22c3cc6e55ac0229ebee2a14e62f481de93b772d00

response w

7476aa5012999980b83726f5a98ce2bd633f9bfc416afcae2291694d564dae7e

080538b21d4e8bb9e0a8cf58262c4e98fdbb320ecc1808606128e4eff1adafe3

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5922c17e4caecdb54e6d42bcb1ddbef6a46b5aa19339239af91d13b8d610a50

Fiat-Shamir context

31cff4068582b652d97b4fe8e49b4ba111ef299da1863207268b2e58bc7150e9

withheld proof term: 27 869 bytes • axioms: propext • kernel replay: yes

ZKP.Sigma.quot_mul theorem, Crypto.Sigma:49

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] (a b c d : G),
Eq.{u_1 + 1}
  (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} a b)
    (Inv.inv.{u_1} (HMul.hMul.{u_1, u_1, u_1} c d)))
  (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} a (Inv.inv.{u_1} c))
    (HMul.hMul.{u_1, u_1, u_1} b (Inv.inv.{u_1} d)))
```

SHA-256 of the statement above

28d64ae96f4ee57bececcd41b1fab10e93c7454e969a6c892a3bbe15b97aca8

SHA-256 of the kernel type

2913dc029125327828f2ba8ea92437245e07b7002a3502bca967a88dec0a825f
commitment C

ff08b3a775167129be8606413cf591f507483748cf93f74b5b5ef2c9398aa525
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32c67716b5389df54a45272f4438ba919b5d107e1aa0ec69726ad9b1c1e0bd49
db33ea79eeb3bbcb0cc45a56adade44b96b1dcebd3bfc81c4df4face3845ff57
2a265f3724bb9cdd88cce29c4a3d6787349aa64aa1d3f94a5ac4eff0fdc79c5b
b2617d1d7616ac383af90f6b6bbb0cbf5373ac976feba48fcb51a4d326a23d12
f030ffdc0b1c2230176cde83ca162648f64cbc9f3f4810ac6e5a41a879ff1cde
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first message t

ac479e2c1a9143e32f42dc5e8fadb57b1088cf6272bff0b284a3da0335c69040
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3e28db458ad09b8ea1d28f42c5ee467223dc2ab9b0bd06c6990e03304fe02e9d
9a88c3c11937e82b9bc868c9a00f444209e05df8315c4f80f8fa10ce057805fe
db20766e8ca7c96bb4fb48419979a6d42ef4d715e49319a1ebffd8c07aeb22fe
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11600b2d437715558deee14b50c57945bc1e18709701034579ef3aa6bbaad55b
d201bb3ddff07c018d11d9d25b0fe51254af85c8171dda752aa2a20e9708f2ff
response z

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b7ba1de6c1782e7d40b2cd34719baa08a8a3336592e78d67ef26b0efe2c04d4c
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89c0db833bf5127c606117b105dc02374bb800eb5d5deb6eb154ebcb81e80d52
b16796a1bff97d8c8342a24f9e2811e75b3845f43efcc7efa44c5b0a01932283
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response w

8b422eb09246be121397639d12b6254210d26c3400ab06c912fe36e4475a57e6
8edccca41062e7d51a52fb2e6cc65a5d542a0c0232c0ef5a9c3f0205adae6821
d4bbc2b35a36c461b046b2021b9b8f56eaed1cccd718ecad466a67d7a519c2a3
b8d734f814421f9eab7302fbd4fee5cd969447d7e9c16bd716d15c6a41d5cadd
03984bdc8708734e1e2a22969074af93ab350e97eb292abfad1116ec8b235c0d
69cf1c519017822dd243db0dcff352fe472b65ee761b898c2133e534a3f4ce86
9f9f860abfd1d5ad7da5a0341316f50e8bb67511ebcb387a01293c217784937b
fb901461cede5ba721130fdc9be4ef5cb3a1313f85e128eb266cfafaafb68ce

Fiat-Shamir context

253a7b019b9c5fb2880612b1dd966ca9e98bcd77b275b23c02c93e15e8839955

withheld proof term: 68057 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Sigma.sigma_complete theorem, Crypto.Sigma:137

$\forall \{G : \text{Type } u_1\} [\text{inst} : \text{CommGroup}\{u_1\} G] \{q : \text{Nat}\} [\text{Fact } (\text{Nat.Prime } q)],$
 $(\forall (x : G), \text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} x q) 1) \rightarrow$
 $\forall (g h : G) (m r a b c : \text{ZMod } q),$
 $\text{ZKP.Sigma.Accepts}\{u_1\} g h (\text{ZKP.Sigma.commit}\{u_1\} g h m r) (\text{ZKP.Sigma.commit}\{u_1\}$
 $g h a b)$
 $c (\text{HAdd.hAdd}\{0, 0, 0\} a (\text{HMul.hMul}\{0, 0, 0\} c m))$
 $(\text{HAdd.hAdd}\{0, 0, 0\} b (\text{HMul.hMul}\{0, 0, 0\} c r))$

SHA-256 of the statement above

c98e9789b5a0367c12fd62f401c91dce78c6edc145e8f1d82f0a88320ed4de7f

SHA-256 of the kernel type

c1769349bc9230d5e832f1a8514d51f59ab2783502a65694100cdccba1e220d6

commitment C

ea7430ece471beab0bbc31a5435180e6f2a5977cf23047988c014451e43b35b7
3c38b53bdf12f75a2142fda81cbcac7ff4a518d571f322206d792cdbcdf584c8c
21b016591fb51c5ccb186e8a001a5d6c9057567105d8591ee8b8b775857f9a94
9af381ae420b2e9298a684c2594eef9da2f4804bcc5ecc923cb61acb1dfe7bd4
a1c42cf6d1ca5d2825ca42e8cc0bfcf370a780043914619f7d1b4ed69cf3fd4c
929da40f31ff6a9fbf3b3a811796242b220c2f827de22cdd5d587e19d003224e
51a518613b6fbb44dd03586622a0525c8465438ca5c84b000ee1ec06fda19b52
9a577977b2dbca83e707940347595935e2abed9290dc387c16d8417269a20ba3
first message t

37abd75ba241099af7deec744370f936dfdaeece5c515a42df975ea392f03bc1
3c884728171ada7497e565e78d023cd07274cd9c7c9a0141d827997755fa5dc0
7d858b4aef298c126940bd16dc81b9c42fcb60be6979d1680fcfb80a42bfb035
8a9de7e4c1b35209d4b9e1c3a06eb3e757003a81945ad5e0e0cac2d7207e4713
38d254425cb44e4e53e1f36f749b27bdf0b7f950dcbaae6cd2d068db4697ae88
72665fa52121f093310428d37983284c82a51e75f70ffc13bbe63da04bb8a625
1430cdc28a681e3ffeb6ca4b6b03d1d9d3b3afbacc92757485605d1f02e7aff20
648341058ffc87382c02e81e6cfd1e5964c1d851060471f733217f033bac9555
response z

5a27 added436a79f5a43d45e81a26876c437c5a5705c19b2291d69df1dad92ba77
2ded1c7b5ce86a74f8177afaf297686dcd86cf25a1971f81ff23101a523b6091
2854e5802e219902e66cf69207ac7cf8de4df7a5470ef291a1ff87dd5c65e4fe
bb21a9e016ccd7818b202392c7528caeba8a9691c82a59a2b7ab8d6b4c6a3ce8
daa8984e04024a68246bb95154f3a4599bcb0cf3865f9ecf6571fd280f0aecb3
27a134c6e689c7dda4a828235818c64fd7b3139382a48f6652a2174212ec7d30
3700b049fd55bdf1f24627999d779757de9d83fb8c6bc2226bf1b20136279929
06c25864a2bc4f00a909715a1b6aeef028ee474a60e922a31e5742fa67d4f54
response w

d5250448c1886ec6a607a4dcd4b7b288257925980180dc12294ac833ae5bafd7
c001cca9b4871e0b82140723519f8512bf028149f7c8d687e049f69f4559be89
4f3ce8d2c8595742fa42e5241742c70c8784f7832048bec9d5bb33b936f72262
e5fa0516d6b0c0a2f471aba7a2d3b41689b8b58fcbebe7601c95408a5f205428
b46485dc460b79c71f33cc2bb007686fe898b2a0f2fa53147c184440222cf245
45ce6cc00e1e2e042f5be1697f33f21d90e5913ca88614101aa30a7f6c585902
f38c4be5d790a12f403a2430bc03d5b0a39c8421ee741bc7b0fda0fed6167f33
5219b6c527d7eae8badde71486a6e2447dabb64ae6f8a7559c4497742fad9de

Fiat-Shamir context

85319a0786fddb26a119e7d5fec6bd89d2729ee7158285fe4fc1f80870b60ec9

withheld proof term: 162 266 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Sigma.sigma_hvzk_bijective theorem, Crypto.Sigma:187

```
∀ {q : Nat} (c m r : ZMod q),
  Function.Bijective.{1, 1} fun ab =>
    Prod.mk.{0, 0} (HAdd.hAdd.{0, 0, 0} ab.1 (HMul.hMul.{0, 0, 0} c m))
      (HAdd.hAdd.{0, 0, 0} ab.2 (HMul.hMul.{0, 0, 0} c r))
```

SHA-256 of the statement above

ca6b2f027602b13b6890b19b6529e4515c5cbcad3b52d6f9ddc093380ad42139

SHA-256 of the kernel type

d8a1524d533897223ec14ae76183b47483bbaa25fd4b9e2c11a51d57edfbb3a0
commitment C

32b86bd5a142679c8414e32ed89b966fdab6417eb59b52d7769f24442858dbdf
c4dd0c3b97893ca7b5301ecff586f3677c8b69ad6ab90441336c1c4ccf165a70
4cd6aab6ea87811adad821690f322ef4f7a70bc9a7d0b688b443b4ce91134251
bce02ff10890b45875e32015bc4dcaeb730470dfb0ab048eda02d44359e239bf
ac8df7e704a998767eec0a9bdf17818888b3820094880bf747f3a410a10c9b1f
2b12e05ccd8ef1f3030db4b20bec5c24d0ba37cdeb278b9dfc6983a30574e02c
f0f81277e795675e3e1d0cbd8d8edd8bc6b33238bc6a25de7f1ceeb45d57c844
19e4a74e515a0bbd313ba2ce6430e2e1345ac048cea7304ff2d85c02b5c0a182
first message t

2c5d47d5fa084b54aae96d7abe84a823b540526b98b7891e7f680c6d41d29c32
4cfe38e8502569a8774296c0184bbdeb3be51ce1b58d688f9d20935e94829c3d
d720c8b35b52608e57ac14b3dd90ab6d1a53cf0d0e7311de82c518c705449014
380f05769baa218ba7a41e7f307d2e0a3b86d9b33c6122a7b32d3520ae10850
236e54bf314d600a43f7531d44d59e87b366a092d9c0b0b204384c438c62b918
04e19ed41b4057594fda1028136c4970b7d0b6254344d3b13e351269e74d
311ce95448af0db6c9182ecb8ebc4e29d5e325cdbc34cea3e4322bda1ccfa39e
fdad6a030b59c0a6b970290dcf0fdcf3c70dbd7b0e5ab88ab7a33174ab69f864
response z

3035c1c7f0ee5d83bf40f34b5099b0505e7d952541787cc4af534e775c22d0ab
87ae13f222112f10e7e153f84ec61e83713ee5e29245f4a2566399f1a1d595ea
0a1132f470866593c27862183a3531fc887a6fbd5abb47a6aaf22cc609dbcccf
bb95d2d58aaa0f83b8fac8cb5841936ad28deb094b76af0d0727b721156a9d15
d9e49d0fd2dd0cd4d58b2390b9d12f5a2b4b68f5cdf582d810e67030a17c0d23
79f3a9ebcb11da32d8944041b41715c3ffb0a72bf3317e6148bca3e589af7ae7
ded0bd40984a7f89731fee067bcc38775ff2238e568f6374bef1d5fafa715770
aa166fe7ae569e62f1f8e2f44f9aae0f98a127189f9ca9e327de42599e57ccbf
response w

5b4f4ee6cddb60ece48c01fcdb911ea79f7699ea0e5ade6764c7617520025091
6f4a76ad03f03a4b659d9d5c63b3f5410a6fc9d43c34849687464fea61712386
482c665ef86fea9df096d96cc5ba2111793c7f5ad2764842de232a89e028e494
b62ce7c458e391c966eaaef2aedc095fe12c38f379cd5a0d62e3177b86292b3
4ecbdeaaed34250309dff3cb5ec1daed35852bd2b801476572536cbd7fd2e384
e0bd4e6384201a669c77908d435226c02b5146c4d7d92d4c56518a1d3c0ea226
936a4caa6e96946653be36497a79e6c2a0f0f60ff72f252e80be41da34d61780
7f9eb0e9719555782ad079264df8523725943afb96eda9235aa489d2cda1c855

Fiat-Shamir context

28fc547ee9246652b1b56060caba6c3f91a244d13c63fcf5d080e022875de0f5

withheld proof term: 87 109 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Sigma.sigma_hvzk_simulate theorem, Crypto.Sigma:170

$\forall \{G : \text{Type } u_1\} \text{ [inst : CommGroup.\{u_1\} G] \{q : \text{Nat}\} (g \ h \ C : G) (c \ z \ w : \text{ZMod } q),$
ZKP.Sigma.Accepts.\{u_1\} g h C
(HMod.hMul.\{u_1, u_1, u_1\}
(HMod.hMul.\{u_1, u_1, u_1\} (ZKP.Sigma.gpow.\{u_1\} g z) (ZKP.Sigma.gpow.\{u_1\} h w))
(Inv.inv.\{u_1\} (ZKP.Sigma.gpow.\{u_1\} C c)))
c z w

SHA-256 of the statement above

bbaac9d7466d65f8cfd8eff2c998f4c88f2df04b2882a026355191696a3576c1

SHA-256 of the kernel type

7946d7b2397c19fc35449f4d5e3c44589582c2ccb28b69bcb3e842fd68172043
commitment C

91b44980fb75063b987bbf96e3a332a1798330123c961177e0d336b232b560da
7af868b2c49fb84eacdd56ed7d00b0d669f59316726ac729771f4c8003f84c3
c2a751f31bf171a3e06168344a9172d4dc752a03cc1dc8c5c24a7330ebcd91a3
1d941eae1ef458a34442d32901baa5ef68fd4b476aee8b9c1d63c859f9e9f1b7d
19959af95acaf563f85562498ce9b039b850068bd7712c57b3c1b89e1a3b6f0
9d895a43714b62d8410c7759eed604f95d709e9ce0370635602153a11fcad2f6
a40acd6224af9aec443290f86102392a6acd1f1b255b44cfec7920dd50a21337
64eb407b5a532e7a537b13e0d1ce1907b4524450f79afa093e94547dd1937567
first message t

159552a305ec25011b804d267b65afceb0e3c0e39fb9ec5d0daa16b03aaf8463
f2629db91edc461f99c840795dd7f3f13ba850b1602c472a60bed46c38dd00e8
8a52950cf26ef4e1e1c1d2f84643ca32eb08403cb69c61329d02e62a5d913421
871e3a6522eef61ba84463db8386dea3550f93af057bdf28a7aeae57d1fc156e
3c1c1d6f789df5d104bb52d4c83b08cba56ecd0e302ffc753751938cbd557cf4
fa2a1bc824d233f2b34cf30e12ea30d4d3c4e05827aa9dd5de017557e612c39d
302a27489a7cc0c2c83727f0451efaa11a574b26bf47d97cfaa40ee70cb309cb
286f2150d9c7a868efd34c50891ff9dd540071dfed8f786159c425b5f2f729ca
response z

5a4ea0bfff03601261bd46c728d67b8700cb21cf2b94a1e62f07d8c5c45ac41a
64843a8b5e795ce696f6670975f6d7e9b6c56c59ce1baad98417341f0fa47df6
eac9fc4728abdd42eb5c2811926cc2903acc6282b9841630159e013c75a61e76
5dfb5bcf8c2d56bc5490d91739d042441a24a967c33482f096f50eb0a2a5caab
168521b94541da30b3e51fc38aff78db34bd9ec8a9750a5b9d6e034310e6cc3c
7770d192812cc8e3f6d9eaaab319baf053acb8587e661bdf317c26b015ea929d
afcfde19689887e6148d25ea436c031c1d38171917bca2d571cc1877b53cebaf
6f3d36a4f241c5fcd85c038c4aa902c90c25ab8bbb9f79510c6ed84d55f369cb
response w

57e7ff950a830a231844497c6ed963f10ac81bd2c62c559e6f47ed5588598614
1e441867fdf7cc40b2aa979774f3b701cdd946d5d717c1cb8b4d413f1f53b8d6
4f653f1bf5a73546ca23f44d6184a16ad4e953321a500dd103a6d86c7e616426
e1ed951edef4acd2e0855786fdb57f3fd32e9f5ed2a99503cc8ece04be0da470
5dced1537980b1bf4a55300376e831eae05f3169b7fd0666c4415d593f7c2f65
1b8ccf80d536e893797b14426a98f2a0d7c6b8bb4e9e9dcfce15ff41334ae0ee
aae46307f80944983ded41fe08ed023cfd61985ad233c37aefb84f8614525e2d
94f776a9eed43b714f9c536d70b2758ed9a6073d68c42fd3f4153ce13cf4736
Fiat-Shamir context

37caa653c8971001f5cc8ae1f0f62250577cb8952815b030729a12ccb6bc321c

withheld proof term: 8942 bytes • axioms: propext • kernel replay: yes

ZKP.Sigma.sigma_special_soundness theorem, Crypto.Sigma:146

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {q : Nat} [inst_1 : Fact (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h C t : G) {c_1 c_2 z_1 z_2 w_1 w_2 : ZMod q},
    Ne.{1} c_1 c_2 →
    ZKP.Sigma.Accepts.{u_1} g h C t c_1 z_1 w_1 →
    ZKP.Sigma.Accepts.{u_1} g h C t c_2 z_2 w_2 →
    Eq.{u_1 + 1} C
    (ZKP.Sigma.commit.{u_1} g h
      (HMul.hMul.{0, 0, 0} (HSub.hSub.{0, 0, 0} z_1 z_2)
        (Inv.inv.{0} (HSub.hSub.{0, 0, 0} c_1 c_2))))
    (HMul.hMul.{0, 0, 0} (HSub.hSub.{0, 0, 0} w_1 w_2)
      (Inv.inv.{0} (HSub.hSub.{0, 0, 0} c_1 c_2))))
```

SHA-256 of the statement above

5998a93de8de3a768d9857ec47fd7104628b706ad8b35ec8305de5c9a61ff94e

SHA-256 of the kernel type

b3ca9b610890bc1878cb28606f568d38bac4eb51ae8185f0cff2687544398291

commitment C

54dd76fecb6da2780fc4919eac7a21322778c9b2573c48298d3d33f4dad760a5
fdc6ad4bc6217fcb4feaa31afade30997738b0dd4a887d66825b76dde14c2014
36bdd73b94e36dbaff3d0cacba62f74c070f537665632a5377ae4fb7bda1da7
02a6702ae143d71f0e0d52190377a07d193fb47ac0f9f05b4a911827e9264d32
0e82983c0ff031b83bbb4e3b56cedcaed4f934107aa80267df0a3f7dcff8291b
745e2f462a9ccee6e52b7f52414fc82c56210d4054e4757c681dc79832ccfe39
1c30d8b9eef56ddfd8bb403fa634ca7bd281358a4e71665e78cea0d6df7e33f
197721a41f13364bcf13ba8dd377bf6ff1e0db2e4b3e1f14380785a17290278
first message t

3ad2e523ad67611dfcbf280ee7b351c0335436a4d7a6eaf11b47c3caf2978c78
2dec070735a655165383177df5c0a102cd7dfd57b517c421107b5e9002b74d17
27a45fbf4eac33126e46283c461d04d08a46e9c0e2b3fe6704e61fb887709237
c8343ecbd427e265184a59ec2c6851aba10789be7b241e9bf60198d4c9b76814
f22ee716b753c994863a02d976d117a4641db3c214b40ed10064ea98ace5c566
8b6f03624cf074d12b4c99b06b9f6fbdf1f7b8500463a1267460733b9b5ce55
2f6a83c1e83ec34a5d47d08839f39ea7a9c8946a76847ff7b54f12e619826a29
b02c321af96e66bf1adbae86c3051532888ee6e2463794e8fd13f95f10dfed6d
response z

52145e37b3e5e41b8846a0d21e8da9c5fe01eb4144b686b63f867cbdc9aff8b7
e554ef6e9ac3e904520956d0c46b129969e6d8d659737da63ae9264dbeb3cdae
796c2c85cb18fd991cfbfaf7d7df6df4f67215cdf2b579d02ac526086d14e57d
4064fecdbdd69af4e07f3f21e1077a77af2704d419d0691a9f492ac9350fe97
59769ca671294aaaae053c2aaeb8b80d3a5c5f42a46cd29b2b24d7670ab031da9
0f82294e139b488b53c976e8252b3e6f263a1a5f9231b1f4da4726f68c8e9a19
601e3d53a7ac0e35c81990626d1af96ab4af4e7c9ec83768bd333bc41123b2bb
41d83cd25551a2890b832d82966499690767809896e6661fdd404468177f67dc
response w

57d01ac023262c3906ff15487f78f720c4c77f9083ba28cea62fba7d6de9e487
f025bc5c34a004d0a15c364726a621f5a72a0aba41d81a3b40d5c0040e9d38c0
eff83de6526668f63fcfc6668bcffce72401011de4ce13a9f006ccde24f7ddf0
3d9a62c2519c5dc3cbf840cc30cfc8722818db016dc20fe17475c6beea57ec97
970c12cedc9288f899378fe547a6c575aa6361676a158628afe802d9cc1c6868
38e16e9a21d37b01d82de7479d750671504f0c41ea0c3735b516560d08df6ff1
8063f8bd3e1f595f2d419c6c9d5316a1900387956546775d66fa3f83d56926b6
7369a8b6f45dc973a18ad3a1056456cd648e6f1855907f08ed683d049edb257a

Fiat-Shamir context

fcab5645f14732b2e16ef3fdb0690173b9ad14f767379ef7fd005e23ecdd4f79

withheld proof term: 158 544 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Stego.chunks definition, Crypto.Stego:86

$\text{Nat} \rightarrow \text{List}\{0\} \text{ Nat} \rightarrow \text{List}\{0\} (\text{List}\{0\} \text{ Nat})$

SHA-256 of the statement above

acd31563902fad7dad01bce0eda60d60a644a89e1485fe4fa6113fdefe82f463

SHA-256 of the kernel type

3d7829b0c3a2170235eb3855d4a4285a73a57164155583ad79d170e22a27d544

commitment C

d035b6a577152166991956ddd11bfc95c0b6264714affdaa9d2a7652f6973653

90816767286dac193d05d5cc0ea8ccfe90fb8e60c1b2ac26eaa713aeb1e2aabd

1d18cb50209f1af5e20f077a885d09368177b5138a1c143754e3670654be943b

2af5539d4606fceb55f4bb3ad976c21eede8ae15aa10e17c87c0e54c366a1527

08968bc7a6088c017588acfb64c609c351b5a5ae909bf9930d79427169881

55a23735561e86d907d048914abff151eb4f5f39e044bc831508b045d84fe4fc

291d222d9c104a949b1ddf16f9275806724a28a13b7c5430637eb0d4f68d54db

6760db165fcfc30f58fc3a260666a0f9c0204eb87534dcd0cf225a5cea1a3818

first message t

255a66921d89f1ba884cea702fb3e11b618b325cdfbc345ba01c5b10bf6d4bfb

e231477530928276cc4b2a9c75a3b99752da57203e2f6099dbfba2a05db72eaa

98771e5366d1a82a41da8140a28234b5a20fd63f7a8a899e4430337b05fef484

1556f034ca6e8fc921eb0f97772d40ba65b12db2631b26b332aca9d79d25157c

c6f4ca3fa4a0c4cd75bdd1376cdc51d4285edc3845c1d2d73835b1bb39af5c3f

daae205750141419bb374c65e5d40d83986bd86f6cac0b2969f01cd827296565

e1dbce46c17867d62ad365b7ed82f350b30f3302ec436e3edc11ccf272fa6da1

d6462ef65d17b5d3eab10dc1c668d743f1b1bae337823a398ad4185b2df6422f

response z

67b05ae795bae3278569c9e14fe3854b5cbf83b465a26608487e96cd2cedb19a

4a206a9fd4097f9413c3bf3d1e988358e359b79f3cf6b97ddfcc48fe68d5b358

4da000076bda8eca13ebc5eb43d77c454413f10ae68e330ac5b2b58abf9f1778

5766c49eb738d8c8fb17d936a0c23740c7adec9d1f8de73fa54dfb357ea3be59

c2a026d3ed6c6037544c56741c40727a5e412cbaaa0d4e87af057a4aab9db0a5

6198183073a7069416b97da839f45f181307ada7231502139ee0f0c02820747a

f94ea67c0fbec6c254463ae3d536c06720f54fdfb67ceea4b3f5838219eb9263

bd883d237015358b12153bf70d902115b84c9f3902bc5c60fbc8262b1abf0e3

response w

9cd105aaca7773a553701efc5820eae79bce678810b50ab4ba4dda3b478d356c

369abc275efbcf9fdc98b405fa8661e8881ea2cc2220cd62e0bce38b1bd39c9f

b7816fb2ba11086f1ad3032054ac4baa62a3142000017e6b9008d58c63d6b07f

042b8cdc8930b2c1f1f4a3a26f5eacb83a4ce11b86d1ec1691f429bf193b1201

d7069b23a2bb5879df1ab0ad11e35b5032aee8f8b2e30cc22c7d225ff9ee90b0

81a06bf32693b3b4fafdfa82bfc645ae16c937bf8fcb2311380b84bd3075feb

3c3d1ba5c36c6ddb463e52a64a9fc238512b7aa9f87ba27d86817b28fa93d57c

b496676bd3bdb6fbf01363f5810a02480cc42e01bd37d98712748aa07ab991

Fiat-Shamir context

3d318fb4d81dbaa3158356319abf38624fb5f2d17e731e1f139daa39d3aabf

withheld proof term: 222 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Stego.chunks.eq_1 theorem, Crypto.Stego:0

$\forall (x : \text{Nat}), \text{Eq.}\{1\} (\text{ZKP.Stego.chunks } x \text{ List.nil.}\{0\}) \text{ List.nil.}\{0\}$

SHA-256 of the statement above

e09f61eb6bab42397f0bcf7e59e321605ffbd015b80c983c212e9250594b94b5

SHA-256 of the kernel type

f39cbf53db95f3250d925e2ecea51c16253a01351114164a6b6bc1e7c60a5697

commitment C

992b657731fce6d78e5c5c00387249b90e0290f60adbcebd9f41744af5aadfb5

197469f6db1d496bdd0859cab95c20f48d8dd1f05f25796fb54abf254aa5f128

42eb735b5fb530d4c77602e4143ab819b2e7f29d5b78ce2cead278dba7dadad6

8b5f4c44a55b4979ba22b1fcb7ea522b60a935ff00c925e5172157077db3cbdf

ebc5eb594c21206311c57fa2f0aa38832a019a0bc5cade5035a29374f2a51caf

9c1e363146ac88f31523de97b9086d5a0a02037baeb6d5852edde863dfd1a9d5

279c38e1359b29ab5fbf7f89bb5386bad88ea64215bb2f838580b120060e4238

7feda8039a5183026a3688f0b64a365c579d1ddf82b28e4b22f1816df272b654

first message t

55d12340f6c96bfd4975cc05cd19d76514a82b83b53b548280f724001f732a3d

978beee8f4f7b7454cc53e7025424b7ff99064f045297dca6601647cd56c05d6

0de4ec18ff4c2595f2cedbef782b603e7991c899c493a8d557ea8ce92b97d4ac

783bb186ec425f85839bc7fd80f214463dd11d83f5c33c2535907408eefe1537

c10b95e1fe81528f147782089ec7c7ea12c916cc86cdf37a8cd6cd522583815f

2f1e369902319108401b5106a8d277a0f27c738d0f53692617ca41ae74bb3700

9dc175bfba3c96faab56e73f3e6fdea423544eb97579823e22d745e5f6cde9c0

3e358f6414161b96ed3bab88c949514f0f8e7e734867408d7c817ab33e5dbf83

response z

d9e445b0de8fa3b4793351c33159ca0a8098659abbbb8bce3e770a526d391bc1

b2af04f58857f2673af161e3d4806fba52cc8862326ff51257a6528b720e9c10

53f85ad946f1457cf166e8540bb04e00f3ef4ec3d84839ff4b73a77b6a616098

8225b7dfda3dae1fde3899b1378bbef142833172570f58892801330937bc1f1e

23022d393ae4d711833815d40523592e65ee22249d35b0cdb1a07ab9acd5b549

0485009e7f46bf8019372447a5e67ec67207186deb08739c89ab9cd7a4f287c2

3d30871fed8fc61d0b2888abb8367331b87bd1f7c3248dea573276090e29771f

72e234c112f7f27b3a41a448421f83e52d7e3e34d5ba5a3c63941a73b685de6

response w

60da5336924a00e242702ceb316f9d604b10a8d74d84dd9eb58811af9426c7c2

015e66af17de647015a2ac597f7d76324f82ea7878f6ae173f8bb0004d5f7373

d651aac4bcbcf152a27743693a0ce5586cdd95395f26ead438542695b1421bc45

a51613daff4531acad2cf91e37159de4374d191ee87620f6916e4148d23c5f1e

304c83348e6ecaaa8db9fd14cb62f2feed1f7f5ff76281c031ae24af50b6761f

38c07f78afb6d9f8017271b1f8c39030f7c0cf11dcbec21c25c32b8d454d4f28

9a0e7ae84f205f165761f0e2af559195c21497602bb81fe8becf80c9758bde3a

946e134b834730f89a35cb2fa8070f993e9bb7dc5aefb88f0b1fa43d7d050b42

Fiat-Shamir context

35e0109591a61071380bda1c46e17c852dad4997b12945b2b7af811de4ab9514

withheld proof term: 94 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Stego.chunks.eq_2 theorem, Crypto.Stego:0

∀ (head : Nat) (tail : List.{0} Nat),
Eq.{1} (ZKP.Stego.chunks 0 (List.cons.{0} head tail))
(List.cons.{0} (List.cons.{0} head tail) List.nil.{0}))

SHA-256 of the statement above

46d53e8327ad31f6eae174a48cd82083ca29f761bc6b92f3a1782b08b6be879

SHA-256 of the kernel type

eae869063396c28bf6c32fe5e1792f37b0f31183d6f59586f22289e6b1ac8d5d
commitment C

41c20a7ddcaf4846c063b65dd28d995512e3a32b7384c3ac584de3aa483f5bd7
fb5bb454033599b7eebb64fc89feab5c323aa69b473d682473fdb92cae4a32c0
78db6fb1e68c087f5c33221b42ce807ac3aa68e1a79e2a1b06125a865f0b48ec
ab05a55ff623c4e541908d1d213f2dd826400042c664202c294476cefb8360f3
c4b3fcc7136cd0ad77f997358d9a26007c444a43b5bd4bc3c87ea379c954bd6b
fe7283ecfd130bef6029bd108c759242c68acf61657b8b27fe002afac2468087
7b2f5579bb76a70a885281211d8146043e14e582865b72a0cfacf9f8250d0129
4b78192def2f5625308710eea771d22be4c002e23e9926c157073018222d3803
first message t

c5750f42efc91a1a7a9e69521f928c8c9d473f621ea8834cc514bf498f491c65
4b5d6cbacd1d46f718d6410ced76a784e08b02964ebe0c6164a4d24855f61463
dd543e8c0909d620dcdf5510a8f671c9a158287c40efb1b223034080d8381c9c
8abca6976b6d3aaf4969a56ea56b81cf4792a0037969cd99b57fef7ff6743318
bf27a949421769af5e91c58f8a22075c64435b08f998bc11412a9c39ec6d0f50
ef3bd9f6564831ac4b462b018f9413e58c20cb163c63847cec4842640d0e12c0
9c0779c89236f0e0e68f5d9a2d9208c2490c96a1c20d7dc349b43cf9baa1241c
403d9d7c6efe05eb500d21e0a2be7fc1c1370057bedc093f276e4334c26aeca4
response z

6526caf069a493734a9d2540ecd7e84f18523fea23fc8d90506ba97988ecde73
c1beda6b73a16c15f8f8e7bf4295922834d2e8146bad7e2641a6e46ea8753c7d
04eac8a3816aa1cd341593da95c144a9c06f69925d25dd0f2eb9ad9f58c02995
3f02c6c60278bad2188d11012a742978c27ba700e1024a944164d34e55a954aa
086eadf45164ae657f375b9224643fccccf006e54fec1d2147889f82221bfb0c
67e9a9da47163db512c001d1c38b98306343a9421ae7d2f4be02724810f4bf3c
523947075aef3247e892fdf8098ff3ede194d9fc02f9b34cf222166aeb31229
bd83ec5c5e2f20ce7f5c17aade37fe7285a1722c6d2c384c7d85574988e76209
response w

6c4615124de9f7dcf8efccdf450cea616afafc758bc33cbbc620dcbed22dac85
48c2c515d2411f03c9e6a74bfdce5dd4ba5f146871b48dc1ba0e9e351505101a
0f6cb4b828ccaec5cca19d19f5ba9b623eaf62d74b3c0fe57b243c123e3a7f24
e178036381f6bde87725b0029bf45b6c000ccb4caabede27e4041894c24913b0
453aa996bf7b5b81992311554aa65c43829d54d54fed17033bcb985701ac45ff
f5d8696723159942cdaeb828e3f0227675a1ed5645296059a714ee65190df4e4
2ffb0036b94ad4cd8075599c46ca880b2b2a842c5c2358273aaeea923f4ff7ed
b83af1dbf598ce826211d384c8824dbb573c52da80fe973e62deb9119d0a2288

Fiat-Shamir context

6dfdd77181ccd0199e49236203cd57f9f861d6f2620be4a86fd656c49350975b

withheld proof term: 3312 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Stego.chunks.eq_3 theorem, Crypto.Stego:0

```
∀ (s x_2 : Nat) (xs : List.{0} Nat),
  Eq.{1} (ZKP.Stego.chunks (Nat.succ s) (List.cons.{0} x_2 xs))
    (List.cons.{0} (List.take.{0} (HAdd.hAdd.{0, 0, 0} s 1) (List.cons.{0} x_2 xs))
      (ZKP.Stego.chunks (HAdd.hAdd.{0, 0, 0} s 1)
        (List.drop.{0} (HAdd.hAdd.{0, 0, 0} s 1) (List.cons.{0} x_2 xs))))
```

SHA-256 of the statement above

2f1028340cbc1a79f6ffb3f6c7048274ea0e3eec42aa17d198270ff5c7edcd31

SHA-256 of the kernel type

9f90bd5d636262fafb954b57162848e538d6fccd38b8967387c4e4d54f4b0332

commitment C

b91f42f7f92a4ce0d188d8deee0ed18b1db3db958c513eed34afb24a6ca4a018
d5ae513740c3336eb7df85597ed089cf4cd1ac56486e89130951ef20d08c9a4c
497fb8f5c7988fb193a1f7b36a37d8eec36414afd46ed0c43e72593ef60f680c
35bf8d64371705998ddef229990410dc93ba4e97e2187d767b49bca780447498
7d91e9350a323664b38193a2d7e18d4a904a22df23386638dbb89803c9cbb31c
b896cdec06fb7769ba02eddb26f13024f5d4028a035b4f28fb0f07e160e602bc
b6c6f77288c7a035de70c6ec7d7dd9811860ed5b53e5ec7ec17318c1d4dd361e
18dd04be0848a1e2e338c755ecf3721c1be5f6d0316587bb5b7db25c2dd9f5e7

first message t

6c8eeb6db3e903cb52c16dfcaedd5666f871a3763e05e27ec65b5afe0a451d55
0e43fbc98becccc12ab70700e25273dac23956ee1b0d0b942110506f3b7adb4b
ad8bb9b23b0ba81b4cee01dc892812759af04b70ace3542f07da973675f685d4
f9f8341d9880b12647e16a22ecbf9e6e2f031f398cf85ab2dcc39e3c37f40b5a
8be2a5e30dd2735a28c517dd50ab6791769c42c32955f6612e9c3a2138548b4a
784fb395902b869f8db2270f3556ce93e932155e5dae6346025cd2de0656aa9d
7454a78d18c6d6d05e9185889e7add9c5da82a44d28bc06a411877bf5eae63d
cda5fb5a01e61b790bf454f9cecbc958e150ddf61204ce05f54fcd5c98ca53d5

response z

2855bb480da74e2731774dad3542e392cb938ff95517f8b980f06790b18e4d28
57fc3081782a64765010b709d6399e095b4c05e7a315d2fc2cfa639f1681ccd8
7dfc7ce92cf16244d475615780c421e5b1bb6e298a085c6fd342dae69c3db743
c1e7ffb0427d7a70330dd51ee6aefb8ff3a68fdb7c9352bb2eef8cb749cbc0f
820b22a57c726aee03178929402917fae210cbf337e1d4410b428d873a370131
7bef9a29d01388ab7ea52791f1551b7d106a78b0b56fcc426b5acbf94ab7094
7fe4586e7957210ca8a85d41aed3703953a0164640361adc9555fc236ccef276
6ced1259789b8ea933407cc273300716cfb17c53cb39210d0108f2e75c5fff05

response w

14cac0f63b1d341a8204b6b07eeb5a162a4dfa48520dae00acddf01a5195d58
c292b4b96ec4c8c690e684b69b4c3d55fe43c02e55868da3f235ed2265cdc2d8
e0f35b9ca8deb24fa163c011423c2ff4cbb3e61c335b2d4470be608e9c5998d4
8691d844d40cc05c4c541aa0de1d735283733b9e2353390ed5f27044473d511e
29f773ee8505a6b8be484d22baac8d066eaf03259ff36851733399a5239f11f6
abaa66cea64ba9748ae2b07962c74ba515f736710dbb14b97709d45a9fdec107
6e6463844de13277d632050ecb41b8617e47fcdda532b7d02b2a8dd458d8fea6
45776c20b47030381a1b69e15125cd2a83a942b4204805e7c512aec054f47e84

Fiat-Shamir context

a62870bfcccd385a43ba5c5af19f1f1c70afb0cdf3ca4936576f9a67a6deed877

withheld proof term: 3 569 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Stego.chunks.eq_def theorem, Crypto.Stego:0

```
∀ (x : Nat) (x_1 : List.{0} Nat),
  Eq.{1} (ZKP.Stego.chunks x x_1)
    (match x, x_1 with
      | x, List.nil.{0} => List.nil.{0}
      | 0, l@h:(List.cons.{0} head tail) => List.cons.{0} l List.nil.{0}
      | Nat.succ s, List.cons.{0} x xs =>
        List.cons.{0} (List.take.{0} (HAdd.hAdd.{0, 0, 0} s 1) (List.cons.{0} x xs))
          (ZKP.Stego.chunks (HAdd.hAdd.{0, 0, 0} s 1)
            (List.drop.{0} (HAdd.hAdd.{0, 0, 0} s 1) (List.cons.{0} x xs))))
```

SHA-256 of the statement above

eddb9d29976ff7b495dbf4e36c9d503d43346fcd23146d6bed5d115a73d1a086

SHA-256 of the kernel type

4ed962090d6645d886540d015c8565de5e4b5769606015be7c698354b852bb6b

commitment C

9cf604f1f3fb1d95b8c234a8d1ff684ebeb80f114f4eecbac34453990666ca07
e453e7432a0fd1af321fde4f798e5517a57902580a9a7883e9e850ee3c2195e9
0b57a00d30bbf74719bdb779c220d52bfa5f6b002c3a36956cde509f8e15567e
9c95fc92bea7754e3fe12f0edee439ae6916c17f2477471f8e8350f8151dd8c8
05cbc3acc9d2e9dd0d8797c3162fafe39d33dcfdb9728156e82e1aed80c1d8b5
6dc299ff91ff9a0e21bd9c30b0544a4eba964147f011220c0a70985c5b64b0
09ee695d5bf4d858a43be24a19db0a18b76dc84ac0efb4be7f60cd6b516bebd8
957b11bdd76778ac5c353e659f1d907e85cf6ba12ce8e7b40db0d2eb516c08cd
first message t

9a068341eeff6ea37019b95feac3b1ab1aa8f6624470112e0292ded0ad276d93
3fc16087e5b004441deef1dc9b3fc4ac79c6bf102ae3620783b66f849345628f
dd16a680df48603336274d94bde4181880ed39622080f3b3e6fc80428bad5944
9f5af1c77b76e3e05156f790300c6342252cdc2886ea2ab1a132dc974eefc2c4
be12e06c6ad957e0519c4d2df5df8919b34c959a83aec855c025ede79c03f81b
d8cd721f3348593e1c7363c5823bdb43306df161ef4933c54e7ee31eaf9774b2
fc94edb27cd26aa49f2393a0451ed2c79153abd47684b7a149c590e0de405031
948ee8c875cb9e4ecd9724da102a57a68d4251c69a5b5d1e05a40b754e99c45
response z

1bf96dee0a9a2b8787bb11b97688faf79762cc6288ff9f94aa8cbcb3119e6601
3a5842072b59898d003e04e3b72e79e1787c4baf6da3747aa7a309a3995c0d75
060b4e8f6b49758b379a930b4df0fced9e31c698fcb9e9d074dbb5cc5197598b
33acedfaa264f4cf344475ace45289c879192718a262aa522211fcf876b361a1e
16aa65068ce4bc0a63af9a5cb432aec7a05ccf29799a24ee02c36e9b180191d
a2dbc394405820f63f9cd88ef428942eae59d1d9eb93e55eeb01218a95deed26
eaf226506a21707806f96583ee9fe25bbe10332082852bfc5bd83e1c8dbccbae
07b1ca3c186386a2a6c63fc615674b28655f68aa9ed4c8e1d778f899f0586833
response w

2ce8c1846ce12cb41351a4742d74cb9d2dea8559e39b7b7d42a3c8f8c0d4df87
a8c85bde1091d3394c1fa9de6bf17b5d154606358492c659b7990c76b3eece56
9ffeb5760e0f7c62a89addb22846d1c3e030b0cfed140c056881925220026302
e358eaaa29adba9d964be43e49c872988b0582437b5ef50306f1285abe018230
35a2db6ffbabd41a82422a9be07fdf57161ad0b3b13393c17c7f8719dbb9338b
ad246cba59da6af5bf239a7b97a2aea78d2111723ab5cd3bbfed6e6582e6ae54
81b84fe025914f37980253363983e6ab7d849edc764e0c5419bc21c51d2ba703
71edf68856acadb38d16f40869acd4efbae3e256abb95791725607ad8d3644c

Fiat-Shamir context

654531c2d8d1fbc1b34f6f03335192de4b760e5988aa45a8604a2bb264bd9844

withheld proof term: 2058 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Stego.chunks.match_1 definition, Crypto.Stego:0

```
(motive : Nat → List.{0} Nat → Sort u_1) →
  (x : Nat) →
    (x_1 : List.{0} Nat) →
      ((x : Nat) → motive x List.nil.{0}) →
        ((l : List.{0} Nat) →
          (head : Nat) →
            (tail : List.{0} Nat) →
              (h : Eq.{1} 1 (List.cons.{0} head tail)) →
                motive 0 (namedPattern.{1} 1 (List.cons.{0} head tail) h)) →
                  ((s x : Nat) → (xs : List.{0} Nat) → motive (Nat.succ s) (List.cons.{0} x xs)) →
                    motive x x_1
```

SHA-256 of the statement above

f6037f790f04d2be20f90fe5622d8e8a2b5c694b8062ca5bf24817164cd258c3

SHA-256 of the kernel type

a6c5c48457aa01e8bef436cf430a329cd91c433062dfe9b9831dd65eb8761bb0
commitment C

9681cab3b4239948abad8c3321e9a3b7dcc1ab2f6724201880b77d1ae4598550
8d724e9b6bf7563a2992a49b01974aaa4c908f061894c3e769c452451e888df6
8b8b96cdd123e75c6d355a1b0b767f6f3427daf8e9271d9bf3d1db1438d86c77
5792726d7f8734910def7eb2f90febf2a6cdc8c0092e6e3c87920c935ea84bcb
425134e93e28c34b834dc5aed41fb65d0ced32124a79b5607ec76f21d9d428c5
82acb1cf7134c5c082a519a6f172d044abf794f8a671aa20bf699db039a1d70e
e0c3664d7f3cb4048fe324c0de3b9e8ec761f4eb329e367516b376e8b9570dc8
cea61136ee4c7683f98a59c0520e8f3f597eb3ff5ad6aee95e7159f9bc2fcb07
first message t

5ba96bf39344312a70ac3031bcce8b509589f90184298cd9cc4b614e88d3cbfd
d19a5646f4c8649c9552808dc1880d9c6a6ba9d0769c487dd49584b38a5968af
f86c76cb565d2d4ee06338694c86f5c0ac1bfedc38ec3424732d1c60cad902f6
5a2a3914fcdb31b7699d4f350235d1bf89bc6f5e01e76147302e11387b7a02c
b912fcb3ac448a852a79cf043688e015be11c41a395d244679b19674e0cd1e6
12cde000a8a18308f19954620a944f251cfae06a581ffed9747f423bad81fc1a
122b7c035d9331d030f8ee62c6c39d19791c12537b63aba0dc60dad0597234b2
cdb81565c4fce20958c471355a51d34081bb51e9d565bdf524fb30149c455d13
response z

211fed4174d341aee5705a5b05fd3d170b91b5d9916d771042e8fb6dba48bfc5
ec258e9033619c5e5f9135ac2206d0c82c548377b890f6312657b6a29816ea70
a556eaf53110e8ea20d4a9eba7680b5ecbafa0584225c15641fe3a0828de90cb
3768353ca4a928e534680fc866ad8573ed294fd3d9f92e4368963c2d39a344a5
fe9e22b9eebf71b0214ad8856ceb6ad825db0ed8d99420cf2057dc0961cb7eb9
ab1097f03e7b4a770f81fc11729a55d48248122ca6ee70d46e74d2573a2c8303
1477e68d0a84b5d8a2fedcb8a3763132e250399a0a7201fdb1d1b2271093d850
4e666550c8f37bfd5abc233c5fa793d858f2905db9dc928cb5096a0fd0d5ed90
response w

689878477acb4258aaf4cbdb1e4f7e31d44327b308b5827c0526b6355283047c
4efd8eb8a4c5e9bf1c4334d71439c9dcf09e10e378a56ec5969a0104dd48f20e
3a13a28c7496b3e8fb433bbf4879f5c8a52ec306c492846fhead11c2d29851d9d
6371aec83f5dd047df5a0251abc6a9909ec3c3384fccb55f1f7866c43ba0b191
6374f9774fbc2a927b19474ad50547e739e84f0c84c0dab446ef8ba19e5aca73
2375fb37fcea48a336436d021407eb019bf021bca54d5e922dd4b38d966bdf0
57ad1875e510ffa7e5d5f58ca01eec268829fbee33b1d3e8b0d07e9c761023d4
09a00b5a4ed2620656ede8c59f55a8882312831283ed491dcfbd26cb9d8d0120

Fiat-Shamir context

234ee2562e4993a98a8a4451cd5360c6d6418a0adccfb9db1fdb9ba3a0c3a00f

withheld proof term: 2838 bytes • axioms: • kernel replay: yes

ZKP.Stego.dist_setLSB_le_one theorem, Crypto.Stego:36

$\forall (b : \text{Nat}) (x : \text{Bool}),$
 And (LE.le.{0} (ZKP.Stego.setLSB b x) (HAdd.hAdd.{0, 0, 0} b 1))
 (LE.le.{0} b (HAdd.hAdd.{0, 0, 0} (ZKP.Stego.setLSB b x) 1))

SHA-256 of the statement above

fea73a6a9ba4fbf84ff7f2b0751dc8ebf9673cc681197fd3787bd2fdebd0f024

SHA-256 of the kernel type

134d4f83a99a033e8838f74e67b21db2fae9c98f1d64bbe017f8b7e52b8a8

commitment C

d17bf6987e1c814b0062e8029e6f8acdcdeb9056ef96d54a5facf4c8ee2dd595
2799fa528023329d763836accb7be99d56971ddb0b392453d6819cbd97addef2
d2e93ab6d787d7291bcd8331e3681ca1a3b0f9a03a94fb85d844c3fe802428c3
0a676cbb11d8646e5381d1a6bd9fccc99a65ff1fe2651b1212c7ae861ca43d38
c58f84cd38b22a473c1227f273f00ce978b2f16b6566c020dfc17c973145fe65
522959ede5e85d400ad5d6bddbe4500e8a5ab57c961d350dcf406fbc1278c049
6872a24ad925cb3a96a3e35925837fb323d6b2e0818cbac6383690df0f18cb88
c92acbd4191545cae2517da1a791dc218b7823784fb36e8f38857f54e34721bb

first message t

adc9afd3a20318951219cb8f697a57a0d10e78c39b353db24a36db6cfa8e3a4e
619280325f406eabac65db972321d26b52463a34038bbd25b3ac1495f531364e
0dae61a2399c9966cdf84d6b6e2f5cc7b2528bf8088d90e1749f985f3175f347
2ec79225f12ecb940045a55823a87b8f06500a4d8a3f9a93cb085e66c848ada1
2e713bbd7728758427b8d8f8d746da26ab95aea0856d21f0b1ce0088a3f8c651
e79f4e5df626a4d9b1635c3dee1662cd89b2e587301f9b0f295ab3d444355fe4
4cad260c45f1276eb353f3be8b45e5b712dc973ec56d265bb08845a31db5ff53
2458d70328a901b13d3f2d9c619914af42f1c6f7b77bb492f7737fa99aee68a8

response z

dab667952f57388ef31921260546701444f8853181762efa777044b73acbdbe8
81e423953e2249c4c1398210476646a96cf7392d715508730e5ef40e828ac451
1fe478397cddaedd8f60784b2db7fcf0ab31ee87cc8df6541b5e2d7fca03e257
d708f6bd8d530996743717a83daca3a56a75058c1de312431e78da6d75a7bff
cc3f24d22b5b87188a64d287d900426cfee3bececf17d78e477173a214c036b5
8413e327713719b745d0a26ff497245a7088987bf559cd3de38f1ee61ced9668
eb52719fc7d4ef11c6764336b862e96fb2febe8907b1b2ac193262b04bf24452
05cf5e9a0362f6f54b778ec581b4f68507be17168f51d7af3bfe0904faa2c60

response w

3db08830eb612750b984840ea19278224d25679b40db9fa5e8c290bb89185ded
d452bd6fc25cb697b6c54e1cd1141197225a849366c4e97c37fbcbe5410286d
58cc90c185fd3faba8c1e859908d6d3a05316e6ef541be3f0be34dae5295372b
af4288959a7e334b250fc0366727ac93aea42bda344018615b6549242a84c0cd
ab290475adce7b9ac27dc55b74ad8a432c868a88456d2eedddcdc9404b11a0c7
2d1ad28206e83e122e8ec1c5dd4784878c927706b48ba748477a6b02cb5aee69
d06808ef3f3976f7b5fd047115d0fc00d0c9fb03ac93cfff45d34f222092c50a
1932e1079c318cbfc622ec947eae7a54b65b101c28125af2d3603ac987f7f73d

Fiat-Shamir context

61d146c85985243634b289b9c547dc3b036c697020308214049586637abb71cd

withheld proof term: 26 866 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Stego.embed definition, Crypto.Stego:47

List.{0} Nat → List.{0} Bool → List.{0} Nat

SHA-256 of the statement above

a976bc123b858f0fc2c33ba50a162c6f38629cd2cb3adc45718cedf1326f8abc

SHA-256 of the kernel type

e441cda503fde0e373cdb0c23a611fdb855eda1d713a629f9286045bda0beeff
commitment C

a53f0990013e229ceb2222f377bcf8a9827d61646b8c724973d82af7f3a78317
e6234dac07d0bd3c33a62de04860e15cdda1e7db1709eb9dcac424c9a6069403
a1bdfd4541a8b39e762e803371d8fb0a6727b5a8d5f665a5eed72f4f2fbe196f
ede5c5b4a708ed6c4e7c7745ba4494fd6bb4f87e3d520dd86308ffed0467b1
59c9fc8f919d082f755e9622679765f53351c6fad9769ae37ba88e85b540d144
58ada98c62b6894e863af49a9b2136f0b16fb2e85f39a06bab40b10b29582e6b
b297de6f62e663cb9f05f4f419326dd0a489eb824d2860284db6224c1e9c5da1
df7f6f7aea58043384db0e89aa1e9121c961b9de711e5af8188deca983142272
first message t

8a84906d70da4a2d1818c06c06ba6e61b67d63b2156be189d440f81d8e54ad4e
c5691e2a1985fbebfb44e22312c0c7a107d4e9d54734ceb2c43e4041219cb4345
693d321408b92718b695d59799692c34f85cdd23054b9699910421e98a4d6f15
c37cd471668505eff983b15a9618f1a8f853120a8d063b66c8b7c6e2a691ea99
252d4c0bc2bb1361b0b02af57820214ab50bedb863638c3d7e0e0ee0e1d9cc1a
64beebdd0ff91929d577bfe764f17c68214f79a0e5ddada41bf6bcde1665af37
cff489917e98e026ddf20c4eead077bf4a5d2809fe118e37fb186d035b734c6f
736a667084d3013dc731e6e375827664905b64bd3950a4994de4d04e348d60de
response z

15a3384e066e719d3cffeaf8f38fc790f83f96b8811a42ab343f8af1a4f447ea6
d7ea94081be207aa5ae6bdb0efe24d86a78808c0ff1b52abe13f34af024ffec0
13bd54a7c08bcc25afa9f68b5292356cfba5175c93c816bdf8e8c3810d9a0b0c
9c478938bdeff17a1f9c19f1fd9a2d1ed60869cb201ab1dc5f06c8e2e824fc9e
25c572b079d02f9d7cbc66aa1f0ab679ca82eaf9dfb1ea6f91bd255798dbee45
f753026920ef99de7533f417b542631ec9bb6c9cbd2c113569241f561cc41114
98f471a9e8f618baff2a9f00a3a71bf2516cd6a6038b63cd2abde83d7917cdc2
e9c045b9719ea94d5a1c94f770971cfc1684850b5f805d9c1d166d932da4f4bc
response w

1e1a50be187022a27e456733edc10934e10f8ee017125116a173616165db0fbe
9adf4fdad22320331651cb26e7720fcd4c28f43ab4df7718c46f3f279b843467
30050dba3f24fd4233c35ade161fe12220115ade9759818e146c6d16e29bba3a
600648f912ff4912aed43510813b233be61f6a8cbe65416bf380ddf67d6aec18
d9be7d046f92550653bb949eea2a78e53cb9ae9935557f3b5934d02a40783342
01da521e4296048419463f32890de467fac35967f35296649a47dc7a92f6aa41
5bc181e6014638e36c00aefd07e45998de12efa4ef3ba0b63e1cc5f4d6b79c17
74c85e0ff9f052231fbd4eace8d0494b645a320b967055b77830d6c067e61aa5
Fiat-Shamir context

7c70bbb723e226130494f4299779e9e4ad1ca6d06c794e100c86274068a12b85

withheld proof term: 2 462 bytes • axioms: • kernel replay: yes

ZKP.Stego.embed.eq_1 theorem, Crypto.Stego:0

$\forall (x : \text{List}\{0\} \text{ Bool}), \text{Eq}\{1\} (\text{ZKP.Stego.embed List.nil}\{0\} x) \text{List.nil}\{0\}$

SHA-256 of the statement above

e30f986e7ea7c4969865c266a89b41e0299509cd1cb2f2d0fe8df16fbbc779e0

SHA-256 of the kernel type

07a00905dd3ea9e06d72c28a3890e167619ca101507f677fc61bbfcd3480d42b

commitment C

c378ec7897e7eac76f23b3b2ece12d00b284726618013464cbd62157ddc5d85f

20f1d03806a754993612e81571cf773cd357ef23196b98d3cac1fc549e65bc8a

5fe79e606915487e36d76f8c230d313c15913968c151c0f5baaf7d5044bea634

3d5252f3cac4802cb3f97d88aa99856bbc905da67041472e68dcb1e0f1d00207

840a8ce12e41de9c76bfce54723e01e724d9443f527e7bc468291321caf0de5b

48ca9c4067db7e827812e7e11ccd5b90f361af74110be8dbb53a3077b6b08e92

5830eb55d36dcba227d18341a4b9480ca4cf6d6a5a679e59813ec9e385a97d2b

2078980bfafe3e5c06ebf8d953697f866f420b171030073efe9e1b16b75f0998

first message t

999b577e394986ddde0216bd72a5654420d440d054986ab89f65eb39fa6e3099

2809869828ea0d3a7e5f51329be1e132dfb1b817aea757b591f95d5f775400f3

62f5e5c61b8e53b3baf0dfbdda62bb1837fb16676e4308e943eb0a8b78034d66

1bd638c9072649d9df669118af26f70b7889cddf70f51220a5dff5110dc93424

aad89af522cd7e2a693b44292de271996bdb19abe750eb2f2e5a9104fccb79be

dccdac29e20ad60d82de3af6eb80de093e2709a5c03ba99356d111c42732ca5b

92879dd08421b48f5321dd9542d1ffe63d588b56022122e834aed3efcb0c8d38

224a8f3b3e302d770f444c7590e936cc0002a874f3904dc3bb71862fdd9efc15

response z

261c06a46a95db1a7585d2b9c3b416e163406f7803506d83a470c98ad2c47395

b0f7436bad6e8c0957ead2a847b427b2247ae5ee9e2ee9045cd229a589ec276f

66bd7655d22c14ad287e12e11fce8e1d4041d1e507fa0cad5f9244f971676e88

9b4857112cdd6b6820d9e18b72caea926d53c44fbb09243f1d59e82f48f25548

6bc63aa91ce6fa9d138c5090c0d022e268a121c7dbdfdb253a32bdfc22cc1051

6a36557c993e2ed6c64421d2a86266132626bbd6f3d51814197b7cb0666986af

2943f87282a318236926deb4d237f91e615112a48a6fb34541a95874ff02ab63

ea87c3ed523a04c0e6b540080ef73cc907767dbb58c5d45a93c62e6aa0a5192

response w

3c74547730b461110b013a570b9c3f6078ed15d150656492a1e77bb54ed9162f

cfc29ace0fa17453c6abd4954da769b268455f8e49812c21a9c97f21f0221e61

8661a56c3f1ebc5c623d652ab853f5816f6c9d54f0a4b953bf cab813fd59a22a

4230ac0aa7e038aaa0c09ee6b650429d8fb3696c5178c89022b3f8b24e4ac5c0

ae49a090f830a37be0044b87aeca4c80999a5839e7e893d464d472b241ffb54c

d316052d4f4b24730710963ee5f393bce8256b11ea12f712884482d6bc7e72b1

b29de2f11dc6221d8d60dca42fc81ffe1820ba3c3aae24866db91555566f4ea7

5c3c20948f4c6fc5bdf8ee007c25ade3808a927b1fd439e8e931b424591fff23

Fiat-Shamir context

620cad415f1ff2903a0b6e2e87028d53227dfce0ffe20a14fcf920e326c918ee

withheld proof term: 92 bytes • axioms: • kernel replay: yes

ZKP.Stego.embed.eq_2 theorem, Crypto.Stego:0

$\forall (x : \text{List}\{0\} \text{ Nat}), (\text{Eq}\{1\} \ x \ \text{List.nil}\{0\} \rightarrow \text{False}) \rightarrow \text{Eq}\{1\} \ (\text{ZKP.Stego.embed } x \ \text{List.nil}\{0\}) \ x$

SHA-256 of the statement above

5e9c0e5e3afa71359328408ab8390fcb522583c0e87319477e287e2ab1e047b6

SHA-256 of the kernel type

e744324dd96380a725670b8ff569a4a0d69df448e5fa472236b9e178080cd1dc
commitment C

cf9a1289cf29aef14ff9ba6f9d5441fde9cc1bddc53796a5eaf81470072acf40
bae9815b75c8d5b2570a54d3a9ede2dfe0398f8be5a756c5621c8dd4a19d52492
e1882a830e06833d146219b4c3930fcd180ed7a14f63673aea64b04d671e58a2
dc99c190447d5317b318e04cb0f8048f16c271dec5d4e4bcb0a581ad4ddb7590
d950be5507b9d0d7dc91d9c9881ed250b6d3bed37253ce2fe0ff0ed279bb79d2
339cd6eda8062869fb3e823414fb621095b19bbd510db50b865eaf7fbec607e0
654e1ad27710413560736e48630e74502a153c43121bba65003e99229d7facd8
4b179459130ecf9a77559b73abd77f4efd2f4c9af3779d6e4cf89f92dcbcd1f1
first message t

8cc1dbd9ea97bff75d5be6f5c6afd7fe37c1720fa11b9301374bc4e3a56b162a
bef4e9d71cb96250e91511a7a79956a567c91e776fe37823c755c8a2c98538c9
579724b56cdbf483c8d103aa8778af85603d5d4df4e363646b2b76ca8a92faa8
c3c088da919ca98dfd36e30389660894df111d73c6b01c9365a02cb976e1f3ae
988badf09759aa5285ee90b28a386e856af29cbcd24e0dd5ff0be7772360cb00
8cbca0a77b19dcd2960cc674072158f9b8946d4df997418e1638d026845b7b723
04b59ecbc6ce50b742a8d50d71cc7d591a8b66686ebc55932c36d4cd91ef48f1
e07fe665b4bb48dd202def2428bbfbce4b48fa3ffddffa63d79060684c8ac26
response z

3c7d5112d62baa3de2f6099ecce9663a04cdfd14a1525b6315ce6b027de1233a
b118ee68edcdc4b281200d9b378b720a7924465315c4fb20933540bb8a246f0d
890aea92160d0ec84bea7782af59742d782ceb6c51120fad0650bd1ac17ca9a6
1eee409698d4bcf5b480911c61fef855cdccbb1d61df67aaece64af9a62abc8b
18f4fe9509d914d3fd7e38c1aa112526e87e5c09a313bcbcb68c55b6b0f4ec42c
8a419cfb767e00405c793d79c554717a181618af21565a85203b6325cccb33c4
db99ceea6c80643615f5f5dd4ec82b488f5d862321a99c0f4c8db915cc296b9f
f9e137ac7a586af871b0030c5b553b70efca6fa69345717ef7193851b51f1735
response w

e62d05798f3d3abf98b74d3bfc6d45b717a03383fea56d15e9305d21b4dea1db
36f92bc107b787d49540f98658d1cd4befecb4d8b8957ad051e5b0cea5f6ef0d
269bdc3cb6efb54630bf78fd1766015c017d68f842b8d2e13d6a79c2a6b19b8
227b8b8b41eafa49eb64297d2bf1b697761f7e81fd84de713401c2d3e46329d0
bc22ce2110d25f4efae3f512faac58079590a5fa3e377f4c901a45bd28b63c05
ad5cec66ba5af5c8d6ed54dfd60ba157f8e39dead18b3cad1510fbae068486e2
7c688e44cc97026e515278b861a927e5ef3be8dbccf26a2d1aff699a105ef828
db78bda07a7ed4bb5c713a13375517ac3c255ba6efcc1bde1b26ff27ec8d4b8

Fiat-Shamir context

e5ddfa77975a73969ca39d1cf9970624d2f720cceff0f04057d2b797ba9224a06

withheld proof term: 4188 bytes • axioms: • kernel replay: yes

ZKP.Stego.embed.eq_3 theorem, Crypto.Stego:0

$\forall (c : \text{Nat}) (cs : \text{List}\{0\} \text{Nat}) (x_2 : \text{Bool}) (xs : \text{List}\{0\} \text{Bool}),$
Eq.{1} (ZKP.Stego.embed (List.cons.{0} c cs) (List.cons.{0} x_2 xs))
(List.cons.{0} (ZKP.Stego.setLSB c x_2) (ZKP.Stego.embed cs xs))

SHA-256 of the statement above

e8b391177996f255e4d2a072694e9ec5c356dbfe2c83a66cb771ff19bf416943

SHA-256 of the kernel type

faa99e37f5d101585ad55ac5a9213d8c012f1760627518eb1970503ac94fa89e
commitment C

922a7809893b8202067f6bf6fc0cabd943d1800ee43c5ed0641d47e56d817ef8
64a7c3375bc884b488026e7b45f03a1527c64d4b3ecb4355e7cffffe9ec9b6290
0a747400d8f3585adba9a6d66b3efb8785b778315dce63aaa7f52e303d499200
2d0049614f7827cf23b67b823d95ddcf26db1e53bf1f9446c04e24ce59e1b884
1ac5542cde403012b8d29ea21a7d65e0a27d9c6cae22afeb863db54c6a4d82a6
d0c26414a5dc095bcab1b13f955dde186d21d2211f1e66a56f9eb476402e360c
8fe955e16c5721eca0afa619da88f0ab552125a7aeb75afa44522a824500fb34
f2da5c4f5b0733f7b84486c709fc97bdbb5c71dd48541882570b3580a305bc04
first message t

ec166eb401edce9055124eb6f1d4820ce811276a6f69aab36d8d06297231970d
0da2655ba196b04c92ade362abe1bbeb3132e51ac8e56ca3be4c613f85771d53
084d4ccb2c880d82d957e324b02f3bcfc26ce86a1788102cc431cd72c8893db1
72a2fad5e4403c3d005b499198de55e717c597fb797f14b3b38c4b303c5d1ff8
e6df027fdc3b1f5168caf0a67265ef2963503af996a71a89223235a1030e02aa
8274a4eab12d7c65d7843777074c643a0d566397a3a5395677103a65bca392a5
a760d454f3f1c7303e98ee5e0be998e8623d460bff9b6b1d0d9400aedfad9d51
1c60b617ddf00fd4e9e1a7bcf4e71e6ebbf9421b162e976228fb38df871f795
response z

679373a1834a9422a5351bb4fcd317e981e0818db899a7ce0b206aafa5041841
bc56702f507b59037a3b96be1e661715d9d952330ebcacdf3951a119c9762261
83722a18c0b433d18736b70999966f0e3ac46e96638b53321f55d9d6368e09f2
f7f1d3011a24a1b97a228152e0f5d8054938b87d75c4b6e7fab9dc96ad70125b
d8ce3b36d999f1817d260b01fddd9272cb9b5fd51a4c29d73696852cbd720412
a4c7127c0fc14ca624dfaad001807695d6a791d8d1a601213eae96814b025118
7a3e986361fbcdd856a3741e5b92e796d3be5484f0c6e8121290842ad74c364e
84eac96549f97a4935ffd3aa411dd95e1d473476aa774e473d21659deb338a27
response w

549e08d4ee1005d863dd4212bcc452f8eb16257ae7e038308e3c3a909a9ba08a
32fc83bcc72e18cc9df99327c68f2f19b8e39775f9330ebd2add7041f3b6c7f1
c4d6e7351711e9072c2814245d051580c2a5bb547d05cde33004f7b1a4adf306
f8f99111f903dfce0e266846c9ca5e12b86a36a7bf988af09e6631f2f0f22157
42c01f8abdb7dc7ab230956886e55ca2d4222bfe4627219171fbb1a08062c2f
0823721a55a89bc79f3b4f882922e415249d0e74e2f9d7eddd9da1888fff06ae
43597625a1ab2c54511f5f529e755cab9d377b24b3eb3d24eed1592096749d70
be1e0f1ec296e469912868135297a3995d7f3c46c4e7b4a90a10895b5c99bbff

Fiat-Shamir context

ea8e54ccd6d195afb55ac55d30cf7e51019c766067096160751950f85a28a632

withheld proof term: 168 bytes • axioms: • kernel replay: yes

ZKP.Stego.embed.eq_def theorem, Crypto.Stego:0

```
∀ (x : List.{0} Nat) (x_1 : List.{0} Bool),
  Eq.{1} (ZKP.Stego.embed x x_1)
    (match x, x_1 with
      | List.nil.{0}, x => List.nil.{0}
      | cs, List.nil.{0} => cs
      | List.cons.{0} c cs, List.cons.{0} x xs =>
        List.cons.{0} (ZKP.Stego.setLSB c x) (ZKP.Stego.embed cs xs))
```

SHA-256 of the statement above

2e6f752ef6015d076e5ec6251489da83695e334a170c47a9ca380609ae79e6bd

SHA-256 of the kernel type

17c89c970c16eb0b9a2ce18fc7cd5d3be9cac1f89c43fad0d57d76fa63e2f8d0

commitment C

617e6d62a9b59564337feab868de91d8baddf5c680deb65b50eb70e10f48bf76
f7943c336da3fecb172b8d46310af67f267e8e0870871eea82793be67ecec4c5
ffedf7acc0cc9147744bb68ef82e70a82932baf6d2888a76cf08be44d615ff7f
eeb55eaabb0ebcd1622a0f9f3e2c509877fa5449410a058826f86fd3880ede7b
b1133244f3b47c14d259c0145d9ee1bae6a3ced1945ea4505d9a97df6ba07e87
48e8a89b8d2e82bd238bd082f8036949b24ddd43589e6637b85a0cfc82743c29
d3267377f3a1f1b02d20e801771ea1e818d94913e15afe1173073da62d80e15d
562540691bde863a1a24a824859dd96603b9275fa1d21853bb85a4d3509a1a7e
first message t

dc0cadd64cedd639c8a3edd978460a236f3a18796120ded63f84e1b43909b936
81d0240ffc2d663b43f0b1a1e6e28f7e5b873bed1e48e39550dcb4085b62160c
c7a5a096fae5c3abf0a708251b58b5142213abd4745d973c3a4b52f6163d4815
05338fedcaea03c3aa52bdd7100fca7f2e7716b072d109da759d54868f693380
33e308a909d7ec9eb5c16b3710817ce68e916ba606043672f2dab5d0c77fa87d
bb2a7a11efe6b76a25c87c0d09b0aef51c7a72cb0983e4f990de303cd99a44dd
f31e68ce5d72b9b5687d6b916d738447df58064527dcefe26c2eae99628feb1
3156503d4c8d8782b0b17daeb2080f44b20beb6bb69be22e7bd1cae782eda646
response z

92bcc0e4b67c95f69f7af58a001fc399be09757bdba3be9fe857f34467e49329
a197c89ccb21533f8fc9b7c2b5b1956fd16cf4bdaa82f325d4de80272eecbfcd
1d633ef56a84b1ab4677a8d933e51404b77b01b9c311d5e7ebd0b29f79289435
c54c18010140b7782d96c788711e7d523477e95f32c1f2b1da9260428ea4f22d
fa67ea319019723a4b0a1c0391326500ec9768d57e01f3be04d1aa6f6c14a88a
dd58ab9ddbd96dece53ae39bcdd21dc12a8c85231deca0b94a07c1c21c612320
f5c5fe3f4f63e9b9a0d663fec0d886dcaa3408f41e2a2686d716fe1c690d09a2
98a708528898f6f2218f14c61d9f176822f9d29d7555d380e536d13a9b5fb22
response w

705dbc4c0f4240c99ec8b4147937073f0dbdc74da8e9b05006a1eacd806ffc7a
2f8cd0002df0fb11ae1a47c50266bde93c7fe65d75ba1a8e1568c057ec60739f
8bc940058c0ffc0f43a9f45141aeb34976aba4ac1d98f08ab6195f064758c7e8
0e420d97d01ac52e4e2d36a7d587616e95f469820a6b3606559f3dd179c9046b
fb9c49c1226c369292497badd1c2f48aa14124a4f14fca0c938d6eed38b7ec12
076bad7c94cdd38a54729d8dbd988ecc553fd15064d222eb0fc9fb71d4f1cd6
2c38826bc6f42bbec217a64f7a531f4ffbf72a763da92bcd0e4f5b8b5451269
dd69e841f802f095654f37e966e377832f9399abbdee903821648129a21fddcd0

Fiat-Shamir context

0d409f9c764b8bdd0e4ef45a9d7f5baf651ea64fc0c58ec6287cae868e56734b

withheld proof term: 77 960 bytes • axioms: • kernel replay: yes

ZKP.Stego.embed.match_1 definition, Crypto.Stego:0

```
(motive : List.{0} Nat → List.{0} Bool → Sort u_1) →
  (x : List.{0} Nat) →
    (x_1 : List.{0} Bool) →
      ((x : List.{0} Bool) → motive List.nil.{0} x) →
        ((cs : List.{0} Nat) → motive cs List.nil.{0}) →
          ((c : Nat) →
            (cs : List.{0} Nat) →
              (x : Bool) →
                (xs : List.{0} Bool) → motive (List.cons.{0} c cs) (List.cons.{0} x xs)) →
                motive x x_1
```

SHA-256 of the statement above

1681728292b183da6bfce7f99b35036f714d3f21f9d52ba0ccaa432862254c7a

SHA-256 of the kernel type

ae967885792fb4d25de0dde9eccc1c29be413916a018c03d31606cf574f6b308

commitment C

6ad5b295bf211bd70bd7fadf01565288b5fea7011cef7b6644ac79e9ea32f133

6e7cdc8a25432d95af76d558943533a88e401cdc6f93805a299c388f80c1fb44

7db909feeff7d1a79aacc07cb7b79b9e260e4bd40cc23af2f3ed9b3bfccf71e6

2eac61ec5b136076dc87509f4cfce139e22ce551ee9a42afa212378c1e0fb857

2374ab8d56acee92cc85896a3380bb0ee06891342280b2bd075e3a7848eaac83

62c61ad5b49b8e044d1c97ea22dfd2620a1ad45e3852b70b2bf0ff30024b8919

e0ed8a9fb74faf40efc0a5fb66259aa37159e2eb6291ad2da4e1d406632640a8

441c49c6eb02c16cc44480c7c5c824d9c11d625f0ed2ea684e6561dc4d986dc0

first message t

ea90caa08eafb5d04e3db74ab78f4e2e42ffcf3b94b331b8435e0851d88d841c

bf0277b9397ed5b01c9351da6e9373b3d60ec980e3aeab40cbf80bff1e659845

6eb683925e9c1375d7227e504bbd1720db4fd34dedba0015461ca134d2c105cc

e134a844cab0fd0eb024b5f19e2a03fb785d351011ebef9fd050595e5495c641

9ab2fa6cda7a3a2e3ca13922f8a0b264d797442977e507fac4ea81e221f12b16

46451df5f48b0b3eef60920b5eab26650b57b34f01943c868b2a638b81307415

2966dd2ebc50f78f7aa6d0c8f8a93903336ae4df68c02e28ef07d054c7349352

94b51aa2ac9504ef945ef4e5db2274fefe573d8765f8110078296ed8cf40478d

response z

14d1cdb4711ef0af069dc29eea35abf4c8eaeca50ad4623405114efc24672031

0ffb329e5ae6c9738cea64f85aa4c0e8e203d2b0c3e6f652904a79c05beff04c

e51eb951223237478ed097cc5ad0baa2265e7c3fc38528fbff294478a2af7a34

bf2fcb3e246e49851eee09dedd351457bbf605b2b5f18eb92c1559cad7118565

632bc322823d4fcc532b0e9f9ce4db2edc11ab7d9b5f232fe44d0644136ccc31

87f4f10acaf15befd0f96031ac3c0ba1def2c7030fb210814d5dbecfa9e5c6d5

3a6c404c9322317dd3fcd40ace4b30583f4d5ee3a9c308e65769f2c764c799dc

0f3cfc3270f86aabdb8e28560de7fd7a1a52b87515d66eaa8692f01d0ec532944

response w

395369218c1b6e194d3dd85fcb0eb74cebcb7b127eceab0181618c250ff2380a7

2e4b4cb00f92130ce0ff8b869ef370efc4f16e787a24a081a600b46a1ffa1833

cc9fb12484125515c992562a7adedeaae84a1a6544bbc880405b4e3e6eab3253

8d4f7c8a2e6bfd8476f179fb5791ac97bc47a35cf88b2dd96aee70d2f90625d7

334ff8b44413121375e4d034ec1dda6efb75b5d463a05052f283123c297759db

6f833ee4a21f7fb7675751d48a97489d12fb7d9a1ef62cab3e6e1d959a76ae70

12b1e54828afa9479b7c22ac0e8fb108cd7c956a9d76159d8a2fbde7fdde629e

be6d1188c1345f953e6b040d19adfb9cabfb4b48f61c7fbb585dc5330a2c0b1e

Fiat-Shamir context

b0bf66dac3b4180c2f7bc348196689bed867ac7671ee9dbc71a4d53639693c23

withheld proof term: 2152 bytes • axioms: • kernel replay: yes

ZKP.Stego.extract definition, Crypto.Stego:53

$\text{Nat} \rightarrow \text{List}\{0\}$ $\text{Nat} \rightarrow \text{List}\{0\}$ Bool

SHA-256 of the statement above

a5f55d493173d4fe4550a36dd717060eb389ae71fbac735707bfe25ded820858

SHA-256 of the kernel type

bbd14b0f79dbfe6d952675630a0d1ae4f669587b46e9aa191bbdec0d49e48feb
commitment C

d78b07ba9f643cfa22d94cbb56fcdff722f2d5d82f321e1c96d58659d9ac5b2f
2f961ab625b3e0090cc23128470a6c03acb380a9e13ca856a3f815197a51875e
36ba6f978512177f0e24eba04db199bfee057f5ce1cf83ab229ec9cd25efb1e7
697f75f8ca5717c7907cd028d954e4fd3b92574a24ebf1eb889ff8ff2d1d0d43
dbba09a0d4a080b54a486011cf66ca52558f2961578cb517ca0a88fdc567c086
f8d80335dac51ffb51429e6441a904e991cf715bce845dc010d1ad70f6b72c8
39b3976cf69c5619d3c6d2df2fae0e4367754792aba56f2c8d0fc53aede3ac6e
a4db76695f1dc4cced759ac576d98a1e4207c032cce7b0fc09752038e796ed12
first message t

e3cb89279187880b25d6eebabee072687f280fd4f0eea961bffb04ca5598458
3d275fe0fdda14f8fb4fc3bcb671a1cc832a77c05b7565623a73bca5a5d8863f
ff858eb4a55c17448b50df072c9b8ffa485c6e6d065f225793d78ce4db44cd6a
ce73e274b6e8136c9c611f1bd9d1e65bee9e08312ce1e77485787c0871ca2547
1af373ffe3b7e0bf35c27d1fb9fedfcc88f982ab38c3c7dea90bdf0887b86a95
a9ec0e69e8c801d09d885d55ea44c4f152afea3ce3a74c35aedcd135f9ff8cd0
d1695888bd46e9be8a463a603ea9ac3f26500c2f8e4644a4a21efa4d8008cc19
bedb3ea334c62e9fbd0c676d220492fcb688d34995790260e740c6bfead2bd2e
response z

74510fb2048ba4a2327789f4588d4edb4ce805247dbbb3279b03cae6e7f69725
3c1216c1b87d627f31f07308a632ad5831a26df0843203d54f4b262193ddad03
1eec6d5a9eac5755afacc32479d24ff7b7fcc8160e9fbd3f147699828890017
d4e1a25494ee8cbdc12abc6520b0556dc16b108b4590ae526b867d5030485aa
7bb1cd2f572231d0e75f48f86e7e0b78889114a23c7caf9b3cae874bc88b5aef
fc8bb9027172062d909437bad32cc30c130de2af4e484e3ceae0e1eabfb806ae
3b3fc4831d93448ef1bfdd34e2a3d67dabfe1617200a7f9c8baf16b327e6515d
7542641451d2bf7ba37e37fc2586601ca7e7f5fe54b459d0701476ff64013911
response w

1687c4c4e0889d30a1efafb90a9b61b32ffc8a57db9b84122e3e99eddb213f4b
70a26b9cac365b1f7232ea8f7ae5922c71e6ba913104e18ebf5308acb70e7fd0
29b270467355fa3fea3b410effe01b1370f8c05679dbe9588e6dfb5c7006feef
045fcc27ec412bac805b922a6b295e4f5234dc1b0f5bf88a9b88f893c29e988b
1ddd78e6e5f00e64ea33366a8c004bad46049bafbcf473d864e9bc5987621cc8
1c11326629c4200f91ea9e4888f3693ac1b548a614a9ecf72bd313934bbb1a9a
0e69f38b7540acc4779c2c4d71257dafed651b316f217b54d75a1ca59fd2dc2a
8e95c520de27df10bfa9ad5af351498d6618884818c888d28f46496e6581f50a

Fiat-Shamir context

8a7e56270cc2afe58491de49d478a8f0dd6a8ef9da7b35240d773b03cc9c3db5

withheld proof term: 2 438 bytes • axioms: • kernel replay: yes

ZKP.Stego.extract.eq_1 theorem, Crypto.Stego:0

$\forall (x : \text{List}\{0\} \text{ Nat}), \text{Eq}\{1\} (\text{ZKP.Stego.extract } 0 \ x) \text{List.nil}\{0\}$

SHA-256 of the statement above

16d72dc141d37efa05d114301ea0d5e2c699ad2b851586e2af36b1b386f1ba0f

SHA-256 of the kernel type

68e310922c4621eba1d2d7124c1e55137697abfeff807890a78c6449103fc1f3
commitment C

c953f69afaf5765fa2ecd05d99bbccb27ce262ce94432f08f554edae500794d8
6a36114f4d6148596b3eacaa807dfe5abaa392d206ef20b5d113bb14b3cbce45
4097ae2cf5c3c6c00f9ff8ea4b659dfd08c1e466f024e4559eee4e79add3689f
07fbd772c69cbc155bbc337d9ab3ba0942240d39bb3935cc3708e86cc3559883
54c307634c12884c0bdc0e661591452ce019d3fc6945e41cddb8237888170c96
e5b9320c8d2f8f321cf0cd521cce94ec0acb75b78e0ce79908556a5c9a2f7565
5fe24a8a11b8a33c7f7fa437544df717df628a4a25dd769aff57575eb0cc4db4
60abad5ad98a60b2161ffd21d50b4d999518faaddb658c1f9552589194afcfc80
first message t

afd8a298ffe1b1246b28b87e586861a153ccc0baf2c245f5dcf7a42d3b1ae817
8500525f6cb1160fcf7c31770621c120fac0e37279d0d3425da7567b824c52f4
a0a5f5e5e0a46b43988a445d3a2a8bc1e686d090cbda480abb3d48fed3650a15
8866fa2deb73f71a3456aba08ee8b11402883ef775206cae3be9ff5a0b5d4cb3
113de82fcec6f99b32409112f4360ae6b036eac1428492a8f8f52235d4e87a56
92b558ace00b50f6ad68146ad1f7e33a02eb667e6f3dc3963c5ba9a36748c1d0
7a4986aee726dee95f00191ebf4a76c8dbde8adb026db8eb5e0766221f940adf
60e3eb9f416c93a978a73ae76d04a2304fdff29a13420d3e426568373fb1489d
response z

2c13a5e17ef8f61d13c3deccdd387de76a9d3fc216c677657e53c78099e0f949
1c5468fd1209c93964f363202cd21660e047ebcceaef296ad452e2bebad551a2
361f6a50dcfae6dc94f290725f256717a1773b6c84b33a03fd2c608365147aaf
67ad750000753c1c7486461076a99f2786bc5e797b53a5601680b5ed04a60b67
f26a27d4b80d025ce5a48d18964a8ecca543e307a001ddd307a615df0f4f9b8b
04d7f66bf91b31ad516167aae7d2ce0db0e5496a5685ef49ec1003cc2966e0d5
a4e3b5125ae4dbf2c5136e504f0456f9ac2f4938d6de04bd1d5bfda8291df02c
2ba61bfe1b442861c1258611038860d76f380c7c36923c99ab1f79345f6fc156
response w

3ae2f5157013ed6a371464d20b5470a65335556d9fb372ba56f4b127e05f5793
882524864724741774a9cf1c7c8eb292750437e40ab38211fa137ae3c56a504e
ecf1e825ffbdcb80863049a6a818331f0338d2fef8dcfe0a4aa4b1c08a79208d
00907f8f7952c59b60b5215ba7ba375cca76f5b720e0feb6042a81daf8d78314
f7e8495c935b87b7db09c2f749ab730daf1273ead289ab0519b91215f6d86896
32af7562f5ea324662bdf2ca01d1690e823be54f6cad23bb86e43999608ba4bf
5f2e1d38ada10fd12f663d4c310c98e5cfb3d9d43f042a723cf1cd53fddce667
731831bdf5e255c670c1a256652f157b50d4cfa0d7d39814f6da288d78987c76

Fiat-Shamir context

9a9a50709c10cff921f138e97addfb41d0701af4b542a709c302dcfdda2d68fa

withheld proof term: 116 bytes • axioms: • kernel replay: yes

ZKP.Stego.extract.eq_2 theorem, Crypto.Stego:0

$\forall (n : \text{Nat}), \text{Eq.}\{1\} (\text{ZKP.Stego.extract } (\text{Nat.succ } n) \text{ List.nil.}\{0\}) \text{ List.nil.}\{0\}$

SHA-256 of the statement above

52f8bea934fa8d34eb76a803358f1d03eb29a22fe804ddac1186dfd22288c05f

SHA-256 of the kernel type

6162adc21c34c6b2d9c6a593362d004fec29277b1c86a5ea91714c8f2049b8dd
commitment C

6b315b5e7279a4c9bf9b5a0546e4cd288f19b2253b178eec81d1b9ec3c090e98
8d14fd9600c97193d4007112f1f38244f1b3b455dd99a2380c6cd08ee5d1049f
ee5c583cf04f0bb9ce2ac34decfb8d58309c0035d2719792491d251826b5e936
ff39fc58cf28e9d91bf15520cfe42c087cc60fd2356684e85aa745a396c9b2b7
02bcf59ef83a67c9fe9a3c0ca931573bec03913e0a6051f84134b4c3ecd5abd
db2a0323f11fef1f27157d2fa33d9eb2a2c89b598b5da8728c883b2f29427c0a
31b3145eb99895e20cb92e12fdd63711a4f7ad744a42962e61e6d451f1a42536
742e8ca51f0285c386b515cd0d9e9c8214044c4db7f704300bf805832ca67932
first message t

e8f4c435ceb7100c8b20bc6984effb7b861f0f90dd8c034762f041add80f0477
c1be5600d337dc5ac4ab35cf557e0dabba0c8085572c14261bdbfb108ba421ac
2b16e8355a2e2281950a2c4827e4b2afd7fc2baa2d1ed699712cad14a3f6337a
422e02f672dc1ff63efbeaae87d00fe5b6b5a28c82278cdade575762e6e39c19
6bc9a0411da67cc8f258594fafa2d98ffa3a82000f86b1ee4c68dbaf389e17de
6365b5d3781279ceccce79f7724837a6d380cff0662b382b4c3f5e29dda8a8796
614c6ca28bf18a465c65b7c2dac956cdf5e69bc4980ba0888ae3a6a45e85483e
929c27b823228b7931936f913500e423a22e992f38a8c90747d9961713a51d48
response z

1e40400a66cdf0818b9a9eae7513f197f5de0c6154212f17ed79fff453a51bb6
aa792ac72b455f8eb433df10ee0ec6c74863433e2cd6ca298f838c26f71b6bc7
7381c79f5121df13bdf43575c0c82967f9166dbba6b02b3ea22d5d0ac9b1c175
9b73c72934d8c6edaca53d13ea45f670adbaf9f5fffb99c0eabb7fa5e01a95c0
76e4301ba0d0d538844869ac0dd6750b01f1a9f6c7f2a339af0720554382d2db
c4f3d5e9c70fe28dcf2362536cba53d5ac0ae7e16259272f07ab49898c536fb6
2904573b6501b74b2932ac8f6118fed1be1514acdfc3dd56a07fb446b24910e2
1c5b589818ca4efb36e36cb64568318f5694f2dd7684324f3ad19f90001277b0
response w

4e59d475ff07e99e6347473fdf4d0cb654444ecb91d0ecc2b6a54192a3d95874
852ea7f0a4c74475e7c4a6d7f904475da8aef6aed1405488412b5baecb5266d1
3ff84ec0520488bff1a039248bd95902ece21de970e21bdc4d2adada3562a297
cd9f6554bb615a179813614a9c8ee58090ac38f6e2a78018ba183432cef28097
9b9233bf914bfaf9b6d07b258fe292b4c3cf4d75098af8b8e3fefdefff51ffb2
15c57bae1350084dd3755697c22181fe2cf6f859d27befe0014a981235971b08
3fd8acc44377c1541de631074415e919e71320b6618fffc761b8101d7c2bfe55
eeeb4ccaa5ab81ff4e0d2d468a4cd80745461ca40ede7d0fdaccdd6ca5857469

Fiat-Shamir context

c12d9f8dcb6db358210a1c6ed8ba3a8ec979ff0ac9de212138e8978d00b3f537

withheld proof term: 96 bytes • axioms: • kernel replay: yes

ZKP.Stego.extract.eq_3 theorem, Crypto.Stego:0

$\forall (k\ c : \text{Nat}) (cs : \text{List}\{0\}\ \text{Nat}),$
Eq.{1} (ZKP.Stego.extract (Nat.succ k) (List.cons.{0} c cs))
(List.cons.{0} (ZKP.Stego.getLSB c) (ZKP.Stego.extract k cs))

SHA-256 of the statement above

43690652ffe753435758667741caeb19ba6902c96e155d0c823c7885f75ce08a

SHA-256 of the kernel type

a235ecd003142d82370d2cf29c1d319a593f8af06b07f4bd2b32be658bd63de9
commitment C

2a36b4816511e782264d488f508c618e98ff5d212980bbe11d6d02030e908585
187f121553e847320d0f15232d638c743a93e14337d20149779d54774f108075
f0fbf42ddb1f0e020774a19d4cf0443732e25b0239dc714167e72349572ce121
57d6fa5381083ee10a25e8d423ec616935278c10b3ec98bde708a5705a6da313
859d3685e609ed67101e7bc2d2d37b3621405970f9f130ef531a1a8455f4062c
a42c7f06b0d6b33a74424a712b2c7774dc6b52b72cea65665d6d092278eb0bc1
f8e173817943c1df0277d4d327412c835197de92c547f45d575ce03d727ca555
6f05a0bab173b6134b967f67634eb6673a56a39c54ace5360a83516a6bce9c4d
first message t

4dda12ef5f11982cf6892d3922a7cc57a3f18c3e6bddea15c3b0638a71f74b81
72930565dc89f1fb82cc0a68d337404d295ade6d9307d06ca9748e8ee8846a14
47d2163c08158f51a0ff6bd2f6b6bcb8fd314e8de8565be997a10a3268f1621
2b404132370b04a322604d4919dcea1e2a08b20212cf8c509362dee0ccd242c
8ef5fcaee4580beca283dafa2bee0c04d70cfa346bf79ba7e2e8c5cb461931a4
15c157b698271774b4f12ea1d4dad37486faa56f5d2b88e87095f8701bed39a9
1b03e99f736e9770b581fd46286139665a8f5a7bc48824780018d6ac9ae6f1af
bb6a4d8aaef7f6aea1fc57c8320720a1180be1804e1ddaae72a55f58114df136
response z

3e7fa307c0053608989ff2fa0192b1872cb564ae8893e95936f00b0dd2991b25
2f5f8c4371c66be1e364a83fdaf6393e8caa9b37a16a2904551cfa1a900c3967
ae956f944ff7a2b1bf8dc1aa8ee6b43e51f9283ef7232f59d39e04d0038f297e
194fe3ccdf11b15384090d0045ce32af7ffa59460cf9816aa39fc03c5f9583cc
206d0aac16478894fecb3618d045afe902120acd0f3b15407c0fe5d431e72fdd
e2b84a34f50ef62879b9ed1f86b1452516409af08e996d934564bd056359c7ad
55603adf8ef26c3e0954b2dfdc84073a4e4824fdc7648351a20e363ad0d5a9fe
2b296600433dfd764ee74d53fa8dc4b8f45e87dbe2f45ba912c50be46ea17d3b
response w

1ca41d050cc0f94f92095d2ed769da7f790d0478c7bc8dea5655972b46782186
951837c3b8e5208219ee481a4fedf99742deab9f003c62077a6059c0b1971cbe
186dfa8c8119ef1bb8d419cca4fa3a2b3f163c4cb02ab4c2884b96c3d33d81de
a9ab68c17576d4d21bc4fe686e8fb19bc22537998dbec4f43b405b7b21b2d7cd
0896d97ed96ccbed185e2d3bcabd4ce3d196184da7f33d6f233a8032978fed36
093dcacd8231b86f0cec731fb7fbeb1bbe768d743bc7277a81868e1f39c406c
a0c4fbd397a9c3766ccc6c17ba9cc01508f76ce0d598c85e844487a56ab7d5ea
3b58cdb62f5d90e00f61161ed8d718da63dbd6fa36b92eec7a9818453ab820aa

Fiat-Shamir context

0be39ebd60031e5dc06259ce4695abdf1cb97cdeeb7aeb1c4e0c861f2483ab94

withheld proof term: 132 bytes • axioms: • kernel replay: yes

ZKP.Stego.extract.eq_def theorem, Crypto.Stego:0

```
∀ (x : Nat) (x_1 : List.{0} Nat),
  Eq.{1} (ZKP.Stego.extract x x_1)
    (match x, x_1 with
      | 0, x => List.nil.{0}
      | Nat.succ n, List.nil.{0} => List.nil.{0}
      | Nat.succ k, List.cons.{0} c cs => List.cons.{0} (ZKP.Stego.getLSB c)
        (ZKP.Stego.extract k cs))
```

SHA-256 of the statement above

a1ec95b19e1e8d36b21ca8ad112839416ae38fd9b44d2c7ea55d3504b13977b5

SHA-256 of the kernel type

34a2aab92f3e767f2fb262fab44bb9fde3f5482c25a7bbf8f12352c02a99e21c

commitment C

156ee13befc9524642ad824b06dd3df94089450ce185bbef8cf83e8b17b65970
afbd390456b1009cef6750734b9148c063f3aa202e986e5ed4182f51a5814a8c
ad0a1b12102aa5439b6ee254d97930eeb787dfb666d655d0b9da25938063493c
403b67a149d273e20bf1f69f88516018f9c1a90a55e82bbf959f2860d87a48c2
d5cd1179db326ab61e6bff736798f593dc2185539e1da5298da4eb0f784d104a
b4e226efbeae785e6a3d2f460da47e94991942de9717e642c7c88c36e8f24e96
78693e0028816e06267cdc4310d3cc08993d7541dd1aef151c69bbe30d747e0f
a060a52821e825b55b145b3ae97ee3871c8523202aee44e2863ea5b3b2d194a6
first message t

fb8792a8b2b81d18d036f722e90ddaf95b6a616417610b9ca8daf51306af088a
1f529321444b58c402f6294b7a63339a97da8e8b274b813299d3e5ec701519c5
ea5b0a1ed9e45e3671251bfc50db426cf344fba8884c8a975cbce7f7adf702218
8ecaccf9185601fa770582df143d28967b38f810168230717756a1061f995838
68bc5e0c8ee989fa6ba9068a452594edbca1f6856a4f20bd7e4485491e9ec133
50a76a7de7f589683a104ca2a9cd1089faa97431dead9da9d0a0629049c0ce75
6321a9d31e867a812f4e851bd33508e562cb33392ee597d7ee903e14bdf5b586
62fbadbce80b2d445e1c6bbe2425bfb3c7335e522c31fe9f46c4f00695bec8d
response z

7fab23eaa6bdecc734bbe5872a6b475ff5cea2a1bffce52fdeab47565b0498c8
3c6ef2df6634daf2a56ec30448abc3a49bc2a82b232e668b00e2fd17ecfdb783
2381bb286247f3a617ea937aea03e8da95580abeaa16386747e4f9a3d92abf9d
2242ff57a925dfc39b9a78a089d051d74d74e3a2a03518971fc21fe1b412c21f
952bd5656b40638ca60a6ddcf2c23d971d53eb5a3021b37b57f96709e8ac7880
bad9f73f090025ea299174b6e525e73b58f36de43ebe7182c4e588a56dd202c5
cf3fb427a5d23101aef50117ffe79285a3fb3a1e1d90288c8b69eea535b4e636
7d3b75b0d43ad92499b3f5be32a6556db32002c72e33e04de7160a137122933a
response w

7313b3030a64bb736b172fbfd7951a43673ca19ad7550942020224bc2458b309
be69972e0a032a34a17ec3fcc7070c5cd4e899b63c79c15146d273eb292e2d20
e264e284daa0148e03530b3922723a34a7a65c99cda24aac855d7997736abbc9
292888b6b8491f64d49725566558cfb53ff631910333eff29af7c613cd998f77
4b6a1f37dd8fb6ed5860cfed83d7e80a402689aa53abc9c4c7760e33c57256b9
4f1f9f883cfabc9db890dae88a4138e367874eda874c8ea396b9b82c1410572a
e06d45190767ff9c4d503ada1e78670184cb4c1cf5c9925512732faaf044720b
f93773721033ea273a9cd7445bf10fa438e531335641f51010105f7ae703589b

Fiat-Shamir context

8e7f6ec5e93edb5d5ac72c4b6c55c2733efc43bade07610e24e5e6d79c1e64d3

withheld proof term: 70 692 bytes • axioms: • kernel replay: yes

ZKP.Stego.extract.match_1 definition, Crypto.Stego:0

```
(motive : Nat → List.{0} Nat → Sort u_1) →
(x : Nat) →
(x_1 : List.{0} Nat) →
((x : List.{0} Nat) → motive 0 x) →
((n : Nat) → motive (Nat.succ n) List.nil.{0}) →
((k c : Nat) → (cs : List.{0} Nat) → motive (Nat.succ k) (List.cons.{0} c cs)) →
motive x x_1
```

SHA-256 of the statement above

7a1d41326cc1445e1e9cd204af9424fdd6a535b71726c6f6a00234473bd05b14

SHA-256 of the kernel type

0b7991a322d9b624ffe8a7d8752e84ad41f945ad2fc7a21b3869ced0532ed88d

commitment C

40405ce2dd4654c43a098cd33227d58ad78db98181a043846cc9b7318322f9de
ffc1c9c05356480827a9f1d818671edcec6b45aef4ee57d48cef91ec16b6adb0
7191f289166941210a75c5068f4daf67ef550aa78aae97b2a1fdc7696f55959a
b83c8ccf3b028d72bab0f8010da3bd985d01957ec77d7e7f5806924a83110613
7dbc9964f390fe7dca155176bc00e4e85d37cbbaa8bed1ccf1ff57fabe34005d
08e72566ef8f35e3fa590e999466ce05e2810d24638d51c7d68305ca9b592925
2622ecc54632408625b03348fcb8c18d385704e51f32c6fa78d5e809daa5a71
d04725b595a197146874b42e17ab47245dd524f0746d14373952578b8ac92853
first message t

4a20466de27f6fe271bd58c05aa109196e866da2bfce8608593e5e3c9f982b4e
cf4b2b83c0756a3a4a02fd52af5f25672aad616e26b29b48372c87fb7f9a9840
63b25736ec369c41e192db46733e87dca0b931d188be2dd6cfc97898ec94b283
c46b0004eb28a8aea1ae85949fc46565fff90ec52295b3496ac58a281f4f97df
6be5f2fa8ee6a00b84719e385a33977d055db0af32f49bd1f5b147dc0e46b792
be8b2965b6ee78dfe4cb3d6771218e408974058989838336c808efd40abfeda7
6489506a8c653ca18f8b67a9f1f920adadef48ee1bcd2a0292324197ae1bd05
bdb52a620389454f1e7f3bcef30455efee8f59fab3f40317cb658cc5cda65f75
response z

1a94c410edf3f819d641d718672d43b47c9315a373d29c5f89bcb0e0162348bb
444b0d11697b47b0b96f12a40e033a68c086cf577f5a20e42a370d85d982f150
54c3e9c9c1d989e95be19e64f23fb879a583cbaf0e16082dd272d4bed80a67c9
bedfc4be490cb83fb65c3a27d0275365410dae29532b2ed377165ca3184cd027
80a408e307877d1585a2c892150e2e9e651e22f1992b2bec41b9f7e963ea17cb
cd2e2e65b95eb04497500c22acade2651aec490f3ed0d0369806968e855da9fd
a773c8bb9d5dd150e67a5a60706fd5dcff0c0d2b43ec8b900b3240a9807bc787
bc9d04721d57ca8ec35792982b7dc06aeecd23d4b12b077bfff8dda90814d6c0
response w

3ee998ca24d1902474971dcb75e19edecbe7b67aef521be8a645c5ff382ea2cc
b627f423fe6de8c662faf7810cd39859acb66f2c2699e357f6a6d6d786b333e9
43308e036caf884376b6240c82c8784f7302940d1f63bdaf6a91edc36178537d
70a0081ceecd81463d7b9df69c0c85b11a7a1aed22b2c77815d74a2e529cb966
734f8bf0f8170f41a4d05800607904ab72eb90c0c5123c1d9a04758d79bb571b
8359dbd6aae446328b9a8b346a5435a2a92e721262eac5038513d222e573c697
ae8a6cd24f1a27fd52314b0329315b534dbc0c1266e12e885266f15854042fba
cfee6dd039ed9f2da82b630dde1c8b3d0e9140c5c9ca47a11c079657388b1d24

Fiat-Shamir context

2e4d351c086bd58912dd2ac9b9a56afc0081fbde36242776e2146a80aba57553

withheld proof term: 1898 bytes • axioms: • kernel replay: yes

ZKP.Stego.extract_embed theorem, Crypto.Stego:69

$\forall (cs : \text{List}\{0\} \text{ Nat}) (xs : \text{List}\{0\} \text{ Bool}),$
LE.le. $\{0\}$ (List.length. $\{0\}$ xs) (List.length. $\{0\}$ cs) $\rightarrow$
Eq. $\{1\}$ (ZKP.Stego.extract (List.length. $\{0\}$ xs) (ZKP.Stego.embed cs xs)) xs

SHA-256 of the statement above

53e4d0e9079c8e1de48dfd6313023869d67c6961baea4d73a346f2484cf2397e

SHA-256 of the kernel type

08eceb99a0a6c4630078061e4ce26c0495e2470d784bd1ee639c3fdef11a66af
commitment C

ff163993cea4aa97ef3ee3dd1209f67c2e4b346d3192e7e4db7e59cba7056464
bd8f8ead065d3463942eda131e600e7691d2874592ff6f6e56ca9d4f5705af36
12fc32dff32abf083ac11d69e281fc47b015e8c7e5a8166221cad1cead33690e
394d95ef643ab46dfca029595cdfs63e0006dc3db089c13b7ea019b1050e85df
ba23ef4fc55d9d82c8ad9eff26b4b7bfa4d6f58b5e9216c2e86e89c7c9d39ded
fbf34c11e17ec85c488d9bb1025c9a07c747dd8b90269b6a883c9335b496347e
337b9b9f0eaed54299a8fd87845ac77e09510eb8ba63c81b0ec7450e1a10581a
a1ccc0f75a365a5974611f287d20fc79dabb8e0a6452a7120c97ab18498bee2e
first message t

d0e2472a6f624d9d86f02c8a49c6d7c81400feca677ff7ba3dc1f2cde3872bca
6d8af6c4b6dde6923c394eb31ec0ea7974e3ce2148cd2f463fd9a31e5165996
21854d07386a79cf0d20d307bc86380512ffdb3248319e830e93e4d6563999f1
758c59d1ccc92d35e6a625d85880276532f5439c8b64e55f4c97f4c57026be2c
943562fb188544488f1ce28c4e000f8ba13317a0a7801db8602bc7f0adf615cf
d8af3cc5b49e3a878bdfc7c22b055737c27711995907d4efadb651ff4072e92e
1416276254db93b3eb53800fae6c4512688eb117c81fb0a8fc3ef6aa0cd5ef28
579e49c3e8551e5f91900156a2d1165a626116917137490a8a161d2e9ac0bcf9
response z

55a9bc6b354ce3418988e168838436d20aaf6ec4c1dbe801bf79ef18b17d332d
8fce85e3734cf568b8e9acea664f5e296d859d4820181060df1aaae2d403e492
65284b179d4e45817ada513d9ebfd644fe8438891b88865ca83ab750b300c5bc
7768743de35f27d1c328ab63e1d0822536df562c0cce1d7b9027959d24f7429d
df4ffa4ca046d0dcc44383d42fb71abf07d4bc1abea278544d6dc49152676ef7
deebbaface7639abe13cd96973392a4147654808cf4e582ae3f9c24ee2e7dd9e
feba298ecbafd72847488790ad770211ed480c09ad87bf9519b35fd738bd4bb1
6f8f8587207f7f5ea3dc76ced4bad64f4c2a15e6a04870ff505bf40f32c7c69e
response w

3888975da922578eb724d500245a23f2a98f7404015eac450a901ce229f6bece
59415af2f0c68668be63317980ab11e71c544d055616094ee5086bd7eb4945d5
179b2980e7269df7b842f41fda5cbc3af847f0b6fdded57de719bc0fe7580f314
f9a2e8ba823d465698d127b9df289b79c04f062ecb75bdcb110fa000597cf540
45955302e75a9a7dc2100c932d05f60eacd9887de1daa5db439cdad9c137183f
35dec420df349039473b575d44a5f477fb5ef960761e3813f8bb2f1d343754b
b4838fc9c0cf1fa854b5ed45c0bc3d29540d33215170ef5ff2df3f58bf7fa745
d2573a388d6d3f7c073b04d24c7355b5087d5147753b03bf28836e813c8541ee
Fiat-Shamir context

97a5db7ee9fa2c49899de9c1dc58e928ce90bf50bf1bfe01c4c014b0bb01cd74

withheld proof term: 8 890 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Stego.getLSB definition, Crypto.Stego:27

Nat → **Bool**

SHA-256 of the statement above

c8ba556743a705352d11591b9b0f8bc45d6626725a8003e43a6e911dcb2d78c6

SHA-256 of the kernel type

c4db1316be9d0f3f2eb5f1ef936adb506b5f290cfa5704f07d50f6737a2e1602

commitment C

118e8a0c509d23420e4a34bb7a39ae5e3f57006e07553c06de0e1536dbeb6923

aefe8486abc53993e0ea93533f20dae1062fcb738d1b097fcf8ebc12d82988c8

061ce41edcf0e8fc719b9302498c164054255e4739bc581f35e24a7abab0fd5a

f10809b015bbaf5c104be04a2564ba182de7ec29d8ebf252c29d682b554f77a0

654edf1d95de51c9d494eb3326eded87f76fd6d36f21b76b9ba7427f828c8c07

20986c11e4fb69769b244ccbfb3b27f39b7d288173891fedf1bf46bfb94e9d9259

ff65b27e28de1c8ab83d93496edcb511f29f792c1e068d2abd9ca0e1e5a9833f

d19dd4df55745b3a9891f9078d22f05bd11cfdc824d49de7ba6198cb49c7e770

first message t

19771e45e099d3d7c3eb7724f88a4eaa46acf29f6d6990396cb7473cc61dcafd

87b138b1883b8e736f3761cfaa412ef4ee5fd97b5a3dfe127fd2b5d4315b1e9

6d3e0283affcea6bc190e5fca488d64dd337f05d93fb2e48b39af6ef96cf8627

8e6fe72c1d699f64cd4c1182526568ac90cb0da61dc8200c86b9a537d9f262a5

8a17ec184c8ec5e2774342d9f1d4095049ac0908e2a6ed1522ba12bb3ec2192a

3511fe506d70f1a899e3b5ec396f51a994e87bbac3e4dfcedf13ccbaa8190e4e

9369e7b09128cdad18d8b233d80160b619b571a3f732c9beeb829101915be5ab

c1acea8214215bbb16043230f39da18b19fbfcc182d54787fdbb3a2ed5649

response z

6357c9a5872ede2ac66bd6a33002f6417bd0a80eed4e40f725d1337fb9ba80f0

897aa227af9a4cc3d6b26ba8d742c4ae6131a8d01d2bc4ea132c497ab277c599

90206cafef2426fbacf3a27b365256357baaa3ace2c047c187dd09c72a9d8138

07e546470492f5b06950b03c3776e999007813b902f19622264fbdad08adcd37d

54be26e15010188c9d324a859cd7ce52c24e0265431cca7d06b43c1449a3ecd0

49cc8e8578123d11ee6248d9e9b52b2ba723437332b7caf19c318b8c0ef79349

a2a596960a52da2d516c33c6e466a31442e113323776d6347ebd25e1e43b4972

57c25af79eafb41d82990c0e40bdfdc86f4aecb2065174f5960a2bdfb1e02db0

response w

21e2ceb5dd455cca0b7540a4040293ef848d7c09026a8141490dc7ee3aac527c

da439b7598067ed46ba9612711e50d0abe63196baf186ea577e076bc40eaa47f

f67146a557472a8a27645e976614fa21529bb6c41ad4496056cd4592dca1b994

6cd607a1d6b2da4b3765a3443e953ac777e7f1c1792e4dba627397bbf10eed26

2d5b6d5a7ccd3037c192b373f462ee6eea15c6a993e8ee2128063f070e82ecb9

fde0c199db7ace42c6570f638f90488f53fe133ea4e2b4c8f17deb92d494cc6c

a6356fa412faa43e06e62c4525c73722d51b252b2b2de03ec16e700c1fffa092

96138673df47cdb1c6a9563aeaa43899d9cfcfe433dde0e83a438fc95b03bcf3

Fiat-Shamir context

759f8ba90fd7cc3306f4061363d6aa1a0b0c9d22d5cd1bb4151df5ef98197d03

withheld proof term: 231 bytes • axioms: • kernel replay: yes

ZKP.Stego.getLSB.eq_1 theorem, Crypto.Stego:0

$\forall (b : \text{Nat}), \text{Eq.}\{1\} (\text{ZKP.Stego.getLSB } b) (\text{BEq.beq.}\{0\} (\text{HMod.hMod.}\{0, 0, 0\} b \ 2) \ 1)$

SHA-256 of the statement above

3ba22891ce859ef40f2ed9c7425c8e8474d3884a6349528725c9dcf17755566a

SHA-256 of the kernel type

43ddc3321cf4cf30a589d868bdd51d20b2fae0bd0ea7408a1fc20c18854f1e62

commitment C

cf801e96a8c87e2dbf88abdf367299b5f2afe2f6bcde4f19065134bd7ab82df7

5285448df54cd487e323f918305370eedfb72fb9c61e601513a6e3320cddb466

82ce86491736b665c9b73aec4713c4956a16e7070a1cd5c437f9e2f3391d228e

85564d999194f76f7950cfc361d00c1110a0b6d28137850ab5a6ece07c3e160d

78cd1a1d84ad523f415ae190b28658a8f275158243e19885455dc5959f56e643

30a18455602762bf938a59de603d45f6b4cf6489ca1326643e25e68076264c0d

0e48979bd0ed2a93a4a9616ed8c1132d3c094eaa4d32e290a64a5c81b3f07353

15c378ec3b8a0dd438856d147e179eaf9e92c769e1844ce4094dab5d0a60b939

first message t

bc0134e2fa05d4ee962c758b99efd5b2ea35b11ba7594d44a1ef6927727989e4

0f7fca5f4b6fb6d504e480772e0d6369563c71dee3838ec88824a5229b54217b

2c0593969ad70b54a45758953f67ff1e9ea20bdcf9e2c558030e78ac58e20e8e

a0b6ead5d98b92aa49255b0c5af2461153b7df80036f069e9596a43ae57c74e5

e324eefbe906a472c1b6a637d0e9ca92441296dbb779e655b33b9b019eee5e16

df9e95f6d98f18b6d0efa9abe3707442910ad98f3834f75462c73da04e84817a

275b9d1ce4c2f003e1a558115dc72873db53ec76a3fceca2e77f0d7790cfac3a

9e2bdbb4a87d2a9137c38f87dd59fbd4c5f968186c9fd3bc0b838afba3defe84

response z

295d9479c4e9766e9d67a77e4d334cc285a14a913b05d7d8a8c6729e5e834915

953d498bd1b421c2834d5c73d2c9558c259c1a203def878a231ece342e950679

4dfcaddb8361e79a5dbddbc8cbce308c7452518881e10831fa0405f0521d2712

e17f691ed1a2d9bc7e80fea0602d47750f9e5a3d31886fd4cb6889eef28d58d3

93ee1873c26c8e4aeefeeeb27acc1e12b95297f1b7978f78650457dfac8f55a5a

a46224bafc5e61889bf8c76ec5e412d0bd59a59041bd760fba2238a5b265a209

d2e627264b08a75986715ea05a0542e5a412075d0c41329aac2f43df8f17b438

4272f8420a958adf9aa706ffb7d9e70ae67f32fed9299cf887e58cd80fcad90d

response w

1ed4280b03a9558c17571759f4f777e7a5a7f457c7c31f01f594a7a4a8ef00af

3ab5e9abc7e7d24dc623fbfbc351fd1c439e24b2ec846453e7e6729017d453c1

e91a748cbbed2050f09d68dc7faf13a9be03414d77730e2aee6a32bd0dd7c0d9

198e947f93bd17b258ee1ee8d0c7b2265477df5d5af41fe05c655fa8f8d0dcbb

8d3980535f9bd8f3fab67bc16f495cf0e51fb44977a12132c3c2378eb6de5826

32232b42490cafb2dbefc8a15c5e7ef08bfcc160ae38f87c39edc6ec1769f951

ac2a1c5c52abee86e3b887f714af5a45e3a0b204efde0bc3dedc9002df62a16e

d05644747fdaf41ebafc739671dc2dc2b4926153dfd56dee5ebca68afbb97c52

Fiat-Shamir context

ed3869762e56f288e1397b5c7577928f66a789e569fe99c7948fd7a74984379b

withheld proof term: 54 bytes • axioms: • kernel replay: yes

ZKP.Stego.getLSB_setLSB theorem, Crypto.Stego:30

$\forall (b : \text{Nat}) (x : \text{Bool}), \text{Eq.}\{1\} (\text{ZKP.Stego.getLSB } (\text{ZKP.Stego.setLSB } b \ x)) \ x$

SHA-256 of the statement above

5e28808cf97161d334c623bddbeccc0f7c9014c1f9e29011ef3b86bb646f777d

SHA-256 of the kernel type

65cf2155ec0a52ae3afe911bc74ece6e2a1637596c9b9c169667a96e196272f3
commitment C

c97ac8fe8760370b2ecd47e978f7c9e5d37e9247f4ca58dad54a62c50e0c9262
830a4fd7919cd2c5b641408cc0c2b13d6fe0e5453c282764ed60cb59980773b3
af201823b684358f6a7465e1038e912c18e2163c721293c799196bb2c8cd660f
8457f3837ab04cc4699ce213bb5b9d6bf1d009a1deef673a233f977b18b44ce1
0b0823f39ac03ec60e2ad13bc9a804b597ec5f1c334b6595ae3509ca33e6482e
e672339cc3ab8cdcd24261003a6e09ca2a76fa0743d8dfc1495535052e2f375
d40b69e36dae0f27f94b4779d9ce940a5df205b4d8055f83af5390c1d1ffa8b3
c6c1b8daaea6386cf9c156c2b12f67575bc6934154b39d0faa33101c4da01f29
first message t

b25f5dd0df14b9925a3c936f39b7799bd1cdb9ff6f5f070f89c9348f177d342c
31a26a44a901eab395b7c35b8a265c4f5fe58fd6a43ea07533000add59ac9f8c
cd7ac344a81e599ca6d24130787ef6240583c3f351fcd1cb575e6bc6338510c3
336d4ef07b9ddf9c8d3514981110e81d63ad2f73c01e2505aea4b4b1df0790fa
d4c9e7b98cdb9a993e756ece555ed9f69b9d2f0ce89071fa4972ddc47ebf8c6e
597666add6224af24dc475009749ae9b3ff681ee3dbffd7f1f9bbbca32d354d
a243f87a469c8922d2ea09cfbd708de35b91ed0edea98f2d2f0bdfb7c77fcee1
e8e7f2f84c009c3923352a521c6c9642bacb8b4742fd6d5b63804356ea93f879
response z

4711a04f25d6c7139e99ecd4fd3110826e0a6ce094ca73ccb3642b508e9f0dc5
c148b1dba662e8636b986d9cca85f09a904a3976306df06bc6f21abec1a41e34
12c0ac4f8f1bc4026c3cd32467cc7aaa157cfd450c32057cd26c1a89ad46ca89
83eb763af2a315ec1ad10e39c9e93f499042d1855870e43d18e1557ca6e027fd
4e26894d98542484320e3601cad6b8bf43eb123c3ff4f90d7c9007c6c98db6a0
09b80909fb8b21c14fe502cdd3497b20e136452852e5e3f6a562d80c0c252144
862c3e83602ce1aee7f1350dcdb5d883ab169849df0004affb8d90d04ab51129
3e624bd216752bd6b0f91a065ea918deab72931e770859f712ffd60dab7c63cc
response w

471f8c3aa568a4b09724d891627e96d59eb8df6735e9cde35624d07a67f9609b
462d913785b3032b418c68fc892c055d230d4a7e0633f49ad0fadd4875976d9f
2dd4f32bde9c324085b289fc866dd3bfbfef6871b367d6b631c1803b1e8afe62
58cff735d3397510e5805a2cb7b5d9de76915630bcd93ba83f219d7046ef8d26
db6908c0088860d84b88bb12e6d1ce2135c56ee96282d2973c51a2baa1573d2
e70ee385944a1b597194152f77c6f79e7cd04cebf7f4ae268f19423041db85ba
67bef09aaacdb8ab3e0190c19c3685a5a0fc05f6ee4cccdad6ee8a6b0498045
b9aa55326a94e781f119d4b27aa36328bdf5de1873220c416d4f16acfb76d514

Fiat-Shamir context

8b41ad23a329e552613086061ea2332cccec1c08c68c052dd911c4bc98703ba3b

withheld proof term: 9373 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

ZKP.Stego.join_chunks theorem, Crypto.Stego:96

$\forall$ (size : Nat) (l : List.{0} Nat), Eq.{1} (List.flatten.{0} (ZKP.Stego.chunks size l)) l

SHA-256 of the statement above

6e748756704c94852ad92107f2fb4540be35df297861f1ffcf34cd6c09ae347

SHA-256 of the kernel type

97d846162ae653e58d2a23399018de7bcc16cdaaf3a9133d1df24a9c0181a63a
commitment C

f5609a2d9f221c5abdc41271aceaacc67826ac371084df6439f078b8e3c1ac3
09a3cdc97b641bcd0caa37e63d3c7e9069c28d2559fb82935c96e8fe91517775
0fb731e4bc06e12cebdbe800dc13eae90f56facbacbd8446e391188ebba143
957136eba0acab2ac1710d5783535905308913cb8cc6a2a91c8e290ada98c602
fa1863be37c4743e8aa5bdb42ce850a61d31fd6631e7b8cc933335d7f89a5933
aaeb58b150ed5d4e9a59a6c816c1027f7a1de57aa6c17f60c0b54449fb374093
182237c0e9de76c6d5b98038a4f9826713a1b8c7b5667f4c53ad56f51f332e36
67636667a0cb56452efe38bc95b547920115388eba8c74d065535395bef18f5d
first message t

17cd6c3dc5587bb2a0467eca614ed6951ce1126f50fe125e1084f8df63f76702
10fe63f979e588d8d9e90eb6c56d3a30557f98872382eab6a1937439853010fb
c71fa93736e3f932fe322652fd3c949384bb46f363e7bd5b279d9cc628591737
f39feb09c7fbcbb3b58dc3bd3e4895fc6d26d0758ea7cba4d5fd845b44817d30
ea76e7203bfc39677b13c1da40cc4df634da1600003cf99871448e5316e7ff4
dce0c49ab856f4e11545dc69945fe4c6af50d1b7b3187075a4a56893b079ef83
035f3e071d9ff19476cbfad7b756f2cd206d3a4ae3c29545dbe37b314f183ec3
74f89b22494fac320074cdfdfb15024f8f480f3dcc4f6fae00ca2eca612538fa
response z

551791c119967e1579f5afa274319fb36d347f207695fb2056f3d086bd2f3708
b1f1fcded3c3745022f4d2179690d9ce6d8cb3856598f52d5efb3fa04c4c2b4e
41d4b24c9a0f5bb70fd1f37e92624a30f6d613b0edd25f8f4bf9d9df2ee6a729
6aa2348b06356c1f1b8709247fa90ffb9cf60c8b9c51a705b7c2bddc4e6cac1a
956c011855d9349e9eb66593d8174f71a413b2329fb9b9aa00af6a303c937324
5f310f9a85aa0b924f34888d9a2302316cf0dcadd0c5e17a11fed96b985486da
3fe9fca4873ad91a0d8f47a363696d0f0c47c26f74b47b4ed9c65c41dcbfb49f
f46b486f8b809ad46b6b725ce512f3b2c9e64abda2f2f03a8ef9d975e680d2f9
response w

709110c4f16f1f3216d9aca4678b1b6f8c3152257965e4eaa363691d651cd3a1
07957978b582eb0843643f334403762794793f2afbf62fae588678e13cf1357f
031d4ef5abd6e8bd51ed77597b2aaaf29e902aa195dd3e30855586cc673660bc
4473a8a4fd5b7341c9ad43be27422a249e17e3e5a6f861a2ce08c271e54dbceb
49a67d935cf510dd4098f20fa92218817f01f0e746c725afde58f6035fe70202
874923cc755564014ee0483eb2ae00486329ca33739ed38ccea9cd35bf02bd5
de8a77afbc7bbc40027bd2457770a5b2ca0c85f19e91cb497ff30fbd97300653
6d62a8960f32127c5f51230b6a0eaf5db8ce95e8ff6d4c1762ccb8f4a47885b0

Fiat-Shamir context

361c3fa3addcd7ade7244b77b59bd24adad277bf7df13e4d31661164d7af1e4e

withheld proof term: 132 bytes • axioms: Quot.sound, propext • kernel replay: yes

ZKP.Stego.join_chunks.match_1_1 definition, Crypto.Stego:0

```
∀ (motive : Nat → List.{0} Nat → Prop) (x : Nat) (x_1 : List.{0} Nat),
  (∀ (x : Nat), motive x List.nil.{0}) →
    (∀ (head : Nat) (tail : List.{0} Nat), motive 0 (List.cons.{0} head tail)) →
      (∀ (s x : Nat) (xs : List.{0} Nat), motive (Nat.succ s) (List.cons.{0} x xs)) →
        motive x x_1
```

SHA-256 of the statement above

6bd230e677f000d2f4a89560d6e01e47afc0b0867c1100d58a433b9b8ab1c1b7

SHA-256 of the kernel type

482bab4b9e2ef39a97c583b788e1a69c2a13265785c55d6ea54fae4f70a9a97e
commitment C

ba799f9aa66172f5f836a4485aecfcfe2184e72b777fe9bed8af7a4813d09508
8cb7923aead2b7ee2a3cae2c9841229a9723ef3272dfcdb6bcb47eed9310184d
99f43014734bd5777aed77790122705376bc9e94fc5c58f76c5545a87b73b632
6b69f038c4173c53b3d77d2135ba15ca281dfdc9f69abe4e71c3ffe92d924ee0
7363a9b5edfd93e553f51431deee97b0d548f769e1d04d5b5d60301d231baec5
af635d97ce1b89aa924c51671cc0174920d31226b0756b8a8c84170ca61f1568
da7f28e501d7d0d4cd4b939cfe47dba5ef0d70e04165248aa579ecea909312
bbc616c2be7317569dc923ccb1974c03d9361dd96468087e92a9636c0ca8d7b9
first message t

4f74d206dc9337e91dcd252ccc1a38319b232c8b792f9b91b0bcc7ef0dc4ab26
96d4ffb823db1537f37d7de88da2fe10a3feeb87eee7776b3ced1508a6747d1
27d5036a72f8c882bf5a8fa0753cfa17530672e33b667e2c307c38eaac7f67b0
dc231c9e16146f6f2c2b50c2a72958391b0de07970ac50b4f349f6ea8787afdf
a95adc1c5263afc22e6ef37cb875932410ed8236f2365c857a2e78261c12ec43
98f66890d823f88bf6192b6c4b595bae818e6a4f5b2086c31e470c07be265b80
29ed4412d7cd02b1ebb47418b0b9fd481acf08c55b32924b190609e0e362351f
979d084721a6eb17ab3bbb8bf625e4c6fef4161831ddc16bcb4415fec34dfd11
response z

333f2cbbad94ab5e33e7429a557624e8b72dc4c662455cdb92b392b60ead195a
d2363e3575708302308a7ce0743944458679b5509ad3437755897edb23b698ef
4b901347f06e7216a553b7c3e9750ca532341da59238c3f3212ddb13e401be1e
a436014a1defa708d164396f4516ab4e153e0ffec5d5a0e2c117c2db35167030
f06ba27eb8ef9dd480e6350884e69eef92f1721c5d71605bc83a04741d038802
9d7b1de3a1ceee3537c52503db14e4092b3068c881db259141f15252fea98de8
9b09be9d7bd26fce24639e032dbd399d7ceda00a9b9860c4ad5cf59952134eb0
a188c7f6953f768ecec8591e599264343ddb53f6d3dd215469fc91189e9fef13
response w

43d38dd2511d9b045675884189321ebd9a11d0081a2558463d2580acc900cca7
2de7120fc4250dffabc5e7dd6089c3b405b2c6b089e07ff907f723bc5f0d3dc8
90c2b416d94a10c7f45e16654849bc6ee1b2d2cc562c8b1ae6827d6d3b443865
a502df3d760bbf1c595094425433f8b195fc32d85f6b0ef1f72b3d256f9563ce
63ba585157f838d3229da846b2999b4355c1650cc875686fca3c25c7a2e314cf
0e42b91262a6d66ec264adaf1e36bb8224e86b10a870520de51f4817b7015941
8c5fb350bea1f9670aa9dd1134c49ec77a2ffd305a2bad8d9a9ccfd6a2314f82
83d773965379496ab61734b0c91cba9a82982234dd0429307fb5faf56a6e43af

Fiat-Shamir context

0c414f03f265bc95b53811a5bd97d637c7365b59a1b38c318f6bda96b18f489d

withheld proof term: 2180 bytes • axioms: • kernel replay: yes

ZKP.Stego.length_embed theorem, Crypto.Stego:59

$\forall (cs : \text{List}\{0\} \text{ Nat}) (xs : \text{List}\{0\} \text{ Bool}),$
 $\text{Eq}\{1\} (\text{List.length}\{0\} (\text{ZKP.Stego.embed } cs \ xs)) (\text{List.length}\{0\} \ cs)$

SHA-256 of the statement above

8bd6319f65f47cc13a5e1c3010199c16824bdb32d225ccd6341e2dd891989d9c

SHA-256 of the kernel type

acdc25654a59262e4f5d9fa82774a7d8a03769f7f3e1a46e6addc3e59cfe7fc6

commitment C

8ec33c397bfe02f6dec64b08afabb9a356be6cbdf99b985988e383782580dc13

80895d477f0f410d40c4b88b7ffc56a26824a49498aaa59e24075f2341f11200

0079f16aa8a72a2666a900f3ed877b07b2699bc36b151b669d669862302a46eb

357b69e3005c81eadd760eec38b0101cf43f0455b3b5d907abe39d7f00416917

e4dad741907c0f7ec832b389ec954e019d486a293cd448154ee468f46c29b703

6045dc4540eee4e9a92b79d5b15d7d2a7a0758d1608450695feef00628a12a2c

a64fa52c580035c918f6234f0930fc87646e9a67ecd917e4daf7af1460e45093

113a67d8ea07ff5cbd6df72146dad7cd98a82b437d596ce52c196166f7b4ffb

first message t

f9e80df301e70f1ba2debdff060f658a09c39f27fcde2e78a0c1907f51dbbc26e

1c8e8e374effcfffcd8ed7588675a8de1cdd6a3d9306a0ad5cead96039a97288ae

4a60603977fb464d298e21f84a43ada617b7e88cd3e0320d55c30b05535f6ba7

6a1e889e4df2289ce2477c144052387c9c414357e854aa4e4c3b0c5e8de7baa5

977d5ac6712213b6ecc7fb655211ca7c743bbae540e970ae380e05ac3c442577

ec6567e43f6341884193aaa4070261c42271269c328cce0a3a756975062044ec

62ac4832092b8c0959307b572c2f6da539c210449b4ca50f6614165fc6673511

c58f321a2758622e226ebef6562b6fc31ad2c1ad97569d521c8fbc8d05e2d3a9

response z

200c2a978e6fb94682334ba42dc9b21a792242e31e0b33ceda35cce124816ad7

ce39edf788848f78ef4e891cbee91950ca9ad39aa69cb249fb9cd57969e4e719

dafa76e007d2afcd88b0db01a79cd3122ae910b5b178ccb97771f33250dd4d

97e64abc3a263f9eaf7df9a6ec95a5b46683949ae22bc0f1145dd10de2182bac

16fa5158bccad12d91d5fd085b4443f0c6d6c853c171135cda3951251a343d5

a15999a818d450aa0d362e641e1de7da1581b9abe1c7c6230dd7f78cecec51ce

a6e1dd9dabddf0d82ac9e1abf8e30ccf174ac8fccbc368f7eea35c1c89be7c70

5c50b7258605dda547abcc5b422a485befca2f1a5aca1e4b4dc6ba39656ef5bf

response w

20776aea3365bdf0852724a9bc90067a6fa75ca25bb2ff8bed1938d1f738db26

e341156efe4735c82109cc7e14f163b084fcaacc4edc79deb6108f52f9570f6f

073f679fdd2e02b265010ef53be965d3bc8e05774eae549c45f64ae5b32bbb

24c378045c1b005881c0ccdf260650c2fc3aca39c9fba67a37ec1e2cbb3bd63f

85c6aed220e6f91b7d552d001ff2dd108226f634862dbb275acb642d39abdef0

3e54fea8a799c37c8dafa4facee19feb82b884de3f843208023b7d7aeaa88a18

a795a95b12550ae27797a200278d92effee4d4b1a9feb3cfabf018345ec677c5

0b11d62dc4dd57b91ec5388d91e4025b9752e10a225eba3fd3e72d5268811560

Fiat-Shamir context

e81879a07704bda3832f8089b86e858c5c40c62c9a9e0bdd1db33e724414d8f2

withheld proof term: 4580 bytes • axioms: propext • kernel replay: yes

ZKP.Stego.setLSB definition, Crypto.Stego:24

$\text{Nat} \rightarrow \text{Bool} \rightarrow \text{Nat}$

SHA-256 of the statement above

a4a4179f857b8ca1d4113fece29be13272dd07ffb4d63beda7d4142db24afcdb

SHA-256 of the kernel type

1ca0f9f544e54249076553b629c4a29ce47cc8a5c67a7415593dde9936f1a23e
commitment C

bc46ab064307b2a57b2b6ca3cbec50e58ad89c303d27f2dfa737b874aeb79beb
200196897308c61b029c806e3fc67873b286e9129117674f4661dd7dfdb168e5
d8b7f8f5a0a5fcd2e2e964ff835a98348899ce38d57fa061ff840c72db18a3ca
2544c150c8588b427892086698b299a116f0921ce9ef726e36cf003ce0fcfcc3
2547b64ac959896e7ae952d60e54d53d5f99bf43cbaa890827232b29ac71cec7
7d1ac0e0dd5029c4fcd57da78df8dba1170b7247f8c41281aa6c459feb1fdd3
15f5dfd49a665f243220c1e8d28a1fd50a442ed99c2655b6d46294ce86924441
c89641ca0986973018561d96f7c5ee7385d8922c114af3b3e260db41dfdc477e
first message t

4f50e27e20a0a45f6637604714a44cf71b397131cee7ff847f8842d9615ddb20
8be02c54ae5822f572ccf62545d16523b51c94dcf14bfd2058a696a44581278a
e9f82eb3952c6c5ff0788df055e4f7db1d05ba7c00e3ac9129b120268d7292eb
04b27b6e320ae21ff720fe0a201e29c9117f1f63b1f0a1972d77f6091e3ba295
beade0df9fa1c1ca926ffcd6865e5bffa7893b6f4163dfe7f4e9f1c89f650ef9
bdcafe17f8e432b262922dec8ba0bd1e1c8aad96c270cc38a0d781ef42016f36
d06b2ac061fab62a3ff6c194fa8325372f0e8ebd2d59b95566ae974ea4fe3232
bb6c4c0d47178d586df8af15116cbcf3ab211e018d44baaa5d0af1c86cb3b440
response z

226119e2c576fb99ae2c332cc4fe71b5ab6409554f7c88c903f4e31cb7cc831a
7e26187eac8a3c991a538fb5f35f846b9262565745c853110777dd5d7abdd1c4
36b519669a954577b465e07251f71b200a6520b78e0d0db4f106732a8740fe5d
32d9b6dadcc592d91a07d390c137dc981b11a693124689e7427848893ae409828
b38f57c7fe49b6fe0370541f75b622678d129e5e6b4b7e364b3d1ffca835b7c8
852792645ede9110b5d0ae0097732b5f82294871a385108c0828a6e3c3f10ebd
ace1c93c7244a524bab75bb56a6790076e0502eb1019bcc3a0d68cb454d9699a
29ca97c2d95dad79c0f6cee949975a42c5c077f8e8f47a2240c2962c286fdbd7
response w

4de9527c77559433dd2ad657a05e9cc752f512041ebb43bfdeb42cb93706203f
506d06efe4404701d95d358a8ed6a1bf803576440ad6f3b9e2e5963b50623cbc
78b294601a8eb56c5ad110b53b839b317ac772ad2ad88a04717d42b70d3e023f
6d97e187a6d9b3ce64c893d42c92af8afac3540f3c30d3e74fc273cc439dc9c1
e52ad5b02d71754670e4c6a3860fb72921635e20ea9d92a472bc1dae13cbfc3a
39712b1f36c1fdb159ecc1980ec4adc8eafded51adeeca6b611a9537b9a40b9
94ae233cdc0cb1dbcbce46297990b298692c9d390170d5a8a312bb2e70779d25
2a3c27a39febf7f4d4e22c79760be22cc9ed25f1106f2a6351e002ccf876a27

Fiat-Shamir context

f61e330a99c1f35bf4539b8e33010bf253168b7b353f3c1b565d3fa95e7b3a93

withheld proof term: 458 bytes • axioms: • kernel replay: yes

ZKP.Stego.setLSB.eq_1 theorem, Crypto.Stego:0

$\forall (b : \text{Nat}) (x : \text{Bool}),$
Eq.{1} (ZKP.Stego.setLSB b x)
(HAdd.hAdd.{0, 0, 0} (HMul.hMul.{0, 0, 0} 2 (HDiv.hDiv.{0, 0, 0} b 2))
(ite.{1} (Eq.{1} x Bool.true) 1 0))

SHA-256 of the statement above

0bcbe498810bf67b129e2712e785e0608cff40ee5d4ed55f7499e12ead9e947c

SHA-256 of the kernel type

02533b61e1dc6a3e377a4f923cd41a8d3c50fb238f4405ea7846caed3da5ea46
commitment C

1f23dfcb6a23a68351af0021ef0896a0a1a8ad68ff5aeae066994384930f648f
24f8d4d430d8d5885cee208b49df92faf02e1e4e46554498b4a518a820f00a
15b27d46b00a71abf4e46076866e4edf3a49e40698c24fd15b896971f88c6884
7d214b6ec1936c61d0506d43ea9cc9ea1b98ea1d4872f17e0b87df953501c553
ffe4b8766505ddec695711d9fc89ef3f28153926bf8903add673e90cf88694
601942e89a7cb095b2a15f41725a7e76604e6686233c0bdcc17aceebcad84032
7c996209b10e39ea214a7e8ca6dad988723fbcc9003df1069703b367eaa1279d
99e56a4da44f9556a0ec02f3f20ce18da9f7b3380724c223ea6270b05eef0ed
first message t

4a5bbeaceaae4579fa3f0d3faa0e5382de902c19c6ad9501a7c29d5051925a76
a462dd7aff4d3079680b7806505e9ae2a1fa1e889c5ff34d7065f2e606408eb3
f147d051b2632423572ad824d296f5a529262cdf6b6dd41b33208db5a1df5585
0da63e8f23cce2046b7ea8883abe305734b8a1c09e1fbd4cd8988164418774c0
0c5713f1de0c347e7242b3e93edd968e8d50b3c4a2a96769d0f9709834fd7681
c956eebf06f5942a09daadabfc4ff68fadf35c4eeaa6d67ae158d04035abfc19
f0fbf045e9b4a0b74134e2c51b7ca0d60c6da333a45e94ade610185453e8e482
d68257e12caa6b92f4340887796ba0128e0a2bb3a0dd41a15abe67634529ca7f
response z

14c9889f7292e59ec845a81a9ea97e55d75ac857583be97ac346bf1569963f37
c45da50bd03dee30c21e25bb230f4b6a43ad0421c0a8934aaf7b5e961bb812c8
fc009c953441ed6a2c65caa0bf5b6e73a75f018ebee721a6a43f5442882f7c8
17c83655e900183508a4a828e3c60c07d8fc2c2bbc6625d5ede9c3cbf82ca1aa
ef10a58e682fa4454eda7ec29327c02c4d929361f12551858c102f527400db70
eaf4849cedcfdc9ec8e424b8f206dd95868acc05b39b2f4ac00cbb752f11dc4
748ae8ff534158e820c5a01151fa57e216d1904a679fde708fa20c7a5cfd866
b8d6975f930a8a58417be623daea49759c2bba7288dc5ed1c3a2a455ee7ca368
response w

79ba262d828946ef0bd4a4ba85234b40b3c3e990378c710c443136b4b908dfc8
a082d573c044eab2875543ce485b4911bacaa12f498480535a059a40fb2401fe
21245ebf5121d36082f667643edbf4c7c9d24ec1e43bf98eb34c2c3fcf3ca7da
23fa1a038d83244f790009661ebcecc68c4a94cf628817ef812cd806751bf8a3
e88c188359ff21c00618004ced7fab61272ff0a4dbf5e0038504a3b6548d80b3
833ca01c1ba791d03a0c0566de2274b0d8f64c3ff7053ca12a93de77597daffa
346ac750c8fe628d039b760d020a6fb44f0d10b4f1d49b0b12aa9054fb76d621
5c15107dbf5fdc29f477d6b53120fc0a6aca23ef359ec0c6af43e683fbebfc0

Fiat-Shamir context

2a3845a6dd3b00b324a58702e885e0fc471a253bc5ed9324246bae043f348f90

withheld proof term: 66 bytes • axioms: • kernel replay: yes

ZKP.Stego.setLSB_lt_256 theorem, Crypto.Stego:43

$\forall (b : \text{Nat}) (x : \text{Bool}), \text{LT.lt}\{0\} b \ 256 \rightarrow \text{LT.lt}\{0\} (\text{ZKP.Stego.setLSB } b \ x) \ 256$

SHA-256 of the statement above

a38d836d9a7451e39709e7ebee56958a92c57efa4b4fbc03e3333878d6f0ef82

SHA-256 of the kernel type

a8f64de82cc659399ed79edb275d3a2f503d2f42cf7544ad3f2879daaadae153
commitment C

79cb6c75aff417f8f4492a6991fc0bbdf7e2611bd2b992540f26cdf5dde7826a
fe99101b19693cf30d9647f6751a135d832c03bdeff70fd8b32c5bd68c7652ca
03bbfb6d9be7ba9d132883bd85d1bdd8193382832d1a0e7416d6130472c85c05
c00aefc1b3bb5d75f0772c31c6bb70f5ab924c2e8750870f530ab3385101fc5e
d12e3adc1f0d3132d55d91ff223035023fd9e50000b7c3632611a10dbe4b9916
e6ae4f7a249e9aaa60205b9385399e2ab19daab393a0c8a6055527aec923e5f3
069a799ec8a869857e9a254c4b2e11a19f9c05cd6aaf9ec5cd7cc08079e41a0d
c7fe03c19e4a2d81294a9586fa7187536db3dc2f2c5de4d2a24e7982029427f4
first message t

88856d5796118430bb498195cba509dfbf074c8d0161c6b737cc35014bf20ec9
1f50b7d0abee85e772c6c09e122c6028201913f4969b6a1b774851ed6e46d304
d072e9d10102890c0fb44ec16432f45b578060b2d372163b3872b58ebb72bf04
31bb5e1479f0aad6f2097187316594068538de99374cf517837a248111625bde
3f00185f97c959ea3804381f78ae47512d9ffc6c84bd1cfe35f89fa48e14e8c3
8464c3bbd79b361f3a5d7cea8caa39bcb99dc9b1c85da99ce5dd0a2b7a32fd48
156622427d869a2505abe0fe933301a33562cf2e0b4a019a91f44e6ad98669f0
f6040847a382357fea3633172c1c76fd08ecb9a14178820555f92e995447aa57
response z

3eb87d32cbdabd969fbd43337aa16f94ea064a7454122bdcc3efbbca66f6717d
dca608a62752c48a7149cc1c6c75524854477bf57dc4ef244dadec6609f4aaef
e01100a4fb3b5e6f791fc413c87c27cd18ebb61d2dc03c65d8e092af43d9ff6d1
575b7e9459ce2330caa1c506bc2ffcd9803f96f9f928ecc17842e126e740f138
2e30a081697ac1a78dcf9975d1460d9e246a842f5a87303f9c1df587a764c327
83e5b1f79680283a0852491f7795947c045059359c5148baa24b6012f2ce64b3
e53c3b9c20205bec2c4076d4e58ce928631eb7e7b7896d5736e0566670d3f9eb
9fff6ced40abd7f9605f722fc827a2a846010eebe84f050d11589ad6bbaaaf80
response w

3c44b52e887cf705f3fc54da8a2e2f0946e972fdcf3101a0483484753cb94023
a12dc17e29be179dc92d42cf4a5e2b43b7fe75976237a6b8add099a1d25f97b7
481fd3eab812b06c1dedd440427e54e31b2008e649d55c50b0316d22c651123b
64b7284b38dfe5ed5c012973aea9f3e8a97ba10b6c7f6b558135e23bf3d2b604
33de1169b1fdb954d7bc0dfb87dacf8893d8bd717e97400e0fbc7c08a6423609
3b31380c1a52356e16a2c0e1487ba29fa512d50f77cfabb7d88cca4b7778dce4
2a753a3315ea3d75cea9977112f0e3d5c4936013d0c307a40ee6795a74888d50
47b321cbff1f96c288cf4a5d15de1a49589d1641a32655961a4040716a7221eb

Fiat-Shamir context

2e1e29e2ab0e4638ce746345340ba4726377c16bd91b71c4800e755e231b2474

withheld proof term: 8 359 bytes • axioms: Quot.sound, propext • kernel replay: yes

9 RequestProject.DirectionRing

FBundlePoint.adMatR definition, DirectionRing:259

```
{R : Type u_1} →  
  [inst : CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) R →  
        Module.End.{u_1, u_1} R (Matrix.{0, 0, u_1} (Fin m) (Fin m) R)
```

SHA-256 of the statement above

bceba5489487100a9916b852be5e37413b4f587c410473f0075461ff76c0fb3b

SHA-256 of the kernel type

0cab444a491547561350747e151bf4c4f47647c671a1460a88175a08a245f3c8

commitment C

5c86f9de78aab40392f07c4d8959d837075884652693655e394c4375cd535c8e

8187417bd713ed730d5d7d7f63e23f70b18a042a06dfdc1b4132cfd08faab1d

1b1c43f07728cafd2df5387ed10906301e6b169fa242427ea8d84b4a4c2634c9

310c66581d415513935bfff58ad3496d9e32a89a84ee1f3b043377ce1ddb9fed2

8996b3793c00068f0e1b85091a1284bbc43df59675b7063c4a6c53c07cc2d3c8

10f411167a59ac05b556c5fb990598e2685516b6524591f70664fd4a42c343d7

87e7574451538651866ea312156cee91ca2360ac7250b3086b2c5d57aa29f5aa

6ee4a21ace687dd2a0f0e8a683c1869d258b8b7a1a3b8d447ac894f1907859c6

first message t

f91fee89f5fb31f87332650a8565fe61bcd12bd003a73cea307a2d47b0ade478

336bdde9b4872fdf4c3e9e9ca2dab83ecc01449e92f9f4b870752a5820cd636b

5beb6e22d3022f90e5b8e248bcd70f8f1ded90a5038e98be29cb43da1aa9b20d

f98cc4ba5fea19607ae0156f442d75de91a9a21de5aee54373c129cb9c361486

c775f05129d893e62c1e0faae1f5b20a83d871e3203169df2084b524f72bf99f

4b7eaa8ad5f040456c87fdf5c32cd02d8f73a65110d8b28eb36d89bae99e46de

5f157233f274aec5ba6e23cf63a97a1b97ce1c8c0f618de5132e21df40b793b9

d9d89b09f1820df845a07f0a2e0c58ed95bf71ed3218c0509404b4ef650f86c8

response z

3e586ba16039bda18153ce9c281421a9c4050b1d8d5571c2f7f3604aab2ca292

65971a2f02de24f25d77e478183fda75013acc44c2bc652ea856e2c449162fd0

0768fe7c960fa3fe12db8db57da1ad657ebc93b08a9110fae7ab6f660e60edf

3891c356b640351969947c78ade0148efeaf591abf3da1466771823b19da1af0

91e111b829254e5d3bc24200a1ac5e6470604d7ef1aff08900d7dc8bdd532f12

05c4a441b240e656f1a685c82752f06c0d043aa410327963de8fec5a41f7ad2a

321c48fcbddced5fd605fb91a35ccf59877f849bb9f607be3e0c73b9c70f83e1

46b52b4dc9e714674423b23a1f9e6d1be5aed65a3453a84accbb1c1eda8d93a7

response w

389321118d0ea25f59d2fef4af0316d406a56e19e251d7e1c56ccfa03812b48e

5c951a38b0a95e22ed457e7dd35da550a59f33111eef445e19108592b3f482ae

56be504b90d79e1b61ced3e876f4c3a1900d8faac2a01f44ec89dc3f17f1d040

25c559d8d58b327584e3e2d6e2f0334a692165db6bae9450a23e8b52d5f8d1fe

23bd1ecc11853358ce6b8f31d417b6ac011cbd6d2285fe7f02995467eb3d11d6

561148af7f7bc4dfd245a29dbfb27112f4ffa21110cdf11827b192170fc36fc

5cfaacf2b4b194f93a217e9d975cdd6e3815377c458403f452aaeced9f1a8a95

b00c687d0007c8f3edef6c552817c8fa96134dd976556ed764508caf374ef6b

Fiat-Shamir context

a0a7a23fffc5a5d192b2ed599cd4bb567e2dc102d8151ce7f8f4d172ba4ab4b2f

withheld proof term: 6747 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adMatR.eq_1 theorem, DirectionRing:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin
m) R),
Eq.{u_1 + 1} (FBundlePoint.adMatR.{u_1} A)
  (LinearMap.mk.{u_1, u_1, u_1, u_1}
    (AddHom.mk.{u_1, u_1}
      (fun X =>
        HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} A X) (HMul.hMul.{u_1, u_1,
u_1} X A))
      ...))
  ...)
```

SHA-256 of the statement above

3013e04e6220e73213a8334929b540d180ce9a69a2d8e146720e17a379d0f3a9

SHA-256 of the kernel type

e4cc459e75d823400ef4dc9d0f4424a90ffa2e39ddd120a745fd4d36bd3f0054

commitment C

543efb10e6a294dd8c3ac7bd355d2f958857456cebf1d724769ba813b2bb241
8d70b77ada6b36d47938bd16618450cdd78f96cacca1fe0913e6b63a55a5ce75
19876ff0e60beff517af622edd052ad51caf08fc7b1296f7cbb786ff17fd4cb3
66d6edee59e5d0a3ac2c984ee31a06bf76d5ee3f272d860053ebd6507e9714a9
9a0f2072903e16b1e1ea78d6a725d6592617955fd30549280875426040a2f453
2a7279c5eb94a9ed07067bd177cbb7e70e8e8606de5390e9dcc3ef447d64cef
4412c9d2fafd79b3f248e040339682a5422e82f4c11fddec8f158e1099171dcd
140af29c92a2f02d0538e89080c9e370a503107fa1b3344605a9a0de25d020d6

first message t

e342812998f1df4edfd471f0e90e8780b646f07962ac0a286648d8ce058f912f
302eea2f1f710ec7eef88f5594491157922c1b8236140dca4d1f18731c9ba5b8
3ae3503b2b67b2607d6a367bdd9f3ec68772cb8fa64b8984f724734b2285e0b
b2ba32eeb4c7ca235cd94b2cfb4b84a758d7c89944ec122b4c79bd0ea206e095
db904d85a812f8f709178a72161965bee88a9cee35153bb6a72b49ad602d99c2
173053fdbdb899bb6faa47e5d94611f8d2daa638acac67789388fed87691a5258
128398d2c8ff94fc2725c47077c1fc7267c84a721bb11046a46f1ae2074fe32d
5e2a85a69892abdc19a224166fb19308499927b84255094a9d8b1e117b0b4605

response z

193ac57ebf9f7a38ec7bc18ed5d603242bde301238466b3c3e3746a6184dbb7b
c76b248bce527149ef220bdf0b17d69fd9e344057452738ad43276d89321a2c1
7f7e2101c26f52a11b37ebbd4c0423b9b51a4db08066e21a59f069d79a0408c0
cdf7e832b412d271e7466f61793b3eea2d2fb6945338b92a3130382cffffd5b68
2a0fdb1ec8c31376a378753e2c480349ffa5f972de893e6242528f24c301d43c
43c45dacb740e752dd24d0ecb8adcfafae6ae04881615786fbef1f5bd3fcbe2d
ead8b5b5df10c6e2ebf27ebb016a5286db75b54345c8813dd680de39e3975693
3327293f5fc708807012bc40b575a56ae41e1e7743fcd02075137e6ac9301f1

response w

217db1b95638bcd5ff5f6a60fc3d7c4b3b6ab619cf692e27177c2948bec44710
88203982f90999b6631771ade78515a7ac40b4ca6929f126f107113dc0cbd156
98d8239464d608c7ae07eb705f6e1176255452c1414011da67ea4d3805359eae
d5c55bdfb081603cb2e6536e5981e1ae743e50e1b071e9fc7b0f661cb3c98baa
bb6a3434a222a40f187b09f8c236d6a0f901c89a3d3c354b1fa63e29ceea188f
e1c39725d14442847fad9eb26c8104959d80fab0f62913ce43debce75581c3f4
1976dcf8bdc11d12f8878963071d252085af2455f78766af49ba474118399caf
53328192ad88ee68e25a0da9d673991855abda4726506aa4dcc8c2e8ac4488a

Fiat-Shamir context

fe64f0f2faf3a84ff0523b5bfebfecba34e95ec50f1970ffeed21243ab86d0d2

withheld proof term: 1122 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adMatR_apply theorem, DirectionRing:267

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A X : Matrix.{0, 0, u_1} (Fin m) (Fin
  m) R),
  Eq.{u_1 + 1} ((FBundlePoint.adMatR.{u_1} A) X)
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} A X) (HMul.hMul.{u_1, u_1, u_1} X
      A))
```

SHA-256 of the statement above

79d1a11ee5cb8fdc4d9fbc87bafcd675dccc19c7ec597ae689b34118209412a1

SHA-256 of the kernel type

654d4beea977858522514ead4f081101a0547e6c9da61769cf4ce1c9b056f7ab
commitment C

62b39a5de89d61656f4110d235b36f20a9fbb701cca27e883b2c23cfcb87591a
483df5fccc27a03cfacfc2382fbc5c539ae667e6ce67ff87808fc800ec21bae
74bfdbb49c671b40ebae08b3586423249ad11cb4629e76bf7f3b65f73257b662
eda87d0a15e30f15ce3bbe3d57c2ef9c6220cdb566fbdee8e9eedfa8658bbe23
0749d8b1411bdadbdb6ca6c7f2efef7c0c0c477a2683443835c87e1d56a78cd3
0a442adb5e0ef906d4dffeefb72458a93b61d6919ea5c2498fa1c6d616477263
6ebb9ac9cb8ea747424682c9a811a21afa07d09301af1495e2001bb9dfdbdf5
17459a117f1ab31d3e554c2d5a22bb2d8ccfaad8b24ec91202e2b9bc94e5d723
first message t

a9ea24b74361e44c1459b9079f2d4cb248844ce3074b196dbc7a922678e144fa
a688e90c55cb3c60154ae358da9d1d150dc442c9f34ae35f4afa08526e4932b7
1598c618baf43211f9bf4139a5d2f8aa8a3c6bf794dd23a49b31890ba6c75c63
6d19b3f4528e63a5a541912d4fedaa88c554dba42871f114ccc99b8f6a32552e
1e55a869c501f4c651cf021e5a4bb0bb7f413db343a1805975ae33d60305adb
826a631d0244b20fd749f39d32662ebe2e16ce0eae263b287c9b3324d04822
ac4b4aab9b30cf9200867500c3f9990df04fde19d26763a7e20b26e9a1ddfc9
a0112336f93d68a1b515bdb8fccb5394781c4076c8580148d6f7679e126aeaa
response z

5ed7c63d1002bc2061281b82045db977cb5a7efc001a614914b4b20a40bda48c
17811bc45e446c0c1552f22fbb89b3fd3867f72e20034e7664ba8d085fc65bc7
11618c7b3eef3244f56dd1a0b071bd1134767b11758d615152091ca79c63c30e
26fc4d31aee740681fa813eddbbc2d0bbc7bdce12c57e23975ff44438824a3a3
0d9e03268b0297adec1e465e75cc7257a6478fba7f461cb1e54a8d84ea06c51d
592e2b7419ad6d087764495c16e863262920751877d98a8714251bf8caec33c4
088453c7069b01ec7c2659bc56ff5f583c300e67beb4047ee28c11a9ea5feaf5
7f4c4aadbb725f07f8d13ff90d6192547f0f50591b82908470a5677394fb979
response w

6e4cb90075b2a36b87056e9ade373dd05b0cfe0854f4a7ec9629432b143f3c4c
9983d881f61053a9b1bf54c20cde25c66458a963394e258b2118bc2b35a1e771
2baae2d8a9cee4b4fae9f306903513c6e300bb6c9414c8089ad59b22a7f82583
a6440ba51d2d3fdff647e14426d74adda7075bd61699e405f02963ce89b9dc97
abd42b8191036b75b95d44d8603681661bb9382581687376e9d8af176a8728b4
48d7e5c876bace831f65736a343dd90ed6050cf00a5acc845bd2f76609499ff9
ec2e2cb0eb137d69d603cdc44853f27d8fe7f0f30918e3635508337f11d76cf3
5a1651255ce874013b3a78f49bb048e65c5c4bafa74a364273bc0dc221a6e9a3

Fiat-Shamir context

07d75f6a5a3559f1d628f944482902d57419cdc3b08b58f1ded88adbb732d164

withheld proof term: 4512 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adMatR_neg theorem, DirectionRing:275

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin
  m) R),
  Eq.{u_1 + 1} (FBundlePoint.adMatR.{u_1} (Neg.neg.{u_1} A))
    (Neg.neg.{u_1} (FBundlePoint.adMatR.{u_1} A))
```

SHA-256 of the statement above

dbb5a6f58b2a34282b7e83dd1f53dd2f559c5db12ecee6793987dd045f44ecf8

SHA-256 of the kernel type

237df14ca0ec63d4198fc51a550026f5a61fa26a6ba20fd42c58c89eaba7ffd1
commitment C

6674ac4b4914f51eb3edad2297b17c760190d170742d93f8efe7a030f1484d50
21b987de9293983f8975e09376f6097869c62e2b914a5d90d2fce6bfe7c1fbfb
be4a92e3a3b4a152ffe69d04b4909042e2cf831df2aecfa4d8fdff00b660735a
c2fba079f31586723a62e1be720c52c030177b69b6fb1f14576b35591e5f8a10
7fba5f5a51195bd84adde97b679a36de2024ce5cb037f3a797d760f71b957d39
713586cda3de53a51cc00e68db75c7555e6ee043c85ab4439e104fa5cd04cf63
ad400f3ba93200b932932bc0bf1c0cb5698e0665c8415a8a7a696be675a69a16
1b234aa153b5dcd0e639b4eb8caa40b90174ee51abd0b9736198c50da9144613
first message t

9af570710f2086278d5f3ed21831f67cb6ab636b68ff86158558857676798797
829b655acce219c1c8dfd6bc5ebcb36d98831fbb9652c9c599fd53bdac489932
48b749bb2fe72194319bfff347b14b38334b89d07ddea309b8c496ee5eeab989
052e830f1d329c466c933af8c301274290445bc408dbed81e73cb3f824ad9d90
59cf9ae9e18b4babac8df2289624cb89b70a63c99099948d5c881ed1401dda07f
967979a2210713722526df659a31d0dc0a83c40484d03096f9a4c16e365ca13d
5e951e6db0528ae99ddf50acd386b3ff521d535645d05094c5d6480b2a1acee3
4a97246aac9cfb68e9440c7d319465ecd1fe5923dde809f31a3029f789692f1b
response z

3e9b02a73d029f6edc2d680b414ab6c53bae677a3685aff15e6e620806075516
0594f0fd5e64150c686e88cc0679fc6abe1857278162c828f062aa008f7d78cd
0e26e618d8182294c6034c4d2fd5230a4b1a51127ad66aa7452da1c254934ee6
6684f460b67a669ccfe57e425c0277203108c017cf56221d76bd6382d475ea8f
cb2515a5b3f5b26abbaf23824ff08cc1a794457a4bc1e55f70d271ab1409f82f
c9cd9945d31d26cb00177e494839dc7ede5919c1232fbc7d3fa1e285f9b984e4
a89a35fa4234f76168abb578a40e33245aab9d926b7f274e58e1a456291ca2e0
da28e79aba12e40569c599b4c77334c223521db2d44c73d00acc88780572d2e0
response w

5a11bcff4f0bcad5a8851f00f8200b262ba75aa1dff5c6d39c34a97aa74b3082
1cb5ac55da08ed287662cfc98231c9d5de6ef37b4cfce9fe29bf61d1ebc0a1
423e834fab84b6a89e933c77001f8f2b77bec13f8ac42eba6b855c2518ed06d8
1f936973961337ad4f93fc49703dfcb1cd1983e6eb773ccb50bf86ab1d5f69a1
9d46b464faf5c0b344c10ab51d4de89687bea2900f364e9abca126832dd2b356
fbfaedcf81ff9c5520d1a2ec171a4ad9b2be18e2036d610cfbb5f10706edddf2
c989d02b199fbd9846f896acaa55b5510ea9ad3f7ac2ab1597368b7045835ac
6650639d7b583902c60fd721a1f79763b78d467794cae3dd1af76888c49b8130

Fiat-Shamir context

5a428d5e7f0bc07cce9e4beeee191fed4612c60d15967604513b7544960ff9c0

withheld proof term: 862776 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adMatR_zero theorem, DirectionRing:270

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] \{m : Nat\}, Eq.\{u_1 + 1\}}$
 (FBundlePoint.adMatR.\{u_1\} 0) 0

SHA-256 of the statement above

4dacec35de4ffe29c926b74fbd65caed53c8e4a6335ad98e2b02d1262a292ac9

SHA-256 of the kernel type

2614488c9733a58a3da13318fb5c46a56844aa7cb4e0757aa468036d313d68ae

commitment C

b8531a8988bfbbe1f7575b7f57169744da768adc577421fa638404546d2b0407

d5a964f81bf62147180c7436b213af08acd6b7f6bab2846e9a2af74f455b4abb

9940a2c4f10483d8430002edc21646c4556d7f216c4821d45111a7022af22777

9c9b5786b5fcc9621d4e718c07e35a83a481f44a0894469008fc110c93e6cfba

185c80e7b7b391ba50e49f8d83ebcb19e5339546d5c945eb0993e10e8cc57e2

afe6c1de19371be9542b7c2cef933fd3961942f606f54caa642e9fff72637849

2a4038139d19e72e11e08b2bb0ce2f03a63c8fbafd87e07766fcae31d594591a

93f4ed877cf4834acd663708615c2d9f96920bcb94d861f0849bfc1c26d1ac437

first message t

f96e5d1b7dadaf3c5c147701d94c2dae8cf59c3fc954dd22130cea3676a3f107

565f4aa74862b10a4fd04ea7166240d1b2c09661e52b5e90966bcc6f85642dc0

2de576890dd34a5fc347c05b4cda5f59ee03893eb0df3140663e39e0e3edb447

6af12356a2c9a4620b4cb249f9b4dd7350c77b12fdc6b6f23044fb28b495e0c97

91aa91f021b99c6314bb5755961f919ed6cbc401d20d3f60a9b08c56e231fbee

a1157e8358a8c35311bbe7d804fada3dd3b1241e81600e986a192a4ab66ca4cd

56775d20bb8aa3c5165e2e6d016275afd4c469fa7d99db1cd4ddd21158ad74b7

47461e5f8568004c4a11271af4590589cf27f4075e0a5140b06943b58d4584e9

response z

722e9a5b4b5890560dbe6be7b558c540d5375042e4a9df92f903935013655e53

d9748017bee2e21a3aafb24b5236c116913ff7cb534bf17b35ce94797ea5d897

58c0d9a3d9e4afcd17f69d7a2429f910d3a5a4c80aae82aff368e7bab803f460

417bb6b3b21a97c0e432ec5ce37d070acb7565417925e0269e91d84e1e841643

b87c8af0c24878d879d54eb36e9983a025f6f8cd0ec00ad0818777d7caa5763e

bcc4d5703043db1df0bd6ed5cf91863684cca2e59b652d578e3e071e8655a8dc

5a79eadf9a03bea90c18fe2323e24e8eb26a9c07db702db48494db5ae5b59409

f553d184270a70881b07823af4e6786bee35e6971d9be8cde6bae207fec5944c

response w

3a95076572d3fbc1ff2986c8a740d99098232b3bdd4574ab8ecf112c1f89ac32

d32b68501cc99026ca5d3645bde6f77b86d66365b991881b6b7f8a0bd53f85bf

fdeedf23e27c122a3151a1df2ded06b12b22813e54dd2a770a3a223b22bd2866

3444040d4ee9a98a83fd1f66082c558a29dd7e67e5332ee416aa91b041e4c098

6899d08ffa0967228ec35668454115f87cab24ade867d44a33a97fa2726df41c

ff9cb66677f0ffc87d5263aa4df46f11ba46fc3ae0d9e688d46d649d93e531e2

084c53913cb5e8535e090368fc76b84acd4bc6adc616fa5f8562dc0014a9af6d

8b144d4fc52eb26a08278ce68d6970c9d56f34dbfd1d7c07ff8d8eadf7eeea85

Fiat-Shamir context

892cdb3aef94a6da23cf2663d108a803772afcc8907e540324609cff091226ba5

withheld proof term: 585 184 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR definition, DirectionRing:160

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

f4a8d58c8e6d7d579c906c4ad9ce2e059a8a4fdabc8448689dfb40ec497bc3d1

SHA-256 of the kernel type

b4c603655b56bc71be86915b819032c24d2bce7eb97319d803fbe6c82437e547

commitment C

d6a64b572d92b997f9ef8b6f1dde550759fe2f2c4fabeb738c72ecf150727df3
dd8bdc1e8230ed4e2273bc0f2a1be56e052711eb10df0af3a7405e78745786df
0a41a76efc5a51d5e142b37041e71cdaeef72a43d5e535c2353b554fece949ef
afb46b057ebfe0374c2c8da6686831f48116ed980840a06f92f1507838c3c3d3
2eddfbb90981c5dd6da97438e39c913b98e586b76b5b665c4510b2e61bd5535
6f51f4703d0cb453d91c9af27c9f8ae2e304adbe1268fb1dd119411b1c733f22
8d78bc99a965be027eecd7e142c0706e61dc3c027a9d9e3cb8dea2481649fe15
e2797d22b8e579764825c093980f360b5cc6f7b3678ab0860bf03cf1d1e66047

first message t

b7cfe7750e0ea3d265dceb34b2b3fa6458027b902b3d710b422d182d398f296a
260e3189e432d6f3fb1a2ac1760694793c98d8ed882bd841e0044f400cdcc20e
5c30deee405ea34bffe8ecbbf49f965ca82507e1a3db638b57c9f17e5d284db7
64fa74ff1eeced72d3f4b6117a67e04544634e2f1389cb351ef03fa2720d91fa
438dc0634af8ba3092515262bb79d7c56dcee785445ab3dea7a1ea3d680124d1
d4ed14f83e13566c12f5fdf781a3d351a573e311d05b9a461cb93ca07395121b
d0eaf6f1b07d9be215329ac8b993fba0dd0099219dd04786ee7fae966edba434
fd566a8def1c079e58190b243c1ea7726b7ee9b447e81f16094cb3c41721ac4a

response z

6a41c8f74813f5b223643f9984bb14d210eed77a599b31a9afc56021638165b8
f18fa24ddf29c49cb8c6d27bed2099a28d364a9ed59d72126a740dd1f1b453d6
cda7e72421e693aa331337bbab58112ff7dd95cdd76bf0b475c8e73a1b28d84
2ef22fc73b266bde94a1cec9bacde16eec2ead46a425937b49ac45c5ab966745
273da9e88c92e2a318aa33e26f382d682b85c39ddbeb67c6fa845bb553fb90e
fe44b1532686909450b9631212c1e4cac7812c70b01763c41f11cd6c328a4687
6fc0389065c0dcc21ff7b480fdd08a457829b2da2e276dcfc5089fa4023298a0
ba77c8a3a6ed6d821d45c71a9c65c16421d9cb9f50704a73c1c9b5225f4562ff

response w

47a4c75e3f5e77114aa2f4d55b259be26b306944e8e7e9b18438b53b9a17e62d
7f15febb2bf746a2e43d68ed4ff24cd212c59c1d9184e83b290cda83a0d79ff8
1e679d6ff0986d23f7cea87466d2daef7050da3a1c1f9828ca536eee02a496cc
1dc4c2ff7db6b15802c7b9d41d768188dd4a7a9a3050e76745ac83e818e26a4e
e9cd7eeea89a36c06d630edf48b3c4d08376ac956add202a0aaf9923e6a2b0cd
0f89d5e15fa9b4b6793c8f9da7b98ae5823ce82ae7f8d5b6541f1cf0ad6e756e
5299c46eb130e645dd5eeb539c8ec468998489092e94b4f37fb72ba0bd894409
4ae751a922650c83b5b93d947117a0132f5878c6239f4dad2f99d39b8d8f3309

Fiat-Shamir context

5e017447609aabc8b4423f0fc35c152146062188b9fc868fcacdfceac139f2b1

withheld proof term: 6712 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR.eq_1 theorem, DirectionRing:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.derivR.{u_1} Φ)
    (Matrix.of.{u_1, 0, 0} fun i j =>
      PowerSeries.mk.{u_1} fun n =>
        HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑n) 1)
          ((PowerSeries.coeff.{u_1} (HAdd.hAdd.{0, 0, 0} n 1)) (Φ i j)))
```

SHA-256 of the statement above

c745e13b19298ebd30c2f3252fef96f51aea8603dc7840c05289e6e616db5f94

SHA-256 of the kernel type

dc05c1a065fcdeda4c18663c1c408f2094040475848d8b2edd9a145f292d2e01

commitment C

3eaca4b47bd2ba63b5c6c7822682605dfa4afeef0d61181e8d30f4fcc16e7dec
03489cfd08e91ea7582e783bdc5f6d113458b3c01d3153000bfd5f67d0339dfc
1e3ca2d68005ee2fdbab34f5c113849b461cd258567108d43af9d4537fc77456
172ca4cf302dcff304e1194670602fdb26c593461425f4a89f1bb0ea6678c71d
9f84f774fb7b44d9bf10ef6f71bf76bbb58117bb8eafd11f53d61d16407f51a0
969670bde9ad5e928e0bae822f1db54822f3b6726df1173942c7f14a86b72101
f7ea4fc01e3871d9735b590b2655883322c1edda3c24c0c0e888a42dc4e2460a
322bc7562be2bc5ce363594d2fefc9748b398706a9e52922f3037577935db714
first message t

1d1604eab338a2169e7937c5c1cb6a57a950e9cfb8c63f78b4151ff279850320
4c9e9638fab2988acc0da6c9811321944c3b52f3f669d321c060a8942a083ce9
50770c197443c9fae1e68b980a99597ab8fb1ef07eab89d8c7d58dd72b5af4ad
2163b120712071c20b1299f3e27a90ff56e41e1191136bfc5c58ec2073f1123c
fa9aa0fa878d52d3b8322b08074a7ef6972c96baedb6b2bfaa391d6e299c0ef1
903cb2d539e423e9077b55f6e0de9e82fd30de446ecec0f791c5e7da2580150e
8822bfd3ef25011b3440ecae2660ce6715f4c890fd106a867cd982873f87b9b4
8a18d697bfe5e084e33b58ba3b55277dd459284de3f73dcc170a782db0fb279
response z

591a58cf2c37e96a4816c8d5d339dad62de7eab4308eef767582e358ff4fb1d4
74f22dcccff873e1ae663e8648fdbbd89a9f913f9b50d2cff4a41cbea7595d24e
0208020ed20e0086dc97e8cce424279a5af525ec8438a03ce634408d127abaac
291cb7690f03928b1881cb480c9d322cf23781988499a169c453b1e7b2c598bb
dac18da3a4466cb9254dc4f9ea85260b0be28087ba86743d108f101ae839128d
ff0287373b9c5b9e785d16c6a59b789e7debc868350408f651b91d6b487e16b0
9393c57f775d4baa803dd0a3e2cebce9c9e39fe59b89609825218bee07fe0f79
aab88618d959f028abd180f155bda4b68559a0999606036e6d192eb94d73f1a1
response w

3f447b6c1c8013143440c9c24e0a449f75505ddfa4864f9df1938300cb137cd3
36662bb14a58b3873a768b9d43107d26c20d5b1ae071d0c9e337ddd85e64c0e5
c6c12f49ecde5db33d840c2dd8b8f90b5e8fc148e15c2fd9930de0360ac8b378
d645ef2c2cb6108e53674ee4b8cf1715d4026c5c605e57dbf49daf62f69cc78f
afe89fc28d1ef6bb2f3655e45c782b514ee7b8d09d8749619fb2f38d6c0a47d3
bf32aecb7214a881160181854bae014ec4060975f0994eeef930f8868ffea55
7c360f9dfe35f9afc0bcd2d021bf9ecf57085dc3dedf1bf5b792a5c10a629fa3
a9afca9b40d696c45bc4dd77cfbae1b0328330afe77a42756879b9ddc15dff9d

Fiat-Shamir context

914b476ee048d983e4dfe524b00f2411a8c66507fa30e69e3683e6286bec5a8a

withheld proof term: 241 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR_add theorem, DirectionRing:176

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.derivR.{u_1} (HAdd.hAdd.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (FBundlePoint.derivR.{u_1} Φ) (FBundlePoint.derivR.{u_1} Ψ))
```

SHA-256 of the statement above

b3a26d8cbc4f03eb7c63732f12e606fb226420e63e8479d6880a67415a72e0c3

SHA-256 of the kernel type

c0cd17c2bcc58c420995d5fa714cf92bdf53b832e2bc121d81b0bd57ae1e724d

commitment C

f3bdbc0597cbda4e54b043f02c7c887413681698b4c5893ad4cf1ab8423c8695
a5af88ffec275bac070600e935f491f741b7dd3b349c3b005e7c2c5ed546e595
7c6ab23378fcf59d0438e8f038cf73cf2b92353195cf7367b19c58906d236688
4072850dbd791673c041a14b06c03375ea52b116fc309a4b3d9a0938017c1c74
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456b6816cc3d4fd2154c44087407182a72c2382925b4a1c347243b251a282243
bfba2e91deb01bf50c854b29f4fe57c6bb158216b4ec435a775d6a1a46cf5222

first message t

8484ec51a67b6d6a2f07fb35254e8b16f3bffe9affc54f1c64ce87231c0380a0
d11a8a510600ab86199b0575c719888566d0a921e938708e18b13ab4a33a91ae
a9472e34246a28a77c9372e6aba299f8e76df6ad766d2ff9f7fb06bab7f2f251
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3b226a28bb4401179aae2e37c68cfb733c684220bf52d17bd5e794dc8db2d586
975d0d684d01308830a32c730b7a24a10af590f0a581db37940eb9e968a7194c
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431d1083549509328564bdd34a92bd0b2853a7af249f575fe1271befde358d72

response z

6a3d306b17bfa4cd5973aaa73223f35a78637f3f64f85c798e908aae7828dabd
2accbf895de07c70723bd5e3576ea6cae74e1d798d2b5f464c5c65ccbf2445e4
1da2d9ad349989a84b5b03df3e98eeb39b32b682296842b589d733b587451fcb
772d7d91ed37f1e0972aaffb22c65cabb30b4d2d639f1e08526aaf43bc6d99fa
d67862c1719aed8874e10213fef91bf832f2e2b4cf06df3414a7fe238cfa343
adc6cb110d52cef6c61969380d6cd5534282896c611f057fe990f30a69211b9b6
ec09fb575f4d63427b9a43f577ea55d0d6bfd3def3fbdface119fc9b3dff3fe1
009be595e11a70007a81ab48d9df54a73724dd7994f820a98f3c35546b25e4a7

response w

363929123f1e9ce11291854f19a1247225f3c9be8e378620f65dfa50ea2e654b
fab0803e2f5b6aedec124c6d53e2e6394c9a39ce38b7aa0760dfe43ea2a1e988
054266d4fd546288bb831bef5bc33aabe60d4a5a4bbbe8a8264627576d480a2a
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1374699e50061c6e350400409b153d8320b1e95f801e76070795928c845830ed
3902370ed7d9a391894ffdcf43e3a5d5ed42b06d2782a9c98ac36572fc027292
6416942034c5f963f0e093464c5a7b2ba5c4d0ea1ade42deaf703ed4eabd33a4
982f05ea82d594cd1be18835def16b62c080503c9b4be785ccb347d2e7709b0e

Fiat-Shamir context

376256f6a87530cb8a5f8626f035b9a5eda7afb3cd514556a29a9a2811e1e0ce

withheld proof term: 319 954 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR_apply theorem, DirectionRing:169

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\}$
 $(\Phi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)) (i j : \text{Fin } m),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.derivR}\{u_1\} \Phi i j) ((\text{PowerSeries.derivative}\{u_1\} R) (\Phi i j))$

SHA-256 of the statement above

ff7f810e38741def5d86e07d44e391976ccc917c83d6cd2b3cfaf6b189dbab1c

SHA-256 of the kernel type

482538a1dc95f1f4e25ca7475b4df8cc4f51e58cf607eed9c610537e0268baef
commitment C

244d554bc87d97a51bba4b9e5c1ecad87df721d1a88824a72fddc1c4d6f53167
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c83caa04828a0c2169ebc203308e2df06e44e2d6049d81a6c0fcc1fc81397622
e979167d648e88fa2c7489f91bbd864a3fabceec24dced3a9e60eba8aa7da5ef8
bd135d5fe6802141a3a3d4a7987214a054d8273d74189ea02b4bda4f4fec0da5
8dbca116ebc75246527e0b1ad91b382e2e05c05b2b143abc54e3b00681d74214
3d38ee699c06964ba11d0f131865d1486794e696ff5673e7ee23cab51cdd4b5d
ff22cf2d92117747aecb26012bc99a03c853659a5d9694a5ec3348c676f2824
first message t

6ac53f0132181d5bfcc5bc4e6247cb8d234f1cffe36f47191983bc2604ad869d
c425cc310ea04115e092d3172bb9b71150a925a6757a0bef67f410ba0b4fab34
654a1049ab83d089295540ef1508efad3d2ac87ef087f1e9e44b58c343cfd915
80354f00e365f48ce5d97250d9c24ed6b881c340dc49d2c60d487440e652cca2
83a65651c3c54b6d9f1e636f9d3b03de5533aec7a9638100093c8d3652c11715
17e3d5a4c3218944329eff10fb8777649ab071e8c0cc63785dfbcb81a474d60f
b5e3cf6cac4668b5ca4467678a67a4516dd294846ba5d03c72569f0e76887649
73988ab8c1edb0a4a02d244aefbe2678e52b34b18f16fd4e11f09c7b58b447f2
response z

1d6480f780fc9de9ca043a7d7d83bcbcb88255c40e52c28831001b3cd52d4d636
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fef8ce4dc9892963c35e4850c4a2d372382464435cd624304cae8d7d70edcca
68647ebb03f3cc517180b0cd512e3aa82905256ee74e3ec534f86f07312bdfff
602a8557c336192cbb458e555128c3450938229ab462e432c19b65e7b767e63a
f4275b9c111b4d1ded0a0a12239ab549ab8eb13257feb89f4dd4b6f64991836a
0a8d03faa033236dc8067021c8af35a80c555db022eee9f50f0390cfa5524f12
593f596b3e747d3400e1d664c397079c2bbdcaf59b7eff0560b601fc353629d8
response w

4e4a25500a4c6d1c1d80e636eb58a55561cf86c27004b4f4a3d944231e069ca5
e3faf20b26b7f3a52ecf91a633607e7fb251bd57706b89dd00e61633ce8bc287
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a6662ce717d6ad1905d569f28bdea29ee9d8b4a361476717ce8337073266c1e8
10fc0426532d18ee508ba7a51c526fb2e44c51c2ce3d38e4263943ea0b680be7
cbb45a327a7fffffaa74b9995432e636efdee8865d21c707a0cfb80cd8c30a92
b7cf0f7fa7bd8fdcf7f61d29049aef4d8f6d58eb80342a9b98495071cef8a757
691665ec535c3ab8c5f95cabb081a4653982b6b5ee8ce0b8c1cd514e18bbdbbd
Fiat-Shamir context

01a6318103fa158d69d98f4c0539ceac19d497f1118193240f979a1cea1022f6

withheld proof term: 143 878 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR_mul theorem, DirectionRing:182

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.derivR.{u_1} (HMul.hMul.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.derivR.{u_1} Φ) Ψ)
      (HMul.hMul.{u_1, u_1, u_1} Φ (FBundlePoint.derivR.{u_1} Ψ)))
```

SHA-256 of the statement above

e4deeb101bcbf6e708772cba926153ecf2e630018b4f819089108783fd39efac

SHA-256 of the kernel type

f3c22eabbd45902bbd5dbba0654e022d9ef262ad40c703fb68d2d2f399874146

commitment C

a24ba0032fbc5bb606b1c5a5b8218dc64667bc805d1ef36465a1792e2d0b33de
184c5451ae482ac6711fcd050f5a3ea53484c600385067c64860719ee698cd8e
2b25775a230777401b3c622cb0440ce39089aed87a1adc7d139367c7223ee009
ca8cc5739683fd30f67a9c477e8d157e5fb3223bc98e4af5ab7846e7e0975acd
f2339be93fe844cf662b936d4b700de0d6398d31abce89699edb86de7676600a
ebb85c839028e66ea8f98380447c5042869fada2f00b1f7f9d42fc236067d67d
db116aeb0871f56361807fb51e1d8a3546596d33b9b9fce390c93480a3f47ee5
caf8fa80eab3bcfade7713109a5f44a24c46941748dfad8c4719415c29291fb

first message t

5834893c4649cdb156fa62e9fba53710b12d8c09ea898b99047ae3743006b33a
dd1dfb61373e9fafcfabbb628b99b6d844a36914986ea634b48b4b5a84577cd3
f702cfe497308b7ea70a8af9dcad1614cd52110df0f6a7dcf320f2fd11b413bc
048f524012891e12b7f9437002ccc09a60bbd1d42786de359d82e4b6ba47a157
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f2144f6424081798721f32a912dfa524dc450aefc1bd104e64a9582ccc68cbf
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341e3141cccb5aacc0c679d1e94b06fc15337c706d10649bcafffe3a1f2a5b96e

response z

11712de0839d1e6d682af5cad974278562418c6023247289d2b6102623e28baf
c1b0648233c5f327b82c6600d40c880387f0c8900bcadee42533dcfc30d8e16b
dc4803de7ab3e346a648116b0aefdde0244f0b37c5002b85461c29cbcea7057
d7b808d7e3e9c6f58ce5a23cee120ef3c2758418ff1ed207e110c6e1efdd242
72306f5a369cb06705cb2816430c47187f7c30ab46ffeba9f1144e6884848328
d82df82e8ae9fc79b4d2db784a9a692cd7eae12bb68c960252cab6c6433d7410
6c93ac3d901a2dbec6d8f5867ce839581be3b340b03a8a82bdfdb4dce279fe6d9
8ae84ca80c1960dedeac86b469fc688a7a8800bb84538f9f480c6c0d3e3f1c72

response w

13d3f2e882877eb7d4f07061134ed5eabfd0db70e19e26c188759372960c404e
f8c1bc562a9b18a7f2f6363d025de2f7384a2bd17b43e90e5985230c740b4605
ba641d67ec5cf2041d3d23695bcc29e17770b8ab2ef76f915d59e611471045db
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db6123846e91c588997d2f15cde3645ab00204cc0fefc955fbef1366945b465e
fd122b5e25c44aa1097b050b3037c0acde708601e972c6fcad667eb4825e3f2
c04ac231b0c5e4efdd3f01bb4820ff413242da604ce06e7a952d6207a837eff2
8af11460ddea1b9acb774f267ac98274a10cabb0da7a9daf9eb3306039f467ce

Fiat-Shamir context

328f1a7d8c4f7036ff5a6e9927dfcb7e1dcbb368de5873d3bcfc11d2e171bcc

withheld proof term: 1 783 240 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR_one theorem, DirectionRing:201

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\}, \text{Eq}.\{u_1 + 1\}$
 $(\text{FBundlePoint.derivR}.\{u_1\} 1) 0$

SHA-256 of the statement above

09de5d9ef366abf6662e8f869f3d9dfb22630f6e7b7090c204172044e20875b

SHA-256 of the kernel type

064a898bb16b1c84b5d411fec25c343d8ecbb7e0dce59893b9903fd9a53d966c

commitment C

1197220ee2f335043b877133c839ba3202c54677b831cfe695414c2ab3af15a8

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first message t

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response z

2b26d7b960e33303ac74a361f388887492a2c22798fff2cdc80cb495a0da5664

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47fc2c0aa89068e04431133acd184ae59cb17a1260b3bb533fe5b480a5d4a1fb

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response w

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Fiat-Shamir context

f1c2650eee2a67ce670de4426ced53b46e8aa66b0ba9bbb7ce87a2d31a685059

withheld proof term: 57 502 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR_sub theorem, DirectionRing:196

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.derivR.{u_1} (HSub.hSub.{u_1, u_1, u_1} Φ Ψ))
    (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.derivR.{u_1} Φ) (FBundlePoint.derivR.{u_1} Ψ))
```

SHA-256 of the statement above

fb459fa085ebdb2f10f0f9b28c7efe5204a8618bf8ee3263a0d1fa090f5ffba

SHA-256 of the kernel type

93eca35338281ea427996ad3a5175876cdd8fd1a0ee7ee4eec913b644b7011d4

commitment C

2c020aa1059208f0214a0aab44dfd8fb77ac3540ad1596594bc0fee4d6895339
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1845c7955a05110e8511b4f83481ff467e2f27394a4843d88ab918895241185f
1d961dafab5d8c4dd2e1071db8f293391a0b9a1b5f358d54a1ea2ea0de5c96f3
first message t

604d96b8dfea81abbfdb55fc6aca870c044677eb4d3981c34bc9651c2942401
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8193730c693812c65dd8aeb44fe3c38b1bbbe331f7066baeaa177249ec18bd2c
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87a06eea4be91c4d4032bfa062ec1479aed9c9651507df308c706d7c72cdb390
c488b9c2cc867fb6e7ad86f97e40018d805498c2ab10df1d242aa08d989b1399
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response z

59c47b304a05e08c452a48c90d39a891464c92ed34000b65d86247a4968b350e
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response w

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90d058f3bb57dba4e6444b9818b3217ac58ae6230f242b605c2d36c34c6eda4c
68ab42aa4b8fb6c71d7bd925b8847cb62e372a5ea8cdb20c817e00b39541227f
baf30af5f0bef9d0d6718af97922a79e18383da9175f6c8acc7c71a4fcfb7928
6a6a33bdeabd6d9861920fc5e39adcb88dc29f7c61d60a9249fbfcbe542deb14
753ef77be0e722425a2c96dbd28db7194de6eb3c61c3afa168702c60832d3bfa
5fd2e87a8d7b95d8eff20f4ead8b90dadd9ebd15db2f11ccb886bb3b245f186b
9a6c145390f253c38c23c5301e655ff1920a9c521f9c1bf0aec94bc5653b75f6

Fiat-Shamir context

be8e046bbc09783eb9de0a0add5b7906eeb8e7b1738911b7d5f06fff8c0e89b0

withheld proof term: 503519 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR_zero theorem, DirectionRing:223

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\}, \text{Eq}.\{u_1 + 1\}$
 $(\text{FBundlePoint.derivR}.\{u_1\} 0) 0$

SHA-256 of the statement above

337e93fabddbcabb0be9d47032f8c0dbc0db1e15ee16997330ce7a4499c2a5f6

SHA-256 of the kernel type

ff6d7d8e3daede7a2a32f450dde84a32d37d7a68699b42a98404d665b7336f2e
commitment C

6b761935d4506a95e05bb46b5620a1ee63da3650d78ff8aa6aed5816b0366bc4
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cd9f1727f7f4f87648643a38700023af8f1df6f4f5e5988b6a96e029dc679f56
00e0ad4134de242ebfd7be593c92ce858d9ce2d88b33e13075ba273cc33b6bff
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3977c73a5e4fd359d5ea2e9cfe1fdef89c34fd75476ac0ad0677a882489a2612
1c27546e8d04fe5906d3356f3755dee18b3c22f48c801c513a9a1478e8ce86bf
first message t

3e818a2c9e3e9f495ddd564846a45c0f5d5961dd7bfa026abee7435a6380effb
6c396ead924f28332565f4557f4ecb2ae3283dd5c5e24893d84764491d29774a
21c6aaeeb50cd8d06d3604369286e1a1ba3b554ad1abb298eaa654d4b4856b2b
deb3c00c1dea53bdf93d556baef658ae59006b3bb491b582769c2f5cb4337cf
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ab2c031a2b9956a5a2ecc7e55898a8dfd859e97935ce4e984b62cf8ce3aafed
8d74abdf4cbd3609ddf5727958f214a6647b617041162035e766f6e2499c3314
62ec7ee7ceaf40172a0d7fea6e584641f456019787673094eb715e26187a59b0
response z

2d0eff097eddbba97cfff166e7dd3bdd31b485c0c9699d8f5e3db23c685cdf01
914d4eb7e1650713c05fab0b9d46897e0f2e3cb30f6ce81ede0d7ca796d7456f
451ca7cc5c6c81d5be25317703aac644c2fe166a2875b7f69df4179b37f69796
87e1810f5165f74f9db2be9988648b32c50c8a076fd9d5f28d1e950cc695affe
28967cea1d68242888417defc231f28480697f6c1f2c3d5f1a9520c2e6d2511f
7e81d87facb3c58a8e26a47a0d809148d2fd8db3e8e0a052dc1226dd0e073b83
618a1dfdaac43750e46c1d4f605eafcad0978af3deeeebd2a0e8fd09f442b9e9
f00da5466caac82c1b75e3494a9b4aff9c62587e6b09fc39a435253828afaa51
response w

8c389fb4d87991a5e86e10af7015ef12fb3a54992ffdc4773399ae419a89745a
13a58002395c93d31e4d631cea16a79b2ac8c385a5dfff3e4e40aa5771148278
79e793667673252d69126152192820049def77dee2a3053f058d61b394550e3e
07e7b3aa6d78ff8c2d8812bc29fcb5292c14eb38d47b45adc61aae43201ee101
3e71c829a83050415c42fb76b6480f000cc09e52ff6526ba96491e7631c52220
0d833c9dfdbb8321329ab4ecbf91a848c503bfc3bc62caf42a8fcb50766941fa
cf4245664f74041c295224fa0948bd9faef4aeca66247a44affc39681bf7a3c
48f92cf7861e9d491fceb685e0a3b0b8401cca72fa3f3b40dc2cd4ea7d3977

Fiat-Shamir context

25799534453730fcd370022aede4fd172ccc7f2d4d221b604f1eb3843d61c460

withheld proof term: 172 128 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_zero_of_derivR_bracket theorem, DirectionRing:302

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (X : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.derivR.{u_1} (X k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (X (HAdd.hAdd.{0, 0, 0} k
          1))))
        (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} 0 (X k)) 0) →
  ∀ (k : Nat), Eq.{u_1 + 1} (X k) 0
```

SHA-256 of the statement above

4f782e91b141f850170b7929a480438a88ba28e2d83cdafdae357c1a4d5547d7

SHA-256 of the kernel type

c8a0f3043b9c4e2c391459bf08381af00a6271251e658a53a522ffaee1e55f9b
commitment C

4246f1a6f16218bdc759734d8569361ce92afc80955713de44fc26248a16c503
cf567d551ebb23db3691c5c127ee16efbe32800082d738dae8cfe0cf61644989
555cd4bf6ee92a41eb631520c5d9d0dbd0790eb0741922bc5370679626246ba4
2925e4d499651a2df84710f6fe5131c71e30d5be05eb91af12a4231fb4ea5ef4
59bcc0c6a95b5dd61836b5855c2382a45d1179cf717dfd6f0dd7156fb8fe56a
1e129cb21c52937f173428f42c85e5e4ac0dc0e13a3d018051ad1c5a0f37e88e
bc987d6b49e7e371aefd0c39aa3ed7ca31f764928418b7c1593f3a61f94a837e
083448a14ccc97f325a28cbe8a0abd6cabb2c262a477fd5a4bc203349fdd5467
first message t

a431cedfb84214cf2d5975aa9878f7de953a421c142ef68bba930706433cd970
f7fa8e09906cd8c6e49094821fe99c0487ed953ea610d2e636e0df79c0795332
527e4818b281b5fc9a5e9ad122c4fdecfa21133ea80357e14ec644c0c960068b
c5159f67055af1f708f5921eca4a36f4d35c878c43b14e44cc50ae794f5243bd
1999690bf69336e4318d9999d2ebdf9851fabff9f1754196dd1d816057aa45e
8554dbeed011fbbfd18bfb0cd15adb69d7faca6025dae596c5ca801ded127739
314303dfcba99c4ca15c8856af4d30cde6f8ef7d66b791ee65f486191dc8fa69
0d3d87208190594ecab0df4935c69ab169e42d635ca3ca98e350004be725d670
response z

412c38b09a6d078f87db5cc3a65dca6e9d5ac6e6a5c6ca92d40900cfafe45f7c
7ba5613429a55903ccc2c82cee24223b71aca4269b0f3465829d2506f7b59417
c0cd395e3913f50c7d077c61bbf779d9e004544dbfbae303f5955908fd19dfa0
c370b83ce54d96251d2fa9c3cfeab055fa0254d35092999cc21926d5050d07c4
056d34099f76771bd8ff32975db91987dc57037ba3cdf61ddb0efbbbbb1b3ad3
c2e3a5f5be016cc9739ca2fddf672fd21c8f4ce50a7097cce61f222e65119575b
822561a2e6665f328fbbd4b2d15fe5a3606342dca396c7304c5006537c5085aa
57620087826b03fabedc48aaa1ccdd773d97fb2e5ef552cc12915a780cf50403
response w

18cbb41f3bc1848fb3895b2867bc472df9d587f8c90c280ad38152662986a3a7
b117caacd47a3bf782e37582f464522918fc148ce2ccb233ac10a31e923d4ca1
916d96717c8ddfc79d2a59871843f9f471d2018b706b17848c96b06aa40cdc5e
9de7ef83a83a0ee94aaa8c4d2552863b56d8ea7b3941395ae983d4190af237d7
4a499c0335d13690081d67071104f253e9632896eda97c74d912be2ca8baef4c
f14761e0c8ae24d72d5256695c0784c33119b0e2040fa6cfecd8aaa5162243f0
07d7fc4517554ca17db3af087a2c5efe6329f35656a192dbd4a487421799fc70
60b3a31dc06eaf3d5dfe18c95cfa5b67f832d398e00bec798baf2ff15931423

Fiat-Shamir context

646266b2a5112af16594bb3637da9f16461553489f1b5d06c46892c484388ec0

withheld proof term: 129 234 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_zero_of_derivR_bracket.match_1_1 definition, DirectionRing:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (X : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (motive :
    (n : Nat) →
      (∀ (m_1 : Nat),
        LT.lt.{0} m_1 n → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} m_1 (X
          k)) 0) →
        Prop)
    (n : Nat)
  (ih :
    ∀ (m_1 : Nat),
      LT.lt.{0} m_1 n → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} m_1 (X k))
        0),
  (∀
    (ih :
      ∀ (m_1 : Nat),
        LT.lt.{0} m_1 0 → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} m_1 (X
          k)) 0),
    motive 0 ih) →
  (∀ (n : Nat)
    (ih :
      ∀ (m_1 : Nat),
        LT.lt.{0} m_1 (HAdd.hAdd.{0, 0, 0} n 1) →
          ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} m_1 (X k)) 0),
    motive (Nat.succ n) ih) →
  motive n ih
```

SHA-256 of the statement above

d1de05e19bf6fdeaa6a8ace370a0bb1070a429acb48d8cca9512c4cbd2052ab8

SHA-256 of the kernel type

85f8ddb75f5d2d72bfb8083f84b9bde4ab92094034123d3c9783bdfa2e2ecc87
commitment C

84c701c597e821f344f98911cd03baa9a5e0bee3e9badfb55cc4f3657f9f41b1
0c1fc1b4d4af964fcf8952393b1d9032c9db840d84e4780a708c9d228a263a0e
05de8a3a5758c460ce204a49e0c2d8b834c95e584e588a8e37fee2bc22429a7e
1184057d81ac747818344f7b775291188732498b254e38b071dd09e62f23e82
428dfca2f14a8f5894ad1a9931e08a5d478f843ecf7771ecd73a90efb37c8a55
ce9006b1d35fbbf454fa64828b32f37df8d49e46df2a44cf5a7de81cba21c459
97fa5c79ff8a5557c4e62da24bc1f667a7a4f946965badf448aedc26368a2efb
89d532b537e583b7ac1a3644e107cba8d467307e60c988e10542af39b8ea4e12
first message t

b725ab804bed71cda437f9918cc967dfc0d4d4f299065fa3187297a22c8afdf
7c2c19b4936239cd373063d78e350e67a29b0ebb29cf570ecf53e627681f4531
ea2115246d3e8d9979602293441f65b1c06ba9d82b3c435ac816f592d57c021b
3561ef30a18a0cc1bb50b4c16bf99e66e50a8958842ba9d3cdd6a861cde5627b
9fc29f331059dbce610aa78e157c726545856098adf62727985f01b7afdf95eb
88a7659d7d38e1e6cf869958d40f0171c466dc4d7ea9f309644c44a5116fbde1
c7356a927799612ec3c76732e399c3a78670e357e18c6b895ead365c132d595d
cc5b6c08c528dc77d2881546ceae421b6ceae8e7315673057ad3fab5915b813
response z

c585adcf845e0f88727c3585369fb8bfec7fc29f5e186789ed7efd9a5a2988be
934eedd3cef62cab6aefc9fc0f8b0f056ef8bd83b26c3ffb4cdc3f8e8fa3f790
9abd6f16de39d578810e5e5e999f8a2fe3529de19462e183c98316be20ee9267
c51b809064900524575d35952159a0ce55e44fca6bc0c5436cfcd88aca980385
fcb25ba413df3f2cff55ed601af4d3bcffa5f7e33a10dd7954098347a178d17d
73d902d4d924f1aa53cec5b20ff2eb5626a29dc314435cb40e7df5473a62
ce53464614603b50a08284815f7b8efaa276480a0dc5ba72b2bc0cf54c02c0de
e17d36e5a1fb35def90f05386f0a233d6a68dc740071ca79581547b6506ee90

response w

17b517f690842516ef8aa454dafd80edb5efe14eae0f94be4259f898977d3e6b
6fc320f5d1aac591c238bdc1a03203738731ec925c23e677fb4651c63c0d09da
e1bc0e8bdf063cc6b1943d31cf7bf59b614d577699e66640e0587484d8497cd241
31d052cd994dfbf6f4b695126629e507457418e7ec90ba56432577e5bea311bf
2d6cf18c5b302c5d565231922ae2c96c5cf413b2f66e764175651646974d002a

FBundlePoint.eq_zero_of_eulerR_bracket theorem, DirectionRing:344

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} (FBundlePoint.psCoeffR.{u_1} 0 Q)) →
  ∀ (X : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} (X k))
      (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q (X (HAdd.hAdd.{0, 0, 0}
        k 1))))
      (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) Q))) →
  (∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} 0 (X k)) 0) →
  ∀ (k : Nat), Eq.{u_1 + 1} (X k) 0
```

SHA-256 of the statement above

40ae67c9b29bc45f9410308ecac98c03f7a78accd2439067d2d7a3a57ea128e6

SHA-256 of the kernel type

ed4cc78ba616cb2ae646d399fa25343f03da6bf683e13f1edb0e7ab8f2548c8b
commitment C

e9b49a758cb249ecee9254c517d3e8fe8f589d182c6dd910bdb65199a8348395
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ac2aa763bbf411d05d9436abedaa8f2b3bbf0643d5e49bc3f0b131c498d9123c
95b6832b52bb146d335c331ecce5618519688ac30cac92f2b86d703fe0f3f494
f49f2c134957d8c28b659ae7df22fd12073f582c22dedb68331365a2629d51c0
687161b8d6355a206c27167ea579f59769a2f30e11ed10584da8e641ee252796
afab3a4967b9f8020b30870c34b1ff14c81fc91ed85106a12470d59a5154c612
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first message t

ccbbe96ba1d7f4391f3f85e44af299837e75fa9b826c857330b83e2ae662ad9e
d137ad1454e7d446877aeef1fde7d7e1c29789edeaeaf550265b56be2280f34a
e89bc82c00868e18edcbc2102f0b629045db6e6424d966bbb2c7ffcd8d4024fa
23bba6f88b42cfd84244b49ef98309b1a9d21568a8d1ca2448b247f72c301880
322b571e170cbd1f9c02db75e7c7e4b46f26cb78696c1a1841ed371e26335a8b
4a0dede2d8dc45b859c8d7c86664bbb8dd4af9b9aa7238f8b78fcd84400893d5
a7b327c8245ce93bec849c2ff89e49093cef795dd7eecb68c6941140c95fb964
d3c3bab1cf156f8f5ca9aac8bb86e8f76112f5841bee7abff84ad7a1d94b01f4
response z

215c4e7a46b6044301e85fc3af5a987c28081c3811015b808d13b141c71e4dae
cd3492021706e65b94f55caefbe0f621f3e12242154dab5335da15afd5162a40
d0e6e95ce63f484755fdbca479d00fd31378bbfd28ac16842a18e793cb91a6daa
0fb1822f1bf53552a2dbd6f50ab9bb0aab41cb07d61e3b958c2ddcc8f8002a5b
eedc950badbe31e99e2a00729167bf74dd809f9fd2495e7ab637817956a0ecd
1767cf4487cf22f0e1aa3e8bb37ad515fd44c925943171db23eb7cb9646639cf
f018556012df7b21a31e145b4d9c8079ef0fdb76bbd9bfcf0b0fc1dfb4db6bd
5f7ea341ae669682bc0a67217729393b61771cc72d2caf4bd2357e32523afaaf
response w

6735f0d2c5ad2acebc37c89b375e548db549de1d17fc7b347776a269356973ae
91a94892cdee6be42ab76bbdeab59fca0fad3c77b3423944a17083228e84e50b
418efac63062968f82e92b0a1bc290403464c14764af318a9cd5a91af7972116
4cd3fca21051c5a4d4a47dfc509c8ea2c18e77550ddac5c63935333cc4781e1
4cea13a7520b8c7fd90949f6cc87dd753b7cfe2535d4c9562c3b01f1010b1e16
ac4c00fc4c1e5e5c56b9034805abb9dadb200d6758746681b2b9cd71ee3f6c25
8c08fc45fecf07a64db5d1f8902dc7c6cea1e495cc4321c26966ce89712f4b08
5af47cc035d56617da5de6c315d90329d6c141b88a0965fdbacd9bca467c342f

Fiat-Shamir context

f6268abc78bcae921989611a378c5d2bd1d1161786488fc92311a3fb159b21de

withheld proof term: 459 207 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_zero_of_eulerR_bracket.match_1_1 definition, DirectionRing:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (X : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (motive :
    (n : Nat) →
      (∀ (m_1 : Nat),
        LT.lt.{0} m_1 n → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} m_1 (X
          k)) 0) →
        Prop)
    (n : Nat)
  (ih :
    ∀ (m_1 : Nat),
      LT.lt.{0} m_1 n → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} m_1 (X k))
        0),
  (∀
    (ih :
      ∀ (m_1 : Nat),
        LT.lt.{0} m_1 0 → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} m_1 (X
          k)) 0),
    motive 0 ih) →
  (∀ (n : Nat)
    (ih :
      ∀ (m_1 : Nat),
        LT.lt.{0} m_1 (HAdd.hAdd.{0, 0, 0} n 1) →
          ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} m_1 (X k)) 0),
    motive (Nat.succ n) ih) →
  motive n ih
```

SHA-256 of the statement above

d1de05e19bf6fdeaa6a8ace370a0bb1070a429acb48d8cca9512c4cbd2052ab8

SHA-256 of the kernel type

fbfda7e8f96c343b45aa22f72ea132c9ab2e6aee07a0526ec95588cb85a40e68
commitment C

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a8d0012077c5384013e4999b2261ff52a27518563dfe1e255d519a65ca0e873
925bd31ff51fdcc615875db2cd8134d05016e137ce44901ecaaff516894af52b
43c4e815ccb3d436e86cd33baf1554effc0e61b2cca165051afe348f3d64b3a1
8425077829ac7adcc18d0fa228cb4d671cddd4b0683627ef6eb8aee5a02201c5
c182efa401dba4c401e8c30e686d8002775f7ad43053ea0757dee11091fc83d1
d2a3c608dcd924046da13cb91a8a0b0b5593fd4a8656a7f7aff24643053194c
faffb0e7b731ff9ed9f07cc66959c09684a4ac73400a299ff8fe1c38c45aede7
first message t

5cc0586fbf01bde9f25f9cf310886a5a0d138b885d34435b08b4d88a94d62dca
14bf3831eefd0e2fb14d672343b13067ac881767e6b1af59450df4151913a5c4
2933a040ba9d06f888efe1b7b851b3eb35af4773ec6f5903d409a37fccb4920f
2d517f65c9b86c10fea35c0b90be4bcd6d09589271a029200c442b60793f9f2
d4bb3aec67a6dc531be6769b5f122b213e0c166cd68f23b08db9ba6c128b095a
77a42a642726cd2702738cccf7ff0dc7c3d6d0d49feb2cbe01e1aee256e518b8
461e369f5d3ef9d75ccc886efff08cfaea3b14826e0c8d1294b9e983a2ae0f8e
6eb5dee581a8f0462d7e8e57429179879151b111b66182d33ccac7aa7c9bc9b4
response z

3e4f0d0244b646d7c4a0ec2f24f4e4a129d9c5a8161d54a7da23fc52f6ddfdaa
a1a4915bad87babf1852cdcae2de689bdd4982f865c6ff3f0937e000757a90bc
dc58aac10781519c196c1444bbf4a0b5268505cf69af3b6a51c633f64845aefc
039782876b47034ed99c82b1aef93f5483a621e1246eac952bed7f3aa18be787
b8cd696c7cbad60d83f9abdd112b1aa4ddf3026799a2833f1c366836ec31c5e1
3ea0a66f90d29d88e2f2c2013b7dd7f709a0e069e9fce2fae78f3d53351d7a66
60fb2a5ec6b6cd8c74cfd124af62c711987aec0f044ba9e54c4214dcf90ca3ea
ec69dd26e1fde704d6ba4fcd713964b4207c50d26941238aa7a72a2714dd144c
response w

7467592593e0f7cda6cfd73e4e65cc54d6120e573085d3294a06fa17c077e2b9
965657bc443221bc02fe308109f4a0627aa458ff746fa4d4ded3c61c82d08b30
b57eefa30401010ff73636643466c1971cd01c95253e364945d0f9eaa9e088d243
119ab7bfb59da43be79984d096a7d12c653052b43a987378503c8c44bf80a2ab
59509684b6cf2ce0a4c12a9785e3e1537d3db3e0d3bf8ea6388c8701abafcc2f

FBundlePoint.eq_zero_of_isUnit_smul theorem, DirectionRing:47

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{M : \text{Type } u_2\} [\text{inst}_1 : \text{AddCommGroup}\{u_2\} M]$
[inst_2 : $_root_Module.\{u_1, u_2\} R M$] {c : R},
IsUnit. $\{u_1\} c \rightarrow \forall \{x : M\}, \text{Eq}.\{u_2 + 1\} (\text{HSMul}.\text{hSMul}.\{u_1, u_2, u_2\} c x) 0 \rightarrow \text{Eq}.\{u_2 + 1\} x 0$

SHA-256 of the statement above

5b4600ffdb59c3ba6e3d893a17ff0a705c8c801e35eacc55524acd3527209f41

SHA-256 of the kernel type

a47baee2958b6510a98b252fdec8937f333d4da44a6a745dead97c2242249b58

commitment C

b4e18190224ce8d9ae49010f3eb8e2dec7d8a3721139cb013915b65d1e12e0e7
d1ca206e93eccdde2451ec7d14de0deab4b212646a7cd62960edaed6b73aa706
9d4e0660f3fd97ed5e8f5562d3134827a1c1ca2126fe2f7671a5ef918cff3cc0
838638f307857c4fa4d1f4ac889d7d9a1af49618be9eaf791cf75ea03c4656bc
59802626d1aba0e7f075dec573bae6872460cc774930e395231f0b04114359e5
15483815e4f45f26f8c75185a4b9298b3e345e0bf7914c190ea004e447affb29
64d8b00433a416d8840299c777eae1f8efd01456e8b11346028b21f44d0d8d90
bae8bcb11191f40d40643da8e50acaf456064a6a0ccb5d0d25fdac021e028d79
first message t

c7b450c0c098d2b82bede0154058e132d426dde1e5f74948b818920ad4bb228a
d36538fe511768584da24f332aa99c8538598e9ef4dbf56135c28b6dddb55656
2b7faca316ac5cf0941cc8c3a43d81681392ca50c0c458ef5efeb99520efc943
f4e2cfbc95ccc0b426c4d9b2a9795d053f518815d28ddba462fb00421e2432dc
47f3458b3cdc411c7b7203e52a39dab7d9976c132a25b968879b58c488f9e0cb
1089ad79d48d4c6ab1afecdaa76cd95261f506b3269a7bf7825ffc02002966d2
3b968efdd4ef30740e55628bbb0fd1ff2efaf33e3ff1b5bc0d6e8169b47567ae
67d43920ec17800cdef1b2ba474616f73882f1b03def914ce0e98218fb4b8522

response z

310e1aafafbc8477c45508578f3ba8faa72228043e08561b100ccd90ff2aad02
7c563646d0f7eb6fb82480be69f3288ed41d166364e5d195274a0d79c1c2dc8d
96df5dc3d1c79b846dfb780b4d252c46d0c999952101325447c65eeeee6e7ae0
36fe48dda33759311cffe014b2c7f2718b416c4cea637c41ee44613cdb1a28a
686f4a4a1e403c62ad98fd74936920cddbbef64a4e070ff1d8a4da4d91808041
03ecf085811f8f9449a16e69ab73a1e992a86b9bd8936945880ef3546b6f702a
50c615cb0d7b25ae0b95d7f0796ac6f8d614991b96c9dd5b458433f07e0b6de2
c9912c0e97bc3fa6203dc5bc21f0368ad3ff25ebf915ff055ecc711e5dbe1e1b

response w

3ff858a0a8edaa8c6ac131d9819ea1fc0b269e8c8c7a4463b1d6809cdefdf95
df6d069e41e9a69ec62304ba64e7dd64117fb46f20782d73b66c659ebd122470
40f026d46893553ce2f86e4752f591f18041d5e6ecb761491d6e63215a0f6dfe
11eddc92999615d155ab7d4da4f2324851cd36d1720744adeda30a9c6329585
041cd1534878cb427bea4d02dedacc88fd2f81a89899b34aa5c132898ca30652
c0efa559302732deb8a60b86699b93a9cee40da88e70589085ff9a0afcdcabab8
c4b1c3dc7c00475be45a0db93a3b4a8ad6cae5261b3707ecd29d70fd11ef7990
4b3a0fdb6e036858322cdeda414813da80d3a5067a3a19c076b09d8689780a3f

Fiat-Shamir context

6d41d45c59a4685523a4dc5f5da0a4b71ae15691950ddda7770eb7313c73741d

withheld proof term: 83 484 bytes • axioms: propext • kernel replay: yes

FBundlePoint.eulerR definition, DirectionRing:206

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

f4a8d58c8e6d7d579c906c4ad9ce2e059a8a4fdabc8448689dfb40ec497bc3d1

SHA-256 of the kernel type

6141d1d03456db3df1c69d55c090bcad07dc56f9f1ac5d11aed65f8721a345f7

commitment C

7d9fddb71cd3b6423cbdfc964094ac0ed9180915be27591c0300d72c477a256
a53561101117e4c9c5b2cb818a21b30b41188ed7dee1a412799b726ff88b6d2a
fc6c20e5004a7c6afa19a5b6358179e6bb2b4ea254ed8309487923101e80e91a
bc37c076f541c25766a6496596e0a7849717dcd6957797773f0c3803d7a62258
8eeb2ca439387d1a1026f86f905be6c1198f645c150b8effb97f34be3109d122
69a04310f6a6b73ebdf23e72a3fea9c28be5a1153d875458f26be0f321a843b3
75082f8020c9e9eb0cc8a6aa026a57666fff69e1a8351261ca9463db75fe9013
4f8fca4cc0ca8ac2a664e5fba237c2ac55250f8349c0f3984d79a2c99d7a27f7

first message t

fb0caa50ce8c6c5d84a99f84d9d961601603b39641788f8b748ec3917e2a64d
a48b2b08bd73e32fdf000dc500f2094370f1dea0561176cacaef6e9c3c52c70f
f7470e973409413492bfe8730ec7a39f12e123f4078b85d63a7a60e4aa4b3bcf
c027f051eac09f9f6a0d97a0820ba33ca6a14f47f8dd66006feb14ae6d7a1064
3d773a1f767ee3795813d50f5f92e47c465b743f3d0c2338c642de050c0fd9e8
7609bca2cc2ae47fdbde4d154b56db22281869654ba3b3e70fcfc5781b949c09
4ad5e293cffe6d6171e24c86a6d346c9e0c08ab8b2902c4038bcd0b34f6a6745
6255e0751c9b48fd7c00f825a79dfa481b1691f2d4a206ea49d5ff4c5cb93db7

response z

3af811e991f826cffeaa740512f70fc8f4147bda84fad054e312a902bcfdd113
49e1421b252a16133265af64b7fa31f2b3657be71ea68313d54193bcea0a5278
af88b2be9d9ab17ac1939d79a2c27b4aaacce8c8d20358847b454f876343c250
c389ac4dd449ef9c22e8754634b6dd265de1b4301bc8ea96b757c613532c833e
3c88fdec09f59d99b2ac8e8cbef75079177957476662c2e682d646f6dff470f
15e2dba66d4996215a1c7354d1255a0e6573c5c04128caefccf870579f429ead
af367cfad23e9918da13380bc9a4b7ae6b59672b928d95088d35223e68a36ca2
f1ea3e37331673de8bd7380aa889d1bce74d0e4f80ff7b68500792fd8f9bc527

response w

56f612a412db37fbda6909f154ddc61efb6cb086d9c80e7fff07187be970b144
8397a04821b8db7497b137f6d3339b5fa01b4d05ea8ed5821358fdd796db2bc9
4e2d3ef521c4dc4c7b4e6cd5ce70799ca485969cea06ebab15b7d656c051ea65
e28fb9be8adbbacd9c83872b4fb082f2d68fffe1a1fe270344c1f96a390d6f76
b6b6075bd73361e2cdabdf5df78886c666266c3a0cc19e2e995cfc2e72e9e50c
8a4ae42027b0019cba6072811283d8bfda0531351ba6b9597a1a034c10fb6f3b
7a35c3d97ede4c4f95bad27aadca134c06289ddf0705aa6d7f2b0e99f9e457b1
b5213ea033ccffc13658729ab95ec87bdb02c8f25b16a3963caad8ebc872e86

Fiat-Shamir context

ede57183775c6eb3e99f693c8d6c8803a6dd69f69808c629d2b4de175e0e932f

withheld proof term: 5959 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerR.eq_1 theorem, DirectionRing:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} Φ)
    (Matrix.of.{u_1, 0, 0} fun i j =>
      PowerSeries.mk.{u_1} fun n =>
        HMul.hMul.{u_1, u_1, u_1} (↑n) ((PowerSeries.coeff.{u_1} n) (Φ i j)))
```

SHA-256 of the statement above

4f29ea5b43fdd3dc14daa438d026edef7d7d0dbf51f0369d5953613a47bcf416

SHA-256 of the kernel type

3c75339d06bd9616073bf43e17eae390f91008b91ca10479894219e9b1c7f25a
commitment C

761068df79b85a011f4b0c886cdc2d3efe0139d473233f66da22ea76505908d
85327ab2b08ae7220685e569a142e68d10e98a3bb26afbef08fce66f08e54c0c
804ca9127009bd675ed262f61c7ed0644a902bdf641f78a0d1e99b9718b2d1b8
1b0d637a081a084234d1f7cbbffda68bbe678f3dde12c01f6f8b10681fe917b8
8fc455fbc838f7b255711fdb41efff0493b0206add456acf1ea6fe4ec751caa
42ced38fa88aec5eef0cfade082b1e384c92938c6cf5738a89a51b3d522accff
f2e5f83503c7c161dfa9433f3fc96a15c7447e1558a51c0e05fe3ab5e92f27cb
09b9e112e60b1c951e7dece5e0d093b21d2463a98f3cddba6ff1d45ce6e1720b
first message t

473ac89ed338f33102b15121048831abd59902c246a5484ce364aebc3e49d1cc
e98f6c96978a7f6bd7f4ab59758eac8c289fd7092f0f0ddeae216b53fcdfeec
4a0b46dd5dbc40e3f134cd955992a51438f153a5fc92572b7ecdad385fe779ca
95424b13560ca432f75830fc76e7cadd20ccc4f96b41d0c111dabfc3fe621626
3fd9fdc9d57cc3f3eee15640880684ab10db0510d0c8545fd907261986c614
b101427c1692c46eda3f7f3d0bc8dab22cb3cfc8dbccfb6acc95b1a295b7849c
05b47c2cc0d1e7c075fafb062144f8d0b9e8d830afffae5242294bcf50c08a27
0bf3a305b2b5026866a3f1dabe81d8da50c64234b692f6cd0fb3306e17e3df4f
response z

194244d4c80b423098ce35813e1b534b802f3a23b18025c4d304be2100834a5a
c2ed2e90a521d3dbb44850577e0ed4e5a14425d2424191885ea223763ea35655
7fa0422692e823cfff99c68fc22d05d05da6a2b0299c118d28db1a0dec9a0ea4d
2357678e3743a8d16abc20b0dd73542021c7f769aa1f030ca942b4d4a5e68787
3ac66e5f49333bdec81f0abd0dc74d5bf2e48f3ba22912a3b40d1561d87aa2e3
2614e42cbac09d891870a00a061f90284bab3275b0a5fb282978e6153d51c9cf
66c55fba0d3b0369a805c070f4874b2eeed3cd64acc51699a2aec9dca64c354c
f3cde5551dfa7f325602a85d609c87963d51a0e605adedb426cd4124abb43a01
response w

584b3a7943836c26ac707a74fb7fc0c44a0bf663df8d22ffe8ec00d02ffc9061
6898383869917bb21bd9701634b817e45ad30487be62276d7e318beeaf363f24
20ee638045a5acdaffc56c0996e973e6a0fa4025bae90a37886083d3eb656369
0d75e2ba9a9478eb0b97ad1d25f27e0f52156f15e083d99192ed4f760e06b289
475e5b72807153a4086fd16361ce0e34d55b370f4fb091ffd8dbc21e6893c4a6
775a4ffa6cedec02883b758d3251821a3b7840dca435fc658cd088d878b6407
8a1b0c760523e92abd51398a20aafd0636d544b3e2c27195ccb3a6a716f8d34b
ca7b421a0b5024b7730e96705f8e6800a6257419e90252d8298208cd84966509
Fiat-Shamir context

8af631ac18d2b20b105a29feb75069ca824991fa4330e2327392e92ad4a05307

withheld proof term: 241 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerR_add theorem, DirectionRing:242

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} (HAdd.hAdd.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (FBundlePoint.eulerR.{u_1} Φ) (FBundlePoint.eulerR.{u_1} Ψ))
```

SHA-256 of the statement above

5a49f2141d1731824055eab90a91ec72be308c562f4bf31229c182c4a463d8e3

SHA-256 of the kernel type

7affece19869f2940a2ed0c8d1c8912fa4db09683d274818cadd7b759b2e9fe3

commitment C

803f4499feecd5df7e6d903fe647d4327dbe5e7b182e84908a10e5ca0f6afc8
afaeeea1dc3275b99dc34368ea6e850d2208d4370db77b82b72e36a5b54cae37
688bb95094b73b8a93318b16ef77bc8a61656935837b139170722465a77022d3
a021350baae6478a935da7782a8ec565f69b207cef37ca93954ec8b5fb63837e
8928bda01a98bc479dc1a05455abf347bbf6f49534aac9b71ac52087cbc98291
a8e9ddc32facd5d02f0ce15f848435537d3c737d6dad5481465fffad6dd2bb2
e5de172529a03dca96ee60427f65efa36452384792994e503495d51be4c3421a
1a74f8be10a292872d386522709caaa935baf2b4aedf61549b72dc9017b4b7c6
first message t

bcb0057f2c118164e481f2da62c80d784662307090a8db37140854c7a9c103ca
3803a4e62a8b886f54af64dcadd6b42d10e9ecb1879659a7412c02aa1545b4e
c62b0615a38513e63c3b75cb8c96276c42725228a12591be15bc577011e81d81
1a22195603b8cb8f21e823308e32746f8a60df44f0d6672f09d90f780d796017
910d32ea387b12595487d41b5817bcd671bb5a05bb9cafee6847cdd7e8a64db0
918edd95158c9dbf42265cfd3cc67f5456b45c2a200e836679560d309b49e19e
3b57641419c15f70742f1b7ad40904b073531efa07edafa188575c49fc391857
80391c118c110a19573a35992ac9640a5ea692cd282c57b0ca09cd4de2f0cd8f
response z

2bc1cfdacb157f2e48de062aa65e29b616672f2680ba19b112bd91f91a88ff7c
056e0c69abab501e58a2af4aea3c477aa4b90fb1e4e2ee016c83636551a81f72
027c501b0dc5219d8af8c30837f3bce44dadfa3d8900095ee54cb8ce59f1b106
b0b8f7bb4c729269d12b04fd43f09f2fb52c7a10c190c905cf9c41cb9897d370
f256b0d41700bbab477b5d8d2a81037b33a504f315233703c1216e109eeba949
4462d9a592e755537115ff62405e4dcecd472e7780645659732d8d5f104eba00
33556184ced917894229ae2fc75b17a26d3eda46895b47246983e5517cd83061
669ed92be5ce141d30e15e647ee65214b2961cde0c15cf91452a8eb0910d2831
response w

3d68bd8a481a51f27a3d9f96d1e2c7ec1d3eb3f8d82407c52e4676a7cad4730d
e93ee59f542574fb75dde36c8c6988ff95f239c6f8a78d0f9242f7e6ceec68a
9b6f596225c0e7b491b8dfcabfa8d4008b1698646ff68840d74a4065b6981058
5e1176d034df4c846ec619db4308ffd4595ea992622df8f919844c2907708128
b7a2dc8e7fc1a06727509dc7009774544740e935580f7028cf3f1235387a843a
5b3e42b12196928e058708577b5117642c8864db0d64e92f79fb76dafe217f37
3c5c5ad4e0d230b18a1a2672627b72debc88676e1c08ac8a9f8eeb35433ca773
5c75d1497a9616f4234c3e9abeec25d94b3d8187e8c56a0199bd33be6c6ba1dd
Fiat-Shamir context

8b936b235e35b8dcc89cafc4c2f4bf7d07ff2ea5141a49c64e59889e73d35e99

withheld proof term: 151 588 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerR_eq_X_smul_derivR theorem, DirectionRing:231

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} Φ)
    (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (FBundlePoint.derivR.{u_1} Φ))
```

SHA-256 of the statement above

28f7e68e1c9f0c91e645c6fea5981e6c036d112b10b2967f0f0d3609f8373ac8

SHA-256 of the kernel type

da5c155beedc23c60e68edb06e3469c690f709e84810dff900b26180c0681f8b
commitment C

1be2e8e694ad1f5fbf3e085122f5f886b0045027c0ee284869c5a6105c1fee14
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3dee5f7a1047c6cfed359fdf2a9da0e6c09cc1d3edde6d70d8f22d2a1f15b274
7d4636095f4e37786ff19e8b34b7bf6fed20c59ff1a8cad2e78bb77a9e4a0e95
b6a980e365e70cff787c007ae73da76f20247f61e7fc5cb1a3aef50f9bbe14ea
cadd92a8d25c6ecf8bd834f0a7b0e74ed35ec96a7919912a3daa56c04431694
a0504fb57a2ea8f0e5d4b014ab965af363d721f6c52b8c0888633ac808543147
64aac0a1eaa21e8c2e853504410f2edb8c0027d15a2d337cd7451d1624699c14
first message t

7f739475c180d07460de899f89b2a2df7da36fc02587d047497ce6490692b470
c85e04fc769c3e024cd6d9abd0ae9609862a9fa4885498df924cbe65067baaa1
6190397b5c868f202a4060e42ce78df4b6169f34738ec0d014cf3f6623770a7f
50a7c6146ff01767409e8c78bc63a8e1fc688348897097b7f57a0e61b9dc32e1
63f5bdf1bd12195007440789010fe86ec29c2bdb5f17d89b4f9b5a300eb8641f
c564f20ab59858afeae207b073de7c12fee4d37acd0764a56e669f2ca18bf73c
bfc91287d6daf0d9e0101c78b446c0bf9786da100d4e036145c62bb64709491b
e86199f0adf780038b079ff51f1ca605ede7420b296bf20d56e5697d93958077
response z

63b8e804b2d384a44063ec53cab78d1e560aab7f30a79122fbda8471c64f92f4
d6eae53c6b3178b5fb83aa0bbad19888ef87a17afc7674e1e18e3ba1da1929e8
d4c27eba931be9b5f62449fd8217375fa6bbbc30e5cfac8db4bd94c7011cc3927
f6008667f631ddac3c3ecba27b1a7f1f54de624a1bb3911df5e21b4363e7d8b6
d934da5665c50eeb99a915cbf6090dba6f861050c659043e2235304b13c5ba58
98d4c8e888953810925e36b05a6ff31c75cfc161ed47b1b1f2b1a46ebbc08ed0
78ceffcb43f081284805fe45ee223b9a294eb9deb5cafd29b0434c3b960e643
a929fd3ee167be87d4f1de9ddf6518819dffbc53fa2eb1359579db5110cc74ba
response w

4e93c8c9cc11c64a5071b69254d6bb4e3e61aa945b3b38b9d54077f54fed38d5
1dfca1c3c67fb3a7cd30910337a3b9323dcd56ccc86ddf14bb76ce944006a847
3f768b84d1b70b55639245923b391a1d624bf3d507203866445d664e997ffc7
95df8e175e02f2747e4bccc0088dae50ae588eff9dbbb28c1053834b5bcc952
6527b0121e3e55a199059e17ed5b3d5193247d3011769a3f2f103257efd96faf
c623ac9e46d35c642b5e4a1056aecaaf71d303407a1fc4afd57ae0b04b1670
119bc3e1d91c8c5722fe5ed9045c9819918727400950853eda7d37c07396c750
2ba51a5fd122351e396363c514ece43ae9f32c613ff572c15ca6f4006a395af7

Fiat-Shamir context

7535414e1dc4e1f994388f409662342b00cd29e2c14c3b9c47d7749fdf9c7cc5

withheld proof term: 610 049 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerR_mul theorem, DirectionRing:248

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} (HMul.hMul.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.eulerR.{u_1} Φ) Ψ)
      (HMul.hMul.{u_1, u_1, u_1} Φ (FBundlePoint.eulerR.{u_1} Ψ)))
```

SHA-256 of the statement above

f4d50928da08a7c861931b973b6c90a02622d7956ce425a1c1785a38ef23cc3c

SHA-256 of the kernel type

c1dbecd97bbea3de19bbc0c332aba217a119bcb45b35383595e08041a4f14fed
commitment C

a9545a69d367f86a81eb9002bb09086d11a603237ea2db20730579a274cab37a
d30ad89b56db06b5cc950802c2c64abb631840efb6ca5e18837cc4705548648c
33903cb297d9b419cd9bf55af93a3132998eb8a8401464e2c97f7584e0b19280
e124e04d9763cb54bfbcb1ba9e974b07444610bc879d26a286d593d50b352f82b
b8eb4f80172223784d935b5fdb6a4c8a808fa8466d1b846bb3c39637cfb24c0c
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fc216ef4ee8b5f34e1c7c5c2756a61e4b0de7f34f4d1ef268d450ad0a5f92c23
53f06f18821c78c7608536228ea93d310ee4e7ed33271ee64cfccfd95539c50

first message t

5a4c717cd50832a2232c5d01261132d05668faa57cf805bd37e1eb00b52142a6
e5dcfb13fbd5a79dd4eeee65263366e30110b5e4c2ac03a2a3a4f57baaea4433
50823ece0758699e39485d5ce63df728a9cbb64dcdad1b57aa99a32e4ff2de20
63d5327f34fb91297cf87efde22f8f414a9ffc831c6690d13749c5e5680068da
75af34706cb97bb3fbab9631c599c8dedd0797ace41497cfe95a35f974c3cacf
eb7c4b5d9b23d69c0ecde7fbb818a658231e093758b8ae74e5ced4cc3dc9927a
1e4e4cb01ee5fa0368fe527acb395d6a014b05b7dcdfcc454913f838d29d5c27
1d25fbb4cdea0270af11e3939193f960cfd93a86b2560b3e8f728ba8c60d5007

response z

50f501b2534044b67dda9970cef9c351695ed2651e91796063e2c96e72f42d28
cbbe0087dbb7b6119b3cebe151d957f76b1b7beb19e9f0751faab67cd8ad5eae
d400a8d09a6c643047d71bcac54565697caaa6de30149e35aa421f73a6bb56c5
bb2c846cbdec1c53ee051379e9dae660798bbf08cb6f32024fac03d69daa9285
dc3e51c639ade753eb2eb93b12762dda8e4ad035593f3b061380abf342868478
c4d418817b409fb7402d5f3b19f7811b114803e4c8afc9464feb41a8ecca5b5d
0e53862637db10a869f426b160b1cd9e889ddd0057c6e0d0d166187ad89fc400
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response w

7e4ec38ee283a3c9653b5571da6ef258d81bb98f05ef1252afab1b760070a79f
505966ca85f7d3356d49ffcf0ea6596d03b3990fb2620006ba50904ee4ba4693
658a0f0e300e732fe42e271e7638d56e33a74cc0577a995cceb9c2a7ec3299c9
2a3b067e8e088388cbc53539c2e8c8c8046e77bc24a43c9a7286054a1e970acc
3233bc4ab501bb8937c913afee1b80ab1245bbd12c0e15f6b8a7a3247fac78e4
e11b64133bbba7d886a4b97a1148e1ca553437c25a37911d20996e4f60140715
654c41127b1173ee9d3f40f1335327dfe896d5d17f27d51ddc5a2c9ded149412
608ba458253dd46309af7df3a13e2872facfe8aee31a73c53c78ebfec8fae031

Fiat-Shamir context

e9cc9b62151dd91389d24e0782973aff37b84e5a66a572f811cfaa2ec6baf810

withheld proof term: 407 188 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerR_one theorem, DirectionRing:215

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\}, \text{Eq}.\{u_1 + 1\}$
 $(\text{FBundlePoint.eulerR}.\{u_1\} 1) 0$

SHA-256 of the statement above

3e1fa3f377ded2420cb7c796d71bdfe798e1dc5f5dd800393b16eef51cc82b34

SHA-256 of the kernel type

b996b4ddcb0dae315fc63adb4464a45fd9be51577734d9b6067555abce1d1163

commitment C

91923b9c6a179b233707b1275b6db82b354cd149b5f42de4cdf250ed0fd07bd2

85dde760cbcf1eb2f2be21d715131847fda68a4caa31f7517b1d9a549ea72701

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74d492e31d3b504851b5dc2f51d003fc1a2131217f6d1facc9c3a4a0e4f06678

6cfb49be431b5a5c2d4783f68a0f6a9b578e12e7d7c114520cf5101136417554

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0eeefbc39b29a2020135da2090ccb1eebc1a208ce666e31fb77ff3f2d4ff016

first message t

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97f259ca391ba76b6fb98221ec57362a0bfd300b0e95265e7aa0eca83fc2614b

e7dc9a6f7738b6d3b200cbc427ea3a60bd8474691a6de1d4302211e7dd647da6

83262f149d02a1debf610f91b5c0ca8c5123a0de225b8562ffa434491098e159

response z

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57897fb9df2118c984a5c8dcc6b442fd4e82a127bd9c9ffd3de3352b1a5d2eb5

31bd950f593ae7688b8543798a1dc2ca3634edeee22e0ceb1406f56bde809818

response w

795cfbc2658da51ead12ef0d84ae0880f08f3866c09dfb51e6ee19054b74bb93

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c818fc4fb091c3d0df7be59ecb83dad42cc2a6226e21a0ee87fdbbb4dad66111

Fiat-Shamir context

703696904d3896a5daf28e5f85063dabb86bf7ef80f5de4b06a842667f06b1cf

withheld proof term: 87415 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerR_sub theorem, DirectionRing:254

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} (HSub.hSub.{u_1, u_1, u_1} Φ Ψ))
    (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.eulerR.{u_1} Φ) (FBundlePoint.eulerR.{u_1} Ψ))
```

SHA-256 of the statement above

b37d2a0d992c29ce6900deabd3c4e5cb831ea4e6e2505c829d47b4db47848b13

SHA-256 of the kernel type

e7bdb7ded1693139b95874c22758287797f905871f83a3f5b6c22f42b7ede05a
commitment C

bae15b8aa0eab657c32881d20dcb8920ebc4414c2134cda98b8d16f5b8ed8a1f
17e53e75ca2e1271f7876cc4654380f416cfba00589f7d300849f6314cd607b8
bb4ce08bdab22c9cbee6606096df5faf6251e5eacdf9c28c1de52ea028f51c76
ce2d2e1f272b7ae35a11ba4bfdbc8c2c5d23750fe4b71a48139808cbd217a59a
7baa36ad7b946fd722049d5ed581b5e362248a27916caec749522d82edaf35ad
b1dc53e405eea06f7d11536ea0ac54c5ae81f116f9e8dd0595b75a94da1e815b
2c8fa56beb25b9e198713156bdd899266de1bb206346a8bf57e81e57f13b6e1f
1726be945c94d366d75db37aa09896efe2fc30ed8b442747441aef00a07aa700
first message t

236cd9dc3d65b53a53c9cc0f6a0485d1c6254abdfcf3254686bd9a1638f36d89
d72e900f4a554ed0a6eb0aafefafe68fcd23380ca0b0ee72f686fca65a27645e
d38531f494e13126a50536b44ee9ef66321c22cb225e669f1a13302c55d3676
ac76bdbb87818292f0e55369edee348375c0062372b1ed2ca4bdde94ab82396c
7f45af0ad71ff6c36583dd164deeb92d99a1b9402b9aa379d8fbc510f145d73b
e7bb0557ca8ce13ac951eb59f39b1913ade4323a923881a62da469463a58cc97
333ef6c98be0dbff1ba269b0f9da3ab3f71cfbb38b52f449a13274f43fe99438
5005bfef3b3d3213cfb61fa4493c75cd6d8aae8bdb7102ee8ae9667406d291ea6
response z

2122012754f3c9cb7533cbcf91e67f46de2a149cea8fbd6cddb8d52d4a2f6ac
4778523f44a057af128944a11c7b2a4e7b0e6355a96467223f696a1949a29553
6898e4f95bb09470e19eb7ce1bfa89adb55661dc9a9e3ebf1e285f418a4367d4
187de50b352e76bea78264394ef69d7ed68306c1d960aea194d966a2635c2a6c
f40da1271643a7d8f026dcc901a5dd406fbbae744951119a4be991eca4338341
6ef159c8dcc049268ccc623797ea4d59949d2daae50febe9a39c515cde5947d
5ce8ba221ce39bf0f3779dbe6ce92e27e3351df464d38e4b65f73e86b76c3156
229de16f09cc85f4a57dee5646800980bf4e3b68a0980993890830559b527515
response w

434a0d207d2f744611f27269d39ed191d04e0be6bd015a6ecff6d44db7cdd031
cf0e6dfae4e1662302c21b8b93f82eb3ac1bf276c50def56b5d98a4110c69a49
8cbb57e33256e1aded1a8d7bc59169c8ab9a681b0e5d8f581ddfc1d4bf702f97
536076b251ea85910379e111f5cc786ced95b9229adaee8a731fd5ee1b8ef208
d670308a518eab7a515a7929725bd0f267bc0bbf3f1a805e43f7fc880d89b91f
c33f09c853a4a6c2f8711c4bcf9e2a58ae59a42d74e7bbaff656a9b4595f8be5
ffd99fdd7f96c33ee7ac09884da70a81c9f07263f4ab7e679444418bba81dafc
ff7cf6e0a826d1c8e91eb3faf99caf717139b999bb30f3665eeb62b888f98f46

Fiat-Shamir context

17c7a888acc8db9104215a23b4c2dcb656d55ca0d451518e7b8cb5b3ce23ef44

withheld proof term: 455 555 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerR_zero theorem, DirectionRing:227

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\}, \text{Eq}.\{u_1 + 1\}$
(FBundlePoint.eulerR.{u_1} 0) 0

SHA-256 of the statement above

32fd012ec02d1d38be3d8ad65192a01b8877afeef82c51ddc5746ed3d4e0103f

SHA-256 of the kernel type

ec52c8e8d80f5f0a477130c95120c4f6d0330ba600c6fb55e57e81b2fb300344

commitment C

f044a5f2a8e65f399858bed21f64be513eb5692f9a00029365e73e7488e17513

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first message t

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response z

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response w

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ed27b815fc4641acf91676e30f749ebd1195e1c0b1c833c6de019c35ff014d47

Fiat-Shamir context

b9a352122064938bc9adbb9b99136aaa38c622e6633b1c2b3c9373bd32ccace6

withheld proof term: 163 950 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isUnit_natCast_succ theorem, DirectionRing:38

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] [Algebra.\{0, u_1\} Rat R] (n : Nat),}$
IsUnit.\{u_1\} (HAdd.hAdd.\{u_1, u_1, u_1\} ($\uparrow n$) 1)

SHA-256 of the statement above

aa61632f0c4763b102fb7955f99d8396a6b7a2cbcfce230f5d1b696978f2649f1

SHA-256 of the kernel type

9e76d10cdd1808be851e7366764884c1ec37c432ae15617c60b9c5008ad3e89e

commitment C

22c94b97e8ae2b56785e52ca216ce6ba2f653ba04a6e1d20dd410d7a088c13eb

62b4fcadf745de41373de3d472cdaf141bb5c9c0e0abc171791724f502b007e7

f3d14e21c03037d0926a340bf030c33687c7f071623da82a5efd9b0521a06591

6b428beb4a28804fd5549a6543228a6b0e934620626630b8432941caf5532ee1

54d16c36678391d27e752bef5129703e57b55f43cf6499b53b984ba5088b0b2f

5c174dda181bbdd1f99ba533bf4cec09caddb2840391b6b39bbcbdef7267012e

e8603992f4027fa61951dcbe754945c1d2c4750a36e277a8fcb71dd9f2b00451

ceefcd8b03d0675c5241f150671884a24b544b25faa7ffe176656c2bfcacd6f3

first message t

908db6aeaa498151a959799a2bfc6ec00f4c4bef4a3c8a62801175855b8ebc97

c3964c9e3add5fe6c70d0df7c1838e93162a076ce93a0265d07e24b29e24c6ff

3a4ec058b14859fb05a6987ea053fbe83643a8d98fc3d20bb3b5e247308796df

43fcb206f3cbdd69ef7c2e5b22130bf6da54f3f839a5825b77811d8095a3c3e1

6beabbbc085fcacf815532ae6a90223d7e71dacd3ea02caa2996968164f5e3ddc

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b12937b795fe382afc3ed0a48f7742b1aeba9080935bdb3b8b18a72a8b66b245

6fdfc6e5bd8434621cb01779ad6935abbf518c47ba8e3aed7ba5c88605f939b3

response z

4a57891a948703110e0574ba81b38918b61faa82be8d39d7dd54332aac775e1d

8d7b5b4778ab8faa77b2a0fd8a522b87a48ada9747c0d76e5ec3f8846fdd1d7a

5364d4f2f7f607f50315dccc4d732a21ae65f7c91454df66d7dd87400e639df0

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e3bc31ba93a067ed3676da280950aa51e2ecb61d454de76470ac6613995eba07

response w

6ad93805c924510fc766a840e1a54714793bf154a12865f9e66f2d56d5a7fece

7a56a66bbdb899f3e182b24f0d563445e139f545ab4a4cd9613badf76c4477ad

4c0fe032e8855a449fcf5db2e49146bfdd6e3f1bb871bfe4afde9b48e7952c51

b8ec4abf9fba51188f34517e7dbc9f3d8e46205319d23998e3d623d328ae991

71a764a035c3075556ba0b1b1994aae7e426ba338f03362f1bc8a77f3879a74d

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080d60a3de4e60d4dcf1e170bf4d28dcc493fc7d7a039ef1831dd2ab9cf7da06

b25d2882cff6778470305d8b67af2e776965bde4f39e451033cb6922f2e321

Fiat-Shamir context

8ef97bf34a9ad63eb7b883f12d4d623b2703421ca442d436a7c856dfcf06bda9

withheld proof term: 159 382 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR definition, DirectionRing:58

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Nat →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →  
          Matrix.{0, 0, u_1} (Fin m) (Fin m) R
```

SHA-256 of the statement above

a38a8201f9ac7fe6f1c37b6df41187dac5ea09fc13a0c5f8960ca3d3f00af5a9

SHA-256 of the kernel type

5a65964eae02828148043362b21198628668b269a9963e8affc83147e5e8c248
commitment C

f291eeba1bca7e06b476d53f8422044b99990d0a06a953afeabe92b0d8e91fad
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c95907a7807eae7cf6f20259fe32398b1b1175d67b9c4a6ea25f0d01d6a45fb
25235c465ffec951dba7bc7dc99fe8444527eb33b19feb70f8e2a8af3a60495b
ca6f257531ab86cafb77014085080cda905514c6c86d4bd59b072e6af5f2e22a
first message t

b031c9981b81db27d71dce22cc550f2f6ae6be3480b788ad0efa32747282fe61
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69b5cdea0dd02bddc290e4368808ad69d8d5947cb7e63b5133aa3aa336134a5b
cc95a9605cc2ae77ee3e1f0a98f56fa9963fa6c922b7fac195db0bb74e798eee
ce9f13d9ada7d26db6e4cb4e0b8fad4cf7a87ce0cf7dba653ffda74457f36344
979a6e3ca2f7a3ece00c6ef709aa47f93e73afbcc2b67a91de63c5319e8782d9
bbbcfd68e4d918ed6fa8a4589e4196bc9fbe9ee8e4de3518af6ffad56282ac40
5a75241e130afad6c2e833803b2cac550d849dfae296f30903e37846bed5bd1d
response z

1d7bdbe4f9cd51696f24ffb425e73d2d60673540aa75a350e900aec9d488dfbe
5df4cc2ed2955d27da9ff10dc85bbc71d954919aeb21effbd71b290829245690
02a6a30e7ca0b0a99b99fc2f5002961cd89707664bbe4ff85773bf0b136f05f4
6b336f1c4008bc509fc4f623a8500d9d79cb7c46089fe7bbb493e34c303bef9e
4ed56353363a0833ea969cf9ddfd82544184d20f7416f404bcf11bf07348d198
3e7a8324adae0748431a52452149047208b4de6227e438dc0229d2debb978069
807cd688626e56747dd0eec457846bd3683291fca4e1d01ae6a061395302d67d
063063776289f8509be75ec68a6233cba3b7876dd33c05f1887123e7b7dc6c8a
response w

79d42d3923db3e510dd081fa475b6200aaf5f91f598410ba87c85cd6ae15c351
ae7e45fcdcf4b04cc81d5d45d5496e1041af336ffaf3e2a9cf2cbb99d596942d9
36af777fd63e46ab577f0293cb8c6d99559ac7e87ec5aaca0ec16d1000e2df4f
3e272be6e5048707796236a6e4832339548d5c6b697b4e9fab9eab88c653ae19
e73a77eaf28db2639e96d423925ac0c874352b350aac165a0fdce1b5b25edcdf
a6c8a278d1e33506694e6247a1701681bfd7f4485742575b152e357dc940fb53
a1eae0b50a53183b91a24847710c224eceb1e5a3d896ae9e0e734f5d045703d
ef3d2d87273b650533b9eec8a4cda063471531c35aa15c0b627225e4176d271f

Fiat-Shamir context

36acd895e19476115d661fa4545a5c71b44acfcf7513fc6373a05c662d5069ad

withheld proof term: 5061 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR.eq_1 theorem, DirectionRing:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n Φ)
    (Matrix.of.{u_1, 0, 0} fun i j => (PowerSeries.coeff.{u_1} n) (Φ i j))
```

SHA-256 of the statement above

35bf7e3545092af9075673c5ff172c457f229b07effc4fe932d4ac5dbd7601e7

SHA-256 of the kernel type

7e0893b291788615ef52cc76a16d0be9ccdce46b60d386e12569dd33062fc178

commitment C

c7ddeb29150e85298ca1531b0b4a2fced4c741404f8920b94c7654a1eaadbe61
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eebc11ae48bbdbdf7c95e4730cececaf31dbb2ecdaa9ba96ddb011dbc33ae11d
f1fc908fbb34955160ad3998e83f3bdd7532d79bfd43ada7f82769791a9bbc6b
bde13fbc1f49bae6ef31ab3673f6d450a4d85ebd1fbd40c8f62afd893ec33ec2
dc0a70a25dde622b2497c3696029884a6344a2fbbc811db50c38bcc94521cf3c
4a3148697ea6ca4456d98e591b32ef2398ada941dce8996c9602c4f5c1678c43
9172d1ea4c6a78eace77acf7898667b8e40ef0af26f7857ad39256b1bde203

first message t

c5396089d1f4842ad15fdc51ec3a9bedd5361fe666adb9ecf957a607a94bde9e
4ea1f30791d5e4e91f6c3bda2706fa4c87db363be659512872062bc4ffbc2e39
1ec260d6f7bd295d4c29b3f3b0cc44a183b18401b5a0c0a7fe430cdbc066eaac
5208312f1e5070893e3377a6fa12d7ee5a592fcb5e724779b037802de6477351
1ea3377bf50a0499d2baf0fadb5cd3a31e29dff48ed3d733abfcfa0e0d335c0c
f8b34c0e9617d8f588cbda42b365f257a3724885a8884a14c50960507c320454
4d7ab4063194c46108067c1ef32b3ef978809c648b577718ab5474cbccc9d3bb
f331b0a92f4e32e28b1b911cec6cf6d88acdb1da5e08526699ac14620ade28b

response z

467e3d2b33861d253d8a7c6b815b9cfda78303a152658ca2887dab47c5139c24
2406e9ee758e39212f96a491d3fe5c7f265e8c58c04b6e4e60efb84a38361bec
ebbb2896fca1c4591182fcaa9187bf922537f61dca6db9126a891708b397b9d8
21adcf4f71feb3af5e797b60f7aca0899f29858adcb372076185fa056b730726
3f76c499ef57ecede48061331f9aa99b8c323bdfd9c298cf9a15b56a789dced9
d4ef15de5b838432ef9fdd23e5a9218d167690c93f66f777520d592881078d32
ff5d426651dcea545e0b4458d06af676a4250f1c6604d931d30092b581762ba5
a620029df3b9d138f758002d0476530c83accbefd162f70cdc791f096df91f80

response w

43648bf3b04e7aab3dbd96baffeab485caaf4be6e4dba48f51224a93c1f14f33
e2f608a3a95e4f5fce5a6828178b1798e1f3dc0c8fc8ed1dfacd5e0715b6d52c
08801b18847992e9f6e4abc0ee867c597a693eedbbad5fa3801520866e9616c5
f39babbdd192acb22f7f11b45c8c3c1bb7a9eaaa5aa330759b6b2094ea0eae562
b5765fd52c87a78488b522609d0e422d0d91e17ffe6e31dcd974c880ea464397
c2b4f298482450061689544c69997fb00c811f54442251f7546ce96da5ab35ad
06f4822ae20d5322a391a878e98b3716594fcd0fa72e028ad69bd97c3837ca4b
cca6ce353c4a890751b9bcd9a079ffeda7e320cc07a69a7624a66a69cf7ce67f

Fiat-Shamir context

67a5709d893139e12821001ad118d183d08aa2c61ef97d74c780f304663e00f0

withheld proof term: 235 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_add theorem, DirectionRing:76

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n (HAdd.hAdd.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (FBundlePoint.psCoeffR.{u_1} n Φ)
      (FBundlePoint.psCoeffR.{u_1} n Ψ))
```

SHA-256 of the statement above

f6acd7698cdfe49a33879f3757ea711ed360d1be4adece704df78ef6bab22672

SHA-256 of the kernel type

12a1d2aec16dda4038586c493ee394f6787b56420a756782e32a9be7e00400d1

commitment C

37ceb9b98ecfe1f0484754693cc90ecd63d391025410c4ddbcae8d1dcc66ac1c
0086d3439d2245d125b82ce140a5d3c27682c721931268d1673b029e1fa60c2b
88d819107eb2995c49b89da7a38a41068e15a36d55dbee10a8d0083580aad07
bdcf1352125c3bc6239e582ab60b77903856f3f89579a3d15d904b4d4ace7bdd
ab97c32853297ca1a9355b4883f228036be6dc49391909f6ec1263f0e5b0e2c5
9df48b51c88c82c0bf4e8b7a9f857624f13355976f1972a6e6bc2ae6c7333754
5c4ee07b32c6356162b892e83163992d892c61c538156cf8bd67eabf42719fc2
dc559aee42f7aaa00e9c8baf535aada58b4feab6fb5ef673aced01faba12bb2d

first message t

7ca0bf11a0042265494852f3db6ef2a8452719d26abd82d7ccc343d7b751ac83
b434c8f794cd8d307a84e69ab25533caaf110eac1c86b66b69f2284f83eb1942
529e5d93f3edd13219adfcfd5bb06a213c4bea0f21e45462d69a6007dfba937f
d537e25d598329804a36fc1914c9d1ad7860023df3f65164a490cfef33563993
15af2c7dfccddb18ea57efe9ccaaca5e8ba928d2b6a6a1905590e8387c5f1d62
5a80255143f9598079b6afa829b891be9cfb51ad2de589cb9ec3d0af030de2e0
7846accfd6d1883c7a0fffb51ab79e3181fbbe4d75f1a73350bd6b3155c86397
34ed59769a39ef29a09466ced9e58ffa0d8da5e48bd115774cf58a375b2d49c3

response z

64677565636079093d5d0504b6214e86c92e3b3e7107d82d8ffd7eaa031271d6
897ea952880478637e55f51bc9a7cc8753bdae965ceaa894687b2bfc01efa20
be78d74c29de14ecdf3d3ca8d34b13f912b4feead43f5669edb6c8c724e448dd
98a1abb4d9edd55def9e1b74d5e6c577abb7ea3f3d40d62f75a1808403618fe0
69ae8ba5fd9f905397affeb030b89e1cf26b0de2b13a7447eaff84f7272a57c6
16f3bf8d0789eedaf151957e4f8138b4f7051d790205b865b1f98b20bc70ff15
5a4f70c2f99b460ebc5c9a645dcf5f44e9f6c95b44a3d2615b34bc244463a80e
e8685e857d743f6c0f0bbc30a893b0d3ac2bb76a01958d3178876586bae28657

response w

4a1d94d3aa1e16862dfffb27834ff4a1b7863d17c7f55708227c5e65368279964
572da448682c9934e705ce7d94e16a9d25d2e06f9905bc28a8485ad98a4ab3f7
d19f3d3b40d1591f72c878a42464b8d04d0ca7a8f625df731390c2c84c42a289
0ef547a6e0bd7940d921f38eb1fd084a662e94b2b68d5f807219015ac706d471
db266742528c9ea35becb42560164356bde4fd204063cd7e1d92d397b8926a20
bd08826caf25bf0b61513b256185fca9a4fcc2b03249483a684adccb5cc7ef18
11f9d483679f2270a460263981f55a15985cdb5605532550010dc226d220bc95
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Fiat-Shamir context

38c055e18691e4bdb668185cdef82c8da80b06170bbfb167bf2bd728e4b2032

withheld proof term: 158 119 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_apply theorem, DirectionRing;62

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\} (n : \text{Nat})$
 $(\Phi : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} R)) (i j : \text{Fin } m),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeffR}.\{u_1\} n \Phi i j) ((\text{PowerSeries.coeff}.\{u_1\} n) (\Phi i j))$

SHA-256 of the statement above

141303a7655884ede78ae87cd06aa7f69900d6dff5a17f63ed5aa4ac00581608

SHA-256 of the kernel type

eff5e4dff82650191fb180804d981db045c3d73c2010cfd0a2526cd27a73f05
commitment C

73675c571a300bd84c83a983f679313a4f6821dfb784628b30a7615dafc228d5
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dbc680b8ebec7e72116627793d9782c609182f8524b2a2fe78dd6c70030c08d5
27198b98996dde639c2c823012da2d765125cd3f6f403577dda2e91e9e16ca71
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first message t

c983d34e24f098cafb5b7c4c0bcd3b5dc9e2987f3d2510d90cec12bb741b3c89
c4ca1a368154118f53afe54402c8976be70891423cee58296d4a316fce6bbf09
de3107b56583ef7d0fdff1bdda84765d8bd63163d857b5fe235595a4ae833ad6
365c53f9e662d21947110ffe4075a7ac1d8c91217bede5e441ca68789a3af901
2e5bd4fae2fa01ac7030304092d9b83a62cf491a498309c03f8886a21c57abe9
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7778e7d6f9ff48f1635216e363abea6d60b88a42c96d557b42a35ffc77d4ca43

response z

4f4df3ec10a2068f6613c15f9bd6e457ad684b4cdc8a7b928450e3f840de2918
7235bcff3585f3d003ffe4c61e145f4d50929992cd43a136bbf5706f5633b2fd
f28b2255f755b9d904cf051fab76f690b5cc54ff0bdbbcf4c3fb2cf8448e3ebf
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96f062423fc4085c8f6ebc02394248bab4bb1320ce4628bc3f385a366abe3728
49454fd7e2bbe0fc883ed5f2da83bd9a3e499c6925c27a0cb79b4bfe19c3b6c
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response w

3a5f98c8c2c94fce0d5c00e958d0d73abf803a6eeb3fbdc865bb2f7805bd7b2e
b9493d8cabb930833a97987a2ca62b1f419ead545a79655ac06bcacf80069649
d2fc6472d6aaa22ccb6e2cffac0064e6f22c111521ca05b8899a39e3bd2f6a02
1f2e940c4cfad319b6e1135ac96891df11f6d2db79bc73b808b30445fb8a5cc1
85cda5d1b9392a731da71cfad94dd3da66fc297ddcf4e8454409c38ba6c78936
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98b395aeae7f59f0d6204a8565e044fd7c4ad6f122e151dd3cd24145f945a234
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Fiat-Shamir context

a62d723b99ae3ecb5aa84038994e16703f7cfc5e323f44888140ab4394a2507b

withheld proof term: 334 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_derivR theorem, DirectionRing:164

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n (FBundlePoint.derivR.{u_1} Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑n) 1)
      (FBundlePoint.psCoeffR.{u_1} (HAdd.hAdd.{0, 0, 0} n 1) Φ))
```

SHA-256 of the statement above

6866e45d7f7da08c0360d99258e9b033df8204c53275804a1bb6e095b1a5e7e3

SHA-256 of the kernel type

b12ff39cbdaa4231b580297f459ff26a8cafe45187f449b0a76f5998578ce

commitment C

160dbfefa60dba6d9f3e7a0a01040d24b2104e5afb5227ba980694b4cef00290
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ce61dc120550e8559eb3ba43ee40073c0f82109a81e4cbf641f1410c43934e6b
8a30d6a531b2f9b96b0e39240a2ea5f6f9ebd5fbd780fff26503c8335c0e1703
be88d23dc371a8dcd980279c9c924d1087e547d80e72ecba76e657b6bbdd0344
e281919d175e8f7c05221d830c972f5fb61e9d745cdd2d395b00f9f30cf133fd
09722e3947109a856cd152f8318812b8043c2337ff98b252ba0767589965da7f
e175613b33491bf27ddc06d2e784f51646bd32bad71741656e64a7ba1cddd686

first message t

f0c9cc182d754596ef38d41008fff4a4116f01b167a654ff3df58296f52ec864
bcf7b506069d45a8f96d43c6781667903bba7754287b763c46fa059896e7767e
ceba111f6d11c88d9ba08deee4b324f4f5ceb8180ef2b449d2a079803b65bbd9
e566533ae364d3c0efa996c30170439d3c273f0be6574eb5b0efcf768a7f6eae
acdbc3548460fd5b4d1f0fac35b115e0bed869a9dbcb4bd9c6ee5fea447fb380
2a5a253ce9401733684649bd3e5db0a19de732122c3c41bfa9a1709d6c57d87d
ebfe6733fab5327a9b55dcfddaa47b79366cff7281b19a444c45cbc04b34a873
31fafc78383d21b5519c2e20c9c6b61fc52bcd2eb61a261e375f50b364b758

response z

6f3abda75892b00aa3a7ef47ec152175c2d2e842b0215914ca8d834d33cd8d31
1f97b503f5c1547d0f1c5891301c970ab1fce882f55d405412fd7666586e50ad
a9b8ae1fa236135b0a9afcfb21fb137091666c191bf420e087133d2470b87611
52420783f1be74b529569a4221125dff4c4c48d806feb0d4bd5e1b0395800988
5a3335f1d61ab47790b2afede77a6951e506d989df8718a6b43283685d0163f7
a4793143356c46b7d7a1e4b51b69773cbfb47b8bcfd5fa38b5f540a662145f06
5be0f1a357361c33005e085ca58b025742396d9fc314c2f2b2b8b861825ea665
c6ef313bb7d1c6e459bfeefc329accadf990791cffa286c975017ab525b5aa5

response w

5da62d0fb50711709ddc05f57df08899d99baf1dda8a26c41f7a1645b9e05561
3a6a55ad1cfe38f1b8cd8fcca92266cfe0eb04fa14b7d86b707e0000d23f3eae
3db417b8a354e2d54594f9b12c2f11ce71a1f2477fa8f61fab0d668c186b9dc5
1993690589cc82582bd9db1bba09af0d62dcb8cd4357845eef069ca238112542
767d555f8079b8bf4357a860fc92054d0d3bf0046bca07d17bf7b75762ea020f
b24f6eb82778444debb2b57968fb8ee9cbec73dd897d7f466eeb620f6e4b20d6
eb9a23b209b8dfba3129f1d93f517c681215e6f966399a7b6ca089691d55ffc5
d2dbe18ff071ef8dd733a19f3bb12854d2e654e11fdf6df78ecbab18c0b70796

Fiat-Shamir context

592fcb8f4161dba128d31ce9704865c2b3c62199d88b45f02288b9ffb705021c

withheld proof term: 115 279 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_eulerR theorem, DirectionRing:210

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n (FBundlePoint.eulerR.{u_1} Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} (↑n) (FBundlePoint.psCoeffR.{u_1} n Φ))
```

SHA-256 of the statement above

94acde994b4eeb8ae61de289501d0fc7750006fb70a98dd71f2a5f297fa27678

SHA-256 of the kernel type

dbeb555c40ac1bc6fa7cd724647d8d3d01a399e8f1b729a9ad83a765f357adea
commitment C

3cf9577079cfa5cdaa066bab5b86be2929e3a37403e1632efba5c0e663af3252
c3bd39fb1d0d7878c4e90653f912ff5208d1147ce32379c22f8d847ecdffcc473
dde6e616658b992fc78256a220bd0efe01cbd11205c290d6f2cfb0b1f3a42743
abd487a77891754e11a54dbdd0af5c57ce2040015d107b94ae26a1c1eba4f1f5
e3fe8ff8e13c7ee3eda06d7de8ede1644b4466f3503f5bceb2d70961c221c566
f3ef69ec00549603dfee461756f1620eab6e6336ad0be1361253ede4b9ca0654
6abc3447cf5713fda9da87b75693a8c36810c5eb61ee66afa09676070325893b
91eb36519d3f985149c0af8c680a56c6f46ac7b9b6d6dc8df644e59424350cf6
first message t

8f4d061eb3ab6bf05e82752b1485460d564d4272c4a2767abeac45b6a095f957
e37703d5867e9263984891f87040a2ccf7cc371257ec7b39f497fd915eee4062
c8072d7aa39f4fc9dbcf790a298ecb0e37bcd71db9eb9a74b3c24d46611e21e
8c07cf136c7a33b2b1ff2fe9ab409fac4fb8da77d49ddaca5f559bc6988007b6
c4499160a71d243db0ace19485e9a073788ddbaf7b0f0df2a64285be26d66101
3e224198af14e9bffa448a943c2d84056b61b0072fcf665d1b523c4f0c9fb4d1
9e9c481f43eb219cf766114e739570e45e9038491d3aa459574086af03ba4a4d
66e83d7aaf384ac681208dbe157cbf6feea1639668a2f1e3c0d982f59484e56f
response z

10094f674d96a731180c8a0593ab270637f9ac62b1d0cbb1e0f5fce603317f48
aaa94d3bdb40c1f7f9fdc210e383ef9e3e9c268ef90b9c868df361cb54fbbdb7
8057677ac3e3892bc10a94aca86fa5680835836848885b1fc2c2689b9df95204
2d6ba318366d4e158a03624f0f25aaeb68148cf1c22092825e7e008b3faf0e10
282f66ef402fc311478d92fa83dde7212bf4f2930ecd6572f0a0ea4e40512502
64916aea63c4bbb2bf7e403ecb0a6fa487d72d87e804e47455defd0a22b24397
74120d462db6c60cb51242a1a6b1cef7061e2ae09a8aa339035e6f4e502e4925
6e403a1caf12d420686276833e672866190e73aac09c356c2f7762fb4deaa3d7
response w

34026bebb9b77735750ee4284fb2e6c5782ba523ce52b22fe6ee14cca6fd62e6
a7e68ec2ccdd315a9a55897a0268af31d8411b149e9b8cc0d7f1254d91ff46c3
681b29cef7a8423bc161c032fa187ee9ffac96e1782a6df689060b1db00f6f90
14f7cff576a0fcb6302044522a17d154edb91c3b79be32ae99fa914e023140a7
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73bc0d8401a89b8db8b6b76689553be1473e470d150361f63c269f7c36a35373
ea68572b54572361c7a4698c39a27e8eafa62a460688be1357fdc4113e126f84

Fiat-Shamir context

773dc49c6a707cbc43dcd1f23cbe767a21cdc50071eb06fc60127aa908f7b849

withheld proof term: 102 478 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_ext theorem, DirectionRing:66

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  {Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)},
  (∀ (n : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n Φ) (FBundlePoint.psCoeffR.{u_1}
    n Ψ)) →
    Eq.{u_1 + 1} Φ Ψ
```

SHA-256 of the statement above

1e5562dad7b97e52d0f4136a5daea35b6426c9fb674b054126b6974da8bb158f

SHA-256 of the kernel type

25bea673fe41a499a5ea44e30902cc2f2a73b83d7f7d353e35a7e2e2f2cf0f0f
commitment C

bd501e68eaa7fd66197bbd6a7f3aa492a1ece7e3493ebdcbe50ff78dbfd69a09
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f3f032f97508d7028cdbf9b36dd66616c5a2baa1002bba7845162c191cf9bdc
ed7d10f8e7c17c04927364472b0baa54e3a4037380f53d0adc601dc33c74e891
caee57101d257305dae9f11004538c9f4a2a10e0889153481c5ccc8fc05360d1
first message t

8deb5ced2e1f6f098597dddb08bf7fee6335c084ade72a0e93fad4869d505d6a
825718c4ed9fab9c97a2bdb2754ddba38aed2064fcf3b27064581aa0e20c65a80
628d1f0bf2d567ac2f97bbd914dea24c7e20b84ce243ade805956473a2d2c115
91d4ff63b68a820713046a07e2fb5518711013508c40bbe85fbd748e1e222b79
1dbd9d3fe1d9027ce98f32ee83fab90c67e8fcb04f851cdd5de8e3db7d3f944
8803cc703cfc6b0662f06de33317218598d239304ee55c13228a58066fe0a528
0c86f03e7fdd58278a798f15ba69cd53d184e82fd5d393a3fda9ae7ff5e67eda
c35ae3858c995109288bd42912d7aed3093335c6ddb5447d192410a5dfa53158
response z

5a7aea09640c957286bb03503fb753c323079f98114db75e2694dc86bd34abd7
c3e14c6fd0acee100aef8ce4b6723960b4b544fa8c52498a17a9c06fd76f876
a03c4c1937ceff4f9808303ffe3a4aaf4332cc10a66491c501d59cdce483003a
ca245aabca4257a8bc7438399d416ae563c9dc2e0862378365bb09c0e4453c66
a35745e954cef1160cfc44b900ee5828a037281dbd1bae1444f81ea46509f163
f3c4129c464a3e070dadd2a7095b48dc9461e65ca7e9ff53789a44f77314c046
9b98a358e8bc5ad1b5bde086c91ee60faf003e6c8f83206345dd86a2ba488c56
fae463876948ec7bf73e989a26d2aaceaa84b2d12e022dfb1ed8ff9b71e6a50d
response w

3fdb69729c200ff8298f93be5fede2fab9f32d49ff620a53ba6e183321a0dfc6
e85a67da71c34863560c2372b4a92b4bf9417d7a9698e77aa2f259aa5d28c9a1
969603e4023f0d550c0f989ddd60f11284ea26b86ed7989a97b09501631a0f5f
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104de8360e46544ad5a685334b7197babdf0301658a036563fe29e6e706d4111
f1b90c34a20e38967a178fb9ec77fd8ed18f7cbcf5fd6407870b4392bf9a2c5a
0e002c8419b61efcb88f35ff4aea90974c07f8f583fcb90d8f78ecb614c31a5
e69e966bd8046d09d7b63584a008bb4d9134766441ec6c2944d089283d1ef41f

Fiat-Shamir context

d6665bc2279da41235ddf8b74199a76ba09e44410aa5ff6b8a02d5b0fd21f1f7

withheld proof term: 1353 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_mul theorem, DirectionRing:91

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (n : \text{Nat})$
 $(\Phi \Psi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.psCoeffR}\{u_1\} n (\text{HMul.hMul}\{u_1, u_1, u_1\} \Phi \Psi))$
 $(\sum p \in \text{Finset.HasAntidiagonal.antidiagonal}\{0\} n,$
 $\text{HMul.hMul}\{u_1, u_1, u_1\} (\text{FBundlePoint.psCoeffR}\{u_1\} p.1 \Phi)$
 $(\text{FBundlePoint.psCoeffR}\{u_1\} p.2 \Psi))$

SHA-256 of the statement above

59952056253cdbc4ae45a4c430d437dc1d426476dde8505031a62b6bea4ad3b4

SHA-256 of the kernel type

2917937795e622c96db7d9ee2d525885cea496b743f54835672a684573d5f8dd
commitment C

573eb5373375593aec553eb9c16ea6d57392ed4279cf80f505dd2539eeac4fde
7607b1b0ba5c91fc7e4f7d5b89bec50a6260d1f6b1c8e6a9f0991a780b7db4d4
a78333d74dc7e277bad87a507e9889a8b9c11e46767880f128419e805ebb4ad1
2ad86368de16e75d9725109e1554c71468ecffa0410644d1894336ad91efbacc
7f2ea30ad45adc27b0a886873144878bfea6990692816f3b2208d1510ce4e1fe
9e2ef0750484e1314e4bc00459f1d7abe1ee5b97953e4dd18914422516a5214b
48fd1412ed967b3eb6262956cb50d55ae9818a6c85ff63fb19c5819f2dfbf94b
4c030467cd79a85c1211da87880b001e19fa842ffb7d0d265bbc6e0b7033a8a1
first message t

81e1e26192e1ce47222d83475f564dab499b07d3e19400d4814fd56f791fa78b
21715fef832aa8e500219b75d5cdd4c4c1986e18342834b88f6a4f5ca441832e
eee9441261fa5c40c61500db8b11dd3f2a9a85db2f0d11b70c09990b4aa207a2
f680538da1bb97db676bb0aedd28d582c5ec2052db8b7bb0465b17a3e7e44da
8cde90924016230b66fbb77fc4025d78704b6157bbc8d3faedd6d5bad4827ebe
5fbf816859acb9487d920a0e2e810c971045b78612fe2a1db415cddb6e723f1b
2d56052b07d245f49f0b2be41ab0354b861505d35c470fbf1785a06901bf670f
d718e670b04f9ce17efe9d6f767184cf6e5e3f66eef170c2c37c7cdf4f6aee29
response z

6facf0d607c2a198161794c4aa972a1298442077a2c1e7da60a24beae6601a98
bbacc77048c236aa0a99c38522206c60154ae110c2124d19fbad2fff3d9f2ef1
8033f558397d040ac38de5d648e68f27ad56caf860e0826f448d334c93a7fe30
6adf6c65219ceb3dfb18c7590a7ff37d3065c382bc984b0b9abd2de41288f96a
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81bf20c311b07ae95309b5185558f9d8aeb0909d6acbedf5bc5760afeb7426e1
cc962d8d58f612353c28298e01f39903e35f66984bb239fb834225d9c133b536
f561eed6dd41d538ba39c30933974ad8411eaf6be6d674f14c7461722c1b22c6
response w

2bec99426f7b6367b3a6ef5ab4e1a1a9c680da83918425c69c290f5a151ff8a8
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37480503891199490d36180c93c77a5d662a8f8d0cd6dc5cb5a1238cad39328e
37be54cd6b38a41c286fe4e94882764f42a464467bfdbf25ae322fdce89c5d2e
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2a725981fc25ea3c62a56a79e1558511969f13097ea8f4d4f7b7c8e4392d21ca
Fiat-Shamir context

b6db0d4ef3bc7432ae1efa82dac0aef40a63c65452c2d28c1e77510267e09e20

withheld proof term: 301 451 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_mul_eq_zero_of_le theorem, DirectionRing:98

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (r : Nat), LE.le.{0} r n → Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} r Ψ) 0) →
    Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n (HMul.hMul.{u_1, u_1, u_1} Φ Ψ)) 0
```

SHA-256 of the statement above

cf97c6192d3228bfe76d226f74b5d2ac931faa2c03d4bacdcf0b46bff656ecc7

SHA-256 of the kernel type

087612f4c619bc6f3be05799d3051a8a29008b19fe7efcab8a253ad800dc32ff

commitment C

777c3a6b8227415ab28a172fb7f6e4e410b106cdf9bfffcc605ee5811b112612
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e4d66fcabf8937366c97c175c1ddd4dd72d35edf851950a942b4c8f1b2748210
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17c1dade976472db770723755fdad8fc8fd99443f6e386b51b04deb3e2054629
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first message t

105d601430002a8cd643ea0a646f8925bc8b6880e8049d521ee0813123f9a977
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2e24e6b8692bf9945a6c6beb7be1084529b94227e98077329a9f3deab2185b2d
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response z

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c2839af1f4cacb4178bc7ee8a50368de362ac9589d9d2932d1a4c2509bf52374
response w

59a4e84be3f2357067fced16450a7693d75edb2cc7cae4b444a6021a3dec0ef
30e72cd5e0bed36fe85ad1372b92b5f43ad0bda5918846569d1bcf6251eed037
0cc80d4cc6abe029f49358722e9128741999614da7f5f387bfbcb68ffc99cbc95
2419d00cb3dbeac7c7134259f34d72a3f12055975d97f50339698e7ddc2e0afb
db24b40a510447eaf14069267e79c637fa35660b84c21d34974408cf0a2ea856
dbd6252b5cfeed80efba021799a31e8ffac2e93999930f9a979fa3ee48d78b2c
3935f1bf05cc384083d899c2d5b1abd53a7777839763fa9b8ecdc8210e3b286
d53a537e4d82264b5b4d00c4cd68dd0ba3666863de425eb3d46e286c0a099463

Fiat-Shamir context

c56795042eaf393b65107199c858707fde6f951607d41dac18dc48a1cc149c2f

withheld proof term: 39811 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_mul_eq_zero_of_le' theorem, DirectionRing:107

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (r : Nat), LE.le.{0} r n → Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} r Ψ) 0) →
    Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n (HMul.hMul.{u_1, u_1, u_1} Ψ Φ)) 0
```

SHA-256 of the statement above

7db1831bf16bca28bdaf8c227a3c1e445cee1d2227ef98970d3699b7a68f46a

SHA-256 of the kernel type

0570ceb72bdcc014bab99eee7b56dd50b48385c7342fd7644ebfe486c7d9bcc8

commitment C

20a62cd72ce223864e6af9717f8fd24953d2e138add902150e70fa6d3634d4ed
82d4b9e99c967484dbc1ba563ef113f6592b8b761b2ba159e4a9beae058d4e14
9b1b40cbe8b119b90b9415530cc0d8bdaeff2c93cc273cd77f87241650d74775
32ce9302493c8fc35cbd7b438470f4ebf1e725f253acaec5b2f50cef7a240c6d
70f8f6ff3c4fb0b6cded6c79fc8cae4db5c71543204684b3cf4501d377682f05
cc6375f18505c306b46a3508b6bbb409e40fd353dd8ddec2c897f244c11d95ec
0c116efb277d3cd435e063c581157ff7387bd5d22b7490afdf280315e2c0bb8c
4005504fc8be3ef6d2a92ab14f78222be7418988d2be926548ad8f8ad83a1757
first message t

66b00a93cf19c541f0a3dc7491dfe0dda1cf112669d3cb87c5c2f3108b0446ed
2a252ee04375dae2b31e1c5bb007658b73c7305a00d0c82bd9c69e68283c0c03
e7de82f452c24952f6fcd7e97179e3c166788dc434d860dc56dc45629f22d417
9ac927be808de15b9b61a4ef513938734282777cce7ec434cbd2ccb42357cc56
6c466f81f4139b225eb5480c26ec32a4386aaba93e665c0236d0f694b3e2af8c
c8bd683c6ddec6b10fe6277cb008a73dd66e105876d7b747e82b5c578ada14f9e
695c18adf0c4621044826d0143fb761519d3ed79a0e932d0afcf87d612dbdaca
a3265308a267bb4fc2fb59af5f119ee05bfa584bd4bc400f8677db4da9cdb15e
response z

66d82eabe02a032db28b6a9b1ef90af4044a27e703cbce2d667e234dafc7da57
19ab2c38e98032cc5fddedf4f0a766c702580279510b31f60c5c2671410b4ac18
d480d7aace62fb4256d37c96c7214d6107c13eefcfe6397a04e43323fbaec37f
20fe97be650764f6ad6ceee7ba53623b785ac84834c2cd59719fefeba9a6b912
1a35b0f65113c4bb69194fe2b9e9068e17f3d0a5b6715153418d76037c452faa
6b7889361e8e02bbeeac5f004c385e978b4a5db716d57c330d642f80c39ce73b
3b43bbbd3803b43e850f9ec39810753fb003590dcebfadfd287b71a3a9d46bba
c06d726879cddf979dc033e3c74bc21658381d92e343b07fc2d694458f84b23c
response w

6032017cab35abbe407772e7cb8966fdfa8045564f26e19402984cf9c9a115f3
8fbc657db837e800d068aae2e788ac4272252190aafdb1870b168b97f6d02b22
42229de7fb3cb61eeaa4979e955956c2d170a7b961e9b4c2286f21447fb87092
2f97000eaa7b0b5ba52373c2d38e0aaeeca1cddf3b93e9c927dc887ee1e2ee22
852e527d30d4cc4055dc84d91efcabcd9b5a2e45869b0a1128fa7b5bfcade76
391afbc3a2be5592f02889f040cf8c8645b0db3aa01cbef3db72b1d6793e1ba3
5bc06217167755d76720d905c19d120928c2b81f0c1b890b2188b3ba3dd72359
83d56b100c819aa63975c509f83cfe4301c1c5617f6dd4f483bd15233d72adb9

Fiat-Shamir context

efae611711f4fb878fb0dda3e004058d7bd8ca3ba2ad3da318eacef0250e703c

withheld proof term: 39812 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_mul_of_lower theorem, DirectionRing:116

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (n : \text{Nat})$
 $(\Phi \Psi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $(\forall (r : \text{Nat}), \text{LT.lt}\{0\} r n \rightarrow \text{Eq}\{u_1 + 1\} (\text{FBundlePoint.psCoeffR}\{u_1\} r \Psi) 0) \rightarrow$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.psCoeffR}\{u_1\} n (\text{HMul.hMul}\{u_1, u_1, u_1\} \Phi \Psi))$
 $(\text{HMul.hMul}\{u_1, u_1, u_1\} (\text{FBundlePoint.psCoeffR}\{u_1\} 0 \Phi))$
 $(\text{FBundlePoint.psCoeffR}\{u_1\} n \Psi))$

SHA-256 of the statement above

edcef61f76d1ffbf55ced75adcc9df175598d8c51640848692b01fbc2f7e6c26

SHA-256 of the kernel type

82dc9018187524cd498213b9228ffaa291a5d916454831d683aeade5a78f84f0
commitment C

b002de327a032f4d7167dee2dc67331d8659a1d9be1f2d32df897addf5a7257c
2d1c6bdb84680da48a27a0b4f4bfa51b849f592a727358bf15efb44d2a3ffd7a
2c237c753acba12bac1004fa0d991c21826535ede7c8b85cbb4e46609b0b91
6b5312a9e57f076b06e632f0441fa0e7eae4ce37742fd05094a98b5bb323468
48784457097ae77e4244a302f9bd65af7c79d1b301f89628d05d6e86177916c1
833774c3cb4fe74b04104c608c1140d7bfc2ce3df11b60a8872ba24b4968b235
77058891a7d2929cf54e58c20ffbc62dfc4e100aff42f53d82245e44e9eb65df
1d703c3271e8ea72bb557896edaf1199d719aa718ce88d1fd8d00f261daed4a6
first message t

e09187dac90e755a4174bf9d0b689c83937bbb1c6f322b60876c7dbdf77899df
5cd133cb5b5c6fe8a9b1ade1b403520713afbe14dd86ac4398089072942adf50
ce7493c54ad0ca06b28259142cf77f40af07517b55a2ee31616f6b2032873bbc
7925647099fbce1b1b7f93fc00246a3d4122a91a0c4202a6aa7c52b4617d39f0
b721b9ed7ea4b0751f29958a155f65eb3842c9d662b9e9a54abf101039e9efa6
6e342caa2a3590a138ca195f86dcce5e31c64e059e43ba23ff8ef56d9b271e5e
8ad9141942eaacf7a8e704202c2b2d40208a172f6c115c652eb48e220b3cbb88
62fc797eee5add2083657edba8e599ba585b52501a0396c72ae80f59d5ea20c0
response z

79aa61304601ac055dfe5f629c7f60170b2c654f51cb1836408a78ead6ab757a
f1c1bc20bead041289ba51f730c2e23d19844a61d558c812bdf7a30268685570
24993538ab5094dca4cc32a532d5dbfbf8825554c16764e3464cb715ec0768e2
9fb69d682ae24a977071f6401b198fc2191b62b42e3cd31048624da84854ddbd
a938b605431ad883c3fdf0984c6ff2a99cb1b8e9ee678d4ec8ada62b74b476dc
84c74b8190df55d5445c013cef5a0a0ba70cd6ee82869e0e1c4709b20df2e7db
b44067d3e1449877ca5abe9c857a2732350888c4cb60fae83e5b260cf179f206
058aacd6b5cb2876df118d729da40a0479eadec43e045881ef11f2887690b101
response w

6a2f3a75f70f3acef8f7cf982b661388304ed022d239f123019365eb081ecd9f
d7a135ca1ee2416e0ca260e857d9a7465b18accb0fefc87e6cb8f5b151ed5078
32f6df68552ce8498c1e047fc5a326df23b8677abae072106e10cf7e795aec35
ab30b396dd6851caf18f98801279c0a5e8dada09e76d32edf743ee08ad5fd0d
e9136061753a3450a17565faaab98526fde5c833ae157003ce11be7079524b19
67746268abd2586845f5df22524884cfd36801a5503229345e28d1f8c6f716eb
ef6c05206ffd48e7ad2b3806e5b9c9d5fda85c2e4d2e8ccfbefea725eda93a94
fcfbe80ff1934057ad4a3fca92b1cb445c361f5f0cf2979f752cf86e02af8b20

Fiat-Shamir context

1fd8529bc85d998fdb983df8b33bf5c8ba95a11e09c9ab0e5dbaff9b8c18852f

withheld proof term: 48514 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_mul_of_lower' theorem, DirectionRing:131

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (r : Nat), LT.lt.{0} r n → Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} r Ψ) 0) →
    Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n (HMul.hMul.{u_1, u_1, u_1} Ψ Φ))
      (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeffR.{u_1} n Ψ)
        (FBundlePoint.psCoeffR.{u_1} 0 Φ))
```

SHA-256 of the statement above

22378c9e05c24dbe6e01889dc566755ea51d2e7b2eb2db3105ad08a723e140

SHA-256 of the kernel type

df28dc3a52793061b4bf4b3ef7e65f05e33a5a54d885fa6dc9e23dcb4c6b228a
commitment C

111a37972ab4fdacfa4b97c51e08417fdf34b92f02075b501003727b35092a
3b0731a098bf68abbe92e4552df09a9b847463d7c4d3e3b977b74e3fa0cbe6a4
6777a393ed928c845a5d0fa01841adef260d9a2ec9ff1284634c757f0d902e5e
623d2c6f9fbc48bef4564d8f3ab6a0ade06b81efa019deed0963a9d59f032eb0
6d9c95a9c3297ca7b7cf9fdb1cd6345587cd0aea9ff67a4810cb7f6e257e115f
a25c46475c9487072940ea4dad503a23356f94f4f1b5c8083a55815305a90a7
058c4b4fa048931872c3e3e4815b86bfd798aa8e5d178a18f50247bff0f4bc5d
54801e54a6e32b372b8c8f1f3c693389baf7fd53fc97c34fe4f9295e76997116
first message t

ca2fa96a71cf12faeb3333c59a87b0f8dcf268e99afc33c8f54904c84ce02d46
24c0d42653d9dae7330fff32f67ca94ccd33279026466eb2b7841164f5f9fb84
fe7da260c3d8b1bd59f8107b8f829af1e019515b5ae7882f9c0bb547d51e79f3
a1db26c325ecfb2ba40c7eafe6776358e876e37f740d0cf407c26184678ef94b
139245b26cbe25f58e1afe3632c4c40c871988e539b5ff26b648e6a05273cf8c
05b5e1dbc85784468e706d0c0c6a51c53d94ee5fefabbe158c8d6f7dc506624b
e0fd9b661007221816d132d82648e0a283851c48ff210e1e519840c96352e81d
617047a5d6e2cfeac8d5b8b1a901e441d2503859247f2bcbff0103b1920dc9d6
response z

71abd093d54aaf43bdc5be13685b55005e7cc5ac5542139e211d51e1b70afc08
86cc826986cc0e256050a50fb987247c6704fa02a90b82136ee0a567bbaee96c
af1b865e487b31f567e046b6d49f2d269366d539ac7cf28426d0850c0f9a7a77
a8e65c2826dd1bb5ef8f4a86e3326c4f3491e2a65fe8db56d6212ddcdc925b8b
9baf388ad3a13443ef27e6e2d0298b9c218e655c7df7b1a19b786cd24a7fb5b0
929de97ac0d3b68179574fc92a7e41b61202a42a7d235fd8ea0d9f13a0183626
62a379f5be5c022abd140f162aa2fc54607e14effd688eea83f9f67289004e31
ed157fce9f14c4ee82ffb38b8dd97def1b8ea6a53638cf7eb2bddf435467582b
response w

53fe1a059007e4ba4a0ea5e79364209de3ea7c9bc70491d0dbfe40b54150fd0f
5a3c5351298292e5dd3533b7f1082612e0e1db5b3acc320b951c0c6ff3c574d8
1a87dfe6f2b59a3e04bfab850cab1c3514329af3a51f45c3c4bdfb04feb7fe14
ae2eb31e36efffade7ec81c3918ed01754c6334fc5f67514a62fcd43d1f6ab90
ee1963bcb924312a38d1e2234ac21bffee487f16bb7b679e08a43c7129f42792
41b936786043dd36613f8a62bd5c34a233b05151be33c78a4aea9ffe09c728fa
80d05e72da6a1ea3f6270dcb0fff000a9c56f00c74a112407b207cb8489e515d
a50deaa2dc5c9f96e107efd9cccb356873231129c69e92064e3b307f5051dd4
Fiat-Shamir context

ae319b4a30c48963139e8470dff2675b87179e0bb0cbe95798e5ccc8aab27c32

withheld proof term: 48516 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_neg theorem, DirectionRing:81

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (n : \text{Nat})$
 $(\Phi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.psCoeffR}\{u_1\} n (\text{Neg.neg}\{u_1\} \Phi))$
 $(\text{Neg.neg}\{u_1\} (\text{FBundlePoint.psCoeffR}\{u_1\} n \Phi))$

SHA-256 of the statement above

14b8dde169f878cf446781954a7f2f75c28a77dee8a521bf0d66d8729e9acd6

SHA-256 of the kernel type

e4caba672baa64b774f2dd7b8a0f33f63065fb04207006a4582f185c4b596b0d

commitment C

77f8f259d21ad209fb5956494d7bf25f4b76ee1b9cf66f5651cfb6d18d50b0b9
f9fd0dff63c61fbf31035cd541c2b8cfbf013ebbee5982f41e8df5a3b4ab84a
a81301ae834f4a921035e1f76aac1352a5f2187cb9f1f0a2329614efe9aa5a31
ce2363838071c427222dfc161d41134d6f79fcb2200ffca3ae42fe69c55ca478
16ed221a23ad7dfefabb9e388bb2f5e4113bbbe7635914667f699128d4b3d0a0
3ce586856059543839678e98bcf16744178b4eab7acb0183a0f0be463a4b1ad7
9312813e698fd2884dc8a492bbbacbc42b879ac3176584b65eeaeabfee53b0a6
a2589770f0cf67dac031679515f7e6c90dbb69167db4e14e4ced9b1b6b150e41
first message t

9a3f3f344600436559a56a370af88fc3f76139f72461c13290a8a37cbb54add9
96a530b84f06e59ffeff22bf392258936d1b999b2280bfe367316ba0dafd178a
ab91a9781a87aff96e246c04b9114e0b8c8142d07c2bb8ac1b1a798134210d55
64bb6d6dc35f27893231573e9f20a98820580a060fbf2ea9db238b8cc3c2488c
55c75d34e5e146bac8c9a0571a96a962f7e861d1ba5fa67f0c61502e9a66db14
a9de1eb9a717849a57b8448286af3e5b8e52a427099945dfafbf93da77350486
16123a22fba08797fade7564633341c1501b0a6ee9478b6873bbcaf561869365
6152681eb0f761904f02e9cc14a7985c9615880759820fb96918e7d6a74351fe
response z

697466a5f59fc659e4fa40e695bf9fa44a664b0f316b27e72a69ebd6b0e6a05b
d07ff4ff675a2aaf1f588a2e3ee35d3711af66fcbec38d23973c2a7ca7be23d3
85c3594ea0ea6fabd709f9ca262cd1e4fcd03dfc931af496c92445e276c69460
07809d825ac96a2cb4e95277581247f5fe7db95bb8f52c3c61661e757022ab34
0a32766d035ff0ea93989d7bc1b22f308eb25b83eac3d05f1cf2cb01bde2246e
477ca3894030446f1f812c78a3a96c43ba6e8353546454db36da1c3b555b41ef
82347e0b0e211dcecab5de5157b66506b6a197847d531477dc36b4b372ed0e3d
66dee87ebaa88b150372ac0d53e475a877881bd401a017c11deb943f8308812b
response w

7b86c6b599b4348b65d693ab225c5273f9fd83410426c51d693fa2ae549dbe54
069bbc7044f9d2a9b01b79d27ae000b026395f1e8280df2ffc4da3900364c634
5c7bd08de62a3413d190b2bfa0b0504ea1741e800eff084163d4cf5dca2d8b95
63ac7e9b787b668686737cf3a1ef5abb385261e989bd0a7e318637ff4bee32bf
1144034234a3102ed8482de74c523594838773b02a97a4321d1e32c0bca23d3e
b70270e1c4cc0a0c58cbf902bfc25d4e3c11b296b1d0ed6c024f409a4542bffa
bc45d23bbe8f30bebe72c9dd224d370c3b599c4d6ec8bc5e207765565acf721e
c3968dd3f892657d2aafe107e66eed04f5df20075b2e48c142a1410db793e79
Fiat-Shamir context

7fee317cf6123881c9f68d23ec58f5aba2301a22cd586b36967deca78e7d8289

withheld proof term: 108 769 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_sub theorem, DirectionRing:86

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} n (HSub.hSub.{u_1, u_1, u_1} Φ Ψ))
    (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.psCoeffR.{u_1} n Φ)
      (FBundlePoint.psCoeffR.{u_1} n Ψ))
```

SHA-256 of the statement above

fbdb69668c9f1ff1403db4f30019a854ca822ed225dd4aa5f6f705f78d0de5f05

SHA-256 of the kernel type

eba0f2cdef5d1e995a212c66a6a7adb60ca53f73c10829180cf7c2c755c1031c

commitment C

a4ef91895ab38ff5f57ad3f2db119f077158ab0667f2780733681878795a2e0d
ce7e74ffa7a42c087ab0a2e571460748c8c86d2d6f54896c9c7cf340f4f4ce9a
013434b9b3423b4f8251df15da1f41a5aa974ce0efe15967244bd691865735f2
ced55e59b553ffa3a462a6e1c1d0fab36ac490b447ed4f9a91e2a0b394144963
da632a586152695d154802f0c9f88511cbb05a27da9be6d6fa55f8eef4b90ef9
90f8f0a366af226b31f5e2035a309c59f6f4985103e602de916c3872a096c69a
a4155b7f3f3e104837fd56d054adefbf30c7d5793176cc55303bc780797c18
8fff1d1dbabb66183510a5a7b23fd211e64e1fbfc5f5507522947add31b773c

first message t

599f04ea01186e1c0e7ef1d309f46358fa2e05421fc46a646fa798a85dc9763d
dd988523c5524cb7b579f2a6ca4f311d848c933e0ea0653639aea182c3505e52
661f2b0cb857f051e0361400120eac9d391aa16039ea4246bdc9e789627b93d2
3d48dccbe1b8d2707d2466c973b5d4282db29f2243771dfac5e036e3a40beea
783b1fa436b4e67bf9ec57c7659377adb885533d22b2c8ca4af31c7a79dc617e
d3122a79b84003d172269b5be1f7163aae272d4575ab7da4045016de5cca859c
fd2a1c275a77459b1b186b8d84db6042aec41eeb0eabe4eec2a908d14c24c2d1
8cfeab7420e7a684dd61923ed575c954c0fc7d8a2173d038c52adc0a82273e9c

response z

d2fcad5e7f383db17268b7abcbac3474aeb6877e309ec55e55204d966243d1e
2e5b04764f63d7d82879bc16df72d9c91de05e2e33db66ad761a41f4db79a403
b975d3453c50dd63079c89086ad7d0352bc146bdc304e6e6d86b96b8843c7201
28db25f7a1d6e69a0a2a2486dfd59629d7996106c3f4e63517a5d1911b257104
7b6d88e5474fa14e272ca3cfd12d39036ee9a4684e7e3c6b5bf83e60e10d5548
297b236dcd3c547cf5448a264a00af3342e732d40a5f14af99e812c051da10a2
4c40635cbac58966f74d2cca4ba0e21b2492987df509422d1b0f9eefc13d5f67
ee37889ac6f0d333e076661194f566459dbd581887e2729af60edd4e82a6316

response w

567b2058ecd6ca6ed3bcd0e66dfad0909a4ca03ea441454e9a9a7c63484017d
4d4d855d2ebb36ccc3ca4c67f4abe5323de02677c0c4c9be33a81900aa038495
97f13a843bee0ab5c27568bbd1db2f91ebc0dc380c995fad7b4019610ef0271b
457ef3bea417016e07edeabd3cdb1bdc69630365ce2d527b0f8dd254480b450f
233f4ac71cee64621e32bf37b209a6c41b17c793e161a98141fa68113bea9270
af2c9adafcdc3f0489e34fb8ad9bc1a0daa0280e90fd494c6ae7fa07b9667f83
fa813e7f8cba4f68e6f45a2c0c12b79d6da91407885e92d2393ccb9971f4dd7d
97ee8d8644e56d556ff987a261632dd2a7640925eafc9c522b83307ef6584f42

Fiat-Shamir context

48df25fe27a12f4538d6d97d99e70a3e47791d114196c470676adef1a223025

withheld proof term: 162 527 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_succ_one theorem, DirectionRing:152

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\} (n : \text{Nat}),$
Eq. $\{u_1 + 1\}$ (FBundlePoint.psCoeffR. $\{u_1\}$ (HAdd.hAdd. $\{0, 0, 0\}$ n 1) 1) 0

SHA-256 of the statement above

e0703a77eb5d95cf0db501f1f9b9cecc2f7813c0d58ecb0b6d80c46b9f81ac68

SHA-256 of the kernel type

8d6375a6b295ad79551b31c7da241035a02f17a857ac7612e56dc5eacb9a3a00

commitment C

348d6db3198a4ff45a0eafe64e1902848c69eb87f51a55e578d7b8a111339041

01255009d3380f1eb9d3fc0545d06adfb34bb43518f45d6a3a3b0464f9cedc3b

f02a2c66b97c5232900a98f497d3661f6c3d29f9825699ed61e5976155a9de9f

12bc9e85d778d6f4d7253b2fdeb43815f2445ec8a8072744ac8f8e45f248da0d

f1f6843fb3ab57f52b9737f61eb83468371c3802c76057fbc1bd8b4b79c441f4

771b4e7f8e05b20be034d46b249d3107ad056af958c59d1016b858981c89c290

c8f25220d646c09ce74c5e2616df17f943a523ccae73651d5eff805545e3495d

37578963bf8a499bcc28a9372bc218bd38a11bf6863c11307eacfd0dbbee2ed41

first message t

da474f63f3211b0f0787ac3680a9b0ca46db394202bdbb066b09751447ecf91

6ee528e901e3f51ca020744028c6ac078f74d4df17de0d21c93f249321acc431

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0b8956bce9774a32b6372af9ecf3166f3630e67333eeced25e291166def7975

33c112deabde96fce7e1e24327af24f789f9af4607fa82f6cca548e6ade3709f

fafb87669f17c29271279226b5a6b54207e135c36334db26e97981eb09f7a5db

faa6883298293e2456bab5b53bfb7d38b923441a95b4b337e555c6d169d8dd82

f673ea6ce3cb1e3d1027bf6624da1a6562fd66f7b61f65fedb01962026f38ac5

response z

41aa7d08d58d82818108a0fd2b52ca0ae77773aa037053c779e6de5f6c2afbba

1c81023a7ca6d93a0cb8eb0077411b910cf068f4ad932a3f545f3b05526848

30526c1d16519fc9720a989148312dae3352f8543f4ac013b23463b9fd35070a

507e421769a1ae43743453fd2a2016d8d981e2936b2998b6165b18a90924bcc0

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c67019ce5e306283ce4a6c8aa51af36cae7096bb255d4b0e8cd96cf2b7d18c3d

37c056e4241bcd91203eb8591337ad14b75198a8b1bd4a3092d366ac14e3a249

a5407e683b8ca2f19dbff514deee54b47cf71273d5436c05fca351db391a0b79

response w

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27267fe2a310e0c019d3a303fb7eae9f9cb9a627b15dafa03fadf1f6c11bbffc

88c3c4f7d96d21dd53f88189b08297c7e2c603ba7951edf4bd124ec065092858

460c2a9e135d5aeb5d73223d1523b9d2c9748bcc6ce79e4c3d2e9c91796a47e1

36741a424381db1a6f3098ec80ce1450a46d041e7301cdc76665b05c05ce544a

4a043e934c2a6ce940f8bca2de250f8dea52f0ec795f48cac737ba3158c06655

fc15297d1279ad029ad91c0c7529b7142017ffefa7a8cd577b6a685ff5062f7c

657f9f48a568dc96ccf43b65dbf656901aeabc1b290c37ffdb91533b64127788

Fiat-Shamir context

e3e6a897d071f3204468d778d1b2e61dddd82ca9a0dd9c7ff2978f788c7521bb

withheld proof term: 113975 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_zero_matrix theorem, DirectionRing:71

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (n : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.psCoeffR}\{u_1\} n 0) 0$

SHA-256 of the statement above

8968d130a5c290e2cab6d639cd833038062af5f182daed22990b005bb0dc86e0

SHA-256 of the kernel type

94534795031c345aa2f3f36a06eec7dc4a88250a3a52c68dddbabfa54529f6f0

commitment C

bff24ffade935d7c81df0f400e7b3e91043ed56a521eac83984e7fdbdc7f18
aed8134d43b32aaadfa731c28da2aa7cca7f442040085be3d9d24804c897c566
e2f2b4223095e8244b2286a24e598404ed257eb6773e37178bfce689374fcda3
32d79f6681be87df270cce482011886b2a1ee2ba34fc0ebe4ef1c87f8961ed58
f51114cd59a7a3abbe9deac998ad4ddc87a12267bb045fd9b70d95f93bdba2e6
c0d914b89965e36af55e3f9019d9f7bbe2856688885f1b27d477be91bf3ee0eb
971ed239df446a409f29ee542185a5924a615fe8eb3e6bd21586f3772a1bed6d
2e02e252fd3d335180d08a3db8109ffad3352a488279d522f0b4a63c532f7cb

first message t

15a4dc7fc9866e48f995a904e8bb8cf727bee9dccb08ed2dbb58efc911f5817
558a255b8fbc247e765b31c26637ce72266f9d2906b54270e48c531ba3d89d8a
b8154fd0edb2513e308591d739c547f777c9c5b516aefdff14470fe4551d9d6f
cf349541b435847a53e630f7a77ee26f9cb6047a89e9e182158c8292b93b92eb
8b75c80403d0b6249a27ae3d2cb70676b931816670af0573dfd488370c1921f3
1f4237cb6bf9d80cea24255528ce77437df65e5b51a01dc9def59b2dfe341c85
4f1788df5a805b596edcedae4afcc9561621e526011a6f7008f47766f86804c8
13925c149b471705ecb8b4956ad1967abdd7fa7c31b452631160914e62ddf1dd

response z

3d2ba23aba045c38966410bb7cb90fa75066c1f1255aab6f703dbf6f1433ff4b
681df46e811f025812fa8001811b63a2c8d065d528468cffcaadf1557afdbe84
246082a13fffb43056093a5ca60d4377f48ac4800b4d1ab455161955e1fb4c001
47b86dec3f8b30b5a411a882ac49c69ede9c46602dc34e4a04c0043d61567f7c
e7d04504654b3d07c39c39e73c84a7fa6ddfc19f24d40ab6d0279d5b63ea0ae6
f383f30065b624ce5b80f148fed3ad37c8fc803098f825c16299dae40da09322
e512fe404d3cb0ae2b20f4cc710563eff826fcdde7e293bd86c6aabbce7e65e
dc2fa60c6154b1e614fa14537e5e5190c81a4a1198250f714c8e297aed37ce9

response w

2dedeffa0a0d91bf8a28a6de88e7b37a19facb088ee622af7c5f5bcc83ba9496
3723330be73437cb3d127f2981884eb86738a9e9bd980175c4ad6d1b75fdb158
6eb3db3a2c63019a8e10feaec1bb62a7de46d792d3c4e012ff6c2c66f42fe34f
d45bebb119a8a662c54c342c2b236f8bdc7480b5bb6fcfc6142cf89d32b8a284
b04d5ba148643e1f3be0dc94d54fa66d30a23690a12cece6d8d211ba8229af0b
895284668a7c5b8dd1b68a66133a6f4946c9b350d456144d76bfa6f042560cc0
607cc9c975f20a8658c87bf527b50cdd47e779c3d50c0e5d3f1225cb39a4f7fa
0155f6402497bd2234c2a01d6b5be767e62f9202dc139e904d642607a0dd3a30

Fiat-Shamir context

5a36519941d2cd3865b3a6fba86d35f28ecc912c5d005f0d0880ed11d72aac96

withheld proof term: 62 089 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffR_zero_one theorem, DirectionRing:146

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\},$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeffR}.\{u_1\} 0 1) 1$

SHA-256 of the statement above

52025074c44044d0c53d15191b75eb20aa77117497dc933691233c63ac996449

SHA-256 of the kernel type

f6d2b2c1c3d8d6aed09011cad3dcca33ce766a159d2e7f372332858ef06e1f92

commitment C

8c1bfaeacc214acb566b432320be1eed002dd8393533e8c102c7770a2a406d56

a7f1fa85bab57afdb124f26b4f1320173df6f5988d6171a79cfb688c82e5006b

06e7fdd293d35821f5975d612744606712735733cc1ce1d93609169cd5f2d910

6e52c0f84ba9397611adac173a8f51f3b144f38d8f2ca44c882c81534480c274

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867b94c37d1e7656c8b797ab8f9bccdce7b0fe5deb8ae5e9eb72ba405f95eeaf

711afc4bf7799fc8d6b01693567e7230a047b74bd345c7bbb64aa489248aec79

1464537b0b76b753e6906ccf46d868f2151c245524e5b26fe32cdbf34482a1a4

first message t

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981296f967475fd9ca7d5d65bc96618af1491fd0f525cd7957b20dc8d593f65b

bf5761a95f11aa806825bb98bcff03742275c449170eced08734944bcf10a0f8

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f561e454180a4194eabefa6aa7cd70c64e1c4e81a824ccc59f35604c42e9f075

fe1f1f86cabdc10f1256a778eacb6dc3c03571c4d400c3066982789f9a675643

6ce79232745d24b6dd4590b8f4998e5b4ef2ca953f23c6f5c9c099e0ef2931d1

response z

7d331715140fa00e584c64d85b411666107d4b0bca2663cbfddfc61046bb7de

9b5415c12220bb2bff8549f6ad528fab51ae0e1bd2aad083128bf4994ee81503

242741f57484263b044b2a0735edc55f5cd64a0805b6d3144b2216daa17cc61

674006a297f7644ae0c8fa65a185d555602a926528e88afdec0e233be8272968

919da23000f435c0b31b14979c2ad42edbdff3950c5a1ee1b101791c694196b2f

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response w

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45e799cca5e78bcd7bc2cc800df1e0cbd593aa2f0ec0a55926627cc5581f3b58

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56254620b2cc55cde5cbe5de79ed11c4008c1323a4d4abdb9ca6a02f565b4877

ec94985b55a9a58a235a30d4ef7f913e48b287b70532984724338e313e8e77af

bdcblcb4233a6b0a82db7d9afc8b59841748066746c3f05382356790164f8e93

9d22c041ecb47d7fba75f67678e7cd22cce2ebf0108734f496023a9ab776780f

Fiat-Shamir context

683b8acef04196bb29d19ca9cf9ff95ebf9fd497701eb5dbd422a56c988c2226

withheld proof term: 138 489 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.smul_pow_eq_adMatR_pow_of_rec theorem, DirectionRing:282

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin
  m) R)
  (c : R) (Y : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (HSMul.hSMul.{u_1, u_1, u_1} c (Y k))
      ((FBundlePoint.adMatR.{u_1} A) (Y (HAdd.hAdd.{0, 0, 0} k 1)))) →
  ∀ (n k : Nat),
    Eq.{u_1 + 1} (HSMul.hSMul.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} c n) (Y k))
      ((HPow.hPow.{u_1, 0, u_1} (FBundlePoint.adMatR.{u_1} A) n) (Y (HAdd.hAdd.{0, 0, 0}
        k n))))
```

SHA-256 of the statement above

04e8572e7dae48ae0ce583ea3e3b0897d265968d3d3ce8685c44626f085f9264

SHA-256 of the kernel type

ddf31d365de13b36af8d51dcf51efcb4a69994d9a4e8f92976b8cb086224e5e4

commitment C

c18b98284d07ccb981011c544c56f3d6bf03f03a72d48633dbbb4e3bf5ab2560
ca91aaed637fdd53f02f1b86ad55ec18f990c2c89bf0145e413cfd9146885499
9c8e128be3e634ce023090e6f2e319441f63e3464de59462c82b5e1a5dff10df
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0112b6d13c0714adc14e67d104b3bea84718baf74cfe52c76301b56d565ff6ec
c0bd614e4c2ca104437cd3b693387905109ed63ffe0e9c196d4d50465b4e0b6f
30446d73676f3b1a18590184d862ed20bd42e371cb685e46a9d807b50748df35
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first message t

98847e6ff942b20dcb49630089d01376c57cd9a9bb37f214cadf140120e136db
6f244aa6a4487d19ba571fec1837028831e23abb419332da207e0508d7b81201
274d74b3b9f4de1e6e1c065cadf659f10fd2b843dda2f7710c9ee3db1970e786
b2a2cda7630c946d10345b2f0bb658093ccf817505f8e06a7ad909db0d243480
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91c505c017bf9597a8408000b8d36da5e5e2fbb220ed32bca748f6dfb2ba0e4
6c30952502861f2e652712b42a16b7f7bc741471a845c136f54bc7101998bb72
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response z

40e171a54838533eb98db748cc0e2fcf27029e5e72e588f2e75af75d2964f6c9
3c420d864121efe2038ac9d38cd0d15e5eae62b3f741de5ee320e0d634bae600
9703ea4236f27fb779dd0f9b53d8715bf6e6a2a5b0befb6cd4abb9968f5e3b24
4929da5b36b1a0497cc0ddef112f28d3463c3ab63e144d9f8d4145c3cefc9190
69efccf68f1fa29fefb8e82870c21d9d4965ab65c6576197a9c2f0b7e8decf46
09621cd915a7eb3255f7f7fa45fb0f14705384a13f15cf5edcc1ac33413c470f
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response w

2bc697314f86fdb3f855c952a6c69f75664e19c9b2330ec493eea559c777729c
ed43e9df160335c43587717214c3f8a964a41395d5251aff286f16d416945d1f
63fb673d670bc8967ba705b6e59423496f2c26ab9693f23d140145e12f9ae3b0
3fd67398d26596e1585627ed923d187a62d9630336947b0f73fd0b39f3d0b100
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77f398389087e0614b4d589970bcb1f35e55aba82ac7ee321da3d9a62eeabd9a
5f2065b118b37618e42d629379fd8280def7cfc25c8e28c75a93412512e993bf
525093d339c1e3e922427e7bcbabf973f45f3ea09e28447a58baf2bd4ff3243

Fiat-Shamir context

252f548125090e8352c01d2fa55ca4954f1bd86133fdbf05aee91f12f7ccf481

withheld proof term: 1020178 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.unique_of_derivR_bracket theorem, DirectionRing:326

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (X Y : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.derivR.{u_1} (X k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (X (HAdd.hAdd.{0, 0, 0} k
          1))))
        (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.derivR.{u_1} (Y k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (Y (HAdd.hAdd.{0, 0, 0}
          k 1))))
        (HMul.hMul.{u_1, u_1, u_1} (Y (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} 0 (X k))
      (FBundlePoint.psCoeffR.{u_1} 0 (Y k))) →
    Eq.{u_1 + 1} X Y
```

SHA-256 of the statement above

9f054fcbef2df04c96b94e68843f7480d14b9604fa6ab5bbb2389e5f44de15a

SHA-256 of the kernel type

039fcc2f6a7efe80c3c130571fa1dfdba5c5885e6624518e7de05f76f80d16d5

commitment C

44e2ea20f088e81a2b6e27f9a63eb98fe24e7747f55bae76be476e8e322495ba
cc71fd02e109ed44ba5c290f63554af6382ab3a6ecc623329798683b8c7fef51
565640877a736f23b86db53ef59d5d809996587beb657314f43ee1763262f3ef
0a335ea7c461619fa7fa147c81d2920b38546dfb7637ef2c4ba1eb63c2ae5e29
36fc77b69dc5aa40c9ebf9778fb7473c6d850523ac01f69b7b134c829efb9224
284495e849be39ae72ecd009fb481d790223427a1dbd1ce7716a1680fff915e0
d6c41bb684e574c4118718b8de15ebc5596f5720c784c4f650fee12a5855fe85
78fa1247a47a4a3fde8db78052b4e15b56f7ee763c253d90aec44ea6b380fdb3
first message t

e4544135e0aaaf5d846e41b9bfaa4b78f28af40a7f59d5600daba0fdd8b12778
99fa3a48526d6bee44dae088967097032790be973d3277cd5ddcd844998f0e91
6b52b60e0a65bb771a781d4d8868fa1cace23e9d5db6a30cc4f32667c7f93550
49256ab0c6846411f29cc23e3de264b9d17878ad49907c64f91616eb2ec0c7cc
0368cef5c6323203ab6236bc19793ae758be236da115becae5505e5cc232394e
dc5be0f79eef491abd5af61c51a969b4a0c221526f7b6e123ca5c179e5a1de81
06bcd2487add1dcc1860878c6350d1fe831ed77fa84646e6327769295569b28a
d8ca5c9290191e198a048771f8392957310a6f8e64fd468fd7352bba6e1bb306
response z

62674ab6f80ad094c8fe07436acc204a9aeeb766078426d28eec8de7739c2894
d7c64e052e14e2ad65b60fb0653553d03f4f9a6342110e4c57f9c3ed7ceaab06
41a94f661a6bcb00a65d786e8a00ee570851286b37ff9a6f7571a8a0dc1bc255
8320b991d1811cfe08407d53a2f5d0cddbbee9f9b42a49c52752cb2c0e092f31b
4d54cb76fdd182a884436cb5edae806a807bd9430458331734e71861473a9e9e
ae6faa95a1c17a58ee475ba855241b1e6126c863fbd0271d0659084e2d384d96
3392da79237a26cee1bb04353939b59355dd910804e62db33b9f3384eded3b37
cf5741c1967b038b7197b916c168c22cb45fe9c6f9903c49c87d8907d6d6910
response w

4c98dac4be370b44dd250e31fb707317a9fac1953e148291aab8e73e76f40e8b
8c67532c72196a6a221b36d96794aaba178ea5d1283d07d47e4cdb9428453052
defd71534766e1c086396bbd8f75a87db4107b7fd7ff637de1f5e5ffdeacc39f
1ff912b3008d50ad29f5a44e26b8015f704bbcf0681b4c537ec768ba344ad1fb
a19bbe3724cd6573072c8316c57d68340fef0d5594f69f5433721dc991823232
c3109d8281d8c81e8f10d1c117fba32deb13495cbf478490eb3951a679f3eee2
2b4d8f0c008309c854d488cb6d6e9598d1e3fef1a180bd3cb2528f86ac26e3442
3a69f519a52bd0e71e933103838ef260bba2fba04763a5dbc822495e0fe9a71
Fiat-Shamir context

9c6a907e3b14260a975622b7f573e3ff0bf8c8f0fdceb74d7af018803c6fb3b272
withheld proof term: 693134 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.unique_of_eulerR_bracket theorem, DirectionRing:380

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} (FBundlePoint.psCoeffR.{u_1} 0 Q)) →
  ∀ (X Y : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} (X k))
      (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q (X (HAdd.hAdd.{0, 0, 0}
        k 1))))
      (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) Q))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} (Y k))
      (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q (Y (HAdd.hAdd.{0, 0, 0}
        k 1))))
      (HMul.hMul.{u_1, u_1, u_1} (Y (HAdd.hAdd.{0, 0, 0} k 1)) Q))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.psCoeffR.{u_1} 0 (X k))
      (FBundlePoint.psCoeffR.{u_1} 0 (Y k))) →
  Eq.{u_1 + 1} X Y
```

SHA-256 of the statement above

9a3204ba932c7971df19d69e7a89973989ed00f818b8fcb816af9ee22d61a739

SHA-256 of the kernel type

93242fd42eb722151510d03c38d91c9d970251bcb7177bf0b807160bc083598f

commitment C

e4fa4428c485e737713ec25994815aae4bc79cda482f85c8d13d09ca7ff9d3fe
066b9b0517b3c345323c33ab3d9e7b16f34418459479b08640dd84400136bbd8
0a19027943cec63d555921bc8354496b306eda8439cd6d6fd72aaa2067a82012
858951e80d09b533cccf1460759c81e9fa691d1955f96273f25fe7c834ca5e6c
82db680538d343fabe990a4babeb715b567fe9088136a0768d63fb79dbb2115a
e6c4cd94536639a09c599a2f87d7ab85aef42bc7794366b4c33c6ba55905032f
c26b55837ce8ce7651e8b0841638ee3229fdc023747890dfd4ccfa0e38d644d5
185016bf1145e0b6cb2fca8be1a20c0b3c0811ce72f4edd1e153eb5777a1453f

first message t

9976c0b1e8e947a8e3123a2eea468e9d2428a74e2233cf58e8c0e553ff66625b
47e13cc4dba5aef0160d4a51ac52f293312f58b2a31a864c054f11bb6515c238
07b0ee533528b7d675a65207461c3f0312fca7b6f080fe2c35dfb011e3730577
e4341b11046e8d99dc7c8e0be26431e9881c29a7331046da88e15ba9dea6d1e6
dc899316f05311ab78ccbe0353050f626552c7fc27e0602209eb1f700ede81b9
9be9b286ffc4e9f8dec3346357e952609b1c20ab1fc744a67fba18d0db4532d
2604de810acd8dd101d3ac0feec3ddf0b2b1a610f69c5cb1b4a0a6f4619f8d7
e1430ddee4deea09cdc3f1c0914d07d2244055b18d5b24d5682139f1177d631a

response z

573c150edaaa9dbb9a5ac302a2eb8b8ffdd2c61da2cebdcb7b9787a576f1ce29f
e633159b88d9d131c5700e5bbc45c1bf78ae3304d49c75ffee7e8b4427c60be7
d30b35cec42d8724dbafcf291b40ce0932925024cd59a4a268c542f8702c88cb
098bddb227300435e444ce5b0d498d20632b7494555ff950c6f6fa8de92e7ab4
555f99fefb07a36f2d31ea75b27bba8c4b68aa831f12e644a8114a9425af54f
649694140725e5140597a08225a8ffded59dc8b9360b68b5e1f0599b2c4c0c35
86267e06e043ccc83ffcb4d884b2b290b98ad83fdc636a0197752b104285f4df
4e212edfab04252318e4f6de5420dd7bbbefb13d524806c3e8d470099d3721d

response w

3049a34beec4cc6a5ec0a87f773f6fea6583b86424b85ed3df52cb737f1f84ab
8e8d6159802a5bd5d53116e656e6a17c7033f32fc05b6deca09f7204b121a4a6
b26c14cb950f74eac7a0da81fdb006697926b58b3acd6c7e3ecfb60d450331f
c58ec8e5241e45e7b75cc52a565dc6d581493958f3435c16d6c5dd89021de6c2
f878f05be75d1101deb165326f7b60601c7011b3ea20a28ae75f1275403a80c1
e88bb336b63ae67f5ae056b7420cf58e747b3196ea4b6562808fe638b7a421c8
3e9dffaa8c52f8d749779bca47aedb5237a5ed818f4bfd6b89aa208446ca1d2b
98ee5d2ef505d3dc922ecd8440288435401249f73c0ce65468398927d16a4d94

Fiat-Shamir context

cb5f9cfb574c53fbf6b4eea214cfde1ee21f99f88078217a3b0c568533693e7e

withheld proof term: 561 256 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.Fmap definition, FBundlePoint:141

```

{k : Type u_1} →
  [Field.{u_1} k] →
    {m : Nat} →
      Matrix.{0, 0, u_1} (Fin m) (Fin m) k →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) k →
          Matrix.{0, 0, u_1} (Fin m) (Fin m) k → Matrix.{0, 0, u_1} (Fin m) (Fin m) k

```

SHA-256 of the statement above

9ba6fb243e5891c48158a1b95d7908839d4781c6c625efca97d89f8541b0f040

SHA-256 of the kernel type

fa9737903d3877b74370d7ab9dbce031e65047e9d197a4ffe0cdeb0dd647538b

commitment C

```

9ec5ff2e77696160c6dc9ca61017b9a828bddd16301380a3dfdb7534ff8fdd2f
cfd7761b3b4c71aafd95e8321d0a3357d8cd55c83b520a0f580bc136746599a0
e3b1ab087ba71f7262c0487b22233753a0933f4d1546251ffa45e7d9147ffc51
d698acd2e0681053613dd4d31008936ec642e17bb179f1581286fe0a6e05172f
5af140ce19b24e581f059393d75a063e147d6f3eaf43baecc5884b73084c67a7
4f79cedbfa8a345e1afe549696b543e69b7ff0009a0c06843b212801f5007a3a
0ef6a46a74787759befb7f6e7849fb86119021d7656b2255d3056563dfbdec53
77da57cf90816554b023776ce516e3fcdbebb7d55dd9bc32f12ea364808c7867

```

first message t

```

4c80f8b952a4509063d270c19080462ea8115354bc3d3267bc7f28ca43ed6794
a3fa89c12508a551123f9a02170eb5a1fc9418c678de0509ba88f6c402ea05e3
14e66296baece7d5b2d013c8a0255b5c2f07bc5911e514add5255faccfafd113
4236972e9d5e39f3eeaad965c0d6d5c99a191ce8b3deee8cc07afd078e1bb32c
214a1185c39eddac994c610c278fe61d95dfadc116bc939ff2fead206649cd39
42f0b130c518f8be94c0e50440f915f48812a5b9d8326342264bfff427f8426ea
678e50c00c7321e54f0ab149f90ff85d8ab0a49149babe3cb7fe4c204db41fc2
665702c6e0b19e73f29698b5b62e836a557c973e805df35af7285ba4172409f8

```

response z

```

3ae415e13fd6e4ca5866df7ec88ee8f737725f4865e22bd880d517cdfc27d519
da6fcb78b9ea020ad311a725d1e533304b66ef1f0057b9fd4518440cd6f27782
2479ba30a7ec14ddeaa439cd7a06b34e830c584afa273a3da6861c9d8f67f2c
554142bcd076a9c812bd6b59a00bad1b179536e531b74c5bc6ed387261e941c
8d212ed5e98ea76fd59f5394e99571208dfd17a9249d816a059dc4330b1c2bb6
0d1fa2407cbf6cded4dcc4f9d6234c594e1b43fec352df311ee1edd809070d77
017e4c7b8076ca52718c02854619790fe970ba69ea7327f17ced90cef991f563
a13102c101c664903e4678055f3996f6310687cafc23bcadf05caa401657f515

```

response w

```

ec4bb0628e98f599396eff706627f713eca5b28848ee78656b6ca2a9d9d92cbe
49975d61714a10b5fd8dfe44051a6805f550384207c4c91d2794b7446939bc0
8aa9cc6c3202da71afca7e9dac3a7c0d673df2dce585c9dbebb27ccbbbf7bbb
6b713b98813a92641f579207fbac80dd03552d4846f38c2257b29933f231c7a2
d5350d4b6f00c0deadce102d142db5898b568080d0d111fac7bbe54c47335013
5ddf36b7f974a2137d342972dc72c3348f5f648c690f346103d55cef9c9d44e4
7d35c0ee71d1c6aff4920936885ae005254b5ce0c2414c5da0323c417ea2963d
ac5aa9cbd5ccf13eb19dc6a882278810d8b3e824a96ed5af5e0d9510fafa57d

```

Fiat-Shamir context

c2d6bc9f8b0f090409b119a4957efc0dbcf04c913b863a30b9221cfc419c02ac

withheld proof term: 3042 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.Fmap.eq_1 theorem, FBundlePoint:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (G G" M : Matrix.{0, 0, u_1} (Fin m)
  (Fin m) k),
  Eq.{u_1 + 1} (FBundlePoint.Fmap.{u_1} G G" M)
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} M G") (HMul.hMul.{u_1, u_1, u_1}
      G M))
```

SHA-256 of the statement above

792618f1f9641912c5f06e5bb112f3c82dee50375313e64473eff3ea082a67f0

SHA-256 of the kernel type

f388b7c5c4b5cf30c88e4a660e89775cf961ef258b86130ebb63661b2358aff6
commitment C

6dfd938fe05424916f186764ca6626bbdd7a490a760b90d0d402ce4bf62c22cf
7932dad8395602b399010a1a7412f0e3684c41a2624aa9b498da1490325808bb
f9426a7866b1ca0275537f45e3c8fb2dd8bc085b28c869a17ca3c83fe74fa358
9877995099663f20b40ea91ef8b834ce3b92a6441a389bdc63349c811d53af3b
04431b91a0c1c195639787d2b49b9a22b3c357dabaf3ab052a09b049c78fc3e6
2149c9bdcda26e290c0171fd9c44e02520eb697e5aea2b9ddac78cf60466d45
b15b3582e87afecfa31b9aa201e1abb2b71a591d7f9afe07bf503801833d534c
ef9242b0a4bb6ade4e7f5d1f57cfa2f3995e45b4107a28b76d62c97514b87545
first message t

65214327cd4b2b2bcd5279014389ed28d4d28712e56565e896bd1c3ec698d06
3346be8c54523de96c69283d0aaebc1b89433ca369f357ad18f926f16780e85
4b1cc1618e128ce03b95d560cecab3101eaf9fa293c36129a1c2ecc90ef835c6
220bc56a98ae695f025e0c81103b80ea64629cf4bc70cbb598bbe17f0d3a9269
b536632c84fb6aeaba62958ce410b9540b9b055f2648b9ae981961213b206add
2e2b29177466f1d40d8734025b3d0da87c9ec637693d7607ff50dd62f617952a
1975bad497b61d1184044b5bd92266666e33019bc9428027f017597e05b6023
545586c907073df267899e77be74d7f440ef8df144277205f87bad26a57903ea
response z

3d84be367d011967acd84d388c102c33ebd87526bdf2b5ddb400000e6627887
9d6fb1bbc39bb86d42b58b01a0cc5f85c8208a6acdbf606c358159c17375e536
894bd3b4b4d131053b7f877c8b64681cbd4a325db3416a5ea64b5931d4777be4
47c42aa18e23c2629cbadec7c36fa66a1ca7d0c95994edeba8fb7ab14b345af5
4e205a8be17bc911dd4eebd4967c1f50fd7dfc470f87ee318aa11121b3460d22
f7352d6eec589ecd6da7abb55d2cda16fe862085599ca6e1c224a298fa0f84c8
05cdea4cd3fefb4468078886ed0592ac5a731c8b5eea74fcc6a2f8d7c130a8
1d23439291190886bcb73a1be8d42326c60474e8ec7ee620d95ac3f00052fed7
response w

270cde0d4c744fa45fbd1f3741e374867132bafefd10a114609bcd975b0104b4
3e63c48b94fc3e9ad2b7619b0d5a565795d90ed4a4f02e9230b530c4e2ec4d40
ba8e8d5d4bee08bfb55fec78a910dda27681e0f416b365fbbf6c9c6acaae0ede
5a3c3afed4868bc09d1a848338c5b1699a17ab697c638726586c8c7159e84de8
5ab328350a692d48cf8809045783a36e3ef057b0c0866038051590a48c3b95fb
df84b33b74244f981897f0bbaea477c60acd320070353fa0bc1e2375fbeb4a11
37439050d516d0c014d36210319fbb85a275f0173e8814b753be771206e8ce61
c59d71d518d231e7acaba4893863c20fed795dd65b2ae99bf24e738ff7c46ce1
Fiat-Shamir context

f98726fd05fdd048983d5be0b04a7fbb1f24baadcc095baf23f12fa27897125

withheld proof term: 309 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.Fmap_add theorem, FBundlePoint:151

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat}
  (G G" M N : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  Eq.{u_1 + 1} (FBundlePoint.Fmap.{u_1} G G" (HAdd.hAdd.{u_1, u_1, u_1} M N))
    (HAdd.hAdd.{u_1, u_1, u_1} (FBundlePoint.Fmap.{u_1} G G" M) (FBundlePoint.Fmap.{u_1} G
      G" N))
```

SHA-256 of the statement above

037a9745e55a456db9f6951fdf7f6fde58a436c6d14c42b249c85e3add2df43

SHA-256 of the kernel type

4b229515ff329675439770b538faa426cdfc96a2d6b76266785606be9355fa4c

commitment C

af8ad797d56f320a4a80b37f32236bedf0eb5f341d622078708470f07c1cf808
aec7f5e5d8d52bc694c42cc6eaf8178f5b0dc7592b0e3c3c37b50eec9c4b7fd2
7e7e80cad4340f015cf6aef4184292d14e90189ef73b71ee6598ba073aaef84a
4be6a6801589c692dd707d2c46c2c3352a08f32d27bfa70e154923c7dee03bc3
f8d78cbcc71094531f8c3cbb044651f7f9c2ec95cb892d922fbc907522d6a54e
6f16d9e57827b10133251146e3825a618654affe317ab5a25fe393a3bcd32278
1d741d440c635608ba1792570b42d086f6475d20f243df43903641544e99f0eb
4628bb10b626a6245c3aae51940129e45abbc970c44c9470735a0930823b276a

first message t

67ad374eed2ff54a443aa1778bcb7e02b6720bd1fb226641bfd9637a9997174
359892de475117c0fc8cf3856dce46ab7d143681479f1c7dd86ff13441e4ff2f
54940d1a60edd77a9e4e961d4ff7da48fe179081a70e58204c50a0d538ced814
d2593ee128caf6e9955777ae0ccc11a3dc0adabc7ed7fd812526d5b178f7ac12
31a298a4cd7b25b50e24a1c7aee81c35b67991ccc47932db711595e24e09b9ac
b990a505253cb72e5a2ae211da6b6475a7ca3b05263d4a5765daea297a5d1c5b
def3c9d8ec6589b7cd2843c37f30b53ca189784f4bf9e9cbff1fa42ad9dd0fc4
412e11198e8d38f195057d6140c34e050e06ff735e8559a57d5b8af924a6116c

response z

610adcf8e745f206e5f2e81b02b0b3411e4b7be3495257f03481e2f153fa451c
0b0c34765a4467c1cc80729a089f5875d6c4428824e3c476e02cbc3eda5405de
acedb3617f4600cdfef0bd0eb8f5535cb39b9e7ea9e14a49a214bfe000366033
2ae743d62f300923912af6d1508dc90d7df0919234fd53ec2b4fd6c1eb8ecad7
3cc15d6675403c2c2053b38b316bbef758db5bf27466c5fe10815a22a4805fb7
bd95a4305fda22a88115ac35e189934226f161d279606fcfa75e043209ee548b
40e84bfa0b76f9230c8fe9bfb6884c8da579095cf470111fbbf60f255a32993
9fabd607e11be313fbb94916ec6708e74edf945035a79ee011382f6e3234e759

response w

e5f82a1f2ee76f3f3e9bb8c0fdd942ae4bbb12c3657558b30576857b5f8912ef
64382e88118e5dfc00c4411b3f3e5ee6bbd970de0ab4aabb4891ccdedbefb6c8
17939213b2a5939875a1aae224560454084874ca987ff316b6969a54a19a6dc7
94d0c9ceeeba267aa8ed8988a171c4d011b439e332f4673260b73e7c0b18e1e2
a6b395d4e4bd8a463aa7f2dd35732cc0769acae06e8ebb0041916a5b1dc87f33
c21b15aba6237be814c85bfb1953ccac059feb9050c03bff390ae78312365e32
2e61a5d29cf5f0005654ff7ef834d4d33906336743c0831e594880ea9f4242e9
a8799bdbf96233059489a0bf32345a64337d99b15cd7bd7ff8a930f95438757

Fiat-Shamir context

ab1d869399cebd379418290d0fa87642d415c321f1052aac3536469b7b05ac9d

withheld proof term: 356 905 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.Fmap_diagonal_diag theorem, FBundlePoint:144

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ } k] \{m : \text{Nat}\} (G \text{ } G'' : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \text{ } k),$
 $(\forall (i : \text{Fin } m), \text{Eq}.\{u_1 + 1\} (G \text{ } i \text{ } i) (G'' \text{ } i \text{ } i)) \rightarrow$
 $\forall (d : \text{Fin } m \rightarrow k) (i : \text{Fin } m),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.Fmap}.\{u_1\} G \text{ } G'' (\text{Matrix.diagonal}.\{u_1, 0\} d) \text{ } i \text{ } i) 0$

SHA-256 of the statement above

425461a3867f383e734f6156b111c18b7ac778f931a5b1d69d4854dda6e9a40a

SHA-256 of the kernel type

9b27027d98e0dd62974041dd3784985b9fcacd9fb11d3e9bcefc6755d7280690

commitment C

d0f4e51ed956505a03016594c45d6ef178b84569724c1ff3e8a84601f686ae07
3c86addd929148fb5b42d80305b987332aebb6b46709359b0f4162f2daab40fd
c5b5c8167a20a7e7c86e16a94964068af631d460eaaae6a5b8f729d000477e69
63b4630e669307b8e36f867396698f7e0137cc43e84f5765af7a57dc9ef68884
a3c6bb50e6442b2a41c6625276d005c9240bd7a7fb7ea9944c4961dccc47471b
b57463ebda35a2c808011134ee439f4f0197769c334867940e17ea625af63975
fe41ddce823cea5bb1b455904df5d6226a568c63ab4677bbb84f9cc200c7aaed
4f5363048df0a30a1f49fa2946144c277e330c0fc7a1b048d058131f9775498f
first message t

433555d3c9f6319739c98f19a05fe3a8a1c29f239567ad708c2d8563077fa384
01fad1274ddb9146ec2b70c27c2e40ab86aaf5ca5a031b7a8a808bec9267a5a
b6e41f6013a352d28a22e13eb335750c6f093ae60ef0a3f615a14b775238955f
b085050ea6aa35358cb311403937da3ac381d6673c9608eed6700233395e3e26
29430278edc4cacf0c5799a9ce54b84bf771e89ada2f973705266b1b10b0e6f1
9e14048ad82b2275b670d1aa30e9946bbb797fc293e746c3bd63068ed130aa8d
910c8c1b681979ec7187e0781bc8cdb87ffb7c6f29ca2b3243318f32ea61f8de
ce5433eeceb44e1814a5adba54063c4deaa248a65382ff5cbe4043609c41911c2
response z

47fb925d3e2dc22004b7f7d2bf81a7e62a55be245333be9e89d60d18021353fb
433832803d92d363b5f058b09ab48586e9612221b34e7bad19b5277c164167a1
544d1e31c3185ce67710d0dea28201b796659d44859346cdb035b81ac559c80
986f0d657485a843fa5fcd825b202df5a295fb8893d95a74fe7d32a638acc37
5560d8083b29322c67bb920db962da44fece770f0c60369d443aad9df9922a02
bfa029b9d1f9a2453858c1338a541ce837973c6be5cf3140f604c16a22a60d04
a0189a30aa84e925856958d2d5f4b4f8036e76279e52f4c5e719e7982cf2d433
468208beac7ce74ab96757a2d0f24f5fb7ef726e886e9b584698203e39777561
response w

28be797f80bc9da91583f443db773f8cdc16bd780d8c242f4bea5d493bddac5d
dbe7410a8d9b4c54312ff02cead16a437fd10e46eec943057dbf493d90f5cd85
cbc0bfe8a553e1ce1b8121438ce2d6d34e4862f3eb989396bc185517ea62fc0a
1dc743af5f341d8b71a7c89f592cc329f96443630ff48a8de450ff2bac359251
dda8ceda92f6b24064489582ecb0a8f12c9af4a384e09fb8b7b4adc7f502432a
ece13f8c6319f177cdbe94e921535215c8ecd87597ca52ed2b30bb7534051692
c1ff0f11a896b32b1426b7f90635fd4ad16e259347c0a9fcf180b95a77a97b06
a3e3436fa517115afa82eeb351f4fba0d64878def66eaab1cbe77aa4d8cf9b60

Fiat-Shamir context

e11ccfb53e2861381395790932ebc430581e1334dec9556128e2dbdec83ded35

withheld proof term: 166 380 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.Fmap_one theorem, FBundlePoint:161

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \ k] \{m : \text{Nat}\} (G \ G'' : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \ k),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.Fmap}.\{u_1\} \ G \ G'' \ 1) (\text{HSub.hSub}.\{u_1, u_1, u_1\} \ G'' \ G)$

SHA-256 of the statement above

0a4cea56d5b7039ee1bc28f9f1e3446e48f4355eb26137a88279a66443ef1d44

SHA-256 of the kernel type

e01c986d4e2354a0b35455a4bc158c4d25d2f52e07b829b64a8b219e5c2bb2e6
commitment C

746287fe85515f3d9b5c13381e7e76028dcd02c03db850c6ae90cba31c1778a7
31086444bbcb9fd102b9628a2e780aef8916d56107f56379adb4531e9a44e19
6d7f62ad52d592ce261822c1aa65cf60189df6237b013e8ff3b28ed6d3522b73
cbdfc93c83f1132ef091bec0f2b73fab62db1f4df28a9edcddf2b9b55973c1b1
f11a0ff658114a26bf5679a66b2f4148bb1a97998a9ca7faa7ff7474fac7f322
b6cdf566e7c37112f97e949e1ee0f063d359e4fab784b1990515ebc7dc827391
7c2ec395f1a6da294ea6cce7b209d788add8a41eff8015a86ba398bd95046d94
e018f2210d9de07641ebc577fd201f7268ca2ee4de23891972e0ddc41b7dc6b9
first message t

afef4f591c0bf3bd79b3efb059f0ad15406a482f40550b18b971a2f91cfe9703
22d44c9c4b72737c20f556e82e579bb7a646c4402adf2fc33358093efa4e94d2
5c841ae0fce80216e733dd73fc21268ff57447886b13ed1afec2f2244886a524
7402e49d0e839b10e75e03adc25fe39577ea16c2002683cf745a1e674e1aa758
e55dd9b5ee0c67275ac99282a67df6062779946e78dcb881210102384d27037a
a0f80b3e7b23b53cab04d4475c7bc3a8c519731c4e210bf5e443c54a966a1528
724a8c0345575e54cd09615289e6c5fdef48c12065962ac3e2e9f5907fc7f772
9515680b179766ea91047721307fa4b0b831418868b54276243e268c448f3b7e
response z

2cc840c4a9c402d6c49dcb84082d7553826566dbb803ed3f4a15ad5ec965a0dc
5ceb12330f37fccdae40a15f4a97de66c53d12626613df31a44604be4b0db1d9
fc573915699c58c8ed5365efb1de5e6cf98f22c5ef3cb14c9ccd514f3ff5e021
536df490888bce72b112cc6510d7eccfaa72202f9e78eedda7f55446eb3591f
9a87cdafdb649047138423c30d5086a934bc4fa0cbb5108d8df928a6b3b808d6
bebd965e3ceadf351875828cfbdeef5baac86fe40adbaadb6bd4598a13be8804
594ec4e03572b527cdba4f13f61d6330a6057487718e7d9cd43506c0729003cf
5a5ec27095c10595cc2814a69dd50f3b198b6657e6f14ddfaece705ffa86c325
response w

129e8391d6a49b9906f2d0acd99b4d0cd4eb6f94ae28669516ea21946293db77
86ba0ae0f87fa837f60a45091e45450a30bf612b8253a18bbeac2368d1568754
e8b8a69d89fa2c57008b9d886bd1517630fc1fd00443753497afd2b53f114d4f
b6649e1a2a33176a8637a148d3b2c563896a6fd7c4cad1c4fa714c05568a18c9
e935456d47081f0980fe8252b988eec6cd56e1e99d5e5bf33f186cde6e7bfd30
c7f51de48a8e5579e5f93cb558bda6540ce2904430656e34b9f7602792add028
5b2e216d72aa1cb136d2ea6ddc78cc6d5dfcc7f32ae82cbd026688cd833e4460
867c4075291a4aaba8c848ea42ba8d709b63177c7afb76ed5f851bec14266232
Fiat-Shamir context

f27ac361d4631358fe8ce129a326d9383f15fba9681a877ae2900bfeca34a9b9

withheld proof term: 18111 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.Fmap_sub theorem, FBundlePoint:156

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat}
  (G G" M N : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  Eq.{u_1 + 1} (FBundlePoint.Fmap.{u_1} G G" (HSub.hSub.{u_1, u_1, u_1} M N))
    (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.Fmap.{u_1} G G" M) (FBundlePoint.Fmap.{u_1} G
      G" N))
```

SHA-256 of the statement above

1f25417732482b6ceee589cefb7ac803dcac8bd5baba8d2bd6e80ab5f9bc6deb

SHA-256 of the kernel type

d3aa3a60ee3f0eeeba20b82c90ea6f12b9d28642a87cff6a9806a670d0e49221

commitment C

242a8063405d8444d26bcec531fe627d4f5b4222bc9e3c815a1ad6728ec27cd6

1e07bc85a5881eabd0aa8ec438abe6ebf2a4ebfec909c21a3c8230d9ef50d982

08e7d5b7b1167d702494391e9c3e69b41ca221246204eb50bae58f5e3b3203cb

e273831b6d5978f8dbe50984d5bc17075d62fb03e546150344f3c453d1d1b9dbc

2c265698297f0cf2d917e5410611f4a52782593429807c5e3b3ab53a987d2113

261bd83c14a9799129009cc11ccf5801fbbd8bf08c31e4f74d314530ced091b0

019b856aa87382f9a5ecb3cdd780d004883457b1ce0894146b9936a2e9416ba4

fb098e427ad033e0d8f62c5e82351cb8245e31d881bad3ba42777df42a26f74

first message t

317637541726ff78dc336071a9fd966c2d6577f06fc69267c68597b67cf402ab

900718c32df7ffd5713e3b9df5627daa8f5b616d33701952ac1f07dc14ebae8c

dcba3daf956528a453013b2fa8e022cca33c76e917399568886e067d04be8d9

877c7bf0bf5279ba556c43275a90fdb83dd1ae197d4f688ced84e93a5a884173

d66c8836872d562d1b2f41f76e1808ce397e29ec0da12cbabd8a62a51221d502

2c35db9f07701f69f11ec1e0478006d0c10c03314a6227ec4f2068985735cc1f

8a68c4a6f7f6805b60d6596922a734c72484001416d326d83e4744ae72268e56

fed46ee7822dc21af3007b06f82f554ef80161f0b97f3e7191b3c8b2704a8d8a

response z

3a1eadbf728587e0074e4fc2cf129d6da77f25128d80d1151bd598024054b183

b9cf9690baddc5f1a88aa0e17bd2e659f345c2fb8d67c8098f95948e0d59bb1b

43153f5824b6f46961e7108d368461e0c7e7db7f75de52abb397f150be0ca6c0

d8e27118eb713b5591b8840538dacc698aa7b692beb521f0a40c209456a3668c

6f08664538036eda29f4f1eeb9f43bcd23096ad34d0654f1d9842b845ce87256

c57a280c048401f4b2d87bff63363a52fb78ef5c38110c84d92b4e77ace9a8d4

d2846c9ae7113f00493d9b5c0f134ed91d80371cfc789f6c207c7d3355d76b7e

e446acfa5096c6ea3ae1c997641fb1571e75fb9fad241a1bbd846378ef280a5b

response w

39769b84f62f8adca79d722663575a6519ea1a72deafe8493da5ef28c01ef72c

db45dafc1447714ff35a07b34be547a3a81c44f11a8746d014915d3c6fad09d7

6c14525c5e155d6a3c5d4cd51e59775f8ca7a2936cdb2dd864f8b54df35ba620

91e1f825572ce70d995d343bc400ea00c6e8f307d268c0574e61b9bb44305cf9

c3439fd5631fd9cb242c5dfabc5b0575e86b6725295b6f6f14e06789c25a7f02

52f0803f7ffb5f40db51a406c71f4cc94b628ec7437094fe82da97c75bb5e0bf

b70ba1435d565a695917e4487e9f20a8f8981b4b8a01a562a426b451fad8ae5

3b54fb1b8cf0b57704f8b041b1419e46f48be96fc67b9a2f93cff5b4b5a8984e

Fiat-Shamir context

b8ea240c7d6f2ff8f9cba5c1f1b29ee590e1bdd5835eced3f988845def2efc19

withheld proof term: 367 928 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.GaugeSystem definition, FBundlePoint:272

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) k →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) k →  
          Matrix.{0, 0, u_1} (Fin m) (Fin m) k →  
            Matrix.{0, 0, u_1} (Fin m) (Fin m) k →  
              (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) → Prop
```

SHA-256 of the statement above

17797ce1123ca8c85f350cac6f2182aa5a105589e2bb087a8b526fbb70af6419

SHA-256 of the kernel type

2a8e2a56a55670536ac85478dfd81e921de858d923be69ba73d1ca65cb615486

commitment C

6bc7c38a3ccc2e179d3260673f0d410bb3ae69b997e7ce0463b0038d16eaa948

108860e496a5d9c86fc04d1f445d1e483e69a3f156b321afe23bb7f70172cc95

18a19f3a61a0349af7ce37b13f471ecdaea46ddea23d7b21c067717ecc7d5bcf

2984b6ae30e30e1f76826aa0b772a7aac550ca9dd974fef82fff5cbfb462715d

d301da0630f68130542c76e0276035f50080ef234cd3e0c6761743d0268f2b42

0e73c6c1c7fab586993dbf0cf01689d1901d4f37770ebf294c5280a307fdd6e2

9a1c79643273688de1c60f79b27ab6fbc366601c5fb6df85482c36136d157ec0

37e5f3b2ad360dca7dd5245e9e79ddcf18525fc41099d559b07722d3c14af8c

first message t

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4fd778f013ffa3269fc4e3c549375ca0716daca481d77cb9b0b60e0590f4c4c

c3d486b89bfff682ed128f95c5f11ed0ad734140c58ce73ba3d0ca70addde6fff9

900a1e3c5d8f1ccf09403882ce77c605c442fa71f0e10243a21ffe5956534098

d8b7d72f2486015c9f29bd2ede0779b1efa8f86f85fff2a110b47b6fc21bdede

6e03c12c26cb1b2da3118574efda51e9e5f34a67614c759be779ef5f497db8ab

cba079feb357c1060c4a4af071707453bfd0e6a0f4872d9733ab501df8775c22

4657566e13b3fce73a0bcbe805cf475535fcb5cac2bf829127d773b5a4eba437

response z

27a075c85b03923f501ea3c2881624803670f415852805933bdde87c6424b709

f594124f20f2eb41294c74cb3ca83124dd42a0a079b3f5c18aa5c99a3198cdcd

13e1ee51228576a9de1e9f69a92d486782640925bc17ccd8a54d2a84726d71c6

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4db93d0d35aab8a97ea37815a18242dedbb3111739858dc4ed72ced714deddb7

68ec20250b8d56cd471a61fae6f80ab58bb29b674c14845e053c9d011bc9ce7f

response w

67f4ba2ecc1b178737b26f9682a323b4d6669c4650852ee5de599e4aa6f4089f

e8045859b403ed3f8394c67aece8d59b1ae5f8c6b3411a1364dcec63b32272b9

c595b7f1f6e00c8e7809617250f7685657208e13f51f3962af7e20ef3d6e1560

bf909ce2e4852ace0f0e365b9f0a298a5e41d01c225801cced85ed63205df1f

8f0e421d4f4ecb36b12aebd45471c670be24686cec4413dd1b3ad91771c9ecd3

1ad4d19d9bea49c50661435878180da623e17d1d8c2721a9f72081b878841661

3461e548f7c1e3120474226529cd5bde9b6f2a4df5b67d4daae8caac1649e826

730ea3c6e4aed20fa5ec132796bc5203b15db1da3ce513cd852719c585a2170

Fiat-Shamir context

eadfdc8a17200ec1fb6f1bd08e1a94731fb090fa3ded31b3b8831138d9426437

withheld proof term: 10 685 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.NormalizedRec definition, FBundlePoint:167

```
{k : Type u_1} →
  [Field.{u_1} k] →
    {m : Nat} →
      (Fin m → k) →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) k →
          Matrix.{0, 0, u_1} (Fin m) (Fin m) k → (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m)
            k) → Prop
```

SHA-256 of the statement above

c820e21dfd254945954ef3017fda04b638e01671829fde22e8aa6840920f7257

SHA-256 of the kernel type

d3bb8b47577f6a033c61962c49a7aed912609ed038d4ac40664db5aa2cb458f7

commitment C

8419e72e48451065a388fb3ab6b4da65dc448cebee6283b42c79b565463c32ab
71f44c9c47bdb45bd73c03f22abd6b15fa460bb55abc9bd0df4f074ca052705a
6225ca3c574715b0ca2fb79715da3966c60b75529e0ce2b3cb30f6527ffcd6e8
370f3bbea3cb3c64b11655f70adf023fc40932b98f63f0043afcdf6ab196e103
1777de8f43f46bd76b521764b5f7292d3b9ff1f04fcb287a87efd8a8031300c6
eb8a3f469c78aa358ee2f07d5216bd93bd96a7d822031c83744f192e679c47ad
eef763ee46220ebbcc1549df87faa3f7e60e50aaa32add19fb04a856f5e555d2
5bde9ba7b8672c79892b0760ad1f4eac4f9f7a3a01ca09c24109edae88f0443e
first message t

9d1fa6a07f1b2ef451257ef84cd4a6513b93ae2ddeee67662f5944dd9c57a53d
8a3b21f42b004a16007ac94964ad60de1aa2cd3671ca8925487a9187b0a15492
2a8a93ee4c3c694277acf850ac01cde7c8861b9aa1f0d9736f39151aedf1eaca
78ec037af9edcfc1c510957b9b1b9e6b7250ca70deb6ae266bf0345367eb4f8bd
5bee82c5d9c16b13221013c40390b83253688f64361bee02e5667fedfd416018
817a3c61837ad1452641c874441d6056a9aab686fa7a706c2af55e84bb3c49a5
6de6daadfe5ecbab24fdbc5df01927c9ec7911a40c959fb6b82a1a340eedf0
f7dc38253f01c782aae7b0af0395cc05aa5cbfbd114b4b4fd81a0865ab3d98b0
response z

36485faa66edb3be9ea6bd45b23a2c2b63d7e43ed8a70752c435a9c9cfa33270
ff60344501d26ad975bde0071c711d6ccc887c2c4e62979648776c711508db62
a63c1d8cac5ff042538c8f0ef13a069ebd2c51ac1286a2a77c52780d7c305f86
538c66db96f89da2f6f1e2a024d35025ade64dc98520a31444133f97b8ee1a54
aa51ca8fb736fb9d7dd06dbbc729aae7290143a13d67ca7d6831a570aeba4535
c2c099c47f0f902ab90b09ab683ce2b0655af95bda691c211e275b1c4fab21e0
cb5febd4999e92aab6a08c227dc5ae6bdf3c2dac41fac1345e8c84b473e34d04
0c91df1828d87c991c028406a2a1e3b3d87e0413d8e6d129d53785939ee6d0c
response w

403af2a560dda79f92ba932f9a85c36b6a7537dca8456ec3c1c955fbc15bb954
2a6558c3b3985c88cd310b5f59ddab41c552796d4213bf1f9866924d0fa52cb5
1e4676bbad30c4eeb529d567d967073215f949be6e320a269b9cfad9aea5838b
37d3337d8870ef56d151f9984f2f2090449542b65bd4e9ca8faa928aefec2b2f
5a10a1c91ca3b11afe5078cf209312a4abfb40062dcc5c7a3adf122e1eb6f28
419308d0085adf77eee74895a1894f2b69033fea59d3471139886dce8bacfb87
ea3e136324a1204190120f2ac8dbc82a6039e69b00d13efa0c2c502600dce4b2
6b9098c6471c0885efb6866138e1be46f05b6485f7ead4b2d6bb308570772b84

Fiat-Shamir context

9f97ed7db46b5a37d7e18e233ca9ebcbb33c705941006f7adf6a60d1612a2b4f

withheld proof term: 2237 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiag definition, FBundlePoint:58

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {m : Nat} →  
      (Fin m → k) → Matrix.{0, 0, u_1} (Fin m) (Fin m) k → Matrix.{0, 0, u_1} (Fin m) (Fin  
        m) k
```

SHA-256 of the statement above

9cbc346c080d87efb61ee3a1125d891a31e461094bfd01f2f0fed472add01dfe

SHA-256 of the kernel type

e509338755ca3c1291c0722804f31a7cd5d03c9123259e3dac72a663d793851e

commitment C

b8737087f9ab238d95279f15190887c8b004e0c489dc87da6b5397ccacafbede

ddb06c0d29162178248938812d4ccb7b8c42ea9df6702f68554043a4f6421302

127e4cb82b95015588d9c29aac758a48a75c32954657ecc168ae7a77c2907293

439f39b844a0beeb75d9f319b571ad9c7baf5e4ec7856902ce1d51c2cd48e20d

21adb77fe8bd3a0148b522d01cdb3656c5e6a6a9a95f24a380300d0ce1da479e

ea1dea7c20ce78841478004596021fe914adb09367d392f92a4f7cf93a4165d4

ea551cf88caf6db74cd1aa718f2c67708cdce73194e97c04700ce6fb45b5e098

631429d61653c7f5aaf14ab981860f89184549c5efa1cf48a5de994a9bd99bc9

first message t

4c47644d98532df53a21c996019acf59a2faa6fe9f0553e6259cf460cfbd0b94

a4394800bbfe868e159a705d350bd733eb4e136e770999d66dce6165ff152c42

fbf26bc557390fb68fd7935e2d4a2504d2829dc7d4253a07b61c5d160d847f96

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b2958cc29bb642f1e7626979c70f7a0dffba1d1336fc7837239317701c175045

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c0e4f64c6c7b6558d0cfd70b2ddbc4e04f2407045109e3f770b871cdb736e1ef

response z

64fb21e2c872bc0a588ad5c0ccba7ba64cf2df8264d3e9812c12af295e1b864a

a2500f4876a23e0f0bda5719ac3097bfe46d9f1fa28c9b0b4b3ae37deb7a96fe

31110f7987ef5f7b77954f7cd54351e9ab71ee74ebcaf69b1c7ebd130a7e384

98323c6f7d273c1d295dc8d7edd8b548bba3d7ec2e09f27229f9baab7c682472

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response w

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396a6df6dc9d7406582886fd7f00dc1703d6fe0b4d24b6d77e65b04c0040711b

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c3810519cf8f749dc59292d4142d7ffc16e07b0b94acb6e29e1156f8fd4fa36e

09f9d55aa989b6179a06e5d2438c591af6478246747aff43835fb758e309f325

b78b8a7328a5ca12fd409a5dcc92bf1169c7883709d2137415e4c8a761eab9e

b445832248272888f5bbaf37a21b716509388c163de6aec14f97a817466d51a

5b1a082324d8aca1a901c7bba8ea5a774eb05bf81a7f409b96c93db5ea03db46

Fiat-Shamir context

072c725f1fb388acd9bb413766abc8cc686d13614328b3804c76f45f8481e9a0

withheld proof term: 4 009 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiag.eq_1 theorem, FBundlePoint:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (ξ : Fin m → k)
  (M : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  Eq.{u_1 + 1} (FBundlePoint.adDiag.{u_1} ξ M)
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (Matrix.diagonal.{u_1, 0} ξ) M)
      (HMul.hMul.{u_1, u_1, u_1} M (Matrix.diagonal.{u_1, 0} ξ)))
```

SHA-256 of the statement above

a9df9043b2d50c17616733d835dbb94f2134aafffa1d51a6e8891e6534329b88

SHA-256 of the kernel type

308f2a02712caf843937d25e210282d9f8b100dc3609b9d8cc88d84fc325b9ff

commitment C

410b0ccdfca0bc68fedcb7369c1ce6a3726895d02a69f8a523c3419737000628

4cdc188b85f62b661d82737353f1d83f56c849b9ce8e9b86102749c14fb9e987

77d0017344e6e5a15169ed32e3c5cf7e8ddd9e51a528dba331d4afeee1112c2b

e4a1c3d8e009b2760bb8b30d8cb8d56a938a05b770c7cf0f8ed6a536cb03b3af

393c3b6c07e5531e4c78a823d188520b38cdac0a6e9e2c94fd11a2aae362087

3072b0ffec9a0f93c21cb6ba469157dd0ef2aa1a96bc6ab782492216bebad42f

a69cec153c047c26ef4e7e866dccc853ded8d66120d9aafbf0ba6ea7bc0fdb53

969594cadd6f8603e7f2015bade1fd01cb380b4761c55374ad0121bcf0bdda12

first message t

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24630688f353b4747a25e2b31a73e1b654c0a7b25504f004efe088690d9b7e76

99f0bba83c66c4c0ffbd50153bdf21af7a08393caa852b45b9089f64974fa2f

b9857783a02a855dbc95cecf81e7dab747e24e29b68dcb39e24b8a05f883f287

4edf19144419605266e2f66aa12611d3825f935374f0085552001d22f89b535b

dfa1490d6856298f439b6ffae7ad0e671e6cd543dde75a5922b8e8f12941f8a9

286f53a2ac38e8efc1399f62644d1f64ad304aae4f37dec70269637b9ef0130c

120d97fbfb806eab39310977dbe297ded5e3ae07f93d9dae308e68706d6f4e61

response z

7424e3a4840c72ae7ef60a79c9181e5c9915f01720c07af2809b70aa9f0755b9

44f0cc71593eb4d91521a01b2fb071483de938f654bda60d75456731b365ab80

0735f96126ffecdda1465020d36eec244f95af6b7354c5ab2d8ac2faf248fbf

68adf9eface0e0f89741a35c07611d0668d0e8de5b66856076f18bf3b41abce6

a316585cbde2bd953af0ab4347156507f6386f90a6a27c60614e3a5d58dfe224

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74ea6421aa02f8396caf03bf92d1bef608bfc7d5d375d3cf910477cd633f9b2b

c4119494344c4059f9260571b3709cacd290c06568cd90041ca09851bf6f0f09

response w

3e3adf271dc4f998ada08930a375a65db07a656b08f4c023e9e48f60ccd47199

b8f7b6e8379bb86a1a1e554188fa5feab03d15fb08f4bdac585ae331460ad9be

edb00e2fa2ac4f4de31f858f54eb2e138556d2150f03698f51555a60ea8b47d8

cc62d0e67134309e8965fadea27496480abc2324d9e3665477f51b252aed17d1

62a77e2431e9f4339ce46a0a86f322389da13b0b7c2f8b68a4a88eddb4c2e093

9a979e45d3104a61c7575cb99ea5a9c6bcf5c12bddb0f579545a447841c8fd50

486b9813fef020a688cbea0173d26afe52fc3367ea717263494cd0d85673b4cf

0671202ee2bd5cb1c4936c068b28fefac24fe8ca972d342fe303d8305722d985

Fiat-Shamir context

99d793bcc720836862b3e6517cf32af68a70954f7e7fadec138dcb423a7a3157

withheld proof term: 237 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiagInv definition, FBundlePoint:90

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {m : Nat} →  
      (Fin m → k) → Matrix.{0, 0, u_1} (Fin m) (Fin m) k → Matrix.{0, 0, u_1} (Fin m) (Fin  
        m) k
```

SHA-256 of the statement above

9cbc346c080d87efb61ee3a1125d891a31e461094bfd01f2f0fed472add01dfe

SHA-256 of the kernel type

12675770dcb81136f8b697404230639769ceaaa5196320a9f53f5d4f7de1e9e8

commitment C

3f434743ce5e0140ca73852bff68fe42cfbd6933429edd9e76ce7d0d495919b8

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cd886fc7e661b7133aa056b8cf152171de085ae8d71ff6a187830f26707ec66d

ac30e1f9d754a2b2563f68423945b9b77d36b723c628ada09b60b49320721ea5

eb50df638516d437fb1989066088a70ff934faf0586f0ac6f9d2e0660589b3df

first message t

86e4f49adace1168e05fe75954c7c140104ee2e7ee186e9c470bd186cf10c149

51b14cdab2690d0e28bc592d02878a9437c6bef939685e326fbabb141e9dcae

30363769dc058e1dcec69cdd53c5432d728431e6515053fd93d596908fd8b862

519702f7c550b14a11ff50aa0bf1e8c4c92bd210a30fb67bc6f3711ae0efb32f

781b2cbc4db97f995a8576eac9bf20f4376946a54f61378302f25fda92f7009d

4719c780acc7a260d31311d4800d4015d0e68b839c5537e5084a852216399f0f

c2c01e43c603709477aa9267a6cb84855f6207179f6519dc445f145ff490d051

3daa1399149a4b4cc9922fb1efae5f3924ae921d0a4ace87926d9ab0686115ca

response z

44f02fc2577673e24a75c8efb87cfb65e628d5d810254249a14b6023915d2407

93d60fbf764af60f04ca471fa00e5c7d449abcc271b37f3c115b851e7a5c916c

d552e62fddc6caf5c236dc71f325b8f4f33ceb58da894cdd2b898d0b18564ef5

63d478eb10c47a40199f3e8c4420692f22d7e51bc769b2f1ff6d88b8602a6a15

78963b0e36a93da10b5899eb027a41fcef47e7fa9f7b066246605f60547569da

4163afa083751f9ce05c00e8f65f283f613f5b564ad951dd55763df9c66453d6

699551f595f8999f8dd2be1bd2b5d1f2467c972391ddd85e0eedbd0609d57e42

6c269c4f32dca58d813b05b6c8f629535b5fa4e7af29a08a33fe600716916342

response w

5c52d838fc2b48af4ba95c80682b917d67089072749f1e1bfc89c2def9818a74

279ecaaf628021e29fa3a16e98094232eabc1189c92f29c5c47ad1135d206879

b6ce0432d4347b3d2651aac7e51f9ab096d206d3cd2c28914550931d5b3e6889

3cc11e6b6565391b787a75bcc0f20f9cd94597bd26cf633d955ef0afe13ae23e

710d022f6b327496bdaaef1a701ce06fa5f7dd87b9b3d71cd8c26a7757daa4e3

9800e6b78376f78af45c5fb3d98e62e31b9d70944f7e3c98643ec7e71a4bbac8

3c5d019d32e0f9d20e8e13d45771fd17d6c0b82c1cd83b0bcc9d50b54cc6e468

b94194c6b4e3f3143bda8523cea4603b663fcb158d30fdad0fa42f9b8c8496a9

Fiat-Shamir context

16017f1a95060a47cbe5f2840f344f0a7b22ec2e5efbef5ce79162304b4331da

withheld proof term: 2178 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiagInv.eq_1 theorem, FBundlePoint:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (ξ : Fin m → k)
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  Eq.{u_1 + 1} (FBundlePoint.adDiagInv.{u_1} ξ A)
    (Matrix.of.{u_1, 0, 0} fun i j =>
      ite.{u_1 + 1} (Eq.{1} i j) 0
        (HDiv.hDiv.{u_1, u_1, u_1} (A i j) (HSub.hSub.{u_1, u_1, u_1} (ξ i) (ξ j))))
```

SHA-256 of the statement above

1e1e07d9c7171780c160e77ac52b4366705f31cc682adef878ba0d204763824a

SHA-256 of the kernel type

5d6c6f81d02352964e23597a3b249756d1c9590d5063f8775da05d2af5a3c409
commitment C

c8631c65e063d653284fab779dea094c9a3d175bc5d486261638905f0e6fa1a9
f8e99081bfcfb095f57e07c98f0fe0efb45d7c30a8c01a5b70412d4950756f86
cfd17a75c8d99fb2ae459cb44843d5c6a5ce3e3adc5d7e8ac937c88bdaf6f90c
6d0c86b9dcabdc16f8cf8a3a1d522a1f042910eef02051a22a930f230c6a3f4
13f877e914b59507b7d349ac36c516548ea8df20baa63e1e815c51185bf77766
1b72b5d9a1b8a9476560ab59b190e364702d123732aab1f733417f69e1b1f418
9e4444f38f7ce06f567d10d115877dc13c8ffdea3fb01dcb160cb24299c57da5
2e1bc09df375cb53292352e7efc419cca9669393dd84bbf027ef03c871051b22
first message t

1362731ddf119d5d3b04244c1b8f691f5209a418a81abb69c17776cfd7550d2
126e7c785d8b3a696da9a84c03a9f3cdc312739876c432a06a328a50f099f0fe
09e582eb2fe540ee94f41a90aad2150905656e6948137bb08492252dcb312c8b
b34c1a77cdeef53f54c304259fdef15cd2224e8831c7f318ce09e794b806b430
cccdab81222c1de0909f7e040f7fe09de4208c2437ce0779a798853a1a0a0924
70f42181c09a9547b38cc9ab803e5456a288b636a81ad4b514db16c2a1605921
900eba4c88242837492e4e282298c5f948ccc74ffbe1c9ec30b3a653f23075ea
dc3a061992e83e1231cc9ea9a2c17fde8123726db6ae0dcd63540663f8c5780b
response z

37f09e2b1b88aaa2c89ccbe650b1347af8c92447eebfe8cd8d49de9f2caa6408
ac4a26cfd516798a728c8c1efc78e4c399141564c4f90fb8177247b17e422b91
00ea886367cd36d426a21855fd51ad8390374bc3701528e8004bc2166da17b05
150232a17a988c9b91ea6cd1e64ff6ffe0df6fba96227a084ff33c5fac04e264
10c0ce8fc2082280560621a9bd9e89f70414db7ef16f3674b1c539ee96d1f362
8706f341e13d1973c6ead9a0c12107c133273e40c0dbc42b4337b98f55374f35
a04cfafa713dc8e36055e6ce23a02d374315c734f811094e878919de7defb07b
68adc46ff479b4deba01b7fde469f029f5fdce49f4966ed86abcbe512f1c0ad8
response w

56cc633c44ed8864c4b388864ceee14aba34d8bfa48476206a4fe5f875c64798
14daee8f08dffaaa03287c77e9a9aa90a3cc3e724a370d3e463c2d7485115f39
2a1eaff375b15690ada7459380634695943c273573b1b3e29cc91029eee5e449
1535eb19c43bc9fc13994f192d2b75da61a63ce7c41d95338a7c415683d5d704
69b3f43512358997938dd551a6bdd6a50ae7d47d9dedd8b0d6a829d24cace767
f442c141cf29e6f567d21a3e142ee8e2f91b9ae182b30c231a29133c9bee8079
459752224ad2af38012022fd0b38bdd941c001b686f9c28d094db612f056774
331731da6f69b92940a5d4f603852de942f5d28a0b4d836c89cf5e6a885c09b1
Fiat-Shamir context

c8d0dfd0345c61623d88bc954c1f536d542da424acd1e43c5b85139816026c7b

withheld proof term: 240 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiagInv_diag theorem, FBundlePoint:95

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ } k] \{m : \text{Nat}\} (\xi : \text{Fin } m \rightarrow k)$
 $(A : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) k) (i : \text{Fin } m),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.adDiagInv}.\{u_1\} \xi A i i) 0$

SHA-256 of the statement above

857593290595c0c8e2c1ceefdf4b51842d5926c02ff45d14728253df052db827

SHA-256 of the kernel type

d1465192929da6d5fd961d1bfd65f2542885b25587fc9b2f8aa4c03978a494e9
commitment C

347debaeaf54be387583514ab54aaaa118507457ade8984c5a94c209f4d4cd1b
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f904edfd9b452c0cd0df9c7cf624c98e001708ff50a067ff5fe37b655a456d1
6476befc4ae3f607df205ee7c6a3d1db8f1f1b387f94d5e213f33b3e7b2ee129
796a78d13c7b36353d766cf11a62d97942ed5571396031aeceea34393be1e4e
93641ffc56fa2658f16aa3a265a5508e1112e300e94eb7910f41d653839ef497
ff025dface1ba65a57000197af7766ed4d6c46444a536c844e0e7e8658151
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first message t

19e89ddb32272f93da6e259ec95e5c3f6528730e5ab668b975af36391f67187
a94e3743d24913914217c3fbb093885ac1cebd5ff15142cae5d7fea3112b879
1cf36a1deb54edbe471cfef70cc35b6bab1bab3aa057220127ad41202ea48700
db2bcef2599fba678555f6157d8cc180477a2ef66bd82d4209cade170c98f0a9
9df4113e964f9b3b9a17f8b3294b9b65d58b073bbe3ac614fdf6512b7be060e5
46f127dfa6285a45af83c6c7e0269a8fbf177c86e6cc17bb43a9e143223ffe3
8baf30f61cfce9f38a1c84bbd8fd32d3be8d2ab5f2d0154919710979bac7244c
885a7e7f4048a1badf5e1af64f21fa07083490b2f1a4a788e288cb12ab74841
response z

2f1652024d25a8871a2431b6f1960d525294c5610059e314f6a6b1fe98759da1
e9930406a087b360a54329c851396b6dc40bb0a440bf6e63b91fc902654d9699
0042ceba432bcee096b2014ee6b066dc89b7df8f648fc1a05e71044922e472b7
dc66054620906f914b7fee36080f3d4b2303add29f75cee815904330adab54c3
66f4c7737c8d69409a165f8aefe756adfffaef63031ce30f172e768eb698f0b2
5eccc8b9746afa5ec223d57b89b4813be5ae520d6a2485fbc71cc3351f1ab9d0
eb1806da81230f0e42f4167875836a65ece55e8afcac638a9b21b61ba51f352
92e43bf865451be185d801bdf7fa6f2c077025d372104143c7abb8aead689e08
response w

1d92848e065bc34caa9b2281500c75df16de03795fb2481241d65aeb8ed7adb0
7a8c5bd3c44a185372b0c6dc2a384a78f57c1dd2c0b6010eb2086e5af2aa5bce
50ce61da5f25770ce582299d48061ec512cba081985b8e826eabab1a94f3ac84
a196415f0528c15b5b9547c21b528816b64a4a8fb535d6ac1f6c0ee2aea3e81c
c42c96b06993e06db1f41bb0c0400801f4b32a01f9ffc625ad3c648546e20a2f
a6787ce99154bd1494b08f170fd9fd9b9c668575c4c69452113730fbd3a7a13f
1bd79e795dbc0a632e4d44d377cc60eec31f9cda27bd98fcbde2f343a742c4a4
5859656276302ea7e35eb9b847c617f9d9753a41bf3099a341622e137bafbf84
Fiat-Shamir context

f75b3a957d8a02369c0c0441f4a51ac717ca0cc8cd5108f015d3ebfeb27ccb71

withheld proof term: 6177 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiag_adDiagInv theorem, FBundlePoint:100

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \mathbb{k}] \{m : \text{Nat}\} (\xi : \text{Fin } m \rightarrow \mathbb{k}),$
Function.Injective.{1, $u_1 + 1$ } $\xi \rightarrow$
 $\forall (A : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \mathbb{k}),$
 $(\forall (i : \text{Fin } m), \text{Eq}.\{u_1 + 1\} (A \ i \ i) \ 0) \rightarrow$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.adDiag}.\{u_1\} \ \xi \ (\text{FBundlePoint.adDiagInv}.\{u_1\} \ \xi \ A)) \ A$

SHA-256 of the statement above

3ab3061a2f4bd32fe191752c975501c360ae3bdb53f7069fd35c8f326ba2bb63

SHA-256 of the kernel type

6f6ca27da5513f22d9e09817f4b41d968c26045d4aca87519440743dfa47ee03

commitment C

108c2b77a8912d4f9f0b57e76229b0b42e346763b120044e5d76997cb09ba96d

88cb5c1d8fc2c7685390825a3f5d4be4c13ae04e952d6f5edf78ddee3305e6ad

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8ea6a66b779282c987da75fdce5de33b1f6cbf644e65819ffa0b7717a574c16c

f5857701f7f5163a5fc76086b54e3810a1cddd373da1463b573cc6cfda400886

e8d4d13c195b4176beae1bfe7f42df6ebb9f0c0d44bba4328050d26eb0890ac3

first message t

177d1326a41067e8af953f9149446f28b75b00119e34443fed3f3dfd318de594

e9f68e0be88131872a734fcb49a213467fb27ee2bc9018562fd7efe6d0159e32

67233bca0e0d902ab17715b208532da76bebf0ba3fe2d211f0346e04dac0b9f

09b0c7fb4cfdaaeabecd99a2a78873a633b3f20efa3ae53fb4fb479b81a586f3e

231f242d1e6785a85c9320766ec3b79dc53af69fc71dab642692f09523cfe4d0

4f7effda0ce34143067d9a8032e7c6d475f37b92fde5a0d6d8e84b4e40a75f05

f1cd6c9f8ef24ba8ba4c020d8684ad0d62aada2c82f3cebf18b672ab9e50737

b61389c9599d7945476cc9b84d627c8a9222f3e7d03e78f3d78477c555685380

response z

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785d5b318b5eab17eebc39d6285448cc5873d3300d090cf3c5bbd128168c853

36e1e27373b7a0f9ecd3009aa8ad9f26eaaa71ff91c0c271aa85d4d16911f7ce

61ca80f674664b7d4948e38abb2e19cb123f8ccd1f5655184c966d5629645361

4a73dfa831e9b065a9b70ba517c798bb93186ea240d114f3393c587f7bf9a8ac

a74850fa406c6b1d94d8b4bce98ee33474f7fed5663715f8fc8326d83e04c5d0

89ffa3ebfabdcea0f8e1017aed506242e4ca799e2eee997ba93c582bf1cb3cee

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response w

f01f7a443a2ef8a9495d56ea55f932202ff19cf606710c66d5be1ec5c1adf69b

e3b151e2edd5642b13ac2d61ac538bfb3002e12a44e9cc360d7f5338ea91ae2f

513f5361155cec8851a9e2ed93622c2036bcc8333ffa8893f5ad55fb4dc48f1f

4be8abc3652b3a57dce5419978044631efdca85c1d62c617e6369bc57df1f55c

37b08ff6eda7f4b0feae3330245648b4b3c764af2b96e3a5ed03e80198794d1a

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aa1d2c24ae51d89d5e7c3927320a52076d3d958651a5424547500e8adaa2b911

a09c8fdbfa4ee29b36db84b69631e1b90b782095335d6cb6521c05f287f03c3

Fiat-Shamir context

78c21414f7748c800d53070eb4b2d85002b9baa8269fd05f951c40c7ce29cfab

withheld proof term: 34666 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiag_add theorem, FBundlePoint:80

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \mathbb{k}] \{m : \text{Nat}\} (\xi : \text{Fin } m \rightarrow \mathbb{k})$
 $(M N : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \mathbb{k}),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.adDiag}.\{u_1\} \xi (\text{HAdd.hAdd}.\{u_1, u_1, u_1\} M N))$
 $(\text{HAdd.hAdd}.\{u_1, u_1, u_1\} (\text{FBundlePoint.adDiag}.\{u_1\} \xi M) (\text{FBundlePoint.adDiag}.\{u_1\} \xi$
 $N))$

SHA-256 of the statement above

dd132b2edce0e6c62ed9d200899db739fda5c10087b5906784ecb597513d9270

SHA-256 of the kernel type

36fe86c647524c17074f3a407648df6b145b5701c153c860ae60b92f218bbd8c

commitment C

7a04c64f25bf3101930604ecbf7d2c63acb83d7a409e36dc5218e5574fa5d4

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a7ba4df30f0c98a4bf6a52c4c29c90e8fefe2afde4e3e146211fcf143ec5375e

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b8053b4c6cb542af0f9d481d435e4c661b4130fb9119600abd668f599d644434

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first message t

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db7bb16b62d320643c87f8071219ad0b99c816af82c16aaab6fd7c8aa514e237

2b548dfc123ccd9065442bafab8a24d8afa0578991e7ab607b108edaede9b5b

efd268322ef0a4fe47564089a7044891de2b59515aa85bdd22a648541e02f131

fcce0fa80caf3b495e0b52b1c08a5845c93f86ad7ae68e0bf41ee1217e3fec2a

response z

6285699f5eba3a6eae8d68b6fad15b838aa368e1ae06c34803f1e946fc247fd3

f0eed6b33fc1a3715a1cadfe4988f94f6d420b05be251cdd69f89db5cb53df22

52d7490559edc020304c4d2e4dc79a2f3077de6cf972119332e1f99310d0bdb6

b6c34ecef99675731ba14d592274e4744222dc3ed4dc368e4e5034472e1f710d

bf27c17d9e293c5cf231809e5dd71c03b52fd2083edd912a02a237f9c17cdfc4

022be27f455403b9c92bb4114098e5189fc94620a7134c0a8152358557c0de6a

d21f4207081875cfce412affdfb75e0b25c0c6a576c0e3073f6f406169f033e

38923299de49920617b3d2eec1e4e10a02c916245d37a2dc048b457eaca058f2

response w

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d40c72e2e2a3d597f1c4aec6d3b64ba51b4391d746cf8e775df57f569f876399

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0f8915f3021e634f7de4288de8e07d29df37cd848f6015a6f243d28191c16b7a

7f38dcd14d4944200bc4bedc487a54440e20d0bc97f9664904907cf5cb3be794

a3676343cc9e835f3b443c3ef1aff58cb1f46de8ede6ffe69c5375cef534fe3b

Fiat-Shamir context

0649a0812b93933b03b13c3af31aebb9b36c9ea11330debcb9a1f1f825540ae1

withheld proof term: 33662 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiag_apply theorem, FBundlePoint:62

$$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \mathbb{k}] \{m : \text{Nat}\} (\xi : \text{Fin } m \rightarrow \mathbb{k})$$
$$(M : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \mathbb{k}) (i j : \text{Fin } m),$$
$$\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.adDiag}.\{u_1\} \xi M i j)$$
$$(\text{HMul.hMul}.\{u_1, u_1, u_1\} (\text{HSub.hSub}.\{u_1, u_1, u_1\} (\xi i) (\xi j)) (M i j))$$

SHA-256 of the statement above

4148df33b83e423a512867d1f08417aaa2fa5f447469aa2b5c7833e3c411215c

SHA-256 of the kernel type

48aad6736e1f40aa519ea059f84a95991c0884d4cf15eb251ee342a698dc007

commitment C

ab4e080f193fd593f8765593034885e375a62a65fa4d17d31bc71398f1a9dd03
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53401f236c05b1c2bf4e5214b1dc540479d07ea55dc0fc140bf88b47e09983b5
6a2c66fe49496ae0b7ebbcc1ab2628329de7d995e3735b323f404d0f4fb51837
25b9e4142b5472086c48dd01e00dcdcd52690b36a0cb6cd53f9db8b18e3c38ea
7e66b4a8e71f59511bf85d6e8d1bd365144aa2e10b4c6deb934b5eb24cbc5313
4ba2f52158b7168cd7a87a92bd947ce74a53f0ffb221b17e07c59a33a552e1bf
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first message t

eb57b1cc019980a8ab490c4b80e335e6e075accbf18a85fee5b92faa9aa935b2
756066734c149f71d8e20cde343b29b4f9cdcd075c03e744c944a5725a90f7e4
a5c11058d66f1eb9d879395e203df68abdb92703e49bb4c900a8342858e46f8e
52eac36d0b1b4b082840c5cc14a9b5bdfce20ba22962411f6be81be3183b6415
b7fc957fe0f946d066603600e85c2953bcb4a8c47f954c3abc47bed6df156629
1a4a1db066a088f090192f857ac35832fa7898f907768fb2eabacef4c1f8038a
b7f360724e551b4f2cec8d8f488054a4a5a2effee9a2909572c193ec70282f3f
cb99f7b61db8ed085cbb4cfc533b94a7d44a84e58a714f41bc94dff7596ac8d1
response z

1990c8d628cc190a6a6b99aeb11b8b3e06e2b73c1728ca472dec4becc126a726
c51fcbbf5ddd5c6aacd9c11aa423df8c0c408e14a0d5875fbdcc31419f94da2e
f5f2b641a9fb0dff42c38f3d0fbd052a232d1e820ef28ecf723be9a7472bf3ec
50770b814adbed915c7bbfda2964e3b19d4f309433f61eace3051eb7e68675d
5e723a4d021391d8d943e32e713b25fda97a1eeb8b27598abfb830d2966e7056
ba0edb7f48f897f39d85afec56912e817a7e1fcb0b834eb4afb3ba6bdf47bab7
4b84f630b229cc6433d6ea9326008306f63c42c7960d9b1faab4cff70555b4bb
2df5f8407515e6ced8b0ec0a693f43558d152b8ab1825378d1a0676009f3b36e
response w

7c758e13240f7404020a8b42e925469b205fe44e172ff83da681e30375f5c8ef
3e06ab93e47088d35cfb56349fc46329051b5789b2c1910fbb974364133d4652
32a1456be6c59c7d50cf81c31410503155696d41d52c3c60a76c71be236c2e18
42c7aa35c81751152e817ccd2f1cbabd51d4d7c5fd04434ef24c0120319577e4
cc90a3f6b5edceec2bf6f888308ef06ccd6e75e9caa7bedb5e0a1d8c075e5c58
7a5b428fee294af18caef2db0cd68c50d9f54b0954b2eee926a7ffb2f4eddf86
4b3fdf01d2af5ef2dd4379f6c25e1ce516c0a506c1eff071ca7ce99328093cb8
65d4a77946c60a2cebc6f7e24fff0dd621de6ea3a9c9c6f1fef15f10dad5d3a6

Fiat-Shamir context

a0cc2d60209e957d32e7807701f19bddb3e984ab02c4d1f405ae82b004761683

withheld proof term: 298910 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiag_diag theorem, FBundlePoint:68

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \mathbb{k}] \{m : \text{Nat}\} (\xi : \text{Fin } m \rightarrow \mathbb{k})$
 $(M : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \mathbb{k}) (i : \text{Fin } m),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.adDiag}.\{u_1\} \xi M i i) 0$

SHA-256 of the statement above

043d3c49b6b392e066ef8a76c259372c7060a78a34118514c80d1345e9ab2f5f

SHA-256 of the kernel type

3fa0a23938d93c0c02ca33b16648e0761dc3cb20a86ab28e7a6c6b3f5f82a735
commitment C

c6b031fe05101ec9de037e5a3fcd9ced65ebca366c9b597c9b0a9d90b0ddcd3
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b0be1d561cbb47d26645b45924fbd99b6ae1e1f787d0324664fb1832009a80a
cb22232ab13ce45f53d81ea565bbdb1dee13c4a3eb2b435b866ae1983e7d4707
ff34c54e744ae6332309844effc6ff8bb9162d65ef3ca61d097059fa8fb22a5b
732823b73dca0c111395d0a5749699fb7aad4c1c9c9684c0df56e7256c2798be
95f49f4147a5c8ae6c9404812f9e9214c69e6a48687cb44a40f041618a199538
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first message t

b3eb3ce5eeaae5dfbe7cc85c2fc75fccf929c9e423b362587046b7860434a5cf
809d74430f3ec91bd3be699336ece45a3162005abd50d085a6d5943353f48cda
6888d4afa034eff8d74710cfe04fdc52fb754f585d0a99f1d0934f9e930ca5e7
dfe6d04814d61aed2592f25196fb0a134b4968e94fa16699b4979a3f93479ec
a5decc4493c9be1cc0859c0ede9e47b2e6908109b25e35f3a4f7954c9c41b9cb
2109b59c60510de4f5c36c6c895647b9ecd2286d6b8db7819c569618cb83d36f
e6057dee1bb7c7323439822fd851bbc49b5e31f1a0053b8fe059514d98ebc1ba
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response z

2b9884732284413f19333750a4c2a3192a5fbd2390b2743cd570372702f58dc3
59ca1444efdb5d5387815846f3ea0683ba63e00c298054320fd5a3213c4aa150
38c23c5a37fb9165caf7d5d22be5297686640d2156654508d0c8ec608154cbf2
ffa1c390edfc8501e8ffcfb62d4e7b3ecedf9f00e11ac1f280c58bf02968d882
ca841d2e516f253b8e7ca214131e01a0405a044fa1302ddeec05b9a67a7552cd
d7aeca845e401473ac977ecdb70179ca2c8abeda83f22a26581ac68ca0eb652f
79d7282004fa0ad46752e14986a9995a0eca44f06653257fd950857c9e78c0a2
e59f1466935aa951236c811974b2545a053003fba0858677ae49d5ac3055d0e7
response w

3445099f32529cd3c5c9377c3a921bcb24671a88c0309719bf364f2f91763856
eea0ab0462db670ed7ea60b3ab6e57bb30e46171a3e6f35878471631c702c9c2
add7ec2ee7645881562116ac66929339a4292ddab8a593cc6adbf5c5c4051b2d
7c12fd0745755c4688bfa8ccdc85d633b9e04ef3ef5b23aaa7a45a0d20f577d
a3ed12fd054b5ccae52b3acf8760280920df589fd6fa8462dd3778c227f23d14
c44cc4989c5ecdd01195ad73e431476d09625e8dca01478c3c69be9714d1aebd
d4e4a437c073225312fe88a38190f818e76b21b7c662413bb933a82f60434b6c
bef772c58a2c09f89fabff291a9144ef4cec110a81bf72b5c1a6f30da49ee27a

Fiat-Shamir context

78a02a7b6c9230af5f5d8099f1a2751fa2ac01be6456fc43e0a948fefb0c9389

withheld proof term: 12828 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiag_diagonal theorem, FBundlePoint:73

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field.}\{u_1\} \mathbb{k}] \{m : \text{Nat}\} (\xi d : \text{Fin } m \rightarrow \mathbb{k}),$
 $\text{Eq.}\{u_1 + 1\} (\text{FBundlePoint.adDiag.}\{u_1\} \xi (\text{Matrix.diagonal.}\{u_1, 0\} d)) 0$

SHA-256 of the statement above

6b189b38e9ee2288d11d46042af9470cbdaf8b0a86dee7c00b5aa0364e10ddd7

SHA-256 of the kernel type

64932a89f8df95db0cf31d3a2b92e1b7e82f226d91abaff758209d3cc3037837

commitment C

ea5e9da780751c20083017d1de5aae9a83952a6e855d7455dd171a12e5a84958
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9e5849546fb9548af6bdd6c6a6f1a6948c3c67998ced65ca829f08044a18983e
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df27ac60e56d4e98f64eb89f9d21dc90ec43052e5ff437d1ae8f03bc9b1f257e
935e936e80fc8d6a9cbbe540997bb34659b95166bf7aba888741fa30d829db96
0c49955da54df5ca6aa7804de2d41abed9e3362c8e804c7d8798146b9a10a8ac
first message t

b06e76f1c0b871bb890c2b6c788a1390274a46cba4bbd28793eaaad2b597f09
bbc47e07ac2ea5ca45ab78cba0aa17308bbc0bfc209dd1bf13bf3abcc1a70174
c3f6ffb2b3edd9d7497fed6f666b50fd2f9eacede1d7cd0364d9dc874d800b28
d33be4f1ec5d60ddb05abc710d6350e4ea42199f6941f632a6970b48434e8
bfe23e5906470b2d14c09363008419ac3e09980ea918bccbc9b84eb6c5a4db73
fd7f1e0796c61647519377307fa6cf161cb1a74c7198508a13d022330d80afd1
cb9506d556dc385a3df682c557169a4361657c43515c15e9d10deb5f99c57a52
9e1a1e75a09cc8ddbc85d8300bd3ff08781afb21ffe12dd14bd38c7d652d616c
response z

29a5d82225683217987489070379029d08b92b01e10d6843dda551de45d979e4
da520f375a80ac829641c2246cb3e275b5ad63af7892e4f7b4f9f5fbb355675a
422278ae065be726f1e487dc7f245ff21e7ba2f583305d8779fdc3dc46552b9a
01e98b910727eccfd5ed09ec8692a5ecb86442ef32fa21bce7ea108fcf30d2cf
4acc2427022b5420ab8f8c4ca064ab31c758d173204419f8e06679b9294760a2
4db349f5c59369fd77bbd64e1880e8a5606a70281279ac35994b48bfaccabee30
5a3c7e8504fe9338bb711df1c17807fc296e395932933fd955c3276cfc4cc8cb
137b2145b784d9682a192476bb95158a20cfe2e4e2fb2a3f657a737833433b6
response w

685e68b8d1405dfb95309a94a921dbc42debdab46fdf813134d98d3bb1cee1aa
fe313e1fb272f825601962e5474f92e38755781498b9dacacf549434883ea7f9
3ae51ec0e789be197a2fb5bbe4564ae7091dfbefaf79a606a98cf095fdf63c62
0bf3449cbd0cc30f6d3aac4f3c3db695a5420329497bb2a1323079a9e2c4aca
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e6806aa234d34abbeda499ed88ece84fff7e0022238f1e720b597e2fa7c916af
c37baa21c94831b7b304f525ce40e621a03e1eec87a499222c5097a5e32109fb
0952c293e17d78f1f365bf96933f94c9a6435a6ac90df0d8e3a77150b674ae9c

Fiat-Shamir context

13830930a48b53da1e83904a6dd013a831ecc156cc3e35a8bce111cfb38b172e

withheld proof term: 38 800 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adDiag_sub theorem, FBundlePoint:85

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (ξ : Fin m → k)
  (M N : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  Eq.{u_1 + 1} (FBundlePoint.adDiag.{u_1} ξ (HSub.hSub.{u_1, u_1, u_1} M N))
    (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.adDiag.{u_1} ξ M) (FBundlePoint.adDiag.{u_1} ξ
      N))
```

SHA-256 of the statement above

b9c1dd5b65d898c69398905809eaf1403a6b0e1d8ee4b66d4f0ae699ac7f6d24

SHA-256 of the kernel type

5591d2091d9eb827f8976b5483a45f6b143cd8be844bac5c23ff5667db9bf4c3

commitment C

c68fdff83e70017866cdb4664f8f41bf9a78e5cc155a74880c3020ae282d653e
fc97466bce34b22433e9214d103cf4ddaee893b151902ff73de68944a3cc863c
41bbce144da8b0fb461b70b81d6c13b63bcd1145cb7a878ede8a4057e3c1574e
a7bdaa66e1d8cdceb2d119d43e57b71cb6503105395ed4a3bc42c5ae899f90e5
b6788d2162742bc0b58c754d24af8356467904cccd8882a3943390b9d25fe57a
767f8f1ebee64358795172a7a3d9bfb01a1b5461a65adbda67d2a1d69b9140d4
f6130557f1a8990454879c5f8816fd0ccc6d38414003568bfb95a95f67139860
5812190d221e5a99408a891910b67db5e44b78f0d784042eac79dfc6e16e9a25

first message t

ae9aecc4c92c3ae3242809a3c2e5aa3841e69b7ce5f381ae1bb29b21053fdbcb
14c535afed580f8cca5a6d704e1af680e47951c50338475192ee0efdac241003
328b1a489903a6e51c1669c305055f450ed55487951047714d2081e070d32f59
52780caada9c4fb3e177c2d8d8baac8bf37fc8bfd45cafbff0427c0c785cfb46
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d0fe8375591376daf1f73f8c89bde8d39ba106a228e728eb40858a041e6512d6
d449977d4f2299810b300c9150598bc3da522a78e52507e02b79d55291dc85a5
9f66dfd19ef6b0714ee0bebaa581dfa28c8d46afe5fcd22c45aac7ceb7ac67ee

response z

437f3d626d96287c10eeb98d81fc9152f07195214af6daa73ae74c18be32b1ef
2e395e4eb6a4e5e7318570e01df1493775176647bff6ad1314e014c51aef3870
b09161d6d7e92c69f4c5754a78d7fae53caf7e8bb88f19c6e6421a2b93618dec
ba28727d877a42cef1bdb4bb3b5f032b659d106beafe0dc959118159bdcef8ad
c0572c4a5ec938ab8885cc04a096da3ad6a88ed428a5f4732853b19b284f4ca0
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25d1516b77a4c6a2ff85522b4da80870b8611140654e902056a523d81cb09739
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response w

7c1c873fe6537ecb79f66c70e124f6dfe73456dec3b7834a22dfb2e175995ad
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48d001ef4cc4fcd70a5bafbe6fd054e3abdf1d1f7b49355ea41e8d7cbcd094ffd
332454cad591e048b2274de043e0d28cb17caca7bc3bd7ea45c04c6891573463
061416edae5e44b23c6c354d17c7ef722c435254e74eed3d972879fee4e9a76f
39b35116b3716e0a1898bac1afab2c72a60cc4c5b93639142bdd9a1475e6c007
2af8f9385e5323eed4ef42c0c45e70df2bb837380fd81e394133c299b77211dc
01a434f60eb5eda0bc018206159907047635ee5031355ccfc35ccf2f2a25ccf6

Fiat-Shamir context

cb6ac34b11c76ddb195ee9906715ba8e5983ad2d3bd63adfd99297e11970358f

withheld proof term: 29 857 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_diagonal_of_offDiag_eq_zero theorem, FBundlePoint:111

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}. \{u_1\} \ k] \{m : \text{Nat}\} (M : \text{Matrix}. \{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \ k),$
 $(\forall (i j : \text{Fin } m), \text{Ne}. \{1\} \ i \ j \rightarrow \text{Eq}. \{u_1 + 1\} (M \ i \ j) \ 0) \rightarrow$
 $\text{Eq}. \{u_1 + 1\} \ M \ (\text{Matrix}. \text{diagonal}. \{u_1, 0\} \ \text{fun } i \Rightarrow M \ i \ i)$

SHA-256 of the statement above

93caddbb62c08425eda21ccb60f3da279dc2f9ae90225a6cab6aa5c66a32dd3a

SHA-256 of the kernel type

b32892fed94d2cb5128f861181e9adcee4f9551e5ba8f174b7c8657e7bf70a47

commitment C

d0845d968b21d5b95a4fdb2c1ab4e7e76693fd5f526362e2777fab0d12875fa8

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6c14c60c1ac7f0aea0d8c3c77fe7a0f8d6cd284d18c83d56d1b1fcc2c0036699

da295f06973305c02e66a704ab2b50b207e777dd99663128ba05d5dee5c6cd7d

888f4eef6a805842dc317df5003eabab4a0debf8b1c619477025bf5c2a98c79e

first message t

6f9498f9cecc245b756b31ad8b27810002d5c55c854796623a13a2e9e8cccf2c1

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6290b2e5a62870811889d7eb349f638e4ef562d00e05844f78aae71af021b8aa

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76519dcce9fc59d19cca8a3ca9b67dea77b81891b7573fa86714929b97da66cf

da46c85bd600a6b953c3837d89aa5772dd273df0cd41df1675486e9ce22f8365

fb318cdc72044b3809d29e2f5c704afb512028d5e42cceb50c1b836227f7f539

fc23f96d481d3ce326b94df042e499bfd990e376c1b81b573661a70a9ce2a6c3

response z

13ee867438ecbdf9e68b2e9a6357e01205f49678ba9123d55390615a58cc6cc

0315a6a01a83b4a1c40ede1dc401295144148b09f8c2e0f75fe2e10db425eabb

bfd5a405bd31d390ac7a7f30c414f7cc16fb948765e7145c7d8cef19686882dc

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response w

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cac3e4663bdc530bcc5d9b7d00de3fcde9f3fc09637624d452144a1905004fd

Fiat-Shamir context

cc0321407b68e3a3e47ae647d38be572879d624dbc34c619e8f30cd5ff42750f

withheld proof term: 11075 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_zero_of_adDiag_eq_zero_of_diag theorem, FBundlePoint:119

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \mathbb{k}] \{m : \text{Nat}\} (\xi : \text{Fin } m \rightarrow \mathbb{k}),$
Function.Injective. $\{1, u_1 + 1\} \xi \rightarrow$
 $\forall (M : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \mathbb{k}),$
Eq. $\{u_1 + 1\} (\text{FBundlePoint.adDiag}.\{u_1\} \xi M) 0 \rightarrow$
 $(\forall (i : \text{Fin } m), \text{Eq}.\{u_1 + 1\} (M i i) 0) \rightarrow \text{Eq}.\{u_1 + 1\} M 0$

SHA-256 of the statement above

3b42db338aba8f9bdd37fbb2b3d457559e57b5992d418dc5aac931976de36311

SHA-256 of the kernel type

c9f54d2f9b088798537027e0eccbfd31a284afef09212c921f3699dba9ec06d8

commitment C

4c05106f46ab4524d3d2ad03e160766644057f2e691f5507ceb046c7ecfa5e9f

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first message t

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08ae847caa7b2442a53c2a1eeb982e51584d90ecc88d605e7f06ccd33b8ede25

response z

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3a6b7ea90fdbf1ef100ca36ca3c977a38898fc4fdaaf47953c10f550381c6bf3

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1f9c4e00b16c4fd83ed1cf875418b60ee9e39f30a386317bdbf4126a63c9e69d

response w

3f793602f2632780dc5931820feb6e63225a2e63a4ff028a883d514270f382ae

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3d9070ac1569b8a4a8849f76926b2d81608c3cb261cea829ee9a49c6b1ec625f

Fiat-Shamir context

850f9c61854364d2f07fe99c84ddb6b2a059d702893c8872cdd248e927a8ba00

withheld proof term: 26 065 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.existsUnique_gauge theorem, FBundlePoint:317

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} [CharZero.{u_1} k] (ξ : Fin m → k),
Function.Injective.{1, u_1 + 1} ξ →
  ∀ (G K' G' Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  IsUnit.{u_1} (Matrix.det.{u_1, 0} Q) →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (Matrix.diagonal.{u_1, 0} ξ) Q)
    (HMul.hMul.{u_1, u_1, u_1} Q K') →
  (∀ (i : Fin m),
    Eq.{u_1 + 1} (G i i)
      (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q G') (Inv.inv.{u_1}
        Q) i
        i)) →
  ExistsUnique.{u_1 + 1} fun P =>
    And (Eq.{u_1 + 1} (P 0) Q)
      (FBundlePoint.GaugeSystem.{u_1} (Matrix.diagonal.{u_1, 0} ξ) G K' G' P)
```

SHA-256 of the statement above

cdcd5acc4c12c9e74c30fd8ba2b748a1567cd1b3d74668776a612c7a7501d7a5

SHA-256 of the kernel type

813b1914764d80220f2fbcc793e9403db722a22af3ab3f7c7daca2fe3cc7adbb

commitment C

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first message t

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f805188bcee4b64119938b19506beba1dcc16482bf28c52762b8e80046de11ff
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effb35e46cc897fddb9ccede08c4e9b1fd0677eb67ac1214521848a00b90efa4
a344b8ab59dfff082aac77c2bda379ada1decffe5da6c329e5646697f25e953a
response z

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bc6fee2f35d132258e08e3ee91fcf30c028acc0c3ab373923ee36b879c7791e4
b85b559ad18fa0e3f11daec38b991c32375b20b77de3ca07d17b664aa57f658b
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6f133eac3f7b878028d517af4837930c32db880ee4267a857d27265b7639f2bf
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response w

6831c3784ec96335961431615ffe90d100acdf259ae29206ae041c9ce22a1ff7
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6f2f978d849e4f2ab47178f7b17424e472b80fb4960124b645976d620d01c5
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Fiat-Shamir context

84fc32cc36dcae7c4d91a83abfdaa5150868d0dd8b41af7da35188c530b6f022

withheld proof term: 347519 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.existsUnique_normalizedRec theorem, FBundlePoint:218

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} [CharZero.{u_1} k] (ξ : Fin m → k),
  Function.Injective.{1, u_1 + 1} ξ →
    ∀ (G G" : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
      (∀ (i : Fin m), Eq.{u_1 + 1} (G i i) (G" i i)) →
        ExistsUnique.{u_1 + 1} fun R =>
          And (Eq.{u_1 + 1} (R 0) 1) (FBundlePoint.NormalizedRec.{u_1} ξ G G" R)
```

SHA-256 of the statement above

bfa3efc741c9d598e9a66d819ef0765dde165cd5ef2dee15fb9a361d6e8ab2a1

SHA-256 of the kernel type

9a70668f637c6c39971faece2c0a9134402d7e02c6aa81d9f1028ade4442c7ef
commitment C

ff1c5b232c3b564586bff9300b4b8018acc2861fcc04798fdf7f3d67799f2119
8af6dece8120239af91ee7f40edd15e3c0a771dab7a6ba485b7e764dc796d1be
e64859a65b1e314c56becadda88b3f3f093d6b603debce3b0126a528bc055232
e5b935827b59e831ff7d7bf74ba828a377994bacb904b0320c5881faa2b4c148
98eb4524d7084184314f253b67ee448acf09101878ff2dba6d28e872a8ea8502
93f481e97bff84380a0cf4a5fc578f7059e0744e91dc1545b884b25ec84482bd
d28cc274647b5fb51d2d52f75b52cfa238bd7dcb7a0506cc086295bcfb4d55fb
513cae5e48d6a7b2f9fb2af354879be8dcd2f8bf8ad3da184e58db1a140e2023
first message t

60ebc34c25ccb850d737c7d7ee2c0f3d0087a13940b130057c94e907ee796045
611dc9a46a73cfe46f893476ef90b16b64cd04bac68d54e70968a6c6eae623a0
fce3c7f8e4a740024e4f21732847ec7e6c3722c78c1b930fa0a44eeae8ce9197
5319eccd7c1fbd7ad6b01f176d27377cc8f5e6a50cc7fcd3daeb4655a3a85430
1183f1394c259ea29af427b3776f64c5dd637afd4f326aa33f74ffaaf73f6e30
6e347bfa5dbe889e276913bcac9ff19d204da85e1e3dace674c2bb6d929b5f0e
a082e89e5c48121f6cb18e6026a4979f428e15161fcb32597519cacdfb5871c3
40667dcdc39dd168decc10de0578593d1d54a624bce700882a84586960674bea
response z

f8cba89cccae0bb317718e4f8d99a670980559617c9eb6e1b7fd658d7876c663
1d104902a9c2b47e8d9a7d8774c0967c0a0fbef65da674c16b3945cd420d508d
2db1a5c491a6b81fceb640020373053c75fa6a27d00ca2a0dfee6b6acc32131
91268284bfc49ef655f278b8694db04ca50b2cb913433d5c819938a32b0006c6
cccd38bf19799c428cdc58033bcc5779d37527c8d57c20e629b70cbc43543333
e10aeb02a4170f3dc2e55eeabf15ca5c8afd8e7aa244808aa38bbab8cc2d985f
80aa8b5d7c0ebc0ed9d1489b1c337fd268b7d75de5602c1d6d782b718134d716
2723ad6fa846a3739c5b94c18e6c61b510259e3976b3fc7598f52cf87cb7ffa
response w

78794f4bf3377f9dd53cb595bf1c8bc53355816a26b054614c4956fdb7506dc
f2d73e2371904f57c08d302eee1d011ec72da4a456350244945cb083d24dd8ea
e2819011b92242afe265b776fc7cf1f5b0c7412fe4612061dc34802f1c53bfbd
6a27519d39f1bd2f57fe3238345b5f9f50fc6fd169b5a0a795d59f8a670c6c95
9fa07a3dc9616dba1495bd0a74eab80fd291819dcaa84743c77ea480e2f38fa5
f93aec686da518fe19bb05cc10dc6faaaa9ad7371e4beed62627ec07b8394c4d
9a96a19e60d281b0f07e7a167c2d6d7c8fe3bda9b7dd5a55eb3cb247a57015c7
2799aebc64a2252f8168cc133d8d11fd25fe39681da66c2957f08c635d5cbcfce
Fiat-Shamir context

6c767fd74abc2656620441e0d3b2695bb67050f735b05834931fb45fdb8d7fc2

withheld proof term: 1 763 785 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.gaugeSystem_iff_exists_Q theorem, FBundlePoint:368

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} [CharZero.{u_1} k] (ξ : Fin m → k),
  Function.Injective.{1, u_1 + 1} ξ →
    ∀ (G K' G' : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
      Iff
        (∃ P,
          And (IsUnit.{u_1} (Matrix.det.{u_1, 0} (P 0)))
              (FBundlePoint.GaugeSystem.{u_1} (Matrix.diagonal.{u_1, 0} ξ) G K' G' P))
        (∃ Q,
          And (IsUnit.{u_1} (Matrix.det.{u_1, 0} Q))
              (And
                (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (Matrix.diagonal.{u_1, 0} ξ) Q)
                  (HMul.hMul.{u_1, u_1, u_1} Q K'))
                (∀ (i : Fin m),
                  Eq.{u_1 + 1} (G i i)
                    (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q G')
                      (Inv.inv.{u_1} Q) i
                      i))))))
```

SHA-256 of the statement above

0b8c37729f7e963d226b60eb4d0eca2ef0ccf448404d7e58aade853e8f8053b9

SHA-256 of the kernel type

2ff27b489fff1fb6748fd98130191604aeacfbce0dc35bc55f661db13cb83ce8

commitment C

988604ae620ac66d1f89862020ccd439a21c098df62adf70e9696b062694a597
4add7935ca9bd729d97490f4c88f9e94ed0a3d52f1f3313699d7b089228480b5
9ac5d6283a3227add2ab6533d80fd86ad9e9cba8da961d9448cdafa40efc5063
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d24f7b88512a315bd44faf06072db98a03ad335203fa8709a7687802ec294b3
86059d2af0c2606f4c5979266dce58e5d2e4c7d113e44adf9e32daf365822553
9bed079573eaa6c4e991e164d00e66d7ecb7457de341e5f858d48181640dbca3
20ffc98ea7e358bd26a75ccc9a2174cee151c4530e82636d690245d4fd3e31c1

first message t

f4a3ff7ac3ab6835de33006779aa994548b8d07b27dc499a66d46c8321cbe36a
e8664c34ee157a7025879cbcb00038e2bee881e2373366e36161d0f0ea108f0e
adb26ffddb8518cdea63d0a1b4b61ab44ede7a0691569aec56990139917cbe6e
02be9c17f5b7f69e97830ace379df5ae6b5e6058867b50f00793099dc8780098
1e055b4ea690236b11a11466d9e99bfd8077077fbea7c84c2d6363a85dc3a7f
1d7c20e62fd6f7ed8deab8141df81900c733299541de6e423724b7a53b994c72
67f42014ed3722fc8aad875ffe46daa3d34a88f61978c06e963caf995d458819
b38a99b457708b29409779a036f8a1e98a870725b7beacb8b1e1d921351a312e

response z

5ec34b1baacb3b7822b3610f1b69cb6975d4c13d08dcec22441f027222144437
f5cef787015311a90de321d51316734265f784388c7a715b200415efcac0f839
7d5741917a865ba437dbad93362042e86b0745e54b32ebedaff1fd1e56f1d600
f25823cb413fa970cd4d9a37b225dba18cbdde66bf7f3a7b930cbaf4eff9dd32
1c90fdc180c4e59a721530a1184f0c07591b4d7defb7efd8aec6aa53f47fb49a
54417bc418aea46a88fc9b65b3b40450c905e533f28b5f5694d98ea5f5fb550d
b1b046a933cc9bcab40f544f595ae2bb073cf53407d67412e001f75179a5e300
1517a9141941b72503be1bb3181f15b563b635100dda045616fb03b64b6683f9

response w

363cac4dcccda76d1b1922a2fca62586c2a59d9c1d02a592427279da811eecf0e
34485970e17a88ab44a16b88d47324468b90d2e01e6b7598dd72d073ded3cca1
050c2451e5d894e32ad12fdea71269b829a2f1cd16f10eb60e63c97c06d613d3
5962df7b4292af2f982899d581e05887a0d69db2c234776ec77cbbb6407b0363
184655e4d8696495bd8628a28daf640263205cdcc03fd87d392956c05e94610
6dcb987a2f33e624760403d6220fd34c0e65b0429911196ddb51496495a4b539
5a27833f92ff9bfde9d9dcb1b26d072685393f50fd44d1e6b1798f21d6b805f1
dbba81dcd72450cbfffad64767d2c04ae9bcd89ba22ef98338d403bf5af043cb

Fiat-Shamir context

9efeb45cd1ad0c4490a80b4e15197c940be41ac4718262b721153b1ec9c31c1f

withheld proof term: 125515 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.gaugeSystem_iff_normalizedRec theorem, FBundlePoint:287

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (ξ : Fin m → k)
  (G K' G' Q" : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  IsUnit.{u_1} (Matrix.det.{u_1, 0} Q) →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (Matrix.diagonal.{u_1, 0} ξ) Q)
    (HMul.hMul.{u_1, u_1, u_1} Q K') →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} G" Q) (HMul.hMul.{u_1, u_1, u_1} Q G') →
  ∀ (S : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  Eq.{u_1 + 1} (S 0) 1 →
  Iff
    (FBundlePoint.GaugeSystem.{u_1} (Matrix.diagonal.{u_1, 0} ξ) G K' G' fun k =>
      HMul.hMul.{u_1, u_1, u_1} (S k) Q)
    (FBundlePoint.NormalizedRec.{u_1} ξ G G" S)
```

SHA-256 of the statement above

29d1ec09fa6ad4b353146d3e1501e08b8a2f4b220330bf0f3359bdf4a48898dc

SHA-256 of the kernel type

861bdbe78d2966f3591383e31454bbf4dfeb3d1d015bbced2c4c1dcbbcc82e2a

commitment C

2da78d02154c9c818ea7f96a01663b0f85cdcdcb24c9711a3fd182bfc4c44157
ab17379cca3d6bc1e8ed48030317bf6f84c31a26b6b165e439963834763bc64f
ac2041b4edab6b8bba73c5a040b0c6896bee453784ae991e17b92de2b4b4fabbb
140c2eb131cc6c89ba995e94d622eec36b17f2bd3a25ae9fb478aa95c995bd86
897d5eabe1a42c474ad096c32b55b8c4cfb09d80b0369e6cb926fdf0404ca728
c2b60b89f2bb2e75391e1245668fc5aa7b28d2c13ccd50bc42bf546285208af5
a5a999735cfb1742f472a60236e50311dc1ff3d9de402c72fc793ddb544f2280
a5a999735cfb1742f472a60236e50311dc1ff3d9de402c72fc793ddb544f2280
86b6b4961f62e5f5987a79d50b7d731de70ad450cb5cb3d0982a49b694f5813c

first message t

40d6d210b423455bd46ce140ce09307d4ee4bc09ba78697773259ba05f4d25e8
a2e8163292c04bec79575cb3ba2dc3f89058de4a618796141235a21d8155e508
f3bd246bba892e602842ae366b1efc267229da3bf1ebcd0cf4fb35246ed21b08
af03436419244224e063dbf99fab84d0a0b44be090cd7cb647d3b36d598fed3
7a93b19f4ee3b1f2b5507ba73b4ed6522f09bb8c6a578bacbf0756726bbe5a4b
9c8f20793829bca4e63979d9d6677f02525479858ecace338ab22e29b1adc1dc
0c0fafd609fdc7b7581836133a7d39a641e29f3430f2f97d5ae0f243c02a7b5f
de405731170ca7190da25f864336ecb0573afbbf10e9689689b1a1c00aaff04b

response z

6f165caf3cd36ea81b5a761c7890b7099dfe0709b733207e06d31ab1bfcf6fa3
9325e207944d95725192c9227ddc81e4e05a3eb99dc5e5fd3a5e5237141aa779
f01710afbed5435b634d124d34214bf50f4c9dab074818067a08cdac41474fce
182b941e20236b025271d60fa05bdce75af0ceb729c8d79b38d3e1da6dde36dc
bc17c0e7da7543c3de7fd960fb76be900b449c2937fa6aff228a15acbb62b8b2
5218afd38cb60aa3bbd78338c3240894139cd417c4183542c6b3a56ae7342916
d61592b196fa12ce69cf7e57d46388bd0cb3f0032323f75c6bf64488055e8ab9
6a2a7439bd423398dfacbb3aee1d709132231e97249b6b77dfda3820b85f74ea6

response w

58d65bbdb71cef3717987414312aead8a97ae44960e28e3483dc701f6abbfb8c
efead6a456bc1f551db73cde3936cc3e031c41c37609ee1128a87990254bc38e
a51c8312aac82acbe90e22d21ea96d754f5cd5bc6e540eb26d9548a114c8332b
655fb41159f14ad5d3a6c63c299020029f43207f3ad7758b30c4bdf0910668b7
af975c6c1b4d31f237cb48c5e663c228e089b0ddc19f12ef38d60f680421f406
cb9b27dc76991ff14f41813532c3f93a99b0ac8304957d4edfb119399cd817a0
b8f5c9fea6a74d3f6e24a6331a20a838424074bd73277256faec70c55b57130c
973e6c2bcd184bb961aee157d0d6c9239db424c5b96a26f2a3f553ff2b4c8234

Fiat-Shamir context

75656138560d9ee0ae86259134b0e11de755669e5c3d5b1e3b8a6ed9a5b5195f

withheld proof term: 3394298 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.gauge_conditions_of_gaugeSystem theorem, FBundlePoint:343

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (ξ : Fin m → k)
  (G K' G' : Matrix.{0, 0, u_1} (Fin m) (Fin m) k) (P : Nat → Matrix.{0, 0, u_1} (Fin m)
    (Fin m) k),
  IsUnit.{u_1} (Matrix.det.{u_1, 0} (P 0)) →
  FBundlePoint.GaugeSystem.{u_1} (Matrix.diagonal.{u_1, 0} ξ) G K' G' P →
  And
    (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (Matrix.diagonal.{u_1, 0} ξ) (P 0))
      (HMul.hMul.{u_1, u_1, u_1} (P 0) K'))
    (∀ (i : Fin m),
      Eq.{u_1 + 1} (G i i)
        (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (P 0) G') (Inv.inv.{u_1}
          (P 0)) i
          i))
```

SHA-256 of the statement above

ace2885744d998edfdff5cf79ec76fff8e9a9401c288007c4a9349c987799f45

SHA-256 of the kernel type

63dc9731f2ac4ddbc89f8587e0ec1a81d1831a8dc172c8ce45e402cc18a3f614
commitment C

f32c6011706745cbeef6cc8f5408f99ead5773d18a46e82cb6644f1386e36fa6
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8529cd7827e3ea8601c81e285328f45f9f08db0491ee145c2d9b5cf435104e4a
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9514b926568101cb722ac6267e409321492abbf0a219004634606dd21684b723
e5a1636d3592e491595ecb9f0639f2c0d11280771db2c4582fed45ae11a4af05
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first message t

7a1b1c984e245bce617ed85cbc4b0966a6358bb1004de34e585c61e31921a464
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fc401d16b9b59939109184f055be98cc2dce20854f50804e4e29f221f0f1f7e4
78585a5b61e883b6a514695162cebb463b7185ad25f40082cafb4b915b07d67
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5527bd5bf89595d53e62b8d5a29a8fb22b4fb0372984e1dc3c7e78e44fdcfcb2
6f7b5f9dd37ad6ba16f5f89c04b9732a8ab2eaf28ae47597fc362bd5a565bf3
9bfd045e9ae5320407a51e43e159398e522e4c1ecb449279a73be950ba79c31
response z

253979e6a976a217b82745292ea1ebfabf031babb0d1f612c6d6833f3b5becb3
cbaa9d54ebad893873c8d85c0b26e87b4550566a977fe1d14ed422d83d86afd3
be8435aa217d1bfbdd59a75f1283a06e768c8d3583f409dc65704c9d60832fa0
d45dc185fedbae6736563e54e239b86e07ca2d926870cad2510eea2ccbf43ccd
a0adf4926011d596e3f93b0d523125cf646f100d2f6ba21d53027311c5b40a0b
3d88cfb0ca1d14f0cb6b77830f2c4570c2d3ba2e84d6f9b91fae08c3366c7ce4
39b2c757b2210dafdf38c5cfd774ead8081af946a1d760a79fb8dc19abefbd8e
64880b0af5cdc2027939b4e2b43ec670afc5161eb28547ccd53e445079ab9fa
response w

17687df9ff566106a0a16d0273636d0275efca52559ac8640d2f9612237e30f3
870e9df8c87edca8ee05fd0a02c57aae16b21795887424e98fa044fe7e7aaea
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d617876b8e343a7d4a6f0b7c414eee570bb6e1cb3b7f06742ab54455e767c9
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c26f8c577a33f2143e86ce70daffbc38b79b9a229e073cd4f2dc1c2f32eald2
e71eacc6d85b841f910b48c0b1f653f30b6275e36228f89b211dfcd874f58e0a
Fiat-Shamir context

030c37ce0671f2726160a8425c52bc963648af265527bbfa5210a1ffb24bfc9b

withheld proof term: 370819 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isDiagonal_of_adDiag_eq_zero theorem, FBundlePoint:130

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (ξ : Fin m → k),
  Function.Injective.{1, u_1 + 1} ξ →
    ∀ (M : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
      Eq.{u_1 + 1} (FBundlePoint.adDiag.{u_1} ξ M) 0 →
        Eq.{u_1 + 1} M (Matrix.diagonal.{u_1, 0} fun i => M i i)
```

SHA-256 of the statement above

20bc5a192da37acce9d53d4fe42fc60496720b73e38687fa2614c108d29284d0

SHA-256 of the kernel type

a82e31e59daab5d93c752799527d7ee37adda4d4350f6a38ebf335d66cdf03b

commitment C

4d845464bdc552b66c8093d078ba1eb054b23e5e5a8da6479c91ab0a6dbdbc17

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first message t

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27023c897a7d22b2ff70fe586c45fd8abb2ef2f5ae8129673e444b489b883354

39db9cf0a43d1590792a16ae5f27f93030cccf516351621109dddbf5e512fba1

fb42288c448634799e5595be976dad626e9f99528befbb5ca09d3b985bc5e466

4c203a4bf20d95f1354f769cadfa4b9ebc5448058ec12cd7a56aa7e502bc9869

adc8475dd026b7709abf162a4316b41edbe7cc1f0da8ed5da03fd87f3bd6148f

response z

4874fa021d3fb17edfa731200c5c559685f7f716597ad238d30d1ee60ce50828

96a66e2ba59a1442a9fb9d3be5783dfe807494325cc15d586d5b1d6adfa7aba9

489a4c6311d7a40d708b3d049ebf2266425ccb42fe9a450ed26e3f44b13893cf

b447ef61c6ae68f86cf6532046d905d516573b973400903fa85a945a886dce52

55a1acf20e4c49fddc584628ce5d96e262a1f7fd924e086487afdce28afb47a8

53532818e2f89a1321e68dada02c23646e16067a945daebeae1845d509d0d0f

96f0c8601b1a603090f56386c8ec7ee43b6489243174e00780bf8e77ca3b2e93

02b43677da7561c5bcd89c3a24126602b511663e06719fdf5bbef82d992542e1

response w

13280a100ed476e36b5ddeb9ad096e24820b956099b19f6759d7d0c27b03d887

74bce09f8372f236a41b4ca544a6e3de5471918a472c15b13d806c9d1f1d9107

618463d888dd07a9fb23d7a0eafec43a2929275f5765d9956f2b383bf9671181

ab3f5f173da558a045be1517f14c5880596c4e89dab56f6b2161667a4060e362

dc05606a43dc2509c9ad0e0ba8a3b3b9f5e55aaf41fdc930d8fee673211ddd59

529162ae0aa4ebc2de60e76f9d0065a7453f4ae9a0094a11edfbc159a212309d

24855f31b1461712b8c90f32027e82eef528d0be07b6f195a7739688dc31394

b241275969dcb7be729bf7f079da3d8a60b3a63cfffdd3a6c3e24515b2d7fec94

Fiat-Shamir context

d941519609b1796930c8cd2c1a2aec5db77bc92e970b42fc9c45b96145078481

withheld proof term: 21 909 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mul_right_cancel_of_isUnit_det theorem, FBundlePoint:280

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}. \{u_1\} \text{ } k] \{m : \text{Nat}\} \{Q \ A \ B : \text{Matrix}. \{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \text{ } k\},$
IsUnit. $\{u_1\} (\text{Matrix}. \text{det}. \{u_1, 0\} \ Q) \rightarrow$
Eq. $\{u_1 + 1\} (\text{HMul}. \text{hMul}. \{u_1, u_1, u_1\} \ A \ Q) (\text{HMul}. \text{hMul}. \{u_1, u_1, u_1\} \ B \ Q) \rightarrow \text{Eq}. \{u_1 + 1\} \ A \ B$

SHA-256 of the statement above

ccd39a0d37b5f4dc63e55b09b0cb5619d6388478c895e44dd9158d7fbc49bcff

SHA-256 of the kernel type

2d8dce7333256a487ac5b8fe30ab05ee55a82b5569d551b51baabb6c4247fe2b

commitment C

b4e83467451748988e7df3a4deb7a4326cc4f8a4d1100dd34485bcb8a399ee9a
b48defb9b2d4aa32c0ad3bc9f0e0492952f0f82d726e56b09d4b72e50891f034
3b8ee1fada8e940ecbd38233ce224fd09557f4566980c4e868d96231aa58f0a8
8fe7dc08e4a71e5699c74e389673ffe9c28054c0e6351c79f862409999884803
14202c9ef0d0400dbad5f108ef1f408005a49bdc63aee6510154c6973683f1f
9dec8f2e7e31146202ecaed28152f5cd42283560fc5958e5ddcb5d4bbfb440e
fa52e3c07f643252917245f671986771f5422a67f3e8b5445737d37e24b3f72e
cfedc91187065ab6ef9cddb4ac1a98eac2f07de33be0318738525b6a34d3353

first message t

dd6eb61c8cee8a61664b2ec4957fa8394826cb7ccb746aa362d459bdbda08406
1e1d7ce6ee2f33f00ad59462a5752f717f257df1abadaa0bc3c63fc9a2264626
c27f7cd8dd0244e45cbf99c2bbefc4438a7f939f0c0900a2191d9aa0845b2752
644556d59a119daa5f56e3ceb728c5d35ef983208b2cfaa91fd1e373514ddb8f
83a08627f2ea8ca43396b9595ead034fbc4fd4f527229142e619ef3a1a295a6
82423f37821fb6306c9699cec0b181c47517bfa9f0d740774fd3d3fc385aed1f
f298eff47c98bc441c0f025b2832adb22ac719933f6eb314c4f808e5809f48c4
af8c85331803a122f451f1306fcde826ee028541131b6f24d21197fdc20dcac
response z

635727104683c263ed763cb265f17fdc5fe0c73d52959ada88936fd76924da97
630601b427869d44feeb640bb7253c41d92aecf70021477281a9296224911008
38b367d7f478091a3d63636b56f9e0fa266ed1494effccfdedfd5884122e6e
91dec89ab64591bb5f7f1e7eabd2eaaccd8058b7092045150bb01dbfc44f1d4
48d3eb3c136f7c2798909765b305befdeb5c36558e59ec6d245964be20e05b86
c328b73029b87a1b51c7df908c50dae93ccba17e403e47bc65b7d35c04e83900
528b468a80ba337d6ffee796a3b6e7d9da0546b9f36a6f95c013b0498bd445ce
09f962c32d8cdd821f3eb4cfe45a35421f5734caa5dc91f93c3fb3ad033ea69e
response w

4334bfa0e50d4d56d7d6fe86339f1e3994e5479ec5a39ed6c26d130a0ecadf1
5c329f1ecacfb2494c4f758f639533e8d9ad7ba2ac268c33cc9dc50ec47b4304
9abdb78abe5bbcf08f6e0f4a72c14f370892a4d89251e8f51c1ae12c7f33207f
fd0be14664ae92656cfbd1810ad123760487ceba44bf71b22105c43bdeb90faa
6e9dbba12d5e22e41bc7b40cfc95c155794bbe242ca956e32fb120b696c36c8e
d9ca21c1f6ec3046113fd7602c8dc79661d2fa8ff5bd184b4d6f66504ed4a52
90f4d9113f1931848ba89e9ded0174d91693de54bd71238171f7cb47e6a6a60e
f9d8d1e6b60b24358471749be1c52097cf4fa943dd70b3f2c98b4fdd2426a31

Fiat-Shamir context

46e4f3ced652b889e67bd035d95e10fd644851597f4a5437206a39069edde12e

withheld proof term: 54489 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.normSol definition, FBundlePoint:173

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {m : Nat} →  
      (Fin m → k) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) k →  
          Matrix.{0, 0, u_1} (Fin m) (Fin m) k → Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k
```

SHA-256 of the statement above

36a6e87deb2d8a1e2db8053f1428bf4887b3012267623edca76df02933288baf

SHA-256 of the kernel type

00df7de1c7d217b2be3365100e8adbb3af0ee12ab88ab397bcc6882c1af51af3
commitment C

d367ee7e3764d5d6509d53a8b378e787ad1be33b76e0af7fa042cb9842ac2448
6b3171cc741808b7009240dc2a1c19b53110439b20172d8077ea7f466d620cb9
3bfc1fae5fae99bb38d5cc70e396244e4a9c5edbe169df1e870b827bfd0a8654
9949c05c3456943858420237b20545f62a7f1aa837dd4383836c3baf48721973
ca31601367d21e0d3a7d061586be60ec8f42aad58cd89e56a6e0bb2f6392edc8
ba5d9238089be425d80825e11b4ca053952e3c5e72f82d8875420d5c24b277c4
6a09075b52029707cdb8c1767725465038b49ec4a4f78b134239ccb858a6fc90
ccdb794fbb68f215f86289db4cb491b57a219f38909f125725c50371135ebd10

first message t

b961300bac036025654592c21de092459338f8b52c3256d0758f71422adafc44
a5672ec829d58a6580bb6deac26c61f7af99730b0345cc1f8369b4f36d243f33
f10da7891fda5690ac23577f9d8cd507e53b1fa75323e5fe2ac4de52e6780b53
6af861abc631e9b70b951ea9b9e98afbcd76bef05e0ccc4fd5dc3cc5354aa625
a43565d373b77f8ab8d7a89ec3c8fa7189eb5009a7e41d2a6de1696d8ce6f044
ce2469411cd37795212c527644c86d396f4d4f3cf83994f89e6b49acbaaf0602
cfc206beee982690ec7d76b0fdb8cb4c725ffae2083efd6929d4a62a135a31d7
b9281c699c98a6fb80faa782f0cf33b5f22c2c25bebfd2293c281a661791e24b

response z

16e9872d357846212688989201c214f12572c5360357efa28797eb2910c0cf67
ee7f8b87cad2aec9bc5eb95ab71144194d4f64c21d2bef900d1d659cd82c35c4
efc623c92325cde1c54317b814eb5a79af5c3365d6541e651695a3a258f1f3d0
1feb45f923b9f207bb01636b63f32fad1812c7bc97aeddca6383ed869c2bd5ef
27119c0102b5750db21eeb977a3033d63bd18d8c58d231727accb8863f097c63
49ba36570e8be416eae4141112e085967a85973d12bd9bafef17292b1327a1d87
54ae3dfa9bf70ac740a78b7a0b28bff7cc4ffd257253472390ebb4a6899e20e5
72ba31656f89ad83139d9ce7a9e864f2313960683d9e3ec48d735e71d6c04da4

response w

13a0d769d84dd1c0bd5537f0486b3143347614a9f6bdced2cea4f982297c63ac
cb49e3e0599afc391cc9e3a0a33993d8a59078a006a42bbb0455b4b8268c4e81
8e30772904c42ba9504fa457d896390e23c95cddab1b4329e9995252de4877c5
4e189832fb28b4a25c4ca9a18b53a263beb11cbcdc61c5f6103babfb0329b045
28b82aa421352b6191d0955ec051830e200da2e843c9c3b11c903a66933d2c02
58dde193f5c86aabfe92525fd35c4e68baa7566a2332c16bbbbbde7bd0e08282e
130f55ce5eb7afb8634514f67a5eaae00f52685b558f3ee00dc17bc5ecbe3c61
278328ce243e908888fb222d06c4773a5a9db685b39eee4879793a193c6981c2

Fiat-Shamir context

a4fe28579337877201c30c7edbfeaaafdc21e290f534237d80d5344bdfb163a5

withheld proof term: 7 283 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.normSol.match_1 definition, FBundlePoint:0

(motive : Nat → Sort u_1) →
(x : Nat) → (Unit → motive 0) → ((k : Nat) → motive (Nat.succ k)) → motive x

SHA-256 of the statement above

e24e03d60f53a6c058b653b20b41aff674a44c44a810b816e23592b63f683138

SHA-256 of the kernel type

c575818fb7282ae35329c0a453d2f55f1e416ae936fb114ace6088089d15f6f5

commitment C

bd0d3a25c585e29c9f90b97e7381ecd4961c531703e118e70da137cf7ebb1bba

16697d3caece316a84b6d9afb06fd8a27a8231494300d967fdf4561ecd18e6a4

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75e96bb1263fbb3f08a25464a6fd57888255a5707af24cfa2684befe8c195168

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f589fb4d413ed090a38bd2bb2fe64bcca1459948c2767328cfa8e9329f09af5c

4cc0fec444e30048ee67ab30235d4246895f24b3ed70324fe6e8ec00068262a

da35dc5f1020026c85972150e53cf197418f3ef4244ecb15f1fbb6c8059b62b

first message t

c945a7b23867f0c8930916ee0d160a8f0e9cda3e6a9673698891f07489577c99

61c5665c5ab72754a1db76435bf267e95df59afeff7b9083e9921c8d074cf9d8

91e0a18a6a15872efae5618d8fd15e8ed1df852f9add462ad4328683f95116fd

d6e43f9f92e5f81f7e42a132c90fa22325873e120e94d523a38844bd7d085c17

2416aaafe531a372273c492b127f41322331ecd0bd22a5271d013060c216a057

eb6c30da058eae9fa649d53cf21a91502f0302eba07cd65df5172b3c527b87d7

6f76eb21ed69b4eaf002019a1ca3fb85442213fd36253a8f50c4a48110ef70c7

3817c4328234d17090a241dcfcf9a497e7c2b757f8562104555ba373297690b3

response z

1b2abc3aee5366ce753c76b5c06fb69b0deaa047519fa6f503f6f9b9f8e40bf0

3ff47af7941f3b5f361c209f54cd2ee5264488f4a0103e4848d75e7799c757ca

192bacf71c3ca8da502f4277fe53cd13ba9a8aa703d925dd43c0152e2932efca

2b3a5e20ff401b3b674a4e69f381f21328168f8b4c1071458744e2e26e3698f8

88258605d2cba2f5fa8cdd7326310ff0472353c09e7db937097570fb107ed84

14ccd6d8ad4ef54c87f8378cfe5c0cde9b69d46f5cb6186104825e77643fc614

ec3c342d42708829bd491eaa729dc61debf7b17a152fb91863a5a1326f82b7d

7cc5361e8d070ee194aef62e76f50b5cca895fd85ae19ae8efc6df0ef29d9d30

response w

3bb47cc3f5a72d90fdc8a8977a755c93ad3fff68d5a521cb2abfbcd84018ad70

397f377612078146cde8d6a208e7dc999298d940690193a53cce62b82346a951

f9f0971c6a87f8447bca260d21688b9cf51bdd321a6a2f77714617cf43d84020

b3ff3915b47590896d67c6a9c588b319f25d1e2f4445931cb0181521303e6f06

6b3c5a7e682f46fecdc1c00fad8de5cc2b30b80789186bcfe897783236affbf34

4215d67aa30586849e890682a81cdd8cf9df34e8d4b497e73efdc839da20560f

115ef39f183754ef116599a16a8cbc2e9ebd5ab6d8dfb8b55d45cba714bf5b29

928716a703ae2a8d4243d48b5393de4a3f8a3ce526069ad01bd052282dadd646

Fiat-Shamir context

07aa4ca3cbaeb68d16539c81fa7bff0e0223dd733106845e7fbe694c3b4d3465

withheld proof term: 580 bytes • axioms: • kernel replay: yes

FBundlePoint.normSol_rhs_diag theorem, FBundlePoint:186

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} [CharZero.{u_1} k] (ξ : Fin m → k)
  (G G" : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  (∀ (i : Fin m), Eq.{u_1 + 1} (G i i) (G" i i)) →
  ∀ (k : Nat) (i : Fin m),
    Eq.{u_1 + 1}
      (HSub.hSub.{u_1, u_1, u_1}
        (FBundlePoint.Fmap.{u_1} G G" (FBundlePoint.normSol.{u_1} ξ G G" k))
        (HSMul.hSMul.{u_1, u_1, u_1} (↑k) (FBundlePoint.normSol.{u_1} ξ G G" k)) i i)
    0
```

SHA-256 of the statement above

16d1c823ba8afbd2f73343839b782276df0cccf140474180df61fc5a46e74360

SHA-256 of the kernel type

a2efabf946a95997236d2e8c431042a2dee933868040b2fc45361ac94717add4

commitment C

6fdf8e325eea541672b3114b4d5ebd9e153e05685bbeb602ff626f00a9c30bfc
dcddcab0ccfd9bfcf11bc0e54e2b8019d324efcbecdfbb43aa1b797bb71ea827
ec2771bc2e8cf547319e701ea36d770a2d3164a109d0a14e60049e1585358c11
ab46ef1b0fc7bd30997733e3c413762da42650ffa64fc10cc5b5cbe8a77b257f
1a09ccc1c5e304c65c9f373ef6e07463b126f52f4405eb69f677b0039e23fbfb
a57f4456876df985ca2a33aa32e9bda8068d3ac6ba0e49d6f1be41bfda57e516
8786b992defddfd8fe1a833580968acc1d68b189277cda9f9d5bca646c3dc3af
29b50097a779338f16e256be15e26a52d7f23fbd5b84acc9d3b6296afab4571

first message t

65153eccd0716b22c67193bd2945fd45ca70ee89068933f9bb068b137ee08c0c
b30a9b6f74b58995710c3277a885e7c12ecd67477c0c33127476002548b62050
a8563e481f193f7935cc6ae67725c270f9d36cea93d4bbdb9f7c5306a234f2c3
23c0a31200eeeb0d8ded9770c65880c00c3ac8a942c65a7c96c9a8ce12dc369e
aa99b7ad5ff4dd1892832a83d1dad2c1f4ffcfaf17bc903b687cf893911dd68e1
732c9138f941e3f1c5d69b15d3af4e7708eedbc7fade447f51a00eb1cd326e1d
45a7d85b6edc7c5187d93853135466f867277352d3e870ce6f7494f0c8c93622
98d480b72cfbb9c59522d32be45531099934dc7bdec77ce9c69bebee14ec7907

response z

4de9b20d4b5ef2a63fc49c1281e062c962b0db93fc3277258a9cfbe2e319bd20
c101965e5eecbaa61af4cb6326d5fb2b03ead5528796c566fa579e1f72277403
3661241bb7155d2dde95d81e5e176be57dbdc3315f836b43c3f774347df598
b7959514ccac72c5c73b9ce9daaac0b24c036d8eb801bd4b1d16bf4dda5e261f
c32efe561279d7529bc9bbf441748b0ea1d8a9038a7458892e63f51ad911ba54
97a59ed7a1914c0cfa944b1334babe5b6d0fd36c75560814e4bf181ba1456868
1fddb32288cb5bb883cbbec6355e520f264569b39b7bc830f11cabed1f80cdd3
604ec60c779d4d99b0678de3da5701d195a4c24fe93715d19bcb8294c269f7d

response w

471a762c6a4898d3dd2af746623c9f0b4e1f17818a4b9c5840357fc174cdb635
84bb62f8b8c75dd528fec4ecc6c01636f84206c430fba2cef3ad894e732c622e
3340549fd8a71f7dabb0422219a8313ebfebd8943fc4b4e7ab285ca0b73c173
58846612958bb5ab33abbed5e22e95d31499a54019920597d34a2241226f241d
7ef38279dafbe455c7996ef804a7470eb803c8bedaabc8f6e364c3fb325a9d4d
3295385ec3e2ea1de89f45a455705ffad5b22bf12d3bcf41515e9aff7af0ca83
fc1815430754a40478c55a592e40ebd67f2f64479208cb85aa442e203cef0c1f
b0b8332c74ce15977dec9d6549c3baff936d76f46f1341e22c711c628a5bf29

Fiat-Shamir context

ce37522528b35b478cfa1872280e6df1649f5f629014f43fb982129d9ce5421

withheld proof term: 2 208 885 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.normSol_spec theorem, FBundlePoint:207

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ } k] \{m : \text{Nat}\} [\text{CharZero}.\{u_1\} \text{ } k] (\xi : \text{Fin } m \rightarrow k),$
Function.Injective.{1, $u_1 + 1$ } $\xi \rightarrow$
 $\forall (G \text{ " " : Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) k),$
 $(\forall (i : \text{Fin } m), \text{Eq}.\{u_1 + 1\} (G \text{ } i \text{ } i) (G \text{ " } i \text{ } i)) \rightarrow$
FBundlePoint.NormalizedRec. $\{u_1\} \xi G \text{ " " (FBundlePoint.normSol}.\{u_1\} \xi G \text{ " ")}$

SHA-256 of the statement above

4c03f97f79e618a0c957cc020c01bf24a3355bb74c17fbd24610b496d70e9daa

SHA-256 of the kernel type

d213fd4c1af017e5418bb1e7028ab1068fba6aebdf3672124d677a3b84a3955a

commitment C

35484e8f3083101649e03ad47c3ab7cfb8bd5ef782fe1d92eeaf74b59813af21
0b9c28c70342e06af08cf5858eaf0398dcbdbed880e36c3328d8e57e68c23d0f
1a64a2a01ff59dc390ae0680ebcee624f5c393dedc40015ecb7c087a2ca3cb44
93df3daff69a6a8870ae9e9e6a72ea395ecb850e7015e54f37e02d78589ff265
ed4c0ecf55f79806f9d39022732b8716cdbf8bb516e244bd1930e33be78a3f03
897e004cd76784fa4bc12e365669a4e1c91e8bb200328321626ed80cce0400ef
699db4f24b65ce618af3042603e17d44b6601ab9476ddbf988fcedac59a52c06
be7fe25b07acb27c4cc693fb3e0ffedfe882584e200ddb293e26b0b753aebca
first message t

e2c879b3ba46365e7779b8faf6d507c3966665783c02bdde346933451f12b88f
e0d56054dc433890c8fe3ef3245ef1e552769c1302301e38ce258a157f8aac9f
31e16c8593ffdd840997667ca9a8821fffb925b4cf06b89d53bfd63d6604d035
f12f2501e6a6759822c1dd4faf4a12e06cb64805626217f778ed234259613e5
290beeb7c312d6d4c561f77a5cf98a0008eb5febad264886064f4231f19728e8
e36048644655942ef763a578ef317e5792aca157ffb23fc090ee9edda6da1b6e
bbb8056c2aadac875b9e30464732d1423861bce53efab2ba46d445b46f21a5e0
3ec1aaa1a294f68deec9cda7b38d4e2593f086c8481941f258ce93a1c6ac772d
response z

2fd4640619e611e117b47a947d1e96edd96343fb22b977920abd21fbfbc00adc
a8ead9f93432d6be7b6c52a656ac637da02b3a63a16e67d644f6bbc5587e6e69
bd643dd06a4ce21c34855a8df720de4490d492dadea5f73993d00e4ac708a450
05bad9aba5b6e08ff4e62fed9e1789dbc2cf9b188aa748c6f97a6db85d6b3450
8652d63f61074d6546a34f2eeb60df1932fb0d85608f3bfb6902ce698baa936a
5acd0139c59a5692bb0ec05941e096c0159d0a4de0742116a573c422a8c4339c
bd45f063d0c1d1f65f43023db5f45e13314b5975188b8047d7487bed547d5741
25fc263e81c64e1fcf474d728c116f03335c5015725f10c3998817417a64d4fc
response w

7151d7e8e5b8fc6774c5c26541f8366a490043b15a0982874c3a2b5ef9eb7711
476e64f1bc3d96c6e1e75e9dac9a1492c9f50183e2d00cf527d43296483a7baa
c861217856b02e407335c73c528aa908dc6148b02aba1fd2cca178a94c5d32c6
4209620e3cb3d96f201d576215a974b0bc1e152204bd5d9cfbe99b76d0c34bfa
cc07ae0f5e649521fee35bc74b6a2047c4716b68d45a9402ab243ab2bf377fbd
fd5e64af51c37f358f868ad3b6589f41fb3b305e00580bbd25ed01198cafcb98
a0385ceef3229e1411e6115ca5df192a1ce96bae9cab6f8dcd83b83038209d42
32274d0e9840db57af6ab9d74251b8694737b1a92e4a7e5306b1988a5fea6460

Fiat-Shamir context

d491a50fa911fc18287430a7ee13f0e48f8abb189f83818c80a540724cf6ef9f

withheld proof term: 136 720 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.normSol_zero theorem, FBundlePoint:182

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}. \{u_1\} \text{ k}] \{m : \text{Nat}\} (\xi : \text{Fin } m \rightarrow \text{k})$
 $(G \text{ G}'' : \text{Matrix}. \{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \text{ k}),$
 $\text{Eq}. \{u_1 + 1\} (\text{FBundlePoint}. \text{normSol}. \{u_1\} \xi \text{ G G}'' 0) 1$

SHA-256 of the statement above

40ab4b43b38f1d0eeaaed05caa01bb395ccbe4c6714381d91a27c745b599de57

SHA-256 of the kernel type

d72a5f6d2a66b6f77e8f3ce297f78261c116fd51e53db988013a5863924034bd
commitment C

bc4919d5839f24e47f540a4a6d270ed165565a43908dc33629560c83885e6a24
5a2f68a92ebd8e1ba2ac35bb75800db18a4320f30cd70742ba6ab1d4e6d91a12
7e1bd9a6e63c951b1f7740b49fac34aa37d64d25bad1c90c79fa3ba137c2cc90
6be40b418ce5c171be7d5994f8fac2b1540fdef2298bf99ba1f7058060c62446
866f5cdfcec085bda8e42faecbfbe29a6c9472a32ba337276e0f8eb192961137
aab9cb2a89b32643b336e0f9a1ca440f2f3c5185502230c1557ca4635bb88281
63e643eb1e3ed1dd044969c7003c05ca8a4aa628bef4289cb7683b6d4ee89868
b3d8646e86f0563df24c5c9163024f4aea90f3fa188575b23deb02a9bf8d2844
first message t

72076f1e78f065e1cfdbe07c0893a3ca64286a4cfceebdca92399a3854cb01ca
cca17cce1800d863bf82e75dbab30cfd3fba3a78d31bb2703e79bb96ab73a266
289cce42bebec6274a3aa24ee148a257da66b05b568fc74e59c7763f154ee1cd
2ee87e745e7cf1e9da37b8efbc06c88f8f35ffcd3b654909341fcf2854001c96
7efa4581b60c3203f2a7cbea693122007b2a7955272cc30050c2123eeea7ae8b
c6b0d8b67799636525499e2cb5ed8670e9498f9e6fb3d0ccc9bf9ef620a10290
8dce5651bc86d8db28c845b6f86a23ae28271fd9a03d5fb390dce4f339585da1
fabd3175a9bb92fe0db978c51f088bda028eb2e0b5cc990e72fc8ccbc760dff
response z

71b420372ac7d012a7b9e00c1467169e44bcfbc5be7b2e7154e08ec17dda669c
61edf19b713b6a93ef49771f3bd16698420bf75ee8a93e6d5f00e6a69becfe0d
755731aa2c974673f117dfc50ae54921de28e14eed462872e0b80507735f1d79
9493b89c447238508666695cab493ee96e67361e823a2f60a28d4d46ce94cd0a
ebf37f569c09a8cae9f6967ad8fc6e258f9901323bcb82a0bab2869fd115abc4
f030d2f7fad0b9fdd2dcb487ec2e405ef9db88f2c353b3b91ead8b312d1350ae
98de427c5880f333ff90d4f8ae7cde7096b4213f875e46fec74b5038bfc82ab9
f5e0958d70a2fb27c62e9b4a59e179eb99eeb1ae25b9d66b18301e99681babae
response w

8af97bc29e4e2056ecdf910cda706d890fe8e9f1881150b2b32ecf2de2b6b3d2
7f127becb1e0cf40e06bca4add0e8b7e36000051a29c46dde08a7278d11d2d5f
772cd2f8eecad65f3ec5b839888160fb4269da996ea590d129d77c6417cf7beb
8fd71797fabcc96b2a45c08c40e8d3e377c54706765361cebd7525e59b906ef6
2d0c9c613bc02db89285add9ffe0279035f9ab808885f8b4596b70d48b14df7
5a4b9b97949864f2c9902f563165b3d4fafecc4174dba18b279636da406e32bf
3deec423bc04c403721cb252675a7c55fe99deb67c4bd19ab4719719197fc6b4
fe9bc87bf01935f9437129be979cc80e86211ea0f21ca0541c49e485671dd0c

Fiat-Shamir context

e47d840c1122497aa140d54512cf6ee1250f2b4435d6981949cea03f40f0be58

withheld proof term: 441 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

11 RequestProject.FBundlePointPowerSeries

FBundlePoint.IsGaugeEquiv definition, FBundlePointPowerSeries:106

```
{m : Nat} →
  {F : Type u_2} →
    [Field.{u_2} F] →
      Matrix.{0, 0, u_2} (Fin m) (Fin m) F →
        Matrix.{0, 0, u_2} (Fin m) (Fin m) F →
          Matrix.{0, 0, u_2} (Fin m) (Fin m) F →
            Matrix.{0, 0, u_2} (Fin m) (Fin m) F →
              Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} F) → Prop
```

SHA-256 of the statement above

57ebeeecd91ec635000a00e23e0f06978323dc2b896c89f32cc419719ea2f1b

SHA-256 of the kernel type

071605c82450f6f97ee590fd528c611862f4a8ca00ced1ffbfced87584517891
commitment C

3c54e7e9f52d4790887352f53ef9fa5d4f86f80f9e716940a869026e6a70f0a6
50fabd464b81e4fc218babd0e258861efd1713a6e53ff8c8200b7d8fca40e13b
cc279cab76b7bfe4af7ed3b6adf9a4bdad70e4b9a0ffdc9bd482e22b73994bdc
65d0e9d548a6f1bf6d691496a3d1b51f0864a168834598db7ce137c0d13226e7
8fa72414b7f1946fde7643d54e7a157b53d9739cc192555fa15f6d72d14ceab9
c387d766d9dd73c724a42193ebfac03ad4749d77f4a473d72541fd8e552bade1
2b1634859d6105848a8f2024c30e3afa53c7a89722b0f5c27664b36c31075e6f
52347889e323a8cc823780016d136fe518b654df5f4677c37595bb4647f100

first message t

f76ffa5c7031e00531eb1ad60cf8bd53a941153d6672b9a9c33724924f8da068
2d31155b32b487568c4a5527fb36be0afababc78201665ce20440ba65660f54b
889c1706eedcb3abd08a0fa62d5aa61189f025859b961aee683fb4920e06f25c
725641004fb4af83a637733ce1e33e864f68177c9173f5150867742f323d1ec8
112f5cb04911440d2afcf9f39433f02e34ee57e9a44820bb8a779d90d82b1f0ce
11aa138eaaecb13cbf22203c5fad71259ca0bdf22efb2da5380f8cc0f455f6ae
9bc417d9e21dbb8ae231ab809aa972781477f2d8812db624d13aebf385b08f5d
c7d0abd29b654dc11f8c05c58a82940744f8e14ec64da735525a73937373bd01

response z

4f225e8898ccbe11d961fed8afb2498f5a20180dfe4175b276b9c05b4972002e
5f152b41910ef274a7d2ad6f2b01e111d7c1622a9f47b817e831f3bc85c41669
a0f8f30582507dd259ab92533cc8f3957c51edbed09de465c40f6dea420f07a1
130d557bd365eb95962b8ebdb23b950b6dc5953e2e0a9ca8eace1ff25e78345b
a715953b2d3855453a4f2b5c05fd68a781cc62ab4be806798c9a30b1f07e3ed9
cc890de6c7227593ce22023ec6dc542c0dbadc7f2b12822f9ad3e67d1cd2f0f5
c6e3ac5fd6cb347d4e3fb92e79e318e839145e97e1d3f7330e57252078f22302
12b46ea5fcf2c5825c3762b44b8503e8562f35dca95342f739aaaca5b3663068

response w

2224c6cc5850dcd5540992066b6b1e7940c58617e978c285e6eb93b8097b11bc
c8d2135ae842a61a4b2baab9137481a567a3ae0a96c41d317a9b581d09576536
cfb4a7e85cb6a50bd1099c9791aa7619fc8b3a35a22ecb233557a20d78eb204b
00144d9c65b2df55852373356e41c55ed71308ed49ae562f6100b43a87fd299c
84408ba7032df6371c43ba6dc2d5c26d1bcba7cdf3c4b05212e6cd5c8a483713
b5c458e2f03e5b07241bc93ec7a1035f5806e3d4e20c3662d766d376f8e36b43
6b756453a5b02cee11f4f295942a3608e181c5107407f0d9f879778e37eee587
ec63b3368620df997b4204eb5c388fd5546211b0b4cd2dac60049baa69968fec

Fiat-Shamir context

64df1806b0b2b5a0e3a89e6d8fb21ee3f4aae49829d477626eeaf5e7c442f19a

withheld proof term: 11 881 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cst definition, FBundlePointPowerSeries:39

```
{k : Type u_1} →  
  [CommRing.{u_1} k] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) k →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)
```

SHA-256 of the statement above

77117c4a8a61e4ce4c558db56fa5a39f8f36a17f3776e79222ad23337676c199

SHA-256 of the kernel type

d78756c1b97c44ac18fcdfea1c1a5f593b7e8753530c3bdc9c611f09c43039a
commitment C

32038216bc6baf09e98724f1a5343d4bc5993ad172e2a9051ba0b22a88bcd75c
b163dac91a092545987e4a8dfbaccb646e83c9c670beb2b947d5ee12df3815ac
1f397751f0822eb0b7d1dac03e59ab26a1ce9d81a937b7d3590a136d4ca03b87
728524d2debea72eba68e04c49ac03a8f58df56fcdad3caf5cd13eca13b39ca6
010e59c20f7c525394bc09f534b81887c73f4c621e8e7650f8d2a04e494bcb95
234d3347fdbb5fdb5c59e7ba97b693c522112aac00e08cb9674e7e495d40a5fc
e09f5585b89d796fd8b8af11ed0191dfa41f61feee6caabb7b152c3adc4ad47c
f5851647d8fc482acd04d39c57ea88ecac52af3c2fe884327cbc0af0a491b20
first message t

897b0b9d8c2f50a3b612e7cb8fd53fa5030e1aaae0d2a836419180286905bc23
769e42edaa59112c7f6ef6e162518cf08635c9d76fd4bed119114729408c55a3
e995b39e7e7203495352dde5c88673fd2f0b2cf321b36d59d326ba0c367ded7d
3f1baa433cee5b49d946baadb7b56deea20a1e333768a85539d7c6b38a63881a
812249cc8d740e35d552f6ab10fd913709dd78b31581654754e116855acff74d
99e23cfe35314197e51f2b1b866f97de077bca8655f9344abef1a1d6eefa0e
48f77cb4b56d74f578422d85ea03028b580a14044eb0681065671ece62ad80bd
bcdcf3e3a07b2eb2fa280aed907cb51172b04f554d6e067fc3b4f2c71a261b067
response z

4a377d458610f04d11f8c9f2805dbe75fff6a2b71530e2152cacd12fbc891fba
a7767f00f099730b40dcb4a39b8e0923df1f8bdb4a44a0098e44a301d148e6bd
3f7f8674c291cd2243a30d7d23ef13f7f943df0c7ace1021f6bb0602398e98dc
c63de815628f23c126924d2919e93c9e31f320c53a7a23058a4f4b5fade632bd
d50b859841ec24b3b1e821f78fccb05be175067d6a806ea4b86b05e69676c1e3
cf63c25ae0a97d64442db1e47b612da5a4eb0f9a174304e5b0d98c85aa4b6a33
c068899e7c1e81bccb6f756297453d4533346429404374e282ab92349d7eb88e
eb4a2a130bd73d6e108cd786f89232bea0a9bd9c8319389a943d881d7fab96ac
response w

19ae54c0303298370263953fe4383617dfdbfb62fd38a7e9e744efea065ff9b9
6ff474059f5197f1ea82dba86a8b1cdd4831aa5e0c5befea6a46d4c00eb1055c
e519ec3c9dbd98bda2c8c057fdf1ffcc0596d7f83665f7d10c73f70aef83ecf1
a974d22b7c7a03198cdad9a733b644367a79f65003ad11d3db01d92281815694
5d62ec9055e735677c0f893113640715ced86d30d27bba2070a7a5b79e8c01c5
875676eff1efacbef72797931603ed5443ff616b75e450abd0f6dbac289ad5eb
0b3c665bc542ff50420cfd2df264ff5aec417136e4a57f7b8aafacbd8ec3b2d
7068729ca7094559a48ed0d4b0b52675ea6c9a7d239ab2c92fda061db055771d

Fiat-Shamir context

eccbf725e3a69f2b6346a7de056bc3d651ffaf1a794d66c76469e355cef9a986

withheld proof term: 1562 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cst.eq_1 theorem, FBundlePointPowerSeries:0

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} \ k] \{m : \text{Nat}\} (M : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \ k),$
Eq. $\{u_1 + 1\}$ (FBundlePoint.cst. $\{u_1\}$ M) (Matrix.map. $\{u_1, u_1, 0, 0\}$ M
↑↑PowerSeries.C. $\{u_1\}$)

SHA-256 of the statement above

6f56125e26273f92ed5f2f4f70e8233eebb22647f9e082121036e9cd96b990fb

SHA-256 of the kernel type

2c3653cfa6d7b8fb04aad74afd9932acd00ff2a3ad96fe194a27d42be1d305c
commitment C

b191d54cc866a8e5ca59bda9732e966d723c97f5031020312a698c6e8c33d16b
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9362c59051cc063346a6c61dc43766bcc15e280be930ac91100e011d0803a5c5
a1d809e7d1c95fd256ca7eacbff8dd5c084d9b5a1b39cee0beddbe538d6d409
4917ff04ae7488231a1a46b2ccd4e5350b1de72d86112dd6bd6ecfb8d4572131
3946590adecce77f0095a8f61cca2be27f1f652889e2bf2fff2bc215e2251e8f
372a736a6abbaf7256f69583d71cff4fb8af368b2a58978b844d2546ef3756f8
e817d3355325ca0cae8bd25c244010cc4ac163bb675ad9d5f6da473ab20c62e7

first message t

396ccf62c3c68ca9cfc5f55327cd2df2649445199b8270142ac82baec83230f4
62c5f71b9e1181ac27f095cf9e8425a6450345284af93fea28d5a6854822a2c4
dc0dc94685cc7008f534097369c29f4151ccb42882640cc05591f9b53821f37
3f643d4fdc2c27bd3e27d26f86f789a17f64a9c0165eb5bf369f682af468c20d
afbf76ecbfff0fd993d53e759d708f311d373363e917d591e4d9241aa5d1e73f
b3ede37a1e455e569ff47d5a9ce672414347cc3fa37bcca7841f5deedaa0805a
0fbb148351135c5590ef011e23617dc6127242c2223864d6e666dead261f6e9b
690e51ac7e536782fa72a4b97c873abd160c80de8ac6ed2e6921840090a2c5b7

response z

779353085ba63c79247039b780359a4797d07ea00bd0940e40a69bba932cb254
1f24bd63f715f9c6ed34683f2331af27797bbc201ce58a8681a5db17bab4b16
37582cd8b4bc8050e6781053f355e387193fd19070586028b4fd03a754c2305c
31d3a62431ba19d9cf11a2c6b612dd731b116c7b4cedbaa71fee41b255de4228
1849f1e5fc1f35f0eafbd8d1b3e2c1d39c2f99e5d523743f1921c1c39497866
ca0b6bf0d07ab73d1f2e49a6230640c1fab9341f2a92c9b81b7d9b92430632ff
d2177c302f322715e5cc3a477663d6626db1dce1f27bec91e633eb446185c2e2
f5e75bec30b7b0c77f3bac5e34ed21f4151fe4ee6034a9cbb9a5a041e88c6091

response w

3a9a323e2ce14393f89e7fcb4212f5fb15302bd86fb20d53ce8093fe4d2d6972
ac8e765cdaa15d58998ceaab6a4ad0fd77b01fbbf3d0d2c58e7b195f49d3296b
6322999d7887a27fb9de9ad1ed534f63cb5e3888494057cef07bedb61f2f6db9
9ae01ce95ef41178d09a582f01c5618b91b5ecb3036ea970d523b754af859813
d6e677eed0ac67d80891ae9a5b8aade272ef6377991bd2f2417d9c43dbd32b21
22536b605590e436e4b5f7df0eb6f4ab5b909ed36f8c5e0295ed48c54aa3cec4
fa793d13346d5a8f0bec32e93bb9aa9fc4a7df3909b14928a3d561d75895aa1a
c372de02f574c6b79364fd78aaa85d589ae3636bf6259ca45da35696ee499ab8

Fiat-Shamir context

e86f027595455e7dd623a2cefee666a86b990efa0cfe13f0b864a88416f326e

withheld proof term: 231 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isGaugeEquiv_iff_exists_Q theorem, FBundlePointPowerSeries:151

```
∀ {m : Nat} {F : Type u_2} [inst : Field.{u_2} F] [CharZero.{u_2} F] (ξ : Fin m → F),
  Function.Injective.{1, u_2 + 1} ξ →
    ∀ (G K' G' : Matrix.{0, 0, u_2} (Fin m) (Fin m) F),
      Iff
        (∃ Φ,
          And (IsUnit.{u_2} (Matrix.det.{u_2, 0} Φ))
            (FBundlePoint.IsGaugeEquiv.{u_2} (Matrix.diagonal.{u_2, 0} ξ) G K' G' Φ))
        (∃ Q,
          And (IsUnit.{u_2} (Matrix.det.{u_2, 0} Q))
            (And
              (Eq.{u_2 + 1} (hMul.hMul.{u_2, u_2, u_2} (Matrix.diagonal.{u_2, 0} ξ) Q)
                (hMul.hMul.{u_2, u_2, u_2} Q K'))
              (∀ (i : Fin m),
                Eq.{u_2 + 1} (G i i)
                  (hMul.hMul.{u_2, u_2, u_2} (hMul.hMul.{u_2, u_2, u_2} Q G')
                    (Inv.inv.{u_2} Q) i
                    i))))))
```

SHA-256 of the statement above

edcd50bece77b15d88e9594487f6728714e8ae2a893f95fc678591a5bfa9ee17

SHA-256 of the kernel type

e37c492915675047b281fda0791cf287ee81ac701d0347698e39a37ec895535f

commitment C

415a975371a1a21b6a7f35cdc169aedef28219f847d07ff4799be4e210244765f
617ba0c4cd168f6c93d742f20dc216dc45350ed61219b61dae37f1980326941f
4b0ab565227f97a0c9b43e046860f5af4d2c24891ea208c39d1303d6e5e56a
7718d24be63c4028ffbec9c2f79e7d5956abdd4a0561837baf4d4eb5af9efe51
02ec5edb30d053d1392961eff1d02ee0e00cc97aa29813535166b28d8faeaf7c
cf55d64c476ac313217cebb98d54415b1240a17f5a4dc19b76b73322b9e432b6
07f9a9baf6fb21227b9ce0ac06e965ee81ea1f81490c490bb6d5bde5c386e245
611f5580ee82efb68981f7b56bf3481e00f01dee5d71c3e9c43ba0b0d1d8e1ba
first message t

9156db5d3117a2fa7e19e95c9a87d8910fa365f05014f2cd90bdb46a1f591716
973aa284c6637a83b647f612682464f454ddf03f3e374328f1ea570c285354e1
c40e3ee357fe73c680cce57f07ffaa1de86c728195ac68a33efb134fa9a30c79
d2c27f300752907eaa06cb36175a58034bd9192e31f39253df1a5d1a2f4887b4
749d14661175620fc5896b3427bce43242011e973f2e4cf344805c4d77970b94
1190988b4464e1c9f203b3b1f3d5f3fe8ac87722bed2a4efebefbf7c03109d6
58160adcf13f03456235d0407f8aa2d3d223352081eca758beb61ff5995a3b39
f56b871bd351e92fc1933d67a73d35cafa437f76c6d8fa48d1a4878b400e112e
response z

5042f63b57482e52405ab18cc0c9461a9a8107e1c75b23d213a7c85a4a3211b9
fbbd88a3f83c4aeb32ca6a38265914aaf1f0d78da94aacf9e108c0576e0adc04
c09f47df3fe1dde3efbfe2a785182fba1eda7f423cbb146898ea24f9cce5d317
ada21e093597aca8ba39e83b3c83a3e8d529327e2518cc36a39d69c7ef040fd7
759d5b6aaa88c17a60d0652203608e19dc1c6093043bdcbb8b9b359bbeb7e4b0
8eebf05d03af90b9523e52a37e103cfe3042322e9e0973ccbbfdff4448f319d0
3dfb9669c978480515093297878ce015f7b58e65ea2f279e70f89346cfd73f0
ac5cb60541f8d76c5e41c741e9fd7fdfb69952328ea7deafbb0164b4fbbff38
response w

1056677b848b6b66a647b38d789c3090f1de1aaf2772be3d228e03517a82564b
ab79b1b7ed40ca39c56c2edc8a79e5ee4439c136bbc0bc962faa3512b08134d6
f00b9b54426d07b2bc3c3bc6b8150c53aaa68fc4fe72155fa1c528a4314bbc27
769fc68f68e35717c6f96ff0200ddce54e9a74892b30490ec925a439037815e3
f5eec1d2d89f116418b29f9b7dab4557cbe42252d3c561adc4344c830580a465
c98f808731bd7a9fbad88158299baafb9fddb9f2f4049e7c75237ee0942d6661
75f59e9b48859c2ac0c134fe088736efe1bfe1276a386f94e35e944e874c5ebc
2d08c1cdc049f714850e5cc90dd88a2b385845a212db0d7282f709907ee359fa
Fiat-Shamir context

1783480053524bc6ca89a615fe922f6b4259ff02d831264ec7aab032273a32d

withheld proof term: 101012 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isGaugeEquiv_iff_gaugeSystem theorem, FBundlePointPowerSeries:113

```
∀ {m : Nat} {F : Type u_2} [inst : Field.{u_2} F] (K G K' G' : Matrix.{0, 0, u_2} (Fin m)
  (Fin m) F)
  (Φ : Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} F)),
  Iff (FBundlePoint.IsGaugeEquiv.{u_2} K G K' G' Φ)
    (FBundlePoint.GaugeSystem.{u_2} K G K' G' fun n => FBundlePoint.pscCoeff.{u_2} n Φ)
```

SHA-256 of the statement above

5bbdcf918761d0265b06083e87e87c2bc831a150a3d16fa794a93dc8397a9fb9

SHA-256 of the kernel type

30a60444cf49752b41cb5a8a18f621bc91f541d0fe642fc4814a935fd3d66f19

commitment C

b447398d14c84beb01e5df9c6020d69e82b68a496a0de5a9aef7e66a7159d116
caaf9339ff31d70dd248928331b2de8fa051ebf91d6e1087b55f559e8e263074
ee603cdf53023b466e5d7ec0309a9f70cbccdd7b07473fa6df621b2f2e7dd8ef
affca1f8b2d7e17b4b31fb3b4efd1033f5dd5240276ee9ed3806f18628ee6dc0
86f8910347bbc913923f1cc565dc05d32d20ede9f288173a84ae0305c8ebb110
186f07ede12d7b74c310f001dfb4cce7f8077927c17b3c85d389a3d11908dca3
42d3741b8db53d5eb56f875d65317f9f34f8bb42fb7a442959fd211c9b15cd56
366005ab39b8cc9372900941a78864b66973b7b9dcc94e3263e14e83f867b87

first message t

f6b3d97749ab1952508a6b0171870c13b51a6794cb43998f2ceb00cddb558282
4397015033123508c37ce146da459c04c3745033d16df9ce124ccb1a44df9e68
1e8eee2c9062f790a18cf502e9b7d72b6f1bbdb6e527ab6b16d06e03f0ea33e0
98ef8bfd74d3ed9076f8e454b77d5a6c95d042c0db056cad5d75c8e4599c4f97
8d8ec45f2ad83d489852a187915ae51023ece02afa884d306a15af0ce1319eb6
21dee7c689866333c84319cca4ba299d9ca82f2b75a126cc9d970c657826574
532c47414d10c7d3ff602db01031b5066c542e58db4ba36c12b0815efe1c849e
68075d0acb62e69798f4c4776d9c55aa31ba30874f9aa85000772affb1be3cdf

response z

4d21ed12dfc4a79b151ca548da5ca6d2585c8aaa6e21ee8f5002ed13a99302e8
9ef0857e6bc7829371c4c6dbac1ead79182258ea3dfe25c1f7d6ea11d1b0ce6
24bdefb6568db928a250e8151b9c6643e81a4080a9207c219d9e45a664d01295
2edd5e4dcb792cea58a05742879612d0742ce9c77e7dee25e2c1a8a68367866d
a22c7ce46a46562c16b1538fc019f34035b46ac570526c8470a482a631979212
04760e051ee81776e68b3200ce2aaa395d410b9767b1508935a463cdb8a67df0
f3b3af610c23c8188e6fc44be5c0a0657be5b0fd2c682b58a09f7b183efb762a
de05283543a392625e44b267590506527609fd36a99aa3bec437e857460ff71d

response w

68dc21c4e93add015c3afb1474f825493bcfa256340814bc512a7592cf80f63f
3310e8caf825e8f0afa8bda798fa106710e84df2e8725377f2247d8fb2294817
f101532d793f89e9ee767c2a0f626611b3e3ef15cb6711783dc0e5fe4932e005
616c572b946546c276b2e52726d0d212c71daed707b67dae80660f97a15d3cda
a3a68850e0b2d7c3088f61f0b099d0d713605005ed041b0652f5a1831e97b08d
a358c1b8869f17c8a6a995b7aa57e6e72b88695d52de11b592f3dcd2c9dd3bca
fc7081299b5d6932e08029c87b4d55ac2cbeafdf98bbdde101d3e4ca6171adb7
5aedae324ead64449bcea5d1908b3a69e5fdfe869ca5aa94a74384198d43d5f2

Fiat-Shamir context

52444e044ee4078446b3d0b03ce7d38ed03e89e8fba42261d9f5a8920c9dfe8

withheld proof term: 1 102 284 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isUnit_det_iff theorem, FBundlePointPowerSeries:136

```
∀ {m : Nat} {F : Type u_2} [inst : Field.{u_2} F]
  (Φ : Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} F)),
  Iff (IsUnit.{u_2} (Matrix.det.{u_2, 0} Φ))
      (IsUnit.{u_2} (Matrix.det.{u_2, 0} (FBundlePoint.psCoeff.{u_2} 0 Φ)))
```

SHA-256 of the statement above

fd7afe9dbfbccf702e04e72c8d8f519318af2a065d84aa9f986c4dd557af169a

SHA-256 of the kernel type

3aad60452f92e13eba145bad77503be9ec6213bf3345a8ce9c19b0955e98e761
commitment C

b93edf6879f0afac77b56ea5c16a299dd4233f15377780513fb2861e5a7e166c
f2aba0cde97df7705d2bd8066d5b0120c3617cf27f87985d9a4f1fe5cdec0735
cc0eeac733febe89d0275ed6c5901dbe86a1843dd700ff1fa8908872831f1627
25ffd8d0f7351ee1d15e3c18ba3df9931c65680dacdb1784d21e1e392a1bd712
2930e4e88edccb2032b1ef548d4ff2a1a39d544e9ba87247f14be6c32adb767
0c9c560140c542c9f6383ba961b1c32b5120eb83bef424bf7cb3a06d2db402bc
73506c571098d935a5849c828de13818dcebb65aaa4ff1826dcc946e114af48b
8eee2f530435c4a984b929f1265944ed4ab2bc6d8f7838ec1f9d2e57292335c9c
first message t

b3c58d1e5532d98a6cebd5dbc411d5b09c0be4315cc9768a47ddb805864ba41a
ecc09e0332e1984213b0d51fd65cdd898192bf8780343e406283ff31d85f463a
2f896afa693e166f0a8a741aa075e6ef6453845f9e44d633947021df4563b8d0
af7d1082277d69d1dd6f5986d885dedaaff7373e4130a0512174eb25b6341eb3
af432ac79a27f46fa322ec346d81f4ace4fd14bf0c5bd55191e2b8dfbbcccf54
e7ac12d98004de416f55532252a74e58f5f378983865bd7ab59ab6382ea2fd90
c2049e16cbfbfbf878d622ab317a867f9103d6078a0d01d325dde0f88b72f256aa
7ad56c3b6b309555dc9a3cbd0106c8baf7430299d80b69620911e8fecdd76c2b
response z

56f2ade42b5e2c62c3d9f58d47d8364b7f741f641440950dba4a839dff14513e
5d0bd82050dbb0a0fda5419ee5162c0ef7c9d7b64954b925ce8cd4a79c870675
4b3c557a7650deb0c09f90466f1020d86f3ed31aa9e9dd286baefa1384627eeac
38a48078d518324bec47e60f7789cc5331964bd081459a5e0a01f483ae90b706
e22a34f57de81d3abf6201938602f8feb5d936662397ecd674354ed94d078578
ab897b8d05ba9963a15160b6dd20e9bd3e5da85a9d35c044a259afb32dae7c6f
a4ad86996b741f3de09c02c3532e3c252e340c16a2c42270863068a8257d095f
89168f622fd15c902df4ec5f76996fe88450c6a23868ce46ae307382e042edac
response w

715a5604a11f67d630afd9ee43377fe3c202ba101571173e67af2af212369dc6
32f51ac8321b3896612c218fd2f8d21500eccdba101a591bee30f46b2e7a47bd
46fcff5e78027a06e66ed76807395763e3a8b68b23304f9a00db668f54f6ae81
28a97e658f368b9782296d74c1dabd1ed07115d41d8b466c1e3fd7280bed2b9a
235c4959818e0e233f11b049308e429db91b5f63d9346418e3f18d6176a1a2eb
fcac85e3a947299ace1d90b54f61c18581df9945c016c277c83726280010a1e3
53c676f78ffa12032f33216128c62cf8357988c42468c15e50b5f0ccd2b0e8c1
49d7f6e9ae87a1e34f573cf74df3a30c69d431984e57e9f8c4cd1109e43a5dee
Fiat-Shamir context

4744ff9cf8ac671992dcf8fa6491c058de2c73813b264002eff71b34f15ffd21

withheld proof term: 119 799 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.ofCoeffs definition, FBundlePointPowerSeries:48

```
{k : Type u_1} →
{m : Nat} →
(Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) →
Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)
```

SHA-256 of the statement above

babd8c3839c611817a92c9941551242514b42ac17809377ea40f85d35f4f3fff

SHA-256 of the kernel type

5bebf946cd507cfb3b6f762d8dc62a0295541a686933d447784c8c4926e7bee4
commitment C

9f89a22b6d1a9693c201a6a0819a120661f58cf7a6b1689b44230ba68727fa64
1bb028c2667752be8325fdcd1e9da4055055cabb0b2fbb0caad29652a743c36
38da3eafbaa96f5bc917f30f2c146b21f70dbf6d9f4e480c9640464273100a0e
565ee95a5b5fd6aab2f4f6a9f8fcfc9cad57cb87e1e51279d327724bd45c0c6f
2e9a6b16097a26a832e4f33faa3b927d299b91266ce89ad4658c72e461b74d0c
24c763d39eea4317bf6b87584be48da1849694bd0a7140a69e83698138352f90
ca45ef48726623534df854e68a6b6c13eae1e94c668098af42fc4c326fa86e70
c35600c38e2fdd5cd42c265f8cc4e39684a4d85ddcf09af42808d14738c857f
first message t

af6ebcddd29d93779a2ccf770544846fae2063ea24764b5bc39e6b8bb8fc76d3
34591914553a977c366ef173de9cee7f21d28b7de7dcbe18eac86eaa0ce69e52
6e2d6d69b661c9e0fb74081205ca5fc194be7069ec06279ff0866a6337b28bd0
a7dba58928331aefe92bbca281da2a4065b95d255dee628de68aff30ddcd5de1
d2afd6a27db93a8099ea7debb9c97178c470e80828499f7e15ca1c6d351eaf7b2
b97d305ab829258459fc18a2d357051d965cc7156a3847780aa4a2c83ceb03c7
02c028db25549c608f9bb8226ab302894c300ccc8781d8f605dc6c2d3f9bd9c1
1e94d049a9cdf3eecabc405eb19b8dda13f37b215f604693ba2cce189855e9a2
response z

79eb01a681b448bbad1effb2c76f45940b7b144d987fa046026c66afadc1dc12
7ad4f23298a09c745ef1be4287d588bb1786e170ecff6ee38163bbfd35cc6892
834c850cf5e087d309f66c639941dccf82c671c95a9e8140c802ae5ae351c365
2384d0250a785d4d02606e1daa4ef8e34af58de2ba33c0c0ab40d3f7b9192bce
a06a2143057ecbe1328822a720c909d87aab3a7de2755f7722e2099816ed917f
b3cdf3a19b3bc19906513f5ec683f4b814587ed9f04a90f404cba14c77077b9
a112bae66a3c77c0a5d13bbd8062b8de14978a8e055bfd1ef2a380dec6fe8189
f497e3a335bf34081d02d7bf147e5ee3a71514add5c9db2732bce709cd6dd
response w

4df3ad4890735d09c2135f2a1f0b47f359853f1b6fc87543255fdb7f74f5bbf4
7f50f9b09102b825ca0f00b9f8e485948c76355328ce06475ed50ee6a1096a0c
813ce8d2befad0daffdc865b840e7989cc9a1802dcec4476861920dc85abd0b6
e109aa1e7bcd9a6af5afafb5de32e1bef0ae06d5add02d8408a8898cc016836
a4ae56a1b907fb4644111d5069a32ae9be965235ca9232a5e060f92d5b105874
f6451db4a9435e4dcd50635f13043e8295d12796c346706b923922003473a4eb
916fd6dde5c018042954af65e7f56bba1b149af919b38b93d64f993fd5d3049e
43c58d46dac5f331055f659eb18df0aecf93a57ddcea83e6a5c1384f48cca461

Fiat-Shamir context

2e7ac20fadd441512bd1cfe96e966810db89d4f4c2d71af659d7de644edcb91

withheld proof term: 1 208 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.ofCoeffs.eq_1 theorem, FBundlePointPowerSeries:0

$\forall \{k : \text{Type } u_1\} \{m : \text{Nat}\} (P : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \mathbb{k}),$
Eq. $\{u_1 + 1\}$ (FBundlePoint.ofCoeffs. $\{u_1\}$ P)
(Matrix.of. $\{u_1, 0, 0\}$ fun i j => PowerSeries.mk. $\{u_1\}$ fun k => P k i j)

SHA-256 of the statement above

af24915a4540b80648e35b8db99523d189e4194d689d5cf55ea316fa8e7f7db2

SHA-256 of the kernel type

b24bd076f6c4b9ea45eeb54aaaeaba639c2d79411e34fc7ebac063caff77276b
commitment C

2f9e99887c7b218c97e7d7e011bc93c004126bfd298c971750c3a815fd8b322f
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0b210aa70d05e517074d079070de0650b3a52c094253d46738c586b42030ee78
df16940be3d7439219446d3fcea986395e37a4bb076978e13b2b303b11b9459f
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84dc7c6a061cb5c89713e9f280b3dab504d1d682729c4e98d7d7772d4c8eea65
eb4abb5e401dedc2cd7172a7adcec7840c69a7730c4210d941c15fdef77e8cbf
08b5e7627e0b0f0b47ac1641da4f71f1a3c39f1b15f6103e521f97c85f00569d
first message t

572adf4950fa74232a83bde23fb60e26c5fa63fa2c4f341e40a61e30e9146984
5ee8d2dbe16e4f9e69ec091606823097ac23420a3c7f4c4b0b0239e37311d253
bb39755b21f8c1018926ae76249c34a5f6a909d8f3184df9005024ec028324ff
d2ecf0219b4670fcea8bda7989f900be94b89d4f380ff43d91644b92268c43a2
31ec2c9ebaa4c66ed363cf8a56831cdb759af1710489e3a2b2a4e40113f74879
547179a57dfab3f476ecb00a7e7dc42dea2deec3871c8b2c86aa79bc8e186976
39ae0c97e97390be7145c78f0e33ab088a63a073c2950075a942e5665ade522f
dd79fdb12487178d5666b30fa46ef7a576e8f61f3e18dc19b46be50672a11c37
response z

286fede5e0704bacae1c5beedafb3416cb98a915a2ac01c9bfbacf3daa32492d
cc4d58f63399370c1c04dc051f30dd6aace66df48777e13918cf5b2ff38879b5
399b9bed1ac28f667d9505de228cc5007708c1b5bce57668676c279dab7cfd69
364bba9ba01dcc40ebaf8533b4f6317538cc952c73a1950acc6c171a19c38ded
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f3cdc9dd3053cddca740806c39639a730aede715c778ca5b6ad4e0c9217334fe
a1afb5856865c033f4ef3b4b2e579cda86eebe73cc22a95b568be61d9f19c21d
f2aa103d084ee3bde4b064021072236df92e05747ae1c82e1e025acc1373688a
response w

2ce6967ec1f8bd7d0d0201a31d54676ee1f640b4966ac13eb0ff895c9d0fde21
2c74a5e41c6599ed8b8770bef4f329e0f4b30d8fe422a556311799a521593b75
59c5e9aa4359809c18cabb65181eb1964cbdb46f342e7dab2d0a4ee03178e1d0
6cae6f2e13cc997fb70210e62a557c0af02ab15b00902b41bba3624f5d4a5b43
d10dc6bbe6b4cee0b5a37582934c0c98aceb790c447457335be0eaea6b651cbd
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d64595f0841fdf0ecbbb38a243ce6f70fcad362f2e69c8d771fe643da3d7d042

Fiat-Shamir context

d525e3a1d8b816ba33e64f8fd42a98317ec7729516eb2866273585b2e448d7ee

withheld proof term: 211 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.ofCoeffs_psCoeff theorem, FBundlePointPowerSeries:58

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat}
  (Φ : Matrix.(0, 0, u_1) (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.ofCoeffs.{u_1} fun n => FBundlePoint.psCoeff.{u_1} n Φ) Φ
```

SHA-256 of the statement above

8abf44701e5b1321fba4d2c0b797912204e69c8a66c34686f763939782fe29c9

SHA-256 of the kernel type

5d1b55f12d209fba3cd12a6339ea4817360f2a66026f860b51d1effe9262602f
commitment C

3132b626e787da06fb657eeb9aee9791b590103de49d78e4136bfe5fa98bcf40
74ff792fb93a1885144e7b943b751be3143b0bb51e433614f7ee7b1a5e741f8a
f885214805ab140daebe1560d57921fff882231f82c8810cdcd2784f400f5a50
4583ebb914969dcdade015bb9b7fe02cdca442ec423b3ad85504fa3c41c29974
4acb3ba400b5b6e469d20d25a872ad4382f291c04ade082018f02537da4b560d
3c3e145d95fc2687d4a5c0a91565f1ac61b257d8f0eca5689d6c89a4502528f4
2cfba658bc2344d305e2fc4da6c86420bae1bdf538caffb9d6e44f8621a7d9c4
a6f6f115345523d317e4c385bd051bbec28eb598f56462656a6f1b69b092e47
first message t

ace7cc0237a0f370d8e9932ef1a7bd70a8d88981e7454baabec011c1f0ee2131
46158634ccdef06218d45b0cc48c074f6a2c79514f987323b8e62c07636f0048
4d7e90d99178ab29f74f020e5a4475a7b5ffe6b3ce2293a0b64a756aed5ed87c
ad05e481e3f6b2f0709d859f56adcea8dcfe28436ba635832fdbb82ff89f4746
8dd5963efff6ef64ff031c7801c8b357b54baca534f8c5ae367a941d7eadee1b
316809bb5eadf496ded7c7e7592dcdad5fc67fadf16a325a1a722ffac209ad5
cf8d6369e43646a76820213f82a6fba3b63decad6e77fe7f0d0bad1add8119f0
53b385e5099407a5be992ad19c2094ae6f4977fec4ee29aa5dc96755e353e205
response z

28ab852926464956bb61305edd40133603336f3683a529ccc5cad52ae6f25344
62f1e1b90e29c62014fa9eec495d6fd79a2c27daa681f1dc3ab6e4dd1b7d3fe0
470bb93d027875b23ffc4e646924048389165c72cf50462cded4ddd01ae95ea2
9e9a6557c97df95b7de24389efb8d7a334d03c6408b50c166372ad3c21fbc43d
acad7c6af8b5b8e693d3691286f6a8301537aa89cde22e74454286fb9aa052a2
63cce44f4bf3acce137c14fb5e4304a7f57535767bb99f17d4e81bb351c3252e
94b6cd3b689b88cedba545c70881e4fa06cd15cc46371ee69770f96c23c3e8a5
797d7d4d457ce51a33f316d25f03b27f1ea6243f50097966462616742e61bfba
response w

7ea909a1fd118050edd855257ece57ea27e1f34f75d04ac29c93bce02809b0bd
5546ad983598528f256308dd211ecaf78d9632b88323cb2f202431a1072218da
09270fed6d474279421110b4017211fd04b7390a6d9247d23950dfe7fa44fa84
f8eaa6d4316fa1d01996447ea6c91f951dac5b9effe810fdb744e640111c2bb0
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caa0e9d77fb8b3e76bfaf6d06e89c302504fbad33c47f92fe24083ae5a8c79d8
775d02a983f194a6a36c27d9ff41b990f767ae8d998a43ac1914e2cd6621a4e7
76a6ae1273a6d1a21162a1326d686348e98e38fcf102a3256897fe30829a241f

Fiat-Shamir context

16e734d297c6150e422176dfd1065b079eb49e78d81b4fb2235daff6724bb595

withheld proof term: 60474 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff definition, FBundlePointPowerSeries:30

```
{k : Type u_1} →  
  [CommRing.{u_1} k] →  
    {m : Nat} →  
      Nat →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k) →  
          Matrix.{0, 0, u_1} (Fin m) (Fin m) k
```

SHA-256 of the statement above

a740baba8f807818312c9288ae03bf752e8b6162ff249e2a115a81e9212843cd

SHA-256 of the kernel type

236be20c473fd8bdfdd467548288b7b82f05c4ccb82e7830e73c879ac621c3f4
commitment C

2cb0ec4c96636a6256b630ab64b8b5438d69a688b65c75693b49f60d540d8c19
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c58befb593984f9bcc1474dbf315688b10a876895c5e59f907cb445f02b42c4b
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6f7aa436e8fc5d9a418b41bd9ced5d106967c736aa95dd9375fa5e7be7a8985a
1f0dfe05dcd37cd69baef7ed53c264e5ad873e2df6a5fd5c20af249e24fdfd74
85eabf640da691ae6d862edbd6860e189eca59369216de792cf630722eed6ec5
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first message t

26b4d8c6a37f19b25ee0345ab09c30442c4424ace12bc27c6a21475efa1df00d
857f5df38484e1b39d5d1991eba9417564cd7128ef81885fe354848f3d424764
6bfcdfaada0a02fe5f3865a18b1c90a4e2d2e34f6e672ea05a05cd34c759de74
93a96452080ad992e5174e6960aa1401b156b5aa764bd7ccea778b846d2fb803
5cf3b4ebad89208485f28a9d639b13dbc39d19575e5ed0132a6cc69d4a045bfb
199dcb701f575e40166173bbd342dd78b5fea17ecb8b7edec69421c0ff916a63
302e1e4673ec163b9e0145b7f81aab2c38948fa67e15201848a291cc48d66e07
bb36888a9b1e5bd1269fd0e8cd9eb19e96aeafffa91fcf9d2ce6c8f24f15b31
response z

3ac8a8c3b13464cfe7aaaa90757f0d95cfcbcef73950ce838abb7d4882fe11fb
14d47c834863dfd5d2c8594cc5d85f0c08b19a415c697a452bfd65ee8f1b4590
d2ab4d04346cd7fe6b8c8e56018373f58e2e1a98e231cef6cfdb7773266ca665
640144ab90412aee6b9896863bc13a22129af1e0d2c6c393650be654f0840b2b
bc9481de34cbf1ed2b1d97355ec281fe588109b86c72d4c19242126cdf35c5d4
ed3e819cf8bd7c414c7b6934a5c30d23a2eabea8dd4ed933bf6b20b94ca70d3b
c99fed5d093129948551de85878458804fc612396eba50ad45a1774bcc89b1be
9f09ce269e6c3c2e312b0ee99ac1ebd9c951a76a9fd0705102285ce43427315f
response w

22738cfbfba7a8638adc5882fb1ed689ecd27e5284de3134c17b4340f08a372b
1ab727c37901dbf9e60cd5ace7f7fac2fa90ceba8aaebf23e326d3450a145663
4c3d0baa4f5ae0abbe013c072c709af951678669d846ba8711d38bb35398ff66
60c7264d4f06b2d7701811a758ddf1ac27e59f45c28bd6b75ebfd47a5b05694d
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974b6a081ade8a658565a1ab3e8a96468e5ea97fa04fb15dd3d5590baaf062e6
993f912e5cdaa99427e62a9a2cfd072814e4231b1ee798175f3d0bf84756f280

Fiat-Shamir context

a24567d03b127add2000fdd67b8067f53d87cbfd54409ef4175a7d33f653c4f1

withheld proof term: 5552 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff.eq_1 theorem, FBundlePointPowerSeries:0

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat} (n : Nat)
  (Φ : Matrix.(0, 0, u_1) (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n Φ)
    (Matrix.of.{u_1, 0, 0} fun i j => (PowerSeries.coeff.{u_1} n) (Φ i j))
```

SHA-256 of the statement above

629727836b0ee143eb7a1315b3e48cc7e8d634d34d1b58e921618b9c727a53b6

SHA-256 of the kernel type

2f497048eaf2bfe447cff4ea04bcd7cc36a33035ed6b158812f3c73592de9ed
commitment C

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f7ab8f840ce0dc6d1ec0c0320329ac9facc9bdf71151922c9b2219442a187f6
6f44e7b70442c4ce7ebc4041accfe8ddd46ad1cbb8f01ffacf25a8853cc5f882
3324cf769d006135fe78516a35993909edbe42d5718b1ad2f137376b8b419603
2fb2fa52154823bd21edb7a19afd7cd348eebaf46887c48bbfe9d0a61b598375
57f1b24248b4f588fd8be77391b3deed5fb32ebbecaf35ea21201df4ca513ea8
658a664db76ba1c6091842e700c89bd93e86b1e87588c0bc8936685019aff933
8d67718264b15772215a3042203937df1e990d2829e2aad472e95a067926f1be
first message t

8008df209c2386043a714fa9f406f687b731593855c8fd90d59dd54d77ae5e0a
4a5d32516aee51ecba8457283ae883c269cfba7c20587582aa9e8180279b68e
aed04b83fc8ee036a9f868d655c0b23203e07fd12bf96d7c5fb488dc74e42e4f
f2bf2f03164c477043b11d0d1fbeeee20dcf7d40c3a8a8b2d0582d1cbd89404d
e7cd70f1a383761408bcf249755469851c6f8117c90875d3fde38919a5eccd63
a2d8a2449cdd38758082d24028f34fce858fa1d412809c45aa69b99f86c981f
458900c473aa405e9bd5e21a469ea85e1beafc4f0fd251be2ec35b6c60858d88
e2a4e454b9c571e0a1099de54ddc12124b507b3345af9cf30efbff712e3cb6fc
response z

30a8006a5e2820d571a3642eef853dee9865d0e978217449e08cd49c908ce9d9
2f0af3dc390dd14b128d1001d9936289a42ff023621128b06b97234e0ad244c3
e61dcdd5e77e36afd2d2473529883e014d433a65ac5c18275d963d42399b79cf
8bf3e6fd5e87fde857994617ad4e556a6936e23b43ac6870f7a2829f259ee084
3940e65a83bb4dbdb4688748fe0dec08d6d3460689c0b7849cf589d95a0dde62
47d057339516fa83f50c4a101c8b3435310fe9b66d207766abc88563b3fe9ded
a149ad1739ce0f272109bec5ffad8b9b831e88278684def228f3212ef1cf0161
61b58028033a20193e567468c839d766a2557818108dfa27c11e894c0ca0bc94
response w

504210441a00c93df4b19348195eeb746bc92c6f772591566699fe76387418f
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c58dd12c97ede7284cc7f10ff165353315c992f92419d74e521081059855a7c6
532fa59df6bc3fcb1fd2fc39bc05d1501c5e41edada93f5d9a78a82c15f1ae20
f1b38fdf57d683db298fc1caf377742609f8b1fbac1a33c8638d93382fd58d4
cbcc44b08624d1d575235db1536180144e32b8aa35ec01e9c8217e11f49882fb
8a87b548c4c35561e050363bb1a775ab381aa24d3e1663b718eb82c03490ea30
3524f384a39d25dc8cc098080af03d0a1f67b0f85452436c5cb3dc3fdae1f7a2

Fiat-Shamir context

0fd64c1facf3ecd966141bf2f4ec377958f9611d60123b97970d70b07d25d700

withheld proof term: 249 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_add theorem, FBundlePointPowerSeries:68

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat} (n : Nat)
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n (HAdd.hAdd.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} n Φ) (FBundlePoint.psCoeff.{u_1}
      n Ψ))
```

SHA-256 of the statement above

bb57ea8add05fe95c6062a84a890df3f5f8f14d2587e2f27fa8be3bc7e20046c

SHA-256 of the kernel type

1f79f55305c993e94ad4af7cf1524542b9030be9b9f498b0e777ea490a7d567d

commitment C

9cb3aca2b283762c4fb902d6541665841606696419a51a9dff51e70cf570bd6f
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ac6d0a5466d30fdd4700d2a0781c4944be62119c568d7840f207bd1b14ffcd3
dd93f05336b8fd8b14a8bac5113b18a0673732c11621bfe1c0ddb6ba31c8528e
3736fdc104ae5659567340a8f16d375564b0bc52387b8ce1839157382c932e86
f42cf662005ec8271cf9e7f5ffa7c7a140f1ad021315c45b4e239add57340ebf
caa2709b5e2d72a53d8149481ad6028c3482fc9e5f2335a2ddbc278f2ada6529
d506b91e44b07bd30eb9282be4c11e81ec3ab8e7a805c585ce8c848a1a2a766e
first message t

5d28fd05e240f47a1951f55e77da5a8b10c7295ec1138880132dd675e432265d
8dc922fe60c2db230a14f585bed7ae2ff5bea776f5d36bc51891152f16207f9a
176be8d7dec1146ef4aa791e4eb20c5cbe21522e3b665f05d92e6e01022c2f13
cd1679a3a0537a35b19a4293465eefe09f925209abb5d3255c94e74adad521b5
29b510ec02b6971a7102caa161407a1ac176a5a31415e9132ea2a58f2b54411b
1b5c0f6749af81bcd71d7dba5924f48e21d4610f3150215fe715654cf92727e2
b379cfc1a44d496bbc7e787b548149b11742ccc9a8fd67b625f9d5d46500b5694
20a3e4fe2f75788069c70504c4a25a2eb994a01b66df4eb3187d1f7fb0bba802
response z

145834d2ff4a9058154f87890d30d9313bf5c57ba30da230bfddfc100e43981c
a375294441d5bcddd39057391f21297e18b0fb6f1f16c5827fe74f45b7293368
dec74f60c74f876b8b849eed7793f06283939cf7365993344155a26ff780484b
9ac43e827546e7ce70794f5f8b39b97ad6d125ec0b3e948a0dabc50967b80db5
d9388160a0ceca36ec1c4b3ee86879657a5b10076f34918ccec3c0436c13ada6
932be1fc353c130e27c3b264b7819c7c7bbb2b9de7999454077d314e8b32ad21
6767a79c8ece26689300df43127461dcc7ef6aa23bf38f24905cfed517bf5247
fe86e163ff57dd8d1c95568d1c8c4b55dece479e02cf5fe1ac085480452ba4f8
response w

72387fa3f83a55f75be317d2abe42613d283d7056d3372ec29a3ebc85f536225
8c251f5b515363b63c89f4be52b9706b060a571167bc1ae2307eb7d218e00769
f58506d4b786bac46a1322e4b6de2e1ece98633194a3c6351ebb600da4816d51
a979d3fbff58e7a691fab68fcd3d6f5d04be3e0a79df1842f968676aa274a6c
1d771630827a911837506de60cc6ef149ce7ffb5d874ad15bbd8015d0b502598
93b5d8aae4ca2f0cc3c101f41a461b0d236fd072f9bbd6e5e6a810f9ccc4372
74f26d7ef60107d8cfa77686562503746e7d44e7beddc263c8e52c882e291305
1530bb6dee2374c5ee734570013145025f70883cca77f39dccc330df65a92cf9

Fiat-Shamir context

4666651d7c7ab0ff0da31bf79605a35175e0c8453e1447b057a8bb6897766c77

withheld proof term: 174 297 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_apply theorem, FBundlePointPowerSeries:35

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} \ k] \{m : \text{Nat}\} (n : \text{Nat})$
 $(\Phi : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} \ k)) (i \ j : \text{Fin } m),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} \ n \ \Phi \ i \ j) ((\text{PowerSeries.coeff}.\{u_1\} \ n) (\Phi \ i \ j))$

SHA-256 of the statement above

31455921a01cbb0ad4dc91c8f27a22a6a9c516a6d9d53a93bfff9dfaae38e4ed7

SHA-256 of the kernel type

0ecacfaee697e5d6dc4516fc9eb54972fc4a0bb0627cf9ba596b34ac3a159aeb
commitment C

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81647d2a5ae9ba7aa7e2bdc4cd6fe9ea58857a80298be7b9a79a127b3ea174cf
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first message t

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eed7f4758ca25a7d8d6534a599440ac4016379f49839c2e5d40d0caac5c1c43d
086842a853fd8d43fdb1af50a048b63c380dd8a9d46e29b5f9efc2dd15903c69
70a6618199b1d60aeae12165bf97812275dcb9b112c1968ff3099aa6b5fb09e
f449ed64ec32d0442b0852ac04735c5e684ec7f0dfb070e2bbfcbdb62617177f3
13f514070e92d9118cec25db963b9c2632dfb296614186f2c885ee40aa65d8ad
fb58d800d40f4c7c8564662d5af3081d61e9dddad9a1c8ae6ea7b2f0fb7b9c43
response z

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be182b9d543ffde149d02460e283729ecc1f9af57b41c6f55822dbdb746ba74b
20328e43c43a8c567ea4ccc3c1c290ad348181715b9ec14b28be61aaa497d011
91d5155d865ce55cfd3c9aefe1202f1c8ea42684ca5ce285a341691319e5c8d2
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response w

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Fiat-Shamir context

50816fac1fd5d55124b619853eece47d2f5084b019bf4fc98bd9b9278e1e4c7c

withheld proof term: 368 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_cst_mul theorem, FBundlePointPowerSeries:73

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat} (n : Nat)
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) k)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1}
    (FBundlePoint.psCoeff.{u_1} n (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cst.{u_1} A) Φ))
    (HMul.hMul.{u_1, u_1, u_1} A (FBundlePoint.psCoeff.{u_1} n Φ))
```

SHA-256 of the statement above

d361fd0e5aadf374884f70974cf3df42af895bede4c465a8071e11b376bc5d64

SHA-256 of the kernel type

ff95b0e0720b2344452d529d5cac85574e48770dcf8ad2ed674647389ebb8929
commitment C

286094a870577a6010f4d1cf8978c4da560aaf6b87862e5ef780ae4931b9400c
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a7bdcc29b5b3bfb4f3881b013cea462d47931def56f2b7558880a12762bc957
72c11bf2af8306a1ff10f210e090bd9b87c79e1a31e620cbf6987b6caa69cc70
ec82809bc54911cabcca57978e4e05efba74c30aae9d151d696b7dc2af2a2d5
fbaff5a99a9ef69723b690705d357249222cbc7ea555b64775f07ca276ee085e
d9a4cc949a8def907a57098a2b97af72cbefc41c4973520ac3639ce3d4f05e5b
first message t

8aca3630bea9d4a545933489f424e8dba8e7f30d692b142db8527667e4d8fb62
642e5123fb8a88bf575fc31e17724469e7edce36f616ed1dbc8b023994ba6270
55682d71f4775bf9f2d4646aaa4371ea4711e8f249a0f8511a9be68369f71afa
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57827f6ae54ae93545e9ef2425f065ec22ef992751be033af4ea9c3fd09f17b0
e1876b1ed00ad61de17825182c0edcf7517046f46e4349ab942e7aa1bc269f36
c67a53c5e3b7ce45c3f973f0f991a6c0effd2c8d54c97f6c0b3ed12dae1c0dc7
dbad792d22ea9a42c32aeec06fd085f5cbee95ca70aa1e9129d447722049219
response z

7bc841248239602aad96ba8654b67b473ac07edbc6003db92e8ae821c1dbc82a
437b2fabfa7e1fc511a3a0e52d1c31123929ca59d11fd78dece67e90e107fec0
1495baf212cde55a4cb95715e67950b917dfb85d01cb22018a7c02d018fa76c9
3a3103db46bc778e1ed57c0cf31e7d245caedd22e89099e8a96291a858de2061
c90158da06bc1ac08e07642e5b7b2e676efbb770f86db6319a2abe42274c643a
27b7473557a46a8ff0cc8d46c743540b67735d8fe0c87fda970b370df02f0aa7
6fb5f88a85eb9455eb92232fdd20c940bbae128715d046778e3edec883003bd0
a9edaac688e1e3e26ae5cdd29a6c879faa75529160efd7e3cdaace277a4425e8
response w

3d69a32a789aec065d6429d7067a7a18a4ac26df2087e151bda10a1310ed7aa0
60cb530c12c630488223158e78018044bbc78cdb79e8e6d561296e7f14ac5a6d
bdbace702918448a38394bc25a97d3e08fcf9870ca8ad96ac85f66052a6c051a
9e324c78f61a5cdea36776710c714ac25cb43d276dc0079738c8ef3d3f4ccda8
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Fiat-Shamir context

c62a6fe325816a79d3878384746170e41e95b09db960698294efe89f286f980b

withheld proof term: 204674 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_ext theorem, FBundlePointPowerSeries:63

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat}
  {Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)},
  (∀ (n : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n Φ) (FBundlePoint.psCoeff.{u_1} n
    Ψ)) →
    Eq.{u_1 + 1} Φ Ψ
```

SHA-256 of the statement above

108f0d159c26acd3b85d956e8c8106069a641fe12cc3c14b839d04c564f7a066

SHA-256 of the kernel type

b2578fdcf74239037bf3288ae3c78b56cda08be1e7813f1f386bc12d31cea458

commitment C

9d98e5f66a1b35dd47b5dd0637c5cec8f3f045518dc2f90285b40cdf373f9b1
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c988a5b07be557a0ca5fc6ab15c2d64efcfc043ed32a32a3fab576b827b00ec2
898971d9ce4a2501a61783cd24312915ff6d8a1d25db7ec658269184bb049cb9
1868e3c8ab609beda2d8a98490ab3a96d65e1c62ff14212bfb0629f02c401961

first message t

dc63db6ed922d55ddc51027849dca3b23e6f165958d29d36536cd1b3c9e24aef
0fbde34827b0ac1b168fee104b099b207576d64e238ad5fc2dce6d97db9e83fd
af21e4319ab3a57aa4bc2a263540619df18d684e4ce82edcef277b82eb4bfc8b
7c0f4f5f640057420edddf887f457c6742a2a6a727cef168011c1f53c8ba9c39
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e4bb01a2b3a0956fa91696c8084588bdb676fb6284b3cd3d350532d9da9d99db
4c541d6d7b97d4dc8e5e22fb106102cc1c3e0f579adb92578737880f99b197b
d7dc8d373bb7dbfa63176c1b4b04018f31f38e4d29a4a5a715d6fb49b5a5810

response z

628df0f73e93e3b928f4bf327099f9d89f6acca2216f0431fdc49d9f36c2992e
d011b2e3ffe310e4bbd4fdc05dff484a37b1317cbd1b2753b7ffd1dca8e82e24
d59e8b38cd43605be80e47d3f3e0759c1fbc5abd0fc391cd1bbdb6b6d321b712
fc2a22bd84e8654c1357387314a9a7ec12cb486f0f56c250943bd3ab70897e29
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acb3adfcab714f6250ea1aef2c9f19c535215d8e3cb941afcb8db8b41a2cb6e5
1d2a123f51f8495c0d215cf9f07293660f083dc73f933135c575789b6715f240
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response w

7393571a672cf2908c6531d4805938727fe6f418a1a56b90963ce2dd51e2ec9e
ec6a98c6b4c7c39f1bdff9665342b95061f9343a4edf7f2753563ade29254712
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Fiat-Shamir context

715a2966a054012e0aa53fa6eeeb2d6bfaa079f3e7aa187c8535cafaa013ada7

withheld proof term: 1 486 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_mul_cst theorem, FBundlePointPowerSeries:80

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat} (n : Nat)
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) k)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1}
    (FBundlePoint.psCoeff.{u_1} n (HMul.hMul.{u_1, u_1, u_1} Φ (FBundlePoint.cst.{u_1} A)))
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} n Φ) A)
```

SHA-256 of the statement above

455975fb45b82c8222edc71fc0ac72bd4f38cd2402b00ce94db86e04b32d9ae5

SHA-256 of the kernel type

a1b5f276f93c7f3eef9e0faba903e92b060427468498d562d8a329697a3da1f9
commitment C

dc3db6211593d632c0867f78b4c50716e8bfcad45e15164b0c0bbc011354984f
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eacc91bb7370d254fae0bdaf9d10c4855b636c60022ee07643396ee9e3539023
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97d4b9591721fee42857e9a5a398dd9ae42728d21ce21e8981f8044bc92b9ce6
c7b66598b671c0c41251f3f4237e57751d213908affb7fdc5fa3e232b4002bdb
first message t

e244d0914732d19cc781c5ef2e3ccb94956ecda220224c11cb08a506e55d0a39
92c0acf016999f26c88026ea2c6ba44cdc0736c3ed559bacea933bc7cb65348b
d0fce4419ef1dc3f10854de660cc026099dbfda364d400e8e53dc6c7499e16d7
7323291798431c2ea3c76b324372c300058ac6953387a24a843241cafb4c38ed
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d884a4e262b0e0646bf3194953ca630f05f7d3270515cac60a7fbd2eaa70ede1
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195cc18f43dae90610aafd29b3762e7b21cacbba026c49d0cf8e62742b63104c
response z

35dd681271c9530a4c2b76f306b43d9e88b4bfd04550c946c7ce23005aa1a3d5
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a89ce48611b8e1ac3e0a37f93e2e981140332fb6a6a0063052cfe30de36eb27d
001d5f6fa9e18d94ba28399e2a6d165217600d583a8737a30bb98d6ee08b8204
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fc80ba28e47cce9c30604580c0b794cc17407aaffecf5cdf05c51c3bf8574c36
8d14639f36214aa6a7405e653f36313645a14346fdd47784ac5abce117b57a4d
response w

a0015b40ce48b1267daf637bc1f034f4c84377aef7cbbe4751a973e9c4f06b35
fea2bf5420f9c82586c037864a3d6841ef2720b9c76fb90910cae7d9ab00a032
f305676a48515382de20af32ed830eede2fa8e46905425ff2a7a1a2a9f186483
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ee303975a008c673f41141388ea1178e05509103d48a41584e3a8cb197949b32
937c5c819dd799f016cca6e096ef17058e2aaf807e986906ce36b12fc51ba5b
Fiat-Shamir context

d2bf4b5b525df270578d8b5aa49ce35319b7d08c29889084bb506cafedfbd2d6

withheld proof term: 203 903 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_ofCoeffs theorem, FBundlePointPowerSeries:53

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} k] \{m : \text{Nat}\}$
 $(P : \text{Nat} \rightarrow \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) k) (n : \text{Nat}),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} n (\text{FBundlePoint.ofCoeffs}.\{u_1\} P)) (P n)$

SHA-256 of the statement above

07426bdd609356f9349be045b4655b9170ea8342c29fe87687b244ad472d57f6

SHA-256 of the kernel type

48d195d49620602d33a796561851a77a41d6b18f408299bb0241e0517771eaca
commitment C

f8a142befa1fe24816958a5ad2d596e4bddc4f8618b6e0913b368b25221dd566
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b2ac483acdf363b02c573917a72f0dfc03f8f5f6b93216eca313350a6e5bfd05
100205fb73f8e00f10a01b4c23666253ae5077c478dca3576diff6faceb6824b
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first message t

ef3cea1445dfeab98e7c5cda4fa4dd5897ad7c1312f6202dc112a18770d29371
7dda9a4c8b09d7d474ccaf8042e4109168c3250d8dad22a68f14f118066729ef
12b54ed92f77cf0d16da2cb99b77b5346345eff9297f53f7e47290991200e48c
36a0be07c129a9cda3b7b3d9b5a69b8e2ede4960e41d55a09caeb716575c6fca
760dc53cc8fd48e6b5b9184604069b0cb0f6ea2e31134e00ff5ed82c502d1808
7465f52bab0381a41af84249063181a6f9e49c23cac21052b4143cd36b0cbf2f
33fd5d2cb05290e3a831045a70f63713cbd50ae57be7d3b75bd3569e7e23c664
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response z

5f1c44145a81f3fa78afa89f2cceb0da04ffa54f7ca5ba6b18137351b0a372ce
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2696b08d3af37e00aa907ecb2c0308692184487ffde72dbeab127995dc06c1be
6e0125aedd38d878522b23bb5e8bfc794c4c80c7e95ab3fa52180d3c4b18863f
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0c3b04f4dd94c1d247f7ed9af2427ce6300b497e40c5d5bad5311bca33f0c61c
1aa748b506d27987cd762b568c4d7be8f9bb3eef1dd8c6085e9417d4e6b44f88

response w

39bd0b2193b4c4cace76e406d1a966b45d62c36baeb76f4ce030b55a3a0274b9
f18de2ba1b34bd8f643c471b24bca8ed4f157ef9692c37dff129afeb83df9e11
90054801afa1dfe1a548b609d705d9d0e526f8c031f519d8f260f4d7d02a3f27
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Fiat-Shamir context

e560d99a3f11b9ea34afe0fd90beb6a8bf9d961dc36b1026bf8cbedbaabb100e

withheld proof term: 28 541 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_succ_X_smul theorem, FBundlePointPowerSeries:92

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat} (n : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1}
    (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} n 1)
      (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} Φ))
    (FBundlePoint.psCoeff.{u_1} n Φ)
```

SHA-256 of the statement above

084a8e23c5de919587744dd6dd76ab31496b6cdb080dc3166cabfe8df16c8813

SHA-256 of the kernel type

6060211416d16a86d2d27cdfa6c772f794e1413020bacce15d83704247cff517
commitment C

1a5ce70334b9e0841c12f95c5bad3f0be172592bb080526a4d9d7a73acbc1dd8
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bef24cc89627389c89cb9a7989fa7e316335c5125f2e289f863c06cdd577e52d
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first message t

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response z

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response w

6a197b34237c9893557c61c34bbe96437385946e8599f88ffeabdb9b926c0e2e
7233dbc7cc66fccfd9adeafb54ed21852d8fc6016adfab4f1daf6670485f050
77276ae38045ad46639a6c74f1be30428883211b00530d1bb6d43012c41ff540
945529f741ca52ed5e78e8873556d1d688d349eeffb5cee7865d975af4433642
aed628e9363e778f01fb3ec0a8ad386d1b4db5898c7ce4b4317695f00501524d
06046bab4bec7791d76f55fcdaa8e73e08b5894c2b35e2ea8f835a111811150b
8b9e8dc1379795e59207f5c635b05143055dd13429092a66f594dee384635cbb
6410ab3f694f2d43f75bfad0e11195ddd60dbfd63de76c04fa2a75969d5e1b2b

Fiat-Shamir context

31dffc40d96ff0c9cf91a8c0b79e400cd2ec05be9a403285c36f1509b8407320

withheld proof term: 98589 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_uEuler theorem, FBundlePointPowerSeries:97

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat} (n : Nat)
  (Φ : Matrix.0, 0, u_1) (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n (FBundlePoint.uEuler.{u_1} Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} (↑n) (FBundlePoint.psCoeff.{u_1} n Φ))
```

SHA-256 of the statement above

d018cd17ccc2182b8a09588ec166d9ea55829df1a7ad18f0729057242000e6bb

SHA-256 of the kernel type

33319ec018182547fe645d261c7f8d98387840b38a1d4ebd0ad468d691601038

commitment C

b1ea53c0288225abf52f6556210bfb21d412b77ffaa6f76c02a39a83950d64cf
21b20ad826013934155f64dea885ade0d0e4aeaaafa8c85e2873f607f1fb900b
7f040261fe4b6cb2c5cad45588a3754d27277baed25030b1852c678504f20d91
1cc266f14dc97c6be5dc91c6bc81b6dbc27d065c23cf49a0c36c0555b1d85b4c
3be511bdf6ac48579b07159cd852d2178c6192637021d09ffe453ba01a88d0d6
a10e6dd304310b163646d203660b3a5bea251d3ddf64c81c32ddcbe6fd8eecae
ff98e6613c722705696e190ff1b6982e28184d7bf293a78f45b4765bf33d41ed
f23ff27c8776784bee59495f1bb2e90472afea13f1f1be0f05b53d8afed38f0a
first message t

787ec826883d2371cc46074ad004b0d15cfaa76c38044b6fbc59f5a576177cf
3e5704ed655e892cfc9246917ac2896b8cca7da248f22d5485e33d7012baff6b
9e23663412cfc1d0258bce345227ab30d535264272e6edc7f44a30ee1db201f8
af92120042fb226e6e030fe424bb8c10a0b4cedaac6d7c69bda6c4db4501cb23
ed5823a492a9074325fdec93606821d1d85a60017c31301107ec85d7db1d9271
5e68bad3780f4e05792e5c12fb054d407f06e53ae6afa09a8d4a0d09ce36efa4
94e6f292324a15cbaa02502cd9287ca89f1080881d1d941b572455b93832c77a
f1cd080be29d8bc20cd6b7772b7d40195b5656e11d40b5744dbeac52b6e15cb4
response z

78b4d0aed8352e4028ff3429f61e08043fe41faba89140be3c577f791d2fb5c2
a388e90ba657b756981f8f6966e71f39de9a8daf3442bed0bb66e5334f4b6289
e2ff79ffc7a3f0f0410ac24207bab728bfdbc81e6bc894c53367505d4b697c3
88e5ef672b71cf12067cfac9f683de8bc7c12665bc78355d841937782a3b5c51
7503b1d0951e1ee691aa203fc312e2d255f7aacc4f85c56d5dbf3165fe548957
068e6d7bfc8b7e6e5079731c6937a91907b767923b689922da56cbb808aec227
c6510904b0616aaad6a63420ca70c2cb17bb249231fa49a1a149c07ee3744aab
d56a8f4dfba9f3d928708baa5b1924e862b6545cbd82bba76b15894cd7ef7286
response w

16bee15ab97fec2054cee12e9abf0fd816a9ddfa68d60960e0dae7079cf65cda
9bb8975f1f2fd61deeb8add239150bc9a362aa076f5efc4deef803029e7f87e6
ba5b414350b6c788db4f94d222ce6006a1b368efa243ab92e1a987fd80024802
0577d9ec993e8f577929a82942a89c95de9cbf72c017d235e5ae50d6ad2af75b
61ec17cbe4db497adc790ac3f94fb810df9776f2d4bc024d05a2d8287341b971
45c9729035e743d983064a59d7f82fa81a2dcbd245f808b7815694991f74aa3f
037b6e79122be7ac10056cc26d196a4bbd578ab4095ecf490ea1ad113e87c793
630304c8cc88cb5b962172d5d2167f9276a2904f6bd327eea8e902fbb2871508

Fiat-Shamir context

365d19256268f2071af2c9c56c1a0b012cd48b50e9f1ee611b770e1bd7236ae3

withheld proof term: 112 320 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_zero_X_smul theorem, FBundlePointPowerSeries:87

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat}
  (Φ : Matrix.(0, 0, u_1) (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} 0 (HSMul.hSMul.{u_1, u_1, u_1}
    PowerSeries.X.{u_1} Φ)) 0
```

SHA-256 of the statement above

fbfff130ebe1195ef4ccdefbaa275aa4f4451addbbd718bb7b9fc2082ed1aed0

SHA-256 of the kernel type

13ee7bfc9909618feeb1bb99dd619f0b64dc2a1ba70c594533b02f0b1294b3a4
commitment C

d2c6500c3b03c29df8e655c4cef53802c7b0a66c286fa897560e1f76d356f00
c160803d1d083bf8bbe3071ae3debeb099a955be30c7be9fdbcb08fdcc5cfdffb
096a1601a6c87d171d9a8404db2148d765e9976968edeb4916ca0eaeccacaf0c
6e2090d21625083f4852758426fdeb66deddc73abc2848abf148acddc11670c8
1cc10d5fe36a9e0f767c3a6ddbfeadfc960c3a96a352f779cfe6de53feb57c82
cbbba132e6f4f3ad540f1399da52e764e3e6c5546e210af1e464674dfc48da2b4
a3f7f4913099190f33991e223991c60b22647772d2645405d7d13d0780388cf4
697f057168eb1ced18a47e8ff9103ff8cc96aa958b114e95d4a5a86bd7482354
first message t

e1ad3c4af5fc33e5bfdddbf11e9e2ea517fd22a8b60ce3b315876df7317d79aa
12f9be05284f88f53474c57239ae629f604b820f2792b989619fd15faca1b81e
e9a271c74b7e970a4b36cb287be3cf900ad15ceb9b4cb7400baf3ff77fe3385
556df30b6171d034b2119c08007c79f837070aeeb07fa689589be1ae5d03fd3e
aa9b7dd7fe1088a186e783cc02a280c0bf774ffd1c929ffe74c8c5123f0554dd
7b5f9f11f51113a7a4ddb4d01a49a011448277b442abd5e3ee753c80504c17a8
8b5fd43d1e0b22c25c4c1d34c5c040fcd6adaa5a14abc0358ee50085895d9da0
7c3506a29f395e93855a97f3df705206a6df579641a7d782f92e47264fb97488
response z

7101f8b1083b8a4d651d3c679e017c360ba70ff07f83f4a45c9d60248d977246
d137337a2f040155cf6da0e208ad89e7c6e3f440197c5b540e78a25590f78c3d
287ea86d5ec193a4d9ae564fa961164bd56c2af5a05674bf98c8559713b0c4d3
854686f2c79fee7b4b6ebf2d717719f4f7eacc76278cb658d9089b11c7e0d109
1a01bf831c35fef7e4c5d8dcf9de828e4a8a1ad0c989c9afd5043eb090198388
614ffadf7535620987114fa94a8b6147abfebac83dddad6baf64f44de2886e15
d6f1b4d5453e783a4e6419ec36a88e1f19be3bd7d1211d162e087f68ed8a7ae6
2b429913703f38bab6577a02fff9c6329a1bb2e54d881729e9b5743253a7b142
response w

4f9a332b80d8a7e7ea6de9938d2270a0703f227d85280f2d7b20faa67c756e78
4c555c68796dd7ce90ff568c995cac827f6c6fe72a6bdc590181cc3db30b601
15ff98b013f85e675be250702162f36ec85c9d2201a5dd6f36add28b8dbc10e5
8ae070a012af3410f4d2464240b698b6de81579a472c870a0e12e079503d6f2d
99c71c43ba3d22cd1837dcf344dab46422ea976b68baaf88210739a4fd3c293
a17f5dd77da96fb9a2df749b785b77285e32ae4214339a0976ba2ded3fb2b962
cb10be10bac6fe6535a17f29bb03c42358cc0f1617d10268216dd1adbb4e0879
fff7564e5d77fd22c1c07872d1c6dcde01d117577e7a2d1391ad3dd89d83ba2a

Fiat-Shamir context

88f12329d53d24b5e034104893c7455faf9bf557d8586eb2b4ab63ab6d0650af

withheld proof term: 86 830 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uEuler definition, FBundlePointPowerSeries:43

```
{k : Type u_1} →  
  [CommRing.{u_1} k] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)
```

SHA-256 of the statement above

d850dd658d7f608ecc6607ad8c08d69ae1f830ce689bf57ee0199927a3aecfd3

SHA-256 of the kernel type

02f8513ba0e5b2ad1540c9719c7d80bc98eaebe1a10e2c93e6053aaaf69cdb91

commitment C

ccd87ce5f4a803cdcffc53210f8d7cc84132b518c0648d74ca8be14f98e27d12
a71f1ea81f67da079874dcb3302497fbbb4747f91a794787293db90cbb2dd86f
af7c15fe0b663417c0442412cf367ccf3d2706507df4002dddbd1a57b53651c3
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16d9f78fca2a102fd9cdf88c647aad784d9bdfcb686afbdba6ff883159add340
51c6bc353aa8c522fbd38317a2acb57b5f8c386ce3902da3c92d9d5c89580657
324d591b50416e92d460c1b11abf0c9eb40ddd382f4b009eba300eaa084cc43e
962c0f21be81765ad07559e2137724651442bab084b0fc722a75ae01155a0748

first message t

17d1ab8a21063738486944803ae0febaa5c26ec957b6283d45334fab2cc3824a
5b2675d64379cf58c725d16b7d164e7aab86fab5ebdcbcb2a1c0f2d0a4788884
51a95f29b5974f783393a648a83ff5fa9a88c701d0fba103ee4ce9c20f3e428c
1bd6086394ef24b4b598897b5b97e850018835061eaf6c3ae2e394aec5f7f89f
7e5102104ab74770adb5460382744c78aaaf9eb1605788e8243c2105cee14947
bf8bd6fd74a397fd27e0daced418bc26faa145e21c9c433933c254ee938834c9
df0f82edeb88ce8f1afeafbc9ed298f8fd25d464a7b796ef1fe51a22f0bf60a6
d25583d17e37666b67129b1f506b36b568c2dc42c6009e3b59085ecf715f51da

response z

5cedccf4d4a7ba19b3a2a9887a0698b3edf8edb605d0e82da816fa9710a845d6
1704111a2ac96fa63e4c2de25e3f6dbb16a3ed70b9f1a7a6d62c7fcae8353484
a922cf80ac7f1737877c415863e654e02ffe10800aa7d95ec26ff3d4f33bdf46
dc39455b7cd672e59dc6b7f7fb11c6b4431ccad24ea42ab669fc6b40019fe55
de3a5964c1ff5516deadf546470382e8fe2edc78eb528b5970e3a2c2d40d5948
b5d5995264024536674632db0d0d39c89be59ccd40df1330746fafbb53236ba6
c9ee67a855c571ba642b523d0b263c2a045d290df3f774b054658ae615259c2e
10c9e76218db09ca34141f11bb1b873c666831fa26817f8a85dd8e16c7f30e0

response w

45196dd8829289b5916cb931dc2c4719833929d7030cf3ac74f71c9451147e90
a6dd7ff7989d2388adb7c97d215ebc283d6964d2b94825c09b3d5194f361f795
b02f46111a0275da66e935421128c2bf2add61cac085988c7e9da58c20954c0d
412a9535a492166b00788ef9c65f1dca7dbfcc87ad77a10325bae08d7c2e29d8
1521236c1a227d1e230daa5a27babcd3af5120cd1b301d8c4037aa8b9c899e5d
467c2cc48961c07ba0b856f38525bc35fafae30f8759419e662f02aca14794eb
5e58378d259c81708f5ef02a84b56857f71fa1fa9bacc98b04642fdb133a2
cf0af0433e7987d1e653639f49ddc34eb812e32c5d5b56f8b80027774a9d48cc

Fiat-Shamir context

3dd44c8ea68bdf0fb9cdc7aa29359d0768ee169549aac204fc32109ec707e9c2

withheld proof term: 6518 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uEuler.eq_1 theorem, FBundlePointPowerSeries:0

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.uEuler.{u_1} Φ)
    (Matrix.of.{u_1, 0, 0} fun i j =>
      PowerSeries.mk.{u_1} fun k =>
        HMul.hMul.{u_1, u_1, u_1} (↑k) ((PowerSeries.coeff.{u_1} k) (Φ i j)))
```

SHA-256 of the statement above

efdbf8d2d219d066f6d16a677e92845f61762c7750935db1be66dd1e44a7c213

SHA-256 of the kernel type

8f5a926d7275d0ee6032a7909eb0a4c6d8a66dc33d1717a8eb4747a209210fd2
commitment C

25fd2b06d0af572d6307415006e06cea7d3da8ada1e3937cdf0ecc00b5a2b7a6
ee937c5d125c86e7e576ea19ab8e0faedf5b26b2c2d25f2114c814eaaeb1f9e9
48862dbe8ae10f1523c6f1cb10496d326bc2e0504c5b95226ba87ec0a57e90c1
868cacb907f7ece2e60ca351dd18272401f6bfab59a55bf27b7c03031fb9265e
0bbef6eb6c0bade9a28bb10c667a4bc6535ecb1bcd6f84cca5b279eb3c1dbcf
02abe76cf6fa60c04c3c3e88a857cb438fc7d767c67ffccf41c2587e9a887085
b836a8e4fc9eb2fee7e516d5af98a3a6cca7b3314390aa0d3b8d6e647d76ff86
e523c72f1d78603154bef2cb8bcfac5d8ecbb9fdbd087f55ae5b413c282ca3d0
first message t

6df2b6bcc7b87ee9cdee56bac3e2da513f36040706426fe6c73fdd09590f16f0
c60dfbb3da01615659a249b6615b6d17d0f8eb646d6fb68312f5380862051e30
309d01dcf1bb12212d7171436894c4deea8e1581137d03f197c75eca6107a92d
0af75227deda6b211bb570e39a19d295d54e05799621875dc26d5b76c10a33379
be91c611e87824539922e4a9f464515886a08cc5c0ba37086588209718e51c49
1641959af9a196fe7b9b261c85cd269c2db21e7a13a4bfdc935eb263b1c10c71
bcaf6ada2d341616519d056879d528e860d2c85b2f231ca6b1158a7a2aab97d8
78a103d5aa5a36071e8e8a2f53a16c96c1c22d43fe3ce067e1cbb9881d4a1207
response z

166b098f05263c1483561b8d98417d15dd4c6befc399520b1fa80e4c60ce28fe
bc8f77fedd5b152f3881b334cee61651d1e05957390ee031787629a94bdd137b
a33895b268f297b66bd5cf8e43294357cb485ca19e24d7e33d932ee26dc660aa
956d07c152d92169e1dacf51d20a42a3365f923a76154b09a340a74bd5a1ed26
6c50fc72133c8fbd6c886ae5eb80350821bd4a35f49d68a75354054ba966f14
25948ee61fc89a95fe673084661004afe327408778303a5d2531e48d93b653
e1aa41ef7782ec08b9605fa93c315c5b59d4b8a6d8ffa10b94267c4c6d184eb1
df8677bbc2466724692cc760118a1d77c2c70b92f8f510a57eb45aa4d5d88ffd
response w

2aa3487a0d5f230fec4ebc42a19491dfb4dd702e0f762247fbb72329aea717fa
1469c94e1b48dacce797c83cc040cea6af4cbb393560ef66e2e712055763dd9e
db97c48f4d8cc80cc51158291cbb7a8b732444ae630fa2ded707545ce197f92
dc9f159ded5a9311b2496dd208ef387303a2bc3e1fcf75725b64292508dcde9f
f1c93c6c56b5c10cd172e17761f6b3e4c8d75c0ecb827761ed3183c9ca3d162a
ed6d42c09fbc9224a5b256601fcf85d34b7335c5d7cf4d30dd02ab5929016e1b
be3257cdaf3ae72d3f70409c4990626e35e919c41173417c484673a4d9e0a609
a7c09d39f8086ca6681779613ce69bbd847c41a4bf6c4b007306f51a744b6444
Fiat-Shamir context

dd39a88da8d660c10b98e860da08f8b2bc2202d9088aacdddea3fd4674ca00a82

withheld proof term: 256 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

12 RequestProject.FlatnessGauge

FBundlePoint.cst_zero theorem, FlatnessGauge:293

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\}, \text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.cst}.\{u_1\} 0) 0$

SHA-256 of the statement above

41436d013ea5272f6a94628792e23fa1b2ad0247c10f8d29ed1f8b8cc710321c

SHA-256 of the kernel type

564a89b0c8cc64abb88f4a04ef153f1b587fb89a4a5f6874e29f29d6d106a5b4

commitment C

5029f6fc7afe1290d690f54d3699344ebab81fd312b7bb86078d758b53a3cead

f86abc6e819fdec4d0cb9a289ba00d228a114d9ef266065a0aad1ef164a877bb

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first message t

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e798c6eff1aa195517f3ee569abae30021e8e3fc01d22a0823e2f14abf8f5503

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8e1e3ba71917563192faced29ee9066a5c6a88d195f6a54d38d3642af5b04a4a

5e19df71350271a95d6c28a7f3ef2c344d2efa48498335c377be8fddf81b4b5d

e35fd30b6ffc36dd1e4847b509cdf17d1ad047657ef251af1d75a248883b06d9

response z

256736201b6772cc4e2c2010aa48719f81ae69e3310ef6e0bee5123fd3e8087a

5e4fc32c41382ecee89df8b53ce3fc8d32eb0fb6766436a6582fd051baebfab

a690bfdb32a64ab26f0b096545324cbf488a75fa06bb6d7abc6e3c990f9f10f0

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6391d83165dbfff45d6e936cb79dadcd423f062287675ef7a56ca3af992eece9c

a5a15c924bea84c9c6611ff6df6204ceb7e8bad4ff0fc6747779badb0062e101

b04cbc411835d01142510233ebe2ebee7fc45cec3b9e79a3e11650e9c30396b3

80cdc1aa8eb2b8020bbdaad4c02423be7515c5b3db98423a379884469a13673e

response w

60524852105bec1ea4870a857b222744e9cce584d7ae27396cfd33638d6283ef

85b195deb5342b7178e8a0c3685bd3829a45318bce90c59653b4a7fe878b57bd

9eb8fe59656ca98bea294b274a1e368bfa1162fb823d637e0fbad58bb3f668a5

277706c3e1eac0ddd43cf3708ff447e52d0721293f62eb4f484a74c179c53301

55af7b455e5a121850d5aa5a5a86257090712806f8b7a26b2650fbcf467671d4

f625e647b5134aba6621554633b6ec99806c704f28fb83f1aca6631b36debad2

9a3d7282b82a2ca82847410fda14d6e9658425b142d2c700b5f4517a89fb67d8

7c20b12e6c8b2ddddd32448c24022f3cfadbe1e9f35bb54fd6821b1e8ada70e50

Fiat-Shamir context

aaa82949e4b8ee7187f37156d40b6f0261fd29bd72d72b74b13be1c389132cd1

withheld proof term: 108 951 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.curvature_gauge_uDerivT theorem, FlatnessGauge:137

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
(d₂ :
  AddMonoidHom.{u_1, u_1}
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
(∀ (x y : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (d₂ (HMul.hMul.{u_1, u_1, u_1} x y))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₂ x) y)
      (HMul.hMul.{u_1, u_1, u_1} x (d₂ y)))) →
(∀ (x : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uDerivT.{u_1} (d₂ x)) (d₂ (FBundlePoint.uDerivT.{u_1}
    x))) →
∀ {P Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1}
  R))},
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
  ∀ (T A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1}
    (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1}
      (FBundlePoint.curvature.{u_1} FBundlePoint.uDerivT.{u_1} d₂
        (FBundlePoint.gauge.{u_1} FBundlePoint.uDerivT.{u_1} P Q T)
        (FBundlePoint.gauge.{u_1} d₂ P Q A))
      (HMul.hMul.{u_1, u_1, u_1}
        (HMul.hMul.{u_1, u_1, u_1} Q
          (FBundlePoint.curvature.{u_1} FBundlePoint.uDerivT.{u_1} d₂ T A))
        P)
```

SHA-256 of the statement above

23e91d92456cef2b7befd2d6ba3f643e3c25dd420574028bb691774168794f82

SHA-256 of the kernel type

a46f400b0b2a319bcee7aaa7f908695bce3b5cf9b6e0909d5df2bf957cf688ca
commitment C

c94ed92506f52cd80db09110c42ac2a54fa5389ff4a6dfc68415519d77b902e9
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36c348b5ab06a6638a6e5d9cc2032cf02172a98fe1de8145160bc8abab8bd56b
d19662553d7b9534e5415073a6db60036df3fc052ea502c7a8c5f24cfdde0e0c
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first message t

8f2a47f3a59b979bae8939504ccbd0cf6343dbf92cc0cca103929703ac698604
90ed5b8455d5ecc382c270630d6ec1b324424f3ff6519c17781c2e181a47c8f0
464b63cd5c29974a55539f272c61db14b0b553a9aaa41578bdb51e1213b19c47
ff5a9b07f6db5d6f08318e7160b5e5b614a99314a34e0344475529b331cc92c6
4797e0714ac957a0ca1b30b78716d9d2f19cd006ab6309474e363db8838d42ee
2dba34f3bcc79a6441b51448cde4624d4ea48b2c05fcb2dd7940095ebc81848b
374eb66e5304dd90ca52ddce793c73c242790239034481e97d923aa6f3a156ef
f0ff224ceb7406dec06dd3fee3831efd7b1cd5bfc0473ba88dcf6835bbebfc8
response z

1737da99db660187a870870c679056a100fb3a772908f7d2fd42f0388a9d9f0c
e377afc651fed69bb15d2553718d573e83d495a69be07d4eabe5011adfa2a46f
1fa48bc9d2c416ddf3ba2b2d7c746a3bc916b1f98c24a2aeac582bd45acc8907
bbf31ce46cec9cb368d555c91b483949bd82888266b1833d9c6761de0bac7afd
7071f7f9a0a0b5a4bbc6b177a9afea293b8bfafff4c164c9eec1e1394aa42268
ff46e44d97a61cdf25c6711aa23db6091de15d0065430cbcf1126bfbe4c0c8
b465374a0340db5c4e0e8cd6006250b64fcf651f74cee1db8c667648adbc5f64
1f18be806ecb244905c8c9eafb565349b126928c8b3bf6724909bd364d417f2f
response w
9feca575d7c09c50855541ca7d75d970b48c9a8031a2087ee3fd3b56495ec0b0
173a7b2b18a1f8ca4d22bf57c60912eebcde8b292c884cb55cfcc77bd3eb0f95
5d3a94373c75f1de5afb3bb785cc75df7b47e88fb46404caba5b4b2a122bd2a330
15ee7b3e363ad7ab1c0d94ff60e9a51f76810249cddf3265c5f1f7697b310819
dc31dc3abce5dc42e58352703e6fb48420e922c7f475489927c26790b8a2234b

FBundlePoint.curvature_gauge_uDerivT_eq_zero theorem, FlatnessGauge:153

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
(d₂ :
  AddMonoidHom.{u_1, u_1}
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
(∀ (x y : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (d₂ (HMul.hMul.{u_1, u_1, u_1} x y))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₂ x) y)
      (HMul.hMul.{u_1, u_1, u_1} x (d₂ y)))) →
(∀ (x : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uDerivT.{u_1} (d₂ x)) (d₂ (FBundlePoint.uDerivT.{u_1}
    x))) →
∀ {P Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1}
  R))},
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
  ∀ {T A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1}
    (PowerSeries.{u_1} R))},
    Eq.{u_1 + 1} (FBundlePoint.curvature.{u_1} FBundlePoint.uDerivT.{u_1} d₂ T A)
      0 →
      Eq.{u_1 + 1}
        (FBundlePoint.curvature.{u_1} FBundlePoint.uDerivT.{u_1} d₂
          (FBundlePoint.gauge.{u_1} FBundlePoint.uDerivT.{u_1} P Q T)
            (FBundlePoint.gauge.{u_1} d₂ P Q A))
        0
```

SHA-256 of the statement above

253dcd4fdf7f1c4f967e5ed70dd1038fe3922f443b7759dbbbb33de2b51da8e1

SHA-256 of the kernel type

35a9acce2cec6e14f5c148578a37731b84aa001b69797c7eb78be09c20cd50b4

commitment C

f141cc7f58363bd2fc6f0b186e5744f31c5da7d4cbbb00a2d53f03c32634a9f
e66f588f5b99d3cc826bbe9186948f36fcbf07e9fd67b6c2af6e15aa76a654ad
a40a9289986d857288f384529937e1b22b583ee95552ddebfd7c19219f3d94a6
0f7b9bd9a2c7ed58d4962e4e4640b2e98eb5b98afd18c9d4445a187ae7e4fdd3
f896ed2e8dcfa85348c300907419192064250b4cd2bd8a2031a1dfb92516c20f
a28f7cc8cdb592f37778f822e26a729973296569173a4574a762a9e71f7ccedf
4ac9f270aa2bf38081488dc64ebd0da248c85d72d1ceace61857ac74d1777d91
cdfcdf0239a60661fd3b8b940d947afa8d12b1e64dbeb4c81e76db95c1eaac

first message t

258b2effe4efe33df3664e822c9f1953dda03d36a652bd21e05a41b65f56c1de
af133f5ad01022892817cb8894c83b12f31285e704ec7ee4ff990f6fd1da7d60
50d6006a708d2ee1db3b5040bd655962aea6d231caab57124e9f452b9a5f78f69
a4f50d9f36cd27f6d6d812dd7dea411b7db1a9dff2e98cfb7ae19061fca1b538
b575eece714cc8a149cdd45bc5cd0589a806e5812abc65785c2c2d04fbcf0e2f
3edd07b5846ca3374c5421af9a23663d3fe1a78b98cbb0d0b4b1ce103e53c157
1574317b4c486801a29249bf984f6e6551feb5948b77997d0ba7e142bd7c5124
21d59c835920daefa535ca414bf780628a34b943a45e17b586809a883cae81ab

response z

7df06556a70157bc55b44b8e0710b30d369a9cd414975f67c22acabbd6e3327d
1bab623fe9930002302b78afbae0760abc637c15b9d7235fcbdd949344579b12
9c74cb71ad224ba8a9953255d9eb5180f66fa86ae82fb221a39e0bc1a50322cf
bc7ed88670faae5e032b3cc88886f5db2a8d2a97c2a3dd3e20c2fbde1e9a442
4cae23d033959ed383b574f0b66befaf226e97e27e39958007d922bd76580f0c
8c3cffff1fbba20dddec54a119fbf390a4e22ab0d755a6d8b4880183b96b6a6a2
f6e4c4ae0f1f059054ea31a21230cd4e0c43aaf3ebc37474f01e3a68068469cd
afdee5e21736ef352d5369192e4ad4f086bd50f377f70deed00df3af340eeefc

response w

40ee7bb74910ef2207ab23038632a73ff75b5ed31b15f43c8f691dc47c34b4e8
f655f9ae75e6299e05fb2c2a8c7f24da93df4bfc503949228482902a1dcd44c3
437f57b49896d57d987923f6cb0251442610ebef735430722bf5341729a4b752
1232458912515221c1e53abe3a642a1d544090caa0b34a97bacfd5dfa8a2d6f831
b74093df81babf8bbe34ecdcc55e781b7220f67fe412962674fdc3ff1571621e
c0e8b9c6516aea6916293eef873ff21e230152512374de9002f6ff2a5a5c0668

FBundlePoint.curvature_gauge_uEulerQ theorem, FlatnessGauge:166

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
(d₂ :
  AddMonoidHom.{u_1, u_1}
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
(∀ (x y : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (d₂ (HMul.hMul.{u_1, u_1, u_1} x y))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₂ x) y)
      (HMul.hMul.{u_1, u_1, u_1} x (d₂ y)))) →
(∀ (x : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uEulerQ.{u_1} (d₂ x)) (d₂ (FBundlePoint.uEulerQ.{u_1}
    x))) →
∀ {P Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1}
  R))},
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
  ∀ (T A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1}
    (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1}
      (FBundlePoint.curvature.{u_1} FBundlePoint.uEulerQ.{u_1} d₂
        (FBundlePoint.gauge.{u_1} FBundlePoint.uEulerQ.{u_1} P Q T)
        (FBundlePoint.gauge.{u_1} d₂ P Q A))
      (HMul.hMul.{u_1, u_1, u_1}
        (HMul.hMul.{u_1, u_1, u_1} Q
          (FBundlePoint.curvature.{u_1} FBundlePoint.uEulerQ.{u_1} d₂ T A))
        P)
    )
```

SHA-256 of the statement above

c63d3de17f2391e31cbb133384bcba707f577bd9e5e03ad5dd5dbd925bf3d223

SHA-256 of the kernel type

a78f6a7b5f1e4dfeca86f9aba858b04b60f3cc1ff9b599297888c2b43fe28a38

commitment C

5e84d4b7c11e92403c69782a80fa9e530fadd10df488669c882568d4e49219f7
be1959c7c8d676922195709a8a5cf26b4eb90651961ae19c2cc0d66c0461754a
97401283610689110ca02d7950771743b9e2f8ce7b20841d66fe4276b3d8f336
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f358fe89f3bc95f4e07d7d57a4525999cb0f979900668032fbc65846d59814ce
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first message t

e3f02e3bff31a57395b00dfbb7cae416cf81c5fbab04d99647eb5f0a82fdc310
4a0d04c268a40fe51b2f0aabe8ccb218368dfe0fc03e76f04286b2ad708a7575
c64f2738473431d41ead5bee3a66c7f263b5c8cb93cd8dc11787c06c591d19f2
8e90640b017f61e758dc83421d983a07939113a9e168dc5868f0b1b771b34722
b2497d582fbff0b3aad8a9cfd88b97410d62e4f82fdf93ae5f0e0e53bbac90c
cbb47fd5c86dffca5d6ada7d50370626e88df612d2be82e6016e32f7f62a5
cd35c1d8384c707d473ffa51b583e51fb7b4532c18738230cc7a50ec17f888c3
6b24b44dbfc518787e4d8e60736b8ca14088b4ae29da132e77f2b4a8c6f8e754

response z

784ec532f9f5f478239e89102510a721413adb4bf66618aed81952d2241fc321
8f229d269ae3bf0bd60db51499c5056071ddc9e9eb8b929ce2417f39365dc3e8
79afb5efef62eceb2453494ed2ff7807ffe1510625486ad7ee5a450205f2adec
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527655be4e3118c618d833b4b6f3bff4104594e45d9bab4fb4ff13b87b92ee71
258ca65ed557e485c4c4ef5ac93057dbfa006444d6cd3093b5429ce2d6f270c4
cd4f46d0b47c30f85c4f981506addb43de91b7e6793f6dbccc2ec5e17deddfa5

response w

714ee5014c742d9c2d47e3a8809233cffb3db9cdd702203f04bdc5a2644fb067
bbcf1ad9a9cb33ea4fb54115623f1269982da31b61b7fafbad0d4cd455650bab
3e108077eea90bc6be3dce9840236cb113984947fd15db03df6977767d331d5832
e0d2c755c18cc8ffbb57bb3a6e68bfc4012e2a67701748ec09baa1d5abe42551
948b68bca3686fcc541852906d5c8d31b2541793a75c37fbc00b2decffff1a53

FBundlePoint.curvature_gauge_uEulerQ_eq_zero theorem, FlatnessGauge:180

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
(d₂ :
  AddMonoidHom.{u_1, u_1}
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
(∀ (x y : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (d₂ (HMul.hMul.{u_1, u_1, u_1} x y))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₂ x) y)
      (HMul.hMul.{u_1, u_1, u_1} x (d₂ y)))) →
(∀ (x : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uEulerQ.{u_1} (d₂ x)) (d₂ (FBundlePoint.uEulerQ.{u_1}
    x))) →
∀ {P Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1}
  R))},
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
  ∀ {T A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1}
    (PowerSeries.{u_1} R))},
    Eq.{u_1 + 1} (FBundlePoint.curvature.{u_1} FBundlePoint.uEulerQ.{u_1} d₂ T A)
      0 →
      Eq.{u_1 + 1}
        (FBundlePoint.curvature.{u_1} FBundlePoint.uEulerQ.{u_1} d₂
          (FBundlePoint.gauge.{u_1} FBundlePoint.uEulerQ.{u_1} P Q T)
            (FBundlePoint.gauge.{u_1} d₂ P Q A))
        0
```

SHA-256 of the statement above

6447bc6d3f663902cb89b8d390253261711ea3b6db61cf59e6afb6c3121b4e

SHA-256 of the kernel type

a5ad98f4c44b4fab50a384cb0b164c23b1ec50185c5570735152a11d66b65c74

commitment C

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first message t

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497c59739c6f37d87a46c9f65885081de9656de2cf61f8028887b016593fca15

414874774c210a7a879e55062e461f6261fb19db75566d3dbb8c23262c5ff7c7

b9f10f591b6ba9a6f3e79a8ca0870308130198543be6f90bf50237837bd73d46

8cc641102ddb4347d9f9547bfc39ae8086be482b3153db51a9d3fbc3e848ec8

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response z

7854c5de61c1ccb3f07699b48932fc4967d938b550abef5aca0e16c888ec738b

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response w

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e6a47ab8fb0cfb35f6a23b7232194d34ab3ba496a64fb2513ae4bcba6efd6304

FBundlePoint.derivTu_mul theorem, FlatnessGauge:51

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.derivTu.{u_1} (HMul.hMul.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.derivTu.{u_1} Φ) Ψ)
      (HMul.hMul.{u_1, u_1, u_1} Φ (FBundlePoint.derivTu.{u_1} Ψ)))
```

SHA-256 of the statement above

5d037563f438fbd00ea85c9f18a6f070ae0b98a8855b9cde7ea8677585fd61bb

SHA-256 of the kernel type

1c2c89cf8bb05e9a8c9917b6c329ac786906d2a7089e238b9b83b3152f63907a

commitment C

6ae4f2a322e777aa9a86ea4454257d1f4fe11d845de7b848a31b327b242d8af0

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first message t

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response z

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response w

9b759dd693363269bf48d4ae09bb0921048ffffadc49eb063efbc636589ec532c

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Fiat-Shamir context

c4ebfb406205355f134d1b44ee89dbff33cce024d87a3c11632fea3247c0fc9d

withheld proof term: 729 434 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivTu_one theorem, FlatnessGauge:305

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\}, \text{Eq}\{u_1 + 1\}$
 $(\text{FBundlePoint.derivTu}\{u_1\} 1) 0$

SHA-256 of the statement above

3af10282da6514d602ec30ee7b20ce2d107e3ce01a68490c34a7cc555615fb32

SHA-256 of the kernel type

4ab74661e15ec17c3ee03ff43d061628b93d33d6c1b460cff968c08671c4d517

commitment C

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first message t

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response z

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response w

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Fiat-Shamir context

376014fdf3721ed44a1efe51a48b41b66e0ac80590b2de94c730171a10ad79b1

withheld proof term: 76 292 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivTu_zero theorem, FlatnessGauge:298

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\}, \text{Eq}\{u_1 + 1\}$
 $(\text{FBundlePoint.derivTu}\{u_1\} 0) 0$

SHA-256 of the statement above

dca218cdba541fb63f5e819ce353a5718b53814436c02ecdea9ce37b6564227f

SHA-256 of the kernel type

0656984b5267d3aeae1697fdd88b0ce9855383c198cae640a1354fe22f60a1d5

commitment C

d79a6e623d10a0f1b3e924f0a0a49ad9f795285d61f573f24c6a55bd652b8003

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first message t

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response z

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response w

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Fiat-Shamir context

a5711986d279121ecc2687cf885e253288bc260965c1566f996c8e926cde5535

withheld proof term: 32165 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_cst_gauge_of_curvature_eq_zero theorem, FlatnessGauge:219

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
(d₂ :
  AddMonoidHom.{u_1, u_1}
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
(∀ (x y : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (d₂ (HMul.hMul.{u_1, u_1, u_1} x y))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₂ x) y)
      (HMul.hMul.{u_1, u_1, u_1} x (d₂ y)))) →
(∀ (x : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uDerivT.{u_1} (d₂ x)) (d₂ (FBundlePoint.uDerivT.{u_1}
    x))) →
∀ {P Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1}
  R))},
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
  ∀ {T' A' : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1}
    (PowerSeries.{u_1} R))},
    {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)},
    Eq.{u_1 + 1} (FBundlePoint.gauge.{u_1} FBundlePoint.uDerivT.{u_1} P Q T')
      (FBundlePoint.cst.{u_1} T) →
    Eq.{u_1 + 1} (d₂ (FBundlePoint.cst.{u_1} T))
      (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1}
        (FBundlePoint.cst.{u_1} B)) →
    Eq.{u_1 + 1} (FBundlePoint.curvature.{u_1} FBundlePoint.uDerivT.{u_1} d₂
      T' A')
      0 →
    (FBundlePoint.IsFramedDirAt.{u_1} fun k =>
      FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1)
        (FBundlePoint.gauge.{u_1} d₂ P Q A')) →
    Eq.{u_1 + 1} (FBundlePoint.gauge.{u_1} d₂ P Q A')
      (FBundlePoint.cst.{u_1}
        (FBundlePoint.psCoeff.{u_1} 0 (FBundlePoint.gauge.{u_1} d₂ P Q
          A'))))
```

SHA-256 of the statement above

70ebce949ff8f423e701f1fd18722a5300e21445c6ff699537383c77016bab54

SHA-256 of the kernel type

672d1344fb8df44270a5aa921230ddc9817a30dec5068ebf9c43f7c3eeac144f

commitment C

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first message t

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response z

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FBundlePoint.eq_cst_gauge_of_curvature_eq_zero_q theorem, FlatnessGauge:261

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
(d₂ :
  AddMonoidHom.{u_1, u_1}
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
    (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
  (∀ (x y : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1} (d₂ (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₂ x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d₂ y)))) →
  (∀ (x : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1} (FBundlePoint.uEulerQ.{u_1} (d₂ x)) (d₂ (FBundlePoint.uEulerQ.{u_1}
      x))) →
  ∀ {P Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1}
    R))},
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
    ∀ {T' A' : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1}
      (PowerSeries.{u_1} R))},
      {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)},
      IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} (FBundlePoint.psCoeffR.{u_1} 0
        T)) →
      Eq.{u_1 + 1} (FBundlePoint.gauge.{u_1} FBundlePoint.uEulerQ.{u_1} P Q T')
        (FBundlePoint.cst.{u_1} T) →
      Eq.{u_1 + 1} (d₂ (FBundlePoint.cst.{u_1} T))
        (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1}
          (FBundlePoint.cst.{u_1} B)) →
      Eq.{u_1 + 1} (FBundlePoint.curvature.{u_1} FBundlePoint.uEulerQ.{u_1}
        d₂ T' A')
        0 →
      (FBundlePoint.IsFramedDirAt.{u_1} fun k =>
        FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1)
          (FBundlePoint.gauge.{u_1} d₂ P Q A')) →
      Eq.{u_1 + 1} (FBundlePoint.gauge.{u_1} d₂ P Q A')
        (FBundlePoint.cst.{u_1}
          (FBundlePoint.psCoeff.{u_1} 0 (FBundlePoint.gauge.{u_1} d₂ P Q
            A'))))
```

SHA-256 of the statement above

f504593f5d0f21133c04b6e8a20635649b6c881a1422bdcc7f7a8e7cf9247886

SHA-256 of the kernel type

59579079aad2c0c0f833377d3b3751ec79ede432e30f3a12ee41513a1970f841

commitment C

14f0088ac39bb3ff733575869dfc81a05d0aac87dc8c8364c7b2e2f18a08dfb5

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73cce832c86e6139149ae2cf1610953a013278f374c0b08db64a11a3978d4279

c88c777df47a144af4487c4f06ee11ae183dd37e761d92d391dd62f4d9b21e45

411d90e697355bca27c41ab140c380b69c35a70535ac5d0586e99354b71d4ced

first message t

37ca1b49e475895a5f917cb2c1b08ef1bc805bfa6a0e90b42d803726e03bba84

fdafde352b2b70224eed98f69e08db5deb5586d35e27e5a08b7c5ddfb5e5d78ea

6381687ae6458cdd1f658a362ed5bb79ced97338df5a70ff01193ef198457310

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fa1ecfb64ed1de8f399ebdc49499fa3343d6b5080b66038d78f4ae2b556f864a

691beca58ed22fb0b42579a4642b79dbdbb829ccfb5d281c53b99c07d1cdf6db

c0eae284e6a8afb4c3e80140b5102fea9614f64d3f025ea16adb1ed82523cfd3

2bb6ae6b536d08b3f59af91e4c6f1b09db62d056f24046e8e18ad01542cb4db9

response z

61a765aaa76aa8d633cb69b8898b2bc6c8ff5fa75ed909f46496423ff2ae4a43

09dca69be8f16f97199504b24c19981247a992275e16904f567c883dcde0aa8338

d8dee817514ef73b4882166ccde79b876a970cc94526a493f58316bd1a23dd07

7eaacec6a9ce41b26e118d59d082707e74583106e2bd1172e697371d95b3ea7d

FBundlePoint.eulerQu_mul theorem, FlatnessGauge:67

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.eulerQu.{u_1} (HMul.hMul.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.eulerQu.{u_1} Φ) Ψ)
      (HMul.hMul.{u_1, u_1, u_1} Φ (FBundlePoint.eulerQu.{u_1} Ψ)))
```

SHA-256 of the statement above

1aaeecb21b1e4e63484f619dd003ea7d4eefa1b6d72d9be7425524b48619ce54

SHA-256 of the kernel type

8d055be6744526a699a4c5aa04f0e90131a9dec7c12fe0b893e05f70dba4c843

commitment C

22df085a1c9651a2652c2a6026a5218502473178a643cf513981ad6dea8e0c06

86e2f82333a47e8bba03ca62561d160c198a40c6583e3b4d0a2454e86c0f12f7

4b73117be104b69b4fa136f93d42d92c0ff92c9753ac388014edd992b78c1f2e

1d7c9014663250f9878b907026b1adb078875e3d85a80289afe07043a6a8041e

f146b0882383b61c226e913e226b26fbc04836c3a955e80ab9e3e375a847546

c6c2e1382f0cbf24eabfd1ed4c7bda17b2e2d37c415bdd86705a01060c2e9f11

d0dbc10e347426942aa843040ff2008befadd23225ec7dcc755ec30c54074d93

9dfe9cc8597451668d198e607e737f1d5933081ccd21eede795cbc747bb574aa

first message t

cf9cf5fbbccaa8599a262cfe9d9f08fd8691144a2ed2bbbdd92ac7a318332730

2f945d81c2cd6b2cbdfad1e262b881bf7b36c8000be8bbc5f1a398a8ce4968f6

29f9d08bc08c821f03e3d9a7663316b99338b9bc5d3ef83cb99031711120e4b7

cdefdfd17f66125aabcf629c30bc38f881a4863faaa8b724a54d32dc6ea8759d

b8ad0eb579c31b2ac222cf079abdc726a5b8a6b66d832171d15c9c09e84e522

e1e86a8daa739831725bacee45b36ede8d703437715bb7b15d8ca943aa49ecec

2879f0ef422169896c2aa85a7e943585c0173ea4bb6dc1a36a253b808051f6b8

cc255c4b8818c6147ecbd0fb58894e92427c91149353a102edbb40e94b1280f2

response z

79a9c2a323c1eee720c69620f9f1d927bd648b03fcdc256b1114bc4d531134aa

a345aabf0c26801fc02484d7f0982bb095364cdc1f477dbcf490eae1ca64ccca

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afa0b45231317a4633425e33408c440479e74b59f4d65c999bc4b28ef9f42ae4

46765eb000705ba358e786409d467bade3957ac61d1b19c8258eb499469e1fe2

31fce9bdf30b69be3ad39e58d7330e7e8bd3e5978f7811d5055a20ac8d247f99

response w

2ec4ab4ef48d3b98a32c14cd5955f49f8adead70ca7240b7f0895c03e3475bbe

44eac3c61d80de94f0e47b8738ae7b3721765008863d25d4f01ca77b3b91f567

be9dfdb165ba8626a339cf264e7a966c569be0cb03e06c39858d14a1b3004d44

be348bd7c5d9da0afdd7444663960dc4803a5570038b73e7e945d63dd40999ea

77d71f17079acbb9595f693cc88466c1ba388c2c0bc809f33b0787087428bf08

6614d6d082e0c4249e15c6ef02715170f7fd960b8f2a961b2d8c630967c7bbe3

29fc6e170b33cca9551995be411ae2521ce94bcd406e0b6d933dedae559f01e0

563d0bc5bc7b371c006d50e5bd89376c32ddb41939cd12135a894eba32d21831

Fiat-Shamir context

a4dda005f74272f6663e65de351fa435a6fc31ec18d3f267335f3a2e05340eab

withheld proof term: 729 434 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFlatAgainstFramedQ_iff_curvature_eq_zero theorem, FlatnessGauge:244

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)}
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
  (d₂ :
    AddMonoidHom.{u_1, u_1}
      (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
      (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
  Eq.{u_1 + 1} (d₂ (FBundlePoint.cst.{u_1} T))
    (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (FBundlePoint.cst.{u_1} B)) →
  Iff (FBundlePoint.IsFlatAgainstFramedQ.{u_1} T B A)
    (Eq.{u_1 + 1}
      (FBundlePoint.curvature.{u_1} FBundlePoint.uEulerQ.{u_1} d₂ (FBundlePoint.cst.{u_1}
        T) A) 0)
```

SHA-256 of the statement above

57f796112935cefc60ff9721602a69c14b7ac92db60e8cbc27322d125f689199

SHA-256 of the kernel type

8fcc7da5e97ee294ab56c424342fea35fb178eee83bce91711f330a48f083da8
commitment C

654f4b53329b31b1bd760876000366a01daeb3feab221b7617a22ec9b4d7295a
6708ac8107e7ce3bf1a5c76e0c14cbea029110350a5cfbb2dcb08e47d74d5b05
a2d2e11c3a0b28cecc032665063ffb198176c6a8378b1ab28d42852eab9a7bf3
9449e348a18544a23819fba111eb233b70b22012e55fccfca16037bb23c517f5
13ca8cf2e827fefb12d54341d21d0d7675707917e9f9dc6c5b9afa9baeb4b55e
5370aeaf356f12033ccc569f24ff3accec6d4ee212b0ec729ee5d8abdbbcc9613
c05579e76bbbc0aac6f6695a2bb129f7cfc348ff93865589cc4484cf6ceb2d98
b41ab33dcc7a285d70561631222b6a434d8b20c7c91a6b32e60ca4db468c0561

first message t

fab56622426d6a74c0e9e53b3ed15dd5d9014ae013409f6f74fb03a490367c14
b3e3f533bb4a207a2d441fb44ff37bf1aaf51f19ef5084b49bd62900f705210f
f030a86905c8e13cf0e80ca58fa6060a7d12498af35ca14b483feed036ed0adb
a3da5efe67a862d7d487ed4c9711118698f813842fc853cabe2c73ef6bd9dc76
c2880904b0850fd404fa2f6576e7cdb924842dbb59ee0af40645915dfa5bd79b
b86c38419538664c8dc4fa4cf842afa53b858fd989e2bae2176ad06c03a5b091
b53131a986d8d7abafb1beea834490b86f45596d3e8f15818d5bfb57cbd52d6b
be4717d24c032e0cce5264acf49e732f7907d6babf36d78c2a34c87ab86703d

response z

5e9ed4c3f4cc599e046efc839a53954ca3b84cc633018e3fb31beb45811257ae
0943be17e5369aaf7131a380afcc17f3490fb74f54edb701115a3033ff357c96
4d7a98124b7232393241b04cb7a72f87c32fa742635f5146d1d0a12e8c723411
39eefea73c63e6622d26c7fc4efd04dcf363be0f1ac1e0d5ae987a7c58b27b6d
bfe0bd648e87983805ecd8804d210b1f5c426656ee180dd99f94702f48c12ed0
bd7c0f2fedd2ff98c8243efb0c4f0664b866c171154499995bd5556f2cf82cb7
07ab2243417fe80522642581b2b563d325487bb728ed89453abe0acfb0afc456
4858332bd96a29dacfa71360fd6748d8c28481a1d1dbc3aa8485c52e4e71856d

response w

78e7ff35db67bdc80be81ec76b77238d4cd173aaa8dd15d6dcf2a16601231840
fe96ac526649cc32ef51b81e3d15435670bf9970ccada9b3202e9a64b7361c15
b54d02b0c818f0e629c8befce6087b6f6d37a5565a82f563081148dcc53deff8
8b8838967620d1b36193e6f902210df77c9d8d4c350fd1e11b07de91d39670a2
0c6b0b63a5557a91e29160e2540d0797013710eb744f7f3ea6be8fe7c551ae3b
cd6a780ee2b10da754332e78fecbab6b4b0f4c44eb6772bf43edd7b4097b97c9
b3f17e06b4640e34ba0dbd46c835fd67502e30c93d334563d898d0eec65784d6
51378f62e947a095101c97510e9a067db9f704647a1e5c0ee563983cca88cb5a

Fiat-Shamir context

b21692cf1cd0ec973ec05455965a805fd13cb43dcff60693674e6f830ea15161

withheld proof term: 639 594 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFlatAgainstFramedT_iff_curvature_eq_zero theorem, FlatnessGauge:200

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)}
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
  (d₂ :
    AddMonoidHom.{u_1, u_1}
      (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
      (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
  Eq.{u_1 + 1} (d₂ (FBundlePoint.cst.{u_1} T))
    (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (FBundlePoint.cst.{u_1} B)) →
  Iff (FBundlePoint.IsFlatAgainstFramedT.{u_1} T B A)
    (Eq.{u_1 + 1}
      (FBundlePoint.curvature.{u_1} FBundlePoint.uDerivT.{u_1} d₂ (FBundlePoint.cst.{u_1}
        T) A) 0)
```

SHA-256 of the statement above

4a36fe673de496e026fbfbfbefafe051485fa9b4f8bc1e1e8abb4db101e0b6b0

SHA-256 of the kernel type

3ddb6947f5f9cf421b5d6c5363726e226ed59a107c103dae664209086614079
commitment C

c1a5fce599a788cca86b487c3fe840fda8d0855c8291cd789307739881b4961a
4cdfd9357bad2346fecdf3ffea9443eb183e6b8adbfeef87a957d4fa8fce8ede
eb1ffb6f0a26530ee3c280563cbb8248da84f7ff2ed208e38162d8130069e0e2
f97739d85bb93e912ec9e780a67679e26e379f86a7944ba9c9d46017490fe951
924cab712746b7891b15cf930a37f743f4b37b0aa693e52bc541be0082224ffc
5fcf6505d87fdbcf457f1c5d82312910379d0c7aad70a571b035a2ada9e24332
16ab71bd12abc2e0e42763cf8bfb4ba7ebdff4fa52e6cca7af53f40ad3055fa1
839c603134b572f6742a05eae6776f292614d070eedafa7061bac8716dc3cb37
first message t

4105e37d3564ded3edd5078551d93f86e2ab8535f89bb7e0c43c2057c31d6c6d
1c87f36dcaa4d0cd8b695919a926c133f0b02b83661eb560ca4cdf7ccea1a77b
6eb9b88cf1fd1248820b2d76314bcbf23b6a03acd35c5058da2c916c53ba9943
2a4db35aafad1db6c81fa962290b99154dce6c95bed55bf200fa3c4dfa48e2a3
c073685267ea1ae0b8b1091fe8f59cdb384b093d65052fe16cd55bf3747d38d7
3d7519d7c5358f2d80848b2525f5e0f640d6a2cad37ac87358140db67be6864d
0c4194ea9654c0e508ceb59c05ebc729a4d8d9a6c33e2c278ab2ce6dc1807fa3
d98b4e6413a2fd1b54d788b35b3f5affce3fc428f6c5c45c63091c3e57e38087
response z

8882a0fa2f78d2729bbbf2847f5e769c98805dff2033f29c0e2d0cb3d0549eb
b5b2353cb0ce3c73e429e4900c7b48f646d69061e619fa5157d9f1be01c660df
b12790fc22ae7daa1930e6b03240b6fea1b2253cba88e5763a8690c2c5e390b3
9b99d17bf64323202ec8e6b2dba20a90a9bd5cf022c94b8428b693104422f151
2b36436c9129e5aa30a2c2b40f0f8ffa44b7fd952696e5e841572b46a071ffa0
09347b2056fcf13a249a29a59d208569fc77ba2b6eb4211e09ed87b9442a35de
8577dda20e5f1a4ae9b0fbc0711a16a14dc0a70a950dc8450de7cebdfd21b0bd
1842e23d1e93eabb79874897f873e2544ae0be3141817438b0047ca01d57dc7
response w

1bb9324ca1c54cc666905b034b38c9dd22d9619749f8d7aa831179907fc17a91
e574883ce51f9c37929acc2bceccce535e56ad103a9df8a9e27c143b476e4ec18
6753fb8c30f4a3283781bd9283cef0a0722ac803c8f97fdc3ec9ed2248456a00
a419e50b63705f2b381878b454d2e62d82f6f855d9c684ca9254126020a1ddda
85dd457bbe0b8dbf06dccc56d890187cc82588d4628940e6a3333b6cd1e7878
2475dfb14f6b3b9794cb2a698738645cc259bb3f241ac9d33d064815f2ad0544
e566c2129006d06e1f969c719547580116826b582f21088b0c9ad640fbbca255
a2aeedde85881d92bd22e74cc724d9ea87bcabdf0e272d41e5c231f83488ff41

Fiat-Shamir context

1407d239311bfce3b534eec6574348530875401d074f05a9a333f880d0c0e0e5

withheld proof term: 639 594 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_succ_one theorem, FlatnessGauge:286

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\} (n : \text{Nat}),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} (\text{HAdd.hAdd}.\{0, 0, 0\} n 1) 1) 0$

SHA-256 of the statement above

c864c6231eb7b5853e4aab36d20538ff3a0c135c58aa6eb1882451817c891202

SHA-256 of the kernel type

ab15011f4df634aa98c07dea629eb9c876cd88bf16b02c03ff73e09e59c88922

commitment C

901ff684359effa8275ba0ac59afe608eadc0738109a0847d7712cef9e2bb08e

06d807f00c6ca139ae3a572a74a096b2c5fe5bd46cce37e150bdb2bc02c5f40f

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88da1cf32eacadd3b28323691a207d02f76d1510d40e8a81727330c9e97e419b

fb524a6250162c640bab3f02843e546be790049bfc09096c2eaf91f1413ed1d1

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first message t

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53de7f261ac6604a17265b6cf948aea09c4111807424ab80dbf4e4a1bbb0e10e

e617eff01b73d8bc9b262a41926b44170a7d5cd1cd2c6ec22d6400309f462543

b8a1de3885dbecde02d7e9fc5d1a69ea2eca0e7dcf699d201b5dc90717c9bd2d

537a7328e2a244dfc72e227e302a0e0c5bf41afc5696613c77c6b7efb2ca8f06

451246af78751514e33b39a30b4f3f5c73fb7e92482c475cb175da86f011c2c7

ad7a3e3a0168b7c3d9150a22ff11e63b93654375f102b022cf4ce0e62d32f83b

98563e71972171d6d9a63de3235298a8d32d707676fa54cad6f41c3d5d18002a

response z

7d3255ab1e9897a22c55c8fab4fedbce37278bb2371d7af277fd60e98ac5e11c

30c517a7affb77e334eb8821a9e98d8985fbf1933e42fa8e387e04efe5881c8e

0b3451a24cf211377eb3826eb6654795152a084d66861034a9bee6423b415a78

28204fa4cb27779c12028cca52be6340052fed71c70d549644935ee3a5d07052

d8ce1e8f5db507f6d19be12bd939ef6648dc4a163aca2e87bc3581e0ede41426

6453be5c61dabc29de23c4ab80fe47e1a12aa3ce0d77048517ac3b17144d4b99

dc43c22bfc36605059bc74073f3e5a9e9bc7f3d095ea013c5ef2c3ea23c29ba4

e03342367bfa63bbaffd882660eb6cde0733023428d1e98fb0baa5cdf748be74

response w

69269a4375eb32179fbf5db80b00c01ad2551b01b381e30b1e00714dea49f12c

d898776d4e72595152a8973a47635d2757b968210fe2549343750960c35f049f

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c8200f2251a2e098df337b161bc6db4032fbfaa3beb581dd0bfff93959a52cad

821b4b27a1900129b3f6eddcba41a6ad6b0f13c318ffe2f6e702386ca6160f958

53e578c62628b3a64b9a9a1a06c053e2a9b299057da7c01f96df6dacd406e4a9

50c1d819e98521f8b6b9eb61cf4068dd91f2ce18ae4ab8c5cd7a5f97bd3f3cd5

Fiat-Shamir context

9afcdf558164e5ede6499118cf2ca66091ffd3abfad59ccbe5ace3a77d61b8d6

withheld proof term: 113965 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uDerivT definition, FlatnessGauge:85

```
{R : Type u_1} →  
  [inst : CommRing.{u_1} R] →  
    {m : Nat} →  
      AddMonoidHom.{u_1, u_1}  
        (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))  
        (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
```

SHA-256 of the statement above

4ffb2966621e5f317b75a7080e5bbba364c52f145d25ddfaa6558e122aab653

SHA-256 of the kernel type

f846bea5318d9f3e5266b162a71793955c9e82c0568a3f05533ac5d285ff408f
commitment C

da7eed68d515cecb790852ec8ea8ce4d934912dbd215cabd64201c843127d239
2f790e9af3315f663c1fe72e7d9474e7940ff9964bed977c183b5c802f61016d
dbee89495473112259ce914cc5a2568e3c2c90f132668d69784f253dea9ca666
51096d27b9d3c469365d1646de467fd7b5af8217cb8c0b148950969d69529d0a
573a0f4744c5f0c94572e6ea2d1f596985f1c1c7624653f2b3f342437c8ad0dc
195b016b4dade2069a69001c7b11b476b438feba66bdf5786d16ca84e0a5dc5
f2b2e81e8df8e74fbff6d336d02bfc881b95e30ace5c49724928b412a1272311
27c49fccb3dfd22235ba40159c276b84abbc56c71e29ea3b82a2688fa7b92f9c
first message t

1e92f47aaefc177106f81705f43c574442e48fd96885702da6523c77d2befdfc
30d523581864ef3674e0cee4990f8ccd1c79b35d597d53fa2f3601d9d02776e9
e9da9e008b43d71eb4f949021c80076b687aed74f379ce7fa32bc999fcd8f69
be278995941a4ad06dc8bdbbc493aaa894154dedb7db52d9a1bb442f7cef8e3
7c259e68bac2c72ebac176d958ae1fae07ddb58203be55dd3ce2da2a730d4a38
5a9b6e1fbf21e07572f60ce898bf4f63e7cbb371ec210314cd332132593b17af
c6ca3251714a38507a6e55ae1b283afc2ab1f42bed9c234dd7ac8890f0ed58c1
783c03250f1fd5295a7084485d785944159d248a699c9ac4b59a46861a063d1b
response z

449d21f936d54312dc651b11450020d128403d6b158b87df61880797c60c8e52
91bc24b92cfc129da99a45ed1b2f069f0972ed0972351f0558110d5cccad0ff8
671caa8af05a67f910ed613a9549449c747e1e1cec6ad865cf521bb0d0893acb
dc9dee1372f6f048edc656b4178d1d78a6eff173bda3169887c3f529bdb7bf6b
478b43b74e58df451035bd60ebb8f594448ea5fc6cea71f15f0cb5c1e08b0d57
d8ffca5efb268f1006bf4644d0b239f50d7499b6762a47cb782590189cb40ce9
02e79fc6f39fe72270e5924442304d01ac547f5771ad1afd315f62eac656eae6
01086cf2122aacd731117b3514640c8641ffbbcbac87cb8ba7385918b5f6fe4b
response w

1166b47609867eed260ba9cbd61a134895c079b0a8cc3386755befa75fd147e0
aef3ba9363c3952fc07e9c417712ac11c2e72934d02485d1645a1ffaed9797c3
06ca20285a07b63291cef7df195cec844fdfe4deaec5e2aa0c0581196f466d27
0734e4cdd4aae10f4b61fef4f2ce3c17ab1e2623132dd4550e124d7f3399aaef
6120b3448c30308d68b8c0c7dee95cb6b321591e1803182709b6e647385033da
a5522fae2f6300b148003252826abfc2360affebe11f77bb288280e23b985fc0
fd9cba9af62dbd7be10d2ed5aa457f7f30b2c86aacceaf6c438f7470a6b0a12d
e4f1b8c835a20ad4bcf80d2acd223af43cb221880e95dbfb3be7d5c1d61d50e9
Fiat-Shamir context

3b3c763e26e1e8e805fb974c374a16429410d798ae0a471ba4080ad2fa9b9fb1

withheld proof term: 4935 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uDerivT_apply theorem, FlatnessGauge:115

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uDerivT.{u_1} Φ)
    (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (FBundlePoint.derivTu.{u_1} Φ))
```

SHA-256 of the statement above

ea789b5487f13dbba20e896567df0804b446026176b3f78f28221cd071138493

SHA-256 of the kernel type

0491d82a3f9f04fd660a88b0652d3618e17591708a231c4c5a7d39cded45a905

commitment C

2730bcc27965cbea8c3c0672994db06d6463ae7c44dda2ed766dacb3bb7af446
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37c57a597656e2c6bf79c8bad79d8504dd1005d1ce98741e1b6d1d633bda2cb9
c01bac3dec1dede7235b1c1d9ec53e3d48fe7e47bac8ab0689e79a00bc11b6bb
4f815f5f4c3208e4a0f63c6f783c05cc1503ad9f01a543abcf8e90c4776ddd81
09693bef763236f9493f114607b2f1fd2e3cae6be9a69de0d8daddde98d51213
first message t

628785b525bcc72ad4a8bd28baeb0f6f8bfaf02517c58f9111f4a7294c5cf50b
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ff5b893f8f16f81a73503b8618bbc0f6991c75f1d51a185e8989b48aea921a95
ed2a20f278960b2cc891c2e07e2ae8aaf74220ce9346c2839d28b858f924685b
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d2ae3daaa211b550e9fc439e4527e0e5f16c935005d08ddab6759cd5aaf925d1
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response z

42f2bc1e4e5fc51123c12f269b5a34c47e112b4c1122acbf2276e9cb1c02623b
1bb664f8f20c0dae255d3eebf409565f93cdef4d368b28decebbbeba545cc7b2d
e9ad7ded3d68f13683e63681889b309acdde0535b68485a6c29da9df614e5b10
d479b8b190d658ffead2cf75d48789da46a12ffc1ee7b32d4c3184df8bfb59a4
5000dafc1b4eee37bd55e5c6c0d1a774c067e18e8c52ce81ac257d2dbacf7c2a
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87372f9669b7fddb711fe5a1939b35013e673023aad5acfb1e8fe47d3a2b9a5
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response w

6a915d5664cbeeb33d96b4feb6d042ec0ac94d658eca711edee0bb0cacb62246
117d467e220b710025a494b20a648a8283b51979212d7ec29fddadab74c30a3f
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b99e2cdeee1cc1ccd575e69fa7dd54624e02fa31ceec9f08ba989d88c19ccc7c
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009df710bdeea7f203a6866643bc9fe68cf11ae7c8b40a4b226964e937fa58af
5361223f71c65fd0334df9688a0640fdc5e7d24fad0d1a8b368fd9be1c7c3c7a

Fiat-Shamir context

5fa60d687497aff654121df866473da9ebb98467758d1233d3eadfbd65f2eced

withheld proof term: 3 425 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uDerivT_leibniz theorem, FlatnessGauge:123

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uDerivT.{u_1} (HMul.hMul.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.uDerivT.{u_1} Φ) Ψ)
      (HMul.hMul.{u_1, u_1, u_1} Φ (FBundlePoint.uDerivT.{u_1} Ψ)))
```

SHA-256 of the statement above

ff4b2955b0a3188b5da1a05fb217da01b68079ec4e2006111c7b01d6afia1217

SHA-256 of the kernel type

01520728de7e90a2beda88e9bfc43ac02e72a2bfd130ba5aaf30c6a2a859994c
commitment C

212e7767a9a49486c3182f0d79f2dca6170dcdf792489938a8d7d4bbe4d6ea58
ee01f0aeaa8e56c38105a5859558bdce6f0290a3381f200314ee1ea31ebf0d89
f602a10621dd4073f5ea4c171c9d7b765bdf92542c79eda61d4737d35d4b93cc
94db1b238b3589a1deb59c647919d777b15df781c76df50b9ee8b1c4680957eb
4b2382501243396913f18c37dbd091ad9a1c0f72366dabf52f3dc3f2fc368b98
2e1b251b7410faed2cb3ef073b25f422f7dc5f9f25f75dcc0c43c0efdc4a26d8
8765e14dfe070f7d7f6ab1bf4ced06a464bbf966ced1b1d7efdf7c5ce771e282
489a4e507fc9cb0a7a17aecac5c5d37e5242838ff72792cb11567709bbba25c8
first message t

123c4478ca020e10490b3a317ea94fa46d94aa2b837cf1501f9bf8b6306a00d5
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7b063a2fd3cfe6b0505c785bc0030b969727c1d67dc2662d2323ccf6742b96a2
02eab11a8d3383bb5aaa19ae1f6b8f32f52a047959d82656844a74862882de23
3cb247e3bb53d77841bdfe7539217e0ae5ae8838dfa5cfeb4ee277bf272d97a91
d43bc8fa7524762ca4674593f3af465f9748b021912cc2ead1813aa0ed249a0c
de9db04ffcb59acafb65b4b613d0b6e1ad75cdaeeef29b308015c2057d1164164
84cd341acf436ae0d473acb7c05b6593e9072764c5475b7f882e3fb21e15c689
response z

534fb544c85beb0703254b82b742ed1b8ba1a15f83155c508627388dd8846bd9
15696c8254a747b9009c3bae9acd551ede6782d91279a25f870af1b21b4b6d43
d17b69e9f144c82cc59f09c5e2e6bc063dcc7c102b3cd00bb395cbda000a0d78
b0db874340e6693470b11d4ac3778cb00fe1ab64aea039b473426812b6e29bfa
a33ca17a061214970d1c1669e259817c73d0d9870e9d626633601b9bec17f03f
d40b1a2ab566f733fae18a39523b369fd320ba909a8b07091d6c3a36c3c351a8
6e7592f612669f4f7f21f92f5a8a06a030c1ff37f22f061c9fb9dddb6a8cf869
9d274cd2613e5c556adcc0eb61961c48f99558a67233b2f88602e7d4c52cb1c9
response w

6ebfc459709fef4384bd2646247651e979729120dc5555f04d09d9b0ceea43ec
7d1002e9bc0b6c5a733edcad52ecfcc0fe5c2418c60486fb44d39a11258b41e9
cecabdf019315dd42f67d065dd3fcd54c43912a47e9be9fc5310814cab1d085e
4d958462897858f716001c9a337e3ea32611fda84b13ce332f3a9d01c005622
2eadd95656287c0e4a6e28ff052389daf8c06712197bb585851c345f50b72590
53a9d469f2f8045aa957cb0860619a8f1f88a2288b2891485d30c7b2d5df1a10
65fff057dec36a3a82ad810f6f741abfffd646e28abeef4ff05ab9d7766eb741
11573c4f079db347afc090bfcee8cecc4eb602a7131563bfb0498b3d0ace562d

Fiat-Shamir context

938a0366abb7513fd2f1a3650854407ff368d937c461f90fca95cdb4bb703663

withheld proof term: 595 328 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uEulerQ definition, FlatnessGauge:100

```
{R : Type u_1} →  
  [inst : CommRing.{u_1} R] →  
    {m : Nat} →  
      AddMonoidHom.{u_1, u_1}  
        (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))  
        (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
```

SHA-256 of the statement above

4ffb2966621e5f317b75a7080e5bbba364c52f145d25ddfaa6558e122aab653

SHA-256 of the kernel type

853c1bbc37df17bd68a9a6e985dbd5f8bc128aba481dba3f237a144f517a273b
commitment C

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9361f9e04325ab21df65b4203ad56b73565bb42a6ac1f4c245f51e23656d1468
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first message t

86a539ed157f6a594aa91ab67ddf1955793b8f14df43d47a3eb7d28cd61025fd
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c51636e202d1bdd8f41d385756331ba55d08ced634709a56e8720552ef301a4c
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11313b8c33658280e0575ad024ea8c433452b8dbc593caa484a533fbadc54f28
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response z

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fcc6a0df3f69faad57b017315ea8890f1ecba15730bc242f8b8a5b7a8321c76c
5c5407d17f9c148dac6052b91b821d7cd950c39cf940c0f6bf6abfa83e4c4cc8
7aaa41c75ca483e17f9e4d856c039d71eb4ec6ffdd8c4581039d5f4694291e8
105fe12d6b0091ded385b03906f505d7e5848f3f5f2fc028d20526d6a0d8c469
a4b816743e608cdc48f66caaa4d5bb5dfc91309b3dfd4c2ebfcf1ef339a63bcb
d4362a32bcbaf1f8684e1a51ae2b07ebee849e18e984ed1716d9f08d4ebbad728
c102448080459c439b05036dec698b2494b85fd0b2fe797b47abda60433c086
response w

6c232342fbf8beacc8ede723ca9ac13e73aee90132e32e381f8a4614a8f9e88d
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a5aab887d738d5f2ddc9aed8811e3f332b3c22763fde9ad09384517c95168a5b
e0ef2a36737bd454fe511c4e691ddcdb2f56c0d561b84bd2c6ed193897504eb2
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Fiat-Shamir context

062b8cbcaff7e98db72ca9c8eb76eb3659c24cf516a94ee5641acb0205755abf

withheld proof term: 4935 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uEulerQ_apply theorem, FlatnessGauge:119

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uEulerQ.{u_1} Φ)
    (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (FBundlePoint.eulerQu.{u_1} Φ))
```

SHA-256 of the statement above

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SHA-256 of the kernel type

610f218011cc93c034f6087cc02a4c3bbeda7534673a368c8cf6f934b4363f3b

commitment C

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f13a4e9784a8c2c3ce8d5a2d5312313daf98bed55a1443d792f6c752f30ed9bc
first message t

32c1f56a8848a9186a111767e2e41eb1b90ddd17f5b29dc19ca42d4b5de90ff6
fe4d22a26fc66eedd52e92cf119c53cef0667163a2db4a7912883d1ccb45f2c3
d8eb3f472de486a2b1bab2f85c7792dc22894d5a26380b8616be7bea07e6993a
e9da49873c331d631c20dea185f386b86a16fd353c01c09229faee9ef2f01ef5
3326bdef6a7fa6ad36f02c55362ae3e4e8a31c804d2ade8943e5549d8d479ce0
fbc180548bc2cf9a9b246e14f3ce3dd511f07d57c0d51049dc28d7c4813178ce
05c2eb8d4f575900fa3bc2e36f6f152c17080d35bf9d1227322caf3a038b6a8b
d128d65b01aea346ad1e003e7bb8b24b9ad3de38856999f9c4ccfa91223f598c
response z

5283696b0f2e49de5be3d9ab9fca3a21cc4ee1d3bad0f68188542793a0b1ee2b
97b5072f8833d221685182593dd53071fbfe764f815c5ea327d154347240a258
a1e85f09ebbe2ff0f3fbbd56443b40c9e6de2e1e4730b210952e4cfc067cid53
a71ed7bb626774780fe275527c73b6bf81cbdfba69bf9508806bb024d7de95ad
68529b2c8a3610fa4c4f14a1c260e5ee4a788f1a7ad96240a4f2237bb3e772ec
55d80a7b04adaceb31e8108f039a5dbc1baaaa1e418d828cf0e5c48c492b13f4
e2a200f5c7398402cbb43f168d65fc095c4c7af015fe1c9bec381bf8358ef390
bb0e3cf4c0638685b3d9201e38a4cc163f0c2e1a079a9d69a9f74b71deee800f
response w

213626348b57b692478f02391a31bfeeda2e98e2c3054ec612de4de42ea0ac99
88b4a7773893446be159b0818efc42066d0ff3353c1944be32450596c6f84db6
84513984d03060751196495803051fb592fc175eea29d6a0bd15fec6b571ae15
f654e98faeeb02c20444f5941589e05cee6862eb2c7634ee5f776f8fd0bcfb62
e67492622d4ea552bbee308bbeaa6da2231c5f766737a61a1b7b4077eaaaf2b52
ad2690df50156fd96d3604d23a68cdfcfc744cd08dd509b215764fd5b0807bf0
3a8507662caacc9c444391d8eb1c66f1ed75f10b2c3621991cac8e7a6bb4df00
ec463040d8cf12635a7c30322499e8023197f02012b760c714ef232745d0509a

Fiat-Shamir context

47f69792ea34c9d6834868f26c167f5bd5fad67388361db1c492f418b5cf3afb

withheld proof term: 3 425 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uEulerQ_leibniz theorem, FlatnessGauge:129

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.uEulerQ.{u_1} (HMul.hMul.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.uEulerQ.{u_1} Φ) Ψ)
      (HMul.hMul.{u_1, u_1, u_1} Φ (FBundlePoint.uEulerQ.{u_1} Ψ)))
```

SHA-256 of the statement above

525d00e8adf59f6c6c0111547670f8d6276d69f9d948408afcb7f8b36a136d36

SHA-256 of the kernel type

9c6c0f9a8705c126608051b46582bd5b5540db3847bebd19b55eae8a512dbef3

commitment C

198d646a963580ca9e55162ce5db4d267f77f5c895ca9d7830dd38a9788eb028

c35075864eca9929fb2d2c3945d9bebd616d21dc86ac92384e01163a89a92152

171bb7116a481a33f4c8c456c9b92b69fd263b4cb6abd696ef277bfedbd9c7fa

d5684fa84cf75aa1bbb33cc8e30d471a07fb9803bbd0b612f59befc71d6ea800

1da916757580232e57db24d25ed1fc0813a71eeaae579c42c7fce61dcdeb9c74

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96c0572ecb28be4d37d8df242b4951a55d47ddf58c02d5a04ceba3a136c2a974

first message t

af4c56593aa72461744a9a77f6cc734680303b94c91405d5d4dbe3172fc736b

efbde10556d651de3aedeab252cba8625e46c3337904e99ddbf25bc3c8f9b168

3603600f34038e773cc1d1f20e60c285b0db7cde42ec462b452986beaea67747

8c163c9f52791f9c9c5276bada2d502ef71621054c4a93c04013981d36cbe774

a97522b776220da54fdeaeb8193310444a4bf43ada79043b0dc69966cef1518b

ae0a97529b525dc960bbb0486dc97686b1e2fef9cd2aaf4ed13b5f3c6f285c83

f49cf16b4494411873f666f347f96636b52b0fa694e02b384dad2bbb0a035b5a

96afa6855abce9a23e69dd993f7279b5398a9d83d829aec5913cb31c345e12d3

response z

4b61c9c983ac64ee0d75fecea17d0dd2a72a2bda7ae3563a1690dcd22250087b

ce53b97e2c08e298e10df2b1d8f0c4c38ff526c25c39193b66ced4516d3d1272

b432bd7fee10791e5a8bf5f559f2d11baa55ef5e183d52b8955be6526b559136

3248a6984197139bef31e024f308fa2b19f77598cf287262fd1a3ea2fe20d9ab

15b7121d80f3307ef198ba238bf9727e0e46e461334b8af1ac9ed5ac9e5a1f7

32d0ab40672abfaffc6ca40a5bb052700daa62fc49a31ec503cf438df82ebb19

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response w

2a5f7cc2af90d31aab049152d1df13ee4bc500819f9d8f4516cb920c08ac9f4e

8d5a6a82b27026705e9178d838ae6bbd4817ff4add8c044eba0b7b84ebf4fade

c039c46664cdd195057ea903ade7d8f6be512aac08c86215a538a53214665f56

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0ebea2d19431736f2e927220d3a4c1cd9a46a8144a5f4f229af8cab7a991cd64

38cf2627e5e0aaeb06815aff3840534e4497ee2028c55f905c08a0c2aa457f0e

8185d1f6b7e4212e647028e383eaa17442861b50fe2a8f83ec25cd4fd12b7379

d11846f828a4ab849517b11324500681b717f8e039c39802b6fd475c91c87f6c

Fiat-Shamir context

8a29df81de8d8f3a1171f7483ee4006b6782b6eb9667c059128ed7c541d0074a

withheld proof term: 595 328 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

13 RequestProject.FlatnessPropagation

FBundlePoint.IsFlatAgainstFramedQ definition, FlatnessPropagation:88

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →  
          Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →  
            Prop
```

SHA-256 of the statement above

39f0d13308a4da71a7df9f5df7f9565f5af6f0a916f451eb1df8d29ea59f2346

SHA-256 of the kernel type

e6d118d087e6fddedf0de1cdce441d7c06a94bc6df4fee70e54ddf1de15262962

commitment C

33fa70b17ed869c1588bb8803e0cbf7cfe5a0ca84904c683f634838777a614a1

da325569cf01b7e6d11a3b7f1c8efaeca5d06f86c774ecd4d2188e22ebf9d08c

9297d0cf1c8bbd0b27ae72441e544d138ce4b4f7646886edcfea5e9971e49354

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424101787af217f1409394ca5395294e8a7553acfa0f9bdd4c34938df786bfa3

48829be173aeafef2e390bbda71b92af84c26aa76db8131d11361ef40521217

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4a096487d89620b247f20f3c357246f96aab8f98f8e4e0a4c27b449eb6154a9d

first message t

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9b231bbeb67e1c06b1d0d9e85d0cbc864b76db99b993fc98239046f5ce84e146

e569c8b24bafcef895ae8c9da5ba739583def59d6cbc184b5efb131a90281a0b

be064f8330c2d32194d89fa0dcee65af2aeef7b5f2c92d6818b86562176ba6bd

104e809acc4e381cfef155018b6eed1ac75eb0efed63c835e28ea6eb3cafb3ae

2626f949c81ed2796ab07d411dcf67319cb09e5331695f28ef3fc10ebb573efd

49f603efcedeeef19a3d613516d42a26911d1e28a3df7839833bd1bc9a742ab21

bc857cb0730c81048d61d21b4cc225ad3b17443b1e5176d17ee950231bed39b3

response z

711574fd59b8d5b027ea6017d149fc5496470e3bf456e0995e652084208ef9da

dea7cbcfcdcd9492061c5ab952c542f04294ed77969c939ca46c34cbda525164c

704462c7e5f99d5a5eb40781d52e0cbc756b2eed4e8ad496b5528d4bb15d7e8a

9011618effcd2a95a31c132524123ed112ce3cf1449c02b0742cda296647ef53

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1b23db70e70b3c307ccc172b3bfc30b646e3543f1e92dd79eae9f19c7922f3fa

12f580e782ff925d419d86f5c36b83816bf32cabda418b261125f29808c35af1

response w

3d609391a4028af10ba98ee513abc0127a552f9924ae1bec4b19d931ff9349c1

6257d6ee5686da6e3eafb22a2a0daf6251ae715304a14fe93cf7d3263360dd27

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9b173ed0dfd1ad25e61f0f2b6dc5da3e11ba18ade19ea8972056440e82f44f64

59ecbf1314fc93cdf3f8455508a387bc7c76cf4847dde9fa8979a59d81c36bdf

Fiat-Shamir context

8ea78a96ac9cff1c81f7e045b28f9220670c8e1eab1a5217619999641089252a

withheld proof term: 8 264 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.IsFlatAgainstFramedT definition, FlatnessPropagation:80

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →
          Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →
            Prop
```

SHA-256 of the statement above

39f0d13308a4da71a7df9f5df7f9565f5af6f0a916f451eb1df8d29ea59f2346

SHA-256 of the kernel type

62299878efaed854e1d60a7d3ff84154970d14fb8960b71e00f46d581c92cd15

commitment C

57897d1dbead584495a3035cba9e9285fd9caf6773a2b4b5b792d6635e99df3d
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0b9d1fb921d516ebf6c75874890e657d02f36d72b522eeafca0b595e49cf073b
986b74365cf0ce651b59471bc819cddbdf44972f0b02ac783995a11c43f385976
317a6e19c3f1b13723b5cd46ebd80f003d7765020bf75f557415c51f1d5abd2f
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0f743514d4be91e7574457ec0b4b4ce431c2708b17a21dd9ebba35f66946e18a
948d10c14c6c35647869fd3fcf5ae9921a68e1f331687e8b57e8f73ebdc798b9
first message t

d634abfdd8ced7803c555894dc02e65794a02fc7ac9b79721a62bbe0e41bd66c
7028267868484ec587a5aabc76bde7c9ae2db985bdda0e303444c81a62d2b97f
0a1620eb3e455f5a86afc428889d6575ca350e4ef8dbd0b0ac92fc9c2a0791d1
7c6634acadb8e0bf0ab6ce4c7a192a036d3206cd35d0b3f062b4156e612fc73ff
d30ddda37c30e76f6acd8a294983087267b27a7ff3f9519f7fb1c3c3f85f3b01
7beabcc443f033d879ff879c39667b5b4aec9cc582771390fa62688d28e4f964
8a9872f2c0173520bd6bb50031fcd6f1524950e3c324981d106da82841f5b426
5ce0ee47380d19e049cd3705c7526b2645a1c67cde2ac4ad5c244ee4cb9c7775
response z

5c61b93f3a3a04d7a581245c2c8628e580b2e53a971ae343ed485738b42ad4dd
1009ff69c786288b757671c2402ac8faecf0e412427e0811f188f3a15db5a124
d66adae34bea95d6e6d0fd5bcd2d1ee4f945ea7ff77829e966b4e5bf090935b1
cdab1341bf918bf136bc615ab237fbc77a4be9ce9b2f940e0d908f1c391b53a7
3da33367b5c55b4fc94f603eb5ebc4f22d5fa3be5cf78d1368e8a721134c844f
51477cac0bc2a64766b0fa9ded5801bc6d19a696d290fa3b9270429261b3ee57
50fc38084c190560aa8676090dfd9882ffb94a87c8cb94aea788ce99628fb9e3
f9f3c46e13e681d0fb22fd63db46a27c63b025895c3ccc75d63271c908d255c6
response w

5671bb7ad8ff096d6f396e132756d2a8b087163bc7b590d09d5dcf68fb9e44a7
9dc3c14f116dc1e7d408b6f7f4b27efe268d4862b5b56a8574286f8b349a0e50
fb190e924dbb6eb903ae5dfc097d2be05c296ad2c90a45292194f7fcfcde0bd2
ee0fa0c699f943391fd73260b6fdc9d9164a7a256b4b9bb8ea6132f51729fcf3
8814d9c6386811c0f42598029b8279dadaa4a35d8ffcece3f4e6ceae92eb985d
315f06b480af5d2fb61052c25b294dfd015c5bb74c7aa192d6a57c533c1ede52
557c54aa27e4ab305e6a357e9d50a58948885d6a16049ab68112e0cd6225bc8a
3c1a0acd1e3a6de8a7929b2bb528eef61c6f1e46ce3c66b7666fb33dec35d491

Fiat-Shamir context

2aae18047f65c1b1f4a788fddd79fadb424b96ba3aafeb24835c2d231b5a6b96

withheld proof term: 8 264 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivR_psCoeff_of_isFlatAgainstFramedT theorem, FlatnessPropagation:98

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)}
  {A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))},
  FBundlePoint.IsFlatAgainstFramedT.{u_1} T B A →
  ∀ (k : Nat),
    Eq.{u_1 + 1}
      (FBundlePoint.derivR.{u_1} (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1) A))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1}
          (HMul.hMul.{u_1, u_1, u_1} T (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k
            2) A))
          (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k
            2) A) T)))
```

SHA-256 of the statement above

6681ebc66c51b3cc9fece1a14aa1626284a8c4add04959f595f16e41cc479bad

SHA-256 of the kernel type

aa3158fe824e0c1547033ce8c5b419863fda24386abdc243caf66161886ed8ce
commitment C

e4a16f5342b26a5c8c3373636bd23491812d414e55d8bc7a790ef1425e0946b8
bbe399b3b902dad0b096025d81dcf2e938cf42ac87227e57653963d30436756
b4bd8b1263d5114163279907d1d69b590065a2afc914396d24bc5a5f0774659e
2906ae45ee9a3cae76b34f9206fe4dd79afb81e9350e20f525516d21403d866b
debcf785dc9b33107a859684c6360774993fc37729f0d9c4fc5fd7f879f63845
a60a014c0e06a643931a1e7ae268bf00041c4cbe5eb60e383547ad8d5696624b
9f16a355b0fe689e5f23d266d456f122f323ef695743d2e7ee9b4e3a8acc150b
d72c31776792e5d62c104ba559e32ba88dacb3f35e7eafdf592d85f50ef2c47
first message t

94775d0c9d8cacf5a089d15343560904bdca82719d31578940e31c8753d7274f
267d411b927f699515c1b17b9e6c4119157e442cdee207bfd99fb3dad67d7eda
95961b3ba23826a3b20e0c2e4906fade96a2ab4043d00bb504005c5e96fcc3fe
53c65d66abfd29a15a5915664b11cd31ba2ed57fe0a4968803701eb49131cce7
6903fe0c49ad6e17852c32b6d15494413e2a621651e185680f035f4ea9ee857b
7c482d33e681aae0a7a24856749f5b4e91bc51bd7fc7ea214bc4216be3d2c2d6
5c74f0f8b909014d21e9db102f3ea42941d9e634c51d457d85b53330f16082f5
1def065b591298a0b0c288779e28e720ed2ca0797048f8bc2470fb54db2ea1a3
response z

6c803b828270a8a392b50673cb1b9643b0f42571d4d785908afb9536715a25e3
c8502910f270aab9c207aa7ed0bdcd6fc97ab54075a6ad6d0268bea7afde754
e6ba50d565ad403bd340f8597090dcd061a329c12c0c4f43690fa723f9abcb61
72e383649b7a1c3e3d2713cca076f6be8bf797376f81724702e6d8176bc7654a
e9de64d0998c571e40019b366b1816b3a5d61b2aa69aa0240e972d9a07f5523d
93a1766ea915b7dac6a7093f0e57923abcd603c6dd1de98bb1fe10ad958b8e91
5ab37f268d090a86fcd68661b5217c9b5fffa8215567c2036dec0652bcea7e40
bad9b9aca86d383b5f03f94e1a1914f1b59144d41ef688d39c68a9c7991a438b
response w

d1a5d73b7e0aa84d8139424c89655a8883fd64281456b756435570e675a3341b
924b853aa5d06f2516c12f538f2ccb5822045ce9e97515307ddd7917808ea2
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1fdea0e5fbfe4e862e2ed90cebb6a69a22f8873f8c85068a791c7b6fa33ec3de
858c0345b37d6aa238bd1b28f8e7a8f3049df62cdb1203ecf4e96f90c7f6c

Fiat-Shamir context

f6a15bec264a9f9aa68b61bfae9c11dac8423ecc5d8085163fe65baa11c5bde8

withheld proof term: 245 616 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivTu definition, FlatnessPropagation:53

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))
```

SHA-256 of the statement above

63b22e1e9b85ccc9c6721aa4785df23ef7ce10638842e2c0db8a0146368c52b1

SHA-256 of the kernel type

fa8ca14bf55d16c199c48489c32fd9e79115369def8072526f8fd183f7ecdd01
commitment C

54a7cc2adba654047e1f3c142bc48d1c2d77bfea2f98de49b94139293499366e
0abcbda6f9ab6ed2ceadbfbf75e9ff00978231cd1fbd51a29533ea6fb881f97ab
e1461458cb47a05f226b88692134a61b0bb545dbb04b7695495ae60d4793106c
fa4d7713650327347c61c551b8833226fa86a736359d0ad6b13debb1c8ce566f
784a2547489b9af68c24490ec15da5967805b0917590cbdb0a697978a1e11255
36f9c95fe70b3c14fdd27c804565be3f1f2c867fad6abc2fd0a00f4a014f439e
9fd6adb73e53493ade0d3b7582c0255007c73e95713cd7cf24290bf333269ab9
a03895ec5b1ab351ecd73d6f60ccdf3be96d02f712b28bfb13791327767fd0b4
first message t

493818424b34dab2d1bfca82f316d849e7b4edc27c5fd5e58af6f1bf429b6bb8
22b7ea880e56ab25bbe2b257575b43b007b1859e2ff3007cfe37c0703855ed88
c952c3cd82b9f88666646d94688a15750da88230efd3532ea3f31f96520b4cce
c332eda5c4402d1bc6583cdb3b0197475bf9ddeda22b96b2244500eea2216d73
2b88e8f6a6cda993673cbf711b51cd475e5c4f2dedf5c9e314c5444c10dcf9f1
42057ddf37247d9aef26f3af1b4d310b59e2c625cf6fa94ed728bd5d649487d
2dda06bcba9826164f8e0d4e9776e8ac528c5d8db2b0a3514a25b52566e12f0
c95df19660e5a77f440ef91ea2a07b4aa2851ee4bcd7b3bb87c81a13412acdbb
response z

170af9fca157289b9470a9b78d09c0871d31469e3219fe5af25942cab2d4aba6
32f6f4ab77b1c5ec6e115e46f178366bd52e0e2f6ef93fc8fee6b55f2c03e056
4e2a6b0c3be0dac37793b4484c9c9047963ba95347b64b700932aea9b8cb1d4f
16fb3714b3756fcc4ab79d26b84ef82778e84688fbf6bb95c927d36b5a425548
f965b4bcce08b41bcb2da8ee827d74215cc568fb4ffa845400145fa524d1d8c
f7110dea060464b8f774cbf016556301c602753fb80bbd148e25b424e819f0af
1bea2528e79a289c7c4a78dde7387280ece5a8278154c8820cdda75c36600558
ff2ef8fdd99ae3db6e50d921565295f5c59d739c5073f2fa767b36d2d9b998d4
response w

66c99d29a95b35164f592cf54b2594f90cea1d847c11ec900014ad6aa85f5ab2
0aafea3479d760fa045d90b968fdf622ae351f8fd8e7736dd7147875fa769d2f
23f0faa5cdb57f383ce089062aa001d7f5c413d816561d5b0359dd32fff7cfb3
c347f69654dfbf5116382e87b83fc3e0fcfedce4a694e4ac0fda1b449a43d106
b38147893b616f08e019d84d97c2ef8cfe25f98deafac6fe2ea17562034b7319
16de9ea65747729e1bd10506296d263008906b0c0fa7f32b1ea938923bbc787d
205cbd6c83f1b92a6d9a349db9b7be106aa6c07c80eba2ca7ee5b6d87490ccda
9e01e4c427daada3593390db0ded1547bff0bdbef13c8a349555d67c540dcfb6

Fiat-Shamir context

2ba94c55cd206e4da69d42878d1f10695e3e7b194671a8b70f6627e62883cd1f

withheld proof term: 537 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivTu.eq_1 theorem, FlatnessPropagation:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.derivTu.{u_1} A)
    (FBundlePoint.ofCoeffs.{u_1} fun k =>
      FBundlePoint.derivR.{u_1} (FBundlePoint.psCoeff.{u_1} k A))
```

SHA-256 of the statement above

fffe56f407748e9bcdeb335121c809f2749afdb0ea3509781702e3b4c8df80d6

SHA-256 of the kernel type

f3fedefe6fc6d234d197c0751a6b27347014f0dcd32ca6762f7685c3c5e85cee

commitment C

9bc4cccf89f2af1f4cb9fb88b301225a1bd213d64822c8e4375c170e64d7ff68
9e4d6322d5b12eac12955d487d655de95eab2a0141a35e9b78d50a4964a6d56a
0fcec05e5bcb0cc67104b5382f2decee60ad9c6020ebc734ad72681d75f80f1a
b76148e3755cb0440e9720bc63e0620e70e70a8972a7417e1bd97f7c1d8e8174
8590e41a958a2a174dd8eb7cef0bce76b78bb58a941c5ffa17bd3e75f258a26b
9d9dcce57e6bddcb3eda8edbe6160801e9b677e7fbade376fc632b901c2f18c7
2138fa4d7259a22daf6fc0b4320313fa4608e9a319e26e5fca7f56153f6b0c49
eea80814f097c253632447f386f46546e09a50d6371b86d3047034bc4a7787f2

first message t

ad4dd56504228483cda614e09325018b57f4966bf2ee305b764eb78675607032
228535af12d4e4f92177f8f4410b743b53a6f36c7950fe5a6e859bcd723c249f
c1975753e86dbfc17bba4030ffdbc74efbca9bcde4bfd41cdd588f272723a36b
4d1861c9b9230596dca98f224d1471ad969dc5700dcafb1a8aeaedc90d46eef
a5f6d34939ccc50f897607d22305b167903ebad3aba0f4f0fae5d916b3c222b8
663cc37a0a13dbede969926faa8b2f4ff7b960f6dc96e82860e21dd82fdc3bf
d9f7c26214518718f7e449e5ab850cee362912d062c2e05503287ca854f5e278
f1c7facc09d2df437aa4e7a57bedcadcbacf1486d940758056ce59ef8db40056

response z

15e300b07609b86faecaa7a2df6721794baff1afc9566fe7d71273b29d19a31b
280409b847b4f6ef5edce1bbe51dbff621d74f4140ebfb834d0be63e2d1cc0df
0065f421f1915a1a3d0fccb451f385f7137deb040e1ce02d53d56b9a9feac587
f8715a5124ea539221bb38caa68e5632eb89c0ef5f00f4ecfdb26ae55314b40
3ddd3e18137ef300b8e3e313baea8fc535c15d2a15c00c2f527a318204fc2e67
32f4dd23ae1680edbc4f3c4208b80ab895ac97db9e4173d658aa92a2fe283cd8
5b621582bb68c466c6f0af910dbe04fbc4e02f722007fcf43befac481466fd42
5182a8c43f8da09666f2d74aade84daa79f97b286a100961d18663d98bf23dc7

response w

569e4096af9cc6e46cbbe176baa893cb2790b7400a04e95c7e299d4e6c0bcdcb
1a9b2d23d378ed0f49c77d9bb5740cdf67883d492daac419060542613b6a4628
7ae1d4946bac855e56c3b29069f4ca27cd91be08fad3e8da77cd17ee2e1eded6
951ef28f7dcd722298c815e843580c63c3e53bb9b5847127705d0895612174b7
d22b534df778999ca45b465e27e49198b06d6291330196ef32d87847c7f914f3
ecb5baf14a6d2b52bcd4f0e25b2bbc1ae8d3302ec03b2894266e91458cbd02dd
b4c497afb22e8c69bc82c4ed8897459f1b1ac84def7a29dd082d71501e0f5cb7
1ef969d5afaae31135af6e8f1d305d275b9c1eae5c8a8486b4d770e60dacc9b7

Fiat-Shamir context

1e4415554218b5238ecec960d476246c1c5cbbf48cebfe18b5bf200e5cd661bc

withheld proof term: 280 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_cst_of_isFlatAgainstFramedQ theorem, FlatnessPropagation:168

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] [Algebra.{0, u_1} Rat R] {m : Nat}
  {Q B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)}
  {A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))},
  IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} (FBundlePoint.psCoeffR.{u_1} 0 Q)) →
  FBundlePoint.IsFlatAgainstFramedQ.{u_1} Q B A →
  (FBundlePoint.IsFramedDirAt.{u_1} fun k =>
    FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1) A) →
  Eq.{u_1 + 1} A (FBundlePoint.cst.{u_1} (FBundlePoint.psCoeff.{u_1} 0 A))
```

SHA-256 of the statement above

4c39441a6fffb6e0ed5c02442d0d84be6e20f1b7b21cf7e128471126d59fe1328

SHA-256 of the kernel type

7c01e7a9f569f77315a3a9fd66bb52f190dbcdfa819f5d76c6fee66e3362a998

commitment C

f69f3e1e5f6243d81ed360784d0203bd8d2ba9178a0b83fae4fae24034231af2
a612c311d66eeba8e0bfffef3a946712ba1d322e399ef4e4b46db2a1d7416ac7e
12aae8cbb76d6023cdc76a9444e7206f90329020d7b607e063cf5b0b447999e8
d93fd7191e749930446ba29fac8eb3a89dbada03a295a078f209c2fb1b159fcf
ad557b27d09830a1d99b040a756d14a5a35a29a37bc8da0650b9561d5b47c299
2145a7e831e5c0d6cdeb8a93f9c0b9f8cdef37bfe90297a5c99a626e6f3246e5
1e51f54cb086045d3785f4715118da7ba26e3578719fb7dd9bb846af79abcbde
794496d30cc475de736c94ce1460f4d5e995bbc08d5c46ce8cd52032ae1cdf15

first message t

8eec685c9c62a33b414d5d782fefbdd10db5c09ea67cbbefef376af0aa4adb70
f8cd62c0c460f6a0d0cb827dd3a8138573f36706ca100ce7544bd8912045d0bd
2a29c5abe844366d96abefca4051f7ce0902a5749126c3055bcdcca664a75b04
ade581e8ca5e1a70d1cea5332cd94180d8e739230b73e42c97d144ad77c2e402
a4e58649f018cecc1eabc7bb05636ca90aad719da6f8fa9e3406f33384542dc2
25e879cc36d3c9389e3efd6c1f820643e4ae97ae107b565e6fac8b9fccaf0fd2
d058fa850c767ea032c80c95c1f9c651f2726866a1f9c6c5ca4b87818f486455
628edff90853d5de37705475932011c651576149288e13c2d30b48e48f8964d1

response z

12a968f517254ea9612f85650b7461f0e34926eec9ede214314ec8a3ff149a69
5a84359632cb610a9f9ebc8cd9dc32421981534e40765d1f548b5a5a840d5fb21
6801b684ff4b85e6e0cea89a1f8c5f9d3cdce2c32f13c3ed13829620c6f85462
9bf842c03c9ff0e1a89fd686689860bba6a95d82e67206a39bb57bf02b799ebc
5a4276321d6909e32d6e2bfa9e51517a65881396b32400705281ad7c63e7ebd7
d8fda6a167c09b2b21fcd03001dc1217f072cb7bf5fcf235a24968ef9fe9291
a17cd45b52b42adcf1a7feacdb89c4af441846a6932f6d9d4002684df3cb95bc
02938e2dfc429de531161838f9a1f095ce814167d927d49e00f7a0b3fdf3326f

response w

50bfd508cf0569f9b37ef1c798b853c17de5ca1ccd6423204cd9934261437430
b5143c223ac2717a1b9aecee0fdadde87e975eb9a0f31d5c5401c710470982bc
8e8d5f3f41cf1f7605d9a16b13e9924c86bf2d86f6a060e56cae178419943d0c
44cc055e76be833e09f289e99915111d30182ce300faffe87c6de2f77e3cffc1
23c32af4f695493794ec5bd9c065b20b2e0572d16cfc49f440969759529a00b7
78e45e63f97a94692ebab8b9812d8bd4cded54f3e1b150ed48c9bf43bfff7d90a
8f6c752f1c309bf1721e6890e2c7946e828896eced4d70afd1f4b3c71e6219d1
6da380cabfd18835b92403eddbd4198e39bc05cd8e9567f36b41d7bda7917f6

Fiat-Shamir context

327f6fc7ec7e1f2a435c44ac259666e941054fa7729f32b864334f6d0a0da35

withheld proof term: 8343 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_cst_of_isFlatAgainstFramedT theorem, FlatnessPropagation:158

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] [Algebra.{0, u_1} Rat R] {m : Nat}
  {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)}
  {A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))},
  FBundlePoint.IsFlatAgainstFramedT.{u_1} T B A →
    (FBundlePoint.IsFramedDirAt.{u_1} fun k =>
      FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1) A) →
      Eq.{u_1 + 1} A (FBundlePoint.cst.{u_1} (FBundlePoint.psCoeff.{u_1} 0 A))
```

SHA-256 of the statement above

e6668cce5ada1a2613287d2b1692345cd7c359fd7dcbec222b1d7c17f541bd3f

SHA-256 of the kernel type

663d541918e68b93ec53df5cec84cb8960ae1ff49ba538b823a80488612ceb6f

commitment C

53136ddcea90bd45e762e9d0c7fc56ea66aab2106790062da9d7356d547b55e3
9add11d85c0337f2fcf87351b1dedffcd77ce47bf4e97a09cdbc2594691cd49
417d3cd82d32071b69b8333fb784cbf05521a25e92bfc8f0932478fc9f9f24df2
73ef4572cc90160b43bb730cff730864aadab4d57ea3b33fec1ab612b4cfcaee
f652bd53fb9b5961f95a9d04b9734f483c82d507d9c3da7e1c48c1eca941b59d
a0a659a451ea87dbb80a0b19e4abc196f85a1ed7c739bfc61d8a76bf7cbbec9e
ea7cea015479afd46eefa81b6243f42a7ae076a73bc8a22934f3b75ea84f231d
60b78d7ec7eda8336dc3ac47cf0f6d70f45cfbdd38e85989afbd493878e30318
first message t

22b6a295bfe99ae335bdaed44c1b243315d6ce40cc585b26abb6387a5afb1ccf
50db4658679c8fd65c3d4c65bbd5366c0d16d9b2fa27e862d0e18104984e2985
daf4c50f4ae720f63918f415166108a77f1bcd746aca84a4e21c55d9feb30c07
cf1c3572c7d1dcc059509ed5c2876d5549e5eef9b95718d6409090fcf3f075df
2bf26c5eec955cb108f600dde64a8433b649bc10f98e2baa6f6c6fe4f03fa6b4
02e3a0a6e848c379da6fbba06f2c0da42014a8cb8fc20aeb00e163631cb80d72
f38f40e7ed9e295b2ed23b7b495c637a8afabeffcb6de5f746ecb8fe774e17cb
1128dabe2844b44742f3bb34d3d04abdd048bc0faf91772fea12b305c640fad0
response z

1e3b3a2c2692498b2ddd02bc023e7a5f38bb0a4911d2cc212ffc4c051f8853dd
d907d57484b22e784750eae03a6772541633c204b0c3ac1f726a40c02c61159
fc26b476db5b35f3f5dce7409141d93ba08311a1aad33c569e12984e6a4afe0
23ec432c93ff65e4c062fe87fe407152b16b67ddfe583c79ee5039f0159c5ff5
af22f156bbe37b1b8971f7d34d9473a5cb8d524151d9bdd2b88711f37ed76011
dab0b6c109bc7af0d5c8a55f6c04b94dba9ffcaf1b144c4dbee9232de0b58000
c3c50d6ad4792a9247e4226a016f9d811a52978c090bb08cac0d5392a87f01de
6ca0a4f9ac04778097b6c1a8e91852d6bf80ad3862029add082f5dc011e43046
response w

55e631f23efdd34c7eba6a9d864ee95255717199a3760327a2f76dba9ca06894
d4be4b3bb7d2587e1a27848877dd4d1973b0ecc3ddaf08e34306db79f30342ea
38a4985231641001bf384ef528caf8bcf6a5af625c4afae0a215010f0acba793
911c7a806dd6cdab2aa881cb6efdd9c51794ab3481b304ffdb84647c70aaaadb
d836d80d67152ea9b443ec3ec47c8af023f2cbc72c3bc680b957c20400418d08
c1d217830c6eb8bb678dc3c0a4333b826d09d223357249c06ec3a2a4123b3892
00407a9e949cf73f566f021c36d29280fe22bb4b16ce6d0bc0537727e037ddd
195498923b1c9aad4d419e39055499a6f704a18ae7b34bc535fd54828038c1b9

Fiat-Shamir context

90932a86649e1724d2af23b3c5ec444dfa0b8cddb22ee9249aa92551bdb6ba36

withheld proof term: 1 429 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

`FBundlePoint.eq_cst_of_isFramedDir` theorem, FlatnessPropagation:147

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  {A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))},
  (FBundlePoint.IsFramedDir.{u_1} fun k => FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0}
    k 1) A) →
    Eq.{u_1 + 1} A (FBundlePoint.cst.{u_1} (FBundlePoint.psCoeff.{u_1} 0 A))
```

SHA-256 of the statement above

0aabc60148ebae2cfb92ad5ffc4db33be0361a591d6f14b7d0ba944050f4369

SHA-256 of the kernel type

00c4dfe8614a5f92ff0cd509ebfd9cd5e1efad14f110e484284a9371ef3ae7d4

commitment C

cac00dc9bebb6ede12ac9602d2d20e70b258e91519f71e43ecc3c1529d2ceb10
7fd3cec8ac8f65d6b6f5b88f831f1915846dbedb128cd05e18e63d2137359618
e62bf6e3192d261ec483fd2dd38b169fe9430742134497b6e8390b8cf454fa1b
57bcedfb03de60bc135cd7b02cbb901f8b1d881b6733ff999acd9b7ed2fe4ecc
c71a7fb97512fdbef048f57335d64c129f83f28bf2dc05d8719c2802a05a267f
ce2d518452d38d4a8104af0e04be00f1f64e150fa83233e55160bd1956cdf94
f95c1f163fe0d1055bda47bbe8337c6004520561951dac4f3eec2f9c44001b
638fadb2ac7efde6fc35213dcbeab694b8f2f8ffe280f35d13c9a18864516db
first message t

bb454ae87081de169195e8995b89715b9f15fdaadeb02c035f03c4ff5a5ec857
c3a17f04b6c04c74d9e730833b3c6f47a9aab44ed216b37efef88b6c2908bdae
5e76f0bda7b8caad0922ead51c17b41dded7355fe7236ce939a7109d5bd50f8a
ca720c8f7e57df2911bb4e9275b8f88e2c7b14b4e234c05c42b627a212803782
903c52f5232b2f56bf1d0e75201c02cd1e28bf66c8dd8f340967dad5ff45d53f
b3300f3b0bc58d78733d00b4eeb686decce31a0e09f2745b58273d0bdf78863
343377dc3a45d5f0d39075879257a04520eb766a9253415f0ae99ea1711026ba
c345c6103aae438732d7171a4af1525435e264c1111c7c69bd030335b13fb733
response z

7809675d8c8fb9bf15a02bdd11303286eab4b7523e651645e6dd6bed4acda3e9
897dd953aa0edb1a329e2087ca585b75f25c4094389b513fe55a60a5ad44d6be
723bfa29252bfbfe289ded6e4eb1bab5a560812bba4b6d9689b4f26d35eb33256
eb42b025ee13930903081aefb8266e2ebd0800135b25ccd530aff335970fda67
43b3a5b7bd210e57beb1d5c7388b15f0267ccd32c75102d8471dab7ab441617f
84e1f15e37cd304403952994ee008ec3d887c6db798121217d511a3eab1f8f8b
5b0fa30727ca95f76b961cf476362706108435b74a6af82825340cdc46806f262
fc800b8ad67d9b256998974e089a66c6f9ec65911fd298c6451eae64b2bf832c
response w

7ca67528d337578f80f4ff26e6f6dae1a362dadf71083b42d2ee7dbdb94b7d2f
bc70a01999e15d3441b48a124a7da068713588647a3cbdd0b0c8f07d48318e55
bad5c8111fc68572ef2e32f5d05fe6fd601036fbd973e0ce56cadd077d2951c6
0f1064dcae573233c9efdf4c6babaebef9d4a661c1d3c5d8f0460f2042c0d260
a8cbc825af35c3608d95909fdf3c17a754bc102e80262f3b65f5774e249c94dc
4ff9613d29de2ab1b9c156edc25ae5b93b83854941abf29da1d95f7c0a10ef8f
5a8f1a5a7dabe251e3f5045a32b306b71e5e7c8c97fdcf855db5fd11a87371e
d0245dc752216bf592a8bb7bcfaad21d804ef1cd2a56f4de536a71479546803a

Fiat-Shamir context

3751e629758f90fac2294997080c08d8c2d5d3c845b3aefd5ab8c632324cc362

withheld proof term: 15 307 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerQu definition, FlatnessPropagation:65

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))
```

SHA-256 of the statement above

63b22e1e9b85ccc9c6721aa4785df23ef7ce10638842e2c0db8a0146368c52b1

SHA-256 of the kernel type

c8fef278c513636e13f217208a0794306c32e332cc2c2ab1f22afd9eb7cf4600

commitment C

131aeeecbdba9d2b56729c2af69cfeabfc7b89e7c23bae7d484936ef2b17a9ece
20680d6cd4a2d76e67b3e748d23e29c7d4ad63a1bd90e52c0311e5ccb6e909c0
69379477e4e329f9d18d518d2fc092c328dcad221af607601766663f0f43621c
18f89468e120869f2dd152260835665da3690cb9d7cf09e2281c6f025f3246df
ce4fc815a1160608939b2d0af38744f17de3da02420aa2f3b1c58f4ddee41564
8077c4c4853ac97125c5c7011238601efc5b206fc910fbd88622ad5e6c717a0a
9dcf5a670880edf71d6809211b8b8a4007db8879b8afb0081bc2fa20b4a2e1ba
bc6aee47f476754f66986750ddeb9414d6bb336ba4f59ac75c000c79a04316dd
first message t

850a40de62400fe1c13fe3d72f67e03a6ba49b5280c5039cb9967c5e83972966
aebc70167bae70ed6ee1db2dcfe6a291146ff2995d95440f60650adbee565a3d
bde35633c496b2fd2555799a538de8f03dc35a1b4a545365598c0baa1071e01b
52c770263586dd87e1cb1c60e4de83942b8c9787a78455d9cbbd45d27e228881
aeae1047bb2045a58b5f27b44e1676c8047ee196d2632efdc858031782aa6998
36d37d634a5979a3a8defb3032607e37b81a0f894679020b378934b7be215db4
6b098c215589538c0204fa006e35cb2d01adecf70eb649e4c381be1dec3fdec5
0c92142331bb79ce11e8b39558862b5c9d3d9c41fc53a1edec00d0df8cb45c

response z

45068ebbb33e48ae000b82019da6474f0946c441fdf833eff800746a58ab92e3
1796e4a709d46f909a41b20c51e291c55fc2ed0ebbe91b91b24cc7a9abbbbc04
eb89188578ff3df07917ab99f16b2cb7c4d54cd2ebdb3399527f0706910217aa
68dd0181ad0ea51251c18714e6e64ec1fa2ebc00555e40144565b33c9f9a06ba
f41d985fd253424db2c182eb84206e1c5238a4b0cc474e84c35d76fd6fc4d7d7
1882e8c410ef9055f751f25aefcdac01644a20585a5f8013b5397973d01fd770
314d8da7a312a5b303ee0024d2823b3b030dbaf9859dd8b756f275a51372c953
76a3400c63940abc2be240d510f0e3926e3c1c9af8acc43fa79a2030bbe40c

response w

59f598924b32fb21efb0ea00df8f274e1d7a507b3e89f67ec52af859584defa2
53c5a5c2e86d537a1ebf1da90c466578509481f9db752dd6a7fc312ead9fc8fb
15828f31d8b77e36693fbae1186a241da8d2a9ebc9c037f5a43b42e9c1e153d7
16de8317e1668c1a08f38e9a377d0d467054bcd663bb082cd32eb797218cc45c
64adf6840760a40ecee64e11837dfe9414a43dc36c2f13a2dd55df2da42b9969
98449579d02f032c19780e840236058135a326487ff898fd1915c25670c90802
7811d5f1d0dff93ff9adb4a03c20376f4a056f702c6cbf556fb511ba0464dcc7
a6b783dc06edb28fd2dcb54cd71996b6b690d3ff208a4e63b43323645770b3de

Fiat-Shamir context

04644993a1af3fa9065e0360e6d85ec73e2139a89566bad028c92e4556a8ad78

withheld proof term: 537 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerQu.eq_1 theorem, FlatnessPropagation:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.eulerQu.{u_1} A)
    (FBundlePoint.ofCoeffs.{u_1} fun k =>
      FBundlePoint.eulerR.{u_1} (FBundlePoint.psCoeff.{u_1} k A))
```

SHA-256 of the statement above

55750186b3ce575cbdf14ffa4cbfe8c9230992b82ca367ca17e8a280dceffd2b

SHA-256 of the kernel type

3a3bfd195412909f61449cfe0149b6a87e0e118d94a8ff1f460bd20f4f8be331

commitment C

59a9e4f6e5036c8b26be3e6ccc65a525cd7681e56409e757506f01015d8292ae

2882aacff1ff0523a7f7910426ceac79120fe50403fe3bba428a2a5bfd553e58c

2f1634c66985ff5b24620f66bc3f258abe1a2618cc5952ab6af68438a76f20b1

9658e635659a017991dda477649ac25267c0d41db2ef891448506647c4640f3c

3aacbf6cdbc06cf14e7a266281bd4570563e5ea099f6b3ae3697fef860001d

2126fb5cdc361fd6ea8175e911ba1b3eaa1985564114011196a0a15e227ebfe0

73bbfa0426f2f4afe49247ffe1c72ae662b292eed0ded984131579fb71f65575

9dcef087ce19419bdd855140ef57691198f0552a4c74664a8d4f418838785d46

first message t

1588e17c3049b23d4acabb1c17593eeb9b6b9c4df12531dec893f71221cdad2

8fcb3c9bf8b0a4ce395371d13a036a4006ffcf86c14e1872ba49d466dcf7a79e

dfc592ed6e5712a43faafc0b1946234436c1ecf88c15170c6cc8810a43163275

b22697b5568eb9216244ea14f1340025f9c4c8f2d96f44bd3682309ed251478

27a2dbb77f874f6856820d1bb1a8d4c02ca56ca1fdf97080364c03ee54e183f8

0a0bedbd5d903efa4e3285c8ba57b1bbd0924b6c50b2e4a1143015b9fbd62462

3cb215bdcd48d877a65a3ea094c24adb7f5e3a8d0038f4bc1de4a9aca86c5b07

20ef3b5eb96bc8a17ee6a1852ceace217e1d169c2164e8b52c68a9048a34d168

response z

163321eed51f5684eb68ba4870856b213b959e02e8a5a126d93edec9523a479a

719931c4df5e7b16d7caf33236b9fa5fe271b616faeb72ecb6a52bf7cabfd884

86a8764a585ebb6950efbbc19ae4f3458baa1c8c2ade23585db6b534d880c416

5b6c27df742e1cbf11bbe8e30ecbc3f2f01f3fb6d8ca28ac8ecbdcd3a181cdf1

397cd44405b308f6b4dfbfd1858a62c72e6c9b3fa2bcc44e287add1436b11c87

a3af99fef7c8b5ef98e61cec37e14ca7076dc618c64e4041f936527534c0d394

740e7c2f89bdb4fb5369886d9e247e3c21ce1e8b5508ecf71bfc81d87546cb71

48164088719af6f5c8cbf9f24ca06a45c15e58f13b1f46a43cedb7abbffe99b3

response w

e35a22750782944434b4b2fdaa3e0f1aa2df49ecf848c37bd680ac4e0aaf0c31

d756ef982b5ae49c80244c1c98dd7085b6e92e11ff31226ba035ba53dc5f5405

869f558ba2d720602f41c424dfbe61c3d32ac25874a599c42a4c59086ed5326d

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5d31cff1a97eb046f15177bcb4e36802b7ce975b29454b7a02478687c7abda59

f909ea4a963fa0ebd4090c8f8f1791e02652da006055f79d70ff0edd0e96e170

659358f71ecc97e3d4fe721141d1a6522100a9006d5c35054e3e4c332c847d50

95ba584577b3d2d520edd9c39bd3dbc7e58028d849dd9c8b596ca89c697e3ba

Fiat-Shamir context

5ca1ddc4bbc0468515c53487c969d247e74a10f113a7b1557872e918c71a16db

withheld proof term: 280 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerR_psCoeff_of_isFlatAgainstFramedQ theorem, FlatnessPropagation:112

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  {Q B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)}
  {A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))},
  FBundlePoint.IsFlatAgainstFramedQ.{u_1} Q B A →
  ∀ (k : Nat),
  Eq.{u_1 + 1}
    (FBundlePoint.eulerR.{u_1} (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1) A))
    (Neg.neg.{u_1}
      (HSub.hSub.{u_1, u_1, u_1}
        (HMul.hMul.{u_1, u_1, u_1} Q (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k
          2) A))
        (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k
          2) A) Q)))
```

SHA-256 of the statement above

6e51ee741f148e8e32a194295c851e62ec011694860a63c6c446f82937d5902d

SHA-256 of the kernel type

7183aa2dab27858bace3e969a9d8a25aa567639ccc2e35e31f8402df5ac24c49
commitment C

c385177d9ba0d2eb3f5fb935c8b6917818eac370aa88dfbc504e88b2e8947a18
03853c6c1473a7ba187cc32a6f9a8d773d2cfcfbc4693b80f58714311c09e905
0be87e563adf2517394d3e3ddc05d00b5fd038d42098129c1b7293360cdcba5f
cb7604dbd00321a2d1b81a741497b91806e2b433dc518888810404ef69ccb8ca
a76a324708b15a0be88cd76658a9828fb4f8c800df2febf5a3f362dbfcd8ff0
e384fc79696d5d657f47168fba7f52d3106ca9772cd43cb78446659c8d1258a0
e1b6d58b99f3dcca4fbb5c1c0870eff04a3016f9395ae8c7dcac95aa57ff1e8e
81f1adfea3a3275a658a4fb9f93f8d61d17ba2899023a95b028a2644e744bcba
first message t

afa70c948f3e4a167edde039cda3a018d15911f621c22a27f36382fe8d590292
cc99725e0153e12640cce31e51eeabfae30a8cb0337dbc2c4fe000c8d62424ba
bb0e19b2fd65a41615362ab4404855fe980c00a278c27a5b0346b00229057dca
4a1ce5b2faa6a1d5c78065f1947bea2626371f465592ab02fb0990524b934011
75ca8d239854f48963180eed08ab48ead90980f367e0767eccdf90a7b7a428f
095f78f06f704d255c94eafe854b0552934544c9ef8493644eca170fa08cbadf
2ea13282a3758d6f6fe34c0c165a37606c4bf75da3bdde25d84e603fc11c90a3
7dd31585135c10b7f50bb341f8a5810c2c0ad7cd231d766436d0a1505fbbbb11
response z

674a6bd5d36087d35311bcb3dc8dfc015153c2267b6fc23c711d4fc8a36e32f1
d5cafc6c54a705d4a6947dec81b7d5afaa214bc5272460c66350a95547a9ba63
db828af8b9a597eba6b55d8cdc20397778d9e74065b7104cbd81048c42acb98
d1c61ba2a295f185b75156e210442fe230745becf2d1b29ec251dcfbe15ba88a
56701e423a5d16cb61888d56f6cf149527ffe18b5e0e696e839067736abf13f9
8258d6e744d9f81e60e008d872e1e0ec9dba05006c87dffda9e426dd9930d38c
05451040c877e04c0cf48483474e1ec8aeb3d3f8ea8045ac439f92df66529852
c67b347dca2244cf61d3b7f77f46a888b6c9a6556f9e25860c39b9d0162f4a
response w

360ad67eb3f3c9ef1a51cbc97d638de4d715dfabcb97211bd97519d30281656a
18eb6026c3fea64a9ade4102f8c117b3971aa979cf62f86e2ef46bfd3763ac98
3601df241f0a2067b82ffb1c1a0b048e453fb20760645394b0208fce23d7344c
7ff08ebceb3154a8c5a483866d766a1cb081ab66ff013ef351df6319b2caff17
fafec3eb4b0c45310736fad3cea1bc8cee8effc2cebfbc039a60efce5558bc3d
51d8bbdd9e9b85301324998bc72cc232bbe05b1fb8ce170d1beb04ada548f3c
245c709c6b0693e7f09a04ed06b2ffbf51a084214afa3a6d664fe29ddc3e1292
6ed9b2afd0a4f8a6fe4107bcd33cb8e4d8d36dfb4664630579b4528314d1dc2f
Fiat-Shamir context

ea5dd8a5b3e5dea330be985a07b8783464e97ead2ccaa9c6f6a67cfb73136c79

withheld proof term: 245 616 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFramedDir_of_isFlatAgainstFramedQ theorem, FlatnessPropagation:136

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] [Algebra.{0, u_1} Rat R] {m : Nat}
  {Q B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)}
  {A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))},
  IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} (FBundlePoint.psCoeffR.{u_1} 0 Q)) →
  FBundlePoint.IsFlatAgainstFramedQ.{u_1} Q B A →
  (FBundlePoint.IsFramedDirAt.{u_1} fun k =>
    FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1) A) →
  FBundlePoint.IsFramedDir.{u_1} fun k =>
    FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1) A
```

SHA-256 of the statement above

5021ed6872c37a7f05ab12449ee85c87570634d5e14c3391e98805a9010b57ff

SHA-256 of the kernel type

d277d490029ea6ed50b99f8ce8547f6529028ebdb8e465783c7c78c5a43bfcd2

commitment C

b59a2c0601d9edff80e986b299f5f791d6d110b2343238e48fec848bdc2e3a3
36a3fcb338c3520cb01f305204c378d86fetc216f6474a861ceae3a878642c7b
093223d6eec99d4bcbf7152064a1b8a65fa7f5631c125263d11689253488a6f6
1282d293db36c6f1b6e5a41aa387b214e722efe556b05ea36f6873764256d195
0500915c17d830c77470eff38de367c0784cf88582cb3f8323a709d907116621
594c8d7b1e562155fec9ffe8d3385a946e79e3868e541151c151c8d386606780
e2649d9c576394f7ac5cb0277ebfdcc90e369783bb3c0baddff0b94855de2503
ae2868696ab5e7bdfb80175f4d88d3ef4c82357907f870697d3388797a7e8656
first message t

1a714cc7e864d05b16e4fb47d6ef525dadbf9e4801a2742bd18e810cdf02c1dc
8387c1f27aaeac52fa1be4fbd08dcd23680d4bc0e7366bc2bc7df8a3f11da69f
eddb62a01182e7ac690d85e08b81e61fa4da2548beb3cdc1e31b9fc885a5fb6a
51ee45c26a789bb7966e56bdece78b168dd0e8b65affa59f676353baa89d09
e57463fef96da311d38446080eb8df2e565d5842caff40697a350b94caddfe75
1768d8b2f5e5ce8c2d431b4f12d35856ed6a4e28ab583f535577c057bcb9b38
62d4906762cce9794da297206a9c77f347f4f95d4bbbf3d5f4d9f54d14f7c510
fa9f5b8e3ffba6c3e49823e3a82d38adf2f6cb44d3aa39632c43bf719dc82644
response z

68a261ed97b0271e712da54046252c868a6f79747a3cf7ce51a9372cdb70ae37
5f7618420d05f40b68f62894e5f6050060b0e4f5cf824b00ecdc17f0ce0b638f
fd9024e1e64a7f912f2b2d82d45ca262b456c33ab907db335270c77f87ab3ae1
6026d28709589c00b20c8dd87e8b800e94f22d41c51093c70db4f31244c5c91b
922c3868d4bbac55a514686bdfc6fbdef955ea36408f414dedff4d7e2454bb7a
501c072184d7589776a5eaddb379fc31e678c76f4e26f3875e6bf5babaddb09b
a5ef4a9203eb8d01280624d826dac9e37f0fc3c2165e4cc4506b49fbceab13d5
4fe2b117e1813cf67974f8307b60770145c5fdf5b0cfd36702999bc5bfd74e01
response w

593cc843b2ffa3aed67d7dda558a00b28da3f9e0ac08ce332a3fa538e531faf
3e8d7366ad06ea18cb07a6e6a36a557650f34d626359282ea7a6bf0f982cff55
36cab5dc3d77da5db1bea29f276990520188446b5f3d208b3963c1fdc64daf0c
0d06786fd3645e53a461d1363d5a8d344480d79e04fa360d14ff582e590393cc
0b4e5e63c2559d5a0592b57dc9fcd23bd544cdad521b8b807c6cea72f1074c7e
55cb350adcbd00a5d0e3e2bc4b8609949fb5c538c2df751fe3a1ce1b0f95e683
d56cc9c1b3ed0807f1c3a9034c4057af818ce1cf25d712f16aa878bf5f3eeca
c8f84ed356541e6b45565527587d7b6a3e221e897d869df93da842c6c826e851

Fiat-Shamir context

622762f3212b49495cc3eaa388a7e99bdc899cf8375b42f03992e458e44256c6

withheld proof term: 8629 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFramedDir_of_isFlatAgainstFramedT theorem, FlatnessPropagation:127

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] [Algebra.{0, u_1} Rat R] {m : Nat}
  {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)}
  {A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))},
  FBundlePoint.IsFlatAgainstFramedT.{u_1} T B A →
    (FBundlePoint.IsFramedDirAt.{u_1} fun k =>
      FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1) A) →
      FBundlePoint.IsFramedDir.{u_1} fun k => FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0,
        0} k 1) A
```

SHA-256 of the statement above

f53855b09353e4770a857fdd21c5fcf099e86a5a6ddfea7423b86b1c1a11a3a5

SHA-256 of the kernel type

6dbc1e6e3c3076442b3bcf1f3ae6b01274eef626257325944105cac1c201f589

commitment C

56121d642dce12dbff2d029ee94319e2572ebf1ff6cf05c50da1d9fd3902d6fb
afeb64deff4168406e4e516a476a0eec5a862359e4f4f113a64e0133d9d0ad
746cd21ae5c6319d0916872e755a1e1b22fae29a5e9b2914409d87ed86ecd7d1
88ec2513d5ee1441351e95fce19c20d637d5c84fe174fb3daa9f5378eb47171e
a09e43d35989bb8e804f2df5c7cc99527849f0860f473973f2d9ca31136e73da
d8069114462bdd4be5b4aa7bb33d1afc022e8b246113d5209b7d9425f5b87293
0365814ffad6bd1dd5c6fed64372942b3b1c558569898983e48a21871fb996a4
a2894f133afc617a10a703fc417ebf80addb381f0f8b5c50157ee6b9611ec3b0

first message t

ce7976d8c5731a0381ea19a896a0bff658c1e495640406c677ed90b8426a00a0
1de4b72726033b6fed6aa68a0a6d51dba9bc7b8080b201a49b098809faefa134
292d46b00f695fe8e4e33165cbd85da74a1c0259383691a39c4c351f5d67dd8f
8bc55ba2e8113fe3374a196b87d9c2e6c23daac27a0b8a886b32b686e3caf125
9e3b947d7d820681d0c6559d56830d211e65f2715c862508d707994aa99e14f7
a5570cbb3f3f1e1559b7d8907cd01bf8d2cfd94ea30dab06b24eb31db959195
52f960aa788b5d9e39260991eaae5fd4fb64464c16145d3f2c6e3dc8812d3eca
8d5421ef72a1f6903c1063873f4962a2aa815ae3987252f7a6d041935456a59a

response z

c4dafb67b74a901821c0deda558dfcb598c20ded1f968bcd810e037797e200e6
506ebd08bf7bb559e84d503f930268b7111e81f05523aa859069f8ba75f66aa6
8dec4e3c90437de3fe9d48548ae82343b8c60dc3577f8b8167312d262863f53d
fb3a0975a1cd1118a3a88f0896ea24ec17e063e414677c1e972bb2e083c3a55a
cdf62ffc68724eb240d792872dcafc4a9550f41eafe24fddd8fbdfec2d7a82af
91b1b78d3be4719fd14dea79822dbbddd5b1b38454cce26b7ae6c48bbf638513
94a5be3f9d40b73e27888d940a7b86798ab4cff26596ad6928c0ffb4a5705960
94aa55b299675ee372d79150ddc5030a8be19c1384047daba59a1964a78a3d0

response w

5ce32a595867a91580d94b3d258bd14fbebabeab157e525de3fa4c874892224443
470238b35da4f048bc860292b64d7adbaa4d9612f7d1f2afa1be7f96827fc1e7
598da40effbd0276b1b77eef28fbb5ef1e06f08cfee3ecf79e80a90c52232375
078d212ed8bba737ee8035e8c85dea21cb59a213431a3fe7f5175681c202ef48
fcb46680934de133930287987d357431133101884cfc6e03b894c8d3f3aaca2f
8cfb5b7dd2034d1a846f1e58157ea451babe5c44cdf264e929e627cc2f046efe
06ee68cd2c2e1b97f7cb7ca2f5d82482e6fc3723bd12f45d8024d62eae7460d5
cd7d79762cdedeb3abd5ce024a411dc57c9afde88fcd213d18521ae749fd89a2

Fiat-Shamir context

fd0ae50832932dc0771e7afbc42e0d20b8ea9bd4437d01843e32cfea8efd89c

withheld proof term: 1705 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_derivTu theorem, FlatnessPropagation:60

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (k : Nat)
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} k (FBundlePoint.derivTu.{u_1} A))
    (FBundlePoint.derivR.{u_1} (FBundlePoint.psCoeff.{u_1} k A))
```

SHA-256 of the statement above

99c643be342b579f68df7d7fe11ee5422e141a94706369153a34fa89825b928f

SHA-256 of the kernel type

2779ba535d91759c140907d28a0cb5ac117f474629c1106b020cd01f730eb951

commitment C

85f8d25169afeee480646f1eece9dd3b4cd00b37461079daf654092ae245e0e9
72758a4c48e5122cd02a23b123df69533442b0ace1d3a9ee5a7f14a8a5431b24
0f17d96230d553f549b7e09069747c56dbeb9a37f4c81108cc79a659ddd43c5b
deb3fe0fc8a3aa5a6005c0155bb143e628c3364fd5e568e4b404a7eda17436e1
ec1b5d98166610ac215700aaf97f060395fc2884a6b1bbd42eae8264e93f3186
ddc2bc691aeb257dfbb87066a3db74faac6b817a2ea9b581191938cd0d451781
8f5dbbd757d8fb5398f61bb7c0657ae9d88adf2d2054938405b03659e82ee0b9
f56135ee46b95eb79707b253677a11decb3e60f1c8ce4b01c79cda4505f3c02
first message t

886cdd34606bb33e13ec52ed67ffa5708a2471ceca0ae31180ad029e6c446447
e80b357edb3be7400bf05344e07e78819a65a681f8455dcbd96461fa6a99e5f0
eb3d8e5715baea2eef1d1404d43d02d9d3d8fd0184a28c93a6dd488f2551041
501f5cf33f48be5db8df10c4ea68a45e8fe7aa5b905b4aa9fca289604c10fbf2
935f8512935246ff01474830c1f85e7a35ada1b9377ed2ca38c78e1f285fc24f
a0991912f0270d606a662b04dd837615ed23e708142f679fc746a26cb11016b4
1f64dfbb9e653ad29566ddf57c4cfd607a5501a03f939817ea97cfd6b749a1c6
a58aa82194f197cab66ba67de834c58f3ad07ad00785dc84c1db58752725e27b
response z

5e01e3f9cd8b93ab6a85b538585e10332b8751f6c9ff198b794ca8e1cdf5bbb2
c3aa92e2fcdca4c6d80246ec4a4b33d31bb14166869335d684db5e1e278795ab
6f7410254a17e39334059e9da48c05a338bf253c888c2c83f56967a6b36865c7
f6615b7a694a1b78a1c802bbc8d0bd0a145c5429e3f5a3e5637f5fecfd6774df
d7d730030a0b6f42c853b077dbec4a3a0d7318eaf1f4efe6024fceb98a0c42f
bbdf82a66a48fa2f8e6b5176a89df12ffe6cc15a9de20338a7fc460fe1055eb1
09f3663985a324e1ec4dedbc464274ae86040df78216e7420a4f000c20d752fc
6982f137499fe3bb201481cbf065024e684421c648bce7bcfd435aaa7d75185b
response w

61210fb6c2fabe60153b1ae011074af895767bf0918c258567dfe2d737ff21da
92ba8838de64eefc10727c98ddeb45beaaf861a7c9e1c80809619c89faefa348
543514013e2a523e7940c72d3f245e0666f1c6064ed2be858f9b92c5a57fdbba
b72ae5a68cc892385e4412dbed0b54b457c1dbd7bc6501d9f99692c83d413142
85dbde5f4b035aa119d2fe34e58542fb0bfbb261cb4af55ba1f460f735fb53e3
f09443256ad05dcaa2acbf31d1fe87e3ab87b3ae67065ac56acea57c58dca59f
ef536e6a808900860738af92a4acdef5279e7316c21447f7ae16953005d3720f
3e1d50ba6b0a2b7c1a34672cd8ba5e06bde46d4ce707b3f6dbda26364bdaf4de

Fiat-Shamir context

b47b69049481b7efa8230cbb456b8042549b38b89b96dc8d6eb2c4e0278cf7f1

withheld proof term: 4627 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_eulerQu theorem, FlatnessPropagation:72

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (k : Nat)
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} k (FBundlePoint.eulerQu.{u_1} A))
    (FBundlePoint.eulerR.{u_1} (FBundlePoint.psCoeff.{u_1} k A))
```

SHA-256 of the statement above

77a6ae31016ba522cd0b1ae4a391c6d96873047eae1bbe8f5d941d0e29850f74

SHA-256 of the kernel type

7f64b53d565521f31476f6c5660ceae6ac44c0189d3435d048141134959e0160

commitment C

e909b017ee033da8bafa0374eb46e236e58cab58e73dd8244abb9cbccb197fed
94b5a3edfa2f986348e6d8e540ecae25d8fb4b33204cd02cf5fb470d141ee4c4
05f9432ca8af0297ee06601712137c770dbc6c0c77be5ca14bb08b3ef06fb090
d208d8d8ea018dcc50e7287e06c73824822b6812309bec0fe3a323c5b446b8cb
3991ea540b79215926163c779b1bd124ab9fce4024594f557b58fe94b81bcb82
c6e10803afa0ece082fe32b1ec05d0154517a0d7f56e3b92e4499bf5d0ddfec5
9a24a675272c94c93fae4fa9366b29082cca08af69e976470cccec5a180f74936
00a949c8362a5b7808f018fc144aed21bba8d2ec5d3fd9674f81b2d29a782959

first message t

bc3976001bfc87caba93a5908097de8ffdf9a8f7b9349e90f62bea818543320
197a525a2139376213d2f76e40b78b39d7d0e39208ab55b4caaf18fdd61b6f17
6d04a63d78eba7113d69e38711480d238a6391979e8808dcfb879bd34020510b
3abe5f6e0c628d1a1a0cc791fdd096cc012e039b9eeb93195ac952874d16526b
891a8ebf4a15008e07150d711d82a30e0b357a572aad537020e91f4f8dbe94af
91e12c8945815050cbe45eb8d4b38b28607d32faf1a4d12c7962aef4006c5fa5
3f1bce0282f919824e88eaca33b7b94be9f022f4b4684d3852b1b567338b2a99
6eb0f3bf7b134aae642a392a96dfcee688e331a22a6fafd5f4c4fee63574f590

response z

6b1a6bdbb034509136f5656869287bb2fa98171d9d6cb6b697401123808a84c3
f68f3787bc3d79805a48a5b9314d8c893bc3572b398cfba3450df311416a6824
493aaefabcf5557793a65310dc555781d5fedaeaf20079a7dc306912ce6ceed
9825629285ce2d708fc241e649337a87f0e1ed567470a7aaeb8f7f2920142338
1c222c8332d301e7c5dddda13c944e25b881b6258b408c31fb60c9aba8bf22cb
b2561463011e92acd07c4da1bf414abce843faf243e9960127193d9af9f34ee9
ea6493f182f48ce79102b5673854f1713713dea63d62c0d8beb9a79022229a5b
8a29586861fedaf2b286dfdb0061e8cd77a2a4bd522db516b58757c45481e18

response w

17e9199bfdb6dd806c770ed2c66e7683a298662462d8703d27cda55af59fc0ff
b54d9e7287efdbf1e5c2535be32af4a63d50e9dac11c8f363a8bead12d99f964
9d5003ad47e65f1473c42f761c2580cc4c4eb0c3b2e4c4638d5e05e5b958721d
ad1798777ffa3d7628c0d37c5675156060532d3c260f8081f1ed15882ea2aae5
1d461f92eb88ff1ae4897879e77464b54e312e4ba2c343c35797fffcbb0bd74
0fc747228948ed66657adba67dc11b257b0fb26259ac642baebe29a8939e8e33
877a6fb1ca616713b4ef2f2d6697bc84f9cae83a848f746a21fc0e0b3b805691
f5e5e49118cf1257cd3e5c16d1a71cc451271a3d8a92d125ce28f14988982972

Fiat-Shamir context

65aca001dd5bb67b92d51b4dc49ea43a5281631a43b7b6663150fc2e91a59969

withheld proof term: 4599 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

14 RequestProject.FramingPropagation

FBundlePoint.IsFramedDir definition, FramingPropagation:57

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} → (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) → Prop
```

SHA-256 of the statement above

75875f7d566658921498760973c6083166a148c822be29873ddc23c308a6bc04

SHA-256 of the kernel type

52e2aff2f3b3f8ec4197a88431a5bb852d0f8fbc0c6ec298347633824102a3be
commitment C

71f3fb9a28dee4aecef1c52b48afcc9a75a9b6a555a9c1657c8500d6b660eba9
e1a0cce105d912dab4f4395b9c9643474789207c670eb72f67b7bc18c35c3d29
0241cf9b7f67052a672a143a958d3befafed162f0f9852f55ea20373ef072ffd
c0fdf70eb985b98a7c848b1595c0eed1759bb6d9627bdaac5cead2fd69135023
e888a1c4fb15166aeec21c320977c9ef2239dfbe3eada96b82b42d0ae8e14a1
1cf8e39d862c5a2d37c2259c93d4fd5e2c76ccd280f7b4d2044783c6dc23a991
47260fa6d94be9753ec07031011df9edf4b51d58badb61027e9b37dbb3b720cc
3156afe239a7ab9e7fefb6f4199d521ae2e50956a5318e1e2ec3f0de64d94416
first message t

926f102b4979b9dc9c38a4d477bea2d15fb3220a594f8ce98d0753085fcae7c4
a1bc495106d19c81902e71d01ce9bd2f507eb0a8eb97c5a528f1c91a250245b9
c5e3f69ce786399cb1ef302730c89caf3ee87df166775510faa87e0d37837be7
d827024aa47410750ee7b759956a0e9550cbcc19da1c9e07491fc3ac7561d61f
2fb8bd4fd6e7d44b4901bb68b6112f056be8fdbbd58cbf7355056b884a109fc1
f8b4cc08922ad29dbe28ea15062973fd40894f0d8c33fe80c8bf9a7a9102ff01
ca68cc6ed91e28792dcaa8e318ad51859ec65b48a73305bf20b1b84bf3a6e900
058ece7d2054d418e11fe108e1421ec03f767255bf1f810475a55d86fd44c425
response z

38f7fbd13d4671ac291358eb352314229e5590a391e158066444652a992ab6dd
0b9ed1cf3de1a2a1f7990888ff92e25e21b0cb99e8adb3381e944822ff909761
50ed9dcd7582c25e345608ab1dd49a66b6d3fd6140c71d936f5f4d2b2fd9c536
07792e42d7cea8a96649387c51f75203c50c2311ec9434c29ac9497b4e9f9e3e
0b283ededde3df94f9f11ab47591c24efac384ba6d90d198bef2623e3b87e8d2
64068bd61ceaae31aaa3b71df16c612b3333653cb684135f78fdeb49db3ca6cf
3fbb2ea507f5096ca5778f72ca05b0ad168ac5441ab485d299c4313531cad3e3
b0868de5662980bfe3c67aafd2aa6fe3e64a250198b5bb8d5653e65769ac900

response w

f004f0dc13ef5ed4a4f907129169ab64068343270c36e83718b25d7492dbff25
9e6eac533420e37d8b0ad9fcec797d4ee01d6693fbb8de03b16d67820d7f3fc2b
5c833e8db8c92907e70d32b8752c6aacf41bc11c6a1fa46f126e19ca0a61156f
9937f69c0b0818dc6867fd7a57b878641c143a6ccbb7d680c5f942f9d0fab800
ece114c8a69810ff8b22bd32750ff2110753ff8732dd3abefcb5883c673ef09b
d1e511c93cf82d4b9152cb0201366b98006878c53d6e6165d17d5a30bc118028
1f2f69e5b034713d7c9345c8baac3b332b20c18e59e17a3c9862fbc748222d9b
648c16d2f0848c5ac89a8c89d0ed6346b9fc493a58d30bfc3a30462aabb5d59e

Fiat-Shamir context

97033a581a4e907fb3dec1d715b179452d81ead5894c876b9a33c4a58172057a

withheld proof term: 912 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.IsFramedDirAt definition, FramingPropagation:61

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} → (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) → Prop
```

SHA-256 of the statement above

75875f7d566658921498760973c6083166a148c822be29873ddc23c308a6bc04

SHA-256 of the kernel type

da56da073d8b9f238ff85e2c3d985028764ae6becaaa9615e93bce1d683e74db
commitment C

7d2c54d68322fe60d1d20b032c0dc23bcc8a397415355a736299e4e6d5145d1c
ae057c119b2916ae33374c84b46e2a687d45ab8d2137d66fb190b256bf4cc4d8
d8dc2b266917442dcf7bc8598e058a0b8c81a8fd4a34e1c386d22f97f0e2d132
b42dc4d0ea5abf69fb3871fe4c571b6713640f00b9ffd462bfc15b2b0f88b88b
76b16382a4db703eb9834192e1c3d0c96696841d639133d3c4d1346e5d0b1f13
a4e4e484f445ec23e43fb792b6bcb22fe02376f6cac9c2b902a6e9dbe06df86f
e0c5a3c1041cf8de63e563789987c3c327e678c188dde8a3c1ccdad88e625f7e
a6a19769fa812cb731bf3818b5ea68ad5abdc02c4841d95eee224fc86c5e3773
first message t

8d9a81222f2dce0058ab4eea5b728dd6de9ae2b2cbc435175a3640114d7b2baa
f07a8dce42351e72e1ff808693b46d3a2485369017811d94eaedb2db044e67b6
6fc7bae2203cfbef2188821020a823766f7d1bebf27401b9805c814a4333e0dc
44749a0f7671d3ce94b29875bb3da214652ac3c2ab5ef71ad4277a2e092ee372
d49384d010ca1cccfac312a0c452bb6e3bf0cb89944c4479c6556b231ae4335a
7573640aebaec8339bcaab425fbf60a036f8181fcef04f0422cddce266819f8
a3691714ddc86510dc6f8653fc9afa0493ada8852136e5bc3f1cf8a418a7fb7
4f442bcbdbf2c1ea08a58da2abc4175436c03e0368ad84408ec8ad16c7fafefb9
response z

6d6e65c0eb851d1ecbe6fc6ee71051187687a92a365de84ddc56b2ec3eb6bdb2
22a72360243082a44818567bea3af159102fcce01c90508130b16d8c32985589
50afca865dcff00c564a25fecab52e666bcb1dec7a9011213f01afd33daaa6c2
0bc77c1a676de3b8e09cd57c16883d6e5cbab237e3ea42853e4686723455d5ee
faabc460a6450b972d313c9a464f2f3caf9c99ff3a8fa4b113b03fc05ae3affc
ccb1320a342ea020d4cd8dbfaeb5b90b6a78e2ef5a01983cff86275f107a1ed4
c7188bf8a4e2a6132b9124d991d9c9ea166f5c651a727140ff6879b92329c6f3
92400053fd77b1e60685caf98fc8382135d1656e1b3b14496d9f59c5a27dcb4e
response w

4ef0efc60e94b307fae96289fe075519150d676eb502acbe5c3e8c803d58a873
8eff4e0a2cfa8fa02fecbbc495b01c20051b8f301ab8f0e1cf92f595ff52cae1
0e9692fe1cca9b9a04523ebcd37ee60451a4334a66a8ba51f29ab1d2f40d5eb5
ff8f5287a03509d2461b28bfa93e617c3fbda2bbfbcd5ccc14e65319a4f1555
967b7bf8fbde4d9aa66130c678a953f5213fb2783ebb61a5162104d58767d326
a2bc8c9d2ea66ec3f4f31f3f1805db1ba69a839c77abe9937c7c45d2fe36f5a8
0dfa3eae4202821f2f1202eee855f88658da0dcb07c010fec6aa4d05c78991b2
6ccf3440627c79c686de5fef4bafd126832556e1e2a5c41c06cd8e3196bd748f

Fiat-Shamir context

8b08a76d530cd49a739d6b2163ad4415211b50685bc442e97fdb96b452ddeaaa

withheld proof term: 944 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFramedDir_of_flat_q theorem, FramingPropagation:83

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] [Algebra.{0, u_1} Rat R] {m : Nat}
  (Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} (FBundlePoint.psCoeffR.{u_1} 0 Q)) →
  ∀ (X : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.eulerR.{u_1} (X k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q (X (HAdd.hAdd.{0, 0,
          0} k 1))))
        (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) Q)))) →
    FBundlePoint.IsFramedDirAt.{u_1} X → FBundlePoint.IsFramedDir.{u_1} X
```

SHA-256 of the statement above

3e0d700fed2bfabff80059387bf28d2aa1e092e6052be09c061bedad0c82d86d

SHA-256 of the kernel type

463ee2ed2ba17cabd4679815d8b087fd5c37c442a4f8542a5a1c7f125f427115
commitment C

dba94952911a8d76bc295fe6454b814d6c1e112f8d1ebdb5713da8484f7ecf54
b86c7ba0c19e6da55d72750df5d292db51b98e3d2a0998e2df031d036117b64d
1fb0bf98aaf9159b658c30874601e5fd484b4ac76920292fb014c97a303b626a
825115b20135edd137ccc5d0016119c3a501a821cb7cb3ed5c1e63dad3748ac3
e385c6a87308b5b0fe8685f3e1b5feb18a65a028ea9e8d72d5fab752719faf83
f484fbc5477231565a5aa218a6b0925cc72521ac4194b356ba4ce0e3b5e4dca6
e6eae2ac1a72c71936c707d265ee624a734ced04ec3ff66440628db98d4b6aa0
a90085399688a5e7aae229309c389194eb889ab2970f43c91263b64832bd50a
first message t

27820cec3296faaddca684e6717827690722415ea7613dbecf9b1a0a16c159e2
059e8f7b6621a82a07efb7b437fd77e2e2c4319e8649d50a1dae28e9d2e8f0cc
a0efa18063441a6052088308d5f6fdb7aef89290978cfaa514cfe3f099c69f18
adcb00e639f9f8b79b07f524ccdb727ef953bdcf1fccd7cf4a7ed422bdb1647
e813e29a3a0fe6f3c9db1f524ede83ecce88a7e921a1be9935e757a229b1f9f0
e2e770757a9de8df6263c075a54646afded01e949308bb5187828a43f0df79f7
d9b79999ff8554952566c4a3934a4e50dc8f46847cde4a7584579787b6d4fc9e
5d3a1e7db48781f60319e2f25a6a7087c2b5fd89b74208c03163a2d42316b081
response z

4bff8e4f88f2a43a9bd205f28fc2664339b394f2375c6f918cc49eb34ae5cc19
ee395b16c020464928c9cbb83ad4a335231cb570a392eede13658100d1790f0b
c6ade8a1bbba5acca58074f8f434719ceffa939b9ce8764a8b33138b300e18a
8378e131ea660204c53ac424eae928a516f3642f720d0c53e8177e38ee149ac2
ad6c7f90989c41c5fbeda7079fcaa79ed57e79a512e242c0c8bed30709890023
6cb1dccc0a90f9db263047869108976413980b52a5570a482ad52b5d189414df
d1c190b39f948204467c49aa5ec6d32643185398dd10d8be5600d5cccfb76e0f
657913eebbcd34a7dcea9987bd928d243142e5d225f52f0fd9989bb10efb427
response w

47e074c995262bea8ab20cdcb3f3278464bd51749bb88cb6c09ec5d6424a2dee
1c3095a1784a7faa14e18a37c9f8e912c78728f1d7f8a07d8bb08986c1e9c7df
0c52637b4d49a989c8edb777d27c55fffe66f953abb896549a57d1b64e5bd6e
cc3fa00f932e498c7e0d7ea739dfe3b10e0dec3484df0266b6e0a6a73e57f066
faebdb8663bdc7462f62839060914f5e2580024c722175c5b517dace102a92e
a0bd75d435097c1c3443f73f5c50f913259277f3d6967dd6d0f84746d0758002
03bb8bc75693919247639da5fe98359cd5b15ba4b94c3c820a2fa4ad42574b36
6fcbdb6114ec8dbd722a193a596d869b1490344f6db1e4582f3a6ee2eee6326ce

Fiat-Shamir context

b482b8b9181baa9e643b5a006a9a23bd359d640d3d24c05c5c4cb24c0b84feaa

withheld proof term: 384499 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFramedDir_of_flat_t theorem, FramingPropagation:68

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] [Algebra.{0, u_1} Rat R] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (X : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.derivR.{u_1} (X k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (X (HAdd.hAdd.{0, 0, 0} k
          1))))
        (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
    FBundlePoint.IsFramedDirAt.{u_1} X → FBundlePoint.IsFramedDir.{u_1} X
```

SHA-256 of the statement above

54c0ddaa72ae7ae59c92aeb0dede016f3aae60426c42ef6f562fb91fadd5184

SHA-256 of the kernel type

15ada36b1df9ddf7e9c25f0092796ad7159f935c641517b5900ed7796458e818

commitment C

934c0b5a360ccb8bec73713c66d764cd3741741cc3bff6a64ddfc27bde928f9
aa14b4ffc75a0885e11af8b7aca622a448872d097e9df9eb17040ee5dcf9592ae
cbc7ede48a52f748ff661b064d0f050486cbb9845d2c26ad0660a6108a5cc867
71fec4e4b6d74a18f74aabb38dc73397d55835d8b77ac8791dd90541f7931045
354926841baa28786bec439207398631455841d9956763ee78a703ecde37bc6e
2a18de68259c3e917b09d4f71feb5d15533097f4a8af914b1fe45afcfac0c2b5f
1a147bcc472e7abf59970f0871693453a35935ac89b4314d0e08b6b889d33516
f4822d03a64a8873d6a652829de503accb89eae951e92eb1ae1cadcbe0117026

first message t

81514e14675badef680b5fa426d23d402be9c20008d42608e87d686239833bb7
1bb64657e546eb570bf31d6007cd2740f6e60c1913ced95b1968eadda0c9b876
1679708b0545b03bb6a5e682ab9db328a6f8b23f9acc8a8d72046f63b09b6714
6edf873b06e3460905941c44d0865239c6b934eb5bacbc8da665a6271d79dfa7
e8cdebff373e3b3201b028630f25210680caa14df20bdc5a955539526f0a56d94
02b72e09944c6c28280c8273040911bd373f0bc6ff7375d863ac98486e10d26f
7873b0561784daa5a5e88cbd0fac852f28b722668ab3643600b62e80fd07624d
14bc07a45784145476c473db62f26848055ce1c726f44304088bef3897d72d71

response z

6c0490e993e1524bc4638261a6bea983601a4890e81d64912d99c8aad4dfbdfc
027e1df3ad93489642b01071ae86dd3123f2c3f42134409a04566f681545c0b1
cce2269501d89c316f5f079454e5721ce66c8b2eb8b14528ec136200a76cf237
aaa4cc930c26454a86a25d49d6023b9ec305f82be8db5c06004c6feabff2c971
5aa36e852eb0496b505a94e86d2cff70c05dc830ec5cfa4179c35c631be45696
8a1fcf2ad260a09c02f62f05e442fc634d58051f199819712a96f762ee29b650
771e4149ce88484d1e739aa35e44fe8f42201d0b94f3af79c4c278838217d51
782bdbd092487308eab33762d20972b2109e65f977808639199d7a3e6b87373e

response w

df5e4a23fe94192720200c1696111bb4f9f416bc3be3e6591e4e2adaca1d3fc3
c622aa39d10e61324d318c3b0d8bf70cd314cc6fd68007ddd38f13e654addc9a
d63ba9c5d089f0102e9044344c756de8cb5ab031ce798d70cc88c19cd3993274
78b406ba64f9635938d37ae7723b963af4c8342871d409b642b078f20b49821d
0d8e2cef5c215b1cc73b2443cd0722fc2ba27a6a5df8242a3ff5c14174bc2509
b13170fe93e33c068a20e48d4d9d7aad1441c9bc33d042c271c72c24a8a04b4
f1ea78168aca082ae37a96dc6fe249407e3b883debb417d5c334ea0d281effe6
8d9121ce6e509eb2e361913021ec09e68beefea55d69bf15d14de03c01f5430

Fiat-Shamir context

d39f9effe05d9bbfeaf9115e1e235acce0b0bab3b5062464279f0e0f685a9623

withheld proof term: 4261 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFramedDir_u_of_framed_q_and_t theorem, FramingPropagation:104

```
∀ {m : Nat} {k : Type u_2} [inst : Field.{u_2} k] [CharZero.{u_2} k]
  (Q T : Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} (PowerSeries.{u_2} k)))
  (U : Nat → Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} (PowerSeries.{u_2} k))),
  IsNilpotent.{u_2}
    (FBundlePoint.adMatR.{u_2}
      (FBundlePoint.psCoeffR.{u_2} 0 (FBundlePoint.psCoeffR.{u_2} 0 Q))) →
  (∀ (k : Nat),
    Eq.{u_2 + 1} (FBundlePoint.eulerR.{u_2} (FBundlePoint.psCoeffR.{u_2} 0 (U k)))
      (Neg.neg.{u_2}
        (HSub.hSub.{u_2, u_2, u_2}
          (HMul.hMul.{u_2, u_2, u_2} (FBundlePoint.psCoeffR.{u_2} 0 Q)
            (FBundlePoint.psCoeffR.{u_2} 0 (U (HAdd.hAdd.{0, 0, 0} k 1)))))
          (HMul.hMul.{u_2, u_2, u_2}
            (FBundlePoint.psCoeffR.{u_2} 0 (U (HAdd.hAdd.{0, 0, 0} k 1)))
            (FBundlePoint.psCoeffR.{u_2} 0 Q)))))) →
  (∀ (k : Nat),
    Eq.{u_2 + 1} (FBundlePoint.derivR.{u_2} (U k))
      (Neg.neg.{u_2}
        (HSub.hSub.{u_2, u_2, u_2} (HMul.hMul.{u_2, u_2, u_2} T (U (HAdd.hAdd.{0, 0, 0} k 1)))
          (HMul.hMul.{u_2, u_2, u_2} (U (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat),
    Eq.{u_2 + 1} (FBundlePoint.psCoeffR.{u_2} 0 (FBundlePoint.psCoeffR.{u_2} 0 (U
      k))) 0) →
    FBundlePoint.IsFramedDir.{u_2} U
```

SHA-256 of the statement above

4108dec8b85731b6ecc4d0341cc947b6bdf140a74a80d68ace73cd092e144a3

SHA-256 of the kernel type

075d6afa9819eed9794f0c58d192e2fac2fad6a2b7a814f2b5f34464e84dca4

commitment C

814a1c781987524be31a61e730159d9274c73ded42d4727fefc84217d16d063e
d3096adb8592c711b25b59769a306c7aa99a0b36b78e086983de9c7026d71203
9b4a48b4fa71250101cc5cefa180078ce480fa21bc34be9b7bb3d0e4e68774e2
c9e9ba243af4d81c672054abcacd85c2c5f447ca6046d96df26b26b49d468a5c
b18c2a1c66290fca44350e364118f68ec42a7f7975c873d2ed45d8f6a65cf82b
090c26cbb0d11a9dde570d458eeca06765e14794e3909fb43b1a889bd47e6426
2ac483bdda48970fc127d606d8f06d16b12c981701b6b333d0013cae4d468123
ecf65035767cd9df14f42d6e239006c0d4d98815bd2579444b057a0f99902a91

first message t

36e3eeb2c40ed9c263029890868caad9d9ffa8b6c813ac91cdbfaf9000bab621
71550c7ee055c1608dbb32c676d54bd9e5a7365b38c82896d21061717c3b848d
20dac8ae80648753039e5e84af9c1de63879efbeb9ed6069db83f3adfa4230bf
ea4b3b367b7d989ae647c65d8747077b6f941ed1980fd479e6767ccbd3b6412
ca99948c2bfcdb277b7813efacc9429ebd41a117a1cbfdebcd628825dacb428d
136f4cc0d4f609ee25a53435a800b5d342bb5bb72050b50f98ca56fcd4f4bb1e
aeb26b36c067da3f25d1e5e7b5a70c7de482406aae5acb07ce0b0d6baea37428
db49bcecdc7e334b8014fea3529a9f3a40c433e06ae5f71dcfe4f9e815cc0f

response z

39fcd0489a7551204b8130b198676f6f1ecbf847cb68ddf7aa945aae3e751766
70284bc7c82fd7fc6d0a17109884d28ba5908d22ad9a7b1d5edede9daf18732f
c9155bf8df27d2a793695693f1bb3cb115ae60355732c6897352b1e25e274a16
3daebe0a2dcfb8e5d4994678365c400dc85572448df6ef0307155249bfd27b5d
af3291df16f370a7a49a7ec2bd3c91c77bc84e71dfd30b6f25cc5c84672a2dc0
887d7d6a9b4fdfeb14c2baab3b9facb701c316d1869041c1ec7fae1acf9b7947
dbac013cea2f2ebf46d9c9d92be6fdaba08c821aebf9fc1d9d0ae7d1c8536585
f76d1229d01262599b2adfd34c67e1d0031479c8951bc844a49d11e14adeb285

response w

1a6c80cec0cd72d9586435b707f487fd71adf524d31640394cc1bd761a55026e1
a442aa08dba246b3e8b3bf6a2bf046cc147f6e72b9e3fc9cda33e5f127c02034
5b2cfe98e962d7511da4e1e8c28329141ebaeb50fc59b492003203c6fa7ee6d
8b9ffa45670bcb8abf7ffb51c301f565c8965873d8e274e5ec0e415a121dff368
ee8c03c9a4bbc9161855fd37116f3a0b54f8c4bdc2c27155a65f0a387614e64d
6b09daf2004be6f7c0c1d309660b7769f2c5d16881d9aea9f98a326480fce18d

15 RequestProject.GaugeCurvature

FBundlePoint.curvature definition, GaugeCurvature:70

```
{A : Type u_1} →  
  [inst : Ring.{u_1} A] → AddMonoidHom.{u_1, u_1} A A → AddMonoidHom.{u_1, u_1} A A → A → A  
    → A
```

SHA-256 of the statement above

1684dff1d9cfb6a10eea115aec4927750fe02048d3b61f6ea14f2dbae9b58bd1

SHA-256 of the kernel type

1e5d5ae1161216365ce7a53a80a68e7e4a0a48fae3f0e9e191deb88d1ffa0744
commitment C

6ea65155126d54561265f24a9384b00f527c916bc51ee147d11d0d24f179a82a
9f7be86e30c2261c7a226f0968fd0f4cd53d4237f1d1716676c1b94f2658ed7a
2cf3f01358c109d11c4395160a93621102ffd74b78824738f970fea11d8a51a7
e547750456c642e80eab1d993eff71fd58e97f8cd64fa4e7cc631b32d0f85125
7a90dc75c5942e25a9d69c53e197176417d5cc6fe3c45623883949fa4cd3f681
e4c5bb452c1120185e72d687c683aafae7e5bd2287557cbebfce6b14e0352d254
8945c4d61b879eef0182128f371ccb998dfa5970d12e87966713fd97c956a78c
b1f999c39d5a641d0c9382c74b460238f16b1b8a6a76b19086cf1bb350a246aa
first message t

d45ac451a6fc6631e31f08de2ecf14e2d7e9789dced519d9553ea0c2e5a8d325
e21f723aa0a58ebd4a5cedf07edc2f3fec194f523210292c928b04ed30fce2f4
8489d5af8bf030cd3045d462525152b2fc4659125eb85177569b385a2ee64c1b
87945ecc717a471bc55994ad71b9e8e878ed32ecbfdb3c25858151429ada83
5a609482c31abb950cf6d45763d5f7bb7170e8b770ecb0eb7bd94ba9e897ba87
76ebe10cdd67055951598576567c686d39b8e6c66eea65c6b2b07b53316783e3
95208b08a808cb2c85801460835758abc503e3dc36561c7d8067871fc0326da1
8fca58119565bf0e40f862827387558121f3b3e3371f1063e78eae001a39f60
response z

175e7b3d73a90915ad363c69d1543dc4a945d34af62e6b5d7fab553f22457e56
f8cd71b0d44e2addabc73ee4399008d2656033fcb25f03059474dec009aa6723
cb279cc67be6b7abbcc5986b78e395735fa53e372e68301fc08200ef6f0ed446
5384b3d48304b972ea95432ac1ffdd04fb6d1d516244bfe237ff145d35565579
55de904e4dd794f8be32865aefbe600077c21ff8a70d0ef4dbe0f1c6c73a08b7
a7e79541ea95ae2378df5518d31c93e3440cdc796d29cac5135b158c838d839d
a6c4c2693084e6343a4a953d1393ca6608f1ba4d878e4c675c15e2222955a1bd
f26e8fc39c0afa68615719e80bfa523504f60872dd4461d47597121b114a2102

response w

2939d105f6627406dbfe888d8f5c3f9826474a3659639916e8b9fe00dfe780d0
e1b692a16edfe93fd38024c2af991d84c1887683832b9473fec604aca89122ba
dc0152f7aed7c2e05512276d6f49dd7adc028ebe9270a94ea74c64e4c3ce5e75
dab484138486259d7356b741fc079bb1b5169374c0f43f9e726a7cf20b1ef287
fe6b4b3836e39dd015bff0ba74a88be54c701393ed13c3758ee40b3a59922ec1
3e30a585945a34a85ca57e37781e95b946ca6572237d3017259868acf02a7504
8541635b40de0c222cb1123cb32469c040191ae220611cd19e9f52d789e0d3de
dfb16adb9958c733fa33c55f5c9272ab5a12cf522252ec2852c554abd268b8f2

Fiat-Shamir context

4490b69fe552f9daff7802e3f8419d322282de94130c189a8f0c3f5c432fb0fd

withheld proof term: 5072 bytes • axioms: • kernel replay: yes

FBundlePoint.curvature.eq_1 theorem, GaugeCurvature:0

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{Ring}.\{u_1\} A] (d_1 d_2 : \text{AddMonoidHom}.\{u_1, u_1\} A A) (X_1 X_2 : A),$
Eq. $\{u_1 + 1\}$ (FBundlePoint.curvature. $\{u_1\}$ $d_1 d_2 X_1 X_2$)
(HAdd.hAdd. $\{u_1, u_1, u_1\}$ (HSub.hSub. $\{u_1, u_1, u_1\}$ ($d_1 X_2$) ($d_2 X_1$))
(HSub.hSub. $\{u_1, u_1, u_1\}$ (HMul.hMul. $\{u_1, u_1, u_1\}$ $X_1 X_2$)
(HMul.hMul. $\{u_1, u_1, u_1\}$ $X_2 X_1$)))

SHA-256 of the statement above

9dbcce14d3ed267c729169c06b6ea4e073824473e96de96b2b8ac6b945735352

SHA-256 of the kernel type

ba42c005ee41935cf2b647860b5c6ba816caf35f3abd9520ee1ea13b82dee3d5
commitment C

2a04b63991d168d03c24c3eab815fbf56e40a74e7d8ae590e0c4de1404ca910d
31908a8d6993af457fbc3829f044a3b4e937b46bae813699c1e2244ab7fe4cf1
d97fbbdf046f981f589965b47f70a92a1fc06f5d2401ce0b6c111fcc810b03f31
b79bf4515c57e9cf3e7311b298b7c492753a364460fbecb9ed20e2c204ba35e5
f5c4157efb57656c91e1249d3b35a0e87154dfdbb1d7df48e9d456d9af1bf426
8e7ad154d94f7b0089861ff660b0b081291ad403780a8154c26cdd03432a73eb
7982f34896625035593f53dc032e4eea2c3dce12c1ecbb06c39409919a07d1c0
f9d69f7c268e64eb54cccf0f43e0db76aee7712279d72bef263e01929586cac8
first message t

b62d725438b10073870a90e260a318c117a15fb71be7dca3db341597898e2ee2
dc06ef0cc1903e4dddf01337c56b170b246b2083f951f46001d6b3ad386970be3
3113aa3a5c886560528e3e8e1d7811d2b2a3e757b7816b4f233cf1b97247e08a
9e1e323b3b8d837821a3fd2875dfde27f47295a4d45ca21b6ed8f954c276ce6b
beb4de169cef3af4d555abebfd36f6e76af2e9af63bb4c0e03d7d8c0ca8eaf14
551c9e2c27261b40ec7cb0e2eb5d08bb43e41aa62fffd37f74fe9ce00726b92
d0ecf7b1312b47caabf444400e410b7cad8427b506304e629cdc10a0f42cb526
c7ceed8f0d9ed9384946158ea57940d6c72cfd406edcfaf632c4584c5ba62a0e
response z

55451ab0b9edd928022d652d260d56ee5f7a82143098aec862ce55de9913e4e7
2bff39beea29a44eafb35983c05468097decc716e6ffb22b5eb7145020a7e761
f909255c81e16c5e88262997a9098d36d606f1ffa00ea53ac03f17451deeee3f
3090ef7f269f7d02a81d942ae7891b3bc13cf6bfd813def6a23a6c83cec3daec
febbe561c7e29824121eef8ec6ae71411da4028bbb44d2c8c558e66874eed21f
9f7a0a338f795b23f844738a841a599b4ea097df952f2315fb33f1e49655221
750f0c3d4ef51644ab4dae51079cf4226bc8942560d3741a9ee7c10db2c57e12
68803b0e0f814de3709f8a47025ef9c9819867977848f7bab0092598c9741f5f
response w

55d02bf3e5c3988058d454918d5c4b4e437267d16fc7519923b2f755a19df640
880eb3e9152ec99315e27bb10569c4e80d273883faf1d3d2a66459dfa89010cd
b3a3f6946f677d47b00dc4cb56d8488401a9af8919e0e0ef67575c756d88a273
497d5929d586147fa99156bcb6a46aa4fc042ee7c2487c2066bda3f026b4bf7d
ccea6e900b19d0da8eec63838b92bd0248802fb83aebd68a74ddc6d65c1f0d3f
e9a9c832fef62623e639598322266c8cdcd4d2de53fe3615129d930e87511ec2
b145d8738d64c3452f33c9acf35efb6e625a29f3ca2bd62a450c237666da6781
d0d5b9fbaf3d39cbd7bc48f968120db0df4743c5603cf5ee9e49e1cb54d41f17

Fiat-Shamir context

1a449c0478e69706cdde79411281e0dbb31256303d2705bf3a61c25c506280d6

withheld proof term: 980 bytes • axioms: • kernel replay: yes

FBundlePoint.curvature_gauge theorem, GaugeCurvature:75

```
∀ {A : Type u_1} [inst : Ring.{u_1} A] (d_1 d_2 : AddMonoidHom.{u_1, u_1} A A),
  (∀ (x y : A),
    Eq.{u_1 + 1} (d_1 (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d_1 x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d_1 y)))) →
  (∀ (x y : A),
    Eq.{u_1 + 1} (d_2 (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d_2 x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d_2 y)))) →
  (∀ (x : A), Eq.{u_1 + 1} (d_1 (d_2 x)) (d_2 (d_1 x))) →
  ∀ {p q : A},
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} p q) 1 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} q p) 1 →
    ∀ (X_1 X_2 : A),
      Eq.{u_1 + 1}
        (FBundlePoint.curvature.{u_1} d_1 d_2 (FBundlePoint.gauge.{u_1} d_1 p q X_1)
          (FBundlePoint.gauge.{u_1} d_2 p q X_2))
        (HMul.hMul.{u_1, u_1, u_1}
          (HMul.hMul.{u_1, u_1, u_1} q (FBundlePoint.curvature.{u_1} d_1 d_2 X_1
            X_2)) p)
```

SHA-256 of the statement above

b6084a09f9ce870c3c948a89e268ad2b817b14f0fa58432f758eba198bedccac

SHA-256 of the kernel type

52b2813df359cff696a12861c850eae31719b5ced7a85f2c5998b9d2b8864598

commitment C

85adf761cd86d03c94ccd811b562739cd5b7657eead904c990765f63061c64c2
7a5d3836975dd4151a48fa532db926dda8033e85ad0d4e8f257e1a6e2b73d28b
fdbba0cdd32725d53a415343cf900c9eaaafa43ee2e3e24beed08143e2c690717b
57f02f531e0d595910131576e5e4939f8e39fb428fe51652ede03ccc9dd8150f
68a38e6f51542e2cd6d9b86f765e228fede7cbc48b1acdf555df9591cdfca83a
86f6a081c977ad1d6e0511b60b1bb8665fc19645b370f2c09e06a25482987b1e
e9b4ca43772139798b5882e5387255aee5b4c427a8fde8732f7ab5f5bd834b40
ddc7ea59598a6b1484da5a25a559eb8912d324e176e51efe9d3a9a08271b79c1

first message t

a90c3d2e91a38ae93147c84ef5e434f5b012e6d0315f693c55d58ceb18b85ad2
ac320f3a91b1dd7b2891b2dbe16d80c1b361f779c8a82419ede81f972d419809
c5703f2680d7ad191dd1bc50b93d27857251de8a7ce9adf2ce46fe517a045f49
6b9e96008a49c7c6bee88c9654f5b0457948a175e67ac6bfed6ed669c90c2a17
50988d0eff127fd44c9ac3ab8a751194980fe28ada87d8c2af2b97cbbf116d2
5016f52906ec040499597bafd7a8e98655eb71e96156585c2802f064fc3b1b81
db258c0e11cb7710a7417a158697f1ada131a4e24f8920fea7d1629675132283
e3f23a497cb7970c1444ef4ca1f0e27f605144dec51176c7b54e5b3d476652ff

response z

3eeb6eb3b55f2785c8d5fffe6ef25b71d31575cd53c4379b69808363657aec03
be907ee7d48f30e7ea4ba031db87160a53d683396e650865e359b487509d39dd
b86b5e8e5f263dc3858080987abb2b309c934a03bf61bd2d85cec58c27670ad8
910b57cd7a270fa5c9f7c580f38184c97e90a7e0db5e1d49a6935e3ffaf22b9a
55c1120764f8125e3c5fdae68adfbbbf67549023ff732171629e04114e48e8c3
daca56ed181b84808b4a23650c8f7b0ebf92db8173e73297ae21c7f01da864cc
500ce23f833cd2539c3d2d73672cd06c5109d6c497245538672249dd884d0676
ff1b8ae457fa7c0dcd075ee405929ef41f7802a2d37aed551623d8de03c7d3dc

response w

22380c6f68fd6e9708112b812bb94384c46984f6d991c63c67a683d01e38b8b2
f747c51ef7736a14e7a08c002571671cbf20e485faaa8e4f0a3d84663534fe1f
4ce5e8fcacf7766c19d871d0c45a90c139f1717dca8197306f89a0a4d8cff9d4a
924e6a3a00d73dbcf8d9249e750a703dd88538aa590a70be6d5b7cfcba42d55
9537e7a971e5cb5aa55e5cfa553f7a15222a1e1030ef16f513b103441b0d49f
04b20dfb62e5ef2a50698493aac710028cdf75ea29263896e5c8238743407b3
13e202a91a2f47fd6ed25bfce604058c74350725c19d5ada0bd74e96a2e10eb3
1f05694ff1eb9fbc8f8101cc92a5db8fcf0fad0fd04252f260d7f16cf2eefacb

Fiat-Shamir context

54070a142a1560cf66b9e0a5f6654358675b5509f76d30cdba5866ccb9f5d947

withheld proof term: 3 781 778 bytes • axioms: propext • kernel replay: yes

FBundlePoint.curvature_gauge_eq_zero theorem, GaugeCurvature:92

```
∀ {A : Type u_1} [inst : Ring.{u_1} A] (d_1 d_2 : AddMonoidHom.{u_1, u_1} A A),
  (∀ (x y : A),
    Eq.{u_1 + 1} (d_1 (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d_1 x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d_1 y)))) →
  (∀ (x y : A),
    Eq.{u_1 + 1} (d_2 (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d_2 x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d_2 y)))) →
  (∀ (x : A), Eq.{u_1 + 1} (d_1 (d_2 x)) (d_2 (d_1 x))) →
  ∀ {p q : A},
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} p q) 1 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} q p) 1 →
    ∀ {X_1 X_2 : A},
      Eq.{u_1 + 1} (FBundlePoint.curvature.{u_1} d_1 d_2 X_1 X_2) 0 →
      Eq.{u_1 + 1}
        (FBundlePoint.curvature.{u_1} d_1 d_2 (FBundlePoint.gauge.{u_1} d_1 p q X_1)
          (FBundlePoint.gauge.{u_1} d_2 p q X_2))
    0
```

SHA-256 of the statement above

b514f69f4ccf29ab1d69db792629272148b9fdf1e40e7abb3b72c406b5185f40

SHA-256 of the kernel type

2b6359780196c6f8ce782e7e95460017140f0957378990adca0fcb7eb7aae297

commitment C

a63c297419f750fcace07aed3606745eb4e9bcc9ea96b7721b1a6a0d4dd9cc10
ebd098f22039212504237b151b1826be6d02c9274174b4d12f9ce0b2e8bd4721
4feab56180e0010dbd145d8773b1d0d99d5840734648eb9e939e81d4862e1949
a8cb7e933488582fd7b9f498504c3b17b2322e42026ef78f929c86a5eb56d41f
d3b3ecf8be0eb58f1ff81cd41e4da3e60eb418217d0faed5b45ba3a80721dc6e9
8677b144f2d4b8c10dd893eae52a45d28b9c8c8246c29f288d698def44b404d
4f666e0a27fb29594a9736a140b865abee85e47045d40d49568f1ce3ba464b03
6183aa4064bba3ef2a061da65c1e472f1d1bc36df279c3211ab88838d61bd98b

first message t

f6ee1d3f7e5783bdb877e509164909366520784da8d45f69c796fdefed0ea94b
7817abb70308314d1cb8b4467c3979f4aba4691467f45c33d1e633da2d106f1d
c64b5b708575696f4abfba4e8f90862474194e7319e12669adef5c0fef4776f6
b5a9ba51df240c998e4b307c9ce2255dfec0ffb0ce0b2e0d5b58d34c13af6df6
caf523d16dce9af7a8f27e296bf15f7f20e11e697f5ce3be967fca11a3f67fa0
7a7cff1c62d4797057a65e06ce0522d935c306ac154e5c800146990a2f15ae68
f93eea0ae7e5bed5f1c087ba2094151820865cdd5395cbf757d8bb17ad28d43c
b680f44e21c239b313f1aede48377178b681bdc4d8387ec355d44f3f6665483f

response z

4cbf1d763db40bf2c209a2d1ae638964a833f74b662e72db9c8a85e199dec685
d44d0db17c035ea5be40af24d770af7891e2e1fcd4f844a8389caef6411f5d9c
748e17c89d474b51daba63e83e9a49c22cc41a311fadf7a7ca698c29787e3224
1b9eb58e9e9700bbbfef5cda0ee02b6e291936e1cd0186a8baca8f3be1c8b7b
65930d1a159d625d226c8d0bcf4c526c21a5ba1c8c1806f38c4e5dba4add587e
894a0de525c902a06aec04e79f0eba6d2c503a41e3d39b87b88e726ad4f9d780
f003732c68795999a7f9f7191c9019070f506560455a8b486b0ae9b77266981f
9ac6d5a3fe975d4616101ca56aa4f164bdd6f48cd9147bd4c5dcbc173f7c74ed

response w

24d2bc655e5a2dd8656647ac0573a474081fc905db03b5f35452f4bbbee885ef
7ea393649f8e22800aa474ed96f32232f1869bcb6988b8235b8aa658692f4239
6110912c2c8661f140cc3e551fee6fd418c723641c0268aea02299fdf04e8eae
e45ebdb83719dc3827cb85ac956785faabca883559d74538d2f55e9ffaca4d73
f3c54bea2d2d0c0059d41191e3b666ed184b48a91832f59637de831e24f62b7b
981b6b665614b61f4eb20837c85132b20468aa9bda470c22a95761717a5bdd83
ae2b09d5c04bad73e2ab1bc0ad743df7b313f562e18c54c7d47e2ea57cbe4e3a
b916b1f532fab49197e547d89338660ba4d0344dd635e64ec048e403a4af335d

Fiat-Shamir context

adced1267002de0b53b66168feddf4e0f32ececef0be7fa4c6a9785dc021438872

withheld proof term: 41 392 bytes • axioms: propext • kernel replay: yes

FBundlePoint.deriv_inv_eq theorem, GaugeCurvature:53

```
∀ {A : Type u_1} [inst : Ring.{u_1} A] (d : AddMonoidHom.{u_1, u_1} A A),
  (∀ (x y : A),
    Eq.{u_1 + 1} (d (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d y)))) →
  ∀ {p q : A},
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} p q) 1 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} q p) 1 →
    Eq.{u_1 + 1} (d q)
      (Neg.neg.{u_1} (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} q (d p))
        q))
```

SHA-256 of the statement above

5beea728d78109874c65d7b5fc9a3c31a523eb1b76b38d94432f15c769cb7091

SHA-256 of the kernel type

f6bf29c837356d8634b6f7ef8dc2943b95c3c01e88dad9ceb87c94a494cd7d9
commitment C

ddad6e7bb0d74e008ca105b069b1a160146d227afdac74526954639408c16f41
e3dd11ff767d26dfbecd94e99dc1697149d415716b463cb4bb36e45dbb646667
182d4598bc299525422617c9a43d79e5b138f31e85cefbde357f29bba5f07fa2
dc93643b3da1d7965cdd0cdfded1cccf6d5f8c8da0981d9aeaea0096058eedf
e98178a30fc15ec1659a0c956c8d1c17146a4a56025a701e3ea6fcae2c210100
13c028982a8725b1669388fc7ef399a6139245f76016a7889f8f868f1cfcf488
47d500736e23f762f3c234cdc2ba6e8cb15cb62372f46c2a027f96505cb59e62
9f47a6c33dd671404636bf7011e4e50a09fc14ff3478a13c8baf7f87bece69d1
first message t

29743a14ef5f7e9be5f091763ada24144b5edc64957fa2ebd00c01692d02d2ef
8e536dca6681bd34fb70c61f5437d12f4ac7cbf40d04b706e9196865b3415ff7
f1962c4321e3ef917f843649345d455cfac981591567b84baa6b4ed30a23f27f
651b6ac51602f2f7d046b3eac3c4c0892a838708ce71224efb3617bd3e884148
1a91503c13ccfd4c6f7fd49d0e9e960aa9e6af23243368966fc73c04fcd83d3
228790fc0e852f5d2f97035e3beaeb809933e4b8eff0ae5987996163d610132
3b8618bb7a430b23e1a72fb0295cd38225d6b25ad77e10750aae18d7b0def67a
e9f88d7066347327c9e6af9976b2673e85dcbccc6340f99670103a03f561787d
response z

21b59e6e5cf5c48d67784c9f8eaf92d3b143fed890220d5b87044e9eb9cfb10b2
887e894816b4cad2a09f49c782c235d98db092904bfd32aaf8164d89a438a78
ac779a93a40277392ce29ae1681693346a09085cb2b652fb7a005525022492a2
cc3be52be65d970d917851741729786764b5eac364d5111ea1cea312f97cccee
b0cb6d90a5db4e600e2467137e09ad03c04d61453cade5f3ce899e0c1542ef7a
7c941416157c0afe563a4d1b86a52ee89e0543026c6be2817ec71b6f7331c057
0ef5848b98f7029ad8e7cfdc0953371386fdbee8c9b80952ed75a2d321284fd0
68e291850da0506982c08004a15052c5f57f89259d59cf95dc2fb94c9ec7b6fa
response w

10b9a0db7462ab9e5062830f84d101191c406d686b404ca91e6d6551366b185c
8207d9f4896655db230bc340e7c438df040cb054c54fb5b31188e241d0ef7fa1
0af927f71849e161184967509572946107055112a7f390c3b961358adec83026
735eebcaa3209ab19b08b20089274bc3ee4d64396eb45cc9e822d5d8fa8554b5
a36f7c015b82ad04842cafc0771c6e937b355c031da808a7fdcf50016e5b602b
1fc5f8d9862174c2bef7f46b35667d5ea40a0844b9c789dfcf5969bff1948fa
5d7e6c4147eaa97b633121e63045a63b68c5b6e407373f3b08dccbac0771efab
05de7ce19c4bc6919e63196047025a1dfb1bb6f4f52068302ba88070a1d7ff83

Fiat-Shamir context

9ff21a5752beeb150e65798c6de915d815282c2cefc2d200bd943b3854169654

withheld proof term: 205 940 bytes • axioms: propext • kernel replay: yes

FBundlePoint.gauge definition, GaugeCurvature:66

{A : Type u_1} → [inst : Ring.{u_1} A] → AddMonoidHom.{u_1, u_1} A A → A → A → A → A

SHA-256 of the statement above

cc2a907b50f6531b1dd544721636b8bbb1d4ba3cfa2dcc803cd59092c063fb7b

SHA-256 of the kernel type

0c406da6faa3427860633a1593f4a35ab9e7c291d4ca7bc2e69049b8b48960f5

commitment C

9a8d6f9107a9123966a916b42825fb341f6754d32a4e8b239b6bd31ea7a17768

dd96a0a26ff77de191709a8896ea9819ba7046397748ce00b4a45044d82ed327

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first message t

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42e4ba7fdd368ab67cd2db10d2942cd78b621656a36d4a386a5bd947bc3cbdb

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de647e413a923a202c0331c85d8696f8c4a1b1df545c8b02f191a9c165631ad9

response z

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ea70c7f359cebf15118eb4d9079b2c00b4fa093d181e3e4e876cd23925cb4300

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cb390f63828af645b07e779e470336132298c570f54cffc977e88119375d6d8f

567ae9043566499d583d01847c52c22c7fbe63db64cfe6f48da3bbcfef60d47b

5b593dbf17de8db397711c096afc1af140e09ceb0a01578875141650316920bb

response w

440a6d67510508bbf0bd95fdf855770aaadd66ab4afa67b4bf30df65e2841428

2f0f7f3eabe7406bd17cb7f139ba3251d29de17659502a3c724b63db3e2f4fae

a860429df152252493a97d451b4db1243f846c5efe1c67afd440122516df3112

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5e5a264d20b53608cf1f64356d57441b3952a467b57a46067a846d62dea74d17

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06a216d386db99b1261cb05d2176f76db5e54f4423ac1a3010d50a3f099f153c

Fiat-Shamir context

4773268334a923194d14dd6f6bc5a61f1bf7b9285e4305b2e00d8875a2ae80a7

withheld proof term: 2832 bytes • axioms: • kernel replay: yes

FBundlePoint.gauge.eq_1 theorem, GaugeCurvature:0

$\forall \{A : \text{Type } u_1\} \text{ [inst : Ring.\{u_1\} A] (d : AddMonoidHom.\{u_1, u_1\} A A) (p \ q \ X : A),$
Eq.\{u_1 + 1\} (FBundlePoint.gauge.\{u_1\} d p q X)
(HMul.hMul.\{u_1, u_1\} q (HAdd.hAdd.\{u_1, u_1, u_1\} (d p) (HMul.hMul.\{u_1, u_1, u_1\} X p)))

SHA-256 of the statement above

bf87f3a20d36a79586b0fb9415af37e6cbf5c5ca6a2ce8d1f86865354b38e15a

SHA-256 of the kernel type

dcf43fbd82b604d008291342087ff872337316f804f2d400f3600d5265e87fd5
commitment C

fff35d770ac6a14408cdf91165baa8c40c1eeb5abd52a238721cba117ed8eba
902bc2fd1ebde951c211b369d1789ee5a9e37ff064c78857a186555b69163f2
d4ed400100fa4aafdc4cf6e42a8b2d658e52c14dece2a46a8147c05861d06ecb
c740b28da5265d729e4868cc8d4a5bc158fdce0260bb9586b4fececc9fc47683
ec2a21fee25ce4ab890674824d29f400107e802650eb8c41aeb1c08b11e40804
e26b671174c4e9d4a9fba11968bc5db41bdfdc8e1c1096232303de227301e595
8c47a52fe53d94deea909ccd7e9abdbfe5054430ac71865c712804ab8289bef9
a39fa132d5c99fa26358a8be7528ae7d4b96c3c3a5d75d3bcaaa8c4d2b930d67
first message t

fc734e0a99f2221ab92f06c8358286a92a16b102b12ab90bbbe13225f7e6ee92
4bc21eded5f2c5205115896c97c09821bf6578eb8af81cbc8c875ec5af314037
94b28ca8227e9346ed2c3729a15d0805c315a8e7fab636468a0ab0f8c91e5da3
dcd85bb264533fd305178354cdcad6650d149f5bc0b953ea0cb542582c17c37d
02612348eb98737643d52a96225cfba726085199674e139397d85acb9a43d4be
a602e6f845f1c6d4d32ff6d0e5d55785659d114e8b2819ae0cb3d83bab14000a
b7194264904d33fad8d7e893bbbf048c525bbd751bc845e0b2c5e420e7991210
9c210d16e17b813753bd23497ee5fede803d610f23fbaa8e44d29078f5e33363
response z

225115d012300564a0d9af100bfbaec3f1aaaae15be67cc9543ee2e6ad642a15
6838a587f1816eea765b328151ccab61282b9ed0a7f4d9d4a3d24692ab2d973b
03cab01fe4e1e9775f5fcaaeaa011111446c17065a6a65329b4e8fb287c8e196
00c08d4c2ce93ca26734ccb43271f6a1b4dab5639bc522a4e24ced6e992adf24
a9529cf664424814dbc493ae157d62103a7acc9b521ea9839f7680aa3361364c
cacd648b5850b2ba5cb8d14776cc4bb91b22fccd8e62c671ad4f5704bad5a32
3e0d9cc99fdc8ade087d5c1b14ac38fee8867e8ee422870c913e46241e7434c1
8462dbeab616ed1644e06cdd32d0dd848db392f9efe5a935feed7eef175cab6a
response w

439cf374a7b3fd23538ec4d4eda5f3a497682631325c36194026ccdf9894f3f8
40e571942568d6856a744865e0ac846a581b25025ac4ccd800a239018f8e5963
0fdb165339ff5818f927352316687a6f038fa914a363ea84f5c8800da0d3346a
a7153ebfc26885b3437e0d0ac500aae2dd47dfa5df23f604ae29b6debc20fa5b
3f1c2720b03008e6f290e48ee5fa97459040c962a43fd39d423a1fe8c01d4f50
77ce54101c4ad20bb0a4b66db2a5e45bad1f5afdb7caebcd89efb026599ccdc1
d9f44b0627deb9fc9ddb8839c8b6dfeda5705e96836f695d388923c9f3e5c44e
2c300690bd6eabdd0fade466821fe072ba2a7a643fcebcb88418c988b147c9df

Fiat-Shamir context

9a7ce571906a40235c30b04a3c822491683334c3d8db8b4b2e050af9a7802996

withheld proof term: 546 bytes • axioms: • kernel replay: yes

FBundlePoint.map_one_of_leibniz theorem, GaugeCurvature:45

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{Ring}\{u_1\} A] (d : \text{AddMonoidHom}\{u_1, u_1\} A A),$
 $(\forall (x \ y : A),$
 $\text{Eq}\{u_1 + 1\} (d (\text{HMul.hMul}\{u_1, u_1, u_1\} x \ y))$
 $(\text{HAdd.hAdd}\{u_1, u_1, u_1\} (\text{HMul.hMul}\{u_1, u_1, u_1\} (d \ x) \ y)$
 $(\text{HMul.hMul}\{u_1, u_1, u_1\} x \ (d \ y)))) \rightarrow$
 $\text{Eq}\{u_1 + 1\} (d \ 1) \ 0$

SHA-256 of the statement above

a684d497398156dde4e41cb51c114bf006f1e561b47d5f83d3be4eb9b65ff310

SHA-256 of the kernel type

063a9427c9fe871121eb8e8145c9d05d9a58e284b076fdccff5c1671b3688fca
commitment C

965608df2bde7a0352d44d6f9bb197591b4425a7a0fdcde30108f0bee36094e6
3470e298e1250948d6ae2bd50da51019ee764d069e5044858166912621725cc0
c8c295f246d81d93858c44261e3bcae51733abf608e6b7162213c0ff83923aef
a3eb7b3776ad4e1b93fdc6f1dca655c23c2a1f12a8530a154a04952551780e6f
90fbb4b3c337e944ef738d1650102c507f91faf0dd6e356bb0683dd3c6497fbb
06f0701a7878480e8d282ec6cfc0e3a2c8f86a7c6369064f6c864f174264d23b
f979a222568e3770ad8e1d66fd9539c767848d7735ce01b3bbecd9cbba91caa4
178e2e1e8b4dc7cfe543dc58887f28b53bf2e03d6d5f93df4343ddd74978178c
first message t

bccc984bb9920c11538424cb6abad867e07cce4a04abd498a6151d3e33319801
886f2eed329cd7c6a4eb84fac5f6d1e3cd3857ce95ef9d4bee8e26666a678e8b
f937e0d700f958e3ef629305d1612823804afb65fb4b68dabfd2217ca1738b61
5825eb4a1a61d1399ccde01dfb86ce17c84378c903c45806c6c77c10dfee8980
c7708522f067b6bf3ea2ffce904739283d543de8379852399b39d4d986f490645
2d2fda867207b6802267951eb1ef122085adce241cd4e2c3247990adf19e5a8d
f4450173b30a73ce79cb0af7cae8966555275422a8a1d40aeca8a5c67a0c8c86
8547eab4800e2a6b5110d4b340947819610561cd17c2be879a788dbaa165237f
response z

10fa2a47007b60cd8c05506cade14b4a97096c97b46085e4e05ac87565796d2d
c1983c2823e70e965fd131a8b82f650dab86cf5e8d22df45e2c9dcaa2542eb12
340ba78d24a018c6303e6619e4c18f4f809fc2b57f29c831207c7ac76cf6f9f3
29481221a5f40c99ec21445808c21cf4ff77e61bba82e3883fa524343a5625f1
e044fc20412d746315eaa0babf2ec6c7f926332e9e6a08262a6cff036126bae6
c26f51f1e21e9b2a955c5444b8e0495f42947730d4433c62369777f98a180892
e8f4a3706f100175e2d3e23acae792f42e4e3ccd4ee31c2a15e89fb27b95609d
fc74a02e3b220ec9a24b02e93561aefe55cc673e47ab93693093d543ef1624ce
response w

40d23589c7ed502078fb05d3ed43b477e734722f5e37867bc74c7dd286daad65
80fb7afccd835bbda8d75320e0714bcae331fb7f259091952e4c0d9c52332df
84f2ef9f92956501848a7f87bb61e6d46415b369d778c611a4aadf14f723988a
018dd93d53699e982d3010f98e8f2b54bc84254bc8ec0849349bc0cd00cbdd75
ee467f2cc2a2eb3f4b738753ea3f5182ff052399e8fa166442c5beaa993ce198
f8e6828e1180ef47efaf18ee8cfd461013b81dc22f8bc2e6965b554e0e4e1319
f7ee24068c10943c63d33726313f5d4f341bab3a26d7215b45b284f9d7961987
4cbcd681bb7f062a0cfac2f0c4fdf1601d4d580c856fecbf9a4d988bd87d3e34
Fiat-Shamir context

68d75b931c2a4b4151ca3929683482c27201bb1e54a83967197c45dfc1b8e596

withheld proof term: 133 357 bytes • axioms: • kernel replay: yes

16 RequestProject.MatrixMultiDirection

FBundlePoint.adMat definition, MatrixMultiDirection:280

```
{k : Type u_1} →  
  [inst : Field.{u_1} k] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) k →  
        Module.End.{u_1, u_1} k (Matrix.{0, 0, u_1} (Fin m) (Fin m) k)
```

SHA-256 of the statement above

3a0c0070c6e8261dd5c67207bb7bcef7ebc046b4f1c0f126f42674185878fdeb

SHA-256 of the kernel type

7f2dd1fbd5e99a974a305a9e42e1367b94191fc4dd825ac991726b0e1f2de165
commitment C

f51e9134e5a2ff1a10bb24871f30de0ec06b5f90d226eac6a3e86355327c1a40
ec6b4dcbd87249338ce687b0208a26188ded830c03a95bed7b1d42dfbf619470
1ffc7f8203fa52b5701967aed6c05997f9a6a31f92c83fae4ec5b8e4dc3c8408
407f868343e8ba9a49cee4804f9cb08733d995b81fc84fc47f3199c9b75e21e1
8e5bc39e741e6297487bb859e6280df33af5f11cbd838083a3caa9b5de67ced
ae4a6f044c5c04ac5f023276a8625cbd2ba1f0d56fbbc4cfc5c9eb827871d63b
9017006f4a974f3289884c8742bf5b619f41d171deb6204fdc60ac24fb2d3b02
ca2ff1376ba16d63a63141c52a12b4bdfda2928b660ecdad48968a4cbde24287
first message t

f5791ca952b96bc926ee07b14c69523e327deba16198afc494e3f8feded966af
b0a97f9a6c6bd13950da37f414d2707f49f8569d8e844566dd612900b07a4c786
6b55c65cea50d17641391ef8632069d0df5df208ce76e02deb30034d0719ee5f
b520ea2e3fabae3e9e058df8ae6ccb04a4507bf2e34a084c927e946bbbc54bed
34f4609b4bc7b86f82d2b3a4a9aa7c1488aa1f1782ec8e409efab337c12a3d2b
89239395c5d0cae7bffa85d8f28a82ce29954cb87f076b0d96f09333c076b209
b8b85cdc45a60b75c6fe3b9edb098d90f016fe65055c19ddc80543b9c60ab9bb
fb0b2a152375c9be172f938cb30cccec9a9ab1d3b6b1ba827f330729225e2d3e4
response z

5ae0ed73469da7dfec73ad4e46733a9d5512eec5af8f1c6d050682d14b91bb8b
45f5abedb292e5d9ae67a24d455f0d013c7499c579167a31db63dc1ea3869bc4
5ac95a0fb84c143e57d873a56dbc4be7c16a5424de66bd6be9d01054e4daf57b
a76db6d9fa6ad31a1e15310a43646a61ac964009ef19fb06e5ff81b8d7bc8813
d1990db1064645afe1190ac03b32f65bf083155aa70fc65593f340a449331112
05dd2b8a0cfa8e6882308a12f02e77f4050a0f29dcda800ec6dcf8c6532f45d2
7893d21ed5ebbdd7ed44f639fcf20bac22d26a246b64172ffa56deec2cae2741
3b62415b9affa58262d93f0527f2e6ee6fee321ea041db12c8427dbb0d7a010c
response w

500444c45dece8f65cf55c714dd7b8959d10fb433944261d54e9450264ecceda
997b42c71c5a497e0c842d1347e56d01d91a01e20ed1cac2e36537825098d4f7
a5d91f56b5c24d4ddfb9d11b1f8fc3c2b1d40c24877d5287425dcd541e58c6aa
dd435b06f6952894435a1c078e390c3bac2826856555c9a5dc7e181e4f10f528
b693471dc81e2dd6b3c15f9ec5ad5a5e85b0d869d7ec93d43ced32254f2710e
8cccee3045089784673e85a233fa186fa1eb4c816917e0bbebb771d8d1b742aa
567473833ac77eb6d4e1abe6a4405d0b8db872ab315a11d86efe43106328341f
47c48877a533e935995dd190dfd0e46f1e8c01abee117016f352fc6f840f2dd6
Fiat-Shamir context

93e1e35737841a89ffc83c5a8c1681899ca90d16a7dacd9289559741b5d8cba9

withheld proof term: 8319 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adMat_apply theorem, MatrixMultiDirection:288

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}. \{u_1\} \text{ } k] \{m : \text{Nat}\} (A X : \text{Matrix}. \{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) k),$
Eq. $\{u_1 + 1\} ((\text{FBundlePoint.adMat}. \{u_1\} A) X)$
(HSub.hSub. $\{u_1, u_1, u_1\}$ (HMul.hMul. $\{u_1, u_1, u_1\}$ A X) (HMul.hMul. $\{u_1, u_1, u_1\}$ X A))

SHA-256 of the statement above

bdf23323fa1fd19755b98354849d41f839360d5ddbc0eddb5b1e21d5d2b398b1

SHA-256 of the kernel type

e884a887da484c01d3fb9b46c1d2310fa5400ade1c465d24367a4c771583275b

commitment C

271f6c65aebefdf68bbc0eeea94a904954d27670c658cf894443417cdc584341
4a27f7a4bff59da0cfce9cae8457dd45cb7a95c09a7a39f869c11c225f34a27e
433d0353f7c44cee1116506f2a5456a3638d51b7187d3ec1febfb3b402708751d
05d4cb08356cae21c66b93c12028582c87e642972213720e38ccc13165e8403
c00ece55d06cedd3876102a5a277f0c88f5ed42a9e39b9b97f59dfd3b2ba1c54
e71bb875eca282994130509803af048e9c22c3dab44f995241a3b27eb27262ca
10e9f0a725794fae14cf4d04a840230332b72f7d6f4ff79eae69a225e043c0c
40d255790c9232163aa17a95a95c454e866b9371cf03d22416e49192e219ff7e

first message t

9f23311a0407ae5ec881cbf7f8121b52c27a51cf168eeb41880b0a5f553c19cd
bb19240cbfbd0d0277cae0c4cc18ca1c39c22f761fa0ed7b7c7096848ba9ce63
2f5cb1b903a842aed56ca80a1ab1fb23041b177b73194015a9547db625d1114
a53caeca97189085b985a099d73f8cafba74fc7efd2b7ce58db4e0fe48b7e166
f67d219a5c9feb0c74eb79c966269fe970e36dca122c6554110a110ef3b6754d
ef72b63fb712453520b7870ecb753ed83675684cebb08337e01a7987d31ee76a
3da3acbd7d040f70090aa61d8fddea3b3bc605c3548de093e3c44dfe73347da
ca5c091d3dd42ba7c883615ff3bdbc9ed40f26371605fb58b91cbdced215e8d0

response z

1fdbbfc21f2e03d5b6f636a83266bf422f355a0ba37f3ffa3ea696a0790bce67
838ff8fa80729fe40bf99dda525dfc991eea34548588bc85014ddcf6dd4faaeb
5def717a0da1e6ce5248688625ca99ea5adc8f470395e732b15a27f9040d9538
cd40ce2d9f372a015057100505114eb79a0b7a3878b50803a870116779bfd0dc
31fc3e7c7c8bbcf9dec26cea520e3507098a232707e16d0334eb85d0f94f43e5
a26d1fd649f7304a9a502cb08b3cd630bbc178c3e8eb67958c3aa3a9900870b5
2965d0a1debb015a1f8e8eb17f2bb746aac6e95e40d14880cf8403467e794ed1
d5deee1ed44c0c055b6fd4a707521496f8364cf8aa3377985b33673fe860915a

response w

19fae1fa7618d5f862fe6412ab6c40f58dd9bd652a45fb6eb75c1a52be7d4ff0
885e38ea5437cac44a5de23ee848277179f7aaaf983b43f57ec8095641a1373
6f6bfef75d134c93bfd6fe6bf4b6530e591997cb2ca54bfe3f2eb69fd7760ac6
0a29f69646d39e6f3cfd3c2fa4846acd31d1680665f0c610bd61619dd49c6a0c
e91b20df70b53494f9b283e956c9c6888eb01e2d9b072e785f59ed75f54f4177
e281a65a8a7eb646e79fa07add4b1ac271a5d1ecc23dddec6f0c4d680dff0908f
cf7c9a5a4f90cebe71e96a8e9eb8556ad31631325b7ea8a8bd5be49ff4f9b737
c9e01966d8d5bf51ce3e09d40a70f406ac65005fcb90f82a1b06086965898eed

Fiat-Shamir context

eecb973db8e05f76e52eb832255fe7e32eddd018868e780e2c85fc6348fc08ac

withheld proof term: 5 751 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.degree_pos_of_ne_zero theorem, MatrixMultiDirection:142

$\forall \{\sigma : \text{Type } u_2\} \{d : \text{Finsupp.}\{u_2, 0\} \sigma \text{ Nat}\},$
 $\text{Ne.}\{u_2 + 1\} d \ 0 \rightarrow \text{LT.lt.}\{0\} \ 0 \ (\text{Finsupp.degree.}\{u_2, 0\} d)$

SHA-256 of the statement above

56295a43e5779bb1cae441187fe7ff8049b07706a33952c0a8d9b2b7c761bc4d

SHA-256 of the kernel type

982cb2505cf1427471ba20fd94a31b8ab13a734e40b5be0033f209784119f2d1

commitment C

b7f85cb09dc57c9275be360373720f720874e9af243e946b50d2d3bdf64f744

a10ee2c8bec939900540adaa36a658d3c03b4a8a708c9f778d638798c7d226e5

f2e6e8517046b240cda7914c07aaea6cfa07cea74cc9f8a9c5ed46b3e60c685c

b87006cf588f63f50059eb8deae498dc4ad0732e789084f1221b2d7f04adfac

59f1656e4224c577d3ea622417a617361227d76405aaba25eb26b2613c18559f

aafc80a053ceab1b09ef8ab0279f7ffb2cdc7cfd8dbd67e975acd59ac0bff229

64fac3136a0a380c39235ce246036d9858fde81e252c7d6caf633507bdd213f2

5ea89601ee9da5f39976270d6b334d7287dd293dce93aacdea7b6544dbed3406

first message t

7535cad7aebf0e82abc65a4eb8ce0834193d290da0f9db5eee102831d10794b

e20ff0f033117913540cfa3ae55b48aede0c06f0509f32723387d3fcb03914a6

007750803bb7070a4b84c487b294dbf0fe3a87d29d96f852479fd8c8f416a31e

7d61d34238fc1eaf07016ab530fc1a2d938feeb0f382bbc7f5cc21e52d149f7c

97d848dadfb43cd5334d0ab2fc090cf13a7c3c7ddb39c5511ca8442f3075477b

6da946f28bf04d253a162355f5a002c1a17beb0b4fb6bbde87c40d3148dd8b35

49493a18ef5cb5c0f6e59effede05caa23ec92ce1c7b76ce49338680471fbae0

4f8dde6186dd006f6f2a329152fed9ff906c3213da9ac5c84dbcd3a891bfa2c6

response z

5c416a5d516b33aafb55056a27f4078ac692f9c62dc55e8614e48bcbfda9e9a8

3efa20651e3914fbb2648e58eb5b979e502046bb964850b9927dfc8f3e4bfbd

441cce83a27c903e4e64a8228336b85355ad702d486914d9cd6521925c3971fe

c6284c1ede602e004a50bf43cd441b917e52cc8f3bd4ce066d47fdb2ca74fe5a

86459a594bcf4500567d5392381d3646e6359d4f95934bcdd179257f00f0b4e3

17cb478caa3a8008414605b41ebf475792e50de06f02fd7aa6296125b0628b58

fc9381747cf66c7854c7985e74025e00082d655a9c571822e7d01733357e1307

cf43f80ada9b316198030a44d279a2c22ddb51fab9c64533e68621f067e46396

response w

16491abc3679a70e0b76c59f98507db43bcac1f8e9bc308f8c8ddc71a51db334

5908c09c386736ece24699afadae40a6aae7b47ccd309acc0c7a1e31446fae8

7e1eaa171a7233e2aca1be530e575129c621d2e4fae26cdc0c7897f5d129c380

47725d3da34481098133df9df95c176143b7e8a75e321df3f38d796485fb7f39

a70eaac60758dac6f3ddce2736fee485d0b02193266a9658e6452eb1fcf0d9da

8b26608114573a917274e80feb071bf2b8c08df2e180080c87a1adc035fc3b43

f4dfea488347e0786731c662045d8b25f25a64241cfa16cdaa0a901de67edffe

302235ac8cb9130d7e8fa85aa3d8494f8b565478bf25208e2ecd2061989bae2c

Fiat-Shamir context

8cc4b61b0683c02c6cc1d5e878f134b0bb2d71ccaa84628ee5e12f38fd1b5a08

withheld proof term: 22013 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvEuler definition, MatrixMultiDirection:109

```
{k : Type u_1} →
  [Field.{u_1} k] →
    {σ : Type u_2} →
      {m : Nat} →
        σ →
          Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k) →
            Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)
```

SHA-256 of the statement above

4f3184b863a9faa5c4dc355aeb9b763b977649faa8a2a5abdcdbf82285f73e251

SHA-256 of the kernel type

de581f7e3fd3f936a1c42dc2481c501b1c185410b9f4db3656f87755bf33888f

commitment C

d283ac511284b374da9c186206a18119793458efd6b8b160619539aef530d74d

9ffdbc581fafb9ac0940b03a89c4647d810221bb71874b1b3bccb8beb3297fb1

4e2a18b7f217832dc048a2cdf91bb0cebd28bc3da64e07028b507624ce185d7a

ba0c37d94e70f207d7c4f293f4bc8743f767109f7465e13189a66f1ef59e917b

51a6b8aef182371b64f4132849783395e1bee773291418232c5f0329ee5a7189

84f3b4d7a57f5f5953d5dbffc55174bb441fbf4bb03954d56f83a4ce0ee7219d

4e8e0f2ba3bf60ee2e7c6b14c268c1fa06fd7ac0654dfc30af80e536d013bd56

34e5021f04547b7b301c5345be3409ade3d91af6f95c8724646c32ff04d56bae

first message t

439d5cdd5c8068584d537e527738a3b10a162d807c71cef807fdadab3e3332a4

0769430a5a36a5b80d96dda96c8f4bc41e82bc2d2ec454f202af84505c83221e

4798d67d4a38fda473fff1bffa0dec0317f45f09ab5b8e6575455393f064ceb

53f26e0ac9fbfce74a7628d47200975019a63f48e0d0ffdaf17e462ca61b0e01

a38bd149de069590e8289a2bf45d9b7b05f20f16755e09676d2242ea71904918

97940b04ca401abd611892ad0aa6b1ea9da9533fd24306772df7676c0d5b851e

d87fcc65b2d6ff67bae54d7740fce6aeb9d3351e1ed56d24a94d9c8ba4045fe8

828023480ba20546c0563eccf5303459d4717f96f5c9f62624220f43d0218fc1

response z

785e4a5f7f54bda4d3ea4cadf38dee2f9fa31740f83596ead1fb2bd21ad6a7d9

86a8c06fb0345df0f210c4061017010ecd6a5e81e1408c807c3fde15220d5992

31e035b7261208944ef8c46f890dd2ce2e3a8647b87ec5e634fb7671b3c6b969

532445fb70cc11eb1ff9026af930cf2d08746346d24f3c0c9e8571cbe4392b50

f0484f166539413c7b78ac5ec3085437918e9f81c744cbfc89d763f700d01645

63f18440350a5b7fd3f400e9445f1bd20ee241c077490078fd35b94e3a0fecc6

0476ff85a3cfd6cc04e3aabb7f5558a078cc8fced63e0871a17e961fd870551b

6bff9445b1fd5c84865016e87de5bbac93d23c59aa0a9909f2cdce4b3026f762

response w

2a78ff1b01035e9d75cfa3a5c6654f3095c1adcb5f4dc7be3b7bad4dab17d92f

18746b9f2f6edf16f915a266dca977a85883e18125bf8d03b08071c3f9d2fb9a

6220ac951a540c5ad37313d4c4298472d92cb3055d25e8e73b1fb765fb8884eb

4f7113eaa925157b12d7a7d51c2303509d35ffd9cc21ad6874671c42ffff6400

c74a2afd05c45ee5ec9bfd60dc8462a0efea2cf867621b65a0f050266011f9c3

a821a8e9b048f14b6c2f8f0de08e16dd1e19c92a337b6c1e25db9f32430e514b

a2f47f26188595ed1e5bb096d19a7a7312c33017e7c91d1c687e30e2e579acd3

b0e97b032e9db379746e08d4d68219af512de21ee05782802a4713f71bd7ed47

Fiat-Shamir context

7bdbdc2184a928df75ebcdd4c17526b39e6b9bd283aea98eff7a4824fbaaa74

withheld proof term: 8122 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvEuler.eq_1 theorem, MatrixMultiDirection:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (i : σ)
  (Φ : Matrix.(0, 0, max u_1 u_2) (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{(max u_1 u_2) + 1} (FBundlePoint.mvEuler.{u_1, u_2} i Φ)
    (Matrix.of.{max u_1 u_2, 0, 0} fun a b d =>
      HMul.hMul.{u_1, u_1, u_1} (↑(d i)) ((MvPowerSeries.coeff.{u_2, u_1} d) (Φ a b)))
```

SHA-256 of the statement above

9bf7657362e8643c1f335a6551d39069b267d32e78ce94037c6ead5a59d63050

SHA-256 of the kernel type

4cc62f11df5d442f4f116094852c601bbce5084aba668773ebfa75466d61da8b

commitment C

214e4015af387ba82ca9446d0c95ceefc56819a7261d17c79f3027654714305f
7bbbd048ec02132acc3a83f4285591e8b90adb79fa37b611082b98e62fbffab9
dad6d02f78054177e9d0f7438df32382514a0d77f67b2855caab7e436a23101f
e94e4a941bc4808bdbbbe589f7c6ddf283b91764ea38ae3d72d4c670a961a4cf
af755c572f491acdd8d657ba457a70659c62d9c98d4254b34e2d13e99b7c5d63
ef2175bd53b7e49fa87b14083eeb4bed6df5630f24fb072dc080996b4508d7b0
06e5723ca84832026f666ace55d03a65818f9158223d9e731a0d4a34ab72bcd6
6a2c1c77b414f1e7828f0fe21ce589b4525a2d9dbaa22e74503dc3a26d489a58
first message t

6904166ce89e5500ecd3cca216dc09fd0ed8bf7beb87a5625da749ddb96170fe
8b7902cb97274cd7976e84b9a81545ea7c853de22d714c86c7b78fe95e68ffd0
b4d769ed53c0f89e00284e1b6e1b02db91b3d1df42d224edc0af9ed987ab77cc
79237846f9322f5d820243c085e151399f98e6fa20df5db87b566dfd18841382
add62e085b5d72b18f56a178cfb15368218f5a2837b879feb378c3a3f9fd7a73
36750d2f51c78448a603da0ef23c6c265e615ce7abc86fc0523015a87793cb6c
2d80e8ebb05e588699ca785fc3bbfe384bcd89fd529baeec334ede199d3541d3
213a9f803446c4aa23fdc3be10251cfe708168c1f024981f7db1ac5d8be864b0
response z

4c02eab8498559148479227bfaf7335503a26f625f12d20bfe9c6357c442dee5
553bf1108e998dd4b4c8e13a15889e1b238328b72ffa1601839ed7711ce66278
33260841937b7b88870f0bf95745917d6695a16c60ca93082b5db491a5c9e461
4546d90f26ad9fe17401a3964cc2df513716e215185cea9e1dafd1b8c21f7c53
c2ae894dd63e55fbf5955d02c7e98af414dbae4e3bd1eab5b89e1df0966b5660
0919096f0cbeef735704eef05d86a9ce448334f2a16be84d00f1e53c5e13c484
7140aad7dca968f1443f792ca227d7cff0a9749cd3fce572421b130d39e4cd10
897b83a9ee9e6830ae7005be7b35dcadd46fd481b7e0c3cd63d4aae6169b4768
response w

a884f09d8b494dae7899fa183dfbaf4558c32e988fd279381a11948c9aea1908
7223b5267fe0af6c541f4cdd9ef365db15bccac9ea70c6c3a7f03ac76141526f
36691466e3103b9dd37004af4985470c24946b258b96d2246135611ba107527c
e642defce03082b2d8fea82caeefe05e147de13c02077215a31fd481567487e7
6992c864d550ae16b7610ff99d6b0e2eaf2ec14fd045bc55487e2c5d0d1982df
7001883a292ac16e9b4cb0fdc707e2cc2eff7113f3175606fdff4f6c6e96827f
bd58b2a223cba0b312dcfbd23372339918d0d2989d5d2e68d56a1389a801ea8a
4f80e831b0463fc0d6c3497759ebf0e063782ac5daddc0d17e68f4b21d820cb

Fiat-Shamir context

3f740c01f6b9524d26ad32296c52b3e37b3524836f5bbbc4c8f577c1041d7659b

withheld proof term: 337 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvEuler_smul theorem, MatrixMultiDirection:134

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (i : σ) (c : k)
  (Φ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.mvEuler.{u_1, u_2} i (HSMul.hSMul.{u_1, max u_1 u_2, max u_1 u_2} c Φ))
    (HSMul.hSMul.{u_1, max u_1 u_2, max u_1 u_2} c (FBundlePoint.mvEuler.{u_1, u_2} i Φ))
```

SHA-256 of the statement above

44094141d65928dd54e511b1b8f11178d42545c6e4b01c64bc61d0069639b232

SHA-256 of the kernel type

5b8b6af70b5ca7469cac95ac2743922c6d5ee365ffb6b88f8dc3b2efec7f9537

commitment C

bf2d923100aa2533f96167befe884b2fe4481cee479b2853d54979ef6821a60d
f79742af2983f1a8d92634627bc8d274ba2aa2ae1bab23f58f24d802c9df1acf
9eb1c1e67024643483058215327398beb93a1280e78ff80aad1f68078ff54d02
0061e52dcd607eb4fed2cde3ef60bfd7fbd0c152483c71315850106d92124cd2
c69f4b9af63b53ff45be7a0efce89cbacad27814974597f902620ff346bffb5
eecd0e26b640d444873c1840e188f2be1d57e2ec909fab4019ba3caacd8f341
3b462c776f64e3fdef82c631b543df35b944c0348a443425242db9f8c2ec2699
8b8de9b0732204ec3afb99992aa8a463491a9eb21d04738612adff7ee3a16de1

first message t

9ac8de3bd1f00e4a11e552e69e51dd0844f1b6f385d40c100974ac99d8138fb7
20fe76c87ffb40200ec0aec108f13de145ed799355d4540a1fa208d2a3cb938c
95775f9b75eaabd2c51e2f63a59e47f71316476a9fc06559390d843cef9a4709
93e4cdd412aa10c18837106647bda22b1396369928a76a833d807947f78d5dfd
2fd931de0700eb653743e03feef201967b4e92bd07aa2b6ab67a6036c5f6ff0d
3b96cc7574bf17917312d366f5520addfa8c1bbc656dcdbd2dd72faf73bcdafa8
f7b8ce7523b8e4a53fffd6e02105db6a806bbdbdd20420cada50baa7acf9be7
c2c3b82f788761ce0132bfbe81abaec10104075257de8caf808decfa1901027

response z

1f6fc0f155d85395b78a061fad848e8b3b7cf22f0df00a62811a4343eee91e1e
1cc59fb853ebee0b070efff218e67d9a36e38cf0ea8c60397c24eb755b3cb3c
f24f0a227323bc71e28e37dfafccad8c1d9da38ae31ee728188f7a1fe7ad104f
f23aef7a40a2b947143da9f97386b7f81825916ab7af098f291d19cbc1b71197
a91aa154f0d46a4c84f583d40be981b83e7c46299046d239aa9a62e0e5a1e9b5
246c7b4656fc677e1a9f588d35d088f009265f719209192647fd7960910f79a8
0ea2d8b0c9ad41c569ce244e276d4ab54e3f2bc30af4412e61633bd497e69538
f373dbb381b013979540cfadbd8d6f0335e14507adccd9933acc746c7ec0f4f6

response w

18d10f354151622fdb492d1e954c218625e8fb70862e9e99594ba33b94f57693
5b2c789ff5ae9a0f1580515e4f07caabd66bd3ea3190228b7cf07d7e512be23a
f590923318188ee73e3148b3ff63d9881bf8bd152daa86b78dda020006da3813
5f6a4997543707261c37024147dd9acc70b4c34f45b05db324ebcce3868f3a19
166be778bdc93ac62017f1c427e683739ee9aef9705c600fa4a07324070d82ba
97e8dfa3a3c6f34d31f155fa43e3227cd282dda2d084fd0b5df50feec9c1a03f
c7b905609db3d8f7153793e94b748a74d4be7341b939974087a47cdb6210c7fa
d3c7a916437c6532939d556d94f25f229d7875b781c76308d914a757494b4c68

Fiat-Shamir context

520eb550ea66e2f0af8cf6361d83e38867c3d6eab3c817d8c1e49428adda2ec7

withheld proof term: 121 215 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvEuler_sub theorem, MatrixMultiDirection:128

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (i : σ)
  (Φ Ψ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.mvEuler.{u_1, u_2} i (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2} Φ
      Ψ))
    (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.mvEuler.{u_1, u_2} i Φ)
      (FBundlePoint.mvEuler.{u_1, u_2} i Ψ))
```

SHA-256 of the statement above

4a9a1d00d10259ecafa15a18f9c902165ef3469b425e759bcae0ee68a7d62d79

SHA-256 of the kernel type

54c5f57d804885539428356e97bea660ccb5ff636ef4d6d3e5401032ad63bebb

commitment C

1c01c4da5eafffb6d6e8b2849ab2f248e4789294192a0b865de5745af621fd48
b2fe3f1a149b514fa20fea1f7d6f380e5f59685899c146b495434e77eac0e0b
08b1a9d442aaf698a7c333a6381cb5a5fda82c7f9fde319bb3e8e6a92fad6169
cb81847c9785bc88b3d1a9dff77bd5b2f2036534eb00ac904f7526c8e9f0b1ef
6e41ac80b1646bb5b384803f358b2870a180e6132db35e5a4bcc0d9e6838886
bdbe97b13e345cef0837a8b8e4cc91563103525be37e64239f03750b311fe718
0cbc563ba4bb8a4a4c32c96fa363539277b1a6dc1aae3a2e93a2e3b877a4519
3aeadf9539479fe5aa893ea5c0477ffefdf8f1b5cfe084575746c64bac80dce2
first message t

24a6d47d153f1ca3fd5a9dcf2ce4b98d20d83d3e2af1bcaeab1bf9b0ab7acf739
eccd32389733929188f083fe0697d522092b7670dcb825b26935b19b20baf215
e98c04c1e1b5a3985a847b369741e42536abc09870dfa590106fa4a97d0622b4
ab021ed16ad8907cfbe00f7dbab03bd07d44774bcfad6e74720721dda11c8ee7
a958c3e5ae42e32913fdd154d34c3670e463b8f7d02f4ea8413681879c7b5b33
3f8658ddd275f8ca5d113c0ca7c502379a12dfedb7f170004a6200153d9cfcf
3f83bacc9657f78d6dfc82f9c4b9b5bfc7e91fe84b1c0757bc8193e159005adf
c6cffb572238cea34f3067e4561f18e292bcd18aff416043ba956d47c8592113
response z

4835fcef0d486f16e58e5e474c2c9d59cbd3f256dc391616e72439af3fdd5d49
b07dad63c531bdd9e60e64f1e0f1471db3488dcb181289ec20524f173b111f6a
0007cc929c7d54a2cc3f19cb6f3259c8a6aa27aba1214b0628414e083b2aef06
3c9ef1c0e8ba697fd1b1e207589f56db83297275a33eb15d999a3a70966c0f9b
9ae4f581569b1e368940ab6785ce83916adebe9bce34c6ed1596aa2c9ef6f6dd
e4f20426c085c53693a3070392860455dad5989d9ede5561db30d4061bf9415f
3f047dfd97350e59c4d81f36dc4bcdcd7071200a181e6066e0632f9072f4a98d
61fda5b7b38d5dab7e676640f9a936ef733b1c0c8c56e3634efdbfbc312927ac
response w

3ca8b19f31c1d50f8f1290cc3dee3763c317c20a3eb03f5cab18b22c59adb27c
ecf35e470e2184429b6eff9ad0ef328a846afa99fd34579fc4d6641bfa024239
196648c63158c0ee9183ea3194a8bb825b33c2be05e1e6f40217def8a2caf5e9
a550f02f28901064a07abd80a99ef01cc34714858a3b35623faf973b03a784ed
6680b387be8ad44cab2dc551c50b4fd2b186a4b0fcfa6ab1b7715d7e3809a473
b341e28258a245e1d13bb2a207c0f1b7aaf262bba59546f83a7181fa8b989cc8
05611b387c2dc962042a8f33b279f8d4261333294b1f695d3496867dfc48951d
554580820d18115af34a58dc9f456048a415376d8b9a4b0b4a9f130977e4fe8a

Fiat-Shamir context

28b8258e4329e4251f88551a26f738acb4ffdba1865b2575c947b0769d83a69a

withheld proof term: 104525 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPderiv definition, MatrixMultiDirection:95

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {σ : Type u_2} →  
      {m : Nat} →  
        σ →  
          Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k) →  
            Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)
```

SHA-256 of the statement above

4f3184b863a9faa5c4dc355aeb9b763b977649faa8a2a5abdcdbf82285f73e251

SHA-256 of the kernel type

68daa820a3fa937b7d651accce713030021494421a8e2ac4156d9a5b02eb2ee8

commitment C

81c97e0f61a5c0ee90b9845a2d7023d6235f050e2b4966af81fa22fdf9df5294
c99263d70ba1933f325f2420f2996d13566efe7be078582eeaf01df6c27c0c5
06364c622de470d2a0ed885a1a008e817c1b510663d10f37062148763e192e75
8816c433ef89e26758478ebcda5f09e9b75a995f997fc9ecc80401f59524c9ad
9a291488431c6a70566f6ebd37b3db25f3f10988f1ecc5625d5891b3ebe80c76
17c980ddf7cb2a6862b01a4409c543af6717de85bbcb58cfb28ee8e65c2b0dc
a8d5737ab4258f7c23154223f125bb33f02bee816b7d227714478e99dd0d4b94
e7e69d3ae48565fecfb98ed4e17400c567fb522879771989eef08ac868c0b012
first message t

9beff69d4e2bf2db204828f57f4c38647eed278e36496d5e2f785f3853abe3ec
36bd14811e1ae458d5c63cff5cbb614041449d522a0eb2fb09987ba5aea1a7be
1c25c3a2a473e330d912b1b14ebeace7d497571552ff66c4c85f05ca5b8ef1a5
2f2ccb02d94e8b7e7ec485aadbedcf82829ae9028bf67007fe5f9248219cf6fc
631f082e1b2cf448d6f6e0c0f909cf59a71755cb74498c3a1790de5621316fe6
2984c6661923b229761eb3d8062814577e33934a44f271f5fa4af7339b15d276
cd7f6b9e44f4183c1102c9a73b4156c83598d304c4aa660be12ed8f081266660
ff919f2123054bb51497f93ed1398023b8c7b196bbf7157bfc7345be3c6f674e
response z

3a57145d45686d35292837ad8fc669cf38482b4da646a2fe765f0c333749578b
b7461474f397ec90ce27745b0d39c99f0fcc54e7974b35268ccc362e832a8d79
e7a0973fefda574c57cb83cd28ef3716a8033c1bc3f3f70dbbafdc8ea8cd110b
c513b8282997e6e8f9e4a31d03ca3b3385434355a1ec0c73f9ad5654a24de7b0
c7a064c1999a29598b219a5c3d6fb9c685ebcaf06a27cbafa30573d33e14505b
dfd5d65726077d8328ce76a1f73c4cbfbc21caf9f8b08fd208e5c14bfc71b8
e9a8ffb9d4d890e5f3d7167a7fcfc517e27c61f7d850df75acd71ae60238cb3a
2da5ae70065838230a95a7dabf062a1a38902c7e630e59433d8ad29486846529
response w

a905ce809184c7b841e73d217c2a95c3574d7e770e593aee5525c5be4a1440c4
54583569c0192cf70c90b4d6c17cac5694e681b0660d0fcaa0cab82c001bcc25
05cca21c20580cf9aad8727f3b74cbcb929106ad738b1ee9ca3168f804df370f
382afdd25daad9d2d1ef7e3c00d9a713b194e627826bef17c1567ef104e3e226
99faf8e67dbbcf139b8d3105c7561b432251be1f09379197b3301ca8875d9992
826c7038cc949e2940532e0b3a2d2aedf8b00a40dde1a5496fc873ecd0b94d44
79e6a65b66ceeddbbf85377ddf9adbc39a3c7687c91560bcd584d55fce3866a
10e470a949f32b5c013604450e1977d919aebcb69975582e2e0c1fddfac33bd

Fiat-Shamir context

5491620599b055145f980cbe7401186a698197a0031995cde5666da8bb1dd610

withheld proof term: 9514 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPderiv.eq_1 theorem, MatrixMultiDirection:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (i : σ)
  (Φ : Matrix.(0, 0, max u_1 u_2) (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.((max u_1 u_2) + 1) (FBundlePoint.mvPderiv.{u_1, u_2} i Φ)
  (Matrix.of.{max u_1 u_2, 0, 0} fun a b d =>
    HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑(d i)) 1)
    ((MvPowerSeries.coeff.{u_2, u_1}
      (HAdd.hAdd.{u_2, u_2, u_2} d (Finsupp.single.{u_2, 0} i 1))))
    (Φ a b)))
```

SHA-256 of the statement above

879dbb9775e87c9c57487f267f821eeadece64f0984ed9f03117343942a777ae

SHA-256 of the kernel type

e40a6a7b37f2e93d95f908d7dc2961f12549ded41a4312230085170fc8f0c30c
commitment C

103e8acddfb0af1d52a51734b947fcd9d32f3504c33c4bd084a258fc8b205873
2767d72938bed0131297a7c9685916c2514711d25d75577bb09e451b9be62dfe
2ac8cbb3ebbd8d1cf3e8495a404033454f4f907d660a92daf7d954c3c884c189
a47b0ad1d06c3af27f6bde0aa856eb704f1e551c2da33aaf94a41deb0032357
08376531e183aba3c79703b9b4ec4447bfde4f4f4d9d8567bbb91427cb086604
40dc5ea6a2f333e9ac317f3fa33c83221310438c1c12189e8d62dacb13e818f7
0847b9333dcf3a31cda1cf6eb0bcc7ec20a6bce635c2e8c2fe93ea675de5428b
75a238aaa0d57ceb8a179dabc6b5543df91f522199a5cf402a3ff2706e167961
first message t

e8c700630f2236144f7b53dbf88d8d4cc2547fe39e7888a4054b68148ddc25a7
e951bd6651937594f3692d2c59ec700400f5a0ebdae4753cd77c4cd973f09ac7
01d764a9b6481b6267cb448f2c4bf3527869972c8f5429d01073c92304904e45
d6c9800e1f63df8a97f18ca90315c182a846c84da69851a244c0e9a2046b3508
fb8247ff52d63de01651eb761ed06f151d7fd896cc0a0f650c35982254f8454e
4830f64e6d55947c0d3300667cac996f1753fc0b71c50208356a779cc4d7aae5
4ad0c68483cc84eb391ce298a2f188dec7c0b4930acecf567ef2ed5ea63c6972
9ad55138334cb4622ce64b3c0aff09b133762b9d8e0c1f376730752d98bb9cd3
response z

6bc99120ec96d6803b704b42cc62c316eeaf8f5317112d6b08b3ad84d0fde07f
67dbf3da2ac6188d3effe3fd12ba7067f4dc78c2c7fbc1abfe9a13b6be51601
cb8607a03c6ac672a19f99633275967cc79f418e8f6e2f6ce72905f382813308
36cf4f1fb3835a7e0abd77f463ca1ac40442a1caf6ebc2b8ea7e9eff78914488
bd9dfcff86a6e1d760144ab4789c6c94416d36373a5cbcc741fc69c226ff262c
bc0e5a72c8ef5a70498e25127934fa9d275ccbb6946dcb7f8daa3cc2f53a86a3
f2add92bc95db8cbae64f7d5b85a5515fc642760b798fa35455cb2b509d70d55
fb12ae7ac3cd2cbd6ea46c4f78dfa6e56ddf6b881d07627504a7bd59c3f25820
response w

71473d2b6d93b301732148d88d4a34ee675f1b7c5f258548f94f5d7798328c76
6909a338d0d2aacab969ad2486eb36965b794044c09f2d4f90eb68d3b34c95e7
8671a930049992fe99a1746645cd2a3c22d924441c02887d373f3ef5c3dc3dc8
9381c08311d6fd96e986e50b41d305ba2888836c4442b22fbd4c6c8213f76bd2
29926deecce87feb7ecfc4c72f3dab5c1e4f5eac144e25b08aceb133e6492057
14635a62368fe240bcd36a786007354116452519c9b9861512b2c68fecf1524b
64f3707a388758f8f244e0458858be04d900f02b4f752a07b31a4b5165587e75
c6b07d4638497d9ed624ab0c2ba2d695b7d4a7601f2028dc881705aa8af38690

Fiat-Shamir context

9c9bf3aa03fb6cb07a08aa750b0dbc39004858eb79530c16d37ebd404009a6a2

withheld proof term: 338 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPderiv_sub theorem, MatrixMultiDirection:122

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (i : σ)
  (Φ Ψ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.mvPderiv.{u_1, u_2} i (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      Φ Ψ))
    (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.mvPderiv.{u_1, u_2} i
      Φ)
      (FBundlePoint.mvPderiv.{u_1, u_2} i Ψ))
```

SHA-256 of the statement above

9bd3c2d62a86319280021bf3186174d07a649899e367a06cc3e29083c8f8b24b

SHA-256 of the kernel type

358634a3d382611d336df3dc581976187f41dd39eefe76c092278f844673cfc2
commitment C

402a7877d7700f5e26ccdfafdf98a253b4a1a514bb3f0abcf8763c22e59b160
322a92105d036200520500896c9b8b6a8257536dcefdad849423009c05fec0c6
db0d13e8d8c2211b4935527ef82fb80697c7c8d1531ba0c424684db02142d02c
8d66ad0b6710a40c5d0a291f0b3cd2e8588f7ebf3c265d337f04f892478a781e
e14eaf6ecb3209051469313a8f3f62bb83a76f8f99a31155dd69efcf2032bde
2d53c3c8a0861b4c67590e21835c19f446fce2ad69a641bf13524fb626b3b4e
b374fc90b22cb8e3828faa7e2b08e68d65f32fd0a5b68964fbb8a43110bc89c3
85a202f0059d3ea90d79a9f9818d846d0d24b9ed88a13d887124eb4667f6994
first message t

197d7e3ea46bc7acc81e0d2b09d1b401948786f397c8f5171ff8f6bb84edcd08
7294d362ba14ccf5c91c676904bb82b6ff5ecf2008a74df0f119069c96d16d17
a376fcfbfc999d8a0fe6dfd6f94e52ec056a09aa18623adb516e0bd6bd55427fa
d1a2daebe4c08071860a34dd42e76c6a3044e7e188d6d451a60ec5716d50a89f
af1fa701975015184bbde380885934f258741ae41e0d374922da8025f02e2976
c78cd10e15cb3a1bb4b51276432b1f26003f5e7f05e21c10fd618226552a7138
b8d063ceae311911cd98d73bfb8f589b2f910b3fe492494aef85b10f46d340c
d704a5068b63cc2067d389335e647626d318fd0c90a60c5e1c9140e8c628710b
response z

230772465348884738177519ffad78a8617613457e2765c9f833a6a6aea5c467
001b65992d574e814c97e06e0065f4498f37a1454049443b68878b967ca61ec6
cfbaf45f7e71258f5a2135c8754276a28cd72b0b1f8b8d54318a644d89a4d9ef
4260f056fbc8992187644b4376029bfc33d6bd701dee94c15e54a787bc1e5502
9d8dae7357c26524be07a8faef15a2afe3ee8c89565388d47b8df06a4d734e4b
1bfd452017a0df89af758f95ee5ef9420f115ef60dba0239677c4feb66cacf7c
e39caeacc1b9913ff0b7642b896d1243835c5fa19ad9ea11494c4fa79ef7d8c2
95d5ac4034df1e6270daf2ca7c852cbc4eafbc0a4ad88835173e80983bb1c582
response w

6d5b1881baf0e04fe1664452f653d2de4b528c9f88841afc5d4ff339362071f6
fa5d31d743d921f92fae232641377ddb6d6cae20481b18ba871ecc0dc2195a90
66591c4488d50fd2d55ea39fdab38758a6a14ebec06e3139b6cc722b38d7059b
7dad9c547a97ca34b0e258363fe093da971ff91350274a653d5ce0d4fc7cc23e
5455646ba5b826d8cbcd275695d3e62b059b1c946ae55c73a4fcbe1a2162d7b1
eed5968ea5d7db95945499a8801851c4261b208fa8e7fa642664cca22dfc6a4d
15b24d817123edbe0822dc5f3bb2ea57a4dcba5848a34cdc3420c98d597c9aa6
2259854644422fc971c4f1930b7c9dee8fea5d8967ea4fde65175ed2b70b885c
Fiat-Shamir context

13023243a58c70b41387342798e9a205f1f8913c191546737ac5c6d73a0cb76c

withheld proof term: 143 578 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff definition, MatrixMultiDirection:41

```
{k : Type u_1} →
  [Field.{u_1} k] →
    {σ : Type u_2} →
      {m : Nat} →
        Finsupp.{u_2, 0} σ Nat →
          Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k) →
            Matrix.{0, 0, u_1} (Fin m) (Fin m) k
```

SHA-256 of the statement above

3a7b11f9b63d1efe3cd84ff93b900aec8fc855c3da7372daf78bb01f989d2b4b

SHA-256 of the kernel type

bc8fde87719f59919cbcb630ab1274975b682487d408fd97df755847f829cbff0

commitment C

6d246a8e7e819258a1f0c3f1e64fa0fb31dffe0b75fba0c07b14a0582b4a1f4f
a4ac6273f7d7864d0261b19bbb568cc923ab2ed8366554218b50f81a6ebadb62
aa09a2e570cba47d6d7c2764d38d1561db1437f3b6f6621b4559f1d107d9fd1a
bac06f6f89f2cf5a5bad41b85a760c0d76fb1b7c8b7e7316cfae8ad847316fa
8e608b0ca8cb403d0c218206224db87521673339a048070b1cfd3cf2b2191a38
32bb6ff7204b113a03b1b5d3658d16bef94624d8a2d4b21d03c2b79d476aa549
e6ec78333c0c283554316e9f3beb3adc9ac3f2cc3be2d24c596e3d72d21c9eea
2eb0be170dbfc275738af1740100bad70c5389788f6db45c0cb1c9e6f58c1264
first message t

3e76da06d07254a233fc2ac13839cee4f741fb4b3dad623d826174b45bb9c95a
dc36626d14a08dd7be25dd70b30c5b9d72dfd080cf1b0145bd6571b3152996c9
c32c7f46c4b4304f05b697252e2ecebc241e0fc425a9b38ffd3d8de806d93c0d
476b0e1bbe28319640a1894c1a40cc85784bcd14ee4e0431cff23d9681f307e4
a2e360f822d0e2e8ed88a44750fcf2d61c23476c2684c6baef571177ec980a06
de96e1cb61074a2bdccff1a5d20b026ca5b9e7ae799e219193918d6698ec2355
30c64a335dbc3ce74f86ec9fe6b4f8788e23dca9e19bac8c9ddb22e7b242ae76
1d49a694d8d397f089f0f5c7fdc69769d95478bd6ef98241dfeb1a3e5e04fb1
response z

285ffb1aa6765fc1c136ab58339805324ac42cf5e4b1169e2e3e4be27b7100d3
ce0e818b3da1f250f823e4433ee61115cae2ab6102f9f1a4bb401878f8ae5eb9
403c31ae21a53d51dea9682116b801c3fbfa6085c4fabcf0b49cce038ff709c2
19235493bb5739f49b30b22ed17ca6aa2b8f4baa6d5ab46a7149308a19a2baf3
a245cbd023b07fc20f7b33d9b26d0339fe92430b923eab115982dea6406b90d3
68b62ce1cd094a877130f9d7fc53252ac29e66095175e8bf3d95bc236959f885
ed99ca8141cf5b6c58cb1a64fd88c0a1d91f16d17cef7f2aae90a442f65e4a11
2716c9e5fbab655e14d9ac6525d49d7275e36485e055dcf51c47be7494f40a2e
response w

3f6b0dac8aa9b4bdd9598b8e78cc8b49b57179300baed721fa40ad165e949086
67a7229d3da0b1caba484dbc2086540603d0321e50b360498c5079acb627e089
8e5ee486bcfc7e2beaf07229852a982e7804df0e045f9de2ccb555367cc16311
6d3e88d87e0edfd93c7bd2bd5aa1e830200f5aeeddc28709b23e86e7ccf31600
d2e39b6c17a9c7cb1ec3e790361483d74a4a9bc73501c9b62e8e4050de8f9ee9
84fa61ea6f1801c16e265535c36e8638b5371ec7573c753ff6af5c0c59d6d3ad
69bd688881348fb008cb72777a2b9f2c7538dee69f15df7098087a8c12ea7596
6ed3eaa0efe2446c8bd9e2e094527eb4b08d49a21de10249e3a893282dd7137a

Fiat-Shamir context

0940e7c023da54cd163b16bc384bce1d3c7b6ec4db8789b0c8bdcf0df4a42e0c

withheld proof term: 6 496 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff.eq_1 theorem, MatrixMultiDirection:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (Φ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d Φ)
  (Matrix.of.{u_1, 0, 0} fun a b => (MvPowerSeries.coeff.{u_2, u_1} d) (Φ a b))
```

SHA-256 of the statement above

da940791d5c7bfc212ba4a799f7769275d3102550f891d6c958d2004e15c68d3

SHA-256 of the kernel type

92fa003d5d70eb14d3e582af5b6c66f26d7b77e4b764511155d9e966e0a87fc6

commitment C

acacd989a2fa98cd7acf4f5c895f6d12632bee92202bd01e56332e059a5c79e5
fc5247b2e6c2d1def3cc52928a15a2c578a1b2495872dba3f147a6c8c13ed9ca
b0407d820520cfc4c308c26655bb6d7c822c3fe7c11de64f3f5f00d2ecfcb959
edaefaa43b5e759650d53aeb0ac522cbad8bfe501e8d9940eeabab4ca2ce080c
f8c2c63666f3d578df33fa2685cc9de859022ce303a11ed35fe039a94fa1bc91
a593ecb71189e39faf4253d0de6394070abf1bf75240216b1dceef38873c8a21
57c7a15b181ac4fead3d48f8dca9ed6e9d0df5fe442b2af541f2e5e95f7502b3
12257011065d758b3d50f401ed51de46f24f6bbf8bba89013a4b3661ae25fa1a
first message t

bec1fea69bfb084986ae3b968b3c5f8bb5e8d3a36c8b9f104d540aa16ca26f37
2ff1e4fe793f0618666dae38d6fc3a603ceabdb25fb60646c6678c41c922433
8c5c3a7217048c015787bcac43772704fe11eb6d506c631561553f157934df36
b4c38e275ad03741c475fcac5240f5f2a579943f3eb1ef4cbe89d79f33c9561a
9fdb7ada943f4108b559289e2148d813e594ceac778bcff29d2b2599915bc641
711526f51a453752a65c792692751ae4037d338d3a9d13fabbcc646f46bcf010
74cb2858c49eec9a73587e47aa1f9593fa9d1520f2020a767a3cb03be8677881
7a3e8762c2f3fa7a3e8af65922cdcdf1909912f63908d57dfe1e0ea22dde694

response z

42977b94cb3f6801972fcb5963b7d8383764bd33ed09148f80f12cea3c228927
bfce44945b8712516f762fcee4ed5aff15cc2f9c9a4f902840fe2945f2387a58
abd1e42c0d85359082e726c70a50307d2fc7c703027d602f021bd7af500a881
4d2775e1d44d92cadcee2bf6c9480f2da879b19a0503b13d37f0b99ff269d5c6
9085b2098e4c5ec4ebfe3b3cb5ac4ac5b9754b6707f4e62450e3ece1f2ab37a0
281f5f058062a0def2a39b73317fd55a28df0e133102a16ec179a0465d2fd48c
f1e996fae728c08eb2e76a7d8739a8e1510018275c73e1a5e86e98d1395e2895
9a394e39befc6008933e7602789984e1608fe1017257bdb1296acdd5ea035680

response w

3b65a6dd59c49f50d9987cbe8ce96ce0941b98b44e9a686d928f6532d099cd32
835390e4b4428c92626194c1e6b69d5fba354955af3bfae003e0740dd154f5b3
0dd1de3d10ad19d3f0aa7f89fecf28a24f55f61a617f0533dbee3f1fcc5876ba
9aed48600e0bca486238f8478a35db813ec5ca2edfe2c5a76790a1842eca4db6
797e1f997c2d43a76f92e973f904a9e2993b685683d996778a47546602e76c0b
749f581f131b90703b8e58ec0be25c8a14f2e336b4e41df56103e0f22a61fa45
75e8106f96cdb682fb7563c7c4ff3d39f833d5be3efddffafe23eab8bb93cf9d
ee9149a1303d40bef619e3c4b2605032c281aaa0d0ee8300ae450bc5fe6855ed

Fiat-Shamir context

cac52c2320e20ae4a7d56215f34a676578167310837a0bc58841890c3fa33d8f

withheld proof term: 363 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_apply theorem, MatrixMultiDirection:46

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (Φ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)) (a b :
    Fin m),
  Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d Φ a b)
    ((MvPowerSeries.coeff.{u_2, u_1} d) (Φ a b))
```

SHA-256 of the statement above

5a149235a17c61168e74aa0cbb088e052ebb95a0a6696fdd0f7a189d282f11c9

SHA-256 of the kernel type

6438f7489641cbfca9bab6449689d8eae0bfe41280ab0e0c094ecaf9eebfed3c
commitment C

34c90d77ef507dbeb14ea1971e03cf388b7d9a82048d388a0e4489b7f47042e0
9b4eede97de5ee9c09908ae7b3454d37f8e02d3621e89c9e2df4e265719c0663
31792d3fedc15bfa09e49966dadabb966620880b96bcfa2439d2578489daa88e
c8eb59564ebae40f07f2f1c7be008c50709f0ac4755222f33124f4ac6c9b2bde
2f737e481d818af72f43d174d791c6f76894cb0aba8202bca8a7583774be47f6
438ac0a8ac458835a554e1b4f2304f2b412f57397c5ab9dbcca35949c9ee84e2
740977c86b7b2e132156ed4af6844be73833fd6346a7f82492566c1bb5b3180c
2c6dce3bc3b61bcbafee4383e4e01a9f886a0f273ebcc6338f74ed317ae257b3
first message t

bb67f1c2e7a06b2a9db1288fdbf7cf868c1c8f2f15b1bb503de0dd3d902bfa71
95ba0bd05cecd9d2d3a04042c1f63872431fa775f2f60f9b174e2263c7f346e62
1c76a55c35f152d26f9d110941bc813d5e596a83423322e70a0a116cfa220ab0
46b69fc2b957f36fbb55371687e619c20ea841b8a3540934298c3a412ee89ce4
753f9e396147674ffd6c931ae7c0101d9e001fb85dc56e7cc9c8e9e11fa5882c
0a040350e63c5019075669d4133b1ac6d6a8a43fc4585b809683149e778a462f
2754aae0eb2de9708e424a177bb872aa8432a7ebcd813781751d4532925eaa84
5f23069418f04ca45b9a8e8f3126cfba4aac9a18bc0e2e7558eacae35ada710b
response z

37da8210b37181885f2b025910d10e4db5cb337b6522f5072382337e15ae3fc2
9becdd60973f58c4c5844ee41dd8116577f482635b1eb971646ca043d9995f58
8788838418653d4d56e26449bf5bd9756680f045d745515c804a442667b92afe
eb1b39f94c1b1caf4eb68cedc695c2ff6c6a0be7aaa51885b42cb3301f49423c
58315325b695d80c48d72d639fe4a0a64031c6c6e3c67e874bc1a5658d91a2d7
328bb52f49700eb0fc4a34fe434c5abe59eff1208d153f4a0d5b2d2dd151165a
ce9ced9c470bc30f1a1617eebff997bdd4bfe5b06438c41d09c6fd20195730a7
25e07c5e743f02f834613f1c065b2b8cb53fc80195d0e2e84aa737ff9650204a
response w

215c69562e084da501a24303fba741063fee0529b28500cd2abacc3c3bf2e055
930852428b2cc2b214aecca00d91ab9a017d40de276f9ae5600d3bb3eca2be5e0
4ff0bb59929cab8d3e854aaa46ba4246ed4d0dba8231febb5f2aee2a7d110c78
523abe2099e35d438775466e817f2a70541881a58836b66acb9f12e769c9ea94
95f3335817c01a143bb1ef4e7ebd56744a46394f23e09cffb16e36d9235a9f01
eeeffccadc3ad9cc9edc9c1f4e7424c52141fa3acfddec37d787007c7fc38c361
8d38b9fc59fea6b350cea574b110558ac5267b44cb59ad75d59daf89c64bf681
a7f9c500bc5710b58fb2735196722e36405a349812f54bb5a9859a7ba730b22e
Fiat-Shamir context

716d7e03ec94a56104563af2191dc603fa9de36d4be60a72450b076484d11fbe

withheld proof term: 480 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_ext theorem, MatrixMultiDirection:50

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ } k] \{ \sigma : \text{Type } u_2\} \{m : \text{Nat}\}$
 $\{ \Phi \Psi : \text{Matrix}.\{0, 0, \max u_1 u_2\} (\text{Fin } m) (\text{Fin } m) (\text{MvPowerSeries}.\{u_2, u_1\} \sigma \text{ } k) \},$
 $(\forall (d : \text{Finsupp}.\{u_2, 0\} \sigma \text{ } \text{Nat}),$
 $\quad \text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.mvPsCoeff}.\{u_1, u_2\} d \Phi)$
 $\quad (\text{FBundlePoint.mvPsCoeff}.\{u_1, u_2\} d \Psi)) \rightarrow$
 $\text{Eq}.\{\max (u_1 + 1) (u_2 + 1)\} \Phi \Psi$

SHA-256 of the statement above

4ef0c6ac9df304240509cbe7d29d79a271b45b72f6af99a04178eaa3490754b6

SHA-256 of the kernel type

b958757861a635db8c043d2cfa50d31d2403afc67c78290a84d4f82dc8c4a715
commitment C

fbce01b239c0bcd13e3bbc4a15bb0e553b32643c7c65a31204af3689adf78a71
2cd4a008280041df85d4fa0f2847ca2933d75048a92999016980246cf92a1ccb
498e0c03b85b36fc4eb7c4451a8b6ccd811c5bcea2bdc7b099c0bb56c3a04c3f
fa84639193115a2838602e9a9a54114e4d2bc907a90235ab55c947144f49c464
a9ae9e14b292d9e5eae435d57e6564b719c5db5a8176e5208ec4456ac59c6bc
4d8d858ec660f4fc56f7286a862ccf8435ad1312fe683e6fa93a2679b2d110a4
c75755f8564e126c60cdc82eec534d5176e7bc2a81684a272de938769c5fa165
e35d5f964a3cdd36a29e6b9f3163ebe3396d43a064ad2b620276a6055accfef1
first message t

8f214a46e66e3e997f29269ec1eed06deb2f2ceb9964b778111331ba4bd62eef
ed855b0d0433e1cc29301083f57a1c028c545ae70258546f58a3ea40b7d7ea2b
109b6f92d674b0455d7b4d73604067f30b72d5696d3e214045340c2496272b5d
996846a9a5cfe71ca2344107615d5d45bbe7e0db20b57f8b3383d9a99563617
63bc8ebcabf8dac99ac120a74c7a2558f2807923cbc70276a8889e6a945c693c
6849bc70275d6dbb817357aadd7cbe3943bcf178f7beb5948c0586b0c2194c90
38ed3e206c0f726f0ab2536148962496e10d7e4d47cd0cd5fef1018f93013e5c
f090782c4d15496d0e83fe85af45ef48114d685e49e8e62c9ff90507d49604df
response z

44aaa6bb5b744035eaa1d775a0233ae03eca1d03352a14914fed1a22eee5111
7683685e1ecf372ffa44c7a33edfe79959bb0b1f90943c8eb0053e3a22b424c
42dd0f7444bfafdc509cf08b227c9a122649979708e7764bcf85e06373f975e
a0edbfab46d9ca92d0988b7b9e5adee91865e93295acc2c62b833e5c43ecb8fc
a3b5f98087e36eca152b7b3094d5b3c9f88cbc0d99466458b24c9bd8eeaaaf006
7f0c874905af2ef435701120e1429651f5c409cda35b4f993acb6b030bff37d7
40926965fe2bdf72b899e4fe8d01829de62354c0c98735458253ca6d25097bc
56a0897eb99fa6ea23db72742a6583cb023092b60b8ce63fae3c228f878af1bb
response w

466317d8f4eb72fcf2398620bdbb458c6a8d616acfc45f5dfcf3627b6b062640
e6005c5438415838bfa71baa1687fdd8561739d87239427568d39b3a7bc062e3
f303540de252abede0f8ba51c79a917f7252650f3529e69211ab8d761c6aacd4
43fbbb964767a66b69fac0063d030905cf0a7750a081dca6f7d58c8c12998bee
f9c5824473deceabcccf1d737b32282612994854d92fed4d0fab6f220264c8
87a3c271186fdfb1ac685bd6293e77f3aec32ac9c138db66b7443210213745af
4280144741fdddc080d236b2ca6536cc493d4476221da290724a3e1c91b77aed
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Fiat-Shamir context

45c236581c4d64afbc27d78a7f602ed0754e5020ea49424eed1e6d99c4a7e35e

withheld proof term: 1544 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_mul theorem, MatrixMultiDirection:85

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} [inst_1 : DecidableEq.{u_2}
  + 1} σ]
  (d : Finsupp.{u_2, 0} σ Nat)
  (Φ Ψ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
Eq.{u_1 + 1}
  (FBundlePoint.mvPsCoeff.{u_1, u_2} d (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
    Φ Ψ))
  (∑ p ∈ Finset.HasAntidiagonal.antidiagonal.{u_2} d,
    HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} p.1 Φ)
    (FBundlePoint.mvPsCoeff.{u_1, u_2} p.2 Ψ))
```

SHA-256 of the statement above

5387b38a67b1c9e6ecb02f41c08cde0719eb9246ad1b90dada3ce67a79fc687f

SHA-256 of the kernel type

64985eb628c37e9a7c08401072c08ed1075a787e0887a8aa9ad4a552fbfea90a
commitment C

34b580d874648d415b9ce99fb2e704d4dc0eb9d76a781920ec73bf73d28c148b
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7727ebe670fab517c59bd5f4c120fd68f970f8b126e24ae1a26b3d15f2bbabfe
07dd6fcec011489b1ef0d7876004eec575374c010813226ec60f01840baa0aa
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0c3ef6f798f8db44efc50055471a9b451770833c2f05c778bdf1d9ce2d64d52e
86fb50b68c9d9b26428343cc2c9e511efb667ea8286eb5215ea667b05d2ec48b
8b045eb6fee22f26c475fab8c25879fc6f9e490bff69895216633c7a369f3a0f
first message t

d7f2a729b1a79c53704a342caed3281fe0fb73bf724ca39450d55d54bdfd18f2
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757a64ceb01ae630440e4efc0b54dd41ec8f3e15b4ece7f9136874f6272d9adb
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92bf27b19897d23dbd89cdcb9942e34665f5b5ffc3d38c8e307dc92159b8f2e2
d50c08e26e2158edc6132c998bc80ff35c593bf6d5f027a132738b823b591752
e7b8b1ff7eaa4e3eb9781dede83587c5c3f558ac0a3263f1db4174a50475fae
f511bade6f6eebea065a60ab6e7faa9c637d7c5663f1bf3c18433908f053d09
response z

a23e541754039cfcea369ea9c2a1af33fb97b9e72ed16aa7d3bdc121e443ee86
28e562c8960874457e716b31e787cdbaa9bd4e5c1d49ad12fc4e92c236fe1dfd
9008347227d8a408c6b55e84ce058c4c4118ae09c6f080b3469ae4b483c3e12e
f439b32b089cc23af0cd0ea2a902a88b4a52a555d822555da6d227e6f7c648b4
4f110df5983c63cc0e654fc12cbbaa4689931a6adf35d3d9a8585e6785f0d58c
847c2b8d25ed6daca5e08c4793cfc4b7b92caaf318d6f054afd91405e4238
5a0c9c00c6863ee03aa08064daa8ca654de372fe02b6d29b87fcfbc4454a0c27
2292d82c585634adb14be71a25e85dd6eff6cb89a855ff180c6e0a6f663e455
response w

211f42e2cf8f383572fab6d7e9d6ac459aa22a9659a178eabf3d281eedf7521
c0fd833914b8c298d0dfe9d9602466b3f90c57880e901fd20cac4e1851c36b7a
7ea15d8cc863b6e7eb4621feade5041326caf91d0aa9a14288b5f124332d74d1
bc1f7d5b1e84ca702d34ca8c044aca3c3fd2b28cac10a222e92d1d83d2e2d442
b05a23291e8fd2574f4599fdd9e05c6e66f2d583f3b0794ebd4a6989b7eacd4a
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47ee1b6998f644dae270b94d79662e329a63f6b21a61275c336cc3b81a216d95
a4fe61f4a63f51c98f13ac644b78d7b6cf26ad5dc937c700b95082ae39fc691c

Fiat-Shamir context

6b4ea177cbc656a9a182dce681b85f318281c008de260bf740a36f81802de0d9

withheld proof term: 418049 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_mul_eq_zero_of_le theorem, MatrixMultiDirection:148

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (Φ Ψ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (e : Finsupp.{u_2, 0} σ Nat),
    LE.le.{0} (Finsupp.degree.{u_2, 0} e) (Finsupp.degree.{u_2, 0} d) →
    Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} e Ψ) 0) →
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, u_2} d (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1
      u_2} Φ Ψ))
  0
```

SHA-256 of the statement above

df80961105659f0244358ea2a5c67b7a2cf9cd5004231fa0e20725ddd10c59ec

SHA-256 of the kernel type

024eb8cd4aa7ac9c9bc1769fc0bae367c483054b6b4d09b78e9df26b06873c7a

commitment C

eea703e45241525034c0c7ce61dfea138ab8dba2336a68f746d4adc958db0580
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bb255d254750353ad9b1cb9b3b6b31dda27ff65ddf4b6a7ca7e53dc7424cc3da
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74e299a4ffeef81639abecf328fe5d3c51c2b16b3339fbe78979f577b0201fa3b
first message t

9c4822f2792041f01e4ee316b961f28a9313429ca95124d3fef3b80277fc80e9
6e03b4b6f56650e2613e2d24129ee8d16489209af3bbae9dc92346459d01d26e
35a79e9ea782891b30607fdf80ae33bb8142a69fc6f4334605067ae907e36951
7cea97505cf101d93c1994f8fa2254210a69afb48cbd694e3658d6f9502bbaea
f5c6af441514882b9193761fe5b1e8e9bfe523f19b2d4c926d8222dac2a0f2b
3850cb1164b9d7cbf5dcf51e04983a6cf278fc15185059b9c2d2453838a691d6
b8aa644ad66f4f2c22cf628f1cc469ff1addb5463a4a4ba9f30698a9134355c5
e819445fba79f144525a7565372f2b56622c8b5b4a39171c8c49434d1a97c54a
response z

45507940fd031962dae19ce485aba437986bbf5018ed89f429fa36ee2c7c0655
76389a80946e8905088d4f4023b699ed7261cfd441ab35b83768ce6d3a7140b9
c2086e4ed4ffb15f23aa56ad7e6fbf1462f6290242afd8a29da43d10caa45920
5f7e11b5a4e9195d9cc2641b7116a608d2c6bbccb7a7e8765bab25cb77b3d06b
7b4533b90cf2a6a0b78f437e9327cd168d85e2036a3ac7a1a3fe96c168eea2dd
6f14be433b7b9c24ce655118cb70eea2b472732254ea1c44b9fb9f78b2920a9a
b381b99f91efc357f24620cf19e8528515932d7ab98fb69f14167381dfee2e8f
fd1d87fa8cda42abd9ba19cefbe57f6d4bd2789142866de8d26add32bbcb81c
response w

780375d6dccb12f720c6d16e5b2d8b7e4cf50691285fb6618afb09ad6d46293c
b426a8730e7241416e05e5fa35eecd1e3cf55dacc811f2493f19dd9474cd9c77
25fecddddd6574b6606324e132fdcf4a18913a5344c2138bfdd7adb6b3436156
7d3b33201a825b05b49ad3f9bf3d6885db9eb963669ca4e0009f183d3dd5caca
e4cacb62dca34ebcde8df951a38fcf44b29b068ca34ea2e82558d65c22b3bcf6
b2f98f582b3e39a2d2bf88ee9eec05dfb2eea58616aba1556e3f75a316fec03e
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Fiat-Shamir context

0385119715aa5fc3c6ae8443c53686c28913e064f4c4d7f853b8258933565205

withheld proof term: 145 464 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_mul_eq_zero_of_le' theorem, MatrixMultiDirection:162

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (Φ Ψ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (e : Finsupp.{u_2, 0} σ Nat),
    LE.le.{0} (Finsupp.degree.{u_2, 0} e) (Finsupp.degree.{u_2, 0} d) →
    Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} e Ψ) 0) →
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, u_2} d (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1
      u_2} Ψ Φ))
  0
```

SHA-256 of the statement above

88348fcd974490e110c27ec488a0a8a36751a532c6afaaa42110c3b5dfef599

SHA-256 of the kernel type

0a1ee5c25858794de10eb8f7f6d1018b4a298bb3a73ac01281f4fb397a32e362

commitment C

7a1fdbd471d8ab38a1061b12a98026f1f0e79b436097de394ac83976fa354913
b38f05cf1d9b0b38912076172698b51579b4be83650a8b094f5de6aeb8734393
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3190b373384a99765c95bc97c49b209c1c35e88360074f7d080142d4ec9e4861
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b22ad118ecf0654ab5968c119d8277dd5e0dded6a424c30d7c103e123164abdc
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first message t
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d3dd0b59838f9d33597cf1346f4cc0192c4af7bccbdb19f4a8348dd510429e33
f94f7f94b327dc4893ee71b9055029a790eee023194bf0a918852d82c8b9d08e
b407a1cd6bb5ac0a8399c7ad5d8a3746f4eebd96fcf8653aca2b891930eb4b43
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19e05d08a309c22539196b5cf883b0a7c19aaaff928dc4bd1d480a32c4096059
3830192718a000f7b2db161b4915108a25bf54d762f7e6f57ff83e7915c2de9d
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response z

1f4082ff1a660a4cbe1e1c29f24ac5ede42bd9d03a037d95cd85f56ae59a9ad6
159c17dfa1fe885e26760dea008a0885d8e1db883b6b3b9e00fc6cc83dbd3d48
9629ef267ac350ff1ec03a47fc4b0812ccc51db1ecfe59f13fcc2c0aea38fe60
13757832736f8d16a978c1e4d5f2c50ca3057f8e8a31c5b7f65777f98eeb014b
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5e07d6f370aa1dc0367e62dfb5c093916bb1a24f89cc35bc6300bdfaa71d1ee

response w

646a15ea9875ea1d172b22f4cf3b7cafba2372e8d558038c0969c0912157540a
008b9c16ca8ed5390e98cbdb232ee8f0a94e1fe7bbdd481c92e5766fb984efb9
49c02d1fab5682500e098c903ae689de363b5cc0b73a2fa61bcf784887efbd8f
731adf232eedd505bebee0f9c038ab8d552fffb9f50cbe374877a4cf889b68b1b
42e9200269e3800747bebaf144a1b962b5eb2f8484102b6c87672d41d3f7e991
012d77480e847476e34bb4f8d336ecdce131e42c34398e5de9536d1a47f3254b
1617aae9a5cd622230a289afa88a7df3fd562e7bc9a23afaad05130ad19d5e5
bad89e550942d65a0eb06f36c66c0add830c14f0e21abab830146a2381dea53f

Fiat-Shamir context

9f1a893f38fe5b479cb40121c219e3955f1c434782c7e969196d275d03bc5354

withheld proof term: 145 465 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_mul_of_lower theorem, MatrixMultiDirection:175

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (Φ Ψ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (e : Finsupp.{u_2, 0} σ Nat),
    LT.lt.{0} (Finsupp.degree.{u_2, 0} e) (Finsupp.degree.{u_2, 0} d) →
    Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} e Ψ) 0) →
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, u_2} d (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1
      u_2} Φ Ψ))
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 Φ)
      (FBundlePoint.mvPsCoeff.{u_1, u_2} d Ψ))
```

SHA-256 of the statement above

c11802c59531d6f3cb46ccb424ae9f041674d3a0435efec18f481c4a4ee66c53

SHA-256 of the kernel type

3a4ac5d530b70b97114b9835fc87e8767eb4b91da22faa660d2e852fc308a218
commitment C

4bd6e161a5d2efee893dd84352fd7ca939e37e3463efa75eeab338d965eca6c4
03e38af186afde4d2eb7bb52546396a3b5b56ba6fd45e9dfc0460a4f41effd90
7fe382a9cd66ce7e44f74084cd2506ecd3013370fddb185c724f63b9afb2eec5
ab41e8d5aae90a156ce96ed855303c3a33e387304fe397c1566060e3c1270b56
21cd2bddff55b9f2fbeb83318710f9885cd437fdf2d3dae1b3567b77eda6df8
ca6a2aa390219ecbdc581388cdd5d48960abd8687dc968d2464b50bfa9cff55e
b25bd2f990585c673e79bcc4bc18122e5a1461a0507b26b7ac5cd41422fa6a0d
c20aabbaf1fc0b52c5fe9f5c1e9c9d9693c53b1b49d0058977f1ac23491f9ff01
first message t

19d5bc872cd02761244ae5eab062826515fcb78046cc3d5cfd77bbf38099a69
0b8a3d0ed28863984455b2b56ee71ed4c6a900637098c1d1f8eb9918878adfd3
996b16cf0d633bfeeb7f9d2eb7f6f7b4a7438d0557ff2ed633972a4e05ad6f7
21060cb878589675abc7d5484e9ecfbed6a3bdf80396866f2ce337d0279759bc
d782d676e2e38c3682858ca26d26e6926d220c40f4833bc060440a678b68b13c
1e421ce32f1bfd85145132992a3ef93c38edbbdb064145bbaf1daa7cded48c1
39493ccfdeb3242eb0149541d466ad8bd265cbf05754e7118ca007ca7e00c88b
5459154570c050ced9d50c2f2e4dbac9c0e8a98d867e6864af49767f5363a9e8
response z

7b325d0883c0609f77d8d3d3b5ede9d3c0387b15ac3aa45afa29cadf14878448
0dc656dda4da1f1619c6a433a8279a15e28797e2f84b2ea1ba5d800257df6d85
8dc43b28f9dd7812c7fe629f988da30aa01e402352b335b4ce76e3a24595f21a
f388e4241e5382cc3321750e302c33fff735be0213a597bc77304d29469d828d
92d660acc875d393b4abd9012b6ad441e3edfb366d77f02396bc830688779082
59e56f699cbc6c9667e63ce3791c0cba4c62d7ff236829e3bd983061e93c0cc6
030cc0ea33613adcaf2d062f42e60a1ecae7696be60feba6d10bd149d0867714
cbe9c1cf6987d884d1273b97693579b0ee39f439bfdded25d9f5803c6275bd9ac
response w

3785a1c46c0616ebb011a4d6766c231c824a90c1d19094b6c2a391e7c667734d
455b04a89406103bfd67934dbbe1586ca254a2004fa091294372ea15bf96fae8
74636577b688f07d820d3fda9c41a273ce16662b11f38d29c1d223da8f167f66
7cbd1f696bdf77b401086e980ff913aed59c881a64b74b343e1c6d98caddab50
d15b3ca5dc33ef3378d022ce2706746b5f07b53c8bb5a4f6ea2d87fa8f086442
ac58ee8fe140ed3ab53a08df93691e3e2ec316b296a22925abf1b481285f4a40
44a6d3ea1f9e7faf2ffe91e30cf1a2e2daf4af95da7c64290163503289021e66
556f68d8fb28aa5f45061d43ea81d5ff2aedd62e462dc81a728e1901120fef2

Fiat-Shamir context

bbcd4769d8d0fa2f73985b897c32ea2fc5b87d0278912bc569fcc18997de0fe2

withheld proof term: 192 273 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_mul_of_lower' theorem, MatrixMultiDirection:196

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (Φ Ψ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (e : Finsupp.{u_2, 0} σ Nat),
    LT.lt.{0} (Finsupp.degree.{u_2, 0} e) (Finsupp.degree.{u_2, 0} d) →
    Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} e Ψ) 0) →
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, u_2} d (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1
      u_2} Ψ Φ))
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d Ψ)
      (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 Φ))
```

SHA-256 of the statement above

98a04cbe9454fd45a066b636bdf014bd26aa0d0c50dbbeb1a84394a13141daa0

SHA-256 of the kernel type

d89e1ccebaf6a144765e8f57e20e04db86e640ecc6036a46d65f76569587be4b
commitment C

c69029cb40db79affac57f1f05cc050f454c70ffde67a9b89cb1d09fb340e78a
3e626ca75213acadbf570aee6df39eae6e1767118d0ea37bf99c96c71d94db27
534b4847f3a03831c4bc39938de1eab253cf00656855c8104d4cf9c8390122aa
ef36684080dea3184a198c4d13c60a8555c500476416c4c988eaf9c42e73c438
248e46984e8314d71bc0bfb0a1fee7644fe20f04edd2b6dc5ac23a32f21d5252
4e4073ce5c056e81747c65fdd663840fe0bbe5d08ab8e9edf78f92e92711e666
b850ee35649b93f3a74da3418afef7404066e1c362e86dd332b4712e13b23ae9
1fc94be048434aebf019bc3d90ed8cebd3e3a8d52bbb1e1dc8e5191d8d59a40
first message t

5a0159678070f080b98141961de5c5e50b440e7f0e248d6eca98260c1f8cb5d2
27cce90d482168bf1eb57c86fb3e754f46cb05235fc3c5785cf1655e601c1730
5cc8c2d4ba0776eb8aa830f7261c178e2f51d67081b8ebbc379216a05bc8ef1b
bb7c326e462704c9db28d8d49f410dd8cb48f0e25af3c6f88b1f8efefa45f210
d9f5f93e3dbe6b6b46d8ded5cbe68b2d0572f07294b2b102972c93a869db3d1a
7593a61ce321cd3b85f5f47432849d6a3cac848e7c48f3d16a57c13cc070df39
b7f233e032347688dce92e3322b013131c0ff457335284831337f972e367423b
8154a846b194845d02f57c0013f1f0eaf65132e4bb69f2bc006002f82265448f
response z

569768707d7a957c3784133fe033a550726e84e477cc488bae7e317c365232fb
2aaf8a000851e26f344353f4675d0f06b358b31a33b49698e7b78b895ed16ee7
ac2cd707a5a447d73d62aadbf0d9a0112b5ef559ee2e34356eb1a54f888371f
a80c15f6d0a71bdeed6fe34405a1ff8e089bf32f1a54c3cf4b4b5a1f6dd0ac39f
c53293fc5d7150c53abe06af4112b0afbfc0bd9ddfa76e912955a21faa6c2aad
aea7b6aa62ac7e11bb6b83117d100c4d4e9bcf4e539fb1f2b0d76a8d6560e3a0
c4b19d2787240d3ed8cb0924e094ed62f503a5008dcc22aa1949a8b3a43e4b49
19b0e9001ed3759e6fe28e60488937ab6769e8ee7cac38b5f2f73f166e5b92b
response w

2f69d3ad0dc15af4dba49cab158cf96c2dda9a1a56610e9c9dc0b27b2e99e68e
becdfa12e506f86cc306b7bddca2be0ebd018a3d4863bc8abc0421c840c16342
68ef7cfd15b796276e8e1bf786ae16255a0e622db4e7e56a1680790bbe33d079
84498b7111cfaa40731007826f5151824fb4545a69d08461d14f4ca049eb563e
b237e1f54eb81250ae11fa6fcc20f4060e80eede95c814e7f9fcb4503ca6836c
4a37579d6c8602578c3f46d8a8f92842c04db0f7290c02c2f812c75034dca99c
a45a2371838e56d14999fb4d1b95e47445adb2b16c96e657cec4835bbdb9b0
96b8544c31e40326bdadaf5d7a33181a1c311261cfc8e74e269f54c19c40c581
Fiat-Shamir context

9f994c263d125d59a5dd8a8729b743c0ddfc01f2f2b2e1125c6a28b15533bcb

withheld proof term: 192 274 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_mvEuler theorem, MatrixMultiDirection:115

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (i : σ)
  (d : Finsupp.{u_2, 0} σ Nat)
  (Φ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d (FBundlePoint.mvEuler.{u_1, u_2} i Φ))
    (HSMul.hSMul.{u_1, u_1} (↑(d i)) (FBundlePoint.mvPsCoeff.{u_1, u_2} d Φ))
```

SHA-256 of the statement above

92b60eea2f2f72d62fadaed4629de3a127533d39ef58858e6a8a7c535dec52f4

SHA-256 of the kernel type

0b457f22d30725f4e15611b365ec9571a100d09b71489077d82efea8a770af2b

commitment C

e7253aca5b1914b82b453429b8c7f885a81dea91df2ad5883f02db5d851616af
15014a96dfb452cce5a998a76157507d8550511239074e668de65069b80c6587
f5cff34d379d8b6e364efee25628a62f8298838d776ff8d9d485abbc05709790
c6f29b8f0c9d8812b47a03cfac6185aec7bbba0739ce164d9ac2d7f8ce834638
969d9ba067b7832be0130ed89ed12b2ab2e581df38880b75e3a8563870473aa
d2d4de6a77cda65a972c2cfca89c1d0bc5ce022d1cd8c40167f97603ddd79031
1ec74b593cd4ab9f48f95591b8428471e368ebf4c71366395b993a2006a629e0
b2223359526155118f3b90186691cbeb11194675ea1554005ae304913d5b4324

first message t

679b3fc937cfb83880163674de9e68eb58554a1743a7b932fe9134dd72545658
1bed587a63726776f46bb5dc93c398ace19279f77957364b629b256f6c9bac5a
7d574f964271bdac890715cdd8bf2f4a2ff51c64733b23c267d91d58d44ab539
1b7a323bfefb7e65c8dda14edfdd77f023c306930afbb6e28ed84ea35b87262bb
d531a6301e7859d571217e5b2d0a7ee4319fa7c9ea7235301723ad39d326d3a0
1de1685d5b040cea7889c96d47dd0ccc448afa48030f6b944a8c28b6353e1933
2ca5c080d7fd9fe41524d8e920b4e29379aefc1a77c9e57f079e0197c7816beb
bc17636461b2299c32dcb24104328ba438f1429bdc598a3894d17f767f6d42e0

response z

6d0553f21eee26d4caec3a7ca70156618e5e4a25fe7500abe032cbf1568e8345
a53e932a4252cb6e3e284504bf59c2a19b41d50fb53e25851374e04990b8628a
686c532784d8df11eb8253c30cf6d45fc0bd9104d6399e5189d70033613cb69e
8aa4257456ad03392ab8da21f5d81a6fca203a8c004f59e3a23e9a17f42dc25e
b59aaa1f9bfb0a2def777a0539f5d7746cf6208b56fffb2c50187d2f046246c9f
cff639228ccc5cabffc8723014923342438fcb83f4a2a83032bf6a1e884ac4f8
8073de1cf74df7bb2c4dbfc7007fe772880e922f72a2367018b6153d957f8922
d73464a6ebac3cecd8e123adf2e619385c988786ed329ee691fab1c235f9cfc1

response w

125e0e4925adf7025c556a7b46445596f4e772f6a25c61c80f5d729d63bf337b
3cb298b8e931d7655f0684954541853678385415bb6c092ae8f4a67e34df9297
498289ffe0963084d5b088bd73f2f6658d7c21cbaf99ead2b69e32b922a9dffa
e45b5a4264bd40255f6fe024af53ed93e2a19515ba1ac26e2f36e9ec5f0af4e1
b4fe8262961e9b2fbc70fc8048e825cfcaf3cc59dd92d5015c380d75b8a77b60
3b45202565cbaf9ef27e64e20851d56ca71db967ac8c177567a10e32fff0267d
e08240ac1adaa9ec342736e46ac18ee5420ecba9206ea3999eec48d73061688b
9a6d77d031ba7409dbef61f6c200308d3ba792b5edac0ce2458fe1c49d0aba8

Fiat-Shamir context

9fb1fb39fa4c372ee20f0247d9751f011c0e0ac3aeef07e69f4f9cde2b327279

withheld proof term: 5 449 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_mvPderiv theorem, MatrixMultiDirection:102

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (i : σ)
  (d : Finsupp.{u_2, 0} σ Nat)
  (Φ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d (FBundlePoint.mvPderiv.{u_1, u_2} i Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑(d i)) 1)
      (FBundlePoint.mvPsCoeff.{u_1, u_2} (HAdd.hAdd.{u_2, u_2, u_2} d (Finsupp.single.{u_2,
        0} i 1)))
      Φ))
```

SHA-256 of the statement above

b327685d4f262c542d1d354af57fd1a90051b895517bdb0ed94490154c1c98fd

SHA-256 of the kernel type

d1fc4d38d7d040e4025af57ce5effd128d08b186de2f16548eac6871d62ad843

commitment C

6d76a502ee3db2a042d58ac3039273069a2bd5ee44d5a150a93f9a2dd7a07652

5de1a5f8a6b1dee82537fd628d9acf5d869f86705f5112565a14d7875f6037dd

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16bd7175c543c4431a7a95f54bbb1f64c16e425fe426492fa23ac138dad2210e

first message t

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75f25a47d473c2ba88d638a7db1f2cee530e7b9231d5628821c414660ac78a01

eee219c40ca934969bc0186da9e9a102c11a8873f21dee40db05b6cbf4f50af4

response z

3e32ea7583dd100636f2b6a1bd696766f35e0095f5ad9678b7780d6d5a05df90

99810c0a81c7f5e87e8dd8e30509a0fbe2c3d707c4e04c582a367b7cf7c7d42

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bc960529e4477fb31f0f0f61c73f900e9659f15bac430edfddfb2f956b236012

f2e68a237577ed720f930fc8e01d6f3094f49163cd1ce8b5b64033befb43e7c3

7d019e9bef1322e4da146570e34107761d0f788318130b27b56e6cf3f7fcb33

response w

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de1d5072e2aadf1468c4b7ed84dd25a9d7aefdb0ed50495a28c089f13e8ae465

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df0649e52e5feb57edded7699165c57e5212cec9c4f6f5d8bf675edc4991935c

Fiat-Shamir context

9d26a7dc39046887d9e25176220faf34410fa6e8fad33ce125198adea4e85be1

withheld proof term: 10 996 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_neg theorem, MatrixMultiDirection:61

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (Φ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d (Neg.neg.{max u_1 u_2} Φ))
  (Neg.neg.{u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d Φ))
```

SHA-256 of the statement above

4d8def8e0b66aaa354ae34890684564530c78953c41dc5f82677cd5ea898e17a

SHA-256 of the kernel type

c5814573b076267718dfe829328b81d8e9a13cc7fd1bb9bed79c67b6fdd4cd08

commitment C

1f701740dbf6ce14ab445e10a12544c64554cebf97642d5732809b6adc0950df

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55ca2ae4e524eeda35396f50b2fc65bbe2868439f38c740c42b1b80f06111c2

d657a49edff14030846141b4df82c2bc6a9625168bbf17aee6514de4dedce45

84ec746761896e5b4de6041fbd77a676ba7b6a8610b25822e2806b40671c8ea7

3c65c4178aa790cf3d828ab5f87b0af08e3be09dfaf5e486af10cc96c2b1fb0

19d577872b69444cabebc14ba6946f9c5f553ee452f5ea5325d06f13fe63bfb6

704e937cb27fb7af0cc4f1b56f526d32ea258a642bbd55b945caf02d0c799d06

first message t

f9aea7bd468f6547e7b21879c2d8ce4641a40f4bfc2a9ce8fc6db6d2727be647

4f618e1267891cb64fc8c91fb83007bc2c11ec9e2f1736afb1c2a0ae070337c3

f694101910a98fa48f617c51425247e99ff90fda2c9c627a1a77dfc7dd723296

3350f32319015b4ec8617ca6d4b2286e73260437fff664ab7bbb4a06778e2b6

1cebb26b06fcc8c8ec251f05a7e2f3abac530aa4cf244c707ee1a4108426010b

4e0a6b399726f153895a8d8a5a6c0fd558fb8cbe2a5a82771cf5aad5b16b78cd

539cb73f34f532cc437e0e37673ce775f46a8d5749b9e48b9a7d3f34b44c65f6

749cea0fb9ff4a4f17b32f1f659e4b7b9438f0d98d237e418a96cc029c06441b

response z

2861296327ba8e22e3b64cd6770442650d6361e14787f6ba6cc5413724ac638d

2cd0e532a46c684316a1c914f0588d9b61f97526cedd33aefcf0e4de72d4954

4b81041d864463b2aed808f044b7f81285304a659e4fd68be13bd3900526085f

bf7c814aef5cd964c2f27b2be8ef313c3e1f30cdb54ef8208967e46218c0aa9e

151a9de2ee8f04c064e48daeeafd0e279ec992b3386c68d166c5d6997322113b

06cd9d9df5d60e2e1e08e638b22e1be622486866c78f672eb923045ff985d112

96e7409843bc9f656fab53ee328170158e09bb77c5640254619319e91d847df3

433d5b864964f27ae4c84eced75eda990c1d72cee6aeafc38afbd5dbd25c17bc

response w

1ed2d30b8d58e4ad982a9987d7d2337bba599f9b00c5aa28e6f808c70c99cb54

8e014c9420b61f955e161a4fd279b4d5010b9b7d14ee263e5b9de48235dece94

ee294b44b4c763244e007fd68cf6d411f3e98a2475075dda43574f5ebed9d4c9

58f7a029df2abcd9e70bdc06b58cbbaac61b571283ccf9a825582ee82646a0b

5293e06da94f9380eb8f783677d1b4a5468e1d284274f94acc21e4b68f5bb13c

ebc318c71e6d80fa6cecf624b51d87395e81cb10d11a24a99772f1b5e0c50cc1

d5fd617909d1cc5faab8fa8487159e353e519e6a65dbfd91a1b841a0aaee94a0

ca5d6cc54969f5541266434f0ab4f39ba484a11a37cfe8315761a1573c309ce4

Fiat-Shamir context

430881e19cc36ce2f5008064d81c3faa61c20db6a1f982f7570be96b01e6b7c8

withheld proof term: 141 245 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_one_of_ne theorem, MatrixMultiDirection:77

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ } k] \{\sigma : \text{Type } u_2\} \{m : \text{Nat}\} \{d : \text{Finsupp}.\{u_2, 0\} \sigma \text{ } \text{Nat}\},$
 $\text{Ne}.\{u_2 + 1\} d 0 \rightarrow \text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.mvPsCoeff}.\{u_1, u_2\} d 1) 0$

SHA-256 of the statement above

351577b63ede0b45cf4f218cbea053b80e96345b555e8f9698e92caa14c38c9d

SHA-256 of the kernel type

eb0821996753d7b9727678456806ebe8dfb44123d792083215430a519119ab75
commitment C

b0f57402098b9de84ff8d5f17fa65ef982ed4dd0ede57dc546391d46b024dcd7
8770d6f3e989f5719dbdc9e9bf5753a13e4fa0d87cfa1b82d207c8cf035fe77c
9f1f0e14a326e3e66c00b4153f0c2e0e49bee01bab31852e8c232b2d46f1be3a
be92a3ea918cc07531881096ebf681c5f9f4f548004bb4c89ccab149a30034f1
e38704152ae7b54d4c3bdc5ef34f1b682e99c4363e451a8c5c3e39b76fa287ae
f3d6c2b292f202dac81edca930e390cd1eaa99a777672e3d6ea7e4defbffa65
29d3caeb836c6039d84a012455a406c3b430d2e98319cb0da467be6d50d08532
88d0974c626e77def733b7b28ebf170027362c0adb9b082ba2f0490eb7847047
first message t

19d4b677e8979ff35a5dc022d05bfb11dcb7634ca86debda94cb5475d533b435
168a1ebf8dda77da524d8c4b6c688f45d642dd917d98e2dcb43e135d0884826c
559a8bd25cdc81ebee7547918fd3c1074a987b53cdb0dbae6584adce8332465b
85274e8abffe0a8186ce8037697f59989402001fba0a25c76560e0e5f3101dbf
4d91d74b4dd8932d035df04ea1cd8ae489d20f3cd0a64fb5d49583e9541c9c52
fa4a9e578e915c0f5e6ab334db1c8ca3dad34f4aa943b4e71ed38511bc2aea4e
0dd052eae6b3cece060e2e7b78083eac1e8f3d90ec073d31e31c053975c6348a
684633818e9aaee9370ab25d7f0f558911d831812984ebbf301ff1d0c466e4cc
response z

25ed0ce47875566875bae4bc802a48a82327028a5d9437c90f64c5e19d99284c
e81c025bbbd86f5e3b4e86b3649dc58fc67267865d9335b68e77fb038ebe60ac
4f14963b1def5bb0b91c747628065e5b1b4b0993b9cbd5c59e0c14d5666a668d
cdfed5b4415e7bfd56f2b46dd90db4c06ea45fb6a3cd9f91f206e1e07ffadda4
56950c826ffbdf32b37c79fb31cd1389602f6cb2667907dc9b993b430d1c23d0
f61e5030c61e9bda7a5351facc9a84ab9149b04a7d11cfcab00be3b84cd404
3ceb3ecfc1bdad595c0b7a4f7092bc1ba9d93f54d988740c88a9e4d3071edb3e
fb04a8127203ed04ab00f8bdc6a0e5aa90f6b39e6bee795ac59724505ca73534
response w

755ecbfefe54dd313d59647fff195753f2c6270c2a9ee3772dbd0e48dce8b326
fd3a930fcacaf7aacb1af02d5e32281da50a44fd3666f62531d4fb52fd6b971d
f60eadac1f78b2727cdeace7ec6bee37d209a3992f3dc3a92fc676faf77c70f9
2be623df95f95b9e41455931896c0c1ac82a7ba239eb06a39e09605310240203
0d967e894650772b108bb4deb5726b52605745383c5f006796113d8f38256fe6
5c7de8c3e71fa9d735c73301896dae171088bd20f7f19b8b4eed7c78174512cd
d1806704c2b99ab4e28f1338a4c4cdae9986d982e5b4a7c44a2f5fde77be81c8
d44936aa618033b717a11db6eef43f33deba5ef558aafcf8d67161e9c421b8ed

Fiat-Shamir context

56644e7e37fd945c9dd18e7a7b3585f800b08551bd239bcff5817474645492c5

withheld proof term: 111 862 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_smul theorem, MatrixMultiDirection:71

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (c : k)
  (d : Finsupp.{u_2, 0} σ Nat)
  (Φ : Matrix.(0, 0, max u_1 u_2) (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, u_2} d (HSMul.hSMul.{u_1, max u_1 u_2, max u_1 u_2} c Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} c (FBundlePoint.mvPsCoeff.{u_1, u_2} d Φ))
```

SHA-256 of the statement above

981a0462f1aab0b70bc567a079382b8f8a35f542ce1567e35ffae31f4cdb3ddd

SHA-256 of the kernel type

e7133706f7c45bd789f91e0502c188b694f401365d22e2e2b8948e689823fbf9
commitment C

d6ec2d428fba6e4a9921d92c2946c8c10489094010496a402cf516b43f47caf8
53ed67a9eaaeee9ecfc5d0b4fe5fdbfeef11cb7f59ab2494372b5e4fe1112e13
44bb0e0feddf6361fd39f277e551739bfbf2a9e2ee3e836e3d6b40346a58f0459
9c4b2f4950cf92730e60205e60e692ac6246f9d3f6514bb4e8b3659946bfbf352
ab8f1bb5b8242dba91c11bbbed8ca2bea2968524287de39bc17a91a2fa3edcdf9
29eb6a0b20054d02f16c63b05561199ac3e078214a0dc8d66949ddf37e709e5e
e8074988c210960c3914e1cb8ee6a8404b0c2245703b75ea27977002d000801
db8ae76fe7ccc47fcec852ec92260e1fb6dd2ee84ce92d107663c58aacb9fd

first message t

295a537953f77518f8383d5a81c86ff38607aac7f656e61918c5f688cf60c419
f2a224a06d8257c0af2023cab0b4b381b5057fc71b8f5d66c0e41b3a9476011e
7d2de3140724c62083d5e4ab028f54c42fb42b5b88e7c89da41884c19f8a3add
4eb4c6422e45d16c4747aa0bb976cd14c6c52d26e764c829d09850370dc44866
20bcca4de9ad13dc0c7150612bde04e0557f4b24e59525347e2afb9cb7e04a7
66339f0bbdd240971eec62dc89c8baf89a21ccfdd627dae098f5376f50096586
264071fb6bc17695a0dfce4e8b0617c21c57912ff80bf622d5085517c85b011
a05cca226f0344ce5373e8064706db1e0554e3c4ec2d54e71ba35d56ee446d0d

response z

b76a4a99a197035dde752148533507b8838828ddc0d9fd39854ac12f23a70f0f
d14d73784de8cfdbfcd095da122a955b532000d3f61c6e5ef2330ce81ad5791e
764b6efcbc8598e879f7e909d63a6dea3b5beaa06394ec1db11599461958a2a0
75a951716efe1ba6eacebf017e0267db62d8ffab9547d3eb83c12a51aa15fa714
d72c97b30df992eeb3c41300e5d29bb8ca603076a243fc5662c685ce5325efe3
d20269e540c0c632feda3d1f98fef1ddd8f6003c6751f4e1f7eb169b7e99a62b
b30acd646f8bc40a210b55fc7bae29532b953e86785ba99f6453247500358d5c
4db5206cc40a9be92a1ca20658b3685d674129eb2a945d8e9e63267962c664f

response w

18a023a56e8e01eaf5d4159ab1c302b9a67654720ce6d87d615d924971627293
01d095abc7d16ad99e175e3a0aec63d04c4b4475cef27f6225e68845b20625b3
ae0e91bb7f3aff87db903b06c08887ae7eb157abab0ecad45458903460c68208
caf22e18fe36c89411dfd10535ea3fb376abcd3c026012c85b884db04ca1b83
bf5ccf8d9c5397b67bd42dd56c48ac57e333e31e847b58609f2fe09fcc022481
92b5f60d2e2b29d3ca527a8bf885a984211f1220f361d8d143b1881558c5ee0c
36e96ff6d788cb91daa326d152798f1aee71e08a82b366095e17b8f311e959be
c2a4ad9c1009b7710a63ce22818f3ff360255389689e6bb6facb2f2772a017d1

Fiat-Shamir context

76cfb3d4bb70c19271941594d04f39c7d0d18ef4397951a022ecba6518e63a6d

withheld proof term: 138 765 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_sub theorem, MatrixMultiDirection:66

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (Φ Ψ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, u_2} d (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      Φ Ψ))
    (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d Φ)
      (FBundlePoint.mvPsCoeff.{u_1, u_2} d Ψ))
```

SHA-256 of the statement above

d0ed789b98676c7bb530dadfc3417f0ef74381b2104f17904c872da2d4247ea0

SHA-256 of the kernel type

d828502ded3e01cd2cf31a9b31e035bb002215fb73066c0a1d4e232b9e1f8fcf
commitment C

d9c0812aa0a03f56b132b67036dd4f0b36d0caf129c6f6a6680c095844646d86
fe3d37a69db5c9e37a24555d6623641dbc0b241900e793503e35bfb38fd6e027
8542e35d994a96a3754ed366a59f430a5d1c4fcab160f6266040cf1913a92900
0453400a625dba2c3f3e97ed7fb11aed2b94e94cbf07604622c47c732cdf6971
51c6eda4c10a180a6fa9ecd4eb3b95f169828f3d8b4f3d409f951f72aaed6a08
bbd72bdda4c6d244bc49e254d6617823f9bec44ba32f90212ddbeb91eddbaed8
d001a979a24512db0fd89cb8a69da974655ab0c460923df5670a7a42a63a47f2
4bcb1bde37bb5a4b930a03f791ab0449256b13af812170275e58e2cfa79aeb72
first message t

579989b19190c9feeb4e9012a5e4ace4520156a057a7f8347788e71a55438db7
98d10acfeaba94832d6af381ed307b214332878a73b46d460bd7ad5686403ae0
3f47bc0f5cfff4d7a5138b82e58e11c3fc2fa9b268438caed78d94b392fadb202
d897122d3a155c24624ec2184c1ed39ecfdd6f20398123fc1ced425922331f79
c733af7c29bd54ba3172473ea33c2618262865dfbf991a63ab7d5ab65fc40909
891b10ef69ab81740e1a77c0cf3d28e2ebc04f3b8b5d1541986a68262635ac66
7e6cb7e4693f51ea39ebf0a13b34c2bc4c8ae59bd915e7d2bb52e503a61717d2
6ff2d8c947fc408f574964ddff2672f473c8ed3af9e00431467e1d924667f501
response z

28b0a9ad552d9edbb79da4fcea7c4697945642f1d30748afa340a8e43fafd9f3
3c2c755c08b3e17eb9e0cae19468f464af7a4bc87a09e8880417ec727f4368aa
385de60d9781b8b5c4b9cf08e996d95a9e4c04b64755e5de4e88fc1be7b7d91c
2f3c2658d9ca86412fc77c5294ca0f8c935add5ccc069b828ea8ceaa5df06f5
df32694278257c33b497d19fddb36ae4c7aa4a6859300a8684b9e9dfefd9b062
2a917dc50de37911cb35ecd249c594bcb8bf2b3d6e7074c34eae2e6fd87690cb
b82c4d78d486a424791e2e28ab3267fffdafc432484184c6c82c40028547d944
e7984d2336951f80315f6fa2f615155f9a119c8832f22e4d6898788ee5716ec
response w

14d8fbabdb128931a0e87b54071440ee24ded9d5f1886bd85aa8f811e06c72d6
103c4618ec2ac74c41a52d674caa66ca9c80a03888c7a5984dc38906ff1d1ef6
25287123903cc0bddf2819615dedb5e168eb4539cabaf742e22f30f308cb6912
65d1e0c4fd01227ca2bee2cef3719c1563eea88d6d5b41d86b54e75608f3c0f2
49afc25330bb9c1d8e7483d8ff3842a318195b9f3969d8aa807a84b5d00f5185
774b1c74fdbda96ca041a0350f708cdd8d2e6b5e8d6ab8a09c7cde91c7c84391
aff9b946c8521055885761961c3ade9e6fd7dcc5dd0a1b8848ee6647c67a2d9
6037b0cbbd5f72a342a3d1ccf0562f3697e399e25d25a4b3f940ca3f6113fdf
Fiat-Shamir context

6e97604025fe956563b60b9b32dc19223ef5e75d640182c91e20c208cbbf108c

withheld proof term: 211 269 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_zero_matrix theorem, MatrixMultiDirection:56

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \mathbb{k}] \{\sigma : \text{Type } u_2\} \{m : \text{Nat}\} (d : \text{Finsupp}.\{u_2, 0\} \sigma \text{Nat}),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.mvPsCoeff}.\{u_1, u_2\} d 0) 0$

SHA-256 of the statement above

1c7f910994fd15f76d4528027b5deeeff9332847220452d09b8b99b926effa2f

SHA-256 of the kernel type

6d4509e224f3cd921479ac589451d4f032ef35b34349a9b84e249f9153e34bfd
commitment C

c3a73935b65643270ca0813881a755818eabaad3895c6bf4f26dbe7f4fcd29a8
d33235a259e7b0834c9b784b776825d5ff3814a3457d922835c4b5d1f86aa41e
e6d1d7fbf15e95b12185f759f229115b252d13295d13c6729b0a7b7b7dd792ed
171165b558cf4b722d653bbc52971d1150148e2676697ccbb791aa00e5412bd7
c5b03d25149ee5302f47fe2ee3f58590f655d0da1f35bc5369dd8e065da8abb8
87a72367798a2ef023827f74d0ccfc1c46a416920af9a37186d707164fc6f224
1e02e48fe6aaaf63ca2701ad9e8b5877a5273332449de2a6d75da701c371e224
eaa4ae393d548b8554e3a52c14437fb770bf4f4c31ee6fb1b6cfec9faf7d72c6
first message t

6834194c821e89ecef96346f4e02accc424591a6a3a53246459a3d6b02426af9
9d1ea5c9a7806ba8ddf4178eaf8a7acdee6d32331d6d06495e1d5897a39ac578
af3341f7098b0fcf0b87cb1a517c492565b171124a779a4d4d6ff60d4181700e
8cc1d24c4f8de24953c2be57c931bbbd70083d370375d55949591005a24f3a0a
d504b1cdc9bfbe53fb3f9e63d0db15200a9c9d65a38a965267ec69d0db8972fb
152aef03f00bc0815689a44c5a4ef67d53393f416911d75ad6390cd2805dabea
fce6ecbe193e8532e900db91d967ffcaefaa9f2e8a78b2a30abab13b5fb953c
9df1aa2f00d7b64934bcb28e6a8020ab873c5a489067bdc7d23de6cf5bb85ba5
response z

37fc731cfd28da8ec7be0bf02f8e6654f6005e4e8436e5f18e79cf69aee0f1cc
7b5a1d05672b38a652b60cddd26469dadffa1fae73076eeeb30d26ebe6c654e7
1bfdcd6a2e07b7f21f4f58b7aaa7143072633d907eca515002e65a187ab71a55
849c6140481ea4dae6c3948a9373382002de5ee3881ef6329110d62aaa1aac48
46651563a99a6d196baf9514a552846b7ac6d34bbe50505cc3327c707f475d03
682dfe780191cbc1662ebecefe77fee6121529a5d9658795fb9e274a4fffb706d
0df59c82dea3da912acbec445d67357ec689df0ed5443ca4a5d1f9b954123f0b
89d6974ebf931cd93a2edf4d3831f0493546a04998a3e88a7c0a8e51b098d42b
response w

5486bfff87fcbdb694d01b50f7a17f57e46b4198640ce8fc9f4bcbf8532edc1b8
635c7c33c710818d1f168751fef9dac1e56acd7ecf20cd1242e516b08245f8dc
26045f80e3356a90e64647e8ecf558f398ba560558b133a24bfdbd5cfdc7d325
47741ad895cc60969842700b25d16c62ef401eeff27202e9e09d7e8738398489
c080bc3fbf32be25f207cb78e0fb702499e67bbcce57f4e7da6dbb780cf79cef
062e4f50727ed7989ed75b699a6de2cdb688af535b9bd5d7e15075d8a79c2f2d
465cfb4b49099f5022f3bb929882d056f7dd580bfec3a34a11846ee328a7325
75591042e745f40e197643f6e0f84905231fa4d85761fbbbb2e768cb8c0dae8c
Fiat-Shamir context

152e4958c017cd0a84d9b6909e3f7403b575061840f4a2c683b4b006d1174e43

withheld proof term: 3 190 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mv_eq_zero_of_mvEuler_bracket theorem, MatrixMultiDirection:309

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} [CharZero.{u_1} k]
  (Q : σ → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (i : σ),
    IsNilpotent.{u_1} (FBundlePoint.adMat.{u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Q
      i)))) →
  (∀ (X : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
    (∀ (i : σ) (k : Nat),
      Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvEuler.{u_1, u_2} i (X k))
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Q i)
            (X (HAdd.hAdd.{0, 0, 0} k 1)))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k
            1))
            (Q i)))) →
    (∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X k)) 0) →
    ∀ (k : Nat), Eq.{max (u_1 + 1) (u_2 + 1)} (X k) 0
```

SHA-256 of the statement above

ccc6a47bddfada124d148b9e747b255df50c6b27d64523ec2b5500fa29bf9d29

SHA-256 of the kernel type

3a10188043b783a8a1a540649b3149ee007335ecf4908ce3e4d02d1e8e81d1e1

commitment C

c82dc1b1204c9304380173aac3e59de4bf2d4cceb3f10ac0cc74838776e3f0d2
a85686857169da5d6461998be4acac18858cc6e93db21d7d33b0effcb0639298
5f38f3ba7828ee115d50a8d5ad299eca4ac880577b2440b3c2428f87e3b6068c
08e8714b8eb2959006b1d69628a9b753be5c2d7ca9559fdfe223ca021bcb9a66
bb3fe9b750435881196d2eeac25bfce63ee8ca90432285bf5128f74d6f36aa
e4e058bba4e140f5b82744a5df8b8cebf61637a139127216bba916bb58b78b32
a58673b3decf7902e11eb5060ac2cb97e9d1e77bdd61c1b19c983645e430d328
be85cf779f00acdccf1357d88ff2a73320af40aa0170b3cec5aa24f570c3aaab

first message t

3f5f20b3945968edfd3455765d8e5a687dd59230fccf1b95668a1f68d6ef5c87
18821ac3299b19493b2a8af1b612798647dcf74c26cda83e57d8a0f3d3663961
39d21620cb8e98444b6d603ee6b1290b6099be3aaa88d7da2243525ac446e96d
208fe03570ae66e45906fb70d350a1632b21a5c60db2fbc7f2067721aac46484
9efc9698dd1c15b879b9681550860aa5a67f3a3051e796b1ac8047b9bad365b6
a0f3eb722b699fa6be062f018c732ce894a1aa7133fcc63ef26ef98159478e33
70f1bc44d382d12d2aeb183fae474857fb3a70302e8cba202ce31fa303653ac7
7ef9609c2c14d04b75c808c6168acc595103258ac0d1b2b1271babb91c5c637

response z

71209a2c472aac0d1db3f03e0eda59dddf51460ddd2951b867f7ecbc1b732ae3f
bb46848bcab76ee8de59bf8a7e3e6930bb05c8693bd35d4117de965a0be8928b
db6417706e96a40fe47213af6f1c5660cac3c69edb3917de3214f8b677c9cd0d
df5135028f6490316d5d2c3aea978219a6378e5b11206b7e49ec49eff13ac24f
e8ebc0ef203e882669b336b655ea534d2b35120a96fe8911c3f08a26cd44248b
749e34fc0ba70485dbf64ec32e71a9109ea405a461cdccf40897dbcb6a6d6be7
b24c9138c49add5a6493c88ac6b460908676fc8d8b81f3d47398e661355f7989
c985daea046e0dbb906c412c3c6482328c18865c5607d0a4998c41f3c0f7c03

response w

4df7992a9ad13dfa0050ca8eec97022df798fae2603b9ab8bcf9d02b001aa86d
0c234071ce33a58d42e84e5db93fd12dfb62ccd48b83b2fe2cfb5d6452d771fb
e03a776921908ba67dba5f2720a53c32bba4aca905b6e55e0c1422b76c4e8271
ff00e9f51dfdf807a413c51d5cc3dfe29e5dcd29881344e3bbe2122aaed77849
58bd288fe3277446706978d24b457ccb8e24d2fb7f6af9a89fd5858821823900
a5214fb04c042caa6cb74b8c64e69cbd958cb938f45503d1faedcc74d534ed0e
5422b946d2ea724a882be0dbbba0f32fba80def9dd258d5fac9d2f553bcd01f
b2651148a8ed46cc34bb4b9d09a9da66150f1ef84c2c162da3f7fa8f6467fb9e

Fiat-Shamir context

0b44ee356cc7c8666098cb6f51d52187c4f6ddaee06fa002ec38983905f3d818

withheld proof term: 368016 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mv_eq_zero_of_mvPderiv_bracket theorem, MatrixMultiDirection:220

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} [CharZero.{u_1} k]
  (T : σ → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k))
  (X : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvPderiv.{u_1, u_2} i (X k))
      (Neg.neg.{max u_1 u_2}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (T i) (X (HAdd.hAdd.{0, 0,
            0} k 1)))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k 1))
            (T i)))))) →
  (∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X k)) 0) →
  ∀ (k : Nat), Eq.{max (u_1 + 1) (u_2 + 1)} (X k) 0
```

SHA-256 of the statement above

95015ad81f23320f2e630ffd342555541d700332365e795bae8df4c87feabef6

SHA-256 of the kernel type

af28a69fa62b52f5dfbfb7223c756cb6f370e92a9d632247c3537014e25388f3
commitment C

a30b36d1bd7fcd7266d5c6624359a45993bf56d7f4fecb3aac504aa40b09b0c
3d2401f7bda1f521d88e64d88a0bbc5efb50449109318ad672dbfb01a1f413b0
4db20584c9f847cd6148e3efff397f3e4251193789b5214dc05334873ee90701
73d5dc3dd214ea87d7625c3497fd30143d2bc85bb656c8b15640fe8c856953d3
40c2e7ec10dd961d55ab2f2cc421720745ecdee6bfd0887295fc47bfff1484e5
3fb171f337aacb77b1e3099bfb798929d8fe8f42e3c372382f571920d3c2c70b
79ec8ceb81cb93ece820bb73a3e62999b43e7f6fff22a65ea443203c60c15c1a
02da2fcef79bb339483a4186449e12bb81e2f699895b336f0d003028518ac054
first message t

26aaba4728dcf83eee3a9d63387012dc908177f479281bb96e5c12f78d425447
ad75c96f5b6f1df341f005a4fef25ba31f0d483857f5fc22dace111e2e2679e3
f3ca62f89e8e1d0069ff75591f0598d0d68e44fb56fd23f51313b66f801aaf6f
9c027399bba95b37a1a9e5507a7499e6ccce9f9fa50ad804c8c8465211ea67f6
2487806299dd6b98c3c2600798a4262a4d21ac69bf2d7d23dec5a33bdb095bee
859ef9bbfc283ee6af836cc7b117ff28244dab9e1c55a2ff414fd397e436e1d3
827cef420e8b9382d1bbe35d5ba01add788b3c002c48d597e0cf58f2313eccea
5df9b1e6430388bae0ab6d6f50bf5b25fa62d70f00966ec0107f0f09070ea41e
response z

69eb7f1ff144716334fb06b76603273e3c1f6ab49fbbd9baf210078b102e9321
c75e0c992ca832358c341077121104b9743f70006369340c0d10af1de18ea0c1
c4efc7b46ca785dcaf6083ec3519e2ddfac6e5a2934a63e4bd1107ceb45cf271
9595c6e09a34c9dd5bea78dad9a7fa7079d9bfc841df0bf3dfd29385e02aee23
78f684719297bb1e8fad87469484f7a09012ced01a8cc80df59237542f1b8c20
8c33b977956f8a4187d5e2fbe2f7d5cccd3aea896be516f86296ffc7e56041a0
725a11cc1bdc0772673d61d3250285ef86c6958bdf58cb537718e9f5a7982e1f
dbe42ace4436f7984ef0d77b5c940f9d0115d434a1d44a7840b873a336243216
response w

52e16dfed060889bb330780f4357f0fddb7b9017401f8507aeaf196f0c678bf8
4c5745c4a448e6e12795945d9d8a9d401d503e7c5cc18937a30bdaf29e256acf
73d23a9d5378d89fede928b47561a06926e0c29e7f1fa5c9f10cad2a20ad9aa7
edd96087397887db384aed896d11074816546877fc0c75a58b9255fe78d78547
812c5837049845e41ac9977df6d5be666417c82bac7a01392bcbd2f945e6629b
9f690628da9392ff55923ff2a436b58f0fbfd80ee68f35f99cde53d56fe4a90eb
7089e0517612fc2bfc3d0c931dc9e7bd03554215c8840c342ca45ed358097632
05b43fd751c3686b2ff96d72ddcbd5d4f15019223c4942738caec09ad810e4

Fiat-Shamir context

6f065b52bc9010e3cbececdc4bf4a8380cd36ba4e73864afd3a43a382239634c

withheld proof term: 338 271 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mv_unique_of_mvEuler_bracket theorem, MatrixMultiDirection:350

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} [CharZero.{u_1} k]
(Q : σ → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
(∀ (i : σ),
  IsNilpotent.{u_1} (FBundlePoint.adMat.{u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Q
    i)))) →
  ∀ (X Y : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ
    k)),
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvEuler.{u_1, u_2} i (X k))
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Q i)
          (X (HAdd.hAdd.{0, 0, 0} k 1)))
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k
          1))
          (Q i)))) →
    (∀ (i : σ) (k : Nat),
      Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvEuler.{u_1, u_2} i (Y k))
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Q i)
            (Y (HAdd.hAdd.{0, 0, 0} k 1)))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Y (HAdd.hAdd.{0, 0, 0}
            k 1))
            (Q i)))) →
      (∀ (k : Nat),
        Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X k))
          (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Y k))) →
        Eq.{max (u_1 + 1) (u_2 + 1)} X Y
```

SHA-256 of the statement above

8d75d12ca8845606effd5c00219e0f6dc502c36814a90d7c3ad08034ead46b85

SHA-256 of the kernel type

acc78ed1df56f28c84898c85e06cffe1d7452a045dcae7e638c3f4c0fbc6f4fa
commitment C

8f8b6f036fe98528a42742e88a6619a9a30438e74444cd7225702947daf7e9c4
1dcd72e16deec59d75af785ba9e97676e0326c9c2636c9e18590b02cf93fd2a9
e8273609f69318315ac97c36ca1c0fdeef99bb015d8e46e8c3de77dc5dc64552
8a5399810c24da0ae84b68862d584ce72c0d45229505e58e326711e1ca29f9ae
add6fdcebfeef89d255c10165e11921928bebb87c602de6110978cd7b992334b
86d45e8f17608e87b40287e56b39c0f78482797de85ea0e2c951f49ab402861c
3264691a58347dc1067de6bec8ddecc973aafbb0abbf87ea47a3778e58c9bcfe
497843a0ba26dfe3a3901ec67fe2e5d0f7c2cc2b5bec37ed71a0028149158eb6
first message t

790930088cadd8090acea394e72214ec0499ca4f1af625219536e7c9fbfd643
9f776c91959a8d3956dbe53cbb7a9bae3e79010c5f70c5976d6d6e556c49e0dd
bac0da41d080fa2e5fc54c9d154a67eb5e8894d7b959eb6f609305e40a94ea37
419beacd240c8ce69f244fe4c3f04f86ba06db999de0b87d5f7fae3efb89438c
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a905fc591fe29c8163afd96d5846f95a296bd757b8ac5d209ddc07f825590dad
0403e7ae90bddfd106f36437a6b52b3e9af82eb47d888f4edefb8be93e7eda0
98bafc2460268fb096425b2b589f78c63d73f09876fb7fa2d0830feadcca395
response z

1a882be98c482c86f37aa6a9a121de514b7d737fc8eb14e905508d4d512e81c0
e00ff8dabfa77786ff4b45d0d97cc3ebd72f0cde723e9606ac458be2171e73cd
077cb4f627efe7fed4254998c554b59f2e37ab1b3719a15902b0ebe7fa10f627
e00d73a199badd5cecaa6e243ce2cbece536cf0428a0fa36eca5654b43608b44
7fdec0f86cc8d31dde2b5e844f91389e2d186683ac4045ab3d68994b12aebcc4
bb2883253e35acd796f8e922ba0b5c067d6aee179673b0a0217fa545cce22fc5
eb51ed63a15325dba8d89da0d735124c1033848d26600b7f8623a14c33b2bf69
de0d00a2fe2dc85f6dc3c3c7a97a415030322971ed0c38f791a2cd4f4a74b1c4
response w

3781f3bdce38ab26a76285679b1d26052d933629bb322826bdc7b5b91a261053
840f07c0f29ae899be9ddb00805200c221fa24fac69a77f3948371f5dc6af37405
efb0e120e8226234271e36e444fcf74cd95a8984617e9a632f829bb2a434913b
aa247a563e76ceee7a8a35c7ece85fecb0499bb110b15999bcadefbfdbb8aab4

FBundlePoint.mv_unique_of_mvPderiv_bracket theorem, MatrixMultiDirection:262

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} [CharZero.{u_1} k]
  (T : σ → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k))
  (X Y : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvPderiv.{u_1, u_2} i (X k))
      (Neg.neg.{max u_1 u_2}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (T i) (X (HAdd.hAdd.{0, 0,
            0} k 1))))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k 1))
            (T i)))))) →
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvPderiv.{u_1, u_2} i (Y k))
      (Neg.neg.{max u_1 u_2}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (T i)
            (Y (HAdd.hAdd.{0, 0, 0} k 1))))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Y (HAdd.hAdd.{0, 0, 0} k
            1))
            (T i)))))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X k))
      (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Y k))) →
    Eq.{max (u_1 + 1) (u_2 + 1)} X Y
```

SHA-256 of the statement above

5f72ab50d60c1e2a000f464eb8b28628f25ec762eca6df893339a3b9ef034056

SHA-256 of the kernel type

7695370609a51489ba9cbe63fa6ed90caf4e2aef533f493872c9976af6ee99dd

commitment C

f3bd7a93e7d0fbb8bc4bcbda77ad7a57b56865c95d66e9754b16355df6e1ea35
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008db7048edf9a627429cea8525a779c222d01f94e2f0f09bde7fd3ee518650c
5ed52d381083de191d2d88b0998af87dc6185f3b2bfc444d9564044fd97927b1
3692f9341bd5ed8ad2152ed762cbc53012b378d38f6351ccafcc7170d90991e9
be5630df61eccb647abebda7744c5f232412ab2e060d6807530b3878cd3476f2
ac471d60a28f4ff1628af240b10361783719f4fb24e4919130ac0736ffb10f4d
5947783ad90bc002fac1e8e4d5db1b6f48c1ee36da8258afd6b9078f4de7c959
first message t

f3bd7a93e7d0fbb8bc4bcbda77ad7a57b56865c95d66e9754b16355df6e1ea35
46b07260fd7ec987b4c457714831c4cd5371bb4611072795cb7d5a550fb1b565
c2840f7bb596c6b5815c5a3d2a325eebe24ed5c15d77073c748376e215018ab9
715812b18db5ca68e47cfaa9f4516eb7195adc5cf6b1a22fb2566508596e4990
368db0d4a4084652c5b2c37434276f9b3f9dc47e34c36b04bc56062748c489fa
0dd2dd8aaf82f0ee058296bf43397077ce526c656e37f7671555352e053edc8a
8030a6eea8589bf86330ce80777954e16888db606ce9df6cc7ba12e33e2e594b
cbc48a5c82e6e440656f53efc8607d6f737029d05ce6aae7ccf444c1b903f588
response z

7acd3c9a2004f9635787186c84c1fe8d387f52c3704fcfb555e5cdf2b3b69b4a
2e7a655c5e63bd3e281e67f226317f817ef5d4103aafdcf6d773593921cf52aa
8bcb72aa76bb92dc48e213c14722a54864407d1ee79e425656c8d474ce79746f
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f3e8bf92c7aa889f96f8f38bf724bbc582d3bd393d2bc55cc860501c4077160e
ff848fe1a99ab11eccf4e49366d4d924fc628ee6628fdd79dc723a31b7a30d76
0d010ac329573863b5e2d2bbd5cef2c8e2c5c685457d9a92d42ce2487030dd7a
response w

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5cd3b293297aeb9e72129ce7b74820785226bc1a117cd22dff4b8d4718be02a1
af690147656e7afb3995d33bdcc6ba7ba846d7d704e2d847758244f79e571c88
7a7008f08aba993ba49bbc48c72c971fcb9ecf6e9a6c6e8c5c3f43702907ac45
9d5d03fb97e35aa080c9a03a6f7ef1fb504ef19291ff6270cc730d9395890ce406
871a5bbcd3baae5320a80a7a91d5158ad556cc5863d60bb2d501f324e2764417
1cbf39480cbaca089a8734655ef5dccc6766ab59ad29d076d31d56abfcf2194d9

FBundlePoint.smul_pow_eq_adMat_pow_of_rec theorem, MatrixMultiDirection:292

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin m)
  k) (c : k)
  (Y : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (HSMul.hSMul.{u_1, u_1, u_1} c (Y k))
      ((FBundlePoint.adMat.{u_1} A) (Y (HAdd.hAdd.{0, 0, 0} k 1)))) →
  ∀ (n k : Nat),
    Eq.{u_1 + 1} (HSMul.hSMul.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} c n) (Y k))
      ((HPow.hPow.{u_1, 0, u_1} (FBundlePoint.adMat.{u_1} A) n) (Y (HAdd.hAdd.{0, 0, 0} k
        n))))
```

SHA-256 of the statement above

1e8f5eae5e25bf843832008ae83b09f1f3e9387121cb4de39e803a60efc0676

SHA-256 of the kernel type

d59495b340d25e59f8875eb40b0b4b01fb44ea0c1ad10ae43ffe8286608a0093

commitment C

76706400880182458638036b40537150110a98f36386608d9b38375532993f18
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ba387f39d624dbd29e350597e29804f4481b06e11c618d40ac5c0cb4c395e53a
787478f68424a363d5ced24ea83ca418524c931228cc7f7e59bad8635e0de7c3
366e493289ccadfac059ce5ae04544a0540456bb0dd20e26057d1c4db8b3d41b
49e07971018584c5a56bda5d2a8b2e0486d3f0844e31981e46e879cab6908517
a2a6d564bd88ddf5b59cb20691051e51b293d89fc1394c38501b424f9b186f2c
533e3023c6079a8039fbb13c5c29e83588e73ca2fdeca8629d509402fb4f2a4f

first message t

3b19289a83d86020774d54dfe824a90b3f502a85ce4b6e1a312832a5ac891049
d61010772739ffca7d21cea8d3967826ed8c72e70a0b1769290d95ababe42666
30674aac8631498210367a22dc52f54db7538f6ef5d25c37de371e4da0ca3680
f59ea6c6e07dc4c3b83ee4b6a8cec633945b4ea187b597d4ff27aa5e37e000c3
58c96d55da84b55d4f306acf412b1a37ad3d5853b690bd2ed79a8a5170640848
51d2d095f5b8ecdeceaab3ebca762c344f91e9af6c432a101433d3707e78af74
179ba9f185be928330ec6a03ced304cbe2dd07db14ce791bc9c8f027166de3f0
d2301970b719483b3a7de467568bc62a13a1c15d23e6bd28736beaa9e224657d

response z

4b24e5b5c72709090d0131d2a39dcb9a36f7d734a4f5e94baee541683caff95d
ba29d248c5572ba7bfa95bd3b7a6c988358a057d27e46e12804148598d157cf4
8493d6e0c632d446f4d09face76aa6f382673eeaa5f34de47b3051925812d5f3
505c522c78f48946985ea4a0eb7afc1a61876cceb5291c5a15ff4f52f07ee58c
f2619f777de7d63ade0532f9786ad342d4106c5b7d7a50a15bf7017d62c8e99d
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8866e51639a2b8bf9a334568da1101297cc32bbca4b201e18c2625fab353b81
f13322a489122961478232d03769f7ad18950ffe0407bb6fecb3c9917b05b145

response w

2961634e021f20f674761eb619e406f0b02b9b776dfd1298283eccee7b0169f7
1d2dc724f3bf68528c0a5124e9dc92de71a661df1a04b0796e1d0bc7d810c34c
c3f787bb356fbc40596fd179e7b998b469b13405bef2030821d8bed029e770f
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fbdae85ffba35f6625fc525407f1b5962c1f7ef78351bee7588bfd60bb8413ea
924fe3d29ce35b0f99aa2eac18474651039962e408aace40e6004fbbfb91d57d

Fiat-Shamir context

95de78cd32195319057bed464293686f629e3bfc691f4b39cb12d0ee65bb70ba

withheld proof term: 1 296 381 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

17 RequestProject.MatrixTDirection

FBundlePoint.derivPS definition, MatrixTDirection:33

```
{k : Type u_1} →  
  [CommRing.{u_1} k] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)
```

SHA-256 of the statement above

d850dd658d7f608ecc6607ad8c08d69ae1f830ce689bf57ee0199927a3aecfd3

SHA-256 of the kernel type

29363b721bdf26c5ea2d157e012dee6933c6a6469c742929622da3a8b5dace90
commitment C

a17a496031cc149d39ea7f007bba3fd7d87622ea99220871a7cb27d2b6994f6e
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a8f57165d31b5bdf01a255dcaaacab4a84a2ad1b809b02b0351ac14450b110c2
17680679a1dd0dd1e25942715793ac3b4fb88ccfa7fb7d2e14e1183f4d184f53
612b4e283b69f6eb5cb51ab8b81f2e9b9deb8ea30516283b3fb83d0621f3be833
08344f0d84fa15593860c503f2f8e89df398e0b1772fcb02f76c570c8689de6
34856891855f21280af5ef78e33b0d49d99003a99543f86c7c1c880e3ef0eecd
94b33e83f0e2af8423ecde508d6d980a277aa9ae0839cc368acb14eca5a3bcb6
first message t

687e9c46b9fb3954626c4ade17bec71757aca3614380df29c0a8128abca21e58
cd332fc6b628e336ca22fdac9b2fb5d5682f5f1ee8a8b128a2afc6a7cc8220c1
f792b734202757e2b5257a8cfcc124e62385b13fa4f2205183d8b9a1eb4eb4bb
5d1c4206664e41ad2e520757ee578465e7fbfd8ba5634befcd5e472d2854f45
c8252baca7c60f01f6478636367310040403acee92a3b24800d1393d50662887
313cb0368e2645b3a5906368cc40e7ab13ecf2ca87038832cb23b92021a297b4
73b786805d2e9c289a9e092eb7d563783d681eefb9aed8cd07926cfc38edab72
c954af52a54337737b65d9e217e9a9f3217ec473ba979a7ed141dd9a5a02f097
response z

3672c6ae77d04b1a9ad240a2ce0365f02624790dd9af448146a9094089a1be93
32d38eacae89dc2707cd02ee51f46b94d16f6eb0dc0e835db5218c85f145f576
597b608f5c4b963b26d1f93831bcd0a24bcd4c9b9077ef67e44b6b494a76255c
7317e4bdf95741d2b6da53bf176e6edc282a1816b644c9fdad1eb753b662605
f8d0a870f3b79334e84412c6818906006ad7ce7e2692a6af2e1b3e2662583c80
84a1eb927a802a6b20f5f197ab1a4e01f2ba8100fa8d380cb794d3aa56572ecf
ce2de1ae92983ce5efe62eee2c7fb81e4e562330e42b72d7f3cacce7288f31f3
a26745063340fdd74022b44b98ebb120685c6ee8cde203327d4a1af9dbabcbdb3
response w

341726c91f8a1621e2e4adecfd5643b5d3bb92186d07c7aa33986fb110e0184d
3da68edace6a39cee06a4af3d9bf55b0e822bbc4f55ab45b7340cf1e5a5aa2fc
1fdf58cf28ebc65fc9a6e72c3bed736db8ec11f51b26f678fe3f9123eacb008a
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6fe3499c4ef23ecd5fd9b166d07311631683daa53677a92da0ae731f677cd48f
263808bbcd221bae44d7957863a6e8d81e95449d5e8c19291ef768ef3c9e17f3
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Fiat-Shamir context

2a32cef618b287fad424551453671b411c1eb75611aec52792bae4964573d3a2

withheld proof term: 7147 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivPS.eq_1 theorem, MatrixTDirection:0

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.derivPS.{u_1} Φ)
    (Matrix.of.{u_1, 0, 0} fun i j =>
      PowerSeries.mk.{u_1} fun k =>
        HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑k) 1)
          ((PowerSeries.coeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1)) (Φ i j)))
```

SHA-256 of the statement above

1e4765f88cbcf8b19ffe8ba5be2b708145e36d04731cbec98c6899d1616e955

SHA-256 of the kernel type

137479251e8847be2fcc459c5c636beb9a5d01507ed7ab5404e25d077bb8ac41

commitment C

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67ce942ca8d576f7c7c1cf955072491e80402af5468e637e26d7ba47baeeec9
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e99b9b9480060c43363d451f3a18fe49ee62a8453d0940b65397e57e14d2e33e
2faaeab320a5905b7892f1ff8d02e05026fb36361276fe0345be4526fd58af14
1af26c79d9468d5ba2b0ac979d6a884977b7b5a07e63c7e34d0796609f0d2bdf
97df7d566cfe5c06708569f03ebc24c732ba9b90e66f6f6538cb202900dba89f
first message t

1e304612c6db7e5ad4b5f6f005b54368d2e0a74a7c1a56c8a97e6b5f044e3303
5b133df14e03ef8945145c1d24d146f86f25451e1e8f1b823102d1f61b24bc93
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ae38b747dd33cfc94869f4f11a7a423b9b2e163cc7d2024fc9cbababacaca6fb0
71cc7cd336037bec2bee26e1747ad22f3f935e88f0a28a6fe50bf07ca091810c
606cf172cd3312ddd8d0054141d7e53f905b1acc707f3dd5b641be59822232d8
e84c46f18b54e2065a9f277dcdea0bf0cca96a3a39ce6589ebb6b7b2cf9bd522
73f1ff5621af375e7011fb0204fa5bd2859eb76f13b8ded17c2daf48d1282584
response z

6969090863ffb8a132c09b6fad0cf2d294299d2065b32d633e27b48d4f417d6f
31da869f6274f8b8638d6a640197704c4b9e4e7f8b1baccb7f16173b03926237
1a62c9dbc218b514c7cb759cfc0d3020d3ca861d143ad0748b3b1359884ceb44
b9444abfb169bdbc9027d220238b10ef28046ac427f86e8c18b4492765e55a90
26186b8ebc0a65c44baf8ab4da5863793e142ffcaa6ee5a290b8a0726ada2551
80ff071c3ba06fcb1d9dc26c68ca69e0016110bda8391db592f0548ddc29fbb6
493e2a23c6483c609317b82b4edee2d5f904b9bcb782c5db74b304a893a7810a
10aba738a425603814c5210eb02d4e90b82b25c9587343a10415a565bce89957
response w

542caa47dd660a28c788afde50976e0244cd90ed6d1f838519bf83b8c293acd9
b7ee86f7399c1f99ad6f037c2ba1639089d3350402c9ea79257d112fb4afcb46
5cdd67269fe8932c1035a2585e4a80c5a0f41e571f2800d90b2b1a95e0874b17
2b5a70d8793a278a0ceeb98cdebdfbc5c4ea8c344554ec634247ec49f22b1ae
e11ba132ccabd1c5ab1a1861a58d4fe651313e4a5570b3ea4cfc4b9a2c048342
d2e82b59a11240c182dce7e03679f742b1b2410d6ac5b4ca80e268f946579921
1f465278f201e640ac7aef3dc8868765df907528eb30e602deb028fc6900ad04
f3f9a444aeeec65ebb26b500247642350bf330ab3371cfc1d67474dbe4871cba2

Fiat-Shamir context

03072e1e975c11383c3a1283ed61d1264fae5f0bffd884e4da287ad1458bd713

withheld proof term: 257 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_zero_of_derivPS_bracket theorem, MatrixTDirection:92

```
∀ {F : Type u_2} [inst : Field.{u_2} F] [CharZero.{u_2} F] {m : Nat}
  (T : Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} F))
  (X : Nat → Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} F)),
  (∀ (k : Nat),
    Eq.{u_2 + 1} (FBundlePoint.derivPS.{u_2} (X k))
      (Neg.neg.{u_2}
        (HSub.hSub.{u_2, u_2, u_2} (HMul.hMul.{u_2, u_2, u_2} T (X (HAdd.hAdd.{0, 0, 0} k
          1))))
        (HMul.hMul.{u_2, u_2, u_2} (X (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat), Eq.{u_2 + 1} (FBundlePoint.psCoeff.{u_2} 0 (X k)) 0) →
  ∀ (k : Nat), Eq.{u_2 + 1} (X k) 0
```

SHA-256 of the statement above

5ad0be2b5bf4ca182f9540b87587a8725561d27e65c9a6b02b99239430e8e79e

SHA-256 of the kernel type

d576d3858cc3714b257b46a2e4a979c95e3f4fff4e67c7f54e1d2e6ed0c22047
commitment C

3fb2efa63d4ccce7693dad3be4876d87e959b21558bbeb1c50ff15f6428400e1
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d6fefbf8a130a7de13c0777f5994d6294e4d6c86571bf8312013dd3cd78c044
9253360918208988a46405f753e65e09e10de2eb0d31ff88ba002fd4f05b5c80
8f6899159a85428d87190b1d31657f9f51ad1291e1a6bf1be2a8812ab5429b78
dc66927a72f7e9123978c445a85ee715d3d6b7a2429fff4e8438bcf5b2ec24d6
5dae82b61d51cea01256125177d9c4b59022e567b355c5f85a98c81f99b749f0
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first message t

d8be6510f9b5f8572741a345196807f45d31df1487c2a9b55eaf6786ad8d3b4f
1817506266239597b0b5fdddef45ef669b3732e7c3c1ed7eed01c80f2649614d1
0dbd871181f1eb00490e73fbf158768a4a4aecde8753a1de91babaa237231013
3741595ee86d392749f114315e438d424b54d299465374f71437468c64a93d81
3c8742a39099a3e0b0928d0cb615bce45a2e9bc47c76979dbe3ae7ed472ab89a
cfdb121becf9a521ba356ac1f2c9573983faf9b944e7f1e35d305b0901ca1e0b
86e0c8abbaf4815ad6d1af7c811b06356cdf9557e2abd7f6576557f0eab63386
d5c3be60357deee2f33be79d18967b3eb14ff21eea4c5d7e43704e2376171a74
response z

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47dcd52948d5324e2892959efc031a46f919d8b9d372f402beb3a4e6c6c798fe
2750a17842b0321afcb6651b5816ec38009558713af5cd8ee9396f6f9f88107b
c5ba3a64c74b09a64aa051be85c26994cab31feb664e1b589d685824d3e85bc9
d6576264219a5c2d113547cc2964aab4cad6d579a2237a25dca11044753b8616
ed5594199a791c192a5b190f6e075d70f0ccaba2c7c22fcad953b05d102469c6
1cab4ccb6f81d6c2d0729925be21ee10abac3d5d00dc445792e8c36b7fd4aa4a
response w

16293abf7836041a6fa0be6fb9748a3401bc5ac200351af5fcef94f3570bf40c
411534f94387456792c84d0cbf95e5fd9c0683ff74a6e5204b035cdb1f6dbf26
5073b6a86bfe1f68db5d693a471864148de7cd0a719429cda50a7e3ef634d6ed
545fa40f376f4bea5b61700e0e38dc51fd5991d75e3f38f90a553e380912cb74
b8ddfac0361f5aaa7925a8b0a69251ce8dcfae61685185ed32bf3640e4c5932a
78e550e0308fc84cb6cddaf6309ec4c586ec697f8ba1d2aaf cbe2979b3d102b6
5d371bea0eaf6e61ac74d59783b80a0a6570a6a5cc54008468c011d004b5e6a4
de97697dfb8635c1d2a46432575312715e8dc9440f74c03cf98a168b7d23ed47

Fiat-Shamir context

4a778a89ba84bc4c8b9c536603a1237c128a784ceadd126157b60d58f8040d0c

withheld proof term: 187 269 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_zero_of_derivPS_bracket.match_1_1 definition, MatrixTDirection:0

```
∀ {F : Type u_1} [inst : Field.{u_1} F] {m : Nat}
  (X : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} F))
  (motive :
    (n : Nat) →
      (∀ (m_1 : Nat),
        LT.lt.{0} m_1 n → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} m_1 (X
          k)) 0) →
        Prop)
    (n : Nat)
  (ih :
    ∀ (m_1 : Nat),
      LT.lt.{0} m_1 n → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} m_1 (X k)) 0),
  (∀
    (ih :
      ∀ (m_1 : Nat),
        LT.lt.{0} m_1 0 → ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} m_1 (X
          k)) 0),
      motive 0 ih) →
    (∀ (n : Nat)
      (ih :
        ∀ (m_1 : Nat),
          LT.lt.{0} m_1 (HAdd.hAdd.{0, 0, 0} n 1) →
            ∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} m_1 (X k)) 0),
        motive (Nat.succ n) ih) →
      motive n ih
```

SHA-256 of the statement above

af482b7b311e61c5702d559b58e6b87ad41021bfe60f110467924c19d93bce40

SHA-256 of the kernel type

394859dc194dc251d8966d0aea4e2fecb0a354faa5de49cba362592bfea3bcbe
commitment C

ddd43936d71619c5716a019d5a050a29f1d4eff95b1e55a754282c3684955358
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074d2f61fd8b35f799959c7a79acd10201c83842712df419ef6dda82822f916b
ef5c928a8b2e42e2d9cf0a87c8c4823af56e6487fb2fbde9359f4507dc379ed5
5f29c70b9c12219f6c5abb8874945b49b186ebae56980cce4efc49be4934c8
a1f3c00f24a36dee845369b539974633fb13152917170f99d4de3a91612453e4
be91efe69a16464e1095611444dcd98e2f45cbe1a2dadde653d8325a33b9de39
3aceb732cb3fb9aad2c6e5116b365e33b73b5d57712772748b1f9ca554402ba5
first message t

17108308d89494d107d5e2fc95b17962d90c6c1023438d4d47c202a26851bc66
36ffee56954efd652b483961841f32598f76f8b781e62654b336883b9b0010be
acbafe506157a8fc2a2eadd81ebc952356e5e97fffd0675a7f0c24eb75381e5
26741b558d1c4b864b681b0d93a7473ea47eff6f16ec9268dc3f74618797f65a
e8afd78947d14d09c6ae82ae659aa8f3d7203bd2b3f4347fe996c3012cc56400
e38cb163b9d9ce18640a4e11e06eeb2efbb6491ee007ee9e2fffe077dd2a06ad
a015c31a17111f178b60680e01fc12834bfb7cbb4708ecfbaae49bea9186d802
015a1d6cb92c9e52a34e3ab155795437f0c1a31fd1c8f2ab9a176eb783b57f82
response z

3a9c118a1f23a6a43d223d805bdb397ef86faf9bc1bfdce69d7c65178581cf30
c585a03ce0a86b9db37f1d918df3a789f7392cab2e25f2728c90cefe6ea116fa
d97753ccfd53c778f6b3b4ab0162bbe666fc597a9b9fe5cb46e2fe6633fb32aa
53778cbecba2e0d91c72cb44ef34bace9073c31ebdf1f47de918bbae00db287f
b7abf28e21a373bf4b80c973b77e7a14a05c681f821855b220d48891a5e2045f
79ec1b1456aea66c07df1014d5a140a5e4d8dd48e2d1766826ee5d9b2958ffa9
06b9b4d61f073c89f740f471d984ea21aa253a5b498de7b0a8c32ebd2f5c63bc
df1764305a6ad8a390533972b2e9f065d7092fa5307cff669f8c5450404518aa
response w

7fd93718a65919a251733590dbb3ee6ec32e7956814d4f4de7ee753920dbfb1a
72a361b2fd8ec742fc3a44f4c81fc69450699937fda0ab766f8ff10ee453fb08
4cac15aa2130a6b4e3292a17e744a38f9dbb3bef81381ef02f2d93991b229847
7590ba523e053cbb2773cf0e52097f8035ee8ecd20c6dabf267ba0076f355f0411
5e121e1f2a95152480e065c5c6b8abfef264dc76d6eb95915a72a56b0af4fcf6
761eebe85f0fcae44af6b85017c838a71dbfdcf46632f0050e5ee335f535cdb1

FBundlePoint.psCoeff_derivPS theorem, MatrixTDirection:38

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat} (n : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n (FBundlePoint.derivPS.{u_1} Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑n) 1)
      (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} n 1) Φ))
```

SHA-256 of the statement above

5b79434907f061bbc4657fc5236a11686fc7165389e03df9bc0e06e61bd7574e

SHA-256 of the kernel type

fcbeb9e78bc8da8aede6eb420c166406693212ed868a6b775580294c48bb5ba7
commitment C

9447fdd79fa236d50a1e7dcf048cab360b3b0575ca686a7da2670fc4f6151b45
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c74b553b6e2a5f4566d24d873dbe9dce1089a6378866fbf0dd10db5f6f61b391
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66741d311c3799a9a77cb5de6d0b1ec5ce37ff0ed76dc739b71e681824be9bd6
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afad5eed99ded071f4dd8a2460c95317dc1aaa83fae988f3744e6eb55e513e6
first message t

c445c26ad4cc88f7795ee6b4d7665f26c0affde528ba94beae439ffd9880e8c1
6b6293f644799a32c6f3c5bac97054047059a9fd470f3c332b1053a1285b3a97
2b94e706ad4d1ee4ec2ddfb6fa80febf465d295b58e228635776533eeaa1a410
1f44c508816a66eacac572fcb37dfdf4f75dd1616b6ac5dd9c71f092fb4e975e
77d763b6ff35cf14b2395de36876da355f059b8527464d0ef1b50f3a7b247415
6e972934820933352984fdf5023f6d10097454074f3a9163e3a239f2d7da24cb
e6e9d9c8dac4d98661370de17e84a39f9b9990b4466ae77c18c4783a74682015
ad0a717de2b447839f48f03d80b4c2d4e9dfa89445a5ce53dc2c94099a9234b7
response z

e1d64090600986f9cd31719a75df9b88ca90d1d51b5da5d1d14db243e7322503
1b0b2da381810fe4f24bb5818670e891944deb417ce8b07ff18c03dac52f40ae
896d283d66c3ed7b1df98aaa6718b7ddedd762335f092906494544192e8f5e46
9057cfdd7eda4a24a4987edf5c1d2820aa5c03720478768b4d253ecd3218747e
ceedcf463421d2918d8c44bb60440cb571331e615392433e3ec5188d3ee652ed
819f59f703747bf552aba944c522be8bac4b6f94b34dc8e03cb8f46767f4af2c
66206009e87be735b0afa5723c195e75b9d69be437bbf8245e362c7d9bd1526a
fdb06490f0f8d74707c62703ab9cbf99f1ec7fa0123bccde2038ae9a8a74bae
response w

2b94079e749706e1646b6721f82f45abe50ce3e38b4c9b8a3f18c9c555994b2a
649e9c3de54a7a5e23fb4cc9393620ce3c01b9cb96796d973d238eef574c62a2
54324ddf9044ec3ebfafdc8e1acc0aadeef7dc7a6e5e592a9e7540fa4fb9e1f2
8ad8fe0591f8d27e4daae58415386a1305fec0646bf5da9cedc7554b77536f1d
8eab4695a84713790a2b3ebb109c08aea67524ce6cdc05ad0f746eacabd72632
e43a1649e164e5d870320c52e9fe779feb1950dcb7b33763b56fe2cbdc8b8efa
1188967f9d688f29e47083f7f6e1b59466f5a7c4be04b3b1e793cfe1ffc1cbf
27c5445bbc91b3984a6c74c0b536644d4133c200202d5b76eefc965026c6d502
Fiat-Shamir context

ff544836b7d7122b296ca1bf02a2f3d39e1874a5390c4e2e4bac74b2a1c2ae2e

withheld proof term: 123 792 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_mul theorem, MatrixTDirection:58

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} \text{ k}] \{m : \text{Nat}\} (n : \text{Nat})$
 $(\Phi \Psi : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} \text{ k})),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} n (\text{HMul.hMul}.\{u_1, u_1, u_1\} \Phi \Psi))$
 $(\sum p \in \text{Finset.HasAntidiagonal.antidiagonal}.\{0\} n,$
 $\text{HMul.hMul}.\{u_1, u_1, u_1\} (\text{FBundlePoint.psCoeff}.\{u_1\} p.1 \Phi)$
 $(\text{FBundlePoint.psCoeff}.\{u_1\} p.2 \Psi))$

SHA-256 of the statement above

fc4bf8765b613d27351bd77417e602731110e78d324e2701b48d62d74fd4bd62

SHA-256 of the kernel type

d1f25d52f63768b1ec62fb60b000d9d8ae2a1426155763b0e7ff8a3fdf2af3e1
commitment C

6e54d6c57e3399501aea4413cc1e66dbd1485ab277bd2b94980cbca383f3a40c
9a5016c962c2d435a0b36408a6083829557f3be111fbaf9fccf8247355340198
245e2ec075a4fed2b091af01626ed77a2c9e611f9c423a1c663b14ec3c55fce3
37a1eb5370d5a266872ba9f3cd9586f7ba311dc76760ab6fb8a8cd63dc0be2de
27ca27da39296c388deb9e5b60b95a54972427330dab4ad6e56b3a5c279f7867
449ace8f07b8605c11e4398f212039926fc05a6518630d5289f72d90630d81fc
c02ee4fab816fdd29e501cdc651a4367312fab6fb00d095b917834945f0727ff
f30fd5b0866be1507194ae0ef31a117e877866b59caea8c629455e7fb91f1215
first message t

3f9b90e940af42367dd8bb4fb7de688887c7426c922de37a31b80e5828804e03
c9e5e71c68af3fe0ace92d096323ea40e03cfaac7a5eac7182cc86a79ca41237
aaf49f07b8205015ef9a9e06feb589ff9688c285a6c30c4a9813cca617f9ad4e
bdb1930cd1e49135691b20bc202fba87348d2a1a7debf0c8f3fe5fa4f7a4c6ae
93a0ecff1b5ce11dbe6db7c86a7e2e24136dbd3e5a01bfd39cec591cfe6073de
36cdeaaa216703a414e421028c7c33f6c4a553f13cff80b4c77caf712411bd6d
95f736e45e2b27571c598f08587c67168be560dbdf0d8849117ffe45ac57f3a1
fc6eb31b30061c54a9e300e72026ce79ca1f955fed5ed6f960fc11fe50b8dafc
response z

1ebd04be858638ef24842e66bf5f59a1854a63eed2cedb8738621af91ca96a45
e339a01c20c22f878c2c955f17a857658c4e1c45265b6926a81955f3c02e8e0f
ee800e32a1195bb6c48f57d36ec6e859a92b1e1795ac25dfab03f0e23e66d75c
04cfcaad06551b1863fe94ff34b7a7235b50055b8648871b1143a49ebb4e95b7
95d67657a92763e3b085a34c6e598c732177c4c62fde83524f7fb8b2134bb23f
42ecc1d87798058200dbee17fb50d97415cd505ddf63c5cfbd3d76615f76b7b5
a1e85686c086efb98c08f4434dd4f649b499c4dfddde48e7ac920009f8268538
1292813e0f5b3db268a6e5f3fe5461315010f18e8c5de991eb397d04fe2f65aa
response w

74b6d74be652d7b64c5c242732bdf7ea9df708bfbed29e6a24e2271e094cfe9
500889ea1f84cc494a8f7a23bfebe377fd7ee2136b7790f7e1c0704aa40e0f60
fbd4ea7e71926f1b3d3170bf3e61445385ee1dc39eaa385a7f0d683a0b9fda9f
fe18dd1a92d5f892050b10869c28ed4dfbc075464340e3a9dd3a5c1483ab39e4
5e7e7df9b46b10cd06836b2f7a95c941f8fd290d6e26f9d58988b87ddee0e5d9
a5e1bad206c25502ffaa999a5a0688f753575127c2d11815d1ac97d410978064
d9f8f0f5a6b581b86c8d4b7982a51248f3fcc5da0948140e703397d8d2aa7b44
eef5ede86fc738ba358dfda94ac790114d0aa865683069dd2f989741af37e6e2
Fiat-Shamir context

c1998dc19165231bc32f68be454576257033d9320d5f69409dab0d37d816f72f

withheld proof term: 321 365 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

`FBundlePoint.psCoeff_mul_eq_zero_of_le` theorem, MatrixTDirection:67

$$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} k] \{m : \text{Nat}\} (n : \text{Nat})$$
$$(\Phi \Psi : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} k)),$$
$$(\forall (r : \text{Nat}), \text{LE.le}.\{0\} r n \rightarrow \text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} r \Psi) 0) \rightarrow$$
$$\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} n (\text{HMul.hMul}.\{u_1, u_1, u_1\} \Phi \Psi)) 0$$

SHA-256 of the statement above

bbc8bbb54ac4274d4c6fd1a4374a3ea08a19eda21df5eeff4abc7f2b76593214

SHA-256 of the kernel type

01a9dc4ac9f5cb208e3d83016d708c82a976dc037d6a8073aebf6d775e43a016

commitment C

a588fac9ffc41a28a919353905586a06d82e2f0799f59a98605672c81893f012
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9d501302217450208565834026be6d53a4e411d6f91457f5758ccf37a1e33b2f
998b7bd505f7350da0ca77dfabb273e8bf2f0fc31efd7082752958e5ebd2d052
8dcc83680ba93b64d2492e3c5054149e23b355c5cd797d119a0d086c0de16a7d
d00ec0110bd1ff4f5ac992824428b6cbb3936c2721aaf54a488a91f4beec83d4
66ba6801e975d6011dc6b8da12b0934e8e26bec802cb06e85f4bcd7be939d50d
439a4dad192c305ac13a20a2efb21035105d7a8639cb520e996f3c647751085
first message t

741857fab392c79c06222f82a1eeea60bf7150c4b1d56654a29187ac010ec405
be079a2f2b0962f98f488042e3a6b58c64d7cbc886b022c354dc612ebb8b2e66
b1bb21901e11a04f9008762484b7b54de92dee90e0d7f255f9e5b5e1b2a726d1
eaf4819f95281433ce173c52d41b5c5c1755f06d2ef681f0038592b853934c5b
f67bfa608bd9376454842e4cb97dbac5443df9b399fc389d65a69a4441b829cc
f2b647daf40f926a497af5905d955bb62a60c7e6c136c41753bd2c0be43c469b
098c099c4227a94f2e17a1223117d0064c1ead06984de718d24f440002ba113b
9d037a695243865bc48e600c89549ed59b3372fa6a1d9ac59e5de8e6cf7c78f6
response z

2cc821247fde7b3b9b7f13053a36e184d4b870070aa5932026f304d61a3b827e
0299d13ee243854ab67969af97fa9e526345770960938a3bb7aa2bcb51a8306f
0c350859678ebcb20ade32ffc55ee7a7b67b173d621c02f2cdc0600df390d886
6520de2ff0baa6bcff5f115eda7fe50161f5a397ec537307719b1877bc59c43c
3f8a2562995a7de3562e524299465448ce59833704dc938dc9c01d52c5c10997
8cc7c600b8fbdc1211e4ecc28c2ed3dcd65ac49eef89698c6c5507a96a52453
16ed28fb3e237b50b14a7bb2e2f4bf07e5643320d647f3acf082f77e4453dcce
82e6dac2c3b616ad16ab9bd6b4f0b584ddf40920b1989ba6347cd86328c0e5b0
response w

5e376f4c2b93c757e28d27a1d9fd56a4da947c0460590433ff8df3d2c2b92b93
0da2dd2c33b4c8dc9db6a4b7a40fb520df2c098e7e8c4dc2cd967c8a982ea380
9b3415eb317ddc0cc451f3e9e91108489d8cd1823ba964e749bdf0cb3ac8557c
171608391f67720641513ccab66a5a07a4dd8ca8aace9f94718a0e4eafdf4f8
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75b9f4b70550dcf9a6db9c594b158d31332aee4a0515aed7eb5dddbfc51249dc
79553846b6ce4e0e8c59dec34836c7d964c055bb8ac2b06f17a9754724c51c53
0f84dc31ab278f8839fbae889f06f611a0290adee2d2941b2b712145aac2ca4c

Fiat-Shamir context

2442650471df6ce4813c7c5afa55ec4d2efa2558ea0c2133dc8419541b369a73

withheld proof term: 41 747 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_mul_eq_zero_of_le' theorem, MatrixTDirection:78

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} \text{ k}] \{m : \text{Nat}\} (n : \text{Nat})$
 $(\Phi \Psi : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} \text{ k})),$
 $(\forall (r : \text{Nat}), \text{LE.le}.\{0\} r n \rightarrow \text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} r \Psi) 0) \rightarrow$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} n (\text{HMul.hMul}.\{u_1, u_1, u_1\} \Psi \Phi)) 0$

SHA-256 of the statement above

a9860df63e6404007d75ec644c141f0bb226958fe998395181ceb2e86adc0165

SHA-256 of the kernel type

f3b0e259efb7a81cac36ce5901b142b5b8ed409f3b799d77dbfdc0de4cc76056

commitment C

c26ac79e3d0452650c02d8e4e9ddfea857ce0eecb35eabe525bd366d682e0aa2
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cfc7f2d7e4b12651c950f09d87e806522c2fb7b645640c7cd1310f1cb3c7a21c
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first message t

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fca7c44e6c8d744740fe01c57e73848cf9cecd23249ecc74bcc1463daed8473f
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479887340f3f12e7b059bec1f5caeb168ed5b56b50714127220c57d31286a6af
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response z

7e6852be03b141f29c723d58170262fe2ebb296344ff675dcd096d8d9fe8e4ba
878f8468385487ce0ed700fe46f292fb6afaa1b5c1eeb67ee530e95a47c0854a
b13a6bb376f933ac958706e25be984b48f8e21c4ee5a266baee5ef2a64ee3966
2bbe57ff211b85b3836400a0a6a0deeaefce8fa755033dad26059832001a9391c
dee20f3d2f6b7e454033a3bd6c8579f5fa91d77afbc817ee49e1e7e4d9b86d79
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806091a867bf212ef6f34fe235531a418700bb9a4c52df0fb8bfa9b627352785
66379c13e9883446d1e028ac6c68c489e8c48e10108fbd0d8217abe2d4080e29

response w

4d5cd35102064cbff4d6236c7f605ac6c2b42dcabf9bce794cef30562a5e8358
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2dba0049514a41b581075690896223dc5bf5d5a1f0e0d01af260aa63744df90e
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909ad0788489a4c52a740d2675f54342fc85697b8e3a4cfb74b4cfa1ee7f89d3
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8768372a8087b1f35485d84faccff18358b5889ce6a2d52d03dbe9a559fe68fb

Fiat-Shamir context

4466fb1eb2348aae9d2c45a51e87a28d9c3b59b177e79a1c0775019afc06b58

withheld proof term: 41 748 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_neg theorem, MatrixTDirection:48

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} \ k] \{m : \text{Nat}\} (n : \text{Nat})$
 $(\Phi : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} \ k)),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} \ n \ (\text{Neg.neg}.\{u_1\} \ \Phi))$
 $(\text{Neg.neg}.\{u_1\} (\text{FBundlePoint.psCoeff}.\{u_1\} \ n \ \Phi))$

SHA-256 of the statement above

a55fae4ee0d689cd0b785d2bfff89be845fd2460757f1338b645f6f26c5a6d6c

SHA-256 of the kernel type

6639b4501e33e621d78c8dddabfd8b9ba1baf502afcd14bc20f4d68e6c17602

commitment C

a2cc141b6319ac6e79d29347e4fd9bf21f1fd6982c9f8c6be06f6738a30d561d
b9d0c4582fc3ef62a0d301a880572f6fa254bc37c4cdaefa2b98b9446cd8223
b2162395566ebce96733bfc641e87498ae02521427e324ef767c52fd6529a69c
38b94e78772012fdb081fab80385a4a29e21b8333ed0a9173c39d75b0e8ab6cd
95500c827b68f10ab1627a9eebf3eb321612f19f3c221868a172d7273b168c46
8479cc4e673761dc3579e4ffed0b601aa5c713d60f8eebf30e27107192ef031
2dbd8d794c3a37f72d3d82faab3ecef36a08ea53696d5d776c6f44a19e8db1a1
f58a0f37f7f600e00e9bd740655f6717d3c9d45a62e4e8348abac9fb42943

first message t

49fa3f92599f86e53cc5c5d3f8fedcc69314cdbf0e833df47b8da6684fde899c
6a42b158988fcf5eb07ec79011661618ebcc2f6874ded70d6786b5f228e7b223
f88fc7e2e1cc9d2cc60c65e4edc7178d60396ca9b7b5a591cf831758faa10a05
9edb171f460274efcd7f17a7864152304463fb4ec22ad547e80b4c9bae397be4
7f86ea0a538a27746e9d04562becdb8a60b283b2a047f009a0706a6f0f7ee5b0
fcc952370c48ecea2b34f8d60645194d73ea7304ace4f8bd41e981bfde9eac6
1cb9e6ef1113ead799c4051e5a613e4faf0403f5b55089433c3113a7fc734b88
f6c07f887a83c6f6b258d4550e78ec3d599588cf1e30c2fe31f9a58d4bd50a6b

response z

144a9ef3a0718773395a3c0254b1914564ee55df53d7e7f672254e5a0f7b2d17
0b32f2a269a76150444c75bd42b880d751a1a0e7bb1c15e19c8315d2e0d0fe90
7527e95b6bb5b14f7e8d2de2dc7e5b2701a0dce2e7c18594bfb845ef5bafed0b
d75a7c8794703f57c96422e8397249c5bee03dc5aa89f5a178b6ad5c5f96c5a6
1d68bd476859706a9024d4cea6a8680c8ff0f60d681d641cfaf0cfb7be84fa89
2af12e54681b66c7371f88b79c8816d2ca5f2603ac33e74023bd3c245bf2c866
af077ac45d36206f69e910884d3e3d26e6a561e8b8635ae4389882f16e06d72e
ffa0ccbbb7f7b9b18d1abfb0dc352707e55a133dd93e5b32b8e630d33dfc625b

response w

45e5a2fbd919d73741ea78429c244765b70fbafceec2a462900fe5752e35aafc
3a2d6ad347ce8032976cc73ab1a840befa6e12829e328570008ca4131efc1ac8
7e63305c054472c4f4c3309077634b6fe005f5281d37ff804812eb8fdf4b5dbb
bf4ba1ff61e6f27d3c2f97bb0d0c15ecf86f83bfd7fad9ea690099a900beaa36
dbefe36c2af6cbd408df98782f3a1081d0e33f26053801a6fb3a65c37635b0be
ac100bd19e4e13f475ae6d87bcd94540c150c28450a3655b1e5e49443a5cb09
ae65c7df28677d728ac24b87eaa94822c1a9dfa29c1e157985b1c05b69a43293
76d3848c1c69d9e00312c9046097cbe6481007931abde70ef92c42b18e55a476

Fiat-Shamir context

0623457a0df411c4f4bba0dcf3bd95333b52205f2e1c32397e39c37cd666b61d

withheld proof term: 116 635 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_sub theorem, MatrixTDirection:53

```
∀ {k : Type u_1} [inst : CommRing.{u_1} k] {m : Nat} (n : Nat)
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n (HSub.hSub.{u_1, u_1, u_1} Φ Ψ))
    (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} n Φ) (FBundlePoint.psCoeff.{u_1}
      n Ψ))
```

SHA-256 of the statement above

aeb77a727d3ef5defafb18a4229615db281ec0fd099f18dc6725eb624c7e5b13

SHA-256 of the kernel type

f7a0219216eab412ec9e4394fac200c7ea6a7a1f841a430eca995f7e64c95e56

commitment C

dfaa39b48391ccdede4d6ebf7b2e23433fa600d80742c660d5b9da97330e05f5
5f02b60fa91ce1819504fdb974b3096f91976eaf96a3abe9f2b7b79b5b24dd6
3e67383d2f270302c1d46452708b476d70839b7e5d2b4786be84db2d3ca823c6
9d030329b3211b1cfe5f7b4d944302d3fdccb5aa61b28a0e94a584ca7f6b7370
344cfb002b91c07d2a086a1c32c6970666761bcc2b7bea39498074027d7b100d
0872f72cc62fbab67f0bb86d37d2cc73bd23d9fd320c44bdc587dd0b02eb4b14
5b811a6f6adf0b57004ac1265598be971e6d9a4a048b86608190e241d340b340
c7c51987d5f9a88876634c98a94c8bb969c75ec2e7c336116830db521dca3e10
first message t

6ad43f11ae5abb79a088dde11ae6bb2a7a96fec0bc080db8322bd5ef1991ae42
737eb7704ddcdcef75ec2bb1ed8f00d480ad7f0a96b43f053f64304330110f62
a0abb84fed9c6d79ef58586ecf73ff8a57aa777815ea6cb60086febd7f97922d
b6b1b0fbac557b95311146e387cd27bb0f2acac76db3f0d76a3869a8f8817976
d0d4f4827a5adec271dac192f844b5af6ed940cd67f97e5b01f5fe1316cc9e65
bf0aca5bc8c5c909d7f792bee398b4cb2a74bd75ca58285893b9485f79a90c3e
ff9d0c46d9e2259741126316f2b86c22921df3d0636b9b55cb0f73332af09dab
ef7fb4f9b13b6dee7074305a7219fffbcb076eda7bf9ab537e1f26f08777e8850
response z

6a31246626a690cf2e7afac2b8e6955cd7e30f43c559abfd79be4a446b180b75
c87281c86c5270ca325ab88a744fba31825ebb29576bdce61fd129edfef53e80
47e8d01b59c6b0e7e906b4b31acafd62bca9e1c37c07ab30a82e791b938e535a
25ddc63f9795a90fb55a2d298f325c1dd773a5fb7ebc838123eda0460aab44d
5e6bce1e74f1ab2181f6760ed8c72a05b5032f8233f433d1b0b2dee658d502d
3197a90f148eb0e9543e1321e493fd6babce4839c7e2627d44708e754b85ac2a
58376daf0066c2e8a3f7a8a20405e2dfcc093cea347c17e2effb2c16940336ae
2cbea04d849d188708277a74d2bb02beaedc1c8eab1390985e808105787e0f29
response w

78fac30970f038c18c0720c73df7d4ddea825564adc3d081cb823916aa3b67e0
49a3923750bab25f49a534ecdffa946e283caa9d10b2b4ca6425a107edb08aac
2057a29e226b5890082a5ca6eed3d43188ad2b5282cea82a812e45c627e21de
137661451e6e0b9ba989cd982ed452e5c3e6008d3142a6ecc43fc675f05e8db4
3693df2e14f8c24ebcc1e3d6754c3823efed062650800d1d65f0b2be5b845172
5e96d0cad014461914990276629d1d37a7ed22a6fb790bc3c83784c6eb476a26
40084879e37175af5f45532827990b24648519276fd3329f190d57deaf1af4ac
198be237dedf189199bc3ad3ca97dd6eb0827c64ea9e47f52b9bb743d51ca223

Fiat-Shamir context

ff09912f3a03ed694270714ff2f84d5173ed0b2da7896f1cd13aa18dadeed01b

withheld proof term: 174478 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_zero_matrix theorem, MatrixTDirection:43

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} \text{ k}] \{m : \text{Nat}\} (n : \text{Nat}),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} n 0) 0$

SHA-256 of the statement above

37a891d84443b02a263d44314e976a68a2511f658acc0e6d23cdc0b5c9b4f18e

SHA-256 of the kernel type

58e0ba2e6df7a1b2dd0f99c13db1f812369495c6d88bfc8f17cf7a990ee36982

commitment C

a7a8dbc92f10f307b3536950c4fd4cbe50c544b7e0d5d4d4a1c8ad4d3d412bdb
d66ec5e87bb7b206b0662aef790230db3dcb5212acb3383c74681bd6eadb9435
9473804947ade8480cda2b5718c39ad1bc9acd443dd8d04dd8693fcfe48ac020
0b4a30007c3edc5fc463b2a6cafe062aede1c830e568a9a1a3880d42c60724e9
a44014712d27e495d4ab0e92fec79e3d2ca39b9e849026d2e0fab23842fd8157
ecba8ceac8b469dfabf39e866eb467c040f4dc70192fd72050647d799aadee50
bf4e6152718d8e5325ba693ac432a531c6b82474f2372289c0323e88d5154976
397f183d06864046dd8f31c3f13087d316094726b0bfe7d835cf976747dd8d62
first message t

6675dba22ecb2af29257474f0ab49fd38518894008a1dbb1b31ee2fd976111c4
640910a3c4cf9b2eb4ad5156348fe63d37ca674f051898c34b1f7bd07bf36e9f
c107cee88da71cc7603e741b9ed8aed49ebaf990533e607f21b0851ca3ca6796
d87bb50554c98be30ac57cbbe943dfef444b1d25522403936c8e30f0ded5bcf8
f9a08217a8bff55d487ff5610f6c89fe2954100c61433f212de175fc3a2621e8
28dba23ba2d32f360b601b039f6a882f5255bdd9d50bde1ea7fbd28189f742dd
95927640931fd430b7337a63c4ce9f0dcaa983547a39ac7d654c430a870fb1f4
a9fb4f1443807b00ba7ffe76e6a5cc3dbac2cf1ab918bf9c115acd8da09b0e47
response z

5206ed2f8efccc7d137cdbcb18d65667324680b8948e0742092038640270098a
a5e2ce59385ea2446947214ffa2daee7fa5c2d5692640349acb8fd558be523e8
1f951cf05f79f027305f17d19c985c0e74d2617a883c0be334d1a5861220047c
1fe07d6351bb97232f9fd37dd967eb7b02431b4adbf9f00f9a00972059f5c096
6e2d00e70e5f1d20568088679092d8173e4fcb9ca07de8eccb03d4afa4942ca5
d3709d2546f0973c9317fab15a41bebe4aba1042a20f760eb147ec3981a0bb4e
d24f9a49a3cb73a8d8289f9229271f6a548a5ed8aa5fe503776aa9e6cb0395f0
8f698471f34b41a38fa7e8003de01eb1815e22d2675bf5851e45d90de50c910c
response w

4f1bf8e0f0ed3b48aab8b8de752198ff75fd0c385ac92c6114fc25fe77b42b8
6330789b692bef06efb4bc8934f5bddf7329de7b43b28f8bcefdc62f38713d2e
692ad027b29fd76e64ed21f63ba06ffab08342aeb8e8234e72fd6438e7e067522
4186de4e5da74d4d5da9281fd0590b74ec1b3d82be4db615e03e554c784f465b
1aa62bfb1c59de38aa8ecc51a8a40929ed9027141d4a04dbac27e4a36df105ab
20fd10eb803c738e45bd973da01e58be97f920cd1c18b06f760b06700fb03c5b
861070c41488c8a2dddbc6a1f71a69fa718466ba0b9662299b05926729ac93bf
bbf7f238e80b3882a3ada2c73580d7788b0562f55ffece322bce13dbd79b2d4d

Fiat-Shamir context

55bc380ab3e5b827e468187472141929ea856c09c9f907566a73fdb64f2f50c0

withheld proof term: 66 170 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.unique_of_derivPS_bracket theorem, MatrixTDirection:123

```
∀ {F : Type u_2} [inst : Field.{u_2} F] [CharZero.{u_2} F] {m : Nat}
  (T : Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} F))
  (X Y : Nat → Matrix.{0, 0, u_2} (Fin m) (Fin m) (PowerSeries.{u_2} F)),
  (∀ (k : Nat),
    Eq.{u_2 + 1} (FBundlePoint.derivPS.{u_2} (X k))
      (Neg.neg.{u_2}
        (HSub.hSub.{u_2, u_2, u_2} (HMul.hMul.{u_2, u_2, u_2} T (X (HAdd.hAdd.{0, 0, 0} k
          1))))
        (HMul.hMul.{u_2, u_2, u_2} (X (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat),
    Eq.{u_2 + 1} (FBundlePoint.derivPS.{u_2} (Y k))
      (Neg.neg.{u_2}
        (HSub.hSub.{u_2, u_2, u_2} (HMul.hMul.{u_2, u_2, u_2} T (Y (HAdd.hAdd.{0, 0, 0}
          k 1))))
        (HMul.hMul.{u_2, u_2, u_2} (Y (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat),
    Eq.{u_2 + 1} (FBundlePoint.psCoeff.{u_2} 0 (X k)) (FBundlePoint.psCoeff.{u_2} 0
      (Y k))) →
    Eq.{u_2 + 1} X Y
```

SHA-256 of the statement above

fc1dcda0a5b1ba32cf05992a0339ef9ef3ab1ceec19a767e9a68f6e4babb8e36

SHA-256 of the kernel type

43fec9ed55449c550d3f95f6d8a0d1c41f15dfa302dc81ed78e17fbd3ff35f45

commitment C

9db76dd9d97bc7c3bd0aba19ab3d0b15cb325f3cb9cbd3e337cd4dcdec188cc8
c81f40bb4e93ad4a0a5a4c20c2f1a5d6208ea790382e3643b92cd0a941260b01
db7ec33b5ed1de29ddd1bd316603250c2a824b6113099d16a8a523b57868a1e4
270b9d3f105173975cb39089afe2e5dce17f132a6f1a3d08ce02c2ed2c3ec86c
bf524829c1c81732a5e7ffc764e4a1a92d945891ec7070bd8888c7adb331f48d
9d35fe1650e87b09fa6eb0cabe8e04c572d0185f7b87bdac05168bfb81fd20ec
4f5a44ab0ed9478132adb3a1015bc49aaa13b0888e2f939fac5aa8e2031e38af
4b915f309fd8c0056fbf693f0357640c0c41b7bf55d54559d653e0f5ecef7701

first message t

3aac6bd825cc0bebd6c9af53e49e35219a867bb25796506291f7d5180ad79290
4b3610485a24bcaa15d118ac02ebbe7f62839c6da99e0127e5e4fff6d91f3d16
d72baa473e5fe845c0804893a53c0048b396c77219e7c64dc3b5205831d400c2
8cf119fb4c73b82d583abd43441ab49d6d75404685d48efabf452fad191548fd
4e9e970a84d3f7c627ba1f7449076acb5745328eace3dd893b853978c357c3ef
793292fbab80c6c3c529141bbce1cd62fdafd276b062da67d64e0fe87d3518a6
5633ffd1258b0a67e8868efc00e7730c361428e45522b93428e9213062bb8767
346575166ee6efb622ec0f56608fb3dacc23d26404ce0030e841db886c7faa1d

response z

26d485437e70468bb75d3a42f844042af373e75f2f8428fc01f63e122237eb09
cfaa52781b2ebb1b89dcd93041a0c15690a08987398d9c5884c424b92d42ec95
4f8dc5e05ab34d01519bdf4943f0953d3d9a0c3f1ea605d8cd3508779938f67d
9c4f0443983612566944e17941d9cf23afd9348bf4338c76b157292baf8792c0
0267157a188661f7ac05f77593cbeb45c9a040eeeb1ffc43b44b73802b7b80ab
7b94b79eee18643facee6bf61724ed62fda46d5e6e59929841496ce1f5f73942
95cf1d98512bc1f24926eb139933fc6301f126b56fcf61b7378f698f3dd69f29
549abc2bb38c04b356ef7824e4e1c0025abc290a7917f432bc52b0e41511c3f

response w

4ad6330148e676a8b667b2eaa5c392d189d90a4c753f4f0598cf35a4a83c091
2b732ffcea03ca7157666473813d216c1bbcf95183bb7dd0fac438499f612690
7a9c004d712445a6dd5e41c7602039dc3c70e23dadf1321c2bd8fee022e55bc5
e85f913888a2f3ea646d7dfc1eca4378838e1aead850e2d79e0d8d282c27fb5b
d239120b68d1e57b8fb033156f8a040b0deda492a23c096ebc03781e6cfae189
3016f9299cd24f5759359b3e3005e7c77ab44d11fed5926485cee7acf0c4afb4
a9f93182c9f5d01a220af712bdaebb63eb53f96ff0379af1812e90432d003453
be670099de445c2a2b800e9a13503e7d6f0ca0c6cee87667a96862ee84c88fb1

Fiat-Shamir context

e7270baab6680b01b7bee1649c3cc4c1365c8c8414d1dd6b0ff0ad3fc6d3126419

withheld proof term: 1407526 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.commute_bracket_of_mvPderiv_bracket theorem, MvIntegrability:68

```

∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat}
  (T : σ → Matrix.{0, 0, u_1} (Fin m) (Fin m) k)
  (X : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvPderiv.{u_1, u_2} i (X k))
      (Neg.neg.{max u_1 u_2}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.cstMv.{u_1,
            u_2} (T i))
            (X (HAdd.hAdd.{0, 0, 0} k 1)))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k 1))
            (FBundlePoint.cstMv.{u_1, u_2} (T i)))))) →
    ∀ (i j : σ) (k : Nat),
      Commute.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (T i) (T j))
          (HMul.hMul.{u_1, u_1, u_1} (T j) (T i)))
        (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X (HAdd.hAdd.{0, 0, 0} k 2))))

```

SHA-256 of the statement above

44bbcc3e240447aaff127c9a23c45492903f801e988828756dd46949bbc81bf0

SHA-256 of the kernel type

c76b134a60c80a6f95df5aef0d08f56caa11b2028fcaadffc179bc48a8917472

commitment C

eca694f4d5feae7037cf6f0487b84e5560117acbb9433fca6150cc31f0a97042
 ef008e7c1fe7a03ac2cef59f22258f97f6ba962ceac2e3ef571e18a45e562aa0
 6a7317698dc1cd2281cad43c6c5b92cd4e1a16137c1296cc4eddf13a0de3440
 6ed9b577caab699bf7d79201f86391f5e09bfc81a67117b0fc555f555cbec291
 3d3ae8b0cbeac13aff2b13c6240d3e559a1bc72949eab5e8696a1dd99fba0f05
 89dd1f044beb887e2b1024a55b702fc9750fa2a3990c7b81abf80028f2a00d81
 c525dfc1f61c878935705a61db8a48fde657fb2228e5932d34a1535f837d5f15
 fdb984199ce4c92f10d241d8a70ef4c7bf5eae6303663c2b3e7aee9e7c1153e0

first message t

4b28456e03d1f0d221e2b69b889e695bd9ddc1855c8de53d0ed9986a0252a7e2
 0fbdbdc4603997d18cc18225777e4bfe539957209ca0d641b45ac42732d26a13
 2e9d7edf071176ffeccdf31a9c382541cf407ce1cdf380d1ee9cf291741970aa
 055b5808f69e048931a953ecef1cdf31da3a32663b2bf3ce28ecb33abdb549cb
 49472519941b4cd91cc9884fcb0d063a8aece63d473e2099e6a5e6fe0e5a5d
 da9ea7b4689bfe8017d0fec2b1a1f780e3f7fcb56187ef0bc7010a3f84409ce5
 651cd26f10d98a87fd9cfd11fd9d731fbad07a67e894aed496285e0b528a7413
 e41e7e0729ce98be03314adf2809b2a45d2e158c030c841c65f8b939ac846d0c

response z

63b7afca9c8c7d0e53254f7ca71e320cc167beca028b53d2878778355e1745d5
 0e59141d21fbb880be37a43d70dcb01355e8f6421833d5684514b3dbe397c81f
 e3a0b436d1dbc2ba9f4aabb51ce9e0199085548728b3e624b1fa52c7676dc74a
 94d4f9c1ac7793cbb103511d81a18956643676d4c31500d8d8359b3e66beca91
 9cf36497adfdee9b0d9bb5037e8a6c98f3c21838bbc0fe2da8a91e6e3f3dc1418
 b6f1aad492d56b1f2d3f6e77337da0fe469a45a670691ac01b5954df8a5c36e5
 2c35e3a534caf7ae8c266406d3c4473ebbbb2e0a7c4a289418206238dad16380
 84ef891acdf77598d241cd694a1befd11a94015059d835f0221c53523c42238

response w

4d2ea3b722e6cce4604f2ad513cba8d35171d5c994de41ebb347319c722b5ecf
 9933bd5dde241805833882042fc7db6f36b1bad3f9416cc7893d518b39f2a6e3
 0b9d71ae8826cd601b2ac403f3f0cb6e81cf0eb7d0e504f8cd8b369a5198c3ad
 a79e47b5c61823b6dac6cff01e456cb6043a6013d3ae940650e8ff56113e0851
 f1346a72715c8b5617eeb46d3fbf4b652254752edd1270120388ead0dce54f1
 d85f37ab0268b96914d697af9c2071b7617fb8f1e0e197d00488acdb5d03bfef
 091b703f61d8a4ff738d4d674b90908f815c2a6ee46d36cba27722bb28779145
 daef9c68bbb3fd52e51b11584be48da476cf9942277a86c6a2fac8fd6aac217e

Fiat-Shamir context

16947300178acab6a1bbe2a0852260f2a05d577fc687eda72601f08b593ad54421

withheld proof term: 4168212 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cstMv definition, MvIntegrability:38

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {σ : Type u_2} →  
      {m : Nat} →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) k →  
          Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)
```

SHA-256 of the statement above

8068042d4f7cd50cd7917d38fb0f2fa870edc0dbf1b10b0d7a9ce9ed8956de35

SHA-256 of the kernel type

46e9aa16f8722f5c5abf3ad4b1cdae48c03625686dd37acf645d86647b2b122c
commitment C

ef7dce1750ebfdd7a15ed5e50bfb62640a8dfcb97e7e389e89018df3d9fd1b89
4e294225004b7c55da71151719391d5f47524ed82b0c295dbd5f19bf90469432
6685e2fe3d64b215ef385a8bdb175c62022b23371e61da2975bd6c170ee963bb
f3008bb49b6d456d4d5c7cac73c20119f9dd9e087ad4ef96be1aa4f0878d907
fba1d0f74757c9f3c3a45b29524abfd6bed73ffc480bc921589620af44b5b417
7731b81b2e389e4038b80241342a4f94ee3dee893f496a196e6cae63f105426d
b575c302f4bb6df77687b1bf99f9ebf2f6f1f43b0a82a11a920ecc863005e30
0059ce8334a3b8d09aa4bcfb14754ed00c662d35a4df8f7f35ab730a6a9b26cf
first message t

3004d3e21698fec21c728b08f582d59833bf424f186517cf2b9cf65436901a8
81eddb385f145c200dcbcfafae9f7e18a3a3fa14048895a9ad5def755c1c48b7
172d41a63f9649ef1e0db602b23c17beaad1c358907dcb03fde64644da970f1c
d25195af436d0cbb7a26f823f376b18021073155a7c0305d4d7ad04717e94ab5
8e3e507a737f3b18ffe43f4f553ab4fa4692aec77954cf23927fc75e09c1debe
c53173548a9d002cf399e01e13bde436293870d2b3c320a4b872e22cb51c9ebc
d4e2996051333955b94d29b8d3867a44c5ece65251981c75f7311c5e9cea221
926bf7122db982ca72b331fa4a7968d4f6a9a9c052b7415d9599b4a28d5d730e
response z

e6c6f278ab6a4854a03ce576057d463ce6dce29d2dcded49303135fa6904c9e4
1b43ae684e2c7ef94fde952c0cf9cc47d4a83e613945c4a46c6714b9d8d32c19
3ed0c7282f1c4589118db6acae53746ed323163d88eee81d0577c6845ba4023d
aacdf0bc6f96ff4415013995bb96b8f389b5257a6594138910a10f19508fdbc
45c8b291b4295b546ab9d0b14294ed09276a628e4e230a502d5038bb8fddf5b3a
15866348b85613958909739ad949b98ec104ca4ccb643b8f166ba25fe31dc2ca
71d712ff5349037b95f9761d4c618a0b8bcec9dca8a5dca9dcf11e1242c03220
e20be4b44995fd0ccf18a8513711284180c7e042807dc6755e67babdbf1c3ef
response w

6130fd32d009de27dc03a7e6f072fee8f1c9da56b56d87111f9f6d6143329199
6f089c1caa1061dd9f603583c35c52f2ffcb3c1ab46aba9bbb1ab423c9ca5f62
4ab8381e4e5d83e1f192296b5c17658f70cc0bfcab736dd5a12004a734ed6180
0833a64e093976f7f62b27ebe097833a5dfe65db765e7f3fd203f57c76563c77
c726c692437cf5a7ddd09cfc1b3ffa5972a956379c3ff2baa9a1180fb43e5c6
6c962897b8c6ddaf82d3c55ff5e97da93a30d5ed3c2fb6369bfbd9c61f0972
8a28d8369411e9db9f53749bb71b9d0e9b2ad83334f6d8d6b0927e8b433991b8
d9183664b34d5929c8709746bf715d846deba58fd01a29a595786be504d5201f

Fiat-Shamir context

11b11adeaa32582599ab02e2cf5a29b1afcc52f53aacc35fb76a8f101594d736

withheld proof term: 1887 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cstMv.eq_1 theorem, MvIntegrability:0

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ k}] \{\sigma : \text{Type } u_2\} \{m : \text{Nat}\}$
 $(A : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \text{ k}),$
 $\text{Eq}.\{(\max u_1 u_2) + 1\} (\text{FBundlePoint.cstMv}.\{u_1, u_2\} A)$
 $(\text{Matrix.map}.\{u_1, \max u_1 u_2, 0, 0\} A \uparrow \text{MvPowerSeries.C}.\{u_2, u_1\})$

SHA-256 of the statement above

ef37eb0e1a4edab0fa8468aa9c5ee53b02f1d4f69d783bcd92f9f27e2452aaff

SHA-256 of the kernel type

5b4c13287e1b196b4b3f79953e2ca322a9989c9df7ab71d687eddbca4658505b7
commitment C

91254ddac43b042120e7d685501d3054846beab2cf5657e3353d6afd51432d56
ab0067c9eb10221d58f8c2e43cc8acb11145f73fde9d412d42c8e0ef9f8c0920
bd5873abdf7bc3af3c1e61d5e98f13f40d8d960104b8de6fa5b690a654859a69
1e0c4f55d61ed3a517be6116a175d5d324b0a3fcdc24da5f7f6a9395c29b5f24
b0991431ab6179eec291aaff83e07d2888b3ce5b56237be698c6a832d2064344
be64d7355fbb8bdf66eb3881d975d4f981989b08a49dab137582b2c2c797b4f0
a63326964b5be99007570871c5da3878f955692670e0520910a174636c710be3
c972fa02963e587a42b7ca751d602e8d6dc72816e821c5793dbaca6a895581c9
first message t

7762d55f6d02d2dbe73c40a93111c018db62a6749fc80ee1cd54f7009eb86c75
56172da2334d244f8980625fa588a00e2217fce7c2210f943d09fc542e073864
ef738d95caf71375972c08a231ac89a6031595f0dd439c4490381876bcbf2bc1
1abe1efb8388b158ce4b9668c5384e6941c6c533e9d5035a6c78de23d804ee6
7b22faef588bb2aec517d80fb5ae31a40f29ee218e6d51bdb02b26bb02dcf8ae
d3fc3db221d4377a38f49cb856b0b87ebae4ad8c1c74d381e264ed87b67ce3b
ee14ebe6ef5279cedea3aa173ffe1bb90ca1105d245e7b782ec7219c8dab7ee1
8d6900e94392514862f93e2375d883ff6d8ca7f6f27fa7c93b298fd1b87db042
response z

52b7fb1a3e0f1673cb8004a96e48910d94005f36592faf94810495be39e66bdd
b34fa2b89177d21e77ab5d894ba15eb8d86b83dd3a5497292720d5f8db006260
5ba33c538d418c303ffcc924e678acc4f83f23d218942fc8440a193559ffd462
de2b1a447c6a5037d4e20b0410fc5f39ea57367232e9945a46dbfe78ddb81f9f
4cf2a4b6f48026c873a39ab430e7669f2535d733f94f63f12d1d95cdc94baf26
c376d8b00fc0dc95080c8ff81dfa4f85958cbe554a8f4a46b1b2f12cb97f995a
d79f49a6540527eb9879ec4db50a0cf3a216f8a2159f91e54e7816ff716de03d
22bbbb84a2815645f1522d1118471e7e008268719cd061ae06661c4e92f5309e
response w

496c1584516b476f6b1ca155edbb9dfdf84d8e86ca1a1c5c3c98d7a834ea06f1
3712d2c63f991a0be668cf29cdd16a0f531fe4c670238dd35f53dbdccc6dbc44
8152dfd568cef5cc8c3d19a178473b95ff58af3c8c4297895d73bbd9db95bc58
70b81e32ab2ca7bd56f305d76ca74f4a9002927fc0ece697deb832921bc7c1bf
7711d3e9da4c92b7e0973844ac2cad85403e6bc29c0ab65ecc5083fdd5bca2ed
e917071be7eaf3b0fd0a1f24b10d467fa2df815a0f209116b7f2d769406bcb9e
2ee31d8086bab181a7dd22c957af822b70434d143dc0c3921014e8aac7a978a4
1c763bc57c0b0f5dba4da5bdd09e27521f7c779451083721687f96a43633119a

Fiat-Shamir context

ba1d3c78a7fc0afff77a2c78218ef5e8a13a1d969ce43071fbde4a8d47aa4136

withheld proof term: 284 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_cstMv_mul theorem, MvIntegrability:42

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) k)
  (Φ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, u_2} d
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_2 u_1} (FBundlePoint.cstMv.{u_1, u_2} A)
        Φ))
    (HMul.hMul.{u_1, u_1, u_1} A (FBundlePoint.mvPsCoeff.{u_1, u_2} d Φ))
```

SHA-256 of the statement above

9e6a456d287e06487b99f2da3977ddf660821d6e45bfb898793e9f05766f2f22

SHA-256 of the kernel type

35d085219d7576232f69309ad35a85866229e88c23574a6fe20fbb327632b1ef

commitment C

dbcb530cf115ec8d93fdec4bcafb068470cf1704bad66bd50cd369b958080ba5
8b1466ece60f8cc01b08ba07c0a837e1185ad84cbdd4b80d06fa88a842a5ce8
28d1651b547eac200ef1d13dc70db7c4bcc0cab32f2a85c6a14cd394d21b5e5c
dc1c49bf0772e8f669c4b0c11a7d4742709467800112b5fd54fc9deede34a903
a1a1a951482b0a4ea017816de89d9c49c2779c27687e617b24e1f75fd4ee15c4
7edc5947cf98d38611e3cfed51639ce75724a2ee0224cb2ec71f86a63a74361f
917ab2d6ae1b6605cc0e15253672d7e30eb58a1102da55823ad72a4d4c79719e
a3d6666935723a9472d4828d0cf87150139a5f2a7295a418a4c4c02a3d373b9
first message t

9ec7d9e9aead07034810113a4b31b8d5afa9acd028738f28a8da1cb6c49de28d
d731fde4d4c4899840396e2266db0a3727a2ee5aaab1245af279699142f992e2
2322286fb3e00478fe166a68b8a5d0bd9b2f3c2caf9cfce8d0338edfb4d38256
115738751902d3d88f7b327075079a41c8184f3fbd1e4308bcfcd35459cce6c2
ea72bae57896981f74d5ca17e069172b57bfa149f7f079b3e51d19afa72f2b1
fe9283d94c951933afa4847fe6e9d8f5605da684c79736ebbb6eba885daebe9f
ae257dedae3313dfc10ce9c821b9bac5cc939347a422b64e2eba772b37ebdf6
9088c2d0755ed5e5fd2ce9d17666c27dd0edf52896e247b9011872fee029683
response z

62fcaa7431ff1fcfa741bab630fe03d70e9df5f79af8b90dfd285faa888b0a84
9be654608e748c9f8ba703232bf57947132d43c3438d1775a645785e47bb12ad
b7962f890b74abb2b4fae4d84c8721556e6cf86c5c9f3e25ceb5dce2e4bb1a04
27d7f4325ed44425f99988c2ad8380a637c5965582c79d9449973b21753f9bec
01cf5eff262c653787db236730062d9567b76e662a0a7e9fd1b903a4616e6d75
be6d30baedd89cc10b16a4acbfcae58c23d21311dee2d3f109832ee51a8813255
27f9374e9a4746d402913f074152bac6c1ef285b064a6530107467ed1dbaab0b
deab8b17b33918310c78a41c2de2e74fe45f8df7f567632f45d783f25c77a3f4
response w

14a2fc3b7b2dae5eed1dbd484d6512cc77cb6173ca4e5d9e9048b77bd77ec083
32892dc416184f3d53cc41b83139aa579892d82b666be168710ce0934faf8f85
576d7cc1316456c8e40d9b03e0dadee7f1be579b78b95a69b7a4cd8267022795
3a7c0177636da405631643441f49979b0bfe006a6984ea977f81cf17a6a98eb3
680d099b168a55d7f026c8227805aa4eea7560e68ddd2a386b66cbdf962bfa2e
0ab9b75df670df4694cd09e56c06fa56c65e3e8767b79076b9258f57927fee64
d71c0f07601c06a3037e42cad0559ab38fd08374ae35df78284de1eed96f1ddc
466841afcd4cdcfa679e552eac74ab45934f36aa186358c0eb23b0dd69728146

Fiat-Shamir context

65913a4de004e3e4475ea2e5051b1415350ac806829634aeec6bebdded6cc8690

withheld proof term: 238 763 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPsCoeff_mul_cstMv theorem, MvIntegrability:49

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat} (d : Finsupp.{u_2, 0} σ
  Nat)
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) k)
  (Φ : Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, u_2} d
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_2 u_1} Φ (FBundlePoint.cstMv.{u_1, u_2}
        A)))
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} d Φ) A)
```

SHA-256 of the statement above

2469da44149fdc4644f3cfbdd8433c34a5d857508e1f2fbc36018fd9ab742067

SHA-256 of the kernel type

3c728b10bd0a7019fe338a5af074d33080d5d7a8d90d2f3257d7f31f06965208

commitment C

1896418605f20c5685ae8e365db70b7c39b95f46b7f1569db1b2d8217c48ac50
b5b46520a3dd24b3c6f7d0ac78cbc7e630df25fe9354751746d25a3b03ee6b8f
dd9767124efac05fe43730f4b262c518b0be2dc66d60575c3088ee6c8bb47d2d
29057cc219f9cd3db300d5f757108ab392511e8a990a83f90f033bdb0f9458c8
6ccc31403b62e7c7e6f57f838e9c68f60a07cfed160d0168fa2401a86ffd5f99
0ee8f728bda3d3b7d3aba056268d74b46c97528eb7f5ec07dbbca0cb86979b36
33d067ad5b62062b5e061ee3cbe0fdc87b05366bdd0d43e9994e100e9f36a100
0ee5386fa4a8dcee2cf220362253f230ef573b293fb01ca0c3fa79a8f5be6d72
first message t

fb47508f8e3b208a15aab9c210869e896fc86f711db98a1be6e5f23ac49decc4
208f9f7232f4f938a541797a1a0732d9a6a2bd961007ce92d162ef74e403bbac
6b8eb95edde93cbace44f5028373cc486f7a04295447e18ed817960650b84ff0
a2f4e485bb53ac0adc0dab11a4ea252582358171e4380e716c4e078135f6117b
97ad2c0e1ab332264d58c7e397448bf0fefcb52ff55f443945186488e203db62
cf58ca551434dfc62e09daac57f0c6454572ab7250f842304bf3c2354e6e411a
d17417ff99ef106b115b9e1097f8203fc8e5a981a292f14552eada82f44c02c0
7cd820deacc0f18b5f776ba647ac600d3f15211122ed72173d556bd7e96a2e58
response z

754ffe32edbb4a63e205538bae8404262dbf09e1610a4dad549fffb95a924e77
7812853c81fd806205ade77180561aab18dca113a9904cc5b5235c9a3f61ae61
f8a4ebe1cfe894c845e4938b025467fb70fe5e27e8e666260dd24c6bd1a2ec2a
de94752bec5ef53913d2393933b245e75064fc9997f233db6147c1b0ae310f0c
e94046c887548eb7249acab459c6a094ac116b2e48338f6e35dcc6e8b73fc677
5469c76dcf8104c66c11d4c20f1e3403263f3cdd90ee7e450c943f9aaacdbea6
6c4465d2e149d64e1b269078dc1d9480f49a5969b5bfd87f6a8393216806d9b8
9a39967184944fcc20c58f1f3747140fb185414da7e1999db749771ef378985f
response w

63db5574550b360fc9079157be28fb66cea92b1481dd166ad873d323f11445da
c2d2781cfee327809d47ef01696affbc9ed94c601cb892458d2abc916524f189
aba73acadd98b91b460121e37fe236d4d243a4c618739ac2f866248fd482f251
8843b3ba61e8012234f31b8a826b602e3610fe9ec36a9f3f4e375bedbd1cda3b
f1ce65a438ad79e4831ff0799834ef3391c142b65b28325b7f851d34de3b3e6b
b4191b903874f014b0cba57c2a4f6e086d85a8893356ab42d133f903d5a47881
7b2e4fdab9a57a3b5c15c3bc1c822b6635e5c3a0fc8a476e5f1325aaa686dea
73d50a08ebb40a9388577897b8a8fb8e4e38d6a9ddda29743a134b3433a1ec15

Fiat-Shamir context

bacb6a80a457c493815ccba16c2f93a0c58504ba69a2760de69a89e9240e6af1

withheld proof term: 239 534 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

```

∀ {k : Type u_1} [inst : Field.{u_1} k] {σ : Type u_2} {m : Nat}
  (T : σ → Matrix.{0, 0, u_1} (Fin m) (Fin m) k)
  (X : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvPderiv.{u_1, u_2} i (X k))
      (Neg.neg.{max u_1 u_2}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.cstMv.{u_1,
            u_2} (T i))
            (X (HAdd.hAdd.{0, 0, 0} k 1))))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k 1))
            (FBundlePoint.cstMv.{u_1, u_2} (T i)))))) →
  ∀ (j : σ) (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} (Finsupp.single.{u_2, 0} j 1) (X k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1}
          (HMul.hMul.{u_1, u_1, u_1} (T j)
            (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X (HAdd.hAdd.{0, 0, 0} k 1))))
          (HMul.hMul.{u_1, u_1, u_1}
            (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X (HAdd.hAdd.{0, 0, 0} k 1))) (T j))))))

```

SHA-256 of the statement above

d9644af6efbcecf2295e2507597f808e2d8f89d143defaffc8188f81fea00ffe

SHA-256 of the kernel type

78e81010466d9c173508daaa8f39c35fb5497c8b734e4893eedaa719da1c9b1d

commitment C

d14a8ce33378d0155750fc55a4f058c14f6dbc146d97fe37b2254ee223c83ca3
b7abba6d585defb46d4816244ac0a13ab56211b10cfa5817f76c1a6398086b42
d46e611cfbc6d5d9f6e4cb78a7d7b0e60025819c3adaf3985cbced3960f7a8cd
8c7c65a930c748a99481c2d3e122b43291e956f9d8f2a255de0224ac1bc55388
cee205df628bc753df2d342b99a59018e2f1559a378c781d1b668aade558efe6
ca42ad3db637e41efc8953de6c60a0d9716537e079bb5db03867ddc80c5018f9
175c557e36d565546d641f51a45565a3a77e7e1fcdf3c310b71d6a4d569aa7cb
4b41465993e7b041bc92a60f105ac1fe060b866e4acb76b7b03ac3d31e4f7b46

first message t

bd653f7568efcd006f10fc393f46203cb732b42358c3091049483bca4f394bb2
d08eec587f5086246ba3c4f3c4af45631d7a68591b6ac5e92f42fcfc12d5e492
85d99ca779f2e45d0bb8034fc521b761b8c1ced5ef48f7840602581328c7c879
ac88a9198610f5ac1df8a390538c4a2e845d4ec4224a387417e75a786cf6eea5
e478bc29cf150c011e1cdd17e59298fc272b915ebd7aba5439685304e2a912c3
7b0248b2a0da7de34533480794f733754087ea38415f238f57fc5646c072fc6a
3960fcb2f65aa56ccbf442712cd07813ae7da1857e6c5f9692d1cfff8ada72
858cdc620af2ed136063d3fbc36fd3eb7447e76a483b56e02ff115c966877e7e

response z

5c607a109863b40a2d75e77243456b182c71e2effd7b9fb70fc94e1598feede2
572f14d80a8efba2e62296881c6c5d66fc53b6caa96057ce4469c676da945d3e
b51b41ade682611a6e81b8deb0474c1708f13709aacefbdb6be216ea40fed256
06733d15766ecfee934a6613223a2d6c265108fe2211f3bc8bc4e0326862c1f1
2c81cd2cd90cbd1fa5e1697141f0ad6eed59703a91b3b8d57b1efb629f6b65a0
a39711a707a64383f800ec3a381c06d474e410cb0b0cca414286ef72bc665f6e
f8707fcd486982de61cb4ae7f818c060dbf4534ef7cad772784feae79624cd19
c871a6dd45001065466f1d27a339398a5ad5e5ba0f6b6a61d0f3c7b82b33710e

response w

6ed98e024eaa92a273fb01f791c8a8e7d2234fddbbba023d5e9748e7d0370d632
b4a1f8dd0b3c2014cb9e90ccf885b7f80c33070e5f1b5ce6924d578aec09d7d3
2039486e439044c33b796fc62cd4166d61b0a03b0414393ab5986d789f3d380f
8fea81afee0ac84b72d56beac8bf0dadab1b99548a6d4740e413dd3aec900a12
8aadd63e2fa808ab6be3b1d9bf410c00bbc73511cec8b732273a21a5459f692e
6bf7daa5917441f364773cba91705a42136d7830b6b7f33602c2aa67dfa7f915
eda8010e19eb9f60b6906b8d8b90a693ffedebaf81f3a9b498f4b5fa60a78aa8
5895b7bb29da7cf330df8e531aa8dcca820741fe21239a08dc45188e3deafeca

Fiat-Shamir context

713494867489e2442b2907ba1794de00049385ccabb7c1656c9d0a14825322d

Not

```

(∃ X,
  And
    (∀ (i : Fin 2) (k : Nat),
      Eq.{1} (FBundlePoint.mvPderiv.{0, 0} i (X k))
        (Neg.neg.{0}
          (HSub.hSub.{0, 0, 0}
            (HMul.hMul.{0, 0, 0}
              (FBundlePoint.cstMv.{0, 0}
                (Matrix.vecCons.{0}
                  (Matrix.of.{0, 0, 0}
                    (Matrix.vecCons.{0}
                      (Matrix.vecCons.{0} 1 (Matrix.vecCons.{0} 0 Matrix.vecEmpty.{0}))
                      (Matrix.vecCons.{0}
                        (Matrix.vecCons.{0} 0 (Matrix.vecCons.{0} 0 Matrix.vecEmpty.{0}))
                        Matrix.vecEmpty.{0}))))
                    (Matrix.vecCons.{0}
                      (Matrix.of.{0, 0, 0}
                        (Matrix.vecCons.{0}
                          (Matrix.vecCons.{0} 0 (Matrix.vecCons.{0} 1 Matrix.vecEmpty.{0}))
                          (Matrix.vecCons.{0}
                            (Matrix.vecCons.{0} 0 (Matrix.vecCons.{0} 0
                              Matrix.vecEmpty.{0}))
                              Matrix.vecEmpty.{0}))))
                          Matrix.vecEmpty.{0}))))
                      i)))
                (X (HAdd.hAdd.{0, 0, 0} k 1)))
              (HMul.hMul.{0, 0, 0} (X (HAdd.hAdd.{0, 0, 0} k 1))
                (FBundlePoint.cstMv.{0, 0}
                  (Matrix.vecCons.{0}
                    (Matrix.of.{0, 0, 0}
                      (Matrix.vecCons.{0}
                        (Matrix.vecCons.{0} 1 (Matrix.vecCons.{0} 0 Matrix.vecEmpty.{0}))
                        (Matrix.vecCons.{0}
                          (Matrix.vecCons.{0} 0 (Matrix.vecCons.{0} 0 Matrix.vecEmpty.{0}))
                          Matrix.vecEmpty.{0}))))
                      (Matrix.vecCons.{0}
                        (Matrix.of.{0, 0, 0}
                          (Matrix.vecCons.{0}
                            (Matrix.vecCons.{0} 0 (Matrix.vecCons.{0} 1 Matrix.vecEmpty.{0}))
                            (Matrix.vecCons.{0}
                              (Matrix.vecCons.{0} 0 (Matrix.vecCons.{0} 0
                                Matrix.vecEmpty.{0}))
                                Matrix.vecEmpty.{0}))))
                              Matrix.vecEmpty.{0}))))
                          i))))))
                  (∀ (k : Nat),
                    Eq.{1} (FBundlePoint.mvPsCoeff.{0, 0} 0 (X k))
                      (Matrix.of.{0, 0, 0}
                        (Matrix.vecCons.{0} (Matrix.vecCons.{0} 1 (Matrix.vecCons.{0} 0
                          Matrix.vecEmpty.{0}))
                          (Matrix.vecCons.{0} (Matrix.vecCons.{0} 0 (Matrix.vecCons.{0} 0
                            Matrix.vecEmpty.{0}))
                            Matrix.vecEmpty.{0}))))))
                    ))))

```

SHA-256 of the statement above

ae568936aa5da421993138b68c5f28d7d48273fc8784d2875d204ed87264141e

SHA-256 of the kernel type

8c893c388acab242798b44254d653d59c334fc39ed5c61774b4900c0e3a5c178

commitment C

c62445535e78df44aa99200b430cef4e54b34ecb17f1f2cdd1978174dc092faf

892c1dbd564c138474a3d1320050a73c6272bc419474714071c90d0aa4ad17dd427

6b26416df40b3bb5cc13ebbad1f5ae95fe7c4af643a5117484c033d888b95f64

24c960e995608344c9ed0bfbdd1f062baa52d17de3e2c55455a1e0538bc08691

19 RequestProject.Nilpotency

FBundlePoint.adMat.eq_1 theorem, Nilpotency:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin m)
  k),
  Eq.{u_1 + 1} (FBundlePoint.adMat.{u_1} A)
    (LinearMap.mk.{u_1, u_1, u_1, u_1}
      (AddHom.mk.{u_1, u_1}
        (fun X =>
          HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} A X) (HMul.hMul.{u_1, u_1,
            u_1} X A))
          ...))
    ...)
```

SHA-256 of the statement above

5b89f3bc8fb8a28afd9aa4211e238e2d15b36a29b5a2ce7251e4838faad60885

SHA-256 of the kernel type

e0e7f6c9f9c3666527a1cf9a1c1c87d01318c5eb320f5985e3a2286779fbf342
commitment C

b027d30c8a5043b1448df226c74898db3780499a0ef967242067608b1f91469f
a1fcc9d194829b0464276b9deb02e990d6e69584bdbaf7f4ddd95a9853378365
37a5d857b00b05ca40dd48e1a73513cb6a1ae8fad9f54a27f8b628b035243a4f
bc2a62e8ce50e11898cd8d99bc82aad0b1e8fe56e0028a463cfa970eaae4e902
90bcae988f329d468a513751a21179924a5db43e0a32289960691fcc909d7d5b
dac9c00b77bef090c382296135ffe4ed4a3183cf9f3f7e84b0c51bd8f5efe7d2
8a9a353fae3050ecd2d8bd11e6b8289f5705291391ec97e82b9879b32fd5a99d
219ee28239931230fe7302da2b85be020ed6d69f9b8297a85c95f31ef7f9f56f
first message t

b8e0ef81c8a67dd02a1dccc945900e5c7e0781ef92244315da70e29a16bd0e02
4b4061a1d2b73fc5c6f8c663c09db73965ec29b5a1bda20fdb4973dbba93e717
70859124d5fccce7f6545be6f805bf5fbbd016962386b6ac8bdeff576b39f57f
012a49fab0327eabcb9e18d85a649032cafe1bcbcb49661a87c0ee3c3051b589
bcb71fc0dfa573421768001f9f18d622a504bdc879617288e7a04ae6f25550ed
bfddd0c059b56d000b49b64d8fb4b14c880f6b23b835304a4600d461d98eea68
f761bd57acb1c3db1efdea91ce8f07548a91ea2b7dc93ae1224e0cffa071c1f5
c6b4c9a62c4df4a913e5eaae5f6892c148dad49d187d8fc2dab8dc6d55bc743e
response z

580fdd377de685b17595c0aea24c756bd3ad367fae6a2b7aa305f06a815a2f86
d169e085a76eebbad025593f211b4f3df683900b953045570585e96de56b516d
583f2f8433a2f7b59904fac198d7ad1245669dfdfb5492d76f6dd472e25da03b
2f614531c781250bc72b66e7356cf9aa26549ad5811e9477fd879340fe9bdecf
cd84adb7fb78a173a91352a594c834052fbf52a8843f1d9b0c1d5c37efbd0108
0c6bc361513c5f302b5900923abb751b176291c8bbb7a46ac3718b46d2e58e30
f24b0028aa4d4820e27c80119dd2c6b8f33017ae4de7d4fb0aab1bcfd2ee61e7
9aa98a152897caac32b6110b4e56b30f281df3445f90f77e07e4ec75b3e1f87c
response w

72db7cb4755debe12a64ab2534aa34a57e4b965dcaa31ce1ea6b1f23c26e3959
0a1c8e386e1a49fe361e77d9b5b6eb663b49980f54bfe8a92e2805919fe72590
af16193b3f691d238b4633c45c5efa8761e646f1952cd9141335bf3f17b929cf
6224c7f1191364082b9a7519f844b26ccedd75f3f2b327f4ce1d845f8aeca46b
592e0ef247b6578c24eabf7e7c0efbaf919ded9aeb90dca3a854dd1eb151bc3
990da79ea926c49eb929706f115a07e6113a850ca8af3675f8adf55765fb8fee
3d1c15c3e0ff320e4e435d1c9ef40b4e2cc3eface4c835e76a69565dcc3b4b4a
4a568cda4fabf4fcdeb5f5950ca431927a98037da8687439fb3728d70b41697
Fiat-Shamir context

fe6c4552d949755772f7ea63ab5aad60ba894e4408b577f98310fbbb4ae402c7

withheld proof term: 1470 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.bracket_pow_eq_nsmul theorem, Nilpotency:54

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (G Q : Matrix.{0, 0, u_1} (Fin m) (Fin m)
  k),
  Eq.{u_1 + 1}
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} G Q) (HMul.hMul.{u_1, u_1, u_1}
      Q G))
    Q →
  ∀ (n : Nat),
    Eq.{u_1 + 1}
      (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} G (HPow.hPow.{u_1, 0, u_1} Q
        n))
        (HMul.hMul.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} Q n) G))
      (HSMul.hSMul.{u_1, u_1, u_1} (↑n) (HPow.hPow.{u_1, 0, u_1} Q n))
```

SHA-256 of the statement above

da66ed335dc3ce2c406eda35ff2a6776e194bd15bf570534c4b5db4206b6ab91

SHA-256 of the kernel type

c1a015ceec12db44ea91ed5fb90901b29b5a2fb031ebca36bfc9d6f1e881cf0c
commitment C

3e711940b720c5df895363658b846671b3bb9ca5a5473323ebd0cb678a31f5c0
3de7dcb9fc3a6deec62fcac0410b687084b3486b6f04aae7efaf40a43c7e7aab
275538fb7729ca47039d6aa154afb073e91290b71010d587ebacd934336f56d9
54f4ff1123abb8852a77ead2d906b1dc7161aa3209a04644e85a8c5947d3fce3
80d6e937a0a202339a2f31afdb620e2d3a556cdc75da9f4dc3e0ea15541351ee
e2f53b2e1d2ff4b225848519374de78569673e8debfb083e27077134e26e2c1b
dad684b2c08fe79b9bdb3c3053ec236877fa5c24d969b0b93ebff4c6de7e6c18
b834f5b97a68aa0427cc61dcf08e9589b1b31b2d7e22c58f10076162c74582fe
first message t

7eddd17e64615f698a68386ae1bc5564b2a26197214016cf596bad4c3f0fc635
d14f4b43ec1c46631cc08963ef7bbd941e691b2517f80e380cc3c82b66d3a1a3
f2aa8f8666268b4318c0507abcb6996dec70f2999cacf3278940613fc7326fa5
0b3eec2191442698fdb6e62489310f2baf4f5cbb54d83e322e1153b5acde1313
7bfd5afd88af0e25ea95881fa77a169a1b40b402dc6bd0edcc48efa9d38d0337
78bc032ee840665491b06eb96b4faeba2dfa0b8672b3cd4d96e88562a3f53138
1231009517c969bb73f48bde3d774165c4e87a62ec72918c03fe062f69a0fda5
d7e8877f582565898095502ac12f2f6d6023fb58f65d6eaa5f361f153e924c05
response z

70471aab5c01e9fc7610dd91f558a232502d5773f3bfdc4a33af52172e486812
26e0f98df6e18edf67d0073ae8befc394e9f7db58d67c7880ed750f755fcf9dc
5c69da1ebe8ac6b8124dae0cdd1ac352f07e0a792f5f755795089999213cd445
a7de2801a5c854be59bfa2ba0ae2054435829cbec7b2baa4554a9ab565fca39
cb2525ecf82ece9a240c9f744f4eea986af437d0b66adc1841148f746e049535
77c05df38bf1c263f80c37d00d922d258765ede6330b08b943ff8c72914dbc61
4922c9ffe85cdd10317fd6aa7a9d4fe8d27a625ad706a8ab05cdf3a52aceaafa
c4e612b0a20bb90edb7ab3885085be98b54efac01811fc23c043b13b3fade8f
response w

791723d48e35caec3f4510ddab45e8ddfe7412f3d03fdd5ca18e2a6d4af2baee
3871fe51314b9f2acaa5563486a22e05a29dbb494673c3f8ca709303960bac60
132cda7cb72f4fa78cc8450990567c3c3fd45d7bc86d6487bc518dc131a6c441
b5b044027b410e3d6e696b8c22cfa1e290d9083046dd399c7c4c8c5db0bcf37
26de67c36e945cab3b8d697b5122570f095869874319a4f9da90d3f69c5925e2
567a8892b24b40368e27340648615014ced252939f941d663cd77259c1fd176f
d486333fd3397e35c88243fa2cf1c200505b6be3efa19978b7e2c165f49386d1
8991c653e49db89175659e6d04c2e76ba52811e9b7eb2353d360d809db2990e0

Fiat-Shamir context

f90ad2298fceaef66e1bf5265ca0183c6c8fc41a3632fa950f3f0b9b5a5afd19

withheld proof term: 1 505 280 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isNilpotent_adMat_of_framing theorem, Nilpotency:123

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} [CharZero.{u_1} k]
  (K G Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  Eq.{u_1 + 1}
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q K) (HMul.hMul.{u_1, u_1, u_1}
      K Q))
    0 →
  Eq.{u_1 + 1}
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q G) (HMul.hMul.{u_1, u_1,
      u_1} G Q))
    (Neg.neg.{u_1} Q) →
  And (IsNilpotent.{u_1} Q) (IsNilpotent.{u_1} (FBundlePoint.adMat.{u_1} Q))
```

SHA-256 of the statement above

9e6135588f2af1df8190ae3b307d3c44ddcc0386eebf8a3a5b949994705989f2

SHA-256 of the kernel type

90f71db2f0783cefd9bd187248b0684b1319622967c78f4f3108197e289d641

commitment C

6ed5f9c1462317c0b0a3e1096827a0db6f5cf1b059ce143a31ba9f5fffb687b91
ab29063342e57bc331cd1759757584e8bf92cc8e9628112757b2d47172c807f8
44a2bb122a32c0a2216aecdd250e08b8766d17e2584b7250c0068d7ed4f2fa147
7b8971e22075fd8bff0c50dc89d8001ff203f573610d9d5be020a4cf0a634a0f
693f3e3f26444c7bad79ba0c49ee8e6362a3c48defc77f2756c8ca642df5430e
41c3f57c9e728eb959892825bdf0c951bc3b956f5a0e93d09c6fd738d6adcab1
1faef863ef54160c447f1390183fcd8e6442b8c3445a97e8ce01dde08975f6b7
ba164d367d0e278b876d56972ff3977176405e87b76351855a187f01a131f11c

first message t

133352b9c9aaf2edbaafcb03fd6a58b7c0ddf3b6e4230091544247df206645d7
1a1dffdc66d77e64073c2652793d8d07e35d4dd1fe3fdc7165a301122808518f
37c97fe886fc7e6b11c45fcfb2887cd7574bda01310730651ff49b6b7028ed65
e18c814f8f286b77d28082556d8b2773404b3fb47c7347abda7c6611912c72e4
0333fb363b5a1b2b5a6e0b93b306a399e4384c745fcfe9bf3f97fd987f0742e0
f78532acb25f00afc05a6316e0a11346c6fb8d5e5573f9aefd66c91ccf12aa9d
263f4e06d25275ec75cd662316b71f01b7eeea76d132c2e58b976c798f7f24e8
be042d6b08d11871838fabd64796982ebe40691491e56bbe6a371a58a24fabeb6

response z

55968d38693a4a1851adf5cf94f8e8c2070197f1d6f601954548c6fa275378ed
b3fde825d240b40e08fb4ad1bdf450c5ad622e669347128e3d56f3067d700a86
9038b5aebcc7d06a52dcad073c73a8db3977487d9ad14aef4349048114b972a47
e93c75fff0b5484c60052c8af57369da1d9b816989ab61558816834f327fb917
2df19e0de75eca4457fab81b618f6e5c1b22717ab96a02d9eb5a4bbaa4c230b0
fb94133a4aa219e869b3a65b3bddd1589159248726a3384ea2edf9e88489f614
9d7fbd904152040694697ba5be717ab2ca5310cbe179982cdce54dd0f4984068
c5760e5c25cbb72d20cc7b2705a6853310070ce6ca31d731a6626b1efe0a5671

response w

4f1402cc993032e52ff4a406f6b1e61ebfe2dd671fb4439c0ad9243182797179
87697b4446f50068a42947cfd2c6933af94bc2ed2c1502ace0c38a70feeee80c
46ebd0e2a0dfcf9d52493a21e3991d4a472d355b50d2a945e6d973b1e6e35635
722941fa5883abd3cbb213c43a58ef04909dd99dd3481985c37a90d8cbd4a08c
c43d8e2ddd26e11cb819def4fdb90f314c30859ba421eafe992d42d27ac16075
c16fb17c9c2873a1e5e84c1b34e4e25095db048f0659eeb3ba862ed4a502ceda
e2595785cc093da19b9414fe6e9a3c7806dc86a38d8a23d47f1db034010a5160
eca0d0ff2b31c5fa71ea6d7b149fd1cfc9ae93646c5339347811f66c71cc56e4

Fiat-Shamir context

6afc4118cc6a5f6c06112a95aeeea0d7e9813ba3f1e3f8010c8d382670d7a2ae

withheld proof term: 54821 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isNilpotent_adMat_of_isNilpotent theorem, Nilpotency:106

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ } k] \{m : \text{Nat}\} \{Q : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) k\},$
 $\text{IsNilpotent}.\{u_1\} Q \rightarrow \text{IsNilpotent}.\{u_1\} (\text{FBundlePoint.adMat}.\{u_1\} Q)$

SHA-256 of the statement above

8ea5ccd9e34e506a71e4f93c0480bcf83fcffdb2bd90d85c873e357d0eb398d7

SHA-256 of the kernel type

075752bc01197d5e8fb3ee35bb542a46bc3dd849678bbc4ad52fb126d17a813b

commitment C

e14cd88321699b9edf171e8a2d0384fdb27bd8439582216131656f9cf68209bc
f42b1671c72fdb40e0d495797dfba7296c2388ee391978b4117069bf4a47e146
8a0ccc954eebf3f94cb886f264a3f71bf569a91b697085a39540dad435f666
00b19b3be4d427761abbbea47ebc7b9ab7e01aed0b44d671f2034b7ed624f203
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b8ec0830076c01fa2710d6b658b913f2f49ef2c82e02a6d19aefb721af5a305a
4a2f90399b3e1216b61d6af2a0de5a431518e6055f9b5b253a2a359e96b2d298

first message t

e2de78b1e54db9db63783d0dd8205ffe95ee18db80d51535c77c33238ef1d6a8
4ab3d1c53552b3d3d59e29f1419b587f4e163007a5de469151b36330976b7197
31e982af92cb2f04eb0f752f4c6cca6b542b42cd740bb34be363d38c1582fcef
21cde2b6d9223e91faabcfa075266f47505d0e11e96278a60768b5e49aab6466
a4c6f5cced661855c630c38228549da060c5ed460733d5d22d16aef548a0207d
ce19e4bf82fa5fb69fe932d873adc822fd7cb5f101ad17cfcf559f42f232f86d
631e2b72f06d68fa7f94b4aa61c02fc508243be0fb310f472e9dc2abd3e6db76
d182c06489a743d60d0f7340b57fb279254825e67700c2171ddc2a38b6fbe845

response z

162eb320f6317e05e199db0061abcb97fba9488b2e5b5ab61f85f6560b28283c
bf197d4e00d45c6aad51952e7608d6e43133e3a418406e842e4ca7c408ac7bc7
99fe867540d54911b0dea917ee40f8296dbeb3f9e23718848840e042699c1474
5cb450169a2167fa635813445fcaa16d930338839172c15400a7b2c9b8f50b82
1eeb438bffcfaf6f313d234908a7bc4a9c0ccd770d20b0c5f2b2adf28036074f
d124f19e7f2c7a1810d0bcd5154653cd8dcf6db15e131bba2589af9ccbe3b8be
f0d8611031b6d2776f59dc336e32b5c32dee1c7fa629bd5ccacb13549b588f52
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response w

b1edf370f00980b8b9c025a6957eda1629e93e5a6a14ce12a8495c68df0dd9e4
304318203c8caf1deffd6a4f22c01d9987e58a38c89372448e6f6f7fea60d090
109bdc2c8ef1226bcf7daf460d992fc16d43fab76dcedc18ef384175e817e108
067812c43d4a576330d49fe5ee19b0453db0d6a740a4bb5397ea3bdb625fdd25
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679f3569ad09be16186120a74f1f11cda701a937e2c6dc031099ddbd98c9b3ab
7ca4829edddf578252c90910ac6462a30dde8bb0007862a4f8b6b9cd172c52a
36dbc9272fcbfa0c48181dc0ace750cd6901c8239be857d79090aee2a7ad076

Fiat-Shamir context

b8587071b4d5fc40362e36ced51b90970799f64667faa7fbfdd33a15420c9bea

withheld proof term: 1201150 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isNilpotent_of_bracket_eq_self theorem, Nilpotency:69

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} [CharZero.{u_1} k]
  (G Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  Eq.{u_1 + 1}
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} G Q) (HMul.hMul.{u_1, u_1, u_1}
      Q G))
    Q →
  IsNilpotent.{u_1} Q
```

SHA-256 of the statement above

f8231d30c2c2fb4413e21a5e5ed0ea29414a09290e065228ea1ac79631faea41

SHA-256 of the kernel type

1170dca4306501ab08db24cee47fdf8ec11cab4c52e95b2ccff3156027379cbc

commitment C

254117705a26bb6cf041dbda00667eda948e29022320b2c102ca4b2bd9d7c743
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9cbe564c0f7b35c68747098fdb1a0398236d8c1a92d3a640311e22a8c4ff6b07
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e8b27ac240ca84c800d093812f7ef9b9d5e2fcc4899b04db86faf67ee8c43cbc
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a9259f365caaba478d1aa373d860a84b43092c5036a520ef417c36302e6106f3
a054a0fca5a01e4329e3980d2b2ed4391c746eca21bd01eac4246e09fa90e999
first message t

d67696445f89b677488f9342ee8f88fd06b8c384fb5a7f468dab42554aade7f0
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3cc8edb4854225c6238e9364252c842a99edbe8d862611263f586df7e85f8652
7d106f87c51d5fed7b3021f4f5e030d8d618a2559b80b4d16e93d460ddb051c3
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85bc6954eb5836fa40e35fcc52cdfd09180d5533b0ae6b96430379a033aa39c
e6b45b40e11286ea6e5e942295cb782cf4660837a5fbb42af778c867a554bd89
fe58800b104474f0a3dd84c77a77449e4338632f7230bb7de0459dc624eecf0b
response z

599acf2aeaf3057fd3ad8a452bb44c4d359c9cd54fad49a07575469dc807860e
08c93b770dc48bac6ac32cbced7b5831156988391226395cdb55214cae322f56
2955892b749fa42271d531a4ca4830a694c59c300d4c9cecd33fe897fb877349
35de1df0c2aff27b19d979321a548e6f3f6092eb8e113a27f63c8f9ef58ecebe
719e40b0bf7629cfaaea191556be76c732391b7dd3c0834def463e7863549697
3a95388560eed4b15e588c43c7ffeceacfd28299e151fef4aca3b5e7bd2f9933
4cc14c0495d31c392c6c3844ee31c5c85e85052b3331ced38246ecc1d3ac4d04
96e1fd45bc0cecf36324bfcff7428a84a604c6373e3a5ceef9c604afc52fd732
response w

5ac8bca45941750ed7bb4d1e822e0abda0319ef10a842269e89c917a3d033ead
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0837cd0108f3664227bef90af0df285242d3f875e24f00aac66120bb8c41b71
f9085e0b44b2b579d2e19a7812a090ab023c8fed3e8d606a17efb24dbab68285
c9284bb5e466fa106e2466d411a1c58a97de663295e5df7fadbcb60dc54041ebc
d1c4bf7491fd01f4edf429c61782b52a800b436583862eed3243538d76648c0
d78739dd1635d2204ad04bf4df0792747ddcbcb8b577b4b110c7340cac1939928
f7d258f7c3b13e3fbdf1d9e407f4130620268c64c4fc07e02a15f036aac62240

Fiat-Shamir context

02714f98d5c84d0828f4d1b4706cfd1aa8cbf21837c8f0cff7840ce9f6100a2

withheld proof term: 142 822 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mulLeft_pow_matrix theorem, Nilpotency:89

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \ k] \{m : \text{Nat}\} (A : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \ k)$
($n : \text{Nat}$),
Eq. $\{u_1 + 1\}$ (HPow.hPow. $\{u_1, 0, u_1\}$ (LinearMap.mulLeft. $\{u_1, u_1\} \ k \ A \ n$)
(LinearMap.mulLeft. $\{u_1, u_1\} \ k$ (HPow.hPow. $\{u_1, 0, u_1\} \ A \ n$)))

SHA-256 of the statement above

615d95c5907d676145069729413d3616bc9b4f12cf1a20709fa300fcc9a6a0dc

SHA-256 of the kernel type

3e23235e99e8233f370a6258fda92ff11d4a7cfb2594c14bfe4704ca5b216470

commitment C

f826179fb37dda1bdb33e6665ffe3399a26a205253e7f4b1af6ae73fe1f883c5
f4fbdb0613fb2edc245ce830922ec09dd89eb70bc62a7473b4f3ed43dea80221
331c739c0b9a85fecae50c93d765e39bdd9fbd1e3d92d70c73b507dd0b778e
6be48e2e03f4bc82488294bf6161725c24eb9e67e9aa817c55a644ba8bfa54d6
ef68be3a7a289b9ef80c9c012e7df66b0b008ad47883ca0da868619dd283fb5
ae7d4fd6e8639f2131d62e15459fb3bd19528475ff39b748c60586d145242e79
c01051be69fc0105a74501bca573189e5fc53b27dc78dd56f2e2b8548f4932bf
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first message t

62cf847582077ab315cea5004a053de1b8d7e39cc5eab6a0eb9ee0d8fd30fd3d
19a4fbfeedd1b5292e0c8afb84869098629161a84f25aa198618fb304322dd5e
55f9f3e365339362504a3da97f21eee66779d93228d553ea17b760a39c05f4ba
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48ef39495bdf6144e14bf2dd878db34dfd2029d4a7b1f723d7154f55f429820d
d9d7b8540b01c596e8dd0ce9d1bad2353faf4211c8f5f21af3dfa0c96fd1f6a5
23269647b39aac3abe65e4ab40ebb1a31a927bb9ec86a8d88507492074ae69d0
e84a6e8a7fbdae815a78225f8ce58943daceebce9cad225d1d49dc0ec82a93cc
response z

5a730a2a29fc6d0e2ef9af42634b88b8bde2a7ca20b1d46d6bcd1f52d19f7a35
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74c8480253fbdab0a55c1c889c7c0b35e9faba2e805a0d9f11466987d51f70f8
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0328b91e8c9d2e0285b75d46bcd35ad593da8650113340a1627442c46498f8f5
9d99247a5aab5900c346f86af7adf7180e4c585eddd077738d641bfa2f33e5ae
88657a3c04784473c8f16159ceb5946983b3e4a9fcfca60707887f788b8470af
af83a5fc49975deec64377af4aaa5f8c629b279a85aa9530a2595ef1ba9f07
response w

12804644f36b6ab53082186669b5dd3452c69c820693da5974402bba9fb981a9
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3b4c302a00172e20060492993aa6bc806b8b5580bee2a34b4b0dfef1db23e2e1
7340d2a01d4a3ccad20e4cc4129f456156c6907c2bd248f1b51cb431a0c8e6b8
540ce031e324ad6488132a10d93220214ca36bf9f6b608ed364eaf15f3f43405
b5187f216a34af6d9d45f84d888e172ed70586188b0d4a97268bd4334f3ca68d
6a2c7cc7d79d6c75b0645f400d0fdc3f970ae8d4189e00064058f57212a7262
1bf059fa6002fb22ceec5a5882364de34d9bdfdbdb54dc4352332f877b7d4011

Fiat-Shamir context

8f838b6a816b61daa5761cc4752949e9f9f3d716c08518f3c6254f4dbc8b9c1

withheld proof term: 1 408 648 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mulRight_pow_matrix theorem, Nilpotency:96

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}. \{u_1\} \ k] \{m : \text{Nat}\} (A : \text{Matrix}. \{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \ k)$
($n : \text{Nat}$),
Eq. $\{u_1 + 1\}$ (HPow.hPow. $\{u_1, 0, u_1\}$ (LinearMap.mulRight. $\{u_1, u_1\} \ k \ A$) n)
(LinearMap.mulRight. $\{u_1, u_1\} \ k$ (HPow.hPow. $\{u_1, 0, u_1\} \ A \ n$))

SHA-256 of the statement above

986be2ae280d0f011666f3b79b369cda51858b7f98894258e768e160c783c938

SHA-256 of the kernel type

87e6a6cad2a54df16183dd5f6021c133f466d9b31d4c1d9bffa935941a054578

commitment C

d93ed35213c5d49009426f1810919eb1e3f019b86a7d654145fc066858c6ea95
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ecf8af0e26ea9bef5b1b1fb59233c2edc14c04509fcf3fd0cf3d1c3bf1889c38
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f16032918111aaf88c0d24f551858543f830768a6d80ee08d5b5b7f03773585d
84a1fb65ad5c1582f9f77f63f80f9c527a746d1a60ac8fc3220ba211bc9fef3a
7755532aa4830c4380d0f4251acbb3601c0410ca0b35c8f62de29b8f8f122226
first message t

96e2510eb1238303c5fee34b067084a9ada95072b153c341178ff685724767d4
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67d0306e573e5a7c83c8a89f247fc821700785f60c4f3187546d9ad4c21a7231
60d3333d83c726bdc6ae4a87e08a6b6a38367b205e5834c69e0d9467f9eaa748
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62cf097a5292d7166ad71f89e1b78bbbf59d0a7bea657c315b55f9cbe645ecc
9cc63ff48989b627c4e972611a3e1ebd68d26d8aa70d062ed900727ab2852891
ab14ee3fc5f9e672dae67468f8b76d560fdf8ab6fb9b3fb9edf6d9b1773f7b06
response z

2ac18afc70ca8ec44764a43824fa16dbd04008163f949aa5910265c2eb21cb24
9fa8d94e5ad95233cce275e188e2cd25792c135dabfc47793870019cfdc5d114
cd26b20cde5c3caa74837e1c6779afb004ab27ce642d43a91027022221b90618
bbd12e46f32ccdc1cbf821e96b0a3aa55fc8749eeceb8749ef4d3ec3904c0215
353bf9b22fed5cb352afc52d26fec9e510f401f65cd2f0da8f79668e9a9e60f2
75358883a841cf689a1d03dda5fc93d7998c20955b07f3d079bc89c10c5606f4
db25988a85f44d4765f9973ba375ab3fb91b8594161030448eb38454d4aa0d6f
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response w

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1fe313d93362105b7ab950271b95e8a61891881f38dd40094ecefdf3f3a8de93
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22530c4aae2c04aad511b97035a43bae4afcf324bc9f4dac160a802c19940705
918ef2cc4bde4e1a387b1ca051f7d3459b3237754ad9c18de758d136e5627c41
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Fiat-Shamir context

3e8a7b5d365f7a9e074220d7e70875722c22deb6aed13525338fd5f4148a8a57

withheld proof term: 1593273 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvEuler_eq_zero_of_framing theorem, Nilpotency:142

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} {σ : Type u_2} [CharZero.{u_1} k]
(K G : Matrix.{0, 0, u_1} (Fin m) (Fin m) k)
(Q : σ → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ k)),
(∀ (i : σ),
  Eq.{u_1 + 1}
    (HSub.hSub.{u_1, u_1, u_1}
      (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Q i)) K)
      (HMul.hMul.{u_1, u_1, u_1} K (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Q i)))))
  0) →
(∀ (i : σ),
  Eq.{u_1 + 1}
    (HSub.hSub.{u_1, u_1, u_1}
      (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Q i)) G)
      (HMul.hMul.{u_1, u_1, u_1} G (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Q i)))))
    (Neg.neg.{u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Q i))))) →
∀ (X : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ
  k)),
(∀ (i : σ) (k : Nat),
  Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvEuler.{u_1, u_2} i (X k))
    (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Q i)
        (X (HAdd.hAdd.{0, 0, 0} k 1)))
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0}
        k 1))
        (Q i))))) →
(∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X k)) 0) →
  ∀ (k : Nat), Eq.{max (u_1 + 1) (u_2 + 1)} (X k) 0
```

SHA-256 of the statement above

59a329775ff2780badb3c0bb9b449be21f78fa10e86754d756ab38d5bc14c1e4

SHA-256 of the kernel type

582465dd3234f4c3ac795604c2857c929c105ed1b40057cf9cdedae71accaf8b

commitment C

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first message t

c73b0b99a9ca70d83c1074b6e996804109b5d883b5cbcd0887b7783c61a47ee5

c421350963e5efc8e7bb69827fcbf997e22e281f209c696c08532cfad3918c

951e53c125391884085d406b6f75d17f3a502d1157adf82ec1ab78478ef55117

a9bc78dc5cccd7b870341af6318af2cab39d5048f3f2010089831f40e82d6f39

304a89d01361973168051b0ece8595454938c8c7bf217a99628b228cba2ef131

afd083e122f99894da240ce163b917c2898b48c599c9bf234c2ae60d25b168ad

71528d440a7c771b447766f6996bfc9df7dcb5cc8e6ac0c94edec5937740499

0716bfd6ec8f13261f556b3a39bb50e759d6ff4b29666483550c4cd6e39da656

response z

1e83f8d815fac165e8271ce6728c0dd11344eb3a6d50ce0dc902b74898459061

87fdadd49a94ca0093311a4ee515dfb34341db30c9396ca67857b71af2a3c5f7

33161ae3ed791edccdd4da85063af0a363fbfb168d9f0d6c439e01677b1788a3

047529bf50ce232f63d1c81c6bf1fc15e51159178d84c74e363b0c3b5b2d29ac

aa453c59c827701fd8982f5a3a4f3202c765470d07c0c55f53f93e93a9c161fa

ad842d0da9918b0e86f3e4058389cfd53896bd0ffd35e82759b063c3f27afb3c

8b095500c9f24a8838044463f0c3b0426f662f864f90f0f360de5106f7218290

ad8a6f597cd51f6c11799d2e47420f1df3c153a0d302a7156f045995eb84f689

response w

22f8180cdb6e607bf1f4a99e3e2b6956f351b0a89dd489ebb144bea106aa9364

e284b8e73edccf6f5458f4b7c54191e5f319e6be6f2a41aa05f684794c35d7b435

e29001a4d82710d1aa02f7e48d560a72ab2d0f41c2d84aefe90ee2848f2fee03

832f14f45f789d9d09791869aa02ec9d60a6a657a0b23c7e1d56a4c01c1af7a5

FBundlePoint.cexB definition, ProjectiveBundle:306

{A : Type u_1} → [CommRing.{u_1} A] → A → Fin 2 → A

SHA-256 of the statement above

dd78cc755d40b750cdc927084f172d4ab1daeef557ba7bb11253606c64bf65b1

SHA-256 of the kernel type

47923ec0455a60c6332da9841fa5e71562ee5e73aebff164c3df1e997817b9c2

commitment C

88af6210d0809a9c82193deb6c8379fd10616fa16ad34cb18ef2847db1b9a130

7577944a8ef8969cc692779a78911191423fe0dd4cba1c0178b324241435f2a7

85b22a483c11102aaa4e792a1c038ef223515486c3b5289b8eb2a73a999a772f

1f57953d022f20e4ffef4ac2aab1ad6851e4c9b2aeebf65d6198817c516d6508

4073782262b0e89665e33551eca4691def7573ff57fd1998d342a09f697a3db9

0bc708d56e939d4840f8384dbb943dc2de441eef4c9d4cd332550be3d56da1fd

baf92cb253f62b597c1a3bd901c48cc38e0b505aca076869d21e91caddb9b59cb

c2400f8139ff443bdf54a5e4a2f26bd99d56413c9df1912f12ba2d428f6e4899

first message t

bc238bc7f3d7f933e24c659a998532eb6f16a94153dd46bc2e667823c2215a83

60281a6f46be062ac326317bbe892e76a9e2e1cd6aac99ba5d132e7644843007

9293e6bd7715c7b115724b47fde7dc601d6733df96d5ddceb43c5dfbd290773a

44799e9944e3076bab5dd5460cf78ef8ca274c9e060dfae1ef5a14019c775f0d

8831adb09be6db05cee215cec03b3296e253e9300e97c98ed2e738679e903832

b3f1eef402752abf7c976f076ac620247caf89c7c13aaf22f5d06543eed98626

420042d65fd29bd5db61feb6cd740c646546e782e629650aa472f9a590389149

243c79da4c72650734f3580a9a082126566175eed3b148e16c237efd980bc3e4

response z

2a07e4b7a075f2dbcb4b4ebef649775553e9f419ffc33e1a33ce57a59eac4d05

f60ddd2fedee012b26b20d23ce96d9d59710fca2f5eef1137b1ed2d3dceb57e93

04469217ed3ebd8e36a0387680030b265335d4ff39a8f6389638158becd35c95

19c95114e1200310d92b4e3825e7c6f25d46e6d678277a81a860aa9fbebcecf0

83ae47a112db223a355cd33d0759bccda03fda25fbc919033c2d2e36990bd905d

f2f4641099a182b2cf5a70b2e7db95309bfc13ee3464d0efff9c24823820af0e

6692509a29ae17f38f9d8cbc0e1081554104b6e2d11399675cff0b5ad1b34003

7969abfb79d21768887eb5c2a20d14fbb891fdf735ce1337444942b87ea1b691

response w

4cb39be636d0dc50db7abff8e679881df31db92602cfcb022d9bb0f6ab347a5f

9c3910e958e5852eb669e308186e31a978da957aa591f8b0c0c85bc802c710e5

15bfa3cc8fda53f268db18f9c2c3e2b5b5fb3d0841803229c5e46ef5d1dce9d2

aaf4265c7605854ad07e74570dac51e16a241634ac8ca4f4f711bd3c82d6567a

2c51e6d1c2c360cf162b561a703451724d10465df3faf8e1cc94bab32402fbd2

bdee4b1179202655536317292a7731d29e3efc491deeb5f27ccca94759df0986

ddedf2b0deb3b2b277ecda771afd9707b976bed95a74af4ad18d096a5fe0e8b0

8f5c5dc573d3cabb7c380da30e17f342817f07fb8ec5b64f0801ff75bbf861c6a

Fiat-Shamir context

955df54eedf84f0c18ca43ff2bdfef21c6aa8695f66da0a10322e85037b111b1

withheld proof term: 684 bytes • axioms: propext • kernel replay: yes

```

FBundlePoint.cexB.eq_1 theorem, ProjectiveBundle:0
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c : A),
  Eq.{u_1 + 1} (FBundlePoint.cexB.{u_1} c)
    (Matrix.vecCons.{u_1} 0 (Matrix.vecCons.{u_1} c Matrix.vecEmpty.{u_1}))

```

SHA-256 of the statement above

7a85c20551cb00b31fe2ec413308b651442c10c08fd4c1407f9ff175f3d11159

SHA-256 of the kernel type

0d2ae4644d3ee7156b95aca181cfe5b6a56afc70463e3ac328eb66d459a0e1f7
commitment C

181171a5167340feb0d1cd7bda810102f643f5b781044c01c4bcfbfd94526edb9
873f368e620004122f2866b0aa7f495456b4575425a551ed8c163432ee82fa2e
3901d83d5f0b060d1860cbfe1d1cb21ed9e6ccade8b956bb79511ccb33125f26
9070f89a89699e97377169e26bb3ad5065f53b367b3304c44cbee02cb0fc571e
3c23c6ff3fa29b7692c5014965908dbb210c1dde4f619d9c4dce51681c613bd7
2cf18afad24f20a8458a8338562544fafb8916ddb132516a0bc79cfa23d4708f
1e80ab9b98586b2da58d63eae0f60469c759a8498b5e8e35bae12055f3705ef2
595e401604c79be862897e4f96d605df9cf0c4e2caba4d57c59df86266f3199c
first message t

349052ba79fb213b191ac186ee85f23121d564bce34a4fd50f65cc4d76b8f7bd
50cbe9f78d4949d95ec0b3eee2fe59096f00c84bb544cead5857daba3b8295a2
3bb3dacf3bcf792b3bf3642c853a63cb8ecfab85a6e0c7494695226bfb22d07a
978c5cc60204e8d9be739f485fc8e712c5b49e81762a69395611a33d65bd19e1
6608c7f610696959c0582bfa4f1c582934c652b72d71db2f3eef55e22edfae9
9e62ed7c37b2cc660aa9384f19cf9d85db642822e777aa5d58d202c4e13c3fd5
b98144b338cd268492be5c214ba8dae996cadf8a5871d2ea6bb561b837ba1a06
764bcd447217bb56f97092148f0b8ea777826e45a609e72fe1a7d4f06dfa256b
response z

404c6effb36ee57504e6b2df76f87e383c7014a7001a74481bbdfe015a313e1f
445646928a27b633a5e94ff522a65e7e3de2a2916c0b763383896cc8108805da
0694784d18683ca4a1107dad847521d64c872fe5d7dfb53d6fd804d59d0f87a6
91d12a916a2ce16e91f3a38554dcab559b1d6d1000d03624d0da735bd957868a
7c6cdf6007fa3e6abf2b4976e6a2a2f319b86b3c1129036f9db6ccaab5a68c29
1ac4dd9018c500867890f0b44e411bcb794c76903c4fa5df92bd4b3279e3b759
d972ae2daa3cad000a0347563335bbfe7ee09db99acbf4b7a25fe8dfac01705a
750eb53ec7cf59313d7dc843b4c90229518f21123d9cc3483fc968f4734a82b9
response w

7d0b3689ee506554fef7aff0c209274a9bf6223ebde91857be9a3223cffd45d7
0a4a061eecb5e8caf611d0cf0ad02876268a795f998d29001102476513eb0c1c
7dc76e2e273d232b1d17fd348891b35124357fa17273eea3677b71eaa89a700d
70c1340c172ee92f9d2033a64c177e45d5f48a5ff0035fa9799c9fba0f1daee7
e9a6d100609fcfeafc6fd951267c183182875e0b1bef80a4f72e1242926d99c1
400a74f77e2dd8e093f18390fe25794738097c9a18e8becdcf553900a0a88f3a
9a5c25cba95947edc743fc78160bbc3352e015ca16ef0c93b24bb9668cf170f7
73bca7c4f705d76c721a92465a85fa89f01d404ee8c373354d208969b92b8393
Fiat-Shamir context

04e0ca0e10ec870d8295bcd6a2d40d5aa232c46c3abdf69d1e7ccf89ef2c0fd5
withheld proof term: 165 bytes • axioms: propext • kernel replay: yes

FBundlePoint.cexB_isNilpotent theorem, ProjectiveBundle:365

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} A] (c : A),$
Eq. $\{u_1 + 1\} (\text{HMul}.\text{hMul}.\{u_1, u_1\} c c) 0 \rightarrow$
 $\forall (k : \text{Fin } 2), \text{IsNilpotent}.\{u_1\} (\text{FBundlePoint.cexB}.\{u_1\} c k)$

SHA-256 of the statement above

11029bd81bc0478afb3a5f0736c8398922064533eaba30da9c084db2be654835

SHA-256 of the kernel type

7a0f3d7147e029e7a05e01cbd9ca972ab984794648f929929426ed4b547c8e20

commitment C

5bd3a6b50335dcd72624624278983b58de4e44c2fe235c403beb20e25ce35d80

256313b88e5628f9ae6c8b44daf4781331032b5b7646cf34c2fca4892a0e885c

e73b87de4831e7ba4a2543cd16ff50d69df57cb2e3f3444d1037e63e9325216f

a529c7c273dc572aa1904104de3d5f827596714e2520d2519686a11cdb6f859e

1f7fbf5ccf522e24597d1cbceb99c7807eb92442a80a2442da249e493b0ff8d4

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216a38693ba6f95a0e0678d05ae504740671f8fcf1f4ef45a27ae759dd66565d

90e65e06a2c0936753ee5627fcdfdb24ab76fc9597d277dcce661b2d34998952

first message t

c12729b22c46810a07be33935205ffffc3ac7179e04791cf3b7557395bf363b3

84088b2ce8f66ee8a6c0908b1ea6f7d1810db3a4950ac6d3bda6436718246d0f

4f5f1a863450977ee76cb8e65b95292c1197aca526cfc614c9e6c99acde7cf96

6c4f9eede0f385b7ca14fed01fd402d2fbac8aa7b020f3c44b7a6b694ea44e58

8953a83408e77fb54d6bd936fc0e09d6536629499d7cd53fb019c8454f388c7b

dbc15860c3de2d45b729398296e7832b75ad59125640666e2f762eb22aa4d4c8

2d9bfff472ad886e5a4ee73c5d613c9f587812d8bec2d12b4ff44b699837aae4b

adf6a45c42bd446a909444e5b0313272bfc7eab9f870971771377bb62e0b01bc

response z

6db846b19afdf6ffc0611edf174231efe4c54feae9b2d3a460ff0565350d902c

77f01ae9dcaace0ad3df13e49faba95da416326b6da83471da6a18a0e1306942

e6d232b8e6a46740511d8808ebd42828db8071506f0c3de1fb03cca79882c8f0

47c89d9260f9436eb010ffea78c0dd611fa1cd56ed9a341d52047b7b1df03d12

9fad42cb9c83624b3123b3c849d3a441f8a99981e513c5174ce67b855cf99670

257548a133ed7e45a52839a17730d51b7217c206dd2fc214eb877c7d7d050bd4

69c93633ebf31ca6884d9ef3c2ad5f207dea7f638c9d0a4730c7675b75c26ac2

ff502724c1a1b8049a923499abe490fca0ab7953198bb40331483e4761970677

response w

6aca245aa07b297b1b788a94ced51fc65a9a310d6fa4a61d0d0b97011769f6a6

40b43a070183f624460895197e337c407622b17e6438168d4944cc7c34417c18

2bcd5ae955d6657f5337648d270188ae1fce65c63c7da4a557e11d8f08751ee9

aa9b5323e043ce683e391942334dcbc3c2cb50d980f03c56b6b2d733495c80f8

973a91e397be451f1a93763f80c7e4205ec3b6f556902dabfd5f0b48d14530f5

39997f7ed9a61f9377c8dc3b4fa709287801af77ea9b1e5404b26c679ca93d4c

01754957691b3d48ec2d1b2799c79338b9251c59235f4622f07351e00a8eb283

80ad5cde8aea121a85c9b278b33b6c9ace7d2cf764c2179adc03cac0ea9a6b6c

Fiat-Shamir context

e1351d3c6d7b88efd2c99659947cf5f13ef443234f7045969a0b432566ebe7fa

withheld proof term: 101 587 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cexLam definition, ProjectiveBundle:316

{A : Type u_1} → [CommRing.{u_1} A] → A → A → Fin 2 → A

SHA-256 of the statement above

d316a8badd44c3f5e9e5ec64d86e48bbeaf24054749cf78421f219070cc20bf7

SHA-256 of the kernel type

8f8a030a471dd9e22d49fc0bae097c29e300013c3388670844f5c400ccb57a92

commitment C

f94dc48a4b6ed9e96c5f8fda8900eb7f4aad0c819d01fe0b5816d16b43ccf4e8

5218b14fd80319f90f7c329458adf3b4165031d9d9ac5cffc5d3b32c592f6564

097de4cb4a0e7dace6d2e743ef6938916dd9e02b486ea18d50dcd9f60843b57e

0f2060ad108333527fa96d460e1eaf5e097e6ca572387a0068f354d5c3551839

3ad627176d1e3ef92e848ec1d157d9144847420198f7215a006faa719569c54c

0f9910275516b00a3bd2a8155ca44fc568804bfda2cb381af407a7b445fc3222

6bbf7c716bbbc99e4ca5a3830f17421707dc73eb0fc84be12922b2b9665eff8b

e9e7b78c2f9aba54feee9364c5c2c4bed0c28c62d127b91c76d62d89a82e1e2f

first message t

af1f883ae44f6e3e3b4d692380bfbfd8fbcc1aaf632c80002a2c2151a9ed0ac7

9ec7868df2ff42b02ac73ca3fd64b382aead9401a92089a4015889a1c0af0cfa

4a57a3129a54696e7eb2a66adf07dd52f9e82c8425cace26fc09ad25b3171ac6

10bce5d34be2a7ebc7dbf3d8ed2562ad44ba19fe6e71e4d4d5a752fddb453646

33ee2ac68a79bf06c71b1eafcecd3f2a5a4d7ef0dc2aca3734d587383c5b9c47

bdbc0397be835e936190f8dc9f42d47fb9612523ef655ef4a31b83949f19f532

44caa49cf4fa0a3ffaf349d0a0e65da3011b745d8dd5b4dd5e047fdf56a2843b

2e36e69cd1a34a73b127574a2f3c7b501cbde15e31b6b40593035b88149bf6e0

response z

1a4e9cdd6df033002070993b235d14d759e97e904e6d3a871e17886cb8e6a1a3

c388e4fdf5b8fdcca68887659086f346750de53cbab9cf2e9acb60ce325090fd

aed10263cfc561a2409e89e73b29bdc22290fcd920e38c743e24357d8387cd6a

720591ed2ebed7c2c3a330832281961761e4cc7f4f4f89c48bf6c59bfd52751e

7321ae14d0e97db882ce70fd70c56db463ae775a4581fdf8fdf0f16ff5f0b009

3ed5fbc22b7a6e50d8e408be68d26ffd2cb7f6086f948fc6b44297bac7d24223

6f63204231a4efb3ac0357546cb6042b55574f053d7c06cc0a97de922d699ca2

f02faf00550028c7b1cb03729ee7c57f30eed87599a976440ef76263fa66a682

response w

5d34f27462ba63a8421cbf836989046ea98ba7af29424121047c6ff746babe9e

bd2dd71867dbba8ecc75f75554ea5fa7de542fd2b0dac9e4efbfc6d7e21cea6a

9f23be82734adae361ad7cf41f7b8309d1f0d809f021892aa5b4d1b1aa1c1897

685ec188bac24091d5e48255f534d85d53a21c8064a6b73a1e0d5c329942c760

762119e11c1f1bf7ce5a732e569d84d92fa54ca90c8639042317b1be6f2291c5

c299cf1b5c72901210fe859e0a124dce6a812cfe1431d9f31525e5d44abb903b

db113c039993d34b48626e91cd12d40709167bc3a10d1836442e97ba0c9cdcc3

0f312f8e961b8eae8591b6272f08b3d7b5d80088d59ebe62abdceeadca7ef9f0

Fiat-Shamir context

bcf108e8797f347c25b63cc4f3f3e2f6d9c717e9c87522599824dc3815a8c3d

withheld proof term: 2511 bytes • axioms: propext • kernel replay: yes

FBundlePoint.cexLam.eq_1 theorem, ProjectiveBundle:0

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
Eq.{u_1 + 1} (FBundlePoint.cexLam.{u_1} c h)
  (Matrix.vecCons.{u_1} (HSub.hSub.{u_1, u_1, u_1} 1 (HMul.hMul.{u_1, u_1, u_1} h c))
    (Matrix.vecCons.{u_1} (HSub.hSub.{u_1, u_1, u_1} (-1) (HMul.hMul.{u_1, u_1, u_1} h c))
      Matrix.vecEmpty.{u_1}))
```

SHA-256 of the statement above

ec348b46b0f362d2939bddaa64df46ac1679b6dc55ac57638f287342215c0c29

SHA-256 of the kernel type

f3ab04023822b8dd79eb207150e7171b876cf2122afb8c5c40568d5fe763199d

commitment C

9f5eade4bb535202c4f81f1d4c3926ecd39b9f3c6979ba039612297bdad5972b
435f64cea3436a35294becd0f7604ba6256b8cef9fe08706408e835b5b84c09e
156e63325e533b4c63dc3fafd7d6ca61ebcc670a3b5ab599d5bd336a25b9fadb
17a12e8c9ac7662b71abfffcacae4a9f94be46a0e302f407820c6752fe229c3bf5
3150e501cbd98caf203a44dde9889e08928784a3df24e30831a7dd95311f1b2
b50130d6cd96d46d29c0fd102483ab31b30044b8707ccbfcae4498b53808df1de
b5be85f78ad958896c62d176d91eb5ac83f05588cd7862bec6cde5eedd1f65b8
18a6ac05266bf5b03802c851f36494bda1c657a7e80fd5469c6aee6381fcef1d
first message t

7b4990ff3c781fe736c75fed0a2386431bc26e57ceadf14f19bf1e88b0b7aa90
b1a575a2fdddbdaabf38dd45ff439eccd0356003beed08045f2f91a5322baf49
738bc8f6cd164bc1bda7202d76cde253337e957e8b02350261b6217247c0b118
4bc6bf73ff53cb51829a41cd37abc83283608282b82c46754f51e39da02a3986
c1a64d3f2fddb061fd4fdc5d37185a7529055525c9cbd6732dec657294b7462
7cd5e328726beef4b722a44bce41d0ee4fa6e5bc26cc4d8ee928a01aec48f473
05ac61a4ab1c2f33fff827a9c2c108fb57d47e5ddd65c666733d6b622a950c2d
8a194895401860a536d6d5014194f07428d5840742896a4c37a748c7adfa4f72
response z

6a9b0e3aa9ac32e19426cf8a45577a42d853d5b34c962f99bb50b766efe585df
30eefdf6f3f26868d84f233a7eacfb8ac0806401fa8539b7346ef20640c083561
517c50ce9648b2e44433859a5db7f3dc93d1e7557dd7ab3f5cdfedcc067680e1
dee5ccac5de1e9be94628ce6924fd185da4a0555f26b7307a70053b0a895960a
352161b3094a99ff170a15e7a323f039ecc337981727fceb46d94763cb34a2b4
789ec7aa057b578b29ab311623ecce7ca0c976f6f83040d199d0965aeb42e608
47de568866c002157ce9ed4c535046f12d928c2b01208b6403021b6a242d61c1
92dbd6f3aa7dab9ace49a6135ec9dafa9027537603271515195bf540147dfc1
response w

668748f99c8ff66a81ef469df17b252db7db4ebaf1087c567ed229a73ee5f044
74c80faf0789d9a229228f1b7d25f6c1a55b760ae3ee5b19fff66a75e5cf3deb
ef8f6caa52f4370e3970f3ef6172f025a4e7284df2fa2be835b86a1967949785
89c0f851322568eef1731e7daa79fef4177b5bb813f38ac3d8ba92b8687cef4f
958a4bdaa8ef5125211fad8f8f2b44961505556709d4276275f85a9f24a29576
12bff194e5226169e1ca84021819da3eca7a154b90386748d0ff6733fbcc06a2
681caf5e639e78db0ae142ee4b88cf265191efb96d83531a538f1bab645e1c7
4598de191c7900e308bb126d6e3359ceed0fa512f0cf48c2359d7abd32f8a351

Fiat-Shamir context

adbef890ed65eda582f83f28b69c6e9239902379a82178234d74f1de4898dde8

withheld proof term: 177 bytes • axioms: propext • kernel replay: yes

FBundlePoint.cexLam_sub_isUnit theorem, ProjectiveBundle:393

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
  Eq.{u_1 + 1} (HAdd.hAdd.{u_1, u_1, u_1} h h) 1 →
    ∀ (i j : Fin 2),
      Ne.{1} i j →
        IsUnit.{u_1}
          (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.cexLam.{u_1} c h i)
            (FBundlePoint.cexLam.{u_1} c h j))
```

SHA-256 of the statement above

b1cbf148244e701361dd63919f00d152e950a9957da32ecd12be58f1da861710

SHA-256 of the kernel type

f119dc7a98e173c33af930ab624f26e80659c55ac6c9b9248a84095d8128a770

commitment C

e0fd00ee0476bcd8187db09645c93ce5d1a8425ce5e0a01cbcd9c0c5e358ea94
7b88b6476ed1844b52c40ebbea9ff8a9bdebbfa4a81321495785429c08f36e39
a9b243d03c77c39e8b9c8bed04329cde2dc9c0585d9e1739b1d84627cf75ff9c
54e43b309517f9e23e5dc1970b4aaa572eeefa6968f01356721531c6a0a4df2fa
0064a0ae918265620e5e4c40e6346c703a9a49ee601c0ab788427cd0bf9b4a6c
2ddcb7662a5eca3ad511071cf012b856ea34a8e7fe30739cca8c3364346102f2
4b169d91e9f3b3e4cfea1b6b1ffe5e20f854e099a3d4d13f0b683c9415f40b98
2b5bf0b98cbb9a1f6879760fd23b0f4ded245b3d39944b8d09e70aca7f5fd0bb
first message t

28e9b26fb15abd4115e674b8916ff7dcd39f7cc72530a59cfc694e931060d12c
0b0612b63bbce95205ef6cf1f3df43ca98464929fd563850a04ad3e35a0ed6f1
e462d18d546c7d4e0cbc0aca2167fbf14c34dde2682adc8e5bb6f03fc0b4cea7
6b131a4b6eee76294e578ad6de2d46d23a804cd4fd509654122aa0957deda0f0
1ecd59d216a8be5418895ab180a0d83483ea2316edd466663573d2c03daf9db7
dea57e76d17241034fe2c42677df6d93efe8eea7da703d751f7ae41f97ad8df3
35d91df019fb8a81f30cabcd32d23daa355b2d2aa0f73274a641ad29114fbb85
68df3df5daee5e117c4be1f2090c387c7f49c88b31ffd38db0b6ea787e8834d8
response z

408afe55a2346e7c784088a7d441bc752dc57aebfdb9c2384bb09567d614f73c
a98646b65c5f4a76a25e412efc34c95674d98d2c59ebfb06b2ac4756b96073f8
dcd3a70554313360c6ad872eb91cf6714d55fe4073a680c15b843604abd8aed6
b5c1c089627f7cd701dc7628120f5b59868a5203c3af9cc5f2a51a1dd171cf38
65b08979186d565c01b9283d335e89c765e65cee51816560ccbfcbfbc223faf09
2de74f5bcc4f3345ed1f2ffa194dd91997f510a930f200a97653594c1346167
775b20e4cd4aca098811208bb9e468ce7ba4e49122dc8d05f6c03f25c3370299
0bd1517516eed229ca712b368d8de2fc4a332af1f713897f7d699340e2e898cc
response w

5d2b09a5f6f8d02db8bf21b37100f1f7ba2a1a036cb8f34256696895e2b3a178
3f9244f8935cb63f7435307dd713b819b9c319f55a26bda623a495f57e9a6d07
137a87154dc1346893d5a125839dc3c51328baa9581e9c35752c97be704ab6be
aa48a3b858c632ea9872dc8357cc9fceb785b7f3e6ccd0ae53054e872dc0a19
df12e9ceb7e6342ab31e753f24166f5c1b5ad2dcfea455f26904478d0062ca93
f1fa73510b4c89a0cb7532ea07d51a7c3dd6433766a51d97184bec1b907fbf15
22d1c454be65f9bd58bd6376137f61d938f51058c447852f774cf88795471b2c
a95473ac234a013d8c0674ade5e18c2a146a2f7a4fe9ac7eeb90d6be7d67c355

Fiat-Shamir context

dfa3d8ae95ba2e4e4f14f91603a40475a171f770024ccc2b0be2c22a3f528fc1

withheld proof term: 587 632 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cexMu definition, ProjectiveBundle:319

$\{A : \text{Type } u_1\} \rightarrow [\text{CommRing}.\{u_1\} A] \rightarrow \text{Fin } 2 \rightarrow A$

SHA-256 of the statement above

d6491a7aa93b1b75520aec5a72bc858a6059e6dabd09dfab8033842532cb8d5d

SHA-256 of the kernel type

b37090fc0844a4d800acc6bc71cc17459a107c45a0298e6a76a731bcb5d3d257
commitment C

f4fad961f4ae5e0955a2e79f5d9b64f5d587862a3ee3d0bfe4e60a10287e3195
e09dcab3e0b144a6e534bc001a250f802c996c2d7b889dad9d467d1df1c11231
f01cee889ec713374fe926990585560358fcb5a62a5b0988e749d5c3be16748c
0403c7e9277d253674f9b2a6402716ca8ab5309fb0d688b6702753d62b777a77
280a1736916c58fab9d6fe2733806bb10ad8efc1d248ebad7b48426b7ef36d51
e681bf0149b232816e8263bee6c6c323e40e2ad3de85526099edeb9b235b1abe
da4213acc21cf82bc31844d8050ca850a0d4e4a8526e0abfcfbab2968359be40
fe9210ff54070427b62366c3142fd5270c1598713795c37284aa092b913ffbaa
first message t

b9599ff1a320422f257633d0041540868a09f2d3966e457708b47d0a3e278b97
69ec7ae8c50547ddb653e5adee0e8a68ba1f7c3ec924b44bd149e3099c47dbc7
2db476a8412c0ee4919012e0078534db1b236b06504132b940993384804dd3c3
11f84237991e860c05ebe50d68eba21ceb47c9c0014a231be8fc5e20fe885be8
3569f931ef4f257b5e9012c42fc5439dd9905197755c28ba103ba367ff744069
d84c7eb1699e13c9039baf7dab5924db1deac8bce936198d3ec042877176f70a
3c9040ad5a8fcd66d6843dc8512a197e3a8fa860f37b36b1f081369433cb34f8
9143e810f119fb0a1fc9bcfca39c804fdf7281b36d5d60564d937ec0bba9aa2f
response z

45727c5806d8d89c91215dd544733a818c2ffa299225c21b50874f5fbc9f5ac2
ef13e2aff7d5afee4fa99f10f49c25d1faeb645d51a5788b130f340567cce1b0
6e0b1578ea7dac23809f23dce63b3e194080e4a85bafdf5ca8eaa51a4f39f8c1
9298be59c25d219a57c79268da8d2ead12291b5f2bc68ea75384c6039ebc9420
a30a4fc7997b423966aa66d72424452759ca26a0b451e2a528028b2342d42733
64f10a49bbb1645838e832186d9426711b6cdf2bf6813d10a9de0890496e3f1
1d349a3b7641de9d5ee204f48ca9ee999d2da91ba6eb142850a3b9e4d83353dd
20010e2fc6f89d47ce45db977c0626307e1722c0b858c0a9206b64285d3ad4b5
response w

dfb57a865e91c9b296e97727009b95a3dcb38067b7df59d3ee9120260266db12
e7eeb233887a6007855a41c02485bd7c485c336a12fdb6340761199ddffa7b7b
92231081cedac8b37fe4c1a8f9429f022f9c44a580310837b5977b6d3d67adc8
40f30b67d126f748e670af52d2a80b8ecddcb00f3a6170a31d45574477aa0586
5817b054028a5e20ac4b5006dc458a3bc3faee8f336f3e942b12ef3be7e80ec3
d98b0270c9202fe2e492d6b97d6df26349053ff1237b820efbe370ee54f7fb6f
ce89879007f595d1b9531f1849d215e22d60e64cd1659f6cd644c25002a067c9
1e3b0ad0466ac15414acd7c6824d3fce52b22e7d9f9245f41768b23418187b5

Fiat-Shamir context

cf3216e276f19ba2204268e96fa3a81cb70e7f19254ca52fa1bef96753c8995e

withheld proof term: 1 119 bytes • axioms: propext • kernel replay: yes

FBundlePoint.cexMu.eq_1 theorem, ProjectiveBundle:0

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} A],$
Eq. $\{u_1 + 1\}$ FBundlePoint.cexMu. $\{u_1\}$
(Matrix.vecCons. $\{u_1\}$ 1 (Matrix.vecCons. $\{u_1\}$ (-1) Matrix.vecEmpty. $\{u_1\}$)))

SHA-256 of the statement above

b0e7034f4fb9e0baf23c9cf79a5b69026fe8038d525de34ae21d27e492473a06

SHA-256 of the kernel type

f5b8b72f6f9e2ba49a651d7dee28a14191d3e151275b7a31b8d835e6049f778c
commitment C

cfab3aefbc2cc0fac521ef993426b72343fd1b02857c8843296fa14a588cc729
918a58d0785f8c3de388318e2d121a64e6c6e589c853f73c7a053ede25c8ca76
8853ae5fc2a6482f6a20b1894e204038c684c781a96f34288a823db37923f458
e22b9e98edfac04663cd280c4736d6c4c1a087cb4f3c8b7b282dac09a8009468
41f330114bf79f17d091a9f460a2c3591a193c364452aa6bb429cf253f43196b
be51b212c0a8e638952f9076e71a3bde537adf38bb09591dc34092e36dad4478
6e718923d2e5087b8095d5b28fbd11a2793bd90b2d0ce68d6a1879d5ca848aa
3e305f62559b2002fa769390283833d89be4275162ca09b4224514ef5c3c9003

first message t

58dc75ce0972813ab568e5953f00d8e945e192a87a69374eef915bc923088788
4b59061aa79f4deda70fca7dceb9caa3cc266fe68c73fcd68343f2598988b80d
9a32b90c2499b534b12bf6f1bb6b345d882676e8fbd9f5eb68fb6a3c0c1a8c1d
4b736f2f6343e92b83d6b4bda59742e044571ce4b71a169f85c7416846fc515a
d66f3c23cc02ed42a2e4c8df016735633f8b194ca4c38930ec3e2808ca3db7f8
984f88ab5d41fcf5023e90fabea4a14ec06057b72659b6c5ac32a8f0d6c6fc0b
3c3f0c83b1f671e443c566cccbe2c6fbf4f79ae7f9a116e09a1f170d1186f64b
e3387e036f43bb23e6040d7dbff132e39b1e95bdd4c3bb63b21b7a75cfc38705

response z

34ef3154a5df7978298f59a40d14a7613725a05438e773959a0812b9c6c4b94f
365406294cd7f25b41e5ab4395ac85504f872b7606be46affb171d2264941a5c
1d411c0a7a64db54fec245ff18bf02de109adef39c07d487ba8a6f3cb925f8d9
e52dcb63324375d0ed8b7f5352ac20d942daa7cad97d2157b946e57d9da08365
25bc6b3c69971dc32196b3721ecc3a172910dbc98ea018d7246a1727bcf32828
302e0bfcf1b1fb4fcd4c154ee8b6af864ae105ae6e0dc5bdc057d11d81fe5bd
a25075ee77cf8443d136b18dca27556e295eb1b24730fbd66957400cd383a28a
ffff145ff9e41c08185ea7d6894f2fdb1c3f43e98d71b781c928bb79afd8baae

response w

4beed1d1c187f5ee6a4dce015e710e3a6d90236fe8e26ca0d7dbee8a5f6dc416
5455dfa248d2ff89cd03f9608c8bbd75c091be9ce43f3feabdf57f5ab0de3db
2d8a1e1853d99ff2272c0b9c262a93888ab96e92c8d6be665219c5ddbbb39bec
5d43914df896aa57269e5cbcc7f27a5c44cae732d63b750a5359ae99e3a46329
685accbf26e963a000dbe1fb7fa9e2f321d9057cdcb99fd9c56f6793adae81ea
097a170a3510fbaee8ac7abb254e8fd817648c3b9a6061dfcf698833a388f5
0bed621aee73404fdb114fce96349b2b403316b2c5a52dc13a23a03475ac72a2
10535657f141940fff3fafeac24457d64edb6fc0d427b2e5162c94d70d0819cf

Fiat-Shamir context

39e05ed6630e7f740566650275abb1be2e0cf81a3c094f92269891e6003761e6

withheld proof term: 156 bytes • axioms: propext • kernel replay: yes

FBundlePoint.cexMu_sum theorem, ProjectiveBundle:361

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} A], \text{Eq}.\{u_1 + 1\} (\sum j, \text{FBundlePoint.cexMu}.\{u_1\} j) 0$

SHA-256 of the statement above

a067195a5940bd18f8deda58d8fd2d2cd0f5aab2c269a0a7b10e2d3367bd29cf

SHA-256 of the kernel type

4edefc7b9de4e86d733e30b6414ae4b8f7368cfb3e4c50d867c17cf3a6c92902
commitment C

29a77de10ec89f6b8687e5fe934cc3cbdc8f7f02b0bb8a3c46dcac51ae0d1353
7fb8f158b33aa058000105a7a19c1e1b158556e8eec7343c2f5770cd2fe8ea38
5e7299a214f30e8a2022300581c49d1de13d64c2c36ee75a1e9a2a1d202cb6cd
37f0e0cfe14e36a40e6a02cda48ee173c55b58d26d6e76a2f3272469d5fe6fbc
d3501e30711beb741b0d76e4cee19dc6d7836665eae33aaca61389022e895cd7
aa3843caf6360f23c3684eee149980ba7016da6b5c07877c134d34bdb4f639cc
62f73ba778139d1caec938b6a04f198934168046c28501141ecd1bdc3d39a7b3
c214bc14a00b125a1241d51be53c7e2170a1d9e4feaf943400a281e316cae236
first message t

7e0b2b1b880dbf94c27f6325c415893e8aa139815dc43a53188861e86037702
b31c100ff1c6cfb21c3abcf86f382ca4611c449ea44b416c666c6872a8cc2e7d
e4d42348b32448c4be38b714eca86833dd4832129c04e894e24c643ee36b67fb
cc5fc95ba270a8bdbcb12226f67db85599d0d6cae81cb29d694dd6983a3a3a06e
22f5640bf755e1f47f1737a4cf558c97eeec49d71f61a148248b22883502e1c7
c02c39373f7f7524423cc3d981176a2ef50a621136ba100f3e5f9dd635dc38d5
36f56f5bea9533abedacd67477ccefab4aa30b1cc154240991ff10746ee53a5
51e232d7eaa5b0a6155f41afb22583b17d0a38c1ee429d042bf903149df5521f
response z

28d3e533ee183761a209e639190128341543bfb8b37e2b711478a9d4159c257c
8b3058ddc2bb4c27ec063f1ce7cf7975d6b72895355c4f7a0d081ce5c1a23553
1028244744177417d4fd8b94ef0460b734362d8ad6cfe70a1646a3fddfbd022f
a64d5006badbac54f5379ec13d366814cb3b56632369e72b55a2ddbfa38615bc
db386178656236e822206baed9f34d9473f3e1932662faf2141b2d29a4af45bb
9bdea0e4d4e25a1b9137317cb93cdf66e09edca1b978b405c5205f430c4adee1
3fbfa01d98b08dee956a862846647cf4445770ef439b3c727c6633a80fd41938
bd55335e7b9c44a89dad16d423b69b2708f628dd14abf6045ceb6758fae658ec
response w

18071482d37541bc1d28551c2f21c339c06f69866babd42ccf206d919278c07
e8db745ed36cfbcdb74903c8bd5ecd47b938bf63573d6a802ac5649cae9176d5
925f7fad002288982b2c132bc13ce69ded36fc6af2a4f8473758eeddccc4b8ab
65d67b6c6372eb374bc5f1a6e17a67f384bb34ed473ed88347467537b6782329
6e133f8e09c50ebc013dc24e05dfe5e881bfd281cc14b24ede80be45ad9a277d
d79b08b3be3f06087358fb271ad9a5311d55a289bc745f08c34bf76cf60a517d
42e005c4b531a751c8d10dd2126a9eabc2c4a03b7d21727aa71073f7ca19f6d8
65a765055c237db60907bc114585fb897461f883fdeb653eaf052a7afa7aa725

Fiat-Shamir context

5e6e964a79aea9cb3e9f0d20a398de617c1061ca30f11ad5840a7005e866984a

withheld proof term: 27 974 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cexP definition, ProjectiveBundle:309

{A : Type u_1} → [CommRing.{u_1} A] → A → A → Matrix.{0, 0, u_1} (Fin 2) (Fin 2) A

SHA-256 of the statement above

210af4b786c3c8f3744ff6f0e3a04a6c52a76b80a11731977f1d956ed20174c7

SHA-256 of the kernel type

44393227c2415ebb4398f292902aa7557e87718482b85cd79854c328fc236cb3

commitment C

5e18179adb8c75d12d00f3131f0915dea97b00733daaf2a2f197ceada54242c

6e677f47ae48db257d135bcb85d2ad02c42b6152e33444748bb540b3e6199484

041c7228ad5de8ec61452379da4312f0517d4708161bb049772bb8fa3c404414

a388ec73d910d8041e495bf22bf9636c139fe53d7afd2fa18330088b6b27fd95

bb0cfbcf4f6f21c60962dba25dcd80c388ba7df5def059cd12c8eb8d30e3395b

e5206863da2cd5dc0356a1080caa3061b5fb245f26d7fe422b734e55315b1037

a2d2c341faa078f530dae413d40355f16bafa25f96783628e939b79b8b4e3af5

05b43f29f4ca90b7398fbb3788753a9834351b10f2fa96d90d53f4c9236e698f

first message t

6db149cc5120e5413f5793c99fb72d1ab5404f949d11a5f8bdd3d8f4f2684230

ac22866905a8a0f4dc00ce7964f599b16cdb4a5282ed43a217336a45ecb3bbd1

89eb8bafb27784dce65d284b90541be317c2c13de3ebb56e060aec82a5b2796f

2c109fb73d3d7da2c80d2d7f3f7ed02959baa6fa6827497364683f59a2cc6c42

0bd1931b0d042784a939de970f8db3cef14a6c8ce3fca08ea19b191a65de1145

0f8a7c4099bc353569ae37fdc9a4f570a79b6d472300536bc78f794cbaadbf7d

bc978f5259cc94b2405c910271f77f9df280409b52179e3fe4fa51de4d5b23a6

14128d5abca3db0eacef2aa04118e38c9bcfbdbb3d888f05d207cea1b1cc5650

response z

6cc3a8b0d6f9a2e2d8d8bdfc62c906e01c910d3e5561bfebede39ab685d5d398

1b78284e2d8b83739ebd196355a628b8a2f3859ba7257d06969eafd885d356f9

02abc93c7493d14e697228df8cec070803e6191fc05c9fd779c0901daa9bccd1

be671245a9a93b56edc7c7b603d53829c088f252f826945368e44e0aae061a2a

efe5ff852ac21510e470105fedb3a5fb4881b8cdce4cf6bfd43ed3dba2be985c

e5c4cf634b8f8029c42a2bb28c06346c49639ce88cbab21a6bfd5d99d4bd0a0b

1da1b525753fa0fd8229a03d8c39e2fd1c2230efc825b9335ad519a22fa10849

559e2a1f39c81244edcbfd8bbefb6442ceca5f25f9209feeeec0a0ae63e3d2f5

response w

74da8b5851ea4528c0cae29a900d7cd7addf94274b62940cdb1e9465b9494341

503309b5ec8882c760bcfa810e10da665de47b4ffc21d40d135bacde3b22ba08

31fa743e5767bd612f24be189ed12dff640c5490c2125d00724d59bdd68b4df1

46eac3080f7e01cdaec14489a5a5f4094dbf2d48c6af8d6ded6993658baa54d5

8ea95622a8579bdda03de78f307f5a122eb945facc8f73de237571a0c11a4375

ebd9029ef8ede8d1313ff0b81a39f15ea8dcf919fd5632a10f16b4c4970abf64

445e2eb44d3901255a8f79ca598558aede230e5840d6887f0bae7049a5045200

dda6aacdc58dcf3d6802940865c8b43c22b0c7b646fbc6d14af811ee88024bc

Fiat-Shamir context

77e85702c013a42533b4a1cce6d0ebda2ee4566a6cfb666adf22cc495f7f0179

withheld proof term: 5332 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.cexP.eq_1 theorem, ProjectiveBundle:0

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
  Eq.{u_1 + 1} (FBundlePoint.cexP.{u_1} c h)
    (Matrix.of.{u_1, 0, 0}
      (Matrix.vecCons.{u_1}
        (Matrix.vecCons.{u_1} (HAdd.hAdd.{u_1, u_1, u_1} 1 (HMul.hMul.{u_1, u_1, u_1} h c))
          (Matrix.vecCons.{u_1} (HAdd.hAdd.{u_1, u_1, u_1} (-1) (HMul.hMul.{u_1, u_1, u_1}
            h c))
            Matrix.vecEmpty.{u_1})))
      (Matrix.vecCons.{u_1}
        (Matrix.vecCons.{u_1} 1 (Matrix.vecCons.{u_1} 1 Matrix.vecEmpty.{u_1})))
        Matrix.vecEmpty.{u_1})))
```

SHA-256 of the statement above

801f50836990b6282af24d03275eca451a98bec7555b7dde108158b31ea65c63

SHA-256 of the kernel type

391f62f2506e2c024969b3aa89c0ff5cb8ada3516ce97da400e219f781f29a43
commitment C

eea8f99532f2075235cb90b502953d353285da7aa809b14568ffb56c2adb5ded
870a7f3c7adb48eb24134f86155a3c0c9e2da75de01b9c0cd7f4a9b1266a8e51
f871c733b4fad7c85d43f4eec88b935b670c51afa01e84388183b69aab89f0bc
e94b255f8945997bb167ce678c371be6eb583870e9910e07779b11f64bdc0610
d8d62d8f19df0b93b42031e9f8e6d07c6414ef132c31ae6fe119aaa7fb693891
893fe0a373daed0bfb76c087a9887f4dbb963e2ede357bc4714c1bd08486c3e
79ce16b3d822fd32c2dc1a8bb0e793661c079e8490e33f2326cd6c42009b26f2
c0c0404ae5a6c13474c87efc03ead856ef5a5e55f0dd87c86bbe7e7bd466bc43
first message t

3fd162bad4731782000c0b089412d03dd0e316ea64a983e604934920d6561730
245d448873be95e5a078df0e12caf9d25a8acfd072f30998f8a8349ef0f837f7
944a5e953522fc549ef7df4b467d7b0be8383df57b359fe1ebc3880c0f956e6d
41ab975a6d41b2fe1bf8e96d8f1a155a77ae8123d01c13d8c7a75c1aaa590e5a
261b53d670852fd07ea21f661228ee65d9995a25ee2598fa5e12ffcd2a4eb21e
cd5409eb91e78984a77a07f9d3905f643c87e8b3fa441e45ffe690ad12dde687
a6e0a1b7a3ee73e1b3de5e1bf99284481d74cfb76638534f4f8c1e74dd6281e7
582be7bba0e025bd1facf23d2522212b37dc77dde7b2c2427707c868faa8b6e7
response z

6378bde52eabb0a1318472e2b647a8363e691fe2eb5b578f0a9a01832b528a32
7f06ae3c7f6a818e1217b1d8b848fe9173c50eae341a2d3ca7815afc6bafd4f
f028d57a1df63bbee4b2430b0430f67ead0fd4d946252d8b4d4a517ec1d27d55
341fc86c4b535af43a23f9e9cda395a15f84e5f3d6d0db06720284f829a9e14b
3c029926839d41c6a3fec9d53573efaaaf12414507b35d89875359bcc16a242b8
8cd60460d002bbff46efd06388cbff7f1ce805ddf21e8ab938771d5f3096f9c2
7b3cd5fc36999be4974cc4295f93389ef782108c85d7fb2914e6a36c41daff2f
2744d0a8bf423a8a4d3ae87e0f42ce6ca4c96f833f80e6d0044aee74a506a79e
response w

566c7305e0873846e467b7fc7a81e3c478074893ceda23db3ee277e4826a216a
9f9d736d96afaa8b5fcd04fbf60b774ebee53a69a7c9638919697a2b985992
d2b8a4f7f22a538eeb920fdd546466b9eca54cd7d6e00efbed2a06c0db48cc80
02bc3e5b70306841ef72948740fb6f2f3d95b78441f154bae94fd9ef20598f70
686835cd818ec00cafc18002df0fdf07e40e922da503c14429260cf9fd71d52b
0cfb2ec01445928e238f59287b86c710898206b7b9c282ff2002c7458a22865
d48c5fe3eed308c99febdef708d53cebb312fc78a0d9592edf0b05cdace8fa9
bd493de1da4fb5af9f5db07537440e1b15ec81bd5cf33200a4dc08f8b69b3229

Fiat-Shamir context

0ad7d80fa32a0975d728a790f32798d8977eccd13d918a99755512fb474e3afc

withheld proof term: 238 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.cexP_mul_cexQ theorem, ProjectiveBundle:331

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
  Eq.{u_1 + 1} (HAdd.hAdd.{u_1, u_1} h h) 1 →
  Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cexP.{u_1} c h) (FBundlePoint.cexQ.{u_1} c
      h)) 1
```

SHA-256 of the statement above

492106c17096cb03f5eff0f9e63415cee7cf49ea35547a7b595fccb37420bad6

SHA-256 of the kernel type

cdc57cada58ddc1ec7ee3f62710deccc218c4c46daf9d4511c4ddef76e099c9c
commitment C

3fad5418df825a5d624d3b242214e3b9356105ffdf0c84a8410b80e6cc07e9fa
b7def3cac194251b61e97a85d4e54431191f9ec2375a375442e192127beb9db1
a39c56efb8a34de96b7999a94b5ba847dad7c1a4d4358e48b31aa7bc02a89843
6ebc0075aab3e6ccbb2be59e51ee8b15fe6d3da2f90e1b1a516e6b08bb910091
b5e2d61a14cec9d5cd57d627ffd016851ac01abac823ab95be3d3d7f89f6ca95
adcb09b66501fd47883d9e60352035bfae61f32b71390afc869c12c2f9d7daf3
dd2ad25f9eaa79d611d35180f35496820c528ff4f2f80866ccab2f70dc1863eb
f3dd101ed49e0c127c11f1744b3c632b3050e5d370a3249111c41a62d1c0538e
first message t

d47d3d2fa0ec7a582cbac58c85367de9c124d5acb7465fc7773318965a91ff4b
667d40b4ffe2f7d940623750fa64c5eb70fbff5afd71712e109899a1e82a6567
16b43a869200b92453eb68cee39a3bcf3b8fa0a9531559708bb44c8621d917a8
9fef8f9cc540e364e75f18bb18c9218e4f782c7a9fc8386528a0a5b7aec99ac3
8bc244c13e7f7d366ab8fbbf35062c02c83ccd52259c0b05e1d876ce84d0fef7
3c28eb95efe58fe48390b83e2f8aa887f8a13eb657b9d96ac0ed5353662c952d
2c63cf9f1840c3eea555df654f31e7fa0548c423b17f5fead9e789db452f1570
9b8b47f57372f430d1a91a8ae83627228f98e92d838453b14bfa3f807a97a706
response z

468b8d5eb921c5fad90f03461b9ce4de6b1ff4cfdba5f8facb176dc995f1d5ff
2f67d648886cf55cb3a17a585393c97ee7255787ca61073f419828bffc5136a3
7dc56e584717f622ce6300cad6e34522c4f1c265abc517a68b312ce411bdbfbb
98d7f583fedaaaf767db926431d0ca9037d94c4cf5eacc647d80a0ef99fe3a1
9f6169e02ab91f957568f046264d6315a7ff0bdecf52b2da44f00fc5b97acb93
95c4b603b4c20b443b2f1f790365d64dec53c1b1fc6ca016cc7d727ea9103d05
8a58c96f8d96eb43f13be9129df14667045864b088d7827994b2e56874929005
4e5b32f8637c616f3200545bbc504dd6db44c6e325630b37b30c195362211560
response w

7da7fdd3994c3f3d796e241302d76094a8a13e83e15e985ecafc7d9b5a93dc00
5e6585210fbf504c9a8e792a17b3f5898aa8eb3d57fe91bc678f3634620537bb
da2f63815c6c73663bd80241fe2ad70c22b193fc1212153dcdc919028011cf88
41b1c8960b29ccb822e116f0ca96ee2cd9302be43f7967c804a2c29319407859
7dbb186a232cfc44c0acf55cf7c04174132c579f7487c7997f41df288cc5b894
500f582c8de0999c01ef806d378b1046e7b6a845a7761cfec871258df0ef047a
028a0c601e86f806ca2dfec5b5804caa806cef160d22a3a4601176e9d33ca8f2
631eff2d57699fb664adad160b44104033f11c6655cd70fdaaae13286e877ecd
Fiat-Shamir context

b3bec1e1385b0d9576c7cde78da6fc777b330e1a15d4608a478465dd037fd862

withheld proof term: 2590591 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cexQ definition, ProjectiveBundle:312

$\{A : \text{Type } u_1\} \rightarrow [\text{CommRing}.\{u_1\} A] \rightarrow A \rightarrow A \rightarrow \text{Matrix}.\{0, 0, u_1\} (\text{Fin } 2) (\text{Fin } 2) A$

SHA-256 of the statement above

210af4b786c3c8f3744ff6f0e3a04a6c52a76b80a11731977f1d956ed20174c7

SHA-256 of the kernel type

c257dc4fc4c0a18d4017863f5938f48bcc1a8a05c5b5e8f1ea13818b65cc8aaf
commitment C

38358e708a744162c1cd62831fd775d1dc010a52bb7c71ab8736af31c5016cd3
66985310a7e9db2b5397cb7661c37432566a8707f2631a00d94e4e20c961688f
c34d2fbd3c5a40b65c8b8319aa8e71c98a0a893a580c1d6336006eba4bad587
0fe5d6568ca24a1ebf91dae4df3d9ec89541c8138b6a0fdb38ff46d0d35d95f4
92d0737618192e5d15ef8cc773d6969e2c14bb5ff9aa5f2c8ddff45036d81fe
f631c695fe5522e67d34c7d3b6946947a923a0c48ed9844bd7e32cdd4ecf8bee
0caa26020d1ffb26ed4d9704ddab5326e3f63e7ede66e993430e4dfa26e715a0
108001e74df3e675f742d2bf5c937246e66934b5ef06ec3e65f0e7f6187ffb1b
first message t

992b665faff26d74afaceb88a9aec7a8244d57df67867805e6fd822fd4d1dd89
fc22175f328ba5e621799de3ad3ac96571529c3ecb7e5c1291194096f11b87cb
af628619440c9fd71135885bf5e744afd53a79d9349636086941e2f4d1f46773
71539e0f5b514e2b5b70109fced4cc02b82911d5af4818321db9fde15719a8fd
b042f69d041df1ddfe61be1a2c32b6fd5ab59378d1096e2a881642f622e18325
5f5c57b406d9dd1a4457878e9c5a1457a788985b29133f84f8790a97dfd5221a
1b85a6c63e2e4f949c5d8980972e0bc0c01a401b315cc1ca6ffef71a8eb9486a
8cf5e48add00a970116ce53f4addb4763eec6a9189b9083264f86584b8aa3bd9
response z

452610947d372a44caf95adb1d63a19ec6fd4efa3fa30ff817fc3a35bd64949
55c1f949d5db04da337dc561cbea95f98fd64b75875d4e0b344ece54944e7f88
df748a719fd910968b6adee751be76e47c3f95a4361485f9c0fa28a79012efdd
f59f1cfb2a0bb7576107c4bbedf1d8b7d9d7a8faadd7e68ef17746be5c74f224
f6386721cbb02ed42abbae8cf80329552f6b2ed2900d1b689e930f301d3c271e
ad9f222b396a7ee1dbbf2a73d733d5a58902abd3c255bf2f71d6e11df0f9ccbb
2caca86a760a5b199a19fb01d4842e04a2d6a624e4ccdd21ad161590d85dd6a3
179855b0100668c4246210d09ec0520dc5d38ae7d822b5d33127806081cd9b87
response w

57cc271b70fbfd19f9fcae69366758b27ba2107157184e82b926b89a668deba
63f2017735f7033ae8a37f369bd4c1f36cf2e50a5b8f37f800e5a02cf08fbe0b
6c0ed961a205c9f55bea7df8936dc26ff6c8264c687e06ad34e89ed1bd73ed69
2d7dd76d726f7e0c61ec879575e681c9c1c7340567ce29e6eb14419b65dd7676
1fed50f793063517ed082860bb05cd838ef716d628d7fd289cf6abecdd4d429b
0ecf8730375813cc05a5105c7244789c3dd42e321f1b28447287b4e0817cfae0
fc6832e40b3ebd5fc1b490d416a9355b42d764d95b39ff66696d5bb5cc5390ee
600525c010c102198dc3a81cc850790a1424a33922ebaafe74b54563cd883bd

Fiat-Shamir context

f5addb7fa85d544e7c851019dae8f267e95e72a7290f9b76fb08345d4dbeba2d

withheld proof term: 5 026 bytes • axioms: Quot.sound, propext • kernel replay: yes

```

FBundlePoint.cexQ.eq_1 theorem, ProjectiveBundle:0
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
  Eq.{u_1 + 1} (FBundlePoint.cexQ.{u_1} c h)
    (Matrix.of.{u_1, 0, 0}
      (Matrix.vecCons.{u_1}
        (Matrix.vecCons.{u_1} h
          (Matrix.vecCons.{u_1}
            (HSub.hSub.{u_1, u_1, u_1} h
              (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} h h) c))
            Matrix.vecEmpty.{u_1})))
        (Matrix.vecCons.{u_1}
          (Matrix.vecCons.{u_1} (Neg.neg.{u_1} h)
            (Matrix.vecCons.{u_1}
              (HAdd.hAdd.{u_1, u_1, u_1} h
                (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} h h) c))
                Matrix.vecEmpty.{u_1})))
            Matrix.vecEmpty.{u_1}))))
  )

```

SHA-256 of the statement above

9d558f4882b08e66a1f1308689a1ac97dde5dfdfd7c7090c4140bfa4ee84eb9e

SHA-256 of the kernel type

5c44c6f3c14c047a5c7d44d732046085d963b4bc014b8898ad794557279abdd7

commitment C

bbbea43be7cc0ca236b0c181bffe0be02521edc3ea70e51f8c706ce858a832
f6293cee63ae090077594f445210c8c73edbf8756cde5943c9e202cfd7526
8cc10b6b6cc01a6fa1b345c556ed08070c9dd8a4bf71c2532857563533633423
ecb96aadcd5e54679cb33a935d5b548edba41a27700e4d178c382b89cf0a0de7
1e7be42836a00371e87b8a9f5de1f2454ef4d0e450356b5ef42cb5785c09e195
f5459d49d4306072fd77b42ce6c58a0924fcd7447d9e5a4ba9c76fc0c5fca970
3c42d1733349a49515ec4ec21947f2c721e618c1b8b6b16e10788ca44e709fce
24b5b366ade8e404463b79ee70b23d4cf0c551e6913981e17c3500517d37a0ac

first message t

ba0126fa9e0585b48c4e4f49e8dbed34294c5cbf33c4e6b927ebe314a021e885
f3414bf2409c426c4e24e2e871c9b44f5261906a340a4512c7f2e1f112b3c787
c87732bcfb3d41d22127f6fd4a14bbbed02f30abbe2fff8b3c17bf03a06263f8b
e54b00d160e0f1aa7f3a0cb31bea964389d4df1be5e037d11019ab146fd2aa4f
2cb48299a4939ffd7d87d7ba8de49fc15e42e3dbd6b16d723935c2750e3275c3
0dd52b783440a35b06a1768821c40247813bdebe6012114a69c4173f95b3295a
c9749cd708d62497123c0af10273f73cba40ba8b5b41cd6e15180196eca15f5c
2fb378eabe87219aa78e83fabfd00d35aaa275a087264a4a2de708878f080a51

response z

71d4b92b89142e39a756fe7e758cfc2c306545249c209d62c6e560d842897eb
d5f585a71b8d61ae8f14cdd296ba4c3388a2acb56d36306955b47a4d618b710f
41f11710905450fc3cf9cf38dd3ed99d079a94df2c356b424d93c6a7d43dfcbe
2a40eaf764197129004774bb386dd96e9aebfe4bee58dade9bee44c9342c9f99
0b22ddb3bce6988245a2b1f78e7df481be2d8e08822f2db28482baebe6f12082
ae4e5544a407dbe1f29266ef1af3b794977882d5c9ea31ff1315167ec15ec5a1
b83b780c695986df25965798a06af7a7ab37a525a3dc7c1fb2f9f67f549462f7
b3954c132634eb9cced434a987da2cf660c3d8a39b18133e2e0be99f7e8894f0

response w

51f7b1eb6056a00cb74124e945315122021758015e93c2be44ccfbfbd7fd928f
0cde2478b36ce64826b3b0a7b22ab6bb5eb9c9c5d612eef3aa952b74a60d86ed
b0035997e3f6782af87b5f15fdd055b48c9bca7736656991dc27c444c18bca5b
372ae2ac3570ea10b7f76881b148d921a999bae0ef336f7f308026eb35a0a05d
6ace975c89744990878030fd09d83cd4b2ec07ff027a2b3bf4d5864cf81311a9
84418dd0f7335fffe61f0029a2c2547dd570f67efc929377607fb81a6003e585
685999ed69c749f27d2e30ddcb4df7922828ba3f82f5c8874e8a4fb0633c3242
5f7528e5b936b0276735c37b1a98dadd7ecda32f4dde20a817db71a5a6524db9

Fiat-Shamir context

6279733f5d1ce04cfff65f646ad9f96dc924bd6d818a49dee247da0cc46ef7160

withheld proof term: 238 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.cexQ_mul_cexP theorem, ProjectiveBundle:322

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
  Eq.{u_1 + 1} (HAdd.hAdd.{u_1, u_1, u_1} h h) 1 →
  Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cexQ.{u_1} c h) (FBundlePoint.cexP.{u_1} c
      h)) 1
```

SHA-256 of the statement above

b6fc4e071083e3e0937cdedc0c37bb5b7ce3d8d5cc37eeb92a346f5bc245a948

SHA-256 of the kernel type

c767cbfcfd2ada2755799a4be67ac4d40d8a00c99d768f1c64535fe651ec8b2a
commitment C

6d5304e1563711a5ff0c48c85b3919ebf0731318c7e5c33d732028c4c9e1526f
5d6616e623a87c157f4ef49e281d426a2018bd518641dadcf3b2b30345c15d9
ca235cd976a1a94f7e1e1bc0e232f17a255b84a277abc6c9800e670f871b95c8
21ad373e33b9c78ec85f82118417f1f52e0952eacb4f357ee49037185bf87e53
972e758bf756bacf8c27c0a52d492dac39bacdb503d62b45e497db33f72bfc79
41afd04ae06b6d49ab133d2f310c3331351af769740238dd9f669fb211b17b31
d6b1b3be15a0eb061c2138e5027d0825bff40440c5cf6e3696bd2953bbabed47
b1df9bfaea715b9ecf3ac709264a33bf42b4791268e5e5450402f63a198a1b0d
first message t

2754b0fa420865f5ff75afcb097fe72a6334b5c368de2aa19650e0a15d1e1f2
b81f8eb5dbaa3d277df0331c776424863723522f79e59814711009b8ec49b9cf
4c34ac58f2c8cf79aca0afe2f13c9bc6c92709973258be2e7647d74a4f3d3c32
39e5664077cc43d7ae1a71051211b432bbd8baf36bef9edc86fc0ee05361a0c0
9a363b5186c373d887b7399988b4baea4ed51774cc70f4e63b13279b3999bba7
a3a7e9ae36442f53ad7e8bd4c99024e3dd85ad51a5edde817e68f0e6460f577b
47ce7747f8223c855e46ccd985d55d448f67a9c2ee6ef43843e761d531645848
395cd68520966eec4d8082d82e4ecffd875b76e9d48b4a1152882a77b486e973
response z

74019d23957bc538101299ac5670d205ef9422d778f9aa362739c1827948e0f3
75fa5ab9ee949b3f44a643d1daf1a61f9aef4cec2e04576a7a155c6e3613abe
16a8ef366d7335f0be63aa8050e824347b697c7256a3ac4eeea300bc9df4dbb
cf158ff54c5fd031ebb50ee2571e9aca66f4abc4d2b7c13a391bd38e19b0694a
f264743b04514c8e587d91cfb475770a3af5e8dc87d65151d2f62ee006b6d18e
1ec3256aeffe52c3e9c97fc9b30ffe6fa7823d3f2ccda4592ed3cddd08a1f00c
496a66d2642db8aafc5c71d32cf57fa5be81af66f867399562088399c096239a
625a68832c5c0bc45428ca0eaeed885fc26abb04d0d41577c25f935af4cb1c06
response w

742031e6546f2aa71716bc65f2add0641a633e2db6f4af229ae383f1dd7899bf
365e269d5d430e87b1060ba5a75749081df4aa349586b42accba15475fcfa6a7
0d476697ec0436c66c0444bdb005a5d7dace8bb3618bcf189ce9380b3204b77c
c206c424dc88a6b5a13bdc2c7f930b887bc752433c4952c970748a6d3a601939
0c3c8393f8cb9fb1d47824659561009852682df94ce7d12be35cb2f7d422800d
b66ad318417b716b1140e5786e4a149dc79b71be89f0ae0398df7afc4d923565
6607bfb56b801cc1412b31565639d05708cf71cbe2f06545a7be1ad59ac5003a
2330117df02a4345c03371f47c1815a0307e99cb261639b71f54f01febce8d70

Fiat-Shamir context

3b36e0dafd5f7d81cd28faae6b87cfb5a5489028c84af8c352f1c42624e7c461

withheld proof term: 2 669 026 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cex_diag_entry theorem, ProjectiveBundle:373

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
  Eq.{u_1 + 1} (HAdd.hAdd.{u_1, u_1, u_1} h h) 1 →
  Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1}
      (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cexQ.{u_1} c h)
        (Matrix.diagonal.{u_1, 0} FBundlePoint.cexMu.{u_1}))
      (FBundlePoint.cexP.{u_1} c h) 0 0)
    (HMul.hMul.{u_1, u_1, u_1} h c)
```

SHA-256 of the statement above

6e540ba15f04dde2d45d746f9be245448c8fab84ea9716e0e161dce65e43b1ae

SHA-256 of the kernel type

dbf170ced2c261841127ee80230f514b43c223bd2c9a8757c5ed0e62b61cd4ba
commitment C

8a85849ca1c9f76f93fc30127a48c31d33b14ec2d39e3f9807ccd4571477451d
71d3f24e7775d5310792acc22b444c5dd19fc8dab4dbf7ba6b1614df47e40fb5
431667b3421c6bb5357f4fffaee5f6f3ac7e3f1ab0b674b9578e12ca7f70cb34
a58b62f21cac0ee135c9247aa5d15ed19a813564c16576f67b46e4eadae3357f
b65e63988f011b4e7b475c36b619f99b31bbc5156e1ced4826ee9f088a02d7fb
4e932f1fa5b2ccbf99a64ce6c51e5acc3aedd0d1b0e35e9f2e8c760d6ed43ed
12e8024782f5c0157709bbdf056fbaaa3709fbf9b85ef874f4e61ab1367c874a
256749a4472a792832dede2f8f453f60fee33e2e8f1b123bfcdbfa52a168bfa4
first message t

1b331a866cb2ec4b4438b3870929efc9661997e39d3a5507e1e92f2ba5285378
49c7b6a3f72c217e10349bc826395f73199a5b2c81b3150adc90e09f6c5d9927
65ca03b13b59b06c10c072a3adfa67a13d1baf8581a4b69135b76f150d1efa80
e737bed87339be17de4629297f0d866ea50b2aeca95b007904855f6a7c233b48
8d849be8d5fb19afe2ed39a497d25840983b03f60bc05b1c183c65ed9c3c8672
2886b5c999c93df2e62f0ba15af1dc34d37fb8af489ca8bf66b5fd0280526426
86286f9e669e9a5c91755b708354b114560118d6a2516034f3f86ee4c19b55c8
384636bfae8bad4bf8f0d3c63f4c69c0cd6c9fecf3d13ec91baf8f36fdca1de0
response z

5889fa90792c1ca5f1b2e6b25d992b41e138927466fa3a4cddebb7359ef3c173
c6a52a2078999567deb4e3a94cd459cc94bd0c77490d22c636f9bf6c9998a7b6
98622dc05321b91e5f27bbdb88cf50db161d3d5119b9e6ec6dcb72a09aa30345
da685879720e6c48b3b3f36b041d3ad928c16c65b43510f12bf9c05248b9174a
7360a254404102c1600a1fa10ae972e2188ee4f5e189793d55ef61be91d3c67b
dbefbb23d271138c825740d26e47e1c2e8544c40e9c8adedda326c5b4e818bf5
067935c6d1294bbf4bf73f01791f6122e7ca620101df88dc39c76b4824e880f8
aa8e73e9a0e043e896dfc51544f08fe8d21a7ed69247f9a9dd896570fb174c52
response w

409f4c99cff3314f4f7515315ca9eac88f6caccba37e054cdfcfd5b28727ad5
2fe8bfbf223db7bb2be6b9a9b8858a4724aaa05e41590d8ea3da9125b22fa242
c8df83eefc7ff4876c223d2973b5a3980dea030bd2bf40e13413263bd6cea7d1
ccfeac310171110effd59f0de3359e65b621f4ecc55d3ff0812fc78e1939f368
e487b3e43fd66349809716349ef3c9ad17c57b91907bc46570d25854786c49fc
3ab4fd073e5257d388b8b7e28fd1156252d7299e78d44fb11e954e49015e5ee3
4199d93ab35f7f4cdfb99931f8c6e5b3ef9034e3ac9c26322a1cbaff23396b1d
e54df72d0e62cbf30c149128f900b9ae7193364d5bf4eeb2fece623a35d34641

Fiat-Shamir context

2eda87b485656180880b1fd3c1c6c1ec793c3dc5b3df8400a124304d2779d5eb

withheld proof term: 1327249 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionK definition, ProjectiveBundle:68

```
{A : Type u_1} →
  [CommRing.{u_1} A] →
    {m : Nat} →
      (Fin (HAdd.hAdd.{0, 0, 0} m 1) → A) →
        Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m 1)) A
```

SHA-256 of the statement above

72b48aef7b589718b35f32352ff5ced3a279a751358cf015ae3e04ef1e7ba816

SHA-256 of the kernel type

31248e14e69da54beed616ae179ca85645715fb2b69e3d9e7801f45416f68611

commitment C

f3b12311085d4f4a650eb6d95be9ee3ca0e84097f4566e5afd89c1f0d12ed761

e280f1faa3acde9bdae57428287654f5b39660edb9506739f343e69ff371d78

903235d284c2e56f82bab1c33861a4f329413369321f0538fb0905311fbf8e8c

63008ec42985f4bbdf3e214c69a8480e00209634819379550e90812605c1be1f

ed08f1caabbb47d047405b5b8c117b6d7169c76768417996db7410f516306ea2

94ccd0a1631ccd6beec98d6d21d5d2eb9ce383c65c3d1d3ace4159374158f234

69a24ff02d055ce8a41084d6fe8bf4b76bb3288bcfb053e04bc5e0a3bd716dfb

913484dc03c68d592610affd33b6062ac45d2d5a470d4c07989e7db2f4cce603

first message t

24f26b65cc9b7adbaccfe4958ee88d7379c5f50fd717a39297215a98fee08c20

d2511d3d71f09b4be764e582dc2b6e4588a7115f0da50fb31fd87b624a3dc3d9

c84827ffcb98aa12cef640b6e8f35cb6348501acfc96f16cd842b63b6ef1d94e

4e2dcbcd4e7c70ad9fa44adee91eb4af942294160a846be7ffefc27b10a9d1bc

d1d9471489c457ee029c6781e446aa063a7c9c06e54ee8bcf3b5e0a3f3ca5b98

f51694edb2b6a6439c51ad71b39e02324204b3603d6c91895a1bb4b308d2fc8d

2ade612bc6de90e30493034bb5216e63e1e4d2ec05f855414316668ca26818b5

5ef9b9c5744148fb4d61600ff8fcfa96a0d621895f6d9d885d2da118494c7e9

response z

4dcc376a974bc1db597a7e21fc9d717919411206bb2d352539e1cbeffb0240edb

2bc6a4d8b3a405134c93ae2b3aeb08c36a2defac77a8a8ceee953eea092f80c

fd1e11bca8c1e2048ff7308e80667e1142ae70cf81c35a0ba98bd3e8847e4834

19d01d7089315c86c7e199b27195bf403baf9f43ed138047fc30b1ba45dd509

73f27dde485c2ce3ef2ca715b2910aa0a01db5a87b3d6c29699e58718118a9af

2b19520826007f5126341649bc2ddd9af03bc8650844399eb461ffa6e7936345

ec92b63428feacfd0a795108e99a58df37fdbcb27581e3326b4361af6f08a4

c72b5b5e72bb20662df1e5e56a5482d0550988814b8c18182aa6ef1faedf29fc

response w

33b4c509250c421545f6fe239c160312a866c1c683a8ea3222c5e3f105ae6779

5e9b622469e932a1aa2133f88bcf097faa59ae9cb7ade20aba30ba8b4a3c255c

1c59dc2ef8cd52706220ff7a802ea3952846b59d2ec2febe6dc4b7782a763e85

32f6f57093c5be341e1178a70c80b2c65211f5f423eab31c667a9e3124642877

3529e5da43fb42d5767019a5a75b6042736b7b386cb47c1eb7ad425ce16836ff

4f69f9e92696d6d4850621b16d3396880776e51f27d3d49ad486976d7277c3bd

25660e54df0cc2cbcf0239c85537c7c88eac083d63d4afe90845a7969b041a9a

2b6bfbf317ee09069fd20f1337ee6a1622fa1c554bf3b6ed6ce0278e557a533

Fiat-Shamir context

d8d4ab36c47d3c6e557254ea8cfab2510479caebb3bb250e759673741e0bd4af

withheld proof term: 1965 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.companionK.eq_1 theorem, ProjectiveBundle:0

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  Eq.{u_1 + 1} (FBundlePoint.companionK.{u_1} b)
    (HSub.hSub.{u_1, u_1, u_1} (FBundlePoint.cyclicShift.{u_1} A m)
      (FBundlePoint.lastCol.{u_1} b))
```

SHA-256 of the statement above

a5dab9cffa9f93ae27a024bf836f1ef756c8a5ff9d0f6296c002c1b8ae4148c5

SHA-256 of the kernel type

ae053510769f87c53e04e3b0faa3696d2de14757d25eed857180bdcf1ec5fddd

commitment C

78d303399b36ef72d443275a3d977e0bb630a58632b1722dfb926923b21d76c2

a136c3cbafdf27bbdcf1b5e8017f1e55a580588a1b10c0536f98099442c23647

0b1e471012e7307b8cdfef7995af7e15c5daadd3c2b307622322d2e8f881a4

b9717b1a387c2cdc833dc06bce6b3a4d32d9f06dc66ccbbf5dc00359f2295366

7c234fc09dcce12895c4302d9cd7484f0eec5f16bd6642a932fe96098c8a2818

10c4aca7d5cf2178b49939f39a7fe40be91c3cf20f06f7d6c1fe200f7d85cde7

421d6a7cd5c91495629f3d93b03491c127c5a361f9590f61065e322aa1075bff

875851aa1a67337634a2cf16fcbb7b0a4473ec8ed52876e325856b3db4d1c8bc

first message t

8149094a6e308ba4c4efcac050a81f3d01b005aa34a60322796a811ec3e063e3

ebf9fe075bf01b344ac5a916c7afac324d0a52b20683f895b2180e3fd553c39f

8658af326edc4709392389405c1699cecea3b781ee8f3832fc03cefc6d4f8d31

c544213fc95913d910555094b1b40dc254c3c7efc7190f355a76d3102d1ab011

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e93db09daf1a83a91cab05c88c50f82856788faf09fe187cc8cf3a1a7a458257

6fbe0bcf288311a66e56dab525f7b0171888fdb7276208d4851dbbabd5a987eb

7f7b89db2204861304b2768e379125996d4e1bc2717991c7a2fdc8020b317476

response z

3463d742f3cf62b7b875d418319b1a9e15cee2de03d5ca72e944cd96eeaf555

44209b8531a03425eec9c3a15facc67e6e5b20f02b5577341fd8c6dce9099323

276e424dac16595bb09fdcef90318494f98d73aae9f0bcb58e77bade5dcca1d

db6b1d37d6575250f8e355ffff38b0af0281caf1f2c3b7f74075a53505f2cb4fd

b1c8355d2db4b08788bc3faba7ea5a36c261a33e3eea8dcf1d8d0b3050e979de

f30085a527146861de7b0471d2d0821cbe79bf07772fabce886b2665ea9feac5

225cac8bf30f468c493d82767546f5c03e444e11090a727a640198371984417b

c8c6698e6e9f8bcc9b12e248395d405e2776eb2fe0f151f11d145d11be41aeb5

response w

5f86792c3b659e5bb077c9a06f21baefdd1350b21d6a78ed4cc9d2050ef66625

c35a588155b28f61b67b17b26f2c781faa135e56a38825a1b0c468f59978f5ee

91deef0c9a3c66a6e380ceec55055b7b36c2183205fe8321f2542a71c251c4d7

40f8f2271e9db450777b10e18375c6ed83f17c2abfe29387fe8aa9e9daf4d96d

ec98530a90ff45b6798ee90ff05681342d32c34f7eccf53f040b2569b1b51d1c

758404789d51712d57d7bcd478db1f067eae5739a0aef7371a240c931fe8302

c3501ecb0398553508e03ec06e6aec62c0e7ea02e8a3384d2a2f3ec1767297c7

44312061833f51b52c9ce804f7ad78b34cf1ca22cef8e5c45c209ca587d5edf4

Fiat-Shamir context

c2602514b63d205478e884f7fd89467afa45c8226ee4c4cebf1dff5fd643c71d

withheld proof term: 494 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.companionK_cexB theorem, ProjectiveBundle:340

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c : A),
  Eq.{u_1 + 1} (FBundlePoint.companionK.{u_1} (FBundlePoint.cexB.{u_1} c))
    (Matrix.of.{u_1, 0, 0}
      (Matrix.vecCons.{u_1} (Matrix.vecCons.{u_1} 0 (Matrix.vecCons.{u_1} 1
        Matrix.vecEmpty.{u_1})))
      (Matrix.vecCons.{u_1}
        (Matrix.vecCons.{u_1} 1 (Matrix.vecCons.{u_1} (Neg.neg.{u_1} c)
          Matrix.vecEmpty.{u_1})))
      Matrix.vecEmpty.{u_1})))
```

SHA-256 of the statement above

134671dd9fdd632e9df1aa1afe7ebb3ef00d443def70d9c932f70b0e2cfce5a2

SHA-256 of the kernel type

116bd5c700d9967bac06096ff721b97f8db0cca3f01c73e63b84214b5b1bfb3e
commitment C

4d965661ba47912c446d512a5de0e114fac938eb98f70dd8b8b114ba250234d4
0c1b5d7f7e5a4717720e20767e270eed82395517c82fd58ffcd560aef7e77cbc
a4e0483b1a0b01d2e968d30d0dac64fdbfca7a09d03359f71959a8f7be594e6c
34530dd766a4c59aa268b7e848d0651f1e0a9410b7d63eb5563e97cdfc9c1535
6bc56942311063e85bb9eafc342b21379582df0a5caa0634d2a970a605cf3a42
2aad079103084d36753080e65182243b71d77567f0dda6ea7e1fd50c451089a8
4f0e237bda0530cc12911ea857de6f82bf18b0f3463ccca81d84849ba55664b3
b9903ff3f794a83e5e011faf37f6f0d825e18ac3159f462e78e5bfe80b95500c
first message t

1268757c556c49cdf4a7833fa92a9014e70fa59020f9872a05a0442abaf1e755
2bd6256012efadd0151f292ab4f738b1fedc022ab4fc4e6b281b899555d5b165
a05c506742f9e606e15e6f4945af46286b896ffa29a22dd54306aec169fcec0c
c60b7574a6166eb67b0852be82beb501b39b73904eec02eabb1498861ac84499
414dbc786d09f4434ea7bf7267babcf45615c0a3c7edb1c29fdbad61b1ebf5
fae54e3982c1fec5fa28296ade4a8d3c5f461d7a235d6937669d281b65b1ddfc
705dbec27e5e1d16700827d260a905b2d49b14083ae49eef5b488c2d6128c2
92e3bf8d1859c94492555229896d08a9740f412d18218454a25795f9ab3a77e0
response z

5f89bf3a5e275054e49630ceb7d1fde964d30bd2061e6f0ae0470aeb98a2bfde
94d99b2a4b309e84bfbe4e32692a23f0152cb51133d863270aa3b008b7540318
d9a107cd00632d1e7ba1b04384ddfe46b1a8119c2460cd3cdf409886af54b
fc8c437297908d9c76cc9cef73e70f38c77ca7acdda634d1248ec312919e24c6
fc880692aab08368be6bd3c10ac662db79ed4c9ba3c736419a4b122c21598acd
fecf54262eaae1e9bae1ec9ae5e0a90cb3d2106a11aed3ef6cc6f97d5dbe7476
5c7d563654456585a7e3438089928f313ab8b9f52fdb193cfc121a2b8b242130
c4928c5847f612bbdea8622e01e5103cf36415798dd6c4a1d1487381d7134c686
response w

36de9469e3c1a494f7f358b5c997eada1c1f2d944665ede5d350e1003ebfd6ee
59db1487f2f6e67c1ed8f130c498a444912601ea66b8c72190f3322112131666
88035b66518a7fe43768c1adfd1dfe3dac44edb8f3ed3ba9d5ccf8eb826ad4c9
bf43fb65c8907e72e09fce5a1b5db97c18c9542c59fb477d9ab48d6e17abe155
d4bd2f6dd5b550987f079d7d44fcc946ca894f6af88835c0f0079800dfb97a43
b29e4671c55eacd8be8ec3ec3bdb53ea2d32abca14ba41e107fabd24524dc7a8
d603bb7f7cb5db5a76d827956ba8191c4ce7dbed312cf79fd19e5f9fd4ae7fc1
6a79422f02c495228aba7d93c8dfdc059b5c485a0e82a82f50d5d73ac455c3c

Fiat-Shamir context

837eccc9933acc5069e6ae7bddf351339297e9b3da2cd73322a5a552526953ce

withheld proof term: 1 294 078 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionK_mul_cexP theorem, ProjectiveBundle:346

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
  Eq.{u_1 + 1} (HAdd.hAdd.{u_1, u_1, u_1} h h) 1 →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} c c) 0 →
  Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.companionK.{u_1} (FBundlePoint.cexB.{u_1}
      c))
      (FBundlePoint.cexP.{u_1} c h))
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cexP.{u_1} c h)
      (Matrix.diagonal.{u_1, 0} (FBundlePoint.cexLam.{u_1} c h)))
```

SHA-256 of the statement above

34ca7f740a6074f0047b9cc8317a718d4b9800d616ad1556e594d533b4c26f1d

SHA-256 of the kernel type

b267f6f23d6dc3b38d33c3f22c2bbae1ad9989e99cb1170b64fba74936d480f6

commitment C

e0cb2114faf5a1bd6589ebc8d147016e6bf4bf4190af7a6252a0f3d939bdc4bc
15bcf6e93e4e3047f52969c40675f1ce5990f6e90a52af4038ab6a8b3afac73f
b1e4af9c581fbd3a430e450c21bb0783e7d94ec6626b4b982dce13c4a9c2afc5
20f9c7e97bacbc75f546ad16daaef8f9442bfee28fdd667d7ee97360bd34e524
5cc679c0e9431cf70c5e3534b8830b182f4056500b6d806c9dbe12497ec5cea0
80b1f5a7011a5945bb59fa6c196b711b04adab618c96007193ad6d59bbc97d69
67bf8b0786cbb8504f291ee1561cf45b4f91eeac7258393faf3fb4ce2330c51b
f5cfe0f9b5c49912725acbcac0b7b0886ec3970523fd6d106fa838e26365fd
first message t

d440e5426f879abdc0d37087d6dfd200e40f19e217dda977e02cc8bb07fa3b2c
05e31b2997aad7c4ab25cfd6d6a7f27495d5531a3f124f593c61dc4104f730a
651fb72672e76495d4511260874bc1ea618ed3a902192a918205f1151b2b36c1
b879036c5024fbd9c8d26107e7e6fcbcca9b5d3e18f4dbc19f63a4b07c12bc37
eb037c41fef222237134012697c4e7ae14f2241a309747ff89bfb5bb8c264bb0
b1d895af47f1d85df0ed98052c0cd3b65830117a637f5ee249a42690195fa2fe
19fd200b4f4f7d690f7dbaac9b30dcaee2de5753b56caf246f73f2ff9c64d012
f42fc7d44ad1dd1b5bf29f3f25e2dda8af02673567a7f57a55c3f9b4fd08134a
response z

754467e4e4a846012be4402dd9c4d7e0779db83c09c4070c263fc07c6e05e213
a7c4a4502abfa72d2e6ec2a26e4141df43626448c31ddcbc125f3e3dee08e1dc
f291ca649b21ca2a24021456240da3fe87286ae2becd61387b713f696b6a5677
14bfcc1528f96683481252f04435cc1fd48ff97cbd4c4834c26fb80c63d6de55
f0d61ee8087b35cff66f2888561bd67d2339121d42ddf417c8af718fb7112684
5e9ed08c62f32a362f267340a9344561d52094573f1e896c2c0ad61e389a8998
22bd45eebf176ec4e1a62f0570d456b57a1bf07abd19b309bd52806a47e140ed
f956342f3a0372042b8f0d714e7f7d808aea2ff5a7cb18e9b8e807125fe966e3
response w

3755742330996b8a8b0d242c878cadfb1ae74c827c8c4116a4f422af60d37530
f13cff49715584ff3ac8f45353c8131babe1a059db31af463ca555b0352d1e79
cf6363e06c009dd65b379c53acfd142e9dad0a12253f0f658c28b8e055728cc5
fb177f5a070b636e05860591d1d20117d95092af6a6a1916f6bea5c969663167
ec007bd072d37bb6680e060dd7b66682efe60946351a12d308151be021d8fea7
ada6bff59ef75aa64ec8d7f9b85cde1b15af02dbcc9476b85e8c861dfc5172ae
60d3b82806ca45705a1430900df1d0a1ec171af0bb62ebf1ad22568f08abc385
9cb10f65901ba485bb6b7d3a3eb298deb18141ae8d04ad7a1d3876cc82670f3d
Fiat-Shamir context

53ce2eaceda267eed70cb0c5a5f808b440cf3dbbd73c51ee1c215d1952211ea2

withheld proof term: 4709546 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.companionK_zero theorem, ProjectiveBundle:73

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} A] \{m : \text{Nat}\},$
Eq. $\{u_1 + 1\}$ (FBundlePoint.companionK. $\{u_1\}$ fun x => 0) (FBundlePoint.cyclicShift. $\{u_1\}$ A
m)

SHA-256 of the statement above

0e476fdb59ecb8787663b0905820f44a8d69bf72382b515bd972125f30960f04

SHA-256 of the kernel type

42e61fbfbaf6547e529c856a8567b4fda2e0895a28bf7738646eef1a0a2b19fd
commitment C

ac6bfe0caf20ad5cb4a683d205efbddd24f6c0e84752bf96c3dfd9cbc70593f
b40e2fb21a189750eb7c9f41f9a4d846a52b5a098b4bcb3248c5fd17585845ae
016675eec21b12cb4f70b9f9d83805c86f100785233e4ca8205032aa3a59f396
bc50388efb3d75f6e139d4b22b8c12ff80c906f37cba04a760402643f17dc646
5be0ea9cb540c25af5ff174279c228879182cb624f1c33546b784b89618e6f8f
c062339c923fd07e4f7160e10228d8c95c35ec684ec80d72e2cf6789fca502ab
06746ec843be4370954c659d1da2704746da00a88d579e1b757a9fe44e6a4613
d6f5e53acf3bf07cf3b81ba22a6aea64e8ea8a7efb0452e77e8353d324fee550

first message t

c7c49e312a1b8462b72ca01408670527fd5c8914c161f7e15ae900270c6b1029
91d8f2b91000d5ff04d0605e24d9613d47b27366af3130220a2e8aaa8c122f39
a6425c4343a1922e1c763979b134a866130a1237cb2266aaeb71c4bd9590f267
e32580dcdf5db2ec3ea28bc3dba0a87d596b0330b1824776fec141561021e95f
db744ba8acd941ac662d7e8a119564a066b83b4c75cbe371111a87fa6474ff6f
ed4d4e993cbf6fbbbad59f3ea542a05dd3299e0cbca5134f9d9e80ca03aafce8
bf80aa488bd8c3c1c5829bb4f75dcd23ed20d586981f34d6e4750f2ec97ae8b6
36938470955fc9d62f9a8fc3c0a0d0a0ee7a0ebd6ae7d2b383333fb24a0c837

response z

1a01f5e9ecce88b0203a777a8e06594167e14481d0d3c536398270771501a655
2fdfa4509f94c434f8271186d68a41f3410764ae33f5ae0652992eff7c200d04
935c146d6d89f4c4bd35a5f58aad876540a35581414dd9bcf945adb42f384f59
6f101ffb5b48be32a9fab0e7b4eb3ced15c463233a3b74e8a2a589d069f03987
7b1c6c0b912a0467c9cfe9cb3e717e2404f81c0eb1274a046471b5071968194b
a8fa55c26ef5627709748304156b128411647a1947bd2a669cb2ba3692a5a478
864cf3d08913133b27a621d99a131fa2585abd632c1404f4c6b5b6469ca09bf8
333f41606bf05c8cd6eb5d5d87e4dcccbe9ad3346ee309387c1070643b4b47f

response w

6bac4d49c8a3989cf982b1520ec6547de3b72cc12e5008736c512f83716be2d6
97f8ecc3b866eb27aa315e7811199be87d102fc3191ac3ffe7351dc21198e98d
028d743465a25dcbe54efc9c3124edf866c8c36cd4fa2cb2ce9e4476ee2ff2a4
190c86c89b4132d0562d46b48a64963b39ea8c324cded9bf1d3dcca0c15c79bf
af482ed81acaff6a5218c396d5e548f5c664ff218df3da20ed8f87b5c60051ea
1281502f98d06334e57d9c17a3fe48a5eea678008e127dd571e5ac7f017e1a8c
0608bd6d7088e3ad61d0f9ada2e0c6d762dea505fc7e5aeaf54deaf0fba12c40
927872fa5d9505b29007193ddfe021cd939f27605a5906cfb1635b591d95a64c

Fiat-Shamir context

98f9296a3eb40f76e1cdd75f81a4c91099efa8b78b2e803228ec5d1fd579a080

withheld proof term: 54494 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.counterexample_diag_conj_diagonal theorem, ProjectiveBundle:380

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
Eq.{u_1 + 1} (HAdd.hAdd.{u_1, u_1, u_1} h h) 1 →
Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} c c) 0 →
And
(Eq.{u_1 + 1}
  (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cexQ.{u_1} c h) (FBundlePoint.cexP.{u_1}
    c h)) 1)
(And
  (Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cexP.{u_1} c h)
      (FBundlePoint.cexQ.{u_1} c h))
    1)
  (And
    (Eq.{u_1 + 1}
      (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.companionK.{u_1}
        (FBundlePoint.cexB.{u_1} c))
        (FBundlePoint.cexP.{u_1} c h))
      (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cexP.{u_1} c h)
        (Matrix.diagonal.{u_1, 0} (FBundlePoint.cexLam.{u_1} c h))))
    (And (∀ (k : Fin 2), IsNilpotent.{u_1} (FBundlePoint.cexB.{u_1} c k))
      (And (Eq.{u_1 + 1} (∑ j, FBundlePoint.cexMu.{u_1} j) 0)
        (Eq.{u_1 + 1}
          (HMul.hMul.{u_1, u_1, u_1}
            (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cexQ.{u_1} c h)
              (Matrix.diagonal.{u_1, 0} FBundlePoint.cexMu.{u_1}))
            (FBundlePoint.cexP.{u_1} c h) 0 0)
            (HMul.hMul.{u_1, u_1, u_1} h c))))))
```

SHA-256 of the statement above

d861c1191c3e73efb6c2fe6db44061434da0dc2b951762155c37100420ed43a7

SHA-256 of the kernel type

3e095179bdbb8dc03ae8311b47165a48405eeba7849cac13a71af0ae873a4707f
commitment C

6efc9952e61b76103317944cc9e000e397185e73994569aad81ee854dac1c4b6
9901dcadf7ff3ad6a884470eaf0f9bd98d41ccc651608663d352316ca7bead9a
358bd72241c7d2ff78c89587adba9ea13967a4b5f6216bbeef736860fe2dad27
55b6be19d252f75de8b1360014031c5b99d7470058d0d2db11b3e0406374edce
484b8439f8bd8fda095046dabf4b04b8a11658d2d2949bd8f46976ed0a352f11
00fccecb2b8e257b4ce9cefff4b0802bbe9a6f3b33ef539a30f5999d3bf1744b
83cf8ff57084a70c0d7dfaad28882f0e85bceedfaa1def665649e6f1cd2212d3
0521008603bcfd9330dec62435938f364e814ea85edc21b83547609f1428431
first message t

766b9581dbe97454cf5019361ba83d9d7aad51dcaa6c8e63514fe3dad5021c79
c2b1c97661eb64f6cb3deef9a38d4bf740c234b7ab4fdc610b3e26d5c1b60e5f
d00b1c031c07f1764e9e52be99c27d9eb6d71334aa505b88a4147affaa702a04
731e9653193921a533d1f7a118f60d0f8eb1f39a7c5127bf92b315a7407fa2d5
8c030bb6d5ec82f2635723b0026d176352dc09a78dc46df6c3e40f7208e2a984
236b40fe739c82a44c6088e955629331f25540cc42c586e94382e54a18ee29ac
ad0544c770af138c6e4b3cb591a8aae3e30e5c2d22215a2e20110131d17a8e6a
9ff06ca5d820f4613c4dc862374c922255aa5b1dd34cf16bcc8e6d23e7b4cf5
response z

12f1577e8b50200d58988b4306d07e4bc90bb80ffe77e4392d42ec83bb06f1bd
1569e6790dd30c6e83eabe449e47bb022742860fd4d7747e4d4aa7da4ea3076f
fa8ad87c56f4579099f5af012d67ced95918e2b4025ec4f6e2805a8fc5e0e08a
8b364aea2f25942a3376c3d5a4ffd657bc52fa88c4f0bf342ff2d407f3f66803
a0d5b0d38f7b23fcbcac23c001386c03afa01dbecdb3df39405ac22e5e3a263e
9f481365d2f148ce47537cee4868b1ed75544549197f95eeac3ea59105f6b16b
1cc03e8520218446d02eb8f56efd667c6cd3063f6d6db985fe63615dfe6fdf38
9bcd421cf4c49a14b0c215c7fc2705fa5b36bd20cb45b6f05e229b734e9bc159
response w

33147adc9828506e0c027a8a782520f99b512297750946d974853b1bca4a1733
b108dc6df6690423b37250286a398490af12f7dbcfb2eeea813de8d15062aa0457
1e3066be4094e05d7e379232773123b6a49abd9439d4e8014215a18c73b2e0f0
3681214bc50948ce523b05ed5a6afa7d8161dbdeb59a1f1339e3a8529bda3968

```

∀ {A : Type u_1} [inst : CommRing.{u_1} A] (c h : A),
Eq.{u_1 + 1} (HAdd.hAdd.{u_1, u_1, u_1} h h) 1 →
Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} c c) 0 →
  ∀ (P Q : Matrix.{0, 0, u_1} (Fin 2) (Fin 2) A),
Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
Eq.{u_1 + 1}
  (HMul.hMul.{u_1, u_1, u_1}
    (FBundlePoint.companionK.{u_1} (FBundlePoint.cexB.{u_1} c)) P
    (HMul.hMul.{u_1, u_1, u_1} P
      (Matrix.diagonal.{u_1, 0} (FBundlePoint.cexLam.{u_1} c h)))) →
Eq.{u_1 + 1}
  (HMul.hMul.{u_1, u_1, u_1}
    (HMul.hMul.{u_1, u_1, u_1} Q (Matrix.diagonal.{u_1, 0}
      FBundlePoint.cexMu.{u_1})))
  P 0 0)
(HMul.hMul.{u_1, u_1, u_1} h c)

```

SHA-256 of the statement above

64786fac10d3459913378436ca9f0424b311307008577cce2b1ec08d06f7e702

SHA-256 of the kernel type

39b7c86fe4481e575f45ac7110c2235ca3e3a55b0841f1444c4417da92963642

commitment C

99ad5939ac84cab8b84573cea451a71b3fbfee8f65800605d4e7f39756283d66
881db97916240a2048a6e3e8c2db9a0b8d9b635c6bc1275bc3c97770e1ebc33b
aed6c1e65a7016b33ed3fc910a801a95e685c30cace22d121eee305a3f8a41dc
7bbb1b6ea168a874aa4734af93b422c7c604346d6b6e34dcefd3ceb39f4a7085
70673dc0770a32ce18d07c1a2653db5f712b335585ddde1de7562600442699cc
c96b48d98877bbe7cc8792a020d2ad3879b12e9fef28a34d81ab51d43bfb846c
bd0adb75f193007f51648162ce61abe69854efa1a80d76b7fb3ba0f562351ef2
72098addcb53ab53b4251cb08cdb59b8f09abdfac3cd11e7c533b23542ddb1f0

first message t

aed5b2540ba9f27d983e8dd0bee8f1a7acd1da1409607fef0d424f75fb9b3be2
494c522f7e825da0e0bf56a075c2632eaae5d7a30f13e81e8f4eab9d22d6ac61
a5a2502b542b2b6ce3844694f7343a0347efc134d4c8f09621512eeda15ff822
73313a3d643bcc30fc0ad25a57c95bfbebf4ae2309d534a462e20e76ff88f3
1494ba5184f93bb8f1225ac4d2b1c1d232f7defb25dbaaa8eedcec8a67e91edf
3445296d1d9096a9046b41c16bcef64c29f5fd849ebc89dc311c435f06d332cf
dee5d314369c890ac0cac6ffb83f5e38a42fe6e7904adb6f5a141f18aa0b7f75
b04424c8db57418a9c1a563f8943ee1d9554d155c146dacf9e99e9dce5f29006

response z

1001978bb58980e0a4f6caf4fb327f6a950ac114d790af321a714b136a193943
a419762688d886916ce30e85c5ae421259561cfaac5c3d2048a29a8b583f951d
14d740c339e58d3acaec6b43ad9d13267a32f429c5d3fdc05d0c858f0e80b3c8
f195fa228da19234e09c4759bf5522e788b7a769b3a526162bcb9c72a6dd8c58
210b6549091952b20ad9642b6aa6909552b4e8b85f3d79a1600726738786b476
33404bd5a12ed881b298afb5da3ceb9cc77d1fb622dbb5cd35f74c0a76b899a1
044e809c0a70a6d43d9c3e042f32eb733ca77fddaa41490afa823ead0928d07d
1ac361be447b92bdb911a403c5b76dfb8e3f2706e2313b2407d181dc5f396a80

response w

2d1ac6dd44455a8da54517bd6a08b5fb28d874121be84b381918f4e3d04abf69
e29c5d4f653aa8bbb1cb069c70941239e9a81694cf9f490d0356ac3752413724
9b29778e19956d36cd323fd78fd52a1a79479f062ddc3f772f793af11d42c5dc
d737f7f7beab7fe8cb560a18e1cda76f8bd29c5f7af47b82f4c6b14ee52611f11
b44238216cf61d42dd9f865ae334290a631b70d3bee3a7911081a586a6ef189a
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87688b5d68e02fc624db3e1c4f2783bbfa313f90d05287e169de1ba2cd1fdd55
432ee08fb544babf2c165143b71f06c06b7565aa5a78aa1bfa381c0ed27cc99a

Fiat-Shamir context

44bab654291c627fb08dad4d51c54b7e98b43442aa82eb926f482c3ac8020198

withheld proof term: 49 376 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cyclicShift definition, ProjectiveBundle:57

```
(A : Type u_1) →  
  [CommRing.{u_1} A] →  
    (m : Nat) → Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0,  
      0} m 1)) A
```

SHA-256 of the statement above

1aa8182b6a9714c5c03a53442960d94e751a3c0d6d62be1bc922d52d23f2c234

SHA-256 of the kernel type

8b7cd0fa1972b6516ecd18a0fe8bd02e3e1d71d9b98640b85d1c5c06bd9e5ddd
commitment C

451d8ee9cbc4b29ea57300872da13ecc3dbaa6dc6b2abc20244e12900ec09be1
751b671d365e8d54950867499b3b4a1c63303ab8663e6f921500f3ec3afb143f
82d7d26c4716ddab24a5f00278aa9e0cd66175bb8ab3949bc6975d080e200fb
a5e91332aeec31fcab5fa7cd05e62f2d64449848f7cb6c5f14c92f1de277847d
b1a3460a7d02c677ffbd0ded19fa72ec05de3dd70ba6d7aeec5337a2fd265aa7
edb9fa236032697db7a6f03b68dbf24588af50e11b8e6cdf33eb9bd5fcabbf0c
fe139b7cf773e902266d9dd0e3e6ef9d63e09caa05332615c1ebff17dd7180a0
237f6d3b88c8ac514af1374e1bc3150d908cde3c92772423d801a755d8794be8
first message t

a09a7819cbe65c3e94daecf9ba99925926f5041bec27f221546404295571cb7a
32ce221a7e27ebd89d7efe92145240f4eabf69a06c5f4463d5fa2a22f9e9993f
ae42af5f1cd4a3791895d887c51814b9f847b58fc19a2524461aafa904ef36fc
85b5cfe83642cbfea291d64113fda611c1f818d4ba95f4b92acfc22a5dbf6a9b
dde7a78ceeb30a7ce08825235f27b86013b862d6551906c31fca36d8b235a703
30acb5cb7acc83523c03d40ddac6763faba3c690a2414c201986aa9e6939bbb3
301b1e4dc8d0fcd8be23a33e20e056453f99298ea40ad9f7ed6e7d3406e02af9
d23b3fdfb0dfad056fd8dbdc32a61ffb28f5a02234555e58e237a88f3e11a4d5
response z

7ad16cdbc3d90c0b623404ec61bee2f379797ef4f7b32443ee8f8a622c6dd5f8
d75867270959a2374972f178f626c9ef8d071f9535d293af9abeb78949aec908
2523045453c94a0c92a0fbb8ae390d7a576ee0c2c8154b5ff0dc1442e2fb480e
3315ab79eaae0bea5408c3d522d0638d516cb15d87b3773aa31af5db8170ea92
511bc58d516f1ef7d2b95f9a48ae6daef361215cb5577a64a77252d6b61762c9
515ecadc9c340e9a3c9fe263a486b23b32800a0568fad3dc5304d22fdd69ba70
028e941bcd8db8cfbcbb044264b19f16d3b90527d603500eb1b2a47b594df8ab1
67a2fed09b68e0d1a18a5eb50aa67ca0260f5bc9839dab8585961d926f7a4fe5
response w

162e39d3c816901e302423947568d7ee6f84165606079f8aed99450354df9e37
fc7d1334bd35e9d5af77be1f40731484f8604db2d84351721d9865c1c06dac53
0bf301afb13cc4bbc3471bb203a0d9b98a4bf5ee29f93e913b9e29b0e15d3f69
2371f1a2e14f03310f8a3fd38e6f870b6f89985249db17952b729e8ecd2a1fd2
e3e6b506ef8dfdfc1856546afcf181f1a66fb9b7243e345b8d05f7886ff21a05
4676a3fb2c98ec513f274c6c68a5e1ef88ac00adbb69a2e968a4ae435d361e0b
1d39e18beabdd547506357038636891c100f969ec93dd25ac6aa44c7baa2b00e
cdbaea7caee6e5b8e69bbc79c033984063a6a05fa2e70c7b88f70b85e0333959

Fiat-Shamir context

44ff19573d59583fb77c2615c34e3fb0666950da3c534ec399a879182c5a77ae

withheld proof term: 6385 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.cyclicShift.eq_1 theorem, ProjectiveBundle:0

```
∀ (A : Type u_1) [inst : CommRing.{u_1} A] (m : Nat),
  Eq.{u_1 + 1} (FBundlePoint.cyclicShift.{u_1} A m)
    (Matrix.of.{u_1, 0, 0} fun k j => ite.{u_1 + 1} (Eq.{1} k (HAdd.hAdd.{0, 0, 0} j 1)) 1
      0)
```

SHA-256 of the statement above

8cacd969dfbe5940ca3fa9eb97c903ec6da16ce45ac113fcea7eace7f5f5289b

SHA-256 of the kernel type

8a6d70259fb11836d66c6ee8293fd4e19c74284c189652194db897545753c700
commitment C

3a9f5fe9f979b29fc8e9fc31672645873eaa39d3c5b1f69e4d28ff3b2037f884
766515434407321a4e785187bde7551fdacc568ba4be8fb443201059d098100
94fd8c6d3a0090a7e11ca5cc0acb391e3841fde2fdeff0eab7f2dc7ee59f858e
5135ad06d5ad531bad997db6ade1ddad14d10055c26814e4e1dd8d20ab3831b0
c51cdfdf6ec76e09af55752d67c5a5ff46976aee9c5e7360def25b0316d419a3
80c8d5192145a66e17887febf8474411c3e3051b49f42fce9efedbddb82197a8
41c222582bbbe3600a590c0d755fe35b4267b5af3fc5346659cdcc8f4441f108
7666a057b2d835e84dbdd432074b1df66f448a6abda0f773ca5e588bfde36120
first message t

84bce4ac92d9684af76dc17d84dc8d1a8fee82fe1b0949c03311a717143b683b
8346f08563adc94ab7d12d57546a56916e627e91e8f1daeff3809692cc74e400
e90bc3bae83e0025f6287c9e3efd75b11211681cbfd1f1ff33b6efce362a3dda
d4a6b88e8827676602ee6313b95511d1e32aa2b77012e0cc8e1c9143513f6dc5
5e7aa3508d4775a0f5cf40c81c3c4d6a848f26c555c4e014227c352fdb03490e
d5c3df3b6445ab161abdf32557ecb12153ccb71bbae96c23133f5325ecedb390
34aeb0fb1878a91b51929bb4e4f2f35f3aefcb7e2f231d0b1ca7a366612bbc29
62cf683be464fb4c6113af87333b6937c2a5c7ba9f9599b08fc584137941676b
response z

66b330602d22afaf4f14dd0ed8f84451e22bc2f294967d3062190cfbb1a7ab5a
115557ff79768634e282eb7bfd60f83928aa6357a9e8d5410cf813d771ad2724
e6e79fa09a39865b7616ff1dbd943d98d4828d80f8f9fe0d852bf873ff58eac2
020fd8a1b6228691c699554896502a80245ea95fc2698861bbbb7596789b2c82
a8b3579c8617d9063b3da980e1f836bf2383c391f8f582fcd0d680959417f7d
fb969d2fb7164f2b9f91887d9a6a09a3d29f6ce1c689db9aacac83d1ee6b78c4
3f3a76088572cb45e168d4472354d792f71aa0063006140c505c99f403837565
4a8e1217669f058b2a9fe3f9880b67121e57cd3dc52f7c4d2e77578a7eb8db62
response w

671f15447e333e3b54adbd4c253bc20bcbbce9d3067e8ff56f56bba0923a7e
f25c989344f9a662f294b6db2149a617e12c1906fbb005da8b1b0d1079c0de11
d8ca3c2354eedec0978030b7fc5479466468c0b74453f1e5a97d647a17eb6891
9a771b874d6c5a916344a0ee8645cbdebb4b190ef03f511234ca366076d6868b
90b7a8ee219c61b00d02615d07203df20d6dcf7608d5c5f3860daa6788c811d3
99a1a273725219745ac872d588bc97a0258c0f34c97a3b4b11036e2589474b27
f04b806c0f949ed291157171798dd02a1b385b1eee9b45b7639a3c1816217b13
73e62ff38037f1e1491991a4134039dac146d7c7a67f869b94a0afa71249383d

Fiat-Shamir context

fd0215ff629129dbddf18e34a205f1193fa56514921ce0a2cacafaa82cf4f1d0

withheld proof term: 369 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.cyclicShift_apply theorem, ProjectiveBundle:85

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} A] \{m : \text{Nat}\} (k \ j : \text{Fin } (\text{HAdd.hAdd}\{0, 0, 0\} m \ 1)),$
Eq. $\{u_1 + 1\}$ (FBundlePoint.cyclicShift. $\{u_1\}$ A m k j)
(ite. $\{u_1 + 1\}$ (Eq. $\{1\}$ k (HAdd.hAdd. $\{0, 0, 0\}$ j 1)) 1 0)

SHA-256 of the statement above

f8b75164d6bc20067916b1cc8588884d24647dfa62bbf65771b03a33b1367cd3

SHA-256 of the kernel type

4b065bc4d04d3db19dce286d2d5b747c42c04e9c3170ef56a82b23e2804e9e7c

commitment C

c3253980409ce6043f4beb6e8d60f1efe4e4b584acec05441c9939a1a3477935
de51b75084357883636f8a0411a288b18d5c9dd6997039e13d3d94e53c38acfc
c67177cd380f218fb9d6e3c1600554b298d91914742f5ad1cc471a79f07184fc
b9dd9a88663b5dc49b605013ec14e66af2d3849b074a930dc47366c28e8934d4
299518140cfff11594da021b8c3cea76456461562297cce4b18b6adb1fea5a39f
24a4d0b934d8d67c6ef86321e30231b6096ed35a746670c890790dccf6de8b9b
2052c2b1e1b90b2add9a06f2fe9f3b734b7ab2a32af0ab5dbfbf5921cdab37b
f1ad8d2819da7c0056fa778bde64bcba97f124531dfc6a074606d17779176d6

first message t

3b6737d50fbc6d3061f47bc988e09a50d8cd682e715875d7fcec777342296284
8ded075a7650335234f459b6617e568a22556b89c30d0d2c89246d2b217e14bc
1d0bf08e7ed3e316d2accacae5661359fb1e7c975034165b6fb7f0728b1a6c724b
df7caddcd8e8c8495c0aa259bc79ada251fadd875a4b07c5aee1f2ea114499e5
c41d831bf072f2dccc6728c7e0e332741e323962c44a423adddff813754a852627
3cb55dee16f29888716d755d77dba12e0f456c611437b339139f1455bd0dc335
97ed39d8e16b6936bcf65184eddc9a5a840aca91bb5ecf7efa595f5874e0b731
aff94ccc6102819fab719ac3f36ee5c1cfa3b084a9bd56b8f45840a70e3d4137

response z

7331565add3961c27c7c1bf1912159172ee72b1675dd99775f5dff67ef113405
216bcbcbdc4d1d4b87e193e87606d1addc3fb5d7732327294d4b95f675c46d79
b3738f32547234f0ff66ba39e265b6cfb221e2bc8e3f8602ea9f17392ac78774
b59ae81d0af435899ef3695c54ec993987a9db54980e55064b8fdb668be911d2
6dc2159a10050ed36992d817f9e2818b5f75a9fc34207edcb9708c08527218ec
55d43c95fc92514702a50b5b8b08c0f9ab3c7ea5d0b378b74d52a600de1c3822
24520a583173b673fd9725c0cf5e4163c71e10ec0d6a5fa7206d2b53756207e5
ade3bf1d5ed779608cbfe410e1507631a735ae37edc5945b7ca65bce868fdcca

response w

3c89daf26cb2a4b3dda60c3f95d0b2b5a2f9127243f0014639b3b6a8f5e34843
611c5c2accdd6c52bb8f8e28d4f6b78e681cabf7c8b39bebef7116358e4995b0a
b3c380f4ea280de93d4ac61c3a372c9c3ec6902cb79ed22e615b755b5f5415c1
e3f9eed05799d4a3d63d1183dbe01bccaaad619e67b9eb85bb0d486ad842bc4
fc7eeb4536e5ef5576dd4d798037a064bb9bf524fa1093f071007982ff1869e0
b47773136d358addcca9a98892e27a9ff74c03f1a70ddc99edf0d0f2cf221942
3f52d177cbc45c2e8e36a0a8a4507ca91ca55991add96dd1973299166a2dfd31
56186f4e265d2f38a0ca2dc1c06ec0d32ecf665adeef2654756b773b0c1143b7

Fiat-Shamir context

ddf0070aaf288bdb5e76afda47aa815dd08c1f587f8aeb72286c42d150a3b0bc

withheld proof term: 478 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.cyclicShift_mul_apply theorem, ProjectiveBundle:89

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat}
  (P : Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m 1)) A)
  (k i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cyclicShift.{u_1} A m) P k i)
  (P (HSub.hSub.{0, 0, 0} k 1) i)
```

SHA-256 of the statement above

cd66c43db6e8d36210269dc4893edc7dd03db36d4c6eccc17de9323a44f601aa

SHA-256 of the kernel type

964bf3c03b7a4448962cce9f407a744b0cf1175c0ca3f0d703eed49b4f6f4291

commitment C

4ebbb49edb55db4f474d315eb9d8a6c398765e48831b399f46c28c8fe4f3184e
acda4b2e957508238b09ce332d40755cfd7fe1778fe32221a7cfff6b516adba7d
e7b89790310ea6ca7201f6567fb8a670dc6afa1df183283cdfa6a16d1335700d
2227b692bdd2329458d9ec47cd11d5c119eec1cf2368271f5149efbadfe3c25c
8d49fa3699ac32eb87523eba4e65826a75fe9a222a71c19fde99690c4ac50938
3430ab8d75e9464987d0ac085333d15d44c519cde00ad3eb46ca334aed140925
54dbdbcb6e686829703c8c2484c0c54f79a7d93604425dad77a72977fea0cd7bf
083ca21a94b791241a1566431bdb9810dc31c9b3c4db945db64c3e3425570a8e

first message t

d018d26fb9f3239ec317bc12220afdaefdae2fb3d8f54da46514ef5ff0a629aa
74996fed3c4806b96159c682a07b45346bf37a854c8ae1386efbc59367ba1a15
5030884a16850ed8d3082d08077ddff7872b57e16f442c78c9383f837022233d
0696c3b67fd2c149129fc40e6d885517fce92f0d7f67ab3558d060e851581bcc
2bed176de762ff70fbd0ea02d2c6cfc9212a12805ffb6384c1145fd49ed99d10
c82709cbaebb73f4a6fcf60f113f3f93c262ad49326e44a063c9ff339962e1e
b61960687f8adffb24dfc947ab6a30a9fe6fc27b33a9417e6c38ba303e895621
f612e4b794aa6032c5a32da130647bc2963593f8547280ac490698092c58d9be

response z

5a99af3d8a84ef2ce5bedf98c0accb85260353127b980ee79135770348d24f74
2961c8a3f85cfe19647d0a0901eecdce8742f898c13c8f0f17ee7e9aa583b13b
a3f323cd6a533199032848f92f4623ab6f858f9f123a7a678c2345207d935268
d6ad4c51cb465d210148e245544956f01d3c4821e9e0d183de0011f491d287d4
1091ddd33e99a897a8fa903cc71eb87a9f3644f4be6a5bc6da793156c83bcc50
b8631217a2eec2138123578c78a813283b9b2c2c5934dff294e8ad17be548c16
c96fd03507e4b9ff0a0e1f18b6097a399dc9a48be563124884952755f30b3e37
a424234a5ed20bf7f4d11bc6f74f8511dfa90661bf6e4103d89c913fc0c1b057

response w

2f70eb588b65248d5da902cb393d7aeb3174f84022b5a2478425e9e05c8cac51
13fb9c7cbfd03ecadde042c5c1d5f66bf35d30decc527054ad977ba99c70fcb4
a5c617297009f2eac357d04ca7ff723dc1885aa87491763772ed381993b3a9c0
fbb7d92c49c4d33524c79cf722a4ba03a0f87216be340b75a79d42f43842e397
43dcdbead9d7665b0574688c9d1c1fb452873bebd345affb4914fb0b6da232f2
114c9963c956e2204907d3db7d0d5e8422d2c435461ea91e2727ceb565009ec
2344fbb503ac4b00a1d6f7ec3f5e74b024c1a19e9e7bec8d7541911b709adc39
902707d743e8159918a72939784e2e5a1269b2b2d8d6c16626f726e1ba86b5f

Fiat-Shamir context

3f4f0d12761780c7c7bd62b0f551f7d30034e7e0f3a04b6d9ced8ba7e28e0c48

withheld proof term: 121 749 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.diag_conj_diagonal_eq_zero_of_cyclicShift theorem, ProjectiveBundle:108

```

∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat}
  (P Q : Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m 1))
    A),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
  ∀ (lam : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cyclicShift.{u_1} A m) P)
    (HMul.hMul.{u_1, u_1, u_1} P (Matrix.diagonal.{u_1, 0} lam)) →
  ∀ (mu : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  Eq.{u_1 + 1} (∑ j, mu j) 0 →
  ∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
  Eq.{u_1 + 1}
    (HMul.hMul.{u_1, u_1, u_1}
      (HMul.hMul.{u_1, u_1, u_1} Q (Matrix.diagonal.{u_1, 0} mu)) P i i)
    0

```

SHA-256 of the statement above

feba15703a2a58476b97e78d45f1a1509accebc2cee3692e83d259a86e3c32f1

SHA-256 of the kernel type

9f2823deea101fff9f53cdba6b1938eeb7b9b64dd0c9de4c58cfac4525c3a4c
commitment C

a2f4ac53921c5d06b7e4b175c748bc852bf3e138cfe2999715e9783516c7ddb9
4bfa52047a56c12261e9ab49a92dbbdcdf4bb12f5e7a876076015632a6cab7
3a3faa3b30b1701e34425ccd0fd1cf4da9177d682c917d71a6b3918377915d09
7074df39e431e8e1b61f96bae847eab11f3c06f1fba08f6fa56bfc3040412be
1b426756e50f541c11ed3b104989e0110ff7d55dd3e28e61a18de9a9b63d9eb0
84a7028ad2c91b4f2417d8042bd93846c7be11b6501eb0ce1c606f223f3f821c
91a0bf490f6ccade2a952d413eb40b0f71511da6d0a17eafc4c5540bc4af1be3
7528c89d8e82f541d04e4689ea7a07dfe22df7add7da62ab7881fede2c83faa8
first message t

f72287729dbde79e14f568ee3c3959f4cd4a157f22f16aa05ccab1864d561ceb
98d8ddcb9b2c73e74bcb043e9ba303badef52e7fe0a2f649e2dbe094835304b6
059d5a07999bcbff06499c70302f7eb9fed0111ec88eec553a0ab2094fa330075
3e8b2de83ef8ab0a5c8308f41131555eead66eab78c3e5c605a736205dc8671
d293b5b242dfc435938513d590321e1221d6e34063389042a414cb1d1075faa7
39492ee70516531a412c4d61532fa0dd7d328ba7533a132cf51c47b95cd7b843
7065da23012b12190e88a343cf77797a1d626cb5664c7781667e497113b11534
a01abcb782c7fb2de2c3477a879f4872c8359435c825df9fd92253ee751c8bad
response z

439531f410adb2f5ee086c9a13f8a0b7e088e42f996a652537a18fa6e359b902
64779c7fe54534925bd7ab03ade607e0c251b4d18578ba0559b6582e1d22dd43
50c0b282b96e78cb8c642817a3d600dd7ec3012de7fa248b88e6613e9f496af8
c8c259c947857a69260311cf07accd29ffc065d60f8eb1415eabc377b391d8da
bab3f2367e913347207a8fed97f3b47f10baf89b74b33a5f6dc39fe119b417cb
6d371c0a5137a8f665d206b672129560d20bb5e3f39cfe1c6e32f13a6594138e
aa912d1bc75538cb8afe618c2b63f2e00ce0644ab172a24631bf363c8f2bcc19
2ea68648fc6362c32631fd5fd87ddfcc749e31217050ba59651da6463755aab3
response w

1202691a070c71db825bfc34966950fc0e8fba0b59bbb061a32d1f6060d7416f
cb01463b8cbdefc904a284592345820e18441304d4ed1d96b8c1289a1ab3bce4
36f80e0db6365aaf5afdb97761f70b9a9e480592ca21fc79eab7c0e1b23c2ce
6e4b0ffc4054d4c13a73f450fd84d799e0dd84004ed4e76ddc61fe5826a64a97
2764cade393e299fe35855c60c82f627922f3ed8a2e8f6252eebb76b23f41c5b
6556a38f89873510f8abede4d59712327581c57a8f2e96c28bfdda6b2af9d6bb
e50877bfeac2ed8db143e287ef3c05183fbca490626328cce45b7d84317ae504
173606673046ef5ac6387b49eaf745b6adfee62d0f73d7cf1526bb820eb4ac09
Fiat-Shamir context

fe7a465453e6ef63c78d4a81256f6d1c113b66fec87f0ef65640dc55b94020f0

withheld proof term: 1 420 387 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.diag_conj_diagonal_mem_of_companionK theorem, ProjectiveBundle:193

```

∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (I : Ideal.{u_1} A)
  (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  (∀ (k : Fin (HAdd.hAdd.{0, 0, 0} m 1)), Membership.mem.{u_1, u_1} I (b k)) →
  ∀ (P Q : Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m
    1)) A),
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
    ∀ (lam : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
      Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.companionK.{u_1} b) P)
        (HMul.hMul.{u_1, u_1, u_1} P (Matrix.diagonal.{u_1, 0} lam)) →
      ∀ (mu : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
        Membership.mem.{u_1, u_1} I (∑ j, mu j) →
        ∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
          Membership.mem.{u_1, u_1} I
            (HMul.hMul.{u_1, u_1, u_1}
              (HMul.hMul.{u_1, u_1, u_1} Q (Matrix.diagonal.{u_1, 0} mu)) P i i)

```

SHA-256 of the statement above

5e503ef060ceff1722eae4555b35c6a873e4fff46d94773c80bf2cef3affb72b

SHA-256 of the kernel type

8d36f37b8cd9cc360c3a640e30b7d24970bc5d0a2958670967af9d5ec94ef92c
commitment C

da62c915b2d27e9189b8165e823f8a3b06fb9b4e06a2871f318cddc7caa72dc3
93f941f0e72b9329c6e4175d7a9410f8a75947eaa75a0d50574800eba03ea2e7
0e68c0d276fb27b742122508caed982dd0a7e462c61c94137d618dcf3b5b7916
a259dc52d13e2b522cff56a4211205be41a3ce372e0415a22462d40e9dc8dd8c
9bdfc220a1abad2b370596a2f0051c9da6a3b0fdf8c69cc04352a685cc75ba49
046c0ad8a98c02f62d64d04e7932969b10f901819b2922a766ba768ec2a7388a
82803e612f9dd18a8992aef36bd75cb79cac992a50d9812f5c52321170bb1944
fceb9e215db6cc967494f47cb7c15f54b657dfd875bc0bbcc255c209df825b00
first message t

b012c19cfa8999160fa31b777a87881cf895f7dcdbf0d4c4d6f6e8e8ac274179
377744d44331799d25f62e931d765b8eb0f8a19eded7e23e21df6dbb9cbfe4f4
6f544d73d3ccc256f7dd3e67e060de4d5432fc6f943f9f56f7c0513774629707
bd8b034fcfb79ad94bde0954938fc51b72a3567a76af4732acea5db4ef0079d9b
f12ef365bec23f51a1217e62503ccfd718a8e4d44f38fa472c6bab0aa5afdf1e
0d8e603f539b568320fd1ab32450aa94d580e8fc687d08e0e0dd43bbc87f6ea9
726fca5ad83fe3130aa652d33cd232e532a95e8a10a6be8c6934e806d5e6b0e7
653cbc67a59900d15bf6746605533e72915c3537a1ba4835d3231974d7138e15
response z

780478aa23c3d9ee40deac3f03220df647b30e2029fdb3da61a5fc44900c0df7
6b322eec1f870fb9c4e9220b59de05468b854868ca945e01b404e1cc6efffb78
3d2ebd8e46c06817bebecf7b786861be0a698246f63c553340334f26c9923809
06f8a258b45f3e091a24aad338c0aee600a415553b347ab52890e656f7e478ac
7f7fc6b1f78becacf323604b48d7f8ec49b59a55205570a067f87e8f85ff8a6
9c7f4295d79bf1a3658f6c936dc9ba7c2cb50c58bc6e4fee389f89f8883c84b2
bb7c870d73d4615b1cbea558a2944bfb820fcfdbfc356b9c79f032a8fb99d56d
31018948962139a1706f117324f5081358d02df08ada6320602ce93f327ec405
response w

7e997c897b222246eb75cca354e6adc272c540c5bff1004fbc1bde57976cd6af
0a19c0e527e7d4f9d7ced325d13011182459be952da91905cfac0c2cf10354c5
4b615c0ecec13af887ff0ddf3dbf44c93e0c76e0a92815b981014dd1f79cacd9
462d5099826b77be9f116911522bd7c0db160eb90af4844b49cc3cbf613158fc
5ada3b5e8ca6beb6995820b11b499beda9ea96e34abd7bc6bd373ccd6efc6f42
c38b8c8092cf1a372393e525608e6de3b5563fb8dfc442a32f2d14b9f51a34d3
766557857a14945dd8fcd05eb06a7746ce80f5d7ede8d3118ba78ca4e59c309
e70df60e9c5f1c982e1674e830e45dac9fb19fd188f96c3cd20dfd80bc3560b5
Fiat-Shamir context

96fbc01d4cb7e549e36a64b81310aa57ed8d1a292373adc6b0901f3912e7adb0

withheld proof term: 2821 122 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.diag_conj_eq_of_diagonalizes theorem, ProjectiveBundle:257

```

∀ {A : Type u_1} [inst : CommRing.{u_1} A] {n : Type u_2} [inst_1 : Fintype.{u_2} n]
  [inst_2 : DecidableEq.{u_2 + 1} n] (K H : Matrix.{u_2, u_2, u_1} n n A) (lam : n → A),
  (∀ (i j : n), Ne.{u_2 + 1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (lam i) (lam
    j))) →
  ∀ (P Q P' Q' : Matrix.{u_2, u_2, u_1} n n A),
    Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} Q P)
      1 →
    Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} P
      Q) 1 →
    Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      Q' P') 1 →
    Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      P' Q')
      1 →
    Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1
      u_2} K P)
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} P
        (Matrix.diagonal.{u_1, u_2} lam)) →
    Eq.{max (u_1 + 1) (u_2 + 1)}
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} K P')
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} P'
        (Matrix.diagonal.{u_1, u_2} lam)) →
  ∀ (i : n),
    Eq.{u_1 + 1}
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} Q' H) P' i i)
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} Q H) P i i)

```

SHA-256 of the statement above

399a017ebb0b53d9d5a88d916aa47b8e2711e74b4de83d8fea83dfbe226019a8

SHA-256 of the kernel type

bde3b35a39773f668bd4074df98908f86eb3ce1060d844de046c42608db98572

commitment C

c9b07e73a5528ac60fb937661fd6c542279b2e0bb1e1c9ca84d593aacfe78806

6443191242f1fa5b6a312e2c65b3a99a6fe8c0fdfe4c6ece3e1a47b37df08da1

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2244e3add4f3cc589de76bb4b140bb614fb34488e4198a7c10e50072a34c8223

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8eaa2f5ed4cc94b4d8649c9994a85fa3617697b0d475c974a5f38546808180c7

first message t

71f613783aa7a0e444057c7e401c11b4b53784eb7a720de91f222570ae2225a7

00c974695d18c77a0bf60d52f3f831a55ab52272c66450fe224df81ccfa32c12

c13841ce993b0b0bed61a1ec172b878964a9c846e3eb5ce662469274e9165ffd

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b03e5309c7ce9a687b913ed545c6cf02bd69d76d8480cfee2d753ab952735764

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65e4a6528411c84a3127d59e3efe062d5cb68ced62c3ad05be7baf60d17e7fa3

response z

1cc56f70378d9943427ed9a04c7b99f1229377864a90ec4f5e44076e1fecdcde

b669b2773e488fa89c52db83e5c7c7700a937c33166607b5e3a46b5655bc969d

daf2db94a14c3c98b3d149b55f55e8045535ad9c3ef2650f1c815b221d3c4203

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23f7b58610c73108cfca6d9b2da1d8d6311920b7e107f8d65d93cf4f779d4099

78c6507f10c4a5078b81c2202d68faeb112c285c6f71f47d35ae50b79919551d

response w

15a335bd0c619a9b45552f6f3b1772a8546914f82d21d5a0dea6c178252fde8465

98cd20a9cb9b66ffa18a8cb4206e3399814895275c9011932e046cf9d5771202

117d3c365e449ae5158aebaba27271d9cdf25e1b0e267c85b74b4f4eb7c41fc8

FBundlePoint.diagonal_mul_inv_of_diagonalizes theorem, ProjectiveBundle:247

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {n : Type u_2} [inst_1 : Fintype.{u_2} n]
  [inst_2 : DecidableEq.{u_2 + 1} n] (K : Matrix.{u_2, u_2, u_1} n n A) (lam : n → A)
  (P Q : Matrix.{u_2, u_2, u_1} n n A),
Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} Q P) 1 →
Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} P Q) 1 →
Eq.{max (u_1 + 1) (u_2 + 1)} (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} K P)
  (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} P (Matrix.diagonal.{u_1, u_2}
    lam)) →
Eq.{max (u_1 + 1) (u_2 + 1)}
  (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Matrix.diagonal.{u_1, u_2}
    lam) Q)
  (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} Q K)
```

SHA-256 of the statement above

3f4314b95afb042c405cebc6c2bf2a470a00bc241f05df87d3607ec6a054e17

SHA-256 of the kernel type

66847a20523bec6eafc333ca4ccb400497fd682f83b9c4545d1ec0874310b2b3

commitment C

9358cf6463f5f471149376c77cff664d03a3d7a8d3389eb42b3eb7a5a22f6392
64854aceb5fb8d1e60f3a1e9304bbf185d44ba1825ae0773cd4f5dae12ec1b40
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6a83af04ebd25def6285cbd5159095e9634120c9a09099c7cb869ed0c9592ac7
d89063c287819ff8ed561a0548f1b805aaea62f1a6d0605812c0fb835b8068a4
88b011b9767e50c91423bdab8878b67e2571ab1119ef5a83859269a2804ab8fc
4448e01d5cafe75595a92f245998cde4cf564978009f94d2f4733dbc1e36cd0d
2215dbb4886662a4040d30652cb93e8d0ad652b2aa2ab1a48106b82a1761189e
first message t

1f3ec8a3d607a538c567989e0232ae19f67d2b0750d426b969ea8726ce72bc0a
d689874bc823fe47e1ddadee681e3b48b18a0217b4a69e95b06f149c4c667d02
4e0cef8872ad276a0688b24cdabb448953f48e208814361afb27502aa5c7da04
0b7c8d3864c7dc1aa1a4583868cdd35e1d4195b12d92c9efc2a4257db61b644a
d42125bb80be720d37c0243bf23d25e719ae6c7fe81d8d02df17bb7e64a6cde8
5604600d059728d0b8e11281adc599f6a595cf383d779981c3e653f076aac991
07bfe5341ef4ba68aa6fcb141cf65dd1cfe087b1b4de558cb8a75e580abd330e
1b1e31865cf03e013662141dcbd5ea7e17b6120172465db0feaf4329df90dc80
response z

74666735d9487d3abbd52d2623a11869cdf46be193dbd7935272ae286009b1ce
0a6f1e50560e398b5dc189e2a11e73186dc19cedc7eccd73667e50c36c99cade
398303cfff5fd6084e58ed081e74a6437247074ea748aaa97416f594ed22aa911
8fd733706a3b033faa9bead8623d01777fbf58461584f08ad481f4ee8fd41007
8d03b6ee1f341b60d5de66cb74630e084e08598cb5e66e7683eb418bd95ca5f9
bc0865967fca9edbadc66c0031f111068d9454b62978c401571f4ab3ed8cfa25
f94593cdbe3df5a4982aa5832a598e2ebc23747f714c18473498212fa5156fd3
412aba7e52b03afe29da836deb6ecc6b0382e58fd944d4aea2fdc3f65d4e134e
response w

64561605d1ef8089c477707bf5ed6280bb10a3bfa7d8da5b9d1f6f3c7bac86d5
ef5d9d303dfef26e66134613c83372038b00b618984c859b5774f0143cc72257
8f0dba3ce89a249c9f7c2e99fe8442eef74645e5e6e10a848d8d124bb6e8bfc
507324ed185fb5db9d44279a19ef256f83d334a80e2c053e94a310df0673f252
b06eacfb8e68fdd380c4d93b0eb965d113355c1aa3e1d0b5674c6afbf1c4248c
186abe7280bc805ed74478ee3d65a74426d06506fc99441ae036733ed401775e
f0d134304249805cbc8b572ead61eacd8703d6d727e11a0ee0ff7342c31e0da1
fa960b071ee2a7dcfa93d680a8130a04233b24f1bcb0a3225fa02baaebf10e0

Fiat-Shamir context

1336a0b2ccc0fac040f68fb11742f664fba178bafd628123b38b1a4abc776b92

withheld proof term: 321 607 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

And

```
(Eq.{1}
  (HMul.hMul.{0, 0, 0} (FBundlePoint.cexQ.{0} DualNumber.eps.{0} FBundlePoint.qHalf)
    (FBundlePoint.cexP.{0} DualNumber.eps.{0} FBundlePoint.qHalf))
  1)
(And
  (Eq.{1}
    (HMul.hMul.{0, 0, 0} (FBundlePoint.cexP.{0} DualNumber.eps.{0} FBundlePoint.qHalf)
      (FBundlePoint.cexQ.{0} DualNumber.eps.{0} FBundlePoint.qHalf))
    1)
  (And
    (Eq.{1}
      (HMul.hMul.{0, 0, 0}
        (FBundlePoint.companionK.{0} (FBundlePoint.cexB.{0} DualNumber.eps.{0}))
        (FBundlePoint.cexP.{0} DualNumber.eps.{0} FBundlePoint.qHalf))
      (HMul.hMul.{0, 0, 0} (FBundlePoint.cexP.{0} DualNumber.eps.{0} FBundlePoint.qHalf)
        (Matrix.diagonal.{0, 0} (FBundlePoint.cexLam.{0} DualNumber.eps.{0}
          FBundlePoint.qHalf))))
      (And (∀ (k : Fin 2), IsNilpotent.{0} (FBundlePoint.cexB.{0} DualNumber.eps.{0} k))
        (And (Eq.{1} (∑ j, FBundlePoint.cexMu.{0} j) 0)
          (Eq.{1}
            (HMul.hMul.{0, 0, 0}
              (HMul.hMul.{0, 0, 0} (FBundlePoint.cexQ.{0} DualNumber.eps.{0}
                FBundlePoint.qHalf)
                (Matrix.diagonal.{0, 0} FBundlePoint.cexMu.{0}))
              (FBundlePoint.cexP.{0} DualNumber.eps.{0} FBundlePoint.qHalf) 0 0)
            (HMul.hMul.{0, 0, 0} FBundlePoint.qHalf DualNumber.eps.{0}))))))
```

SHA-256 of the statement above

f50d59e99ca1da914e118cf725deba60c3724aad5567fb8733a1389206be2e89

SHA-256 of the kernel type

03e674d83c9ec96cba1998680d83f2f68af5fa81109838a0376d61219ee9f384

commitment C

2d32bbe92fec8599e1abe3cfd02eb96f24c9ca1c4f60c3a15f7fc49166d58d0b
 2d6726d77d376de5231d355733abf2715fa9c7fba5568b2da25024dceb11f0dc
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 caca9a00a780f1f037bd5f99a7f72a52aa35a228b1d1b707b13709d3ab5ca8c3
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 882742c78da08742a8dd43d4383298b9227b6eef6646953be0e3db19dc5d3a45
 0a108e07053f004623e4306fdc211f468be36a9cc21bc7bf32da3f32c5a6bf27
 first message t

1678ace3eb968e2912f8fed3981b2896cd84890c20700c55988f0f5a5edff944
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 11e47d4fe6df70246991577b0799f8517e8af4d8dc1776a44a2b1d40447e44ae
 211df2acec02f967fce4e20b10c4bdf4e295478fea53cedfd496e7afdfa13c
 882b1d79633cc4d0b16619048c75aa6ffd5d763c23fc4a8dfe665161bdf76f49
 a75d45e7c42165dd968ef506bf795cc9c6790e2b00f89891b2ec3e1eaf45621c
 cb7085e9260d4808338275ec29303841bac855d6b0976ac30e9d8e33d68aa411
 response z

78cde7508448ae90cdf2584d2fc538d07a9320e1c99f67bd46d877bf689168bf
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 90801d6b3e67d2ae2212cfd80659c1e5b2e7c12b54d0b9484e01dbb88b2fc419
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 0f4dd606de11558680faab8ef89d0090ec24f0e1f522486dc49401330969c075
 4a0fb431c012f3e63109f8692a01e59d6f55fa37d39a805232c1e213d18ac38d
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 1d2a75236ac2eb2b1f995acb54c0503ec38c335fd3d1861f5ac6f6bc7fdc7e0
 response w

400c482d1aff84d197cc325c84affde1a28cf916e4eb5b8e14bae7ca0f5f5769
 db4f582859d81142815e01fb391db7f508f258e9fdf9be18b849ea44ddc3dccc467
 615511e8c579a7699fa755f10852dc8c4922c2fdb33785f249e8861d6a0d5772
 b02d8b463918363294bce4d8b3058bb7e28915b79ab559ee6e8812462f866289

FBundlePoint.dualNumber_diag_conj_diagonal_ne_zero theorem, ProjectiveBundle:459

Ne.{1}

```
(HMul.hMul.{0, 0, 0}
  (HMul.hMul.{0, 0, 0} (FBundlePoint.cexQ.{0} DualNumber.eps.{0} FBundlePoint.qHalf)
    (Matrix.diagonal.{0, 0} FBundlePoint.cexMu.{0}))
  (FBundlePoint.cexP.{0} DualNumber.eps.{0} FBundlePoint.qHalf) 0 0)
0
```

SHA-256 of the statement above

cba2fca9feec724039355f764553dfe4baacfa6d1dfef081c20b8db829c66509

SHA-256 of the kernel type

dd28fbe65a299ee8214d8c62f3fe7e0d79820a6efa623b58ea9ad1e6baa7f457
commitment C

17fea53f85d367eef9415c4a0a6a677663107432d7cfd872cdf2cc0d69f6f046
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first message t

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d68fa08099b80fe46097e591572ef17d6bc9c69f8daa348b73f89aa4c4f61192
ed38cde12930f6fef22da6479db2d9bcca51e119c9e44a6cd77b187ba0248f27
dcdba70a5bed579605aac57ed9a08dd4545bb1516ee26ab73f7cc316b1e1e4e4
response z

27e5f2934ed0527477e8fa38ba9f6677ab4fc8f53b4c0b9af749d35baeed39e0
bc61926fbf4b511cd9567ca4ba26be2dea6d0633ede85cdc20d4f996c03b7d3d
18a87d44ca92aa377a2cf92642fb256c1859f736e0e1b6b14d8b211d8a471d81
9e7a7d7d2c6c9a52f84b72949eb9fa5f7f31967f11c80c923a217bddbed1d4f3
71b3035f1cbfa663857d933f5cdde90b312f89010498b69f7e1d004b763f0f5c
794dee88d591c0fa20bb5eb205f0b33765f067040bbc2c44d3fc29f402f9cee3
6b159e5bf9e8c13497d81717dfb9e2a7b763ff899807d8620fca4453cbf01358
e540f6ebd90b867efed94bd6218aa1e3205356470e52c4c3c8dfc416a9dfd74b
response w

7fd65d8e9e27e2af2e4354212c969e9742402dcb9fa042ce8da13485619d41c4
998e23b757085cedfd2e9554f008ea8c71a2847c594dfaf4439ea682e2542c86
d60a1c207713f86f2a076036bfe397adcb7b8b1f6b04ec7ae34e69186893e639
e21ba9f8e9aae923ae794dcd2f38d9e04b17b0fe08e91c35cb79b3564117e36e
c51645b770f0c606cc609b3ad48170463d579065e8a5f351b5b13558464c0d6d
cf0249bca248f955f586b319301dded98061a81b3ebbf94793d9e341a0374293
93c1707bb839375aaf65b8eef11e78503b23d4dc76ddc2b42cb4c75129950fbc
d9545134dc3fa66563b77980116889b4041edae32fa5af19bff34b9cd6fe7fa5

Fiat-Shamir context

cb86dcaa37908dc4e44aa4061d9b476a922518a43a32125378d48dad641f978c

withheld proof term: 30875 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.dualNumber_diag_conj_diagonal_ne_zero_of_any theorem, ProjectiveBundle:468

```
∀ (P Q : Matrix.{0, 0, 0} (Fin 2) (Fin 2) (DualNumber.{0} Rat)),
  Eq.{1} (HMul.hMul.{0, 0, 0} Q P) 1 →
    Eq.{1} (HMul.hMul.{0, 0, 0} P Q) 1 →
      Eq.{1}
        (HMul.hMul.{0, 0, 0}
          (FBundlePoint.companionK.{0} (FBundlePoint.cexB.{0} DualNumber.eps.{0})) P)
          (HMul.hMul.{0, 0, 0} P
            (Matrix.diagonal.{0, 0}
              (FBundlePoint.cexLam.{0} DualNumber.eps.{0} FBundlePoint.qHalf))) →
        Ne.{1}
          (HMul.hMul.{0, 0, 0}
            (HMul.hMul.{0, 0, 0} Q (Matrix.diagonal.{0, 0} FBundlePoint.cexMu.{0})) P 0 0)
          0
```

SHA-256 of the statement above

aelf1bd4ac9b319b613a0eb162da42e4d8ec289da021a8102d0b3e3375900858

SHA-256 of the kernel type

5bf4f1b1a45397676c2a49fdb5ad1ae2c1b57cc5adc934380f103e9545d17f06
commitment C

4f86d016a0c8f4797ab3e55b0d09cbb87fc029a4f0c79039a128a714842358ff
73be121eeea6333d74aa339486730947cb0a016add80e7ff2624b4f0bb13dc07
0bd4168a8cfaeca75fc7fc50fc5611c253babcf2b6a45b4c0443530cbf906cb6
7d77d55a3b8199c30c507e6cc4e280d6716ca67942ffd58729d2b2a5ddd921ae
d357ccb3629398257cdeefb2f1ee6f8783cec80ad29f8d57b783b4153442ae2a
a5d4c85fac998651112036d226994cc20034a9466f4378209837d09c18fd1723
86b633fffd352aaa28529bc420ee1d5a1f60eb741b6864a5e697da53b7e50e943
a5859c02a019dace0beaa584e0d54c21270a376f596858a5e353726603e5780b
first message t

7379d9f432c812a6af154bc31964e81f7d7f53a3d20dc41706155b1fa4b91176
5aa1aa071587f1bee024bb3340df9adb9a3c49ba90192c7e82e2ec865e6f81ad
9a3010975daee150296384fc0d2dec709744bcf01f729e606cddd2ca845eff43
044ae2408a4a1b285f25e0350b40169d32573c981eebf2b0fd32b1987c71826a
a5d5c48144f041db6d01f9d117c5d0c179395c6cd0830d6af92e842295a1a836
95e9cb9788faf913d415a7a213c8a37ab84000445abf360c99f9e89d4c15f26b
a7341d3da53510ee0413f9a4d4c5ee7a9d9e2d161da12af1d33584f33438e7e7
5aad6ed1fbae5973e9a0394ddad372d05bbbe02c6d58b8f220c70b6d3e832c6d
response z

299ec37d12dea354b8da8859eb7f5c6431fe8aae247c26502c317382d9d2923e
c6 added595ec04c72263381888c01d9cf5da90a699174528e930aa54a4eb1a7df4
2a95ae1fc81008392abfad53b7fed3a0fef1b9140ace2f552a71f49cc2d45759
2c2ffacc839f0565599adecb47904010c3e2ab1cb5582b5d6c89500045bdd3c0
ad95d2a4d9f6cd7d81fc8a9e7dcb9393fcf88e9f584d567fc8532b3a4d9b8929
44dc6991edcc1deba756e59190985c0467b6779b7a08a62aff6d3fa120401ede
79b86aa3a86dc6ae7d22e4568cfa64ec9493fe45956c99f53bd939a7f9cf6cdd
5ecac0a7761a1e58eef3109a09426c8604641192d970a880603fe6f3d720032e
response w

48570907266498d8987287de99a7254cd81d740de85d36a38e23db8542038d90
0db7bdbc2d635e4cb962929b02933ceaf524305653220108a86f75fb7c0b4b08
e29b03affc16463ee00867f37057d95293a8199a4cd200bfd08be20da62215a
fe231e8de7c5cf6a1a7d46014c3635655ef479e69e0e895de63e73ad85b9e7ab
d8f10fff1e698221a146c2e135ca9d02ac6b4f4d6b1281003d3b1a461cedf5a9c
a249df20e3438243aea8f3e2cb6c4ac3efc75c360ed3ba9d472f58a2cd50822f
b8bed7218b2aea56c9d301d0cf31611b9fcc458e297275bd4ed2fbaec4f28b41
26e6cdf4f73665f3dbebda37f911d2560a7b98471cb74a6b58344517954e2416
Fiat-Shamir context

f287bdb87b58ce2251ba526eb31c97497c236e5d58f46696e4ca2fc24f057818

withheld proof term: 39472 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eps_mul_eps_dual theorem, ProjectiveBundle:435

Eq.{1} (HMul.hMul.{0, 0, 0} DualNumber.eps.{0} DualNumber.eps.{0}) 0

SHA-256 of the statement above

506403b79245d4de38b48320ca0fcdc06164d39035978f9e169f3a82a061987e

SHA-256 of the kernel type

5f5adae8dd84ca8c983ead6182c9db1626cf783368656693de40dad696c394a5

commitment C

c048e1eec7b5c818f33edc9cb75d906cd58675b70fae9b36f3c484bfdf6e95d5

7644ca15cecc9d7c77a95bc7a6ebcb5690982c7210809f97fc0c77db9edac232

ee8cd760fa94e1c161fb849ac5d1483d8900032bcd76fc65005d8814511b9b75

fda31b33c5cc0fdbcdb877cbcb410d8e60dd3ac1427d278718d9d7372d7f7404fe

0b63d8c08dcd537889ebfd041146d945956df9d645163089daa323cce7fff377

70e295fa26e049b09805beabb8f9a2ca6d945c2015d7c2ec2765bf82736238ad

7299edef29322aae550569c531766ebed38445055dbc7a9d0ee2c2d40c73526b

5b14b69eeb19663a8c92320057a782d0cf917b60040553f8f550059cb35fa776

first message t

a0e82324f32c67f05897304f97f188e0e3eadb76331d6951e22ffffa1a8d03bd

56ae3e97b38e27ec730605d48beb1c335ef1c91aef5622710012c13d08bed960

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0559dddeee5a56f82925c8a29f0fa41a6a3cbdd7adc5250c1c6c8fb94a3a0511

a3be1f81953642ffdb38b5e499e350d6940a91e92446b06d6e69e5b70447496fc

184508c97c6d9235164e3644d6fa58c0ac5e921b18f5f47e527b0146e3190712

1e02060005289ce25f6fee87285e1120f5190d57c703ca6b45f90d55c102f68

2fbe0cc11d6b42d0e690b0a54eb27358133a593f681269f5b115d32fed193854

response z

2b21651537de54fd2518122baf4b58c2ba7dbbe3af4d4ed4ae7337904b8023cf

7b84f3bcc4929938c67053301643b25cb0b3432be05fa192619c83b8a4f98707

c6ab6c6151674bcd91d6b3eb491e51dc4b582dfe89752d2fe0223783c3c9c6c0

de4c9a996e12bd68346bdd220ea2d4b36707466975319b209099c91c1b4d0b9

8040e25bfb5c3e88af393124e05008c33deff4672e89f42b6efeb7efcf86dfb4

b4fc48d75ceccf2a09a48e65c189951d23aad5147778577a91f7be6673adeb52

3bd96555e56422d875d65b2e29ad2e13dd3ebaebb18b20f54b671c1de7eaa997

853463e22912cf5fa988ab8be321a48fcfee6d5a9eda43f5f14548914a409320

response w

367b206a2c03c23d1c39a45ce92bc423a56ca2cd12ed4c5352266651af77a808

3a5ee4677c0ae429e003e42a1d275a001401ef05d574a07c1ecd5303390464b8

ad05c52d81d9c8d8a782aab949b175f85fe7f5f06a593003ab43f6e91d38abfb

1353944b18169fe88526fbfa07454fab74f7d366edd337811d2db16f52fbfe66

da02e5e95cad795b1334b84670c73d722fa4f9490b02fb2579e0929313f45a42

75f79bb7cc38267f23d869f6245568c903e21a4e0564eed907c739bfe8246e48

6b68063a818aa84b663ca9be43e1ad2951ad3a7a6b5b184fa0f1e854434331e7

deddf6d0ec3ca74f18c77f3758f369d61efde9338f5d2ebb3a3f5ae1978057f3

Fiat-Shamir context

f3fe5b4b4ebf2f2c1893cc7d0ca448494d7fd98bf824dde183765a8a159a3be1

withheld proof term: 12 863 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eps_ne_zero_dual theorem, ProjectiveBundle:439

Ne.{1} DualNumber.eps.{0} 0

SHA-256 of the statement above

cf2c1589993da12031d7d3ae7f847dada1acbaafc156f5b58b2d7a2996a67bec

SHA-256 of the kernel type

47ab6c31321a6264e280d3cac801d7e5aed4341230059f68ad331e50aa534abc
commitment C

90e4b1748ffb29267477fe3341cbb0d5d097dcecd69026826e56156625f757d8
1dab7cacd6ba69e6dcac7cf1c690e2b5c7d7b710cbf088b655a2a1bf6d33f6d8
ba317ea720944fd88000b1b3cb4f58291777d1f00205a6a1617aed6801e8e1cf
69c224f8989bc239f11cb1de1691134baf30878f1e37654ab198a5f4467512ee
27d9d2450b4234d013552707aab7dab6f77ba1cc59c245250f2b15ee36432231
12fbceb8cd1b9c6ec71d2045747011b53a09494d82a304a110c589eae8ca0d26
079597b3b8b993c8bc18b437946c23c1b8d66eb989bba71a71653b957d02a8bc
2b32721cdf82405d6ee788108149ea4850e6051ca709f4c7a5589e42cc18152e
first message t

8bf94a4d459c20a75437721dcf1a7de02b89eccb9a23320ef62f2822d2278788
37977f9e4feb54f03312ece70f91ec3805fad79d53890006cac7eab32f3c1d9
add565a34fe537824093f96a411fbd5dcc9ecca87ecfc100ef53713d6bbe3017
4c0e4a273737408d1f760abc3ab38bc2480b022f1aa0528c336779eaa467c623
0597afa65a713146b96adb4059dc4c1c1bb771e8397300a914690963c57cb8a4
b2f7eaa75d8aa89adf3815abe9dc8b3a2cbef0df9d80e4a53f1927a175409156
52e51b5ca5d623ec470a14e94dd6d81b18c8b5da44af6788c56252f68dd61576
1a17b2df5525debc8bfbb602e84bc17f3762e9192797c6627ab9ebaaf5b48b64
response z

7c960287345f69eda653b19d06a6dfb3e22793a911b1c08efb2702cc934ea244
360b66ad654cfa301806320b53ab8e656bb40d7faac6b0ad91c0ac84bf6ffe17
0f623ffb2c2174d853e1224fb794c75f70abac63e569acd5f1be611b2d776806
c37433ce1593b6c12e0657144a4ae78b9b699a8113fbf50ee48d2922d46ed3b1
81ac2197f5a8887e7210674b988761e67f282a9cc659b635edc7c16b0c5629b2
5abf78fbc431378142497498d7b25555b105d483f85a66c272ea01032b26b5d
794fdd57058d6dff22c8cfbc4b6da733d839319307694d0ba5fd75a22bd1e4b8
96525fd3af34a5ff0156647df8331e803bc71a06e9e0d9c4d7e04e0aff5e9572
response w

26d6bdc9abb09328d801b459f384642f9ada3914e2a7891d28aead6982e86151
7bbfb2e721ff5df1e229512709ab561cff66685e735d526d1f8821525e8ff6cb
812d668d8693739e1c67b13edf5f7b54ef82cfc8afb152c0853197508e35e70f
d84d09d9d27802e6290c2bc274cd76453025e8e0546afe3d98e43566418855fa
f3fcb31d5b2da166f71f9fa170d7fcf1279492c7a624d168c6835127f2c142ae
ebd5c3261cf19d89f9c2ceafecbcf5480c55cc121eff21892437051d638773b6
4354b57aaa901640517cac5c8c9223c693c84cb2fb1e5d8a7c442685698a63f3
c03023b6ff4b633e0ebaa326cfafbef76eed6031163b60ac0ed84343f214ba06

Fiat-Shamir context

26141c25182e733a2f19d394ce9b685f0d3a78a0c835e707cc87abe327239b25

withheld proof term: 12 391 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_diagonal_of_commute_diagonal theorem, ProjectiveBundle:233

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {n : Type u_2} [inst_1 : Fintype.{u_2} n]
  [inst_2 : DecidableEq.{u_2 + 1} n] (lam : n → A),
  (∀ (i j : n), Ne.{u_2 + 1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (lam i) (lam
    j))) →
  ∀ (D : Matrix.{u_2, u_2, u_1} n n A),
    Eq.{max (u_1 + 1) (u_2 + 1)}
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Matrix.diagonal.{u_1, u_2}
        lam) D)
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} D (Matrix.diagonal.{u_1, u_2}
        lam)) →
    Eq.{max (u_1 + 1) (u_2 + 1)} D (Matrix.diagonal.{u_1, u_2} fun i => D i i)
```

SHA-256 of the statement above

d74efcd1c39c2b07a46ef918d58daf5c7a63c6cc2d882e2a011b23865fcad1f6

SHA-256 of the kernel type

2260cceec3f40616b04a9d85d2affbc1467b574828f01204fb6bb65ae3d9b16
commitment C

3e29e3b749db47a24e5947617b4466115d06f7c57b3af654d9b442e10bddb10b
58ae259a273943344fe5ed0bd709835fe666201438f488e5a58374e0ec534f33
cfbd317e3d2246c9174281614b5ba57dc487351f71b4eb90a197385aac60aa4a
77bdadf782c6e8c7440f163f8aa5b227ad6ccd243b1abfd81b5a5595c0f7154d
5ceb7f1df48052d943bccdba12557b3225283204deb2eab7633832b02cff74de
c4d560646808044444c2a80adbada129dfc36296ba61b38803845973953e96915
44ef8e7b9a252d5b3041a7c94282f8742cc6a455f0be448ce28a0b3b3730b16f
bd37bdb64b09e0cbfa965ccc65f405a017166fdbd990e4e9a9c0880dc3316670
first message t

5a1d98d3da2d1016b45d21982109a86b04d1f220db780379f3ac589d752df3a5
8f0aada3b7d060a4b7afa65c93cfc77b0a0ef5955294ff21fb9f058ec53c61f6
e5fe84f364af70081c03fc87e8c500e592c1477512a2ad842d06179b12036428
008ae0343a40362058b5a031b780fd5559262b7f03600eccb3d1b4bfccf446a3
cb35d39b514f8f6f1bf478aacb0f4967ec91edde301559664d9e8bcd3ff8b12d
2949a635c19e0910b6ee039968300128b416cbd433b65c43f934f00aab9d6573
34d59e250dcec65f177a93556f4faac7fa5b5300b3b27e2052fac7443ce18faa
eb54c1428f7c06533a98bf5ae37c5d7fd106d428edc230bef91f90ab8d56a84
response z

4ea74a9f7fbb31164d29085d2a9c0d0b4934eebc651b8f27cda32359c07568bb
4a6377004fc4b092c42ee37f2436faeb9e6fb39f54c9b7cddc279cd2fdbcff65
a8995c1f8b58363bffd6b6d6f4aa2792cefabf5f419c03355946415ca282c350
4417f86a0236134c837e6c4f8769db513083681420c7316b5a0832d85579cd20
5d2275b5fafc6b29a3cd00ce53355c830d45f5d8763150e2c161c1c93a1303ef
d0f323befbe37b7b938e687f6090c17a46585c6062ab6bee29c615681cd13118
35d6ad313cd4c62a108cb5498c5c7f3d0735f2d3a3e35edde1b3dd715ce3669c
ef02e7d13c189fb03c8bce7771b0435985cd3c7cd40f09e2e63b5636eb538908
response w

4ff134cdba7abc3924194774ecbc9bea314c2dda1b0f60cfd849c59c8be4b051
fcb989fbe6363543b51261b75cdc0d6a09eada0d0e6f105da746266e8e30a175
339f9152bd32b740a9e215accc324948a80773d29af90915a3744c0a0b8e12c4
941f492097137aad5dd577c5753c5d85263fb45d1e87a3e3333f940d5fbeb520
0962608ac2b391ef8138d3eb821d71018e3f634b6350f9cae6e680ad44f96692
055ca540624d9289105df20559b4ead16e218e077f72c4308e456e9605e069
931a9932ed739f8e5f8a95f1c9e117a1e65b0e137e1f47b3d6ccfb9e98ab3ffa
e6c09716f9fa98eb7328562fb74249f05151c836ad9b80d40ea5ce6ed137c123

Fiat-Shamir context

eb4f0abe3c4f153b356fb96755cce2e17d763e7c7a99e103b159a05c1f089cb2

withheld proof term: 168 115 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.lastCol definition, ProjectiveBundle:64

```
{A : Type u_1} →
  [CommRing.{u_1} A] →
    {m : Nat} →
      (Fin (HAdd.hAdd.{0, 0, 0} m 1) → A) →
        Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m 1)) A
```

SHA-256 of the statement above

72b48aef7b589718b35f32352ff5ced3a279a751358cf015ae3e04ef1e7ba816

SHA-256 of the kernel type

d8942714e88dbe1a686b3cede2f5fd53b48f2121e3e22900b1122505a83fb08b

commitment C

6ccb209ca61ab1cc434f2ab28ebdd62f8143d27716269debff8bfcc96cfe52d96

a54432a84917d2a1c8108dcab1883165d9a5439d8fa2a330218da22d498e507a

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b807dd8eaa510ea0d82093493dd53ac477be7945f598738fd28f6ce038d7b22b

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5d4beac40925634bf4843606d763cd67f24e6ed7f33b9181c0f9cebf054c4496

e47d41b2360acba7fac7832e495d0d7fbdcd0de9bb1ca41d75f1591c529d24be6

first message t

e57730ed2ebf82859d0de9eab7a0b0caabe7236d0d1ece5c2ccfa8742bc9bedd

b2c638103052a37aed58b150c31bbd23f485ff9f42af266c88bf0ed2862b5bf9

0cdd27d9ce81db05b4d393e286f263df9e9bb453d3086114957dd8085ac6c24d

dd7ae0dd3972850f7a20ea4f1b9075ae9cf2eb0a8896639d2ab3730d3df4f17e

1d158efd051432ce6d3359746caefcde2d59cfda8b3b303ad49c5b1249415b19

0ab9bac5ed95803387e427b09e6815bd85165e285951c8d4a3c5e22d42c69306

15434edbf8b8de8878719a4565e4220c924060eab3c8fc80a98573b1d4ee2f9

3464dc2322ab1c5d2804f9f747870e4ca82d24de106167bb686a98203acbe399

response z

51d0462cadaac67ff59308a79297746b9cd7e304dcaa6810631e144c376f57bc

8cb7820ad8db05f0d1a5bd54df1fa4cabb62d774522bf5c728b0fd23fb7c183c

1adb2d49ae0789df073d2c8958d7a1a2ac2631b8264b7f54f6aeea75b761fe03

00c2a1474db586108225c24ec4c5ab57f5c214628e89a1189bce5be7c061f68f

f005685326314e286d1c955540aa5f4f00a42f44791f9e34cef9c8d8e31b2fa7

ba6a1650fc0583be0b6c33174cf305f754c95870e05017de0f249a5e26f01a71

abd1ad5d8b3d2ce99691f517b75c93388b3836e1efb8c4f170eeb4276195f6b

bf40d0618b5d71b06d050b8c5bf4d35d5413fbf8e4183803ba74bae2ebb45992

response w

2137bed77fd60bf4efa382577b7ba272ac9015dde5657b56e99d19196951955b

ec901ed8c8faaf512d6440d64e939ce3d94f625df28644728ad04779c68786ee

9589e53556fb08529595fc8e6336ba309f54b8eb0b4132cafe48e856a591b414

6ad8ec225e55c3014952666853829e61dcf11c8610a92df69b0f9c2a2507a4ce

055a716b39790b0471465ac673c3611dfe3649a1955ce81a2bd116c2c0cdf89b

b8cb4f3368951a10059a6a447f617f4e1ff6903c14628ee3017e8b3e20da6ebd

06968559fc5d20cd4be70cee2985e3b9da8809f4fa846dcda41d5e64494e61dd

3d77354b3db8c351c15c66df72de79cd4370765ca5b2b27da8bdc3ffd096478

Fiat-Shamir context

43cdb23fff0c0125aaa66791d0e9cd5e19fc55459d95201a54fef96d7d01e25e

withheld proof term: 4 473 bytes • axioms: Quot.sound • kernel replay: yes

FBundlePoint.lastCol.eq_1 theorem, ProjectiveBundle:0

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} A] \{m : \text{Nat}\} (b : \text{Fin } (\text{HAdd.hAdd}\{0, 0, 0\} m 1) \rightarrow A),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.lastCol}\{u_1\} b)$
 $(\text{Matrix.of}\{u_1, 0, 0\} \text{fun } k \text{ } j \Rightarrow \text{ite}\{u_1 + 1\} (\text{Eq}\{1\} j (\text{Fin.last } m)) (b \text{ } k) 0)$

SHA-256 of the statement above

2066fdd74d4beb019bfaa740a53ff9efc25a352ba24819d81bd3cbaabf08200e

SHA-256 of the kernel type

5bd68bdc0443cb45fb184d8f091ef1faa079deb7bca667cd4eb3d4bbd8163733
commitment C

bc44f117716846edf462b272f09eeb7a2b3a51e03c345f7a04fd154c09d96d56
c788d9c655b049498d945f34906910a0536b2064b91b811591d5dee03d69a8d0
5c5c4c01342d4fa709199017b220857da0e1c72473bf941c6dc1f24323455a4a
e89090bc5d19bbcf597f20498b0efcf2600e07b2f0c360bed1045bbd4f7a0af3
9a106e2f79ed21161e960a1cbeb98d0db2168a7aecf14b56043ac24366d96aee
fb3c60e87d6cb76166fd29921791c78aa0aa6b908d56cf6b2a518308fef78e49
5c7f4cbe68e7e93622dfee9a3cc5b9ea8d4b91702f9740855ac6fb0666aee97
137476b1d524b574b0c93d356de1f7ae76c9e481ffc38cef3b6dcf03b4c0727
first message t

4245780360102aafd9a2314d1722cba49cacc607d7ab8fcefdddb6853762ec408
e5ee708366fbbd45e72e9931b217d5731047e61055c2694d437d877e3eb7b3cb
ab42c22f76fe7d8da7f6ec524390eb7baad00804b8ab72b739d5b760f8e71c50
09a0959006969b0d9c0130051a5eede11224ef86ae8fbfbfdba24fbee07f385c3
c7d43f67c04ac9e6ac3f9c6a8856aba09d7ebd6495523d920460d3df1db1c6bf
9cf2df7a58f03e484949998256004e920e9582ce275be01868ce888bb64d64d4
c072873d7753c0b109daf1993e4d12cee6eb289c5e4066099ddd38b057f331b4
cd9f3227ae807312480f230023216e0af23bfd524177b8c8c0e943ebca94dff8
response z

3e8fbde38c644c9ace0ff88d21a7e1035df9f912d8338915dfad44c3520b31df
05e1026ac16a3b5edc2c7fd846d1f372f9447d7a5c6ec91300712b8b3b188f6a
1698029041c28d549c2c43451de6cf1e4a19dd870757a98c58989ba9bc408e41
4de50dd0bc167d58dc55a40b36be4463934995ff678d6fe6b9cc4c63de8d1b30
bae8f2549ad87fe138f24d11a1ac48ad672d20a376f605aef4effba80ec94993
8f7a75c15630e931530a9e533c23424d6167781891ce491cd45f9848d43af6cd
85935d3b9850a16ba1c7739002c11fca65d6a76d762ca4e2bfd4380bc74f7eb0
acd256b9a8ba1eb76d13648bb07545174a6d57cdc32bb9ae822a0762e3728f95
response w

9e0dd44367386d805b111a41ec256f300db2f4ca12c346dfe44dc64d2fe6e59e
8587d7a53c2f47ecc15ce4fdc42142a90fe842c34acd19f7213665421ecb36b
d999ae0a3c7aa89483f40598a00db7103cf46d3752e5af029293521cd32ad237
265e692f8be5ef7911792a637f264f386b1a27b3e320c045bf5a174eed2dda17
432c085208fbc518a1c861ad7932336b09ea8f07f2a8e2cf609146fa0e91c402
a71d0221c2fe850c9759b942a6bae4c089c7866d35443d15100d11ef028f339c
9c53d8ef521f346a80a327ed43137796e3f13b459966d090a73e08811abf02ce
1098178d7cdf607c3a12b1d1e32b75ea489c84b37c5f868cdfb8ba2cf458ab

Fiat-Shamir context

dbd3812d1f084fc2fca7f0825aa6a66d7e9e5c52e0ce679c7598ade62fb6ad31

withheld proof term: 491 bytes • axioms: Quot.sound • kernel replay: yes

FBundlePoint.mul_cyclicShift_apply theorem, ProjectiveBundle:99

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat}
  (P : Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m 1)) A)
  (i j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P (FBundlePoint.cyclicShift.{u_1} A m) i j)
  (P i (HAdd.hAdd.{0, 0, 0} j 1))
```

SHA-256 of the statement above

cc2eb7b2291a728d550e1f3aa76b283c902a88114eb30a378443cdb01687efb5

SHA-256 of the kernel type

26bbdc785a4790ddc6b8e734982fc3ecab69624c6abc2d0a6ab7275f1559ac81

commitment C

8d2142fd994bc0d3e3f9d76290ef4f0c7f62b98885b787543cbaa12cf7dbcdc1
33f6e87bae5abadc99183baf9be046db3b300d8d83a13a08de4af007e984a6d5
34e841aa3dd9b713b457eb36ee5258b2592ad673b1522c06cbc62d5f2450b978
4b76730d5044483640833144d26530fd7f58a276e52a6f95c5131201eabfddb4
d4a1d1c1f44c66061f3051176b02f839e07b0440e4a23e916fa4e42c40922763
d5b357a96f6d4af39e8f53e4dd1de00ca50a355fa1a8d9ccb044cb3b91cb435b
1fc46f4b1df8a8ef59f27471424ceb0fd21c3b0d19ad4496d2a20b1de0e47cd0
17b41dea2e7acbe000370315ee95b358611329f60d00fd76894db655c98e609d

first message t

33a3da08fc04d194ff34d30143d523a1bf4bbb847c0bfeb4b86fef52cdc0f6b5
df8d3cb1203723a99d2bc5424f0557d5d32fc98559b05435300a6fc06d91fe8a
613293f7f1ba6615f708d3c9339d78521999388f781c8d534dc85e6be311e9d7
b0840a95f3d694029e1e360575433c5679d10418808c3bdf10742307a8fee28
e518f75ee65c1b0afb2a6e4e03772331f47dd8d26fb7c11b40b290bf3d6b59a
a84ea1a6783d7c969ad98020de7978868ade952f5f01370fa02ba1b388012904
8d0f83a6d6d3904de241a3b9acb96f85b8e3f0bc9712ed2cf3b0d6d8afa99615
7260ea1c03150280ac26d38e1a34ee353983ffe72487d09db58be503e753f16e

response z

35077a178493434300c8b79b97abab835fb4100a01a9a550f18f867ea4e0c64c
6e2884932cff38a315c50fc2074cf8a5a105a22e55db557a7910c6a1da798ec2
fdb4a4741a01c21a693bec5d3a53e677e6e6d6fc62e58944c4517e11f5077b3b
73175ed32d1a5ef2355ad4d70669e457d6886e51d5ba94cdcfdbd42d8d11c39
69ff3e8e248a4d3de7e6dda4998d377b83a4d05e36eb17a10e8480fb64e5dd15
e6f59bf597aa3829f6e72364842c874d4c505758bd33825921cf040e5daec0d8
b4d48a500481ca298229d30757e49d79cb3b9f342a6d5858ba73ceee9f32a3fc
a5c378d15bb9bce6d257f30e967d16b38fad9869a14ebc4aae311513b07b466

response w

256de142e074eae70fd7394e1599387fd10f938cbc582a7d540d4a6850e11580
ae6c1ede8cf4ea907cc6f17b243ae3e300e3e1e57ce50168d1280c748e685ab0
a15895d33a9f2f3ecbff4d90d1688555152fada1b9a3282c36fb4596caa84646
c3b19e801c2a67ea36c2c8af8a958db297cc5604f771931cff196042528bbe5b
c68e24c040dc8721b2fe56487e865ae6a15ed39ae6b1f4b27f17448c9c7f036f
cd9e9fadaa9b3c9223aae76c0404cd22470fb26dca77729531f96b55b228b9c2
db6fcf32a388fec27e8e69ffc66f332e6b24eb3fbf75119f16a577df1d097404
526224a92870c8502d699460e5a27b4d50e8225ad7c27f1d4284220afa1386c7

Fiat-Shamir context

e62dc65ebd3a57f9047759042e55570d8c17f06cc7dfb5e8c04fd39612bc53c3

withheld proof term: 93 025 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mul_ne_zero_of_add_self_eq_one theorem, ProjectiveBundle:413

$\forall \{A : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} A] \{c\ h : A\},$
 $\text{Eq}\{u_1 + 1\} (\text{HAdd.hAdd}\{u_1, u_1, u_1\} h\ h) 1 \rightarrow$
 $\text{Ne}\{u_1 + 1\} c\ 0 \rightarrow \text{Ne}\{u_1 + 1\} (\text{HMul.hMul}\{u_1, u_1, u_1\} h\ c) 0$

SHA-256 of the statement above

20e45ba6863ca07dd1da94fc6f6b9a82a1f3e55887974218080bac115e322e9a

SHA-256 of the kernel type

25a7ffff7a1cee36761fa408c0525844111720d6e9f52fbd9fb1106edee9ec9
commitment C

bf2d1a2525285f49d51ee89947fb136aad9ee32d770a6bc614bb63cef433845c
1a639ab93d14de3365c3c5716f543397286ab83901e56edc0266f52cd0df5694
378deb7f0102864803bd988c2080d107cdee2a3438030e7b82338d6c95ad5f51
521834a6664f27dc5167cc434bb628684c9c52e675138c5ef47c47f743ea329e
e529f0e637ca976cc23d72761f321905b15b8500ebff3791d22b6fac4eb9898b
1c2b46a70b4b9ffcd703eb3df3bebf8c8e9f7729149491005132d4f1f433b3cf
424da11a24fbcc03ba42162b91e17305a361bbe3473e685f5bda4521d4340e6
49769129806e5a00bc033c79dfc752cf8430a2d5228637c5a99950adeb8f7838
first message t

689e93c0c11a9798cd2a5b0756029df1c01b61ddbb1988d82963b11414b997e4
d004d4881faf2d56295903bd4ad655cb92da31ee99e7b31f8d19149e388e8ebe
f7099546c6e08383e6e18b54764c545d56eff564caa7b0a17f8eeb323ffa91a7
971bcaa9ea8c65f334cdd70fc6427fe886781dec7bf26ebabde35912dd4e6187
9a9505a7ac3d5c5d59c7519c5f7611f3b0c78339232706baa82150a5e43157d3
2969aebfc54237e6a9d33b6c358012d7f9fa5ebbc451eed11b3a73c8570fdf6c
e67a04f99aefb6bd88a74eb7ac5410b2800e4cd90f1d51c8f48ab4af521302484
19f73afb6f74bc29efb690c09a2248165dae967ff1ec7df8462a57da7b81311d
response z

47e36a2956462b0e0c44dedd0e907a5f0a9f4701e8e4b448fc5b05c2214ac28d
f3bfc002671f792171ffb41d1069c87edca500fec5b7cf3121c1f946c5710f48
e570baa80713840099778d383a440799fabaa8d42eb9f6d33210f570bdb6649d
8d6919a3ebcb42686b52a25c9ccfcf55375c9471bf16bbe493ea2256b54987e
f20a5003e007ff6755da8cac7dcb2d0cdf42d6370e3686c0675df61cbc2cc96c
ed9e8b42e1a2dda11219a90b81105f006a3888a332e194560b36973fdf3dec59
71dcc748242b6e39d4e91e719ba5a36e3b92eae5c9aa8fdba2678c683eb770b8
092cd5e0bc2fb462e552e8f0bebbec2dd251d4af1e605edfcd984e2f0fae0503
response w

10d93a3213345be2db184a99ee1d18afb39663a1001760224c247ee8d5ec8682
0226236e685e72aeaeba44ac53ae261d580cc8e43e44bca086505da0deefe17d
34279393c4bb89ed1ae9a19a956a6243770b0b3d13a9e237723bf14355a1d584
3f7a6f2f5ef9458d7ff9422e01856844788ba2abced676e69ecbe6cfb684d0ef
b8e7b87e2c8ddd690c18faad58e1e715fc314d387dc3de4a90dd140925a0a0ef
c30ffd4e0c5b1af23976fdb13e1b0c53a3e747cf19533cd95d002281d7091a18
f3f51c4d6a4641391abb1e25d5f6c0bc44468c80f796367376947bfd2ac08341
18baaa84fb3e124c9f127e7cf9f43cf415684801a8edcfa25dc81f0cdb0d2736

Fiat-Shamir context

4a7661ffb78e9512a09096916a1da25f2af2b97bb8777c933f3bb70f1f199a51

withheld proof term: 189 122 bytes • axioms: propext • kernel replay: yes

FBundlePoint.qHalf definition, ProjectiveBundle:427

DualNumber.{0} Rat

SHA-256 of the statement above

1974ba873f5550ddd2bdc2b45542f9a987214fdcc6e199cccfb812a1020142a

SHA-256 of the kernel type

1974ba873f5550ddd2bdc2b45542f9a987214fdcc6e199cccfb812a1020142a

commitment C

b8a100b3c2d413ecec2010201a653c84245a9e9cda4226051df63a5347463973

1c3c63c931fb5551a4a1b55bf3512c955ee81ca7f8cf285503ee5cf0c7352aad

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c24af25e355272e1d25d6c5df63a3e24bb929f99b2873623639b04c0e9f5c6f9

293bb8ccb003286ce0fb3814dd805637be5358efb84abefe6f1b308cf4775d78

first message t

35b7ac9d4a7d4e4f0cdc928c04e1171256dda6a5315afae036b39675a60e4d31

6e5f6be2eec7b2d0c32c65107359863cca362da8e69647f70fe8f67d7168d484

b2c8164411eaff32420dbe855165762c8befac7fcc4390050dde8dd03fb61758

6af02c8d6a8aa68a7aea1ac3e6517a9f4fa25f58c1ff6dd0222b1980e141a35

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eb8f0eca2c1a65a17e1b70c3c55acfb5bde57296d0cfe9f21c9a07b0751ddbc6

58d9f0999a5108311cfd946e7cdb1972e1aa78b075ea0fdfeab8d23c1d8b6d1b

c40e8cc56f50abc80c72e82dfc9356505e7bb46fd4a55e51760bf784f0becebe8

response z

558c433f5a7498db142faf3cfc893c7b97fe299e4ac0096c3e4c199451a883cf

192148282bbe955ac2c35acadb5de5a797f256c1c9324a612631670cdd37e9e3

4423a0e738fd47a59720b98edf1b682f6fbe89e72b9d08ee598efa114596c3b7

9886f05462bba740537b9c70cd9182a135791d8eb2ff6460ff5651fe9c29b052

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df3fb8f894b0a746377fc9c3ff9b8596651a86b754845b479c532982f30eb58e

ef5f310943729d8fd1b9fc633ac141987d2057548cf04ad7b2165fecaf1bf233

6c6bcb1173899c7c5f2c0c4b2c4ef2aae75434a9e7238ea2368296c9610f334

response w

164bb865f5ab25a13b54d5dd49b258e8725f74b3051cfa7b5b72cbae5fdd153e

98026b3357bef2f7b469db5f8660f846dee572406475092b716093994c4e82ef

2c891bd97a208c90a81e379cdca38354e0b5512c4d7d4cb5baf47c38772304e7

bde958cca5ae32cf048fce674e5ed303718fcbd009f6c1cbea75979ee46d2ffc

65f834940a1408c43204347e8d908c6dc519102fc029615da1d802026c164f8c

f6749cc813c5a7ebd2b547f3f6fa4a6772f89bb614a656ff26a736f213520f61

98c15c95fd2533edb7116f1476b9fe2e77794bc0d49de4e9a75f0ab767201cd9

ac4641c67127a5c102cda3823993870a64eff00549f5f01477fa01dbd39437c3

Fiat-Shamir context

ddce48a72fedecff5126c9fd6877877d190459cdcebc649aecf17ad4af5337d5

withheld proof term: 1 509 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qHalf.eq_1 theorem, ProjectiveBundle:0

Eq.{1} FBundlePoint.qHalf ((algebraMap.{0, 0} Rat (DualNumber.{0} Rat)) (1 / 2))

SHA-256 of the statement above

8af30e9dcf79d548a764dcad82ee6ff23c4f6cfe577ad9203c199cb4b9cc9630

SHA-256 of the kernel type

57a18ca571ca3544848c91d62c13a76fee6875ce9911051be20e497dc76b9375

commitment C

aaee72a4701a81b3a88f8b753347c05deabadbabe3d3bfd2803504ca27c1f8f0
4cf162a8f65a1a702796a1eed480e6498c5c2a2df53fa1f02898ab56d560ff89
fc08ff55e4f8b0840e309008fe811f2de336d7e9e99b5fe2870b4e1d8778f9
a53940155ec9aeca6e3514b594bb201069387cb6a4d9f1db2ca87ad845ef7510
49234828f49dc6f76fd4fddf39a89ca7862ee9c4148584864b78ae58ccbc8e70
afb44e9a1545085823786ead0744f1556f4581bf873527fa86aee38dcae397b8
7057f864e32cac849a1d9bf3d25497e0d884ae443bc289ea56a7db367b54a02e
db2b6487484e1cf5a8a4d09b8ae00b31692f18c6227b39df3a171f587186cd68

first message t

6768aebdbb360f2e8e3a8ce187b0b022d90a73ab96392ef5bef9d0a49b2bc928
81a92093ade6fe498fcc6576c76fddd30b80171cba8d1eda90115976efb6de78
53fe2c518657d1af5aff0953e7839c62bee2a7ce3c338736744b265601cc9244
10c89eef256f062f1fc02dfe64d3d6ede53a5003ed915cb5eadb59775c3b0d1d
37a903722b526e6993cc663e8232c527505893c146c75ae1c4bf41767a5ce559
f18f6da973f46b379edf7a8edfeef12d33fca657083b38f4fd9c5ed47593cdee
9d6419d34fb2d49a034a5b75697de0a01fb48a317c0a180706ee99b8ddab1f90
36eab06739a702320d4947ac145ac632cc3ed6bbb7e12e397cc52370102cdd9e

response z

53a3d7e75e2766d6bc70047a3218e85cb457a1994e1835cf36f41af0b65b3b7d
aabfac770645070103a9cc6c48d91533a855fa122db7e2b2ffd853ffb47f7564
16410de6f9f5dee41452e55a199cbc8a141719478e96f206b374cae525da4b3f
1743d1e73fc3d7d27aac232477f7e63b2c7e0d79b8987567d4903fef4b679be1
49da7b48d91381e88cf9287e2eefd829f13fa65353aea9daed662687998faa6d
847d04bbde2f0497bd632987da15cfa2991b6637f96313de822e95516f56450c
84d23d7b58f23f9e4c31d803b5dc59afb4933978a412c41d3876e2303f3dc67d
bf2535d8ed87d8ab34ffeb3c2ccddb6107a5b8cc6d66512dc884467195534642

response w

407131763ed4235e02789a7919b3fce33821a3416cc00af295975d4afd89414a
5e3f429a17c2177fe009d480b26e141416600df2506391cc90db34be3c911486
cb5e9be61d07ca0996dbf27befd353a165949341d44c3f95ec3e3294615d158e
069a530a5bae35c8318f98b4d0d1f88f32bc5b2a78769a10a4b087554add4480
bc2f33b26e113d1e750dc42cb3c2e597733aa8774d5d49f5cec8878b05db8c02
71c64c249053bd7490dc0890fd6ec8abdb5ffaaffe330cb63ef26c30238c65da
d1d09af56155d1adfe2e50f9c0ee28f17573e7126ce7a1659ad5ff383b2348dd
8ce23ad21eafb35ad4ea50180fb0df235562083f2a7bd8dfefab863251b473d6

Fiat-Shamir context

fb2395522066535e65d8b0bcd7a85cba78a74f5dfcf821e10b87e46e7dda92be

withheld proof term: 51 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qHalf_add_qHalf theorem, ProjectiveBundle:430

Eq.{1} (HAdd.hAdd.{0, 0, 0} FBundlePoint.qHalf FBundlePoint.qHalf) 1

SHA-256 of the statement above

c61971b01127521e660cd8729e407946c33f5e8a2964716ac8e400765c80d322

SHA-256 of the kernel type

31f8047ece8d7a03324bc75a05877a90b301f3c3ec5be8b27eb2695aefc02b8f
commitment C

ea2d732a63d8ee040423fe9d13cba6e9050f1b90446b341e2bef0e6f2896eca1
a0d17df4288360aa7ecfb422db83fbcad328a74c34d1d8fc1017dcc30243b286
d3a713f64038e09e5aefb7e59df8fe5ce1ecc78285cf0a1fe776f58ef099be69
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832475cd13d72d07b635d20a7fe9505252add69a313570a00616429e46b91b0e
eff9b2f9b7da77690b7c39467b1f65e1d436c9e510a9c6519e65d85cc992b2a3
first message t

3deaea8fa5416f9c0460675a5031fd770433809480c203565e41236db8a56db4
beb16796e302397619f27257bae05568429855d968760d1a1578fe880d5e30dd
ace9c156b8ea23a4aab447344da5b47bd5bd2c5c1754ff2d69908afc707c6947
e676fff0c2ee92fcca7e4860671ec4613875f2976f2239922252c2c83f778455
dbc775f159639c568a3bd829618488b95cddfb230fd66cbccc50e8725508fd89
9d26411877da5d226b8ad7b0e47ff502630e82450ca71e877226d0d50057aae7
0b3f00b1035f8b55946fa4c53930d6119737b3efb1dca5a182c3ecff336918e0
d2800b3456ad3e4bfc2afbe8f1269b1a8eec2a27a7b413f7815e5492a8212a06
response z

2ff9ba427ceaae461b95955f66ba4b26b3ca3346e01c2ef94c166973d0bd5a9e
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170d533ca547dff77cdd7298b7f013a183f6b57c9143615d0a8fb75c564f0bad
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1fb7606e0d745f7de4ba6793e17c72f522c123faf24b1c5fc128c5dc2ee71442
0b516cafd5ae404d400c539328122598ffd809e86c86a8cf2c66fe8c4835192a
492f3874ef77caf49a5e7c2f1f24d1c673097cbbdf426b80efab982c59fdd242
response w

3493fc64f624596cb803f337f843ef934bf420de02dd1f3f1a84ec764f5bb119
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82aa9e003f3df3610fa5321d51c702e1709540e10f97f5d295cdaec93998cb49
53c7d8b6f676feabc4f353017de776e771f06872398d4570ba56026c8c9f29ea
1fb7bc652c3a5964664897dc2b408ea8ba577312fd10d23510e9d6cf9cd92165
cf98927cd9d9910b73fa7c068280e6b7b411695e210aeaccd336bd365ed8471f
87975207777319c8bfd945cbb403ea43b3cf5187d48f8eaf5b338b6748c226c
463fb00d32ef6690e634a577503f3f2b7a21ea0ef09158c1e727e71aebcc2b8

Fiat-Shamir context

dd660b98be5e491bfe4348c9fe5ea0015bab07e8ef0de0a9bc64c763acd9f797

withheld proof term: 73 091 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.QFramingSystem definition, QFramingExistence:160

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
          (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) → Prop
```

SHA-256 of the statement above

8d2fe14a199ab4443054ad373394a7d9498bedf70faf40c329c7d39de9975264

SHA-256 of the kernel type

90d9f47fa2d8f5c086932daa32cce2ee2062de5b54893e9a7d4288d8aca0e8ba

commitment C

6c2982b146ec3bdba3d0a8dac004a0e6623b06453b30ff96c6f7ad8693d9a6a4
 3b7a86812391716a4fd67e49daab85d70901acd150451ea30bac23ca160afb82
 bb440c701309bb3db475d87fd7af4aa0f96acf8e0920f597dc6916be19499ecd
 fc1724744b3493ac513c728c9a97f103a00ae256424cccc7d2d913d4bd9c8e9
 4f1f5303e88fba16f6a0ed88ec75eae5282edc124e1890f9d1348e826b6898f1
 e21290ddbf12f5f2b506b25ebb7458ed3bd44ba6bde043d2e40355f81c3c9815
 18e0f825ef0861b1abc23537a96de1aabe7d967318b5e90ef1c3c3175d98466a
 c057f0d2e16968f25bbff6e306d206e04b38416dee745a481f98180c9762f75d

first message t

c18b6338d62f15d54163e6cf4ed35de18b51c86dabb6f8d4ad21dd1e904b8ace
 a91b48977a13980a1e8accca27ca2612dddb1f65819c2b0484a472c9f117150a
 c3a9d0aaf01abc37f70d667782a5a9d373ce0a38e4880d08ac8b747e6d770cc9
 b1f0a73b0c033ef6dc1937e833df9e88f6b964eebde60041d104d9000b2f02ec
 8eb79529d873164d3097db689d812281d7891bb41f6e4c9d29be6b579a72fe95
 4d7900465e01e27abfec59e0700fd8f8224b2729a07f6fac4ce37ea600eff9bc
 0918162562d743d77315163a5969ed69a491457a5ec1b49277d58be4460fe6ac
 9c132313fb4c460523acce1ce2d07e3b4bfcf938b3d065920a2080e2d0cf356

response z

36fedf313c765fd58d3e59a6d7a96de4683e85450441141b373774b5fd1dc84c
 027754b40f2d8dc56b7983b7083d8e94239d029bcb8d5159759c36813698cf8
 5ece22aee2ad01806604f8d79497839172054c1ba4f27bbca73890cca76ccf88
 6b53564fa8c80ec679668e89d70305c2bf63cafcb466a95ca5fb149934706ab8
 b6c5267308dc712bacfa078d40f6472bf1fc6d98df4fe373ffebd92eee12bf72
 3efedbf38e76a21ea07ab60b189f3fa5e7d662bac7ca95f393372e4cf058c35f
 0a098d811ba9c36e90449019c7e7a69a95e4321260a863acdabf37844fd59da6
 b4a3f59e4716bd98ab9d83dd649a3c3129a66b178a632ffca2835a92b2a4e4e6

response w

4efda38afa2535669a9aa3403f7789fc4a461e60338e78d464dee49ff2ea229b
 ec7933bdfbf049f3b5256781f44377a5ac6368a03a11d0f8dda9d8c50a9ca686
 b1ecee6b68513aedad39697c985f0eafdc47092e332e19a08652bf97f180ad623
 89b09f5a5fe2e5340d7f5b0bdace1a8e8ebabec998cb43e5f7a9f462dfc66db
 d660472e2565586e4ce25321939517e2984404518bbd429acfb3093d20f38f94
 0f20eebf846b560e9628748a33ecf783f5ba885355d4a7358c381e8bd64555a9
 392c1d045305f1f08cf9b6e0d738701efb842cf8b9de6d8789fc59fbfcd36be
 e69825731b39065f4e5fa90ea520cdd33b71e22fec0f08189db6f0250a2cfd

Fiat-Shamir context

5dcca6597c9c3d455d22bdbbec06f0410bb0247ce19b1e8533d4aeb3086401b

withheld proof term: 6984 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cst_injective theorem, QFramingExistence:210

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} \{M M' : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R\},$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.cst}\{u_1\} M) (\text{FBundlePoint.cst}\{u_1\} M') \rightarrow \text{Eq}\{u_1 + 1\} M M'$

SHA-256 of the statement above

87a989f38a1c3449a821a78ac2471e8cebc64bd8c2ee1d7049f56ee4bcb2d399

SHA-256 of the kernel type

3df855eb235fdc86edf12f293eb4ee3c5df2cc59d15f78db93ceca297325dd09

commitment C

a0ec8495c6546538f0aee2c4c077b1c06c92ea248e7bdbb9da84a50e805f75d9

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32c734a619c080951bb2d97b7208e62ca0cc35006e3c8c37bcec347cd3bc229a

first message t

8e9d15b245cd3c4ee56b2517d15bee1e88a6902aac2465d10786cd5f83a526ea

62ed830bef974af14b3acbc1eedc8e9874598cdac0f583429755b6e26d0d2793

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5a78f71cd46843f7df7a8addc8159fa47bcf6c3e9d6ca74d029252a8e43d01aa

c1f2ee5a8c07a979eb5e266cb1e35e720e4388e39a1d5da8a843fedfd3d24e3

response z

d24303e13329ac5253247cfc279423b3efedc85e335e91ccfddb23e97c35d912

2060f47f287ace7da864375b7ee9888a982214cc4c1b689a95b0212e4fbf153d

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response w

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6ebf05a545ac209edd396ab5c01ccdc21ff3eed279dfbab45a4db0cda72c5391

Fiat-Shamir context

85af77a9556f2b59eb912197961a7398a41815b9b6ef1fdf24f6c25376b8f734

withheld proof term: 3 408 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_iff_of_sub_eq_sub theorem, QFramingExistence:196

$\forall \{M : \text{Type } u_2\} \text{ [inst : AddCommGroup.}\{u_2\} M] \{a \ b \ c \ d : M\},$
 $\text{Eq.}\{u_2 + 1\} (\text{HSub.hSub.}\{u_2, u_2\} a \ b) (\text{HSub.hSub.}\{u_2, u_2, u_2\} c \ d) \rightarrow$
 $\text{Iff } (\text{Eq.}\{u_2 + 1\} a \ b) (\text{Eq.}\{u_2 + 1\} c \ d)$

SHA-256 of the statement above

e9a0a7abdb93c7a8e094d102429922b5a544b762afc041e6d84bbf05fcbfcc7b

SHA-256 of the kernel type

c597e0dee7e7a8a026a628dfee2c7f50eab36389cd64ddc2b85d156103905c7e
commitment C

c902f0e9c09b86d15d05584f7a6d2a21c6f00755794b39239b98d05bec960c01
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9302cdf4565eb88f4146b4293f617d902500ddd8e6e078f98335ffdb1640e0ec
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first message t

efc1e125ef4662bfda390f5f31e9d2a55a3a5020cf4dd3bb6aec3dc49de074c
9eed664e04b8f06f826d3f7354ad4a9c33f17611337adfa7bc2d34328bc7070
f438d0b3b945750f63632e71c6f11c3ba735e39f87051c083db18a1a3219daeb
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2d3a5bbecdd90f119aeffdc5bfefa776295ddc277be11906804209ae2f6da682b
47c61bd840be458bf31dcef9da82b696966d44cb2650764255d320de7c38a178
540a623e79da5a0ffbe371d19006e4989145a11000f70529bf39fc7be7bc7fc5
response z

1da0207f8801c2eef82ba1db7380cf371c48d8a3d7d65243b5e638e9d3d53f99
f10b7be0e5c74aaeabb583a8c97bcf43f886bbdde87712ba2148c1f17c79da05
6ff2d222b744ed469d6133fda31c0cbe731ce324cd3bd8fd56fe5d8678da941f
284d244e4a3dfc5f598f9dba3aef67207f44894310d1647175df624d67a40530
eef30fedda93099c683aaffaa0559e3da5ed711c8bc03f6e07de2e46c65a25a1
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660b79cc5a9cd695358d0f5f43c8625f2f42533f1b4403e0579dddc44be2ce6b
response w

4cacdd5d9a687fc916e6659897d231681fc1c704685e5a9b6663057ed19883ff
e4352f5ab13bcec2dd2a529cb37989db09583d6b50dcca034782e7311c99cb3d
25f52d6217574a82f2abede301dcb082ac4fd54b84e802d2b659e1a1507dda1c
5e78d172dc287065121253af644a0f823f45759c13687b8a161ba6ca5bde428b
9ef6ffa049ae64eb695a3f170b716f55d7b16f946881b27131c0bcc4d6e62bdd
def71059043da4e4d5f95e46afbdd1f34dacc07f719eafeda8924a7e190c085a
ef51692e7d91dca5e194156759247469ae19ad77c8a6a78666a3037068c66844
fd3ef4033ee850f34dfadb014dfe29df76f0fa2f6d6626ec1e4a2e155e5335fd

Fiat-Shamir context

fae6b3986abd7008ac3e63d1f552c9bd35fa52e8c8d3523dd583c7c73f28ff5

withheld proof term: 16 087 bytes • axioms: propext • kernel replay: yes

FBundlePoint.existsUnique_qFramingSystem theorem, QFramingExistence:480

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) R),
  IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} A) →
  ∀ (Q : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (Q 0) (FBundlePoint.cst.{u_1} A) →
  ExistsUnique.{u_1 + 1} fun PQ =>
    And (Eq.{u_1 + 1} (PQ.1 0) 1) (FBundlePoint.QFramingSystem.{u_1} Q PQ.1 PQ.2)
```

SHA-256 of the statement above

f4809e39054f44eb46762e6232b834d84cc5b0fa29ae994b3af8fcf8845cd256

SHA-256 of the kernel type

25c9979c1e86a4a71f15670ba6175471e539c7ae5e6ee966cf04b431dea1a5df

commitment C

54a71c5ea10a7d09ec662306aa5ffbe392f9257d56910e1ff38b883f7f49af86
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cd9144a9a32d35610cbfee24748ed91a258c3c385ccb0857e437dd11adb2530b
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b587ea5abb8be88627bcc1107362b9939e9900ea8b2e6f91dbc0749a6dcb0f3e
first message t

fc2e59724626665efab6b753a3bba116c11d75fa1835eac10d63a312be1261cc
995df8524a807432dd256132619f2fb4d4f10c745c913b3cb23e9a70d15d20c7
8497d4a8723406f9de8fb4c11936ac1ee54097d5d3834df90fe585a20ee1703b
1773264acalad5fb13f697aced40ff082759e060291579045a89507d2a38582a
0233fd7eee4d2c1b43e0c985ab58ce24435a2fda5df9f5272ac3cc84c5910fc2
1b696af29a983196b5cc25612d1c2b8b30aeeec1f4efbaab73650398419b5b842
4b3f7687841a86a88df40db7416b25d0105069945b78ccd8c15d78bd5c4ee659
359005138a1a122faaaf8c1f0c642a563ca69b3e74f3438475689f9e6e194259
response z

7f1afe687310160cb6a6099edb48733bfaa1ef0e38ffda53d8b2f8f0d4b36996
9f43fd8bfbfab1fe48719440b132b6bb8dde25856a21ab9faeaefff42fcbfc159
bfb012ca71af2d873edb0abd4ffef1081cbfd2248c14cf306270c8dbd5bce358
49bc49ce391bbb9735d5a5539e04cd659a8b9cb3ab06d31268c6a65fb4719b45
b0162b3a34364aaca325d94096684e85dedab609fabaf0299cd028f3a5e055fd
68ff02b02f37387fe8a430ccaa9969d7266e9254ad4811c057c27dbbc533417d
08909a6c5d56c07067b07673361323fad95bd6f0dd537a00f5022eb8d880a8a6
0ef476afd770922ec5bc817403f5b1c89568f509fcae8d0a85881c22d62d8346
response w

481a31bf5fd4ae38017055cb30620366f9f42c02e00ef5f9c8b42b666cffa75c
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2847052a72009c152b1aeb6d061a797550f4ff4a6e7c2d6c7f8ff8db9077ef9e
b97fcec1f00939c93909d2c1e1ae459c81b8c98be00229d4d99dd317a2e8a916
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87fac06d797b1c704c4959eef31947b8c960365e1de84a962643f21d5fe741d2
c6e83ba6b67682c7147de4e735e81cf0c36465f34fd68bfc0ad205ab789f4898
4b7efa441c93c3c859af0be93043355db1188950b146a97495e254aa12f1b23b

Fiat-Shamir context

50b9e79959973c045f621f48fd3aca6762e10869fc8112d010e3118a7d1aca3a

withheld proof term: 76 626 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_succ_cst theorem, QFramingExistence:205

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] \{m : Nat\} (n : Nat)}$
 $(M : \text{Matrix.\{0, 0, u_1\} (Fin m) (Fin m) R}),$
 $\text{Eq.\{u_1 + 1\} (FBundlePoint.psCoeff.\{u_1\} (HAdd.hAdd.\{0, 0, 0\} n 1))}$
 $(\text{FBundlePoint.cst.\{u_1\} M}) 0$

SHA-256 of the statement above

e7693e7fa66648173cabe5bf4210bd2f80695e6f50fff6fba0cab3bc65c58b9a

SHA-256 of the kernel type

dc0e22c036da73bb1d9647d97573fd41da670ad2b81218d76ddc1062a1eed63a
commitment C

c0b3282a4fef9d44199def37c3f05d84268d479ea4cfbd6b3a5d508492faa9c1
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first message t

b533c4cee7dcc042d82884f04abb8868ae59f24eb43ee680531b0873f9b6b119
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8e5f016ed8c039b7d599336a2cada79dc3f297f3acbf055e6e5f847ef6cd6113
6da466decc3811adfc7d775c0f809ba8acedfbf6ea949fc12ea5988cc81707
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d822c7482694bf2cedbbb0254e0f2c7e7e54e44f6152cae67fd79ec47183f0b5
be2abae394b2c7f22119d05601100403d1e85b87d7270d6fc77638d0dbe7f4c
c1e8e1e0648b6cf89c6c63daf7b44adef0238e17840e274bc0f35193acad1933
response z

52a0856cedf4ded1d7cad11c1cc17c2dc81b1700a6a25fba97842425dfe1b7bb
fac2d55b78063455b0e4127d31ec882a9a08aae7fce747518dcfd63887ab9e37
19b896fa12ca51768fe8b8235ee41d54e3bb0e90229155351497ae4414584fff
e7ad52032f0ec6fdbdad4bc3f256e3a41c21ef306a3256db7fd2c35ecc88e257
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response w

21aee81367a9c5850682ea5d1a1250807a88d60ed1ddc73110fef3b83e09eac0
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b5ede7b6d064efacef215c824596ae5a2486056a4d1bc075ba1001679b3fe886
7a75067cd65b3229b349afff8cdee8773fce2e987c799728fa9684efbfaa0520
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Fiat-Shamir context

60f993623c81ed559cfe4a5ff02c7f9b771617d47921d8fcdaa57ac08877623b

withheld proof term: 35 555 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_zero_cst theorem, QFramingExistence:201

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (M : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R),$
Eq. $\{u_1 + 1\}$ (FBundlePoint.psCoeff. $\{u_1\}$ 0 (FBundlePoint.cst. $\{u_1\}$ M)) M

SHA-256 of the statement above

aef5b50ba4b5f7f4b994fe0065380debbd13dfd89d6b0032681c275a5654d830

SHA-256 of the kernel type

8219f3d945934c2ecc5f61c6d6c23fe052a4422355fc00124ddb534ffe3ed3b
commitment C

8cfb938ef0dd00fdbba4bb87eee4c9f5393216940c17d828e708a9e36070c590
87637132ff1c021fda86368ccd0bfe5a7c2ff4c8ef99ab5885801e59c508081c
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86561bc142995a2c5c794d9939e92647a274807007a1c765fdbcad4b38c3b968
49215cd5eb2e0bde05edd76b1a43d34832c7341641afc1c51c65fea09794417a
ff085ce2354a993b96314caf0d50fa5ec560961f053aed81caadd07d84a1ff7
first message t

d26b38e2223569bd9dc6d457703cd07a5d9d205510bf2788d0d91f6f2da6958a
3ee51568ef6b79bdd2c56d749391f7195671ccc6b2e067f58f9a412a2eba8cad
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a7bdd7d33e1a82dec34e131bb71a2c1ac36efee042f17899f7a183712953f8db
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3e79fefe262f4765f5bea40ff115c6885fbdff96785b0cb5303b61675c304a84
bd6940dc8584d4618ac4dc19e54a61c2922a9682354f6992789edf8570a3d7b2
3a0f10b874a7fa9e9dd1c2b1e39ddd9f58ff6234704787d957684ad02aae92a
response z

36c07ce1bfb709db01e2d4aff02eb0594c10642342f76893b45f2d69b21b93c7
e112755d6cbdfb4e0ec2b122d31d778a4915a0e13ff304ed2aba84a092a63129
1e49d36de6c4dabcf8524e8b7341694170a3c83ffaa0e2796cf3c1027c8a8c8f
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6225b8d6861ea632cda4d90b74ba4213e00c9864622c4b66faefab6894027d57
d5039624f0c26fe5ad8a779a7b9522be5eaf74bb20473d7a1154a0f5af40afd8
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6e4dbbaa1d8663e67dc7ccd676e6785928b348fdd40c8664089df27e96d5c5c
response w

78807fb3f98f9fa0cc7a15feed56185f5fca7c4ff0f2607cbf2eea9ad9e843fd
7b756ccfb768fa752f7e85426bcf18b54239b885c17488802c3fe5c44973a1ef
9acda74bcb9fbde06c77af26614d694aad064a85930abdd20127c7967c3ab12
ccd36d68ff0ed0c451c7b42fdab81a824c5fd3d1eb567ced1e3027af50c5931f
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47c94b70305aab47b5f57d9966baebdd816944a20a97e312d3e7d208eef593c7
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Fiat-Shamir context

ceee439642f9bf833d57e0f8a3e6f079b29c6cd61f5564f52b6288f0b9322231

withheld proof term: 29 545 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingAux definition, QFramingExistence:272

```
{R : Type u_1} →
  [inst : CommRing.{u_1} R] →
    {m : Nat} →
      [Algebra.{0, u_1} Rat R] →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) R →
          Nat →
            (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
              Nat →
                Prod.{u_1, u_1} (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
                  (Matrix.{0, 0, u_1} (Fin m) (Fin m) R)
```

SHA-256 of the statement above

fe857bc214455b43ec1c4fd7d134c8cc2bdf0767b73ea95c63cc83ce4111f886

SHA-256 of the kernel type

c38ed0babb626d2645095c31fdc7202de49631dd2239bf6b353a2b167491fa1d

commitment C

a49cf8131008499f08d315a9eae2507ab708690ff4cdb8f7e98ff8ffc22f0c7a
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c101bdab8438509b3cb4699a4bb1faf2bfd54b7fa7800bee52f95db7e63fb35f

first message t

40b598f9d9fcf583315e15461256bdeacbb5a67a26bd272e8495b9eabae533df
6dd99dd70e9228cdd740933193946364365a3975da8fb2e2e498b4ed0574f2d8
a390e7143a4dadfd925125a92d28682ef69d198eafd6ff2a4a86e95ab1d9f945
42fc8badcf8fe67f33979e5399e33520653d91b33b5764c6cdfc992d21c6c01fa
affefe52c466a726c5c575ffbf4235a70855ebb13fc8552091fbdcb43fccce55
b8e8d2922cd9a78e9682dc7bb46f630f96bbfe5b94c7493d50c96513a35f1c16
d18912ad4f920bec03b1526e9cc10acc90c74ac583ca89798f05cafb0d840d4e
871f1987621a55dee3ef1034680ddd37eb394ce17cc44f26342d5631d03b4876

response z

25a6035c4c14fc4f0b34ac8fb10538e6c9c07394c5252a33ef0bfa9eccc6f2e
f8ccebbaeaa5c9d8e8777c835baea772e02035c7b843f66961d4a15ff9404aa
519e6f7a555fa9870ff337cf5e5186714bbcb60c75791199b1ea3b2afe96148b3
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df8a72c910f79f4f48f333e3c8e2093b2298a00e227d95cbd954380d040cdd63
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75f19f2014eaf2c4f35a353e93a0138474170486faa56f91ff2b48df4b528d7f
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response w

4a7fb15c470986d7d933a6bd8d7f3a759ca21d44f3e0f02a0d828456679a40b3
e97785487112146421f992d5fca8471ae2b4601c0ffe66286252b4f90dfc23d3
69ac4c310e6d3a0e012b0e4fbef8112bf8ebac972ba14ecda370828f4ec4376
a491f8eb34453cba0f6468c73b20ecf7d83abbc2a616fbdcd8d5fdcc8fb34225c
9af482378a9d2901066523b32bcc032522add1eed909127580be88a35b69a5fc
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1c972d6e242e66ed103d994689e2c5bd986bc73a8d4a99a1ec4a33d9f340f475
d631c58992708258c9df2b8ef84f8f6b5bd12d3d782f7d96b42ad99c7ffef84e

Fiat-Shamir context

c8876ff50ba03fde1335a78c6e097ff1f4b566a758b08253b206f516f6200474

withheld proof term: 8160 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingAux.eq_1 theorem, QFramingExistence:0

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] \{m : Nat\} [inst_1 : Algebra.\{0, u_1\} Rat R]$
 (A : Matrix.\{0, 0, u_1\} (Fin m) (Fin m) R) (N : Nat)
 (Q : Nat $\rightarrow$ Matrix.\{0, 0, u_1\} (Fin m) (Fin m) (PowerSeries.\{u_1\} R)),
 Eq.\{u_1 + 1\} (FBundlePoint.qFramingAux.\{u_1\} A N Q 0) (Prod.mk.\{u_1, u_1\} 1 A)

SHA-256 of the statement above

421262ec541b7cd490ff6fe925d8d68bf0d8c498e265b922f67191760e13f1ba

SHA-256 of the kernel type

1ea65f189c49ae839cbf4196d72535262db09c6acc65f807d5f1488d19f83439
commitment C

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first message t

a05b7a4d30bbb3425374c344db0a36d9d77c1fd0870ed375ccdee7cac53163f8
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6ca09dec90282876137dc16fcc56c96f3b490ea5ed4a701aef04b52afd474d1
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98f4dc98303c1b6b0773bb6446e30543586ba2822985ca72218da1962831ce29
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response z

671669157f582fddf79c96b89615927120db707ccad58193376df4c71bf83efc
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response w

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Fiat-Shamir context

2130e143e8a0cac8a6a514bc57c51f92cfe3c0197373ad2689e598861593373e

withheld proof term: 539 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingAux.eq_2 theorem, QFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (N : Nat)
  (Q : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (n : Nat),
  Eq.{u_1 + 1} (FBundlePoint.qFramingAux.{u_1} A N Q (Nat.succ n))
    (Prod.mk.{u_1, u_1}
      (FBundlePoint.qStepP.{u_1} A N Q
        (fun j =>
          if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
            (FBundlePoint.qFramingAux.{u_1} A N Q j).1
          else 0)
        (fun j =>
          if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
            (FBundlePoint.qFramingAux.{u_1} A N Q j).2
          else 0)
      n)
    (FBundlePoint.qStepQt.{u_1} A Q
      (fun j =>
        if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
          (FBundlePoint.qFramingAux.{u_1} A N Q j).1
        else 0)
      (fun j =>
        if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
          (FBundlePoint.qFramingAux.{u_1} A N Q j).2
        else 0)
      (FBundlePoint.qStepP.{u_1} A N Q
        (fun j =>
          if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
            (FBundlePoint.qFramingAux.{u_1} A N Q j).1
          else 0)
        (fun j =>
          if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
            (FBundlePoint.qFramingAux.{u_1} A N Q j).2
          else 0)
      n)
    n))
```

SHA-256 of the statement above

a2c09c3261b07060071535e771f5ea2657e54dae7cc7469a947f6066f2c77d6c

SHA-256 of the kernel type

d3389d5721921452d230e8489795ddbfbf99224b7acdac5a4ded9ed524b3d45d

commitment C

5ea65cf02985cbeeb8537d47327212138ba0217e2ed4bf117af5f0fb5955e856
15f346c8bbf2df70fff199a3e6549062b64ab693a6fb2f92ea0a3e187a501600
5891abc404cf8b7990bf0f6c34a498b855fc3fd0a8f6294532d4bf866eb508f3
05ba7e03160f8fdaffc749e4059eb51e4fc88d20ccba8f6ec838d11869f7a0b
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6ff33242924907af5e39e7fdf0bbde546368c437ad64998384ba99cabf4017f9
66c625a030fe91ff21f528efa65ad0db97f41cf1ec480e6b64c016c97d547737
5ce5ef7ea0e7deb4862163ca09282f9d04fa54bec7238856b402050fedce31dd
first message t
70124ace940e7664e20680c757297cd7ab0204ef38839074bfeb1bf15c6c8b62
c2cee42cf617340ee0e397bba5819bfcffd784252a7e32e3d44cfe0d8f39fb84
26677d2aaf4c07732f420251db32314ee8e1b9a0797ce418574010473903c0a8
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d247c0bad34204f0b87dd0770e0672274d0e7f1213222f5aaf01cb86b6a4694f
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f803a5973d02d528fcb37396c905faf0987a1963bb9cfbb6358a820135b2f994
2e6103f0d01b13e2d88b968a17a7af2a3d19ec20c891b78038410401648118fc
response z
186d794ebda3a6e5a434ece3a05e4ac1c97b1a2264406cd24b48ebf4b0cdc41a
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72daefe7f2d4f17f6497e3ea1c3f994e2e91a07aef7ad89b723f784cb2ac566c
33fdde96207aaeb224a3f2acf5a1be80a23a87085617de76bed62c759a1688cb

FBundlePoint.qFramingAux.eq_def theorem, QFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (N : Nat)
  (Q : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (x : Nat),
  Eq.{u_1 + 1} (FBundlePoint.qFramingAux.{u_1} A N Q x)
  (match x with
  | 0 => Prod.mk.{u_1, u_1} 1 A
  | Nat.succ n =>
    have P := fun j =>
      if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
        (FBundlePoint.qFramingAux.{u_1} A N Q j).1
      else 0;
    have Qt := fun j =>
      if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
        (FBundlePoint.qFramingAux.{u_1} A N Q j).2
      else 0;
    have p := FBundlePoint.qStepP.{u_1} A N Q P Qt n;
    Prod.mk.{u_1, u_1} p (FBundlePoint.qStepQt.{u_1} A Q P Qt p n))
```

SHA-256 of the statement above

edde636b77cb4e4d671d76c2583e942829694c98535a1577b06e965d2e9846c9

SHA-256 of the kernel type

f27d2359552ef3e80936e0a39aa906741e788b7c61e2ecbe6da9bbf18f67da46

commitment C

f9a3399b6c5643507f288cae406d421c2bb0cda8cc82816d41d84a300b1c4e64
de7612501b6b7d7c5105893b1f94e006b0a93491fccf32b5485bc37d9b123ade
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ade8048edf9fd97720b75b460cb357644be4ea923829e1b9c8861366ebbd031a
fa203f0be33a4bf4bf002dce7f428efcd090d179079e22247de4fbd1eb6458ea
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first message t

c3fc7bb77cba3712994df634ac0b647c399cbadfa63bbcade54e527fa9282e1d
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response z

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20aba4ba8481ed534b13074ade3a1b02b495760ee90a1bcd8cdee8845848d133
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c84c9b4f158af5cb55a2d281ca01176a042cb58da90f6eafb503d4b45b035284
9baf262231afa3d81a974764f51040611c852d855690e83b5d6e807585968713
9670af0d60279a0f4bbd804482eb2c0e29639ca847f0f8dd2514b1512efe6b1e
0c0d39bad52202b2e825904fd1137c703c945eaa48fd5a9e5d88cfb64fcb352
response w

17b36a21607dcfad9180a1eb130a32063d1ff47fea8fb0d9d67503d5cead5301
d8bac71c4b705644c332955cad26ef54216046e5dd697cb22a2e1ad391c5d7ca
6284d835ace3d49ddd1cc5c3808d615f4f15980e8793d58b0f925017b3a7745b
f12989a73a3dfaea1a5e4818783fb29e7f43b2a9c318dbd4ca5dd9778f2c673a
0caf6bc4430fa9125b0bbb2f0b7e159f80ef45c704d8dd03a5617662bf6896d6
905c8c77183a5de862d16eaa5508fb5aa22bfb71e493f24f2c18f13daa825a23
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c0098d8eb7f3c337465d785f3fdd3b6430abd4102b56e1e2b6e118a8a3b2f12d
Fiat-Shamir context

af18dc63e2137698f8de4c1e9c2d3a40fb3ccba881f6f206c317b09d3c49331d

withheld proof term: 182 898 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingAux.match_1 definition, QFramingExistence:0

(motive : Nat → Sort u_1) →

(x : Nat) → (Unit → motive 0) → ((n : Nat) → motive (Nat.succ n)) → motive x

SHA-256 of the statement above

52d0eb6c42572858dd0767c6141fae90019d12278bdaa57490569441b3aa333b

SHA-256 of the kernel type

33199dd6f4f3fc98e061fa050907a3626b9b333d6ebb731a5a906e8afab9faac

commitment C

c48b56f3c3bd7cd295e3f865827ae23f6bdf888e83dc39c3b8b84db408e3685

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first message t

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27946ab9235e2d4e3598d19a78e39f2c06be78e4a89edc3f37e6da8a6e2334e1

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response z

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77deb75c49275d0cd3caa5baf06ce7f38b21add1547b88f4279eb9f28946b0f3

d4d29f3e868b6c32e174eac8084e5506e84431bfbc388762289ed32d2e2ef241

3130a6f45b0e9c70c80a8f7f32d3672450d18afb43188b376bbdcab73b4cc89c

60850fc9b76fddb10f0e6882d5a9a2c98bacddf6fa482bb5d672efd2b29a974e

6c22c5628928ae9526612d71a221d240cc6f0d0759801b5653a8a88e5ffcba13

response w

2be985f8ed887770880bb67c7eb7eb1b5f12140101a9892c58d8c7dc9ffdfc3

aab55dd3d2dfd33e5faa24904cd0b6077962447d35406f52dcf3117fe99acc12

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17bc2887810c3adeae71b7391a8d2569659b983e284682eba71376b9a542281e

de58cc2a6ca8d7ae113a19b28c13cca96887549061ad321246de869efacec49b

9d6e94f9c304fe6a30dd9206f143c4a707ca33381d0e945ec74e0ff3658acfd8

Fiat-Shamir context

9f6a145cca23eb938a7d6f03e4660c47ad384ad12144d4c1daa8783d35fd07d3

withheld proof term: 610 bytes • axioms: • kernel replay: yes

FBundlePoint.qFramingEq_iff theorem, QFramingExistence:215

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin
  m) R)
  (Q P : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Qt : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (n : Nat),
  Eq.{u_1 + 1} (Q 0) (FBundlePoint.cst.{u_1} A) →
    Eq.{u_1 + 1} (P 0) 1 →
      Eq.{u_1 + 1} (Qt 0) A →
        Iff
          (Eq.{u_1 + 1}
            (HAdd.hAdd.{u_1, u_1, u_1}
              (HSMul.hSMul.{u_1, u_1, u_1} (↑(HAdd.hAdd.{0, 0, 0} n 1))
                (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (P (HAdd.hAdd.{0, 0, 0} n
                  1))))))
            (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} (HAdd.hAdd.{0, 0, 0} n 1),
              HMul.hMul.{u_1, u_1, u_1} (Q q.1) (P q.2)))
            (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} (HAdd.hAdd.{0, 0, 0} n 1),
              HMul.hMul.{u_1, u_1, u_1} (P q.1) (FBundlePoint.cst.{u_1} (Qt q.2))))
            (Eq.{u_1 + 1}
              (HSub.hSub.{u_1, u_1, u_1}
                (HAdd.hAdd.{u_1, u_1, u_1}
                  (HSMul.hSMul.{u_1, u_1, u_1} (↑(HAdd.hAdd.{0, 0, 0} n 1))
                    (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (P (HAdd.hAdd.{0, 0, 0}
                      n 1))))))
                  (HSub.hSub.{u_1, u_1, u_1}
                    (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cst.{u_1} A)
                      (P (HAdd.hAdd.{0, 0, 0} n 1)))
                    (HMul.hMul.{u_1, u_1, u_1} (P (HAdd.hAdd.{0, 0, 0} n 1))
                      (FBundlePoint.cst.{u_1} A))))
                  (FBundlePoint.cst.{u_1} (Qt (HAdd.hAdd.{0, 0, 0} n 1))))
              (FBundlePoint.qV.{u_1} Q P Qt (HAdd.hAdd.{0, 0, 0} n 1))))
```

SHA-256 of the statement above

97c9272ef474c8666955d3d6d1005a5744dfdf42697306b407594490f509ea99

SHA-256 of the kernel type

30de7fe8a3fae0a39433ccbefc38cfc79733e577432a4c62a4f1cc6c9a52b210

commitment C

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first message t

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response z

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response w

811da5e55faeb71c387f9fc81127543b9fd5ba32b8ae7a9d8129befd6fa61982

FBundlePoint.qFramingP definition, QFramingExistence:286

```
{R : Type u_1} →
  [inst : CommRing.{u_1} R] →
    {m : Nat} →
      [Algebra.{0, u_1} Rat R] →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) R →
          Nat →
            (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
              Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

8dcb4499350126b24a4f6d4cd670499fff7bc86bb3e6f875eff2e49fc57f8058

SHA-256 of the kernel type

f3cb0fca24bffb3522be5195f25e2c1cd92771d9744d873544107f3dc9520c0f
commitment C

9994cf762d441799f1676e715fb90a199bd62f69cd02cd8433b58c1c558d5091
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804f4b1c54b36909bebfba638e2b60220b2f0db1c6332f3f149d5cbd6392459a
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969aeab788f9719a45cd0c7edd4c3a6c91b130bf5c7f71e22851078bfa8ca9d8
first message t

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9cf8c837f16a75bbf1d83eace8ed9b7f8b23b0177c726c72de905321eaa6b4f7
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a7ac58798bfbf1229fccbf00e0008a797a402ca9de1fb95af5f5b1b23177810
95953ca99a54a0e1ab60303e1f9ab83ed43daee69518067cf6c596e3a0d15ff
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response z

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6a6120f34a09dcebe3926e9afc3912991fbec24f987bffb62f4b104ad8f71669
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74ef08fb420227ecaeca1ba6d5f191b8c130e883a173dbafd97bca46ecc34197
7f94000179dcd308fada734c89d772ab88270816c0a6ab3739e58d056c3402ec
response w

24a89658329c1f5b65e4bd56df683ca9cc22eb72b5e6615ee468ed603c8a30dc
b8c14de623a98f263ca025f9153463f2fac1526639854c8336171895e6819902
7d7e985c9a8038b25d5c313fdf3c6923643b46dea5215777e5adfdf5a860864a
9db1b4c420d5fab4c17967266cd2e051a8a2cc9cce95f2916a78ec386ea415f8
8d04e535ce3101e87c7690bf5ef80ba0b3f8b6197512a4a2ff9e1a37e1ff36c0
39642b3d134ae933934e170a534af6c90f3d6bf22d54866af748374a620b165d
3f7fe3e8e3bfc7292f2ecc048c96429572352198ff3a34e44bf4ebfd68507f60
2d437c88a9168f060d29f0fd83ab4b86b183a9706dc9abccbb4f2274d31311b1

Fiat-Shamir context

12f2ac00b0871daf31533257e6caa911ea7608690b37e06171b708f4898e52d6

withheld proof term: 799 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingP.eq_1 theorem, QFramingExistence:0

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] \{m : Nat\} [inst_1 : Algebra.\{0, u_1\} \text{Rat } R]$
 (A : Matrix.\{0, 0, u_1\} (Fin m) (Fin m) R) (N : Nat)
 (Q : Nat $\rightarrow$ Matrix.\{0, 0, u_1\} (Fin m) (Fin m) (PowerSeries.\{u_1\} R)) (n : Nat),
 Eq.\{u_1 + 1\} (FBundlePoint.qFramingP.\{u_1\} A N Q n) (FBundlePoint.qFramingAux.\{u_1\} A N Q
 n).1

SHA-256 of the statement above

cbfbdc921f53fbfe0e533534ffdd1df02e06fdd2df2aba79d6fb529ceeb438f5e

SHA-256 of the kernel type

38b1c57a5045ccabad1f04cc4815f541e4875700f56b3e545229a89a57307af0

commitment C

4e94273a394806834a277e8f8ffca30015e63a85fb83135cc40f709cf64c54f
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78d2116ac6a490d0d6732a333966910fccf1d22b6b1a8ad30ceae60a61bfe678
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959324b6664bb03c2ab99e1af1ae687d35d8734c7c87abb0d651fd2bcc0423ef
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0ffb59f70d4f25a6c25742cbdbaf00852c1995ee56a522459395f55e4c91e650
38b44e5fa8015f8b949437f069a214342e8c233eb40af357a676895049f1dcc4

first message t

ac828c854c24d6dbb1d9ea96a6cf7f6607c2bdc687f65d6187eb06d42b1b8465
065758de64729f1d167df711b5e96331e0f297fe0947470fb7b0c949ef5bad18
cf210ea1fa5747fb2ac44bcfdab4f10d06b52e67fa9174b3cad824e025deba83
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dddb44bac8a6533cab99f280704dcfd4ec51d90e7b288cb5edd9fb3264b81579
0ccfeb51f6ef534056ae126f6cd6e1d312711f33bad67504ef862f01fbc513ee
2ae7ae2b6478aaf4d89315371711c869657d0ef326fdb8f61a79311d473daf15

response z

619c9512063202ff6ac4da7622548b0449f3d6e7fdc083b00106b6e56e00c33a
3604b47282a16fa3a37e505f0b713729896a4e6424561dc1f80a3ee704e37311
74803b70e7639408eebbca32e3a6f08f7d78a98ef042a3e8dc4602abc9008c1e
c2e2b53ced614e69e8adb92cec2a1d76e8a50ee3161184ae3e666932d23c32eb
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a42dbdf17038d6d3beba0645b1e164d27152f7cc2200df8579c9ab8ac96d1554

response w

75c7293c223e447168f4db5caa684286d02d99178029bab12bead6d06b80b5e1
4f8527edfdaaf0a9851e7516f2d89c66de1a893526a0f63601e9330452e8fe7b
b1b1d2d5bc3e9239f00ecaef2f0e7efe0a6b41523fd42dbe8942bb608f858363
eff07eb0eac49936fddcd847b83072af9194a389462e60e47a3410d9ac531ab8f
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Fiat-Shamir context

1a439a5e682d11dbe74bc66b66c19a62af6c6d6a6874fc037d8c54655e3ef696

withheld proof term: 451 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingP_succ theorem, QFramingExistence:307

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(A : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R) (N : \text{Nat})$
 $(Q : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)) (n : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.qFramingP}\{u_1\} A N Q (\text{HAdd.hAdd}\{0, 0, 0\} n 1))$
 $(\text{FBundlePoint.qStepP}\{u_1\} A N Q (\text{FBundlePoint.qFramingP}\{u_1\} A N Q))$
 $(\text{FBundlePoint.qFramingQt}\{u_1\} A N Q) n)$

SHA-256 of the statement above

f8d3177fdfff2d0f16f220e088bf056c0a0bd43f7849b76d3e7c792706ae593b

SHA-256 of the kernel type

c54688ef494b3c1412b7d18a67fc75dcc13814b1ccf6b2d8c008519861702673
commitment C

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first message t

340bfc254638271a8f54575998ae0ea3911c29883b484d889b2cfdeb1933845e
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cac6cbb881447ab639043d30ddfd9a996088dc168014f3650cd1a04ad8691f2
77caaf6f7bc670daecbe99a846fd62e7f2aa8413d1c9dfb3159f16bfa03bc52
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7edad7e18c23055badea7a41688bd527f50858052336540197f7622e5d5d02bb
f9c318249eac783965bdce1547da382c9de3069d573943cd1db9abfe8bf04b0f
response z

68cacb8f9b5449daa1b6f00323c4776904eb74d9cc8e30a8f78d152268eeb0d8
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622310183410f6ce771c00cb55981d67610149e15de174d8bd332641749c3b28
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d11eb1550683ee1faed733eb5a768260912deb0ae04bb8bcb3f30f0382ab768
response w

42f37e913011542754e58496c6dccc5d0d330a4589b7c2291d396651b98b04
07ecf804c117651071031d410034abab8f4e407311e7857730120a6baa749ac5
4bf793899ff6c61b771687c445796d0c6089293e4ddcca4a6edb2c865676c192
b19b31011f9fcb40c639328ff3a3f2058dbb832d5995943d46484255a0526e5d
d2e9d9d2f97fc5b4ebfb847b926263b2ebf313605ecc6d9b5dc126de7b2c9515
84967857f1560ef1f0d75d28a8ab91f8d6af440a743057ac1abc57a7af574968
9a1d60afad94cfa948a796bfc0a73a410ff7a2839cee2ad6928ce8080ba6df11
8bd9a83cc96fa83008508d7e38e57d4888a10de2c9dd2aed360b0919585792af
Fiat-Shamir context

b362733484c0cf3b19db07d78241c741a86dd78ab912a8e3ab1aafacdba994cf

withheld proof term: 72 783 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingP_zero theorem, QFramingExistence:297

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(A : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R) (N : \text{Nat})$
 $(Q : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.qFramingP}\{u_1\} A N Q 0) 1$

SHA-256 of the statement above

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SHA-256 of the kernel type

75582dacd470f71c48fa6a7795c14f479f5aa5a1cb67bf2e5b9ef8436abad74a

commitment C

ca258bf10468f6fe0ac1986f28d511284e4560f45bfdbbe35d40a6f33b33192b
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first message t

e82bab5da3315042d96981a861d2e80778d5362b1eb29d3bb68e6123cebb6295
ab054686f36f5feb26caed3f441b6cce0b0aa30390dd9431f89b19f4f10eea4d
c78a149360b3e54b60d041b900919c5074d1ec50407312acf509e019309178b8
c21e42471a3053c544914ab0ace75289b5212988daefbfa21e552debcdc8eb59
5f02429445c8634795fcd90f54891b8d69620e4ee2625edc32251cc01953dde0
0566ff79aaac9e4e2d52b1243c268cd32b599040118b6253c82b8175d6cc44a7
a3a313ae13218d3c016e7d00472a70295b9345f3645d70a3b40c9b965229252d
63508114310cd6236f28f4bc4b55675321a60c58e297595e48121d7694cea5c4
response z

4bc5bf6765dbf3769ddb8206206bd93857242ff2b36249617b5069cd5099a088
618e096dcc56da461ae1b17c7df2a7202df4bd0974be1f89024c8e269c6589a
3fa3a3dfee5af5711d97ab93209f36a7b7b9e32d98afcc37790b12f3d063b960
f757981e948eedd59fd128eeef8cb54af65390948f9a85b94b1ae538ee52ded
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026dfc467dc7f0c4fc054141d52698bcbf881abfff002bd44177b887a5089f97
234eab9f82379a6f4630adfd146ce125a13538631dde89ab0667496f645adff7
eb613b9beba90e376cf748a90154229c4d121edf7f6bf340245ae07d9e54332f
response w

4159b8119946d509a8d07d0b8769f3f0079e67382ab00cde776c3a13854274d5
788a95d633d9f45033f7918955ce996bc484183e182f2ae8d705a31267ed1b23
343c67ce75ac49b1c0a6d7bf3dab07609d32ae8dbf05f05bfbbdc41b4fffbf7e
b7688876658cea5fb455eb7bfa83096b8b781451df7db8657409d030d7e1ab5c
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82ea871988b4ca5c576b081d19e8e53a3bc2f8343904008c4f22ba765ebfc4f2
c532d5b6f56090c0b1b9cd462dc8c73186531a7e8965fb1968348e1b1f080e54
6ca3ac186bc597d3528b233b323432017affc6ddb9d0c987f380f685b4878138

Fiat-Shamir context

203569aa2689ee6a08f4e3a91213a973b785b30c8ff017b41ead4f1b4f9e0306

withheld proof term: 17 025 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingQt definition, QFramingExistence:291

```
{R : Type u_1} →
  [inst : CommRing.{u_1} R] →
    {m : Nat} →
      [Algebra.{0, u_1} Rat R] →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) R →
          Nat →
            (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
              Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R
```

SHA-256 of the statement above

a1e1474db451a0807e20e968e121ac6805053a3e14e1a7451dd03dd8e1c8dffe

SHA-256 of the kernel type

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commitment C

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first message t

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response z

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response w

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Fiat-Shamir context

c5576feb979e33cafa91903f026b5b268cd543eddb894e7e014bbf40a5b08d37

withheld proof term: 799 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingQt.eq_1 theorem, QFramingExistence:0

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(A : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R) (N : \text{Nat})$
 $(Q : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)) (n : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.qFramingQt}\{u_1\} A N Q n) (\text{FBundlePoint.qFramingAux}\{u_1\} A N$
 $Q n).2$

SHA-256 of the statement above

6485993b6e8b80ae2b29a4a98cc0737dfb4d08fbfea9c8d2eb6e3c34f94f246e

SHA-256 of the kernel type

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commitment C

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first message t

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7e0d448d817d542bf9f079bef6dc4b5c1bf39618f168fc58a09e7cfc61c1cff8

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dd729ac92f69259789f5dbefb5bf5878a531ecf212ddf6256480ebe1bdc01c62

response z

2ac53871e10c96e9fa6d0f6adf34256d6734644186312c64442dd72804c45e39

0d5a6700825e8291e7d76009655b52e60ffcdca9899bf2d21593bda93981dba3a

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73c23ec7dfe9c59f65884a4fc364f090182e9d13965c79568ed64700f77138c0

44c3022aaa8654a2c6bf8739a77dac9bd4e3bc8eb5713a12a257a8623832af52

response w

109f16d656c4daca6ec095cc6cc135bc741253b14f4dcc123eaf0934e62d2597

72db91f2e53e719e1d6b24f00dc4189e4c9af4b6fe64eae356641d88cecb49b

77c9b17bcc41b6766e1ab8a300c504ab230ba11ff204ed02eda46fc1866314ff

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3e1829969b61e6c7d5430a35d7fba9042ee22433e52b91268e8b79b0a7d759eb

Fiat-Shamir context

06da4cfd87b028fd20b2a2198a1951069ccfa30b2a1491fed7de3160104c33f5

withheld proof term: 432 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingQt_succ theorem, QFramingExistence:317

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(A : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R) (N : \text{Nat})$
 $(Q : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)) (n : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.qFramingQt}\{u_1\} A N Q (\text{HAdd.hAdd}\{0, 0, 0\} n 1))$
 $(\text{FBundlePoint.qStepQt}\{u_1\} A Q (\text{FBundlePoint.qFramingP}\{u_1\} A N Q))$
 $(\text{FBundlePoint.qFramingQt}\{u_1\} A N Q)$
 $(\text{FBundlePoint.qFramingP}\{u_1\} A N Q (\text{HAdd.hAdd}\{0, 0, 0\} n 1)) n)$

SHA-256 of the statement above

1022d386b6fdeb7d6561545b69158d29459dda6d0b96b28302600ff2454221c9

SHA-256 of the kernel type

7bef06374914c22c779b541caa48d84e3e4827f9069202cf564b1480549084c0

commitment C

98dc7e1d9afe1418135919644ad48d1674f116295371efb091060cb6e125ac32

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first message t

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response z

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db1b5f84283ccf46e86d8bb611572c14dd62d1a6478c5c34ff0cf2218a837b91

response w

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f046aa78f7e92c9f47aca717f2fb4c12ae1ab0d6a9357616a0831065fc99793e

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Fiat-Shamir context

b53707f4e8c0ecc5b84d207d51624be2fd31aa47597159880d23afec8fd894b5

withheld proof term: 213 768 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingQt_zero theorem, QFramingExistence:302

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] \{m : Nat\} [inst_1 : Algebra.\{0, u_1\} \text{Rat } R]$
 $(A : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R) (N : \text{Nat})$
 $(Q : \text{Nat} \rightarrow \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} R)),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.qFramingQt}.\{u_1\} A N Q 0) A$

SHA-256 of the statement above

f1fde4f2345d6bab582935020b95307edf81dbef4a092432c8a4655cd6c90a07

SHA-256 of the kernel type

352c933816e977552fa516c151f5aeb58cc10450ef512af0afed67ac60b3bdbe
commitment C

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first message t

e471b0e38b594766bb2f460ab4cfb7d802aa8e7d02bffb561449f34ee35222f3
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60754d03cc3ca6deb420937a99aeaa68f04f42484a5c2c1a0b4ecd4b1b8857cb
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d249ccc8dc3ec92b67ce55daa2150b9ef07c5f73b9ca97f858620134c90f5338
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82029e0f3d7b3dc3a35c5d5bf1a1cf4861d70ea907ff0e85caea329dbccabb4f
34a25c049054b7d0f3ab0925057f8e151e03feac3e38ae4aea389546c94cc4b8
response z

48c282d91b4557081fbbe773c7798bb0e4c5902135dae8e9270a7cb92c69f2ab
c291b2f84ff213fba07e22881a800864c4167eca7a39cb064e54e9d726804fad
599bf587cb3f2c005ebf999341d4ef8d0e6ab3d0c8bf819510af37a7e9b36773
e2bd66265afb6fb916c96e31e67dec24821c520e9268e8378b081287bb0d3c5
74f29a0ab3cb9e8e801ebecfdd93fbd08317ed1ba29a6654993ab78c67acbe22
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6c6996c30fabd1df5e8b5c646f5f8213e8cb0ba9deec900323f47628b456a3fa
response w

6715bf4b0e1326f5feba5a5cb936245745b57a2e8e489ecd8eba2ea57e976ce0
47981153752a7c434b9a3887689c5cf94b3e6fa5569d2099d97d2427d63045eb
1788d20f0a5aba2ef8f85cd4af2cd38fb49b1ff199a97c99d18ca6b7bac09b1f
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Fiat-Shamir context

89e2b64eb500b9063a0f29982cd51021d78a596971ac2426e4838b99c9a7905a

withheld proof term: 8 739 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingSystem_qFramingP_qFramingQt theorem, QFramingExistence:361

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(A : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R) (N : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} (\text{FBundlePoint.adMatR}\{u_1\} A) N) 0 \rightarrow$
 $\forall (Q : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $\text{Eq}\{u_1 + 1\} (Q 0) (\text{FBundlePoint.cst}\{u_1\} A) \rightarrow$
 $\text{FBundlePoint.QFramingSystem}\{u_1\} Q (\text{FBundlePoint.qFramingP}\{u_1\} A N Q)$
 $(\text{FBundlePoint.qFramingQt}\{u_1\} A N Q)$

SHA-256 of the statement above

4d7047043c9ab7361979eefbe5e12a11b7356b77dcec460331d34a039b3204d5

SHA-256 of the kernel type

722c9c64e1d9c09b241b5ac6d66b74fa9f76c12df1f13fecae624c4f5692eb6a

commitment C

87258fbf16a37e1a18826034642cd63b369a5429f064d2332914fd19c92e7111
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eed5878cedc9516fd06c78ac4585de1c4a850fe83e1613c7983a77a40e98a302
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a2f112724e43df1fc1e345855b2865725ea93825f18d320f4ba2b1f3b3b965be
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first message t

d43cb66e4588a10cecf8e85675b7c9f9e725bf20308af0cc0e46e6bd61548ada
0e702e5e7c1bc7b3a149f9816a46c19538e72dedbd20855b650271856cfd4633
34e6f59b71812dfba11dd16d1f9c57527126f800b21cdeb1add13dee6d009178
047e382ce06b589d958bd5d49acac62003322a3c9f9dc7372ebdb0ee004d69e5
ae98729b523baf0d90f985f81d9cb868a65bea8793e48d459b8ccf5492f9c53c
6eceedf8a4157d057f8bd77f950dec5aa4759487e4c347cf35b96794f9dcb565
14899202e454494b009600087e5794a30f39a8a508696e2d924402fa87ffdf00
9ce337518724ba07650aa3d6ef7423fbe4075e565f95449e821d6b5a0ab1065f
response z

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76b6a3058ef54cf817dfa0d7b40733847abb46c1c0b43674006bac91da250c69
f6e3c7953cf3c96b496d7eb297dd6a1621eb27e83a683e076c5f52df5a6d5027
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4833a8d74365a6bb4648930327c37f8db9a7cb51a9033fe85cd01f8e8965672e
da11cb548d75545ccb3f09a3dd192af27f47587a371b5e3bbeb2bb9802913d38
response w

4a84ec3c605b26e261b01b5a42c780f4c7faaf25d9effaaea2562b58925bb3c2
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Fiat-Shamir context

a9a662104479ca4a909162e0488af4ba9fd560e80dd6a1823b5164147cbbfb2e

withheld proof term: 325 242 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qFramingSystem_zero theorem, QFramingExistence:378

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] \{m : Nat\} (A : Matrix.\{0, 0, u_1\} (Fin m) (Fin m) R)}$
 $(Q P : \text{Nat} \rightarrow \text{Matrix}.\{0, 0, u_1\} (Fin m) (Fin m) (\text{PowerSeries}.\{u_1\} R))$
 $(Qt : \text{Nat} \rightarrow \text{Matrix}.\{0, 0, u_1\} (Fin m) (Fin m) R),$
 $\text{Eq}.\{u_1 + 1\} (Q 0) (\text{FBundlePoint.cst}.\{u_1\} A) \rightarrow$
 $\text{Eq}.\{u_1 + 1\} (P 0) 1 \rightarrow \text{FBundlePoint.QFramingSystem}.\{u_1\} Q P Qt \rightarrow \text{Eq}.\{u_1 + 1\} (Qt 0) A$

SHA-256 of the statement above

5837bc07274dfe07aa7f6e35331c5ffc26f2c7f794b003998cad8a60dacdcfad

SHA-256 of the kernel type

ad02f42a84083e00d81e17c2d79d8beeb1209e6e828b4efae0f4fc11cfb91cb3
commitment C

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52a5456b9c9fab032f377e89c411c9cb38ec8fd15b406e6be1fbf70b3c886f03
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first message t

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802399d65e1b95a872b0c37ae01c779ff515de7e5115211d1428cda0429154f0
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response z

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response w

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388ec34821ac1f079ac9fe150472d1722372ff3eb1b160a62244d558228465dc
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Fiat-Shamir context

9b929508b140c992e4f1d83aee06dceb845ed497f1ba23bc24071453cbf4317e

withheld proof term: 127 448 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qStepP definition, QFramingExistence:239

```
{R : Type u_1} →
  [inst : CommRing.{u_1} R] →
    {m : Nat} →
      [Algebra.{0, u_1} Rat R] →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) R →
          Nat →
            (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
              (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
                (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) →
                  Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

fa96327d6623f227aa73d09ec906c03298efa4ad11554aa84041b08ea0a72e9b

SHA-256 of the kernel type

96f485a146337aa44a41f1e438ea19aa1466d68b6b3216c326fac343946a15ac

commitment C

bc40c5c5f9bcc69eeff1305ee4555e93324978826884228216c1ab00b378c238
99165566797fab077eb1c6c52945e92fee88491bbd6c07c3b3e3fa7bec34a33d
4e835fe45a40f1a13292aba7df3670b7418ac5f83b0a82295e1c6554de926a8e
757849c2201ac5c9ba692315562b13366a2e4dab344ab36cbe0ae0e22c764bb7
54c654a54d4b0d44b8157fd40fb573e7e45a6ef31bd94b1e0bb96a2761fe44f3
ab2b34654766de55ab2d5db2782970dde80dde470551dfa7242e0469abb02a65
3a0d6b89bc54331834bd3546efac879731f49812b7174ebb0333172b60c04068
dcf22947bdbafec63ec43284063d3db09d73238360d54843d913092067e6f287

first message t

82f4b87508e8f5ac7364e73c19dc46d31bc5bc58649845da977ad90e58b6b93c
b614f4521261cdea94021707aebef7f25499e604e08f57c007934a98982b7af48
c9f53ebb8ca89e911950ea5639b1c4af84e9c0793a46b7508b233ddaa96294d2
2d4dc5d07ff619628b57a1f45883a523d005bcb2bcb5cfc9f20b231ae6015c
68b4a57a2755116d97b941095725d7c036c581e0f6d5971c3515fe70b376fc6e
98a21496854ec6cf9863af5b7f849fd39854e88ca9938ae46782d869b758b253
77f053c1e92cca928688b0369cdc254f4e32012e9c4b6befcd146f8c6549d4ec
a6cc7846faaf00cf36018dffd1baacce8befec96275bb754c061cb8c5e1af9e9

response z

854f3882deaa9e51ea2afe49ae6067fabfb61738830ba81b0c0bad1bbef23eca
5f3fce7579861e4b03b35b8bcd78f48c17b778572d4e942a0f53c293f0606499
28d68038eea857568929ea3d147a626341e8f1be4df97faa63363993f860b3ca
cc671e63f43abcfedcf01692a8a57f0929519b92b593f27367715c1856def6ff
67dc5ca2b6f5fdc8dc6e8bfd99e52078258db0c1279a3fd0d430917b98fe5007
f94848f433831d56a6ad575cd4f2af33e8ec438664f2c1c5ac70989de95da1db
7df97a3dc13d234fecbcbf4692312896c99b14fc94baa4f2934b3545f1522f29c
45ece48b0b2ce3bf0109e19f1656d76c628b66c7facde497b063d006fc385e4

response w

5729a4558f57fc196a0be2c06e7ebd0654b7468a724d3e8f0521b05d0bd83f9f
70f6f3ddc698e71ead52740dd01f314a1cc9656a56848b9167806cdad3b211a1
f6635c6b78d3239374ab6f6dfc5f965bab25924cbb7b14e297f453c164bf2e0c
d685a9ada054b046ba10f7fd4e935b67acbff1b689e5d71ad9300fa811bb070b
26fafc453f9d15324290c756f5734107a214bcb7755516353319084884911093
10ae5f65c37c05444526fdd7d6c9026b8dbdfbc1125ba62538e63b1e1c419c59
243a944967c109e3e7831837f0261a87a87a9adfcc90f8fc09ad4b5ced5fe88
e2d31f7b983b225478c174e20c4b05dae54a3a928a8b0be375c4e37dce775084

Fiat-Shamir context

88212e0ebfd060b6af240b41ba21eb09872aa4e8f532552136b06cd2e7614f89

withheld proof term: 1366 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qStepP.eq_1 theorem, QFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (N : Nat)
  (Q P : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Qt : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (n : Nat),
  Eq.{u_1 + 1} (FBundlePoint.qStepP.{u_1} A N Q P Qt n)
    (FBundlePoint.ofCoeffs.{u_1}
      (FBundlePoint.solveShift.{u_1} A N (FBundlePoint.invSucc.{u_1} n) fun k =>
        FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1)
          (FBundlePoint.qV.{u_1} Q P Qt (HAdd.hAdd.{0, 0, 0} n 1))))
```

SHA-256 of the statement above

f0c8c231e31c55e05770486cdf60ec47e6208d17d49a3731290f4cb13c80cd3d

SHA-256 of the kernel type

7183c32111310304324da067ce28e2a0f2fda4420ab6f3ebdf0ca11e4e6e8c84

commitment C

10b4d30cad9d9b4e01e69ce7eaf405580f9e9ec65d3468afb224e3358a17221
b4ed08978c4c52ea03208f6aa7630303b2769102985f418a21e6c8dabfaa638c
74502af0e0581c34afeeffa0d64db7ac007608272b973712c792590bd976a203
ec590e1614dcd348bccbe3b384b4696c8f8ea080ecaed7bcc3ca51005ba12568
83f6e3f1c463242335a1c8e6a9f333208cfba4222e5871feea5f59527d668799
9618e10fcc0822dfb03db3685886ea2e632665abc80b88af532191a9de88cffe
decf2ac6063b8b2f660d97e0db9778926bf679a6442b586d794ad5606e91ffb6
fdaaa727f5b27383d2bcd51c8204c9b9001d4b6cea75740c3e56017cf454784

first message t

767b281b91042cb809f5629dc4ed34c4b646bf7b2d1ca6e9645e99d1f75bb891
a23b2ae30aef0f62ddca0f128548dcc74f4c8109b9276b0df54ab5db812c1846
2d144b1492e46472574d887bc87805d40a9de9965a72847163e6f083fcb3c7f6
5031ae79b2142a134759d3fc5f099c9658b119ae445c380858f45f187a7a801e
02377801662defc4de45cc6a76d78522c8a7b72f58a9a86a4e4ecff4c2fd9946
93b74296a08f7abeedc323cb104374a885c613cb592777ca5493c6bfa2c31907
8891f14e90324d11c8e3e7fcc59c1c38340aaacadf85548a63d2a4c42adc53a4
41e3236cc95a34dc16c1f6c29d6c5797f1db64a9f1e26d46ee62fcc17baa5642

response z

2c6a56323feb84564702462c01255e31f0caed11677fc56002caab495b0d5ba2
17be2105f3378ace4392eb2548767c142b4b85fde9636d0e470924833437e099
cfccc7c90a50f2b4308475b11b1253415273603ad4253c9a15f7a2e3ba9fee17
2b5f7f3ba521f540a38a233d8b34695f24e9fbba10da8e12ce6519627d54c9d2
9b0a57144bddbb3f5bd979d6b6c86264d327d1782e1706b6a614628b7e96c14a
82017195a683bd306aa9af17d8d38759ce6adce2fede3b553a5d593d7988945b
54e0eba0fe99224f80894c5c0d5046eda8af3a66cac5942989db773bae7bac5f
2ffd235db762a696c706be719a425f7fc84d205d51ae903de938d02206bb0810

response w

2905414eec3c4f5cc23683eadc9316fa4c099f7cf1c75ef93259c4a39eaa05b6
22e1986da38f3024f3953f096bcd0338ffe6f6f81f8c2eeb615a8be24c872683
dd9c2a8d42d64f2832fc67659a1fce10da8b76038d0dcff1d02c7ce9a3b5ecb4
32a1cfb2906b07fbeb8aa0fcc3bae15d2469904dfaf4a3ff82d85dc68ce1870e3
fdc58f73646d0eb299b6009f5e18e5945f85148b692b62862f119abc73326761
cb8f8f4935fe44518f52376bfcc0c89419646fe2b56b2d2abc678f7310805103
0e384fab1245de69690e589541de0279d885efb2760742414db1253dbd639b2d
c80990b389b6b46af31e98fa9cbbb98706e078e5deed3549ffc6299f3e2e215b

Fiat-Shamir context

4810e225751fc582ef737d68b305340e59f027774ebd2ecae506f80b0b439e

withheld proof term: 578 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qStepP_congr theorem, QFramingExistence:253

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (N : Nat)
  (Q P P' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Qt Qt' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (n : Nat),
  (∀ (j : Nat), LE.le.{0} j n → Eq.{u_1 + 1} (P j) (P' j)) →
  (∀ (j : Nat), LE.le.{0} j n → Eq.{u_1 + 1} (Qt j) (Qt' j)) →
  Eq.{u_1 + 1} (FBundlePoint.qStepP.{u_1} A N Q P Qt n)
  (FBundlePoint.qStepP.{u_1} A N Q P' Qt' n)
```

SHA-256 of the statement above

6eadd7f3cf383d8463d8ac946a68558b18819d87311547b28109a6610fed576

SHA-256 of the kernel type

fc5603b0825315320362cdfb7c9afdb71b6f59303bf49f459283bf46be036f
commitment C

542221ffd9ad63ef60f6075eaf00019cf76d0193ba9cda3b5c5eb29dc0dd5026
ad66127551d8cbd225b964aba0880abc2f597786bcb86cc814bd4dd7a19a04b
aa3d4998daf654442dd0677f3bc409b45beb4bf9d9162e03a620735b05fb50dd
6d283480a9fcc73f1d02a8991d2a22c575cd63672ae5b16da6fb2c0b2db774c6
5e91698ebb78cb1a1feab86375499561c2357922d1b78c69fbff6b6ece76bfcc
3c8185e8c64d9a12c75987807bf9633bca02afd9551a687b3a3b4a98c018ac25
aa028d50c137df738d21cae025912c644628280457f50f6a3bf88cfd39d908a5
da29df27f708435dc17e319ad51ceee22125c9417e374d9b1e5c99546e8c4c1c
first message t

f21b475bbddf28627271afea944ad2adfaed16f908f69bc5fdf3951d1f9b6a17
0533175e2fa6c8cee6ad9b35f221597ed8b758706b646f0f4d3f8d17ea60b047
e39fd3d6dc157cb8070eb765f1f416d9fd676b12eb87c96ab43d821ee57f6788
4aa3146e7a6a7a764d218d841ed0bea1db423b9a0e5be4734c12289e7ac2be39
f339e1bb69c3a48223914ca0d6c750e0d08b7d6230df67bed227bd4d584f8176
3c44428d0bb479c1b742e9a11b94990fa40091cc903178a7d2f7bf9c16d0d7d6
9591422834045b36fbc8944b48dea302eb64150690550cfe8a848d566a1b2506
e3a694d460d0a7f803eab77aa491987f9a15c765c846f3fc9b6984139b99226c
response z

7300aa08d4189eefb64d782841f81b6d6b7698da0f56a44c6e5e4fa77dfa40f0
5e74d14a118b8cc0825e720e66c865e3d14a0bfae244ceb0e1cc96267b18b11b
d3ce6e91773b496cc9b1a963d3615d402a8c1f5de858665eae25ca9d17f178b9
461e282fdb11794331c35fb9ccba489e962b3bfd2397049c5b7cf9d20a033c7f
9d0d2a96c063e212f6e2c0ef4bf12569915852c41fa98a2122f3d510b732bbe7
f4962e1765746383f4ef991b8b1a116a72d810b60527132e972bbd9152d977f2
c7c2936c338a3878b3664d38ccf7adba8b4c1c0d18080213be5768a7f7fe2527
b6093ea9ed295e95ca4b4f6e9781e5b16c1df6e8192a99b5bab3c7835231d671
response w

371fef6927a12ad2a98b6498ba3116045c51d0b732e7c41fa26e61403c6b2535
2282e9ba2cac4ca1a11b7b4997712cda510d51873f913016288e4447d1edd8e4
5b2b4eee748ef25dcda03c9aa995cc66057ff15c2ce5eae7710640f140787a0d
4bdbbb892b4dd456f2a905cea53e6fd18c7808bc2598b4e02c4386f208f19277
a1217d62016157320cbabe5d529eb07892dd5ddf2005388abc31150418de3358
2f8c50ea6d8e91ceb8fdd394584922bc472af70ad3f09079a4a66bfc37262aa9
724d93040b9fd6779fedce26a364fd09cfa0583b6ade9128e8a05d727119669d
482215bb6f879487db1dfb9ef72299f4a7a279d80718d9eadd59c62eb8111c31

Fiat-Shamir context

43491ce1d3c66f9fc1b71380b2a8991b301733ed91155226d93bc40cb9861e31

withheld proof term: 22912 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qStepQt definition, QFramingExistence:246

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      Matrix.{0, 0, u_1} (Fin m) (Fin m) R →
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
          (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
            (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) →
              Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →
                Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R
```

SHA-256 of the statement above

ad2525cdee9f142b5ab875bc49baa9ec9cf1f4568fdb22ac1d9f4abc172730a1

SHA-256 of the kernel type

287657a43571eff19d68339300278bda70ddd99a32c49a40045f91ee14dc9c4a

commitment C

57299a6d1fe0c87aaf568137bd605b68d046313bd8eaa06d2994e94c6302e4e8
1b71436ed1760f0b08ef5c86a7d0af0babc945dfae681484d159f9e1aac67da0
04f1fedc7eb1651cfb853c42608faafd136e8df53fa3d463e50037c018223259
6e934f1a7c34b07652fd04a545f0c04b15650b5f61f9d01095adc96689883c72
d7b40fc4428a2b0924e397a53db0919af26212792b647c3eef7538029b3200c1
d7a9a3c187fa9aaa7b5de8bf67810a84894d6440a96cd70040e6a3fc8fed1ec7
559850655a425cbcc4f02b313e3840d38c642f60db607a9afa2072d7eead562b
78f5a02810eb692a6744c9bc5bde4bd665dd5f4991ffa2e50064cb0cc1dd077e
first message t

8045c8223ad9eb990b7f1949bb5b684b61639f6a20901c7847393bed2ef4ab5d
248c1cda7ec2ab7139671d70d3a414424ea0dcfdae756d530f51edd799f96344
fd63110f1cbb908e507484032c20a3f4602509eb62ac50e10bdf35b5193976d
cba1a187e68b2ef3012022ddffa791dbd67b8ee88e751335140f30eeb660b45e
8be7011b6dda944b8f257b809f0cef9e00a9a2a2930905f9295368ca59ea954e
2e833cd37524a2599ffe3fae4ffe1d575f871d60e2dbf2d70e43088d9bf6e802
d04585be5cc2ec7e1ecdb3682387dfc32e56ec12da352173a0b844e49e1a8a04
699f761e8abe81c7c9687fad41be8d5b327334f21f31fbddff615295e505e1a6
response z

8a5189d5a10ab0a2ce5aff2239c499c76fd347762abd17ffd2bb7bd39633e2a3
f264dcc57ee497175715c5cc3183f0d8848e492906ce5b4148de639bf6365e22
da0c4b104b75b237f0f161858040895859514e6ea1da2aab59832f32b0c1b7a9
05b59a18e4fd033a86506f633cb15361a5b695486cfc72e5d18a239e989a50e4
59921a0cd99c0019f0d5fc52d68a91c7821e48c8a58ca9923322df9f85315c92
224c7ebdc91d2d9a5eca089314dc249e4e2654553ffbac420efcf5489f307995
fd375603681f7149d830b5e45e2ad63bc32b2e8e648a64a7ea1d844a17722ccc
793a0ac95ce58289662904f5a514fd7d65ed2ba6242d5aa47c49b9667820685
response w

3496ca391339782c669625822ca81f84fb8d6f2f5f989cca3085c88172c16722
64b9f835737a7ea86aca6432efd1da52bc8e2fb7705128804e63052bf0bb0006
439963574be4a2ccf2449905172f9fc0334b603a8b43ef78408c21d0f41ef474
91e9bf5da34128438f24c8c3798c9dc6932b70ca36d8b4a9831898e10a0c2491
fe0b14bcd5ab754fb7accec1a30b20d9a5e748ee67949b16272b38746009d6ea3
b33eafea1385ad84e03068764c7001d09d683adf4d0c7ce81d7751acc22b2f8f
68ba300f93630705e52b228016bace250bb6ad9ca366c6bf95168b349d9375f9
343c7b4411c09b846abbe5b7cf758e410b39e9df487ba9da78fa37a83b4ea34e
Fiat-Shamir context

f4a3a01340ea600c9c530eb42aaa96130ea1aa9dd75ee362f251a79f31b4c19c

withheld proof term: 5734 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qStepQt.eq_1 theorem, QFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin
m) R)
  (Q P : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Qt : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R)
  (p : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (n : Nat),
Eq.{u_1 + 1} (FBundlePoint.qStepQt.{u_1} A Q P Qt p n)
  (HSub.hSub.{u_1, u_1, u_1} ((FBundlePoint.adMatR.{u_1} A) (FBundlePoint.psCoeff.{u_1} 0
p)))
  (FBundlePoint.psCoeff.{u_1} 0 (FBundlePoint.qV.{u_1} Q P Qt (HAdd.hAdd.{0, 0, 0} n
1))))
```

SHA-256 of the statement above

331d2bb7d511d94a6898533ddf96e30601e66420055f1a31b003c1b45d8ae323

SHA-256 of the kernel type

57003a5abc7e2611872d3de1c027d373b9da88acf11d86e6de6a53435b6f0d11

commitment C

4b399e7db2fadcad9d086187b857560d9ed71663196481bf7951f3988b1d0b2f
c67042941d9cde8b06f0da47749555ae59ffe3c786800dacbd6a5915b0bbea4d
17df19e97150a8ac63552b7cdd0d1009a81a403b0bb5fb1c2851d4695cb631eb
a0a53017349825be372ae506930ec7728942f591616a76b88c100be846fa9792
eec1d54b58873c6170cb76993c7824e5f14f9425fa951e24a2b2429a81694316
4fb1495ad4b82a3894eb2bf3b1ee8787ba838edcfc511d207b8d4ab2e2721cba
0158dc3d55c531576b5e30c5c10755bc4012a25514139604663e85a2db2cd801
6fe8c10a756478d7322f87e1e2162f25436ae4f7bc63eaf27261c14fd52cca1b
first message t

c5f03b9387f5f2952e4cd2caa89975ae4f81d4a8fd1bd499699d8cd98142500c
c0c9806525473e52b507a504179e362565f1430ceef99b49bec99ea84b08865e
8e7322a1fc3465f685eb96bfd5b86a54e6708c50e42910ba11e3590f3412b9a3
b4e363c3375b2f4fa1f0f87dae816be5d217c5f0b5d4674b6d5c6e12598203a7
469a4bcd6f6753b0c53a6b6ebc4cfa0a6d3397b9cea3e42ab836bbce48db7308
f8e920b85c8b28a7d99cae1f3769fbc5ea483c44635a55060f867692c758b324
0940ebe8b5ae8253c279e0c6d8cb49a48cb68acc4937a528cb4fab2fc26fe0d3
9a3947720b7814ebfcd1cc20842bb27a2c90b3a5d7003974193790b7b1af280
response z

2f5c46e1ac160e709839c99609378f582d3b650d2283e11fc243e920bb405c27
51e72b50c01f50c69d9d8cdd98ad3285246036b8ea5a2cbaa2fca4005a89dd61
bdfb583a5a1697228c190b089ebe086a5b6ebc841981480fefe000b6a3bf5256
c86f95203acfebd065387cacd301d3cbc59ddfe98d4644ad8d5d54600c71197f
72e9935603c1fd4ac84df302e3b6302b82a424b48e282f25ea09476c0111ea05
7838f48f4c6a8dd15db80d90a4db9d8a16701704baff37b771ac924ac935789d
627342381281298bbcb4e85ea80d9722c9b5132d83e41dd58b757b69242e421a
11149a6f96a8e88469fa8f8a432fd5ba76d686f4a4bf27d7d1f65fc87b2be4e1
response w

3df162312cb0a9506141b5cd6d0fb287f93504b8ec903260fed82882b46458e7
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c648fda8c55c8cbf01b9f80b5e8c5d685e0be422d48b38bd2ac11f65585f8b78
cba96acba1adf72cfed3053487ca53a095868f5b8002a360a0be3b76d9b58dc8
c7781ed88adb6893d027f3ac0657e91ea9146a840c4f651fbd77b470e4e58dc3
9d4919528e8f216fdc4fcdb193de33d2c5989be429e8596421093661327101ce
8e185f9eab8956190f5702e2f7a0ab3c7b0010f597b9054ab0601f553539bf81
87354914c75b63cbf485923b9381f1156c05f54f3242c28bd3d42218845a306c

Fiat-Shamir context

e8155ae06a01ea3cb78aeda2d4e894205ea726e7b71e11d8c8764266b383c034

withheld proof term: 481 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qStepQt_congr theorem, QFramingExistence:263

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin
  m) R)
  (Q P P' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Qt Qt' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R)
  (p : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (n : Nat),
  (∀ (j : Nat), LE.le.{0} j n → Eq.{u_1 + 1} (P j) (P' j)) →
  (∀ (j : Nat), LE.le.{0} j n → Eq.{u_1 + 1} (Qt j) (Qt' j)) →
  Eq.{u_1 + 1} (FBundlePoint.qStepQt.{u_1} A Q P Qt p n)
  (FBundlePoint.qStepQt.{u_1} A Q P' Qt' p n)
```

SHA-256 of the statement above

e2593ad318d34f68926c041173007a003d6b8c5911195b1724bb30e5eabcb9ac

SHA-256 of the kernel type

313dbab3abf41ea3add95fab05c20ae8c10345e1285aa9847d9d7bd09c50dc0
commitment C

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first message t

700fa2ffd91b5ee383f46a671ac37880d9c0c087134b2cbd457b16320f6f21ca
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20c89a623033a26881e39c8d448fc21cf47c164fddc00ab7ab810366538960c8
c88ef157d495e3cf2715e647b73b4325ea998ddfd375a332a648799eadff7dde
512374f226699b51c9a67813d1d3a67ac9a98c7ee516e360f8a54f545a64cff8
c01a8729d2280689c6397b49211bb2d7449d981fdcd7cf724b61290fce225b61
168690d29fd0c94f1dc398a07a31df30ccd68348e3df1f1655ade240c4f89a36
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response z

758cf4551bafdb374fe2b470342e1648c563ca81438d47813800e5d5954feef
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940c40ac155df25c44771431f917038cf9bc34e0e7e7c7a9cc2b18ab13b202
e39028fea47f8a5c6a11b715d1ef056504a11492f3f763c75f3c952d1f1cacbe
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bae5fe9f57f50c54b3e21f811885ada372524c17f6a7821160369b87acbb4bb7
687fa474721eb1313403e9688e9e2c050150f1639e1e04695748b0c0cab62f1c
dd496fbf3e6e381579dc86a0b615177fd6a35c9c55585a624a3881ee4a6b0b61
response w

12e8e7c5d7d8f3a5b9adf43dc378414edb7075a97830bc917e4e0ec35c6a8aeb
9834c43b21b83a6ca5f93109af2245cc81cd9e8f5c37b77c79ce9b994157f286
53d3229b2d5e94366cf36c197b85d2e3551d4a49271e9fe6ffbc0c95d684bea
a48454ff108bafdc0aaf3daa1fe632e8206488d890d47a3abf47c535b4fb62968
3bddfd9da373a727dc5feb532be934e27ff10edc068510cde227bb4b58c4a2f7
380f30d2e4a01e14caabb4e9ec72a9e804f489fd426316b4d19443865a0efe1a
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061eae96ac3d6baa35e603c3af0e36add40748f306d1b36e590c6e78d526bbd2
Fiat-Shamir context

58e48e3d93deea20b0d22237b9779f8d69cf91a307953872e63a55e938d9f95f

withheld proof term: 121 323 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qStep_spec theorem, QFramingExistence:337

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (N : Nat),
  Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} (FBundlePoint.adMatR.{u_1} A) N) 0 →
  ∀ (Q P : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
    (Qt : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (n : Nat),
    Eq.{u_1 + 1}
      (HSub.hSub.{u_1, u_1, u_1}
        (HAdd.hAdd.{u_1, u_1, u_1}
          (HSMul.hSMul.{u_1, u_1, u_1} (↑(HAdd.hAdd.{0, 0, 0} n 1))
            (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1}
              (FBundlePoint.qStepP.{u_1} A N Q P Qt n))))
          (HSub.hSub.{u_1, u_1, u_1}
            (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.cst.{u_1} A)
              (FBundlePoint.qStepP.{u_1} A N Q P Qt n))
            (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.qStepP.{u_1} A N Q P Qt n)
              (FBundlePoint.cst.{u_1} A))))
        (FBundlePoint.cst.{u_1}
          (FBundlePoint.qStepQt.{u_1} A Q P Qt (FBundlePoint.qStepP.{u_1} A N Q P Qt n)
            n))))
      (FBundlePoint.qV.{u_1} Q P Qt (HAdd.hAdd.{0, 0, 0} n 1))
```

SHA-256 of the statement above

723fbfa3d3d7b03901b30f0dc43eca6fdb62588e7c4c62079815655d2aefa1b

SHA-256 of the kernel type

fcafc99370877a6b2571f2eff9e229591298fd37ec8d5c95c24a1bb42f566ea
commitment C

67686c141d4817ba9c1f45596dbe9e3ca6687fdaee00594546e27031659efb14
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c08c6d32124eaa18530b685a76aee9c59d78fa0972dbd6ea2e9d2ccfefce277f
bcdcb01a9e897504a73141c0723be928e137a38c0b841c8ef4dbbcb5b53544e
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first message t

24a875d68e90b36e45e323777d5daf4a1fb416ed984adbebe9b5fb8713e151a8
03be5f8f8d3418099deb7a50b424cd9aae6d5151c9aebd38c4465cb22586815b
932f3b39f8b7326b06491894ee70b1dd325268b65309c2175741b8369a6effe5
6e8ffe1d886c7acca51e9f363f809172fc341efa5a7fae04484daca03e6c455e
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c19d57f6519a74fce2c36c3b875c5b9c219cb914818edab3a4f57d655003ddfd
f9c698ff9aec8f359c7e778dc5f7896f562c0d223ec1ccf5730e444380dbbd84
response z

25c01415331597e70a6dee6c9951fc361f66069a10046ecee84f7d7bebeb5703
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ab14adec810acf79726c630a87872d666edafb0ada6aba5be5b6dac3fdbf1e16
40141546370894d6d624eac7c28410b5cbbaf3d4488017f9fbc1ca5566722dd0
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0073bb9b6e4a49f307d345974f0ad5489f9760662cd34c914152d768591d5b98
ea6aa1257f4664b2742dee9f38d9cc0cefac3924a235cd29622572dfffb57fa8
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response w

4d5ce43251219653837810ad0fb5aea09fcd608502c256a2f1cd1d006c7ff8f
6bd4bb2dfdf0bf22e2289888ce1c7b63d20e89dbadb9f15040803baa606035546
6a7e0a842f89ea0470f4bcd3f91860feaa279f53c3c2d446d70d1f62c483bd98
c89fb8672965062980aa13bb917d250663d5142370e85880547d49d9838a2465
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Fiat-Shamir context

feed4cecaf8025d0ef4eaf64ad5ba33a1e9824aea86fa323b25749857752d789

FBundlePoint.qV definition, QFramingExistence:168

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
          (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) →
            Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

e972f3bcccc5f38d8d4a86f683befe509ca23e3a2a2b3a2362420ca7ed521b8a1

SHA-256 of the kernel type

1ad32f0a040ab103f2abd111035dabc9933829ef82a1d4b3aec9a12cbb16b983

commitment C

a0a8fab6d75ee484e0763dcdde4910d8a4e859b5839c2421970780088a2128bb
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c9d873317c24b54f67eff48367440bf7dec018459a175aa85bb5103ba4131d30
d59d1621d26ff1879399c8cc2c49fea8599b6ddeb1712ff79c909aa8cd1cf636
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09e9f3c70151c7f5a4be1bfbf211ccc9c33297045ffb5f5de42fdc55be84fd0d
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first message t

bc62f6fd474d4edfe37e60239cf7dca137d47dbf7801045c9b7cd989368a75de
9a6d7c3d96a2dfd7e1904074ef17c2a0801be9b9d297092d216db075a207b9df
59ea531dfbc9c9db81a5866b6a5c4d90bf2acb2fe5419c1422574891a0fd74d13
535be220fd4e08d47db2f310340fff968dd442bbe0c6028da0abdc3640b5d0a
b451825a3347dae45dddb15b22b77339ed793fd0212f02548ca3bfa814b09a3b
d3520748d4a36332b2db6548983d1e79d749cba0c6cd67083eccab138b2b893f4
b3fe89cf97e46ab993208f8d2dae1bc0178b0322f20402fb43033b185f7ffc61
4f3de7576bfe064df900ba68a430abd2cd3d6f38063d1a4ce6a418a41d94c310
response z

37e8f2fb3e0ee1a42d4017f2eb4a973311506e1a238fe3c31ff381c0f75f35a8
137ce21a3e83a146bb8b03d381dcbc46cc63e56a3cf94fe29c3ebf6b761628e5
6990ee1f2302efeeec4151e7e89f385df47eb32fa92ccb675ef907407f2d4d69
75bd61eeea89254ca2659ffbd6d1cec17587266abcfe0cb8a8c6f0e9619ed7a9
756aa3040dbc36cf70cae4feb9af7dbb24ee463da4216fbd1c5cc88d54d32017
c8ca522c4564521bc5b37c0cc9deeb7e4f3897e45a19b57a38aca71d6fa92b66
f4581d7099ec3e2e999f0876bf5b4c52aa443aa94b2a444352994a72cf5d9e7c
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response w

600bb2ae36f02041c56a135bbf93553ad6a0bafefb05ae3f809730cda297c3d6
ce5604e62ac52a04d15d711fc4f0e20f7f05e29b2e9b970af6d665fc888533cb
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2274ca7f3d1ebe5a92db2b3e607d6aed79484e3813828e3cd6512b9b4a718b71
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Fiat-Shamir context

84864bbd56404cd80c064dc352630c2b3d1ac1cb7069db99c0757ad9e35d822a

withheld proof term: 5542 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.qV.eq_1 theorem, QFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Q P : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Qt : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (ℓ : Nat),
  Eq.{u_1 + 1} (FBundlePoint.qV.{u_1} Q P Qt ℓ)
    (HSub.hSub.{u_1, u_1, u_1}
      (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} ℓ with And (Ne.{1} q.1 0) (Ne.{1} q.2
        0),
        HMul.hMul.{u_1, u_1, u_1} (P q.1) (FBundlePoint.cst.{u_1} (Qt q.2)))
      (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} ℓ with Ne.{1} q.1 0,
        HMul.hMul.{u_1, u_1, u_1} (Q q.1) (P q.2)))
```

SHA-256 of the statement above

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SHA-256 of the kernel type

e3969f50d67aeb83a40ecaabce6b42088618f0168d7362b090ca1551ac23a457

commitment C

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03111efc88636736ebcb01414287c35e73e7ec7f4f62e965599bc54afbeda89a
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first message t

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response z

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e4f085055d3e293107c745d25f29d6edb78c456ce9a2accbb53283bf34fdc9e6
cf0982e1777d2cdf64513cd6262e3894e3bac029ece1f7f50622abd4c27af6d
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response w

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Fiat-Shamir context

d63db223850303dedec28d7066c11ac68640c49c511ea89ea37e70f4a7146d19

withheld proof term: 390 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundPoint.qV_congr theorem, QFramingExistence:174

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Q P P' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Qt Qt' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (ℓ : Nat),
  (∀ (j : Nat), LT.lt.{0} j ℓ → Eq.{u_1 + 1} (P j) (P' j)) →
  (∀ (j : Nat), LT.lt.{0} j ℓ → Eq.{u_1 + 1} (Qt j) (Qt' j)) →
  Eq.{u_1 + 1} (FBundPoint.qV.{u_1} Q P Qt ℓ) (FBundPoint.qV.{u_1} Q P' Qt' ℓ)
```

SHA-256 of the statement above

6b68c756a51565fa20795b0e968faebe33232cabd5835b1545b82858132ef597

SHA-256 of the kernel type

f97f10294e4dbaff220cd1f5521397d1376d93f826efd886839f2068e7ad4baa
commitment C

50ed769b44922d37fb13597f6f93c6a1840bd3afa39d01f183a74574c7d6c51a
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531ff7acf3d46a93f8c90396c30f10e3b171b49d961dfb95d4e4b75d7204f6df
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d810b9db34daf18bcee152cd382a660be5ae4c5e28e9146b1a6862c94dc8f583
c744abe6d6ec0426cec3d5b57c02f39606cd9c3c4ee902d9a14b9b075070f0cb
b0775445480e79ad7752e6d453ecea164b0ebd06ea8f63eb67a4f3a141343a00

first message t

fde8386e4569bc5757138f59ee1a17a6123506efd3efcc674b70ee8e358c43df
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878a9c2ab8f681e9a7fb61a7310c584bc8551383d65a74036fc9e5dfa9d84b68
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dc856e7b3d206adfe8891bd220de11f70178dc6b946d268bddb4901b5005b377
b0b9735d228e2bdfdeb9d8a35a16194d098627c652ec514707179b89324cc4e85
827944c3049e874c55c02aeb5a8f14f57c4cd54187054815a7ec9a42faa10e7b

response z

1bf2a7c8477ffd189eb7830a33df8254a8bae07a876fa004868608f5fd0e5340
4b7e2e182a9661fad3329685617cf2f668937e2b3cc69a7e7718ff79769fe409
5918e4ebacedde3a68876d4ce2265b14665a585ac9d42b8a1e2686d729c62382
8d5d0e8ee5d34bd21a9d871b78df84acf8d26c82b9bc7e3faa1089c6a3fc8dd6
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28ad3daf7312478102eb1de3c5e3ab4d7ab63ce72c6e24624fbb4b117889306b
8e41e4bb8ffba92c205bbe6218b5596ee8f02f66acdadb152002a8e37e876477

response w

297880b2847aef0d562f93433910f383b9349320460b580af61526783880f2ff
c277067342a8d3602b3e33228fa264c93097cd02001f5b47902949f241183547
76ef6d0289247917abbd093974d67c99073fcb09c162cc869b489a634404a470
c7c0e33596fb4d7eb8921139aae52e9d2c66ba0e0097bacb77d2d8c9613f2acc
5768a0c42ea932438ee01020fbfa4b85aeb99421fa8bafa8cf8f5db8d66dc7ad
8c63ea1e7a58948022f15d110a8dece0682873a30d852d381d9e53a100467d31
97e4bc7e5392383cdfde36e3226bf879aee5570f58c80a683e60e73ea94507be
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Fiat-Shamir context

474a64d2559d5e535c8d273d7389d1b6de971cdabc400f55db0d9f3f2b732ed9

withheld proof term: 243 708 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.solveShift definition, QFramingExistence:118

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      Matrix.{0, 0, u_1} (Fin m) (Fin m) R →
        Nat →
          R →
            (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) →
              Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R
```

SHA-256 of the statement above

991ba9522aa254af24e81af4573eb9d7dcaf72c7dfa51481455efbb3039cedfc

SHA-256 of the kernel type

ee0130107ef815570f55c1827266829a0b7f8b2ff19647cffce25ae821171206

commitment C

dfbb4280bad69ce840e6fd3194b25aa05838e4f29183c602d366d7d2ad2dd64c
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71171da192064026d4e845a2d271be2ce73985dcd6178ee1c03a47084f593cad
a32f767942b15c968453a76ba369c0521028a1103eafc36eabf059c984aaf02f

first message t

e3c3bc7e8822c095f068bee08cbe2d3f7abf9c7a710c504b88db3131acd96ad9
8f4415cf10e82f04a9d689c0ed02874be05ce70bb0391cece62796bf234224bd
54db0a28c59733aa4acc6532700cd450a56093887882b8ccae5b440f20a5de43
0760c63ae848cbf8aa3bbb7aeddff2a716b8eb99305af3e1ce6905690e04362cd
561aa4a7ff13c29254f1d66bc8e17bea8565b18a3b189c2fab8e2cd3b0772e83
32c070f5eb0be711bd263058c7fb29fb853d68bee54627d1d10ff08e96e08855
0dff6c92041d685ba3b394a48098bddc7fa0b1c8af90cb1e0517c6051e65d1bf
460476b92f5301ee8525c69b15478c55d42f68a2706a194cc8d82cebbbbb192d

response z

60a41e03bc14000ba00d283a9bd2b56cee9beacebf7a7253eae2ddaededcf06a
27cbe514e91191f2c516651a9d88ad75f536d62c7e3fe7464e83845c8be5d15d
ee35181de612ac885f27566decc54dc4712aed8969830721ac85fc0259fa2e9f
bb6b2820ae62b428e10adb2da297eb2a98f5e29833430520c33bdda96d70b0bf
85c232b0bed8c7642136b4df761d81324b1bb3307630e3cf0ce5f49eccb0407d
226b591b713c020c0e1ad2e2789563af4dffaf75a67b863b6ecd6b765ce4d2ea
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response w

fb721617e731010c46276443f9e9a062c7e240a6e9bf0d478e968bf6acb64be1
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40dd5fbc02cf825cd0d365a4ac15532ebd0933efa987d9d068b04c85a2e870fd
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c30808755ac0d13a6359127d19c0c6144f54ca58529fdb44234b5b53c9bd478b
a60a52b4d81fdb0c389a298eab30a979db109a7999d9429d60e526869df33

Fiat-Shamir context

1d6eb4382a7d1fb85afc836bd983d1faa9c66140d3e99e14bf27c0293648e869

withheld proof term: 13912 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.solveShift.eq_1 theorem, QFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin
  m) R)
  (N : Nat) (d : R) (Y : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (k : Nat),
  Eq.{u_1 + 1} (FBundlePoint.solveShift.{u_1} A N d Y k)
  (∑ j ∈ Finset.range N,
    HSMul.hSMul.{u_1, u_1, u_1}
      (HMul.hMul.{u_1, u_1, u_1} (HPow.hPow.{u_1, 0, u_1} (-1) j)
        (HPow.hPow.{u_1, 0, u_1} d (HAdd.hAdd.{0, 0, 0} j 1)))
      ((HPow.hPow.{u_1, 0, u_1} (FBundlePoint.adMatR.{u_1} A) j) (Y (HAdd.hAdd.{0, 0, 0}
        k j)))))
```

SHA-256 of the statement above

2dcf55d8330480106c371e8daf01ec6039523027a782610ce0875d979bc70c7f

SHA-256 of the kernel type

0fe3a2593f4895c7bbd5965db16d1dc91cc34726e57687e5b46a4fa14e4a3927

commitment C

67a6feceef1b6409c2b90e93652e088532e193fc93190760ca12faa58824d1f90
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2f5d484bcb4a5cde55389c327930273852a81728ab1e9b8f6a831b00984f7a63
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first message t

43675060629ab50c54b9a14b18689ef53d1cfdd2745593fe2347777dd55f262a
deb9a7278020672ec3d3f47d745dcdd49f204f4d73d50b3d1fe481614ed9201e
aac1e10bf30821cb3841a5db342115c28a8307cecad44af12a3e02aba88d4a25
8eef395aeaf07b528337f3989896a71c69d18494e5a788ca0f37b0d98b06864b
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032fb2d22c875b7bc0d6ba35b2e4a928c26e90b5cf735d78b000e0b9b7b0707
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response z

7306273c863c0220df62da9b2f6229d3ae42f18497c104ba932fb981d9a4db56
d89a9362994651c139fca0a3d268c348a9830bc0f32499a2d6a743133db979fc
4e6a6a0d8a02c0bd2255448c2e30831b47a537f1dffb89819fa580ca2ce74d62
7bf974d723d7ef10d070f9ac6e0cb9b42998cf0f940a9e7d44f3797c3ae35d1e
1222a33c24ca190abf2b355e872dfc609490f877f24769d835b08c8bafd228c8
6ab1d23ce5a90058ce873c050f6b1e9b3532201ea3476ea548ae0be6b1401d80
ceaeblad0c57993fd8dd5b8d2ef3d5b6ae6abb44695e7b57ecb0870716d6c4dc
a396c667a2311e17e9e2598b16e79be3da87f4c3f7448f4526f59de031865996
response w

361ca7650c9d802d964ff8869b70066e7c83dd75aa97c4bda3e17dd722c1f0ce
d4e1e9e17189fb2b47d13e1def0a5b85ff9f500c7a32ea6aa811e5e3b24470b6
56cf933272b274322c7f84a1375b42c96d3a8ebc588569a853cf4615335c8d
38a9a5ed350f689e1095be9af4c5d7358c4be13013ea6c56271f14d35d205aac
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67e452c985607f2c74e6819854d3bcfaad93ceeeaf0700d37e9212a634c6251b
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Fiat-Shamir context

c402d385ac91d9eabffbb101f47d3d47669b55717cc2d4e8932d8003045c53c

withheld proof term: 291 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.solveShift_spec theorem, QFramingExistence:124

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (A : Matrix.{0, 0, u_1} (Fin m) (Fin
  m) R)
  (N : Nat) (c d : R),
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} c d) 1 →
  Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} (FBundlePoint.adMatR.{u_1} A) N) 0 →
  ∀ (Y : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (k : Nat),
  Eq.{u_1 + 1}
    (HAdd.hAdd.{u_1, u_1, u_1}
      (HSMul.hSMul.{u_1, u_1, u_1} c (FBundlePoint.solveShift.{u_1} A N d Y k))
      ((FBundlePoint.adMatR.{u_1} A)
        (FBundlePoint.solveShift.{u_1} A N d Y (HAdd.hAdd.{0, 0, 0} k 1))))
    (Y k)
```

SHA-256 of the statement above

80c98201401e49fc13c60161f00dc55c26f4ba5c09cfd3ef18534bf6bb0f79e3

SHA-256 of the kernel type

1ecfc7c023ecb15c46119a7ebe7d0d3f9ccc1b7e0067915b3cbca5fe0428a37e

commitment C

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c0276855320a43202fa794eb260d989a8517ee12490552b5b4872cfe38e35ed1
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first message t

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f9b4d38bdb39818264047e6733f65858c275bb3554b2c519d8ae231064217158
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ff4cf023717ec7da09a990766799439a8f29e73dded6d01448068b2231bba988
c03ad67cbb48282102e2ef12aca19c2090941e384cc4c32efc89bf41d6cd072

response z

58fb2f5dc2096ad7fe541a291df6f8c85015053ef16fdab9f8b47442026f4eb0
c9394289783b52bd6b9f3633339f7cd17a53d5cc5f4184e26c2e6afe71f19b8e
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8a4a03de61c6bf12c600eaf05d0ef46a187fc19a814435c02c7cd7dd79a473fc
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ff9b9e4093acf888a81b3e6ca80e38426344ff83c3b9e37024da30b8ee2aa3a6
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c1307f97dfa8c32cd67854365bf0b5bc232284334891430481e7b56589218e09

response w

29df990ac392b21ce1459296e1d7c416645f5ffb15c0acee14bcd5874b48bbb2
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e92c35356490aa9be2f901f66acf08648c412857811fa437ee8a523fb79f90f4
d34fb7d8429b3583846424742e84cb7ff92fb316e087b854c8ad62a79ac97db

Fiat-Shamir context

ed3c482689939199aa479ea76f83721984a42c8824a5d756076be1d0147b2a66

withheld proof term: 2955 277 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sum_antidiagonal_snd_splitfst theorem, QFramingExistence:91

$\forall \{M : \text{Type } u_2\} \text{ [inst : AddCommGroup.\{u_2\ } M] (n : \text{Nat}) (f : \text{Prod.\{0, 0\ } Nat Nat} \rightarrow M),$
Eq.\{u_2 + 1\}
 $(\sum q \in \text{Finset.HasAntidiagonal.antidiagonal.\{0\} (HAdd.hAdd.\{0, 0, 0\ } n 1) \text{ with Ne.\{1\}}$
q.2 0, f q)
 $(\text{HAdd.hAdd.\{u_2, u_2, u_2\} (f (Prod.mk.\{0, 0\ } 0 (\text{HAdd.hAdd.\{0, 0, 0\ } n 1)))$
 $(\sum q \in \text{Finset.HasAntidiagonal.antidiagonal.\{0\} (HAdd.hAdd.\{0, 0, 0\ } n 1) \text{ with}$
And (Ne.\{1\} q.1 0) (Ne.\{1\} q.2 0), f q))

SHA-256 of the statement above

157199ba7533219afe3cd76f871fd7566a7017924d825b93a88449c49e3f56a8

SHA-256 of the kernel type

4c7e246ea4fc666c6f81ea1dbdd62921e067e2c2b7215693e521d96795c15f09

commitment C

41be2181fa45f635456e2fddc48e902c5bf5580c5e5c8c7a3ec0174b82027337

159b1cdcd18506774f7fe3f7da0cec6f958994568e73d49fde362c4085316f8f

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da9d9c36c1ce78f425e02cd81674bfc3a39e6b64620f452f4e4a8388d47ac45d

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first message t

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ddb10750a50ef22534283fa5ef60fdccb3462c2cf9910de538a60c371b3bce83

response z

2c44cb765e7c09ee66ec0e79e763d799403bcc9d429224c94135bb114d840c98

e7be11be511a430cfaaf8ea8384df9720b12fb8d7e851446ce71a9ef398cdd0e

4dded8f39d2f7e19e5fec8ef99334ba55d7fe41a15c15fbdeb2712f9fefcb437

323e299806d8b8b2cbac24a82076e4290169cd20bb5181bd720a9a56d76ba74d

578c0830cafdaa6e81563050cb54571b054e23d2d7479f30f35bffa7260751b1

3fdd9d85ae8f300c99ff91c62294a376d0ef6261c883f33d132ce88401d62a79

b9a7b791fe9b576e65e277b25777a54fa8b4ff40c1dee7d30555b46f16f14519

184cc4c032ed31501f1b8798bec6ba15662ba144567086f179a0decf9ae7444e

response w

6bf21f1a3ee0f50d85e6f62912e847377fd003821dc33378fcb44e7cb2dfb4c

f0fd43601b205194259ed7454753ee7f7c19c776f360e8c9e314508c319d044e

946bf7bd2747a74138b0d0a0205d652dc635db69cb7ead70fc42012605edcec

bdf30a6d99562dc1642304a57cbc2a48848c1d056c17fa2900d9192177a6bcc0

6c5e0178267eacce769a1ca76ee6c44b4f7e9add3a7492aa9dabc83ad7b1f440

d7565325f12515f40a9213240fa20d986f5450d2becbd9ebf6b82c148108abd9

304bd5d986365759effb33c047e227014434e27c9264c38441b911923ae7f927

cb7605c1bc67b8961b3afee7f38a49aef561dfb67e39ddc01495239dfd250035

Fiat-Shamir context

14de375ff3afa7c1240f43e83e195e50eb16a70913c86b8fe7567b22e836a884

withheld proof term: 157 981 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sum_antidiagonal_split_fst theorem, QFramingExistence:59

$\forall \{M : \text{Type } u_2\} [\text{inst} : \text{AddCommGroup}.\{u_2\} M] (\ell : \text{Nat}) (f : \text{Prod}.\{0, 0\} \text{Nat Nat} \rightarrow M),$
Eq. $\{u_2 + 1\} (\sum q \in \text{Finset.HasAntidiagonal.antidiagonal}.\{0\} \ell, f q)$
(HAdd.hAdd. $\{u_2, u_2\} (f (\text{Prod.mk}.\{0, 0\} 0 \ell))$
($\sum q \in \text{Finset.HasAntidiagonal.antidiagonal}.\{0\} \ell$ with Ne. $\{1\} q.1 0, f q$))

SHA-256 of the statement above

f5aeadf6505f4c5e47674b54e970b4df291be9767d46ee5557a3c79f86d07fff9

SHA-256 of the kernel type

42cad92e574e482f134e53a60d0c14507af6feaa407a6a62f248cbb0d282c4e1

commitment C

1a1e6cf6b8e32d97b40455138c2afc62f7d34f61628af2211e7424423597f35c
62aa2a8e44244640a7a1075ab65d9c95bf60f0012efefe3d67b77a7576095e35
34c32ea73cc83a218284a98f8b6128ecdd41a3c06c48154693a494ccb1b6af03
7a47dcc8f4de573fac5a9a820b7c314862c761d70bc26f177ed6c9665cff2ad0
a7151b63a6fc64b31a80e568fd90b0630d4b65211a5d78826bf86c76e6489141
e3e75095a7cc953f2d06db13882fb15426a3f51721ccba1c3bbd904a0fa56d43
9f0af5aeca0f75cdb884597c269ae90bea9a49a196a89ac3204b78211283a153
376907bdbfbf01a7636475384392b46e28a0e43866d51557f3be7c8fa3ed1952
first message t

2b688346e184d63ac3adb74a687246013650fedfa962a7eb8978a26d3fa3056a
2166160f1975fd47a508e9c563dbfe3aca99f74af87b04c761e4f8f83a6c59d2
72ff7d99695dbcb0b6214f63138e8d308ca3ee6c1b97449e968442ac3690c359
bdacd20f5188dd620476253c7b00e961bc6baed29eecd03e11dba8ede8bcd8b
3a084b33b40f843e5a0e52da3718402ff382073b3606825eafcb1f6570955b12
411491edfe059a78e867e0cf9ebccc55e32afc3f044211a7a5ae9af5845e9f29
9af6107c29a35b3b3ef71af8438653a35965867e962b586ba7736eal3c67c3e9
9f07e0941ceefb8cb82a06c2d3ef06bb780fc0a2d81106b5859df37dd6c39489
response z

3fc5c5c8f747a15222a39e0b75e8161e7c8d2feb88e535f109d54b854a3c4835
b13627a137c1eb01ccbf001eb6dae9bbff5303681270c8d2abf37731d5b3eca2
da5c906699a11b4cc6bfd7f50db48d4903d6deae494fe0accd9879fce12ec13
defe959b3c047a492ba7ae92c646516dcc4ffc51912ca81102c3b26614d9d39a
3e73c90a670e78093dc7f93670470638462a11d841ee4730c94a928464c84946
5451298a8fd3b347e72ff35e81d5a1a8eccd382e075282f1eccf5c42578e5aa8
1dec3b78702e0e36c100bcb024942ea4d97b58637238a753ff8e5f449d50f5b7
82f60a1ad529261b39feae5cb5480298f1eb3389ff37fc61c83f422b2b817955
response w

c72d335c40eacbf48748d7a93aa0fb01d55b4a0f313badac3ba0d9da0926ff5d
f7697b57abe5808767827f74c11a9a234ab5353f61a5d4dbec5a252689d03a3
698aa7f6243d9697199390c0d42901a9e7067eb1ef23916a7da7b2db925e419f
b3971bd81d29a6b5ae94435e1126bca147f414b8ed0153ae15cd1403ac17524a
7a6d886b89653d62f7458f9e0ba080633bcabaad99795e87dcd345c0707c8c1c
3c2d291eb6247205f6634a85c853d829a0d361932c51cd424b3d2450c6224148
2e69219bc92a022a2b74ea36d043724922dfd0a5115f1db6dd09876138bf787c
36cbd93d0e6cbc45c7d7e79f4d24c9b00debb7d450825aed8a5eae20f3599e5

Fiat-Shamir context

9b63caaa0f3f1519aa21424c9d433f6024b790fa6235c8b5f7b7ac93e86fcd84

withheld proof term: 66389 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sum_antidiagonal_split_snd theorem, QFramingExistence:75

$\forall \{M : \text{Type } u_2\} \text{ [inst : AddCommGroup.}\{u_2\} M] (\ell : \text{Nat}) (f : \text{Prod.}\{0, 0\} \text{Nat Nat} \rightarrow M),$
Eq. $\{u_2 + 1\} (\sum q \in \text{Finset.HasAntidiagonal.antidiagonal.}\{0\} \ell, f q)$
(HAdd.hAdd. $\{u_2, u_2\} (f (\text{Prod.mk.}\{0, 0\} \ell 0))$
($\sum q \in \text{Finset.HasAntidiagonal.antidiagonal.}\{0\} \ell$ with Ne. $\{1\} q.2 0, f q$))

SHA-256 of the statement above

74e04785127ad576eef28be68db6ea50770b1f233398f404a0eb391f5f48f946

SHA-256 of the kernel type

26e57eddc2ebcd269b80423dc1cbe83cd041d65e65cc0fde7456233b3a522dd4
commitment C

85e47c221536740b044a22d793867d8c4db59aa6aeba2f3c78d7b1036706072e
6180894547a4543904489ef1b6020e1660b278bc7a7a154d10901b7da177ed21
68a38adfc43434b5ee4687910276e8d48330e28164bf7fe870caef29bf8068b
870613f3f807de2348034ae5f5507d4299cea81ecd3f837fcf1bbfa112f37451
77a17fac567b7902dfaf49242f410535f7e3a1f5c8aa12fc6d9d7eb63f4998e6
a24d717cf5061862aedc21c8c88b4ceed45ce03c0e1cf18ddb9e94cb1f599518
2dab9fa8eb2b85a2a9fc4fedde4284f59b27a57095552b80480a346f4b0a9f4
414c123c64ddcbdc074c8d6401e7a28e9ee30c3c4bf15faa0d107318572522bd
first message t

13efda25d00b55e08a5c612e9e5d7a1aa34b2aa373561738699da7fc433ade5c
e17ee3a62a19160633d9adbc76b2cdd6a2035393d7c1b8616918d56d4fe4e6b7
b34c9040d1035b204169f3ec23dfb5bd5a090155078c31040cc8204d983da69f
0fb9db2effe56158189f2461c0a383a6c54e6d1426e0eddd2c7fabbbba7974fb0
9eaaed844a793e6a4ccd32f8f41cb0678e919ecd3c4fe1bda7ae42571a690176
64f6066860a27a321db4a25ba1e664a66154aeb1e09b78b598602aee25a4538
03efd1aacc2d0da4615eb0271222ebecb71e35ebb94883b0d607657bd03a777
6aa412f3536852e739792f30968df44e966e992a1bc39d0760faf0bba1a280e9
response z

613b3157c20d230a16c8f8d4b08c674bc5334aca7cdc5dfe605bbed47dc2fa7e
11654b09f283529781899b2aa8613c354251a4c8e45622c9fe8a89d3aea831de
b79737977b3ed6d78dcabf61a64388ed83f85e5aca68dc7db38798a1770265c4
c300accedd8857e112de55c2588a310d2f33d4dda73ad7c0ba38be725c95e991
64317155ca476e6ccc4916da6a8df86341c3b9e151a3381e95255b371815ce6e
0a72287314f5721f60531f0d182744ee87020075faa3132bbf28fef9bc9e3f45
4f337f1529afab575aa16b3e1d9406ee3cf2780257b5c11bb45347640eee0e4c
38b5144cba1618bbe99e9125a3f572dcd5915dd7428031e5315bfebd0c15d0dc
response w

23d27b1a6a9cee0fae2703cfdda8e051dc635659cf3ee95997516fb93e2595b
3a128b623f35e15d2e4b4cb582ae4cb5c598ad0d1c885a676bb7b525f7941ac7
33cbe1d64fac03d0e58688881b602aea7e10ddeb10bc88258d379834c98675f
199b34b7e78e8e2dc21b8337202c22153b4b84f40445af9ca8e4f25b3f07d9a4
2d92655bdbb872263c4158dc20269ad5e6f138550463d1456ca914f399b8b883
9053dc504b719d4e6655dcf0c733ba5f3999dbc2218c5029937fe85336e7d464
25200b74414f9716617a547de2543a8bee355366a352e4dff2b27aac50cb8b61
d0575752ac7207ba1a004ffe65a080977feedf79b5727db1c20681af6b5173b8

Fiat-Shamir context

5248ccce1f7600e8d1e2cb3b5462e7185cbf3e2affdc0358313c13222c3444f4

withheld proof term: 66389 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.unique_qFramingSystem theorem, QFramingExistence:388

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(A : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R) (N : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{HPow.hPow}\{u_1, 0, u_1\} (\text{FBundlePoint.adMatR}\{u_1\} A) N) 0 \rightarrow$
 $\forall (Q \ P \ P' : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R))$
 $(Qt \ Qt' : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) R),$
 $\text{Eq}\{u_1 + 1\} (Q \ 0) (\text{FBundlePoint.cst}\{u_1\} A) \rightarrow$
 $\text{FBundlePoint.QFramingSystem}\{u_1\} Q \ P \ Qt \rightarrow$
 $\text{FBundlePoint.QFramingSystem}\{u_1\} Q \ P' \ Qt' \rightarrow$
 $\text{Eq}\{u_1 + 1\} (P \ 0) 1 \rightarrow$
 $\text{Eq}\{u_1 + 1\} (P' \ 0) 1 \rightarrow \text{And } (\text{Eq}\{u_1 + 1\} P \ P') (\text{Eq}\{u_1 + 1\} Qt \ Qt')$

SHA-256 of the statement above

51fbaf61637f11416488bc52479c4d4d489488f2c4861437a85416982514f03e

SHA-256 of the kernel type

240090e26aed62bd3522da2c8da70f43b176edb14e0696100413e90239bf1af4

commitment C

8caf7d8db7c98061a231aa4f4e2e7211e55d17243d62d551a8e500b83731f51c
3551e68ab41cc0db6d4049f6ee007779bb68076a54dc40cff2183d4f688824ea
142d8cf5a7c05bf68ffbc1c7af4e4d345b2c02bb99e780177705521995df1d7
6abdf65d088bf5a218d6e439a302b9432eda233ceb2ecccb1b9cb10822fa03ad
dd49ca0fc9847fb5a794730c2ae041fd9fd616080593e73803933e07e10cfdc5
72d9242550a629e1f64982d78de87e6836b7be51e665c65b51194b40e34d2b56
e2d4cc53af60c4ff71bf2092aeb9bf178185479a8a0b78386369f2b45c159e5
dc0b033de0fb0f303747d35926b573c4d59d99af03241720e4941224d668a3ac
first message t

7321f0ab7be07607c2700a01daeb248580e5fee6cf39f411f3676e49aa1af1f4
0100a4c70e7f898534175c9bbf3dee6006bc422aeb84407bbd7311345d0ea6e
ce2dcd397fe575bf07e3190df4fcb98969fe6acf14bae73f55fa15b435798019
2dd991f3cf73f945807c32cd6a9e043bf02a1bc830e3e5abbae1d56b9e6d43f0
48fea7defba27dc6a615c412e1e6b4c2ec7f385d758f59ba7339e169cf357d93
d72fd8d520d03b621b257765dcfd80671098aed0dca683432189c838c166f1e
5b11e1e8dd45191f2148b01667e8bb4e48530e93af9528f527eefcf4ce921f8
76621d7191291686ae3bb508cb8a26c138d9ddd5cf703791038f01d323ed335c
response z

5d63d8fe3ce17c302bd99ca3aa7dba2c012e993dda9ea5aa881d7c1440e0e231
3188dda1970b12bd36f7c0656745f1e41da226d2c6593bbeaa4fcac06bbafd22
336601ad95438f6354595cb5c8ebe89662d848b7c218955daf5b23698a537e8b
1c0e4c71708bd758f83b15e6cb9997ce12dbdb5defd0174d60c7a82123cae1a6
3d65b7e00010771aa7ad28004a926763769f5caa6dc232e41dcd843f8ff0dc2d
59c4ac9e4a5fd1621344acc07bf1ca5f370171d0cd5ed1c4d2245381c1b02869
66cf730a24efad24c331cb045e02c5e1f9c2950066f3215e9afe69d83647e4f4
6ca6cc8cfc463cb77c8850a93689db231e3046a6ffd0fd979db0d46630a8ba87
response w

34692ddfca370209e1da77f824c924c29c07d23ba08e68a4cbb67ce88a9bcb39
5bacc66aa6e94afa3cf47fbda66ac97e2968af8205dba18ca9de69f3d63be877
94be7a3d7cf9010c557ad5765896bda7dfd87b279b5c9023f7d69a91a9edfc46
f6207b37d14a01f75ca89b6a7f0597580c1399f820ad40195fbb60fff0f1dcac
4664490e5f951b12d594bbfeb4bb54ffa21ba294121e3d6cd10e24e9d8573973
2abb7e511a47b0d4d7b73f651722b23f3ae5126d66d2cc6f113270430fd6e546
244d5fee5782859ea21f68e7b22dd01d0d53051440ef7372edbb5f486f4689
83420b052799217a60bac5de085118518692a43f35d581279fe191540acaca0a

Fiat-Shamir context

2ea8714415eaa35579b4b0ff5f1fece0f945793b3f3f49dafc320ed505c0a746

withheld proof term: 3 242 208 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.IsFramedQ definition, QFramingSeries:65

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →
          Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) → Prop
```

SHA-256 of the statement above

b2eab65d00299b6a0268b031a49b3b018e7e6055c9c6abf899f306d75587d8c1

SHA-256 of the kernel type

6484e0739f149860f775963a1d72fbff161f4eb221d14ee0f10a708529d7c233

commitment C

c26d3d3aa712be5b472f5b06a2f7d3139ed65b1f1d77f013f66fdde9d25aaf44

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first message t

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response z

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response w

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Fiat-Shamir context

ebf54d219b8140a318b3b45f363ecffafd36bd724c2cf42e31dc8d4fe1699e8c

withheld proof term: 3 993 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eulerQ definition, QFramingSeries:52

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))
```

SHA-256 of the statement above

63b22e1e9b85ccc9c6721aa4785df23ef7ce10638842e2c0db8a0146368c52b1

SHA-256 of the kernel type

ebacfd3298c7a3410a02ee33801b8505612a680755434d2b6c1381a095e90d18

commitment C

c469af5161f7583a6ffaa5ca776d589cc0ef9ffdf640384eaa9aff65a1fc98d6
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9c188c09c70608f2e46d599e049fb16be3fd959c1aae5371a4ae3d36508706e1
457d60cbcdf69b380ba07eef62faf9721d979fa17234267fb096c702e243a254
7fb032af2b352582859e1deab4a2ab765f046daa0374adef8b5021bf98fa2352
1ac50071fbbd064bb7585c53f3a58e7f7bb106c081586c07ca0955dbcbd6c41f
d9129fc71c45f04fd3dffaea76bc9cfc2d09573c9350319bf2866de4e2811bbf
be84e80686239214b67869559661addce0f52c41947562cb5370ac01061684ae

first message t

d608a4492f46a56eebba2dbda4de33c007cb0883bc4b1756dd3731891368cdea
ac499f153d302994388ce04303967a0c9d05c611fecbf4716211ac8c96b2e949
aec90fd3f90cb5e005b63265665ef66ad8e2433910911f619f606c7d832553ca
5eea7d5d8e46e638517dd693f4c36203c9edd81becb116278379a2e6fe6579c0
01176ba4b676cb309991074a83860b2fa289b8fdffb51d2c5801cf650b8d4d82
13b55b8099114d1f98efae115bb83c101b979437d26146af43c2efd0ba2ac56f
ea058215c074bdd8bf18a7cc29b17dfa6886db37fd254a74622e59c2ed997945
65df117ba0b0428fcc45fbaed25e77b3c29113b40f9b6a373d8d6a30dceb71a

response z

29cd82b5b8f2c5c598be6b14f0f15d812999194edfecdd199dd1fd80d545fcb81
4622532a95cfd815ba68289813b2d3e36e4e0b11670121a467e4ac6725cbebf5
ba02b60d2fd009e578c88f1731979b9a37f9c3a955f6698cc60f1806e02e836e
894ca3ba85a5aca862e4735afc26117e89f918b93853ff4227d17242dcd0c04
0f8cac93c640efa0e20839ff353213bca30a9f363f783d136ff71e50b622919e
93b5f9b3527bf32da7bb300d5452b439ffdc192e89377e0e02f899c2ec97b4ce
e806258978d9a1b45e5e2cbcb194316f9c059ca224e4c5633f444e570bc0856b
c6ec6daffb4bd9e17144383de95d58bdfa9a837a2e5ec5e626a069421e46d392

response w

68bdfaa6acf5aa09258d53deb7dee930394b36a8bb2cbe13edd96542777351b8
4a607f9977cd621c9ab19728f2a244092772999149d59a7bef471da5584f3721
e4324375bd9900bbd9fd1c8379b962f7bac691c24d410f8049c606a6d705a1e1
312699ebe0389c383f87690621eab9525a158e62994931e1ba90f99f6fc50198
81c76d1a24e20fca1becce0b861a9058e8eb91778d8fb2d8b74469f4c9e66bbe
17d82d9bcaa19a94bf62411b8d7440c5f8f287deb5461b051341dc8a629f50aa
d368fccbca38ecb8780642aee247db66111026734f9c1627d4f426da78f65ae1
ec9be1d7ee7910d2a493ad7531e8f8888b1df13c2f3566789b8692f816c920b8

Fiat-Shamir context

6ac19fe456bdeedf7f8eda35a7241402870962bbdf8d9c9fcb0c1238ed13115c1

withheld proof term: 8 238 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

`FBundlePoint.eulerQ.eq_1` theorem, `QFramingSeries:0`

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.eulerQ.{u_1} Φ)
    (Matrix.of.{u_1, 0, 0} fun i j =>
      PowerSeries.mk.{u_1} fun ℓ =>
        HSMul.hSMul.{u_1, u_1, u_1} (↑ℓ) ((PowerSeries.coeff.{u_1} ℓ) (Φ i j)))
```

SHA-256 of the statement above

3d45b0d6ac910e206dd7c1692f669c62e19ae5fd9026c58e5519590abf1555a9

SHA-256 of the kernel type

bb2432e3488a0d4ecc13117eea634135ef601f0d0311e6430ded2f06d9315e2b
commitment C

50603ba04d25f6cfbe68ed5f6319a5031cd495db686509c9c8c4c4387402636a
7755ddda71514a98cc25588b024f74bef3f31bd5b621d84ce517ee6c6a5c31da
05d59a7dff3c928875a6932830e7f18d098f263aadd3bbc724bdcc8259cc0625
adb947c8270a53363b403e4071cc34964022b1399894b463f2429c1bbd0f61d5
9d5742302b00b56117b1456c1b6da06662bccd0a2b84848bc6bc7a335738eeb1
e58da9eb8ed7875e9b36ea0b54c0ace67f626cbaf67d723711222fc56cf5d92f
7eb7bea2011045d83b23fb710f61706ed69594cd3ee828425beea694b11c927e
9f9a90d9ddd1fb4adbf341c34156417f65311d1d09f8cf0edfa575a7115f262b
first message t

8c6c5dc3f06c6a5949cf6cdf9011a4ccbf0be1522cee453748ce70542e9aa210
c241d8a09097e6df099c2ffec1a7c744758c85f8e7b56fb70a2ba26fb5ec4f0f
3ae01e83be42b811191278e6e9a5b415ce8823fae5607e05b0c1e61fcf0cba66
7b43eca74fc403909c35b82d204e676425495bb0eeb9baf892b40f908ffe986
9ac324eedd69b67698c0d8a185ef698acbc64bb338b05a61afe139bcfd39f2a2
d34ad061c1ebd3f18825b800280551f4deb78bec8bec2e8f9f606ed4fa3c9a41
ca433e8f63aa58e1d085e5256b09678e174acf1056607a7bb41ce2f2d22bffa8
5f8cc2c07a436f6e46e035bedb021ae05560a10c92cf8e8944e661ea042eed56
response z

173358dd08c3ccf0759ff47074489e4c241289e58fb33019ff53f33105bd07f7
7f2aa84f57d887283181d5b27c27ae3ed42f77e9cb1919e411dff02a180b9f47
372950bc317df10d03fefef1d951ea6a2837d404a06e61f1ed8131e83e26340a
71d51b1a06b23c020cdc9f35c6652d8db9915fddf19d9c48fe5cba80171308fe
f5669e8d1fcd33b3f37b43ceef26315709903069825f605bc99361633e64d2a7
24261405aac9fffa32d9930928c96786aff45d4d9cce1231777b4e18a52e6593
a8a830f5849e4dea768221feab5b4ed1194ccd362a974c250e0eec4614c0bb09
6dcb70a7c0f5342a20a015fbf8afcf191bea19fe9a46f03aa81d9b2408e9e469
response w

3c74b745f138e6f8d31ffb516af9592d4c39db508abcce5f6dbac66db34f6e5c
712546dd595659f23c58d5f37046e5783b42adb13da86ea2bc3dc4bff38998b5
334463524edf0bdc74124b64b3e7e34c41b2d541121ab31750cbc0a0419c7cd4
35f6b08f32f869e0a2dec468ede0d60499d9fa91ead2522b61669508635c5305
6152a2b90c3bf367a1ebd3bfc82354099edcd25cab94119308cfef627ef1b96
df2a3a2d8a5d534fe9b2153e1b3a36c87737975bab5e31808786c2cc6eec774d
0c3be6b1459d8f14f78e448d952f24d7418d501d0c288dcc7184c27bd68285d4
d10ad6105516facb12f493faefc5c33b41f3b5a8b815f2d123a31ccddeec3ad
Fiat-Shamir context

1344414116c54d96b730d96720a9c918053db56b90c4b549b9d52b939d9a7615

withheld proof term: 281 bytes • axioms: `Classical.choice`, `Quot.sound`, `propext` • kernel replay: yes

FBundlePoint.existsUnique_isFramedQ theorem, QFramingSeries:90

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) R),
  IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} A) →
  ∀ (Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} 0 Q) (FBundlePoint.cst.{u_1} A) →
    ExistsUnique.{u_1 + 1} fun PQt =>
      And (Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} 0 PQt.1) 1)
        (FBundlePoint.IsFramedQ.{u_1} Q PQt.1 PQt.2)
```

SHA-256 of the statement above

8bf40f46edb4e31e1e1e51bce481fa105c30437579f20e20bfaf7807e207408

SHA-256 of the kernel type

dbdbd38b190375928c4d6c93d2706c7d2a5a0b00be17c8d8f892faed5f2b1644

commitment C

8722a7ebb01643adca48e0dbaa6ec99e0bab85a46e20a9586db70f39d13c43ac

84eca2b2f3ff218f4c3c7ae439d32bf11610e9448e63afb89feaa1ea42fb0aa

1738cf23025618ae2807d82c3ad980cb8217b66375243a31ede18b25349c3faf

65106ab4f6e25593431187b96b3ae487535f4a8566c2c5de9673052792a1d700

26431b7b9e2f434d6d6b6437a3d5a8ff56bee223b0df488cfa8b4b48abcc38a1

0168b617681814912a314e8e5eea69991ccbf47ae5ba56049cb0d19b605d4075

6dee0f0f21c76cb76fa7b204cb7b29f4ba11a1ad22571401d44466b5d3e940b9

0230f4d6c92e992d59bcd38eb96c53d40fbac1a7759725e019bb42331de367d

first message t

539a2a4c051e8e54fddefa2581dacf88aabda21ca8b15ac5a7a2d184f3927620

1a1ea7b167ec749665e683285d42d0926dce87f8218f304c43117cb7acac2141

8eac43b6293b2376ecae44477ef34f5870f3b6c9fa5348ea7c489b185dcc3ed0

9b4753b4f8236f526fa6475db273e12807d7154d0acb7eea1249b172131eb549

9eb490ef722384a97fa3f27302cde880e037675159b57da11a6e049555f579c0

f0314434547c20fcd37b663c50cbd42080502b8153aab18abe50e0ac332b2acb

400afe3414060b3f20bfea51918a8903370180c6161b01e9ec8186eacc0a1638

e50eddf09181670bb745ae542e225d73336d137b157a4b6c6807db2c59a44524

response z

33aa1f1ee49176bab16256dce7c5138e38837c06a456008c41bfb2c3c34623e3

b0dd3f3356beb5f4ef43c516c0c4d87f1caa0fe80446229c7f32c9c927a64cc8

28abb856bb842831170ea3f4916561b062580df3ea7a9a124594c4ae9eab98ea

15ca1d866cc02d838ba9f8d6d2eae22db0e5c7ecb5e1e0c75e227f75d30a1f1

553a3be5d8e9d16b89da8b17ad8f70aba3e828066b7f8e539d4ef107aee9f476

9872c7f562b7e7ff94fddf3c070183935575a17a6b76ab1f0a0188d1dbd276f1

b3607125b1e490e81729191449085e62941c02dfda944d499fd0959734a956d8

d1b5cf73517d3095d3e1c35c4b81535d73207c361d2678b62f861dee2f1f331e

response w

23bfbfc749ef1b9c1e7bc1b165e339a620ce9098fcfe03cb32d76f0f1f2b5780

fefde942d187685a37c5a4402293807903e442267b95e468261dd81a778443f3

268c6d9b4a2ebe416e0cf8112a96adac9536b3fd83c791a187564233beae95b4

224262d4d839ab4ebbf9bf760ced758c59c4ae44ef1fb84ea74042a4fa4d45cb

03df54a5fc3060f7876cfb62223b89666df8e472201f708f12526e192d7fd85a

242cb7c1ad0079a17911852393b4c69e920a9728e26e35aca23e9672bc8e0e4c

a5d84577347a2de4ca5b36636329ae1a36263e69b80fcd779bba68b7294347b3

61ffdd2fdb2cebe4eeb68b5d61fac3993662438ba9a633b256f0cd86e7936683

Fiat-Shamir context

9f6778189b3bcc499e1caee7eb497cfd146ffc64de5f9ea87c6fc0969dfcf480

withheld proof term: 240 054 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFramedQ_iff_qFramingSystem theorem, QFramingSeries:71

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Q P : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
  (Qt : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Iff (FBundlePoint.IsFramedQ.{u_1} Q P Qt)
    (FBundlePoint.QFramingSystem.{u_1} (fun ℓ => FBundlePoint.tCoeffM.{u_1} ℓ Q)
      (fun ℓ => FBundlePoint.tCoeffM.{u_1} ℓ P) fun ℓ => FBundlePoint.psCoeff.{u_1} ℓ Qt)
```

SHA-256 of the statement above

31565c8a26ceee2da2d135ad15375673fc6da8c7ba84e16119bf04799327c333

SHA-256 of the kernel type

41b7c9f8c1999b4808e5048efea93195bc39f7031a5d66c18347de74a2d11ace
commitment C

f30ca8ae6704b28af1ee61fa0c170abf6556858a24c08d8dcde6380d37587dcb
354ba7ab823b9b3026a03afb67c33291d4efa07fd57a0c97daaff0cdfab85abe
0d24ded86ef11e9aa081a469cb879c2389b075e5cac8972d2c0ac835b6e41e41
d1e103f84b513866b78b6227c4be8e7cc4582d039e8d0b607e747404546e4fcd
2b5fe0dc991f1c7deb00fd837075a5c0246efecc36e44a864d91902fed7815a4
9e00d27b2a20de469985ad0ed83d79ebe68064775a1ef98ecc85062992e12bc7
1d97bbdb656cd1bba2cc74b7a00bb3298d73044ef07020a2950844614eb2009c
c1e2e58a915e3d54bbd9bac72c4ba13fde9fde6d5aa83f6b34c7006d59997871
first message t

6d104da6925484a70aa4205aa67e3423ac506b7e042e2cf2f42978d1a32c5d2e
ebffa06623ee53e2a1161be0b3a1daced159c92ced76429ff07d370a0c0da119
77a22d95f6b5392e1a50ac17ced92fd76ac0388bacc0f44f4f163ff7e7266286
2089a792dadad05aee750eef979fb15d6b0e1e8bf06924adbadb2a866c927722
8728c1a02a1de6108b5d09e2d3634b7e1870b76a13251e0564eea5c89f994d67
2d71e40e227bb59dbf6ee86b869012418ca28b10d012ea2dfd1e0cd8743414b0
e15b6125f03b0d9177afc0dbf8080d293076e18ec34639d06d946d8ffc142f50
9ad09d8668bf2b16e45496d4da2cf10dec7ddf4f72fc327e02a7b77b2d0289c
response z

2ff32a00237cfbc43f290fae2d2a40f013f2ecf12bfdd9617ee13b49157acf5e
79909c5b607d85fe1e52da91207e00958a807a0fac7b46a6fbaeb8b2f43634ef
5075208cffe20356f0febf55b681c8618c04215396034f81d7aee3f1cc6c9a1e
141e1ae1d2cd7f73a9b36b137cec7430dd68239c2250c175bdd48babbd8a825cd
c1b80b6d4b8eba25166df541a7335f29853a0f89fad85f9f163550bb95eda24b
3d7ca2cedea6aa0b2c6078eeafa59a9a8b40e694184b9adc7b50d40294cb7642
a9cb478649385ed26cacf2b4f3456e45135afd8076aeac500dbb5c05d3a6d7e3
25d86dfea2dd39f45e4be4fb54673004526056413f42c6c2c026c8f1b52b7478
response w

4820d2311eed6aee6a1806e9318193181dcac431a8c66d95a5202dafb80160ae
2da053fcbec42cf032e7a3633bd50b7cac65094b526eac236b035a66fa229928
d83826f4af713a1e1ed35150bfe28548fb0c60f9f4d808f2717a3b0a56fd5c8e
fb39653524e6206cb925b68f68dcdbf98f56050a0bad411c631d94b962f06cb2
65ae81da390cdcdcb2886020e1f3e7c36469ef9e6a8452f19eb676e540924e051
5ffa94a629430e9f7cba8df7dc64a954885499ef5f531897a418b4fbff35431a
04d36e1d0f7f55d848c0377882421469eeb9942a646f352781906e8f6b2f850d
8338eedb8602737679feefce11acb297887b1320d213794339c852df5fc4ee0

Fiat-Shamir context

83808a8fea99a7f8053c4a7d6ab6a34bfec2dab6b0479a4a8165e913ca877621

withheld proof term: 373 623 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_eulerQ theorem, QFramingSeries:57

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (\ell : \text{Nat})$
($\Phi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} (\text{PowerSeries}\{u_1\} R))$),
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.tCoeffM}\{u_1\} \ell (\text{FBundlePoint.eulerQ}\{u_1\} \Phi))$
 $(\text{HSMul.hSMul}\{u_1, u_1, u_1\} (\uparrow \ell) (\text{FBundlePoint.tCoeffM}\{u_1\} \ell \Phi))$

SHA-256 of the statement above

6c50490ef09cdc4761bf48157eb92b12368accd3284a8d9244ab443d786e34c6

SHA-256 of the kernel type

29d9bc62079893dc5e9b2a769cf525a774585ddf0a57b3abd95efb7fcbf42775

commitment C

ed14ec567b41c8010d9bcb0fd5f2ebe97c617bdd45fc9d6c071074901c10d528
b00edb26acc30666f7e08a649f0c8f14ed3a9b6bc1206aeb2b53e032aebab
41758e2df5125cb7db81e1efd03c96cb89ce872cef01c82dcf2f80f9df15859b
518c488e3b50194b7718ad50c899f9f67e07d28cc1e320d165b9a11682843ae8
58315ac913c752fcd5de00137552dfe0dae99a2881aac77349108c1ccde72aa
7756ab4bee7cf8280ca4d89aef903bbc8b8ba2d3a5b27f343df75fafa0d14706
a85f2fb98670846d5ff3a05de5fd25c2e366a90d97a9b6006f31849b2d529660
9482b5a51297a9c22f5675240f72ff43a5291fe5b836e8d570eabf497d05d7a4
first message t

2395a1305755c44ed0022bbcf1b91d525c588a89ec4c19fdfa536841eb9734df
2cd5444e3d48328445a2da64b0280080ce756ddbaa06773eaa0e7b0355b93bc6
f84aa9545a9c9231f3df31c25a0ec293f4a72a7cf8f247fbbd26ab09e42066b4
1e4abea3d97e63cb7abedfd79a4dbfb322a14e6f9619c9fba76adaaf8d6606e
09e4f4e48795c40b46476623eb7df59486696d917c4982f881fd6f3c06c9e405
3109f71871e0c8629a51ae166041cd509893bdbb2b41d2c0096cf90eb011f3c1
c93a4c692365e9fc69ddb4cf7ceba577d71cc191af7d9301e67a2f29b2ca5000
abbe57023fb3c9324d0438bfc8555c1cab5254b5f59cbd25bcc227bb549c2c0
response z

44c8f9ff80011dec8210caa6f29823eef72023a97bd951dfbb91a3f54a0d53f7
dfb531971e973fe0d8d3ea85899cd2f5cc41f7a88751ef9d11f3b95f494c9f2a
8ea38ba05330ecd4b3c53c8d8862b9f062c715316c11d994a734198f405b8ab
b5de18287f4196697fa4a0f4e2880b837e88d56eea379bccccf5feec5d7ebe48
0e474efdf8681d4a87690058d1a7c8d7dac7b6bc6b2f2abb8dd4b1e3df79c607
df94824d2319b4c7345be2dab40ed192fa07c587e3899a4ec7b1ca04504e9df8
cdfcd39ec246f18114fa2abc0ee519937242371c4641807407fd84a0a0c06a50
4f311f4d16c821c4e6e1fe4ccd97794f5f2a78783c53c243ba60866c59674b90
response w

4a70e3f32f2417fbf50d59e6873e2bc4a32cc14ed4791790c38a3ca7711b34c7
73b500d375213b847966e4def614e9a5616ab29fd026ffdba84984f6448f6311
57d64f2bdaee6094d0d99535b0348ad964f335af4deaf61d920d3f25e4a19e52
08a17cdcb030795422f821b4bb4b710170ee08b6924a509d0c71449fe40bc390
c955a30e3867a6ddd00708d34b7c1063d0e8bb7217378b019915205b59188ebc
8d3d29c06afac60b2eda281e85bb6eb0cac10fcf5c57ec89992590b1b6fcdcf34
a3d8627370eddd30048888d7ebd77b76aed009aded39e977b5aeeee90d7dad55
e251308ba566b49689a6bef6efff5bba14c58f5fde2e6149eb2bb65216cee85e

Fiat-Shamir context

621267e1eeb7d71f13f2f763fe2e29c9223843a7a42e28984112e708010510fd

withheld proof term: 260 846 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

23 RequestProject.SpectralDecomposition

FBundlePoint.SpectralGaugeSystem definition, SpectralDecomposition:46

```
{k : Type u_1} →
  [Field.{u_1} k] →
    {ι : Type u_2} →
      [Fintype.{u_2} ι] →
        Matrix.{u_2, u_2, u_1} ι ι k →
          (Nat → Matrix.{u_2, u_2, u_1} ι ι k) →
            (Nat → Matrix.{u_2, u_2, u_1} ι ι k) → (Nat → Matrix.{u_2, u_2, u_1} ι ι k) →
              Prop
```

SHA-256 of the statement above

795c00f368b2de9345efb1c49af757adb8c81f0f77747864ca1166d6a6dc1d8a

SHA-256 of the kernel type

fbe6114db212c238fda8abdd450b70226b3f889f301b00a4dd8cebf3ce5fa07
commitment C

33778dfd85af6a35117b3ceda2a668d33d728eed1594714c359aa08761e2fe09
96e699c924f51864393d0c9bd0c46b0f30b55d47426002bd123207a87cd070f3
bd3b334220912bd8c98adfd858f8e4de1c2c7a8676f6b79d952058baf389a8ee
9e58b70ed4c4a4ca8e3a15af9ef91b58f49c1e934f0ffd9b4045f3ba00b4732d
5b12249a6aa549eed50b3b1cc1a85644ea47e0fc1182fce788a5e68832459d0d
b90ec8f4f5e3b99c3c73f2bf6a30d3549474920c5fa9ad93188fb8bf5e02d561
26903a50645b624c3ee785e784d42312b67edc7fd3b0680f05e66faf03b334d6
a3252786f566f3107612ca4c2ba2f82119f9cb3623ee07b7d99c4d04d189d993
first message t

30429f2a86bd6835a848f824cbbb286221499be503f955fc26689c8211150721
9506ece25633c72327a7224fb554a915dc20fd939368960daca5004c292ed0b3
44365b2d595006ad9be57d7189eceeaf c69a5310e1341b6644c4f3ac46e8ec74
afac7bbd15119aea98f3720b97f93a96682f186c780e136fbc676dcc2e313e0c
3eb345126e2149cfb8e7e389c354970124fa22899201338c8a828db53b7a2833
c24104d607aed02ca4bf6473c42544c782d1c1ab5eeaf1bb78fa29c5108f01c7
4d3a084a45b0e46621ecb775ec0b4cf8ad87ca00256d259a8354044f11c88c45
894395b9bc6aa695abb1fc42a6b6d46c0e2051cf026a730c13782dab24ebe50c
response z

5176fd38fb5e1e303e186dc21036ba1343dd0e6069cca06e10c5cf95b86bcfd8
ea6b60e46437dd6941dcb0d8189853cb0103d9923ffd23454377c3f4ba804493
760a1b8f4db60343d8d907c597ab70d12570d86379576c0fd04f5aa5f75a1b6d
8c5eddef3121faec509dec92faadc806399bf3f791a8118d113f0f6037b5e7f2
5a67dc18b06bbc360890da8251bbe6f07424e547efe3e27bb38f3e3df0c79e38
eb3c8e08e2d35fd044232cb7369151fbe47024f72f12d5ebacc53ebc093cb63c
329c6e585f7c4018faf01095e53930f0a33ead90bc546a9939f5fad703d7f173
603f153ae5008ccdd817f4bd243337d585c74077282f846c598f07567b87b41

response w

66f61520f85db395f01c901783a9a382fbf96503efdc14cda45c0361036cb13d
75ca8eb90101e597733461247c2351ad9d859cce99e22a7c5f5767418be4d9e9
d03c3233f9d1aca2f5e32244d52963d39ac0782f06502654d7f0962feed5e272
a8baefeb8248e1b59d5cbf8444e828f3799964a34d0b6aa664c7ff0920043979
00e4111215d84ff012b6c41b34641207621a4b3238f865b637afd98ca6ac052b
3581eb00ec5d6d3aea5693f4651b751bbe75e5544c722c0f0d7a543e8f85cef6
d74d8d788bfe85e00ee4b217e682105d4d4afe41b8d4a6845c3ea51ab66a3ac5
c4f85910f43c13d8b9dd82e1345f9cd472f300808575a303dfc4a739505f4503

Fiat-Shamir context

cd9fd32d89b219a86a8242125f343281186ff5c0806c6bfb463946e3006a857c

withheld proof term: 10 208 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.build definition, SpectralDecomposition:85

```
{k : Type u_1} →
  [inst : Field.{u_1} k] →
    {ι : Type u_2} →
      {κ : Type u_3} →
        [inst_1 : Fintype.{u_2} ι] →
          [inst_2 : DecidableEq.{u_2 + 1} ι] →
            [inst_3 : DecidableEq.{u_3 + 1} κ] →
              {part : ι → κ} →
                {K : Matrix.{u_2, u_2, u_1} ι ι k} →
                  FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
                    (∀ (k 1 : κ),
                      Ne.{u_3 + 1} k 1 →
                        IsCoprime.{u_1}
                          (Matrix.charpoly.{u_1, u_2}
                            (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
                          (Matrix.charpoly.{u_1, u_2}
                            (FBundlePoint.blockOf.{u_1, u_2, u_3} part K 1 1))) →
                      (Nat → Matrix.{u_2, u_2, u_1} ι ι k) →
                        Nat →
                          Prod.{max u_1 u_2, max u_1 u_2} (Nat → Matrix.{u_2, u_2, u_1} ι ι
                            k)
                            (Nat → Matrix.{u_2, u_2, u_1} ι ι k)
```

SHA-256 of the statement above

eb4803dcd79de81bab3aafbde3800dcded2efd837b3e72215810e90faf36895a

SHA-256 of the kernel type

d4c6fbc029482a36530670deab907e2c114a85f956497f98d345eb76c4a32f95

commitment C

cf1de359dfd12ee32d795166ae3640394a2aa3a4cd5395bcd651f6dc12364298
b73da0f314d334f177509e522d36270d235ace2a15fc56e2842c87a461070177
de18984ef22827f015d4aed9bd23339d456df35ff02b644c0f6e535e23878ed4
b48124a72679eccc61590fea49fc1d6a834107a0cb1cba1499b7f99e3433c45e
999f2316c6d5a4b42352a308745f03f1cc13a9c2778e428ee3a4e1c56bfb8cd2
023491455b6ab570505b824e332387a7c2f40a2b1f85e9725d55595c69f793a5
c4088d8d86885a0dc6fb73064afa1e7e887dab9a4ef5f7d084bdb072a02adb14
d2d65443d0807be3b0188b336689c1ebe19b5c4d53d2772a53bd2608a1b005ed
first message t

eb56cccf332fd01502608f3a4a1a548913a510d06113233fb90a2cfaec00b84c
804cc3e2ba95b01912447308029a692bbdf031429c8cd2d45332592ca3d7b4b6
3b99ddf3337e2704f71bb0d3d20d4f763241f96be22c01adf4c554a349fb2ab
eecf131acd8f1bab62def883da5ceb175232b3e452a7729091ad44d72c1f89a9
abc31f97b929554f07b2412385bb6fbdf80ea1104df0cdf97fff5a745b80f185
f3328c8914b9782a755e778cc4ef4303695f71bb2505dfe617c325eef8031f16
845e4c508a5b8c4d55ab962a3c070fd0dba36ccf24449e943812633867e715f4
88899caf4cc9d28f2a51e107e78253212414bdf2cd2768c99011707125857ab
response z

1e4fbcdd11c829ea7c0b5d289c3401a9bac290937c856273d1b59c1cde3697878
81ed189af2704c7f601a7820147327c25084e230d5281718aa7ce446c7acc5c3
1c7c67a5b2f37761866a987935aafbee1377907b14558846be8fdd92c1942dac
edec61df88372bec3f79d2edeb7d6bbf7a41b91d67589e59e017a6b3572333f0
d8d6cf23fa5646dc8b7efd3f588890a7d2ea0b4c1d7c3303e0f519d1a62cc27c
829500deeb54d3115b16076eb6bdee12b4ded6f9f48d6eefbc0586a82edec200
ce62eb8feaf033ee7e3af44223875bc93fad86b3aa72b5f8c152a966679f7c53
48b4e345eef78d7afcd33b06a3579afea04c71f4f686bcd4b35f48aa4299c8ab
response w

40454af3459704b0bd77977c60ff23151aa5e06a6d46bca891f806fe8a44e280
08548020cdece94636677d11908a8938f2cf4a54ba2549ad0066dc1d7b27ad82
2fd0db8f9c2708db43389bfd73744cedbb4c6c050b8ba4a6c8754297f54d0d5f
352b02212f6c1b597b2036f03aad48f06d70a7d1b1c5ec44e0fb99d8fe230690
b5c04f2d13b5b43bfaa872e38fce8d14572002fb60da0b2576d95e44f8bd13c
2487c161cf9e3247d0529e2cdb595b3f139922931abf6425bc38613b7e2052e3
f06231d126b423e7e95f4251291fbfb1e764486f36e8abc74d29a6d7bed9cca526
11ad04fd3bbb4bb8c05a7f9187f758fae0db138fee6b9567456d4d47ba69b786
Fiat-Shamir context

FBundlePoint.build.congr_simp theorem, SpectralDecomposition:0

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] {inst_3 : DecidableEq.{u_3 + 1} κ}
[inst_4 : DecidableEq.{u_3 + 1} κ] {part part_1 : ι → κ}
(e_part : Eq.{max (u_2 + 1) (u_3 + 1)} part part_1) {K K_1 : Matrix.{u_2, u_2, u_1} ι ι κ}
(e_K : Eq.{(max u_2 u_1) + 1} K K_1) (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3}
  part K)
(hco :
  ∀ (k l : κ),
  Ne.{u_3 + 1} k l →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(G G_1 : Nat → Matrix.{u_2, u_2, u_1} ι ι κ),
Eq.{max (u_1 + 1) (u_2 + 1)} G G_1 →
  ∀ (a a_1 : Nat),
  Eq.{1} a a_1 →
  Eq.{(max u_1 u_2) + 1} (FBundlePoint.build.{u_1, u_2, u_3} hK hco G a)
  (FBundlePoint.build.{u_1, u_2, u_3} ... .. G_1 a_1)
```

SHA-256 of the statement above

09c98ce771f385611d4def5d8cc03b6a3c1f71f2ad50681daa8b734d4d62142

SHA-256 of the kernel type

4a506d40e775e31ae0b5ec0cc26bff52d87b3aa7b65d14656f657fd4beb54500

commitment C

f4ccb87add1a1f45ff59826264e2194ad37f9f0c52aae7549ff6062582e66892
bd37868caa0108b5b97160c62023a0de3fdb0b4a70e24e8e63275d2d8264f64a
e83da5587dfdd7dbd59816c43474599e0b35be4b22bc88b65770becfb3f847af
b5cd454dc77b1f4dc7ea52442dd99ed644a984ee2132cc1784ee9a39b5575c7e
8d21468c1a649747be98a7920f001f34e7adfb2060e1c3176c3b262b19490e50
bc13aa0344cc378dea7ed93341285130a3dbf7919421a327f2e089037ff3ae26
5c910290f8b40d6932c594ef36a3929744ae082dbd5fd4e0dbd88312c753b010
6204146071bb04dabccce8d5e33b2ce33fa081091bed64b536fe2fb521821ccd

first message t

d1842520a29c06421c1f6806066821074b381febcbf56d4282cfccb380ca6d084
4366ed5d9f8fe342cf7b171c6e40ca4c400b00daafe239b1659deda575522228
3709aa178264a3eb32eb3aa292750d0ac213ec2fcf9a83daee2cc7c024820829
5ae5951d0d0d552ef8de63e8e0fef19fac843179144c9ca36e8ef57c2fe8185c
3701f54278c6bd7e16679f0b555697eb98b48b4e3682d9e2d9b2a5c94c5d898b
acc27b9fb9a353345f156435ef79ebdeb635e348d9810d6419471c759f4cb5ba
45db99243c71cd4434950bb66956e245da8dcb2dd99e8751a841935134a5d938
4c9a988b2ce90808f020cd83dd87f86e3a496700c9a421129918169bab626c39

response z

adc775cfa7f7a4293963ecf041e1b42fea838038f6e8e60f1b2e0f38ef521678
dd835c5fc43cfff0f647afa477c019b642eaf3433c995865b72167bf231f38
5e1546afe60e2128faed8b1bc5bc7da1fcf985f4551bcf471c71a8a031afd1a4
c3d56a484a991d5b5f219e94a2e7cc1e00376694468785d84125e203a08c4acf
99b2332490197af6d7ec356c4687d1984326fba2485259c842a91650f58ed423
fa4e33f434d063892b6edd6a5f8323451bc61cf509449d00a96854bb11041099
a0e04d10b1790e0eabf03e92c29529b67c0bd2c47d55949dfbe13899bda51852
398a2a971a8a6e1c12c30bedd440c9c5d92f7d930ad9b076f05feb0beaec781

response w

efe5aff25e7597b21c918399268df4d15e782881088bd8e11940b12e9d0da00e
81db6fe8cb034f8e08664add933fc69f5d74ffde2d60399f8583fae23c645d36
d649c2787fed0288ebd7d957c59a604a3c084f6f6293998bec637a733c1a82dd
ea4c3a012e274da642dfc73681d5abf0680b2d729361cb2dba66212bb8ecb862
07111831c66d6239b7b4b78f6fc9ed1fd820e93abe0ebd66847ab3e966d4fa80
931a369c0576186dcc69b96fac85fb15cb4bbb9f79387e53ca379986fb2268b3
6a39a8d86895ea133011488200fbee2e760ab30a2c89e7f27f5c113546d81903
502e1691da4f2a74b7bd11bac335ea5132640a98c6107dd1197d501101b7f52

Fiat-Shamir context

9a200fd1c0d024caa81483ad2a54fb305d6f8a3531b9ae35d8c0ef1f5d2437d527

withheld proof term: 61887 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.build.eq_1 theorem, SpectralDecomposition:0

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι κ} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
  K)
(hco :
  ∀ (k 1 : κ),
  Ne.{u_3 + 1} k 1 →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K 1)))
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι κ),
Eq.{(max u_1 u_2) + 1} (FBundlePoint.build.{u_1, u_2, u_3} hK hco G 0)
(Prod.mk.{max u_1 u_2, max u_1 u_2} (fun i => ite.{max (u_1 + 1) (u_2 + 1)} (Eq.{1} i
  0) 1 0)
  fun x => 0)
```

SHA-256 of the statement above

4a2eaf5a831cc7574e91696428937f184820fd42426b88fff04b2f4408b2dad7

SHA-256 of the kernel type

343ebc5e6bd22524cb4f3214f72a870e8805388d3da88778658e7ed017dfe0b2

commitment C

802686f43b39e7e68ba1c83d2344afee6ad64a02d74af4017b714f7830138080

2625d43696242cbc1b675044eefc24c8fb71133e0b4ce8d24901690949ff6e6d

fd632fca22562facc9a3a9ec20aede0127dade3fa4844568d7cc4060aab3c7f4

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04b6957037debfe2cc13464686d1170976081f9822fbaeee19fc5884548d31ae

db97b2c5e9bc37918258640acef6c05422614a104ca012f10bc103c3c1816051

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2a3f4f0bd94b30bb08afb51d300ac095383f4229919601806c3ddcb559f08561

first message t

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29cf0e69411ff0767625c3c4614aa7616e34caa7c4cafb52c2bb15417bc832f4

80446fae0a8f7d6423253e9208bce36df178e360146fb4f47f31f2cc6e43ed59

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e8b1b888750009fc21f98f141c00b01c5f848ba21344376c82dd0ad4f6314ded

939c349cf2eb9068a1a868b233746d818600a29689d4a3bbae97b1e852fb24a6

9ed866d744f60bd84a5efdf8f8d460b980c20c18a9753ad3f58c68ccee2f436c

response z

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b738c19dbdf7746ae8a5fcc2cdc87065ed1e3414d78c8897bd7c4fda0352c39c

71c3be1eb96c212bc827e5bf6e1eaea13a1164401379718337d6058d5f0d3c75

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79da3151c7d88a51c34222d3b16e4c27c518790ebc8a90411355a49947361e2e

868e35b6151203aaab3cdc686c0afb4ac56679969e045684237eab2686e7a5a7

7d119bc92516658b59ce4ed9a684ad074ec60b269bfedbf37e37c8edcc44ecbb

70dfffc2cbb5fc281f4c2837a2d4bde1a1ec235b17966ccbd6b445e0cb9d5454

response w

7ef8549d1d2b030745a5ccaa2f43c8ad3126bb21fe649d07b18567087c5b092b

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26883a778e21dc0865b528d20f09a31ee5fb6b39b447671fc011efcc0404ace1

a9a3a2e30fc3b47239c3d965c7fd45eb5d86392b5ceeb146402c130d875221a9

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a272fb6cade1c6559709431028c2a0f076484101ca1bf928bdc26101df0d8624

Fiat-Shamir context

89133f5a11d17614a5e74f606d6bec36f1d09ad6b54e2797ac6ee8a4f4584387

withheld proof term: 2 330 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.build.eq_2 theorem, SpectralDecomposition:0

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι κ} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
  K)
(hco :
  ∀ (k l : κ),
  Ne.{u_3 + 1} k l →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι κ) (n : Nat),
Eq.{(max u_1 u_2) + 1} (FBundlePoint.build.{u_1, u_2, u_3} hK hco G (Nat.succ n))
(Prod.mk.{max u_1 u_2, max u_1 u_2}
  (Function.update.{1, max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).1 (HAdd.hAdd.{0, 0, 0} n 1)
    (Neg.neg.{max u_1 u_2}
      (FBundlePoint.commSol.{u_1, u_2, u_3} hK hco
        (FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.build.{u_1, u_2, u_3} hK hco G
          n).1
          (FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).2 n))))
  (Function.update.{1, max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).2 n
    (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      (FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.build.{u_1, u_2, u_3} hK hco G
        n).1
        (FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).2 n)
      (FBundlePoint.offDiagPart.{u_1, u_2, u_3} part
        (FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.build.{u_1, u_2, u_3} hK hco G
          n).1
          (FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).2 n))))))
```

SHA-256 of the statement above

f2cf5e7dd95d951d086cfef425d0ca70773a25bac68bd8faa367b82be7397311

SHA-256 of the kernel type

30a211945dade42b3685f00f7c2e0f61bb68be10f93fcb2fb900b3609906c391

commitment C

210e463e9dcb24f9a1c0db3ad39af7da50422d329cf6f4896e7085b20deb3d65

cf0656e15cb8eb7a7d5393c689be4d90de9ca4e387ec28d9b2204a026771dae

51fab2a811bc896313109ff04d0d56616ff7c0d8fcd6a92eea09d6f537f97992

b51730473c429820ca183ee1f3f8ee1ed6124ce32d9926a7d86ae3f1b6d5cb67

767216186ef06f28fd2ffe22149a98b36b86f64a84a19699c872a40b83507e2

633d3a8b2cfdefccfb38007cb126276dec2b6d2bde6f9b5f550d8033cf4ad8

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ee13955090b529607d28d44ec651dc92fb84fc015fcc97b8572ca074ce0315f7

first message t

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74d3376dd40afb2be0bbacf584661b018106d9c49033b09004e39bb1851e7452

d0a1b37b0b08c961c4e514409115ef1d11d2dd351624d01459e603da31b5c45d

b4d18c3791a8881fd389971cf08ceb76d8ab0e08ef40306a3de9af6301953d1c

cc9c7c4403d87040c6f251b9c3d4d97942ea2ed3465ed64b9431243871406e5d

4f46a0e9e8da1b96ed969ecbbfde72134c80e7e3ac4147f34a0fcf08ba491c18

11b66bd9c178dc06bd86626c68c6045442513c4c5fe4e470f706b1d69a184a85

8569cebb2707bce93430af4a58f23b8959863fafaedbbb5f9676f2fdf5709e2

response z

7f5c4c92230d71817ff8bb278fb0aef51c375bb0975bb20938200b697a07d24c

0049be6140cb0f83b327c2e63c6b8d220e8203b396a9585e1c57b5aa15048152

6374bf8bfa57abb35b5137c3a18a5b7863fea17212722024dcaaa8de012569f

88e6c657cd86e84b4ee49a0ef2b0e6510e98732a5011b4ad77d45f3a51d97e0a

d6a8f02a00cdfded5f0161e986be30fa6bda838e34b1d8890d7f0ab11b122c26

f2f4a960f79f1d60f4cbb38db73745e897a69900fec2cbe0e68ade5e9a1d3308

d7e4806901a49a6578ca9e90fd6cae460efb15eb8b3b2782607dc4e70fbca11529

cdcac01c5a95a8fc4e8519f89870db2b8a470fc20d6ffa4df48d79231196aa10

response w

FBundlePoint.build.eq_def theorem, SpectralDecomposition:0

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι κ} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
  K)
(hco :
  ∀ (k 1 : κ),
  Ne.{u_3 + 1} k 1 →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K 1 1)))
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι κ) (x : Nat),
Eq.{(max u_1 u_2) + 1} (FBundlePoint.build.{u_1, u_2, u_3} hK hco G x)
(match x with
| 0 =>
  Prod.mk.{max u_1 u_2, max u_1 u_2} (fun i => ite.{max (u_1 + 1) (u_2 + 1)} (Eq.{1} i
    0) 1 0)
  fun x => 0
| Nat.succ n =>
  have PQ := FBundlePoint.build.{u_1, u_2, u_3} hK hco G n;
  have M := FBundlePoint.specRhs.{u_1, u_2} G PQ.1 PQ.2 n;
  Prod.mk.{max u_1 u_2, max u_1 u_2}
    (Function.update.{1, max (u_1 + 1) (u_2 + 1)} PQ.1 (HAdd.hAdd.{0, 0, 0} n 1)
      (Neg.neg.{max u_1 u_2} (FBundlePoint.commSol.{u_1, u_2, u_3} hK hco M)))
    (Function.update.{1, max (u_1 + 1) (u_2 + 1)} PQ.2 n
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2} M
        (FBundlePoint.offDiagPart.{u_1, u_2, u_3} part M))))
```

SHA-256 of the statement above

17e6b4808c1af9b243c7416628b01bd9a185db392e2df73f09d7ee375f9a4cdb

SHA-256 of the kernel type

36b233de1df006b400d693110c82a0e711f88e6a4e05ffa6266b8247ce905ab1

commitment C

3ff800f3985bd21f73b621e9d11255316f3d0a74af4ecfef81a26a11e61897d9
af684f9c1c336e9c0d734b2167589e92c3d7d4dd1e8dc261809915f3b7265b1b
25e09230078c2e847376b738fbcca46c0f94a576661b302b7859f6c6dc7a1d6d
17074e24ad9d826ad07c55e48b48a00696662cf391ba4f6d2636ada9007ed2d2
94972af49984960e62b1893f3717bcbf0819365842fffb811cf625bde2e0a62
12a8cb95c2449e2bff33da56dd76a4adeb18f8cdccdcfee0a0f7233b70a14139
db281f2a76b0a5d4395b6f3fd0a48e50ae0f81c4e81ee7ef0046c18da0bdade0
a16e19553c76a09ee351e065b0a3ea1217657720ba8eaa80e50748a3447b37a1

first message t

7cf7e8e900cbd633f4f0dbb71cda88b585b685aafc536be287da63a62e699bb7
69a3c9b7dec942b0ab8adec36553eb823027c53b85e37937a2adb7d07f76fa5c
c21af47390204fd459c01e71b4b19530de2be1596d39855e68494d03929829a7
4de1aa85786b8547d68c426b84ab58be855c95f51d4ac2e5b07064b0aee720f
61883bc1bbc8d3f09a393542981fcf7ee8c5fa75676b8b91be4ee7514044ce92
9ce7bdcc5a1a024e0204e2305c107bca6988b1a59642631e82c26b8b40fc58bc
391cffbd81041b25147355f6546d71c9158cb7a5892712ecd388c54003081c0e
52173b13d8c702c092bf31602756c193d6e4ec238a3954b8f3dd56e6371f78fc
response z

152c34c241e538ed18b709eaf6023a58f2972c8b36d3f7bb53fc3f729b23edb5
ae0fb9daea4cef1d132e4e6352be163d67debbd602439613619dd62fdeebd42
7358425a1b0d1af976cc711a724978f779034931477b30812807dd9c0251e9a8
f4d3ebef9b7a3ca3de70196b640717f03c09f1214f7e1d2d5a87bf1e399e199f
783f5f1fe8706514f2a71db20014836c8ee8eafb17091023d901c9b9e5945292
ac2f57e6c1ac083b9cc9c731e2f77c2220d6f2a0e1e387ec9a85fecd4cf92ec0
76e111da74980e90d887e64c78a31a8d74890a47c70cb4d0e14f07913679c725
1dae2a40dfafdfcc39c37e3799627ea90d3792308fde57c8724ab6ef77df23ef
response w

2b4807ce1e14cdf5439b1fdf5c1494d31a9ce5aa20578657c1b65ef7c13b4b98
fe98e9061ced3d3c99816f4732a0af3a90b4289c3387d10bd29777a200f49b4530
5b825bbe2a485200efcf49c3e978e7c283a9a8a51a1011fc11684a14f5652e21
6ba226c39791a1fe8974367fd674d12e34a1a48f90c12c14b7929b4f9d3a9c43

FBundlePoint.build.match_1 definition, SpectralDecomposition:0

(motive : Nat → Sort u_1) →

(x : Nat) → (Unit → motive 0) → ((n : Nat) → motive (Nat.succ n)) → motive x

SHA-256 of the statement above

52d0eb6c42572858dd0767c6141fae90019d12278bdaa57490569441b3aa333b

SHA-256 of the kernel type

aeec166879790b06482493d75f798a3f3bcb3c3d4b9d99dd9965a7ecf380949d

commitment C

1a1a53d200d89a89d5e9fd2effed4854aca62a3217b8ce0fd4611ffc8a38f0d3

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563b1840603f0c19fd18b57a3fda0cc5db20479b53d4d2d0c43d83ab0b023e57

first message t

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c6b41391424e1d941504539200fe5df85b0569b2d8f21757ef3b3f7888cff640

75fd42c0c9a22dea5b03591d9ac0f7a5be54d4977c6a328c6735c065522895c6

614670cc2fc6df1b7e70c961952771a3c82b1b38bf6dedc91c0b332783010939

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fc5f2d0a6cc52e550010975d94202caae140a78369cbf5928552265cb8b22ad0

5ca624e1fdd0adfd1026ca9fae21deabea65bd2b48752e01393731ae17f3cd5c

response z

5ae03a35253c0e4de042120ebd93c4b35692454bb491ffe267907070dc5d5ed4

ba279c6ddc72a320c0a80120f4af084bf632c4a61299d01126992415187bb878

90fd498cd27ab1072300c39082ae243772bb3abe8443fbedbb10aabbbc90b194

04989c330841129c954fc6da7621811ddb2d2c7e012be286261ca04d5f00fe04

d6cae754eb97291c1dba4bd58fef73e7e188611c1781f6c994b9214ea0f2f4d1

e2b6ae4260963b668802b5a42f45b1dd85adc56a882eab3e64e37a4fd734bb39

fdb8d09fe66890a39a4f882364166719d21d7995c9fc9ffa10a6ab7fea8b4bbe

5943978ac636cfbd6b91b3158ee48c7da8977afa759bed1ed8d6aac49140c050

response w

22ee2ebe5c6040a3188448f695ff8ca8fddd989460323fc23d6eb3bb6cbbeaa5

dc30522f7e93be833f9cb18d5c7827031b18660150b4d8bcbbf72767fa48d474

16e52db0d59f9f1a9793793a0fe7bbeb55f9bf2fd601149c0968cb03c039aee0

00d01261a519a6252f5ba4a13be269b7b3b43b5efa696270aa45cd454f6a1573

e6433a531c5fe449f6000f716a9d49d723f7d9720b6f5e313bc8d9461d2947cf

d1e4b0c5af8b7b7b57f808ad94b8df972211d63db723e5362fc5ebf8c68983f4

3bd4947da66baebfbee6dbdc3e85fdda666e0fcb858371cbe4a0893990ac0be4

2b87f2aa9b630e4da300f8c4e63395ef187c72bbb54c18af5f3bc2fbf85d6943

Fiat-Shamir context

b90e80df9079dc65c25acbd05eaaff828839586add80a3f03507d3c13247b7e

withheld proof term: 636 bytes • axioms: • kernel replay: yes

FBundlePoint.buildfst_eq theorem, SpectralDecomposition:108

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
  K)
(hco :
  ∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι k) (n i : Nat),
LE.le.{0} i n →
  Eq.{max (u_1 + 1) (u_2 + 1)} ((FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).1 i)
  (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G i)
```

SHA-256 of the statement above

888c08ae2183bec7bd6631cb18fe9701d7dec37ade48c246435a5a1a41c1b5a

SHA-256 of the kernel type

6acedcfdb92069835c6cbfe434f41a97314b5a70cc4c6f3c66f9a170c70784fa
commitment C

faaf8c4a765d897a525e477f5745f52ecf8f8dfc726ad3ed6f2c108b7c87c2ce
6ef3374c13fe7bedd7a5b422635e9634d5fe26b39bdd604f6630ed2bb68ff78a
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4344b3a79b38e90e635c970aaf75e60e1586c8b0c92169d8e0c68a912cbe7c5
1c1cc99e2cf7e42f8390a30edbea9c1f3e324e3dc5d52760893f712b0dbb374d
06f91ebcee7e3c5ade40ac9d372fd5b052a3c53ed79b69f88ddd554d3e277308
e1561b76b1c5e87c6aa9a59ee9e1fc4599cc840cb58762d615f79fcb6cbda45
first message t

fc6fe71da7e638b89093de5bcb680017444deec2e274296e92db8168c506dafa
d9b8689466bbdc9bba2fd985fc6ee31d627411a50e6c20d89efadfb4b25c982
ace54468e34ba0c0eae8014aab1b62f46f46b9905d449bfa0692e6c41df6c29
0c2f655ad1ac79b9b513f0351108dec00429323a228d2c2cb40a257d0e46ea39
fc2ca9582f0df0be94cd2778d1e2bafb6f771f20be8f0be50f0433a72db719d1
19b15f72bb80acd841f5d56b72709da0b29b6f616c00813a0ceb7b324f1760e8
1840e9334549999492b4a98b8295e77c3c41091c89a7f1ada45509512ebbf4b
4ad79efcf0e025bd28da8bfc1a9fbf2f77f3e86a0637d2ebe6195f95b2a870c5
response z

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c1e91bc1fe72f6f63d0c83991dba13eff396522fed15622402aa96c2310f2045
c74d8fdc4b9424c51b13ef0122ae6ae826f060f018a15ab123a48f1fb41e12fe
344c173932bf6c91f8aa1474a8769b5ddc0986ccb9865ea2389be4fc3acd8f83
a3c794b4136241e60423dee6196eba548a27bd4330973f8ad9797f83a0014298
5b15b9ee82c0d9eec3a4e2ace1e4b8c6b2c83612c5dd27821db558ec1728ce7c
9743dde2507a144eb223eafecd144e171f7aec95dcf1766add96bba51615bc7a
response w

356574eac1ac2b1a626bf2f15f1752250a32dbe47d82acdb911e25080c00a368
3b5edfae18c4b63dfff35880577b765bed7b8192f9480b54b77bae3042524a8c
aa2c21724c6c0582c54cd8dcbe142f660bcddead34b809fde236593b3f43f72
6567968ae670c9702234a31c74a69902d76fa034a27ab9b34778efc8a6a721bf
ce035941278c5a7b4148e9bd2b9332052d5ec878d2fda89f6df1f363c9fabe38
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b30842349e05512125d4330f9f19e2759db53b9c8d73552084c9f56e7bc37e79
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Fiat-Shamir context

2dc7c5892bbc7a2f7ffbad85604f253a9eb551c2282908ff0e82d392a8ba821

withheld proof term: 35 553 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.build_snd_eq theorem, SpectralDecomposition:123

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}  
{K : Matrix.{u_2, u_2, u_1} ι ι k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
  K)  
(hco :  
  ∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
    IsCoprime.{u_1}  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l l)))  
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι k) (n j : Nat),  
LT.lt.{0} j n →  
Eq.{max (u_1 + 1) (u_2 + 1)} ((FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).2 j)  
(FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G j))
```

SHA-256 of the statement above

fa2fb910d9e1bf8a93a90b099a823e45ce44cb294e0ff8ec1399b35206c9b756

SHA-256 of the kernel type

77008d5f89e00394db9a4eeed9db802619c79f74aa61a66d840dc1f82c446719

commitment C

19d99f9bad0798b1a0d94264c60c205698747b178f8693d364393cf7c3c45f07
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332d92fccf5c0dc9129d5cce856e68a6dc3a37802013c78513599ee601a240f
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f7cc1c79f2afe8c385b80aeb456f46140d0675b1f7a8fe39c319613516f271ec
54b23c4d9b3fc63c19148781a81785fb350d3215eec17121973a19603ce15590
d229e04fb91e8a29b24bd13aee225d4287dbf9aaf8650bb9c52056907961e48a
1724cf4903368da1a5adda09ceacfc54a9bfcab41468d73c8a6637c681fb5cc8
first message t

12c98d3c026ed12d591bc2809015336a68356e8e7ecd9d273cdca05d4e9089bf
fc5b59b807fde16c4723cbd6935124a3bc2a6a25ef7885c04abd1222025aa02e
d0e507a68a43dd59707d9787fee635826d3596905b60c42fa3ab650fa7b16efe
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f60ed039d92f09d00bf659762461148eccd3d030e73de386e6094b2f15a473a1
1ab9ee3e2b7dbbd49dccb384a6f8af99f3830bf11616f9a846ccff9db92292a0
response z

2115e5640f1107cd2470aa6b6d8787daee92a0223babbb557424f8eeebd7385
9d050c723983affeeb4c72f921bd051aca450445b9ad550599551d9a0d8a4d0f
17d5b1cb533cbd08c45856792fa0b0afcc92f599d825c3d3ec8b51e72f9c9678
7ebb15881e508bb2e9eee7aaf28400b37d728701a217cdb49834db7aac21f49f
1e6bfff9168cb19752784b59faa37fec440b25fa64713b38f7d7a4b1a53776e8a
874c43283f107d06089bd4a0211d95ca778931f76c30bad2f3e397b2aa1c159b
524557b4f1801c03ac747e64a9d94a19866397d40b32cd64b28de2499bf0dad0
b796592a0d457552ca06daa259bf6fde549ce8ec0f1cf23f51223d5dbd74300d
response w

5fae84b1300e12793914702d30f36121ec07c0bf6d3d800cb66b15ba621981ca
a6ad4371a57bda20f246172ce43feb6fb56dda928cbc52b2ac4f534637448c03
85adc291511d3767e5c22b1f8b3be02e51cb913a2ff54c4f9bb5f52917b670ee
35fa33dd0695fc0c6fd25100adf4f8e8fcc26eb2ca26db58d9f2a2bde052f3fa
d830f59ee3932b27620f5af763d39433ecb1c4fca359c3d5fb6a1515ecc8af24
f2b9fa31f6aeb51077f1dce6ecc5f34c8af43c1bbce1b9b34b686fa15f2a426
464c8f1b6e6a5a3af276c6996d6af38bd4aaace47830aa2eace7d650623a07b7e
345ce89b8fa12432e9183948b6001a4f17fc5f4a27403e71dbf38d5aa8e1e191
Fiat-Shamir context

f0d4d0b4665b38f7126c39cba1e696386f2b3c57289f23885c1de3299f5361a9

withheld proof term: 35746 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.commSol definition, SpectralDecomposition:57

```
{k : Type u_1} →
  [inst : Field.{u_1} k] →
    {ι : Type u_2} →
      {κ : Type u_3} →
        [inst_1 : Fintype.{u_2} ι] →
          [inst_2 : DecidableEq.{u_2 + 1} ι] →
            [inst_3 : DecidableEq.{u_3 + 1} κ] →
              {part : ι → κ} →
                {K : Matrix.{u_2, u_2, u_1} ι ι k} →
                  FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
                    (∀ (k 1 : κ),
                      Ne.{u_3 + 1} k 1 →
                        IsCoprime.{u_1}
                          (Matrix.charpoly.{u_1, u_2}
                            (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
                          (Matrix.charpoly.{u_1, u_2}
                            (FBundlePoint.blockOf.{u_1, u_2, u_3} part K 1 1))) →
                      Matrix.{u_2, u_2, u_1} ι ι k → Matrix.{u_2, u_2, u_1} ι ι k
```

SHA-256 of the statement above

6f60091ad80ae42d96021e77a2fd1d243984c924303392bd3c58cf7495ba051a

SHA-256 of the kernel type

b06686803483d9af07f879278605f2a2bf51d5aa71d28b12a9b5a5e80f1ce682

commitment C

2f643d3c695287ed3690880ed13afeab40392858f2bb77bccce1606fd994bae9
4da175bad4c0c0afd55737a5a5ccc8b8971c7f87814f20c105b89e6862d5a465
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724fbe4de54ee0e82c89bf6896c8fc1fae5df7a49f45f65d5a2302a2faed09ab
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91306d31990bf7c9a35656e5a878da9404cb1f2421ed1f8bb8d48b29e8103416
4fb6044e04ae6cf8d5cd8bfc258855d12142c48727917ced5a1af0fdc12db5fd
01b9e26849cf9f27759189de1adcc6c61bbece28e0bd7921dc0793f9e2216505

first message t

c8a365f2aa6a482d01cd30c55464ea29322bd45415d8d870d0f2158bcaac7c7a
882685a182abbcead1f3da4339635bcfb1a00c1712093235a9fd99f214c49a84
0996168a807cea525f60433b021e673e309068d3e82a653a31c067a4089f5a02
d3ac1f37b1201ccb5e2807be4e5a3106a60d9b5cdabe16792e6b6b81c8a9c526
669efea25e0c04e6a9b63c7b62701641c8a3320080d0237691100ff2ebdefd8c
ea68c7ba4d64b0192f4932ab8d7abdbe1cc4f5b2f86b59e90dda0dacd5f3a0c3
16b9ff636464c63e3a828778783bc70ff18831fd459ec6a43b4f7aaac2e5f56e
264cf5d4c0ac08038d3f3985c1a792ad8f3e22f48ddc83416102a823eb0e1951

response z

249e64e4eed5a4dea60bb85d0693e3f44d550981db9362ced27a422fe3b7a93e
5bcd71bc0dfa94c78bf3021ecfed48848132419abb651c0b0fd5bde42be7536
36b1c49bdc86ec122134ce26d185eba6c54e16000a155bac2e784f9e43c2d478
3e9bc5bc8cbb925ed61034b9d4172448d69cf97715dd95f7c6c70141a469025c
5c9f4b6b676e00d6ba4e46e3b7a8547fb1b0f324c662efea4d171bb4baf78cbe
a9a2b632b4b7da965715275f2752811f78c1a9b2c9cd32e2f1078b1dba69a685
0e773dd7483f72ba7745c67112af0b6c806e9bcf296e6285c0d195cccf336d35
6e1c5c90cd6fe98d9f377d1d5fc81569f1ef07a85a9fd7b3e39bf54c776f3c66

response w

dc694274aa9c5c2503d82038edf3a889ac6e4ff0fc6fccb3479385e73ed5e033
0894aa3821a70dece2e61868464e45a3db6a98cb12dfc9903c4f154279cd08f3
91171fb5d340a33670fa2f622aff47e200374d7448b69cea6f328fcd2cbc71b
139e2bb9ae6951b76bee0213fb92686bf53c8f7975b83b6bb6db1de0a3774f9e
882d644ecd2ab704c32ecc30a576a9b9546887b8276fc01bf6b0d56df3627ccf
04327be3c25c74baf2da799b5bbe55b794784da9eb33d33f1d364b77280e0c7b
ab94cddd6dae3710ae1a132ae1583e2deaa485c5e04989ea39d7673752d3d478
7ec8462670a51b25c12b9d384386aa7111d4d124d121fac735fab45e32efd1

Fiat-Shamir context

6872ea6480140be966858ab66b1c7d96e47fcf4232a3be55c6f7ff9b1999ae70

withheld proof term: 6939 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.commSol.congr_simp theorem, SpectralDecomposition:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
{inst_2 : DecidableEq.{u_2 + 1} ι} [inst_3 : DecidableEq.{u_2 + 1} ι]
{inst_4 : DecidableEq.{u_3 + 1} κ} [inst_5 : DecidableEq.{u_3 + 1} κ] {part part_1 : ι →
  κ}
(e_part : Eq.{max (u_2 + 1) (u_3 + 1)} part part_1) {K K_1 : Matrix.{u_2, u_2, u_1} ι ι k}
(e_K : Eq.{(max u_2 u_1) + 1} K K_1) (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3}
  part K)
(hco :
  ∀ (k l : κ),
  Ne.{u_3 + 1} k l →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(C C_1 : Matrix.{u_2, u_2, u_1} ι ι k),
Eq.{(max u_2 u_1) + 1} C C_1 →
  ∀ (a a_1 : ι),
  Eq.{u_2 + 1} a a_1 →
  ∀ (a_2 a_3 : ι),
  Eq.{u_2 + 1} a_2 a_3 →
  Eq.{u_1 + 1} (FBundlePoint.commSol.{u_1, u_2, u_3} hK hco C a a_2)
  (FBundlePoint.commSol.{u_1, u_2, u_3} ... C_1 a_1 a_3))
```

SHA-256 of the statement above

a66850523ac483ce97b69096c3711216f5fc3f122a047b04b89269c9fcc6acf5

SHA-256 of the kernel type

b26815276652fcbdb61c731b57084ac2f4298f6b331feae016058d0beaac9a1bf

commitment C

747c36c4d69b7659a018ec82ba7915d6bcee9e1e0e680259b6576fb8f01fbcd5
6348330cc472a50be3bca1ae136ca546067365e435ae5995f8f3da0a47e30ade
5f39a71e7b6558391be07349274c79db67a011347d7e6fb654c5def639908539
3cb0f2a0ef6ac0d8aa5107c9b77b76f16aadadfa28324de60238d20c03e1d238
7ddeb09ce4e005947d3530e3e1bd7385dc003e31d0d2e02a96a92e648ed352a3
9dafddbbab56caaa4029cc6db41c0dd22c1c0f1b9c7e5926f6741cdfc31dcf59
caa49465128a4fa5dbc2f5c017d4e0f0a5ee6137fe7d62cf58b79b798ac437a2
f5869a0b1589368ed4abd6a8e2c89d8c7457042d8d80e4822f6480223a020af1
first message t

f1ebb5b0b26ee1613e2d3e71ece65d0b9b9f1a621fc9112ec2358dde975cc124
1dbead117c84d453960aadf003c6a5ec33d8a12c14cfc4aac098d90002eed497
d45bcb3d9c181f35d5741aaeaabb40803484b3161a7c46e127e13bfd1f63041
bb30252e0eb2e3c94c383443e945a492b01ccd8e80adbe6cc7d8301d92e6d8dc
e6ef83fffa5de6a247224085958f7e109b0568a31850365fef1074909fbd6281
c64a0cf678903624515082e4ebee546c6d7922a75efa6e58475cc65d366ed638
0a4532eb81016b728314997b2119595ecef8b81e628f37c687953c554e5932a7
336bc5df85cef4a3fb29218e29d25cc0eb6b1df802f3b135e4b4f7747e43be89
response z

43a88ddabeeb6966a21e9e6c3127e823213d495fed8b939166022967ad6c86f
d8a08d96faf0537cf0839a13d0782beb211625f4e40a6cecdce1638c7921ec14
c6d8c051976545f5bdc470757427ee65e25d83e932980c9ed260d849a58dd843
5c9e04789fd6bee04950904da4c8957a8f3a1c658c756921d66698e5c970ee84
7da87ab65e05907460714c57d3487ef8838f39935be27788695a34cf04274ac5
86338d3a229e7aa5865ef088a5f2f2a2a966d89f0f46b85dc8d6aa101106eac9
29b2fdac8c394a10f6f158d7044db5cf773bc8cc30e87b992286306115edbed2
37d00d81c04e16b0f3a20cdc76960b613264cee70922815aa39fa4e894e1d2d
response w

3a43614852523d00e2376cde5d82eb1b3e28194b355b3554823e6a17e1f712c8
72e31d3d04f1d18dcd9a9f510c7c683dc1bfff0670cdc98d1255dadd7dd872792
6e729df4ceafc7b06f6b92f027872176b8dde8dfa8409a5f972fba645ff3c84d
e47a889fac1c8f24ec21cb29fc4fd11380279e777c71733c3ece44f31c0d33e4
67e1749832e1d5bf56b95f9b1206c658e294e81f9ec6f2e520611ce0490c021b
eeb01308e4d44bfbae3c11b2bc1888ada8cdc7f78ef155b3ddf5fae8afafa86f6
9d9285f7ca36c22c50c91c16eebd27f978a96e7934d4690124977751beec30c535
f820af56e6b4a8a242156d42c3b9e7a7cd8b159b1907cdcc8d871dc64aae7e67
Fiat-Shamir context

FBundlePoint.commSol_spec theorem, SpectralDecomposition:62

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {l : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} l]  
[inst_2 : DecidableEq.{u_2 + 1} l] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : l → κ}  
{K : Matrix.{u_2, u_2, u_1} l l k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
  K)  
(hco :  
  ∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
    IsCoprime.{u_1}  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))  
(C : Matrix.{u_2, u_2, u_1} l l k),  
Eq.{max (u_1 + 1) (u_2 + 1)}  
(HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}  
  (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} K  
    (FBundlePoint.commSol.{u_1, u_2, u_3} hK hco C))  
  (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}  
    (FBundlePoint.commSol.{u_1, u_2, u_3} hK hco C) K))  
(FBundlePoint.offDiagPart.{u_1, u_2, u_3} part C)
```

SHA-256 of the statement above

54b10a726e18b1354f9f090d1e192d920b48451a16241191d70b93573e018470

SHA-256 of the kernel type

66e7533d098bd7be3532674b244d5f4f6777630c33effbd60c1ca61b21bbf600

commitment C

78e02a55afdd9de494d103b430c705f1ae57db20b45d2670ce128fd4ab62adab
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075dd4888a8af74b99f55fcc77d735ac9ada9deac5a8141c054ca039513bdbc3
6cc8940b788c4ff42676d1d3597ffde183593afc6c93417d8e61380f6fee188c
afac6da0d2055cb1de46fc40869e38171157a44301e6d01a3bd74477a679e994
76dad6e7ffe13ac4f321f0d0fafbce600e2be87933761399bd1ff5175b22ddd9
ced1a9f7f3a1700aac3bf49364b233fdc8d997339482f0c7e65fd12d169a84f8
38199276d8699777116aef16a3b6b719412c8c29851fde332da7bbd1656b43d5
first message t

58ef7443cb45cb9c3b7aa1a63e954143186d4f3f0195fd08fcc6c31d9f9216c9
566d831c049eb06fdc0185460cb6a97a26c835ab2d11fe627dcd40b6ad68fc51
8a8e4f66c6b352cc562aa633caf88ee2b1361a95e4cef8ba33d20c220e0ccadc
b5e2263ddce3b90c2141c24479b669fbcdfb022a7157e93d7d6067faa61918f6
c701c7f5200cbb02286284c7d6272d673a8453397ea9e70440b19b840ad3f531
c08ceba491d8231a007b274289a6836eebf28916f0f52cec6e479f773f6fcede
8942e34f8e79def4e2690e1709a547beaf23f447cb2c26021f3323110c5cae37
07eb351597399e3f5638c651cf68879ecbc517c2b6e691f5e3ccd5eda27ef602
response z

7c3f8eaea66f85b664c312679bba9ba8fa47e3f239fb2f0a94de854f666c12be
d3ffad8dc7d565054f3d1d952adabeec8b61022bb7d66e213d9a711b71910ab0
ff5d8e022b38eec006c9793e57ceff2cc51c2deda82ff8f4f237ed25760000a8
5db33b1ee69e37d0a69f085a56e7132bc33d27d4b7c5b2d9bdd0d595993e60f
4b5ca75e6697a32d4a74fd98648138774334bf259e3dc03b5f3e9ddb398046b6
3b26f7d09205ca20295f504ff445073783223e31739f9d5d7fcf5e6624db3f74
544c3b1006066b3256069fa4a200c1e2c4791bd2fb676dc71a45256c9aad38ff
9917783b130b5ce7360ae11ce789ea981fbd37cedc4e203b05dc631940c58ea5
response w

2ccd50078b6cc061d9b5ccd860ddb0996845cd31d264218ee901120c2cbc343a
887101eddd502ac9192a8d7247ccc6b53baa24960d16f733d098693e30c75632
e9e68474830fe843c5f6dab4f7e7be1c5a4218db7f6bcb2c212c0da2fa8167b6
05616ed098209841c93f6b75598fb6e1dbafbc74e59f83d3ffc7484d22300c5f
9d3f9cd4718dc56291f8cce7dcab33f5121a0f0cfeff49ea571d7bf2ff2c4506
3306371409dd7da2dc2c6eeb7e91071afcf4d10cf4c7d8da42a2bf0852a8f0f5
67360b83cd16fbb2d5961496136d44c137dd9be1993841521b3e59f555c46f1e
f5d9a3e9292ba2b6cca7265d4fa77a9e129cf9d465aa1e91a75247f8a659be33
Fiat-Shamir context

bd0c48b21026318a70d62dedd230ef639223bc1c1657e3a93d71a6dc45d1279536

withheld proof term: 38 272 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_spectralGaugeSystem theorem, SpectralDecomposition:202

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {l : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} l]  
[inst_2 : DecidableEq.{u_2 + 1} l] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : l → κ}  
{K : Matrix.{u_2, u_2, u_1} l l k},  
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →  
  (∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
      IsCoprime.{u_1}  
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))  
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l l)))  
    →  
    ∀ (G : Nat → Matrix.{u_2, u_2, u_1} l l k),  
    ∃ P G',  
    And (Eq.{max (u_1 + 1) (u_2 + 1)} (P 0) 1)  
      (And  
        (∀ (n : Nat),  
          FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part (P (HAdd.hAdd.{0, 0, 0}  
            n 1)))  
        (And (∀ (n : Nat), FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (G' n))  
          (FBundlePoint.SpectralGaugeSystem.{u_1, u_2} K G G' P)))
```

SHA-256 of the statement above

5be18c1e82d0c5bc2d62d6e9d772c3f29f583fb1eb38ffafe785636294f1dc5b

SHA-256 of the kernel type

5d8f7d324d5f63fb8c3fa32fea8e27c5c888dcde32ea91ee4a45b8f4343953c6

commitment C

1eb2ab516bbdb46cdd6b87af7d5bcd3b4c49a344ae1ce8c502c38d4522a6ba28
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6bdd6bf052c3ab032933a0afdfc80c2e301e98bd39ae8e6c8fbb652d0cf507
75fb7c4651191922328be02c75f689f4d1ff83f4732dd6e1b5bcca873d6f3407
f615f8620e61902251d8e0a30d2cd0c2e1f3d57ae7b008290b6b6ecb83185e5d
7f7b431764653fe5182b26b35e142c880853dd5feb3109bdd4ffac4b8e2b08be
e0bbc45b9c3c281adf010e24c101b63b30e03950737b750bed0205f85e30d051
1d97f1d1b1326416a64e931d37175cc7f182ceeb78f16cb6cabeb6b145176f5c

first message t

de7297a5a09ec0885773e94a1cbe9b899d2b7977effcaffb7c4a0611a1560794
b536a579885967cad8b12d4e456347333196442394d808cf8d9594623b1e506
3c98b87e0596068b9f0b46279f101074f75dd4c4b93a7478f92e32d437aa435d
181255dbab4ec3f2ccd48b746353236468daf2cdb778d5110ad08169022344f1
8c7e7500890870e72f88771a3f93b5ce213323575e0a55922e77ae07feed856d
492c547a95f796709a57f548338d38bbcec2ccdcc415ecfa281b20a71c8604d6
c7e9b9865c9b2e97d50e504db74b2ba05ba3dc1eafba79779a0854fc3a530467
c79c9134a5b5a5349a9b7ff74c9edb1a4f7a4cd9642ff9701da0522ad2fb0c4d
response z

4f360d48829ecab1db7d8c1a78f77cc86b135a085310095187c3aefa88f22532
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0675bd1d115c289630b6d579c2665c57a9a3776c6681305710931b189e018cee
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response w

22af77f77d96ff07dfd33937fec93632e8187110aa51fe9c2ba3442decf170da
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d96895dfd673364bd4fbd078e3158a76e962d2c7c219807f57c4366d3d8ff33
6d4fe23d97064c03085d30a437c27df8e91cf2100e340230b58be22f667adee1
cb3290bd6097fcfa11d835b465b53538cbf335bbceaa40de2481f9c87ce5725f

Fiat-Shamir context

efcd7327d7b2d4afd7dd3d1d43fb8158ff45f31622d70212d9548c8c26032b8

withheld proof term: 3 502 127 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_spectralGaugeSystem_of_disjoint_spectrum theorem, SpectralDecomposition:303

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] [IsAlgClosed.{u_1}
  κ]
{part : ι → κ} {K : Matrix.{u_2, u_2, u_1} ι ι κ},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
  (∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      ∀ (x : κ),
        Membership.mem.{u_1, u_1}
          (spectrum.{u_1, max u_1 u_2} κ (FBundlePoint.blockOf.{u_1, u_2, u_3} part K
            k k))
            x →
              x ∉ spectrum.{u_1, max u_1 u_2} κ (FBundlePoint.blockOf.{u_1, u_2, u_3} part
                K l l)) →
        ∀ (G : Nat → Matrix.{u_2, u_2, u_1} ι ι κ),
          ∃ P G',
            And (Eq.{max (u_1 + 1) (u_2 + 1)} (P 0) 1)
              (And
                (∀ (n : Nat),
                  FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part (P (HAdd.hAdd.{0, 0, 0}
                    n 1)))
                (And (∀ (n : Nat), FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (G' n))
                  (FBundlePoint.SpectralGaugeSystem.{u_1, u_2} K G G' P)))
```

SHA-256 of the statement above

3d51cdeee5b180a51cf99b87c0b86d788d024b14eb0da4c02f0a013bc11b4a4b

SHA-256 of the kernel type

78d9221a93433928315942d75bb2684e06d4d4b298d0a4a391d615ce7beb77b7

commitment C

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6acffe3a6ea9619533af94fb9ee3dcca9cc9e528455bfa25743ede2c79685521
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first message t

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response z

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response w
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FBundlePoint.isBlockDiagOn_specG' theorem, SpectralDecomposition:166

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
  K)
(hco :
  ∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι k) (n : Nat),
  FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (FBundlePoint.specG'.{u_1, u_2, u_3} hK
    hco G n)
```

SHA-256 of the statement above

d9562e2f4149151184c043b2cf2ba5b4d9af11d84d2bedc0b67349e0d9edfcbb

SHA-256 of the kernel type

34bec35c16a70aa38f545938043c7ae490e2356f3d1e02f3348a75eddd7f86f

commitment C

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first message t

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e893531903c5cb2460c5b90904e3dd39f67efd7541463fb1ff0059a910f7af0a
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1b143e8f2c4e3f69a60f6c23ead72deeb3ff82908d0fc99e7f69bbc7e9354fbe
response z

4e90a67f5c9eb861bdea4da7a40e80f51f90415dfb58789f4b7cf3e8ad8ebaef
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772c60062e401ba7ea3d629066e81b4b0330773ad6ed72712973494af70790b5
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response w

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Fiat-Shamir context

faf847a5da4bc7eb4476f547bab4c84a52be946a4473625b2b4ebde20546f2ce

withheld proof term: 14 257 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockOffDiagOn_commSol theorem, SpectralDecomposition:187

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
  K)
(hco :
  ∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(C : Matrix.{u_2, u_2, u_1} ι ι k),
FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part (FBundlePoint.commSol.{u_1, u_2, u_3}
  hK hco C)
```

SHA-256 of the statement above

b334fc21ae462f9441143a8f186b8deb03e73da456e0f721527039d7deff8f77

SHA-256 of the kernel type

2571b8a5e50c945bc294625fc27d97fc185740df2082f113a457a7a46d099053

commitment C

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first message t

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response z

5e09c97d763787f2470de4b71a19c3a516bed23b9ce1526b9f7ab77c63694b3e
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response w

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Fiat-Shamir context

932ec7938e87da0d57e921be4af10dbc1af256973cdfd6f9458f569aacc42f8a

withheld proof term: 38 271 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockOffDiagOn_specPhi_succ theorem, SpectralDecomposition:193

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}  
{K : Matrix.{u_2, u_2, u_1} ι ι k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
  K)  
(hco :  
  ∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
    IsCoprime.{u_1}  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))  
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι k) (n : Nat),  
FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part  
(FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G (HAdd.hAdd.{0, 0, 0} n 1))
```

SHA-256 of the statement above

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SHA-256 of the kernel type

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commitment C

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first message t

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response z

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response w

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485d2ed6b5d04cf39e841aa1342c254ee066d268cdc847796858a8edac09621e
68f4abd5f4469748853f752f0298e6fd8e52cc0c0fd0ec938775b5a92c36b884
7f02c0e86257613e1c26e747d697e86ef58b3261553f400df7c646d0e5a0d895
6783a6966f1676667fa52e527f922a80587ea22ff1482c4fe6d5af62924f2c30
dcb3b6214381a763d03843854aaec5635540210a8f23231422face290a512890
495b413895bdc6d960e59b70d4a139aef14b85b7864cbf3cb253c7af9221c048

Fiat-Shamir context

685b8f279b240dba4a840f3ceddc9c2a9a8ccbdb07da83fc59560c81dbbb8b74

withheld proof term: 44979 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specG' definition, SpectralDecomposition:100

```
{k : Type u_1} →
  [inst : Field.{u_1} k] →
    {ι : Type u_2} →
      {κ : Type u_3} →
        [inst_1 : Fintype.{u_2} ι] →
          [inst_2 : DecidableEq.{u_2 + 1} ι] →
            [inst_3 : DecidableEq.{u_3 + 1} κ] →
              {part : ι → κ} →
                {K : Matrix.{u_2, u_2, u_1} ι ι k} →
                  FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
                    (∀ (k 1 : κ),
                      Ne.{u_3 + 1} k 1 →
                        IsCoprime.{u_1}
                          (Matrix.charpoly.{u_1, u_2}
                            (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
                          (Matrix.charpoly.{u_1, u_2}
                            (FBundlePoint.blockOf.{u_1, u_2, u_3} part K 1 1))) →
                      (Nat → Matrix.{u_2, u_2, u_1} ι ι k) → Nat → Matrix.{u_2, u_2, u_1} ι
                        ι k
```

SHA-256 of the statement above

4bc3eba3c91153e9c5392a052a7fe5478772c31a7b86969cb3cc29ebeb603e314

SHA-256 of the kernel type

10972247d780df6ac1746bc1310fd17335b0ab06986bdc2e8d83bfbce88a6341

commitment C

40129be4c4c664a2cbfb99f740c32bf646d9285b5fff91eb637be6dce1fd80a7
a7c5d434d3e23e8b971a746dc49d144837a2bf65cd7f53c4f6dc52c1ba7a4d78
1c87a819e51331f51bf0ba6f42bf0608b9773b199dfc88257ac004a4a451d970
7623e81dd1022e61e0a95eb3f4dd033d0d8698876370b02b5b69d18d6bc1b9d1
5a1733583bec7f431a1a507f6808112e1daa4d06d37cfa96e04cf9982cc495f2
cdb362d68c075a94ff2daca855b88c5de84c95b1eb2e71d6201f67b04b496573
2cdec983868a41631b271cf90613a187d55a29c04e975bcf2a67983dc06213a6
091d897896a68854af57a58291f6d47831152c63ef5e5fc5757dc3683906a59b

first message t

27628f1878507522d1a2bafcdcf10ad30c473745f6844768a752ab022e3fee73
93785a8dabc6b1e62d92a996e2336a12ac37d568ca7f09d757e46f665cef8246
02944ba717a7c7e25ed7d1128e24963d6bbfb9fe78bef647a52706e2c2e61e6a
fe21a42b855072c098e233096cb5e71dc4144b81bba59c0be2dff116f23c930d
45d29e9538fff77e4418fd3fc2570ef7695a76640c86523c261615e55d390080
7cc53b547444d2b9516f192b1648ce615f47e4053611d4f2a99cbc0a188a42ff
a94b13837fcc12b548e04a12924e3940be3135578e6a206d1716d4cf478ff775
9e3104674b17b4dc506d52c520f12942177649636d9fbbb786a6aed5ca930e8a

response z

5e7d307c71bcf89514fef8a0d7ee592d4397a4b7cfc130ef47a2e2d5828e8d30
09ded5a35a7a5dbd0bd963665a93f8bc86844405f55c2fc86240b1e8ca52f2ac
bb735b5a3d43dedc13da578a19dd5a86805d696fe2df85b065a1ee754a6f14eb
48c3d33d4a20f4d0015a04f6d56fd400a0301bcef5356c7f620818e72310e29e
4bbbfb92312d0397c39ec059efcfe4974388fb77ed3e1a02f0ce38654ba15c6cf
3fa54f7ed6c83e5de147b4caf7979a99c90e1c42cb2ef59fbc8ad891e01ff2e2
450218e88d9f59accf646df9899332fb724edcdc5464fc349864842c83f2b724
7886dd1a8a66b478fc80183001699aa5b5a157f5a5958fd66adb1e9fadcc0a83

response w

25a6ea20e12054a78f036e0004d909e3a52371501442730184756cb41557f8bf
ee41a2458f27d1577c65afb9baad197824d48280832fda46ef6fd5d07a3bc179
d8a16f5e702465d2cde859c8f15f473f7d40ec0e92030f2f35c4a97a90dc5872
6e87bbf6ac8d2d458a8ee1c8cc2e3f90654da1f48e908cdcba7373c93922bec4
2dbc28eecdb793e07c633edad83896a340b35fbfc9f9d2198d7b081ee29797c
4acaeb86f137a4da964ddb92f917eae19d00aa0de61f8c6b22a351afec66a19e
f7cc3bdb5b7ca709ed6705162ecf93df87a1995a8fe168859ceadc7e6c7f81f3
c939aa26d089b9efa8238f681139c1bad02fc0605dd0fad3bb49b7c424dd84

Fiat-Shamir context

c82d149147602079e90b78453a00ce58f584c63444de843ce3bb5d59205c1b4642

withheld proof term: 3 619 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specG'.congr_simp theorem, SpectralDecomposition:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
{inst_2 : DecidableEq.{u_2 + 1} ι} [inst_3 : DecidableEq.{u_2 + 1} ι]
{inst_4 : DecidableEq.{u_3 + 1} κ} [inst_5 : DecidableEq.{u_3 + 1} κ] {part part_1 : ι →
  κ}
(e_part : Eq.{max (u_2 + 1) (u_3 + 1)} part part_1) {K K_1 : Matrix.{u_2, u_2, u_1} ι ι k}
(e_K : Eq.{(max u_2 u_1) + 1} K K_1) (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3}
  part K)
(hco :
  ∀ (k_1 : κ),
  Ne.{u_3 + 1} k_1 →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K_1)))
(G G_1 : Nat → Matrix.{u_2, u_2, u_1} ι ι k),
Eq.{max (u_1 + 1) (u_2 + 1)} G G_1 →
  ∀ (n n_1 : Nat),
  Eq.{1} n n_1 →
  ∀ (a a_1 : ι),
  Eq.{u_2 + 1} a a_1 →
  ∀ (a_2 a_3 : ι),
  Eq.{u_2 + 1} a_2 a_3 →
  Eq.{u_1 + 1} (FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G n a a_2)
  (FBundlePoint.specG'.{u_1, u_2, u_3} ... G_1 n_1 a_1 a_3)
```

SHA-256 of the statement above

2892e8043042e3073b8c989ab844b49bef31c337cb7d6b7d954aacad1664e8c6

SHA-256 of the kernel type

91b9e949b8d14a9c39fb2f5bd3e377e4e4719fc451969fdd16203a0dcecb7e7

commitment C

3d174c34b705875fdbae610294d4cfaf670d5d202da37819118b7afe2a7e61f2
6418fb0ce12b2977e02f09f71bbd614cd3e6edc78273354f4d2d1b83f3ff0be5
025b57f31904b65baca2449eb2195e0869a4cd580578a5ce4e3a8b43df8c5f7a
e5b987e6dce938fe7e2ab3d51effa93cf2b359662ba94362db2e4a5661e3c17f
edec8428d2cccff6d6555b2eeae29b8c6d2a279066787162e7eb2d9068c8e3c6
ba72c5d9788491cdc544313745e907f3c6d2404b8e00894a1446f4955eda464d
373780752b98480c6375b209e4c262d1b114abe18ca56d90ef5d6c27b6df11c6
ae65028b4f78180a89289bd15b9eefa8d6ab932321e2624c303d0c964cbdc820

first message t

839b292634b2556dc643344777e0806db11f73e4240618e6d45ba320357d5a83
d708e601859cde89834cbd606a5ef1d8a69331cc3de6e80d8fe733dbe2e8f2d3
81007313ea6ab49a0c731ebc3f7b662cbf05bae34abba5b508a1f172d80276e0
6e19baac2d707828a8034d5925cf0621c33781aede2df4b1aeb533ba0164c16c
544e937654d9165c717adc3b919e5852344282c8c3b2b3adfa58568c6b632d10
c13c224e9db5e04a5f975354369ffa448c2a779e4a2e432dd15a752085fab747
e2526a29a45bba80a57738ad0b2b4508031254f1584687f6663c4339b42436a5
bf9328f863f3290ef3c46cc3643bf2e26c880f961ec2ad6a0121581e2a5611ae

response z

5a2008b05a5ca254115adbf7f4105d58dde51cd995523443542a55f0f3fb3ed
43e169cfc6fc8ad7009e06bc4b200a6b11c3e9124e590b69fa7f004c005effb3
13487de7304ad1a7d10dff3f982f90ad180550300e0f3595402d693f87cc8c90
24cb9c78e0f81263bc6ac8d614b20e1f76815ec8c8fcb67c1a60a4c2abae74a6
3bb7258253b6042da8982b72e5d5d686f0ba8d1216c60aa94df3e427755dee0a
ec730c24b8e32217f3ae9d5a7b2850b9f7e050bed2492805b43f6b66f92e565b
553534493b3c94729d64abce5ad6fd7f3a6c88709db4ef37c48e70acd2206a7f
2035443aae9cc310834e18e2f673b5a68806c07f8164de901dd00f159061f63b

response w

740bab67c89a1a4a79845039ec26288da77db9128a684a2a3acacdfc24838450
c2b43b858db994efc3600ebd05f620a036717081ef6b7b0f4fe73c8b5fcf8f8c
53a79b516bf498d5db20e532743be47f86c82966258eb797cb41a58d8c0ebf32
7ba8abc221f9894f69b516b5223ec635a22ebf98d090a21a2dc0b440e2486910
542carfb66be85a2caef064a2fad49a389e5f2a2d49fe097b272de24b0b9400b543
65476700cbceceda67c580b347239f22d24493a23bcea9c3ba942ec5dae0b43
44271aa211ce34d00a817d5b7b46b4381c8d1f1fd5f54d5b8db164cea2d0e487

FBundlePoint.specG'.eq_1 theorem, SpectralDecomposition:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
  K)
(hco :
  ∀ (k l : κ),
  Ne.{u_3 + 1} k l →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι k) (n : Nat),
Eq.{(max u_2 u_1) + 1} (FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G n)
((FBundlePoint.build.{u_1, u_2, u_3} hK hco G (HAdd.hAdd.{0, 0, 0} n 1)).2 n)
```

SHA-256 of the statement above

183f2fc13f6f1adf898d424e01be0738479b7359393a6321974604111402e29d

SHA-256 of the kernel type

a4ef59400e5ad69bb32a22cfa6bede6b90cc9f21786ff556cba6c1c5aa7e0774

commitment C

c304c114c8fda517fdf00836d270b797b42bb80cec69a293f7d4a73b1e5a264b
22d7ffa62d1b86c461e276df4f67611d3aaa9fb51c8b99f2d00654701823d533
3e9afa98f1b9687932f633041007157485022382d276d342baaf12a45c064009
47680f40833f16c20fbd9b29b7fef7c02f43ba4de22c6eebdda3ebae569156c2
5d1e0087275074e7a5a5d5dec6b153f895509d7e1e2a9ba407abd70c86f60411
266cf6666110bb359d77c35d9ada1de8aac2b0c71fea98fcfdedb21f1790298e
dfc6042fe889ea6b46a6513b1b1b41c8f992389ee0fc7a9ec4fa277b2b67cbc
03952e5bfa9fc108d1c6f8cc376f5978627012ac8de9cb7e317def4b2ebe4bf1

first message t

4a2fc6eea46fc19cc6767867ad1776c25e79633a75f717c6fb2ec218e728e93b
13a1bc42390f313678853dece55768f209b667f918615b35ffb384a514a32d66
9401438f7735fbab5ad64031102110430371879ac2647234a0390662556462e9
550f9073f44ac0f8e2fd58278eb9e0e37dcfb222de0c7dbbd9ab4475cdfd6edb
40baa1c6e3627e09afcb6ce36105ff112326dcf295dfc99ad81197d3f25a7567
cda61e147b37a4a74a8d9636933ccbb8edef7a687b9f6c2bcb546b549df0a586
a7e8a4f288b3c5068df5224755b0d0155cdd7c601331a2c2aeb822fb61c774c1
423dbaccf3a1adfe091e3d0826becc5b27f4de3f0971d8f9c4501ba765683a96

response z

71df0f64428f4ed7ecfa1367d9f0f388ae1e6cc2ea2616eaa82f6aeebfb71f3
506b58ff05119cc9081b41d5657bfb3bc71c5e6c19e1638f02843897d8f6a5ab
68894e813a14c5b2f68e8100dd106034e2ec40acb1b443604805c6b569f326e6
82fc29b6684606f3a4479986b63af80a4237466ea93b86633c25eaebc195947c
1f8c7db970e5458e12a5914776851d03aabb5ff3a53d442238f898bd7da8828c
e0afa4a7d3722e7e21588f2d46ea3467dfb6f78e389514980eee612bb7792c7a
5b54eb9e8ffe137fe45c8e7c6a499a3f98e8ebe0788001525dcd4bc7e64deae0
8a6c480cc566989f4444bc8bc85d5e0b7492aca33f630665bcb9c1175d455024

response w

6b5caaebe2688f67f2e2c1eacd18a20cf1826f79992fe2a422fed483cad697e0
7d62812b75a5f249635c10194762a210069c8f9e6eb3c356aae20685d136def8
8eb5a71bf23e60fa5efb0d1096182482d837e6a7f6fa024e0ca15a0391aa645b
17a7dbe3a08c53c24051d4650eef82c924cc9c624467eca442bcc0ba35d12dcb
c062d211be5b17749955c8dc568f1536ecb3a27acd6360f5eaffc88f46013017
b92174e3f1202d9b8eea86ac1564be0059a6a97cafb8ff4f3584e5e891118778
7c2cbaf0158a448598a47a4bedfa2e7f7d8206d22efd13279c8f77a876863626
7441b6158ede46a2593231a0ae3cedff755a56309f77b39b43477a52c91d6e0d

Fiat-Shamir context

e7f066965215bd8d4a8e59d0c38d6cc8ab69244fd26cdbdda2e51e0415fa202f

withheld proof term: 2214 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specG'_eq theorem, SpectralDecomposition:145

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}  
{K : Matrix.{u_2, u_2, u_1} ι ι κ} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
  K)  
(hco :  
  ∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
    IsCoprime.{u_1}  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))  
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι κ) (n : Nat),  
Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G n)  
(HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}  
  (FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G)  
  (FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G) n)  
  (FBundlePoint.offDiagPart.{u_1, u_2, u_3} part  
  (FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G)  
  (FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G) n)))
```

SHA-256 of the statement above

0a9d0aedc9a8b893c2b0a0bcfb4aa0268030150e9ac6f49dd6da369c57dad321

SHA-256 of the kernel type

3b254e16e7f30d09dbaae9a37a5d976029a6f8a63253551601f1821dfc0375a4

commitment C

a75b4500484961362dabcc608f3b9d1d3e943c1f54f068ab0016baf2a9227cad
a7c8db8f2018d0c05b8d00171c5a65321fdbdf067a6278e35c202ea65e3dc8aa
f4ba1394c5172d8ddc1795f766f436adecef4797a30c33a86432de1624a20a00
223a403d7f3317d20f829fdd95ff2d763089e0cfc9d8e75d1eb4cdc71239b49e
de09b54796c9da567ce2135663701952972812b4f618689053b23502c99bfc76
90d42d983ca538340a66d5fe19f622be0f31cd261a7c3f76450809a9fb86f599
45a42fc765259d30b8cb36bce9f54f465ad19e8667925f47a8e506e0b1c3fd57
a59418153b43f82cfe23254487549344caf264540fe625340baf9e5c61afdfc0
first message t

251601d74fe7f14cf17b4d84ab3090a8f5bb9b027cd0f361b6143465f2050fb1
632d5221c1e2fa5af816cdc5c76ac5fd44e82f1e105c1269506ad5f86f57eec5
cb10ac72bb4bb699199ed9979cb9b5b724886756234992fd7438c8b4c163778f
c0aaffa8f0641ed7e6d9cee3cd8babf6ea9e00cedd3fa8b8522d8ae27e789da8
2541a3d098622968b624fd92523be700df8e6adb75dffa5902dc74ac0facd490
727cd5dba12ea2b9a137842e0a8e043ad663c521d266753d0a7bf22a9f5b122
d3b69c7aac67a46bff0a23fe9ddd3e90904eb814a8bf2d74407c70a227db1bf4
01dec6c947a37e3bbaf2e1a6b51b6080087bb431611b6d59fecdd500004b300df
response z

2088c81e99998249b765d04b887b99d6f5ea164ba4662b3017cdd7094d4d3a64
e5012914ce9a2e8be09bccbed76512ed3113337e5e040f8f138c683a4f5d8381
dd4d5cd61119cfec634d06ba312920825da74a361ae7139d11096c8874811b51
b5fa30b3b7ac24188b984279d3c74f97df6c9d81de742e7f2505da06bcca981b
7d8022001ac99efaad8b3e9565a1fda35741f2d192b4fb44ea57c958a4c76c3a
55f91d3dd33366eb271900d71e61d483d8414c6727ef1b0741dcda8db07468d0
182a16c7eb2a0eb6eb5d2ee20295b4d974a377ddc8d0ec78894e711858acebe7
8c2e639e100535c9688108dc0ddbaa6a28c8b8db00ecbec37cc822642d98ed01
response w

150cd2157e891dfe2d28f08a8affa83c23f4c0f868504599ff23abe598d743a8
757cc84fdd89521bb0cab64ac3e4437031a31c68129a01202b06fc4d760b3038
386ed36f4c54d2062179a5f086d0ac33cb94be6851aaf62b5ab8540612b9a613
ba1252c2f97f5e7021305e8f1685598e297e1315d3edde3a0a30a263ed21d73f
dc2f5f3a502e05da9abf3d5d0d1f08332038e96062006e7ec17595bb0fe51b50
d1d90d3ed7d7da54fea8499bb45c294c37dc07ac7eb934193b4e78c13db9c782
9358f81a7ae7431b0e22cc8631391489fdf42ddebfd2c56dfd1a7fd0316c8558
2eef3ea1adb2920f7e68d6c162768001d0596e2c80ce09092f4393bd5859e388
Fiat-Shamir context

f075273566833e0751cfe6c0f2bd95a5be49b7a46312ed7ae5347573b111c4f545
withheld proof term: 90 459 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specPhi definition, SpectralDecomposition:96

```
{k : Type u_1} →
  [inst : Field.{u_1} k] →
    {ι : Type u_2} →
      {κ : Type u_3} →
        [inst_1 : Fintype.{u_2} ι] →
          [inst_2 : DecidableEq.{u_2 + 1} ι] →
            [inst_3 : DecidableEq.{u_3 + 1} κ] →
              {part : ι → κ} →
                {K : Matrix.{u_2, u_2, u_1} ι ι k} →
                  FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
                    (∀ (k 1 : κ),
                      Ne.{u_3 + 1} k 1 →
                        IsCoprime.{u_1}
                          (Matrix.charpoly.{u_1, u_2}
                            (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
                          (Matrix.charpoly.{u_1, u_2}
                            (FBundlePoint.blockOf.{u_1, u_2, u_3} part K 1 1))) →
                      (Nat → Matrix.{u_2, u_2, u_1} ι ι k) → Nat → Matrix.{u_2, u_2, u_1} ι
                        ι k
```

SHA-256 of the statement above

4bc3eba3c91153e9c5392a052a7fe5478772c31a7b86969cb3cc29ebeb603e314

SHA-256 of the kernel type

34efbb04cc54cc007727010b530015293bef4238637657d9ae761892acd4aaf3

commitment C

b30e8d1c3b9ecbedeaf3516eb1f95619f38bfa19abe715efd2f5b821ae6f7cdf
023c3ec0fa07c75ddfb698de3854a39b59c9086274885ac637058a3fbeb4731b
e3956fd2461949a2f1a65f4b5a55df25b63fd52631eb7c8ece380b9381459f8
815e6096f2ae5f8bad7490703eab1cc02911afe0e324c099510f0f4e5937f22f
874f71a1e03681ebf991693713e251c133cdc7e1c3b6c008867a1e2408e0c1f7
54ba88c04c86421853f0dff09c1c8cd935c81ee3fba0ea8899fd8fffb704ff987
b62caf71e78bb19e5da789b926c15c6d7ec7d5500e7da1b3a22141bc1317f3e2
f10adfe92ea0fce3ad00fe6f132342b738e80f028a75de60e2bf19ce3edbf67d

first message t

42b06e5be70ab75368745142d17ae6ad59aadccb5ada4e4765c339cc364cc30
f7eda1967bad5ac61acbbc488db58fd1dbfabe9967642d11a8b0d76118daad89
26b91c7631a8a6a45a1a54aaeaa0095b29ebfc0e08614a7e60d6c606df742c21
975361a39608cc1cf9fe1f16a7b448d7235f98635f0a20246f2feaf4ff5b83f9
0b6bbdd35d0441bb0739cd310e6e94930d2e606ded24dbfd1acf4b60df3a5821
699a8c1a1716e25053d600a430c7de13bf70060293729b7cf267c68329c034fe
58835635469e66213bcc549a48f6d9f010526b19a482bb8f70ee1c84add2e1ee
c9a31763c070365553201d4fa1a496ab6caf0e982b7c8b4c5730fdd78b71ce53
response z

3611d97678d53f768545de5a269681544ec63119f0ac1522a237125014d8d4e1
fe032abd9eda330061ccf2db0a7a56a272cb397ef239fcd0f7611dfd92afc3bb
60be0ee81a6d83e6694824fbcf27d82f64a1280fc38011c9e27174e2a7f1c931
c3f57c060b28e04fe451652d6d4f14f415cf5e57e9368e607b2c6ae067b6c703
c7a4d61f5a77712815f7ef000711864e04a339796ceef9848e7d4d5ac83a89b2
ec3aa50532ce4eeaddbcb953ee0d6a87c753cc79a9e66b825357e139464630b7
7f48abcd8c48b232c5836ac0a8a7e6fccb7132689e629a4670cf257d7839e4ce
af99ed1d0ffe311e814bfcc0762a8512a90ced2a39b9738bb682095a1d044109

response w

5a5eb4de5824e74a0d67e89b4df6e060b6dc9c34238e07df0786d05cd43e701f
1e3f9bde7d8abce7d6ac919a932cf188d8362ff6d52b779150896754e42f770a
c5d4be9b02f351c941f59868997776fabce0c809b2931dfd4ba72b01c418179c
a0afb8c4c63368152a5b5230ffd619774aa55e24690abe6e5596d622e9ec10d7
fabbab72d90e13722c8ece22a6997d8ff3c84314c78fbb9f2f00729440cd1372
fa43fc19cbc263e9a79e735bb78814ed6bfa099ed7977de4471c4f79502ec7e4
b64f73c51d30990e312007394bd5c228def296e772d17b569b30d27bfb6426d0
999aeb1ed637816826e88e5d48c86e2effa4f1448a551ec11cf2eacc71096265

Fiat-Shamir context

50fcebaca8687eb8dc9151d0b7999eda7a59c7c6f4b0e22e153f494942f0751546

withheld proof term: 3 514 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specPhi.congr_simp theorem, SpectralDecomposition:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
{inst_2 : DecidableEq.{u_2 + 1} ι} [inst_3 : DecidableEq.{u_2 + 1} ι]
{inst_4 : DecidableEq.{u_3 + 1} κ} [inst_5 : DecidableEq.{u_3 + 1} κ] {part part_1 : ι →
  κ}
(e_part : Eq.{max (u_2 + 1) (u_3 + 1)} part part_1) {K K_1 : Matrix.{u_2, u_2, u_1} ι ι k}
(e_K : Eq.{(max u_2 u_1) + 1} K K_1) (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3}
  part K)
(hco :
  ∀ (k l : κ),
  Ne.{u_3 + 1} k l →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(G G_1 : Nat → Matrix.{u_2, u_2, u_1} ι ι k),
Eq.{max (u_1 + 1) (u_2 + 1)} G G_1 →
  ∀ (n n_1 : Nat),
  Eq.{1} n n_1 →
  ∀ (a a_1 : ι),
  Eq.{u_2 + 1} a a_1 →
  ∀ (a_2 a_3 : ι),
  Eq.{u_2 + 1} a_2 a_3 →
  Eq.{u_1 + 1} (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G n a a_2)
  (FBundlePoint.specPhi.{u_1, u_2, u_3} ... G_1 n_1 a_1 a_3)
```

SHA-256 of the statement above

684582eea946f1a0aba1b93805ecc9e88ca29320945ffbaecb02a68ecd30d7ae

SHA-256 of the kernel type

501b94f3ca0daa5be09501b29e400a7022458b72facf27e244d434a64412bb59

commitment C

b29f74dc52de8dfae15bf669cbf9ec4683b334fad801e5867c2589f4b1c907a9
e01f14c903e66c627d54775b92fcf62c717017567f263ee9aab4e30bfac4b443
8137cdf21ee09b918cf2fa1126bafb828024004ad605832a7b39d3b6a845e0e8
33e612e9e33022dea3f8185101c910037b2a1b02da2b94ce70287c7cf50b8cf8
8ae4466350bca040bf7b4de9bb95c9671ba13f4d7ccea55fcd37d6d4f895021
e7f297f42d0a58bb071682f1ba3b74092ceda0e02a969b2d97bb61a0f8f5232d
d6eaf69d502781f1889b74fd3fb711bb261e1349b192b842ddc4d8d203a4fde1
de29e1f2b76ad89b6e0b937c0a7c9847ca225655b706c9094ed1573d736bfa23

first message t

ed376973349b4f80e433d893835d9025103becc77a167cba6ac2ad1ffc46b50c
1e9fb0c45f8e838bc888f357655e03a1f4cefa3c37dd0b91dfde4d0ae9e315f3
4df29560387d7bf08ee8f7d46c935edbf2371d600052ad96cd22c9cf8cc1c935
37765560839afaab9049e9eb60f2eccb21f39945c8785f171f3fd30bd76112b6
c09f393d80e8a3c30de8fa018cabe2b5187a57228c7b8b45e1fd48201ecb48c2
76b5960f8e87d099c64e8f20b69cbdd22f5f9d9f14489f93209763cf9db906744
2b0fc1257df27124521b5dc7d16261d96471ee381fbb8d9c4e0dc67cc5b2883f
386d7c33d2f56877804662f4690f754d0043d97acac0c654ce06a475f0b79d3d

response z

24e3555da1aa3c586532edf8dfc853f8e76356718bed5853b32cb27622bea573
b84d489e6fc38271fbc910ab5fa3aae38686bac0a96dd7f03370622b10dcdf06
bf1a1e278657b8e07ee830b972f600da07b3f54f49d08aaf0d53c06f2e4aca5a
757d0a331e3d1d2ce1085fe4bb1b9db408f8c2c442fab728d682f7fe7a0682e9
cd5ddfbfb1a56741a6d452c1648728e789f7ae21d01fbdab33a22ef950bd5cc8b
3fe1cc236258fec344d441232068a3919de8e87ab370c138625cf8f9a8a1547d
85faac775ebc6f0dbed3f8bf5b658c548862b010600939f82d237928cdf4e21
53bafc37ffa037a497471e24fdebee709d89bebc9bc3de0a7d3b550d53daf557

response w

7de10d14c33685281fe3c44e5d354865d5928ec82388610cbb79e3c6d352dbbe
c50b536fef1a9015cd9e4e166490ff687e61e4c9cea262d83ee2dba3592d0739
b62c0ae28f2e057df3f0a41a3177612ddd3e802bba72de0c91e6172e434dbf35
759c16bf03a506f460ef695c1212b78a2e8b4edf5eb3b30f1f667c680c8d246f
89de575043281a6c7983338e21fdf0584a9bacfceb9fb6645b6fba7df6fb8f3b547
c9079a55b369ebb0439f59cea44070a0c4eb46ae52ffe49935073846ffbb1c95
33387bc91d5196b422267611857dfc2faa92a6b3f8c41839d778140f9fda79d7

FBundlePoint.specPhi.eq_1 theorem, SpectralDecomposition:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}  
{K : Matrix.{u_2, u_2, u_1} ι ι k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
  K)  
(hco :  
  ∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
    IsCoprime.{u_1}  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))  
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι k) (n : Nat),  
Eq.{(max u_2 u_1) + 1} (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G n)  
  ((FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).1 n)
```

SHA-256 of the statement above

a3e5287da5001d90760f17afd2c3b2ec0d8be2304dbbfe29311c970cb7e381c5

SHA-256 of the kernel type

8ecf132a29d74bb0ece5d12bf40eed2912901dea5b84a2146f8dec56086134a1

commitment C

9ac180b1f880126f751ca3d06e69532b07a748064fbbb588190a995742571f0c
2f28d64e1737cd33181129955fa6d2ea40acc963f2e4794c17a7e671f528076f
6dffa34072acb4368a20ea4fc6144b8f4965ea9b462960907f3d75c9730912c9
bd0461c7eac4fb613b09cd950c82f3102a21ab558f75a9ca423d7adcb55a6c52
9a7d88131a487e46cb97baf894b2a65bfb560df6494541fcd7939d4df19ee6a8
1f323547ee88c26101000a5c12407148cddf5b39c5b7c5536c6c4400347b452e
b554f1754c3c642834a3e529b957a8498b9814b312504acbe9d525ba9b49c0ad
f0f27c6c946dae104f350b86f4cc6713cbf466cf41c002d1738a3e2e81188fab

first message t

d82b7ac26637b250b0864b8bfe7d0cad6cc67674bc13b51ccd13a75a523a61b1
31c95a21f162bd2277da17fb0c25d8525ee577587694aa91ab8c7c48faa33c3f
b15b572d49cb683f3c8ba5ed60d2b22d8cc5ca278c2491f34ed9975bbebf8d15
71227d0285bf2ccd71f819e7d0d27a021d3ea8b0169de3c521969c98217c1391
3cd8c8be317503d1801a7e992aca7f7f38e4e45b390e74accd14a74de7cfee48
380fd2e47fd03ba134175f45e4b6ef210ac428700377e218ba04549064feb7b3
f33a28239931a4402602393434728340844b2bcfd1286ecd984d4e8f849624e7
4642b8a361a4634fddb86adf7cdf781937a5457a03efe42234d0b18907387ed1

response z

3be41c60cf3c3ef64a691b0d45d84143481d86e1934da3d29d92148d03634b4
fc03f68f41a838adbc79a96ebe9842ff4c24ac636269a4111e4ca9a09326e95c
49d07ab57bc7e62f44d85006703a5c11fb47bcb149840913ecc4abb1f32c76ef
4b9bd89115afc0fa83e48700ab17954bc16bd992bd4e10b878880431c9b5c843
364df913ce1c9bfe20fc95dc68beb8082b628fc4494b090b16b0f90233bc2848
b5faf5e9e7695dd02cae3586a09eb9e6d6f0fd0799f73e7ca5d857b1c2e8ceeb
615e66f2b0005711f5188785f9254e2cc7f6d09017479b33aa23785dfa29439d
6fca9f1e22b886bc2d320c3c8995f1fcfe19a6763bdf6b997498d3dad4c94315

response w

fe5e6f3d4df46d6a746e8edc59cc21bb4d0fe2bd5959e9eff94435c7bd1ba6dd4
50e4ce843d04cd8ae5eaf75a4d24c94e1d867514b059fef6cd49e30139f53d31
3e376af892aac83d4bab4b228b4e599c51b7018c06e1324e069b57d6af73ab0f
1fe6cf260148f714913e6c0b1d68db1d8abc0c1a53c0a742ad5ab6e5f95056f6
87e19bfb86565017d9d43009220ca23192e3966773c57f7174d8391bcfef8b23
62ec2bd36404d0291324065394c2509af60789985c4ff40c2eaba4eccea7cd29
f7c6f43d104acaa66b7260d55971f3bb050e7e53b6b7cd940b05d8cf6fa3b55f
875ad0bf682860603fe438423330bf12bcb42be25f8b5752e758adb8ce4346b

Fiat-Shamir context

cb50012e356bf5351826d58aa22dfb6fe328fe13eab4fde61b4bc6727de820fd

withheld proof term: 2215 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specPhi_commutator theorem, SpectralDecomposition:172

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι κ} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
  K)
(hco :
  ∀ (k l : κ),
  Ne.{u_3 + 1} k l →
  IsCoprime.{u_1}
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))
    (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι κ) (n : Nat),
Eq.{max (u_1 + 1) (u_2 + 1)}
  (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
    (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G (HAdd.hAdd.{0, 0, 0} n 1)) K)
    (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} K
      (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G (HAdd.hAdd.{0, 0, 0} n 1))))
  (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
    (FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G)
      (FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G) n)
    (FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G n))
```

SHA-256 of the statement above

baaad9167e292eb66301193546d7c7c22398028e388645b01b112368d6edcede

SHA-256 of the kernel type

b0acf15df865be81b36660ab5621979feee17d9d502a970b33ca98ce2ef58efa
commitment C

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859064cf1e91fb9e00f41b8f23d834709b147361c82f16c9e2b499d9eb535f14
e6a57fc84488cbd7b9fb9b94ca278d34e848775ecb80d5898bc6bdb3e48b138a
9cbe97e1868ce62894afa194b853b5e718414099050a52ae1545655fd7f3ef02
e5a9a90c7d089f3ee5b50b8a015a51199f26b96e3c2bab61e009610121f089c1
34d804d134eb3694af3b09d2fc4050d0f025c67feb058db9198fb88ce3cd1821
d08e3162a2cd706b4d1e7c08349b3d299596ed2b389b71e59e5de8816e49e541
first message t

27a47a02e7f25d303083f97357b36416fa0b44e2ac56d16bccf79b9771ace626
d68cd09143ec07e1f46e651351866b7e9beb6b2d7e83240406c84aefc8d7f02
f5dda042227bddb39e22539b54a519e1b2611a999c8f789f1348f05ea20e7724
bdfc875511e52060c0cd8a44da77c33ccbedf2b9da2defa08acd535cda69db01
9ff6eb70b25dd703d64445bcb31f380bd521081692ac2792b0991abaf2d8a032
7b7eeae21e32d54347241c9f93bb068076f7e7421b919f6ae1e89102faa5c2b3
3f60bff6364ba0ee63767c1711a812952a8871a7d7a227c13d96bb87be18bacd
4558d9fd39ea2fdc9c57e6ca33e8b498c7f521ef86a8c3dcb02e7871c45e1025
response z

6a237de0d43a617b48b594d42ca6c29d8228c3a9118bc16d636af56ca8199f31
30fa30cc229aeeac3192b095b89b1cfcf9f025fcd8e7e4479275923e7278442
6393f24f3df95a0ddc53bae016de05a92e2d4dba2817908ba4b74e51a539879
6454ffe70de6311154d0e38a1dc61206f0e365e0305a1e3859d3223d86924297
35fd518cd33cd9ddee13bcfbd08fe2a085fd88b03fb46b8313c89c5252ba35
e7d2f0ebb3e3d1d678e23708eaf903b35e83773bf53854ffe6db4e5a78db9b91
89c4bb2a2969c04e8a28482385ae6d345c6cc37dc58117a7ba68b63d1528cafb
6468e9a3c869eb67012bb66ff4bfb84a239e4fcf4b302f3a45802f521cbe3c1c
response w

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d4801c3952a056aef1d8440c8dfdcfae969bac47907acfafa9d49d9a51b0fbc1
c1857870cc022438a5c6cbf01e730a76a174744b9e8ea8bba088071574f8d190
7b2933a3d519c0b92b476299064deecba782542ed0153b181852ff792b4a3d6d
fc53b28f0e6579a5a8542f269a751641c0d486d8410580c0ebffd2260fecd416
381930fa0c6a1315b82581befeaca68706a0f10c23ba49d402b5410900862952
b29996518ebda088b14f6269f3be38d0e2f31d01c7f317029835aa0806aa282549
1f204846fc3350ba9f9b4bf6510287764f866f9121f24811152983ebb86283ae
Fiat-Shamir context

FBundlePoint.specPhi_succ_eq theorem, SpectralDecomposition:156

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}  
{K : Matrix.{u_2, u_2, u_1} ι ι κ} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
  K)  
(hco :  
  ∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
    IsCoprime.{u_1}  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))  
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι κ) (n : Nat),  
Eq.{max (u_1 + 1) (u_2 + 1)}  
(FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G (HAdd.hAdd.{0, 0, 0} n 1))  
(Neg.neg.{max u_1 u_2}  
(FBundlePoint.commSol.{u_1, u_2, u_3} hK hco  
(FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G)  
(FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G) n)))
```

SHA-256 of the statement above

15db3085dfe6f4f1a2e35395e472a2b6f76d660bc3c404f293162bac660276cf

SHA-256 of the kernel type

b779a9bc085fde51d0b9b2c7f31c6faff3fb90ff5de827f7cecb12c45a7657e1

commitment C

cabddd8b9371d151f07cb7962b662fc3ce6c6f62a25f4a13b293e6cc9a595aa0
33a6f4c9087648965f3ad00d3de847c5816cee45b5570501f0661a0ad309df68
a9a5dabee3284f73eb6cdcbf1ea1f28c1d7f97b721f5c0ebdca8805d34c75a25
a4099708cbf069f16dd9f6af0ddc7b75c703be43c7780231b3556eae5ab2c3d9
e89aff21730e1f86fd78c1a1d74189db3dff982efd687e8837318a9d2760747c
a4dd09713ae84c96fba7edfe71240f710173b876c459d5155f5b892aa7ccf844
bd3d39faac458f5de8b8b4c18442459a51672af9486d68c6b162f55a8a27f1f4
835de1f2620d0a83554e1c1f1ce8347795cdf0795290d77d553f3f633932b2c5

first message t

dd67af44632b161cd567edd1990e6e076727710a5691dedc11039e4798a400cd
f00f9444b0727724f41f03d9efc3e93969cc4693f2dc3446b6455a1443f90f05
a755a8f220537a315f69c69639422e51ef9d36a2ae7621d77470b7e34027d131
13ed55212ac9f4f546917e5829710e78b216e491e9bb240331a036d42257faff
d1c79dca38fb034ee73e39efa7ca40e7c2e6f363f7ab4213cba637e8820179f5
71234d23837fb0676a494468782dc53c77723348903f9e9d762051509e7ddb07
0afe61fcdf77afb3a43776cb8da5f466d4d0a720ea0994751f92720672d9878f
1295b9405e6f8db5a16524c0b57b6497f052fb2b90e1e3bc49a25eaada6ce7ad

response z

1e87aa3cb7dd624e432ca88a9a4ff871b3876240f7ac0c19c32c951ce0b328aa
d31aa6c930afde52265f8d90febaac47aa6ecd746121b1bba36f4cdf0c0eeefd
3e6eb26539cb2f32613f0721e3b9e5916816c46d5bbc01b856b55aa150836805
ec8496e109426b5bd4a92370ffcedff4b08afa7c63129890c96148215fd4820
6b719b0b70659cb392a804436905d3487d74c0aa2590c358dc01f428067c1e89
f62bf3e0a4b708a5db5ecd7cf2b937b5ab3dbf7adc0509e64da11037b7e33650
601173edaa76738e3a8e27c4ef0f4af39ca3d82de06a54e30570f31694461335
2dce944cf40a10c130d77152d0f7bb30ed1140e9974256349262405023e52d64

response w

722417e6ad6ed77cb9dcf326c4bb70247108086b7bbbd7da19fa32a1ac2ad09
70f6ccafbb8f2d4a844e9f6f5f29a2fcbbeab4140ac9007ed6acb8e3db96eec2
53a16a6fc0b894719f880871bd569a6d4b292bcbf3e9cbf63e3272c106abb9ee
2b6216edfbb1a22158fc1a36195a03d10b50f5452211e50b195797e5c48bd2a1
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032f4a4b05c544ac33136ac1b8bce9023fc4cee247297c2a02a6efe13db5d1d3
772e4bba68542d8bc02609f0788fa74c57f66e6772b5f570eb490519c51da72d

Fiat-Shamir context

429c4f5c44d138653ee262da67af9eda40feebe1a65d2809993645407fb47f96

withheld proof term: 65 564 bytes • axioms: Classical.chc550, Quot.sound, propext • kernel replay: yes

FBundlePoint.specPhi_zero theorem, SpectralDecomposition:104

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}  
{K : Matrix.{u_2, u_2, u_1} ι ι k} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
  K)  
(hco :  
  ∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
    IsCoprime.{u_1}  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))  
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι k),  
Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G 0) 1
```

SHA-256 of the statement above

96ec4de7bdec413ac0d68d644b4b422b08c9f8e25bccfbc5b94fe6ce2becd73d

SHA-256 of the kernel type

93b7e79965e6c910d70caf26b2ce097384ebec40d86da3e624e63fed564c6ce2
commitment C

3595478a3a7c0501fbeb92e0c5ae440498da27e466194712c6cf018e533d63ca
79f77a235f6e5a708ae3fc6f83d7ab05c785a4f3c828d43f3b433ff096123101
011b116397abdfb59ccd2ac96ba7ef9ae59141581f38b84ad64c4b3259dfce8b
76762568a56c5011a8b09b6ede9cd9cf3a0bbc6429f652cebb4fa2f7d838cef1
4e59da85d2b12c2eeef2b646fee24b36e398c4a904a3cc4ce61b72554111aa5e
337801f93db122625c03f7d402bad0f55c82713e4f2fc571e3512c38f626b1aa
5f1deed3589e45209cd52bbd93927341b3f16d958e99e6fdc5940f0ebb80c158
767a2c8f25ebff4579fbe76999099d219ae6d760af0ca2403c70411e6588e92f
first message t

b2e236fb8accf9dd22372326c88b173034a3e18bbd5783b0db602dd201be5d49
d43e63175fb5d916e108ef77a0e86e92cbbe37bf25bcc2521788848efe5dbf0c
7f298d0c58c3a9eb8fb2fa51ae3162cf7cbcb01e98c129924cdca0a4ce942412
a5749f280217b7aae09238ffc5f192905db0da9716f2bbe5055292e02acaab28
11c592be7697a8ea2a8b654234d9755be7ae0c79eea3b9515b918386f2c66269
15abf07fa082d9de649608afaa3fde63821948feecb525167a022983ca03d431
73b3396fa4a8e0609469f67a4edcce1df107eb56fd7e2b33ba5162a4a4ca1546
4a2050ea42ef493df8edd09c36c53c510689799c592c273efef3cfc0832e4697
response z

11b0162c16f8d53e97f9052d262521ea89d0c24d604ad72f6cebfc5fab4e8e2d
d1a255d8744c11d8b6d8ad09ea6937e7354a1a6da07b6cb4d0f0b411eea243e6
2c5e9a4390e551eabd689d912adb40825be7ad586590e9c3c8967c8a64be0f58
d9dfb48f583e95af9a591eb4c3fda37dc768b3143867e58d191712e38a5509d4
e90bdaf7364139792be0663bab7d3039625ab0a9f704a13469804c3984a96f02
de9b5ec308c6e8b3f24a3f74ec85c47ac3235d572a37aae6484223af27a2b268
dcb21fcf32da7e6e9800868645efd5ddf9318cf98fdeca8152a2d5c9f4c3f389
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response w

39f930cf91a784ed69f221edc3cf99d60a49a40d28947257389f31ce0fb850c0
328bcd97fac59bcad1d6d31856fece10551bf009c61cf6bc40dcb4845f553be
b0815f8d25ca7c6276c49e549ac6ba57ae12e8baf9652a9e56863bd970df1c58
45418f044227c8d780dd1cf7fb20a7b5ccd9d61a148eb9dc504f42b093f2681c
d87f1c692b48a1d1081c0083b5e636a88322e8b6d76a2372764098334d1e4f5e
bb6b9575e18196491f87e34a1fc8666779f980a702b6dc117bd14bf9a26183d5
9c029494a17124ad0345a2c925af718f781ad9f68e92788aaab7a008760c84d3
6d036b90d51d96c1522eb7ce2b497177ec7932d539a15375840e7d1012c07b11

Fiat-Shamir context

40fbd8292d88d9b783224b8fc534a818a30c201a780d90d0e951f741729882c3

withheld proof term: 14849 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specRhs definition, SpectralDecomposition:68

```
{k : Type u_1} →  
[Field.{u_1} k] →  
{ι : Type u_2} →  
[Fintype.{u_2} ι] →  
  (Nat → Matrix.{u_2, u_2, u_1} ι ι k) →  
  (Nat → Matrix.{u_2, u_2, u_1} ι ι k) →  
    (Nat → Matrix.{u_2, u_2, u_1} ι ι k) → Nat → Matrix.{u_2, u_2, u_1} ι ι k
```

SHA-256 of the statement above

745224ced14eed056d8bc7f71bb5798d941f86c0fc6223d87cedbef30db1022b

SHA-256 of the kernel type

ea503467a2da3ce4d7acee18d998ed710ad89b12217977972aa5a7fd206372b0

commitment C

5009e9883fb7a43d12c69e37ffddd3de81879011dad6191939f921c14f58675b

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25111c4311022756a17bb7a44ca87598e5db989b9557072eb23c2989f61e8e59

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first message t

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dd5ffa0f65502d6c411a930398e2dce636bba2b5d7d149df9c2efd2befb63d59

8c508be7873f8bc831ecef89b5d17d11cce5d43d66760fbfa0f39060576eacb

84645248c4de614d2de5eb4c55639c82e74a8f903a703175a80be2fb92d5d710

e59e08836219aa85800b9de5bfa732fb9fc470419b778c2ea8bdb3e6b24ad638

response z

373d158d318fb7a4a466ef26f837b1d55fcd878a86b487818974ccacaaa8222d

8d7d3620b892d9fe911ee0ae59bb54896450582aaed88bc6e2127f45cdb77a47

30408ce7fd877a28b4fa70f29f54053e43dad384e6eb9e6d26a1adb8406f5a17

76d493af2f7c4011c962a0dfba9b878d09a1ad6f42a1668c138c017d16092906

648adeaaa8178551525de0ed10b6799e419d92aed1497d43a515372e697c086

117c25f44e39b9287ee3f63286a96a4db0a5fff45abc859aad7e40a1c8a4ea10

c380b78171e2277beb7b4c6d5a87ee7a057f4fa03e61bb808a372dff28b51b2a

8e6e0e6bdfc994ebf80ccd9284c2d4885c7f5c5df33c2b47783a632268b75608

response w

5b355a54f1fc6d1d949d46b66e03fdb6084d4c4aec46ea337e8623026cccec2af

7b53612cd72f1fd986396672eb85be0714511ebc2de77f4a6acbd7b1707557c4

0b7e33962ca9caa12ffb51e5b4ca0f9e1dcba34aaa8a45e10c9269930f959862

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917d4764464c7a7f0aab74d062831136548f2f91da0c94355c3d8c9d255e2714

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2b70364976b429aa55c0ddc780652565d8230f53730f3b31cbbf3f92a111f977

e078945d28684dcac11e693190d76b8f2a0de98838b95a84a990f640a531fae8

Fiat-Shamir context

eca32024f84d175d6a7b4e638c1c9072b11348bb6533794d5d71d1e4b7b05616

withheld proof term: 6962 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specRhs_build theorem, SpectralDecomposition:138

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :  
  Fintype.{u_2} ι]  
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}  
{K : Matrix.{u_2, u_2, u_1} ι ι κ} (hK : FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part  
  K)  
(hco :  
  ∀ (k l : κ),  
    Ne.{u_3 + 1} k l →  
    IsCoprime.{u_1}  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k))  
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l)))  
(G : Nat → Matrix.{u_2, u_2, u_1} ι ι κ) (n : Nat),  
Eq.{max (u_1 + 1) (u_2 + 1)}  
  (FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).1  
    (FBundlePoint.build.{u_1, u_2, u_3} hK hco G n).2 n)  
  (FBundlePoint.specRhs.{u_1, u_2} G (FBundlePoint.specPhi.{u_1, u_2, u_3} hK hco G)  
    (FBundlePoint.specG'.{u_1, u_2, u_3} hK hco G) n)
```

SHA-256 of the statement above

33c6d3a5f35d9a03ff33f1772609863978e9e2baf994a8524aa35f29d38382e7

SHA-256 of the kernel type

12f3261c657b9ad1e86c610328b2e8a0ff7afa0f92e4bd5a63f5c42d72d58c3c

commitment C

ab27b4e16a6fb715f07d296611ed6915a0c0679713442bad17a245c98f5c7951
55405d1957e1e20dfb583ff7a77e8f02815c0c15668d43670645256165900707
32c70e3ea9bbfbaa8a992ba2aad7ef90771befdec22ba2a5f06a8c3d3fd9c857
a14295a75fdb3b9d9e64d0a9a7bdda5b780ae54b8e4ad864249b777cf408e5ed
aa4533655ca21b5f69a49e1f23aaf82f09081a0601505034b86428a272e4952d
d8146d4aa9e31295e51a6301e0e6c533f033b62b9ab07a6de2ffce46093c8902
66dc122a0187b3a499448fef321d0acefc8f866491c348f19ef0da3e4df22cf0
7cb07d464b3d73a7a4cd02687ca75cc291a95e0f201e3833eb62b081d366e330
first message t

799ea13b3b91c75488b18db3b3edc829b82647a5cab9259342dcd7a505aa35de
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1c3a8aae4b2f78b10787a8e1e561308bcc41eddda026dd9e283542bd3ccb93da
34597ce42e58036027b4f7cf0dcd384643a475de0c82f255bb10fbd43988e9eb
7faa426d0928843a286925447cec1b703a72dcc3654d750bbe9493f92ee583b3
969a9c09517380a8f6e486fa808ff54d83ef3db4b4b71852dc20150b7e90716e
8ace085cd50462d65f03d5a01ac9758ea25590442fd576cd14d14f7f93d64d2c
15dd6a2e87b2e4ba808912e166113495f94bda003e5f4e9029fbcc32410d7d88
response z

4fd06239060174ffe146da5de40a9567211707820890e13aac3ccde3c1eef91e
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4faacc32df7e2c61d5ef283320b6981ebf3f13da05e36cf5babde34a88eaf4d
9b1c46788193873cc36275132d850e95e710debaca42927435b8a4a3a947066b
21bbbd2add57547fb14fba3e3cba45af5a8f8d81f68b9c41621a3ad1ea72dfd9
faa985cef72e3e76886e41710fe201ea7cc13d8cd5e9c8fdca02207c550a9c87
6bd854bf061643107283fc6103f4975eed969eb2a7569c0f8182183cf12c5ee1
2596409dbc71c1f96d6eb5af837f949409d728f7bc669c32d3224f7ad6978939
response w

50efec2b9ab111cf3b0efd5cbf79d422bd6d5c6355f0500f80e070ebe0360d04
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24d4e96d50d3ca7a9ac155e90a85fa29066aa99e6796730eabcc3bfea95fdb9b
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335320800ee6b79ec7b4b08d937b9d0587e86354405846f8ae2f657d5c5d8bb2
7c35969617f5a4e63b4e45272ed48cdd112353f545e2a9a0ad08c3611b3df239
6884e56d96c4e62df501660ffc188adbb24fdab5a3d2388075945344a69e3460
91e8eef98da42baa7ee8eeae3835cfefa903e4ab57403097826659d57437d4d9
Fiat-Shamir context

0fe5e70e8962ddb3caf5f4e84d87405196e75043aac4e90689e85d9c7b941e8

withheld proof term: 5721 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.specRhs_congr theorem, SpectralDecomposition:75

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} [inst_1 : Fintype.{u_2} ι]
  {G P P' Q Q' : Nat → Matrix.{u_2, u_2, u_1} ι ι k} {n : Nat},
  (∀ (i : Nat), LE.le.{0} i n → Eq.{max (u_1 + 1) (u_2 + 1)} (P i) (P' i)) →
  (∀ (j : Nat), LT.lt.{0} j n → Eq.{max (u_1 + 1) (u_2 + 1)} (Q j) (Q' j)) →
  Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.specRhs.{u_1, u_2} G P Q n)
  (FBundlePoint.specRhs.{u_1, u_2} G P' Q' n)
```

SHA-256 of the statement above

3315e660734ebe2bad7858ced507e87d25b6c8831ffd0775e94b3bafd611dcfe

SHA-256 of the kernel type

95478f963dcb316fbf202d0dd0768f1470ae3a1958f6630adf8282c8a1941934

commitment C

8b64794e43974b006908c73d2ee0696d84ab7cecfbe64950d33ed3330cac19a5
2a098b73f210bac8d49eec944430836e2b95d07ba9448bef0498d5c7b7e9dcbc
6ef9fe89c566c78a20aca5b4992ee91b2aaf8279be058b5805a82bf4118d6dd1
82b1e79710b7750add1e522c548ab14eaca2174d0db4a9ecd56aa763d9c18f55
b61554120efff469c08da99e55615f5a223367280e0bde9c51fa1cbcaabe9afb
43c8e68cfe0f6be379022b2c2be16e2d22e550f5c36fb7f7ebab53d2dae5a25
df9b039eaec2bc3928c0238100759b9cf6769d922572a2e1dde62a492a9f51fa
c8369621d77454bf873ae619f76e81c529cf908fec743a06d29d29e3f802c5a6
first message t

c638750128515e76469eae130370e5effac12a80c179ffceb178c012ca7c63af
4441e1912f32c2629c0ea702c68467da7cbf3a71275fa75b9369720c61a93006
08bb06df6158dc905864f9f04bcbe68fb710537231a91d75e977ecccb6e6b6d2
6d69f96aaa51a0fa1f02c7acff660c974a04603b3fc290408a64262b001cefb9
ada0f1dd2d16271a580a4d041fc430ac949ab9ae8f90fd5d8a95c7e546ec7e72
19d7a8f95577b33cb321fd4d2b77589c078e5d3e774b2a5b66b259d17d2479c
7779c4dc73549e23b8a2bf4a145afff589a26829e96e566299c5746d0d9c0473
37cc25efb3d3981e7a8dc29de203f48e7ef07c8078cee25358c58a8fbff4d651
response z

d12604381b504a66d948ea5bc1a2064c4a1b5671ff0934555d08c82beac874ab
53de540db829a4bc57b9174de0379c7161486aa721652492fc777c8cd86637da
db25ebdbffa77c9a40ea66c2b894dc05ecf950df5541ab6333551c1e3e09d777
1b9b1c3d4643788b436ddd5cfed2292a44d88cae65476296af282103e0315fdf
af568dc536d4104dca05ee669dd57c1ae229c6832cf772c60e0973d33322c351
7a82ae94a373d270bffc88a8e6e1aced7e4be1f1b4cc20ca44ada01fc81ee57e
01ad0c87ce1e3b16460d2e639039349002b4cf410f007f6b91fb22fb76de4a28
fe94d650c2653a2f19f25e27db9da3d44b5ad801c067b34fa30454c09398096
response w

3f1edf04a2871d6daf8783d7b7c93e8a67d5ec1f02a3f26e22256a6b3be4fa19
4743ebc255d0faa3c6d32f64a12379b7bdc499653f44503a5ddbbcb7bce29ab35
7d3e251a2fd2c99c0b6bf2e5f3734a8ddfabdd1ef0fa8e8b951f81a7aa66b64b
3080ffed7a026f172ce853e19842b6e8e2633f50a55ef2e7240db95b38cf8a1d
3af3ba42fd6806ed8a8473bc3fea055799ba11ea3a5119c8d3df9283e904f71e
3fb706525a5383bccf44f2400245f0feb6ae87eca1f1b27511f6db6daac63647
8350480d067e83eae21fb7651f8a991627f1d39943fc59e271856e1d3164d0
ad8c1dcb84455415c60b2163bd6682c917374d8bf188a3cba2ec5c2d873748e4
Fiat-Shamir context

f34687308970371310c00da07c6cbf4ed0f68ca6d6b637971dbb7039da0f7fc3

withheld proof term: 163 379 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.spectralGaugeSystem_unique theorem, SpectralDecomposition:233

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι κ},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
  (∀ (k 1 : κ),
    Ne.{u_3 + 1} k 1 →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K 1 1)))
    →
  ∀ {G P1 G1 P2 G2 : Nat → Matrix.{u_2, u_2, u_1} ι ι κ},
  Eq.{max (u_1 + 1) (u_2 + 1)} (P1 0) 1 →
    (∀ (n : Nat),
      FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part (P1 (HAdd.hAdd.{0, 0, 0} n
        1))) →
      (∀ (n : Nat), FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (G1 n)) →
        FBundlePoint.SpectralGaugeSystem.{u_1, u_2} K G G1 P1 →
          Eq.{max (u_1 + 1) (u_2 + 1)} (P2 0) 1 →
            (∀ (n : Nat),
              FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part
                (P2 (HAdd.hAdd.{0, 0, 0} n 1))) →
                (∀ (n : Nat), FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part (G2 n)) →
                  FBundlePoint.SpectralGaugeSystem.{u_1, u_2} K G G2 P2 →
                    And (Eq.{max (u_1 + 1) (u_2 + 1)} P1 P2)
                      (Eq.{max (u_1 + 1) (u_2 + 1)} G1 G2)
```

SHA-256 of the statement above

dd218ee2a76d3c7f85d3604ea7941a583dff6fa07ec79609e5180b6f145e8473

SHA-256 of the kernel type

189c6edd5cb83e2fef5130f405db1c069c82ba58d60d084f9b13506bd2affe67
commitment C

9e6c6ee688e1e3aa9b80aa0483532194b7245aeae1d744baa99768af8e49befc
e091a1ab4a95dae4bee7b5e2749c0a0b839784f4367cff1d5279fab4a4a79ff
de2b0ed570c8280c8c3c9bf38cd9c7809872014156dea8ec12f7b552461212d7
8aaead0b7110c8f78ae853ffb4e852492f2c8b69980e0144a2080373456d2df3
0a1d7a1bc39740cfba7178eb47185c2a86f899c6fb0042ade9ba75017584101b
42cd796c2235eedbabb791de4bc88d6710e54d66f66163b700a13efbdd623e0c
4170483d0246147ae1274c91eec97aef13d9acf7d8c12484c8df5dbf94b3dce5
d2f6a60eed44309b1c95ba19067b89931a78376f0d061281927c684e003ad982
first message t
c99384ce5caf3507402aeaa20a704ef2a12886d84a712a236d803eb1724c9cf5
1e5bd4419c5289a6762cae68314ffb1739f19f2c20afb3b2c049eb509f16c6d1
28956f3a23c1a92f90ebfea7b090474c2a398ee24cc8bf3222e83c98e712863c
ecbe1286e8fc1987f677779caf275940ce5353c581a55ee2fbbbf3f73b259dde
6e9d1d9d5b6b86dc8ec25624f4d3154456f41ec7234ab4da2dd616ee58a9f2cb
e13150148739e5247fa7322c4ef1f90d46e4f7bd63882b8de6c64afbbf55c789
47ee4be5a74d1802f2c37e60777e292dc0b83787c14dfb3f529ac34627d4ff86
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response z
3d6fc8ac52bd12a74b34a55966279833a398d1d85efec929f3efe187b34aa3b5
e910baefe68ce710fb60942118903aae97a9584163486d6112b54ba6e64b5487
2efcf39123eb984a92281bcb063296556177d6e783404ee5be97242ef9521eee
db15b72607ecce4d66afa7ac48c025fdf991da93f12b2219b170ce90a80dc50c
9bbfbddcc00bd380f2ef6565a2c74679b886efbaab396ad9cb17869e577ab8e0d
0422518b10b24b0bf4055670976b0590c15d21679841763bb44cd2f8a1ac80a3
b4f78c49bdcebc71f613285abd0e04eb1bd2c01f1539fcce3acf233b08232c19
284af2b08d84715580b5a7a1a9d97af2333ef12787de92c2ab2228d15a48f99
response w
1f5235bc41f969d0c7aebfd8e08c736324684c23e7e96c2a703d6cf1526475c4
4db86f878681cd0c11c0e1220bd3e0fdeb22485f00c35d1a1a2536a39152b8b9
e8230aaa0ca2aaf50f013a1acf6829e9fef3348a3fff9452af85acc6b5f790f555
73e85332ea0f42fc01728759396a611bc447b3c823a3a83242a9ca075e6971c
ebda10c817103a19b1154b1edd457d7fa2c1f9f74740e1f663c50f602cc10a1f

FBundlePoint.IsGaugeEquivSeries definition, SpectralPowerSeries:133

```

{k : Type u_1} →
  [Field.{u_1} k] →
    {ι : Type u_2} →
      [Fintype.{u_2} ι] →
        Matrix.{u_2, u_2, u_1} ι ι k →
          Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k) →
            Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k) →
              Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k) → Prop

```

SHA-256 of the statement above

25baa5837717acd053bad691a195ab4fa86d565d3156147b48bb1659d25ed333

SHA-256 of the kernel type

7bddef29a7e4e055bef6f6fc55dff9387230bd27ba34ec342ec3699a4b6365f1
commitment C

```

8bbf4e446238fcfe093d837bd1c75effcd2e229178a35a52fb3e62852c79e2e1
2928163e74a04b5b270f665987c3914a58a44c4590bd105816ed72027432bc5
1a8721d068cdf1acca2d9c3408ba3884c81c1dd47f39cded590002363a60787a
c1acd57728ed7d1f167306b1b1670cbcd768ad04e612bc1ceafccde0e38792b
a7cb083e31ac506e24ba6a43584d500fcaea7c57d5f31464c52a2891d531cd1c
c875878733e4782964188298629ae29fdc74cfcdc8e91e55fb36b8038d1d3479
f0682aeea07d00f776cd0a2e7f91441c3cb1dbbdc7bcb38793463ac0463e9810
6e1ccf7b9e368ca9b1e50d0b2c5e127e23f6a02963ee4feb656ed3468de17cf2
first message t

```

```

2735e63bb2055f7a0ef9f3ff5c6d0c3174fdbb95bd13c401128519a9ffd5594a
2b54d921dbf773d72537a15bea57be1ce2342d09a583f480c73b17a621ad6f33
0d1c3b44c3435cc624e026c158aeb3656170a39f9c93b6b418cbe9c391a38cbe
a9be587973f5c380b9d84165e3daad562f8c528bbecef1e0a93ba51eebe011ca7
4b52bc4b3d5520c1aeffe0a32c07b980d5c0c49267ce78b63920b8199ff11e20
0a9be191a8dd61693f6b7d02f34675c1e7959e8a5bb66035ac5b6266b17721a1
0702d375429a67e817626ec1b2c20501ea1bb63d2b6c5cd0dbc0cab7e7ffde06
77e276cc68bffdcd6918a60527cfb394947a2ec9a21576769ec2d601a636644a7
response z

```

```

4c24ac3279db0d48a0088c2198068f0307b235fe7a7d886b3d50b5a57ab368a5
57d5304e3399de151488a8ef834dd24532f8d43ec1aa9e65bece39b451ad1114
90fd5fa4d4879daeeb363ab5d1ee0535b5d821ea069c5ffbe41038c4c9af06d5
8ff4c4ffe82da58de366f83cfaf75f11fb40c16e53bf58ad6ec21e328adec751
bae28bb35804fd7a7fa21494c157a7b87c369a6dfc75dd7db176f6f48a4325ee
47150a265dd1c2fb13762a3b43cba38a215ce08b8b38cbefa9c27a1f9cc4db41
3cedd53bea6195c01ff48682e808c52d7c9bf685ab91abfa5524d6779615c00c
56c1c05d6c773f5cabb0bb2a2f6bb40084e9f894e8551bec226c38a0f3b8e75b
response w

```

```

329d82fd8ca704de8373f2a33ed49775984c4a1db4e9e6077e805cb1e6a728db
f2b09d5ae3133cc8287dc68389c4f7bc180d79b2126807062ee7817b4e3174d4
3280f50448a514e030c529d06fd5110603fe98b95a7b56b8801a1648344aec3b
9f71ba8a99370c2f884d296b3b958ced77f389427921c30f99b0ea8e5a3b68fd
6a7fe663f094f91f629479045696f203f01c42161abcde8a065b16f6e28f50a7
64ec0f1f5bccc49797ebe58aa1130d961953e6c6b96e5d26a55ba38efe2d69656
a05e508a6187fdc179ca3b854bb2bd7058f6b305ab69d43d641b699dfc0da66b
23218cba78982e4175a0a950bd19b3fb761f4258428d872a34e936f89e40e0fa
Fiat-Shamir context

```

401be0a4bdc31b899c66918f2785e4c05e0e3f1c47b9f9085998c2d0a686fd6f

withheld proof term: 12085 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cstM definition, SpectralPowerSeries:40

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {ι : Type u_2} → Matrix.{u_2, u_2, u_1} ι ι k → Matrix.{u_2, u_2, u_1} ι ι  
      (PowerSeries.{u_1} k)
```

SHA-256 of the statement above

7e7065ec3e758793fd7ab83bd44051b5e9599f64ce874075d375387f441e976a

SHA-256 of the kernel type

3ed5e1b833afed3cad8b5db486b608a7de1af96c2370ac4d981d4cfe88303aee
commitment C

5638661d385da1a163bab0b056a1ef1586d9897d891fb43661bd15f6960f6754
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bb4bdd1801b1b5ea27bfd33e898d72496d7284023e3dc61951327ecc38571657
d3ba36cc34ec6e499b0b7cae52c1e0b388e6ae7faf20d4cdcbb919fd766de018
0b7961f864ee114c8366c336372c88cf2a19689c1a1cef9b5d20d16055d797ba
f3f72410359ecb31632d70826011d42c57a625b23d1e9ee14c21f595c5b0a322
86410d157a9209e9578add30d7bc1b3f2ad853c417aa16b2e3388ce48dfdb0cf
23715f3de22caead0e75720c5f7be80a426cf639195d5cd7400d6eebc2a30379
first message t

cca887702061f3e54712150a06919778101de5461b4ca0c4bf76ea28ad2952de
9fb7b76a3ea0a98db25c05b1bd83b47b7b52a72d5c94b7e4abdb11b014ff77d9
e3a9b8e35e4453e6a848a678c7299fc6db26fae4ca482711db385bb07e928efa
6698500004785571cd4e30f05b30dae23053ebec012b98dd45030129046e1c7c
91ca12020d7288a3a554a55513aafa5519a4bcfdd97a0630f308b0603eead7ac
d68e9cc4d8bcfcdbd6f1ea52e00a2837ddaa76e5129fbab1bad90f9a72368b52b
3902d18555d41761bda70c626af31982b9459556e887a850d56dd7c75bd2b6c5
0a87fad8340644cb24711d4e0e83640a7e668bb02f8a0349ec4fc6c41a2b8789
response z

3fcaa22829ea4610bf13538ea37aacfc229ad0a5814663727a2bb461e97316e9
e24875e3856f34e2e9e4bdc79fc96a3e6fc7f4d387b9950961d699ce401d5bf4
9fefbea586de38cfe7acf107cd65235b55243bbdeb4e552c6142176561bdfa5
887a964e72bc7cf15a602f726f800ddff9d248deb234b359ca3b755698fa272a
5a273eeb0bad7c91a8d17670431691d98577616b59bec9027198bca16a0e9c15
f1c7304de14d21b60ac9a6ee393733921096826cf4d4b2ed29ab9697de684edb
af9d31cd1eb94cd513d8e6dcf0afaf9dd87fd04334f6ed14dcb5d14d3e70f47a
c72f5ddf84511bce00b2c0c2ab87e89ff4d3d4b3650d2c8c624a6c7ea6f6031
response w

237edc8fc3f33017655f9d1643c7e6d2042bd364b3e76752bdf626b9057ae7c9
f89bda587e6c47054ad1f2cb1e5c1441ad52475063f33701035b134e1b72dfec
dcac0c92ff6814807570965932b500ea8a1d4977efdd3f680b42a134714e167f
57e4d44377f6e385e8eb84cc063abdec947adcb6ca2b09b52b80fa986e49591e
e0d1e7a5863ce02a0c3d5c1afa05e691260d56146db50440231b2eac2f3a4a6e
213efd9d00b40005791438a77c746d1ac94fffb0df0643be7753c733f88c4b274
189da9e2bbcac0f5481dcb140bae071039e647274c21ada5645ad926ac07be22
bd59cf225d6bb4a75278a7c6813facd70dca47d2b80dbb438864184fcf969142

Fiat-Shamir context

41438387c33e4ae04cae9b8c7474abbc32b9976c8ade1918b010591edddf4a6f

withheld proof term: 1731 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cstM.eq_1 theorem, SpectralPowerSeries:0

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ } k] \{l : \text{Type } u_2\} (M : \text{Matrix}.\{u_2, u_2, u_1\} \text{ } l \text{ } l),$
Eq. $\{(\max u_2 u_1) + 1\}$ (FBundlePoint.cstM. $\{u_1, u_2\}$ M)
(Matrix.map. $\{u_1, u_1, u_2, u_2\}$ M $\uparrow$ PowerSeries.C. $\{u_1\}$)

SHA-256 of the statement above

5bb06d2e2c21f7dd7a3aa1de7a26dde060ff3e5088f984108613589f551ee648

SHA-256 of the kernel type

849d051c41837d05f617845c41f9d76ec6f788fc1ec848c528b81f27c27a0454

commitment C

b0510af85b54f6674c605005e04fae8e78a687d7f95e023c93284b145e9776d2
d06c0a17fedf8fb6261c30230be97eaeff8e08c521c88429acc6baef27a2b39b
ba514c2b8fa7b23ae03f701b32c493cd90c5aa19b2c5937116a25bad7f957436
9ffa3c3508e43efb4bab5209ad1edcdd185042200ac16f5c637ae948c0433bfb
5c3f399392c12eefff6094d240874b486687f7e970c16363ac804738f594ae60
d7368ec9bb33474f833fbacd845ee905e11fe1026f334dcf8657124c5ea62128
2133f0ec1a80ec94e9ecb375ba4bf7f7c324612ee135f2a794f3a3b14258847ed
200e09e17c633ffbfedabbcf0925468780a86e251ca575aa5870dcfd3603c0b

first message t

9445e95e868bd221879fc5232d4d64089d657f191efe77e6a9947bca63af0fb2
ec98bd1f357fcde3076380f620c1422c777b89f6594a7fb45693d23b06c09944
543d48d1d0bdb210b8c75d08925515683184d915ad95a289a5c892eea59c0ff8
c40e8d19c235f64443668dbd01328339749494b685b8e510f06f2b4f5837ae90
ad52fff021153225dc844755a97a16d983b6b6b6a3fd090c9931812515f62d33
9ed653455c92176ab88cf5306cbf1951f7fc864e5a5574440c3b6d9bf17ea7dc
73dba86b8a536c5095b03f50f8f2f099fd8008c723e92112563521f8ab319ec0
dfd81453adf74df1ac4d78633b88e6f33e808ae19717bc608ab4cc6159ef179

response z

3eb718e6917f02bffb666c04cd524539f628a1fbca1eb8d0c49de00de6d811ed
750bccf5b4d34018e01d5d44560fb8cfeefd3047ba09e18aa5ccb6d8e4028160
8d4a3b70386fb0eb425644921c7ef6f2c4c1ecf7524f51c16ab767922c7c13e7
83f73658c97ae50b27577e68c3d60748e55269fa6a709ecb8d5e0cfb54b0b0c3
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9488bdce2eac1db2db1d3f7a6b91b283899b514db79c0ace2a5d46fabb459ec8
05dbcc8ba18c0e7ff00e322722b3bca2d91525e10df4d0ee670cb9cc88fa0053
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response w

201d2f1b75e4d8db16458ea4aaa5990c0809918a71a18e970cd403f3f9f90b39
46c64d619ba36ed98115b9bb9ef1ff18417de20055ea0daa30ec3148232a8894
ba3840266aabb8ad2196ef02954fb28dfb9597673e738eacc2e5d1fa8f0a0ef3
034f7c7248292e1b3f24f9101a27c8e4b207c8fad6e7d6045a99a646e837840
8ebe91a1511c884ec1618f73a69a08cf9dfb5e4393c7d8b9b1be1a605e362d81
22c956e1d866a538d49835f9782e2eca9ef1838d7b09e26bc77475920c413cd8
0a2ac15a871b266e992dfdf696a47930cd27fa865b39a489f39aafc84fcf38853
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Fiat-Shamir context

e6ac4a4aa4b5809dd4c5f07605e99a870262083c935078080c9f52aac58e0b4f

withheld proof term: 241 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_isGaugeEquivSeries_isBlockDiagOn theorem, SpectralPowerSeries:191

```
∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι κ},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
  (∀ (k 1 : κ),
    Ne.{u_3 + 1} k 1 →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K 1 1)))
    →
  ∀ (A : Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} κ)),
  ∃ Φ A',
  And (Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.psCoeffM.{u_1, u_2} 0 Φ) 1)
    (And (IsUnit.{u_1} (Matrix.det.{u_1, u_2} Φ))
      (And
        (∀ (n : Nat),
          FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part
            (FBundlePoint.psCoeffM.{u_1, u_2} (HAdd.hAdd.{0, 0, 0} n 1) Φ))
        (And
          (∀ (n : Nat),
            FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
              (FBundlePoint.psCoeffM.{u_1, u_2} n A'))
            (FBundlePoint.IsGaugeEquivSeries.{u_1, u_2} K A Φ A')))))
```

SHA-256 of the statement above

34c2a280231d8f6b61fc6b51588b3f8f795d62fbb089c6395302511274318003

SHA-256 of the kernel type

ac7b748c087d6efd035234d5c8cf926cd3f40100fb6c48a0e5626a1debb1bac8

commitment C

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first message t

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response z

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response w

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7c1792928fd8988ab68678e58facaf0e29eb5682c41d3615c7187add9da344e0

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44f839fd8b14b5eebbfe80d2396c64d471b467b8ce3aba647d1014fee1cc45f559

8c6d559d3a63ae0abd2267bf2ecbb0677ea714409b87a6bb70eb1849af1849f5

2ccfb5f9d986f5fd92292013800ba12ca5d3c342fdd6eddf7a66cc265951205f

FBundlePoint.isGaugeEquivSeries_iff theorem, SpectralPowerSeries:141

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} [inst_1 : Fintype.{u_2} ι]
  (K : Matrix.{u_2, u_2, u_1} ι ι k) (A Φ A' : Matrix.{u_2, u_2, u_1} ι ι
    (PowerSeries.{u_1} k)),
  Iff (FBundlePoint.IsGaugeEquivSeries.{u_1, u_2} K A Φ A')
    (And
      (Eq.{max (u_1 + 1) (u_2 + 1)}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} K (FBundlePoint.psCoeffM.{u_1,
          u_2} 0 Φ))
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.psCoeffM.{u_1,
          u_2} 0 Φ)
          K))
      (FBundlePoint.SpectralGaugeSystem.{u_1, u_2} K (fun n => FBundlePoint.psCoeffM.{u_1,
        u_2} n A)
        (fun n => FBundlePoint.psCoeffM.{u_1, u_2} n A') fun n =>
          FBundlePoint.psCoeffM.{u_1, u_2} n Φ))
```

SHA-256 of the statement above

34b7a241577d4c4cf4a9d9ef0746d03a1babcccf72d7addee0274b1857e4ec7d

SHA-256 of the kernel type

46a67a8a8e56ae83fce66218d422f46c86d6eaedd883e0901ce8d74be95c543a
commitment C

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first message t

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response z

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response w

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Fiat-Shamir context

47bc5dcd6549f2d25575645013e0376c2c37149a1f463d051ab817259b4d1d55

withheld proof term: 4871 520 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isGaugeEquivSeries_unique theorem, SpectralPowerSeries:212

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {l : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} l]
[inst_2 : DecidableEq.{u_2 + 1} l] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : l → κ}
{K : Matrix.{u_2, u_2, u_1} l l k},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
  (∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l l)))
    →
  ∀ {A Φ1 A1 Φ2 A2 : Matrix.{u_2, u_2, u_1} l l (PowerSeries.{u_1} k)},
Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.psCoeffM.{u_1, u_2} 0 Φ1) 1 →
  (∀ (n : Nat),
    FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part
      (FBundlePoint.psCoeffM.{u_1, u_2} (HAdd.hAdd.{0, 0, 0} n 1) Φ1)) →
  (∀ (n : Nat),
    FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
      (FBundlePoint.psCoeffM.{u_1, u_2} n A1)) →
    FBundlePoint.IsGaugeEquivSeries.{u_1, u_2} K A Φ1 A1 →
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.psCoeffM.{u_1, u_2} 0 Φ2) 1 →
    (∀ (n : Nat),
      FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part
        (FBundlePoint.psCoeffM.{u_1, u_2} (HAdd.hAdd.{0, 0, 0} n 1) Φ2)) →
      (∀ (n : Nat),
        FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
          (FBundlePoint.psCoeffM.{u_1, u_2} n A2)) →
        FBundlePoint.IsGaugeEquivSeries.{u_1, u_2} K A Φ2 A2 →
        And (Eq.{max (u_1 + 1) (u_2 + 1)} Φ1 Φ2)
          (Eq.{max (u_1 + 1) (u_2 + 1)} A1 A2)
```

SHA-256 of the statement above

22c389691bf534a58ddbef370d5125209189935f6ea4657922b9fc3e885876b8

SHA-256 of the kernel type

a88d0787c2fe2b8e1490bc7b0ed653bce6814bd7775cd82ef28731abb1d18e53

commitment C

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response w

a7a663c793dbb33e5e5746e622d594045710d3e507b7d511b136da2a02fda16c

FBundlePoint.isUnit_det_iff_psCoeffM theorem, SpectralPowerSeries:176

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}\{u_1\} \ k] \{l : \text{Type } u_2\} [\text{inst}_1 : \text{Fintype}\{u_2\} \ l]$
[inst_2 : DecidableEq.{u_2 + 1} l] ($\Phi : \text{Matrix}\{u_2, u_2, u_1\} \ l \ l \ (\text{PowerSeries}\{u_1\} \ k)$),
Iff (IsUnit.{u_1} (Matrix.det.{u_1, u_2} Φ))
(IsUnit.{u_1} (Matrix.det.{u_1, u_2} (FBundlePoint.psCoeffM.{u_1, u_2} 0 Φ)))

SHA-256 of the statement above

6273cf91504ef070857cd5d1bc2e787e89ee67edcb1ee1350a56b5221b750f58

SHA-256 of the kernel type

4326254ffbe07f939f312febf216d57229656318ac1a5e1213cd23d49b1126f5

commitment C

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response z

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response w

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Fiat-Shamir context

2c1d5f73c4a97a2f51f3584f85e1b6af420547f4736767a08815b691e3d405e3

withheld proof term: 125 741 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.ofCoeffsM definition, SpectralPowerSeries:48

```
{k : Type u_1} →  
  {ι : Type u_2} →  
    (Nat → Matrix.{u_2, u_2, u_1} ι ι k) → Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k)
```

SHA-256 of the statement above

ac6efc65f1bbd42a3a6700af6b9003032555502211db5d5d1acdb851f3e36c4f

SHA-256 of the kernel type

943f6b1e611369029c6bd880b947fdab66ced4c0baa9c85d2703033480a21ee7
commitment C

aa28f9ff655b0c658141a1a9243aeba016bce134c34b4086df95086fbb1d8b25
bb8f2e11563e379c274ae94ce4deb94eb9c5f3443fb9baa566d0c15a898ea768
dc9490370e6ccb10fb141d66886121eda4bde441e9b66b2bddc4474bc48411cc
0778caea83086ecf78b5bdf4dc785ecd79b45c0f12fbf1c9d8144a596d67be3c
4e0f5d5bbd8f99ad96d58bb9a88bef4749cf5f10b60a6f97d678cce917185fbe
4e30b0424c57df6904391055b01153912f5f515e5b5a45020a6fdac5a548854
755803d0b08ffc75357889aea32bba48f9809173015e97c26c29e8c6d3d2e0cb
c91ac45af50723dcd809d8554c602e129ed266701c96f69d918bc62c703bafdc
first message t

e5843418ed2bfaeba749fff0e80ec607d24f1f108269cc0659cb874a632c2f19
b32a0c7b1dc1eb6c5245c8560cfd580afa6c730638f27563b52db56b8a29e656
34e1ee8b1f4276d92d94fa7367118462a498045693d63181b5eb751719d4ed75
521d05d490b7681a38ac7d32111a126c42c4fe5637b62496a8534d9e2fca037b
40fbb9f893e53c33c5d22d8f2de42f773ac0ca2a8130bd6d0f9ea48b5062340c
84b7d009cdb716e3d03ad13be31fb26b61de3d9f9ede17614f0f0956e798fb5b
5d4d5d9616492515c4b7683ab760061d52e0bc7cbb29e29b908231221be4948e
7377128b2fb6238cf0b181548f47a55998dbfe710c6ec77fb1c2da32ce22307
response z

40b7fb4ead6f9d75c1b08d1b2fbbc814e98d61a762ae657802f7381737dfce53
73dc13da8d76b6cad40bd8021b457c52f64fbacfcbaeff8ba18e1022b3a463f4
55b015b1857f1a13fe0748c18de04c2733f30d082d7c59781f619d09cd3a2da9
f71faf391cf27e37957f2b277be14d035db472bee22d93c465f66bb8fdc462d4
9a8cf06807d1ca4f1fe300dea7beab8c4d4e7a3c96261416768c3ee7e7445df2
5713a212084d6602a3351813c3606e93ce66cf0f9ed3ef777f9ad3f873be06e5
9670cdc70b5e752060100a9c18b6d2c3018ac58a49c02bcd7b4ae8e9994d4736
4a9ae8b3e319e95eaa4231dfa573ae304cf590f806c17915db311b372ba1832f
response w

6634ba1ea0bb00ad4dd4dcb275b29dd6d38654899b5baffe26fdf9eedc5a35ca
8fb924cfbac665cd926e5398e7f2ae637b389813f37c4ed890327f50b789ef7e
2dbc6348bb45f9963b1cf6dd8168d52dd9bf35281f6c264cd49ea7c6a3a1e9b2
003888ad8044122a88fdecf6fc10feeab0dadf7c97fd714187b7832c08e0c055
fd65eeb1f73073340be70fd164b922fed78faeb34b2f50e8581dc12bc585e37c
f4328d1be352615e0b0f6a5022eb387a6a61093647750ef1ee61d8e75a462f02
d4be1b81521f3cbea86bf17a5b343873d9dd0d9fdeb414975dc8418293ae32f8
5e3712cd2bef69d47d623ee18751a0f9b0f9215237d9867ed8145df54f4a81b7
Fiat-Shamir context

92c8ee25dea64b5732747909a4500b17d8bc89714a12a3fcfb51def1a4aac56c
withheld proof term: 1 216 bytes • axioms: Quot.sound, propext • kernel replay: yes

```

FBundlePoint.ofCoeffsM.eq_1 theorem, SpectralPowerSeries:0
∀ {k : Type u_1} {ι : Type u_2} (P : Nat → Matrix.{u_2, u_2, u_1} ι ι k),
  Eq.{(max u_2 u_1) + 1} (FBundlePoint.ofCoeffsM.{u_1, u_2} P)
    (Matrix.of.{u_1, u_2, u_2} fun i j => PowerSeries.mk.{u_1} fun k => P k i j)

SHA-256 of the statement above
cf49ba5f378eea8bb402dabe87222cf2b3c7bbe5895fcd9003f6589b8bfb3b17
SHA-256 of the kernel type
b26b7896c9465e85709e4ebfdce716be971fda096f044ece1a948f49d7f1d549
commitment C
d99858a0e01029363148c37bfb3393116c4a3c11c43cfd5d64b27ef2b0bbc6a0
29b8cadaf01f63e07a2f0c2d89e9f68d5a907d4bb8651b86cb62848b8a42301e
7b506a58aef8d097c7c082268fda11f1905085c04f9b60b0354030a6ad0b393
6037be0d54ced6e24bbe8832ca9f59523edbaee498b64477f58d44959f63ca
a7f1ef0cc8c74617d36bf34dd5d6cd8f624089e7276d509e5ff8ce68c02406d4
431fe49d3990c8470b6f516d413d8bf2adce8ff722482cde9e781a1af0204b7d
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a929a6d1814fc5974b2d14f71d2ccf3f551c89d3ac8468d0019e00ac75029114
first message t
c65131eafdf3fc4f3e054fd2f6910c3840bcd9e90a211225c388d6129dec232
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e5fe805de9f5fff895eab637de6740c359bac94b5c9306ca81386a85be4c097f
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23a95ed1808ee25784287ff8e63c4d2a147244cbc5b973ecaace43646c8ab4c0
77bbc6604c614f910fbcfb926d7a53817f22bbdb09d6a3386dbe9d10954043df
17dca4525eaffeeceb0069f76b32a549560f789a44e552d0c45ecc5332e0110b
response z
1ebcda0258163aa392e8077335a0aba3eb96c9c22a25c443f03da2c059a6d5be
50671afb0484658d768a1b342794ed629f23754dc757d5843195383faac6f2bb
e487d8409c3c8320e4a17f41a1e207d5acbff221bcd541e5b601ee61e72e6dbd
76bd4e9579506c9130de85d8ca0d4a97e08af89f39f8c60521476df96d4b99f0
f99a98608d5522e14160272d9c0ee43d63060191aaac01399da0a1ece5c2bd3
3ba0547b14683c75d66c2cea099b09452a8d1c2797daf3b3fce522bfc4db0c34
7fa1fe84a5756c22f8e72a135627d60e538558ff65857b4f1e06181f3abf25c3
e8315d72499f2655efde4f9865a3e9e9d288f360d40a11819ff599c7c9127a62
response w
3b6c11f6fb3e6ef2f0a9bd3cd34e42f8323737ef486617b421e548dbcaccd48c
72732c07dc79a1668e583bbee7c04d49255cb96919cd5b83970f3e06cf4f206a
d6bf884b8ca68c01d3b0b030c44b205dabdcf9f2ff7bab70979c119a11f347b5
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b319e859f6c8ab5f0d93f8a8108db267325e75dce2797be94dd750e5d27ba452
ee2c0e452f497e3fd8098d48c5074507f0f2cdeb2ee1418ec08623dc59162fe4
843f2862bd1e9c4be7776746a0be9817dc0795c1435d1e7f786cecccf702bd0e
Fiat-Shamir context
8aedc877b3bcc9c088de584557c850b2765a94cdb37b162a43d7000743d7f54a
withheld proof term: 224 bytes • axioms: Quot.sound, propext • kernel replay: yes

```

FBundlePoint.psCoeffM definition, SpectralPowerSeries:31

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {ι : Type u_2} →  
      Nat → Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k) → Matrix.{u_2, u_2, u_1} ι ι k
```

SHA-256 of the statement above

0d338cb7978dda0ea5509c2314bed9b007702fa6bcf9031c3ca062be4a43e3dc

SHA-256 of the kernel type

f01b0e81a73510d8b642fa9c5fa3ab7b2a6c5e48cfc19cbf66fbc8604b3890b5
commitment C

487860744d6232631064141a547474e7f78994d47ab8d2abb9515ff708209915
4dce70c6bb0ba4bfc49d510bf91bf4a482b89f5105e890458a9f2a4138941803
06f3d5a910df23eab964034046eb9b6a8430cc3817a98a6e25158577f8a80ea
ff47dbb33a9c40da88f3b62f80f23772c7d9e85f4a99efcb61cb4248059fb19b
b118a752c53bf6d0a78fc387e7aef72b73931ef7833deed9974f8d1f3fff77d4
1889b806bb3cbf6775677bc040e9f071a20a9f62e8a5ce62bff86537fdc90be8
58ce5d34aeb8f658e68855b0d08b21d6ef4486c37a12b2063cc532ccc8c4216b
245955f3f27bea606f60da9e534ed82dc691d97927cd675cd50f20dbc416194
first message t

f56a42180d181dffaaa355353ab27bda1ef44d52d92925180cab796342f47f67
446556606b39c2810ff50cf4cfd973cac4905fb058eb6b34e5998a6a91d7169
4e22b02fb7bbbd8d7184d727f25654ed0e37cc63a002d405be77bbeaee790c33
2f1d36ff2c0a5f6c19755bb0fe0101c7675b55a4da62e7cf9bf9dba122d867d8
64d9cf3f5ef8de5b9fe5a576f6b6e493a6ae4e243207ece5630067d54604375d
966074029f7add51092bb5f21cbb0580d73cb666fff66d10b650886915ab4ff
e9f0878c98df407ed184d91a9b079b14dfc37fbad0e4fa0191f2e32e51dbd997
4dc952fcd039ec6428138adc426c60c80e6b827390393c0ffbc7338e9f0e98e9
response z

5b7cfd80ec91328b8dc953ed403b33cb9f5e9820bc1a3cc7f60cf1874cb21ccf
4852d66a804d4049a0d21b14edec942cb705f007e1b6ec8f67f9e9a962bbc5a2
7d92c3b1924643634e9470053dc3cd03c221ea74e641b1fb834534bb97a3e6db
a77e2ec7dd1a026b5cf4dff799159e5ae6c0cbd75c8aeae01ae2b14bc2784c3
5917f2c1a844bfc0d0f923e98f86eb5c2762f752f25986d8f655642d51df5b54
f73a093a88e82254981f07c1075a956e2a04a5c04de5ebae3a4c5a54e3458840
a1e430ec59d61f6569cbf28994cee204641bf4e030f31a0900c37304d216992f
2502cf4a47ca01551579eaf5b46ec2a2a5422ccb1125bb287ebf695ec1bbf9d6
response w

6e6fa06bce486ac80c58e174407217f9a4c2ab51361d18a332b8800debed23b9
c572182f100396dcde4666055f07bef7e7c582377f9209dc90e57b61b17691c1
f02790cce6e38255c7441d18198e85d5a35ac2a2fd348d5f694341108a79497
bf13ab66d67111cea78e3bb9be525fdfaedd6fd593678081cb8d5b14f01abc53
aab2380cd8d4998c2ca244c2aa5317b7621023fe1ab38837ee234bc372e3c075
8f56c30360bd8520eed72f613f0dafcd82a5177c092cc8a98de462f6b5310f5c
389b0fbb5ec73b1d130f2259e46d17faee6f3d0bdc8a9f36add6766117e3eb99
89079227252d9b7e480eb8401775c47189090ae069674bb2a4700e7ecc961495

Fiat-Shamir context

92474f0310902b1fab53ac7dd7993db89739dd7446d5e210486b0f81d9c844a8

withheld proof term: 6237 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM.eq_1 theorem, SpectralPowerSeries:0

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} k] \{l : \text{Type } u_2\} (n : \text{Nat})$
 $(\Phi : \text{Matrix}.\{u_2, u_1\} l l (\text{PowerSeries}.\{u_1\} k)),$
 $\text{Eq}.\{(\max u_2 u_1) + 1\} (\text{FBundlePoint.psCoeffM}.\{u_1, u_2\} n \Phi)$
 $(\text{Matrix.of}.\{u_1, u_2, u_2\} \text{fun } i \text{ } j \Rightarrow (\text{PowerSeries.coeff}.\{u_1\} n) (\Phi \text{ } i \text{ } j))$

SHA-256 of the statement above

7e225569df6b279db4a817f9676d0096b04018df8ed002f726a04a5d34b7d70c

SHA-256 of the kernel type

ddf60985451a50e6135f3b562b477a5758d406d71df1b69e112e59bb235ac9dd

commitment C

7c8dc9bebf20ad14352ec8c10eed6fb5a35774e6613a6cdfebb6ccbcfe65fc58
c19752ca26fc8d156da88c6a69608a9d34495cbdc454a950af406808affacdf2
76f432e604d42b491f8bb240a0e6159c668c175ccfe3ab16204d5e2956aa6640
51369adc64147e23f9b06a30df19deefeeb891ae5ce42e650b2b8245a32a6aac
7b31fb3fe91de515d86c9541227110583d72cc7f143e55400f3ee995270a5c3d
d14d68bbf880ee6ba0df07e4b2749fdaa01bc5aaed9388549b2edb81d1602a92
75067e77474216b04d459f9a58c400e77ff68043bc52d90bce2a2b646880dc17
faf06ac460b8a7ec4f26bcdcb00cb5be419d7180d701da75344d42f51deec83
first message t

dfa617b57edb8abb17cd587932381839fb0c5a923c3817628de4b1f4f5e2bbae
48c0da7b181391d730bf63e7a286b2b9e20aed1aea0fef489b332f4feb39d83d
a162fdae022a1de6538d73cd6a0359409d33c1b5aa29768a583117c24035a4df
61b5ede1cbe3253589501777b421d397b58ec52576c69f63ed157fd9eea22ead
5b5f97fa011e13ffa6af3d9ded5ee53200d4508dc1d6fea2ad2acdb26c35502f
e1fbad95d9dfd1a88c2ae68e86723bb74fda1263a937166a1dfc48570abec26f
16e9ad3bbb89713be092031ea52f9c3b0c96959f9a924933508df66bc0214697
cb2c90f4596bcb5021ff66cbd22a8a9442946904e6de724109528e774917b8d9
response z

f57224467d21a4b58b32930a6e8da27748bb0c87f4095641e764ed01b3c76291
7cc79b866e530dd1e57458edf5b873dec7cf984e1472653ed90293ea7a9b062a
976d8528eece98a3538cfea788901672658aed73bf8ff2c29ed664af6906ed03
6f7bfae04ec3aacb90447b292ac467851b00c042e63bc511b6f18101c9648315
1f7ac4d7fd1becaf3e2a2df2962911908b10a77c7a34cb55e7eb9fc5e083836a
5c926032d7d6c628e6251ee793ab434b69086c2727f85de13d137d2b46c9b1de
9307f413700b4baed7e4798ae0535e6a043ab2d30e08d0c9d9611839ded24a48
22394dd1c4d7d9a373dc08df85fd4a549f4c2d0c06794c52daa96ff85f6dfe4
response w

a4a77801693ec87a756ff833210f93ab9bb8389d685437ef61d95b07f5739e20
8e8d391b10e752a731396d33b0827d413d7d0a765d5f307a3874a737192a2e36
ae12464447268fc2ae436d99826799fe0514bc4060978196e4c4da3311f63670
8e85e4e2769067c6ab1af39be4fb216de0677022ba66c25eb1998372e2ce10e9
60f7c831a8c7d0a7efec24ac9d1cf3a810e79f5cfd768532a91a3d931e1b12af
f5d5ddd3e3e8da305cce9293e1d94100ff4194f47aba1b4741ef1886a1e782c2
75db48020205f378bce32203fc92e03d8d40c9a4756338bf78d400138a60d6dc
d4f2de8680cd4b4117fa42f12fc1b06c9bd0f75063c0fdbe973664d7ccd29f1

Fiat-Shamir context

15d00640fc672b36fc7b5e8ecdf2677ac1a6aa432e2475f16925e8d25741f47e

withheld proof term: 259 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_add theorem, SpectralPowerSeries:65

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} (n : Nat)
  (Φ Ψ : Matrix.{u_2, u_1} ι ι (PowerSeries.{u_1} k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.psCoeffM.{u_1, u_2} n (HAdd.hAdd.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      Φ Ψ))
    (HAdd.hAdd.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.psCoeffM.{u_1, u_2} n
      Φ)
      (FBundlePoint.psCoeffM.{u_1, u_2} n Ψ))
```

SHA-256 of the statement above

e8010acb723d676bdf9c64e20e317dac4d19776c1225f4864a57991fa3aa8d79

SHA-256 of the kernel type

3d025a4e3bd278dbbc8f3cea9aece1b5069dcf2fcd3b8896f50a7d62c61e9417

commitment C

26acfa96f9351603f7145d6f1f8800b7f45df0e3b7075d9c1f8a3d0f94d0d23b
ad0da4536c3b68b5d795ae593527854e9437b6feead2758308909175c087482b
8883c1e254d29799dec4fee8fb2ec43b0d0abf5c0fa1019aeb4d98daa915727
b38c8cb1a4fdd637e5ee023024c1ef150c7e4a0680fdde10376d3b7d11de0de4
a8ef7384177b2cf72b0e6a90f230d0468e84376239e18dfe56723b6872e7a84
bd9315de37fb80b5dacda4a8f367fe856fdc89c9485e73a34ba33813ff71b6f8
008b87ec89ac310d72325abdb2a42d73b0671200a01cabb6d48160256820f681
6807ee08a5c197f1b1569d76f1ac36265072ca4aa7ee3e3c2b255eebcc3d61db

first message t

13168479735ab65e8e88498d4682ca77e97c95cefe0c06804d84e1af2656c0be
b6b5b5ec85a33a38e76f05b78973ced4060404159619d75dc92137969f2bcc28
6f8d8303c87e58ae56b2b77d81ba9449836b855cdca50aa9afebf2a4889ed24a
afd917751b1a761fb04675b453af5ab1a8391e9c935eb608bd15d9e43a9a2925
ad811afd058ce0f168a478dfc74a775bf3793ee804881da78c10f6e5bfbed5b3
46e5b9b06f4eca7b4c3debef960b152c0ce44445917c33bc7229f0de182cf3fa
78c78f6952b3e4186d7b00985014cb241bc107908920e6cd41ec37d553c19d47
e6c7d18904071a8152abe023a1dc85a077ec12f82d8d34845f087419904b187c

response z

356419993fbdf55541c37056005be410699be4d8715710dd61c545b352706857
860abf15acbc5a1228049a95ab1be673ac8440d4d40c458ed63782e3d30a9753
08b196c805408cfd193d51c33bd4831147b972e9f98d1e1b0f6b89daa81d8451
4ca6df904570102a6bed2fb0880da4af239899b87785290319d865e09cc9f0d0
ea74c0a7bfb4d4e64d6340f0aa2995058e885ca5976d2141cf01ec6f718bdc638
58b529291920ec7e2c718274fe6fcb596f8e813cc1ba88b3096f91e1399fb43d
3330eb1ccc3fcb91a8543b42311e95fa37e16f72d2abb7dc0fe8b3772faa8af8
38f66203b42919346872ba65512173b95f0d0377e22e8da5533f0580c0f21a62

response w

493d655774a710e8e3b6ffc2f6d91319e359b8343f78c576a351d63889ff7acb
46e9ad58efd806312adf07202ea497db346cdde8611b69a2cea587ade469f304
c68d48b23c8a85f9d7ab5dbf89b4b5d3ef74e0b6d7a0215bd5e593e35950f5eb
db532a79c66ac45e6e58cab9a66269ac262064730be5bf49fbb893b4eabf69f2
be777f1c1c63f0c9c1f00283c16e9b0270777b25ac259afe1fd4b2fcd8c8abb
e12dd62c5c794440dd21bb492538e2dd1abc28c98f2a41ab329f4e9c3bf10ae3
9328e6adbf96dfb9adf09828e63e62a805b2757482aa125b386d8b7b2f003694
3109c0b1712e119d273b2492b55a78a3d5cabce3e47489a4b3b7708d2a40b016

Fiat-Shamir context

f78432ca309a967869f1d9d5231e15aa734a3359b9f6648b00300d84946b3566

withheld proof term: 199 912 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_apply theorem, SpectralPowerSeries:36

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \text{ k}] \{l : \text{Type } u_2\} (n : \text{Nat})$
 $(\Phi : \text{Matrix}.\{u_2, u_2, u_1\} \iota \iota (\text{PowerSeries}.\{u_1\} \text{ k})) (i j : \iota),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeffM}.\{u_1, u_2\} n \Phi i j) ((\text{PowerSeries}.\text{coeff}.\{u_1\} n) (\Phi i$
 $j))$

SHA-256 of the statement above

c5ce15589a8b4794d13215fb781c063f192743a3625051e3c327769d9926713b

SHA-256 of the kernel type

98a48e2c1421a875c1e6c5e091e3f31e0809e2c59f72bd18031aa0db1e50f94c

commitment C

d3ac7bb7401b5e32fcc2222f3772b432fc7bfcbb10a3df2cb4a1d06b5dad00e2
6b75c12e5aed8df9405454d36f6c0b44b34b713db9fa2525a681bddff9966c28
685f14da229c152a8d5cd85db2512cc35f17be5b28b9509e41f7de11f5c41c2
6c87cc392df7314572290e48b9a0db94a60b9a33831f05cb30865773d167ed92
63dd89454d3a36d16533286cee9d2e1a84e815e752d1d3f7523c1ef517bc9ad7
0bfb5efb7a227fa98a7b615976913ab8635da14edcfb9da4bfa55e3027134b27
646ee64bcd93742648e0abc0acd4d5024838ccd0cc296d1cc5661ae43a54f4
1dc8e02ff8ab5f6676a8d19ce66de4078ad77f95bb0549be915a9ee63602c9bc
first message t

e85c51737f43f1fb31b63035a17e8a50d45820eb248dee0d6c438259b9329f8c
0ff14bc69f8ada1787453dfcf789c1152c2042dcc02022a65fb0666d3db05989
7067cc45cd7652b1f2bcb968a8ff5c1d7fc6b289848777e4b9c97cb349e5f79a
5f3b7ecf6db13e541bdd4301d6df070c21dccc7c7037ef4740a1716c944894c1
a9e4c1bf77648d95a27e7df174b7947c8028bdd29229c2b0132f923bb54cc98d
f058d5ad4fa4c1044d42b2211fbbfbed3fda962d17d1a1c15080bb6bc60989b2
0598e21746a2fa8863a4ec7c23ce816208a6bad8afa8a153973d7582473fe01e
7eadbf3626225db78bc970b55e56e75e492b48d05a5e2437febc8c83e60b4d21
response z

136d88ec7e3b9692118b5756b9e4b4fc69c30bb85d989d410691e9ace62bbd11
b63c85014c68ffa817f00b28e2a7cca260f40118f3ae6313318be83d0783f984
dedf81111f0894470659f48c50d786eb3e7578b5d03621fb5416624f17dc29a1
76dea9099ef5895b923d893c122ad37ae3e59d9af6ef08f33ef9182ee7b5f51d
c04f43cd7a3944b16cc7bfbfbaa695bb4d91f26d20a8a250a4af9037be0830ad
241cee90cb0c52e383164c8a40118bbfe70654e02d66ae44193f9c1c067d9839
0f79080a8007f1d81dd2e7b45f6e295dc3b52d40c19e7db03b28e42f678de90c
1cd441c2c8e06a91da1cecf54ef5e33d55a6df1379f43145c409ef9b6f406645
response w

21a97d496b3d58f02c5f488072eea237abb952e397ec900021e8b26ce52d3da9
f7f8765b30bdba18e322a6cb454c53f1c55d6a4bbe3646a597f007545379fefc
ee815921a3c113b7bbfd33495b0fff995d180f94c84fdcea11a573775186fdfe
51a8343d2cdc07f93ebaa1779b30633cced6609ce8b98f88d55b1ac38dbed93c
03cb2563e0854a30490947d2f4e61f2859bc7068085f80b90b07f6636bbaad9c
c712adc8a13e30d63cd8c5f0a15868133d98762bbba4626b2530d0166e34e9ca
b61a36714d7da3d9705d2ec54328c9e247611485a47a8d10088d78c3275d0aec
227c59c27d63b7373821093b5e7458c38132f24cf6e255f795b1d9229e54ef2

Fiat-Shamir context

84151d55626a300d9f4c54babb51231d6621a0026c66862b65acc9197cd7854d

withheld proof term: 364 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_cst_mul theorem, SpectralPowerSeries:71

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} [inst_1 : Fintype.{u_2} ι] (n : Nat)
  (A : Matrix.{u_2, u_2, u_1} ι ι k) (Φ : Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.psCoeffM.{u_1, u_2} n
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.cstM.{u_1, u_2} A)
        Φ))
    (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} A (FBundlePoint.psCoeffM.{u_1, u_2}
      n Φ))
```

SHA-256 of the statement above

f9c4bd405255316c58d859401a4187f69cf9bf01e6d07c07f7b74e51d5c54ced

SHA-256 of the kernel type

f63e1d3508b0eb4fa3d09316eceece26f8f1d4ea503f5d5431a0e692366b0d6

commitment C

afe4e1af381a84ca1d3aaf27194921a018e4579dd4967097889da41a24b9beb9
521f621c821e90fdc59606319dcb4a8415929a5b8052c604ce6b20b32c2be5a4
e785009f2e41f5ffb4b91273baff62c3b6f8a3f590a5429800ae9e553e29f5e4
1cca4c88eeec27128a08f613aec7189060e744cae1a264c0f6f4b401c30f49d2
2787ae3e1089b12f0d174f9caedd5b328368563c2c9b4b68ba2b75285ca70870
1163c482b654f939eec1ca9ad8e77613393b8e790d96d7ba41127bdc9df950b3
1681eb845af1bba3a9347785cb49acf01b920b758ff311ad7c7d7a73bb41514a
508258f6115e9f9175b232d5ae9ffbb5431b2905a944bcd2ded463301f90ed5
first message t

b7ef05f8b0e77d72f69b5532c1185be3fe6d3cde9638471722c06241bc032587
49b09275bcdde09abb95ea7d8f76745b0c80d8ac217d587432bb56edb4ba5886
7815c8b6531346315779452aaa67fd2cb5cc0f3d02e0580e6acf9d19f2a1e098
0032be9df04712de2fc9359dadf3cde7946caf8297916e9369b6e135bd2bf07a
0cf85afc37cd8a57f1beb5056e871c610d2893043b6ec6e51e8b51431becb32f
294cbdbd8f1521c06cb19aa179e31ea40a618889c3da87f8a0af10694f8732b
1550f01e07270abb8ad6a7d859a73033e959267ed74ebd3d8640eeefd929f3d9
491d6cce86dfe1ab1c71596fb7ff8afb384125d0406722babf5262102a6d01a

response z

7a4d3fa2b946589c275ca7dd6756665033996bd04fa80d62f38cfa0488f17b57
37815f8d48b60650e3e38137be872700e066bfd2d7df6308db8349bdaf5f53c6
e7302b2137ec76bae02731e26b24302f2b3987b67679032a49970fc26bbcd4e4
83d3688b03458616d768b0253ec5aa67d5c677362ef7dded4baf8854102e9fde
649e7f35ffc3419a53b69ff7a6e3a0327d3b00b4e0056bf86c1dfba3ebf7df6c
2b6e54970d6f0e8c06a03a70cb97c6d6ed982102e98c2a4e38c3d53fad8cb046
96435bafb73ff494acbac8a755f49d29dc6da623a51b0de90eb7341b47231e6b
26bb194f307186bfc1c3bcd8a1822d03643c093834f28905436959dfdb668639

response w

6a55d2fcd2786dd551edfc0070ee22af5ee0c3480302c817a1adcd3e5608fa38
d93fd53c5ead182a31fbfd89887875f0b7d570b034034f96060ec3890b3057f6
40c75cb055b8d455d24900a841a94f0b5a0b07b0f197f20948ec02f67326b951
67b438259a4e6cb4ae7d71f3e6e969aa0a5c1c10e9e39199811b4a3a0f678415
0e93f702613a6a9fa16d707ecb3afd1f2a0585162deaeef847f0c39d505167cb1
e51514157df17a55c1f8a3d4f0d17f4b88db157022e28145f833b4f0a2eb2ad8
edb474cead7434c10cad8a40fc8c99f01ddf0ff5ec0a6f5a1c5b30a74de9a7ea
8adeceaa911465ace2c09d6e4685a26674d3d9819b9d201e58e546df6f84facf

Fiat-Shamir context

94a57afd007310440ed026eb9cc63dffe5aac921c56906730be11e0d9b918b98

withheld proof term: 236 462 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_ext theorem, SpectralPowerSeries:59

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2}
  {Φ Ψ : Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k)},
  (∀ (n : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.psCoeffM.{u_1, u_2} n Φ)
      (FBundlePoint.psCoeffM.{u_1, u_2} n Ψ)) →
    Eq.{max (u_1 + 1) (u_2 + 1)} Φ Ψ
```

SHA-256 of the statement above

5d64f6b3add1841097816a6064008600d6179bc9224ee3c9cdf1dbf85a8cddcc

SHA-256 of the kernel type

f5e7b6b78d7cc014f38fdd931f7fb6ed31e71522f793c0f47dda78ddd79e8b7e
commitment C

80874ff14f25615104921095bdd54c395c2bbbd1b72384b51d462affb5c63350
4aad545be25584abfbfc7841a19c8cab222e102b3718bb460b448c1e087e3d1a
165566a43cb28f87ba00093cb74a9946be510010c304157d487ae4a435846303
485d895bc2ede7b81e1a67976ef26cbc2f5fb3f45745c2952901bf65f8e293ba
dbb56fcb72b34634c64fc0cf76e9c914e1269f7c42fb8f6a72a8d3c19144c4b3
331e0a64d1c9c57277d04cec99885559f6e4f3391a3ebaed8126cb95c6fff146
4590486dacdd95f6a3c2aa35bcad25b4439f6b2be5b50ab15fa672a24a5d2ae5
cd30364d87777377e55e59c34b78e9e2dc221b58aa94a17cf17f95d6e324ccad
first message t

197147960ba4538867fcbab1b2cacf26aaefe51f95de10cbb06444a27276ec1a3
60a1baca89d2a94e34ec3553398d384d11f75116c368446695deadec0136c815
a476c83786ba0deb18bba925123d02f8e857c82ef1bb74ea92559acc5e4680e9
19c50ff6fb069c2aa057264a0a32d3ff815ef86de680ada4919e0339dcff43a9
85544f7ecd6a908e4b5c77ef93080a4cc55a9dd0269b3829eef837e3c130088d
2518864542474e1eb5c2f9a667d87ff1b3fdce7750de4c27e62e208da5f427a4
8f919d3eba77e8ae26f0cdd0523c94806b5ee349219712edbd9482f79fbfa937
91cc92034f72ab7fdefd0f199123257675fe330e546e1b336368a69527dff012
response z

1846a47e76bc8c0291db08dbd49e57c5619867c84bb6d90e3d7690c9d2109309
d44a7e68828d45ca7e6814cab5213f64907603c41435fdb380fa62341e366d2a
74e0a9008571fab5df146eda2c666241dbf7d9fc43a120ff3dbb7091efbb291b
43ee34e9765a33e4bf343bbbaec216f7686a3b5f7541018491439bb71d934535
0ee3b5ecaf483140a033d6ade816b618feaffe50447b9b4ef782fca1e255f40d
b0a965a532503a5bf00b5b3d7a27c296894244b362b1ae5099b668d86edceb37
0be37ed153e29812b2ea4a499c476f4ae8d66e92a04301296996458abee1eca7
1448c2aeab221fdb7e697d120a407080ae289c889b450d7213a5da2ecc5d20a6
response w

3f076e81046439d6f67f453380cc98d368e2e4713af5b271a41b1d1c94cc512f
29907ce57fba7fd4a2d43b7e22c00835fe99dabfd2261e7d77ed17bc203cbc36
e66d963cbb027f3a8b3789e935d7fef048f76107360efab59e5688bdc495b6ec
82432df1c3ff19899e7ca077714bd1a7b2a897d3e8eac30712381fab6e3a5f2c
a1db9d58290a64d7a0763e17cafcd3bb69e0e75c14a012b61ee9835edca9a33
9fb1a952208abf59118aae55e43623b82fe8306307aa9bc5b9420356bd6e6ec82
fd82b7f6e6a344817fcf3466297c76ebc71e60113e35510b86f52c18038158ed
a47519f82033f863f7a6ae7fdd57beac0cb2c0340e74595e2bbeec73872fc964
Fiat-Shamir context

51dfff874ce5b512101989172a732ee14f55ea50f758da0fcc0411b3a6463860

withheld proof term: 1538 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_mul theorem, SpectralPowerSeries:113

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} [inst_1 : Fintype.{u_2} ι] (n : Nat)
  (A B : Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.psCoeffM.{u_1, u_2} n (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
      A B))
    (∑ i ∈ Finset.range (HAdd.hAdd.{0, 0, 0} n 1),
      HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (FBundlePoint.psCoeffM.{u_1, u_2} (HSub.hSub.{0, 0, 0} n i) A)
        (FBundlePoint.psCoeffM.{u_1, u_2} i B))
```

SHA-256 of the statement above

f818e3ec262c25f01d6aee8e3614d90f477cee62bf064b77e85fc3750712f9b2

SHA-256 of the kernel type

c33ecf35bac03f2142b399e108a32c9fdff4af2e5c31833ee7e4bb0edc723a5f
commitment C

17d9948588779d14452765ed48e0f3ed0198cab9214d86dd549023a625e41acb
eb3c62a9c43479af0b48993a92b50408a7ee2921d65f406d325e80346cfe0c86
51d67313d17586ca00c8e2ad240224a75e61686143f30ab278bb3e9a536794ff
9b594e1cc2ff5ce9d48710c1985aa5ce1faf74fc4a4df690bfac0266c0a67cd4
005241972e8d01a27c5efd27f78c760b5a5bbe896e4b1d67cdd711c1eb404b95
91b01a42b55cb2929c902bfb809c18141e0eca1766851f4c29a597cd55e27df
0627df84ddb3134056c7d276733f8d0729a98ffdc7876b745a0e10e8a3cbd4c
c4afca793ba4ef4fefaf567528c101a74434db1702b58dc686fb206751da2a49
first message t

7b6731ea5f56249ff45ad19360c0864646558016fb503d953dff5a4f31ff6c68
c5eb6d9e337e61ea9c91dc70cd523e208821341a44563d08ffd185e9d5b7d465
566a5e55577bd6938606c59f2ccefaca1a520902c7e9148a96b8332da518571f0
cc8951690df9e50628bb22567f075b6457535b231dafb7f560af25b5aa43e6a4
980bd89a798f9a57249764f246b932f0a6650e93020376c245ebc3c14e4cb16d
59cc82f99b508918fb08f3eee257592790278831117ca8e5a5c337c625967513
44e3f780c2ff547b6df0b784635b2c30a9ae6204a03a6ce3ec55fdc422a4b043
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response z

37e7f4fde9a846983c719a209884ef570efa91e3e64dc0d0afa785d145a09c9b
eccb88d98fb41df649064ec58667ee796951ed41f19f953f0385ef1d8d8907b1
067d60ad57b3127091740dc4870cd87660e3d84da95127e36e8c5b675df45962
8050ba06dc0e2bf976923e29892e98348c6f0ced573c9807d89cee4c6971ea24
0441da1ad116d0a730a7900a8060dce7e88a253e8940bff0baa2451d53c0d9c8
f1ff7e8d25a9c1a591b9442cbbe02b3b78bea0583c76d0a1206d2d0c9c0a6081
ca62559f114bce45dbab585b01f04246e87e57a933ee9f55368e3e523deea61b
fe1e144c0da02901d9858d93706d5881ace7f62a1e1090a84502d6691fe4883d
response w

5c48795d20efd400d94f16641e53b55281b01c49715c3f9a83debb098ce50442
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1b5890caf512a905f72764ad1861575703e24c4f3e5e796e5a99b9983731bfa3
c94ee7ea829e8ee69fa8baa40829d12cb1db7cf65a1b7adbe6152739883888a5
775ffb7c89acc49b4b4a23235b20815457352376e03ac40a1dd6c1cd059a095b
ec7ca52457dc4954bde6f2525bf7ab13e018d46afe6a4cb37acd1144b200a77b
15f63e65341fa834361348e9a8cd87b363ea9ec2369fb4b08d1476fa30166e27
ca0dc5c79bb731b0d2f23131f9ef11db0c61cd328ad5a86994888b70eb6ef83d
Fiat-Shamir context

8a11bde0242e886ea8b3005b0cc5709896906bcc746cda4de7e92531400aaa16

withheld proof term: 707 067 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_mul_cst theorem, SpectralPowerSeries:78

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} [inst_1 : Fintype.{u_2} ι] (n : Nat)
  (A : Matrix.{u_2, u_1} ι ι k) (Φ : Matrix.{u_2, u_1} ι ι (PowerSeries.{u_1} k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.psCoeffM.{u_1, u_2} n
      (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} Φ (FBundlePoint.cstM.{u_1, u_2}
        A)))
    (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.psCoeffM.{u_1, u_2} n
      Φ) A)
```

SHA-256 of the statement above

00e4c408835496f4c08a049f8c4776916921ae3e8ea94457febcd1e6792fc944

SHA-256 of the kernel type

cf6ca3eff50eb1793f4877458d6b19c855f2d393aea50b7c59f24dc0a65b234d

commitment C

4badcfa7b17a0016245ee286f0a962bf439dbfd1e716e8856dc735f25e3edb49
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be6142cfe24c9cdbfab9dbe74e30bc1d7355f90ce1e46dbc7a5a83c07b9ba17e
dd8cf6fdeece24fa9533003ead69041392b1027ba6f310114db7ebbcf0d988c6
6cac5f9bf579c44f430f5e07582c887d5a565f33664adf05f2dd3496913d2a1a
0b4a86215e0212c710870e8ce1b8389f811f4d45103cce071489fe4ecb33502c
2cdf526234b3d127d5923abc1d5601da503b0fbf3784170d81c5ea05ee0d8633

first message t

ae01ea2035d53313eeab5418fa17ef8b84e8e87fcf3e6aea74660781c76dc0e7
9dff96ba1a39bec3a79001e5bc40c68b95afdadfff690fd79a7e666600ece8f
4d4a17fe5dfc945347680894f765573a99643eb841008123ba560edcd650c629
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452d36e30c92f866ad38b963ca9608307a7ebe808c6d35bead9d90967dbf8575
2c14ef9504383d437f63227152ec952a32d6eee6d7ec316b9cdf95234106e3d9

response z

21c58242ab30b5b079bc45ee8af2ec94c936814e5748c86f489b45b0ff79b7c2
e26d9a05c456d36b24bf290eca96148d38406325f9b119bc9a7009fce538bc53
6a99f6173a4782fdeda5f2ee9a6ffe50514328cb565cdf3a7aaae27227a990dd
0ed37509d707d5e2db0e995ee95b345af3252a87e6a318ff7d869d580ae44d12
93878834e7055b4a0158c8b4d6863bdfb98de6eba78aaf18b40f932cf2f126de
9b6e362252c785243f55a941b4452d617f0288d6014ac750f6cff74cab47f55b
8fca41e3fd19c966efe5433b0e9f3655c22223616389105b5ed1ed840660a091
94ae2dcd8b636af262d42f686ea7b38aa3c3633bea09897d39df25b926d31d84

response w

3013417893754a956d7678686c3bd858d9e340b6fb9fd695388a787720869f1f
e9a0ea9fd41a38006504ff1e5e530a956447f3381022dbc8ee664f0284a541ec
033ab2fe0bd87e273cbf44847fdbb03588fb6892d253fee6b8025602f0c216a0
136872762e5c8b046ca19ad9771ae2d02125e12fc65428b3e1f4358e771a47fa
416bda27968414833e11c6bf9ed3e221d623c781b68acd5751cc20e82f6b6f00
4df67378ca834a51ef53c392515504ca33301b5a7ebb53d0f60c04a6ee0cd75
9c475e4ff8b1dc068cb5c58b2df800880949eaa33fc82ac806db5050ef68ab4d
307c13591f23d8d26d1f30c4be6a5be49f1413d4ae80045b3492299e04ae3813

Fiat-Shamir context

54d702ee3caad768db9341ac8c58b66d2e9c7870c76d91381d7aa3be6035de40

withheld proof term: 235 644 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_ofCoeffsM theorem, SpectralPowerSeries:53

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} (P : Nat → Matrix.{u_2, u_2, u_1} ι
  ι k)
  (n : Nat),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.psCoeffM.{u_1, u_2} n (FBundlePoint.ofCoeffsM.{u_1, u_2} P)) (P n)
```

SHA-256 of the statement above

d81feb1326a99564b0de409a7452360790737ab180abb9124d5c612be02f735

SHA-256 of the kernel type

a598381890225f355cff6567688c619772b6e879d1228096affb642e63934ec0
commitment C

169ecb3d4b3c9731405bedbbd0cca7d6580392294991b42c92f7c0fff46a7b84
7a38f6bbc5dd9c2360975951346ddf658e2a2e03d591994cdba7669d7f27f4c
de3df10ef718b617b57e25f3f514f98f65256b9048bd2592b55201e5f461d871
5c4dea10683fbd9f9c78e13bd5f724109bfff95b1099347f49eda1b5e5041e6
fa332dc8396aeca30e7e8514fda31a44c94db95e8dd0b2c73bf79ccee1ace8e
3210824cc9d6c2741c41f3475e2f22968fa8a7982e914dd0702d9ed3b09d6153
b6aaf9de8d7e2392ed2e93da3aafa2bf189065dac451b9503acbc2214868652c
a4d6b70df25b5e58d9430c1be001e8700364913e37e2aff35babaad738f89d79
first message t

392c5de14ffab45d9e49df90fb802537d2600ca3e50fda82aca17d7e7a1bfc36
461802a30a5448b24e652e3dbf2e025551249590c4b4094f83f119c68ea20e43
77bb88e6ba3188daf9f0a39e1a4f46bed6dd542f0f4861b765c46fa0d0254b5d
0920d87fac37ad244de16062af21a8d224c14fdc5ed99f28e792b61eba525b7b
a802210bf7f1e338f2cc3eb55e1ea48c259def13dc93ed993596d424e91577cf
611f3ff13ac1c0aaa43d6cec38bf207d248a0bf625988137629ca0be235fbc9
a8802b79f7e1c56b819fe2b85f58c2095f51438ca7f4de7ec699072481ff32f1
dc97620e9dd2f2cfbef598aa0a571106310388893fc6a76329bbd3965431bd02
response z

1221b39e9328026db414da8bed42124bf1fa32d18fee91bce62380e13f3e0227
0bab4c47729802b4be901267a16f6a007e318f30a1a626d4601f367ef3ef5410
e93342eac3f2c67919bfc9f6192a85c746fbd6c071de1d226d25f62f03d3aca
7f3dd30fc6db09b43184526cc30fd1baf046a8eccefd06f243907a2cb8167188
9fd5446b8b7e0141aa54fa647afe4f12066b3ee33bd41c1bc774736e1b177a25
6fbe6b163c5e80ee8cd8495d1e245d43a1e89d1c31ab5e2538863cc24133a366
f3fd0c1d9e98e2d3099aeb768a2434aa6bb0e73d7b5dd4f053d62f033033833b
9147bc76db259f7b67cd3a48f05b4bbd1df89a1975cec73f622dc19db85474f5
response w

2293e9643c3bab55d1ee99ce1a65c183d7e9af0ee4c4d308ed4c230f64d83af8
6b7e8c0ae735acd54c0d59f76e03f363993658d2544bf3eacf4df5257e1f9fec
5a0975c6f7412ba88bed1582bb0ebfee5fab414fec945c013e7a74b8856a728
5807ebfc2f82ad1bcc7c53edce0a4c50ad79c6de03dc4a4d5580ae269168a0f
85fcd476e181680590d75e91a851761d90c23cf9288c9cb6091522379ba8ab7b
6fbc6732361c5fdaf05da4dea5d08c89f124dcfe5eaf1ae37482e945f8e4bdb1
a4af91d71d63efcdf1250a617d1990fc80408430cfd261d155e61acf6fb1eeb1
c30bb407f0990ee3e4d1cbe9c015f0976c697455e897821ae67503f717fe0ed3

Fiat-Shamir context

34d04c221039d62a9a0fcb7f2019619f2033968d052c725bd5f60d1d67b62388

withheld proof term: 32577 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_succ_X_smul theorem, SpectralPowerSeries:91

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {l : Type u_2} (n : Nat)
  (Φ : Matrix.{u_2, u_1} l l (PowerSeries.{u_1} k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.psCoeffM.{u_1, u_2} (HAdd.hAdd.{0, 0, 0} n 1)
      (HSMul.hSMul.{u_1, max u_1 u_2, max u_1 u_2} PowerSeries.X.{u_1} Φ))
    (FBundlePoint.psCoeffM.{u_1, u_2} n Φ)
```

SHA-256 of the statement above

d244b54218fc7d0b60e43417a45cacbc291b393d16504cae040313ebbcc43c00

SHA-256 of the kernel type

5932c20fc15dc89a0fbdcb66250bd1227977567555a03fc7adfc509ab4386a59
commitment C

76139fd89899418c53d9fc34bc70b4abbb66d02a0ee55fee1d4006716d8ae1e4
78b5d89fcc6691bae9f87e65a50092f086744c4bd89bd93e26d9db53d7d0e134
110d4d296e87b79a3f9f52be7e3dc15956c32422ba17be009135e3be1a17a191
68413b3717a7c67d1341784d674ee43d97c6570c59ab8a0c4b6c4fc111ea5061
6deb6cc7de917a41c5055e3155849c2351a21225d6a9228c3002dcb36245d38a
5ab0f064f00d2a9f6092178eaf731aa15cf5151577d1539385e4cefd6a0a2169
cb1e9385005094409fed0035f7936b60c6af2f2cd020a7be4b34c06ea3a0d21
a2c052b3d7cbecf37a6c231085b66c9ad4ad2cbab10effd08f23f4c97700ed5a
first message t

b8ca0f063dc922dcefe33aa06b1ad8df7d82bab8233087417a272489a551ae03
d9f8901cabd553108343199637cac1d0750875518be16626f655d68c646ca99b
0b4c7eb6eb7a9b00a7e04c3e6145829ae3d9a8723883eecd02d5920b6ad317b9
b7b22aba78b2ae7a209935e75457063c47886630e2444280ae480052ec5500cb
5e2b52bf30701b4e6ca151825d42910638ef772c962aff8eec7cbf13fe710419
ae77ef5e717881e3cc5a08b5e922e1aacd81d94d8dbd7cab7dbd3541fb668b72
8f3eb0fdd6d3fdcf97048f655197418ec24273e222fab452c6b9417268d5fc6
a8024dae98cd7478e0a6f813789fb135a85009e89606f12675a9dbabf0fcff53
response z

5c5228da33104131c2fb20ec8a48270e0c2ebb2b999808a2ce987102e5d0c61c
2905fb3fff80ae4d8571c630e5aa1c23bd2d6a5c161d3ab8e2dc75c5416b6438
477a2fbd62a5e01a47cde158a5d53fd58deaf9a8742f33bd03631bef6724f4e
45fe617c808adebf09fef4bda874b7c01f1097a0b7c3d4ebaccd4db272bb64a2
22b3ac4f790251a11b0adac38dc9a9341625a963794d795fcef1a4218baf0b58
51f7010e56c66f434f3347c2c9de69efe2a965fc1e76607e5fa1d93d6be61952
345fbb1cc155947cbf68261a53b4c463f5802b9b987b7742da0b90da158028df
1dfc18384d6c86022d4e3ca9fceaafdbee8c968bb0b38083b5952da2982259814
response w

1e15db85d8659a233de5b8bc369e0a48609e6c5d6ca93f52c6baba6741a5648d
dc3970162e4e734279385a846c67ca758ec121ffe56775c5fb9312e06e284d08
3f54c44cf3783e8aed0c9db335b253a63ddf8aeb68f0c0998109d9331d0cc2
f661303eb26ddf61664462900bd6431ec01f985dc2364b655deb151f8f006a3f
6893b336ad7fcb8c0edc742a136529c4b51b7c79168ab555d8f73197d1df0773
af67604b2a87f5947649fc36b42d804648bbe7e15269d2bf4c0cfb4f48056a95
d6d831b4c2c5dbd8f5d80cc1b9ea2f6a1d611965fab896d8f6f0f84610afb187
47cd94b146c356072a8574bcecdc8c3603da30d0b3f070e8445c13783ffe4ccc
Fiat-Shamir context

06898843142ced238eed0179c27630f87e1c60217164c8d263f7dc2abaf95e00

withheld proof term: 112 559 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeffM_uEulerM theorem, SpectralPowerSeries:97

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2} (n : Nat)
  (Φ : Matrix.{u_2, u_1} ι ι (PowerSeries.{u_1} k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.psCoeffM.{u_1, u_2} n (FBundlePoint.uEulerM.{u_1, u_2} Φ))
    (HSMul.hSMul.{u_1, max u_1 u_2, max u_1 u_2} (↑n) (FBundlePoint.psCoeffM.{u_1, u_2} n
      Φ))
```

SHA-256 of the statement above

53a80d613842af08abbef161a328dfbcd704d89cbbdb9631dc79b82b4df4131b

SHA-256 of the kernel type

6c22576aed913f7cd094e08a7bab8a50cc32ee1b86847444a92272ecc01bf74
commitment C

c10d10593f39e8cc956cec0b77f8a7e062ce340fa212b2b0053b91af6168b12c
9c8e486bd968703b2040f1b1117f0270dd760a2380f29a94d814e04cf8b6c38e
a45c5255a64494a24b1a990e622e1fbfae3044e6890a1c607282bed9e66a8047
e8ea437e5d8a597766b3fcd65fd9a113ccbf6b1d80ab278976ccf093a53d95bb
57ad5dbe8540f9e899d2c9b94b3e44a78fb5dd5028863099e3e32d273cb459b7
c55ae79213239cfb89f618d2e42ec2f6c9572732da27d0d4836be17ec9e2d083
599550952d41f04858788545b770ef3b589b0d3c50e78303f912dd0280802174
f87f22a2af2a8928407d301c5ab0e7699846de32788783470f9b9770598850d4
first message t

cf549d0cdfb04647e7012dd122c33f0256760eb60c9011af3cbec61a29918dc3
393a05202e1e7ff613e3cbc75023c4057ff4f4dba2147198bd5a7f71a9cbc8ec
e7ac136b75c948e053d9a04e6627bd3a1e2dcc01be93987b5a8c33ed488e0490
8dfa8da4acad9703ba1727b7cff6fe4291f9fd84c50f02fba494f7e6cfddcd774
0ccca9844573f1f9369bd5f254960fb4a9a593c012af462bcdfa87576a6b6394
6c02a545bb01f69b795beaf11f82dbfc308780e7904325d6c7965ef13701fa34
8cc3b9863213ae4a9899719540be22582053cfa05f44199b7f9c6b69b37fd4ad
67bf7a0fc7971cea3abcc3e3f846634fe40ff3843f7732174a892471a75284a5
response z

3b57569f688dc0592c88c09fd22fac29a68c6968df57f0f365b3a5dd7626eda1
a42c5fe4931b30dd42da0e181ccb5ce7791c79b460c085a1c9c65e28dea8714f
5ae7a52ecc8b8208285131826cfddb28f9ba26b5cbd19daf0de59fb8efbb107
286dd30b00ef7e6dff0f495b723fd070f4d4417023e3f4b1071dee50fcf5fa0b
e6552bab19fc7c66cf36aebf75bd04ba4459a68223b0cc2e4eabe33e9d3b0e0b
99c02fc29b1b7ee9dc550238b6b60294c3f48f0e7b3aa96538a752e5a516878c
f57e7be8505387687e0fb1030f6fa18c0b3e41e274530160ce65a9d8c7789e5e
0e4253221f1e2f2935ea20e04c1c3e2e95797bf2fb0cb2715bdeaa8f6f2789ce
response w

16aa1f0e7567e260de6f2e05cb50a8eb4b8fd1620254baf0b3af9da032dbfa47
85abb95340e0c3e0425ab05f12ef2cdfc269f093ec575dd43dd33145563afe82
00586a9a0fd15ba71040f22eb2e7b6a84d7277de2b6c50f8ef5522983bfba7f9
9c2e350992655fec3e54db78661c040252eeddc4c32d0df9582e70ff91459a6b
146d82c0d8ab4ca69b3e728b934ab2267ca783f74b01c509992f304b478531a2
06ff8858ed65888fdb264899bd48c812a4c34834312102af7f2c0823c3cfd1a
8e203e68186d59c3d83938de3857549b06dff50fa3e2a687102d13a0070f3448
5264c196ef431377117258b9e9e7cf3445b79278c07f7cb54f5023505dd39bac
Fiat-Shamir context

b1a0f48efe4646436326310e246eef96976ed271076af54f0d9c44f6b387ad6f

withheld proof term: 129 814 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

`FBundlePoint.psCoeffM_zero_X_smul` theorem, SpectralPowerSeries:85

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2}
  (Φ : Matrix.{u_2, u_1} ι ι (PowerSeries.{u_1} k)),
  Eq.{max (u_1 + 1) (u_2 + 1)}
    (FBundlePoint.psCoeffM.{u_1, u_2} 0
      (HSMul.hSMul.{u_1, max u_1 u_2, max u_1 u_2} PowerSeries.X.{u_1} Φ))
    0
```

SHA-256 of the statement above

701aaf279d5e6b06da7170cf45abc810fdae88207bbb46baa3885eb213126dec

SHA-256 of the kernel type

bd6f950fc85f6b48053af3da8993047113d3e41f56fa86c41d82cb78e8dd8446
commitment C

713ec6ec01f3115d1f08f9c1a8e03127ee759cf0fd65c307c9b568996a227f26
5630ecf49dfb2247f0a4f70f6836dfa0c67fd3c17138900fa4f2089e09a20352
a0a732d738c63408fd80da65193c431b59d9f474f90f976a67b6486cddfb707c
35913f086cbf7b426a03cbda2a15e0a7ff34d82f06698d6cfb955bcc67c53029
f7f6045e699c495d0a913fdb011bd01726b88986e379db3fe3bc87c8e9142942
e21a566d5b930413ef7e11862089ca52a31644d195c03352242f02f927ed2349
f1a9ffc6190197c503241626c57c4f3b2a599a2d8255361649f57b8e57868857
c70a283d3dc6454d7eff41eb723e0b766a9e634130b373f870570efe11294f4c
first message t

9cc36321f53a22e188ec37ff124dd16d1930727a9e502ba3b586c3a13f2da585
0611e5c048ffddae123b90b9b99cbb60bb08def06111211e5427aca70b807461
6d3b2a7afb495937eeb699407b5061fcf645be0f216f1953664f00be42435576
94bf6bdb077eba722362706b490b194023f6ef9a2fd407b40183c92d82101da1
5e2ecca9ab3eebd0f0dfd3a542489453250edf330a03eadb7ee3c01cc9b641f6
77e26b544215408031eb3095ff3416252db27d9c98e59b3ae4d987b0ca975eea
5332db9022980efb92f5c7211d12925e9106b01d528631e60ddd0b74d742f862
ed0e56c4f26c2363d9b4e4475a9197d9b912710fbc00c933026ea1ff19fd03aa
response z

2ee794abd20cd75e38a7e00ddfd10a6d4559c709f699dc03af7c835fcae3a2c
0de0aa0c3a5288b125a2d4e4a0a075817fdd2245e2487e4335936095b01ca8f4
224f0938daea2ebc06212b98fd7ff8227e8ee2280bd7685ba0e59038f6be5345
337f5817ce676575503bf77d0025a0c923f7fd8fec68aab4f6290cd4dd8361e0
b3ba3e09dd4f202c526933cabb94013b6fbee4396302c9241b5afd614d5e41a5
ca49db982159ca58428e29d6cd83e2d7fa11d01cb168ffb6e4c4291a656ffee3
d4a6c89db8a0b6485de5fb77614d355e79bc19f271081b4c1043dc9ef67a38f4
4a5ac01d3615edbe83e2512ef22a51b904d93b5935bef52d8034cbb15670c8b9
response w

7efd9755a567516c39b58fa48562da9b6e89f8fd67b004c4acce98e949c1fd5d
2fb512ce31b76df399b2d9c573c1d37a9949db20bc97ab6df9599849f5870b43
f791e813e2f44cd9e36425170a6d7045afef1ef0989ea98d0a083b59bc845084
c674a8081e20fc0a44648609016ef5d1b6e8d960ebb71256bf9e2b829c29bd81
c40ca82aa99f60dd750fcd711e797d8deba93570009855c802fe4a623613d9c
44ef67fa8607463e235002d9fe384887a5ec43287ae3c4ac8d38b813cb10f2d3
4c1c7d10afabfa7edcd6f6ae199edd3c60d73bd522ac7eee1030466c0e760eb9
cdb11f3aa0f6be1f8bf00b6c1eb2ef4c5be2c34822fbfbf7fa52a724407cf9b9
Fiat-Shamir context

0b85fddfdb601ffb50480aa99665ab31e572c9524a97cd888e8f1ba556dc8935

withheld proof term: 98532 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sum_range_succ_swap theorem, SpectralPowerSeries:102

$\forall \{M : \text{Type } u_4\} \text{ [inst : AddCommMonoid.\{u_4\ } M] (n : \text{Nat}) (f : \text{Nat} \rightarrow \text{Nat} \rightarrow M),$
Eq. $\{u_4 + 1\} (\sum i \in \text{Finset.range (HAdd.hAdd.\{0, 0\ } n\ 1), f\ i\ (\text{HSub.hSub.\{0, 0\ } n\ i}))$
 $(\sum i \in \text{Finset.range (HAdd.hAdd.\{0, 0, 0\ } n\ 1), f\ (\text{HSub.hSub.\{0, 0, 0\ } n\ i})\ i)$

SHA-256 of the statement above

048158e74dfd047379afed1df35362abdf70e4dde75a2dd645cea989a6faa10

SHA-256 of the kernel type

7657b1490ab10a47f4834b9309b8fb501156f85fd7fb4a4ad96fa6ee40b84d17

commitment C

f19debc67b7b93c4773e641dd967a9e283e8efd13a93add4621cd4714adb227f
817b661b15381a021d895d2b7b698825a726d4cc83a399bb6346e66ca6b6590a
84ef5dbfabdce2338ccf15e92c06511fb98de5635bfdb99e25370dde9f8658b9
dccc6b2eacd9c51933628b96f94e8af7317311525a3255c1ecbae1f61559db4f
9be68d2b01eff1c14f041d01bee904baf3df59071c2464e64769df86abe7c786
bf980415687ac95192402858c9c05f6afbfb0462787e48436e648d3bdf825c61
9ef8d8c9f6534a31b5536c0861ff96af5ca09e74d690d58768edf945aa0964a3
bbb1272dcd1c1a5337455a2b9d3bc6751d899600cca5fc3edd1ba3721bf5818a

first message t

699ffe467077c1ba68cd35d82e54f986559e13d1a82c2e8d905c3b836640fc46
77b26bbeff581213fec8cf0264ac93dcb9fe3e728da7ab8b2b25a21688f4c3e
9837c24625a31564e5caf5538d84bf73eaaad274a79a83a7f799817b1fb63ae29
ac701b046ff0c462bccaa98e10eb3450f2c070a79481b94ac75843d010894015
aebcf4cfc67408bc59fa857d01f0e3be169e308b447a94f6851ed08cabb64ab5
e32dbbe88c0bc0420b6e59fd75586568fbd060c7363c7dc5bf8e37f39eb1c5d1
45956405f46820145129a8d469a818a53efff00b408192b8d0a8e40e2df7d55e
e3f6ab2ea06a6ae3e544d939c00b2ab1ee930b4b0bdd94d291c3dd422c039fdf

response z

3f72d8d03a770fdfe87091affb860f69e3439242c37f4a75d215c73f815f60eb
932a62df389a0e6757f716672dbd9eaaaea43f11f1c85afeaaec5d3a6a3ce1b
01050990d104e216eec5774555c2854ebd05c6991c5425cfd2032d22997f85
443b290541581570db7b81ff04482fb48936ba22b0ae37a8e4bc0c40dea2a425
5bbf3f7ae89867dfab8daf18b86c369ea9854cbb468e24495a9ebe835756f90d
7a18dbb83dd81cc620a1c910124a048653ebaa75d4a8811b1885dc1e660ffae2
69a4470c0678bc1c4a131806feabef89ac40b4809172fe887b645dd138abd053
bdbfafc3b404d72feac8724603d4e738e9598432dfd7f6fdf6f839682a5ffc2

response w

660485b86480be660ac109b6a138570d2a4533e5610ceddac19f608a1f4b0893
d9bd0146d7725c27460331084ecfa35f542e91e3aa6c162f07488a4afcc90a22
93ca03c0be5eae68a0f5d921b8e454fb5614d92b91ef5e27f0e3ac63af36e31
16c8d3496e85737fc59268a02c8a868e23ba0667eab6ffd3b132ded845867e62
5d8b40de0dfe64fc8ebdb00d5705b75ae99f7f1b48476ee4177d450c24ff8150
be6f2b17d0872e7aa32ce6c309baf0b84ad813564a76a3c62df1e658e9320022
f2a0ac00395f51c2dcb93ca69f1f0dd7667675609f4f9e2e744b70663e0306dd
46eb84aedd7babd8ac39dfbef8c8f330803af5af761aa2cb90f729d712d6337

Fiat-Shamir context

a5fc35ebc127c1f88a8b6d375d1dcde1bbf5976650af223248f8d8d37dc001be

withheld proof term: 15 456 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uEulerM definition, SpectralPowerSeries:44

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {ι : Type u_2} →  
      Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k) →  
        Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} k)
```

SHA-256 of the statement above

bbe4c53a46746e439dcc51e765383cb1f82978a3b4dba317d39075fede03cb68

SHA-256 of the kernel type

27f44e02f30da92fa5e14045c2d2558054161cdcaf8be3e372a03a15df06ac6d
commitment C

21eb841b68d8a807d74ca4ce4402b273571dad3c585b66d39bd943af47438f4f
93633c39e1ba3b6c2f94d3dc91ccb749914db9cca8a12a29fe405a7f2d267aa
4fa94c327856ed4a03bb5715dd28692dd4fcdecdd3f24e8bc3b6c92dbb47a3d7f
829cb4d0e7ec3ed6604e337c7931a3e12a69bceeff00ec86f65a57a12f9d396c
9aa602ba227e968979132f262fa2376a5fba9540cc6d101e8232eb57f810e29e
d770e65ded077c7051921d58abaec1d842534229e1a0da9dce39ca040d74bef8
fa985e933273f410ad4853edc8c2b30c862842b5c06fa6023d62d11eb499a518
f156959b3643b7df3749109e8a04c68ec0334dc4dd8f81e1edec26451d71824b
first message t

4fd3eee33f50fb9990744cd41004cd3649cb2bad353550996e3978fce4cc2940
e72237db782718f386e3b2fe4a4647fb8fd09ee0103d86e01a294322b008f39a
0a7d391b33e1cb4db55aedc2a0e33ebc21c05604d7a2d780ed85c3ed06e3e157
bafb0c1a445d445ba0e6b39f441b16ef62f57ab21799606cd93642464e982614
4b23d4d016bbe962edc694755e822168b68654d3d05412e73ac7b28e58d0ff48
394a5e0d7b0756d045545a1dd005a01fe6289fcff432cead2ede38a016ec0a6d
dc60dd31c6f84ede7a3230fc49916723e75ed3338ce41964c3d3a64859fd545f
19d5eb1606bbb77f675bdd30c4d42a059aebcc5838fe22184d1f171b65b35507
response z

34aab088dcb3cd4ee8d1dca4cf9e40afc61ac7b42cfcc9d232cc5c733e69beac
84e69bfd15811913d0419d2427c015dda5c7e64f5c5331dcdb6b0fdaed81c33c
b7662873c44a7b72061286f302a4c04c0744c14319b87aba8e9470ed3ba96556
51a58708bcd52fc247a4f2c5e2b2ebe42e83ac85a2be742ade19416158bd8a2
01e982dab41c3ae17974ef6645b4866b7a471821060dde5b504749fa8c828a51
b88b0b9fdcc268f63c5dc21398105ec6470528c374a97c46e079a612daac63f6
53e687fd2b7a2881f3b250c084d8d12370ec2cbaef45b21a6aa69565b0f8ec70
7d6eafc200ada09653d13413ee435ab8dcd5391f9dc27006bd6f8aa18df156b6
response w

6454688d4554d6c86794ffa4934e8dfd739c35b7ce36dca7448c64644616d0d
1996e733ec447b6906270f13907aaf75e7345d0efc28b36abc325ebe688009ff
4f165c1f9461032c04dd97464c69b772ae3c38c905f5d9e00551383e76d0387c
b9203d42a53e45e55b04e483370024297cc75759ec6bac032bc60b7dc6ab1f30
c550733aa087b6e3341136fa2b763c83d4322ff84b45794516d5df6128a19dfd
304f5e50676aca615300c1c0ad5eea7266b5bb4fa6f32597477c4d8f57942cbf
bbe1bc3e2d112ab2867190f28f42962d08b78da1de3150060a7a0a6f896f3d6f
54283c7a0c9ec5502686a7caf00efba0440f22b8cb448b85aeb92c88a4386585

Fiat-Shamir context

3d26abb3a0d749ef2975a37a049e71b8ab1ef9c703679e977006467bd50ee41b

withheld proof term: 7310 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uEulerM.eq_1 theorem, SpectralPowerSeries:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {ι : Type u_2}
  (Φ : Matrix.{u_2, u_1} ι ι (PowerSeries.{u_1} k)),
  Eq.{(max u_2 u_1) + 1} (FBundlePoint.uEulerM.{u_1, u_2} Φ)
    (Matrix.of.{u_1, u_2, u_2} fun i j =>
      PowerSeries.mk.{u_1} fun k =>
        HMul.hMul.{u_1, u_1, u_1} (↑k) ((PowerSeries.coeff.{u_1} k) (Φ i j)))
```

SHA-256 of the statement above

87be47a55e5308567f3facf7f68e29d7903d0f066cc2c6e660b06f1c50524c9b

SHA-256 of the kernel type

0d7939fa5f682429d29cea16c703ad3c3f32f6d823f2ddcded406edf6c784e56

commitment C

9a947146cd72fd49352e85acf7ad28d3d39c3cf1d88b50a76ea8f1f882cb628e
ec361637353bca5a10d4544d542b274bd00c74f73cf77aa6106f1ad8ad2eeb77
26ad70a0d9ef72ce88885e442a0769bf279346c2999a0200aaa5938604c6bd83
ebdfa1e62fe31cafd22af722a4417e521fd6a3f3cbb4094bc51e0474dbd37470
c2e0744130cdd01d17ff4ab9733c6912e28d2e1bfb3f21de14f30b0f21d58240
9bc02e80a6ba6eddbcfef110c3956f15f5c39af106d8a894cfa5df77516d260f6
9c17d2bc6d182b9f2ee8e21ac1c298eb7afc85e1769e718d8bd9e13590138a07
3d892f27281e2cab430bdfed2585cfeef57b319c99006fdca23933487db09a72
first message t

dda3251348586f4f3374573edb32fb395b13540f3780d6d95d3a07f044599b38
5900372b485ed545162474ea72f181fa0c7fb22c53e448ba076d562a1808760a
deeeaa70d706f89544cde02fab56c0df46ca195d4f84bdd6db0257428f57d7774
0f313635ed1bce4dcb0f509bab0bf6c5a68462f010612f26932c7c79cdd3717
f8ff013de3f9799f46d5dc5cb0de9ee73b12773d69f4f021e99b0b0d1bd269c7
f6aaceb52e76fe064998924b2f03c5d5e88cd027825a4a69a2a8593d3d1dfe73
07cae33adb93b7810f026beb8faaa64f4e5a7872f9b487bd8348f72e0f34de46
7692f63c2ab9997b9cd34296467d115562908923135ff3da333ed6ae440b2073
response z

557ab89c89cea2a2eac18735dd46ead969e1475e1d9510effa031dfc8198685a
d8d3e54d7bfff5b039e57366e6000a523ed37742c431ac276bb8ab96e42cbfbd2
34b18dde01173ba2923a36061f740c0509479518656e2847e4dd1ed16a802062
c940c9c8c1d8f8748d8a1a751ec9759c5a14ca567d856e5b6b654070fe9bcb1b
433bf7f91c99845201ce792111fb3543cd6d5729087968fc54bcb7c656a91bb
3813c36a1f69fd4b952fa8ca10c353a6a2da55329238e5acdab6bd279de8da62
eec43d85775d5ea058c16a5c0ca8e0eb962c769f1fc8c60adee1a2bb3052024e
64cf993e0cf64f345e2ffbf4763deddb8d21fd38f1ed63f28dd348da3df17629
response w

4565b6d3f58c904fbf60d69c967fdc7e6b7149c801f5a3992fe4396e3a27d4e7
773109c458e1dfbdf3b1cec4b22a5ca3c356677726d72b634d334d7d5a483519
c51e31fbaa7abb5de825c447ea2845b57641e348b163925f51eb2f71efbca172
b8916047168cdfc481205c78cbfa79dd0084dad8d0b177e9a6f590be9f10b737
e6215f346a258164017d071fedd24c5f9cc424e8a93ffb636c02663448a57db3
549710fe9158a53f6301eef8c7fe8cb9dfd9ca7e144c04811dcb2bba68d41d6f
c6d63f1148e0cb6bfdeb315adacf9464732627ec0f69489f5f4d9903ed063172
2b66bf9343edf65c1297f11a0c7bb0908e410e8af00affef102b491ddf6af7b

Fiat-Shamir context

f1b27f2eb20e6bd1ecc216f2c772fb99de02e30c1e1183829ce4f59fa75e1045

withheld proof term: 266 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.IsBlockDiag definition, SplitTDirections:41

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Type u_2} →
      {n : Type u_3} →
        Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n) R
          → Prop
```

SHA-256 of the statement above

814219426fe307dd3c4d4fd2a4b5e146669a9db27ea36b31f1f4efdec8b9843f

SHA-256 of the kernel type

8469b57b42d8017fce66e17705343734a6888e7769c5f769204fb900b4ecfd91

commitment C

c9a4a6a61865480cea14f6a78fb10ef8811b472947f135060d202810b06bdca6

c4921d706c6ce29c0fd69ed75d2079d555d7c37c8e4ec908f806c81b7b3f632d

0445a7ca29e5315a00eeb6a18988bea1e4b5d0f96fb5d0610249fb9ac6c331aa

15aabc4083eb4b0d80ae7f0c6893aa21e41eb46ead4e4fd17c644c2d405e8d08

5a8a79168033a853a0796c1f654a5d73ac4ee6f2c52b7d2632dd24377b46e0ab

1dc94579a914f94750591b48f9d64754ef62101f7343d1fb14399609288a1084

083d95e1e1aa2c4541efedf301f0f06136d211b8cb65a03fddc3c4c9582cdfcb

d9ab844ccd5381d29b657410f89056756c6ab3fd95cfb730db9cb5bad0dddec92

first message t

973cd0b6d4a0599673d5a23afa3a8b55d0ba8264a810a8cf96c1adb91b6a274d

0083ef0e8f2acfad802c69eef474ad5f2ae093090768805e04220d57c0508886

54441a06b6bf859a2dcbe8561ee1b1737133f72238cf18a5ea269b139f2739b3

4ce8719d3a7e0ed8647376f67982980d90adc18a80e638fb56d86f97799592be

2b982c13f95b3d63f5039d3c0e288748eb55bd511cd597cd34f0c5fce511536d

2eb0b2ceb25a7a00d6f66e5ffdb5836a3d463d5fb1fe01a10a877eaa416ef387

81f28eeef715bf3972d8fff953f3b7aaae792d0b6b9cdd45a00e212049b58f79

68823eed660036dac464bfee9773c7fb109c10eadcba4d33a983ac99f7b74a63

response z

46ceb37574dbc01bd9b5ec9d7b521b93d431ba36c0f9a0751c2f5bddd257b943

2d3b32f44260e9974d26bef138b04ae766b50273512b7db604df9e067670b191

3a084f454570db04c57c7b800b26988d14ba420995be057b9e5ebda6a19abe97

a39ea2ec52a0332cd6ac10d10d3ea1694e44fe31fe6035ae5b9ababf78bddeaf

8f03f1da57a435ef552826a79b8737dd7a49bdcbdd74ef40e5a560ac8636ea28

dd6aede3bc19877e8ed2f3a0d84db4a5f8ff721040ce8cde3490e72499344e0

f731363ad81ec520a373ac54525fce855c95a5bc5a02389e00d38a56a1e8b869

0a53beb7e0a8a2bda16a733d4a9f3a82ff48bb4fe14f5e2daf9e2d4dd995457

response w

18a33d1c08a01d01111985a2ae270a5466564ce390d6b99175b0ba5af3c68af

eccddd6713b00518b3fff033563ad5cc283dab35590408be8c69c2cbf4bc42d9

f172f773fd9b9182b76bcd4516a1785c3ca2068a5e5adfe3030799dc562689e3

ace1c7e4e2a65b92d4abd89622ca813c74610d14839bc9f63c2b911b6a0850a7

7c79b3d386869ea58545a23f2aef1078f38176e9e1d63d3b742a641f965a0841

f7ab4c93788394bcd9d9ff79c7829ee6ae852486f0894af022933e4c712f973d

c2b14415b8d856956d11650d0187dc4536c12f132d9ca40070f9f0f934fa9e9c

f50945d168cd3b1a155003df6e58cf63395820714f735f3b4776da12b011b3e5

Fiat-Shamir context

c4d6d5cead75bddd2eafee361d7b9f1157f48d5bb3edfcad8269cbbba08ac0ff

withheld proof term: 1 603 bytes • axioms: Quot.sound • kernel replay: yes

FBundlePoint.IsBlockDiag.add theorem, SplitTDirections:63

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3}
  {X Y : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n)
    R},
  FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} X →
    FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} Y →
      FBundlePoint.IsBlockDiag.{u_1, u_2, u_3}
        (HAdd.hAdd.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} X
          Y)
```

SHA-256 of the statement above

182cf5f8ec701d608a195d1acda57fad918ea6aa0713592eb4f42f0624ea2e7c

SHA-256 of the kernel type

0ab465e34a1228026bd68c5e88b824d18794dd62f1aa3ef216c85480c89fcbbd
commitment C

d875a49732f425eae7a20d1f45e97493daa89023744b819164b95f02dd75039e
070f9f286c313ce93d3afa70e13ca2aeb319aaf5220d95a7514e487696b6d422
d899da9a26f0d8979a68e92e9c8ac830f7b902e6a2df22ea6cdb9a46024a6094
43c01d32d15a1360a6d32adb0d5b2f019cdd00e16e2dc61de59de00703be67d
58ac2a1836f5d5e7bca0c2784622ec5c0f140b31432720b1c40d58966007e018
3f82225f503a908e417c9c2ca47e9b20db4406a0f1f1a728ea73a8ff4e89537c
7dd62e6b980bb5e45acd207848b04721e227b0bf5039e81cbb32ade6eb5930f5
e5f35fdc3f1334520f285c4b5f47be11cda344716c9cf542724edcfc9bb7f5f5
first message t

1e8cff9458fddd3832059b53c72b01c86cde4633279c7f54d4adeb8c97f031cc
6d0f5e49b6e5599181f1ea6d444004b733867b8295a8dc60a93dacb0b538ac95
3426c2300dfb195ef0173e6a03baf741dc1f94756f4070c0060ae02a86665121
033d9eddcbcaa549cf014d7f3edee60046cd553e2f894b592cc8e5c52af5c0a4e
c3a277c50a18d99b2da7aee656f2070eb52575a0934e16f4a53ff1bca4322721
0e388d4df246b6f5163c4112ac0dfb844696795ef88a21a835e9dd229c4a6e65
c66906480982eb2b0b02285e608b43bcfe87991d85df4d57ebf5758e2559bd13
ae9d908766068d4c9eff14df4523216f13159e31b65ad13404df8ef9877bf5e9
response z

48ec6edc5210b98e60b5399e835f649f5fc1e1375312331c188e4a21628dbed8
8b018519450b1514943e74afcc4c9348d10616af95e8eb7b4d604e1a18a069a0
e9bd6aad9f4b88b339a3d6887b94be209f8074a3380bba638e7c330ee6e01b32
0390849c4523adba5fbd0707f66c9816a320a1a42cb2885079a5ac8d539eb07e
5f1e17bcf38aaca9f5689992a6dd3995090ea464444195904a40d3649e0ef2f7
0195e13b36cff498b496f4adc1b532bbcf840d7c556b4ea3d0d3f70ca6e0bd2f
5b52952b9b291545c9157630997cdd6f97463b72bb685fc0388b47cfa28bd24d
8e7d302b44249fb5fbef0b4f3181ae6115d97763fe5ef94b39aa7098c1fbb20
response w

1c298d40054864057ca542dc870e81f13346d54e64467d48a719fb43ecbb61d2
cb8d6eae38ee7b02759a746d6b79a7aad010137de0911b31c48a5f0fb358e5e9
ef95822f5e1bd52a3ebcd8412109401b27476804c82304be085bcb4689db44b1
8d0c0c733366dfd9de40eb25df2d0cb7e771848443eb3d9f64c9f3755b63aac0
8059b4c900b7a8dcee1798c6de672d393806c35600e1b0002070d9852f866b26
14ef7ae597fec3c3a93eedf232b9c91deceac49ffaea775708fe1e17ad00c6df
7d71449471e177c1476bc02e07b9dfd4d02a42eee4d2b32980c1a6fed3e24699
091262f9e1f2d001fae013087d7f295225cb935105c008768d6fbd5fa5a8407

Fiat-Shamir context

44ca35423b20bacf1494376511be906033ebc233429ed4090ddc6b5543b55a2f

withheld proof term: 79 841 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.IsBlockDiag.mul theorem, SplitTDirections:85

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
[inst_2 : Fintype.{u_3} n]  
{X Y : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n)  
  R},  
FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} X →  
  FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} Y →  
    FBundlePoint.IsBlockDiag.{u_1, u_2, u_3}  
      (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} X  
        Y)
```

SHA-256 of the statement above

1f19be51d91e493e46efdd35c7138849a864b4613c61c035a5a632a9c5b6fde3

SHA-256 of the kernel type

6168412618e57dfe48990874a752cb1d5d1b1156e296e74334c95b9c51f474ca

commitment C

ac01995f5974bcfa50ff8965973e703a815e5880df08e042a93c735e280247f9
87e281caefd5f66761642a8e80b838015886d8d32691ecda41996a04c725defe
8939fe801735ba0af1576efb9379816afc863de3a21c269cb656c5edcc98a784
bd8335fd2d840c50748254b7377fb2c76031ccea556619c5fcaf02ff3da0f3b3
fc936ea9b78af0b5c705916282f795c69f7e3627a02cc9c5c1a6671eebc8120a
1b5c14fa7f7bcb369620c6d62db1a855badd1b515b6c5a16bfda901be794c262
c06bcf5e71b1311ef1bdb3e1f89c98d657518dab5918f99629e54df600538b8b
4fefb44577776a2883b47bc2e7500de22fc569d439c5e8d64b957efe250a066
first message t

731f6cacbc514498ea14766646d27afa593dbc37b5c50f576be65ad9e673dd11
abaf4bc8ad4ff0c638b5345dad6c22298525bc2d491f94f5f22c40fdb6f24ed4
753bdcbe5e5c40e66df21c695bb99188f759e62db3abf331d1f65306a1ae0c183
99f396c82aff406688eb149a369650321469554379c379003ae486ca13960a59
a3ac4a3309ca6fd1fd85f3f9492b8ce61a565952e2d8651cf68b28f8d32bacd
ff85f17840feba0029434ec8e20521c587f9448042b0a1d3a3b14940750c4de4
4875b70c3035b6ac5c53c540aa1cf72eaa9cf7732a1c87629f123a74704a747b
6f83861f21079a91e7dfb1b046d3a07a71833ffc0939ec33a92e7cd22aad4ed
response z

bc8ccd3b168befc1ad75e7566492d4a66b6d3d0279b174177077098195e64022
0243adb0bb14a21ba69ab428751a46fef7326eeb7b5a6c042be6996f9072a07
6da6caeb2e4192e920270c874baf4feea7aa836292d17dd58b27b24304f29e7a
573096310af905af4a07c36542ef0652705bc0cdd0a6f86a89c5179a4e2cc9bd
20a0528eb7139ff64b678f8eef37ee90fcb356d821e1e81f535bb8ce586096
7f92f09f7b6f2c5ba14a6dadfe8647eb8230b240acdc298168572258620d2e5e
3e6868581bb56273c0338e9217fcce0983543272a66a7e2ba76fbf5655fe9c60
6e5b48443d9d6953821051920522841fbad5735c4db73b4cd8b25714b2d9531

response w

b4cad137f5ed1240771ff1e8470bfe2016a8bcfd23e21532989a109a2e6adc77
9f7607679b1a84f60496db55ffc3e712b4c0875f0f9ba0e569cc18e26873ae01
94bd8dc7067bce46003fb6a1a27197ecb26f0bbb2fb1917d5c70bf80ef194a0c
60cedfa8d7452dff211d4be9a94bedb4f6f7d589d46af24cf4ee843669f9b00d
01022cb95192162cfd4a55f4f99bb4c4b67e31e36625a9e90eafa699cf3063e5
71ae4d2c6c73cb531926e1dcf4a795c6a0d7e9ac5db9ce9177e4217dab3d1145
619445decddcc20bfe7c81c21a4d707a31d939deb3f1e6d4adefc76d15f0d289
71c0da096e98e63c7b2cbbcae826a47f9ae36ac4b251a9b0596acc08cdea4a7

Fiat-Shamir context

85a557a1710a671ef8a22bc8cc154f2369cb422da48b1eb0a883ce735d72b317

withheld proof term: 196 085 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.IsBlockDiag.neg theorem, SplitTDirections:71

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Type } u_2\} \{n : \text{Type } u_3\}$
 $\{X : \text{Matrix}\{ \max u_3 u_2, \max u_3 u_2, u_1\} (\text{Sum}\{u_2, u_3\} m n) (\text{Sum}\{u_2, u_3\} m n) R\},$
 $\text{FBundlePoint.IsBlockDiag}\{u_1, u_2, u_3\} X \rightarrow$
 $\text{FBundlePoint.IsBlockDiag}\{u_1, u_2, u_3\} (\text{Neg.neg}\{ \max (\max u_1 u_2) u_3\} X)$

SHA-256 of the statement above

7e198aad926be2ad4477518566d3c98df25b0b47e7820766aa1224d5b1f9227b

SHA-256 of the kernel type

10048aa88ce4a146f10eebe658deea5ebaa8c25f347fffa6bc6c3eb38feb5a4e
commitment C

abf354a802fe8b9ee4ad40f8e5224778d890a1d9bfe4f95a2204b0e9ad254b33
15a86ee3452ed31640ce53cd0ff14a9e8c58bd36d0b88028f3a5b6873c4a83b6
d5921a02e5dbd1c20217074f68ca8aac7550422e1510c6621105080a0296c56f
4f35b04373691f36d41d412a7c96c96ad38fffe930df769d9f0ae26588c8f618
df40438cb85e202d38e9f546ff8077d1b629a5f7a909ff6a71be0ad33e6f2301
24f9d8b913c34aa76e05c860692a8dfa79635f54fda52831290e73e9a962981e
30e56ff7d7c26bfa2337c2e7d434c75325d715fb1859ea59a5c534931da0d9e0
bef31e473aae3edaa37de036c2cdae0d0be435dbf8c3f1bade31f7d66c075c8
first message t

26e566662794c7225f7d3b941825a8370641ddb4045e826d2200632e3b0032e7
3880815aeb8f47ea49633a1473e8ebe6204999610e86a1c47627ac0bdd61c04b
ddc28b551671c28a0a37a23f5c39c6ddb17db5c51de72ee52506aad239f74d7
2dad5f62504c483e124e6122018eb583567a938b326bf2595189f5aa7c7c41e9
19d1b250ea62e0f99a41e9a035b756c39f1497dba5c24143edafb2d92aac1165
e5eb38a66625e13ae5c404dd32b5672097c8c692a891653f7164302dbc4a9870
1308a36d52e1b0f5e75a369ca22e70ba7e32b82f7fac14fceecd82c3c566ab66
2c333208bd32554d4a0e83f936109c67fd4e777ca051167b52b52cc2c2713b50
response z

12e1768c5af08c186ff4c7b6f829f45cdc3ef1c1d1eade57ea40e2cf92839692
d26a611e58c63a7cb14c72b19f9c04d731c7f06a909ea728e00ae13bcee5c58d
879435fbc35738a9501635c1d8f672d096c34f0336e9a7cc855884c94fb6b8ce
11e06366fa743f450fc906a7807cdf50d683300c66ec1f9aa22ef8e10125c5dc
149e77ac9f50273ddd4446551ba19e3ffd0e5052acedd1cdcc21c7e16e241060
faee7fff402122e164ab610857a030ed5b7e2beb6d6b450642834ca3c4f07697
5162c076cf386a7aaa88f19091ddde6940f5f1c3fa5a3e62b637dbd78e95734b
870cf31dd74184f9632f54b68e6a6e3adfc2f79c288169cda7f816c7a7715a55
response w

1a756ff6e872ce222b5f1fc3cbda515ac52a40681e0709cd35066c80ab7667ec
413bb9cfb543c73058cbe10015d55a5e8d98c766e37721e70b733278283f8188
15edc1cfc661b3ad42e4a59c6a367cabcf321a79134b6a7b6a4a239e47c573df
bb8b411c06128e3ee0a72d4e379de737b3ce502e1726e440e45c651d329aee05
20096c5e14a32c69d8f2eecdcb2f4be3abecd1222ea5ac85f9ffcb0e9260cdc4
0d27326817419162c5de765e8ca2d20cf18858a86dad65750e5e985aa2c4588d
b218d3faff31afe9be3b9ff44ece087357265cd4889df5add4df8aac92d8c27f
1561e28058f9726da364b957e4e45bb58bd5aad2bfd0675c2e2018a45d84bacf

Fiat-Shamir context

dd189211d4b3951c3898c90f211ce3ed0336e44837a4f2c8ecbd336a423c58dc

withheld proof term: 56 850 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.IsBlockDiag.sub theorem, SplitTDirections:78

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3}
  {X Y : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n)
    R},
  FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} X →
    FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} Y →
      FBundlePoint.IsBlockDiag.{u_1, u_2, u_3}
        (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} X
          Y)
```

SHA-256 of the statement above

b4d54455c2206c91c2cc4a7b0194fef059485f0f924927f3f0566a2d60e10a51

SHA-256 of the kernel type

6e5a131ffcb3825cb45c41584f02986c23ec25c851b15d029098659dadf9ee05
commitment C

dcf6e8bb5241449e4a692f0a7fdafa5636a97ed1e7fca226feb625fde052cc7
ca41540069ee5f9b88d88686a970f4392bffc1c28a6c75c33576792d74dac036
f79ef546dc07f8b8dd00d7541f90152d3c3840c524966bd07cdefd9e6f8f536e
d53a2717f314899e2066ba617d1808d7ea278bcaae799b3169c4340b7db31c21
a773bbd6090a54e44c7613f3b73bd156ac868c8f39820e0a2195f8e3975cb884
df5932f998d54622ed2be241bcd15fed0e3e14dd178c5a4980e68abda0ed59a
1d8a26d83f68c7edffddcebf0b805e99b6dfc53dcf5e68ff9ac2c283160bcbbe
a88586360da6cbee1fb1df4e71d5adbc6fbcbfd552d71fca61bfed77d4b99c71

first message t

5e0b090b63541133fc80ab54479848aadac008cbc1a3d0650d66e63213ed7fef
2813836da7b21d150fef69aed42c0a7c6be481d10da9609e22db96071719fe2a
375f81806acbedad4f24c762f831909b9f14c741f8c75539b3da3f0d19c78009
144f49b429e84b492c86e943da0a58d428bb0d3b113d7c48821a89b73bcd784
948245c9f9bd86fe1f3b172a57652cd9a0b95346f3e2bc52227b146f18560474
ca12865a61f2da4071c4e984d46d4ebbaaf8cb42d970c1afa460f42f1e5e3e7
aee25017150887424ab58beb94bc6903dddec03182b0582e2b52e981c9569fb31
cd627daf8d1367fa7590e92c73f8f686b4dbcad88c6c23270c3b8e95140358eb

response z

5d0a4e541d3eea5380e39c6ee857d74d71259581d453a6928b0920e415569adf
bd5ae78e17fcb8e43d1c41ecb64bb87b940dc1abcbcb8e2e2a97eb19032a391
cb6101e67de3c318e4ac3b9c1d8c6ce6dbc95a502255a449c1dcaf382d32b411
5a387348ca55a17a3e652392daa75bd35c887af1f0dec6d80dc9b72707e1faf4
61b87cfa51fccaf46dcd0a6e991d615d312b19199c2b880d582f506d8140ae0e
eea71ba6d83594563b8d4671547dab3df94d150d606b79face7caea09798eae9
a7f88b89a349f08f5b8065564be0ad92bb7ebfe52ac632c41de68b452d30d4ce
3ea5596df359c4cc8f5decfe5ebd3184f12ae63961a991ee486e7b4f6a4b8bcd

response w

2577920fcf2c57e1b935c038b762d90004ceef23fd8b290969c7a51c5dae1944
ea50ebe9f576cbcc272ffe23699c42b0122619f64d3398cef836ded76752e37b
5e87985a8607483b7e4a81b1440968bd8d9c7ab619e1ac24d8f1611d5c897128
476c186c898c9d0aa71704bd73f70b9441603aa39953225686332a3523743a89
efb4f5f6330264ba278b2cd366a5ecf295ad1ab32d392e6d3adc1fed2006c621
b1ed49d89680641e4c36a155ab163701cc30c3dbd74761aeed2aef46b21c76da
3fd35d79cfadf70368e0420e91319395def6fba9f6a090f6b894e46ac119edd8
ae9c6bfac53339993ba376740ba2c7884454d6596baaeecf96e6f3208c15043f

Fiat-Shamir context

1dec52fc2a88061622633827320c74c5169e69b86ef25c684fb664799d191586

withheld proof term: 11401 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.IsBlockDiag.sum theorem, SplitTDirections:94

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} {ι : Type u_4}
  {s : Finset.{u_4} ι}
  {X : ι → Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n)
    R},
  (∀ (i : ι), Membership.mem.{u_4, u_4} s i → FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} (X
    i)) →
    FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} (∑ i ∈ s, X i)
```

SHA-256 of the statement above

2bdc27882cfe87a87347464cbd736a483e80553de9a106664d71423d698fe0f6

SHA-256 of the kernel type

25d3b26a447979c291327fa8233b5a1e3aa7eb63bdb889fba1cc491734464116

commitment C

d1570da8ffa795fbd188d4dd9b098948f81ab8ba7a50f82588e98990de33f8fd
0a399efe001fa0304a93773e35dac88dc709e34519af64a021cfab12f59bc0a1
d8504b290952263d3cfbdc46ec03d71d6454ebccb1822e062fb002c20cc54357
3dbde837dde4c9ae1119231545bd348ad40d04855ed64d34b284001cbfd371e2
9b1ca449ea3608620334bedf9e3b49dc8dc253a47ddda40ef60d49b53b53cfa
90ea9876924d5e56c373dd3fc52adb36d2fb8f181da114cdd387567cbcd86855
d99889e03ee54d1cee2175f9a660abf94f4898f9326e8ad665d3f0fc57791b94
84ef4e72af1aa5d9ee76dea9f3f2ff0b083478dd1eb051f99d573d4a12a407c4
first message t

2f38cd6297a44f516631154ecd34beb5aa28bb6fb3dc57d2652ae70501e3cf0c
42f12c3df907b933610f5515c65a8e60f492da5e1878c5096afddcefb20a9f8
1da5119df8c463c1bba74f619bd945e0ff07de46270acc83af525ea1c78153a
71cd58d8afd406a616287a5131984e6e03b8c0afb476c45582668c636f60ba1a
bf3a4dddc529736461dcaf92af3a90a815fbc7b0adfae31e83a6e4a6aaa2cd1f
f09a4eba33a3f5a6ef93c7f796353add123316b2ebaf6b5231de9c01f0c4b0d1
194b070b9f4da535dc3f54b19dd220ebaa97ef1ee79e9cce1505948af9eb3bb0
5d8a25ea2143a69a07ccf34d9d2efcc39fdf0943769f13ccb2805bd759e494d
response z

3107921e504dd3e53cc17078133efb69e25c3a813781849ae785a183f457c9b0
1e0d1678164ec71a9fd2072739fb28f56db6190bff7039ee2cae0bb04abeb50
d80a7e9a3ad472e79c768cd9144d74f4a1726ee9c47742206d27e0726f84076f
3ff01bb3f574e3ef27299d4f831fc2bbad3b125b3962e6e5a9ac916415c81e2f
1fd99731fb97a37cc1d6a0d8b7f98ba8be044ce8d6892b88254ab4d15369346b
7e23e8b65ddde42b3ad60ca200617f73b2e7b871ca4f6e6643db898f0b3956f2
b52b29d70a2252e3c9df16138c258e5f3c415961a4c1a8342ab541b94a1fbb7f
72f14684c9eb6921e13f4bcd1d0d64d2564934f6b8462a0b781eecb82fb7a463
response w

35e1e140e24e1ab3d9b0db5863fd26bdfdf469c036ceebcd60ae3714db159465f
5cd77a0de1419ca2b70c08a43173305a757025f8f8f402d0bb21c7d2c921c8f4
0ae35cd2e4d11a44d6252c8f28fa5470c937ef0859fcf761cf68d9bfd3fe1401
81737dcb361b3f8e63698272607d1b0b32245aa82d571c201c62e0eff2803ef3
1d7494ae7788b61e880ea6db69264f0b4ac35fa71a0c9e3aa73dcc1ae3d5d154
0efecda34b122686db39e31517dd71a51d0afdd6b7021488fe015975ad04f3b1
4c38110194d7f31137120670e7845b3f365cc01fd95a03e5c4fffe862e17c9ae
6abb5015032797706d7edf45bf6a2a6107fb9669766a50e264f2450140a7b241

Fiat-Shamir context

ecbab8bbe96a7e82fd4c623582f678360444a322388b396ab953de84229ba68c

withheld proof term: 15161 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_sub_commutator_isBlockDiag theorem, SplitTDirections:154

```
∀ {k : Type u_4} [inst : Field.{u_4} k] {m : Type u_5} {n : Type u_6} [inst_1 :
  Fintype.{u_5} m]
  [inst_2 : Fintype.{u_6} n] [inst_3 : DecidableEq.{u_5 + 1} m] [inst_4 : DecidableEq.{u_6
    + 1} n]
  {A : Matrix.{u_5, u_5, u_4} m m k} {B : Matrix.{u_6, u_6, u_4} n n k},
  IsCoprime.{u_4} (Matrix.charpoly.{u_4, u_5} A) (Matrix.charpoly.{u_4, u_6} B) →
  ∀ (V : Matrix.{max u_6 u_5, max u_6 u_5, u_4} (Sum.{u_5, u_6} m n) (Sum.{u_5, u_6} m n)
    k),
  ∃ T,
    FBundlePoint.IsBlockDiag.{u_4, u_5, u_6}
      (HSub.hSub.{max (max u_4 u_5) u_6, max (max u_4 u_5) u_6, max (max u_4 u_5) u_6} V
        (HSub.hSub.{max (max u_4 u_5) u_6, max (max u_4 u_5) u_6, max (max u_4 u_5) u_6}
          (HMul.hMul.{max (max u_4 u_5) u_6, max (max u_4 u_5) u_6, max (max u_4 u_5)
            u_6} T
            (FBundlePoint.blockDiag2.{u_4, u_5, u_6} A B))
          (HMul.hMul.{max (max u_4 u_5) u_6, max (max u_4 u_5) u_6, max (max u_4 u_5)
            u_6}
            (FBundlePoint.blockDiag2.{u_4, u_5, u_6} A B) T)))
```

SHA-256 of the statement above

c27fe57da794eca94321cb1e0ad6a38c6341ce6e3195ae3abe361f8aac007e23

SHA-256 of the kernel type

201917d97136035db21cfd9ed2b7c04055f9d83ce17f64eca172657237c6d8e9

commitment C

1bfbbaae78309f4c1610f4e17c6b20563e89a90a083bfe13976ee492ffbcfe3b
f57416a9ff4e16c9f6798c4f2389157187400bf246a45f776ad64a5fc53c226a
7a6ef26a1f48446944c8c11e1c2e6732ad202f1f12e5bc4ec997f44328cc869d
4e32cffe846008965f8bdb7c556eb03cf3b91b5ea9e9f631e26ad4013efa3e1
ef3093dfe522a583015511d4d836e2b5f512a99755121b9541ef048ce0ee325e
dcf02bbdeda95672f9b821d1a54e8d4061cfd622afb2f4a4e6222cd19b5ebe2
1139852d56c00a9d3064c3f242dc6181b29e2388536a613abefed56ab2e15505
e0b217544cd58b773849ffeb36a703e8b1864d4656639f93a372eb4d63000f1d

first message t

6420ba0585d5efdc7f3066920bd82f23230483008c619b7f44851d32231096d
fb1b787a6eacafbc87c5aa593d6e2932e0bf14dbb39de5790cd96a044bc3007c
971dc8130c781f73374a617813385aa706e9c79a1fb650c0343c472f01203063
9e46ec6e4b6dbfc2bc2b74e394bd168ce884aa465abebf9d24e6140651f06ae
991bca875d42a17ab1a3f7aed2a81c05984fe55b0b3151da7d69a30b8a1ecfff
b1ead0f804619afb0475a2315c78e1444223e49443c27954ea8d7fa382f89e5f
19732e40e5e4fc85760e12f88442b38821d50d4c8e838b29a2daec1bd54794ee
f2cd3dfa7c305ecf9758f0bc4ac07aa07341a4b3fb8d472224733ab5e8ed744

response z

2f002273aa5b3ce37c6172f7c3d7c6a97e3c5b73644bc2b8ff4ca24e087abb3d
1fad98c0c644541f82fc1febd22df18415f30b40d1fa555e4b83e2b0574a80c7
c353370a00cd18df9a660cef5f80a83114030a34e14f9424b1c62c9e738c58
5ab32aa20408d25ddd2b0cc0d970bc773e8f2dd88c92e461fe1011745f95f9d
e3de41266fef4f7bfbe6ba2dedf49d8be0a164f365cc94a277aac4af3918836
5ff189c53bd5e76342cb0db4edfe716d728c089a59379914d576cbb961dee5e9
99738f5f1b15283dee3f0d1c819208daa890a3a861732219f56bb396d873ac51
e435d2366a52f20d496f76dcd791fe8f6d06c4cbda331de341b86ff1c42bb37f

response w

7d6276cfa9ee9ea620c94aaa8b029cab1990df9d88347cc3d177baf7a9132c80
61dd11784cad0a8f07bbc185c70d8fb55fccc0971553ff7152bb2f097d0ce88e
a1ce8ccc0a87341ec9b033a4f6bc697899ca2c656fece018efcdfe3d8858df8
9d9808ef0817137f947895933e377ee0f314d565dab2abf4df2f2e67482d5173
5e5430f0f9af277d0e77fc2d56596f967d5f4210cffd485fb4185d417ac60730
4a1611dd18e7de61dfb013f4d221580ebe1b53e6f67fa86315d854ba0036444e
4e1d700df61f08a4e3b5bfc4776b9eb1d5c4a0d4acb58749cb3615ce5a139b17
74a79380338a535ef784d7bd9e7dc370768c5efdcfbfe589f68dfd9bccf14f2b

Fiat-Shamir context

c762f0b41e67260b494827c0f26d083672e4e955ee46b4e4fe78dc980653790d

withheld proof term: 1677535 bytes • axioms: Classical.586ce, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiag_iff theorem, SplitTDirections:46

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Type } u_2\} \{n : \text{Type } u_3\}$
 $\{X : \text{Matrix}\{ \max u_3 u_2, \max u_3 u_2, u_1 \} (\text{Sum}\{u_2, u_3\} m n) (\text{Sum}\{u_2, u_3\} m n) R\},$
Iff (FBundlePoint.IsBlockDiag. $\{u_1, u_2, u_3\}$ X)
($\exists A B,$
Eq. $\{ \max (\max (u_1 + 1) (u_2 + 1)) (u_3 + 1) \} X$ (FBundlePoint.blockDiag2. $\{u_1, u_2,$
u_3 $\} A B$))

SHA-256 of the statement above

0fd16ec91298b2c2428655d4cdaa8612fb98e2a711bea748b508d24dacb88f88

SHA-256 of the kernel type

e8527cec0495048a2b1ae966cf0c4051594e3a2817348b13b9dff277e8886094

commitment C

d663e783759d2e556b2e1e732c317fb753b7e5fe9551daa11f70d0bae94f37c1
e228db8b92d0b16dd59da71ad8bf7d9ba2c67604d03a8f03bb503cddbcae0197
00987a44496d22a89e14111a60107c83f407c89001d28c97291f2179c72cd708
51c9073972b5e69858c42764fa0038a62deae52cdf684e5192e66c1e1e4cf39b
a88bf1bc34a2162f6a7e5583d31b70afeaa2257003879dec3cf536d4138a30d1
abf784d8e24d63f86f16e2aeb0dc2222f1837619c4135700723adbcd8b3a726
995e58b13412b896f16c2abb6f7947d11dfcf898da43198fca70dfd517213a7
3e75a3448818e198c6337056e7b546e09855045293b99667b934552d3870ed51

first message t

ef6af8f02a88ebb93637ec4d88041f15c4183908b8cbad0e22d38f4b54f4b41d
91be4b02787397a79940ac9f4108a6a5c7f979112458a0a7884e5a3fc7be5fa4
ea1913dcba3b51aaf008a750ed1dd646aee98aff3ba433cb8821c55835f49ae8
8f67820466acb352bb5984ff48afe595309a8719da101e155eb73321fd2a3c
08e5ca3c82ae91fd6cfd19ac6469ba7ee01949f1d8970ab9e5765a83e0e44b9
61f7be4f39570c5d3de737a3ad494201fecb08f22ef928bb2032fd3e446ab08e
6a5e348f97d9cb22c3c3e49b0dc21b0ec7b1a0a4045b544b0db4e776e8062142
b0ce7e25e37f4833c6446996fb652c5b1e8ff9e04ee84b5b82912702177a4ab4

response z

4c443fae1db80ad3bc1067b8b97e915317eade303a2e5d60bfcf1a7911d7e017
d6537c8fe244cb9ad0de23fb9aff0b7b369b657f3810926177ba0bb5078bc002
22441e3890feb34ef814069df10e500d307e9913d1288004fadf4cccd5c98186
92e60d62f862869e663eafae40520eb0318c052b113db0bf75c1595584eab67
d09f55e24935fe58312685444bb5dc04bd5ad57cfa82e75829bdfec2ad6f24d
be3c17f892c7069103864464743791bc210761de332d22678828b169ff2ec9ae
7d420ed8aefbf1ffd7ecc898ccc504e15a97f89a167b16670ad53029d46ffaaa
2babcf06fbb7aa7f039554e0190827aa4f8bac1ecc2a4f8a7e4424e08ffb683b

response w

4c91a35983ee0718878934cfbf6903315d6ffffab12e4e7a007d59715a19ce562
4d7c37f0326df4c4f0b5e90b605c3e4ed3fe55410c31a0495248dbfab5455644
e8e042ac5564664980e9ac1c851f9528ecd725221c91311ae3b4abab0ef642c8
9387a047beec729e27fce9074770a59480d323430d3be41af8d077a0629363d5
9deaba9d3f2ab66a3b1c8b2150dac4c7143956f93f15ff5ed62fe1c9cc7b0cc5
1a8908779ebbfba86f3fdd5195f3f666c5b56204f9975faed29e5ff6a1df2e3b
42be9d4312c9eb9f9e3b5fbd180bef9e6a4a592dbe69c14b2ec4c85234e52ccf
4e738bee3f68316aef876963ca218a7b35c2b065205d38c5995700f0fe397cb3

Fiat-Shamir context

9c727d003bd70886c3bb87821f856f084e7d6e2f9e074e6b9f376a70efdab8bd

withheld proof term: 36 202 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiag_of_commutator theorem, SplitTDirections:105

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :
  Fintype.{u_2} m]
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3
    + 1} n]
  {A : Matrix.{u_2, u_2, u_1} m m R} {B : Matrix.{u_3, u_3, u_1} n n R},
  IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B) →
  ∀ (X : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n)
    R),
  FBundlePoint.IsBlockDiag.{u_1, u_2, u_3}
    (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
      (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2)
        u_3} X
        (FBundlePoint.blockDiag₂.{u_1, u_2, u_3} A B))
      (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
        (FBundlePoint.blockDiag₂.{u_1, u_2, u_3} A B) X)) →
  FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} X
```

SHA-256 of the statement above

ff5a554e795a70da0ec43d2706bde9fdb9da70ed33a911154cb3f211402709f

SHA-256 of the kernel type

aa04ce116a41ee8604228d2394119b7d5758e8d55f0acb5bb3772372a612986f
commitment C

4c126d00d0c4d0640a7a7ee4318acf9fece90adf58a8416843441f934dcc970
bda81875866489895e16ec844bab20d99ef978d2b94ca60b42287f8c584f89f3
adf1bfd68c55957dc4cc8af6c91c9b59c28b28db98f5f1111bea28580b052a7c
25c2eed30325cc6f65468732e064dfae83d39151a060ee8f57eed9913e6fbcf1
211943924fec1d0d82abc355f8a7eab63b7cc7f2373a6818069513d35a676c58
e6142f65f31d74b67fc4d40b5c69b4b94c1a3807c0d7e0fe57410e2380497b1c
a5d74e10aeeab6f16a8988b6f548057f1ec01fd42bfb92ad01d09866f413c98a
18fe58db472482c891d43e577ed9eac5d0752c64c5584b0c7037e2d621478137
first message t

382b1ee2ae01a7780db72d5849080eaa140364c2d7c81a1ac25398f58a101f3c
df88f8c7b0adb77f67dd6877c0739352f7b493f6050a62965396044e4cbcd6fd
add347d49f1a30d5bdb763cd8c3dccb9d4c320d5e93b6f15167c6e22a8736cca
2e42d278571b50a51f8a3099a41e49179e744e49698a605292120090ddaf94b8
19b55140575867e667fa09f65b23751588fdf1ea69dd4695b0ff92062338c9d8
06e4289d85a7e9f41308db58a973ff412af0bae11b5137050b342b486fe8997
a5cd4901cb17f9e47a72a25b147235bf218078149cc468f2fd4c56a014dcc3c6
ea46e2b47b11189b5e5b5849f2a192cab66eed1fd65e12d0ea3b0d3c0c7949a

response z

8f3db1e0aa57a07c5c06ca280ecbf528827099645fc114a8be3ad7fc5a02273b
15f957c1deb8073990bf15bf737cc7e4d671f53b0c92f46b59691559a2e8a14b
401cc45eea3922767976ce83ae75c06c2d41194f26c6a489e48ab28395282e28
503241657f0410b34cf8e4ac91c3d064eb872189f414fb333a9452740890b51d
c283619c1cde6fb20fb011eb800fa821b451834b0263517cf3dd34d3ae5ee94c
455807e83519f010146dd3ce84a0fe01d3a1ae720784455b574139471b24996a
cf468211aef31a5f53ac7715f41839ddc6bc1eabdc678b9ce59c82954616e3d3
d62c14ef98c50d8d15fbd8c04fab6e011b904efbf8751b31d7c240b86a282f2

response w

6f19c7de84596c7aea98d0f0271d75fece01b0d3ef6c5faa7b223c4078edd96a
f2b25e2fbf352f1e19794913a4f32469a9f3774649ad530a02536cc38b3019ee
8b0740166777747caec126bcd123d57780aac8c689579e45ddd3ee14a964bc03
41a929753d5abbff491bc796c01316699e22ec0b1fcf3c4720b91a580ed3ad5f
0ab119b3898dd8d9a2f54b9d56ecf3f40f7dac208dd73b60b6d9bdd76c3ce474
8edf4da5efaf6a945eb19714043e420a808833ab95afcaaf0b710ebfa28cccec
3c22a5160effcfefb03212d0fa15d6fe549a1d7d57a553a4634063463a140ba33
8c25b87e9e6094b0b7b2925d298e687f9203c9a8ea7d746635d4da4feb50367

Fiat-Shamir context

ae51acbaa8f5e16f520840bc143146fc4fdae821516f17d7f9a7d5619422ed9

withheld proof term: 208134 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isBlockDiag_of_flatness theorem, SplitTDirections:116

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :
  Fintype.{u_2} m]
[inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3
  + 1} n]
{A : Matrix.{u_2, u_2, u_1} m m R} {B : Matrix.{u_3, u_3, u_1} n n R},
IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B) →
  ∀
  (U T S :
    Nat → Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m
      n) R),
Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)} (U 0)
  (FBundlePoint.blockDiag₂.{u_1, u_2, u_3} A B) →
  (∀ (l : Nat), FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} (U l)) →
  (∀ (k : Nat), FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} (S k)) →
  (∀ (k : Nat),
    Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
      (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1
        u_2) u_3}
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1
          u_2) u_3}
          (T k) (U 0)))
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1
          u_2) u_3}
          (U 0) (T k)))
        (HAdd.hAdd.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1
          u_2) u_3}
          (S k)
          (∑ j ∈ Finset.range k,
            HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3,
              max (max u_1 u_2) u_3}
            (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3,
              max (max u_1 u_2) u_3}
              (T j) (U (HSub.hSub.{0, 0, 0} k j))))
            (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3,
              max (max u_1 u_2) u_3}
              (U (HSub.hSub.{0, 0, 0} k j)) (T j)))))) →
    ∀ (k : Nat), FBundlePoint.IsBlockDiag.{u_1, u_2, u_3} (T k)
```

SHA-256 of the statement above

b1909e23d3e56e9592329fdc465a98065837cc5eb6d5183a03d56b5d7654a360

SHA-256 of the kernel type

48d40c3ac3f977513925259a8461b311bb85d9f08bd243880c66d5e52167a329

commitment C

d08c60c67bb467797c29be823f986030ad1e1ae2c7258db9dbc1d5ec982cac87
4232a09b35a774f9062e87d02f8a44d457b19731f39b953299dc62378ed4e0bc
a3aecfd120d1a4a48647a89bdb8d48e4fc0770f7356bfba890ff736cec98a720
b8be708f3506ea7b5afb8048aa0216595afc3ae2f28c901a4fefcd926a89624
3be931cd29b1e1f683df726efc8967b08386041bedd35e5cd7e88874c96b6fab
c8d3630d954f46e600e4ac543b42e6d39bb53e271c6cbcdad137d3b9efd57b51
548a2d8540a64b0294bc51c8fd46807f4fa950ac7ed78a20dde78b719269f3ea
c7fd2383545340c47002c69eaaaa583f1e0cabbb50576b3e503866c689d4d387
first message t

28c5f207920c771ab916a18dba97fa1c71dc4cb20812935c4c9bcc4cb0da3d6b
9a78f0fb97c4eb2593dc5b80131f532c8d6da1dca811f737ddb5fbda8a1eaba2
6b1bf10dd0689bea113faa53b283849cddf7002e13dd653b4c605395a9512f84
4ca32f05d42ded127edbd94f20e674ea91a8e2fb1239dc350f18a0c930b4cf87
68dd63640f41ff520a700da31195c5a5db28df7da23d6a51a99ce8467011735e
c74591619da138b29a13db7262d77f4993ec2b718d95707f73c224c70d74a391
e1caf32dadd0a2df25a1ac927daf2b3ba7350883501653e0b2434d07cde710ab
ac38b106039bebcab3d67386c6be81efe794a63e8ae78758d33316608bb10ca1
response z

1dc3fd0efe0866f64a03d0655fbeddb9c24ac819005761440f01f1411c6586c89

2d3fa8668476eeb54d5fd799a93b113950dc57b003f3c58ba5ae575de6a26bf4

2cbcd9a99e9fce6b9269d0b9e228baf93e63697e8a540ec15b4f855330df7eb5

FBundlePoint.isBlockDiag_zero theorem, SplitTDirections:58

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Type } u_2\} \{n : \text{Type } u_3\},$
FBundlePoint.IsBlockDiag. $\{u_1, u_2, u_3\} 0$

SHA-256 of the statement above

3f06a6e3d9cb7f5b6160f4f22256e90f435b5254fb1adfd10e5de9b28479677b

SHA-256 of the kernel type

04328803f05bf23efe6fc6a3028f1d71ef41fe3117b5ad82a5cd04151791fd49

commitment C

caffc5d5e85ce981d7e937ecee4663093e181511b7c0eefaecf6fac30e12c22

300b279a373d9e8f85c1f65450b2f1be4b9900f2c2e9abe89ee02512ab2df1ea

83746d97b3b492b3438c0bc14adf84089a67e1e3d01d91191126c8ab1cf9e0b2

4ecfe4339cdf52d5c477d5f71d57831d5933b31df17034487faff26852ab3db7

d8866a735d4a815084131ee4077b1820dfc55538e15498045e66ce307f976004

f83f0f654aa06c60db0b50935a3c61e70fc85db161e25ea8f83ffc793b4702a2

d0157e3b8874554bb83231ad71373a6ca159e551c7e0b9addab65ad074d06d78

b54b1904a5f45b556429ec056a15f2f37406bf1c8514e946bb61d6927368353e

first message t

84bc282c783bb9e2fbb0fa236cdf7d3bc3c357ee85cf8ce31b2e5972f68e829

92ae3c1d365b42d411ea9bdb9181ea485550a5d9d065d087c4554564c99c2e48

b3813b1aa5298d27a2da4b04f1fc41544d220e702eab3e41f78978be26017f7f

f59fd5deffbd9d5e2c2da9df1ead7c02d5b4259b7e2ecd1cf12cc00729c57cbf

e7f99d5afec5dc319d29edb8d372d09bac31b3bb30c541b9765f331598b95597

690dcabffd91cd85c06f7b6da7ad91dd3da0af8e0c34799a250fffe6098b4b5b

a1fcdeea62e215fe0fa945a45b9b9e99bf6216c1ca44d6e061fe01994e463dc1

718ce13e50dbe9034c8f980281729858c72ed77fb0c1f343526135f29c9556eb

response z

2d5af4943861fd62662609b11971a205d321b32b8ad4d69701a689aa010fb7f4

121fab7eb18bbd144f1ca73c57722f441379e4f0f0e09c1855465c60b61e68ea

63d41484896098ef352515540178f945b354b93eb59bac8040df2fe9024e6dc0

28a1c846bf57c73b66522216b2fb34d73a81f8f023045918eb0312818a15067d

5fc0592ffcf8c83620d69d2556180a75add918eec9e6bbeb1ddeef713d156997

1d1d1bd89734ee50ca6ac87c964d0a436e02972b712d0f17c3fd8bde6e20d513

0f38d2e3608a1262e1adb28d02347f70ef8b98ffcac5f95b0113713cfc9c0332

278da09f36cfa3c1e42e0e1a15aad3b161b9837e6511d04a7c3d665b7bad60b1

response w

22ec45426586d08b502019652103ebae6fcf025ed0cafe701cad3b779433ed8b

427c750c52786387dd0301f3dd5adce997bf51014b2401c46fc47dbb487c8571

2e8ef8f5c4c1782b1bfeae90b546ee53bdb9f4fa0157bcd029ebc8462a9530d

4b516e1ac3081edf354286e55344195743f9a85f91abfbffe43a8b2fbafc182d

abae0a5f6f042a4e6111766955fbb9f73ae4f56273ef19cae35ed2bdeccee9d

1aa3fdea985eb7a1573e7d6fae9c076645bf70bb676bac56a8bb826e227d3f29

75469f7c212e9900902f801b8c59b4b9253f7b7dfec4209090df9d6cd7cd902d

9c328cff45396cd14ff62ffdd4aa2a99b4a49612d33ccfceb4402d4562cac3e7

Fiat-Shamir context

e55d5f057000c1c4cb954e9ea6f1bd86fc91e87c15797e027f53e655aee0a2c4

withheld proof term: 16 519 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.toBlocks_{12_sub} theorem, SplitTDirections:145

```
∀ {k : Type u_4} [inst : Field.{u_4} k] {m : Type u_5} {n : Type u_6}
  (X Y : Matrix.{max u_6 u_5, max u_6 u_5, u_4} (Sum.{u_5, u_6} m n) (Sum.{u_5, u_6} m n)
    k),
  Eq.{max (max (u_4 + 1) (u_5 + 1)) (u_6 + 1)}
    (Matrix.toBlocks12.{u_5, u_6, u_5, u_6, u_4}
      (HSub.hSub.{max (max u_4 u_5) u_6, max (max u_4 u_5) u_6, max (max u_4 u_5) u_6} X Y))
    (HSub.hSub.{max (max u_4 u_5) u_6, max (max u_4 u_5) u_6, max (max u_4 u_5) u_6}
      (Matrix.toBlocks12.{u_5, u_6, u_5, u_6, u_4} X)
      (Matrix.toBlocks12.{u_5, u_6, u_5, u_6, u_4} Y))
```

SHA-256 of the statement above

7b592eb56055713f17ea0415e91cc504991b46ea2ddc1fdc41aef2f311f2b9c8

SHA-256 of the kernel type

88f8cfe0e0624e5e4399bd7844162f6bf2ea2bee8770552fdd201f437078cd11

commitment C

d805e2cc608b43311a3dff0da5edefeb4327dd43dea8eb48f5ff64e656f06741
afbf108e6ddd799b0c6a3a6125985bf4e1f9e1f8b5a230c9c7627abfbc39ead
a3bcf816aa84929be278b05d8b88a4463b4c803d7ff1e21b20e4a443324ee880
8b58e9f34b3f6ab7c04a3afd2a5a036b5fb1a32779725b3a1d7302a3d6bfa68b
474451240706087269466c3fda8e1a4403ccdc7d7682812daf346d20f39bae7
2244bd68a54ceab328ba751bb824e5cf59978e571342cc41470fab4d1ccd0c8d
9843d03fdf9d5ec0736c2f178532443b8efe124374db2b4ec563f4ae4808ec6d
8ea2ada2b8677c2d7f4d9d73c88fd4e842d808a5c7ebb069c392f562e2a39c1e
first message t

62aff892a6ec31b84b82558f3afbfcad220ea7ecb2bc36521e22f09d63edafa7
ba5dbfe25513fe1edb08f9e36fb5e2c90a5bb0886b3d609c6f54ea1010e8fb0a
62be9a6194b0dafa8b4ca9ef98c0af0c357f73e3de89a3615d4792bb8ce8247e
b20df01125bd3ce72e6ad3db2dab98ca65ccbf8affe4fb7000a3f5ed90dfd7
af40165245bd1a1addaed5babfb50806605f0cc1d85cdb80cc954968e01e1f4f
38f06c1e5eb1d8adefaf36cc1438c898deac03abf0d3fdb651119910e4099af8
a7e15e5f3485fc34ef56143e155393c106ad5091c249f5fabf7bf33324e9f5b3
64dfce6f4dea02c25d6521a42715d39a02b9c144dfe8eb9cbadce432fbca0101
response z

7d58f083dc002617ab7c7be2aeb1c14702c7e37d0abcf4bb4dbb7732660e618f
14f3349f887f020f85d490af04c9cbd5f64fb3163045374d60fa15b6e49f6113
8ef7f773e97a2ab03470be8f6d015ec995b25760abaa9c32cb9d0889effad2ac3
0785adb4e7b7730d236e575fece6c22661b1e0187a46d09d1a1627a0139bf5a
eabdcf442162d114fb17bbd07b853d1bd526b6173765bc893c158597f6665042
3d89543be64e5869cc587ce738b9a68988036e89acd6abc48e1c48dd3ea82d0d
87532e5ea5c03cc44c58b11e163505381fddabb88c3e5b1dca81a9e4b7b6ed8d
a835dfe73f08cf652991d37284824c4712e9329b65015b0c4f299972e6fc62
response w

315284076b76b6ed16a57deccf6b95328aaddc7363596893e842f7e58b387f1d
52b21f53cf70adc1659aee55f658a4d4a24d3d1220ac24534c5db82158ce379d
3a710369901815a5aa959de6ed8b89fbb215dac29f1f361a058a98160c1b65be
7bac6aceb4e90519d285fa2d38d294002022525e5be03a4f24025835984276fe
5507f2e174a5f60a37d12bd653256f245fdb456b97a6b108892ce1ad4d5683ec
ef082878820e807f9e2dbe0562a6e82dbf30c985d66ec28c4545729c365ec166
378398219a906a802dc48bc246240b578e5bad17a5c47a57b0f19ac0f5a43948
92525e1a5aa135a750f994e6b5af4ef7dc77cca3ad52ae9ea48076e6d808f44

Fiat-Shamir context

7235f724a4641a8ba5c70117eb8254a5887b0d0227a4528f2a1f275a811f8337

withheld proof term: 1 322 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.toBlocks_{21_sub} theorem, SplitTDirections:150

```
∀ {k : Type u_4} [inst : Field.{u_4} k] {m : Type u_5} {n : Type u_6}
  (X Y : Matrix.{max u_6 u_5, max u_6 u_5, u_4} (Sum.{u_5, u_6} m n) (Sum.{u_5, u_6} m n)
    k),
  Eq.{max (max (u_4 + 1) (u_5 + 1)) (u_6 + 1)}
    (Matrix.toBlocks21.{u_5, u_6, u_5, u_6, u_4}
      (HSub.hSub.{max (max u_4 u_5) u_6, max (max u_4 u_5) u_6, max (max u_4 u_5) u_6} X Y))
    (HSub.hSub.{max (max u_4 u_5) u_6, max (max u_4 u_5) u_6, max (max u_4 u_5) u_6}
      (Matrix.toBlocks21.{u_5, u_6, u_5, u_6, u_4} X)
      (Matrix.toBlocks21.{u_5, u_6, u_5, u_6, u_4} Y))
```

SHA-256 of the statement above

1fa316308ca222dd601f5bcd6c6177db8a878dbdd868d9bfb138b1fc7d1493eda

SHA-256 of the kernel type

a407e65469b82c6eaa1711e6e3e3fc37ee7d9fd779ebde39397dfdf0823c7fc7
commitment C

fadda8690169f89be6ed6e0fe49bea22f60ce52df8c8b40e206b61be862eada
6bb8f3913024fd8fd4f4a094bfead83e666eb20b98ebf20699344a802082c926
d1ae07c04ec3b6a1d5b457f71571a885ecb0b86377e0fb95b1cf2c669bc52c3
3122b6c21191bd44b919bb0a0d3ec91766d0c5bb60e7b913ae1adf2184a0a4fe
da44eb4a9ffb629e90eae9c998850cf9a7253e2fbf8ea51bf0af010fb6e337c0
9c1014fcf18166b0f896c85345798de6fc8962544c2e47da1f9bd80593da0497
669e1e86ff421ae624630ba77ddc56a935813d70dc03ec650ce3984eb31adf2a
075caeabb8aeabd855ea80e3a8b948e7a2ddd8aff3a0309c6877abb73567bda39
first message t

8d646b13f4c362a7809bb8a7ec5bc82499d70dfd34f66e6e0823c86b2261fb3e
bba136c0826de4191eeb9de770be2b3dbfb6871b4a61509d458721d0fd74c014
e6576e154925100fd6039e202fcf7ac01c6103fad4ff935619f31a9e02f9b014
4c281b4eaccf54ecd11c70d93c49f4cd3f4b47d372ba1ce1ab37d7382dbc351e
5b7f84b596792386bb4b0e2c2ad62b8675b25b083cfad46e82f05a85daff259a
c7af043f6bb8519adef048edd9dc06d3ceb3cdf65c5b158e4ae36c9fc441196f
bf37bea35b10894291cc38ae6a59453b9d04417d6afa0df4873b3a3a71ed74ae
0e94c3aac237ae4df0c2ddf51259da4dc8a482367a0bf44cbdfbddcaf72769af
response z

3f9bbc6d0a67c3766199a3339bd0083daf49d49b40489472943e0b6c2a61e5c7
39dc6a02d04b8615e391260d0f66800852798ac93bae615b76285eee57ca67b4
f48e06a5812874110dec86bd4fffb9a20cb64945537eacaf1b5a76d196651369f
370b5bbccf5f630f6526e6ddf05b3b44fd99dee60858231c19ceeb898dff6c3
1a4e9290ec60244b6f3649fd7f21c9a4b0df087bfe384212b7aebe8ef38ba1d
86667d37cf37642029497e070413315f7b9e843f41106956a11355bda9ec003e
e3c2c4cb31ce32aea9381f861f371ca568207bbcd7cae569c0315e3788093668
c7b9ba55d3146d523c8e6f48e5227bb0ae9c75239670ffc44a83f70d5db8e0c9
response w

37937f7ce7c5c1acbddedd210b70708737128695424d96f7076d9799a97ee33d0
2bf584f6a429a9ec02efdcfb7837e2f18fcb7f01b7c476afbce1006257dc8ef1
b5286a56b1f4803b65dfad57edb750683ba56828809f402714701db79da6ecea
64788371cf69b7e8d2351c51ad8a03e1e20a3697416adb207837880569c1fbf1
76a5326ef59c68a3d98e0ab9960b964af152e00d2e2965c319c47c3f615e7a25
7af8e581138314184e4866fd0b8f1ad04fe88e20f72cc27a65719df39e2e00b1
2cd98e2a89aca7d30f5e3b0f81af832eb04c353d14077ebc06648e9e5720bf6f
4cbaa67d78547a08d651239ca9958d8e96ddf66a0cf5e110b40d80cff6763dfe

Fiat-Shamir context

ba88b37bccb4ca72e3a2a356a822484282d257082ab75495042b3d7270efda9c

withheld proof term: 1 322 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.adOffDiag definition, Sylvester:197

```
{k : Type u_1} →
  [inst : Field.{u_1} k] →
    {m : Type u_2} →
      {n : Type u_3} →
        [Fintype.{u_2} m] →
          [Fintype.{u_3} n] →
            Matrix.{u_2, u_2, u_1} m m k →
              Matrix.{u_3, u_3, u_1} n n k →
                LinearMap.{u_1, u_1, max (max (max u_1 u_2) u_3) (max u_1 u_3) u_2,
                  max (max (max u_1 u_2) u_3) (max u_1 u_3) u_2}
                  (RingHom.id.{u_1} k)
                (Prod.{max (max u_1 u_3) u_2, max (max u_1 u_2) u_3}
                  (Matrix.{u_2, u_3, u_1} m n k) (Matrix.{u_3, u_2, u_1} n m k))
                (Prod.{max (max u_1 u_3) u_2, max (max u_1 u_2) u_3}
                  (Matrix.{u_2, u_3, u_1} m n k) (Matrix.{u_3, u_2, u_1} n m k))
```

SHA-256 of the statement above

5f4d8e3471372928ced11c5277418800760efc69ff7f846d9da5e33b22464f5d

SHA-256 of the kernel type

ba5b7848ec077924c8b5f16d1d677df4e9df05ca93c95128ed078c94d4d857a4

commitment C

c79130d1bff5d5992d26528af3f8553c9a72658a570a894f8fd36a25892610ce
 d1b1f78dca386667003646ae3bec3bf76c968ce4a7576cc7fed122aab25843c2
 72fab2f12762fc65d0c90ac0c4aefbbf835de811d3831d08b633dff607d2e8b3
 4070d1f090e80327275611d3fd21600e64420dc9c1e1773e7764c1e208095b84
 4411feb5890fd8753c66d4939310120b8365f0a9ba70931969e0b88a394243d9
 59a079f768a46c64c3c08f51f9f2b87921dc2ef7a89c260ba9d868e69a11acb1
 d81a9ef4f678f9146d27c7d625e66d0ea45ea9dbfefd9ab3617ac5405f279cbc
 a3488df539e1d905a9a6e1b06fee33c808619cd9d2d420823fd4dc28c746fb2
 first message t

4e3efde68c9e83c543b0b456e8da83d2fb1b287a293430f083ef6a08efd95c0a
 b0ad41e5da6a2730d89057562452f00075ecdcd37a5e4747c622657ce4cd03e5
 5e1526480424d131c7cc8886636e3a1d7c605d390b36073b7dd13131b19d4753
 12b04546dba3183d133894523c87dfbd27742134bde2209ceba06cf066eb4926
 b0c5958d755d56e22572b7b51077d47b25ad6e55ab20b6861d13ca363844c3be
 4418dee1d9ec84942c899019c9ac8f21a79a503db72d84d23fc01ca10f54de26
 61b5aad8cb736433892bf37a61a2ba559f3103d039143e68d7c61735406a2688
 0cb08097c59b526f4e4921948df8c25394ddee5fc645dca8942ec3505339d632
 response z

7a504c030ba6a549c10c3072fba22b7a00dd29dd6349ffc76ac6c3a7818dbae
 7ae2b450f7fc4bacc328571abb816fc92918f410b2ab97dfba44254ce3ad7cb6
 845fb6337f3079b77546787941506c5e0647bc1be184ebc5d4a7b2ba35290c33
 e7123deddfc6676adeb3f2008fd4e09651e3509a59e725f85219905cfe787c4b
 12d96c8f8f3f71130a3d5f72f82e5eafe41a0ede0f98ad96ba79f3ddd3c8786d
 33e1f9b5bfec84ea1fe2215686afd99ef0279d13026ffa0543647a50f06115d1
 2fc188f28eb6cb9d49ad58612b380f0e294930ab309cbddbcba3d26e1032ce6
 985c45ab2f39d4242f5632839b635a25492e13f2b9b86c452735b4ba273edb6c
 response w

7c585879c741b1edc18ca4ff1e8f03384d4dea3c76402482498282fa43b10c4b
 ee23c905f7a9fc916f86e1534ab84d06837242210af10cd639b069b3b66090c4
 5b2f8ab588f3dd2e24240f3bde059f4e7b9ba07fe88ff44f9b7b015bbcf321e3
 d156f200b159e9b9b67fa668b4a6c5df9dab826c97402f4db3567e17a8b418a9
 edbd8ebfe13e3080de53dfe713d525dd900a37ceb9ba803fc20cb2f4da368337
 60cbd64a8788b2682500eb85823a04c5143ce6893f1168e710a70b9ead68e71a
 1f6468f5a4a7c939c2f3fb883788da31878c943d186b4dd8356f9628674e7542
 786059faebb3838ef5489965a401694d4bfd1a5090fc3d7dfb8124a199bf1f09
 Fiat-Shamir context

0bfcdd0f765d8e230223caef57aa48354ba1acf81529f64227fa38b8d2751143

withheld proof term: 6 862 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adOffDiag_apply theorem, Sylvester:204

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Type u_2} {n : Type u_3} [inst_1 :
  Fintype.{u_2} m]
  [inst_2 : Fintype.{u_3} n] (A : Matrix.{u_2, u_2, u_1} m m k) (B : Matrix.{u_3, u_3, u_1}
    n n k)
(P :
  Prod.{max (max u_1 u_3) u_2, max (max u_1 u_2) u_3} (Matrix.{u_2, u_3, u_1} m n k)
    (Matrix.{u_3, u_2, u_1} n m k)),
Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)} ((FBundlePoint.adOffDiag.{u_1, u_2, u_3} A
  B) P)
(Prod.mk.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
  (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
    (HMul.hMul.{max u_1 u_2, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} A P.1)
    (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_3, max (max u_1 u_2) u_3} P.1 B))
  (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
    (HMul.hMul.{max u_1 u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} B P.2)
    (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_2, max (max u_1 u_2) u_3} P.2 A)))
```

SHA-256 of the statement above

e3ad6a53137e7aef8a5da951f606233b5b04613141ba1008c00fab69f8e925cc

SHA-256 of the kernel type

16051ad2f3968fd3c4205d477954a731441cd4dfec5437319f4c6eecdafef1de
commitment C

50a8ab2b51aba0b22ebb000413dc31fb9512ca2f7942474aa346635b9f41b7aa
c0667c93896ea6743505811c2ccb9445d7478e9445f55e2adf451cfad596d5f4
0e40998f9d1eebf3958286b23ebf4dbfed0f196430804d537ef28463bfc435da
8236a76dfae4029206f62a0a2c843bc3641e508ffc3201b1057166440c3c4d18
80fa30915e5212fb8dc00c00b3f3bbd363de8e3559cf01bfc5d2f002f7c98fc8
bfdadfd4621b8a50958a4194af7587654831c7c30df0a8f2fd6fe01e5ca63cb8
9df31350fc7c9b6bd386dda6591bee5266f9df5cfcbfca3b637ce834e8d9126c
0515807fe1fbae32ef3069d3b0a52697b384ae44d53b9ee89292f7d63b1af
first message t

9c9f825261d3b6c0c9625d094f5bddd276b853b72ad59c5b33e26bf4aefe90b2
8b073d6125e2cb77b8706ef9a853d181e6bbd43ec52aa525a4e6063a9e0d92fd
5d1c6a02313d2c2fb42841514a4823d639e014b33b3d7c35bae706ab715e6202
7fcd1b7e0fe01180f5bb98277eaf58a8a068aaa1725a5742aff6b5d9794753bb
94f8579ecf591d997de935ce686833324afd2f77279245155e07f800d9cd8783
421b961361aafccc301da696b5b92df8f539591a610b042e3ccc6fbc94f06c25
e38f6f2815d9b711fcc4b58a0ccea81e63007d6d45208e9a846d4aa1cee8c0d1
bd4f5e6174ea6b9eaf1ac17f7e1c6c4688a0381fd677d6d6b56884129f6f5d4e
response z

6ae4a0551c758e8860133dcd01d17c430d1d6424c06e9d5883c4a822d155c54d
db91bd4ddcbad69973656385d61b739a6a5de92cf3c013946a0ceae19d9d92c0
61e63cc8da2fd80526674b729622f995c67201bf19fa768e10cfae3ff5c9f7ce
2586d153b3af3aec94541c637342a270d9543760576589c2c71cd0512d2bdd25
a3aa32a0c04aa129d7730c0fc450d3b5fe7b6d0a4681fa4d4f880a5e0d728d11
0735af0b508f0c22479c6e87ebc7343cd396aca5d2db6ae3a43c22b30178f5c5
e3f950d0a4ecc806db704374508ef6f9ea62c4f23e2afbf3f92cd1d6d9477948
0bab2567975d6da808a669d39510d1b905dc0e6316bbd9d8211cba077c7c8ac2
response w

40e51b28befe029e5b3b1b462c9a2f03defb7d1b7e44f62c9344c1c0fe2e7eda
1fa0619e71a763f8515d86358cfae5042d09645e1bf41d75519f5065fb9bb19f
9f710d3758f4a54674d040411544215fe0db4bca672556603cee4c34d08e3220
4e9d5e37611ce003338f0f2ee2821bdc7f90968e4d332e5bf47fd3c6f7628553
1bc843714daeb2686c80747fefa4c67d6604a411df0a8cb8b891ee5b25bc102c
1f8e658ea9f992c52369420b24449f65a9d3dfbcd7e8a686d0e01efca6eebd9e
b5cf05aebedd36a80483c6d6b28a88b2fcf78cb18667c659b1e9b9b33fe3adfb
9992d1c6a72ce1275d2b53cedad7df719ae2fb1c12aa734bdc8d5f0c3c2a31e1
Fiat-Shamir context

0c16214675363bb757aacf14821dd1d3d91d4feaea981c23e102a7c763169194

withheld proof term: 19410 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adOffDiag_bijective theorem, Sylvester:209

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}\{u_1\} k] \{m : \text{Type } u_2\} \{n : \text{Type } u_3\} [\text{inst}_1 : \text{Fintype}\{u_2\} m] \\ [\text{inst}_2 : \text{Fintype}\{u_3\} n] [\text{inst}_3 : \text{DecidableEq}\{u_2 + 1\} m] [\text{inst}_4 : \text{DecidableEq}\{u_3 + 1\} n] \\ \{A : \text{Matrix}\{u_2, u_2, u_1\} m m k\} \{B : \text{Matrix}\{u_3, u_3, u_1\} n n k\}, \\ \text{IsCoprime}\{u_1\} (\text{Matrix.charpoly}\{u_1, u_2\} A) (\text{Matrix.charpoly}\{u_1, u_3\} B) \rightarrow \\ \text{Function.Bijective}\{\max (\max (u_1 + 1) (u_2 + 1)) (u_3 + 1), \\ \max (\max (u_1 + 1) (u_2 + 1)) (u_3 + 1)\} \\ \uparrow (\text{FBundlePoint.adOffDiag}\{u_1, u_2, u_3\} A B)$

SHA-256 of the statement above

2ccf71e55566ed77087519aee0d4856f0f07397ba95da687729ad2e6c257634e

SHA-256 of the kernel type

2c667a1ce1bb03f432ff160a53a25c24840167512558299b316deb98825b63cb

commitment C

650ebe3af5b8c3fa8136881eda099e89d629f6a68630e6e6134931416fffb5bba
e133a2c96ef525468531730c38d5345191c170fad15b0a3482f79b41c16cd4e6
0ec8dff9fa97936a57d55dd6fa274e264266b635d83d6e82b9ac9a6822680457
bda736466c9c7f54afc7a9d57979dfc14001f83e6a3a5dc44fa315f348c7fb4c
e19139e6dc742432ca131e90670b29b7b4b399de71afb8cd0d639f9459e15a9b
de6ec4e3b4c3a4c6e335139282783fa56b1c8918616a1ced28821c314bac7946
987629ebe823f791951c4a0f36a1f35e9a52fdd9931d0ffdfad174f72c2cac38
368e24ae179a407a9edf7f72a21e6d2771f18fce9520d635fa9ad4a60a80d520
first message t

c0df682b789359b73f12d28f762588319788f20d37ffc0fb5b5b59e36fde0621
59147501d5fb41d0b20500a6114d5dd42d0b20709bd1e8d29342771d8601b487
42e5f1ab5775d59c6e3a02a1882ed08fdd38a8d0bd272d6e331a81df5ff2c168
5da4d36e27f9f102d3d074226faa3042598a3b5e9aad1aecb4a0898064865a3b
9e0b1df10dac789b313b3d2dad11ed2e0f960c2842bbe29a68e4c4c9afe64949
27665d5e46b13fc0e1db7d3f848f35e5220917b4d74cf704a745a842950134ee
50186bb23a1b3a3f08f1e88c093f3e81e4420f3021b0adc3eae36dd224ca2577
e8d08151f6f2d53f48c91d6b8294942abf7f2391bce58f48c69cfd8e8d4a192
response z

25f7bd7f33e84b9ff2297027feadc86fff1c749c21f3fc9bf13d1588a80b2010
100ef828caa25b1eb04546a4af08c1c25103c96ec3339f789c311a0a5625ccb3
220463a57aef7d8587708da5fb33a8c8209e36a7a4446a80e5e69a31e8da1214
df7639add4bf62791ab183c55584dd26d29bd7cd87e7cdb2297cd9e7cacc0ff
ccdb1fde58c814261f283c35e9d2dbd11c7e22d682c9a834729c8f3e95db605e
a1f7a5262af5d3eb8292f97cf6df4c660107a819703e8a53e920d98104c7ff4d
9bef27ae6869ad80763fd336e01066677c00a82cee45cc4895b5b4ff2c068d64
daf33b551f9effccd37e3f507a31ad97af17c0d9c513bfebf8872601f01b53b
response w

4f8f1afefa2df802c05354f00f116aaa2a901c6051970d8cb5e3bad0457a789
0ae6ebf648e0d4721676244b67561559780fbf4b28008f5441cd4349b0cf88e8
c732142a4577eced9c7abf39647b5496bc110e6745a1528cf0b45b49d7275c38
dc0e3e93be341ca321305e5cfbe144dd0da5e272ee7e41aa9dacefc5973f2e28
01a9a2ef3e599ff984edab8997db1ed1e3d01c633e5db4a551dfcece810802914
4979695baa616521726dd9c2440caf4c25102ff6461bd91487619e221ab0b0fc
a31ec88b620a8da71e279a982b16cce5049a3936db725761be3276c020e6e0e5
405e80875647db37b1f8c552e12564b03e3adc218d31b6f6c8301983ebe88ac3
Fiat-Shamir context

249f5f55d6b8e5b44468ec5b8c9982079cdf1f76a812b94da4031f8fba0e64ea

withheld proof term: 433 090 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.adOffDiag_bijective_of_disjoint_spectrum theorem, Sylvester:263

```
∀ {k : Type u_1} [inst : Field.{u_1} k] [IsAlgClosed.{u_1} k] {m : Type u_2} {n : Type u_3}
  [inst_2 : Fintype.{u_2} m] [inst_3 : Fintype.{u_3} n] [inst_4 : DecidableEq.{u_2 + 1} m]
  [inst_5 : DecidableEq.{u_3 + 1} n] {A : Matrix.{u_2, u_2, u_1} m m k}
  {B : Matrix.{u_3, u_3, u_1} n n k},
  (∀ (x : k),
    Membership.mem.{u_1, u_1} (spectrum.{u_1, max u_1 u_2} k A) x →
      x ∉ spectrum.{u_1, max u_1 u_3} k B) →
    Function.Bijective.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1),
      max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
      ↑(FBundlePoint.adOffDiag.{u_1, u_2, u_3} A B)
```

SHA-256 of the statement above

e12a45491f11405fb9bcca5af6fa9165977fa49a3e431d2e263094f9780c389b

SHA-256 of the kernel type

f4adc351c29ccf298992ca5de71b110e64b389f5b941e276a8b633d9a0341ddb

commitment C

f854357b50311278e4967beb6d710eba3f3cbf4945e0d91648fba9b07ee00baa
2afbeda4cc2f170645c4ef59f4a527cc49f2e40ed1c788921b3ae76526e9c6ff
5d99ad558697e78979b64d168203255c2bf2fb84869c12c1fb2256c289e13a94
caaf7d2d7bc7facbea199bd0b580f04ba78da2b6a49455ff3b1dc318e41fe1ac
f59cc66d8cfd4afb3bb1291b0ad77db6261ecce5fe907d1a622836be61e91cb9
e24e06181dc65f79236e6e4c992a66323c7cc5f484d76dc9aeb6ce9c102b921f
e06327d64e9b07a0c1c3b3e29606b743b86c517f201c4e05adc8f91d0b060bea
b0f8e8bafac94347e40a86ce18ffcef1db6213e554041292c7826eb291af9983
first message t

7a75410ffe72c8ca11cc910e6a4d9686752ee602778cd36bd8a228b18d7e4749
311fbc91bcf57d49a3bcc5d550607c38bae19f1e25422ababbabb8fa363967fc
b270c982798d884e4ea16d4323190fe26151522ac92c3a32c67e3cbe4d8102a4
9797aca61e9af4e004f45e076b17d64c55bb18163e0e3d8881b2ebd91abb3224
95f66c377ab92e1fd9eee55d3b9bef83544f113122e7682cc67d7b4e10e1f405
c29bb9e9d177404f03bf386b536b293089ec2070234f4ef124569919395a1e3d
ac06ad1af779cab83148e16826fbb054d541873fe781be922de1ed552e141292
1734c4d840551bb6cac2e8cda659a0ab8c1a20ca037763484ef44e0d72edbf60
response z

7be4d4883081272a7c835803ad0d0cdee5a5ad65530dd6c4cf931f71d5f214d5
121b0d1995617c63587c82492c37b0670d664dd691d959bb686e868a9b62b4af
49f905c719538e45ad65adcbde21e59b1b10853205ee0aeeb1c9bdaa28b831db
1f30832c11071dba9a7b95825c533ddb03c4dceb45fc82e03c3aac2a7a4bb8d8
5a34f57efalc45d1ea5d7a70825126a59f0f5b8c7e1ec15738c3d730c77c5fe9
3c9256bf1b797f35664477979189c316b61b50743d973868e2c278b4aa0efac9
daf3c0bf0c20fcff16e37ba28636eca47b2b3933752541a03f73828329c0c194
570f652ccc7d114982b10688422be85c309f33999df0016adeacb06dc4e179c3
response w

7a6dbb9e7b9b688e03b07ea8e86626e2160f9dd3a1c7046850305708f9c1230b
3365e1ba0c2b99c2c9f4297668318a6342f1c4b5cdaaf43d63c465f812ae2983
a649b981a74adda01a09cd1d20331352c174b9808f5f54b3b5bcf1438c21958b
32a4d6fffc9e8872902b7012e545a6db4250f6bd80b811a41f4adce9b7246c2e
4a13fe33dac6e8aeb2c3e941f311d21f4209d6a0c1b94acee8dc2a039a030d14
196fcccb7b43fd481a28d4cd8fa2be1cd391d40ef09d66086cd42491d30cf524
88ec3ae2483c10390efb5988200bf3670ed4e82ffac1e449ff8be78d2c19c815
7c6cc627d9dc88b88f00bb4d1f9548d2f7dce80c291d2ca9c54112b2a576e8fd

Fiat-Shamir context

aa30440b2457400090acb5e0ae01184ce7deda7ef8be330b24f151428df68535

withheld proof term: 7 268 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.aeval_mul_of_intertwine theorem, Sylvester:61

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3  
    + 1} n]  
  {A : Matrix.{u_2, u_2, u_1} m m R} {B : Matrix.{u_3, u_3, u_1} n n R}  
  {X : Matrix.{u_2, u_3, u_1} m n R},  
Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}  
  (HMul.hMul.{max u_1 u_2, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} A X)  
  (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_3, max (max u_1 u_2) u_3} X B) →  
  ∀ (f : Polynomial.{u_1} R),  
Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}  
  (HMul.hMul.{max u_1 u_2, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}  
    ((Polynomial.aeval.{u_1, max u_1 u_2} A) f) X)  
  (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_3, max (max u_1 u_2) u_3} X  
    ((Polynomial.aeval.{u_1, max u_1 u_3} B) f)))
```

SHA-256 of the statement above

2acbd707ceabeaa7c2424f30c600f3a64d340092d1d864b96a6982925543e7df

SHA-256 of the kernel type

85d7b01e7ca8ee06178d757388bfd64e119daf305c8ff27d347f0493d1c60a58
commitment C

f52f90c33c1f625a9669332375f4afd21d7d133cfe9fb57312305c07bad3a861
aeba70ac321b499a634a3ca1137e6e99d2b72943ff43abbc093a87ad117db697
8e73647e20af856b2628a005e0d0a089e066857fb831de387c671f2998405018
0f91b6fd38a30311f829c3048135322486fe79d481fa1d607176d35e3154643e
40fb881a557a488a666760ca5ecaaef0dd26f685a03be84b9764c175c4066b9
19087b1319a5aa5b3e95c1bb0ddb7bee4c8bf5d9ab823efb4eebc264c371fe2d
445c59090c08caf029fd1ac823a2046f90124a03ef5d344f31713f7d165d82fd
b632adb3a9df97f1b65ed455059e68b012caab50c5b10408b0c55603a6cac88c
first message t

89aa2a43400cc9a988c4fbd9dcb7f5ee638034c383b674f3d304cf7c3c9f8574
acc0d33a3bfc2d06bb525f37fedd43532761b612dddb2d427428b4e67585c86af
009dcf32567c4af1a034ece4b10e3186a7973e35299ddc553c54f58b8c00670d
5c7dc73f9201435060c8cf0373e9cd3069b82cb7af9416c289a61d775093d72f
505cabdf8a280caed4e69a31ac1515720527cc24bbdc01ee20da06565a269578
a52ae307eee008a90d6b92b513d06d1748e8bf59add1f0767454bd80b21b2d4
6a4bf9a7bb16b78890f516730c39978ad8e01e6716775ea59e8946472b967c80
0f6a581a33fe018675508abe0a5611b007f74e9c56d8046a3ece87aa01d154f1
response z

1b6a3949c98526fd17b473d8f0b3678e6a45f97ed0c910daba21a722d754d1ff
a52cb3e5a720407f0b9bc1080ff9f5e1514faa4c2c837b9ebc089cc693e77718
46eff672eab5f44b4ba441becbe83adf4a0a5275c25d43a089d51e858ed3fafd
b05e49f0b097039e8012bc49e6c1b8ae9b80b947672282a032c1b90be83dc5e1
21d4882a339f96b9d17a159df98be73b8a990f5a28120fe848faa6eeab1162f8
a350bf77f53a64cff5853f6c60118f4ed5dd93e6af32e744066d93b3b82e47dc
1602445c3c66b5b01c980ade1ae6e389bc0229a2172d0462e5d4b4973248dee4
623216549455d1c64e55d7095c6b7dcfafa93be71b49ea2f6f0052756bce4565
response w

333632d8ca0c44edfcab2c45a8f81903c4b6a73f203647f19ac589d5569f78dc
2a023ce8e06a8456da6b65a315b0fa5e759994f5028e4c64ba5abc777d469ff2
4c87bb2ab5adcd7e32511e9716a6300f3b9a93612f0c95854e690cf7199c21b5
63f9ef82e9062b50de1f854820c34c6edd3c215847c38ed5affd77e45f03c53e
044afb3dce273c3e7ddce69b87e3ee49fd29c60fa76d97561a1e454e84825a9e
2633eff29f2aeb98f0ac42a10a158429e656c12c078e536123cc443bad856d41
b1e6fc6bde0f0e5d7b6f53d88ca37af44df1ebfb228d6b6b79bebe9de085b4d
2cb4a4c4ea7be5f104c7c1b5872d4284a5a1e9b8a6055b27ec31bf4864287901

Fiat-Shamir context

45b22147ae4de7829127633dbcbc64014132b72dc7bf466722dbb0c18d68103d

withheld proof term: 714175 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.blockDiag₂ definition, Sylvester:131

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Type u_2} →
      {n : Type u_3} →
        Matrix.{u_2, u_2, u_1} m m R →
          Matrix.{u_3, u_3, u_1} n n R →
            Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m
              n) R
```

SHA-256 of the statement above

cccb9b2b6ba9a926e6fa8c89ced013c939d42b0d2ea34992d1f2e884e1060622

SHA-256 of the kernel type

7934ccba237be31f590d82c97a9b81e3d20278504fac6908b7281bedb99c6c1b

commitment C

e88a42ef622612757c4676d97fa697201db06c8a2ef529fa3bd958889b7fd393
d746ec3a0bff006d9d2c2db0b4ecfe541a4d23983822331394e263b56b214641
dd0bc28aae43cb1446cb2f646180fc2448b4c1eeba898569da69ec6099c123b3
8d3f4bc64eb1bf7dc272bc6774490040183eb1c1ded6c7dddc8a6f422708b787
4bf522891201e49c047b98adf0177ffe234aec4fd23c2dd4d898b8141cfcf69d
784a5bd7a2393dce47b8f85de5e2226c82543320f64a0d05ec69c4e292827546
5989fa4e75e35060a8a94d0f41ef827fa6f6b5bcd588a062b993719fc13c82b7
92215199ed539a31b192eb0c5fef423abfd081b1c5d0aa4180686c1ba5539124

first message t

d4109c39395a12ef2fedd9b5d986f18b039c81ed4a38675927a8d18dbbb6a0ec
c7f2ec045853d2aecc6e2c58ec667773b1d196f31a9db130a4124ceb0c58bee6
4393b40a72dfec29d3d890b5dca932c02ed3cb2790788ffcd39ac5248aceb95
b99d4f02a2e05171d4a62c74ff191357686b485dbd6f9333ea460ce5f1339419
0a15f6a42a6205629b88fb07409a668ada8cf0f2bf58f59dcdd7a187d6f4aded
70e222a13232064945cafbdbb64b347a541568bdcaad3f8c97edf5139f73c56b
ed9923d2fb6adfe57dbb9f442b4ec5de42d559de76e99369caf06ad8248f2fc9
7d8001db28af53c74027fa17d4b19933053bcd959ef5ac8818c32c8e86005d07

response z

2ee0855b1ab22596c623c1fe75bb9f7b2048849f9e971c0fb21d767676c30270
475535c63894245f478fbac2d9972db961ed55beb13683dca1430ae6b431f413
283f8e1c068e9a9196ae3269190378ef727bd9595501a67e8fe249d5592963f5
58321dcc64975fb49c9be7b3759afa717593a28b4cf2c69e44ff1c907a00f9a7
3e58f5fd3ea882d79820fca1bf3cd23e34efaecb1afcb96d503186f3d08470d3
8a22661123464deea19331574ac6bcfa1c256c35d642e6fb77299493df3ff646
74a7bfa77e5ce7bbd0360dfb1e321a0cba260b9a0ebbc9f37c63dd75c46802b
62a654dc7e78563ea0c281d1789f80d3d712071c5d625fbff9fb2c081c301001

response w

5c2fe6b4b81ae78f29a478fced63da78bbd6321ad66968e9e1c6c965f094aae8
b1845fc6d9da6a17872f8bedeefc867145935a6e6f432dec3450e341930cd1f
59c592ccfe751afa7412a4728c3ad0f549375febd4449c1a6645d98e46a2ef6e
3b9c50e4a77f9f8636e0f36e92390f2c499294dcbed77f8f614aaaa5faf8a9
5a08f156dee5485a41fd119f51842288e973d1afd83a52ac0ea6a5f0ddcb7051
b493c91acefe2d06989e136bc2cbaa3b55ebefb8deedeaa890485394f670fff69
45905a1562379afdbc6abd66e181a1b1326ca94c19d8dc7f36f9df630d7f6f00
381294d189a567202ce38f5798d012ad936ead4a87a2755c8c77bb413e1ddcaa

Fiat-Shamir context

643c804bf1ef909acf5389d050945f175059d590bf2b51699a5e72a9023c5306

withheld proof term: 1331 bytes • axioms: Quot.sound • kernel replay: yes

FBundlePoint.eq_blockDiag₂_of_commute theorem, Sylvester:180

```
∀ {R : Type u1} [inst : CommRing.{u1} R] {m : Type u2} {n : Type u3} [inst1 :  
  Fintype.{u2} m]  
[inst2 : Fintype.{u3} n] [inst3 : DecidableEq.{u2 + 1} m] [inst4 : DecidableEq.{u3  
  + 1} n]  
{A : Matrix.{u2, u2, u1} m m R} {B : Matrix.{u3, u3, u1} n n R},  
IsCoprime.{u1} (Matrix.charpoly.{u1, u2} A) (Matrix.charpoly.{u1, u3} B) →  
  ∀ (X : Matrix.{max u3 u2, max u3 u2, u1} (Sum.{u2, u3} m n) (Sum.{u2, u3} m n)  
    R),  
  Eq.{max (max (u1 + 1) (u2 + 1)) (u3 + 1)}  
    (HMul.hMul.{max (max u1 u2) u3, max (max u1 u2) u3, max (max u1 u2) u3}  
      (FBundlePoint.blockDiag2.{u1, u2, u3} A B) X)  
    (HMul.hMul.{max (max u1 u2) u3, max (max u1 u2) u3, max (max u1 u2) u3} X  
      (FBundlePoint.blockDiag2.{u1, u2, u3} A B)) →  
  Eq.{max (max (u1 + 1) (u2 + 1)) (u3 + 1)} X  
    (FBundlePoint.blockDiag2.{u1, u2, u3} (Matrix.toBlocks11.{u2, u3, u2, u3,  
      u1} X)  
      (Matrix.toBlocks22.{u2, u3, u2, u3, u1} X))
```

SHA-256 of the statement above

bcd649c5cb66e2488efb4a34e6d16e598c4abc6daacb97f6903d0bc51eefa396

SHA-256 of the kernel type

2bd34f86722487934e14881b01d22ec9ded9efcb9f9dc965c138a92224c424325

commitment C

678e16e9d693025ed427720e3b57cb22cf75686f71b8895361df9d59883d5245
99059beb1fd9bf1814b043ea61c646729e68e81ac91fb4f9668ac59a9ed13f4e
d8f1bd8706c78978cdf51c0ec98b3cd565f3349cf4a3259067bcc66107bbe2d6
bf40eba7338c028b1d259e3e9d301d1e02d930d0eab2dad3c1768dcf581f4a6f
6221287dea637e0a359502d15e01666c33d2d6c945f7baea3d2ffc5427e612bd
2c20c33db080ce5ec654e4015dcd0daa7b4d003e75a856b8f5daf872223ed3c6
e1abaf27d732baf242e859c5d44a0160f085a2f4dbed917b8c704ec915dd6397
024ed2e664718a8740fff054d5a2ba062964e59f59fb36c11df00c7cdcae47c4
first message t

44febdc3a77bac038aa27affa1479a09e002cd55fb1d62f738de16602010b6b
8e7e8a857654e3630b855acb2c4698420163533c029b4f8a961e7206f1365c54
470871126de4bf2f55f3e2f6774313f8363caed24c99fc8d80881b98c7fc3354
d1eca1b85327d337128bb55b174aa79189df8a5d4181e501711eaaedc2148bb8
1713d172201a0f8edb69aa354ba7b8c6180c2ba7f9f8f8585f1b5ac95895a7a
cdde82ba467af8d0af360cff88919aa5e26aa8da6b1117024d7c104ef2ff253
5270a9a51a060deac68d629b92391b8f9a1fc95c2da677f9365b24697039e563
38c825fe64cfe9dccc2a89a7eb0c57245c820c6b2b33fd3e7fb082e797f37fc03
response z

2efe0e2f07925d6a5a7c1ee23f42b3d6f846171365138792f2cc6ad45d4ca460
dcf5f175d2cc4e763152cc6ccca069874ad34be98fdd04b35fe01e89a507e35b
14472fb90af5c78e9f244c1f2aa8b08bb1d7b26f7462a21e933eb100750394bb
b364df53490b36c9f3bf8b509f14367979b1285635bd7be1a648e62541abe9de
99532dd76ade3a508e29e909f7a31bdbb50c741eef3fb82b0c37f6d215bd982b
99b3974306c14d0bedeb106e5ab22bc6f7d4fbc882f5f705d25956a4762fcbb
97a2e07f911055f776795c720193d55a2e2b1c4a4b66948699c2a5c291f7ab61
b2a53a1b112523e950ce04337aa4d97fd2720051204075e5f7a360ce719695b6
response w

b90c7e555d26970205f0e02eb03b44b5ed7768979b7d5228e38a4e99fc670ac1
e8d4193eefcb7a3da490e574f5d80697938c9d68faa62d334b60ddef05356c9b
71d35067aee95f388cf6bec4dc4202a90bfb509120e747002016529583ce9d33
68226d2d88bdef173435b1718ee87df4ec297ea47b5b2f071013e64ad776d544
ef1cccc93acedcf6a0cc991666293fa596d984f914ce533f9431e262413d1aed
6f1078525fabccdd0bffa7251157f4f08e4ce808f22803fa58f1f5d27f218ee2
47e0400e2f91ddced5e5a74cd8e807d8b871e2615ea82032204313aed84d8f964
f274238c65d3393d1cd3e7f19f378857e0f95864a03107fc76480f036041de0

Fiat-Shamir context

2f3dec32b58a84b9c04453225dd35f0926d847dc546550a7ccbd882c625d3509

withheld proof term: 29 667 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.existsUnique_sylvester theorem, Sylvester:113

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Type u_2} {n : Type u_3} [inst_1 :
  Fintype.{u_2} m]
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3
    + 1} n]
  {A : Matrix.{u_2, u_2, u_1} m m k} {B : Matrix.{u_3, u_3, u_1} n n k},
  IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B) →
  ∀ (C : Matrix.{u_2, u_3, u_1} m n k),
    ExistsUnique.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)} fun X =>
      Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
        (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
          (HMul.hMul.{max u_1 u_2, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} A X)
          (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_3, max (max u_1 u_2) u_3} X B))
    C
```

SHA-256 of the statement above

39616398e33ec309444299e38ab6bf0dde593bbf761bb55f395c330584ca1e3

SHA-256 of the kernel type

110bfe48bd25676884cce70f6034eca234ec08439559c6bb1b54e0c1643d0fd3
commitment C

6f72de03b635dd906d272291c84d87c7e463a038d905da1903345547bd256585
276ebb7981f6844cc7b0a95277193205c233ad6d85cb0cd00812cc4f006fef09
09a223a20398dc19c38356e728d384cc9276bb1a9fcc00b31544dff446881868
fd458f8cc818ed466b6e9c50b060afb91fbd890011c0bf93318334f674f6ee6f
0cee96564d9e1a84e29d6ecb61cb7c2c1095b0cec697e1998f0f4046181584ea
68474938b51767661009c70a4a7d0fca91affe108a54b142c2a1f5154c703282
736ca72e8f6e40b3cae2165af3e23ba12d8c9681f267314cd6229d27fb6c636a
3e15d5f59b536419983b7bbd84c81daf369a5c5b7b05fcbc3060f2db3c56e81b
first message t

20b6b1bf8bef91d2716d49ea874ca83268c4187ba482800c08fe7b72649a8783
fe9aadb8aaba0b9e30dd922da2844797faf18ad173493c8ed9c2475ba3f106a5
73df1b8f013c9a65039993ec167639a8931bb1fd5b070d3bab1c837ea2b939b6
adc8fa739d9060808899fd68828680252adc913c5ba77f2cc10e649357bbf45c
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ad75c9a6b77377c642a61efc38dc672ae8f8ef12cb92da37fc15b7ae75ee91f0
e51ffcc491aa8f09918bc08a4422666b8723ab1090be1a98fe97001163669617
5890615787a9f37de1c329f73d7cb9c2d6a14d1fb3ea97cad45ddcf9a8d15fa1
response z

3a51f0ed55ca90ccab5b19244a627ad38f446d24488a53e208560ed50c40dde8
9cec5fa78f37f81fd45259666509ae09d794661ce0bdc9bfc9e18d599ab2ae48
1577ef417830cc21200d869ebb6576624cb5c3ea47ff2c9680f995b627a85e48
b29ac4699f81229fd4a2e13c40fbb07e06e8d1764b496a1d73a4cc2ab8725307
bea71069babb8443d838f519be3ebd9fbf2fdad82a621d3dc59e2400a108f580
b202a96ce1e5953a4faad7e9aa0ea5b5ebe5b5ab91697280003e9f133c3f5fb6
462a525e66fc0386de1491d66b67cde2a2aceacad85c1b44be2004b283225e48
0c3d1fe9021fc35e8837d21d40f9e281d2b02f03e068bc54252390706d22ffaf
response w

49c3ef73df7d6a92138ee6524081a4c80d018836adec80fe9164284717f78720
601b6f645c7eb61a0525199ca7742fd1bd034fe37ab139357362fa85c33fb87f
0c681284f7f8ccaf270f0e68fccfed11cc7372fc96dba5a98111b7570ce28437
6566cb94893b68e721d93df3643129ea107d811ecd2b4bb18afcea3c6f147afd
f35b62b7722d1715cb2cd3c2c89460d849fdc58201599d1e9be9163f1a748a5d
0b23ca71c2095d82f3ebc933ef9cf1ec1b99877d9e306606cffd719107113dac
56e88d5d327ccdb6907847d42390b3082abb5a0aeb01e6579ab76784c24453a1
2897e1de983362e17f45a1a02d2362e082a8abc8edfa7b25ed39a86af0d133d8
Fiat-Shamir context

3d9e212c348b9223bdb899e7a00f11d001b198e054c77ad5d489828fa87efa33

withheld proof term: 152040 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isCoprime_charpoly_iff_disjoint_spectrum theorem, Sylvester:233

```
∀ {k : Type u_1} [inst : Field.{u_1} k] [IsAlgClosed.{u_1} k] {m : Type u_2} {n : Type u_3}
  [inst_2 : Fintype.{u_2} m] [inst_3 : Fintype.{u_3} n] [inst_4 : DecidableEq.{u_2 + 1} m]
  [inst_5 : DecidableEq.{u_3 + 1} n] (A : Matrix.{u_2, u_2, u_1} m m k)
  (B : Matrix.{u_3, u_3, u_1} n n k),
  Iff (IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B))
    (∀ (x : k),
      Membership.mem.{u_1, u_1} (spectrum.{u_1, max u_1 u_2} k A) x →
        x ∉ spectrum.{u_1, max u_1 u_3} k B)
```

SHA-256 of the statement above

189efd7521ead4ed96123b7229e0d26e848a7fe537b35bf193b6146400471f52

SHA-256 of the kernel type

e0f681be9c3f1800e2bee7da109d7d86e02d7abe20c65455138f86180b4a92f5

commitment C

4478952a950d958399cc6e617ccc1ae1ec0c7772849d2665371068297eefb515

4617318089e2790273c7644396fb6eda845a9f57661d1b7c1d38c259293846b4

7e54e1b011e34484194dfe669c761f70367f45b5ff1e33f432cfa2fddbda4f5

e8b8afc2f817d819ccd66852d4a64720397d63e908e04c9f2da8d77201ace587

525294dd9a541adb27eee4b56546e6e5df419697640236f1836a00e0850f2c0

909a3c442d55ce63d39cc330350d789f1ca2bc0a3036844562c9ecb257ff6937

eb3336ce0585ec159a23df0a327f8002feffec273144bae3e05b7f86b2d80b04

c9b3a1c6ea3ddc5d28e5595886943d18c2b57a3fd35aa8f8e3c9be566f5596ac

first message t

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7d3275320936215a4b34f3a9492d1ca73764d120db63a0d5f2d0d3a20d8bb034

c864fbe55a82720c2b4c443fa2928429f16e8cdf22b8431340048e9a1788c995

679b1cf528286f4a82f6678812439c679c20f11639d2feb901991726ff2a9072

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a2735c4faeed0254ed84d424b816da3a90f448e5e010d109382d3de1f870409b

d5d56d77c24ee29fe24f4761177ba118b8a819a7aa6840ad2d6f8efc8b48cdcb

response z

1a738d4739bdab6e45db91241b01310eefc3e7dc42502318e30fd92d17a3b2

5c8cd30e76bf60aacbb33cd095f415615a94fcb05a3bbc1d64309640e67996ea

749c8ba8de84b98a12227ec491575703d835895c93a3fb4c2a1e2249f2338112

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1370b2245143d2b362f0c2b23e473e90e35fcbd2c137bcc019138c3101b5c50b

86b16990d102ee49b47a7ea08ee2f48aa7d5b7da73028a802364f44c93f18a0b

48f865807830e094d99590fa79aff5f71f150d610e3f759878b8ac2d890ff750

129da338e8446ce2ba259466fee38b731b107b1c1a8a9355b74775c5d6d5d9bd

response w

c906531e5170f3305ca35d1d990c54889140271bb912faede9551b6c152cd84e

ccfb6990a91cf4182cb4fffd0bfbe022443336d678abe2b1115c5e4a75df75ab

3e0285181b2200ed651ac3b8b3ea247aeb5b71a1f31f3872d74e82812fb8785e

a21601a3e6394a488a4a7fd818b5c3f0903b2adc1c053d8e1205321740af123b

dfc5faebdc0da04b88a5f745e8c1abee964e6ed4413a49943c716112e381f6b0

5e9de567ac4f03a6b59a22bdd4ff3316c6c1f74a23973b883313b9c3503c0cbc

8b4b77277a4f865b338208d7ce6ef5818cc62c121f73de5c7f9bbba5942ba5ea

d26cb653ae88adb4967f7a846d24b672799dc0c59d39eb11e2f5df7af1e515

Fiat-Shamir context

1b43d3d6de428989baf85ee8a01d969b990eddbdbd62f87a9e339ccb0c7c3434

withheld proof term: 184970 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isUnit_aeval_charpoly theorem, Sylvester:70

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3  
    + 1} n]  
  {A : Matrix.{u_2, u_2, u_1} m m R} {B : Matrix.{u_3, u_3, u_1} n n R},  
  IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B) →  
    IsUnit.{max u_1 u_2} ((Polynomial.aeval.{u_1, max u_1 u_2} A) (Matrix.charpoly.{u_1,  
      u_3} B))
```

SHA-256 of the statement above

65c5ea55eb87b167f06454c5180f2970c91c9596dc2849bf5b62463dd651d728

SHA-256 of the kernel type

0241f935910d1842c2786454aaf665281f70f36294a467bf5bc1061c1b52cc67

commitment C

4a7c0daa971eb68a805df826a91ab15672b02c331fdfa30a07320579dec496fe
a966f0ec0a357f0680df733f127912e9c6b1ae7958e167706bde69c7706a6af5
74bacd4683c50a680cad6e6ad3b346267b5d69d13c83798d0f718d5dfa40ae8f
499b7df4d9587df18e649147a3902ad5f76d9c9331fea79a09f59da39448eee7
c3022425f960a4c7e4fa923fd19a9390b2481bdd1a7e59172882c7f2ea7296a0
aef0a5947a68675c53289d9df2fcdd50560916f2afa281df7b416d43187fa200
2840acc0043aebdd6b6e147b4ecea0f9ce0c3eb789dff0abcf39ac96cf61be66
e0fe640c888922cb0766bd0a8958dc1da7a637994196e11051f02ba5768ff921

first message t

801bd8ac0eb6846921ec10986b7a5df55efeb2901d8e469f54c8aec07859fe6f
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3e48ba8cb312e0d1c20571a4605a8796222852aedda34f51dfca824b6dc8bb0c
1301b31c7675a0252fd9ff97ee7e8fc099020861ecffad6a98b956e29e5c8d05
359cb8d9cfa8c2e3549635f3febbdd7b5b3f2bd4c7d7e6edf34dc0ca6a52bcd7
90dd28c4549cf6177ad5a7f2e42872347f3a3b2f73ea30632a58afe35e574a93
8f3d6c6637dd5813caf5c68b03c0359030f052b4f6ae99738e218465d121ce48
05d068a06b742f3ab6c70c1209bd563d0120097324ee882c25a52778d673609c

response z

76a13a5e29f800c015002ead3a2bdba6b1cc64b6bb131161292983d511ad04f5
f2b1768ad5ea1748ba316807ab5f097926e70983a659c236d46041d7816f449f
c99bb815c058515579a0b356cb1639c311312b0a27b1f88876d5a33900b38c00
9cabbb52d3ae239ba039c83b90081e327517d07d41fca1069cfa71b7ca696482
ac7160b110bc09a2972c68c0fd935958f85a4ef118486a38707f174caf539893
11206f051bcfccd294e5081c8a8e1bda2156c381b210b8162c6ed97c0aa61028
bb90927f304a5f0e1746b60478e5d90ca99515008cb077c3cd5017f942884a73
55296729461ebe57c3189712168e6679f8229399f71ef71345d53cd5354165f2

response w

7d8d0c5afc995c0e42ec2f3a1ec08866b3763a058aa2de438f542637f5bde6f3
08452c8672ed35986c333b530daf78d574d7544c3468500ce62bcac1ff71326b
02cd376479a55a6d8198465f4d8e26b28f289d4418d31437650f78cc48044424
f7b1301a30ee6ec8d62b6a49508e8df0ad529b66978441620fdcfa0e3a214994
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ae3ec3592324042da265e6e0cfe946f9dfd19d2d381d352e24ee62a9eda2e23
7b3c3992dec1921e5d347b196f98c8bad106fc4565bfec9cdae38ff6aa6f5afb

Fiat-Shamir context

7b14d36a76b773dd5235c2c691416a1507967cfb1d57d6271119ab6ab1571df5

withheld proof term: 465 317 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.pow_mul_of_intertwine theorem, Sylvester:52

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3  
    + 1} n]  
  {A : Matrix.{u_2, u_2, u_1} m m R} {B : Matrix.{u_3, u_3, u_1} n n R}  
  {X : Matrix.{u_2, u_3, u_1} m n R},  
Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}  
  (HMul.hMul.{max u_1 u_2, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} A X)  
  (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_3, max (max u_1 u_2) u_3} X B) →  
  ∀ (k : Nat),  
    Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}  
      (HMul.hMul.{max u_1 u_2, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}  
        (HPow.hPow.{max u_1 u_2, 0, max u_1 u_2} A k) X)  
      (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_3, max (max u_1 u_2) u_3} X  
        (HPow.hPow.{max u_1 u_3, 0, max u_1 u_3} B k))
```

SHA-256 of the statement above

cf00075a537c537fc92b8262c7c3f6a1e94516d8b09d6a87eb965d62ae1a658e

SHA-256 of the kernel type

f97ee777493f9a7902664e0fde0e0af0176b3f088e8cf7da0e3f0bda56e07a0d
commitment C

b21e298fc2a67821a78c7f22815d29d97aa0721f50dcdddf602639a15e89ffe2
8fa4a0632552e6029de7cb889c146c2d21a424fe81f14486eb31042c6027a8ec
6de57f8a25370a5b5a8b60ea1a92d1481c48dabd01e8bba25534cbd2ab6d1674
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e749162b8f3c7ee3c0170d97359bfedbd09deb8dc751f7c6be69154a367a3903
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2ce54546c32e4d5efbba725a61144069615f574db0e9b597ef694c93318e8f0a
first message t

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d611176154da774c79d7771486f92d69b142268eaff374cbcf9a25db2535b2c2
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response z

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0def02ec00a905bcc390f9dcf3d1611235b6ef3c79d8f2241f034c35f730e205
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c09e03a7f2c6b0b30646bcac832e6e272b8de3cce557f1e8960cd5bf997e85e2
9cefe6ee59c6f2157d5198533191824e45b35af5762e2981bfd87200dfc37fee
df12b62732ce43a0820f8d55593081f8a86a5a1de321b9cf190dec5db36e29ac
response w

3d1cc18053b28d2f7c7c9d577ef77b22db3142153068cfa79af2316b06085b66
a6dda79e46a17e7d322e0975c17382d27318febac01d0163604d226d50683ca4
432c9a897b58338b30095fa1b7af03cf0b7d562cf0788aa128ad7dfc6e893120
065f77d58c663682b5f74d8abee17369136b1c7fa0e441e8d613a555b32fa8c
38747b58e214050bf418eb221a6c52c2685d86badd759db08361d070381c1d66
be992bf8a02985478ae2aa9f523d77bad7c6623faa4dfabe3fcb5cb4bc5e4b9e
11d5ebf281c389605f5fe97c9c8a1452a33805b0038b61757e8576f40c4d0c9c
b72e3da41d6a822ac7d4bdfd8c463e9182f7a952da4ace477150ed980a17dcc1
Fiat-Shamir context

3f0beb64e4b2ff0fe699a34671f75cf89bcd4eb239eb6ef2888e44292a7adbcd

withheld proof term: 234375 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sylvesterMap definition, Sylvester:37

```
{R : Type u_1} →
  [inst : CommRing.{u_1} R] →
    {m : Type u_2} →
      {n : Type u_3} →
        [Fintype.{u_2} m] →
          [Fintype.{u_3} n] →
            Matrix.{u_2, u_2, u_1} m m R →
              Matrix.{u_3, u_3, u_1} n n R →
                LinearMap.{u_1, u_1, max (max u_1 u_3) u_2, max (max u_1 u_3) u_2}
                  (RingHom.id.{u_1} R) (Matrix.{u_2, u_3, u_1} m n R) (Matrix.{u_2, u_3,
                    u_1} m n R)
```

SHA-256 of the statement above

b211c53acdaefc4deeb9f297d05f9cc9eeabe0ab3a24554899e45f1e88be6a85

SHA-256 of the kernel type

942c2288ba2b2deaac1a66da0a0332531b2052da791724964981975045d149b0
commitment C

a53db64ec4439fee7024a2aec9700f142acc71bc14dddec641c02145299707f60
815f4e699b035b07d5ad427bc7a38057310a45259b0b53a37484f4c1bd6798d3
8e5f7b67a054d50c4de5f119e790b94a15c32fee295b8112aee072db574e9c8b
80e80887d9ddb8f416b2cb61cf49bfa2fea8c934346d1ebe88781e1054f1084
e18f649cb5cb4697f9fe66f1de72935523056157e54316cd8b767e9715de0dad
8d4e0a6fd07e26374e44675f065c3f2ddf8f6846a67e5ae910539b9e25a2c027
283d75a1138cfe6a579e3b80eff886ca57a9cbda73102a84cd6b2094a5f58a70
d59e123c21a98ba313b8f43158be0c6f0d006b084f353044a9111b2476b270da
first message t

b79059e257c37de36a460782178ce1b8f859ae3d28880ce9ecc78f2a562a926c
3112c911db63aaed0dac6adda0362c18b28055a503938185616b8172c3b5998a
17f47570809a41e89400735ca6909798f78c532379b3c40f117657ee33a33f5d
fb45aab003391e461056d36ac88b0c95c8f0f6bea9b7a298cc13380afd5c1203
ddf38faa25d0edb32fd945b54514536972211f8c1eba3e09e011fe0f8384e862
9466b5158a24801248a81efbab7b541c9f3c9d762839e1f469ca3449eb46cd8a
9985a019b84dc71bfbe43b871b58622fc4a1a0f722d871819efb672a840d6218
5dbd5c40f393a71a4c6d0b568ec033c210080af0a4272df46373067f5d6c9da9
response z

42311eac13c88c11c4185c989228d0d00045127cde1cc4b25fc7330d7e86b803
6ea2289508f28712eec5e0507ffb6fe3fc196a88ebd5179fc1e77890be9afbac
df412d703bbef64d91fe6d784f3d7cd871e1a8e9dc6627d5a9600092265cbeec
561810601ee27f071d749efa34cd567a0d5ea959d6883c0a5f80f4a9726e177e
3eea1b47ae335f7e377c6c0fc3168facaddb930bd991440e0c40781c2ba3181b
6de3eb2d846b23698e66b99758f78c99ba5caf1207bfc524cb6b7ea4c7431f79
19aee1a40aabf7bda4781f55bac9d673b50bc32c9bdccc994f73a81152c5ca6c
a2479a62f2e2e746ca9478c458dc9875da52b507551fbf598b50e9d9bb034b9f
response w

1dc5a1c7cedba9fc709858214a2dacd9ad4addb1d833297b343999a53e0f12fc
55674eb2277d982c2c7e3bcc9e7d930d7434b6fe47f54137d693cef87585d3eb
be0bce32601f8b330b655eadd9f382a10d4a72bd3c7c31815a13a51a4dec841
640eb6a4b3128e6494c26d16eb99c10db849e26d9bc3075a2cef379d30a9efb4
558515cd25e58f2623d4e5a8c7b012de34cdc52788cbe65421395a52e45f6f0
0d9ca9dab5386a02226502361b7a35b01c77f3acb797fb1a19fed1edfeed8d6d
3a3a233d2c78aa5ab66796ad10b22ecc1959c5b535d6e0ca30b7a841a64f07ba
9b4ca002aef37b3c8de35eef619c19806c46a093930f326fcc1ad65a595751b1

Fiat-Shamir context

3229faa457eb2e001dc98cc7743b165817fbabb652a613bf6e443db1b73beac1

withheld proof term: 7301 bytes • axioms: Quot.sound, propext • kernel replay: yes

`FBundlePoint.sylvesterMap.eq_1` theorem, Sylvester:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :
  Fintype.{u_2} m]
  [inst_2 : Fintype.{u_3} n] (A : Matrix.{u_2, u_2, u_1} m m R) (B : Matrix.{u_3, u_3, u_1}
    n n R),
Eq.{(max (max u_1 u_3) u_2) + 1} (FBundlePoint.sylvesterMap.{u_1, u_2, u_3} A B)
  (LinearMap.mk.{u_1, u_1, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
    (AddHom.mk.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
      (fun X =>
        HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
          (HMul.hMul.{max u_1 u_2, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} A X)
          (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_3, max (max u_1 u_2) u_3} X B))
        ...))
  ...)
```

SHA-256 of the statement above

2499cb4c1ce9fd274c444da4bd9eb7f16fa01923b07bad497d775ca249cbfbee1

SHA-256 of the kernel type

6f33da946963793a46407bb08e57a8a19062cb24f3c64df38824df4b5a00c681
commitment C

26b3880b60ebd4e3d369bd521451ca0113225921ffda479f0b8812d809281d92
068ff3bfebbef88f9b058b93d1fd2b206dcda2bd7ba65d89f95a39fd2e0750b7
64643d88d81fe624c45b5663f7dc903bc99acf27cdafeb4e6d2f610db5c05a8b
92f82800e2bf749113b202a0f8af9bc97aeafaac68f757ef3ae36d861f827b
9b13ff5fb93db2b22109ca5efb4a9ac0211ee272f4963452d4f48de0d7fb5b62
f661bc39b749a650100f777d1d2a54295fe25c03d0653d0dfd71f8a8db28f079
937ec51a5c33ad8fa03b259af61aa08ff7c97605bb4723008165e8052a621c21
73966a39aae166768ee06f087aa8ba49988e687757e5105e00d401b584ea584f
first message t

234e8789fbf8e479ea4a761785f1785782509e71dc02130b7639d52485b4f65d
776d8c83ee60bd39a1b87fadda35d9a067a64db9fd0087fe0b0d44a778a02379
0c67600bc567032ec3d775ccd04d29259708fd7e73b3584f6f3fad32346fdc8a
7e5e5a796fe8805efc5520677f326f8dbe094bb53aecf8b9541f003112dab371
bc4c6c47c6baaf757a44d84e1329c676147f16f620d7e57964cbbc2dfb5b0002
c419986e72970af4fb494b9f9327739b89d9440130f317fa6dc5050a60305c23
05846b2b2f60d10b922f9e1325f16cd619daafe303fb595d435a3e5b870283dd
cf5113b5279aad4daca644403ca59b9073f86259ccbfcbab8c3cb204521794
response z

5bba24c0d70fdf9112e032fd76b1c5f272e43b313336742fffe1ad329838c27b
83fb340675ec074bcd9183dcd43fdff1b01b25b469caab1f394b4ae378533aa
0ad85a93f1cacd9324ca600f93e54f7c6d24563fa56bc8c226bbf1bbde935553
c1d1ffb7895c0f0987b490888b6d6d56a055498ae33dfc9cdf38a9661d24be90
44eb790a5f72efad0b08909dec2bc8b25d6984db2d90d6b8acddf05bae4b40ed
e70c1688e5cb4978a09bde1011b9d77b839c73a2c61891d266e6ad5afeffa78e
93613112707b52a68b6f013daedd29d57bc32645eac3e5d779c85e76a77c0989
6ad1fa33ca4829202a3f918975a0921bc9a33c20d42a5be39c3e578ce546528e
response w

3ae2604a91e364c428c4b30f1b8b25dc9f8c8f9ffa73b8935e4f97b59506cdcf
7f2f6a5046338c23bacb4cfa33ef51a29353d43559bab9bbf895c8d0b0a58dae
2e621d262cdcaa3f058cc35d0bab73f3ede85604b6813d61942e26c23348e7eb
a9471a0a1b4408a910b26e76716baadc561a5facb5fc9fda5939fb9e8d3bd6b3
e7b8c30547525c5cba4079efaec16624993f9cb14a86c2b38e22d3f3956ef507
6872cba68e7d1fc80d7637c823dd87f38dd6f3422c892904e75a0c9ba02eceb8
323a984cef4efcfb57185b18a94684aac65a9da6c2f34148fa4e73efc327c2ff
8e736686a624b7ebaa5b96631f9f6a4d9bd4ff0066f6e257deea6521dd34958e

Fiat-Shamir context

09728b05e404488f0c5fb67f5e840612b7abbd077ae9f219018e562317921a39

withheld proof term: 2348 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.sylvesterMap_apply theorem, Sylvester:48

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
  [inst_2 : Fintype.{u_3} n] (A : Matrix.{u_2, u_2, u_1} m m R) (B : Matrix.{u_3, u_3, u_1}  
    n n R)  
  (X : Matrix.{u_2, u_3, u_1} m n R),  
Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)} ((FBundlePoint.sylvesterMap.{u_1, u_2, u_3}  
  A B) X)  
  (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}  
    (HMul.hMul.{max u_1 u_2, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} A X)  
    (HMul.hMul.{max (max u_1 u_2) u_3, max u_1 u_3, max (max u_1 u_2) u_3} X B))
```

SHA-256 of the statement above

7f70b5a0740851d76243a095c36a2967a2c571a0036df53f4985d7701c65e480

SHA-256 of the kernel type

0e07b3328c281712856672c050f8b6175cd6366ed8936e8bf1d36142d96212df

commitment C

f724afdc90c2fa8b9146af733d0ef1c627f72e9eaecca9064c9d24e3b3bf7f92
774343d39c73f33150887cbaabd3690593c8c0264710a287f38f9244dc5f7815
379502b93b3e3a6cd4c2ebcb971e138f6e3c507da4aae9c9473314f9530714ef
f4cf9e15330845faac52d48aa356bb9f04cf071450ea2ac1c824b903b1149d36
dbc655d36594c8815eb90edcac9335563bc49c5180792ec8404cf375d4298440
9fa34560df175c4aa34a7d369d116e19e650ffde804862b3002c9df1f5c7a704
cfaadc0674a7cf786c34a1186b02022e966d67c37848df42a8208a8cc44d717b
1b6664beae41a6c944e2122d9cac9bfb543023ef1cdceda08cd20137b857024

first message t

560623e1f613fa87b9010ed319b739a26bcd4c0714d3dd946cb076958046bd53
f2c3bbe12e3ff75188e81b4cb4a27d2b24d9736489a78afd3bafec5edbf5de
39689b331b490e43bec990e1393be60ad6d33e0a2059d2f1642065d66a1a112e
fdf1fb2e3d2c8cad1f2a4a74ac1d55a3c25ef89dc6bf33de8e5d73b443cc2009
5f4ec117db1f76361a5ee9d26912e68e61575c654bd87f2b19af5e0ea5ee4541
6d1587506bd93b384ad57f9e3980c1af62275b800fe294f9e42c65cbeaf30abc
f9d8e9d08f124f71f82059a3b9c31dc3f4c536c198ec7141c0eb447064f7b6fb
40c53998ec1bf9f9af001d87da490a46eb6cf2773cc478163377d6152d95e80a

response z

20e100b20f49757cc23e249d97ad10bf9398472c7824d011e93ad1c55a0cacc0
9dba66f039d534e5c01fab37aee8e71a09adf78018be962e4e881b0200139ca8
a51b24cfe1cf1eae008e8885c1a558aae71208b514105762884d46f7b0205099
42241357c22d304150811066795d352bbdc1134fe137db5e541110bcd620eaf
4fe6c06714c01169ae469d1dc626aedbe30605efc20522897e02198afe9f8ce8
7dcd6030f2e61bbfaedf56ad98765457e49d962fba84719ebfe9223dd8dd7da4
8f1a21b6c361447b2f53cd80228192428c5e4ff60545f82772863dfdd264f5ac
1783d6a065d626feb326268e60bb3af8ea85c46771631629bdb14e2cc217ff2a

response w

486be838ab8cba738b122f3ad7e0b09776701828637ca15c96f32d9bc195e033
d2cfd5e119ed9be32253b2a0988b10bf43fa15d11dac225ff274d6deef829680
97b29a2412b20b81bb29657bb53941d5f290381927d484d577dfc34f1f29c90a
b8f08a18b0c0294d9e73be645c642bf76d170ac755007c0dfcf9228a7e6b4507
84d521455f0db53183f86cb5c02a7c9c862916ef5a5f9182433a50d616d53d1b
e7cf26b19d07270ac63d7c1ef7f097d526d32b5613be7d0bcca685acd184a4f
6b0b29022e070ba3a1d22e5e021f07391ba589344a3ae6c7ae1a02f1938ba76b
00787543078317f01602a8e6fffd32137ec5dae1abcd3750e35da04b0fdd8f1f

Fiat-Shamir context

7d9c15c672ecc3adfc78bac6942ccb1f753ac5936352834276aa1f0057d54d5

withheld proof term: 6 282 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.sylvesterMap_bijective theorem, Sylvester:107

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3  
    + 1} n]  
  {A : Matrix.{u_2, u_2, u_1} m m k} {B : Matrix.{u_3, u_3, u_1} n n k},  
  IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B) →  
    Function.Bijective.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1),  
      max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}  
      ↑(FBundlePoint.sylvesterMap.{u_1, u_2, u_3} A B)
```

SHA-256 of the statement above

4ac0ee478018e3a21a6c00cd9733af2649f537b333bf7b8a98269d2e6b1ee679

SHA-256 of the kernel type

510846d8cb74e1a08aab2bd4418d93ac72b996e0e428e360b1ae8081e40f361f

commitment C

ea1d0fda37eddd82350da6fbbba75bcb73158b083dfc0f3d35b2cf7cc18d7c792
f90c7c506c4117588f43cf4aac1c28e162938fa66cc5ccd2d18f4d0d8ddc0a73
b0a73a0d52e543fe3dde602988d67e4733da69e7e2abde121553d994c3ccbb02
6333b5a9dd650ed7ab99b3c1ac6ee78e7dfb6d65295f7a7e80c1deb770be9007
012590581bbba9b28365a00960f39175cbc350040ef4d89aa32010ed64788c2
d324a3c8feaa6e6135092ffd5aled171758571856f706bc0cce7f597c2dc2af7
3950a7949ca1e0bb1cf984b7ad27ae3928c656a21c0b21dd7e4b72805c48f359
38fe359cdb617b1ebe23e6d844dd9dfe2f2f334954ab794f1a48b0c2e4b84d27
first message t

212a70807405c25fdb3c1add632dcd5616b04c7d8da71b66e4caadc208395e99
03adaf361eb4535fdfed86099a6ee83ff626eeefccb3902988a0942fb5b3914
a476b8fbfa74ff05271cfc40fe6c42a8806679f21f1d027c8dd68ea0c41a3a75
3eb9c480b35e437726da754a8f5840c577d5a4ad6d82a628b553acf291898678
f7a865274699d813f76596a1359c8504bcf18690a10bb527ed8778b938282e00
bed6a816fc067626543fc39076c63540a5f3dd7354f37222f45466e40b920725
11deb8169e2390418be68f60f1c5b0986862da50624b13fe9b627724b7fe6612
cf58cb3ed65fd1c0da584989b8308708fb142ddf019e30533729b7dc5f2fb703
response z

77dfd81c12d7ead469667576b54b5b7c7acb523b50083fbd4e0562199db68c3
bda148f71174f88d5879171e93095a73bc1718a53e3f6e858dd2452b538a85cb
8a2385bdefeed7a52024696fc3f34f1217568b5b03d65d7f6b30699b7a3f700d
7848f58a922877a89ceb4591adfc816b60ad9d688fc17bf44751bd1cb3c83d32
5db824605a31a91d25a354bbcf372f94f48382ee4680fb41d616525acc010c7
482c368ab2a47226458a6d68f57e9ee0b5369dd44cac2a1be90ef0b5b7fee676
30b59143468b2d01134b8dea5f8bcd300ca28695ef99d911f347fcb03cb2ee6b
7ab701d0e625bda040fb1112ef45b56f833e94565979355b5546543537e603ec
response w

296d80c03e8702d227e86dfbeefe162de0936c644732c0a5bf4f3eced3db766f
cf69a17598b9b9ccce84fd6c9314579cee097e3499c7ec6fa04f30ed72a6a1d4
9f5a4d5e8b20d6d253dc3bf4b8ef4f90a1db8d9cb118282f65d5bbf4ed3523ec
5ecf842cecf7fa5fda54475b292664aaa3a991cb1856f20e9b43270e9ea2357c
0ce671cf6ce356331f3703f93dc61c818f18929f3919e25f479c9f09919b987f
96c265b5fe1127a356bcabd7f944eb66455c22998ead888f7e4b4e79fc8962b
819260710932b22bfe43501823f52a090f79319cf22c140b91d830ac90c69a0f
36f2e4fa28ee71141be343d57b586f7b01f87b28a9e766094c8e835b48bf742

Fiat-Shamir context

3cbbafbc8d4124aa5c8efb9de84435e17d76e92acb28beb1b3498fef8de40f52

withheld proof term: 45 211 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.sylvesterMap_injective theorem, Sylvester:80

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3  
    + 1} n]  
  {A : Matrix.{u_2, u_2, u_1} m m R} {B : Matrix.{u_3, u_3, u_1} n n R},  
  IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B) →  
    Function.Injective.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1),  
      max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}  
      ↑(FBundlePoint.sylvesterMap.{u_1, u_2, u_3} A B)
```

SHA-256 of the statement above

40f05c464becbd23acdabf4ee0c96521871b5b03f72e9f3b62dda31af07d80a8

SHA-256 of the kernel type

64e7be28bc1a145dae8c490c2216566d3d9668d68cc3806ade34edbb28b33be1

commitment C

b1b4ff432d2f3d869760fba775d3855e2e59c8535647fdc1d77432bf58d88e81
e069ddaa922d40bbdd60505e04980f4d314c72546cad24017e78e0368254dab
f4e39bf3db2b63431f84220c9ca02e234091d110e00cd271d976e6979b4917c3
0a18ff13919b0bd51ac0fe1e701e4cf315b951fe6855dcbbcd504da71da117f4
9783437de78cb0ba32e541efafebf7ff9f522a3f6e9f6b87729aa86017d05c89
58eac375f19cf108aacf051fa91c2bc82b1314f871e9a9bb5809ed2501fa6e36
bfc42f065842457f71897a2f1072641922c041f7f88b058cf942b151a94de2c6
1b476ec09eeef6fb52685e34826de50e82a4ab7f97280e8e5eff720abc7676
first message t

2148a17141fc79990cc77080b6c187a8401f3e90d677cafde1dd4fea2c72b1b4
61c18cf720605dc9ca61606fb18e431b6de9afc5b23b4abb9eef0dd1427305b
5c8a6c9702e553c10ce827c036930b6cd15445e179e5ea3b0dbdbcf51d336c93
bbffe9d9ae9711881f361dd3dd0b9cd433473e7a119df5679cc3fe8b4d1ed7ba
2791a36a1efb39c3a8acfe6c3457fb81ef79c5614edcb665e9d75eea3ec28e34
716f11f0ee2d24a8aa0508f30566862f98bd27bad2ec42e0a5e028792dea9c53
62ef37244812bb6acale2abc1620f7d453039e48549075151364ed7615bae94e
8edca2f694866e7ea2cac40ba4dde0c1a8cc9256c163f66e672be09f8aca8a34
response z

786fef70320ba61307a74d780e84cb79552b9fd534a7fa54b208daf645075538
7ca06771b9e6d10f1f098fff0bfdeb74e0176171e6508317d14c98caa35bd6b5
abb4a5ae792412a2e6d5c9e2b6feb6e16d0c7ced84aaa2bd1a7a9646a9a9694f
c14474cfaf199a38818c5a3667182e6ad6978ab4a0cccb483ec0857df832be3f
f5cf85f5071dff0471611e4169d2eb4c6b2970b3d107d38a397a281242389426
0f2a55a2f4dc2a67212f264d4009e8f28c26140293871313c239bc94bd09f8cb
b71f5918759f187dff27a8c6a5a6195ea4a9e0a82e2f954544843fd4355f9f69
7840dfbb855d46363e2a8f0663f5c9eb03c777ca278afcc6b4c152125f0dafad
response w

7664d0ec775057068c1f9c82eb897362e73a24c14685e63120238105b8b25a1b
ab8068ccb714e0b713724102a27b22d007cf5a8589d17d2aefbb1298908ae597
a1632431d109ba22b45eedc55ef0ea152f44ce5dc505c6bdb84128f5a6c5d7f5
33b9081fb2c492b896254ca6029f36b50e7bc2139bf82eb1fc561371bfa29125
7e55ca846e2cac78e29b4b5807c49b6bfdabe54e3aafbefc9201bdff5b4019ea
9353d2d870614474121c2fb88a4d921659efc1129be1916325bb1fb837dc9065
9d1d1ea52388b63cc0d2f4a512fdad7ff1414ccd3683fab430ecccc5befe3f2
993424569fadd4b5922c8a429f85392c4b1b508926b0c1c2fd64e7d124c28faa
Fiat-Shamir context

520bfc98bb350881e3236b1f1441a6e28b89a73353325c72c8131a0a939ee38b

withheld proof term: 363 774 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.toBlocks_eq_zero_of_commutator_offDiag theorem, Sylvester:155

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :
  Fintype.{u_2} m]
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3
    + 1} n]
  {A : Matrix.{u_2, u_2, u_1} m m R} {B : Matrix.{u_3, u_3, u_1} n n R},
  IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B) →
  ∀ (X : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n)
    R),
  Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
    (Matrix.toBlocks12.{u_2, u_3, u_2, u_3, u_1}
      (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2)
          u_3}
          (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B) X)
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2)
          u_3} X
          (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B))))
    0 →
  Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
    (Matrix.toBlocks21.{u_2, u_3, u_2, u_3, u_1}
      (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2)
          u_3}
          (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B) X)
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2)
          u_3} X
          (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B))))
    0 →
  And
    (Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
      (Matrix.toBlocks12.{u_2, u_3, u_2, u_3, u_1} X) 0)
    (Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
      (Matrix.toBlocks21.{u_2, u_3, u_2, u_3, u_1} X) 0)
```

SHA-256 of the statement above

27f02e53d586c26d27f5a732790118979072e7de96342cb0fd47c4e81f6bf190

SHA-256 of the kernel type

76dd681aa719ac6e64ac8a8222e121a7ee1497456cf89fc5fbc212999ab6175a

commitment C

40586f1c5589cb59c767da704d9888af91c3ddb539df384cad59d84d1a0d5667
2b4d83872c3a2fcc456f131c911e01654d65cb3e7ac345c89e8a09c64f405308
9a6bc9de687f8d8755b84e9fc04ae156bcc6c27921704ce7e81a427d19860c47
ca88329789e1be8c7aed59c01caae1b14ba27bbca75b72a5fe7a6424cf1b0338
cffbbbfab824a0f99685428a0cecb7201291288958173425e6c6a581c8f01393
835933e93370134772156a2218cea57818e2610f571d216d79fff9e141a81aa6
d219e1bc278c822723547a9ca29784f390e1fdd80262fa95797af22d8a38656b
d8e5ac42d71d3767cdd9849d759fbbe3e135919f8d50992e4c33c353e3d8fa14
first message t
d08c0b37bf3e42fb56ea59b311a09a318270a18ddbf85c570dac29ce00b5d9a8
f348e2c6262b3f87347702aad8399b1e9fb53797c4128a44e522a665d64f2d17
6601812b6ed999ff7a7e5e016c92e655b50151bc0ffb1f00a44ccf24fa6978e0
261377f1e47358d51c202ebc24cebc808deef3a5766abb79212928879790d25b
9ed0545627f07ec08080c82359d7658c9128b012ff3039ee85c245e4ac7ca669
28f3d0122b1a612fd2d3313366e25bc135d94b4a98d2dcfcce695b0348e4df8b
b1c043977d40339e9b64ea8e5db5d78bcb0212b7547f5004ce59e32aab8c4ff
6e36d3be00c5a32503f62d55d52e10ad696fd977458784913fcabf00350fa23b
response z
78fdbc3ffa457453f9387d6f72773d2a0753083b175412c71ea11ace7f93ba60
81c0635a8a2ba74a870eb5e2d6ddcdf81b1689ed56737a52104cfd0d24c98a14
7a6069427d670772a2e6b21cd93ae1d12e5aab5e7da85b03ea411ed41280776
4da9df1db628e519652703c15569251c5e78dc88b2d9babd39b8f81980edc7f610
a8b40aa7797b73c77e815da5fce110adb8dcbb327731aa897b44fcad9b58fdd3
c6e71b549617d11767ed423ad5b4d1aed11af9adee65304643c97c48ab9b7677

FBundlePoint.toBlocks_eq_zero_of_commute theorem, Sylvester:169

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :
  Fintype.{u_2} m]
  [inst_2 : Fintype.{u_3} n] [inst_3 : DecidableEq.{u_2 + 1} m] [inst_4 : DecidableEq.{u_3
    + 1} n]
  {A : Matrix.{u_2, u_2, u_1} m m R} {B : Matrix.{u_3, u_3, u_1} n n R},
  IsCoprime.{u_1} (Matrix.charpoly.{u_1, u_2} A) (Matrix.charpoly.{u_1, u_3} B) →
  ∀ (X : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n)
    R),
  Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
    (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
      (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B) X)
    (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} X
      (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B)) →
  And
    (Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
      (Matrix.toBlocks12.{u_2, u_3, u_2, u_3, u_1} X) 0)
    (Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
      (Matrix.toBlocks21.{u_2, u_3, u_2, u_3, u_1} X) 0)
```

SHA-256 of the statement above

bf0ba462a00bc03be0eff7591e2c6f09b7588c2d7a297161886b79143d31d15b

SHA-256 of the kernel type

89866e9d16cdc1d2443e21b60c92d04354d5e7047f40933dab8d6518984456c6

commitment C

51aef9cb9fed4307686ce345104f8816ea31c4a3ac53ce0def146ed6af9def03
e11c3dd6e43b779548f23cd54a08976949f33c998aa2300431c41c73144a793b
1aa91009b1a267f13e84e20b4bf3a61b1b7e0c8d612851e4f18267ceb9791f44
a0fcb0cd9a72f2cc15fe84437ac96860d990429ac302094d6449964e474ddcd5
28e5be4752dfd68774700ae246010bc8e62e497c93cd6ee9ef4b586321fba7e0
5678752d2dc53b8475aa32ce717f512282182846bc1a519f995f0d0b47e055ee
261e096870a62181d9475037095e69ee93aaa872b0266c7549334702fb6ed7e1
e1577809b4a8565fd1705547e9afd8705c8d4f7bd4bc3d571d40cf104de44b4a

first message t

7a906f4f693b46894b025dcb7e3fa1f7117deb915457d5d780aeb72410bd5a2
cff8ef39fdd15b8c22ecef5d3b32ebb4bd993681a05955f91a152dbab166358
2570b00ec5f71a77352974207403d3c3222b19b87612c4d3301986df5a73f828
5144243072b672b6ff3c72e2e4c027735b0b8271ca4e9eeff701197b77fda441
50fc7f38516a040bd221a470c7bafec6e2d465f2220360f85cbf4d5dba8b59ee
32c777936311b8683d9eb8e68490d9ce302d331309b002ee8813dd4c6fc3c9b7
afe5b0cefaee37f3e58e785e8bb7e4e281d289ba10e180e256fbd3fd72b76407
6125912c7b6beb16527672790434360c882f2e11c291797d482402cdac2e57e5

response z

1fb600537fa037486143d4c11ea8d4e3418bfea365c4968ce86d726259a2ba40
d2e92ece9abc0bd9fe2e2e62cce696c7b399462878f72d71e928d55b7c9b08c9
06b921eb85274c91897c566ee3b70e229e4baa446b7b51d77c9cad78935ff9ed
cd679aa17f7fe7ad56ddea0740ce0bdad42845f135bb7db1d80e5f154a75d1ab
5ad2615cac2a803a8697ed6ee8c28916681331f369250440a053f9692a9f9392
1689d1727dd730e7e95a32b5401e6a8348ecb62398479ec12884ded0fc702af7
3a453f184bd0af0e89b3044379f352ff5417ae207f42624ce4ab950b8a103f22
02b978865a7f46cc0c963ee76f15d05347b175a8452767f16f283813ede52775

response w

49260aad2eeb1d86c6d8521f36770d57f484e1da689317e61f9dbc6f1036db2c
c6515561e4e1fb6ac369a0b4f763467b79cc7386d22e2487e2712ca27e846441
d60445675a685ea1b32056a42aebae32a74d4d9c391b08abdf03eca96627ccc14
728c3ddcdcf03d9b0bf3623b51abffe037e7518b84d3e33bdc7445e392b84a39
50b73534fe527c9a669c33eb7760f61dd3c33aafecf8280b2f839cf41d73e70e
3d15b0cae2cb55c5e65b1a852095c792cf4e10d2408e42c1fb045b97ea48da99
a485f249e2f8e53f0518c9d1985a1f01a11f770f27eba686c1fd63ceb5680b16
5929e75458b9e5a5559f8dbb8bd488ad999aeae15403f53f0de6edd5c996988

Fiat-Shamir context

2ab5813bd857075705c1ba16bffe5cd77338358e75a5cd29da9b91e4c4fd79e

withheld proof term: 97 095 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.toBlocks_eq_zero_of_disjoint_spectrum theorem, Sylvester:252

```
∀ {k : Type u_1} [inst : Field.{u_1} k] [IsAlgClosed.{u_1} k] {m : Type u_2} {n : Type u_3}
  [inst_2 : Fintype.{u_2} m] [inst_3 : Fintype.{u_3} n] [inst_4 : DecidableEq.{u_2 + 1} m]
  [inst_5 : DecidableEq.{u_3 + 1} n] {A : Matrix.{u_2, u_2, u_1} m m k}
  {B : Matrix.{u_3, u_3, u_1} n n k},
  (∀ (x : k),
    Membership.mem.{u_1, u_1} (spectrum.{u_1, max u_1 u_2} k A) x →
      x ∉ spectrum.{u_1, max u_1 u_3} k B) →
  ∀ (X : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n)
    k),
  Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
    (Matrix.toBlocks12.{u_2, u_3, u_2, u_3, u_1}
      (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
          (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B) X)
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} X
          (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B))))
  0 →
  Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
    (Matrix.toBlocks21.{u_2, u_3, u_2, u_3, u_1}
      (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}
          (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B) X)
        (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} X
          (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B))))
  0 →
  And
    (Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
      (Matrix.toBlocks12.{u_2, u_3, u_2, u_3, u_1} X) 0)
    (Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}
      (Matrix.toBlocks21.{u_2, u_3, u_2, u_3, u_1} X) 0)
```

SHA-256 of the statement above

7c6b07a609ac117c1215a29848d0102fc187a8ec17c463a8ada70f0cae04aecb

SHA-256 of the kernel type

784ab35e107d8e22be316873acdfd1789fdf88fb122b86aefbe006a0a71e8bf0

commitment C

ed38cfdf380924337828f9dc418c93df0e48dda160b62ada609f385b08b13ed
ba277aa548bdc7ee0d433a6524b827c9951bee3d43c18ce8e95509e4c5a0be33
7451dcdb59baa2c2e7ac3289b7321390aae02d4ef8a40e5ef194c8800f1520b0
07fb8fa3fe06ad8eb7ecb791581e09bb9062a2ebd4f7f05a3964d923665d7391
c5db2e3d0922ddd474751824e7f55eb6b29836f2c77093b9e2c3bd09f92f2eea
47f99206ad70e6442b5ace8a181dac9bd55b89e8655aaf68692ca0ca07122c91
dfffb96587ced2f1ec7d4136f867b149134a63cebc552c30d4c085024b715d888
88c6ea6474aa94daae82b8e098f95dcd9bb914da508f61cc8f4aa297f32c0b111
first message t
dd040248f49964be41734ee2f9349e0b8bde511c44fd9d4c9ee33a10498f520
32caf8eb7066bbe63d06fc57a283eff102c5710ca21dd9f46b150114b4cda425
aa754bd8ccffbbcdab7dfca78565f5f503fc9cd52858437c341f03e428d83cce
bb1383b66c7a39c29cd7c4b761aa43b271b2728f5370e64c5a13388ceaaec786
199510220f77450b6b54c9e9a64b7c27118c5faf71f1f7e78ed480763373846a
af5692e3f94d04ad9ed21c92815961363a0eb0fdc688a68b0b62897ca94c8295
4489b94cc185d210ba58e080b9e3dd9c0268cd6519521e53234d65caecf2e261
d866079b395a94bafa77e1775b1d918a41ecf682fc34d253fc5d5b2ffcb9fd23
response z
1608a660b160d1e87640caf1d9c2962ff0d55fcaa7175990f2ce18ec93dcec17
88f05515052e24aa51fa2eb710d796041e16df284e84cbd0e735e24b5c9922fa
65ac311cf1ac9195d2b07bd6be81aa25a833c4ad44459519d4d80718d845c39612
b301b52bbe344a42eb9cbbac8dae584851ad4e18366cedb4518c7022df8d3534
2fe942614e2dc9e905a5174e63a9f86521b37add3d350e10975c48680e49e4d3

FBundlePoint.toBlocks₁₂_commutator theorem, Sylvester:136

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
[inst_2 : Fintype.{u_3} n] (A : Matrix.{u_2, u_2, u_1} m m R) (B : Matrix.{u_3, u_3, u_1}  
  n n R)  
(X : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n) R),  
Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}  
  (Matrix.toBlocks12.{u_2, u_3, u_2, u_3, u_1}  
    (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}  
      (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}  
        (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B) X)  
      (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} X  
        (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B))))  
    ((FBundlePoint.sylvesterMap.{u_1, u_2, u_3} A B)  
      (Matrix.toBlocks12.{u_2, u_3, u_2, u_3, u_1} X))
```

SHA-256 of the statement above

333050e2f2e52dbaf1905a6631327e8c1db4f7bd7c6881554f170d0353edf297

SHA-256 of the kernel type

fa4623a36c1115c9b4b2cb12e9a1974ebabf7e6558d07c07b36c7b2f3b96aa02
commitment C

2424618c4061c85d5b299bd85f590769ae1923f03a10fd3092967f7c2c7f46f4
c63aec82f951719292d62813424a0fbd4033b7ca24ad89812350910617cdd8c5
a35c99f17ada0f6257abf8440629d6cb7993b3d39a4182452b2c1571d7192a7a
8a2754453b14a6250440f33f879a3c97fff78b6609aeffd8cb9489111ab372ca
594f4cac46abd7736f6569098137b790cc6a9efd2dfcf1c6fa77f672f66bc2f4
00113e146a5c15677408bab6d3a6f0c76c06760ca3ed08ebb05cab789686b909
ef9a170ed0e659d1618f881ec21d24f0dfa2e9ddd1c8eef0e285ee4aa0d2441
2c01ffaeda6843011dad0abbfe6d1a30cd4e7f849774b3bb89c05a4c6368c25e
first message t

60f2cb95668baaca241e7958ee922440d0663026d3fcd527539152bdc9596ca
60f8610095b3d8fe4799b851b94ff97785a6633c1728c2331ff34691b1d61ad9
8d069db97cedf53758f91aa565b4b66909f05b03b3dae1c7c1a725862551c31e
f495a7ed25533b9ee43c2d50f085737197a4dc7d4d3c29e7026e38c850acabe5
e15766a1f6bf8124eee259e31842fb0a2f41be70ee82f30091f362bcf4cddc9
2752a84cd6e429a52fcd1735a13951a08beb4abc1d67188da947d5f28796af7
065b28a4739b7d7f852e89f97ab5f9afb617a1253ceaa30f3f34c012366069f1
63ad68100409b296d03fa5d9358c85a6bfa89b0ef626001ac204a291e06a41dc
response z

3eb49d43499b0ec7e21a03540580f8e098ae77d29758431227158fea9aeb2b87
50d8cb97c16e47a5b81280f869c43dc44e636a8d77b2e84dd4d49cbd7bdd239b
c41e4a0f35afc5223759d9f281718d5692ceb3073146475b572d044e5fd33b93
5c44683e1ccfd3ad9b4126316d9d524c3ac15d1eb16d8242faf9f62cc3f70c82
6ba560b8275c6f06d98be42a717f35e0e1a3ec8b32328822073885b301b739ed
b9e0d719ccb99db3740410b93cb06e8d72bd7c05e8dd4722da067bb9234bfeae
eb611b046f57f0bfdd10479832804995b89bec6580c634ea10ae275554f1399
b4b2e814bc2e2fbd08f0d8e18f8e0155f536615778892f8f212b309ca144ecae
response w

31f2c8ee12b42daa7c3da115cc5ee86da1e063b4eda0102b62fa0a2f7c3a9f3f
3fc2138e4eae53130e27a3ec3b1a1c470b0b49f9e9d36fadbb9c1c3d855c3998c
bf5892d24c396432fdfa7822430722ffdd540f81a5afea877ea66290cb8548e8
4eb3eae19b2a725cec21a5fbc133b59816fa1ddb982b67a08e7265bf0eb339a9
658fbc9bc1b87c662460bfab3b2d062f9c1555d54184ca5f1d1ccf7b0ef22e72
1dba3a2db2e941a93114997e2b2cc12c66dde6481bf4c7a5571ca057a63b2496
41608a3955da5d44fd0eb4d5a3e11484eb5e3cc7eb87f007510572e07210a4e3
f77c831dd223f9cb8cf50d1aff941e6ba10606505a327ef713d4b8d7b413900f
Fiat-Shamir context

78c5014cb07febee347802ad1090fc61245a8c02e400884dd8937b92b6f0f1ad

withheld proof term: 198 761 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.toBlocks₂₁_commutator theorem, Sylvester:146

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Type u_2} {n : Type u_3} [inst_1 :  
  Fintype.{u_2} m]  
[inst_2 : Fintype.{u_3} n] (A : Matrix.{u_2, u_2, u_1} m m R) (B : Matrix.{u_3, u_3, u_1}  
  n n R)  
(X : Matrix.{max u_3 u_2, max u_3 u_2, u_1} (Sum.{u_2, u_3} m n) (Sum.{u_2, u_3} m n) R),  
Eq.{max (max (u_1 + 1) (u_2 + 1)) (u_3 + 1)}  
  (Matrix.toBlocks21.{u_2, u_3, u_2, u_3, u_1}  
    (HSub.hSub.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}  
      (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3}  
        (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B) X)  
      (HMul.hMul.{max (max u_1 u_2) u_3, max (max u_1 u_2) u_3, max (max u_1 u_2) u_3} X  
        (FBundlePoint.blockDiag2.{u_1, u_2, u_3} A B))))  
    ((FBundlePoint.sylvesterMap.{u_1, u_3, u_2} B A)  
      (Matrix.toBlocks21.{u_2, u_3, u_2, u_3, u_1} X))
```

SHA-256 of the statement above

46d4eb0ea30b03fc4d04862c8f9c547b3033d533463471d1e63a7e0f01993a80

SHA-256 of the kernel type

7f839e66bbdc99778b016764f3b54763309330426b01bbf4c7b2c605b089f781

commitment C

cfadbc3d6568b20c0876300f056d93c6f701e77d2f9ce9aff6b6169488cd65d
199783ba34cb7a8322968e635f597f18db224d03c7a49cd19f325de0f03d86e5
4e31751e5c7583c3115516d68a124d1e16021c546d49d6ccb63197d1f16f0a5d
d06c33ea05fdd489236c914e58aef0bc8502c3ad6c69248216bf6373b5c8424
05cc8104260cdb9d69cc22d7d1e4996b2405b8ac3342822a24a59cb52f569ab4
19383fc27074090c080bfa7f57c5cc3369ff853c20235851915febfe2cd26406
2eb9426d39004a188aec22c01e98ecaa7468b7deec2c8e281a117015ea40a7c0
f203f8dd46db3960199fff1b1f538919eedb58a6f28bdd9d1cfa0d59b14e1288
first message t

58ba004e5ea8427854e783f04b045a701d2ade7148364e5ecd980d6277a7e4fb
176b7d392f200c1e26d077c6f3fa3a1f3a831245084777d729cc0afb15e81b61
c5d428f6257c8583ae0a80c8453fad9d8cb3d7b624e3ede5dce20529362cc90f
504ab17e99a6a34aaef5a28a1266530e92da52deec28306d60d2d10002d7e6c1
786041fdb5e9ae838a431234715906fd7bf7c29a2a46598f965bf291a6c55049
a561a7b7a091ad13ad00c2e872e2af821ee3e5fce61c7d30e1f1bf0b0af0c329
926b774985449b043ed186644cb96c654e766531686831fbf19671194ec3cb9b
9814374b78e66acf825978bc5ab85210614bbdf9eff91bd3c92161f416a8d4
response z

5f253eb1f3a0c8fc0687e1e35d71f387c95de3b03895e75f318d5fe6b27a1d6e
ae2fcca3708c9c068710d7051746fafa6faedb0ea2682913dbc3a31321c66408
c97494e95edb4088d9f7b164e86ff990c7cecec7cd3d139f8a5bf75e79a2137b
b1b745e6d5650e1cef2fc54109640e72b99b598b453f3885f3abce01a18bb300
14ea729ea0ba26e05c76d3d3a9659e916baa02d9f9a097cce4a35f945eb4d13
94e3a5fb4916fcd39f3f2dd03dd59b1c79f65871188d936fbaf609f30f5bb5bb
639a73237e2821054af0f9340c18a0fd41362bda0f2b09be24ed833a64bf74a5
3a01e3d3ab3210de2d16369701180188e939b3a281b76af1bb85184164024b2c
response w

1fafd2ca3726b931c68eea08cbdb76ca13e269d3344eae6daae6dc5e211750f
27af51f8e68f958c91f33abde88865de2cee32d3fb26df2e771f9d820e2a8f63
ad89a361508406c2ebb22b881634115faad7d140eb8c63977a14669b7c2213f1
ce304d7a63d79b6c4f94dba62b34770026f1fcde4b015fba334b5909de441497
9cfa27c01dbcf3b39771cfea7ddca99efe5c3790f6e7cf47b2d68a47389c16e
819b407695ef172e6dc9ae7942ac980e9856f33f657801a86fddb87fa75a8700
97de8c4dbbddd09f26308ef71a5098bdf5d77817a4db5fe48440fa37bd937a27f
3993eb91208e0661fd37154bd7e980406eda60481bb59bef499afa15d6148f80

Fiat-Shamir context

131cc0d572ec509977bdcdb284d2716723c500d4bbd4095f8e97fdf9e3961cfe7

withheld proof term: 198 195 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.existsUnique_derivPS_bracket theorem, TDirectionExistence:95

```

∀ {k : Type u_1} [inst : Field.{u_1} k] [CharZero.{u_1} k] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (C : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  ExistsUnique.{u_1 + 1} fun X =>
    And
      (∀ (k : Nat),
        Eq.{u_1 + 1} (FBundlePoint.derivPS.{u_1} (X k))
          (Neg.neg.{u_1}
            (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (X (HAdd.hAdd.{0, 0, 0}
              k 1))))
            (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) T))))
      (∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.pcoeff.{u_1} 0 (X k)) (C k))

```

SHA-256 of the statement above

65cffa963e101fcb95cbfdd0550a17cef2b9b07c0bfbcb8d1bc7bf9a9e19478e9

SHA-256 of the kernel type

7473fc03ea7d99205dacc0aae522940d8a4981db78202bc53299f7624e47e1fc

commitment C

f045d9260f656d14df2be4ddf29f0b586ab03f6131873f6d34b342288c4d86e8
 fa430a415da9c9b3f6a907298d5f344a3ba178f6502b64d28380b4629987a0ed
 c0c5d8f820f89fc0591a9a13373285b12f5311fe9a71d5b40f22157f71d44bda
 750d7008c473f921a641dfac0af60b23382553ca855b094a1fbf70d8cb0a3370
 f117eb268c4487d50a9e7de3dba4cfe444d4c1a73eaaeb51368afbe4144f1fa6
 984368ecca3df46223cd2cd550064862bac0e5ef3531e50119ee7a4bcd42f0e9
 404f16d56832c57ddfa5b509f021f5797cb5cf1cdda7311558db695899933276
 78e87c3dc4149b1ec32b7e46b0ac8906a7ee494ec1e1c615a1cb192e5d7dc454

first message t

9ca9e7605012543d6a71c2def396a317b2544616aa291f062ba89742c404093b
 fae69fae137246f03d0e5a02d48184d40279e1423727a2f62a954a2f0526df32
 f894f6142c3ac3cd1ccad2a8187eea572ee3204c2728492abc6e4344dd659cb7
 94e00c466cfa0bc3d90e9e74b5736258abc52eed4e80fedb215ad47166847987
 3add3ed8904702ca1993cc6dc20d2362569fc980bcb1b7427d70e19ca82b1bd4
 cfe4f549aff539d8590ea1e5c79a53e3a688174a63d53294114932aaa1744e43
 44f331a6e5d43272b175f3a81780f180640b2d1add49d9027f08d33c980b32f8
 65bceb76e720f96027b2b58cb3c3f1173bd189e7201ef0e8f91221351ec2c486

response z

393acadf6bef4034600f534fa6d6e7166db7b4a8bce727766e65665137798494
 cb790c2402c5bf4d27499412317e2fd28d5eb8c139ac1d6a7d0044920809fc38
 793194f1bb360c7de47430de649070ca7dda0cbfd013c5dbccd4132431e7cfd5
 dcd3371ded7fa3bb2edea4d4a7fc906bc17dde825ca024d64e91b91f061bdcf0
 ae43bb118e5803e3ba41c34aacc58dfb872f561d7387067babc3668d421dbf3
 ffe973614942be587bd32c58673ce0781f76a88962f56ac35ed022bbdb7fc5bd
 a5977ef181bfdb82b7a97d97837fde41e6b1c98a1f613f02e6a3289fa2b4424c
 743871908ef21883e7b1be8d4b2c7aebd243ceb0d53e09361f407262269a855

response w

5a31505535c6ac4b46f37b8fbfc7379e3c6e6cd53d59f863783181c506e48d33
 86e2b26fac0128d9d7f1e1dfd01d911f558ed2f990bc5d87c989c2e3309d2e49
 f28b8c6c2f02c68e54bd4963b26e089718a07b6497a8027a39f74c3d516bd53b
 761a0c155350b076a9851ab7339ebd3fe0119d4240108d0db75d2c665519f55
 014a9c0853d2b9aff2c3594876c08324afbb021831764549c19be1502da21799
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 89722a57ded9d71b7c46c2ca0084289ef1741cf17ec20464b35bf9f42c09f3a
 0b08ee66eb7fedcd6228e24119af0981b0bd1cecaac719268ded689918746c69

Fiat-Shamir context

b69550880e8dc74611ef92d0745e459b3c27499cfc0c0e8ab893a7b0b69303af

withheld proof term: 75 553 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.exists_derivPS_bracket theorem, TDirectionExistence:79

```
∀ {k : Type u_1} [inst : Field.{u_1} k] [CharZero.{u_1} k] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (C : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k),
  ∃ X,
    And
      (∀ (k : Nat),
        Eq.{u_1 + 1} (FBundlePoint.derivPS.{u_1} (X k))
          (Neg.neg.{u_1}
            (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (X (HAdd.hAdd.{0, 0, 0}
              k 1)))
              (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) T))))
        (∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} 0 (X k)) (C k))
```

SHA-256 of the statement above

00cbd671a9d8f142118c9c90d62c8dcd13284e0f96ad10fdbbb29cb61c54229

SHA-256 of the kernel type

6c7951f4da0779e01740d1bdf9484a275424922161548ffffbfa6417d8c51dbc
commitment C

245b1c941e9c0e187b6ec6226cabba1d77bda6727347a3aa4dce3913e15751c4
8ad4363e6815698c4ecbd8a18626ebc93ad51909d98df7a143e1f695d81224ee
6735e38d9aefc02c3411b360b976d3bd8e7b90c8a7628544a7e55b8e44f93ba7
9e79f65bf69d7d4501ab25cf53b065b8379790e698ef93c68507db205abdb65d
fb8bf1edffbf349f8e06b23cc9de38ff4400472e34f75bbe84bdea92f13f651
ab09b28a37289bee04a8d1581d4073313a061fb213fcdc789d8f031330169b6e
599a55aa2dd65e971e3849d92afd6c170cd53315fc10b257cdac7de4755800bf
bb2af6fd55456c988b219cea5c050ffc40d8fa53f5527af89d5c7fb90c5f57
first message t

6dc5eacca3b3119383b257d5a58e2a6a7736f3dafbc85706689d47a3eb79c795
eb917cef89caa958f9ea6309bd7dfabb4f8294f8dd73f9991fabcfdaaf874d43
557564be0392dfb38dccc73bf90474b7d10edc6948d064cd9c7b283340c7e8e
7459f6ed8e7abdb954a585a4d0760f32f6e3265a992ede1f6d34bba266ea60a9
3f37d89d6440e95c4d7de49e885e1cab46d48bd0b10f8ea964c563da0d063a1b
0ec0cc5592981285562e4e3d5e2bdcf1ac337235e4789c69ec8c1a275c0a4fae
a5a9324198206038a92833cf0e8a7b232bd953bb8dfe50527f5cbbd7961fab14
3f56a14a7884abfe0eff9c790530bfb32051f55a5f3a8d6839386e22d35fb0c
response z

6a97c24da48959a6ee8f10553f9b4c2aed14b3cf3e08bd534141743d9981c30d
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d3e8656a417c8651b7cc088d7c625de8a559962005382bffc6b3ec14124edc90
8b7262ab008ae9cdac96abd86257eeb1d36b78ca63842699ff714c224caf6644
b9d243386a1b65248cad3ddce6f227caffae886e648acc20b53df38050ed4ca3
4539a4ce17fee6d44fb8855dc88ecbd39ff27edb4e25096c039afdc74c032f8
9f22c99969836be44074c76ed3a8be8d9dff9a5d11711536d007c31c9a96ad4f
3c997b94992d865608d49d2785e174ee54da8d2dd2267e5016a55823989f6ef4
response w

6806bd8f3946556dd5847825fbefbea73ba4b2f54abac5752ac0b3d9d00007b8
739bbd37b2638f15a0a07ad5b4c8643c008730b5c224b1809a93e056e15250a7
a8f5d4f9b95e305362ecbf9ea7b3e930aafac5e2f17af0f623d385d167511252
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c15be760ecc092c73a670602f9981c8f288f19215f0b18831a305d2ced7fcd90
24ef9ad951b2312efebbc604ec039eb17ed1e8bb204ae1c6a68544bcf93b180
0f30ece1ba27f995dbf5fd5dd88750523594b6b2e82c4719051774314ccc0e97
2971e763dddec6d18d631bd390a4c57df9b4899be32b05b939a1e2d7aae4e0b3c
Fiat-Shamir context

869c696943df44749e48c4bb1ddaa88cc66ea03e013080c25e69493328323e86

withheld proof term: 497 565 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_tSolution theorem, TDirectionExistence:53

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (C : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) (n k : Nat),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n (FBundlePoint.tSolution.{u_1} T C k))
    (FBundlePoint.tCoeff.{u_1} T C n k)
```

SHA-256 of the statement above

cc026bc9e4c1e8ed3dbbb3ce1081c681c079384aa92087bccb040549a44f377b

SHA-256 of the kernel type

ca2f470ba2ca4120a1aa1048466c924aae184e6a9756b3beeff9441602f013a4

commitment C

75c3affafe392a1cdc31e91d1fffb6ac4dba8d49e013fbb69fefb10e24d7cfa3
e6e70175d70e2379aa7e895f5e57149160a683cf66debf8e08e9cee3749ea64c
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7d00fc7cdddea37f1021f88d2349a5fb4e9e6e1ab94ccc3925a2865133e34bb2
dd04e1466f0638c84aec6afbed9bf6885ee952c8a93fd6c661bd5fa88990443

first message t

68ba0d1d70fe5848e4da441c4cb1d6c8741528964fa9b22e3321e6868477a173
f99d1986eaf85a03e71004eab9c658048ab9b3e4e7a2a2102c14433dddb35d77
d012ca32f49ccaa1bdd33615f5e5ccdd2da134859ff13de9088aa5736e46084b
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1453a64a80a45648fc8ae01442ee9147010aefb428eeb5bf4f1d0b6c67b3799e
564514b594630b56a2cf6247760a8e5ca54fa2b29157acf718ae9a7e1e8632fa
80dd676b231a4bbbb6a4d4ea626f050d0c665b099bab01e1be080e4bde2cb6

response z

4b7e07de5038f87e698f9cb13d0ca30db7a5ec9b2113a47c5f042b6a273a7a32
d02e941cc367fd1e0a353246c504fdb01d2b732172421ccc5242dacf89243a7
1209ee35ec1b91ee151223f70de33b71f0ee195e8447a39c287cafd6a2ff797a
af40ad795f2692bd3333166f9ce41e16cbd14b1f22853c416c63c57e9a05f1d7
6de31632e9a7a90ff8e6f713a1f6b1dbcce8772cc99197cf4c4f4926fe05d1ae
8f7478eca6a7a72bf29b59cedd659620a987515d3bb9f38c34559eef9a722040
e7d4e9e0aa42f5b43f87767602791ee6a82af79921e1b9fea2c14071f7b0ad86
9735f15e150728c72b2a00ec3537794ac8b0cf62437f4578c945bd73ee1df455

response w

78262a174bd7c1a25c396eef7151aadb7f4a398412958e9866bf451292ac63c0
c2373b9039aec9dca8ad750c51a764f881a79abd0deb1f249e32ab8572828cd8
064bf3e043d0c7f9f96fe2e417d94a34af66f024368ec5204ba1a71db4082f71
2f0e51c082eb4d5828513b5703c738836bdf21a92d169c529306c487b4a1f600
43a97e9de9c2fae32fd6413537113d2d1b3832de88694cb66e75abf21fea4250
27efd9618d1ecd618f32ba4b08a22175e5aacb874ad0a39a06fa729a3f36ab49
3ca5ceafa36652b9b6b0f6094ef0d167bf4f57228f87fd78a0f4a19a64a12b6e
2e1276462a7e064812298e9d3679eb129d3f610ca5ebb3d4620539821bdc3499

Fiat-Shamir context

7cbaf49594bb101c458d6d4d1302997310dfe9dd01194f1e1b41fe6cddb8f7a1

withheld proof term: 3319 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_zero_tSolution theorem, TDirectionExistence:60

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}.\{u_1\} \ k] \{m : \text{Nat}\}$
 $(T : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} \ k))$
 $(C : \text{Nat} \rightarrow \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \ k) (k : \text{Nat}),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} \ 0 (\text{FBundlePoint.tSolution}.\{u_1\} \ T \ C \ k)) (C \ k)$

SHA-256 of the statement above

d311d0212bd941496c48709c8880a5650ab938f8a6c367c5eb9b18b040cf7cb6

SHA-256 of the kernel type

3f9b859c089e36717bfe6675d3f477f07ddb4f171b854d54541fcbe10b860f0d

commitment C

d3c94832a037f7d5c8bd43abc3a05b08ea5e1137cd132e9fe8da3dfd1d1e2bfa6

4900aa4e53cbe981073798c4835d7e83d32e9ca7dab71d52e90113fc9badd872

81feb182664ce2f6b8c88da8cc1ac5b90f8877e2819b1651cdb53cc2a01d00ca

370bf815dbd5030183eb6199e2608cd2c30a7d5e5838106b2d089864294d6a75

80657fe99598f6ed99818c7f47e8a6a18ad9809124e79f485bb8a80854bfafa8

2cb81b19a43a6afb62697c3a1149a573e0e94b45bb67b46d68910de6ca9ffa30

9134beba05f0b5f08d8529cb5b8e840e275c8ab3d98c7f765a210db1e777f59f

ff95ef08e8b540d3f7206c91fdaa9e1defe89bcc9e91cc610ed33cccc3b24874

first message t

f3e0e24754360f8efa2e6d1f84880431a467aec0d6294784430d553f813470c5

132f70ed71c3cf3bd66d24eb743cd635279d4caffee52b3198146109fae4f87d

9c84d33300884ac74e8d59a6858b9863c06e12434f6154d57317ea27142b4674

b32e376144e934b5eb7395fd3a131f48b1c10e524db07d8c2688df364a453282

42e9bacd1f59d33262a2e96a08ccabacf8db868a0f7957f138feb896b75886fb

42ecc0499387ce7babc7d48d979d90190c9325042e30fcfb46a6a30d57729c1

5951538840e966fc910ac2fb739da513e0ca6ebab56fafb0de11090cf18281b8

323a244f533497c4387c231f454cd236f8fe43b848d6f56a29ecdcc1bcc1b1b

response z

2dde89c879d94943f0ae6e90f41d634d7bc3c285cc57fd254105bfa22ea8add0

7f8329a205cc27db8684bb33f6668c87e8e321c4f84c1d3fbede7cba207d0c51

800acecf9fdefb791f4f4c9924929ed57b0559ef3a44e59160451dd0ade09dd2

9799d05058beb6202f4bd2bb5281c3be16600dc1d7b693c18f26f2b8975a900

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5269fc848a509bbf425927bb00739cdf68146c04ea8b126fd514b223d4c7ed6

response w

3c45a5f6a48b0fb0615e5219ef5a31dbeddae00cedc6f4c3e2b7be63fef1ab6e

b33474a9c761f9b375157aa91a4a2c75bb3b6ca2b76f226c655ddbc76c179c9c

b55755601f9bf8318df9c3c507b5f2d4caca8f8899ade5e5db11b17b04ba70f8

93a3cd91c6dff551361cce39f33265233df741ac88a24f2077511259279631ac

eab2bee395454293c78cd00621c49614659f64626ec15981619a2747ffcdac0a

7b949ab18bccf3b032c03c948ad73705fe4b2a3e1a9c13fb9ebbe6878092a7be

39bc52372dd332c3f41b5824c95f08435c4a2edb672e07ecc3e66748ceb7651f

7126a4a3cb856994ec1ed9c9f26ccedc66e2143909118e84b2d074d7263f2e06

Fiat-Shamir context

b81600870cd1d3497bd36e19879813e07e8dacd06f468fcd4e354d0467f788cb

withheld proof term: 3 488 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeff definition, TDirectionExistence:35

```
{k : Type u_1} →
  [Field.{u_1} k] →
    {m : Nat} →
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k) →
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) →
          Nat → Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k
```

SHA-256 of the statement above

b6d42119081bf99c9642d7716cf6597cdfc41efd07b01814bbbbce414c0f66f2

SHA-256 of the kernel type

f52d51e459555e808be4a14833281c6edfa9513a55514e74d7d5190deb8977e3
commitment C

22d81d28a1fdaf22861deb3bc820fa34054e5b9b1980d5dcf1050f343910510d
5dbfefa2eb643341a02f3e8c47bb7244c04ce6ba29bfd451146af15046568855
d1a8612993cca238b1101bd7d76c14d881f5d1bfd8883e6a668c2f1c8549ac45
a47965a73532653a6e12916a3fadfab6501edecbdc686d9e3ca53baaef4c829b
ab718c493c93e24783179b0229f8ade50a3b05ab89951ed9a8d383b23ef07354
28d1726fc2382c1c784ea8b085382762f408dd85d9dead2af9ffe564308beb7b
d804afc51b0ebb8b946d2ed276574a4d4ce37ea34c9ac30da9d350d9fc6c1b9e
0b27e29b1f2a993076f50ba1485b660237bc4df7e4162760185872ae6e33b600
first message t

5d33fb4bbe5f47ca1c6d38f8379dbc5650c4cfca9650879057f7920be99709a5
66baed653a390ceb998531625ffaa54ace45e23d95dcdee01dccbe45d56ed6a9
20ab1b403258c9e9a3c9f6a1d74d9700174eb042a7087476e2882697f30a7861
cbdf5fb8d30a81d4bfb6db9c5a7b4d037f4d49d60cf3b5ad2a71caa34afac6293
f3069c57b00c94905127f8f151b2d74d91b21340242e9904fb0cd92a6d5faa9f
a599b07b0267b3f1678a55ff16164eca3113446d16f8c7147eaa3c8a4e630361
efe1a91cd50f64e7318d560794bd8d24b469a20054e6cd214b3d44b2793556b3
3f85ef4f8bbce895a35daf7cef69acecc41421808fe251d8bc1156dcdcc9f037
response z

1f6f1b966b3e62b4d92de15d12af30a4a7a0f9276051c219c726be7ad4b17b3f
731f2c0cfacdf11c434d38bb384260faed09687ee1edc6ae5b232d1c99a196e5
9965b0486eb4c975c0d2650bc919f08535af568b0a27ea715a92e07d8ee300c8
4199726595045a56328308f700c5451b84e9966aca150bbf70cc7aae9efb00ee
eab77210fad2086c2d527eaeba2fdaf89c205c5e07e7e8de75fe293bb1132134
dd871f9ebd8ce3d02c512f7a255fa87ef37e11769b2a6f602740808ff0544be1
c264715a0707e50d1ccf7c5ad722cf45255e73f3ad42a3267efaf3a6e5b455a1
e91dc7c9e4bfcaafcb1c5e1b73850afe94671452e5d17d850f7e850a87e63412
response w

6226a13333be547c2471cb5ab133def16db642b4844059fc51376dedd1255293
ee922727851cd3c60c19665c3dce95c5d9faefc1fc049c1afb3128fb4ea4ffae
725c6f47b1086b42a4757dcb8b68a46f459114ff57858689da49e5e0ac10554b
e1f880ba50bc64760316b0a97b2ae4e4facd4f91b7919ef567320640f63828a3
7062851c3de1fef258523faa22c99c9b9fafa27c0b0ef826300862d6f5110175
2274f7990c38076afc34ab674ceb8b99cda0a2d934a817a6190a2eba6088fdd9
fd0482381db4ca642210942c1b733d58b032515cb0da9728afc52154149a68db
b76e29758deb8da0a09373a7ce798927ef68a6be5ab46ea24184ae91ae738783

Fiat-Shamir context

88886120848f0c55f5f0d082854826214c97b7b12e921fe53c8d1985faf42664

withheld proof term: 534 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeff.eq_1 theorem, TDirectionExistence:0

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field}. \{u_1\} \ k] \{m : \text{Nat}\}$
 ($T : \text{Matrix}. \{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}. \{u_1\} \ k)$)
 ($C : \text{Nat} \rightarrow \text{Matrix}. \{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) \ k$) ($x : \text{Nat}$),
 $\text{Eq}. \{u_1 + 1\} (\text{FBundlePoint.tCoeff}. \{u_1\} \ T \ C \ 0 \ x) (C \ x)$)

SHA-256 of the statement above

2bb87427b1c18cce0b4f117985583873b592b5b99755f0d2cc9905657bd1ac1f

SHA-256 of the kernel type

9b7e4de12d88072fb39a4c9bd481fab5342667d6d6038bfe17870d235a5cd80d

commitment C

52e308ee8c9f0871d921ac941088f91c818d759ae8516ea3ec14177bbdaa0efd
9675ab1b4f0b8779781c3f17342c104a72b64ed51793a4d04051aec532f52e6f
f758cf92a91c3b3c24f9779e8642ee9ee5bf4c18c5dde23c2c2059e91174b5ae
f21161229f41992c20033211a8db2ae51c3701759e693e0f27bc31526336449d
d9cf48f682c5477163b61c00b6b7b70a7416187bdb020b15cdf2f922dd0380f6
b882417cdd65895ea9e06abd978d7d5fe39687a25641555a82f644447f54a240
a140ece5dc2f91c84d8b27e4dee8ffb137b8f14127091a7f9c45d5687225e238
417ab86fbc28fdd9297396511e5aa8bc78dd6db92af5eb0085acbe799d56563

first message t

a447c35a609f0bb3bbc6dce9ff7df7646072ce94636ddeb529d157a3957342a
4a1acd9c996544f24f739a0e260ddbf113e0b1c4dbd79718be3ff41b607ae98c
8d984b30245c96d78a483f9df8e2bc752925bba10b8de0e8446e26eb35f6605a
3a3bbc6c6cf0255c1209d054b8dabf58b5709ffee2f7f55dedcfde7f428f6903
e79e80b3315b58d42dda987f7c96da233ffc0cd377d818a2e6947d5c62f2c029
ccc8bc5dcc08c9ea9f09cb3c75c5c4c63cb072c7b2c1687d2c18d4dd7accd440
61a448903b720bb127f31609ec13912b443b92049f3df80979ea63a8c02b872c
4943d4ebd2a2ebb63af88bc3cbbb945f01e19e7d94bbcfb01ed1b43f19bddcae

response z

1d42ed7400d7731c49bea5474ff2e3794f435469d4a8005038f1a003f0dd4c1e
b0b05070f108189e8a26d6e1a58998b659847bf07db6a646c370168933345f01
ef063b1aa666086829d5b5063a65de438fdeaf71a7fc499745a3a25c1ad4ed9f
1ef52e0e498e7831d8939eccc730e0d0bee43cf9e9f1f755ece2ab0dc511831
a094a7685dba4ece533a820342a416f984782f4c1049536d015e5bb606339b39
aea243e646c75739a0ce9a62c988107474130c3cb4114feb060615aaa3fcfe46
66accdd53b62eb2b423bf3972eb6cfc89df9156907f6b897962ed0257cccc4f
7aef802951e7b744d12ef0bc43a2d0af5213cd5f974619e340e5cb6432f198f3

response w

4b91144dc3bc035e2b5b2d0a468e83e3ad048febd62638ce22b4c66c7d0089e5
7078227db2e049aba36e304641b2dc4b517a2648f19ca0d6a35224b8e1d5f34d
168892c95adda7b27e612001f65c03dabe5ce195aaa2adb373fdc62c89b68d2c
bbb73bcad3b8a840bf2adcec87d8348bb0375858e159cd182439126c9255443f
5be828505e6860f1df5b43d9aac2afc6d31d2c5fb00f1e29826c6cb624ae68ae
ac6a7ef28ab7ca08648ec91d7b292b0bc6579d523ce72c435fe35810905a77d8
675932a7420c1c24cb5611e3b34e0b9b7b297df8c8947b535348e374326d0040
49f597242cc70fe8d102c56ad8503dc26d6f8e92e75f14c9ff8812065c7d0f31

Fiat-Shamir context

65a1f3b64b6d81949fb4f3b40c960ab617f58e4e7aaa344743f80b23cb554a29

withheld proof term: 341 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeff.eq_2 theorem, TDirectionExistence:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (C : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) (x n : Nat),
  Eq.{u_1 + 1} (FBundlePoint.tCoeff.{u_1} T C (Nat.succ n) x)
    (HSMul.hSMul.{u_1, u_1, u_1} (Inv.inv.{u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑n) 1))
      (Neg.neg.{u_1}
        (∑ r,
          HSub.hSub.{u_1, u_1, u_1}
            (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} (↑r) T)
              (FBundlePoint.tCoeff.{u_1} T C (HSub.hSub.{0, 0, 0} n ↑r) (HAdd.hAdd.{0, 0,
                0} x 1))))
            (HMul.hMul.{u_1, u_1, u_1}
              (FBundlePoint.tCoeff.{u_1} T C (HSub.hSub.{0, 0, 0} n ↑r) (HAdd.hAdd.{0, 0,
                0} x 1))
              (FBundlePoint.psCoeff.{u_1} (↑r) T))))))
```

SHA-256 of the statement above

051b84034687553e3e16d6e6a66223c93e42f6cc7823d7b1c508696f8f98c629

SHA-256 of the kernel type

c9a3a559bc66d87ad887ff296608ac9d572fceda37f25f16b6e12a255119bb9d
commitment C

a9539d58b0b0f8cf172d09c748b6f0c8e5be4a1ce5203fd54354f076bb4dca8e
30c8d4855cd7e93fd7a3f23078e816f5958bf42017b925facadac675eb97b18e
7b34a8a57d824cb508e31ec2c31948bf15d8d4d5f3cc502583eb5cf7463eb4a5
69a584f977a92c6814938408c5865ce7e59f889e4d478a9f3eab09f0e5f53a83
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affe1ed439663f41bfd19fc85f0a862d48ae5b0c7020d1c4df24eabbb8c71b9c
79ff0ea4a42465a040ca559a47eba92caf1d38b7c82a3d25c40a32d0cdeb5798
eba8bba368c36893e2d06d5401c2d1e974bf11d511ed4a89643fb73f0e769a8d
first message t

ac3174a45d2b6eab52fc10ef1d13187a248f9063871728ac2880baf0f5d05c8f
4c368a5b3dd485537174e10cca3ef7d33208a949af589c89f0ca4b1b6092d1d1
f0724068539131aef9866eec3f7f49e1af5c2cc2611d62c1e8aae8c6807d660c
a52ffeeb38a793ae81261b97c03b45349798a4e190e8a2a7549d63fc8541a4e5
7ee9a08661e8281f948ab7fddad03581a23ba9c3ae701fff198c69001408166
4d9abb08d030f3335dcca92739f29aa873226503edc6d4d32d12e346055d93
c71344d2c026efc09e17d2a6c1831183b28be6b9febcd42e8b283f399d1985ba
e60717e8331f1a2d4167bb7d3c7b09293b544cd997a50de6bf97e29656aa97af
response z

79232031a21f19639960de6ae9ffab5dcbe7b583cc05db138882bc9e553ef289
b45d827f546d32894aced23368432508be9401a0cf13ad13cd346a0f4078d03d
50c521f1677ed40631b55fae65ae0c791e13cf3792568d750114c2ed36df19dc
d00426a82391c4567fddb6f8276202ca81ad08d6871c42448873ff90482e609b
746f960b054a3240ba781fd3128244cfc221826a82044a5f4dd90500f58a4b
f6dfdd0cbc99ff1bc4acc5acf56a6026ab1edc5f7eae1fefe4bf5d09a7431d7
60ba67aec81479dff0e3b9d5c901b1a12c5b43731ae6a7ebd0bb42c61f7f230d
f3ca27c2d9aa7fa48ed682148fc6d79a939bd434f261e0dd94041f5f5f36752f
response w

2d12b9bf50cfe1b24e8999869c8a48338b9c515f4e75c0cc6b267c14a4ca6cff
9d31dfd3e80daa7a9b1d5c7801f2140f99d012dc45ff5723f3e20300f7bcb5f0
04c272223afc0b48628c10613307e0f5b0663ab1e672d5d5c787f2177640c88a
2c1bed9f95dfde68e693b7ae6eeb9597e5e0f46acdf9d375a94002d4373b3c92
297d1b01364c4a10406c49c728bb70937761df9bab3d6e1ed8946671e332735a
81168aab3fb96373d15d80f00341567fa695a71e4ee575fb915603ed598bbe06
3d62f72f6c8e5b2e887558df524ac8a17be0a0c092039bed8efc10df7ad04070
6c4ffc9b97d218b2ccbd1cce0927bbb79696d4a8f72f070bcb75480fb64e19edf
Fiat-Shamir context

be34435a165f5c7ce0f1be18272065e1bad412284c7386f4fdd6ea2808121ab3

withheld proof term: 21 880 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeff.eq_def theorem, TDirectionExistence:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (C : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) (x x_1 : Nat),
Eq.{u_1 + 1} (FBundlePoint.tCoeff.{u_1} T C x x_1)
  (match x, x_1 with
  | 0, k => C k
  | Nat.succ n, k =>
    HSMul.hSMul.{u_1, u_1, u_1} (Inv.inv.{u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑n) 1))
    (Neg.neg.{u_1}
      (∑ r,
        HSub.hSub.{u_1, u_1, u_1}
          (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} (↑r) T)
            (FBundlePoint.tCoeff.{u_1} T C (HSub.hSub.{0, 0, 0} n ↑r)
              (HAdd.hAdd.{0, 0, 0} k 1))))
          (HMul.hMul.{u_1, u_1, u_1}
            (FBundlePoint.tCoeff.{u_1} T C (HSub.hSub.{0, 0, 0} n ↑r) (HAdd.hAdd.{0, 0, 0} k 1))
            (FBundlePoint.psCoeff.{u_1} (↑r) T))))))
```

SHA-256 of the statement above

3ef5b6f60f54d026f2c771433d934bd8f5644f081871ea509b8fcdd22fe9d2c4

SHA-256 of the kernel type

d8af0a9efb108fae9acc309164df9f6ac365a9fd438e81331c7cb150dfa5137e

commitment C

1c3ab9a5c8ac5371a64168751870e67f95713d5464c0ac2a226c7fbb43123f21
887c76f0b49e60b7fe192c0fe38b5d0122f8415517b1bc293c5f477fe67c4555
ee2fd31a2d7a291b8c039d2d82cc894e4b370a62c1728f7f8e97825a08519ab2
1dcd6378be5d5e0ef1f711ab5ef7269737e8fc15a53dac6839e147fcfa638555
45d5b77ee3e55c3f39773ade6a62f0abee8c624aacce72de34e9080ff5c29b6a
81afc5d26d5b68a7f79e827f017f455ba90b159ace5af904a6d9550bdd86bd18
35163e79c6c111c26b9a4c73a770e8b8240ebc0f1c235491626e228eaaddb25b
39a78d278c7c0a9839c2f0dda9d7d142b311fb3761f2e57dbd1812932faef5c

first message t

ba8634c2ab39f73e799b9db7b14dd4219738595820af5eb0e31ae114ce08c9207
ecf0bc29b93837cf70d3e0427db6d8b236a15328ea29aef9749dd2105487eaeef
3a69d69e400ec773db0bfc66d97023326218694f21eade84583181a40b48a1e
e55b20e72e4be154315651d1923be744fdef7083b366c245941e8975fac7344c
c61c1b2093363fa94224082032e5ad8ee89489da1f9975e549d8a52ef65c9617
e709fb165e0bc5c23fb49be0ed73771b35340413d29a0a2351ce053d3ff0ea93
c920569b7afa9c745d212577c6759a13e50a9837ae4d5a3f8cf04222ae3de43
c8eb42342a78800c7c7e064a46233752ad9efd5c1476bd033da825df9d4fe85b

response z

e8c05a0854d2c3ea8d868b348788b444d6ce629b46c78f091aa305b792701db6
2dd965a938fef245cdfb01cfbfc96b35d11cbda8e4702bd3a3cb6904270af51
6c5fe0d7a4576231c099c191be998e11ba25cd206106dcb8374cd07d9477235
2d318a882faa189ab87b8e4f4b9ddb8f2e0b95476862e80968dee9e5c9da5875
bfa8428c8a662be1bb93924c5b9c70641be3ea1274b64d896f7a9ef559ae4dd1
09f4e0c435207738bea4514cb02012adecb3dac99eb6628607371a2836579d0c
f16ca734bca3bef8252d3a3e087b6c2aef66f2f04fbd4897544a3c8bbd2aeaa
12374365ee95e47b1a49a0fb65a63db7882b141578202d3b29f6c74c8c32c

response w

3d5f6f64cce4a147e58ed9299cdb774cf729e9ad0e9a2b8a50eaa3cd945bf46
7064cf1786869538216efe6915b3bcc452b9a42ad1015012ff3957603f1b101a
be890111763c62544b50334f573f9839120b171260cb1d9d236ef21abf2a091c
5e1d51336fa0648ad1baced3c69fe8f3b91993e92c2255374edb4da0bcf6055f
d951d59633d2dead90285d225002a7aba59a9ac0112f6cf879a0ac4dabb49111
c0c96ce9528489ac0a5d21144abd9767b4a6501c3d4da5cdcc431f3fe6964b3e
6258fc08ff62f8786dbe307cfbc74a29633fd94a91937f85a795e6dc05543d23
7224637bf639eb5d86910ef05363c31fb5ccad384d1f523405d370d936712c04

Fiat-Shamir context

837fec36b498a7b56dbf4dacf2e8e0e3a154758c667d40a9201d414a7d198f42

withheld proof term: 9579 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeff.match_1 definition, TDDirectionExistence:0

(motive : Nat → Nat → Sort u_1) →
(x x_1 : Nat) → ((k : Nat) → motive 0 k) → ((n k : Nat) → motive (Nat.succ n) k) → motive
x x_1

SHA-256 of the statement above

a53dbddf92ca72478fd3ca3018af758758e87f16cb081960f2a91904554cbeab

SHA-256 of the kernel type

97336e76d112726fe7e4a8f43d5e1f7a137b3ae6c005d0dc282c6410b3db45bd
commitment C

c6cf465707625287287c060c49e48f934edcabc37b9536f83da06872cdf40018
db801d356a14f2e1acd8394c55da3f76d9e1a63fa17600d19db63981c40fe047
d136084806e3d063bd97db94c058c67131102ef4895b7f28350dd0781a004d08
97fe718f931c6d8388a45edff35d905829325ab9c663761f3c3b44fd48339aad
d85faa4eee317f9fa5c2f1eb87286a2881e63fa083d5cff03fcbc122a173bb9d
1703e7d2135b5ab56099b5883cdd6be0f61682e74cd16251601e44dfa60592a6
2ea4b72b3f85d9e80fab22337f13ebafd44f0d0491fffb071b21cb6a8bca078c
fe80d3ee6e3742ec0d40826b4b7f49f423e6754e2b8a81cd4208282cc1bfd77d
first message t

d397c26d9c20ad1acdbf926607cf30e94c4a32a764c2a5ef7af320c726aba34f
28f96a762a1f5c0cc66cb665a60517ca2795de708561f11f55797ba53d4844ac
54d4dec317c5f86b9552719fe74d44ceffd66fbaeae9856a4748eabf64cc7203
4fd27027e9570fcee27323ea1e8bd04e57ab0e7c5fe6d47d9862db8144321c53
bb17ace603c840f215b5c050468565ad99f9525afc494208302b7fde58854b40
3f04101f98f114c23594b34ebbfd025af192e7c8f98cd5b3abc6a686a464452
ed2dfef91c6f53976901db9709e89fbdade1ef36e9a0c42a2e7f8ef28905fc2e
8b3f3a317beea153ec1d963b3bd6feceacfbf97426efe1e6df339b16e84fa48a
response z

740a059d1ec77c7bf340b90df38efdd715a4a0c1f018aac2eacbc6dca6447a12
79b929c7025e87db90f34d9556ffcec74ae7d45098d939098f711e88b32c6f32
655aee2f9eda54135e559def85a8117eafa6b1a6d1a66c5c2816f0ff7dacbe91
0cbb03704c7e71f4c451bc00017433034af88167fc404a1b0e3d1c0b0abe348f
c5e3a4e43840a2420d8c8045f8c1b519f9fd454e0d1317fe120981706d38498d
9602e9ac96330262297385968bb0ceb0a0542a74493ac29bb789ee416ecfb321
42f51d6558b0f3f422e3381d2833672d0926bf460cd61238d66838fb2c98d90b
99781d1f40e11c2625adced224e71a61f6667e0794cf05c593faf1468e4b6919
response w

2829045b5a076650836dfcf327b16ec28828012fa0fd5cad829cac146f5fa28f
0bb2dfc9540d30bc7531429ae07b4d3363068dca054ec57e2d4eec01e708f8a7
960d2564cc887b253a7bef514971c2e488a3de25dc9c5a28ba13cd0ba1e989fb
43f361a80e6e636f660a908bfb8a6519131c5642130e2a397de465aa1a1b9bac
eb0f7670282985a374d37f90740853402e1ad0ca0c66ee8b889041775abbf42c
22e5b8af72b4a33865361883db2498d4f428cc1147ed9ab869984358e991b5a5
1c691346b555c432128b73edfe4b351b625c743ab37602bac538c333e3673b25
592051b8168b6f8940e2945cec8fd0c499694e1f568228c7fa6e87dc0a132b46

Fiat-Shamir context

0e0a77e16f49a03f1e033951f20e4d2f68f211fdc438960865bd4ad66dd34c4f

withheld proof term: 1 145 bytes • axioms: • kernel replay: yes

FBundlePoint.tCoeff_succ theorem, TDirectionExistence:66

```
∀ {k : Type u_1} [inst : Field.{u_1} k] [CharZero.{u_1} k] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (C : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) (n k : Nat),
  Eq.{u_1 + 1}
    (HSMul.hSMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑n) 1)
      (FBundlePoint.tCoeff.{u_1} T C (HAdd.hAdd.{0, 0, 0} n 1) k))
    (Neg.neg.{u_1}
      (∑ p ∈ Finset.HasAntidiagonal.antidiagonal.{0} n,
        HSub.hSub.{u_1, u_1, u_1}
          (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} p.1 T)
            (FBundlePoint.tCoeff.{u_1} T C p.2 (HAdd.hAdd.{0, 0, 0} k 1)))
          (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.tCoeff.{u_1} T C p.2 (HAdd.hAdd.{0, 0, 0} k 1))
            (FBundlePoint.psCoeff.{u_1} p.1 T))))
```

SHA-256 of the statement above

4f518183994ec362d240fdc866c4ef48585250fbdef563910135f7061e60fea1

SHA-256 of the kernel type

b05405d0c17a62495257dcba3a12240f6f0e6f5809a8a1f1de9ce87ddf1f4385

commitment C

cfabfb939431c8fd6dc8e3d5e543efa15300796be84be27965f3fd2d3d9cd42f
264bd15e9b0cf33207a3de7c856201b55dad17985b10078c56e6647a4e9d1509
772dbe45fd1b2f368750e343d49b508f76dd9631eee09b6ce751c91490bb1373
24ef1f6fcc013adb6c4ee19791c6205837d3792ec4a0fb326efdb9e3b38d1c1c
566e9705b804067bea7a921bc3078869399e14367c08a2e49098dc0e414c6fad
5f2c9f0baa21aa99177357d96f1f49b474ca365b0b5ec2b9893cf0d82df7a187
fc06ec8e66c21a0a4bb02bb61caca0cbcf8d2ca2a8a68edab569cb6126d9203
1fd73addaa063d5309247b04f08f1d9bfff79263af9068a618cfc2cd255a89f6e

first message t

4dd1b8f35ec37faebe30567fab3bb14d5018655816f25b03a67a851bbb9781d
87d1c2940d0bf54fadf675107f08de1a7c9674d7e8e3c7ac7b111ffa78c0e741
c284f5afe1f6e4d7b1faf33175a5298910fe580169b80d6832f930d2f76c0ca5
f39a34c774c0975223098deb36283ed1a3dbfbc155904c5134cfa9f4c1aa0ef1
eb26f0208acab5a3fbacc7d7a32bc29133424abeff193726c4dea2eac6a26cba
f5fac81d467e65303c143b2b3693879afe38a52a9512facb19b14e7359b924c7
c541cdd5d247c8e85233d001ca0226fe13f382124b4afc8c4223cb19563e6c92
40f731644e53802078985999911e38efdb3e2d4a42703f4a2daec481fdda3016

response z

7b40b0d24b82cc1ea89cb2a2e8394eae985a8b22a93dc6279292a350972070f0
bc4390360c7edd9305e97dc4733e8f2d97dab3eca39e8b3ea6dabccbd8c393de
bcfba17db415d409fd54458fe2342aff87b4b29ab17e4cf10f2845d0d2d3b6c7
4d2e75640f0e2f835aa2050bc56bd5a393e7240688cf5c52685039435e0df531
611cd50d7eb58164e3eb05c3889f1aa2c378fb0de7abc66e56d6fbc655add1f2
f35802807df978defb239e2f65e3eb72d9de69a42f04509aece000da95265ea0
8e1b5345e191b33603896f3bb9444926c0fc94ac36f1e2070b5258bf3ba1e9f6
d1a58a7a616fa51ae5dae3e95ae2bfa1583ecdef8b637cf89be6b486a5234e4d

response w

71ae697437baa1399e637847c169863f5570b099deed306d3c702025061d06a4
20766fadbf0fbb11c93c63a9b58ed7ffcd0286b8584a0a19d2b865456b3485580
312fe9a9e8d0f7520ec628a033924856607dd22905edc809ba8e3978117d9b0a
bb44fbb3c0d7a89bc4a2fa19c65ad0532aac8e49b2f2eaf517117d3dcfd03ff9
76956e8253b884a265570b19aca8926c744da5837ed253bf805abcec14e7a935
423caeeef9e4c70a01746e37ca82755dde9f21034a5e0a94308e055260fe08
e13daa4ae7f4793d325b822f7ce0451514dae734f09875079975ae9f1acc41a
1fd5d296d6e2981132d6aa2a4f809847445b67e059b8188b59b97c7525b78838

Fiat-Shamir context

dd47160234b91145da4fd5932f611fff783c2650948988d5dc52e37029c23d2c1

withheld proof term: 294 503 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tSolution definition, TDirectionExistence:47

```
{k : Type u_1} →  
  [Field.{u_1} k] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k) →  
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) →  
          Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)
```

SHA-256 of the statement above

71a576c2369ebff95c9bb4b75925ea420d713ffd8c90060a29a67901d8582071

SHA-256 of the kernel type

02f15bd72c539096b5b7b436750bd9697a2b9736faee075c7ba6c01a2127280e
commitment C

78a97963eecca18fde04a976582ad28422b12f858dcc839b1873670ffe1e0445
6d62dbc6b80f3ca4480463417eedb6da99de7338cadd06d4954dcd9f0510b6a2
f4e7d857480499313b8cf3de2149d7a0de15e3bd93c737dffffc6571cf1c6f
45a89e94cdbe58e14b726b66e09f0af9b051ba90755baa438d867a596948dde9
5e04f4559bc981f3c7391d9a85ad90c2193e5efb448b412e919b4fe7accfc2a9
51e14a79f6ca788feeb447f0e4416c8145e5c5b4badcbdbb48e910407f402cb6
56c2dd4e7acfec1ba0405b2ccae923f512f144f5ceb5152a0767461b9cf1f84c
534738f7fa431592edfbbad91938dfc87e1cbb9cf97462aeb7d0719d21cae97
first message t

1dd9e2edf63a03673bbebdd822a20bd8e0df8cf4c9bde49ab176a2ba6b46ecc7
8217c4c39d4587c8d93fb51e0e95624feaed9a5056f967e85bdf6a6f8b684580
5fd0424307c0336032346ece5fbd5799308c452751cd4fa87325b6476c5be762
205c8836b0bcf2e3d439630e0b2b6b743ab0dd8d402e0eb4665f17e502bb9c97
bacefc6269f3c76a7ba12511801b2ad5fe873b5ce3b20307f16467d812228170
4f205ebffacfc3f15adf4b9aa874cf360ffe1520c429cfd03b8aa1495828e30a
1edac69dd8480d4894ba800dff5f31de6b305a7c26ff78d75ab7419010efaab
3483bdb5b4c18a8567b9d6954ffa728a21c470c751e079232a771f70f87828f4
response z

505d901eb636f73fdeb1679c98bc88439fc4a08dc87a5d289191b557ae75221a
06d89b1d74d9b0dda8bdef82ad6f0726cddcacb8c1d646a3b82e9034f28494c
6945e7b8e5c194fc21d0c914f360a94616379d40fbee9bb9b76cf426c00bfd26
2c9db18097ef1838c038954e8d7617ab0b274d5db86eae08f27e47ac8ed9f297
2820819d2361f42ded4fdef80c47b669836053f96dbdb7be10c982fe4b7f26e9
113ab95d412957f10ca83ad5e1012cdd5f0caf3022b17321e58f33670b022239
2340ebbc5027f051644ec7408878d822a24a94a20b6240b527cf1057bb081e33
4874d6d9fa66378268453ba15c4d26d1fb0dfe63cdda6f4d71b0a79bba10256
response w

1916ecb875d33925c654eae0b4baae57311b893e7a9093f6a25bd16f3dd75ce0
553b860775ca66e30767fe4414d2f7780417dab17af2f930dba1a2dd87497c96
c6259a565e4ea351b56f675e41dcd2711fce2753fc2fd8c3ce96c4e142267b7c
793d8965ea65482c440b7ec3f018b28f3d535253059ce4dbe9d251acc183b8be
b41784c973e429e0ca88ad2ca0c1e18ed396db4ef432613eec87c82811a4eed7
12c6a77513e2dbc6e50f66c4d633ba7a631cd1e24007135f68235f435a133fdb
a9e7534a618db9895e65b0183302046a24af4b1f2c43c0d8928e706eb6827397
65f6e707fdc53f3f77d2be7d68058207acdf88ebc03ce2b2522cfb320bfe957

Fiat-Shamir context

5c47b33210b96854a0d900aebaa25dd0bd2f0ca2a32c7851a9319f2660969144

withheld proof term: 421 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tSolution.eq_1 theorem, TDirectionExistence:0

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (C : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) k) (k : Nat),
  Eq.{u_1 + 1} (FBundlePoint.tSolution.{u_1} T C k)
    (FBundlePoint.ofCoeffs.{u_1} fun n => FBundlePoint.tCoeff.{u_1} T C n k)
```

SHA-256 of the statement above

e54cee6461d4df0de67722b78747c34ef46e3fc89b519873473f33238cebd772

SHA-256 of the kernel type

0cfdbb1a36016c45db4176a5c74bf628d6e707e44ae3be96617c277f066e2465

commitment C

271ff16b7154c6a5b4ec6c5dfc3fa5ad356c430241c61fae15b6914455a6620a

1e1cc95e778e6e9414f340770744ca51b5893246328349a8bea33f057d4d5123

ca1b8c3c7120299610508d461482767ae9525b9177f0a9abe9862ff3e10c73c1

653fcc79a14599b139bfe16e92b50a1dd5b8cbd676c0708d82bd1099fce3892

db380eca85687a82ee24a5d4c4c623d47799b6fde860c7f6d94ae4d1940a6766

2f9ce1a3fb380a6f828f78d3173b90408ae848c89976ccc84e253fc02d14700e

bda357c41c72e8cc441faf53883ef57953a000428a09bebfba1cd7c3c6cef078

9405d54d2d02ac24c755c1f900b42d4a2ac3c961ee877cd54f36b85536cf81bb

first message t

1917469c49977a30f285a31ebcfbb201dd663e65fc2d397620089991e2b2918b

8b2d5d6f58c37ac0933ad3886063f12640fcc5e4ff7fda020cd93ada373befd6

3ece0e1b757c3cfa51c393a3700721deda22cfc45a334e00074cbcb43d899b8e

c8fdb5c55dfe083484c3fac862691bd5af847fc217713737a44642dbe4fd6708

e418f4ddaf52dd89732b2a92720ea4964132f8662534ae44e9a9a5f7e3a787f

cfbab841da5789a0f415c2573b54e3d8aef074766e886b62e40a1066535e0ec9

729e47d6ed3f08e4417a174b28bd703ee740a94bac1221e5e71305eeb8b43428

17252ba3758b5325a6b541f79d428588aa8b609d5730dc63f22522dd8512394e

response z

32812d8bd82653023bddcb444242a395bd3682b082bbb7186dbbbc5d12d00255

54042a0722c9137d651b95cde8b1758e9d0feef5ddebecd433fc67d11f574fc4

d0ef2af96390ac6a3510d9bee64889aedb1d52b449dc06ca49b839bee5299368

8eb98afaf50ff29254c7dc1a0ae947d4a787eae7df99fd4e07398aa0bd70b6ff

216e555441eafa2c696bf968104a53189accdd15838a197c1b9346d5224f67b49

1c6f0e237949490ee14af8d2ee7873e65f27b9aca201819e7e57876bbc30c0db

f801b9ebaaae74f179ee0e32895d957b9b738f2375829ddc429fbfb2dc93b393

cb4f56f72a7841d5779bfbcdbc58775cc58c15a453e5ab5f09556be2ce1676c6

response w

1cdf819ce4a91369c4cce983b7900cb612babb6541dc5b288ed7680430db1fb2

84a1ecc590c785ef11bb614f4214a62a95d1cd2035c129e40ac4262ef0f1856e

1a234e348d53b67bd41798f39ec3d9e8b89a67fed2c22b7da353c5a2a6f16b2f

91209869d61fa91d95aa41bdb09cbdb98b20488878484e9e17cbb126131c1cef

082d39df7aecca78e677ecfcc55645bdc8f0d969c54f1cd2a9674a8d781bb927

2036eb77b56b215e70be24e6403eeb5a4c05c2a5c55efe5f7ce786af1a294605

9053205f1e612bfccc3a7e3e6eef6f634ddb297df5795cc1016773e7ea3b63c6

0a40da9b1dac5525f0e2bfd7a94db1b6d09074a84356466cb8c1e2827fd6e3bf

Fiat-Shamir context

fb5d14dc440e9cbf7f045e4db7dc85c87ac3c8c64af6eacc927798359578a6d1

withheld proof term: 323 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.eq_zero_of_derivPS_bracket_of_mv theorem, TDirectionMv:93

```

∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} [CharZero.{u_1} k]
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (X : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.derivPS.{u_1} (X k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (X (HAdd.hAdd.{0, 0, 0} k
          1))))
        (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} 0 (X k)) 0) →
  ∀ (k : Nat), Eq.{u_1 + 1} (X k) 0

```

SHA-256 of the statement above

cdd74267cb7877b19cf4eba63e96cef330b7c783b605100201adde304585e76d

SHA-256 of the kernel type

28662292a2a53f5c8e24e4caecc669c6beed9e0bbcd9a882a2ea6e7ecc0f8347

commitment C

fe7a07d98e741d6d405ccf671b63d1e478ff23ea0c3638e8bcf8cfe8c15984d3
 44e6744d3728c39cb33bf8187773742cea0d040c59af1bd6c13a68e8e49b0112
 9790985e559ab3bbc231b3a215c81ead5dfc6789c9aa9c63ead8c92775f22ac8
 efa240fa92c22695d174bd9d770d0f151efc23beea4b4a11193c3c4d23a75a92
 becc7276401868cbbf5c44f9ca0393f3abb459e2a5d5e15f477e86561a01ca92
 e5a5e479b778020e9f6b8a34df279e1db6eb3da73f43f9548a967328c251df35
 039424d65288b5897bb98eb5cdacc48e76fee8c0f6e10ac91af71719111bc075
 8ef45aff5464dd5ec6f8a27cc762f653201291b31c048edf3f8f8ac91102e1b7

first message t

fda5ac31b702862c9cafe8c33449f012ace0f0120e9effc9cd132baf3640aff1
 83b694f1200b3fbd39a1fe251bb72d58bd62132972dafa8e0c5e1f634ca0d0bc
 16cbe26a4d39cc550759800f76ad97478b770eb0b7cd677b6ad13d88e72d6d11
 85a266edc53e241efbd9a92c407ee3829fd3934f51ae78632a9f625680d10940
 f2817882dacac04d18391919b0204d6a9053452645048fe5a4a6a72707695f4d
 f804ce23ab92da1f800d5697cf8ec8fe33f9bd870666a952bfdd34718643efd6
 7fa186c8009e6b24be263f04a03a41c718310330d8ceefa9761ec1463b35666
 06cdb5e39b650fe1a3480a753999d3d948e52fd28d4c2228a5f35e78f26afd34

response z

10b8bb3008229d802810f4ae9620bb998dcd9fba5d849bc2743c6cbb7729e048
 db2e2a9610b747bbdf2bdf82b77b01f07f588a2db075c27529ec85ffbc9e7f41
 1e5c7c3d42f386d6da767bc16627dd99274e30cb020c49f1dfebf2740864fbc9
 fb9794d9a4ebca9bf529ed4b57dc01446c92ff55b09f78c81782aacb8201298a
 eefe4c513a000e9873c35c31a365196c83be4641d53e3d588cc6a98e6406f2df
 bad604bd5173179608217e6a9e687c3b5a59d92f8e65caed9d69f6ed7ac16f3c
 6862b52436122b6fa7caa5aba9bc600cf220a2b335215c0dafb7e03c7789a855
 ff89bd1222111e6494a8ca75b5f2929db9c531570f691dbbaeef30c357d2c4ef

response w

16dafb37cd5ded1704ee96b2ca9684e687179f908a2896cf1d028003a966e017
 74034664822acf6da76bba2a0ba7f11d16259d500a75ac7df29ef0403ea4da54
 671211d8d21c0f3e73d1eca995c47be1cfe2f13950a02ce7c742c14fb9ae839d
 a167672e39aaff5671078bd88a77d41321a5cb028a89e047733851a51763dad
 00b69c31ff6afb73d60b7c0023eaed6ef7d5a0812f94f27eb33045268257e8cf
 b865300bec4772ca5387361aabf0005fb7fc69a57c00bf4f7a493a71e51eff49
 c102183ccc55511b5f7195fd1134471b79505b83b32a81dcff73e5c523a512ad
 7cdde531575d87af1cbd915140a80c95ed37a38d53edde44df5487a412609338

Fiat-Shamir context

43d49f9348adacb9c62c2dad67fa9628a086cca50a1b3fe8c023dcfe70057de0

withheld proof term: 101 168 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.finsupp_unit_eq_single theorem, TDirectionMv:67

$\forall (d : \text{Finsupp}\{0, 0\} \text{Unit Nat}), \text{Eq}\{1\} d (\text{Finsupp.single}\{0, 0\} \text{Unit.unit } (d \text{Unit.unit}))$

SHA-256 of the statement above

e6b9c962f9826f7ef646c2cf24780f982abe22ae0aa18a805a244d83d4d19c12

SHA-256 of the kernel type

4d80e65ee0efb3378f9e2a5f6558162040f5e491b54d80ac657cf9e6adc3f
commitment C

44315a956a5e703b88881176b33e7b35a955cebb889abc6a31dd84e7db8e0091
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1a42de87ef0e37eab0226b7f1121f66316dd8ef8095cce3ebd829b713d8c2721
9a9f7d5a6c8aba55f812dec7b2d56803c0cd9fa7d3d80655fec8b40b6da38609
8b8482e294cb39d00527f0d7b97f30231de30ec991bdf950246ebcd4d75adb11
20e295757e8d709f16e5dd5a7c2d3deaaff1cc2c81de664a09a5b701a425864ff
80eba406be0e46e8737bfc1cf640c058a663771da62ba72d8d6588021b3529e2
0221589a784d2125acf6f407547cd4486a1eb3bf6c3d411b34e05221071e4cd9
first message t

f7a7654be2e001e847b90681b4e71ad9b22277cdaf8d31e80338d24795995f82
4b72abd9c676c1ce20d57157948d785b6b9aa6999b9ba2ad5538ae6b2bd403bf
1d0c037f859ab1850036b0eb09434202f709df145ee213cb4a49575a6610aa8a
1ceef39647a0eaedcd7a3a82911ad9d936a9e6af64a8ffe2a373d0b710734272
5153fd3f184ff4ed453edda78bbca901dd98a69a86708f5e85ee8e5276533f33
e0e8fd0e495ce37fe0f50f3f23af88985786c77520254d4ce5315934ea13d4d8
f7e1ae9ae19a51452fcf044bf1821aa1288f8ba3eda1603ed6ba59209d74b62f
8a9b9e5986b18a4ce98e7824ca7fb5557f3df327bb9256facebd8636cd23b5bd
response z

44ce7d03ce2f28475d40479f75bf2518c9b64dcb2a170c004f894f4b6c4cf1cc
bfa707b09ce09111d19e6ca17b3de90f6ab447a057eaf18144e113a4ac0a022
d39787d466bd80ede175dc515a9680154f8e42cead66f4bd9db18f806c79a374
b072e42ecfd0ad04b9b5ce0af8447fc46ce1b15d5e47c41ab4a5db239207c73b
35a4dd2331c88fa2c8d525972339fb5bed9cf25334f9867aa5506b7c10274b3d
64f4ef4370d972405167c81b05cca59e7a010f07b09950651199e20792b5f42d
aa01270ca2596bb0270d597d6f19e9274d8540e9ee3fd0744c82c28c4f2bf663
8f92423f192b3d38411931c7bdb12fc31d642ff34d83c4e8b4e53f36cfa3789f
response w

18ce7f24ed54f688df3b080c163881c206eea9d2d6a1a27f26af3e0dfc0a0c37
a5a9a7c945f8984aefccea9be37f87c0d995327041eca13085f741caa8553d4
d4433e715b42e62fab7bf5ee61f3d05fa371772afd3412b8b49a106070627ec8
5cbcdcade6772779af265fa0fa0957cde8616d0847c8526b96825bc8e9fa981
e792df9358c7291ea1078bdbad53b78607c226dc28bef0fd48157709f659de0c
e5eb1f7d5042a88afd0914a4f9e6378d030b44fb79329028c5c9d6b627844f2c
f45ef0d4313d2b9c751b81c38c30ac0b76bf6700f2dd970418167c9f8b236a50
3c8c27a265ef7f393ad4ffd24c46cddb716b656bca7fa8fb2500fadedf4fb52fe9

Fiat-Shamir context

9a5f24d75efcf0c241220cff628f4a4a73dd75d9fbde9bc973e9003a2db3a809

withheld proof term: 7542 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.mvPderiv_unit_eq_derivPS theorem, TDirectionMv:71

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat}
  (Φ : Matrix.(0, 0, u_1) (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1} (FBundlePoint.mvPderiv.{u_1, 0} Unit.unit (FBundlePoint.toMvUnit.{u_1} Φ))
    (FBundlePoint.toMvUnit.{u_1} (FBundlePoint.derivPS.{u_1} Φ))
```

SHA-256 of the statement above

0456d5e10449554d0faa2fe74207b6496c199dfd14963f9379acf663c3e57558

SHA-256 of the kernel type

ed36fid01f2ce649115c74726154f40fa4a3a546d83c2d2d383e27ac8cdbcfbfe
commitment C

9f0905b8e761f219047df2f0a2ae00351cf085eb936ce66774f432e5fa837259
2684611b28868d6743e9656bbe28af7def38a534622123214c7407836d5f65ad
65e1d2deb287fd3b8bf8572fe9c10ee985c82b69422c7e6e543bcd3f02e97c0
24d80faa25b98f6656bc14feb95fc7a58cf41f165d9c056ccc0dcd1594c614a3
339a3139ed02fa76a6edfac710c8d13282d9af5f5bbc00dc7db6a05319a8c5ec
a2c2cc8fc7d9390eadc17a095d30eccdc2df67d5850178634d4dfe0d2645031
42f6b174df2e0b35e3ffe5037753bdabf0733eea53a32cc7f7df839ebdd25a89
3cbd0b8e79b2e42619b4731877e705f015792c4db3800feb17325d9d21d3fbe5
first message t

3fadaa1269cfcb055f9aa309fb0618692df78e49b0dee0fd9a47bbd57fbcf9aa
081000e56dc3a3a97c91c74045112cd4532a04a590dfc95b15f6498d2f694466
00c32f95ed1057d789e44d588dba488d43a45ce037c6f8f69b1dd4041dd7894f
108721b61b38e9d02ce469bbd2918aa0ab7e64da128ce146f261d12456101f57
b502d8bd584f8d0c312505c827b820766d81a787357c40f859cbe82ed20af2e8
1816f1bc60a65c85374e7563590bbea806396f4777ab90c4660e30c4765e9fe9
ad68605a54458bcd178fc0f832a7b66ef12e6004b9bed10f9a476d3c70bcf6d3
e298a6d884d86b68de10f063c3c7962f6852eb8bd0628a05f021bbb9289b2543
response z

2864f0e76cde1de0392e4752bd6322e36ac3a881ab07f3ca14916f631ad27f81
3d170584243901a15c73d259fa7b7ba49dedc895126ef1f6c5db1041b50f73c3
05e0bac384663fd628b08447e35b8134d416ce71ebae1ec9c203a9bed52394a6
295984175ef517b06c4ee0940e58fbba6c82c6c4716a9c8135cd6dd60f4d9b4c
d788492d84a9af16b77c8e3c1e8bad0fc1ecb9b050a7dbf174760a676e31caa5
44c13be728842fc7f18cc54322bc6f7398050f1c27cd2251793f838a3c551f00
cbaca558dfe1e1c436b9c5d8cf185e0dcfa78860e22c2dfcb105a8ff916b0f43
0868211a7add3771edb835f24575e60a3602ce39303d068fb8be316d0877a79b
response w

4adad3fd5471582083a9face9202b5d0a6710ec1f1018203022fffc672906b0d
8b01453195c6e0e34d7dd3250d3823556331a73e46e933194943f8e7ba6dfb6b
8da20442f86f19f61bda302bceea1a32ef140a19f00d1608fc5dd1f765edb8f5
c5c3f70d0c2d4057189024e8b349e38c96ab5ad99cfd52e2fad5562c4cbcb63e
f803310eb3ab1953296e2eb5f0aacb90fb8976008af09ce3a16e3895ce8f34b0
222c9e921169352b1db03fed5a4f79b242baa0ff0fea510407de213ccb92332f
8892b20e16df646e23277caf502b668b08350dbc6e5a440a3c27d61cea97f3a1
ae9e396bf01bd81663cc84d283f3e6582d7896ba92b438413c107fc738a2b62d

Fiat-Shamir context

e986fc9314b84d5912a7623ce6dbf3c31b0b30a0ae9fb859925e87012d8b6915

withheld proof term: 177 670 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_eq_mvPsCoeff_single theorem, TDirectionMv:60

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} (n : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  Eq.{u_1 + 1}
    (FBundlePoint.mvPsCoeff.{u_1, 0} (Finsupp.single.{0, 0} Unit.unit n)
      (FBundlePoint.toMvUnit.{u_1} Φ))
    (FBundlePoint.psCoeff.{u_1} n Φ)
```

SHA-256 of the statement above

37055350e41818a608e8cb15ec5be4fc5af19cf3303e51d8db99aba509000fc5

SHA-256 of the kernel type

e259b4c2b044524ded11dc78046d8a5032a7546022d6e80eac1cf4ce1d51d309
commitment C

184223157fbbb6d758301c05b0f781d8612689b0eeca8a91419b0a9a23e0ca70
1460c5b7341210d81038506bbca98a0433af163579227cea529947402250273b
2e76c1b8140f92bc838c7eaa569bbb4b520c4735d6fcd2c71a0d4c5ca6eb86c8
3cdf373a8f8549188ef68f915e1db879a477db28bd587b64af213c2d9af4e133
da2d0501f538fc6b2f4ec882c1ac25a09c20733cb8b560457669350d04207908
d3dd383d71c8dea82c741fd06549deb2419673dea2a4ce0c127b77a6a3f8a4d1
e0cbc82c15a04faeba7bffa17a63f51148189428d74a019af738e2ccdd1fbb0db
d1d902f8f6c2c28391e70abd81ec8cf295d860b2d7bc0395b298cd93b303bf75
first message t

bea32fab857e758d52e2195d350c9487e5347bf3d763bd5676fa0b525a7558a1
ea67cdce306b0ce932a9f8c4a0ee2743dbf87503734484c26299ace3b3fe2849
d4f8305f563b3a85f1540b435829ab710d4b4709bb0958346985f9c43941cc8a
3774bacad1f856f495b606829367d4418ff41550ca009ba917c7b56001d61474
5b5d85d3ffabb86db562bc59bcd445d82d9189c5151f8f502ae519179a05f5c9
1cfa13077857a882b5f7072a1119babe508c9270cc489242eecca230349c8b65
69d1bd9484353a9a2845f832c5ab2c75890b53a6e82c7831b05ccd8498b69dc4
9b47873df4ce40e51386ed2039b319313d6c742e4142e60da07691b879ec1ba4
response z

7e4f2c5e9c4670dcd3e0a9e342e6b51046837dce3a49d9e6e1d8f82c381a0c08
7485bb840225568a7340cdd14e55c7a9425228ef1dcb131bc490d6b01298354
828bcdd14eb74f1f81eb710c7f2f36e31d1927846b11bd56d9ab0745e5ac78b6
a26b00e7422851b04f3d5be6010231c204bf8b9f46f737f6f958a981a90111e1
db5c2b1a61aaa116764ecf423a2547e73e7e0400b1330475c5232097e6181c3c
d26b907bfe6c20b8f3f760c0a6d24684de3e9c6b55be1d54ac388af0edde3498
27effad81db8bf00ff9053f6c6cbef09839b01abdc1269a09f8bd1ea08e0a081
b79590c8104ebadb39b7062cffd2510e1c3eb714c5b8be63da59b76a82ec18f
response w

659be28eefd28075240762b6b29c6964b702af20c426fcb90ff9dfea4fea3f8d
96ac6067db03c927907b67623d8299c040fdc0b6ebbfd2944b7325a5efa5161d
316c1903f41171cf01271a6334bb15cb3cb95fa61c906f9f7bd8e12174bbb1f6
898b5437cacdd0efd228e1c589413d76b0d0e6b6c39907fc0017ac6ef907580
59a7ae09b1706e2179833646f3b87f695e938f8a7ebf12c6591b3855d04d9578
6b23ef643bbb0c1e5fed65287c38c3831d2a11f02b3f975c05edb835b96ea858
c32239d1db2f847261c44990474cc20e29b39cdb3188702aa3db5ce28977536d
449fa1c524bd2dbfbede74c2614e6e5d9285c3582f7c4d2974fbee3356e77ac2
Fiat-Shamir context

3ae388fc0e9c9c9336ec4cc8c516c60f2e8e42353a32e448c5edb25180b16912

withheld proof term: 6 425 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.toMvUnit definition, TDirectionMv:40

```
{k : Type u_1} →  
  {m : Nat} →  
    Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k) →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (MvPowerSeries.{0, u_1} Unit k)
```

SHA-256 of the statement above

f0f388fcc3418260df7fcc1cb138c3e293c0ace9046c965e0745381dc8dc8fe7

SHA-256 of the kernel type

4ee8586c08b286f9b7869500650cc01e8d79906e3764617ab38a0d062347b290

commitment C

f68003ca1fd893a8a04d4802f38440b7dcba2d10fb29e2892f7a074bb17ba88e
85ec0a8e7f1df2806c2d1bb75c8e470a9e1b364b8b458c349ad13ccbffd28c69
fa0148266e6dd3c60fd727968ba3abae8e9bac0274dc1eef424755589dddcd3f
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1be7fe18130e7578202ea5f20b61a36026c589aa317614b6285afb54bde8e342
35dcd6a7424fe53c8751ad83191b3a24840f2af128fdca03d03d1ebcff928d33
856b0014ce8c30d4ac430aed3a0b74dfb68f0e8824ae60fd2298bf7539bea919
479ec6554690151f9959d54883b3bac5e84c42f34c4917ddc9b529bd42ab94b5
first message t

89a6466c5f11cd67fcd73ad64a860fc1441d77b111e49a8b68d9e3d0c6b1276c
74a4685ba7bd725bcb55f66472009c9d060c35a50b675ccdcf1b5b8e35d1fdd
72dab018f51ee32f8da51d4ffba9a70d63c64f46f52e51611c5a93537fc08731
91ce05a7bc3edcb332f320a47c3f22e410c7b6c52e4a52a899295787f1d0c07a
de7b4e1f218cb46aac0926636ed8645bd5e89d6bb0c6e08d95464999040cd7de
6e74ca54eaf5bc43f0f3bede20cb8c8cbf5b4243ebb232d04256a2aaba73606
3e6ef5d74a6cc4c8fbb815a628392e69d73120ae09cef605c5bd0d4fc30f66c6
c3ab7160f12fa93fc2568951cf82aaade3fd2c149953cc86de8c008bb9f143d0
response z

3005a56bea3c84a45b8e9e49fdb84ad51aaa2b2ebf5cd094aef205db2e1c180d
e308f3a56c2966753d8fc273d98ba262cdba030e8e5b4ce1284fc8d98c0bf3a1
84109579d62e450fe35c25b22dbb41370bee88d8178dc76e27fdd0bc6df83779
1a94f09ac8a3c678f4bf47a9012960dd20fe7ac7b167424365c5de7d67b313ba
8626ced34e4e67c95f4e797e29787c3045f540275c04a6dba5da37cfd25a5200
72caf5b1f1505f88e41738d5631e1d76d72cfbd0c185009c870565e5adede73a
8138c23ffccc7e314ff76df9619d93e20d78787bdd0e6a0d6beb1051c62866f7
8abd1ebc4975b91c7d78f680ae9b8e8d553e6a20d8b4ced5728423ece448423c
response w

2c42b5f5b3c0340c97c48c0136e84f0c3cf0f747f7b75b9fcd796d38fe99fcda
22bcad2a56e06d04611618eea28f7e3f74699adca00633beb76b00962be29e4f
cd8722dbf3d54499722250aaaf7f4c3af63fd31bf7f70dc2dafd14c88397cf9c
1ff08e38af20f7d22b90269d8e2f94cc79264268f4d69ed6dc9174ed30ad0746
53d3e4dac4b08746403051375d24963a1ef81be1e3a25ab532c9f9d753e38539
6c3760f29ceb9a11c2755cf552f1b08f2c3f4e1c65ccd1916d69684091387694
91d774383d10200b291d6fc3083edcfbd72ac015f680bddde2638134fc2c878d
c5ff42285f2d192f6e022eb8003817c1af723cb7cc266c68ec946362b6c84364

Fiat-Shamir context

a30ffecd455173a00f61d815287f3334298d0399f24b576ec6b98ad6298d0543

withheld proof term: 106 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.toMvUnit_apply theorem, TDirectionMv:46

$\forall \{k : \text{Type } u_1\} \{m : \text{Nat}\} (\Phi : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} \mathbb{k}))$
 $(i\ j : \text{Fin } m), \text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.toMvUnit}.\{u_1\} \Phi\ i\ j) (\Phi\ i\ j)$

SHA-256 of the statement above

6a59b6d00bb54f2475732c03278329aaf8a25ab279fc817c9a6e7d64628fca7a

SHA-256 of the kernel type

7f010d979c21376b76d055e384bba55484e581f739a2a985fdc2a9ff2a292c56

commitment C

c2eac8aa74680a977c4ad4a36630e483e7b808471f232c211b89401667151a42

6f0b38c6e0c57cdaba42fe5f7384e78e4145e94d894716852a3ea1bcff75ab53

6e77a49b8a1699a48068f832dc43bb840b05367a121888c5ec76aeeccf22e104b

cde876d6f7186432ce21f9309af5c6b13b00674aa3c5887d0580980a2a619196

b754cf55eb7cef0dc7f2fd73fa27a68fd4206f8e2ecf006e5a6f83c3e1cb1486

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f4872f17fbcf21efef97c7c00c1d1032b62759ed9759e88e7741ef54aa695954

first message t

71489ba57f6e3b64ca32e893bf0a09523b329288f50d9bbcaf46e2619c908558

3392a9cdf1efc1cdc885e38fb08cf6e7f74bdc022d8584ccde14073cc8c6a7bb

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7c4f1ff4f6b96333e04132055b614b3ee19e79abad6c7ab1bcb6ddc4aa9e2234

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872480267e16a20c927af9c4881a05997f7389311de9dbd660f29408a5e91ce7

3b3e77ffffb407950bd86b96f6a2de7d7846748dc79c98dab46392c91831fee9

response z

a02f0efdffd6b78596001e8b9bd33a825ae359e23795562d093e0d55b489b7d0c

fcacb93466331b4edc9e45b179a32f87d05b381e29b5edcf0775b085581c6073

6e7e9636004f3e08e898ca5ac25bf91f29f2ff5a84300cc2cc4e733c4efdb5f1

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3a968f2821141916f3f0cece52c91be7b01022290bce5f98ce5ca1ccaf7d8c8

119dc4a41dad560e6b227a7978df37ff8a55adceca6cffed372b3e59a7175e3

response w

25cb79e0e5206453778cfeefa57c888218b736d5aff5e507fe9238076d5eb239

5561694dc68f1e12cef22f888e49b485d340362eca81700cdd52a69a3b552950

3322f1386d06e504f1d4bbb2996f5396c0e33c7d707cbec62e8926a3803b93ff

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61a549cbb039e3e85962dd6545efb268993c1ec69f1753cbe9e6cea94424ffd0

db9d4236a526b9bd7749782ba4c9754608930d496125c6e4ce1e44eb8e824aff

48372633a330e3abfba984b3584831bea4892c20c24493fbe850e52849bb9c89

a7d9c7cb53da9b64d7b593e9a261b70add62efdafd269173a170030953df5d54

Fiat-Shamir context

7fd9d9c12da5b05083e7248ea6a1e20d47c8e1934b5b7cdcb69f4832e4d1e6a4

withheld proof term: 221 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.toMvUnit_injective theorem, TDirectionMv:55

$\forall \{k : \text{Type } u_1\} \{m : \text{Nat}\}, \text{Function.Injective}.\{u_1 + 1, u_1 + 1\}$
FBundlePoint.toMvUnit.{u_1}

SHA-256 of the statement above

2ca4bef7e32d40cdad0615d49451ba6c44bc5b43b65b61197b244a37381808a0

SHA-256 of the kernel type

aa52f0dfa2c6c9bf5cfdb465c0f74f9d4d62fdfccea97d1d679e246291ce3415
commitment C

27bb334c583ab82d80e5a96f6040e47863668f8b179c90a709fbb8872314807b
6c33eedcda22ae655b15d1950d0ace9cfd747ecca6ca90db201ed904e2f264a
8811c3d33ce43246ccb541c66746949a8a6db7906ddd2eff28a91f6199978a07
083145f3b0f1bf12a4dd88142c11031dd49967c05d0e8b9750cba13d7510aedb
dde06f18ccce220d84f9fd3aa1ca8a713dbf091b114769f84f7f1a2167bdf507
8fb0e8ecc23a654d08736365d0db824e3b05913debc5d97ea566c4b7e7ac1e4f
b0b6063ee89ecd6d8d4fcf28b3f9221a169b06fe0671d11a6ddf174d3e16cafd
2f76d709666ca33f33b8f733d1dfcbc8ce1df3a5725908284923c6170dce39df
first message t

9125976d7c6e80c2e905750ebecc639d9c95ba9ac2f483e4aa7630ab0d12b591
a13447441785ccc91e35375f614fa727521d5b4c78fcf5aa2073cef1064b86aa
8a435064c18e1e0795325b6ee810801f2db8f6ee435477690c9f9b8bc48b06ea
9099b293b4014fd0d62c0da541225c042a162573246d30d352c712c445f79553
d277ff4de9bd998a3e743c9a145afa0faae3d5edafa2f938093ae8b95f485d1
ea1119a6752be32224423625517fff9bc4abe470da8fda3e41b0ac268a79f21a
cfbcc4f99bc7557dccc4971bb80efa765e7a0614c382c44bc70732e375020ee4
e6100e71905c1d7768962f6604650e32c68a48842b9357ffc95fbdb11a2b38b0
response z

741e0f262a80fbb43de3b7bb875525fac4af205694d4ce6a77e4239c628294ba
24e259c29a519d2e889b3eeb6e3733b9ff8d74b266e10503baa99e164fbf393c
650c9515f2e8c3420c4b424c2c7442d252179849541c70d4c6ef7e35bb1d2dda
44b95b6e86d220a7808383ebae520a44e8df2d891cf87d2c8dcda6078bf1ecbd
b0b296e8366725b33a2b8f11fa9fde3c484c73ee29a189f9b5e348ac1025bb46
03797508f319c27b15485c5295dc71652c6bbd5e3277a9585d8d9ca58f9ba789
8e232f05a547220dec169d150b7a994a6a98f5afef87ad2219fbd4ad1009d7ac
8451102180f49c4bff06171608e044cd6d2c369bad846585faddb3087df9be1a
response w

4a3eefe0abdb68ac82904b3e5fffb2b506320cccf3a0c2ebd03ad120e0fd5d1
0609f6e28beca401b73972a2e946969d1386719a03d9cab0232e3d9ef3b7747b
cde036b2c84799f5981e031727232352a7598fb3128c0406e677d655e21b91db
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9eb93d04e6c22c62552805c64da5991dd18da87ed0f8f636816543911073bcde
fef5fc8504746dca528585cbb9b4f61796a67a997b942329108e3b60f161cea
d6ae86cc7b304f0044b8b541a788505b5227cdfa36f8b899a8b4e6b51b395ae3
977fc0a23e62319dba8cb664b0d3fdb4319a995b4f10a38e64577f8873968d52

Fiat-Shamir context

68549af149c12f79f3fda531561bb48b4a7d06b7b05f7d6b8f8718387ee29448

withheld proof term: 564 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.toMvUnit_zero theorem, TDirectionMv:50

$\forall \{k : \text{Type } u_1\} [\text{inst} : \text{Field.}\{u_1\} \text{ } k] \{m : \text{Nat}\}, \text{Eq.}\{u_1 + 1\}$
(FBundlePoint.toMvUnit.{u_1} 0) 0

SHA-256 of the statement above

642d2b48cb189942f64e03a93eceb5e6644ce3ac99a9e353d0aed2c9eed28bbe

SHA-256 of the kernel type

3ef8e11ecee9e5e66cffa5977285f4a97db3d5088b5534f920c10ee17a83828d
commitment C

4513b0cfdedd5d45b9685d9fe0402f049da4847dc84bd362d72a476c9db35f22
37eccbd9f9abd7b8096d5b3f3e03c43bd1e4e8b571da74128d10d8b71ed09d5f
d7f7f4d352862bb6e08b0d76d02ef78ba85e250cf7ad518365cdb9007844a0d
c83566315197f1ff72ef2d8c4a2d471656ca991c5a431b8372b4707e18012141
2afe5ffc980f79097cb435b37b8be6c94284c336b6e6a19390927041dd379e6e
90474abe61278c5ea6b5f4f2e3a95c74a88d57b152814699868dca3dc46a4758
8a1e501215e119ee062edd3a2ee0658a1b84b420321cb358bc01e306208c4f31
ee7d4ad74db8d24961d42b8d5a5d5121d551f87ec7e03f9d6ce0734c9cbaa63b
first message t

92ecbd9767152bbc98a656ef484bb2308395330486d5f95f350f98ca99cd0053
df1a0dea66269b86eb7eda94b5b3aa10ea3ec76fb66b65d7df89656bf740b26c
1850e1801cddc7f911b65c348698660d4053fa269ed7e865a3bf62a54d4b313d8
603d3d2eab9fcaac16e6167d075294c7a5d490eaf461e6d7261f9d08f8625b30
1ca847a378e65b023bfcc9d7a8f0c324875c1077203cf0cf2b13c58bb6a33b37
b814497b55d876bf1bb39fd6255e4a87a7f275fa29b51aadde970b3af4a58915
ca71cf67b9546a86502d3e5ebfbc7f0348e41af9c719a359b015f74441d2d4d7
0d8adea8e1fc4f855991396477ff5d4095d7f33258271cd32e734f5ce208eeb1
response z

59b68ff86baf18b3371dbbef41d8583f560d64fa7c721141f959c39a1be6ad40
fb615eefc5e2d92db49dc7e1d3a25299bbe2073d56539b74cc01ef208ae3c508
e51fc2aa19cd52ec3d3bce912c237f05ad58c97ad780a20cfab87ca72ca7581f
2e8ac94b4b2b4ceaaf172b1e70359339ad0e8e44157fdad15e8d32cc20d18a0c
5af50fdd1736f3c491bfe3cbf9e787e68888331cbbd366f8185e2b147d6e403
3b7fbfaa96cc9f3a8ac10e26248757ccd287c5b3ac43b5650004620df94fd7ba
d7dc6cbdde6e8f53132153e773b3e439fa332a25c72a442dd9784bd032ff5ed7
c46b156dda2ca33c71f08a6609a928327940a161a4cd2ce13a073d3810e5a6d5
response w

7c7a4914e6c8e74d884661596d3699bfc8c1f80962f81d56f05d1676f138cf55
baf6031ca3a894de908dc5efaf4fa7ca7c3d33cb9c3cf3b9ca44dfc31770befc
822c33617f8fa974b1f74b4ae67f9152250d7aa46378c386e7f8ca41392a41fa
5d71e60866c1d4c82a189e5e9035bcf6c7a4441d04d27d8798c791d22c8aaac
523976db3ba73fcd0e2917e1fb868bcc92dd817b507b628f459cd60a3f485d9
4f96e292db0d4a7f7a4c8e24cac758f366cb95dab4956fe152b4c080850fee58
3b83f49b38017f05adf5f909f36997334d5e2e54de70b2d0835769d55e90ffd2
9dae0c946bc257b985097013159d0332187b2d341ebd7686e09a921e9b5aeff3

Fiat-Shamir context

d9d6a4c4464239cc3747db01d9427fd3b22474703f2bd2876b20405e9f164af9

withheld proof term: 965 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.unique_of_derivPS_bracket_of_mv theorem, TDirectionMv:110

```
∀ {k : Type u_1} [inst : Field.{u_1} k] {m : Nat} [CharZero.{u_1} k]
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k))
  (X Y : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} k)),
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.derivPS.{u_1} (X k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (X (HAdd.hAdd.{0, 0, 0} k
          1))))
        (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.derivPS.{u_1} (Y k))
      (Neg.neg.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (Y (HAdd.hAdd.{0, 0, 0}
          k 1))))
        (HMul.hMul.{u_1, u_1, u_1} (Y (HAdd.hAdd.{0, 0, 0} k 1)) T)))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} 0 (X k)) (FBundlePoint.psCoeff.{u_1} 0
      (Y k))) →
    Eq.{u_1 + 1} X Y
```

SHA-256 of the statement above

bb7299357d12e99ce9565d69e8d5bf94762b15d6d4d6d8c450aee2a230ab9094

SHA-256 of the kernel type

5c06ce6d38be3492ffff16a8f3d875e0f051b819c7c043d6f3ce1a025bfa5a20

commitment C

f44dfc80c2583a5381317dfce46b199f09783ff1f861614be29683e92e6b3d49
fcb7884974e982f3d88b1de55365763493d2ec903a7213c18cbfbbdee866eced
ea95349fc8d0348ba4c78ddd8d9a503cf7826763e59bacaba64b34eaf30f5cc0
3f05d20252600dccc83a55b6d971f68d3f514ef61d76bd6e5d487e0facb5658d
f5d58e1dcf86e0f0c1080a13425dfb28502277186cf0e1d3493cc5abb58c75e8
e13afcdf2b50a77fe8fa63edb3047d045aa7d22988faa53b81329cc2b019d9a5
e5ac6d32948048509f1b3c910945a920313bad61ad8a6dab3f7adc3d4f54ceeb
ec750d380c6ce3188e2458cbf73b257b28d1f20229029ea139854fd7df4f7596
first message t

4d4dc7975497944c099d8a470e204ae35f2c4b067d97c52b1c91598e94e111a0
9092efd2e416c2c4918e43a7362085b405b47bee02d0561a66c8b928aa1d1f5d
3740c7ce2549c5269bb118a893568b90d73afdb92fdee19e815910a2c7e8e0e3
51e9858e0a8d5479cc32d3315312c1eac857458e1b6f9511559c6ebae8f05a80
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076d3998f1e698116d578a09a9807f5e8df7410ed5d252b335007623be4269fc
c2a3ec45235da6cb1c99c07b3d00a0ec76441da5d4e16492c2528080ff6bf270
e313e91e82ecc28b705b760f05db6b8f496c7c1506def11fbc350f4ae84081a1
response z

302683900d5c5b09cc068c76f4dc9052d49de6361940d89f760de634134cfa3f
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bfa96867360ce0a6f14ae8e61d8e3108ef21f1130c44d0838e15229533ec0238
5791e49c0f01c1124b092a459ed6b576607c7765848229c99bc51f68771b1b6a
5a1701295e71ec0d081e10cfe81f18005ab9270de4d382e89b60a903517bc2e9
f94c2727acb9bed1119e0f74eeaf2e9e41f6a5133a669e932b2fd2786087ed6b
cc4403c0de11b1c3b368e147f1e0f1d431df59bd6ee954d254afd8bfa202558c
7c581a5d2324756f0b3b1c3e989855fcb4b9386246f81e782a0c53b613a6844
response w

1f8f3132ed9bef241b47fefbcfba488f69167c464aa4026b216786b3eda6a3cf
f1f1cc2f316c3f54893fbd56d13dac52cc9c77a76fbee9fe3f59cf21d7d678d0
feeb8e534107c48b4897efc117fea0291b78e4c57fdc29ddc8a7f3ba4a2df38
c1736d1906087fc7421a4bfcd887329dbb4fee683dfef0eb0db8fd38abe05612
3be310b78431bb63c69184bac1dc27baa8d463422051beb06cfdc82852cc3777
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9e686bf31d1c7a55656b447071c4acc4cd9bc2536f8e6f0c474ef229579ad47c
Fiat-Shamir context

97e04aa268fe284cec2d4564145e7747a6d283ac23b4d4586f5756af5ab4f04635

withheld proof term: 170 463 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.FramingSystem definition, TFramingExistence:97

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
          (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) → Prop
```

SHA-256 of the statement above

8d2fe14a199ab4443054ad373394a7d9498bedf70faf40c329c7d39de9975264

SHA-256 of the kernel type

d5414f4189b0971700a90e57cb3b07d98959989066206fc6a0aa6385103a06d6

commitment C

e85625b90e66ce0215625a380929958c0b92ba32f63e4a37d56df260fd7d37fb

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eb750a2e10e969405b1b0eacafa7f802aa8f5f62ac4ce0e7b29ce277632f589dd

04bb77b529d8caf8c8d2130203c39761f14e9d6cb4ceefd6b651b33c43c7b3d2

84ee1ac29b326ce09a1100aa4de4ce244f676ce07d2699cb2852d93213d261c

first message t

4e86c55e6317b3ef0f2c3e909970460f89ef9f660e9168e70cf02a422e36f547

a714f1212a490e9e1ca9addec8bbc237de6284736185e2ef611ce7f7eb16c1d1

3a0a372b6e31054f51bcaa43a61d79141d58eb2fe3def316c51aa732eeb685f7

7d1aace37e85e64f7dc5d28c7c7e034efd6ca057a95c265f125e76790ba4ad90

f64198cf71d22864df09ec7cd9afff762226cef15ac75c0ff854b92409f5ef49

4d505a55e957c88ee84c5b8542e59b82e94d703f415ec9392801f20ee2563deb

604abe57bbe9eb79d45321a919ffc20869d365ec22ccf42bace817b163a3a7ff

7b4f3b37d11a3da9fd5132a53edce9b91ae817cdddf8579815c042f99f83f57a

response z

7c4fb1ef26fa5261f7d52d12e2414a342d02efbbab94792fcbcd7440d0b5e521

5565d8d73333e877e1b76e98139154a4f9fce23b185c1f6cb3fce6a2d2ff3864

15aa17ba35d67658ef3c92dc3f1d82279cfefe0f887bbc26418a38f4f974d99

820bc17e4c4355016e3bd7fd4dfda705919c33cf71ec15bcfc819230db2aec9b

a7fc747a6633596922ba1ccac3e453a55401dd61bb49aa9b97b939a4d4e2311a

ffd4d14cb2d1a26ee8b50106f7960b2902d801023a6f102daf1c76f4f3b154a64

9d3ee5b24cc154c09b45f83547a5cf5fc3b60555b08370d63f5ae24ea3a88d89

dd442cd899508d560b187facf1f256fe29437a7fce51f5bd2666c13ccf3c1783

response w

1ef981c5b5f7e986b28b338051f4b71fdc380c8e1a120ba8d2c058e51463f4d5

91a54f36563d4499147c50ac5d45b1df55500805ec45e95f886a0e3a679581a2

68b14a293302b7a2edd5cdf9734aeafe8cf69096e60b35a1c5e4c045c402e949

1dbc0fcf4925c1b178544b7a5059ca5153e33fc6b4fc9ac80d3751053f72495e

1b09eda47756dfe7a895253647791e71c6d16e476f1806c4734a118f56b938c3

454e4691c185e4e20a24a36b94fe1ef21d42600df389a35ca941eac5247516b6

2dd5ddd7166bdbb62bae2adc969145d761723b19f02d25bd503c17a17969ac5

5527f3c2e22c0224f0975d4eba945321be6cca8c7eb0140813511d3c01b5a54a

Fiat-Shamir context

c8c940bbade7300b2a1d2580112cf0b82fc92618243a6a5287a085f3398a798c

withheld proof term: 7745 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.X_smul_divU theorem, TFramingExistence:71

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  {Φ : Matrix.(0, 0, u_1) (Fin m) (Fin m) (PowerSeries.{u_1} R)},
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} 0 Φ) 0 →
    Eq.{u_1 + 1} (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (FBundlePoint.divU.{u_1}
      Φ)) Φ
```

SHA-256 of the statement above

ecaf94cf88aa846c925495d1d1e1ba293210fd9d35bdbb96c0cbc9409f4bda6d

SHA-256 of the kernel type

1e1d8f72ac4e073e06e65c45baafa50cc16f0df885f55348982fb6be1fd925e1

commitment C

e13903f6ad22c7ba7bcdde0749b74da3cf3017fc35c518b531420c96d6445f75

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first message t

3fda494bdd49bfc4548961abc86b9f4dc632fc1451074b1f500428a91bbffa0f

b6e5290f88b04c128144621e3752a74c1e8cad5dd5477597f0c00d31f43dc4f4

112e5352406c42ceba845a8b45ef63557b664429886d420c23ba9ac96b6ab1d1

367d792af14f7ed9e06962a9394171003bc2b63571105c97bf7a4566ee0e06a0

180c7381f4a434380133df317931cb1cf8fb4b51159e0b7bdc0a119997bb72de

4a8715950ef3e5c116b7807e02b2f1eab69a8525658558a554f985076deba6b0

c256bedaf259ace2453dadd7ee4b4e15b817ede94cdc076855f24041a4fef549

f92d8a286c7ff4bc80e1667fdc6476055ea733364a490a9d6869495e875b5fa2

response z

3bac91a43ee204331831db8151a8c6f69c4e8e84a8b8cc0fc2f1c2a1b1594545

c172fce13c127678cfe683c089f6eccdb306a4d6dda67150a8a5cc9da75ab8e

1ecd65f429729d3293af40e5459c58745675892796e5e3e43b3fb53f40a24a01

4329a6ec9c192d2b14fe41183235b2c3926a398d0028d53b76a1b7ff0cb4fea5

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response w

309a7f75a351ba614423d3783ed8e4c8e8aa05faba0c6f4d299f6728f1b9097d

4e66ad57d5e82a10ce7fe3982c0c80fbb9faf35935f4bc8b7a217277f0af5370

027481fbba5bcfb7326143c5809ce507c54af1ad1edf9036edfee5f646c09813

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b99ed517d306b07f434d892acec6e4e2e7f9918a83d5b4b3340ef5fa221bd862

05a2617e2862788c3945e0a6982bfdaf8cec936a99bf0fcf511acc695c46535c

267ed29bfcfb89509cf19ef8818e2bcd45e97acad14ae783b5af17f36303665

Fiat-Shamir context

9112e5b291852bceb6feb4f72da7fb9ac00979bf6e9a06ca4b0777a1b207102c

withheld proof term: 27814 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.X_smul_injective theorem, TFramingExistence:79

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\}$
 $\{\Phi \Psi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)\},$
 $\text{Eq}\{u_1 + 1\} (\text{HSMul.hSMul}\{u_1, u_1, u_1\} \text{PowerSeries.X}\{u_1\} \Phi)$
 $(\text{HSMul.hSMul}\{u_1, u_1, u_1\} \text{PowerSeries.X}\{u_1\} \Psi) \rightarrow$
 $\text{Eq}\{u_1 + 1\} \Phi \Psi$

SHA-256 of the statement above

4aa4852d5235accd50dcd54e5c7dbbca05837cfb5988bb5f00380aaf8d2a58cd

SHA-256 of the kernel type

e34ef7612ad73830992f534b2321b44c4e84cfb8bc3f7103bab873a4443ea56a
commitment C

96a95621664d27675672c60faa5f930389f7e9c298bf7c89525ca6fee553fc55
7be8971b3b592c32c5afd70ee4f7ba1a725a11a3ff4e218cb6bc14ebc61e329a
acbfbc7b4f98b73a763427304a8840888cf10a2e01ea76e909c720c996650bd3
4248c3f5cc3c6f4b728925832bba57eff3f969cf6be2618cee2d372d0c48922a
5f339550b6b9ff8af2df875482ac5f359dd9df8f0eb303ffb0ae6167379c6854
2e2f715dec216281af8abb833822321f6000e589bc646240b0e19e25690d082
eb8cf7b9205411458241f8a9d6298ce46efbc3f51dd0689eb84d6290883d956c
4bda569811dc7b9152d34aa20e31a4df403d06e2d72a184968f7479fcb881a2d7
first message t

914b48f7949144ce17bef8c85732009e78283dc8ead31a607bee896ecf8f8154
76d982a96b13fd4be8082455031a9e77f53fdf7e3538045b5845534879ea43d1
b200eceb4747689c58faae0a22a02e1df6fe9818a19ad69334e6ee0a44adc7c4
ee3aa44612a790f10c9c3a08bcdcb76476472c7c971d181e27ed4073d464006b
4757c08d83c1e3860e4b8e8df233b18e81c8e69455c4f27214be0da5cbf17661
e5c4b48124fc057b5a68791e29c82ea13ba6d1c4e58ad09f1e51a47aa4937de4
c36642e8d8b9e743c8365a5d2a27b83788b76d158fa1915ee776c9b4c0820f62
397592dec5c9c6d0f5b5739efd4bceec38f65ef5f49ea066127e1845100bb6a4
response z

66ec7d147f0d6d69de2d83d3d10cd776259114b1aaf6ab786bd3958ee188699a
8a57781f6242ac1f2ff956a3ba247a60d749aa11bad7dceb1c3b8d6c53fa410c
6ea71c6d1678b2474c8a90f99c1d8c5e2dd38a92e972142a6ae7f13a3b290260
4834d6471f2b9df2bb6e2f552660f25ad092dce1250dc69db421271379b101fc
123e01ee7201bde017cc3c7071f2841341e5faab59b1b90280cf86d184b13839
e09df84ab222df0d6f8ec6b307a8118913747a4e089ee76d70f7b1d5060c3154
53bdf0065f89bfa68182f4ed0b9c108d45166f68cebfca43b206c810a8c8c86d
6ae57e0ed473dc36b279fdd1859fff4a80a5483f149f14673ad701d068735abd
response w

64239b04e908e5d9e11b9f6b0ad1263c92ec2503ac64f82dec28253570a49532
a73da359e337115d729abf479c35854da289f5e7a727bdae2d708af3bc63a50d
d2011f95cc10ab35123139844f7c006661538dac4382ac6796e4a7165e51e171
919ed8e6de639b1c93f4f1274cecffa515a04a9b374600eb25b4efb315e58eea
91a395bb317c953b977337d58327b3323823109f233ef2a43b0c21ee2fc450fc
5644cfefbdebee106c5748e48e6b4a96736810e7ae7c7a10e1564bfba027f1a
14d09154eb57870b0aeb376643109f7e981dcca9b1e09caf59c8fbb6c0276344
79159d49bac3f2b4d9168c05cd6c5bb876a880de0e6fedb1131fc4092625ce2e

Fiat-Shamir context

fe3826ddd4e3d5a39eb335f0a844f2d65a57a323c1a74ef62c75bdc56e3b5faa

withheld proof term: 15 677 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.divU definition, TFramingExistence:61

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

f4a8d58c8e6d7d579c906c4ad9ce2e059a8a4fdabc8448689dfb40ec497bc3d1

SHA-256 of the kernel type

d3c1b2dc12d3107353bfcc052e7902820e01dc7eef95cb7275923b9fb4d51f0

commitment C

1b26a4761cf7906384e6458a26f732322ce33c631f24b4d1188ec0fc386b456b
cac3c498c783c2e49f2d3c3da61845e137cc7b0482f7504ee19a28a746d97e27
caf9a5fa1b44a58b02b06bdf8dc8df685ce3baf2b63b0a00739deb91ecee3cab
4522e737b9fe66f3f8177830c479423bf02b7b21cc1cc7b8bd9dca8be0daae9a
ca10b2fcc92a065faf2dc7090f3547000b5db77922b88d1ce224d882c65c3506
7b88f0b573d75abc31468649b58e8bb4b53b0bbfa843804d87b30e7a32575373
fb7c52d141ada58740cbcaf88378d723834e608c2809bfc5556e2bccbf929be1
cc63067df40594335cc01640efd67ad7bf0da1b2863a5142831de61d8bf019ce
first message t

18c9ef66ab64b84150ce69360d0f885cd5ac7ef24f1583b6fc22ca6b366aa391
dc6cfb24a6f4545619f1e449db4c1460d8e3dc61eb381ef05b0e3e7837c7ff85
e269470decece7f54e7c43482b39225c620685de9c9470d2707561a324d15229
9c5da7960dd17274e0b252453398916cd50be3cb77591388adf105db1ba6e057
d4686385d4b857d657cce6ce759f00bf67bea940a670c2ba22b8c4f164c209ad
6e7b47d78f725111cce689ae954957fe1a1f5df3291a322d1337dfc9f67a71ab
1f76a09d0fb8d7d74ff35ec51e4b707a4094ceea106ea1e68f74ee06f7ad3129
1d883742a3083059379b988eef1b00131ce468714ec0fd76bd459d456d3b9fad
response z

50b34cf339d860ff7a974254a27307fcd69a33958765da196f942e5b4f0d82b8
2fdd767eb003cc25ad8d9265a324a471e0c3d22fa3e6e60c8edfa2baf8dd4f80
1c6fc1761fa0c3b64eab891ae3909dbed43c57cddd3776b0d9c8b0b2b0ff10d3
6016615fd1a88d2659e6b915f96c8460423bd5af06e10d0b3167b07a52e3ffa
6a537b65cba8f48497b7ec54e0336cb9be0092afb6e7f6242709dc346079a989
93b25f06150db72b8d721bcccc93d4314b553f232267720a52b0d5690e4435d5
9cd6f955aa1b9565102a86ec0807ec32374c6259c89221d5f1d94c203fabd037
ac9698491112e2d8f480f76e405a43307c4cb0dfdc428faac89b0e0a1dddb2f3
response w

25e253a4a57002e5fb142851f0bb298fc610a8adf2cd1964b00cebe90f2a898c
32796bbe712d05a2f8b7243170115c41bd015f65f19152c9c8ea137ffd8701f8
2596f94c4e7c8cface7f819735154c8ba663637c202a2db76ef2684a5127c80d
77b1d84a37e0592c8d35aebb9b06a5f5e4c91c3870f1cc05213218f9ff17c051
23d0257fc2a7944c10bed33aa4606dbcb728877d71b844112242852cd61cf852
dd32513db3cdbc10059087bb8f85c1d92f0092f2193fda0d441f7aea7419a9d
3ea7420d774d38263f99eee9e834c4e32e6a024bc76c960277196d7d2eb512d3
4f7866023429d71d3791e0b468a015fa546f7ecd1a9c9abd3118883a817e08a1

Fiat-Shamir context

7861603148b93eb0c010b4e1494928b2e6cd77c8228af9a70388bfa49c5e5c9d

withheld proof term: 444 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.divU.eq_1 theorem, TFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.divU.{u_1} Φ)
    (FBundlePoint.ofCoeffs.{u_1} fun n => FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} n
      1) Φ)
```

SHA-256 of the statement above

ac86bc813fa55972fff41d79ad2a6f03cfc0cacd45d8cde4ecf4f76771c618b8

SHA-256 of the kernel type

79d0b0e679d0097aad2af97c92ac1202207f104ca0254ff615bd00a8623bbe9d

commitment C

3421dbed9a0de6c7641b653b7deb31a39c546350a2a3a266c0e38692ca12e4da
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9814aa31feb9b0570018c6ad24252a687c9c5b17b937f3503bdfa53257813c7a
8176e92c91e69187e58683225015d43fa838fb8afc96c00ad7dc6c2409515fbc
8781caa80b8ad155ea80a752eaa0c36b4e7acac090ce6424a1d7c645a14a866
72c06106d52bde0a1f20696738862f7afbe1a13d0cea3e261e8498884aa35b4e

first message t

bfb970dc474a4272f9478959744631006d8c67717384014ff8dfd868a89a9b9
272eab5868af0e77c5743ac33ccc287e9b047f0a78e4cb8922799033a168acb8
614b668c8c7e19987c01ca5c7707d17e1160e8f3607595147ba5aba7ff5e1a8d
a2f8660293b7dc27130a4f914dab97e130c12e558c534ae4bbe1faeff3404e3a
833606980647069fc65f511bf6173ac9b95e846421573be5af704e8790c8731f
05f55007e76bd4052f03352c71dbf187c8954c9bd7a5e84a741773399618914e
bd33af7023d107390110fe02842a5caf0b6017fc550b61542ef3bda059dc0cc9
5959af85c5ca3cf68ae664d51024424c2d2b0947b7c4654f7fb839f2051b982d

response z

79d5a6008b86cdf250754277397cbbb7c2c14de043c78ac8d7a1b191c15aed4b
b218558c95f9656c059c803fe6b07611ab56a0919dbb1964785a345bfe7eda3
d1d777f560eb3fe51015a40e694263dc8b4cfc977a5d157efbebfed41bccfc
234b38b8210a4db78a20351cc77f15825f89da35d4cbe087ac1ba9bc3bc81065
4c0611e29df62ccc70ecd9438c03658e3294e4a9208252378d043db4b3652386
f15b4ab26d708eee7457a56fd5ceb06f9cea77bf1bb859e5a6e13c57af5ab889
2263490ef7481feb1e1026cb7ac2c8b1be016d6d9089a7432d2be34dcf66a720
17e5fa109fd7e44cc5a8fa2f828ff16fa435c1a21e9adff71de359972d9dae12

response w

5bd9b6fea0196ba367c3c4b559e31c2388511b996347ff3c80d1b9c3ed7571a5
ad1605a67602b225b3641f104d3c16e2f183e2001c10c3ff49cf8621737087fb
663a95bb00d6a3ce45aa013957011c74899081d4fa857f079578f9d60d51d840
3d3fa3798f3298c36b3be330c78743eb68dca78cdfb7cf1cd59f27b5815f8e95
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7ffe4a96bbf72c4a53ef6c9fdcd61ee524117e3e04aa1dbefbc0760cc6b0bddb
9867ae728db2cada0172654c51e2e69571cd384b9f1902268150d1ad1457cdf3
77c7d1286375f072f181acdcd0b26a3199309e592045ee03baf78b5220524e9

Fiat-Shamir context

06adf07c44b4145d343902da491a916da24b88ae50951cd89e0cbae92eb4941c

withheld proof term: 239 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.existsUnique_framingSystem theorem, TFramingExistence:416

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (T : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  ExistsUnique.{u_1 + 1} fun PT =>
    And (Eq.{u_1 + 1} (PT.1 0) 1) (FBundlePoint.FramingSystem.{u_1} T PT.1 PT.2)
```

SHA-256 of the statement above

34096b0fb13b5b82aff115922b0c96d44d6622c95569d8e68ce4820b1f0c7512

SHA-256 of the kernel type

1ca59e9683d0581ebd71f326cf00c24d3cd53e84acb6883838a7211e728febbd
commitment C

ff4e6ec91f0003a134b52ae98d031555e50119232d0bbd9849cba04f3de07541
28e7b9c1d2fbc0fd68579e9cb5cb5abf66149938602656cbcf2575510a24cd45
7db2602741477954e6d7661e6e41d85a7b3392475756eac1f28c126f68ee7850
ca067fc5231850e0099853d6c4589d4eae76d69c9014495a100b7087798c4c
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b9b15ff9b2dfcc04b09a0f6d05fca187a406182e50fe3e7fbc14460e56b4286a
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933c5b273849ff04277e89cd09ba325def36efb4a3be285836e58f24caaa8381
first message t

8543e22b2f91e94736375d0c9bc99e1493fc1c8cf7088fe9845d2e2ac484d886
2dff7d367d15e887c5501e4dc4c83bda2450b4d5af44f96ac83b462cf686af05
cae9d5861a3ac3a46805ec9d39e620fd5ac219d9fd5852b0da96fb357920fddb
f318f9f9815608d42d718f21aaf7fd1fa6801ced3419d8b2e446f4a81096fccb
9e22c186d5414f453e5878ae188aafcc9798579cf32016870c67b75581e2af75
478c56326881b37a2ecfe76b6982c530fe4c9e3be3bedf2686264981b46ff10a
ecbb9ac4439527d991e3dcb2f2a5fc8baa1dcb01b26ac56259d9bcc0495ede4e
49f5316ff2a720d986987c93b03e40d0806424798ba1bc40810bad3cadfe2c09
response z

7e6b71fd5b30ec40d56cf4e4a0afba4729aee2d01885d8f872d2b183a05991ac
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9f44356e85a6e1c41fdb8528507fcf27f17c5d7a270042f1a9706c5b62dd8865
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00bc98a7ffcceb6543aec0b84f4bf81ea7e5a21ce73e70a0c751f6fb6271d86e
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response w

61467b5b27c4acd78d551e9a53db2f9f2aafa7eba885f22677f8ae1f0517ad97
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514374e8f69da7fc74534b47bb8cbf66aa9639fe13bf19bb536399574bd4be85
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67283d3bc89544be97df361e2316b831f3cb729aab967d7bf485491a6dfb8a9c
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b237aa371009cbfff398dc8aa07c2a96a05fae02060da5f81e55288f23295aaf
Fiat-Shamir context

720c2f873f06961d48119ef3d875494387a3d3278eb1555cb4a2d5095b24107e

withheld proof term: 34472 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingAux definition, TFramingExistence:187

```
{R : Type u_1} →
  [inst : CommRing.{u_1} R] →
    {m : Nat} →
      [Algebra.{0, u_1} Rat R] →
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
          Nat →
            Prod.{u_1, u_1} (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
              (Matrix.{0, 0, u_1} (Fin m) (Fin m) R)
```

SHA-256 of the statement above

c5cc871ed07e7cf725455c328214a681e830e888c7d1cd7674bee0ec589bdbba

SHA-256 of the kernel type

f2ebb54fd3e0418b84fac86ca9e74a20ac501bc60eb795efe26f8e16cdab384e

commitment C

965e94f29ad6855f5e91f5e50b45b07cc877b55acc29c930870ef4c7a8112608

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first message t

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response z

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response w

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2b994db6c960fb1cdfc22a49899ab87eebecfc7af552f446d57bd5ad2d1d64

2fe0e2f9e66a6a026fbd67d9cee671812dd4cc99969f39f1c1e79e0f7ca44297

Fiat-Shamir context

ac1b96f0b09e9a366bd1788ecb747cdd272226e567466697e27e298f8595e7

withheld proof term: 9 478 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingAux.eq_1 theorem, TFramingExistence:0

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(T : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.framingAux}\{u_1\} T \ 0)$
 $(\text{Prod.mk}\{u_1, u_1\} \ 1 \ (\text{FBundlePoint.psCoeff}\{u_1\} \ 0 \ (T \ 0)))$

SHA-256 of the statement above

0a76f375d57619367914d07b50a588bf1ef0d426f2ac3a050aa24d9ac4e58076

SHA-256 of the kernel type

6fad021cb4e40a7a050a70493db6b1bfd17ce9f6a8cdc28317ef96572a0a893e
commitment C

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ed55e153b573c3163a322bfadade5716b6b28fcf841887e1283f4ce56e2b2697
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first message t

e3d2d410b0f0164b9f6623a726a289d4a85750a0b1d40e330f4000967acad68d
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0dd6c7f66983cdd8817f11e1103adf709fa02595722298c62e453cb57b65b07e
4b150c2bb2c2e63048e4f7d476c0511a1ef4fca541b43c8b96a7c9c056240d
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925dc4070bdcc1032fa866469d95c5a46661d12d541f44115ace9cc8b146b67
d8b6eed2406c2053ea183d37bb8706e997021747d27832f55f61bf2330d7ef2
9abbcc68bf12cde3d25287414ffa2512c03345b76b063b7a55b81171a2ed829f0
response z

77da0c6a4027c4beb63e089f42f163904a4ef5bca6c68e4c0ddbffc91c6f068a
5128cfb5eae82fa5e575557e48b6d0a983986173003c814e988eeea10f71ca7
755494fadc3c1115299435ce5eb0083e06f38019f7d1d3acc5824642a9bba7d0
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be6a99246ebff01da94aae51f512b962e83c612058fd6db9f43756aa7d90ab0a
62bae6dc1f7b66b25b083a4ef22110fec9ffe95c986cc185ec804e0053f55fd1
384e03518efadfadaae1875bdac0545fee0801b96350f670bd1da391c9bf9cbf
6b760c9e26adf6dd1a53f63d8551031861f66ae63c33b47fdd093c34c660b7f1
response w

70938940bf8bd20e2149dfe1db8e6a50d347e0f2e374091816193c9eea020ff8
a3c6e7c549e08142c83ddc3bb0b22243d60f055a804586b3938ca64c4521f980
180917ebd9987714db1caf1753679148732e1e055544de77e558edfa1ed8c01a
b57fd8544bd88da98d9c5c327ab5088a39a768ec9e474c686cd7b13900633255
5c8ab013e5e34f0f60dcf2bbb079520f10804370e062fa9808a4bb0693dc097f
5228f059bddc046088b492ea6a284a8958aa011367bbda4cb980aa896e71b493
50d090017093bfe2235651a22118440ddeb88bb9cb19da59f0508a8d1f08cf80
d8b239b589b5905428861cc6a0c8d1d61c7f85bb4bfd5aa331d49218c33873fe

Fiat-Shamir context

5fd4ffdc4249c9b5dd437ce0f751db6cf1b233e79a2c2fb0b3114c91650a2ed4

withheld proof term: 481 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

```

FBundlePoint.framingAux.eq_2 theorem, TFramingExistence:0
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (T : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (n : Nat),
Eq.{u_1 + 1} (FBundlePoint.framingAux.{u_1} T (Nat.succ n))
  (Prod.mk.{u_1, u_1}
    (HSMul.hSMul.{u_1, u_1, u_1} (FBundlePoint.invSucc.{u_1} n)
      (FBundlePoint.divU.{u_1}
        (FBundlePoint.framingD.{u_1} T
          (fun j =>
            if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
              (FBundlePoint.framingAux.{u_1} T j).1
            else 0)
          (fun j =>
            if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
              (FBundlePoint.framingAux.{u_1} T j).2
            else 0)
          n)))
    (FBundlePoint.framingTtStep.{u_1} T
      (fun j =>
        if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then (FBundlePoint.framingAux.{u_1}
          T j).1
        else 0)
      (fun j =>
        if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then (FBundlePoint.framingAux.{u_1}
          T j).2
        else 0)
      (HSMul.hSMul.{u_1, u_1, u_1} (FBundlePoint.invSucc.{u_1} n)
        (FBundlePoint.divU.{u_1}
          (FBundlePoint.framingD.{u_1} T
            (fun j =>
              if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
                (FBundlePoint.framingAux.{u_1} T j).1
              else 0)
            (fun j =>
              if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then
                (FBundlePoint.framingAux.{u_1} T j).2
              else 0)
            n)))
        n)))
    n)))

```

SHA-256 of the statement above

6c26177d324997e43d594dc81a27cc61a5527547eaac77e58b0b56ac2e292460

SHA-256 of the kernel type

6d5defbcf810bd33a169efac3e47dc77fdd76f149f2c7dc35918751b1338f9be
commitment C

38155fcfdea66bd8f2653eeb891d6c30e313a04091580ac59054040f0e643fcb
97ccde895c499f2cd05b2c8d5814f3c15f73739a748d8f96b29fcf1a54e06c01
45c81a4937340f701c6f10777d7fddf1ce1d8a81dae3f1c119f07392e8ee06da
bdae127585b2f07d48fa55b84ca61768158d7f4ca76cce45d0a9cc027b76647
bd67a1b4983a3b9b09dc023114f5fc7e6611767af9962656341a6b6480488026
072e54d6e678360aa548761f5b8fa27a7f7957fdc4f6adab3b84bc9510cdb688
0dac5a0e44d5ecf64c688a25a8670985d84c696b82b76cb6823feaa397631521
b2a1234cda313b24716602b09b3a4f4d6b3016c7f06225982cec3a6becd6de62
first message t
8b84ecdde5c9dcfbb7d3d969bb23e891543ed6691056259dd77ada21dab0c4d2
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5323d42d6f7e20275c987676184bea19bef82037bff84644d277ae274c928937
3f38d46d7afc0e94d0fb784ccc2f6d912638caa35dc03a3033cf050a377b6293
d6b0c43fa0272b1e792e0e604503cab8a8e0830f49d59ebb557df663fetc37a4
e696153d50faa4186aac0f835a139088f2e4024ea0c70a2fd99c57bf137391b7
ff3c3acb45d70b834f8cb7bce6da596e9cefbbeb9f7bb756cd76aea51fbdedb9
d96a5a1c0cad986514fda989440402edcf17b277885d90f3497f1f0af7255df9
response z

41a7b71b6962400ab6933d90c2576d39bc5522f740c41dc900fae04e3a714548
cd06675abf8a3a1ace65baee0c6fcf3c4b73bba404bfa97fdd08be4d1f28ec90

FBundlePoint.framingAux.eq_def theorem, TFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (T : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (x : Nat),
  Eq.{u_1 + 1} (FBundlePoint.framingAux.{u_1} T x)
    (match x with
    | 0 => Prod.mk.{u_1, u_1} 1 (FBundlePoint.psCoeff.{u_1} 0 (T 0))
    | Nat.succ n =>
      have P := fun j =>
        if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then (FBundlePoint.framingAux.{u_1} T
          j).1
        else 0;
      have Tt := fun j =>
        if _h : LT.lt.{0} j (HAdd.hAdd.{0, 0, 0} n 1) then (FBundlePoint.framingAux.{u_1} T
          j).2
        else 0;
      have p :=
        HSMul.hSMul.{u_1, u_1, u_1} (FBundlePoint.invSucc.{u_1} n)
          (FBundlePoint.divU.{u_1} (FBundlePoint.framingD.{u_1} T P Tt n));
      Prod.mk.{u_1, u_1} p (FBundlePoint.framingTtStep.{u_1} T P Tt p n))
```

SHA-256 of the statement above

0e4e3a1cb9132007dc3096b18b4fa8d139ef8d5a01b562e5a5daa7a567bd95db

SHA-256 of the kernel type

afa1da2ee0b0f6d96ebac02ee9e345a8c7d49ca6055a633ce4efb57550f24228

commitment C

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47b1414a06a40e77951831c3804e2a3e812497f16dfbcc83e72142d784cd6884
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first message t

25a936157f5e58453c5c8ce7bf79c8563edca6ce3b6b04c892a442bccabf2ec0
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36a8780e241f49bb10bd4b72e3093ed3b535e8442cdf1a4c5fb1067ab4b6ef60
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response z

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response w

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eededa50fae9509cdb39ad398825e1a962d2424ec61a8f939f87c9f4475fbd48
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6a5ab5b75c4f9a6b40fda32fc44ddc8a7b1416b4be7192cc200da74d126f4a6a
f08facd9480f8a9761357f28db1c9683b61353c09a377e0423d44860a034d5bf
9d3735ddb32fdf5f534de1861901ce68bda54e0791d5e32c9dc6449cccd3ef87
70345584f459ab1151ad525bbf96bb520da9a67af5b26c087a9f1d9b9f79d1d

Fiat-Shamir context

812c5cbb7e13bc138b7a2b1a8a44c9fe3adb037c791a1f0956d208ce7a196992

withheld proof term: 220935 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingAux.match_1 definition, TFramingExistence:0

(motive : Nat → Sort u_1) →
(x : Nat) → (Unit → motive 0) → ((n : Nat) → motive (Nat.succ n)) → motive x

SHA-256 of the statement above

52d0eb6c42572858dd0767c6141fae90019d12278bdaa57490569441b3aa333b

SHA-256 of the kernel type

759e6dc2b66d4dd20fa430d5f81a29765c6337b051c240b97805445f1dd27411

commitment C

a4dfa2a5e1a17c2fe962b2b687f9af3f92dcf057697f09af18fa41e0a7877de7
7f11f0b8e9cc524ad94654cd0b75877747bba6e4ce7df999acf4fd99d47e1e8
5700347dc9e0e7b2df4f3b1ffb812abfb1e924d7c82df3e6d1f6aca4bbc73220
b34c439d30f36eb39f460221e4bc8a578452bf81cc845079288070ea7cd0629e
bc30c7088b20bcb48bc50fbb476995f901a0c4c96eff1148a2237f29196b5280
6f9364a9f4135e39511f47579a7f50ec1117adf2bb773fef28ffc80d305b04af
10ef03602b581c4c15438b10e682173b068a128adc409ae9dce20b9aa0841638
93d7e19ceb725381ad2b9474cceacf232d89b0e7870d9e27858ad50d2e3e3e8d
first message t

dee1f9b96a9104f0d87ce232c04e13c6af4531fbadbb439cc2c0afc33469f964
d25fcf3ca15cc5387ce5974e4cc72acb4041cc5fdbd49f817e0d1915d48bba9f
e8f41c8a727bc9ec48a32722d5a2fb13c07aefc571b8c3fd75bc07940084e2b4
e50f4cc38170037cdd58ddae77d574799085dbc89c8f5fb727f01857c71c37c3
24370256db28f44258b1c1ae22250b4125bb1b51d2e2307d19bf7f3dd60c936c
c6369b8273fb190b302c2fa7d7f1f33b557777fee0e635d1819924f7de529ca7
087870aa199a47d068561efcaa73feb04cff613612a9a47d767278e822c5a177
ed461b6020044437a94fe4546e5bb7764479b1f046bba171de6826248c47efdf
response z

1ea64d00472d1d5e9cf4606c5055ef9213cb585a38cfdb347969ac7917b4ed5e
c715f781a4779bbd0e9bede7f108b33731c0f2dbf091f78e1830f04394aad2e
cdf0b0684ba0ba3fab58624feb6613e81bf9b89692bd83e56aac490328179ff3
aeeffbbfbab82c1c2e47544e745ce2eb375360cb1d76011d47e6f61f822729d29
3665bfe7fccad319c4ae9ffc1b4673cfd59b071224fb4e8516d158c0410eef63
a1b1b9a6d453defd99338ce95bc435fd0b34d4f8c9c680a95ac6b492c4fc20e3
d7d87d5aa52622643a3a253217a4b3f145fd8d84b3a6a1df5500f8aadd01c8a5
84477c8040a022e02c1c5a6229a284972829146dac9a7027b1fe41e061602a2f
response w

e646b923835b15d878335b6e90388182d27151889ddca4ac133d55ae12a63470
f4dafc7d6616a39cbdd94a7a4c07f7a8a40420ef2e8d35db79ba28e2cb8a1919
408efef3a83a1520aa32e5897a3b8017e15914b5c45bd3953e69986a943c0f2a
14979454fb124463b9f8684b133bf4b7b53da1675d374c431a2de886b6c29ed4
0885c1ab6c3060d84f29926d8689a7d61ca92bad92c2845d462fd54ce347e0b0
a8453ce285772319f3e85f4f9981a55962dde9d6aa0a9d80b1df0f22d9705086
51598ba8165f1d44a3448ad1f133b21531e17780db512b0423210c5becc170e2
bb2650c48c52d5eaa74ab5155b350e4b426f3b11488a6f8943b873525b30bd7

Fiat-Shamir context

db8ca1d71f3b5f081a5c17d72a9b82029420fecb7c2eb92b7c0033cec74821f0

withheld proof term: 604 bytes • axioms: • kernel replay: yes

FBundlePoint.framingD definition, TFramingExistence:106

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
          (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) →
            Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

e972f3bccc5f38d8d4a86f683befe509ca23e3a2a2b3a2362420ca7ed521b8a1

SHA-256 of the kernel type

382b1706050c21ffd3172c53345933d013d4f745db0170e32a50995b02bb2b40

commitment C

8b716daf9868350e7946e79fa551f6aa634a33ca50e837f508224740903bfc2d
163b55c217b4d7c13bb4a1d186f4dccfa743a86c9fd22b83527c3b93dbb7e0a4
fdc1567f7769bfb9de2138347fd04957f5bc71a4f213abaa26f6701a91f9df
3d0c91d6b0959ae365df20b9531f74cd25a379d7e17729dd7661c02bf520d77b
e06e0b76d488fe31cf8eedfdea7e3ab687d99cd1345a8d426ad16b56b0b0aa2f
46efc75f9b9d1c4b58ff111bf74b7f1f6378aa8d902756f87db948a235854ce1
490d89e07221d54ae198e1d9c7ad420e1935bf5782875e07b613dddf17a759aae
025470ffd18fd8f3b6989fd092fd0281481f9a79af1f9536ecac10b56a7633a5
first message t

8962b9264926c80b56140e5da2ebe138f2c6c27664ad6532a1370e621828bc7e
6fc9e714afb291e8f853ca04706ac9d7ef1efc33b2d37ae340957a422b049e80
33f4ec0b3ee97a18ce64e9e081054a9d5951aa99eb8614cd0eae39ea15e86d05
f0b1658d14e087fe550b52ba4e27e35b7cdef987708ae1a671bb5370302482c0
d5a8f5604be8a1b6d2f4036f9b250ec785c9b6a4e9a09d9f6cb101a8fd9ed2b5
512754a4a16983b25f78fd7e5021524a21658737c3c649c5c82faa5a837ef0c5
afb2b34a6c3b31019bbfa375ba20bdfd020ac735f651c883d13d26f2318dcc01
b776bcd40078039467a97de31ba58e370e3db7391a6675746288263c82438ea9
response z

68cd10cebdfa4cc047f03daac417fbabb0c612ad92a29facd334850441dc78f9
974b6d4556c202527b83063a9746e40947a940cddb5efe9e1d286a2c5e10b890
3018943c78a5b3cfe01311634de29f7f61bbca0dea83cf77d9e3c688a45497a1
fc25381ee3b496d969897edb168a8881a0b833e479fa7bf02769bf75b5cdcc65
7b918b3d1d34c06cb1791af8601d597c9285d938926d3b262d6c67d08f95f3b7
347384c313d684766ff260a869e572f22b7b1cfc4e0055987e477f8c6e1f7584
278c169fdb0c83dc9c54733eacd84e1f7e5ffa49b73cc2de52ded39bfbde9fea
638d4bea833eb059a0ce6c9e296bd43c94515caefd48db7d7226945f883ddfd5
response w

38c7ea693fb148628915ee803152d90820ba473590120de5c6d4f199b1f67e64
084aa9c45ae13f4d1e4e7293b3b8bd486d73659d7c298c622eae591c9a4cb643
c8f85eb27984ecddbef7ce44314fefa059520d4760392899562ff08228e487f1
34de594e2ca44e5ce2bca65ee4d7e0041ddb829e9575bf681bfac47351da79bf
2c4170d6ccf1fbccc60461e291bdfac005f9a3ef0e4034f3d0cee3285759442d
daf357aab5ac7e228a3ee2bed893fab6bc2c9184802ff3f088088ec1f2861df8
29b8d079f3e3335faa7893b5c5c7f080be7bfc36e2496a268a266491b201e162
83bb81d6e02616e2a5e502094382ce3d1d95180f6ce257e5d9b1d2df79d82912

Fiat-Shamir context

5364e8a0b972fd6dbc7d26c6626d7cfa50b40ae46fbc316ddcd30fa64a931c0a

withheld proof term: 4318 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingD.eq_1 theorem, TFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (T P : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Tt : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (ℓ : Nat),
  Eq.{u_1 + 1} (FBundlePoint.framingD.{u_1} T P Tt ℓ)
  (HSub.hSub.{u_1, u_1, u_1}
    (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} ℓ,
      HMul.hMul.{u_1, u_1, u_1} (P q.1) (FBundlePoint.cst.{u_1} (Tt q.2)))
    (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} ℓ, HMul.hMul.{u_1, u_1, u_1} (T q.1)
      (P q.2)))
```

SHA-256 of the statement above

f63bcb0bd6640a6f54e130e1c2011464fb6f76cd34a2f3ad0c02d5986c6bd9b

SHA-256 of the kernel type

5473bb19ef8b298c7241356eb0629263d6e4f1db64a0c000ec8708ead1d48e6e

commitment C

a137696634d5259cdb07c924409128a52a32ae04f79dd1dd37bf702930ea703c
20c77aa3df24985874e7d068d38d03dc8eab232661af8adcb33e417c1ea7c48b
feaa0a983b86a66d46dc35ae6b810cfb00a7b80fef1259fad984afd4e22e7c32
f8ce91ba2e8e91ee6cd17ba209ca98767866ba02127bc63e46a6af9c859e5d07
bbfa705f977f7b6f883d5ac0c370c4d7f91e81d7a9f1c71487c49074075e86d9
0a6ec52474979b0976aaee3ae7110b66078491277051f9ee92505e25c483a806
0b7e5c41cb135ce17696fa058084ddfdac6edd44b4d481a88ff8717b4fb91bfc
14e904110b5dafbfa87f2a53b6be49bd6ec6f746fa62166470de17a4a129ec67
first message t

506a82dc0c8e3d5fa363ed49e51cef6cde6cad92ad8c929fc336d1bebfa2dd45
dad3fb8a8d5d8274fc063698ffa48d11c9d3693eed62a5b75eab30f79db16ae8
41006d56aa3b736065a4aca89a8dca70f652e95f2acaea849997618de2d23029
94315a5744698d5caa02ca0675712e850abba9d423de2652246754981cf3cf09
74b0510470e5991dd90379d579ffe6eef4f869093f6c2ee480d519842ce8fac0
03ef7284011249a9c831fa66cdc721f65d52e72e74d535140d508dc6cab0b052
1359c528631f184407ed4e157c5cc5e80930d15229ad270c9aae62821086a400
efb011cb2eb12ad167d5ae2e8dd35f8cc3b8c11e8491eb0d44d0ef29935b1e8

response z

7cd342c5a59aec170d7eea9766d460827a33cec079d7e4e6ab42e80c5445cf8c
843554f5fd7b4d97f7d643207594e38fa1f58e3d384ce910f411d9e2610497e3
2062714191b9045d297e5ee46e6d22c4c1883f6dba6bf6116334f0c730d320e2
383ed20f3eee6eee9b5a1232af869247a07c7b6d56e3ed10c1d29d8e597194ee
520e22c42e8bedc812e6d95f9ae6c879344ca24da63ca23f741932f1a28563e5
345e28aa3782a58639f35068488f561044181f6d4595226528a886890e4490ae
7c4c93d6b757f6ecf62bdaf41a00c51c4a9e57ab115786051ed4955da8824b6f
73c6050789bb43242f5de3d34b0bdcff20ab09dfe406be60b29aea533830f0b3

response w

49f8fd9ea4adac8e983bb2f6ddb4fc4e064dbec775cbb432128387f84e00f426
d1ace03ee2b7bb447edeeb8067c34ef96e83218571cf5c452d1027b0cafbed0a
32ec9409779a2eb1003875719e7c67fee22a5f00b1a62d9f6e50a14afc79c1b9
441f379ceeda2fd918b70eb7c2112589dafe40450a6289348ec06554a892053e
dc49c1d3f1b2acdb2a4171230ab09956291d1b4bddd37a65b99398ed00962fe9
59b1fe49160e957940e3517c401c54279697fca6943894df1bca570d81f5c339
01108ccfa4a6fb46cdb4b7f69084ad56733ff98af4a3064af9430933a2f9942d
5920e247ae1ed779647d0de9597d4bae21373ab9afef27d526c2f59ea6261a4e

Fiat-Shamir context

615656f87d61343d32c3829ec65f687633ec059342727f620d875610b90f5da5

withheld proof term: 396 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingD_congr theorem, TFramingExistence:124

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (T P P' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Tt Tt' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (ℓ : Nat),
  (∀ (j : Nat), LE.le.{0} j ℓ → Eq.{u_1 + 1} (P j) (P' j)) →
  (∀ (j : Nat), LE.le.{0} j ℓ → Eq.{u_1 + 1} (Tt j) (Tt' j)) →
  Eq.{u_1 + 1} (FBundlePoint.framingD.{u_1} T P Tt ℓ) (FBundlePoint.framingD.{u_1} T P'
    Tt' ℓ)
```

SHA-256 of the statement above

4b5fc755ac68226be8522cf757cad2b28ed5272fa77cb2b7fc2ac6a68ac8eb56

SHA-256 of the kernel type

8c7e621f3a1b39c90405d55a1eeb0f5bf838f68d17a8234daad9a5fe2d987bd0

commitment C

c58972fd2c159e760bad7d63cea9c7d57216943b726e14fd07e86e22518886d8
c94dcc2a2dfa77492d03aae824dbb444f3baf72f2670abcf86bf7029321037f1
fbaca7a123cfbd405ac0d792462f8651254bc6462d07f50af4d7b5411af8c764
37b1f9261862802f4404d55c49ce68104310dd435a15c2de2e79cbb111b713d1
5f44abf1a8c2714a699e6d57224ff58863610969c7e0b2e4920cbe5754b84045
645922017ee256689897b98c1cb82ba309719e754984de444736f567c5dca408
fb1609020b9b12f32ee865ddd7a5d007d80cf5cd99fa388e3d3a73e6b874fef0
ac07ef90100de69d4ddcbff52ba2e18de998edb75b18a76bb004ed3b2018681
first message t

53e5c56e94f013bed57071387c94db489ab3870613a323e0d2cecc836206aae4
a60cb554e915ec366e571de0bf97fcd62d5e98d442c23501e3b1fad9f23ae88f
91269521ea84488a3fdd30882359149bf2a24ef0ea37418575224fec5479441
e7043ea6e28f0b2407a827da4ac90c20ca7ebe11324ac6c6c18db064db5db9f5
36f7bdc927a75e7366575e2a4af3092dbc734f740ff4a133686b2c081c494add
ccd4d7d83b7c55d23e162367729cc5bfd4972592199cbb2ef7b315dba74148b4
3b11990e04a73eaeac9524371f839035bc69d5e0ef62a1f1b1a270b45ab164cb
18e464fd35158b593628d88c43c9308bb0f76b9310430058f1a36735eb8171f0
response z

6b8e3386cfbabe1e62e1bf15fa04c1f613c0697fed041ea123d05dbc61f9e02b
f4d1964fbbb8c28d1144fe491d1a6ca8645dbdf59b6b2c9388272038d263d5ad
302aafb7d7c64b2ca4bd80056e9c3162111e8a5511c5b4ecfa177d80f6f7c769c
a0e12f0bb25b06aa3e2feb31d69bc64035862a426e3cf8e24efc9543efc5b262
f2dba35f0c77f7a6f4941a68428b38f17fc7e140b323e4e44a862aae1176a23c
d3222ad21f4fa3ca337be86747b1cea2633107d2e4ac068e21fc7c8ec8f82c4c
49d858216b76b1d147943fd1dd6dd7d9926b19316b04e5df87b670c7630eb778
98ded64a1630d7ecf41513d28c4facea5f724d54e7a74f920a77e9ce8e65ede6
response w

30a37addf09a5f93f06c5d4ed80cabaa74e214431065a676cce9dfe26c086347
89e0c56114e59fb10972c994863f499e762cbf72a257ed6aa0b2619d5e1b0e2d
7ebb7a6d7762a1b22b2f8e950a2fd927013c1888fe5860e59f1b3e9d39673cdd
a690d9e66c00fcc3876ea535a403a8e49823753ae59a4e47e907b0e874ba7b44
0ed453aec0a5a032bdb3e6f42e48a49c987dd3c93b10b84cf75132c546698042
9379b285a6cff4a25b79ef2ef8a0fe353bf372ece5bcc9f675da0244c85f9f03
9f9cf8a9fb76aaede6ccd3fc7106189852f236f9b489168bc1011a3e04df432d
f862ef3f9a5fa7b106417b7afb0e0f3bb77b5edf321230af513fbf80b207ae8f

Fiat-Shamir context

69d11abcb03d2ddf76a6d2b0fc73c4af4315acdcc552b592d209a27f0f01600d

withheld proof term: 52748 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingP definition, TFramingExistence:201

```
{R : Type u_1} →  
  [inst : CommRing.{u_1} R] →  
    {m : Nat} →  
      [Algebra.{0, u_1} Rat R] →  
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →  
          Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

33a642df9e986d92a4331b12bac96cad689b461298ce2eafcc39b05381172295

SHA-256 of the kernel type

013b5ade926eaadac65f08343356f82333ad6ffa53c50503c92b47f82473c0a3
commitment C

2e5e1df9c010289e31ea90c3ea9d5a85a6ecd9dfd5576b956ae59c4f24713746
62acf5e883e9eb9b8b4790b57c6d9f39c53ef51526eb1651338d033dada9846a
f21953bbc7023bc0ba7e24cba25cb5b13d168ad0743d7395f4f77678f2ba1ebe
3d54643c5c6c542d99e63aa7e874669879bf8f56a903a9fe34a2280770efde0a
6719d8e96241e1bc8eaf91c382763f2d6ca64e0444ff80f91575b1395df90239
111eca86bbc72f932fe9ef963e84679402f2282120a5cfe260e605db18394455
2e0022a43b394bdb9dff7c2cec1108a999c2fdc5e9f4f9d0faeef2b8a68e8201
af25ba7a0ef3ec12a96b0215a021b3843f69f1b4455dd9cc9dd5bdb84b8bcc1b
first message t

b1f729c663dd3305b0dce830ba2dcef2124a0b53100dc12147729af413048fd9
2ffedd6500a9d29cd88c9ffb51e063535fcbccc7223d8878b53af19ca12f821
0b9864e1ad95ca46939f434cd705d2d06d4f8772fba40eb20a75afe65c67106d
71dc7e9b3a432c5763f0b496fe041b856f289f1d445183c48681d7799fddf2a5
18e9b6ec85a3f6be7da8767af11c496937c54f9b2fcc100fe23409ec93e6c56d
37f1dbb3676c0b6ecf19be2727ac1e196d916a6cc1286f1485d0b0839b3e9b58
f3d44cb7a71950b7c061574c14e33e6f6f9923b127d90f7994d607fc3040307a
ebc8155f06062e752f73f9745cc477b34aec33e5b8b2a21b781f7bb2d548443
response z

4e8b13cbcf477b381fe8e11d130d51fadce0cfc2c920467d252a420daf603433
f96aa895061880dac615a5c346d1e60585a73e75396fe0c8d105a5678a58e3d1
57d00630f60dbfe75adf223194d0acab59cd63a7cdce8a38885239c85a22d226
848201535babe1d8c7c9c29f7141a4f9d2406ec65e1e9bb32f3905603a5351b4
75946ef54bd3a0c6adf2fdee45db70f7df1294c574259d7a82e084cf247f949f
23e694c2fdde3f16fab027f095e60e2ef83f3d7fd3d82e6589c0cda686460e05
91e22dfd8fdeba131892e843c39080ea79f831ff2e1250679504381c0ff58b64
46a94f08d587d62ac97be5e2bcf86acf2128f1070f4604ce0b01771312db6a6b
response w

23bbfdd4cec7c6d54d1d705ce535e5f9605cb5913c309bfffac4517e7e3ba8153
6ca073942618212d4afba4efc91686515008c021e680da1e90d8f215b567f5af
52e63979bc2d3031a8573977f9fd774716ebf3a909f67eeec696d1b73cf1d024
f24d55600cac90f1e1c611a96243f10c8a121ddbc1039b189b8d327ed8abb6aa
4432254d85485c339256d314fb669bc6dfe40c4a3583d369ca578888c0457316
6ffe789a9cf9028b741817f98c2aa4eaf0747a33e073a2d5577fee25b2b53db8
517f199d8d11b7dac0afe220181fe41d95ba0526b3f14d55dc5ef68f22d20eb6
3ce34b6356a40e5f11981df39765560e23e73db67f93c253f5e25a9340e263b1
Fiat-Shamir context

515c36695070bb50762ec2fb88d717666058591fbc380275cab3072ba396c76e

withheld proof term: 741 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingP.eq_1 theorem, TFramingExistence:0

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(T : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)) (n : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.framingP}\{u_1\} T n) (\text{FBundlePoint.framingAux}\{u_1\} T n).1$

SHA-256 of the statement above

3d50c4fe4772fad0f607bb3dac922b2a241b4a3d491d1877f009cc74aae2667b

SHA-256 of the kernel type

bdd4abed625a068683318380dac81d2851125b6e4367a23aa2bfa6b92570171c
commitment C

a31986fa45a421861d45a85dc1d9f88f8e7f0c5d5120a7b1bb19d5b4bc2f77e8
1cc54c653702f70bbff399264e9fad9558172b250a2b9bf75a9f31179ae97577
61bd44dc63de4a78decfe7af10e62268d2400c3ddf06604633826cd81d4e8386
4653e28adc3ddbf45dd6214a55537257e61bc91c1b09e60911eedf18923791a6
002a575b19d0b6c235426d217db7023a29ad44a1fbddeb45106f7666710607b6
42ffea9e5cb12882b3f89bb01e44739d1b714a8a16c91080df4de333f26dc1c8
f2bd8d1fb6dfa858b2aa882ddca8f7987402262b20d4941e42d28f62408bc4ca
e8703b5a530b52044b386b9fc99801d3b219cc259e6f598db37482eee0f0c754

first message t

bcd579b0edefaea21293efa9f3a9929c700e7e4ab4779525376ec90a0f02a73b
5b69cbcd51fb701993b509433bae415e6d130792e17fb59cec61dd203f8afe0a
3665ed997ae1623b176e0af38fbaf5952e80540825bef68f532ff626463d0f76
7c80922582a23f4cccef646dc7d8404ef0c38267cf576a5a8ecf41e1a0fa5d7d
64acbe11a184222a7e641ed58aaef843c35cea778cee27548e4a6b0d0d98d5d7
f1e238e1ef1275d0223479d663bf9f19676fae461e3ac51279ba796c60a00359
9c8e02845a120d26b660e90056a31fb7bc97f48e3881eafa94f6eb10a87909e7
c91cba6e8bdc1fcc530e64e38ea76c71658eaa76e0ae42e65e51f28dd685a65e

response z

5da95e8e6ae7280df8784582b7f818c0f7b0a570e91f6d59f9a4c4f89142f037
e33f06d9b1a4747f46ba61b48d41e9022a827ff881ba6d951732062096ab07e9
3ede357109ecdad79a42cf93f2ff51d76c138481edc80dd7f1cf09c10d14daa7
877282eb5737c443dc5497ef8e34792325a86ade21bd097e9e257509a588f314
867e4651684ac9827e62c2c3a605010cb8ea157f62e9e3410343ad4993f55cfe
41e0a14be03c2989bf601c4909aff240a7c0e96863db17771671157cae079c29
ecf74df77c1925901b300eddcc49e2acde3d9c390225382e2ce32d8a669102e2
da1933264b4141933608679fc15a204b589a1d3fdbf1b15110738d4112e48439

response w

b931774c53d8a1a1ef07e5451311baa9ec5f2f1efa18bda5143a10f9908f06e0
ff16b75ced6b343333cb59111246ba0d16cdc70e6f3884fd9d86a9553b2b76
3a7a0f8d74263fd72cf39ceefbfb76a381e22fe237d3dccc355b84d19a6abf0
fbb18cd952e85641bbb93474cfce497adc9c73b6f5a04f915eb9e65b67ead660
1b6bf0fc9baf8d52595eaafb3c456332a9c41e007b6bcc03c70f1476debe948
b34c889bf6591e576eef970ce3ebb8e084c0aa97dae90636ce7c6c5d67e30e0e
5af7ba6c8c2e51d5bdbac0d45487b7fe4380ec1d3b5460f7cd4b7008c6190d17
7950a02178a2566bf0f70ad4b8f816de7cef0f40d54fe1398688bc8f4c3ff13

Fiat-Shamir context

025f8ce610f13ea98ae5d80a0d8ae8db651771744bc9273c1e5448240b33f13b

withheld proof term: 393 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingP_succ theorem, TFramingExistence:218

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (T : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (n : Nat),
  Eq.{u_1 + 1} (FBundlePoint.framingP.{u_1} T (HAdd.hAdd.{0, 0, 0} n 1))
    (HSMul.hSMul.{u_1, u_1, u_1} (FBundlePoint.invSucc.{u_1} n)
      (FBundlePoint.divU.{u_1}
        (FBundlePoint.framingD.{u_1} T (FBundlePoint.framingP.{u_1} T)
          (FBundlePoint.framingT.{u_1} T) n)))
```

SHA-256 of the statement above

7fcfaaf2991e11682e2ea6bb57a7b0a4579c779291266172f4bfca1bbe3394b0

SHA-256 of the kernel type

4177b180272e4d963ea1a224f5b48b942b3f12aa90b1e54a71e3362818ed25b8

commitment C

330264dacb359221f19b2aa7bf678628b8d72878639407429d48096691f6f832
c683bfa091ecf0f66652617f0e775fe535538b7b225ff0de817ae67ecf23b6fa
4045665ec80a2e64b1d3d5aee18522f89b33a0fb27a413964e5d2498b8327d49
86bc7b7a98e0520afb7f14ab5174b932471cb8a27334f46d92ded1ffd7a561b5
59f7b6bd2211dc85316a100ff5bd7b93fddd49586bbf76d246adc093d3f961b8
79fa8ec0406e2c8a0a2a35f54d859a2ca9dac38c2eceab91b5c2092419bfac1d
539a13a718739f2bb23706e3fe229c5559b82d980d8e792f66c68363cebf26ca
0f71de0f702d4f1e0f7db99905ebbe9238aaa4e11d2e96ad45465e7a1a1ff79e
first message t

caf9e96c6ba7c5411abdd1f3c7a225d0422d2ea6882eeba875d125324417fce7
25d4d9074e0a88babcaed863c3c2014ce81ee2b12b7548ea68c565defbcf3c35
8fdd13610202c11ebcb4e6d3f68aac49f46ee47daa7c61f21d29144fb677823a
9ff6b3b37f5512c288c81c3ee9d77a4534f2d2ca8687ef62cf0d1bedc47b6741
07d3a88a2768f25b5cbc56df71580250bcd55335bbed38571e2fffb65aa981a
00833abbc10a3135512c24e59a37b07cbefa57ba1b0672d426f572f1e7408ec6
2ef1b6853af6429208165933157a73018a4fbdf178db4b5bf0feac5f7d0923ee
7b9ba1f71b51737743e077ca22f8afa617c0a76d95c38b29798c646c4f40ae68
response z

78ba3f060d2e0c80a3712eebda9f840059d0e78ba791ff14ef261bf90a1ad502
60a224b5b7ee5e0f9beb40c24d2039c240cad97998a9476d0e0ff291edb8ab3d
d1e27428d4c7c9cee7efc2bbaf47d94a2dc280ba58c6d835dd6ff526b874082f
f8b925728d94b76ce69c010653621988c6d8bde908894c7d8c21ff4cdc04dc83
e554bacd74422c5732a5d3ac9ef779823f2d057782b81524ef0d163202945c70
40ce96a0bb746da39b0d3fb39a915ca93d655b3be931f4222bcc0465d6a24fa6
993371f71a2ce96dc35808e20d0ecf33b8293995c223e4e7a20eae002598ada
e2ee0754bca7f40e5b6cc3b27320617cbefdc915a19b6b9d90adda6136df17e
response w

3f9514942271bb0d57f3688cfef4eca862205967fb40ad5834e9a9cc6cab0452
21cf7c53200d47469623049cb120479199025a05290657b5587a8362836db3c6
083a1c11dfb6792fb20f552b03d5f5e72d7491d942265716bd2749bc77e4162a
9e7463ec53131222a624a63b25de891fcc539bc47fd4998a00cfb459879dd180
41f83c771132fd2928cef56d4627208ba944bb00b5603df528971bc1ea407e47
b018db2afebbad05f6fa8d6330476301a79d1d442a70bb664035a985cd5f7077
8a3385dfa953b0110e02e7ea53ccf7dae0c42c27fc8fbac8d394cc44ad9e776b
19f9bfb2e6dcae9bef7275d8cb4484425278ab543aedfb691d16b8a4697dc2f2

Fiat-Shamir context

6fa0b043c6f265202982adaee6af7491097ea298e7c7ecdeef1f7f5c5d3b2c68

withheld proof term: 106 946 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingP_zero theorem, TFramingExistence:210

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}.\{0, u_1\} \text{Rat } R]$
 $(T : \text{Nat} \rightarrow \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} R)),$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.framingP}.\{u_1\} T 0) 1$

SHA-256 of the statement above

5dcc1cf50c5dad31158a91d75a64a22ef13b155d6f58a6f55d725b2140a03b96

SHA-256 of the kernel type

8b9e01ae11e68db6c2b7bd0eac80c416b1df62085d650a04dbf997ad25817f90

commitment C

f26e5b3d627280101c3e811cdeaaa7b790718ff36b496667e9817cb2fa149e11
e7c11d2f5ca298debc0608883ca0420df17dd0b457d794f6249e6c15a8875700
680903dd89e02f3cdf21803c40e95e38aab3f63b837b85fae6197d4426bdcc21
b47d20d0b2c25d4af8f3823f4d8a9cedc0f130fd9badc2071bbedb3a0bfee330
ecc44165e1d7ab611eda17c73028212347eef75a9e3a7702d8de8985c8fd0297
56c6aa4091c068f1e5b00902ee7667f463ab028b82aae84bf5e514600fc37500
36e5d2e2b1c483dd4e93f8c007164252a127b12307a8e75244b7c9bb4d70fe16
0344d4b6beb4877c7d61f4996ca23dad8996477cf7631653d82c2b0127979561

first message t

bc00bfce54971f2d9c9543358f5831a2bff9c4a40c8ef56c465f6ab632b9ac16
4d4f6732e54d74682f341736ab4158d43a8c0a1c16f254c0d55c79330b99de9d
285ba3749a0e19c42b6ed83d6c98612587d4a941d19d1a74d473b0dd26887943
37b375f8b4197eb5bff4066654c653acf90707e075cffd936fce3c990d11d618
d72148fcb7b62497b87b685c5521917e0e97df5ba47166bffe0719377d8c4e31
03c5878eaae8590b40d7788b62c666c3bc6ca78d56f0e858ddb9410e1d6fbf25
5ca1f9896d28064fbffe204f075f2613aed6201105eb88f5f6607cfc9cac680e
2f53ccf54d1d35b7beb99aebf98d31f1966f40ce089f1c977313bc25003662ef

response z

53c906471f95e24cf0c2aeebb9827ee0d28a36b12427f15fc9e87d58eb7931d4
61fc20615ea988dd0a3f1e3e9c89e9db7d1d5476593b6abc8ce6a574b8850261
e30e8906d628246b877c59f84c9a94d406cb5e50ce7923f71ba49e3f0863f186
1cc3db2b12cd1f6effba34e303024726d4763c117b9143edf37764867ee63d3a
9b6d1959a04407c51e8092d60905c6c44276242ae619fdae26c270677b010359
c79cc18174f1f2ece11393103df6d173ff1dc307f4851a0ab3c46e3a73956ad6
7195e9608c3af1a3b316919c4edbf8917c92afd6ae532f02cb3ce709854f19dd
a75936a09b6d4ffa3aa63dd3d4bd1b5721a2bae1caa2e61315194d3b988ad717

response w

4dc83524caa43a9bb8003f6477ddb8db5ce398fa01fca420be096b8921e63afd
f50bda060f02cb816b4e66b5e4f1d146ee233e73ff8ba1fc7e91a68041f7a650
ea7062f13fd54aaf2fffb4ebc54f65b9e75de659a13c629a65bd89a79a2ec8925
fb64d798b0541970adcd3aeae8435d2649e35d0fbf8a5a1c3ce0a3342dca233a
a5d19e69f76676508d0ae899ee5656754005cc7a5b78a3692c5a67210b4020cb
1dd5ebff6898986d2a287a8d67e3c609dfb405f05f5e57bbbd3f40b0c3c57b6d
7d9550dedc9e2e63718f6328423b8cf405157d50e9ddab0c1c2dc9fd36a68637
ad26f170c25bbeff63c3c863d3687cf26c837361a4049a55212e1b26da59fe21

Fiat-Shamir context

8638e901dac056cc30160165e70f803bea585429848f4549a46f70e2226f8b26

withheld proof term: 17 539 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingSystem_framingP_framingT theorem, TFramingExistence:341

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
($T : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)$),
FBundlePoint.FramingSystem. $\{u_1\}$ T (FBundlePoint.framingP. $\{u_1\}$ T)
(FBundlePoint.framingT. $\{u_1\}$ T)

SHA-256 of the statement above

28a8fdb2c4e7ea9a327d07e7606f0a70bd30aba01f72a4da09f7bec68c288ca3

SHA-256 of the kernel type

a0db2baa0ad8346f987227f35d00ff2d95a4381313d69c5026880afa8ea2ca07
commitment C

a1db4b402b228e57b38f1b222455f1bbdac4bbb862627d7df04102a5c8bd4b2c
cc08c5ee084027bb0a9a4f4d73dc93aacf3ba48ea2f5310ab8340e5aef5fb752
af79da6cb2bca6f9addda38f368b2dabc3ac6ae38c6901ecba5e454ffc6b42d0
d6e13070003c04ca16ea0bfc902939231b92e3781c1270bd8d43001072c20518
86061b1b79a16b1c4983e3635e0b4f3ecf82d3f89e215d6535ddc23866fb1e82
2617b0d774917ac149717db4fd8ca09aeba0f3998eb2042727260a21c95fad30
e97522174b8b39523c75eb5e8bdc733ff0a8c7326bdd2b54bdc14949c0afee53
7e0a636730d57b0edce0cb4f758981d55a73130f0118a0e993945aebbf867351
first message t

5c68549aab9f26bd7b74aeddff73578360ec0cb536db8fbb08be9044e789c40c4
69142f8c3af98ee188a412ffe80fa4abe31334ce100496c32b73cc97a1d21d3e
3cbe63e57bf7e47d76e5ba4ba921e6812e7bc8afbd56b334243adfc7735cf469
426456cc2c5c8a7c3a90b2e30c174ef121d810fc7ea15fa6bd9f2fce841df45d
345d587b9e8da574594f3d313dfea7fd484a4b6b5f6d394629abef3ad2f9849e
7c21f17ead7571ef95567ea5b6620fd7e28ea64d3e3efefbc4bbf606767d8514
828af5c27f524031d6431cabd8eb5cd5f616da5c77092c12f8b813be0c969ea4
bbfa7e78255d235f5b8a409148a212e6e7502f2dededd96031e7a689f85260b0
response z

127768fb078d50c0113ca8a7aad94fbefc9d7dce58e7fff2e6d938b51a76b8a6
a11ac74feadcf7411424b4dbd808c8aca7bb7e2b546f95578a6d3f18bbd24876
08e3ddc29e2e77fe267b154fd91e2668fe619d90feabd870309c340639d2e521
d75ba42445f850d4e74169fcd8a1d26a093a1fa3bb1e074a103129dfedc58f5
4aef6c2589d9b182b2551e60ed388e27da1f543bb3ed242e9c513aedd4cb89e2
3553122d31d93f69ce9b4401750800392101d92152e4a86f6b02c8cc490d69e0
b62c38982728a800931087fa223371f496beed776012bb959223de4406a034fe
fb04a8f39782d2616f8adc9bd3ba396e15be9fef02922ae0aae228ced97f477b
response w

117cca9455aaf0466e5d154687a77e760a69081a2e0f37eb331805e86c8deb02
7e1aaa3961523a7d22e8ff4b8c798220dc668ef3b9dda8d26ff6ca2e484fa6e4
fd4faa2720da49bfa625ae2aef281f71d4b9d7f9c3e2f892ee98bf18ffc0a8ac
88ead9b7dcc98e44d3c990dfeba063d85fdbd496001f40550d5a47c71bb6e279
0b4ce56282eaca0c273dcfa57606cf3c0f0ab71e9cdb2047d313d6c0e8ba457d
dcab3193931d41e9323f335e1d74e313814648b62300b72541d8c72fa77c53d2
8fde49e6f7f7123edbd727d078778b0c91fc78f898a0fd0732c00925042d5e5c
6be9bd4907982d2b72f7900a4a3c838f7440d2f3b10e0b2b2526189ccb7455e5
Fiat-Shamir context

bec22d24a9d04569c1dc9001f5f037b2db257b64183d3948b8a84f4f536a98b6

withheld proof term: 305105 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingT definition, TFramingExistence:205

```
{R : Type u_1} →  
  [inst : CommRing.{u_1} R] →  
    {m : Nat} →  
      [Algebra.{0, u_1} Rat R] →  
        (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →  
          Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R
```

SHA-256 of the statement above

27117864f6c1c871d136ba801791f1b690fcb283354b1e40f5b8d808073ecf5a

SHA-256 of the kernel type

a0c106903a6cddbcbccc0fd65adbc6cf509c589fc0b329dfd920c4c5995a1e19f
commitment C

1b908566c46f36ba983baef12033c930adf5312dee40abb3179f1cdfc506f6f7
232343c9c6a045bb9492ec8db89501d7e75e04b53ff784589025fed67f837db7
6e3f144c35911fb4750283688053f3542374864cc66a41dc499e4297387e1f3a
03178762a2fdc5b50884b64f7b0c9adff5855158915b83ec21057a90201f9d25
2ec47d817c60a047f4f754f9f53c3441de15d570a138e20aa3d49ef208b1e4be
d631d0be0428d700eb8329e654d997e461ab567bbbe86bd86075800148aa4924
933a56bca83aebaa4f80d1e77714a47e3eb73f85fecbac7a70636b8691605ccc6
71d853b0b716c0fb182d48248d2314bd7b05fa5b76be015500f357fd41f479ab
first message t

409aa6614bf38a89ff1d2cee9693a9c2bf7007dbb667e8315386ec1228171955
a01abcca6afc788633e94337b459552d79a2080e48f4c3e8c2fe97dfda8eeb5
4803526ec127626620969239c6775ac547760d7c3cd1d0972d6651d3d2324d3b
d1a2846f318f2935078c815a6400559bb40d992137e27b93a7d214553155fd4f
7d77840582c26f356132e37a24577ceef08141e7e8f56797d6bdb7e7376a7041
b0cba31e2bdb669c1bbdd30c57c08386f25651d42dc4da59595b03cef69ba7fd
66a21d692865ad6f9406f228367f8a83521e77d26a8fc2fd8051e415a509a489
85490185ac80773ca4facd7300fe1e2dec21bb27d5ad63c3d816669d16791659
response z

5a90d176d3acaf088359cb9d8ada79b1e790b663f54af0a70d42ddb3af3822d9
210d570c78852c0a13b8875332faf9ca6215f394ca1145f42303c414e45529b9
6aa96b8d52835fd2e565d89d7dd72538a00ab3aa09012ba4720020c0b5dd3d24
26037a0461ded9d5d093e8f0207acc9de896784033329f756edbf830c806c97c
cccaec6fb9eddd6defc2779118cc8e474bb5a3beba27a621f48fd5deb432c068
d79907375611dead7cebe73c2c49d8b71fd5dea216527e8cd6188d632f4b748f
b003f5975259ac63be03a8c1b25967a7324a3865f82df7b279ab3823421e96fb
3a065b7f8fed8b236159db7477790d3f306955586a2e88da67300a7b581ba9e1
response w

74e56031ccb50f0f6dcf8551baa0bba5960a3be5b506dfe5a90c52c78ba55467
0bcbdc9ecff6ca91f4377b12c0558906aa4e00396c572eb5023fd22b3d3a8b35
0d48fdb240367d5c9bf86615b88b961b3f774cc19d6a5435826ec8465636898b
a1aa0beaca71d4d9b9cd79fd75e2e7a7ba5f0c6ff72b4fb23d3963fb480efd4b
ce76a50021ac8a787afcb4c679b00437aea401f17d0397a8907fb7d43f4b587
1a4d1b6350c7d742ed1222af584dbaa607d5a6269a0600d69efa7c682dd1a9be
1997e807af4a4d6426c8c67c3076a8d0cfe7838097478137387f4384499dd755
c2490e20c5acc4aee86cd75de491bdbbea7746846a060f7c4beb82c2d34427e4
Fiat-Shamir context

72cd4ea74864d996dad1ccac1edc9d2189d922b876ca8defc787184ad0781ce2

withheld proof term: 736 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingT.eq_1 theorem, TFramingExistence:0

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(T : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)) (n : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.framingT}\{u_1\} T n) (\text{FBundlePoint.framingAux}\{u_1\} T n).2$

SHA-256 of the statement above

c8a43e31b98ac567e117a4487c6ac11c0bee946f59a7109ec05f7462456116e6

SHA-256 of the kernel type

41d5dfb5de4d2cdced8510373bad6f563cae545eb2ff37437f6ac7567945ee00

commitment C

cc10d27292397bc479ff2aedd2012973840220fde3ed697011b848dfbe637dc1
fef685cb570648c99d87f5193700892367f5b9dcb4b78900f91249a4c78f5add
2a52842fea2754a42a795902adc04508435259bdf1b1f17282043ad372740de
3428543c3ebee34501d2ed17fea4f078265b73f61c1d4f366959d29ebaa086e1
bbe5d28e64c97874cde3c6d59e9596bf191c2c12380479d1f63173eb2ee574b8
368130e889803455d098efac0d551ee827993e4b6118838f437139c82b144276
fbb68a071b4e17d006e6e52ee4d6e7e4defb1fbb78ec0a2d0950fe83921ed1db
084a3fc4c2685fb75a9ccc512c948f4d3049fedad178bd435307935af167b2ac
first message t

abb7115fdc3a4ea0a98695f1c5c838c83e5829e920e129235683f8c49a86e2af
ccac0ee4c5c2e0c6f4ea097c14abb8cdc94e5557465f2385ed24a5e6763cc985
b40027cd5f08219850d92a3537ed5b8a90e898208d05123def0e8a2e777a92b7
fce17e6ec246a040a2df05c74a1b7ea2db5fd4f2df90a6223463fb6ffa93c807
5777c3d050619a40e4851979e5c558d030b8be69f2d9fe2d6c882881c42938a5
6d075785584e99b9376fe5e2a7ffb33b751116d9ab0f326dc743dc318d0e8b2c
7d9e2d085f19bafb5b16c277fd8fb1618c4eeebdbf1130fa6d88c4fe539f0050
af955d72cb30a4a7ffd6d08635c67dae0bc2114c43a6d7cf00ec3672f3730afa
response z

6ee53cc254571bcd5a6af568fac9a18db57297d319a54a250295c06cb8732b6
0d87db099c938f33463b0dcb4cc77a5a23ae84f96dfb6c8df142a1c4c68c8df3
c00bf1553b9974c0bce9d54f87a0bd6e28d2a7843619b514ec171b9b2c7e0535
0c7522eee58f668220efe8185f63c10bf13da4a18e9ced5bf4621d77519f3d95
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211436a0f252018a781ad1bc2b227a0237b8f9ea04b2eca809333171683863bb
ab2d3bcd87545899ef05042699f6043e21d9a0f292ceb82849b69d2fd8c626e9
0e85c0e42297977898a36f27c733fa393485473737b33aa594977ff55b9fb7cc
response w

5234cd3bd564864126de7173cca7496d9051fd01aa72e05934080146a98f9205
c09cbd3acc74ae81b56bbec3af09297146fb81c1dc8dd1e01fe960ccf73fed34
acf2e26d0fda6493ba7ea0cf55cb00caa8232310f120c73fca25f2f0e0018045
eb6bb11bc05d80290461a9e9563ec6705eef9b86256f2110681f2397faa45c25
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a57846b55d8760d02fbd6151e3ed1026882777b4ce39f36155b500c2e2113ae5
4fe19606c13d1b0864527dc05580c883bc0329544d23d60d55a63ad96831b0a7
75f16395d9b4de76bfd84cf1f8e10c28dc5c43652c006f0291d8da21c6f6c5a5

Fiat-Shamir context

2c6ae75ce9cf3e3836e5b49edf2568bd864e2bcb46ce0046bc6786d8a62305e6

withheld proof term: 373 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingT_succ theorem, TFramingExistence:229

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(T : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)) (n : \text{Nat}),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.framingT}\{u_1\} T (\text{HAdd.hAdd}\{0, 0, 0\} n 1))$
 $(\text{FBundlePoint.framingTtStep}\{u_1\} T (\text{FBundlePoint.framingP}\{u_1\} T)$
 $(\text{FBundlePoint.framingT}\{u_1\} T) (\text{FBundlePoint.framingP}\{u_1\} T (\text{HAdd.hAdd}\{0, 0, 0\} n$
 $1)) n)$

SHA-256 of the statement above

4d59249f3521403ecc95e58581565e341c2d9419e30980276dbe795cfda9063d

SHA-256 of the kernel type

d5b4d11d949260e2d948a2ea22e4ac93ec1613dcd8ec80277e6284876d916c78

commitment C

b33e93b55f221133338c8e4740a17755921f646179baa0380319001fcf74e470

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3d17ebcf33ecc770e5e4f2a67463c29197cc0ae9248f7218eb09305b034a0444

eca8bf11e99bc213c1c284507156d23e9e91a11944d428a3ba3eaaac899d757a

first message t

32e26fc75bd087ae95a5a9a1c12a65fcc9c07cc245708117c6ada156d8b15c95

524d5de3a3b54bf79a0d12f148e33c67eeb8d49de9abed84d6986bc25464cde6

d878a002d87a2714cd9771162d2c60bbc4f9a113a7f1a91f878927d202206e5b

c2e99b96f6e3d37b56069eb32237b42f825dc597d3a39e7f429c888745201154

55034fc02accf6d4cd84da827887bcfbdd55fc8c843d757d6a39d2b683a5f6873

072b904ad2ec49bd697055df0fae9351e7d362d0ba4038e44efca3fd818a2ea2

32729bb7dbd2c6a16ab4bb5640c1ae8c7eabf044fa26f11edb71d02710450108

4ab6ede3a96d5ae24247a092dbaaf6cfba5005bb882b9b93838bbaaf48135bc2

response z

71481dd081a0d66d2dd9b8b3c2d1215af95755cb79b4094788cc74e2278bd7d

b920592c524a95e95409ddb5257e9a1da2652066d12a157657ede80f38681339

21260e78831a8239c01b0e4d47f5045b1755dd00249f2ee82c36547f6736ecd2

b56a258112055a8c3ec6654f70b18f83238900eb745884a53e2a4c409d388e2c

7af34a7ed427d92463a74cd4b1732ad0725c4e1d74e1f2f5413e981c217c96c9

736459bf31aae439dab081265fce3879b197a97f21785f7ec0e4f6910e699bc9

af370d6ab51ef699dbcee1356a683aa0a1f60123df0b4029225a1833f3d0b2b3

2d4d11c988dd440885963d7d20f68dc2df8e0ea86d50ad378c7dcecac6324924

response w

14e7dc899f4ddd01978b89d0bd47f986109323f5aa7b212ad0c1c42545499bc4

66c09e524d3149e51e234d0d96d7fe81fc38e3e8f145b4c056432dfdec79a654

1f352deca955b3c6b2aed8070d41f751781ee7b4d5723569162df374b4e3d2e1

2b4d5d31e0df8379122228b03b45f545dda09f26ed8821f2141f123d5d71a59a

f33d73832b91dc0530045a499f31c27219b8fef584d40412b47835936bcb33d1

ff48b4d3c7afe53272bfd5bb23aecb7be838ddfa82ef39de1c137a9e26d9463b

7eb976a920e004129f2d2a65ad88d32591ca50d9b4228a442c2f08186f2021e9

ea0475a56f7f01c17f25488391f51c49243248330dcb0231c2335cb987187226

Fiat-Shamir context

89ff42821a389738db58693b01865253fde43ce4e152a2644f022c3ae3668a3f

withheld proof term: 268 171 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingT_zero theorem, TFramingExistence:214

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R]$
 $(T : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.framingT}\{u_1\} T 0) (\text{FBundlePoint.psCoeff}\{u_1\} 0 (T 0))$

SHA-256 of the statement above

0162ab05133572d51c699cdd6429129b145f8e88baff2315ff39c6203be1afec

SHA-256 of the kernel type

82e4741356b34caabe7407f62a50229f6a05bbffc4c89657fb25ba06472d85ba
commitment C

71cc679358a2a48fa44516c903a44af62b799c2170cf5a4edb57955fc3192269
3eddef361d3fed88ec8515391fddb7ed379f292ad1105a3ed12be5b7b2a27e8e
1a474680dfe9c605c6f2c2484cf8eb94d63ed3889105500c0dc7243569c3ad3f
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3a65ee3a23080f54b82d5a84d4fb2fa3b3cfa0d01975262b440af0ba7df93cb2
b88c2141c2ab756393e71a8fce40189b7345f1f2969207a857c110baee0b1997
5e137671abee7a53af8f1a14de4916c6515dad045d61864acde2132ffff14f3e
606ad74555a4d59293af727868a6beb078f9bafac5dc2a6857918f403ec5786d
first message t

d5d9c9e1afdec1a8574305f3af839f9582f52256e5dced31d79969aebab92b3e
a87688a0cd9abbe46b5435bc383c9fd102405fc958d6e613ad97729ae9f8b7a4
62f6e434bc14069a2e03bb78267f2d1fe775a2cf9043c2a9a77cbd018e0d9f8a
f51eff7d4856ff98bf1f95cae40b33b98d74e167582d0d6c03a13612cf7c66a7
74f70bb1d739ee6261becdfc5b2bc60755a5349b6b55b3afeb1504972d1c2401
91231486a90bda586d066dbc0ffb984d9fe611bd202fae2391c3538ab3c93e14
e942e8d8a379482d1e2e5d2d42799b693a26578b6966e841a6c78d2068817283
59a315cb4a65c9f8489e119b253b612825ec56cc54d282a736c7e340228bd6a
response z

5089c184c3808d66750ec74b04669243e80eb3dcd70b333c5a5679ce55bf1854
e22ba4a4d7644889f6e3ebe8c79a9405aa7f91d04c9a94badc385c8b832089b4
ef167d33b95f94388ce18e236d7db4f46ff80cc338363d75f6564ade87b26441
a0d36c62e7f17c25787c70d044b7ea5503dc29da28546bec1fb948b47c9a417a
1b297bd01b7a268de092c70f87c59345dbfbd011f49722b8a7458cf128f87946
7546a74e0a8b720ab7b13d50c090c6ead93e0840f4e646bd0f3bd4b9ba7cd068
53bfaf5b9472d3ce4787ce39f81bc299884a2b6bd9b0040cfaf89f41b66fcfa6
6605a4e2cf82f0fb2c8807e18f0e531360267b8b00f7fdfe5cb97e7575606bfa
response w

4a325e0f63e5959aa93937585a8703a59fc626fcbbb68740aa5b703e576df32e
2464fa43a7634451d27a4194843790c384e7f5daa04c063ceb8be0ca2cc3e165
f206dfcbdb43e102fa2645dced65dd168b3fdc458cc8465a3b3642c05b3ba2564
300777168896d237a9e68278c97516f6ffd86a7a57cfa01bcee2e7c65017afeb
2f7479eff60433c4bfda8f6f855e204cbf74a92c84a53bda2bb90e5fb3e6429e
9d164f14e0023be522d5832fb8110d9ae723becf938b5e2152627027b19276e0
e873f8271acabb89f32e84c677818402acf7c6cc66e143e226e1560de151216d
4c63d6ad430d9d518a02e3a585b9bdad2adac739b109ff78edc0d15a6aa8c42f

Fiat-Shamir context

159222b820d4ca04db893efe6569934fe4b8b9c2a4e8bd4733fa98296d8b88f3

withheld proof term: 11203 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingTtStep definition, TFramingExistence:113

```
{R : Type u_1} →
[CommRing.{u_1} R] →
{m : Nat} →
(Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
(Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →
(Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) →
Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →
Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R
```

SHA-256 of the statement above

edbb5dc9a3f1fbb1865288656e13420430f0c4d0f8cdbc755e5dbd67e73ec29f

SHA-256 of the kernel type

8a3ebef875925da3a719e800055ad096c559d3189fb8d3c1cade7714faa44c93
commitment C

b456a6d1b55429a1732defcafafa694b338fe26806c73d3f14cd54d3ea983739
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8b7187380cf954c7ec41ff714c798656c8a67668be144997c66afa2e1a5d9e52
465f667ecce70ecb7a81e6ecef42b520304130006e8810447e174fce30533215
4945dd9df390d2b7bc224d380f47426e56a02373f49c9419e3f599415e94b6d
bd16a07bb6c3bb2813c2749acc3472de7c4db3b4d10d19b07c3dc3e2a8d52353
dc452e24fc4b1bce9f99ca44c3f67fab07340c0eb5995997fa4d464ca0bc9896
41e9ddc92b2813227eb6705f306d8d1cd3b95ec3c5e55e2bc55b36e23ab94a2a
first message t

a4e5ba2c0c4ab1bf433e0b29277a3883c52cc9aff74d6faeeb90d8337189dd8b
fe740684ddc60dbe381066d14718ccb56c97e774a21222f90def44e4c9ac1e7b
d94e57efa73ebc82ff444d7c3b1819613b63648cfa35f65829a3eb069752e949
3382114152f4e8247048f88454b90791d05c65ce39828a5f26d878140db4d293
e9aafddfd9707bae2e6eba4d187d29e793093ba15026ea1b7fdc47b64dfdb73
3dfab6d4252be2cfab0d9324d8867a951f80516f1c7cfe288d309b41f41e5942
6174b5e368ec9f7ea90c61cdcd96e3b6b644ce79e607c5265a7290b284983db8
a741a25daeb88fe7f2943cae70f6b532fa4d7f3774e3a306719a88c945d6825e
response z

5d6719c9aabbef1f003cdbc9ad9f9a771fb67f4ea773d8984a04953dc3852c14
563ccbe884f0f1a08edef8bb2e19d54b1bc0611b775e04c4049607a14dc0279
6527c93549f4b895268ae16afc1cd85e31faf5d5b29c38bbac40aa23ce93acb3
1deb2bbd9c28eb19349babc9973aa1c5807106a6964d84337b78200a49703907
a092cfd0a90ea3e9c61ed32d8c4944d9bcef8bedd186eacee440c883df4d8174
44efaf5ab989da8480fe046e7adb893337e244806b2c66d61011c3c89f0fc5d6
8618847fa8edbf7bf5ac3d653819b275c3d4485a40b1ab711de0295029aa9f00
0589f4d304edd42c5d365760bdf9765f41e66574657adcd04f44585a31ba0b9
response w

112b4103affdff59e9b7f30fa73b97f0ef148a38b615ee1209779f4c767d1680
f27da18711c4ff5cc9b3feb62e5ec5aaae953e44bbcd39d9b52534f80630894a
bdde86eb37dd6764f3bb3589edc0ac8c40e1396a0a8d089b27016a40e4287173
a1c01871f48ee3cae89184142f8bf099c92e02dfd4fac5e7c1cc48a28b062fe1
fc13cb2d2292aa6acfb486c5af1f684f2f89d329b857a4994b460081ac808c8
009019fd67cfa821f5e1a925f0b2522db23eb6b873a22dfdd160c78b67246acb
8c9321780245b151a3178a5875e2423ce27c978e954d398b6c3574aabcd6653c
d1753f7c6a21d56b2848dbf8cc061c828e5e7abf96b444dea91611788d230d4a
Fiat-Shamir context

cc296c2047f7e28efe6ecf1ee371de3f642d6f0a37b75b02cc95ae1e2b726caf

withheld proof term: 9219 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingTtStep.eq_1 theorem, TFramingExistence:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (T P : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Tt : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R)
  (p : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (n : Nat),
Eq.{u_1 + 1} (FBundlePoint.framingTtStep.{u_1} T P Tt p n)
  (HSub.hSub.{u_1, u_1, u_1}
    (HSub.hSub.{u_1, u_1, u_1}
      (HAdd.hAdd.{u_1, u_1, u_1}
        (FBundlePoint.psCoeff.{u_1} 0 (HMul.hMul.{u_1, u_1, u_1} (T 0) p))
        (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} (HAdd.hAdd.{0, 0, 0} n 1) with
          Ne.{1} q.1 0, FBundlePoint.psCoeff.{u_1} 0 (HMul.hMul.{u_1, u_1, u_1} (T q.1)
            (P q.2))))
        (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} 0 p) (Tt 0))))
      (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} (HAdd.hAdd.{0, 0, 0} n 1) with
        And (Ne.{1} q.1 0) (Ne.{1} q.1 (HAdd.hAdd.{0, 0, 0} n 1)),
        HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.psCoeff.{u_1} 0 (P q.1)) (Tt q.2))))
```

SHA-256 of the statement above

2ee2772f3c76910119356af7441a9bd381a7b4f04db6d0585cb0443157d3c097

SHA-256 of the kernel type

52e425d4fd029ac35c546ce76f5bd7b6b79a02bd84b883e0b973055940e87d20

commitment C

e0de2cbec0ae977093ee68ec3237c21cf1f8299d0c7c2f45e38353a6c053609a
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1c4809cfc3dc3e0355e87326b29bc431fbfa4af0eae895256e1774fe68d440d72
a9e2b391dc81271178898cb91587669baca94941803282a7280a0db4f3a66b2f
306f2ca1fbbeb77b123aaa7279ae9f67a18e0367ba7b7343c6e996aaeb0319900
7e98d186549263b5edb3f24e00f514008726c11521164a5771d3c8aaeb6efad
first message t

be3411b7fba5261bc428ddcfe015dba23873a45d32732f051998cae4393f3964
aa1dc2ec2f45c5adf491d9d25d4762e4d41a34e1aaafc7016ae736fb93fdeea4
9d4af67ae4b79771af9332e911f09c7ae4e212cc50d4259de651dd19a4fa76ad
a3445295cd379e38a30f0a062c2cf828893622d01177c77dc5642a9ccace9ede
7febb2ac35741bdb9d5df1c5d39476b5105ef9f9fb581cdc6ae313d91472079ea
4a5a127555406c2ab84d7691d230cf224303b6f734a03bc737dddf2b6e1a891
bf56130e1ba14ea97f12918f1dc5d4925ff3429acb9baee7777bca12b53ccaa3
1c69ef1168f0b1b1a7df43e21dd39188e8926f6578aa1233786279147fbffe96

response z

528105fb7cb47b40264159735b382baf4609b3750f7342221d72bd5c349eb030
4f308070d6fdf3fa6f05f6e2640b372cd2f2e06a9305cc8bf822598fbc10a1ac
434a68391ea78a4c2e7f5f3fcb1e58922b47f52841a2790c1cdeff19d91df493
b14938586cae6b07dac9e742b080ce352741950e2ba09c0bd5424089a023ed11
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361eaff20d4beed43fd3833af3f9dac9fdda81b93ad967ab58b1e34c5bba7024
151b46107f3ada65f1dd573ce60ba25a810fd7f0d9acc40880e0333108de3fc
cc4cdd4d8138c1a43941b1e0ec978af04a74a931df38f083776ee36d9002100a

response w

9b3c358b2d0645f11e4890ce239f9358455c88810fd3dbcccd5fafdd5b06d24
8091e68771574fef2c0361a9b4e81f7ca53417835674157d688635b77a539dec
9439c63c5a6d5092939870b652b845919887ceede0629346fdc551ee98c6f824
45c2d6ee7a465687f4aec324be74057b8417adb08e0d29e489bb5f6681caba7
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fe560f22082632035797c85b198de3c73d76b8251c261e80dfc87a26f1f57637
15dc19bd625baa0079600f4981b61e0f0dc142062b692450c285db1ff094f66

Fiat-Shamir context

ed5c40384bf05634531f3eda5bc021c13b8af61cec7222a51d2dd1cd28fb4e03

withheld proof term: 442 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.framingTtStep_congr theorem, TFramingExistence:137

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (T P P' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Tt Tt' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R)
  (p : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (n : Nat),
  (∀ (j : Nat), LE.le.{0} j n → Eq.{u_1 + 1} (P j) (P' j)) →
  (∀ (j : Nat), LE.le.{0} j n → Eq.{u_1 + 1} (Tt j) (Tt' j)) →
  Eq.{u_1 + 1} (FBundlePoint.framingTtStep.{u_1} T P Tt p n)
  (FBundlePoint.framingTtStep.{u_1} T P' Tt' p n)
```

SHA-256 of the statement above

bde31c820dfe060f839607c5e16b146b8407428ffa5c0524f2e42d0b84e1e9

SHA-256 of the kernel type

95c191757105503356fb79f4ba79151dee22648a1b612abb91923377b3ebed95

commitment C

69ff6994c7489af985f7844b3af28664a5c797bd6d1a017dade77a76c7400897
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7371838e85bf9c4e950c89a5869d4a8c344f326e477cb2b20469d8a655318b1a
bb2578a57e70ccd10ef5676cb3ff7acefb93113814c781f418d9c9f4d80abcf0

first message t

264f0a974807ba5762da59ac68508b416481525023431b570f8d06c7a494b5d7
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d28c59a4ec9c3c31137727f6c87f05f7470cd047cd4324edcc6adc934559988
894111ccd1da1142a02972d8f1faf8725bf80a7bc86faa1e5d021fe02eafa4d3
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1e2bf373517faaf5f64d05c2107b70e56dee819c64a4bd99b6a65a4bdb081b55
e604241949d0f4e9cd97d5c56f4ffd0c7a993a2e5c6c62d7a6b7a95dd1f8ddb5
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response z

264383906296bb08f21adeab4c8de49f844d26f36549b0985cb62bce928c173
2976d3af453e49f48b7b95f7e14df524cb8e7dc6e23e21d29a7a73c9565c7182
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45c9fa49fab0111d092ef9eb7c401a3defab4e8d160056105baa853ec9fe4aa
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db2d4857c4e4e875fad98e5a9a0c3eb09031e37d90f8f2093c228af7c7e47d16

response w

53e6be32e097be2052ff876b1b1391a3ab716c205b6d9a00598221ff1cdcc444
df28d21a18b549d447cd262e2132d4f35f02e535c6e75e65227401ef830020ec
7d3ab6c1eb7f25ebd58f1a34ecb29f2ff798e4f0855a44acbe62f1abae61bf88
97611564c415db914748a25f0634b8c1ae083e2236381949600583f129d3cea3
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5229782a05c0a804ffbd8532cd537fb64bce0378e945693735da94ac0f7ef593
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Fiat-Shamir context

a65ed66850e97eca845ab9363b4ef5697a2696d093d50c2db3b3156b9d65a195

withheld proof term: 360 555 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.invSucc definition, TFramingExistence:166

{R : Type u_1} → [inst : CommRing.{u_1} R] → [Algebra.{0, u_1} Rat R] → Nat → R

SHA-256 of the statement above

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SHA-256 of the kernel type

39c55a4e71890ab404c70699d9e03acc59a5a948afc2ff7032d4c17b6cb9252b
commitment C

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first message t

70d6a16df8599fa5be5d8e66f7d34210cf101b3368480aa1db1c97fc44174905
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0ec0dafae05c0daf9bf844eca1569ce4918d9ea1f5d87b21d8affc084b698f95
4394324857b7dda085493e8172b4d97dcb322088e326283204b238309dcfc2f1
88c7470b35e3d7b98c8d37c8308ad1193ac6ed0b730dbe13be879cc2d30e82d1
6db3cf82063b287ce433606f1c10de7475fc62cdb00ac322956064aa19d50271
aa3600656d6b304a524a63cc36eb206932df5dd53618d3a3a66b6b8a5348fd0c
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response z

99ef17171065025dd8aada46c061e8f374ba07615adbe51ce233f5359b400907
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057dacf58d178bb48671e54a4f3c3f39ed2bb9b12a467cd9c6de014beb54e71a
26ae7d0ae6fb0a6a058ec11d45ac0bd1be294b37f4c81e2d74e2ecdb42bb758f
ed47099d18b871909d4da997d10e8f9e5bf18f36a00f4eb197f6e8567bea6b47
a6fe4bc6816083a2ae337d855852231533c3c99af3c288d2af9fa0de1c3dfa6d
3ad2a4d17a9071ba9fd51b990efc2ecff45975f42cab0097daebbbdaa4541526
61b211ada7523e5fcea07337cb7bd6f3fdf84836d694cb97e2f4bfab201aabd
response w

31963eaa5fad15c5e07cac5dd90c6bc476f1952765cc87bc82bc1d7190f3e0d5
03df18774032f74fc8181225b73825aa1c4441aaf4c44494f65a0ab5d54aac4a
adcc09464d68eb16db10e8402efc38c806e1825af89f539768966ed4e679c30b
04eeb04485bd5bab6dc65a2c347f8c7c338d03ac2adf1a0b79a1f18a98e673c9
9adc890dc34ad5605c3f3f79bc34c5efc020b41a43e040cba0781a06e49af9f7
c85bc9092d28eaa466e50d28828417f1e7eece14da85951c8a91b9a08e0a9095
e83ba6edd78dc6e40eceb2142b5c2d3f011f192605c5cdb40e75e216aca0f24e
1358c7f00e9192cadd330ae564229a0ad1aeb29630861ae7b342b61ddbba192a

Fiat-Shamir context

434cd86311e76a3c6a455806afb97856c4a9af5c72e5f403c5fc4732e2037ffe

withheld proof term: 1 493 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.invSucc.eq_1 theorem, TFramingExistence:0

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] [\text{inst}_1 : \text{Algebra}\{0, u_1\} \text{Rat } R] (n : \text{Nat}),$
Eq. $\{u_1 + 1\}$ (FBundlePoint.invSucc. $\{u_1\}$ n)
((algebraMap. $\{0, u_1\}$ Rat R) (Inv.inv. $\{0\}$ (HAdd.hAdd. $\{0, 0, 0\}$ ($\uparrow n$) 1)))

SHA-256 of the statement above

935d9f6091648ef89b8f22b063dee8183cc261e1dfc9a07798027260909292757

SHA-256 of the kernel type

ab891e0f568d4f786fbbd0d7f463000a5eb3aed7f64cd3e199b42bd4fb383fc7
commitment C

eb48856c968aa5db2792567d64b19e0efad574f7ad3f699838fd9c19904c1662
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34e38e9ecc6bd464ae4d714dcb013bb22c26ca212aa17297b382461fa2a6dc06
d609ee9074c4086102d06f83abafc90b9c20b5c4c29e0b1332921a2ff99149c3
first message t

cf82f262966c1f0a49f3e4e230ae3028518054ea6ae23d6a84364be3af1fd227
4d6806e204d3c6f0e67cbe80411945da8fbd970209deb7984c52b701b6367cd9
7a2880f05344af42b1766202a9cebe3c56013bbbcdf4942de7471c67367286e0
5bc7facd768b021c2ab1c0265e837373aa4146ae72aca4b417d68a4e12f4e51e
ac8e594e8f830059154cd41a2407b558b144caebeacd137d7baf17f69be81b40d
9f3b34f8665f739542ac25e30e86d2465b2632afa87a956732cb524394903696
e93e56bef484a04ee181a89677f42265a0211d0ee881699dc91a93b99b3b806d
d31675ae01a163345dc4b97bdb11e7be5b449a8b4bb6319b50cb64bb3e3487c0
response z

555b272ea0159092de70f00b8a7704fbc969e85f0b0354801a741338abf2cb62
b539692fa1baa7d8988a95da642b980831972b302acf84b9281c5de882f6f5a6
eb69e63113963e7843052f161ff4af593a67c758d1b12c058aaeb2c9c2f11e31
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response w

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635ad5a6ec9e01dc0b9af447350569b9675366455e6f9625d352658e91c7e074
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Fiat-Shamir context

da5eaf6d002320d7388b5dc0c9cac9d2b6ffc103ee349db2073fbccf63997148
withheld proof term: 249 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_divU theorem, TFramingExistence:66

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (n : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n (FBundlePoint.divU.{u_1} Φ))
    (FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} n 1) Φ)
```

SHA-256 of the statement above

af932b5635a8161ba6f37adcb5ad0f479d4e00bb04808111c4e3c2cb7cd17e12

SHA-256 of the kernel type

ceebe3ca4d393a0b6b81d8282147f7eb8203e9899351e33038a7f02d6a16c57c

commitment C

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069bc1d814d471f93f87775658e679e363884149540d6dbcd4223e26a0fe5898
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first message t

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response z

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response w

79bbdff8cf4003c5e15ee18f011c300ee934b6d53f04ab2030651d56c04ec2a8
e1d8cc26043c5336e6d502789d1e4b238b07a3e58bc0df1119c3fd71f9412f2a
1fe8888d6d2f72afae90008cd9952b5a930565a0758edf557f1e40bad7c6726
1a9db18ec4cd1c2965f6f1379927a151679e3328632a4d61ef04a227ebf57d9b
227fd3f059154842e709b41132dc3b26d41b3126a04cc147953d670b08775e25
04dd5c234da02fdd4dcf3083d89624c961d0275c012a7578b168867915a4b4df
fd835b01c30200721d87b2eb9d89a6638705869410dadbc2a14975a5c2b07df8
278e761ba1f0b2bb549a48ce8f96ed2c53f2ce0efdfc47fea62d40140608c444

Fiat-Shamir context

bc190a98782c51735d4162d89e8da94ee838df22d7079272042d71a6e6244f1c

withheld proof term: 3618 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_smul theorem, TFramingExistence:257

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (c : R) (n : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} n (HSMul.hSMul.{u_1, u_1, u_1} c Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} c (FBundlePoint.psCoeff.{u_1} n Φ))
```

SHA-256 of the statement above

2d4ae8b7cf828b7eb5a5f06bc5325855c3be8849e189c17527c643e7ccd7477f

SHA-256 of the kernel type

ce5206e65cecca4c0513af960d0719c01a60dda8be1507b3cedebb11f21336a4
commitment C

5c1b1f082eb847cfc5534e0df309549d07390354be0850178cedbc9260abc03e
aabe4b0e70816d5ef1fe2e9a5649364798684528cf41662affeb0ceb4a754006
4fef5f1c9b85fd5bb26acb163949ee6d84b03948a78984f79b047ba325606723
f2b185eab993d3e4ddd7950272c53d6167d25a3db3688634072e1036b660349e
73098af4f8f1a15f6b0361321163f292d17dd817c4847d2baf46e353ef20b6bd
46ebe07b87991c04dfac8054eb03ea6eec46886ba0ef6a0cfadf5f2630cf67e0
4d92b1d6b1f034a2f871f4079018287c187da7363ff8bbeebaf29a9433fa3c7
964ef27fab0cad117f6a1a95a2c8612ee0849dc2822e8f6b917396ff10306f1
first message t

768fb5236404e891bb409eae18c9cf880937ff8b13fd021217271a8aa91207de
060081700e2872bb8fb5fd4d6f26f5a7c1cfc03d12f2aa1394ed094fb0ca240c
a6c8de4424bd4f6d15512f4b5bd50c13db7dbca97114df2b7d6b3e473b44e341
c73ceec9a97ef0767278ecfcc5d48460a0561597851709b7e213d2ff1169aed3
2a59c8b269b62016a11d109a1e649dc183a1b3575fde79990105df382d3ffdba
9383830f6b227035f20d69b1bc5cd436ceb6eb5a64a6c155cb998e5951d5f98c
c258d22647769e6e4be41d383d39c2e65d6eab7ec4d35f21ed7e005ff357bec8
dc83be88fd01f3325b313214766e8af442d3062582820d5dfc7a4bcd53797efe
response z

5cbd85ca6c2186a50eedd347818c29793c8875b67c3f912be0ca9f868627e12
691805aabb0b5c3110f69786e1f2ef996fd634b7df53854947659d77fa7ab393
91d9ba2fe75744fa115d05728b8a2ada8dc66af40aead44b49351dd71db4e35d
64603b71584a3839e8e9b10e6f47777182c2b2f504cbe30bec372fb49b1d5b8
8e0942982bd70fb8e951d1fb20cf2d8c283362223a0ec7973c95470822839713
e0d8b162e3ca7febd32f36d44ae6cf2d5f316bf10a54bf6f06194116806ed0cd
79bfc5bb4b6aae45df3ad1a68f81fe071b14380392a67dda39ff79c8b1b7da62
7a5c549cbe333b7272cf89594151c997873216dcc92a983a64bf4e8626f24004
response w

101c2739a0cea7d620daae13031ff5fb48aa302e26ae2ab0a0cd0061a5284be3
a93c8f7ee87e5cf60c9f11fea4fd518c97d3fa09c77270c2f7e6f695436f988d
4e1ac3eb83f5c06d167adcc51d0a990d80b38c933c0bfbdbfd0235eca7a13a51
f488e619d63dcede7e1cbc309e0b0d16b4ac5536005b5421343fc0b060090d98
80e5c3dceb8c5f732f4c433b5474cc68bb79a02c55f7dc5b630263e921668b9a
1b81f3d14e9b1d183731bf475b939b578355ec79f564c62b198b5060aa08dc60
cd59dbff88bff69bcf165f3038a7d0857d5bafed0d30a0283f31d7eff4ae83873
d9ba4aa359ae667cd4d1d774cc8aaac656aaac0f7aa1c9ee558029b358ec848

Fiat-Shamir context

9a85b28caf4f61e42bd79d49658902de565ffe716726e8cf6a898c4776e2afff

withheld proof term: 108188 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_sum theorem, TFramingExistence:86

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} \{\iota : \text{Type } u_2\} (n : \text{Nat}) (s : \text{Finset}\{u_2\} \iota)$
 $(f : \iota \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.psCoeff}\{u_1\} n (\sum i \in s, f i))$
 $(\sum i \in s, \text{FBundlePoint.psCoeff}\{u_1\} n (f i))$

SHA-256 of the statement above

efd344f2c6a64ec738f413aab35899c24b94cccf536ae226022f46073847b9bb

SHA-256 of the kernel type

c28347f679d15a9348df3068122c05a0f9910c72f20bd1a1bde10447be9ebb86

commitment C

7e592ed80e2bf9097194256770eb1e6d10d4187a95e69ffcdffbc9cdc901b8e

a00e2dbd6491f22c95b55ffbd180d7f5b3beeabfc61b0395bb17efda89f6a77b

ddcf7743773357ae78bc75813234d5d043bccb5db53085ce9b73ebfacdd4ca5b

421cecc9451fac008d2b8cb079f4da137c1e635033237b23548033015ac0ca51

7aab0b205fcff7a551b0828860618c4aa671e8878eeaf6983d1a7c3ab89931a6

d3bea7b64921b966e2f72337f0306a0ac57281ff7d3b044487b3b6ef34c7af8d

759abd698273ac8e96e7765c205a880c63a3b3276b0e2aaee88904d579d7c69a

c3af522375abcbac8c5734d9bccb2c646ffa7f1a27b9f4400a68eda63e7aaaae45

first message t

21fba5a9b12fb27440b9ee97f470719b2c222789ae7c5ce6eb151d9a767559da

20789da9815bfb9ba819dc36996a92f0a6b892850e47c766e716ee2d9cf54b46

c4609e7884650d241cb631fa799c381935b9e39a8b0c14e0ea640f17ba2df23c

30081d860bf05fd12c21eb789a3f7daca5d4ffe1039ee6677f7846ee47dea112

0965d088cd6ee272641fdb7a5c4b7f9a5e263b83d22ccc0920ef1aaaae4c9f1b0

1752feea2e57a36f2614b601d1f740edd21a675a18c7b89fa859a0ff8f3a3659

bd2ae7853b3a51fd38128605dbdcddb3ede263ab273f96655c33e8031dbc1ac9

c9d60be447f585d81cdbe31d16c6d53ab1e0275489529e27017d31ba2500f0c

response z

2952325713751db4a9a08e1abe6b4ab7cf481a1d15a4e232d9bc02cbc1e661b

ef68f77a1d1343166ceae1041aab9453c9e8fb9098daef0083045c5186e13f14

094802c7679650db1871aed2aec45f9b855d07f817f42737107dea392f265d0e

affa92a900afa8dd0806dcd0eeca9242b3369964ca9ad9f4da1fe5e1a2affb93

d5889d4b04af73c079d7ee9c57925936dc36a8ac8123087390817f56e3053174

4e01df2c4add487ec723ef7d2134effc77ef8916e29cc2bfc27be3810688de6

2067feb394a8bc76b1bb2316bc78c4670a96f00f67729fe6ad4fb261a2975c04

9f89d299a239f6e2961c8ce165bc86fd826edce8d6c82f2cf8965b70040e9352

response w

224454a010594a6c2e995ce33d49dd77e570463154405e0ff66a0756a2982171

59eaa6e33e600a7f359599188c5b01ef54e809d3166592865314e1ec8939886b

574e227fda3c0b422371432f257b33a2c5ec67335f8c4c5fa146dadfdbf25ef9

99272879ae913d2b58e2d996e4e482e748756bc224ee176a3b6772a1edf4ed35

2238e5f6b6a7c00cfc9dc891672f53ed8fd126c561b6b82801227eac1e149e652

535162a0ec764b1f45326c5c4b4351e7ff42d3874adfc44dc99707575fde2d34

2ab109a805e0f25b27282448fbfc05e93e236564b876aafa5889296f0e253fef

9e79d7bb04363e019a9c245a1e030093087b28abcd148271e27d57cd338621

Fiat-Shamir context

a233cc62277c712c28656cdb62a147811b3efc1f344aad58379f507d16a6e9fb

withheld proof term: 75 864 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_zero_framingD theorem, TFramingExistence:290

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [inst_1 : Algebra.{0, u_1} Rat R]
  (T : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) (ℓ : Nat),
  Eq.{u_1 + 1}
    (FBundlePoint.psCoeff.{u_1} 0
      (FBundlePoint.framingD.{u_1} T (FBundlePoint.framingP.{u_1} T)
        (FBundlePoint.framingT.{u_1} T)
          ℓ))
    0
```

SHA-256 of the statement above

8e3f1db5d8d1c02a67628888df840f3d245866f07fe159a22520a847ab0b98b9

SHA-256 of the kernel type

c7609e83018f060d39c22a056bb172c6117b518451675f4bd4ae700e70268879

commitment C

bfad7b2691453e193399a98a2c863f64452c238e5d16f7c89f99af02ab42ce49
fbc028debf62c0f4d731d239adc3c21d7ee42a173ba1d84501849c2f87c014b4
6d8b2007005e5affbd9c72560d5d350711e7f017a5dd942c0f9eab26dcc68043
51b2adeb80b4db2f9ded8159fe467c35a19cbb15fafef9f464780dcb92ec9183
9830f2dc5a622848e52c864fefafa602b43f94e7128532a0a3326fd03cee8c60
8edecc57e66d14648586b3b11ca2a218e2e26a94db34904fcf4545db5248e64d
3638c296e42a3cae5ca76443207eac01f31d0a018712bf24eb5a334045737e30
39f2606d8a59223a854b27c2a4bda12c9d107176393aad0f6389cebe5661e966

first message t

8300a01f871d58001804c5a1e8725ae56a6ed08d7f76baeae43e24be62322760
8bf1e77a5cf35675fccf3bf7b1a898b29e3d3b8ae3175b933ee922dbc3e3a3eb
ac5ad3ac992e58bf933e73acbb93e685433b55cbcb29fe18cbabc5a36a6c6270
b5a83410729f071b10fc36425d9d0566111bb6d29a9f0d5a1c4c74720e3ffe41
f248bd3439b3fd4267c899b4c8249e9786e1bfa2febb1336810428156460852b
fad5f0c49b04073684d1738665e75813f913a2e011ba46e633ada7e50d40b720
e001548d049a4f6862cce36122e5d5e24b159088159a7969b4b939f5090b10f5
08b7ee8652205b2ed738b7379b97c9976232ae5d161afbf7b107885a0f9bd320

response z

47f79bf8082e7294b891becd98f05936e5def9f8743eb876131888de0073643a
0ddad8cd4870fa83689cc0d45b22d025e5242d5979921c9f4a9fc919a8e6c4b3
07cb9bf7b3c08796cf414d0026802f394bc8e4a147d70e8fea64c9c399891c33
9e6ce960736f292f23057ba66af69e5711b5cd071529dc6418c38ad13cdce927
5ab9683652240757ce5d1b9f5d2a3fd1f58d5e460dc5c0c8991c36a541a298b5
0ae217d2151e204ea743d34c0d0e923150123f399f7314194b7371a80afdc8cc
a002406f76a3473341bdd291fcc06f29941749212f13c7df60acdd67fbc8dced
cd8bd157e94265e82e33adb70489ff37e3ce58118084891c51ebfcbb51aecdf1

response w

42a8137104bcd31e0bede84b522ab4d7637505fb470708c03616b47e2ccafa4a
2962119b83632bf28094889a80818a01546fd6c5dac8f6203ce4389b99cddb18
3150e7fe2b10029ba74c612059e8c03e828e89956ef96c1d0fd3cbbff9c62f3
fbb30ffcb4e91fcb0af26bea4ceddd2655a7c584c42a392e27a5c208634155ad
50bd79cf9c9e9c9e0981a2cd599bc2a83fa3d316a461e703c764f4c7ae058931
7fe36adc5324306bdfb54235dda0514c5104e90486444cb9db895933c5f72f0a
816f77d7dd19458bd99906c0fc01dd34e2481613286b91001602f869a17098ee
6807ebb21bdc135a8eaa2a3de53f7daefd71723ae4472c3ee4c052d654f54f59

Fiat-Shamir context

746382915965c9b10853fb21ca2556b80e5fa01c005fa5a0db2c49af25091754

withheld proof term: 1579822 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.psCoeff_zero_one theorem, TFramingExistence:250

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\},$
 $\text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.psCoeff}.\{u_1\} 0 1) 1$

SHA-256 of the statement above

c76a527cbc87f4e06dec9dc3976a2e51e0368611c9d3516612ced8eca3b86b66

SHA-256 of the kernel type

989eeaab26e6730011b4514a7ec6fb24da119d18e6442908f87a55f7aa341bdb

commitment C

a465a21c120532eb5e2fb39e91cbc8ab2e15ef14de108ae86fb161d034475ffa
63b8ef460dea8b4d1fed69f2c72f6669e87afb4f4bda8094306ae5f755c0667a
1d00376306ceb02ab079d001fe4f2e8f110574892e670f20649dcd1bf87c9e49
6731cb5272952465f75431edc1b59fec25c194869c0726d3a5ca54d13cf981ea
2a48c98867491ee97049957bf4a2e19fdc8a7edb88a9b5b623b24f7a8a024e54
e8b6d29e915bfff623aaedea0596b2507892a79f5f743f7eea9c9c28653f4ffd6
143ad34f269b3418ed0750cd579f026555ba0ce213c42cfa0199130b2cc67071
904f0c24c659e7b1ea48b6d1a940127894af7fc21ceada01aa6e1c90a8554032

first message t

48423130effba2829080d92746a35353d561f51aca3a5f04629a221bfeea194d
33bdb1b93bd786334fca72eec61c61b6eb16753baa8a8a361dde17695ba3f830
9abc40329f8fcd5eca7e7979c394fe7d78d74a8b6437dbf1daa6de23398829b
75be7ae04e9c123d1144f5cfd0062e2342dbb4040f6339c03a2dc4b858350046
17b952e301db3329b98af95e305dd6e64ba70c9cc5cf4ff81ec7ae9c10e1c7
f59f89b0105a6b68496d421af192e33b936bbe6659d5ad075da859c4ffbe4127
7c0a684b5acfb84384f6ba04323f14899b8cf6ba50f80ab132d938ea281a66168
6848dbdefc2776f85635c1c8741b4ffb19a7ea292a5983893ad94797de9ee5a1

response z

6258879dae75f78d31aacd6349bd874fbd219e5a7545fd03033024fbae17cab8
f4575b9ce4ae5cef93996691edbb5b65a659af0430b1da96b2fa2022ebc518e8
d90318e1bac1c59b72b7214731bfc4aa8a5610a782f2e6e36321f408e1d6f230
c89abc9fb95ab6968b848bb84803f1dcc96582d9bbdcaa2c3afa63e1571b7ccc
141ad679f6ebd42ab77a93d4c66029bf5cbc079ef63bdc13eb05acc2e6c96fc9
d4f4a097268132d0cd511a524c4aaaf4c5e34629c498d1c26a8131f3ff14eefb
f44011677071bf3d40fb8df8774889f12045831b652b761cb6c4e3fe4a05af91
c5740bd65c741dbe7ed5ca8c22f82d556450a13080881c81e9030f5a6b200f49

response w

7bedcd2bdd0592e4deb7c131a9e799e01baf853881eba82ca87049ffe5a91b08
ceaf015e620f55e7b64a2baa0b03921877291493aa330333a5cce2b68317bcb2
f98f0802a3c12616f84943ab228157b8dba5369cf2b07a614a95cc629074c36b
084575b02ea433b352b78b630f3219e4b679411d951a27739289397ef36bf96b
893af799f8a050f5e7c93f05d72b1a1637a031559dc1e7f28662bce2466d2bb6
5541af8ca24645b5c0ba4d421ebcaef16471d4ab6ae17e7734397c7cffffed2d
2154bbc2e8e880a83c3fafa85daf7deb35bfe9d8d9d0d03161e052e66ac0559e
772d2853920d31d4eba2e7e420d98c3e84b645b834573a3eb384699350324d0a

Fiat-Shamir context

f9c3c79cb2c6fd58f546b78c568381a9acb7477faeaeedc5482eb1362750e9c

withheld proof term: 140 435 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundPoint.psCoeff_zero_sum_mul_cst theorem, TFramingExistence:264

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (P : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Tt : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R) (ℓ : Nat),
  Eq.{u_1 + 1} (P 0) 1 →
  Eq.{u_1 + 1}
    (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} ℓ,
      FBundPoint.psCoeff.{u_1} 0
        (HMul.hMul.{u_1, u_1, u_1} (P q.1) (FBundPoint.cst.{u_1} (Tt q.2))))
    (HAdd.hAdd.{u_1, u_1, u_1} (Tt ℓ)
      (∑ q ∈ Finset.HasAntidiagonal.antidiagonal.{0} ℓ with Ne.{1} q.1 0,
        HMul.hMul.{u_1, u_1, u_1} (FBundPoint.psCoeff.{u_1} 0 (P q.1)) (Tt q.2))))
```

SHA-256 of the statement above

adc5c230746bdaef929c791e035a9820bde74e1ea632b9e22b74245fbef314c5

SHA-256 of the kernel type

8d508fccdde42e4cc1a58db44a3883ece43d9cbb7e9584b0677f171ac7767a57

commitment C

abeabb9d174f714d535f12a615bd4692251605298c953c9f38bc813623011b27
bdbb726b72d807e4f4a33ad0005394332928ec4e1e37aa3110ea4671f3a0ae87
48bb666a6e6a5f92123775be0ad25d75cec4519540d2cef1f9dbdc52ca81c71
14bfcf3e9f2df297a630ec31e2965dddc298a09a46727973211ca6668eeeb1
075c1a1581b0ba88a76524ba96b1103eddbf6f35ab4fd40039b49183c7166db
29d0ad7594174a729638f030b4dba5b06fa03983eba424d53ffd443c37f5597c
77e23e7ed36292ca45e2e653c593df657f594b5fd126f8197322a8a8ca68aab6
1b51697e8199c9692fc2fdf0c9cccfdc7a8de96de38e6895c7afb54604bb1941
first message t

945e734e3cb8bcaa03d705ebf4c84de97bc55204a65f736fe8abfdc959a2e95e
041b1d764126fcca5cea69fde1c042937ac7e62cd41ecada38e7321e22e5c7a6
d0b32e9b74672bef52e3185238ae061c9c4d9e83de4bc0a77e03b0f77489739b
db4984f8d18e60911afe06b3418f60b45f69c3170b4f51028229e1b4533ddc85
bfb277fc0dff314237fceb33ba54b0414ac75c2cb360f01329837dcb2361d17
72fe66d08423772f286fababeff4339cd93c0975d8bafee680a0dd5750d36d56
95cd05f01feb0ec308cea8f0d240b2948181a1d5f8109a68512843d81facf9e
ff2b19c94c40b90492181cce1ddbbc9de1f78ab46bd8c63603470918679a06d2
response z

79dbd90bdebb4c12f25cb3bd701af1f8ddbad52db5edc5b0b49b46d77277164
7b709ef430eef009019d24dc6517f93d956875e53190b19a33b0565a6046c7d7
3c223618d3017288a60de0bf0815474cf6e8b0ad8dfefc84d42c4ae476c02dde
47ea25d84f5822d5eadf9011f90474d00188ab885a6af390951f33e24274f41c
44dba80e8005030b8c51f0c0fe159de5713685e41d1c0edfe90197c81cfa0385
e7c6f239346a233b9f601a6d399b6751258a90d6b9ef1f6de3d03e968680cb3e
08f44163aca0358cc2c12d2405e50d025091990ae6811be8e09d4b9e48e2e8f0
00f004d06cbeb4038163df60862487fadc4f1ba63c1231c007323910847fb75b
response w

4d1f48c8167da22e4739a4a2a56436a69a042456cd4695ba3037edaad088ba00
1689ed52d82a795e335b1b1911bbc44e94bd316536580355984ecc35bd317a13
f551e494f9877775c73765e519326e5a2a4c2f7f915949c4d96b815250aed587
c6bc69418784e730606d5b761b44b6aa75227855b108967342b54d23ad981741
04e22e637e1c63d1a73956715706e81666b8a41e58fe624962a45f419fd70d34
eeb6ef0001c74014ae1034eab49b98994ae9779ed534a92d3b2e6b3317d0c539
e95b2ad944d6bcf984a99bc99feb3143747ee5e936a70c2fbc9c61446ff8e23c
5ad079619856d3ff8a877a6785b8b4fa1205eba9d96f7e6caef9fa633135ef87

Fiat-Shamir context

e1040c697f59bc052173115c9527baa886ea6951ea9c38acfad09d7163e39b01

withheld proof term: 249 937 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.smul_succ_injective theorem, TFramingExistence:181

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] [\text{Algebra}\{0, u_1\} \text{Rat } R] \{M : \text{Type } u_2\}$
[inst_2 : AddCommGroup.{u_2} M] [inst_3 : _root_.Module.{u_1, u_2} R M] (n : Nat) {x y :
M},
Eq.{u_2 + 1} (HSMul.hSMul.{u_1, u_2, u_2} (HAdd.hAdd.{u_1, u_1, u_1} ($\uparrow$ n) 1) x)
(HSMul.hSMul.{u_1, u_2, u_2} (HAdd.hAdd.{u_1, u_1, u_1} ($\uparrow$ n) 1) y) $\rightarrow$
Eq.{u_2 + 1} x y

SHA-256 of the statement above

2fc79f37dc66b82cf25d4d6bdb2e9c6626f026e19818ef8fd434b82df7e51cd2

SHA-256 of the kernel type

36ba8b54c87995d8a54a984ed5c66dcb69e58a812e00ebff9d53ec75cd7de84e
commitment C

54bdfbd12052059deea9d17ed2886bb8c1d304e6304692a4a3d045926f765c89
c9197804be68434f5ec0f0497025ba2429f21dc9e62a1e942f54d7d9bf13036b
f8c4c7de5830ca1450d42aa9cb9b83ae80f3ff5162ee97895ce503cb4c45f44f
18674916bfc41d36bc60d2bc2add0213080295d882d8fab729acab635986cd0
64859101fc50e7f76484950c802ccb5207544b8da8f5c07a904918ae7765005b
d59903a20ac31a35b900da1352cc1e1696093045eaff5c792fab01dae3447587
c2eaf680960a7a3517236d3733a295c9162f15490eb032eaf5cbada4329c9505
05a709c5c1aab894215d79785456daf5f06bcd944ef470c7c4a0437cfdcfcb20

first message t

e31b482e018de360c81c4f62b20eabe65829119654050d85ed599266bcad3ab0
43757f01d4f3ae377736a5b3ae548e5751a055fa3b1b7da52dd3afbaedd89fc7
86012f38bc9d61b54d594ef6ca344b5f5e678003339289e37e83948259c7022f
b6ad61a97f437a6fe383d849769f470ea9c62f71f9cf9ba1003b899c4a1728e7
13b8f355f78e6dbadf24504e93850d91606ead07e0fd0589748daab4f9edbe04
544cb213f7b27e8d375b09d59d2f18249c4fa33528c70addc088f4a19e61c3fe
d4cc5d8d844f46ecd7afbeb4e977768bf6aeb456823c1f257876db9675bdc09
1e1d84c31d8d64e9827536168928bd058763ceb39a7118a78359d1d9096c6134

response z

2307eec9c46250d74707a9ab6a6e91b03df07ecbabd686a354de0431d246e1a
1d108b1a87b2721a9ea235861656c708af4e784f69ae968464d55e7f1ac49db5
0b3633e38771aaf114a1dde194f27999fae429568b6736c5fd429fa14d4cee05
7e577ca6834b08cd058a7dcd63378b6cc076febd67ea8a70427f8b1f298ba05
621a3a98405b87b665d7d4f3fc4f8a904f8b35980ac09730f5f0acecb7bfe1d5
0d7202327aa4b6d3eb54b05bc45f6b4c021158e71b2d52282b044ccd1dafacbc
891d1c8b53a47842ce71c06ab14344faaee38203c61bef31f9ac1c6610082caf
d7207dc5fa0df25e11a0a9ff87796a76b68b862a65d2a06b74cc1a27777406ae

response w

6a469e46a481f7ca7f8102abd670474f40f12ce6692151e3a1a34e9db7740842
c2dde6f6b537c802c4b3b0f80e3a4081a32fc4a33d0f9bb01d9e971c858c5148
e6189c2e46346bf5832eb561123861e60ec3d9a72b419d1f40711ec154c7963c
4e0abd8b09745c40b05dd6e6c1bd6670c09c2a950bf8e5e4378c6e040608da46
7a578e8298cdd9e68f68c714de6ae91a374802e5bc88a11826470ba965bc25c4
d00fad0e9c43e6fd1f55c4bd7dabdf5fce1afe70f4dc5138f210a367452b4b4b
21c3844001379694de8c922e7e02c95c0f32d288540c22715b77999c9dd18838
7af80ff074aed2d4ce83ec87aa588f75c5dde5a669e27eed0a137bfb3a6ac54d

Fiat-Shamir context

f1e401dadf136de1d68ce2db3a58b99630717d067f23fdd70242bdf6b5f7b36c

withheld proof term: 83 506 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.succ_mul_invSucc theorem, TFramingExistence:169

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\ } R] [inst_1 : Algebra.\{0, u_1\ } Rat R] (n : Nat),$
Eq.\{u_1 + 1}
(HMul.hMul.\{u_1, u_1, u_1\} (HAdd.hAdd.\{u_1, u_1, u_1\} ($\uparrow$ n) 1)
(FBundlePoint.invSucc.\{u_1\} n)) 1

SHA-256 of the statement above

5cb67726420999e9da279464d60f74c39be28b6bf9d847dbc8b3d2e5be4aa2af

SHA-256 of the kernel type

c0ff55bf592822e50b9232f1dc6992b270e709d23ca1936c9f61ecae2460bf9e
commitment C

151e4e8703aee66c2ede946952ffa17ec541679566da34fcb4a2a9f34f15c275
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2f4d6643e4f65ee1a46eb7d50ddbf8d6c9d4194a391cf6110df47be36b9b88a7
4320776e48f916eed2aacfa0f6de520788f48a960602a730c4fc1cfffdf2f2ca2
acca036471dde5713726bc0ba01bd26c9b923ad4d5da6610eb60d66980107c46
4b4e9261668302bda08abcea1f5a7a50aa4bed3e29fce2603cc2e63380912955
429680ed69240a014c35e3bf4f9face78966fcd4e5a0aebbee4f880e9d617d7ba
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first message t

618555b7db80c6674114885883a93fafcac22a7bf913b8bef19b8493b33a7e54
af59e7a2d003766830de36bab42b07c06dc8500d7423ce3d2821fd12a86ad9d1
3718668cbd80a95cc52696cd8c5e6af5d9cf6e0a8779b61372c71492fdb371d1
d1143b070cb617c80f76bf78590160f061fa213464414e440bdd98800dc6c783
ee502e22d2b3337ed0ac7847c8e8bd2ab33076b6f270b8d88a7136363b26b1df
2fa1e9b62437b1b0b87b6fd3401a0530e027b8c12326d65701dc8c895732d145
8c993daed8ada02698601ff22b21f942ee45e1d594b4c5c19e53d4ec51032b8f
d41ba6142e252cbcd6d772bbf56943ddb4a6346d09acc225a5fad7d411ecb6f7
response z

4b8941accc944828cf08701fbc794a27b7ef3f45ca4aa21a70068e3edb0b6963
9a47f98c1e2e0a0273bf471ddd99407658aad77c0b9eaf134d22db14846de11d
fd473a43b8c372d4808fcd1463e374708d659f501e3280f6a1d45e8ec7bde414
b07ed56f03f44a50b94fad62faff99c96a0d36defa387f84916821c9cdf7876a
3d805623384f658ecd950dbcc04109cf0424c2393871e4de5754328a3df9ce60
fd0bef4a916af5a158b0c7e572b9fd8a39f32cdb5a540709370fd8e9cc13fa10
becdef607135b0fadd77fd7983b10486a626ec2bd1a8eb94c3f08353dcfdb51
6d3e493d7f685179dbd127e3b8eae784605fe6cc8f73c042a49f9a11a7bb6ecc
response w

5d1d28e8449b931d5a5a58beec8184f9bd78b96891af985dd9e9089aea3fdcfb
1f425ab877cbb34d324d48849b471aad75eb226730f8a8b929d04780e1f521cb
8d8ef186b48761698fc73ea6506c038180cda5aaed10c7954bbb38b99574de58
69e0c720f56cfcdd67548ca970b63bb78f54abad76cdfc96e5f0a4f8a0a7d9
cc4463d849f009e633224272d1c626d9b836492bc4116cd0559dc932c54b2e52
9d90de65de6b91d75fac3fb1bae6dc81ce7f0a578960c56f3bc510f50bd06bc7
4f6444b118f609eae87234b3ef9de2df5dfd5058a8a18186e24e6a5b24922861
e0a27378dddc9a08390fcd8f101e474713372c658b7c428f38a1e0e55f6eccef

Fiat-Shamir context

d562dca5b0c9fa17871f71ea773d11f3bcb18f27ecb590753212f832f8fc7805

withheld proof term: 128 867 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.succ_smul_invSucc_smul theorem, TFramingExistence:176

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] [inst_1 : Algebra.{0, u_1} Rat R] {M : Type u_2}
  [inst_2 : AddCommGroup.{u_2} M] [inst_3 : _root_.Module.{u_1, u_2} R M] (n : Nat) (x : M),
  Eq.{u_2 + 1}
    (HSMul.hSMul.{u_1, u_2, u_2} (HAdd.hAdd.{u_1, u_1, u_1} (↑n) 1)
      (HSMul.hSMul.{u_1, u_2, u_2} (FBundlePoint.invSucc.{u_1} n) x))
    x
```

SHA-256 of the statement above

75ce30a713f73d2801f3aa51974587643dd140fed795b71d5d74fbe63bc2c85b

SHA-256 of the kernel type

7dfdef9d4543220a21ca649618d5d7da87be9b76b531ef929ea706dda36c4356
commitment C

73e62eabdf9ba3f10969eadfbca6d04ba8249d2cfea9c10370d88eeadbe456a
0d3dbc47a0fa70e3cb37e8ed6403361fb1fcc27093d29cd2498c575991775022
2f555e2cdf4bf05ae0197be5a222835bf6c8c940c0527a522e62cf72247f5df
3a20156205b0833b82cd24e1c5095cdea7b5f4a356b32d84f9641639687a5ba0
c96d995ea881016cc4932b2837fdd4b7b769dc48310b2f3b7ed1403dc0172fb3
b6b0a279874a97c6cb5d66e3a78f018cfa06e722ca076e9f7ba729aacc4b425c
63fbeddb3bf59da1dd65f2e1ff2dbfee5400ba559e6be704afc130994f9f9f85
c786f4437b8b6b9cf070c345783f08b7dba54c22670986c8c9c51dd3bfeeb587
first message t

83e46801cb944390425c6d74607124dbf6c7ddd825b40efe925ea0de651bfd7f
71074a46edc6f990ad392524fbfb792934bcee09ab89fdb717ae190a77910417
3a8b16378e29707d85055daf427502652cbddba2c5e5bc39a0b86c210b1047a4
32c45a30f9e328812ea7bd78f7c6560d95b4cc18efbaed420b859ce7d9dbda82
f734f1b08ed699703eccdac349603616cddc11bd2a348849ccd9a3f3418a6456
5e99eff2156518d1c43b7ff29f238b87b8fbb5e36197944a20820707848f4817
ce32e47a100ce678dc91d059a737db586297f8103b5facc805cb0011278b533
64d6a4ba38aa274d424bf849adab2f5505d794e1b71355277389311d4edb540a
response z

170b222be6afdc10f80e510a7d8965707f75ca9f59052623b5f8aa33c04f1b14
f28766895ff55bb20ddf5010e318c171ce68ff7921272cddb65d7380676504f4
6bc59a234b9c7c31082e527443d2703cc3dd4a7139e034dc0aaa8b5839fd774b
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a0258505ed3b7b2a5c12434223014c76d3c5192cf2fc637979957ba7ff9e4469
d18784118fda4105267e6c8ce35854a236f13b99835e5046e91ec29624444567
7efe5e38610fbf0424aeb5f9dcb86a2a22bcb92d058c4b65a60da72f27d707b6
d2fe7ba378e06aeb8990e9f68f5f8bf73af85ad8066d5cd2370e2215905255d0
response w

434c1185aa5e374b2a0f29eb4fbaf7c390d63ac73ec604922669b3e199593513
159047dbba67181806cdbe672f67707f6e2aa0a00e0254c685304612594308ba
a2ebd983e3f9ac75dca420352c51dedbf03aa8eac8f265a0155e477e04eb6764
d2e08316a3fbc15ea68fcba6191cfc88f4acaad892ca95d1b17e8bc5b521616e
dbc6c7ffcabef46774e681883d01e22f83826eeea1546f2d8a6e1ad15665e1b7
2993e0fdb754874ea5aed907197d8f9790217f34fada827cae915fc761d774ee
f9ad1ba25e8ffc90fac4fa92e717724a0bc9967901b6b9137152e5751f3410da
a75bcc71890ce8829c0af17de89a957bd9b95971f23ae9acbb3e4f24b11286f0
Fiat-Shamir context

a554d7b0906123783d8cdb918dd4c3a2fd922c75b9a4ca4a904119d1adc817c6

withheld proof term: 40 520 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.unique_framingSystem theorem, TFramingExistence:356

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (T P P' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R))
  (Tt Tt' : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) R),
  FBundlePoint.FramingSystem.{u_1} T P Tt →
  FBundlePoint.FramingSystem.{u_1} T P' Tt' →
  Eq.{u_1 + 1} (P 0) 1 → Eq.{u_1 + 1} (P' 0) 1 → And (Eq.{u_1 + 1} P P') (Eq.{u_1 + 1}
    Tt Tt')
```

SHA-256 of the statement above

8b8ae4b912c5341680aaa9fdff7f5dd5192b72ae1327e0f10c006c35c5f0e5d1

SHA-256 of the kernel type

901290da4a61e5eeb681bfd730d445315f5476b2bcadf849a18df863a1fc4d4f

commitment C

e1cbb156e9f11f29a01850321f14f3e27c0187603f3507f5ac0fd6f96aa216dc
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527ce3cc2cb62c0b285d7e95a1c12d86a678a4b4f5f0568a19f7dd1061d1dc43
97ea7466db35a43105aacb9a4d418362310bed2f10d1c5484cfccd2f0ffcfb59
cbf2b07b55c2c6d6be3ee4fd2d13574b92fae21e5ac1e9aab010bad2f5437866
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first message t

f913cc75081ef6b6fce127c98631870127c9ec7ccd047489c97f19a4ce6aadcb
c99c2ac98935e237b93e2988e3c2a6ec8bac411cfb6295724579d62c910d6bc7
5afad84e7f94a6167f5e1a00be16b69edc70a602d239c8b0ab1007e62d3a6915
59507f629d18030df960dc4b27ec93e40a126df01e8a37622d5fb9f1e260b401
16c5d8f344b9b051c1e358b1a15e1ecf2a169d9672e6eecd8e4998bf28f4c8f2
5cfd992fb915173571eb81abd5652b4a0bffc043ae2c4de2157eba5caad4393
d9a8f66fa94aef09a4680ed77339d03e904fbc323ef6f9822e020ade5da4295
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response z

10a00b9a93effc1b616f0e80d3aaa599b158422c09c4e8f5c8e0c05f7d71519b
cc89bd31632a912b43e0ecb2d96843e1aa088700de5b49351dd8b42f7e44f77b
5af04cf885d65dc41875a39e4157183b951bbd11ee8e203e21d7badb5c46720e
87eba510ec492da8ba02c5f5b9f3713e8be27a280bdfb87f39b025e6f046e581
aa52915dfa69c7eafdf634fd79ded661f2422cf5f2aa3cb82dcd74b74796fd16
fefae59d086e1c6d2d93e8bb8e22be966d4b60aa1303be439d87766ce3311c54
690ece1e2ebf51114b281a2ed3a612bbf35aa7646e266ae6f602e3e5ea3f6cdb
047c748e6d3f288e9f09e56defa929309dc8e2bb7e7b2c39a84ec8cc76425dea
response w

1bd6f0f1dd0794f51a8d7721347c71cf65418994b35b1bce22c051c5ecc04170
fb803d560f362ff9c34252fd27099dd38b07c56291d25a60cf55b38616d006ff
5397fdeb5ef235df01aa3e34426b43e1398d1fbbba6e3b55957bd8e0800a9a0c5
34350daa87bccfa6957324fa445f6151c3dba7f35f7f337af2233dfdd4a707d9e
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8ca548b6a08269963692ce4481dc3e0718304f357b1cfefb125027a687482da3a
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Fiat-Shamir context

2001729a89ae14c3ae49355453ef4945e82221fc77869d099eebef1ca8fb1150

withheld proof term: 1051669 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.IsFramedT definition, TFramingSeries:151

```
{R : Type u_1} →
  [CommRing.{u_1} R] →
    {m : Nat} →
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →
          Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) → Prop
```

SHA-256 of the statement above

b2eab65d00299b6a0268b031a49b3b018e7e6055c9c6abf899f306d75587d8c1

SHA-256 of the kernel type

5965058722f7fce9b2291cb55b28bbc255ee2c23a4eb521e9f5f85aba9ca2a45

commitment C

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322424027475a0e051108d6c3e8c7858e1d1ef5ce228266d024a188a983abe1

5bc6c5c0455874ab6b74bcd3801a6f3ff737eba1c91d660e49202ee530890a37

c278baf621ee1fdfa931374a723e9cb82285cfea7e5ab37d244b2bec81a7969e

7994fd769df4bdf0a68bedce237d975eec36bb940899d955e730be865472873f

a20526037d632da6ee547e98acdc1b8faf4725a1b67edcd3913ea21373e275ef

first message t

479f1bc6ca676ef800df923e526856e87ef8f2e827e6db99f3ee8062f34efa87

0f12cef1a292bee773444f684c0d8fedc1336baba48419eaca99a865fa98cc88

798b1c6808b1fcff665424138ba8c5ad1ec4fa7db996abd4f9f3defe2236840a

ccb9c07759a7f8ce0f917a9b48d0db3c19d2de17a2b66ee2c3001d2785f1169e

2b5a0298adfc3cfa67b4de4d1eb3b8c6849cc98e4dd8da665475c28148c16b99

248b532cbf4570b866fae2bd518e90c295e51cd1b23f7cf797a77a3e4f4a7999

4ddde0680dc8d20c9657c2e9cf379ae2161fa5d8014cd4e0d6d03578d89dc62f

9cbf8ca679411c859bb3fa96ae76dae1567a067d47ead74cd4b079aa5581c32

response z

6a33a51d7b95c55c3332683443ba9e0a6f80a99c752349ce4f0a5c12a2906821

43e7e54d64ecac432953f4284151e97326c626a9fd8e2d75c638145e8ca8a5da

af7e697f1ddc396a6ceae4e47e03e75ae40aad2970360948669f46c972c714a3

2cce04314bd0d6b08b243fd7a1e7421739f5f8ae8f0c07540373ae5e8d0a81fc

8ffbff7ad45cff4c8bbdfb5e71a31daed48ef2cd176f54ed0091b381d34cf564

0b4b676b7b9d225e33124c764f3b48e09b18cab6043e4e696acb2a59c3bfb4b

90e770e3215c6c84dcf33f4f49aa914b7d82c5a04b11de17708dd6c2e141baa8

c9cf51e41ff6f01ab6983a0faae1423f6e47e401f657712cdf83a472bbe58f1

response w

26475e33dec13f6fcdc1a1964efae4e120d3239815ea71a9e3d3182a5374acc

9867a6850a027037c3993d1a2c3246087bff0f0b867369386b77a0346f8f3386

9253483360e5217006e41e0fa6caf60f6dc0ea0fdd05bfad6cfe192ef60dff5

98ee17e0bdfc4af0f3427b142fb0c7d937fc7e92b6c57f7a6cd733fa25b24158

98a151a507065108b6257fdaf48eb65390c78d70e7a3cce93123f86b03251ecd

989bec57d67431c1efa8500348572099dbb3b8371ad6fe09149a6e1e9f4c2e03

469e268bb7ff3aaeebfadad9d26151929773371c5cc5050af3fdc865a5bbea4ac

dec734b7551a37df6dd6caa6e2ae713d32bd39670dd3e55278801a166aeaf57

Fiat-Shamir context

558e6c997c0ed14946b9b04b930c364a6eba2f315fcd43f60befbffa05ed5c59

withheld proof term: 3 993 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cstT definition, TFramingSeries:115

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))
```

SHA-256 of the statement above

45475d8fe1e9a250f2d6ac8e92034c9bebdcd5e31ceab2bea11d7f4844ccc798

SHA-256 of the kernel type

a41e5e69d7bc050fa71944b0a444ecbce82880020c485ce630c591076ffbf108

commitment C

f9165f7c99917ae6a5b3d6d4d9ee3fcabc083e636bafb4e88b8d62890e6d7bc0
203a85a24d00a260d2a2eca520d8134519cc0ab44414926a30d94d7313bef4e7
a7aecd176aba73f728a0cebd339572095c9469e70e1869e8795ac6603c008fb8
2605da593bafab3e6b5c47e746a995ad3c340f05d5e46ac415fce9fa173e9588
1897b7d9dedec676a42cc63d5d53ce5b1f48c32db3312487fd5192a13e848994
c84a85a23d393f92e3a157a7f72bc1107d34374554925955c046e6099cce1dca
afe03de810120ab3792fe3c60b66c07ed59604161bdda14e9dd67b7f8b08a26
c035f3a07b2fe7e1abd152b4fb39a3525bbdede8b21e0e7f045097d64d7827e3
first message t

f2c8f92366b790a3d44f808b10dfc0f08bd6a0ac4f110a6efb561f9283930c16
69712c060280f9ce553d3fd5dfa49c047f9b91bb881b0fde0fffcdbf96f1603c
3c87ccae4b8f738452e0e40de84d44e633c1260213c9d4905657c8a24410f93c
ad8fa5bee393203b2e936932a9bf0dd2e5547263d209e49f2b42aebf5d81fbd0
66d715407c2155d6dfc139ad71d6b3db0aa3425543df4b3a2208be22f1d9c4e1
a86459b0fb2ce9cbb46fa570de86aaa25a1eda9b16c618924527c9a0e3dfefb3b
889902730aa1dd0d505b1013ee9d280b0cb830457c8440e4ce81d4efdd715016
2e1cc8a83760ea4d0eb52dc1b3b3105856234f7943e78ed67ff89698f8008c0f
response z

6a45979cc8e7fb96a8c1241952cba3c6a514643071e4bfb1055895605389bfa6
9d883e96aee488a6ad255bef315ed43969306a5a1a031719635524e963b2dca8
a07600fc77a2b9af4e41f73fc8b7e16917b239673a27e967f6813b2f1e1542b5
f31dfa6a1f27cdd8b721030bffc0efcfbe68cddb4372fa6a808dcd4fe431aea6
3cc12c13d74caea9f1365c2dc5e05e7784811bd5f0d3c8fd245903106a54ff61
07a8907e8a35e6ec48aaac7aec707f907994b39d8bbae0982aac729e44e39be7
7f663cea0901eec499d47e670ad9f9e42a71f544c6678661c33d70bc70177556
2bbf8322d2189ee478d32718dede4a819b9950c3c1360cfb55faadfc3deb1528
response w

40585f6b23f507d4618407e319b6fd2b9ebefd6121c09832a075839902174a0b
75ca60155b2826608a1c18a588a990ed824f1d259d9312fbd5a0f689efde8df
64107c16e2fbc5a5c8224f2265775a2ac47c90f97495030010739c47e8af0a83
82722a4f8b14c37f9cd683f0ef81c578d0d8fd9288b92fcc5b6c3bbdb0f86f92
c09df887f0c6d2da923bea62fc8f8f54aa2587597e26606e172f7192cac345fd
7d8251a3490bf30f1a47eddac95b79233f00dab3fcfb988cdc692216f7da563a
4451877089a0fba5ac77fd549787b250ee8136db768c91aad66a9b944f7abacd
0c3c2644b7a8fafa117716380a0ed1d6aa53daf57ae0b1f5823a745edff6771e

Fiat-Shamir context

bd86bdcd863547d2b84533f625fa3981f8d58bf9c5efc221d31a2d12f0930e95

withheld proof term: 3387 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.cstT.eq_1 theorem, TFramingSeries:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Eq.{u_1 + 1} (FBundlePoint.cstT.{u_1} A)
    (Matrix.of.{u_1, 0, 0} fun i j => (PowerSeries.map.{u_1, u_1} PowerSeries.C.{u_1}) (A i
      j))
```

SHA-256 of the statement above

a0fcdc53d02e27785302d863a3635c59859c33ca06a6f117db9cc80225fa7753

SHA-256 of the kernel type

c083389c1b191f7b272c4b7cb89c8541eabf57f91e1caa65a1c1578d80bfd0d7

commitment C

9ad15ec926f9e210d4c080452954e7ed525aa11091ec0ca9763e654627646c96
ad3e231979603fc675487e3c0fa52c2deb329446c05175b37a680e15aee70a0a
dda5cd330e6426672f2028637ca0d187cc834940055a4f24e94be9a57288ff1d
f434f578aebd434d2eeb5258d3190a3e925760504ffdde9ebf1490ee5f941b58
bb1732eef2606342d4190de60d620aa36e8eb4c5dc4caa64b33176ea248aaa7b
a1f39eeac59e9801250555274a104e81d25eedae5d4dfbe5274a8b9fec044e
ac3b91c73a0f56fbd03ed2ca44a4f02049a20cd25b3018d7ae8eb7ba11d750ae
383c2ce309f3dbdf7e0fc7d5f64130d2ebd91903e2310a3283c2d5b32fb0dce6

first message t

3c0911d65d010624501ea954ab71a6dab82fd79c01bf9ac8d3853d1c643a2e9e
dcdf2373d12e8879e579b7b032205fa5f658671e65710e066fffe8eb9c9bcdad
0537c6576868d45b0e40c1ef5fe65244fe6bcaea596629ab82c8c32f1411cc0f
0218f6af0ddeb54fd86933c3c29441f4faa0e11f42fe97558414312fe585c198
0dd3b7de66fc7f88e2ded6b324a52233bbc049641e109d286ee8982f6721e590
205e1145ae40116685447c977b4d4aff022cf0b91913de373c88420f79b40875
bd377ff985cb6ff1caa9f674f4203e69ce8df0301f73faaf798fdd0fb1d57d4d
6b22876e59f18e1d66bc3bcc0e75cf8c9b7e4c80f61eb2f0f43ae5bad4094dd5

response z

65267bd9cedb05f785f443c74b29eaaaf13b829442cabf26f92828dd916ffa2e
2b8e8df20ca8fd56aa4fcfc8ea625197291bb17e5cf3409253627b193dd09878
dc077a449efddc5367dc8f76d22497a5db5aa879f804693c007d6477451781e4
103925cd4359a37e745b51917e8640cd0164117b15831d7b62691ea9872b136e
c20d6509826e55f7e0e6c6adeee27ae5d3a3ec8a1b783e6aef78b63a2a323d95
dc0b7ea7b440bf6d59a306980ff6b5a05de0da5d2d4cbcc170f9ecc0cda40bc7
82c40f5d11f4cf4d302295683dd7fa30a92db8e0be20a310dd3cc36e099c5e6b
441caad4960df0758f94855ed02fc45e5ea866f462b339fae7349c603da38978

response w

5070560fb3c0b034361857b331c4defb50f1cda6700ddfa3f01f768284ec8abc
104ccf08c23f0411aba80068b73b702b0e47cde43e5331a035559528445a4e0c
ed036da87d699fdb99fed134432e629f259256d52d80716def02388efe67279f
697596a07a68a025d84102082ce0f0388e83f43c052b87e32eb85ee76e47cf62
37a47658b03aed4d049e6a2d87c8ee8e7ddbe8829654b574a197024e005518aa
74ae24a3dd58314446134c4a88a79ad6643f25455a17c1a8708fab0593fd228
0ed5a2af17102311cd538466a6dff902b89c7326c9ec28eccde514b1aed467c4
9e720cc6a07f9d02bb3f777fe37b1cc9887c9895501bab0aceb8ce49833ca838

Fiat-Shamir context

a4a8054c66271cfc7663750a4125df17f820311955b20515dd20022792cd9fcd

withheld proof term: 257 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivT definition, TFramingSeries:92

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))
```

SHA-256 of the statement above

63b22e1e9b85ccc9c6721aa4785df23ef7ce10638842e2c0db8a0146368c52b1

SHA-256 of the kernel type

98466d32d1df8b1922f38233b905b94ccb364a970668ecbf77b95230e2656216

commitment C

acb7a052fe2636a5c45f7084117e532cf16bf464241abff8c61b06f5c8f7afd5
0364e9f726dc68f3d5f1ea661be69c1779e9abb0a70bf3774c46fe10ca5e4944
3354c7f2ec7a0a968f85ef57422cd63bb517c3dc8d299a01d383412d3d1f714b
6841e025f90b358c01baa885891fb60fe8bbe91a7a7b3f0b8f0ae9083f3bd722
371a16d83a31cc72f31b0968d0216aea02ebf792b2ad787525374c9055a09466
b490ecb5074285e3d7789ffc504f9dbe571414f03e48c02cfad25901bd7dcf94
6bce930c8251b565bb075d0f7d33de7b7e40b9afbbbf7ea8b6073e2b7fc19a1
874b2c255fd9292af51ebbbcc475901691c6be240e606a4f644c39fb9dece88

first message t

88782b2aeae93983a41009bc7040e6ee0da34607fe101d2ce4af93823885f663
4d7b377079b03f95756c01b6cd23c58c37873504556126c75220328f1f2b0396
0753758463f133ced936a326b9db5676751d3e28fc439c230996b8274c190894
bdc5ec08de49651721dd676aded28398f0d852e2b9b7350f2d4f000786523929
a1073ff7a4c4c6b34c8ac317072bdf2e4c75213a9788040e5a257e1f4e8beeea
247c750dd8fe91215101e35e33f3f8e0a38c47f8bfbdca7257e7cbb2e467f067
60ec57ab1c3a7b70f7b8a25d3e1ae3e8add841861321c0c4a3f032ed793a2758
ea041d572e700105db27813deaf362c2d72d1e173c8f70c6b872036f558f643c

response z

5d185b59ac1f76c8d7eeb3d4bfa88478075101ea6c483b88c6faaf36ae993539
e8f48a92ff3a305c2265ab6175a5983e10b1e9fc90a17056d8bb5c2e85c1c5fd
d722205f600541e2adf2cf6ca3dc7b4497dee25ae1cd906ef136e38e3b8bcfa0
194a9877db51d1518f6849c28d8877e38e1fee0f61c865b32a841191d9656842
af4b8f3b79435a6c68beb434f694ef066c1a60d288439cd94588ca741f62c36b
cf11160cd29e1b6b5a98d66ada488701e466e3ae9541c24edb1f6d80e248635c
5b386cc2c19a0e69415cb22285f82fc89ca14cf9b5c843d830c7fa83c730b658
0c1e3c134466a7e6b7aee785dd49a1e14bd6ccc042ecc4228f1a98d891e0a4d5

response w

7c93f570998468987e1cde130ff892b82e596fa41816dc11245e60c36921c41e
44c2f6f3ce49d500dabedbd054a916aaff1787654b61915c2dba0b39b0465d63
d7144a0af03da04983f09c7435516a824cac73a6a40309696ec73b4fb4357faa
8aa60e4f4decad915a541c9ce59a9b08e43614a5831f5694c6a397864a862dc9
4a8f22ec47bfaafbba08bb24b40707ef272cc7d153f7e5edaf78f21e9baf1507
7206f523a32012a7822548895859cb4ae2a01da14a58a43ffe9d883922c61d60
95691494e7c775add58ed109e3a525c9f229436ced5747ff10010633026609df
312336186fd86bbb42cc6bcb5c9d451698d52d6fa98671b3873d1297dd8777ab

Fiat-Shamir context

2c1b52d8eb4c82910eb5db528b13d5b636fff502cb911366e180afd9b6b38c48

withheld proof term: 8965 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.derivT.eq_1 theorem, TFramingSeries:0

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.derivT.{u_1} Φ)
    (Matrix.of.{u_1, 0, 0} fun i j =>
      PowerSeries.mk.{u_1} fun ℓ =>
        HSMul.hSMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑ℓ) 1)
          ((PowerSeries.coeff.{u_1} (HAdd.hAdd.{0, 0, 0} ℓ 1)) (Φ i j)))
```

SHA-256 of the statement above

b3368a703919266cfc74c14b6f85e43d6c6f96467972bce1420abbe6d6e8059

SHA-256 of the kernel type

e13d5521f307253a7451ae664c4e9e9cb64f30314073d546db733b5ce9043c45

commitment C

678e18e2a50c926afa8dad8c257d8d36d9e8d5307ab0bd1c2e0fb67940402d1a
ba2a6dac08ce1aa3e0562cf21de8dd344bf6a0b538299f90a5883c267f5da9c9
9679f8687845f55e7434b9144b19bc421782f090fee7c0187e34e309ac1768d0
2bcd2c7c0ed6343226524c7b0079ae5a99a29a7810bf1de00c64899ee980ab29
94e55dd4ad8b075e13dca1709efc049f427fb677ab1fe1727a955fd1166588ce
f4ceb8d00f5d6d2334bc1f2b4aee5d4876f96a9b4fdf14313c460c90690279a6
7c7a1be75748891f0b4251c7e378dcbc6f0e8182e31cc8309bc2fa29865fb75c
e689fad715ccd44d4a37ef1e01065664a726f1a8428bc17149ab4b11c5aa0cb9
first message t

820a4ecf9886db9a66e7cde819f0340a8814205a3c99cdba89adaeacff857aa5
6a00c8f78d60eb5ddfc0f2e19bf324b24a87d0332af1967558dbae979e0718c5
47cfc3247c78b45871749c8d4af1a9a24bdb9fe758dea849a797005b7733776b
75d582206a9cf4f430b294c112f9f06eec1c5c69a735c1971793755e37b2d6d9
684b4b3074250c48921ba01744fc5c0709192f5471ef5f37efeba66761e2a184
0a1135ebaf2720111dfd7e14eecbcbf18dbdd80d70888daac43d5f83b05c244
c478166efb9d5c844c768f6289ec9c691f4ae5ec9bd9d386960ceff4ab4e920
235992a49c3a7645a0208a1318bfada4fcf6a155a315aa91c8c028f24578a29b
response z

412c4109cc08e326a11da166109184578ef6596f3a91594612a242d34d6db829
4e6ab306a5d7f2f6b50743b0455fe527d03faf321f62b31a4d69712757b8fc6a
287c163e2ec2f958115cb41d4974102c0a54c00ce69c605a23b1ca213234f296
8a52435dd71cf7fa1d6c3914d8227b8e9a1398dc9751e69973a613658c5a4347
b4d7382c24efd2ee88a04200df776a5ff27d7e788a98bde2a225b9bb339c7198
bdfd2104d79d8c57e8a2f5aa9529ab0079b4d8ff64a782ad81af6009c81d03b7
10db879dae4e70a552630b6169255749ea20d5acca00a5da5131f7cb8ecdffc6d
bf277921390b98008eb196bbf7b2258e312190fd5c1e516c835baeabbdad0a21
response w

5fef6028dee0420ffcdf8292d5697a9c19f3f6aff5c469d58c4f5e3ea6f4c376
f959a16cd701b3e566c3d5a2273d7530c16b7e6aac1b76dbd9aac4db2202e982
30774dda5872fff8179df6a2bb72f6947345c8dff51f2ac7f04a453f879b51f
cb6f9f6d7da8c26f05d8b79dc4c6a42ca8520cd6b485828021f88dc2102693fe
5d751853533ef2705d735e95efb1e45604cc9473dba80d205a8ea08fcc5316f1
219831699fa80a80d88f069aa0b90b5712516e1ae62072f79313109a9d26cda4
091aac048953ec188e8c426eb9a1de3572aa12bb261049a6204e7a4b52cd757c
c53d89f79fecf8f8f3347ab36a034a0fab9b7762dc653c9108597917ffb3a64d

Fiat-Shamir context

5a794a355138fc81b305dad668ec377d6348e3c0df365dc55af594047e328f60

withheld proof term: 281 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.existsUnique_isFramedT theorem, TFramingSeries:177

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  ExistsUnique.{u_1 + 1} fun PTt =>
    And (Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} 0 PTt.1) 1)
      (FBundlePoint.IsFramedT.{u_1} T PTt.1 PTt.2)
```

SHA-256 of the statement above

f602c4a06a1f754e26b273535cc054803334058b3c0d19b35d4499d478abb1a5

SHA-256 of the kernel type

8426b7f139b4110dd787b810e81378cff064b508ef6f344288424c58fd1858f5

commitment C

71ce6ba2f9643364f4410b0d3db8fa35b9c60b8873b926e54d813f7133a7d5dd

46a24def971f00cc779305607bbd3c3e8404899a801d152bfd525c0859f5511

c361ee5636ea4c963b2221571aff50c254400a0e10bf1994dd1804804363a93d

5f86693b3b22965156530bd4a0effbd0c86317332b114031930eb336e6ad3caa

c9b11870a1228834d08ab2d5675bb8264533de9af1cc253002a4013b8b695300

9299e2f763467402ec08e25c51255a6025ddb300bfe9d1d6249e2ccf09cbc0d8

5d665ee5b723bb6bf2bfd9e1c7da085b92ee2774abc20e86b5ae4f3d17120c

a1b8d190934dc83fd637767fe8553b2235bac2743053681800c02160156d8db5

first message t

af88e9f8dec0a3c50e4609a81d89c65b323ec8ee4949ac9a1c37e2c6cb8cd8b

432d996807d52192d68daf22c876c2599b602e7dde9cbb81f9eb650b5e8a3ad0

e837076d68beff70c25b0676bf2414b38f668614b94fdf8430de37f358b916aa

f055884eab6c3bec510a7f1f03fae97bcb0a8273e8d4600649448602dceafdf5

6cf4673557c5cddbea63f37d747addec0682c6fef31ceecab4ecae8fdd6df853

39815c6a8c7cb382cdae1a4ac99acfd670deeda97ca3fae74f9a500e6a5bf0f

fac6a93461ce3d3540b06639d89e286a3c81b65ee72d693b41686da8eacc17f5

9c0bcb108cd83414a310c2660a707958bfa4bad48b059e16fe00db6048fbfd78

response z

fb3a0b2bd36dd721ccf7722dd06ddf56f8d014155d19266eccb4a384bc324c52

6b9058de19677581e5e8ba715564171a222418d2c5314f13c701286aa8c8fe20

172aacf793a18cd9fad3a81485a282f04991b2657a8e72b185153a114b3a95ab

a0905675fa5860c605012a56606783c7c571ea40cb1b38e98b037489808c79c5

3f005eed6421c06ec07b7ed3d41b52657a1cc9263632ed84221c7afa53039806

c6e5f8c5409277590c296fe8554aea72708009ff3bf02c38037ad2cac09ab8bd

41870e6aee5ed7850c5adc9193f174444c79cc412002be79b505835bfdfd1240

61eea9ac0e947371e77cca5d7ec72e8991e49757da4f590c0823659e332bd29

response w

6376f88299c0401df67bf3a718f3bd8035bd7d5eb14bc95863da53366b7986ef

1a661567719dca07b1903588d8eca20ca2c2571c3689deb20c8771914d914100

eb655db620ced2722923f8c8bc2447490419f0f11116295262c0c33fc0d48c5b

606af9249912e55e098d2e549c6a2dcc19d2d8b24a084c1f6b5404b9131f41bd

7001cfca7d9e6a9e82e1c4d9175a838739201bcd9ab5eafa0655cb0a368eb27

66ae9b2c72afd0e0a29c66ebc30f7e85e4414e58cb681a577760aade44ad39be

cdaa923da61fc54bfd155074f7cfe95b021b30c27565cb8ca429a33c69848e2d

eb2910bb897b5b6899dfcea2b7a9af0b56d93ecb83bf99595b46321522469388

Fiat-Shamir context

ed431a8f6fbf8cd86160bba58071f883b80cb440cf11fbbc87c4e31680694551

withheld proof term: 207 174 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isFramedT_iff_framingSystem theorem, TFramingSeries:157

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (T P : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
  (Tt : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)),
  Iff (FBundlePoint.IsFramedT.{u_1} T P Tt)
    (FBundlePoint.FramingSystem.{u_1} (fun ℓ => FBundlePoint.tCoeffM.{u_1} ℓ T)
      (fun ℓ => FBundlePoint.tCoeffM.{u_1} ℓ P) fun ℓ => FBundlePoint.psCoeff.{u_1} ℓ Tt)
```

SHA-256 of the statement above

18fce3e0ae320ce7250aa0c3856a2a2bb00b455b589f195c9c3d3180cc662606

SHA-256 of the kernel type

ec12005863ee7a3ecf1c62f1b7d309c62d4e97b564bd79727cb874e18943334a
commitment C

b7182dd168fdd9cd73ecc8e0d3ff8a2899602594b432cb61aa0f9d28260505c1
3c2974f93465c1d4aff15d664b8c6c9a5205dcd78f950647786e14337a60a83
1b5d054e077a96cfd4ed93425e9e40528f92cff8f46c08ee05493c5cc4f756ae
dca187f674ae36769db8d008db2d405907b9c95147bbf1e42217946ac9cf590b
1e15a1ad7ff0f38ecfd916fe2af7bed4d0113057d03f54cae6c1770f62f4d584
16601a94a23982cd587e8ed4728f8a895188d7f157eefaa4339bb6db622805a4
224a56798b92641be512783b0010e2dcbc593a03b4e6a99a042e724c93208f5a
fac3f2e81a8cc02fe3c551ee2e30162fd6b970b2770964eedd8d40d0da2fa375
first message t

7b3a0f2091ec363f227b8cee47d53e6214aa3fa0a8f5a19efc47740d1496f7f6
a7ac557244b746621d31b5e09cd7e9126e997439da2e2ddf36e8e3b3d3c31a95
20de3a2f532e0af39f7f12305cabec5a349ae9b6e6630830e92c045a4daa7f44
ea4eff64f17bee554b82e66bde657f38127c12a8f38f2bb8e9e9806f516d86b0
db95458046e8b187077f54f9cfa55755655f4e5434b126af3bf8810615f29eaf
593ec34f5cf18398dce18f075c39b6c0fefb3352ae8575df9215872d376a105e
21dbd05eb418d44b00729a0db7d0dcba29b06c1efbe0df705d89b33322021495
6177d08e35e7048971d27b0e26fe9b248a7e9c4ad68cd6c273d04a96222870f6
response z

1f36895f3dc870250a2b67c7c38e1849519fbaa1bb23304288a7c0029b7585ed
f29280e818d72c807c5f1b75a1957562b40707fd684bedf5f6b2699dd40d7442
2a9a5bc1023dc86aac513599c3c5efb9cf4e1ea86ca777649af7a5b2054c5588
5dd099e31fa654d019121ff9de41fa34cd4d25be67fa7aa996c3fdd0d372a6ec
2f180763fc88078172622b383231ef7077e93f19b3becf32c6f72e9b7ae0584c
f0fd6bd64f45f6772f4138bd4be5537573bdcf4807b6463b6a46f5961abc0381
2586405a6e4aa80d63c3268f754f154817e21497b5365b67eb745de72908a9f0
68ba943f752b4388dae08fd734716c0e6b5f97620b0858ca472120ac3ea65106
response w

3b6bf90f79da61c69b11ffb5d44fa955c5a2876f91174c297a82b24c135bb288
eda62053d9c92214629d98597282950df0109cbfd4647eb27a17f545b80c87a9
aa7a99c49dc1b5f09fa96861d0f0ccd92536c155eec6999fb5c50c8b1b1a6319
4de66b7d3428df773584ef1572f1c2506f6e11126a7ddedc742790317336b918
cd81a77f5138685f0db98698126d9edd36528b01e1e023c35847cb7f17386183
823e890a466d27a950b1cc0523df106361b7f737b58d737c626f72c4f5dc6ff2
59aac58c31d743c6d9fea0dff1647c2c0a659579c256e87001efc1526be34613
8a8b4b1612bcd86677b2893bc17ce78aa93730a730a5b6b80a83b7ab309beec
Fiat-Shamir context

e39e764fac79f05e6dc3c48b81010a5655ff8891d8a6b00ea04561851955a339

withheld proof term: 402 050 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isUnit_det_of_tCoeffM_zero_eq_one theorem, TFramingSeries:142

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] \{m : Nat\}}$
 $\{\Phi : \text{Matrix.}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries.\{u_1\} } (\text{PowerSeries.\{u_1\} R}))\},$
 $\text{Eq.\{u_1 + 1\} (FBundlePoint.tCoeffM.\{u_1\} 0 \Phi) 1 \rightarrow \text{IsUnit.\{u_1\} (Matrix.det.\{u_1, 0\} \Phi)}$

SHA-256 of the statement above

a41826ffbabc1693561efac03bd5823ac9e65ec2a2e48879b46a2a357bd453d5

SHA-256 of the kernel type

1209f771ceba62650fb5a09b536d5668a5bffeac13c72454d396d3f8acabf49a
commitment C

289d0c7703344a7f7e8e62374c9a3ebc39a2381d5c1a41b81f94704e964ffae1
5b886557ff66325f894e6f24d3b241c975e8eedbd26d84c2d8287175fb643c53
d5720bcb16241e72fcd22496c7f96688657576306e2b3959a8481c61350f9bed
90bd4b672dfd9f6e068d2fe06fee8399c7e6f4f971f761604528425754ac7eb9
1db4ff3237bea42bb8747f768e826a3e7fcd0a4e3e3ef1d473d182310e6e5509
efff4527fcd178468054478d30a4909b9000a6758574bd584a2fc1e0c0ffa44
1403df4abdfc1c3949630a65b86c41525e4aff39ef79b7b2cf848f2bf42d9bb3
1b2acaf33624528956118b4927a8dc9a7371f84289073096965a9e6176193ae9
first message t

e3c59ea8c2fa3d5fbd95b3aa815c5a69d8da0809e38f83bf3e54875f952423cd
0153a5a8024aea3a871d68f80592bf044e738f5bda19c6f428a8e88e068a9de8
0b4a8a632e9e42bc86a44fe3869ba8dbce656df1d58e0f71a3b3f5b33164dedf
84a540734c62bfed837722f551c122abd0d89a698f6040cff7f7c10052c42e4
031b8c8e1ee95e34f525ff7b469e8afc186764f24ff7ae6f92fb740d26ea9245
3f8875e99fcd3826abf0b18fbc03a7507c783623358495cd427753df08d29c01
b758c37dff6a5e323730e4f42b4459fec4f608ba85ff7ffb2250f8d427a60425
e37e75da903c28be9dedfcbeb7248ed861ea3084b1485bf9e22e300c47b0b913
response z

41a2bd2bc05b068005f60f8e6f363aab4f5c85124dbbc30b45896e4fc66a86e0
a8232c814ce2b76926aa9f86e37aace4d8a158c176c99e3d3a8bf0cdc5223e9e
c0b97904fd26ab39d263710467c24b4d800065bd5a227ea52f11c79c2d1f8913
4b5eeee62bec533400d47bf09321b8c1b359efbbcc7758438c6e8b2075920bdf9
5d538237e1372acc9b4b0e35e1a739deed58ba99f0937554676a89b502d6054e
e4f1ca23879efc54b22af85d8b66f52ed615d2eb74585dad6d13d910293c0559
d396814f4d8397f6019f945e9feeffbb2c4a1533d13f11cb4bdbc22f96da531
8ffc1585c1c79612f4ead0e58726bd139759830927ee372a8fc6b1d786734538
response w

6ab6d15f31c2fa78018bf7e3f083e1882dfd162808e6ee71728c16a42b8049bc
1269b92f7a2b64c05f608db5a225433229c94d011a004dab9bf8598962c5f6ef
b3f03c3ca02765ae4d57617926a60e9ad83b9505158ff7039d3f209e585d34a4
698720a846c3e30b4a802ea83e7deb80becc4e29d4be8df23f75ad05e7d85728
8c994eac116da1bb3ffd06ceefa8e1b55560f639c4474ee6a57cb890dfd2aff3
869f41883c7ddb9b359e6501d99d1c461964d69aeb091168703e14f25e0cefd
1c9da74da1554625c6245b8d68a8165dc01d3ba5042c39cdfac69b07216b7f62
d992808c5a2c0d7a77d0ce31d4359946727fc34701e3640e47652b5e1b18d865
Fiat-Shamir context

663de09dbde54dc6152d16d53a9dd159b26e90e5fa01977806fa971ff70ebb85

withheld proof term: 17980 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.isUnit_det_tCoeffM_iff theorem, TFramingSeries:129

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat}
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Iff (IsUnit.{u_1} (Matrix.det.{u_1, 0} Φ))
    (IsUnit.{u_1} (Matrix.det.{u_1, 0} (FBundlePoint.tCoeffM.{u_1} 0 Φ)))
```

SHA-256 of the statement above

fe160d8ffba4f148895070785b114402e9aa2c8995249f3646a32f8a257e4f35

SHA-256 of the kernel type

df0e9ffdaf725023cc7a49ecebe8cea2abd7474df42c0ea27862ca4e2ed26a76

commitment C

424cd130efea63ecad743d8434be1851331413056a8c7f6d07a98afcb4bb8cdd
fb6f5258c3d79fd2c185234e3abcb7235c701cb391870a8b43688451c3138578
971d55fd3921692d8751f325322f966e88ca4ea220ffad97f4fd3cedcd3495cf
79f19ee12d3f0a5b3baf3f6a3eb31a3390cc16683c136af23220034a9bcbdb6b8
9f0754ebcf368317452797028ba039d85afad967d87e085e883bc200be9ba3ff
515ca01816281acf3d73e44cd77c724e422f0335a87e7d797565b1c03e0d347d
c9c2b2b03f976d6e41b263b5bd4b9da94c2694e8862645883dc4f2074cb0bb88
623f3a3f689738fcc317ffe44ac486e6bceeb2ad55394d09262fe3a687b3faec
first message t

39d8657904a2daa49d0a8e1d65ded2ba391e57337eb840f65f190b52a48d7372
9777788aa30b8eee336ebffa0bef249b9c5c8e7b183ab74921cd05f621158f12
d2783c93c921bdf916390dddb24918afd1b0c3b8eb93117a11b082269601d038
351120c2eeadb1dce044d4ddc18f74b4c0a5ad2849597996bc1d3cca0578f86c
19e621b27df3c7916c17f6984caa3dabd7af751c0d500605857148ece59a69
86dc704a0a47cde86582ce9dbca245911e6d235facc86e5fdf3bf89c6740ce36
ae15b32a507f3aa18501f6c8e3e8318f817dc8caf1773a81317ede50bc99412d
4a42f21b43581bc4b8770ae2190787e193cd1847b02957ff6c081b6f01f4a80e
response z

2213d41fd026137431626c75686bdfb75ee12fc3f76195b81028605166343cda
c5374513ac1120c882b1a0698aa6b8516041e62174dc8b04bd809753f3e1c531
cede05cb9a0e77013fdf0e9c96fa75c2db05d0f4bf5638e14414230ff1d46754
dfe030513220fa3ca6bff9c60ee64866dafa013cc34c6e50fab8c511c435074a
10fb9890ad67dc225c68ed55f56c21c01feeb34e018cb2ddd5dc8623bfd700cd
f3d1f551c9afc1e354cdce303b76b075b44be05365588412db25f5c0f8e284d0
38f24112ec38043d7b090f334e7da908f046d282a3c1e6ca23289d8b99d48e15
da7ae3d359dd649c443ec72f1acb122bc5b24fd3184f208d6070642e0f39f479
response w

1d4c80be53a3db872c69ed5da42a7f61cc952d85c24d9e6ac1ada3be649013d4
3ff1c9941bdaf1f14ac31172106f2710ed6e2680e96595948c4f2f5b04cc0b58
96482eb7701da46ce2eaca9c824d93a5a0934175b30a7868258e5c0bbae7b98e
a0cb868843c370b1f97312322c18cbdef942ce46aa1e53b79912a856ad5588e3
082e5f9380c8b944a60f8ece52479dd064c59bb0be599e6d87589f4ec0427064
31e7e1e4791f410adea4af344c89fd62b30ee8f9ce40e574adafa93b8aad83ff
ecf9480caccfbdc6de3cfc3ecbe841606bbf984a6a75e556e83beee456d9b2b
05e2e8d90bdd3098397b7b2fc4273ebfb9a26d2ce7d5044308d9f770904a1537

Fiat-Shamir context

4e5be3c13973764bc16ec443d756ef2d3466a677b9861e97431805ad27ce1173

withheld proof term: 127 526 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.ofTCoeffs definition, TFramingSeries:59

```
{R : Type u_1} →  
  {m : Nat} →  
    (Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)) →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))
```

SHA-256 of the statement above

932ce7e0a37797bb3f8e45963e2c50054ff7162ef375993683d1c7aae1eabbaf

SHA-256 of the kernel type

b9740651964ca74858e79f9a06f268d0d276049e6703371ebb2845b59e45c22b
commitment C

2a8d2a39329508a23b03f9537d91242026969a7ff01311f41e1441fc0f56fb96
12c4fb6cc0e222677315d01524b2934b1cc9d7d618eb7b9fdf15993e78c55ec2
6c78c9972942e981f4d7352a0dafc05f771ddad46c4d82a84a257b0f0004ed55
bb6cafb5daa4a7667c9b26848e7284118c62c0d090f31a2add4202d24c05b919
270b1eaa289ccc4f9f0b16004914e6ab6450e19fc7d7a71a915ea88aa56f3227
86c0093435b7f8ad5a05ac7809ec3cb653ab64b03109869b89019e74a98c4309
2b2ae9ac234517e8e890c28166e29a8eea38c16fdb467adc8ced7f897231e38
810ba2f7452ea8972450c7460b397365b0ae8d389e0fb3e6f23ec69a1004d65e
first message t

7f8bd34b5f5186e9994c06887f747f7c16e2c42965f9b7f66bf0e241adcf1587
69936f43f3fc4c1c4758305ef7ddaa602eb11275eb7e8f42032ead6699250d9
081597c38b8602a7b1667f74fb3477ceb9ad980b7a1094e0759871adf357348c
734c09f46b18c14a6bfb7d61ebb2325b0ef54de253c457a7c7846b18c8803e34
0b409d9d6ed35121c9c24c86ac67e6d1fe76b342a13ce8e7c3ee307d8454676b
069deb3f3584777006a1e346888eb4fa1a75ebe5c9de56ffc92457f1be8f3421
0876d0b7aa98277c08bde8eae36c9b07c8b1c194de42fe91889d4416a58cd9
3ef858a3ad4ea9d790b136b1f3ace7f215dbf07cbde25eeb60741da458fe70ca
response z

29ca02c9327f5fffb0e70994bbddae41f58a6a058f549a41032de2821937b39fa
16dc0ee15407f983fc1efb352f01cf9b47cfa9d8eb8c0c5fe709a96d174d6cda
d6dff4c8efb1d3b7015f7cafed4062be547e1a8b3f6c64b64c3dd5aebad39f64
a1bcba24bf58e84454d98778f8745e9ddada622f9a83cf1964e604bb60c0c367
788bcd8cd25a857f8cf4ffa4f054417cbceb65a52d92851448a55ca7dad5936b
1bdd9eea46171c91fc82a5f612cad7c2eb8f84475446ed496f25436e83b71f1e
142ebde83990d9d742ca7d858daaaa63717d23286914781ee5753d270a85a952
7d410dce453ed363359c3725d3ef255b61a795838756b490466aeee7a3feff8
response w

48d13e25a6e89270553789114dd2ddf39e23dbde6ebb9fe613c87cb267adeb99
f5391cef5ad74e4ebfcc72f8f186995fc10e8d2d132f642a567bfcbe01eb5a2d
e072da5b710d502365cb3578df350f19c2471ab39ccd8264727e4f1d68f714ae
b9cc14dd35359433207057cf7e989ff9da225fb8ffdec2779b0a5fb1fb9772d2
2c5d33159226b1632ebd4598d94d7c59c48e6f9e8ebf8d2fae6568ada90b396f
1b3d3901d9b87e0d3258f4ec12c90a832dc683721859a59ffde33b926c542b40
ad85f8d1c34a582ee0f1965f020f000bbf1bc0306e351c28a3738732a5f01cce
18cac1f9c96704c5bc66410d4e4d1038962807c5158a89ec33390270579808a5
Fiat-Shamir context

9d424b385e1a220369e4f5be4d7f2ffa5e9bc5bebd3a38c28babd886d1c9ba8b

withheld proof term: 1 447 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.ofTCoeffs.eq_1 theorem, TFramingSeries:0

$\forall \{R : \text{Type } u_1\} \{m : \text{Nat}\} (P : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
Eq. $\{u_1 + 1\}$ (FBundlePoint.ofTCoeffs. $\{u_1\}$ P)
(Matrix.of. $\{u_1, 0, 0\}$ fun i j => PowerSeries.mk. $\{u_1\}$ fun ℓ => P ℓ i j)

SHA-256 of the statement above

80b720e2b55dbf00ef4c111294053c370a0827f07cd3cd93224c72844b35b1f9

SHA-256 of the kernel type

da2d7a2b0d56c6dd1c054f73471a82271f3aa1fd81770f023d64966c6addbe4d
commitment C

745f37649a2d96bc1b39b5a4de0912e0ea836e602b5d7fa5841bbe80a835ca83
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first message t

6ddef2631695093b5acf53abe28ba997148dd29f1ba2f205d152657f2cc44c4f
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61c0cf2fb58fa3c015fcd1c60ca872ea7552a9e51184f955dca1a784470f5fa
09ce295ae1e41d660898b661478c1e332ab2f467335cfc874fe636e0a746be08
2c08fe9d05e70e70daa748dec0d78bf15eb73c97ff673e847e73610d22dff1b
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response z

a5cc7111abba21adb9a32812fcaef3cb0d4f7cd0db8d584dfc79ab3511bb5de0
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800ca78a846036be4fb872785452f0df29067086fb0dd08c80c495967033dcc1
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fd6d4a71c93bec76ea2e0579325c201a59dbc42c772df2e60f75089f0b3d22b5
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response w

49fd772b2bc60f8401964185566d105437c07c3fc7a2bc7931e858b9634f5e47
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Fiat-Shamir context

a86de66c66f12e15ae5c47e55de224c0714f231b0dc908af60b20b261123d866

withheld proof term: 240 bytes • axioms: Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM definition, TFramingSeries:50

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Nat →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →  
          Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)
```

SHA-256 of the statement above

937338866d69cdf7652ee0121613253529fb89ec8e4a538b4bfac1c326009300

SHA-256 of the kernel type

73c46e3c16a3b010554bb0a923041c832c24094811bb53e9834c16cbec70b641
commitment C

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first message t

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4ad17f2519ca4ce8d7b55be93a517850b924edb236b6ddc19cf14e76cbac5ac9
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a714d653b65e7685d829ff48c742c6963a8a1760695e0a9698e11169d41830a
response z

54cd5bc472ab7455ee1683ecc933eb6488b6da72e5780f7cf38be0752afac448
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response w

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Fiat-Shamir context

c787d3b7dcf7808e39c1f514a1237466f212ed50551c929557c3ddbb98921dce

withheld proof term: 6888 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM.eq_1 theorem, TFramingSeries:0

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (\ell : \text{Nat})$
 $(\Phi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} (\text{PowerSeries}\{u_1\} R)))$,
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.tCoeffM}\{u_1\} \ell \Phi)$
 $(\text{Matrix.of}\{u_1, 0, 0\} \text{fun } i \text{ } j \Rightarrow (\text{PowerSeries.coeff}\{u_1\} \ell) (\Phi \text{ } i \text{ } j))$

SHA-256 of the statement above

53b206d3e41294484a5d297ee88b1bd86d174fd17e37a8e715ff46a5a91aaacd

SHA-256 of the kernel type

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commitment C

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first message t

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d06b642a5a286b32ead9be5d3da40e55e80de75f3a131070839ddca4750120ff
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response z

38a9d2295453ad6fa774ea2875bf90f30622310098e0c45a436bbbc66a31f651
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b644731d86be8009f80eae982e99ffea1cf12ab31fe97039092c7ec7aab1417e
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response w

4d587af5e29d3a32f6ee180cd51905af27d8ca2f7e74fa654ab176039acac349
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1d10ee9173ed43b0c03cb46f339547a5c48fa0e8a6a98abda9231484287cc273
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f4c469a80c49f54f1b4adbc3cf16f0805455510f8e809e7a3cb3e42e5e3598a8
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Fiat-Shamir context

57b56b97ba0b0cc5feb632a5b33714d144a8aecfa1443126789964d421c4c24e

withheld proof term: 278 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_add theorem, TFramingSeries:78

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (ℓ : Nat)
  (Φ Ψ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} ℓ (HAdd.hAdd.{u_1, u_1, u_1} Φ Ψ))
    (HAdd.hAdd.{u_1, u_1, u_1} (FBundlePoint.tCoeffM.{u_1} ℓ Φ) (FBundlePoint.tCoeffM.{u_1}
      ℓ Ψ))
```

SHA-256 of the statement above

9f51cedc5697afd050b556f6df5cbf0c28fee277a0c1a1e0de388ddd4900ec18

SHA-256 of the kernel type

cfde39107222a7f45f84bbfb3ab6230ce9315b6bde28216f16a9a65f6b8f05ce
commitment C

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578af11783c172ef443e88d4ad7a2085481586c956f60edd033e91902ecbaeab
first message t

d0bb26bb7d956ace2dd7dd4527170fa8e3c05a11f1be12e155f415f0b25ee4d00
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73c490036c3b8b292fd951e7b67436c6f4c9993e9f4f11be648c03dde2308410
8d3609be4a8f067a2324775ba6e72b996433b2531c5b2ea730af9a1871bd8748
response z

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d30b6827c02259e2cfec7ffc5b9bc6529d5cf4eb60c096f6a30314dce0d88a82
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response w

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Fiat-Shamir context

884bab1e0667550bae4b0d34feda9ec4924b05a1a64014f44a92062cc88cbbe

withheld proof term: 384585 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_apply theorem, TFramingSeries:55

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (\ell : \text{Nat})$
 $(\Phi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} (\text{PowerSeries}\{u_1\} R))) (i j :$
 $\text{Fin } m),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.tCoeffM}\{u_1\} \ell \Phi i j) ((\text{PowerSeries.coeff}\{u_1\} \ell) (\Phi i j))$

SHA-256 of the statement above

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SHA-256 of the kernel type

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commitment C

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first message t

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71aed1933902b7e62f74f6912755f8658a0fb1c84ea0d26a88116ca55edb6efd
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response z

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response w

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f368587fec16e6884c1ef71723c34d47f2c3491bb68e6116ed2e45a73cf9dd88
375bb4cb3715c70a61f3a25ddd9111b082a14103be721a2b63bdacbb63503900
c302d423b0076786fa894118f8c3be77826465708fb3f5ca429c9bfe16d31ff2
51927ba5bf52e68d88f791bf77f132783d3c748ad65c1f73767cef3e079aa974
43ffa72283d63c49e48e3db5a1cdcc5f903783e02c641857c65e0c10daefa149
ea21a0494f39ee7c3c1383070c5e97f4971c94969e21914f2cbe36f063d2e1b9
c720478dd45bedff6b25e12a0449b0d181b5c458b04dccbb41ec52358f5c72cf0

Fiat-Shamir context

a8f9d03e4f9a3e61e8dac081c0e9cde6f93d5d458f01e77e375cf4a21eb4bc2f

withheld proof term: 383 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_cstT theorem, TFramingSeries:120

$\forall \{R : \text{Type } u_1\} \text{ [inst : CommRing.\{u_1\} R] \{m : Nat\} (\ell : Nat)$
 $(A : \text{Matrix.\{0, 0, u_1\} (Fin m) (PowerSeries.\{u_1\} R))$,
 $\text{Eq.\{u_1 + 1\} (FBundlePoint.tCoeffM.\{u_1\} \ell (FBundlePoint.cstT.\{u_1\} A))$
 $(\text{FBundlePoint.cst.\{u_1\} (FBundlePoint.psCoeff.\{u_1\} \ell A))$

SHA-256 of the statement above

5cb895daa1dfb733c9b080a2e608bba8b1866a24a1babf793cff3dc1df2c1fe6

SHA-256 of the kernel type

9abc8a761ef2d963b1cb7c3d4e05c271d9b2f023f2c186f08f2d689acfe895dc
commitment C

16a451b16d08247703b73b95d9589b7e8f3f06d5af34bd924dcb7a5f3a0a16e4
ad03ff079f73c8b56ce029e082fbd62bec1d341a4d9f55d8ca6a13c807981ac5
012e9ae8248a30139a7af82097d9b7c4504b2b68c014c0d2fb988f2ddc6fb204
a39bdfb58036d797f39ec04f75eaa0b53b10f269532fad99600d885a265a7e3
07548b2da243a6e69e2385a92fce47fe5d1c1adde7135d93a68ee78d86e67806
e08f06aefcd149f1696f87f3de5202a58018edff580c358d390c5c78ba95740f
c928e5c2e3d7c419bdd7d0715b3dd102c5e364669b347eb7a11e3b832785fc8e
f1bf02eef8452f7f03ca0e5649f564439aacfcde80ecc4f74ccf6ba92524f553
first message t

5667dde644f1d728387485a0c0adcaad86a2b36d071f4caafb4217678054ee41
8ba9a7d6df45b16eaa88ff727b9bb27df9da8da69d9132f389fe7908cdfead50
4f429296acbcde6027de13838dbe92b7d0861d1138d8952dfa0aa7512a5088a1
2a0ec61fcd31bb50223b968235a0402a2cf7f1f6f0fedae9a0609948504f76a
591076ecb23ddeae8865b4fdae831f6c9c7dc85f6887196ce86ffaa6457b1d63
a10fb5682b144ca02abd6fc3ef9c20098177f1f5bb859ae0992a9bcaa5792d9e
70cd108d647dbc40aa03cc5ef275a7a59b62dbd4282bc5e1a020fa3a134b1336
cd2fff02ffc943c661a3cb2988a9c92511813b03026fd264c157518372707b44
response z

5a4b262f6894f50a7f65b2e3da065dbbbd98b34f565bd23533f87b64a52f9f6e
81d562eb67007d6155597fb20d179109af67423626579b9a67b1fa43d26c4e40
83a3bce71da5b7d2c7062e5444fa3b6068818691a399fa3a462fe7958ec4b741
c52d71f9eda832d735da6aa608cecd6ff654f2b5dd591752aaa341fc89f89911
e58c1d6c83449c24fe589bb3039def9f1ca60ffdaaf125ec0d7a7e9d1bfb4fa5
40d327e6d433166c58eae501a63592f047d2c0f6a0fb63099201d970c3aafbc
299370f0881fc80f7393c306ba99ca15a4ce9d7366bffb640140c3e620fefbc
3c9e10ccd6a788b2d20ba635256f7a1e4687c2793e49bfc41a34ece47828f379
response w

3c72c8b125fbe49e7e317189deb26766498671b7888bdc92f1e4485c3d5acb
b3da54a392ddf8a67fd6def1184d5b4bd888f3c1f225f8f93accf661b1ac70ec
d123c4c165d2db8a77e4cf98478e0cf49469513e45d897e1075c8e744338c63d
fcb911b4d7f3e0650c68cd1d159290e446a4eb7da9312fc0ad48a67c887e5eb2
8e979ae842d8d6a55f36b2d49f72ec929470f3629425bb589426fd3e9e2d7ed6
a1bac5933c212437cd9b6ee70c07f2d1ef9ce8c96d18419358647eec60200bb3
28f5b7e4d03962e62030e3f7c9f6a626f4b19594d56d7d68cfaad5730f3ca12b
e76aace9c6aca4eab5ab79f24dd8dba89fae68e1d676a63d9e33b8f35bb02518

Fiat-Shamir context

8f2ef047f144227ae379f8bfd36ef20943cfbc4280ad52af8cd6fee0c8264a89

withheld proof term: 29 712 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_derivT theorem, TFramingSeries:98

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (ℓ : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} ℓ (FBundlePoint.derivT.{u_1} Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑ℓ) 1)
      (FBundlePoint.tCoeffM.{u_1} (HAdd.hAdd.{0, 0, 0} ℓ 1) Φ))
```

SHA-256 of the statement above

8cfe07cee3293da91d3a9aeae09cde9c9345ea32f3ad422cf324c4b676ed8e19

SHA-256 of the kernel type

470456a83b9e67073909df55c0f14f628f2d151dde0774732a165ce13ea42b73

commitment C

aff7a4a82934cc15263002a7c51053bffa56071a06325e66bdd496835aa98327
de4ad987a2f45bf0a764aa87c186752552130231b2517672131a4da145b3c94e
9f7ba0602fa10f545d74817ca99cfae866be882080634fd7573d79b3ebd8dc90
3f3cc14650a9f7cc9b10aa272cbcf4129cbd476cc5fd77ced7b230123a9a33f6
877a08ca829e19fb8c2282f641282259a1f7f66859c947f75850442aab44e40
38be2a94f984d6419d0378ebc6ab01c5aec5c60e82fb97f87550bf4a3cf92a27
86fbde202fed0add37d962373cb3029ff92f4157d788786000759767a5ec72f
c2d8a6d816bb3b0f971590101c7688b7f2cd5368d4e7c0d9ffb0506533d72eda

first message t

472cf08d48f4a2c380289e4133477ff620ca5758ce9ef3fa57a494004b82d159
f17870658361e90a27a4d88840c52e18f67f69c081d90c28fa5f6ca2d5bc732f
2ee3ad2d6763a77f3b20574a6ed0fb9eed3e2b70be95318ac51d956cf0fae9e1
8a387b2c861acccc347d6110eea64c1b6d5c3a2e54a1584444a8b4cd44572e8c
78d83db6eed4a2a66bf6fa5117ec0a87b697841227295c1f6a085e3be4f92d8d
c838b918896fcd66ee6b11f84fe5452b5b1c61a214e42407db6270eed63d221
c44aaa67fe48fa538a7df76a59a722bd4d67c50732a1116331e466b839ba361c
32efff84d2aa16312863760cb687aec6b31fb6710c1abcb9138d19fac4f83182

response z

286413b4dc12b817cbe2d3bfe280b2bf745156d41f72f451df2ad2bff9c2edb0
50b499b3b3c612ba75f35acfbcbdefdeda5981369d1dcb3c5651e4d079056a61
16ffdb8fb7028c2bd063423746769fd81affe700e34fb84e70d79bd34d3ce1bf
d67c9fb33beb87a18e71e92310e4c85c1be53b154c3b89fa0da3bad73ea633f5
c27e328f469411f5b2e1c24dd44cb24c9e6f38906cc8eeb707623c4712af025c
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2f08c30fa132c1d49f40393d2eeb44651e7535330fd283dc73f21e25552eb01e
d27f14d58c9d5a2e3826b0db10b11b48f749e19874f236e1af48c812f9097b28

response w

32bbf493e2bbcf5f83b7e738877a05eed25c0bf6362eaa01729dead44a234866
f754375e7668324fb69211f85d78300f3d74846caf1c84f68ab3e14de0456626
5c220762af4f7051d77ae2d84a0c4f8e394cff49b78fa1da72c37f3e3f96066d
1eccc0f7f39b5bece1c80b36746a44161e1c40eadbec2ba0ae494bc566d062c0
b224f89bd93a4cc2fbdd50fb6633ed05e59da650b67eadac1933beaee951b3a
454f2cd02bbaeb3b2044c7cd059925104fef8fae939fd81914c40512be818bcf
0dce874ea71dc75f87cd448ce3df07e4be31608899709e8653e0ba5d41887de6
c74913c3a85f4dda94591ee9a67114b5a62492b8086a0c727f6a16d1cb5b8dd2

Fiat-Shamir context

d02e12eaca55523e6c1075d7be051cac59beeb679ee788bb884b8966440b664a

withheld proof term: 280 026 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_ext theorem, TFramingSeries:70

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}.\{u_1\} R] \{m : \text{Nat}\}$
 $\{\Phi \Psi : \text{Matrix}.\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}.\{u_1\} (\text{PowerSeries}.\{u_1\} R))\},$
 $(\forall (\ell : \text{Nat}), \text{Eq}.\{u_1 + 1\} (\text{FBundlePoint.tCoeffM}.\{u_1\} \ell \Phi) (\text{FBundlePoint.tCoeffM}.\{u_1\} \ell$
 $\Psi)) \rightarrow$
 $\text{Eq}.\{u_1 + 1\} \Phi \Psi$

SHA-256 of the statement above

ff787f775938c35b1afc7566fc1da4d4baf30fc7e536ce23370eeebf9a00279c

SHA-256 of the kernel type

d0be9952e27bda0f39c3cc45b8b5db7160f0e245ce72eb90709b551be66146a6

commitment C

733ac5dc8b3721d087aaf418d9fdb7b50f41f1caab4fed055f1ff6a474e8ab86

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d6996b7ee089b8643092bef5469a11adc1ae389494b6f26a8908dccbfa9e8d

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first message t

8b77f6f93cc867549d52be085847d15960b60b217831e35510cdac27503263c4

77f8e91986826f0379b4733f2c3bee47ac1d510fc75de100bbd6eb7af5c3817e

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787a0de8037f790b804322faeb49fdddfa48a59d5286133504176be93301e45b

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f3ff07491340073f53af53823c3fb55fc3c4f0514c00988c625357072cb43150

32e30342e06e4bffd446432cda2402861abca5d860cd0d24bd9df399fe6a7c79

response z

698e06257ec836b9db1e8ee990674b89c7b24658dcdf05d4ae15382230089244

db78727fe88608129069bee8c23690eefa3f63d319422df24db6b1e0ab32bd8d

78ff7e9e071e4b70f504cf9654dadeafae5af3620e8c6989a90d6a8d20c998d5

bcbf3250f03fa62147ed6d231c39e3eff172b0eb6a6e13e2d6ecc4d612c5d952f

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52a058b0c580cf393709150f339759e9a6e00bb56ad7f8d168c4022f9e10f431

c57d763427ee0af2cad79da46882a42406e9aa22752e7f99a64d5b71fc97a73a

response w

552a96d2201914c66146a7ea46207a8447712255d492c2441153df421ca839d7

f15e164bcd160bea66ef9b15814e60c63b967787a6442ed2f0fb84ded71ba144

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b9127d19a6146d0d44866c80308bd499e7f252334c6da767161dd879c6e7da7c

d073f629c10c4729b788738c2de3b4365ca1c51112cec973bf8eba8a0c08ba11

9d4be7c4ac011be6dd6590734381299817ad614ee7c8fa4e2dba6c2aa9cd9117

Fiat-Shamir context

e56b7de09d6bcfe6c68b0303a600a3d8f45461e96710f16e66715f0db84b9a02

withheld proof term: 127 163 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_mul theorem, TFramingSeries:83

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (\ell : \text{Nat})$
 $(\Phi \Psi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} (\text{PowerSeries}\{u_1\} R)))$,
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.tCoeffM}\{u_1\} \ell (\text{HMul.hMul}\{u_1, u_1, u_1\} \Phi \Psi))$
 $(\sum q \in \text{Finset.HasAntidiagonal.antidiagonal}\{0\} \ell,$
 $\quad \text{HMul.hMul}\{u_1, u_1, u_1\} (\text{FBundlePoint.tCoeffM}\{u_1\} q.1 \Phi)$
 $\quad (\text{FBundlePoint.tCoeffM}\{u_1\} q.2 \Psi))$

SHA-256 of the statement above

d7a52a514a93f6756db09a55bce5f9c270d9c3d6b1af152d7b7424c2092c3369

SHA-256 of the kernel type

d5750b85131e5b6da169f0ec937237895c3f35701ac9a35afbabe957b5f9433b
commitment C

26b5be56434cff4b2d4044c0798016a154c4029041919a2fd880ff223da9d7e3
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a184ff743d0a6126188316dc51778cabe3190ddd3d092ab647736431ef0f355e
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75188de57e7af71ce52e243445644f8dcb0e23adde9855b3e31bc0f26c7ecb57
first message t

d6d13ae176fe5ade60de9f226bc395b893a00b951b51c0aeafb5d8eb9669ec92
ce4ca809991faefbc5f264ee90e08dea6b3573117deff0e371aca5db3cca32c3
7d9f44926d91e6c51045cf8c5a4a6271ff7351fe386d87b532ee4448a5c82aa2
c4ab1efe0a855defd8327059024d3eaea1f57020960285e2440628d7fff7cbca
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1795fe32078bd72d3d4fcec6c5ecc471f23eef931bfd492b0846bc02fef6182b
250306f847651c32caf500c23b4f97847b4638d976ccabdb8d4f788ac8d126bf
e413918a54e0e122b2cd380d0991409cc1dcef8d1764ea33453b31d607c948e1
response z

1737940bae23cdcc53c5f3f1bc394fd7a9b62107a38ee21059138c0db46ac568
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f0c47de3cc695a172be9718133de628c7dcc9ee0d96cf6799c512b39a8fc19b3
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bbe9f4db9aeef7bc32a859226123440caa35071539dd5777e4d25ac68b62f9f
70726f7f140e1516b0931904170fcd89a60a2d42ce4b6bc0657a0fa5c63c3b
response w

5468998b67d97c6f8a139766a0285cf9f17d930fdfbabeb25464caa29d266016b
0df7e77542904afe8fe19fedbd3d7f6f3d18ad1d8c5dbb77d5f5f389cd816adc
1756986b8d816a2d623e212a38b2151114667faacf1bfb6d1f8692f93c0dfcd4
93b928923e11000c15168583afc641a5b26b226abc618bb05eef05c8483f144c
8b5bf87a1d9ecbb4bd1480e777fcded0300b47770d4d9179a1a32f8a49148bff
da154310b9317a85930fc518a1ba99fd6f6187545bfb339dc12a70679beeb5dbd
e19a02433ac4a448b56fba40b79ae57da1a40c4aa84c0e247b8243a1317f2870
cc58a75223164fb5105247d4a759564a2f9e689020ba4d1055e04442ea70aeab
Fiat-Shamir context

637e629f821b021f7fe500a268a1d541d3cfa7347504c120c09bf1acf8ac733a

withheld proof term: 1 220 833 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_ofTCoeffs theorem, TFramingSeries:64

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\} (\ell : \text{Nat})$
 $(P : \text{Nat} \rightarrow \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} R)),$
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.tCoeffM}\{u_1\} \ell (\text{FBundlePoint.ofTCoeffs}\{u_1\} P)) (P \ell)$

SHA-256 of the statement above

b5ee7a5d4ceb52f45b233e6643cbe032c6461f5f5436ba1a8ebb97227629aaec

SHA-256 of the kernel type

ff42de9bf47aeb1f035224f93fb469b75f14c10ef2bc362ea547c89b05234995
commitment C

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6b7b71d68c1c2c8f3c04253d631b004ca440d47043d45cfdddcce3876a558e4
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first message t

91c5f0e6885c406a4e05d3e1a088a6723b9ad75bcf6db747b153cad69878de5
b752fd5ca0feac0bf4df975380f057ba2455ccc11df20e5aa369f480f6a6b190
72cc2d36dec8175273b60a39e6af3c2d8dc2e6a1140edcfb2f813c4e985f00fb
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7e98c71425fb7d6e0d7f57b5b9c6458dcb8fa84a2ed61b2feb61373c325e9e67
3649694ee97f2789d024180412881bfb19dc7e65f37f45186943b08b0617eeb4
f4435de7c9f6b9fee52afbb0d53a386f0a8505919212ca9493cd04f64b883d5d
91f35c3c90eb0c026018048a61566bb818f42f2e992791a84df2c35d6850aec7
response z

bedbb1130964fcd20f012860f782fb8ac59576129fe6e4042823f362f36c501f
7fd3b4bc283f9fdb9692e8b0058d41e643589080d447927a3a7f836502989e5f
3ffcff61e47813a7b4a51b07709aef99924e7a647f46ad2bdb44147502d6de16
99ce715660f11e2186f8714e290bf13aefdf434010ddb41acc3dff6fd64dfde6
782f6a550d196e4cd08e438b70b662a7ad8d25a6824b515c5c83ab71120d70e6
c5ca091eb1d14f72d83f3a01785405434e96ac8daec392001b6a9eb3fb7e4b03
fd7bdfa6111bd57788fe9a7cc2ddcdefca39973ccdb2ff68b4f91c1d158395f6
09772a284f8712877e4e40f093b82a33a18ad843b2009b13037b35d2283ccec
response w

6356572fdf3996bd9329c32125de08e561fdc8fd44271b91d98fc1ca9fc8c81f
34039eb777a6e2424307cb06dffb8aebd448b8137a10804cdf77a6ddd1d933f0
5e8c738a252a4c28423f50c83880e458a89bd207a76c4784ebd36babba697fe0
1b052af6b1cbc99a3f5c66f701287cb84ed7ed3210116ab5bbded8f57007993b
e7089e16a43530da5bc884cbf35ab7b77c226d91a5a6a5882b37dc195dd79450
8fd9d4092e92d2ccb16ffb64ae6ac838379153bcd2611433082e574bf95add77
ccaf15f3d75776945f684459f9960d669dc4338eef42bbc16b3f68ca4be9c942
bfb518929caf536f31b361b01f231e78a2328c79add96e7162d04107cefe4a7a

Fiat-Shamir context

25c34385c11d525b2de101f67b87c115f6a80d6a4c2762604f742de1e76cf11e

withheld proof term: 89 574 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.tCoeffM_uMul theorem, TFramingSeries:109

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} (ℓ : Nat)
  (Φ : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} ℓ (FBundlePoint.uMul.{u_1} Φ))
    (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1} (FBundlePoint.tCoeffM.{u_1} ℓ Φ))
```

SHA-256 of the statement above

6d077fb8a8ece9216140969fb05a0d8041fdd8279e78ff87a4d4b9132e3b1a61

SHA-256 of the kernel type

dbd21e8fd71c3eb39c432ba48cfeaba30fee31030f666feda0813adbb03b2809

commitment C

b42f00f45c5bf1a03ec1ec1c99382b2a5efab3a4f28510054397900ffd7a06f7c
cd493c04d511597771e5668cb346754bbc206f561f93af174be17f83e5cb7fa8
fbb1f41bcb7d7daef23f8a9bcd6f3dcbe8d0c88659944715ab4e27e404c73ff2
ba1bf88ce2e549d5f8602bc6da4e56764928d57f137661d702d20dc4101573c0
3664f00438dd83c0915c8dcd6fd8f279df16196e4f3ad5b831c1e1b1f4c52b0
c0cccfb8500472746e9f3b689778b738f73db63f2fa536ee00e4bf8429ccd2e9
2f8851febc924a0fd04ed8768033294b2e04eaad4747afad4665b08c0798fac0
3b3e5ecfb05950a61552cdd0c8312b8f4ac43c4f51da38d51dbee1134b6830e
first message t

af506916b48f3e8d615683544f5f7e4e74dd236caa33d18a76115b53f5bc0f41
06ac5b08ff560afda0861312476a4d99f4fdaa2dd25dd6aa8c8cfb27f8240bcc
72a29cfbebc928307397023041df863f10baebddd7af6ed644f0ab0d5a845008
be05c65059ec79b5d90fd8eff3ea2f060d6aba4875fa5537b1da4e3dcf3ef13
2a2b6c35a03245eae99f3123d08355d2e110170d65da5fc8ad7c9c8904c57fdc
e320005a721a77ebd47e5311e67a39f86bbdd0794544910302b3c480d473df0a
684ce7a45eea2dd1095928ac8dcb227934477ef532bb081399f08a027aa58eb1
7d84c42cbeab45d8108a751d71f577dccc1a6885420659c1d41c90e66d525e1
response z

5cb88f9d1fff5d9e64b1e15c10716723729906a7d462ffb6a13a4b402f7a5966
45c595d51b2822abcf77f84ec338436bd30a39c38b4d2901acfb26f1defecde
c7858c521cddc811a32139f3515d491b17b1df9cd13a12c6519905f38dddbb51
b4bed952d00299c8a199a99ae7d7ed252b214a85eab19a03d42b99dfe9859c37
7e4d89ec2a476c131b9c995386888a59829d3e38dd6c29a0e9e7e6b19264376d
3d3ec012a8df29d2c3f2faec70b9ad3e678521f93baf714944d8b1fed4faa9a
923e7f394f9e2fb73d62f8e03105dd6555777a92d51c659cff90923d5d2907ed
9a3d4418f89d5413e555e4631e6032762220c88e4555115b7350c041d78ea4f6
response w

2aa972023abbe8cad4bbf6ee73e8c937af9a12d3364285805899acf5e83e4f7b
f4d833f166941bba90240f371d784ae35b72affca09265f5e41c1b56b84bead2
219ed803755a6ba063b1aeb929843a734d26d9c1514f02bc7c995ad3018fc909
fb54cb864b4f15f052991f692469104b22533015efd784d5ae33b91c5d03d3f9
b3efb7f24a41bc5bd89591eef71696ff0db7761e68f8f39fed2e3f138fb6ffc0
a51e3864789a365958ddba2a24656ad60a0bdacac6ca2fe8cb767aaa5d2adcc7
fd261627bd54a63d35c016d4e5e331f9e646625f8e2130adda6b3ba090bd78fe
4699e43d1d3180d8be8ea6e50aaf16300e686d95a3a42ff9c780a830ba5091e7

Fiat-Shamir context

628d179dc4999a35888340ad4b84a300b037ce589f6099238b44a92b60993d27

withheld proof term: 184184 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uMul definition, TFramingSeries:104

```
{R : Type u_1} →  
  [CommRing.{u_1} R] →  
    {m : Nat} →  
      Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)) →  
        Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))
```

SHA-256 of the statement above

63b22e1e9b85ccc9c6721aa4785df23ef7ce10638842e2c0db8a0146368c52b1

SHA-256 of the kernel type

a32882ae0cb3966a17b285cca46fc419ce81dba06c922e20050ece864fb14725

commitment C

4b00e90c21ca0cfce214192689b5c0ca4305154b1b8c9b2283dad63e0e7e0358

974c419f490f153cf7e902728eace67313b7e115a26a8a228a471dabfbc3719d

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3cd039ca1b99bc9b2d1ad5d82a10c8adf38f2c707fc95d791d6093f984cb08a8

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33640391edf16bb0cc4cf96be78411f29d0d851832a8f29bb800b59b02d6dc45

first message t

7bf4613a485d7edf8dd0bdd46762658c0172b458cac9bb2ebc833bf3e406ce5d

65fa53ae41b6923f36e62db24155a1b690fb6d6fe793c036ca6503208e18b9f9

34984b6e21a44ae526c2938705c04c42940b48e292db6ccfac2b3f8b3d49ca9c

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48c5ecde10997e118b101f97d7dc83b5e91eea918c54f8a4ef934f46f1cd296f

4692b83394b46f7452145a934c5d07afdacd5eb56e3bd686087fa742bab867f6

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response z

645d480b3cf28a3a59f761bfa46302826208754a9d2efd31a962a2df2014f600

85a740652a2a09e56e8ed2a3ffd238dd85bea2bbf22c653e92099ec28f6cb26f

e493c49f8bfc23b8f44a33cf881f104f845dc8e6ec8656612b1b5414bb62ba32

c526c231958d8ee1a34e65286523e3abfcd02d5cb7ab0a1fc621d19ab24cd97

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951cd8b2d50d302970d7cad78c44ae111cea945799c963c04f639f03b543e93e

580ed3403bfc1510c85e00eeca4a91f4b071f2eadb558063cebd01320fd16f46

9d62da928c2642b1609a0bf915b7687117d59f83bb0fcc942fa0d0f4a3ef6d24

response w

45ad3472e7baf4209357be94e9c8eb43658903e3a0925d54c1048174729ccf35

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66030d3624e97323a3aba61ed3363c0e99faf25956df9c7f2a836883a6060b5f

e06e0cb0ad77bbb30c0569b735047d6e8c6157f92090537222414a15374938b

Fiat-Shamir context

9a3b5112d2605a3fa83b79e6ac4aa544f6777a82b770333f63f83da2dd9da549

withheld proof term: 3342 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes

FBundlePoint.uMul.eq_1 theorem, TFramingSeries:0

$\forall \{R : \text{Type } u_1\} [\text{inst} : \text{CommRing}\{u_1\} R] \{m : \text{Nat}\}$
 $(\Phi : \text{Matrix}\{0, 0, u_1\} (\text{Fin } m) (\text{Fin } m) (\text{PowerSeries}\{u_1\} (\text{PowerSeries}\{u_1\} R)))$,
 $\text{Eq}\{u_1 + 1\} (\text{FBundlePoint.uMul}\{u_1\} \Phi)$
 $(\text{HSMul.hSMul}\{u_1, u_1, u_1\} (\text{PowerSeries.C}\{u_1\} \text{PowerSeries.X}\{u_1\}) \Phi)$

SHA-256 of the statement above

7aac91d46e26581225d377f5f0b6f6509fd9d601e47eb3cd68d7349f455462b3

SHA-256 of the kernel type

b26efb7f11e831748c9a54cc48da9ec01c9ae7195bf281ab88e9e11cf8fcae45
commitment C

11561fa42f1b4ea6c0dbd0b8a47fb48eb2a7cde68f93f7ee3f1ff24ada03fe0d
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91c7889f9c00958edf8d6b09384631216f1e823ce9f2efbb54e0372ee6636b6a
e3881afc2cee4cce6f1055bb75dedc7c6f74aa6720486f1a21b2b2602b1aa633
c60a78b23d1986607b84940783fe28fca151065b874823c2cad87c653766b329
dd5eba1595e9323b470aa35c5a705e6c6cb80c2a8e76e1f203fecb9f3e217d89
ff5a61498a52054d6729ea30f1cdf525ae6aad20cd8ab34e7a96b90b45249796
a763e0650feb729b654063c0227ff2778dbb477dbc12f6a08b3ec141091a562
first message t

c9510a573409e8d3a76a59099a9a94d5df7efab1bcacfc7fc360ada4e91f1fce
6647a0b81ed5ba58c546eac6d8043c17c8769a8aa42096785ba630d00cc5251c
738a64d477432af5e982298e04454715cad7fffd7d4fe2d98f8a82e8d14d2d61
2ee9074188e4a672fe80b9490cad74a48241f64aa3b02497503a1e20dbccfcfc
0fd46ad41aa87d6e0cb6187ff663f8c87a55f6546e7a5a5361dc4d1b0b9222f1
f956d25154713f3979a34d5ae056ea04319db1f2afd5cd844b4403813d0e9609
a1d44db6a055c67092b83a1201f402d2a0f20e969b13f9cb59a47245b5d5f249
ca7deb5e4490f40f1ff2535d463c077cdd5293871b93050d33ff5d7ff42c06eb
response z

1e9f0565d511e9e8c981c33dd7bcc604143067d900ed3cbd9212e010b74d99ee
ba3723fc82347ccbeea1317c8ee146f9aceb518495e0e12babf54c849763c71a
18e8d920da8ca46fb79f01365f9c05a90f484de6753d5e6903bdb0913ff9b23
6afb63b8d16120a4261f5ec5f5d1895f6677bbace776f8139add0836c09807b
44f4293dde39295df3719b7cc9f39be6da228223b779c7d3e10412d097aaf60d
9eaa85ae85292fd9d704bca718a30bc7466b0efd8b8ab27a46ef594eeef1a3f
96e2f56bf744ab7ad171a99902ccab5771baa1dfb4c9e3a0f0714562df65aa95
9af48fdc053025c75ea3b833ec5ded648ad86bf4ed70013f9ce87ff97ae4054b
response w

59972361b7814b5fc5a2cf451274d924744258c6ea7209108dd16ed32eb4e7c1
2dfc39a7bc2e2e5e3f79f661dc6a0272db551b06cdfcc4f75bfa9763cbb6163f
be21ebfd4b785417352563f5b625d400e3dd9a474cabf8f02e1bcc646b3a7641
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Fiat-Shamir context

6a13321903a1bef10026b300b07b0cfb452a5aaffdea713e91a6d93d8ce82858

withheld proof term: 279 bytes • axioms: Classical.choice, Quot.sound, propext • kernel replay: yes